The Claimed Complex Structure on S^6 Validate, Repair, or Disprove

Working mathematical notebook following the three-stream proof audit

6 September 2026

Abstract

6 September: explicit spectral and umbral maps, and the original family's parameter identifications. For the literal common lattice $\mathcal{I} = \{\sqrt{2}(a,a,a) : a \in D_4\}$, write $a = a_0 + v$, $t^2 = |v|^2$, $Q = a_0^2 + t^2$ and $P = a_0(a_0^2 - 3t^2)$. Its complete cubic image is

$$\tau(\mathcal{I}) = 2\sqrt{2}\big(18\mathbb{Z} \cup (6\mathbb{Z}+2) \cup (6\mathbb{Z}-2)\big).$$

Three explicit integral affine lines give a section of this value map. The scalar-retaining Albert matrix $X = \mu I + Y(a)$, with cyclic off-diagonal entries $\sqrt{2}a$, has invariants and labelled spectrum

$$(\operatorname{tr} X, \operatorname{tr} X^2, \det X) = (3\mu, 3\mu^2 + 12Q, \mu^3 - 6\mu Q + 4\sqrt{2}P),$$
$$\operatorname{Spec} Y = (2\sqrt{2}a_0, -\sqrt{2}a_0 - \sqrt{6}t, -\sqrt{2}a_0 + \sqrt{6}t).$$

For $t > 0$, the explicit labelled eigenprojector satisfies $(\Pi_1)_{12} = -1/6 - \sqrt{3}(v/t)/6$; thus the frame and spectrum recover a, not only its norm and cubic. The full discriminant is $864t^2(3a_0^2 - t^2)^2$. The full Cartan family retains the scalar μ, radius r, and angle θ, with $r^2 = 12Q$ and $r^3\cos 3\theta = 24\sqrt{3}P$ on this subfamily.

For the same marked root system $A_5^4D_4$, the Cheng–Duncan–Harvey prescription gives $S^X = (5I_{12} + J)S_6$, where $J_{rs} = \mathbf{1}_{s=5r\ (12)}$ and $S_{6,r} = \sum_{k\equiv r\ (12)} kq^{k^2/24}$. The coefficient map to the retained Fricke function is

$$T_6 = \frac{\eta(\tau)^5\eta(3\tau)}{\eta(2\tau)\eta(6\tau)^5}, \qquad f^X = -q\frac{d}{dq}\log T_6, \qquad T_6\left(-\frac{1}{6\tau}\right) = \frac{72}{T_6(\tau)}.$$

All twelve theta indices, the multiplier determinant, the original glue-group action and its twisted shadows are computed in the proofs. This is an explicit use and comparison of established umbral machinery, not a claim to a new moonshine module.

In the original analytic family $\beta_c = \beta^{\mathrm{part}} + c$, $\operatorname{Im} c < -M$, the actual global identifications are exactly $X_c \simeq X_{c'} \iff c' - c \in 2\mathbb{Z}$. The middle lattices agree for every integer shift; the multiplicity-four filling retains an order-two obstruction to odd shifts. Even shifts have explicit finite-filling and toric cusp maps and a proved global additive gluing correction. The resulting parameter quotient is $w = e^{-\pi i c}$, $0 < |w| < e^{-\pi M}$. This classifies the retained analytic family without using the claimed identification of its total space with S^6.

6 September: cubic arithmetic, integral lattice maps, and the canonical ring. For the retained $A_5^4 \perp D_4$ Niemeier lattice N and its original index-six Leech neighbour Λ, fix the Cayley–Dickson frame $(1, i, j, k, \ell, i\ell, j\ell, k\ell)$ used in the complete cubic proof. Put $\tau(q) = \operatorname{Re}((q_1q_2)q_3)$ and $F = \tau \circ \mathcal{L}$ on the original Niemeier span, and $\mathcal{V}_F(L) = \operatorname{span}_{\mathbb{Z}}\{F(y) : y \in L\}$. The exact additive groups are

$$\mathcal{V}_F(N) = \tfrac{1}{27}\mathbb{Z} \oplus \tfrac{\sqrt{2}}{54}\mathbb{Z} \oplus \tfrac{\sqrt{3}}{18}\mathbb{Z},$$
$$\mathcal{V}_F(\Lambda) = \tfrac{1}{27}\mathbb{Z} \oplus \tfrac{\sqrt{2}}{216}\mathbb{Z} \oplus \tfrac{\sqrt{3}}{54}\mathbb{Z}, \qquad \mathcal{V}_F(\Lambda)/\mathcal{V}_F(N) \cong \mathbb{Z}/12.$$

The Albert determinant is $\lambda^3 - \lambda(y,y) + 2F(y)$. The quotient concerns the displayed value-generated groups in this fixed frame, not an inclusion $N \subset \Lambda$. The actual residue $r = \rho(361)$ cycles three

Leech neighbours $\Lambda_j = r^j\Lambda$. Their pairwise intersection is one integral lattice J, with $\det J = 81$ and $[\Lambda_j : J] = 9$. The fixed fourth isotropic plane in $J^\vee/J$ gives M with exactly 72 roots of type A_2^{12}, $M \cap \Lambda_j = J$, and $[M : J] = 9$.

$$\mathcal{V}_F(J) = \mathcal{V}_F(M) = \tfrac{1}{27}\mathbb{Z} \oplus \tfrac{\sqrt{2}}{216}\mathbb{Z} \oplus \tfrac{\sqrt{3}}{18}\mathbb{Z}.$$

The successive value-group inclusions have matrices $\mathrm{diag}(1,4,1)$ and $\mathrm{diag}(1,1,3)$, giving quotients $\mathbb{Z}/4$ and $\mathbb{Z}/3$. Both containments and their reverse generator certificates are calculated from the full cubic polarization, not inferred from lattice indices.

Wilson's construction [81], transported by the explicit isometry $\iota(x) = x/\sqrt{2}$, gives Λ_{cyc} in the original triality metric. Its cyclic fixed Gram matrix is $3G_{E_8}$; its orthogonal complement is explicitly isometric to $A_2 \otimes E_8$, and their sum has index 3^8, with quotient $\mathbb{F}_3^8$. The directions remain $c(q_1,q_2,q_3) = (q_3,q_1,q_2)$ and $c_{\mathrm{W}} = c^{-1}$. Its literal intersections are

$$\Lambda_{\mathrm{cyc}} \cap \mathcal{L}(N) = \Lambda_{\mathrm{cyc}} \cap \mathcal{L}(\Lambda) = \Delta := \{\sqrt{2}(a,a,a) : a \in D_{\mathbb{H}}\},$$

where $D_{\mathbb{H}}$ is the even-coordinate-sum integer lattice in the first four stated Cayley coordinates. Thus Δ has rank 4, Gram matrix $6G_{D_4}$, determinant 5184, minimum squared norm 12, and value-generated subgroup $4\sqrt{2}\mathbb{Z}$ for $\tau|_\Delta$, or $8\sqrt{2}\mathbb{Z}$ for $2\tau|_\Delta$. This is not the actual image: $12\sqrt{2}$ is not attained by τ on Δ. The preceding proofs retain the triality products and spectral boundary graph, the flag's degree-$+1$ collapse, and the retained cusp class's nontrivial three-plane reduction; these exact maps do not supply an interacting quantum mass-gap result.

For the original analytic fibration $f : X \to \mathbb{P}^1$, the full canonical-divisor proof gives $\omega_X \simeq \mathcal{O}_X(-2S_2)$ and the graded anticanonical ring $\bigoplus_{m\geq 0} H^0(X, \omega_X^{-m}) \cong \mathbb{C}[U,V]$, with $\deg U = 1$, $\deg V = 2$. The complete system $|-2K_X|$ recovers f with the specified projective coordinates. This calculation uses the original analytic charts and finite fillings, not an identification of X with S^6, and draws no sphere-topology conclusion.

The retained constant $\beta_c = \beta^{\mathrm{part}} + c$ now gives a proper holomorphic deformation over $\mathrm{Im}\, c < -M$, locally on relatively compact parameter discs. On a fixed regular fibre set $T = \mathrm{Im}\,\tau$, $m = \mathrm{Im}\,\mu$, $D = \mathrm{Im}\,\beta_{c_*} - 6m^2/T < 0$ and $\delta = c - c_*$. In the plus-graph convention its exact Beltrami matrix is

$$\nu_\delta = \frac{\delta}{2iD + \delta}\begin{pmatrix} 0 & 0 \\ -m/T & 1 \end{pmatrix}.$$

The connecting Kodaira–Spencer class is $-[\partial_{\zeta_2} \otimes (d\bar{\zeta}_2 - (m/T)d\bar{\zeta}_1)/(2iD)] \neq 0$. The intrinsic pencil recovers f also over the dual numbers; its three fixed critical values prohibit an infinitesimal projective base change. Consequently this deformation of the whole X_c is nontrivial, not only its marked regular-fibre deformation. The proof uses no sphere recognition or CDP conclusion.

The 5 September continuation adds the higher-torus-wrap and octonionic comparison in Section 30. Its equal-diagonal positive octonionic cone has compactified pair (B^{25}, S^{24}). The flag $F_4/\mathrm{Spin}(8)$ has tangent module $8_v \oplus 8_s^+ \oplus 8_s^-$ and integral Poincaré polynomial $1 + 2t^8 + 2t^{16} + t^{24}$. The peripheral character has values $(-1,4,-3)$ in $\mathbb{Z}/12$, and sixfold suspension identifies $\pi_6(S^3) \cong \mathbb{Z}/12$ with the even subgroup of $\pi_{12}(S^9) \cong \mathbb{Z}/24$. The particular cusp map now has coefficient zero: its entire regular inverse image lies in one original toric chart and its actual normal framing bounds an explicit $D^2 \times T^2$ in that chart times an interval. This proves $g : X \to S^3$ based null-homotopic before any sphere identification; see Theorem 30.49. Retaining the angle in the same conical quotient gives $H = p_3 q_2 : X \to S^4$ with $[H] = \eta_4\eta_5 \neq 0$. Theorem 30.51 computes its entire inverse torus, its literal normal four-frame, a periodic $SU(3)$ contraction, and Arf invariant one. Hence q_2 is nonnull. The exact Hopf–cusp map $\eta_2 \wedge H : S^9 \to S^6$ represents $12\nu_6 \neq 0$ in $\mathbb{Z}/24$; after three suspensions it represents $12\nu_9$ in $\pi_{12}(S^9)$. Its peripheral composite has values $(12,0,12)$ and kernel $\chi^{-1}(2\mathbb{Z}/12)$, retaining parity rather than lifting χ under reduction modulo twelve. Section 9 also gives the full map from the fifteen marked $p = 1201$ selector rays to a three-dimensional real Jordan subalgebra of $\mathrm{Herm}_3(\mathbb{H})$, with exact census certificates.

Section 31 constructs the modulo-840 lattice $R_{\mathrm{ES}} = A_5^4 \perp D_4 \subset N$ with index 72, $\det N = 1$, and exactly the original 144 roots. Its coordinate/code maps give $O(N) = W(R_{\mathrm{ES}}) \rtimes L(G)$, $G \cong GL_2(\mathbb{F}_3)$, realizing the Weyl quotient recorded by Cheng–Duncan–Harvey [74, Table 3]. The residue six-cycle lies in $W(R_{\mathrm{ES}})$ and centralizes this chosen $L(G)$ (Theorems 31.15 and 31.16). The marked D_4 dual classes recover the vector and two half-spin weight sets of the flag tangent modules. Exact shell counts

retain the prime and shell: at $R = 3$ the same residue-one primes $2521, 3361$ have raw counts $0, 6$ and unordered counts $0, 3$, respectively.

The smooth literal scalar–Gauss continuation gives $\mathcal{S}_{\mathrm{sg}} \cong \{1,-1\}^2 \times (0,\infty) \times \mathbb{R}^2 \times \mathbb{T}^3$, with the original coupling $t = g^2$ as the positive coordinate. Its four components are embedded and six-dimensional; every fixed positive coupling has four five-dimensional components $\mathbb{R}^2 \times \mathbb{T}^3$ (Corollary 24.75). Both spatial equations remain outside this domain, and this refinement gives no quantum gap. For the cusp-regular periods, Proposition 10.3 gives the exact admissible constants $\operatorname{Im} c_0 < -\max_{B\setminus\{p_0\}} D_{\mathrm{part}}$. The cusp-free affine solution $\widetilde{\beta} = \beta + i\,t \circ \pi$ has $\widetilde{D} = D + \operatorname{Re} t$ unbounded above. The cusp edge maps retain the multipliers $e^{-2\pi i \mu_0}$, $e^{12\pi i \mu_0}$ and $e^{-2\pi i(7\mu_0 + b_0)}$ in their stated character conventions (equation (20)). Restricted chart domains and closed-polydisc inequalities are explicit in Lemma 10.5. These are period and local-quotient refinements, with no additional sphere or CDP conclusion inferred.

This paper is the consolidated successor to the former 56-page workbench: it incorporates the nonduplicate proofs from the subsequent 18-page repair branch and the later Reviewer B/Codex validation work. It validates, repairs, or attempts to disprove a reconstructed manuscript claiming a compact complex threefold diffeomorphic to S^6 and of algebraic dimension one. In its present form the workbench gives complete proofs of the simultaneous elliptic-orientation normalization. Let $\Delta \simeq (\mathbb{Z}/3) * (\mathbb{Z}/4)$ act on $\mathfrak{h}_z$, let g_1, g_2 have fixed points z_1, z_2, put $g_0 = (g_1 g_2)^{-1}$, and let $\tau : \mathfrak{h}_z \to \mathfrak{h}$ be the normalized period lift. Define

$$\varphi : \Delta \longrightarrow \mathrm{PSL}_2(\mathbb{Z}), \qquad \tau \circ g = \varphi(g) \circ \tau,$$

and

$$\mathfrak{S}_1(w) = \frac{w-1}{w}, \qquad \mathfrak{S}(w) = -\frac{1}{w}, \qquad \mathfrak{P}(w) = w - 1.$$

Then

$$g_1'(z_1) = e^{-2\pi i/3}, \qquad g_2'(z_2) = -i, \qquad g_0 \text{ is parabolic},$$

and

$$\varphi(g_1) = \mathfrak{S}_1, \qquad \varphi(g_2) = \mathfrak{S}, \qquad \varphi(g_0) = \mathfrak{P}, \qquad \mathfrak{S}_1 \mathfrak{S} \mathfrak{P} = 1.$$

The sign at z_2 is fixed by the simultaneous parabolic normalization; the quadratic ramification calculation alone gives only $(g_2'(z_2))^2 = -1$. The resulting nondegenerate period family descends, and the middle quotient is a proper holomorphic submersion with compact torus fibres. The toric cusp filling, both finite logarithmic fillings, all three punctured-overlap biholomorphisms, and the global analytic gluing yield a connected compact Hausdorff smooth complex threefold and a proper surjective holomorphic map $f : X \to \mathbb{P}^1$ with connected fibres. Its full cusp fibre W is reduced, irreducible, nonnormal, and normal crossings, with normalization dP_6; the other two full scheme-theoretic fibres have multiplicities three and four. The workbench also gives complete proofs of integral nearby-cycle specialization, the global Leray page and three primitive transgressions, the equivariant toric collapse, its integral cellular map and its equality with specialization, the based van Kampen presentation, sphere recognition, and the very-general-fibre Néron–Severi and meromorphic-descent calculation. For the constructed threefold it therefore verifies

$$X \cong_{\mathrm{diff}} S^6, \qquad \langle c_3(T_X), [X] \rangle = 2, \qquad \mathcal{M}(X) = f^*\mathbb{C}(t), \qquad a(X) = 1.$$

The same verified object satisfies the stated hypotheses of Campana–Demailly–Peternell Theorem 2.2 and contradicts its conclusions, so the workbench records a counterexample to that theorem as stated as well as two exact defects in its printed dependency chain. It also proves the strongest corrected conclusion that survives those defects: existence of one L_0 with $H^2(X, T_X \otimes L_0) = 0$ implies generic off-degree-one vanishing and $c_3(X) \leq 0$; for reduced full fibres the invalid singular-fibre shortcut is replaced by injectivity of the exact conductor connecting map. Because $c_3(X) = 2$, contraposition gives $H^2(X, T_X \otimes L_0) \neq 0$ for every line bundle L_0; independently, the conductor calculation gives $R^2 f_*(T_X \otimes L) \neq 0$ for every L, so the stronger relative sufficient condition also fails. The independent alternate certificate is now complete as well: the actual finite-band maps are $\alpha_q = (\pi_{1*}, -\pi_{2*})$ in primitive integral bases, the cusp attachment reduces to blocks with determinant $\pm p$, and the resulting obstruction squares commute with all three Leray transgressions. The candidate's Proposition 10.8 is also verified: the structure-sheaf direct images, degree-one base change, Picard line, exact finite normal characters, and uniform period/countability argument are reconstructed in full without using

the alternate collapse/Mayer–Vietoris route. Independently of the global topology calculation, the conductor construction also proves the required direct-image nonvanishing for every $L \in \mathrm{Pic}^0(X)$. On the cusp model the invariant form

$$\Omega = \frac{dx_1 \wedge dx_2 \wedge dt}{x_1 x_2}$$

first proves intrinsically that $\omega_{N_0} \simeq \mathcal{O}_{N_0}$. After global gluing, the open embedding $N_0 \hookrightarrow X$ gives

$$\omega_X|_{N_0} \simeq \omega_{N_0} \simeq \mathcal{O}_{N_0}.$$

Write $\nu : \widetilde{W} \simeq \mathrm{dP}_6 \to W$ for the normalization. For arbitrary line bundles the exact obstruction is the restriction homomorphism $\nu^* i^* : \mathrm{Pic}(X) \to \mathrm{Pic}(\mathrm{dP}_6)$. A direct Leray-edge calculation gives $e(c_1(L)) \in E_\infty^{0,2} \subseteq \ker d_2^{0,2} = 0$; proper-base-change naturality then makes $\nu^* i^* L$ trivial for every L. This avoids invoking the packaged theorem $H^2(X;\mathbb{Z}) = 0$ but reuses its decisive integral Leray transgression. A second all-Pic proof now avoids every global Leray differential. Put $q := uw + 6\gamma\delta \in \bigwedge^2 V = H^2(F;\mathbb{Z})$. If $E = c_1(L|_F)$, the order-four filling forces $E = 2nq$. The pair-of-pants defect classes satisfy

$$\bar{d}_1 + \bar{d}_2 + \bar{d}_0 = 0, \qquad \bar{d}_1 = 4n[\delta], \quad \bar{d}_2 = -3n[\delta], \quad \bar{d}_0 = 0,$$

where the first two values come from the exact logarithmic-cover norm maps and the last from a literal cusp solid torus. Hence $n = 0$; local integral specialization then gives $\nu^* i^* L \simeq \mathcal{O}$. The cusp-local map is nevertheless nonzero: a descended line bundle on the cusp neighbourhood has generic Chern class $-2q$ and normalized central degree vector $(-12, 2, 0, -12, 2, 0)$. The third global certificate therefore incorporates both finite fillings analytically. For $c_1(L|_F) = kq$, its $\mathcal{O}^*$-factor defects satisfy

$$(\bar{d}_1, \bar{d}_2, \bar{d}_0) = (2k, -\tfrac{3}{2}k, 0)[\delta], \qquad \bar{d}_1 + \bar{d}_2 + \bar{d}_0 = 0,$$

The finite norm identity is computed in the typed affine deck group

$$\Gamma_j^{\log} = \langle \tau_\lambda, Q_{j,\log} : \lambda \in \Lambda \rangle$$

Its extended defect satisfies

$$\widetilde{d}_j(Q_{j,\log}) = d_j + (T_j - I)a_j, \qquad \widetilde{d}_j(\tau_v) = \iota_v E,$$

and therefore

$$N_{T_j} d_j = \widetilde{d}_j((Q_{j,\log})^{m_j}) = \widetilde{d}_j(\tau_{v_j}) = \iota_{v_j} E.$$

Thus the displayed balance gives $k = 0$; the cusp equality comes from the argument principle in a Stein chart. The comparison uses the actual toric-open direction $U_{\rho_v} = D(z_w z_u) \subset U_\sigma$ and a contractible Stein chart S_σ, so adjacent central frames have zero horizontal winding. If μ_λ^+ is multiplication by χ^{m_λ} on $\mathcal{O}(D_a)$, the required map on the negative divisor is the typed inverse-dual

$$\mu_\lambda^- = ((\mu_\lambda^+)^{-1})^\vee : \Psi_\lambda^* \mathcal{O}(-D_a) \longrightarrow \mathcal{O}(-D_a), \qquad s \longmapsto \chi^{-m_\lambda} s.$$

Put $M = L|_{N_0}$. If (d_1, d_2, d_3) are the independent normalized central degrees, then

$$B_0 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad B = \begin{pmatrix} d_2 + d_3 & d_3 \\ d_3 & d_1 + d_3 \end{pmatrix},$$

$$E_M((\lambda, 0), (0, m)) = -B(B_0\lambda, m).$$

The preceding balance gives $k = 0$, hence $E_M = c_1(L|_F) = 0$. Unimodularity of B_0 now forces $B = 0$ and all six degrees to vanish. This third proof avoids both global integral Leray and local integral specialization/bounding-chain input. The workbench also completes the Hodge calculation left open in the manuscript. The torsion-free absolute cotangent direct images are

$$R^0 f_* \Omega_X^1 \simeq \mathcal{O}(-2), \quad R^1 f_* \Omega_X^1 \simeq \mathcal{O} \oplus \mathcal{O}(-1)^{\oplus 2}, \quad R^2 f_* \Omega_X^1 \simeq \mathcal{O} \oplus \mathcal{O}(-1).$$

Hence $h^{1,2} = h^{2,1} = 1$, $h^{1,1} = h^{2,2} = 2$, and every Hodge number is known. The Frölicher spectral sequence does not degenerate at E_1: its two middle $d_1 = \partial$ rows are exact

$$0 \longrightarrow \mathbb{C} \longrightarrow \mathbb{C}^2 \longrightarrow \mathbb{C} \longrightarrow 0.$$

Thus $\dim_{\mathbb{C}} E_1 = 10$, while $E_2 = E_\infty$ and $\dim_{\mathbb{C}} E_2 = 2$. Put

$$\Sigma = p_0 + p_1 + p_2, \quad D = f^{-1}(\Sigma)_{\mathrm{red}}, \quad \mathcal{L} = \Omega^1_{\mathbb{P}^1}(\log \Sigma), \quad \mathcal{E} = \Omega^1_X(\log D)/f^*\mathcal{L}, \quad \mathcal{G} = R^1 f_* \mathcal{O}_X.$$

Then

$$f_*\mathcal{E} \simeq \mathcal{O} \oplus \mathcal{O}(-1), \quad R^1 f_* \mathcal{E} \simeq \mathcal{O}^{\oplus 2} \oplus \mathcal{O}(-1)^{\oplus 2}, \quad R^2 f_* \mathcal{E} \simeq \mathcal{O} \oplus \mathcal{O}(-1),$$

and all three sheaves are locally free and commute with base change. The global calculation uses the actual cyclic invariant lattices, the simple divisor $p_1 + p_2$ of the logarithmic Kodaira–Spencer determinant, and the normalized global scalar at the order-four point. At a finite fibre write $t = s^m$, with $(m, a) = (3, 2)$ or $(4, 1)$, and, after removing the common ds from $D_{\mathrm{KS}} = (d_{rs})$, put

$$A = \frac{s}{m} d_{00}, \quad B = \frac{s^{1-a}}{m} d_{01}, \quad C = \frac{s^{1-a}}{m} d_{10}, \quad D_0 = \frac{s^{m+1-2a}}{m} d_{11}.$$

In the invariant source basis $(e_0, s^{m-a}e_1)$, target basis $(b_0, s^a b_1)\ell$, and ordered degree-one source basis $e_0 b_0, s^a e_0 b_1, s^{m-a} e_1 b_0, e_1 b_1$,

$$[\kappa_0] = \begin{pmatrix} A & tB \\ C & D_0 \end{pmatrix}, \qquad [\kappa_1] = (-C, A, -D_0, B).$$

The exact finite residue cancellation is

$$\operatorname{res}_{1,j}(x b_0 + y s^a b_1) = m_j x [b_0], \qquad \operatorname{res}_{1,j} \kappa_0 = 0,$$

$$\ker \operatorname{res}_{1,j} = \mathbb{C}\, s^a b_1 = \operatorname{im}(\kappa_0 \bmod t).$$

At the cusp put

$$Q_{\mathrm{cusp}} = \nu_* \mathcal{O}_{\mathrm{dP}_6}/\mathcal{O}_W, \qquad \eta_i = d \log x_i, \qquad q_1 = (-1, 0, 1), \qquad q_2 = (0, -1, -1),$$

and, for the normalization connecting maps, set

$$\alpha_i = \delta_0(q_i), \qquad \theta = -\delta_1^{-1}(\alpha_1 \smile \alpha_2).$$

The standard Čech sign gives

$$\delta_1\big(\rho_*(\alpha \otimes \eta_i)\big) = -\alpha \smile \alpha_i,$$

hence the exact row $(0, 1, -1, 0)$ with target $\mathbb{C}\theta$. Let

$$\xi = \frac{\partial}{\partial \zeta_2}, \qquad h = (a(z), 1) \begin{pmatrix} d\zeta_1 \\ d\zeta_2 \end{pmatrix}, \qquad h(\xi) = 1$$

be the unique global lift. At the order-four point, in the genuine invariant frame $\omega_2 = s\, u_2(s)(\gamma \wedge u)\ell$, $u_2(0) = 1$, one has $a(z_2) = \frac{1}{2}$, $\dot\tau(z_2) = 0$, $K(z_2) = 6\dot\mu(z_2)^2 \neq 0$, and

$$\kappa_1(h \otimes \gamma)(p_2) = -\frac{1}{4}\big(a(z_2)\dot\tau(z_2) + \dot\mu(z_2)\big) = -\tfrac{1}{4}\dot\mu(z_2) \neq 0.$$

The stalks at $t = xy$ and $t = xyz$ show that $\Omega^1_{X/B}$ is torsion-free but non-locally-free at the cusp. On a finite cover $t = s^{m_j}$, its exact local module is

$$R\, dy_1 \oplus R\, dy_2 \oplus \big(R/(s^{m_j - 1})\big) ds,$$

so its torsion is supported exactly at the two multiple fibres. No step assumes that direct image commutes with duality across a multiple fibre. McHugh's compact-threefold identities then determine the remaining Bott–Chern and Aeppli invariants. In total degrees 0 through 6,

$$(h^k_{BC}) = (1, 0, 2, 2, 2, 2, 1), \qquad (h^k_A) = (1, 2, 2, 2, 2, 0, 1),$$

and the non-$\partial\bar\partial$ degrees are $(0, 2, 4, 4, 4, 2, 0)$. Thus no Bott–Chern or Aeppli dimension remains open. On a marked smooth fibre,

$$\eta = u \wedge w + 6\gamma \wedge \delta, \qquad \eta^2 = 12\, \gamma \wedge u \wedge w \wedge \delta.$$

If Q_F is its displayed Dolbeault representative, then

$$\int_{(F_b,\mathrm{or}_{\mathrm{mark}})} Q_F^2 = 12, \qquad \int_{(F_b,\mathrm{or}_{\mathbb{C}})} Q_F^2 = -12;$$

the marked and complex orientations are opposite because $\det_{\mathbb{R}} \Pi < 0$.

Let $\mathfrak{g} \in H_{\bar\partial}^{0,1}(X)$ be the distinguished degree-one class and let $\mathfrak{q} \in H_{\bar\partial}^{1,1}(X)$ be a chosen coherent-Leray lift of η. The latter is determined only modulo $\mathbb{C}\, d_1\mathfrak{g}$. Choose global $\bar\partial$-closed smooth representatives

$$G \in \mathcal{A}^{0,1}(X), \qquad Q \in \mathcal{A}^{1,1}(X)$$

of $\mathfrak{g}, \mathfrak{q}$. Their marked smooth-fibre cohomology classes have the exact period-coordinate formulas proved below; no literal equality of an arbitrary global Q with the displayed coefficient formula is used. With $\omega = c_1(\mathcal{O}_{\mathbb{P}^1}(1))$,

$$\exp(12\mathfrak{g}) = f^*\mathcal{O}_{\mathbb{P}^1}(1), \qquad c_{1,BC}(\exp(z\mathfrak{g})) = z\mathsf{A} + \bar z\,\mathsf{B},$$
$$c_{1,BC}(f^*\mathcal{O}_{\mathbb{P}^1}(1)) = 12(\mathsf{A}+\mathsf{B}).$$

Put $\mathbf{1} = [1]_{BC}$, and choose a smooth top form ϑ with $\int_X \vartheta = 1$, so $\mathsf{T} = [\vartheta]_{BC}$. Then

$$\mathsf{A} = [\partial G],\ \mathsf{B} = [\bar\partial \bar G],\ \mathsf{C} = [\partial Q],\ \overline{\mathsf{C}},\ \mathsf{D} = [\partial(G\wedge Q)],\ \overline{\mathsf{D}},\ \mathsf{E} = [\partial(Q^2)],\ \overline{\mathsf{E}},\ \mathsf{T}$$

are bases of all nonzero positive-degree Bott–Chern groups; together with $\mathbf{1}$, they give every nonzero group. The two load-bearing fibre pairings in the complex orientation are

$$\int_{F_b} Q_F^2 = -12, \qquad \int_{F_b} Q_F \wedge \Gamma \wedge d\zeta_2 = -1.$$

They prove the required Dolbeault nonvanishing, while the constant fibre coefficients give the typed identity

$$i_b^*(d_1[G\wedge Q]_{\bar\partial}) = [\partial_F(\Gamma\wedge Q_F)]_{\bar\partial} = 0.$$

If N_{BC} is the span of the nine displayed positive-degree classes, then

$$H_{BC}^{*,*}(X) = \mathbb{C}\mathbf{1} \oplus N_{\mathrm{BC}}, \qquad N_{\mathrm{BC}}^2 = 0.$$

Thus every product of two positive-degree Bott–Chern classes vanishes; see theorems 57.4 and 57.5. The result does not select a preferred literal global coefficient atlas for G, Q through the finite fillings and cusp; that presentation choice is not an open mathematical premise. The marked positive radial cusp also carries an exact auxiliary flat-metric calculation. With

$$L_T = T + \operatorname{Im} h_T, \qquad q_T = L_T - \operatorname{Im} b_T, \qquad m_T = \operatorname{Im}\mu_T, \qquad D_T = L_T q_T + 6m_T^2,$$

where $H_{\mathrm{cusp}} = \operatorname{Im} h(0)$ and $B_{\mathrm{cusp}} = \operatorname{Im} b(0)$, the complete dual lattice is $P_T^{-\top}\mathbb{Z}^4$, and its shortest nonzero vector gives

$$\lambda_1(F_T, g_T) = \frac{4\pi^2}{T^2}\left(1 + \frac{-2H_{\mathrm{cusp}} + 2\min\{B_{\mathrm{cusp}}, 0\}}{T} + O(T^{-2})\right),$$

while the unit-volume metric $\widehat{g}_T = D_T^{-1/2} g_T$ gives

$$\lambda_1(F_T, \widehat{g}_T) = \frac{4\pi^2}{T}\big(1 + O(T^{-1})\big) \longrightarrow 0.$$

For fixed $E \geq 0$, the scalar eigenvalue count with Fourier multiplicity is

$$N_T(E) := \#\{k \in \mathbb{Z}^4 : 4\pi^2 \|P_T^{-\top}k\|^2 \leq E\} = \pi D_T \sum_{n\in\mathbb{Z}^2} \left(\frac{E}{4\pi^2} - \|n\|^2\right)_+ + O_E(T).$$

In particular, for $0 < E < 4\pi^2$, $N_T(E) = ED_T/(4\pi) + O_E(T)$: the number of soft modes grows quadratically, rather than only one eigenvalue tending to zero. The mean-zero vertical direct integral is a single injective self-adjoint operator with 0 in its essential spectrum. The unmodified regular four-torus

fibres F_T have native pointed limit exactly $\mathbb{T}^2 \times \mathbb{R}^2$. A separate ancillary construction changes both the fibre lattice and the cusp base: for an integer $N \geq 1$, after the added operations

$$t_c = v^N, \qquad \Lambda^{(N)} = \mathbb{Z}\langle \widehat{\gamma}, \widehat{u}, N\widehat{w}, N\widehat{\delta} \rangle,$$

the punctured degree-N^2 fibre covers have pointed local limit $\mathbb{R}^4$ and momentum lattices becoming dense in all four directions. The extension across the central fibre is explicit. With source residual term $C_N(v) = C(v^N)/N$, put

$$f_N : N_0^{\langle N \rangle} \longrightarrow \Delta_v, \qquad W_N = (f_N^{-1}(0))_{\mathrm{red}}.$$

Toric multiplication descends to

$$\kappa_N : N_0^{\langle N \rangle} \longrightarrow N_0, \qquad \deg \kappa_N = N^3, \qquad \kappa_N^* W = N W_N, \qquad \mathrm{Ram}(\kappa_N) = (N-1)W_N.$$

Factoring through the normal base change $N_0[N] = N_0 \times_{\Delta_t} \Delta_v$ gives a generic degree-N^2 map Q_N; for $N > 1$ it is quasi-étale but nonflat exactly over the double and triple strata. On the normalized central fibre it is the toric map $[N] : \mathrm{dP}_6 \to \mathrm{dP}_6$. This is a completed *local* cusp morphism, not a cover of the entire base or of X.

Independently, the compact cusp lattice $K_\Lambda = \mathbb{Z}\langle \widehat{w}, \widehat{\delta} \rangle$ is not preserved by the finite monodromies, but its invariant closure is $L_\Lambda = \mathbb{Z}\langle \widehat{u}, \widehat{w}, \widehat{\delta} \rangle$. The invariant character lattice is $\mathbb{Z}\varepsilon_\gamma$; fix the displayed primitive generator. Its real group quotient has exact circle circumference $\ell_T = D_T/\sqrt{L_T^2 + m_T^2} = T + H_{\mathrm{cusp}} - B_{\mathrm{cusp}} + O(T^{-1})$. Four synchronized copies therefore give a genuine $\mathbb{T}^4$-quotient over the full regular base whose pointed cusp limit is $\mathbb{R}^4$, and four copies are minimal for this invariant-kernel mechanism. The two finite fillings extend to explicit Seifert targets after simultaneous-root normalization, with central degrees 27 and 64. At the cusp, two paths with the same limit and different character values still prove that the *literal fixed-target* quotient cannot extend; no finite base cover repairs the native two-copy kernel.

A different, exact compactification changes the codomain rather than the character values. After choosing a base point, collapsing the inner and outer cusp cross-sections gives a based quotient (denoted q_2 in the detailed construction, and written $\rho_{2,\mathrm{cone}}$ here to avoid collision with an earlier Čech vector)

$$\rho_{2,\mathrm{cone}} : (X, x_0) \longrightarrow (\mathcal{A}_{\mathrm{cone}}, *) = \big(\mathrm{Susp}(S^1 \times \overline{\mathbb{T}}^2), *\big).$$

Its Cartesian square restricts on synchronized radial fibres to $\pi_T^{\times 2} : F_T^{\times 2} \to Q_T^{\times 2}$, and basedness induces $\rho_{2,\mathrm{cone}}^{\wedge 2} : X^{\wedge 2} \to \mathcal{A}_{\mathrm{cone}}^{\wedge 2}$. Likewise the regular and finite-filled circle targets glue to a compact three-manifold $\mathcal{M}_{\mathrm{Seif}}$ with boundary $\mathbb{T}^2$; coning that boundary gives a based quotient (denoted q_1 in the detailed construction) $\rho_{1,\mathrm{cone}} : (X, x_0) \to (\mathcal{C}_{\mathrm{cone}}, *)$ whose fourth Cartesian power restricts to $\chi_{\gamma,T}^{\times 4} : F_T^{\times 4} \to C_T^{\times 4}$, and hence induces a fourth smash power.

Separately, composing $\rho_{2,\mathrm{cone}}$ and the cusp-local circle quotient $q_{1,\mathrm{cusp}}$ with their respective base-angle-forgetting maps gives continuous surjective quotient maps

$$\widehat{\rho}_3^{\wedge 2} : S^{12} \twoheadrightarrow S^6, \qquad \widehat{\rho}_\gamma^{\wedge 4} : S^{24} \twoheadrightarrow S^8,$$

after a based choice $X \cong_{\mathrm{diff}} S^6$. Both full sphere carriers are based null-homotopic, by two distinct calculations. In the detailed notation $\widehat{\rho}_\gamma = \widehat{q}_\gamma$ and $\widehat{\rho}_3 = \widehat{q}_3$. For any based map $f : S^6 \to S^2$, the smash square has the exact factorization

$$S^{12} = S^6 \wedge S^6 \xrightarrow{f \wedge \mathrm{id}_{S^6} = \Sigma^6 f} S^2 \wedge S^6 = S^8 \xrightarrow{\mathrm{id}_{S^2} \wedge f} S^2 \wedge S^2 = S^4.$$

Freudenthal stability and the vanishing fourth stable stem give $\pi_{12}(S^8) \cong \pi_4^{\mathrm{S}} = 0$, so $f \wedge f$ is based null-homotopic. Associativity then writes $f^{\wedge r} = (f \wedge f) \wedge f^{\wedge (r-2)}$, and hence $[f^{\wedge r}] = 0 \in \pi_{6r}(S^{2r})$ for every $r \geq 2$. In particular, the entire quotient map $\widehat{\rho}_\gamma^{\wedge 4} : S^{24} \to S^8$, not merely its moving four-sphere restriction, is based null-homotopic.

For the other carrier, let $g : S^6 \to S^3$ be any based map. The exact smash factorization is

$$g \wedge g = (\Sigma^3 g) \circ (\Sigma^6 g) : S^{12} \longrightarrow S^9 \longrightarrow S^6.$$

Write st for stabilization. The exact low-stem data are

$$\pi_6(S^3) \cong \mathbb{Z}/12, \qquad \pi_3^{\mathrm{S}} = \mathbb{Z}/24\langle \nu \rangle, \qquad \pi_6^{\mathrm{S}} = \mathbb{Z}/2\langle \nu^2 \rangle,$$

$$\pi_{12}(S^6) = \mathbb{Z}/2\langle \nu_6 \circ \nu_9 \rangle, \qquad \mathrm{st} : \pi_{12}(S^6) \xrightarrow{\cong} \pi_6^{\mathrm{S}}.$$

The full groups are tabulated in [10, p. 339]; the signed smash formula and the unstable stabilization calculation are [13, Propositions 3.1 and 5.11]. Since $12[g] = 0$, its stable image has the form $\mathrm{st}([g]) = 2a\nu$ for some $a \in \mathbb{Z}/12$. Stable smash multiplication therefore gives

$$\mathrm{st}([g \wedge g]) = \mathrm{st}([g])^2 = 4a^2\nu^2 = 0 \in \pi_6^{\mathrm{S}}.$$

The displayed stabilization isomorphism now forces $[g \wedge g] = 0 \in \pi_{12}(S^6)$. Hence every $g^{\wedge r}$ with $r \geq 2$ is based null-homotopic as well. In particular,

$$[\widehat{\rho}_3^{\wedge 2}] = [\widehat{q}_3^{\wedge 2}] = 0 \in \pi_{12}(S^6), \qquad [\widehat{\rho}_\gamma^{\wedge 4}] = [\widehat{q}_\gamma^{\wedge 4}] = 0 \in \pi_{24}(S^8).$$

The one-copy class $[\widehat{q}_3] \in \pi_6(S^3)$ is zero by the explicit framed filling in Theorem 30.49. The one-copy class $[\widehat{q}_\gamma] \in \pi_6(S^2)$ remains open. The unstable classes of the conical-target maps $\rho_{2,\mathrm{cone}}^{\wedge 2}$, $q_{1,\mathrm{cusp}}^{\wedge 4}$, and $\rho_{1,\mathrm{cone}}^{\wedge 4}$ also remain **[OPEN]**; nullity after postcomposition to a sphere does not prove nullity before that postcomposition. See theorems 11.21, 11.22, 11.25 and 11.27.

Their targets contain moving topologically embedded four-spheres, equipped separately with their intrinsic quotient length metrics, $S_T^2 \wedge S_T^2$ and $C_T^{\wedge 4}$. With their quotient length metrics and midpoint base points, every radius-R ball is exactly Euclidean once $T - C_{\mathrm{sys}} > 4R$, respectively once $\ell_T > 4R$; thus both moving four-spheres converge pointed local-isometrically and Gromov–Hausdorff to $\mathbb{R}^4$, with smooth convergence on those Euclidean neighborhoods. Deleting the common smash basepoint gives literal nonclosed topological four-cells

$$(Q_T \setminus K_T^{(1)})^2 \cong \mathbb{R}^4 \hookrightarrow S^6, \qquad (C_T \setminus \{0\})^4 \cong \mathbb{R}^4 \hookrightarrow S^8.$$

The based inclusion maps

$$S_T^2 \wedge S_T^2 \hookrightarrow S^6, \qquad C_T^{\wedge 4} \hookrightarrow S^8$$

are null-homotopic. These slice-inclusion null-homotopies are separate from the global equalities $[\widehat{\rho}_3^{\wedge 2}] = 0 \in \pi_{12}(S^6)$ and $[\widehat{\rho}_\gamma^{\wedge 4}] = 0 \in \pi_{24}(S^8)$. Two copies are rank-minimal for the native cusp-local quotient, while four are rank-minimal for a primitive full-monodromy-invariant quotient with four expanding directions. Here $C_{\mathrm{sys}} > 0$ is a fixed bound for the holomorphic error in the radial period lattice.

Exact five-manifolds with boundary interpolate the positive four-torus slices with a boundary fibre $\mathbb{R}^4$. There are continuous surjective comparison maps to the compact cone carriers which are homeomorphisms away from that boundary and collapse it to the cone vertex; however, these maps are nonproper and nonquotient. Thus they are not topological blow-downs or resolutions, and the compact carriers are not the quotients of the five-manifolds by their $\mathbb{R}^4$ boundary fibres via these maps. For every unrescaled fixed compatible target metric the radial slices collapse to the cone point. A fixed cusp-adapted metric can recover the slice limits only after the blow-up factors T/r_T and ℓ_T/r_T; the corresponding full ambient blow-ups are $\mathbb{R}^8$ and $\mathbb{R}^{12}$, not $\mathbb{R}^4$. Thus the sphere constructions yield exact carrier slices with intrinsic $\mathbb{R}^4$ limits and a decompactification boundary, but not $\mathbb{R}^4$ as the whole compact sphere or compact target. Moreover $X^{\times 4} \cong (S^6)^{\times 4} \not\simeq S^{24}$, and Borel–Serre still excludes an almost-complex structure on either S^{12} or S^{24} [63]. The distinct first-coordinate holomorphic quotient is now classified exactly: $\mathcal{A}_z \to \mathbb{C}/(\mathbb{Z}\tau + \mathbb{Z})$ exists precisely when $6\mu = a\tau + b$ for $(a, b) \in \mathbb{Z}^2$, has connected kernel $\mathbb{C}/(\mathbb{Z} + \mathbb{Z}(a\mu - \beta))$, and its marked labels obey $g_0(a, b) = (a, a + b)$, $g_2(a, b) = (-b, 6 + a)$. This locus has finite image in the completed orbifold base and misses a punctured cusp neighborhood, so it cannot be the cusp gap-closing mechanism; the theorem does not exclude other elliptic quotients or smooth integral magnetic fluxes. The reduced classical Yang–Mills Hessian has the same scalar factor g_{YM}^{-2}, and the finite cubical model is diagonalized in every mesh regime. On the degree-M_{cov}^2 cover, an independent N_{mesh}^4-vertex cubical subdivision gives a correctly typed holonomy map to $SU(2)^{E_{N_{\mathrm{mesh}}}}$. For $H_{\mathrm{link}} = \mathrm{diag}(i, -i)$, the explicit unit-flux link field has

$$U_{12}(n) = \exp(2\pi H_{\mathrm{link}}/N_{\mathrm{mesh}}^2), \qquad S^{\log}_{T, M_{\mathrm{cov}}, N_{\mathrm{mesh}}} = \frac{2\pi^2 M_{\mathrm{cov}}^2}{g_{\mathrm{YM}}^2 D_T}.$$

On the reduced finite Coulomb space, the Gaussian oscillator has gap

$$\sqrt{\min_{k \neq 0} \frac{1}{g_{\mathrm{YM}}^2} \left\| P_T^{-\top} S_{M_{\mathrm{cov}}}^{-1} \mathfrak{d}_{N_{\mathrm{mesh}}}(k) \right\|^2}.$$

The joint sequence $T_j^{\mathrm{rad}} = j^2,\ M_{\mathrm{cov},j} = j,\ N_{\mathrm{mesh},j} = j^3$ simultaneously gives the pointed cover limit $\mathbb{R}^4$, mesh diameter $O(j^{-1})$, Gaussian gap $O(j^{-2})$, and logarithmic action $O(j^{-2})$. This oscillator lives on a linear Coulomb space with Lebesgue measure; it is not the compact-link Haar/Gauss-law Hilbert space. For a nondegenerate integral magnetic form $\mathsf{E}_{\mathrm{mag}}$, the exact oscillator frequencies satisfy

$$b_1 b_2 = \frac{4\pi^2 |\operatorname{Pf}(\mathsf{E}_{\mathrm{mag}})|}{D_T}, \qquad \operatorname{gap}(\Delta_{\mathsf{E}_{\mathrm{mag}},T}) = 2\min\{b_1, b_2\} \le 4\pi\sqrt{\frac{|\operatorname{Pf}(\mathsf{E}_{\mathrm{mag}})|}{D_T}} = O(T^{-1}),$$

for the Friedrichs magnetic Laplacian on the corresponding Hermitian line bundle over the marked four-torus. Independently, the explicit reducible $SU(2)$ critical point has

$$\mathrm{YM}_T = \frac{2\pi^2}{g_{\mathrm{YM}}^2 D_T}, \qquad \mathcal{J}_T a = -\frac{4\pi}{D_T} a.$$

These are geometric and classical operator theorems. No positive-metric physical Yang–Mills Hilbert space, reconstructed Hamiltonian, or scale-controlled state-and-energy maps have been constructed, so the interacting four-dimensional mass-gap implication is **[OPEN]**.

The same native full-flux construction extends exactly to every nonzero trace-zero integral weight vector $\mathbf{m} = (m_1, \dots, m_r)$ in a split Cartan of $SU(r)$. Writing $s_2(\mathbf{m}) = \sum_i m_i^2$, the resulting reducible bundle and connection satisfy

$$c_2(\mathscr{E}_{\mathbf{m},T}) = -\frac{s_2(\mathbf{m})}{2}[\eta_T]^2, \qquad \langle c_2, [F_T]_{x\text{-complex}} \rangle = 6 s_2(\mathbf{m}),$$

and

$$\mathrm{YM}_T(\nabla^{\mathbf{m},T}) = \frac{12\pi^2 s_2(\mathbf{m})}{g_{\mathrm{YM}}^2} + \frac{\pi^2 s_2(\mathbf{m}) S_T^2}{g_{\mathrm{YM}}^2 D_T}.$$

Its charged Coulomb–Jacobi bands, coincident-band multiplicities, wall nullities, and Morse index are computed root by root; on the deep cusp tail the real index is $36 r s_2(\mathbf{m})$. These are fibrewise classical Yang–Mills theorems. The nonzero c_2 also proves that the bundles do not extend over any proposed carrier with $H^4 = 0$, including spheres of dimensions (6), (12), and (24); changing the gauge group therefore does not create a four-dimensional interacting quantum theory.

Finally, for $r = (r_1, r_2, r_3) \in \mathbb{R}^3$, put

$$Q_K(r) = \frac{2}{3}(r_1 + r_2 + r_3)^2 - (r_1^2 + r_2^2 + r_3^2) = h^2 - x^2 - y^2,$$

where

$$h = \frac{r_1 + r_2 + r_3}{\sqrt{3}}, \qquad x = \frac{r_1 - r_3}{\sqrt{2}}, \qquad y = \frac{r_1 - 2r_2 + r_3}{\sqrt{6}}.$$

The real-linear coefficient isomorphism

$$(h, x, y) \longmapsto (t, m, \varrho) = \left(h + y, \frac{x}{\sqrt{6}}, h - y\right)$$

identifies the Koide cone $Q_K = 0$ with $t\varrho - 6m^2 = 0$ inside the three-dimensional imaginary-coefficient space of the canonical period ansatz. It does not identify that cone with the full real period discriminant, whose three real coefficients remain free. For every choice of those real parts, $\det P = t\varrho - 6m^2 = Q_K(r)$. Hence each Koide-cone point gives $\det P = 0$, whereas the actual positive radial cusp has $\det P_T = -D_T < 0$ and $\det P_T \sim -T^2$; the Koide cone does not parametrize regular fibres.

Independently, write

$$\Delta(x, y, z) = 4(xy + xz + yz) - (x^2 + y^2 + z^2)$$

and, for $Y \ge 0$, define

$$R_{\mathrm{K}}(Y) = \#\{(x, y, z) \in \mathbb{Z}_{>0}^3 : x \le y \le z \le Y,\ \Delta(x, y, z) = -1,\ z \ge 4(x + y)\}.$$

The integral coordinates

$$n = z - 4(x + y), \qquad q = y - x, \qquad a = x + y - 4n$$

give a bijection onto

$$\{(a,q,n)\in\mathbb{Z}^3 : a>0,\ q,n\ge 0,\ a+4n>q,\ a^2-3q^2-18n^2=-2\},$$

with height $z=4a+17n$. Calibrated homogeneous counting gives unconditionally

$$R_{\mathrm{K}}(Y)\sim C_0 Y,\qquad C_0=\frac{9}{2\pi^2}\log(2+\sqrt{3})\log\!\left(\frac{3}{2\sqrt{2}}\right).$$

The stronger estimate $R_{\mathrm{K}}(Y)-C_0Y=O_\delta(Y^{1/2+\delta})$ for every $\delta>0$ remains **[OPEN]**.

There is also an exact arithmetic bridge to the earlier Erdős–Straus divisor-shell construction. Fix a prime $p\equiv 1 \pmod{12}$ and $\varepsilon\in\{0,1\}$, and put

$$L=\frac{p-1}{4},\qquad 0\le j<L,\qquad R_j=4j+3,\qquad a_j^{\mathrm{ES}}=\frac{p+R_j}{4}.$$

For $u\in\mathbb{Z}_{>0}$ satisfying $u\mid (a_j^{\mathrm{ES}})^2$ and $R_j\mid(4u+p^\varepsilon)$, set

$$T_{\mathrm{sel}}=\frac{4u+p^\varepsilon}{R_j},\qquad \ell_0=\frac{T_{\mathrm{sel}}+1}{4},\qquad \ell_1=u-jT_{\mathrm{sel}},\qquad \ell_2=1.$$

Then $T_{\mathrm{sel}}\in\mathbb{Z}_{>0}$, $12\ell_0-4\ell_1-3=p^\varepsilon$, and

$$\begin{pmatrix}1 & -j\\ 0 & 1\end{pmatrix}\begin{pmatrix}R_j & -u\\ 4 & -T_{\mathrm{sel}}\end{pmatrix}=\begin{pmatrix}3 & -\ell_1\\ 4 & 1-4\ell_0\end{pmatrix}.$$

For fixed (p,ε), the marked assignment $(j,u)\mapsto(j,\ell_0,\ell_1)$ is a bijection onto the triples satisfying

$$0\le j<L,\quad 12\ell_0-4\ell_1-3=p^\varepsilon,\quad T_{\mathrm{sel}}=4\ell_0-1>0,\quad u=\ell_1+jT_{\mathrm{sel}}>0,\quad u\mid(a_j^{\mathrm{ES}})^2.$$

This is an integral-relator equivalence, not a geometric map between the reciprocal cubic and X. The distinct residue-lattice and count maps are proved in Section 31: the fixed-shell divisor bijection and finite-sample Fourier pushforward retain the actual prime and shell, rather than replacing the divisor gate by a residue modulo 840.

The higher-rung calculation now has several exact, sharply separated outputs. For two complex columns, left Gram reduction gives

$$\mathrm{Mat}_{3\times 2}(\mathbb{C})/SU(3)\cong\mathrm{Herm}_2(\mathbb{C})_{\ge 0},\qquad S^{12}_{\mathrm{col}}/SU(3)\cong B^4,$$

whose orbit-type boundary is S^3. Here S^{12}_{col} is the one-point compactification of the column representation; it is not the cusp-smash S^{12} constructed elsewhere in the workbench. For $\mathscr{V}_3=\mathrm{Mat}_{3\times 3}(\mathbb{C})$, left Gram reduction and the alternating determinant give

$$\mathscr{V}_3/U(3)\cong\mathrm{Herm}_3(\mathbb{C})_{\ge 0},\qquad \mathscr{V}_3/SU(3)\cong\{(H,z): H\ge 0,\ |z|^2=\det H\}\cong\mathbb{R}^{10}.$$

Consequently

$$S^{17}/U(3)\cong B^8,\quad \mathscr{V}_3^+/U(3)\cong B^9,\quad S^{17}/SU(3)\cong S^9,\quad \mathscr{V}_3^+/SU(3)\cong S^{10}.$$

The compactified rank-deficient locus is the standard locally flat $S^8\subset S^{10}$, and $S^{10}\setminus S^8\cong\mathbb{R}^8\times\mathbb{C}^*\simeq S^1$, with generator $\det A/|\det A|$. The full rank-three cubic Jordan cone is not itself the rank-two quadratic Lorentz cone; the marked angular quotient nevertheless supplies a split positive observation onto that cone while retaining all five complementary directions.

For

$$P=\begin{pmatrix}1 & -1 & 0\\ 0 & 1 & -1\end{pmatrix},\quad u=(1,1,1)^{\mathsf{T}},\quad R=P^*(PP^*)^{-1},\quad \Pi=uu^*/3,$$

define

$$\mathcal{O}_P(H)=PHP^*,\qquad \jmath_P(G)=RGR^*.$$

Then

$$\mathcal{O}_P\jmath_P=\mathrm{id},\qquad \jmath_P\mathcal{O}_P(H)=(I_3-\Pi)H(I_3-\Pi),$$

and the positive zero fibre is exactly $\{\lambda uu^* : \lambda \geq 0\}$. With $X = (R\ u)$, every Hermitian H has unique coordinates

$$H = X \begin{pmatrix} G & \psi \\ \psi^* & \lambda \end{pmatrix} X^*, \qquad \det H = \lambda \det G - \psi^* \operatorname{adj}(G)\psi,$$

where

$$H \succeq 0 \iff G \succeq 0, \quad \psi \in \operatorname{im} G, \quad \lambda \geq \psi^* G^+ \psi.$$

Thus

$$\operatorname{Herm}_3(\mathbb{C}) \cong \operatorname{Herm}_2(\mathbb{C}) \oplus \mathbb{C}^2 \oplus \mathbb{R}, \qquad 9 = 4 + 4 + 1.$$

The faithful embedding $T_P(g) = RgP + \Pi = X(g \oplus 1)X^{-1}$ acts by $(G, \psi, \lambda) \mapsto (gGg^*, g\psi, \lambda)$. The three summands are the Lorentz four-vector, the defining complex spin representation, and a real scalar; the vector action kills $\{\pm I_2\}$, whereas the full action retains the spin double cover because $-I_2$ negates ψ. On positive cones, $\jmath_P$ properly compactifies to $B^4 \hookrightarrow S^8$, while the reverse boundary observation is nonproper. The determinant-phase equation is

$$|\omega|^2 = \lambda \det G - \psi^* \operatorname{adj}(G)\psi,$$

and $\omega = 0$ is the rank-loss locus compactifying to $S^8 \subset S^{10}$. The four-branch chart supplies the exact seed

$$K(z) = \operatorname{diag}(z_0 z_3, z_1 z_2), \qquad \det K = z_0 z_1 z_2 z_3,$$

whose positive $SL_2(\mathbb{C})$-orbit fills the future causal cone. Finally, with retained $\alpha, c > 0$,

$$p_\alpha(Z) = \alpha Z^* Z, \qquad m^2_{\alpha,c}(Z) = \frac{\alpha^2}{c^2} \sum_{a<b} |Z_{a1} Z_{b2} - Z_{a2} Z_{b1}|^2$$

is the complete $SU(3)$-orbit momentum, and $m^2_{\alpha,c}(rZ) = r^4 m^2_{\alpha,c}(Z)$. Hence the unrestricted pointwise carrier is gapless. The complete presymplectic reduction of the displayed worldline action is now proved. For $m_0 > 0$, the rank-two colour action is free, $DP_{\alpha,Z}$ is surjective with kernel $\mathfrak{su}(3) \cdot Z$, and the characteristic quotient is

$$\big(\mathcal{J}_2 \times \mathcal{H}^+_{m_0,c}\big)/\big((x,p) \sim (x + sp, p)\big) \cong_{\text{symp}} T^*\mathcal{H}^+_{m_0,c};$$

thus this stratum is exactly the future-timelike spinless relativistic particle. For $m_0 = 0$, the intrinsic nonzero rank-one quotient similarly gives $T^*\mathcal{N}^\times_+$, but the written multiplier action is irregular because

$$\det(\alpha Z^* Z) = \alpha^2 \sum_{a<b} |\Delta_{ab}(Z)|^2, \qquad d(\det \circ P_\alpha)_Z = 0 \quad (\operatorname{rank}_{\mathbb{C}} Z \leq 1).$$

Its rank-one equation is $\dot{x} = \lambda p$, with no equation $\lambda = e$; four real transverse rank-opening directions are first-order null but are not colour gauge, and the apex $Z = 0$ admits arbitrary $x(\tau)$. On the later hard-constraint carrier with $\omega_0 \neq 0$, the momentum projection has rank four for $w \neq 0$ and rank three at $w = 0$; direct variation forces $\dot{x} = 0$ strictly above the sharp threshold and gives $\dot{x} \in \mathbb{R}p$ only at the threshold. Its determinant bound and multiplication/Fock spectra therefore remain exact algebraic and free-operator results, but the multiplication operator is not the Hamiltonian generating the displayed worldline action and the threshold is not a dynamically generated interacting gap. See theorems 23.2 to 23.7, 27.6, 27.9, 27.12 and 27.16.

The same marked data now carry an exact field-theoretic continuation on the affine normal slice

$$\mathcal{N}_P = \{Y \in \operatorname{Herm}_3(\mathbb{C}) : Yu = 0\} = \jmath_P\big(\operatorname{Herm}_2(\mathbb{C})\big).$$

Polarization of $\det(uu^* + \varepsilon \jmath_P(G)) = \varepsilon^2 \det G$ gives a constant metric of signature $(1,3)$, and the congruence action induces the spin cover $SL_2(\mathbb{C}) \to SO^+(1,3)$. The standard Moore–Penrose metrics select

$$W_P = R^* R = C^{-1}, \qquad K_P = \{a \in SL_2(\mathbb{C}) : a^* W_P a = W_P\} \cong SU(2),$$

acting on the real normal field as $\mathbf{1} \oplus \mathbf{3}$. Gauging this pointwise action yields an explicit nonnegative classical gauge–scalar action; the centre acts trivially on the normal field, so it cannot screen a

fundamental centre charge without the retained centre-nontrivial spinor or another added field. The candidate inverse-dual matrix remains outside the positive Gram momentum cone because

$$\det G_{\eta,T} = -\frac{2}{3} D_T < 0, \qquad \operatorname{im} p_\alpha = \mathrm{Herm}_2(\mathbb{C})_{\geq 0}.$$

On this same Lorentzian host, the Klein–Gordon/Fock construction has exact gap m_{KG} for an independently supplied $m_{\mathrm{KG}} > 0$, is gapless at $m_{\mathrm{KG}} = 0$, and the massless compact gauge Hessian has lowest nonzero Gaussian frequency $2\pi/L \to 0$. Thus neither the determinant identity, the Gram cone, nor entanglement alone chooses an interacting mass gap. A full-domain comparison satisfying a uniform norm lower bound and energy-form upper bound would instead imply the quantitative estimate $\mathrm{gap}(A_{\mathrm{phys}}) \geq \gamma c_R^2 / C_R$; an intertwiner on only one gapped subspace does not. See theorems 28.2, 28.5, 28.6, 28.8, 28.10, 28.12 and 28.16.

The selectively corrected Jordan determinant/Euler audit identifies an exact family of Lorentz sections inside the marked Hermitian cone. With $[a]_\times v = a \times v$, one has for every real symmetric S

$$\det\bigl(S + i[a]_\times\bigr) = \det S - a^{\mathsf{T}} S a.$$

Hence every marked positive monodromy form B defines

$$\ell_B(\lambda, a) = \lambda B + i[a]_\times, \qquad \ell_B^{-1}(\mathrm{Herm}_3(\mathbb{C})_{\geq 0}) = \{\lambda \geq 0 : a^{\mathsf{T}} B a \leq \lambda^2 \det B\}.$$

There is also a real-linear order-cone bijection from $\mathrm{Herm}_2(\mathbb{C})_{\geq 0}$ onto the positive part of $\mathbb{R} I_3 \oplus i\mathfrak{so}(3)$ for which

$$\det_3 \Theta(H) = \frac{\operatorname{tr} H}{2} \det_2 H;$$

it sends positive-semidefinite ranks $2, 1, 0$ to $3, 2, 0$. If P is a rank-one projection, then $\Theta(P)$ has eigenvalues $0, 1/2, 1$ and is not idempotent, which separately proves that Θ is not a Jordan homomorphism. These are exact algebraic and order-theoretic maps, not holomorphic toric, period-lattice, or filling maps. In particular, the toric limit $z_0 z_1 z_2 z_3 = t \to 0$ keeps $B = I_3$ fixed and is not the operation $\det B \to 0$.

The corresponding rank-three toroidal degeneration is now constructed as a separate analytic object. Its smooth fibre is E_τ^3, its deepest local equation is $t = z_0 z_1 z_2 z_3$, and its irreducible central fibre has stratum counts

$$(f_0, f_1, f_2, f_3) = (1, 7, 12, 6), \qquad \chi(W_3) = 6.$$

Its normalization has orbit vector $(1, 14, 36, 24)$; the strata of positive codimension map with degrees $2, 3, 4$ onto the $7, 12, 6$ quotient strata. Thus any real S^8-total-space fibration retaining W_3 must have its remaining singular fibres contribute Euler sum -4. More generally, an injective integral shear $\mathcal{B} : \bar{L}_3 \hookrightarrow L_3$ gives

$$\chi(W_{3,\mathcal{B}}) = 3!\,[L_3 : \mathcal{B}\bar{L}_3] = 3!\,|\det \mathcal{B}|.$$

The toroidal object is not merely analogous to the reconstructed candidate cusp: the explicit epimorphisms

$$P = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix}, \qquad Q = \begin{pmatrix} 0 & -1 & 1 \\ 1 & -1 & 0 \end{pmatrix}$$

give $\Phi = P \oplus Q : H_1(T^6) \twoheadrightarrow H_1(T^4)$ with

$$\Phi M_3 = M_F \Phi, \qquad c_F \Phi = Q c_3, \qquad P Z_3(t) = Z_2(t) Q.$$

This extends to a holomorphic map of local cusp families with connected elliptic kernel. An explicit unimodular change of basis identifies the rank-three cusp matrix with $M_F \oplus \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$, and compatible order-three and order-four affine peripheral actions exist on the added elliptic factor. Under the marked A_2-Cartan identification, the six Kuhn tetrahedron orbits map equivariantly to the six diagonal Weyl chambers. This last correspondence is fan-theoretic: it neither identifies the four-branch toroidal germ with the quadratic Gram-rank germ nor supplies the still-open analytic gluing over both finite special fibres.

The free rank-six object native to this construction is now identified without conflating it with a cyclic quotient or with the different defect group on the original regular locus:

$$\Lambda_{\mathrm{tor}} = H_1(T^6; \mathbb{Z}) = \mathbb{Z}^3_{\mathrm{angular}} \oplus \mathbb{Z}^3_{\mathrm{deck}}.$$

With diagonal coordinate permutations it fits into the exact sequence of integral S_3-modules

$$0 \longrightarrow \mathbb{Z}(k,0) \oplus \mathbb{Z}(0,k) \longrightarrow \Lambda_{\mathrm{tor}} \xrightarrow{P \oplus Q} A_F \oplus B_F \longrightarrow 0, \qquad k = (1,1,1).$$

The sequence splits integrally with monodromy but not over $\mathbb{Z}[S_3]$; its quotient copies are A_2 weight lattices, with the root lattices sitting at index three. The rank identity $\operatorname{rk} \Lambda_{\mathrm{tor}} = 6$, the chamber count $|S_3| = 3! = 6$, and $\chi(W_{3,\mathcal{B}}) = 3! |\det \mathcal{B}|$ are thereby separated: the six homology basis vectors form two three-element S_3-orbits, whereas the six Kuhn/Weyl chambers form one regular orbit. That regular orbit does occur exactly in the associated rank-nine carrier:

$$\operatorname{Hom}(\overline{L}_3, L_3) = \bigoplus_i \mathbb{Z} E_{ii} \oplus \bigoplus_{i \neq j} \mathbb{Z} E_{ij},$$

where the off-diagonal matrix units form $\mathbb{Z}[S_3]$, and the chamber $\pi = (\pi_1 \pi_2 \pi_3)$ corresponds to E_{π_1, π_3}. Equivalently, the chambers are the six mixed angular–deck basis pairs with zero alternating pairing; the three excluded diagonal pairs have pairing one. The real-linear S_3-isomorphism

$$X \longmapsto \frac{X + X^{\mathsf{T}}}{2} + i \frac{X - X^{\mathsf{T}}}{2}$$

identifies the ambient matrix representation with $\mathrm{Herm}_3(\mathbb{C})$, but has integral index 2^3 in the symmetric/skew coordinates and sends every off-diagonal E_{ij} to an indefinite Hermitian matrix with eigenvalues $2^{-1/2}, -2^{-1/2}, 0$. It is therefore an exact representation and chamber-incidence bridge, not a positive, Jordan, determinant, or toric morphism. The separate map into the Gram boundary is only the nonadditive piecewise-linear chamber map $\Phi_{\mathrm{G}} \circ \jmath \circ P$ on the angular cocharacters; it ignores the deck half and does not descend to the torus. See theorems 21.1 to 21.3, 22.1 and 22.4. This successful lift is distinct from the diagonal local-system candidate

$$M_1 = I_3 \otimes S, \qquad M_2 = I_3 \otimes Q, \qquad M_0 = I_3 \otimes T, \qquad M_1 M_2 M_0 = I.$$

Although its finite factors have orders 4 and 6, each $I - M_j : T^6 \to T^6$ is an isogeny for $j = 1, 2$; consequently every affine lift of either finite action has a fixed point.

A literal complex-fourfold analogue on S^8 is impossible: an almost complex tangent bundle would have $c_1 = c_2 = c_3 = 0$, $c_4 = 2u$, hence $p_2 = 4u$, while the signature formula forces $p_2 = 0$. A nonsingular T^6-bundle over S^2 is likewise impossible; any real singular replacement must derive its actual filling maps and satisfy the displayed Euler, van Kampen, Wang, and Mayer–Vietoris tests. For its regular-piece monodromy ρ, the corrected homological Serre filtration is

$$0 \longrightarrow (\textstyle\bigwedge^q \Lambda)_\rho \longrightarrow H_q(E; \mathbb{Z}) \longrightarrow H_1(B^\circ; \bigwedge^{q-1} \Lambda) \longrightarrow 0,$$

while the dual edge map is $H^q(E; \mathbb{Z}) \twoheadrightarrow (\bigwedge^q \Lambda^\vee)^\rho$. Thus the filling maps must preserve the coinvariant subobject and induce the required map on the base-one-cycle quotient; using the reverse filtration invalidates the map-level test. Every effective torus action on S^8 has rank at most four, and the bound is sharp for the coordinate action

$$S^8 / T^4 \cong \mathrm{Susp}(\Delta^3) \cong B^4.$$

This is a real two-pole singular compactification with generic T^4 orbit, not a three-special-fibre T^6-fibration over S^2.

The exact bundle-theoretic alternative occurs on the regular locus $U = X \setminus (W \sqcup S_1 \sqcup S_2)$:

$$H^4(U; \mathbb{Z}) \cong H_1(W; \mathbb{Z}) \oplus H_1(S_1; \mathbb{Z}) \oplus H_1(S_2; \mathbb{Z}) \cong \mathbb{Z}^2 \oplus \mathbb{Z}^2 \oplus \mathbb{Z}^2.$$

Principal $SU(3)$-bundles on U are classified by this complete c_2-class and extend across the three deleted fibres exactly when it vanishes. This is a free rank-six defect lattice, not a cyclic group of order six. The formulas from this point through the normal-holonomy and common-centre comparison are exact results of the dated fibre-restriction, defect-lattice, and common-centre auxiliary audits. Their full theorem–proof source has not yet been integrated into a substantive research fragment of this workbench, so they are recorded here without a **[VERIFIED]** status claim. In the marked basis

$$(a_0, b_0, a_1, b_1, a_2, b_2) = (\bar{\gamma}, \bar{u}, g_1, h_1, g_2, h_2),$$

the actual collar matrix gives the primitive fibre-degree functional

$$[\deg_F] = (12, 0, 4, 0, -3, 0), \qquad \ker \deg_F = \mathbb{Z}\langle b_0, b_1, b_2, a_0 - 3a_1, 3a_1 + 4a_2\rangle.$$

Define

$$q_{\mathrm{col}} : L \longrightarrow H_1(X^\circ; \mathbb{Z}), \qquad a_0 \mapsto c, \quad a_1 \mapsto \bar{g}_1, \quad a_2 \mapsto \bar{g}_2, \quad b_0, b_1, b_2 \mapsto 0,$$

and let $\Phi_1 : H_1(X^\circ; \mathbb{Z}) \to \mathbb{Z}$ have row $(12, 4\ell_1, 3\ell_2)$ in the basis $(c, \bar{g}_1, \bar{g}_2)$. Then the typed identity is $\deg_F = \Phi_1 \circ q_{\mathrm{col}}$, and $\Phi_1(j_*[\widetilde{a}_0]) = p$. The order-three and order-four normal holonomies combine, with the candidate's relative sign, to characters

$$\chi_{12} = \rho_{12} \circ \deg_F, \qquad \chi_6 = \chi_{12}^2 = \rho_6 \circ \deg_F, \qquad \rho_m(k) = e^{-2\pi i k/m}.$$

Thus the natural finite quotients are $H^4(U)/\ker \chi_{12} \cong \mathbb{Z}/12$ and $H^4(U)/\ker \chi_6 \cong \mathbb{Z}/6$; the latter is not inferred from the rank. Ordinary coefficient reduction instead gives $H^4(U; \mathbb{Z}/6) \cong (\mathbb{Z}/6)^6$; this is not the single distinguished cyclic quotient selected by χ_6. Equivalently, for a principal $SU(3)$-bundle $P \to U$,

$$\chi_6(c_2(P)) = \exp\left(-\frac{2\pi i}{6}\langle c_2(P)|_F, [F]\rangle\right).$$

Tensor product supplies a separate exact centre statement:

$$SU(2) \times SU(3) \hookrightarrow SU(6), \qquad (A, B) \longmapsto A \otimes B, \qquad \mu_2 \times \mu_3 \xrightarrow{\sim} Z(SU(6)) = \mu_6.$$

On the five-dimensional CW model of U, c_2 classifies $SU(n)$-bundles for $n \geq 3$, and $c_2(E_2 \otimes E_3) = 3c_2(E_2) + 2c_2(E_3)$. Hence every chosen $h \in H^4(U; \mathbb{Z})$ is represented by an $SU(6)$-bundle admitting a tensor-subgroup reduction; its c_2-factorizations are

$$(e, f) = (h + 2t, -h - 3t), \qquad t \in H^4(U; \mathbb{Z}),$$

so the geometry selects neither a nonzero h nor a preferred reduction. Separately,

$$H^5(U; \mathbb{Z}) \cong \ker\{\mathbb{Z}^3 \xrightarrow{(1,1,1)} \mathbb{Z}\} \cong Q(A_2)$$

for the combinatorial form making the three support components orthonormal; this is not a cup-product form and does not identify $\ker \deg_F$ with A_5. The finite-group identification $H^4(U)/\ker \chi_6 \cong Z(SU(6))$ is not an $SU(6)$-connection or a gauge-transition cocycle. The tensor map is injective and does not factor through $(SU(2) \times SU(3))/\mu_6$, because its central μ_6 maps to nontrivial scalars. The distinct common-centre map is

$$\Gamma_6 = \{((\lambda\mu)^{-1}, \lambda I_2, \mu I_3) : \lambda \in \mu_2,\ \mu \in \mu_3\}, \qquad \frac{U(1) \times SU(2) \times SU(3)}{\Gamma_6} \hookrightarrow U(6), \quad [z, A, B] \longmapsto z(A \otimes B),$$

where $\Gamma_6 \cong \mu_6$. Its determinant is z^6, and its image meets $SU(6)$ in exactly the tensor subgroup $\iota(SU(2) \times SU(3))$; it is not generally contained in $SU(6)$.

Fibre evaluation therefore gives a canonical surjection

$$\bar{\kappa}_F : H^4(U; \mathbb{Z}) \twoheadrightarrow H^4(T^4; \mathbb{Z}) \twoheadrightarrow \mathbb{Z}/6\mathbb{Z}.$$

Identifying this degree-four quotient with a degree-two common-centre bundle obstruction remains unproved. Global $SU(3)$-bundles on S^6 instead carry the separate clutching class $c_3 = \pm 2ku$. The Lorentz-observation module and its exact presymplectic reduction identify the massive particle, the punctured massless quotient, the irregular massless multiplier/apex branches, and the strict-interior versus threshold dynamics of the hard-constraint packet. These carrier, reduction, and bundle theorems do not supply a Yang–Mills mass-gap conclusion: the required field connection, interacting operator and domain, gauge-invariant spectral sector, and continuum comparison morphism remain to be constructed.

Reviewer B independently reviewed the reconstruction existing up to its evidentiary cutoff and provisionally agreed on the non-collapse route; the later Codex proofs here are separate. In a 26 August follow-up, Reviewer B also independently supplied the finite-complex/UCT proof of top-degree base

change and confirmed the conductor–duality nonvanishing from the three named candidate inputs; see [36]. Throughout, **[VERIFIED]** means that a complete auditable proof is written in this document, not that specialist replication or community acceptance has already occurred. The candidate construction and its original claims remain attributed to their reported source. This paper is an AI-produced verification aid: its own contribution is to expose the relevant morphisms, signs, lattices, comparison maps, and singular-fibre arguments as complete checkable proofs, thereby reducing the reconstruction work required of human experts. It makes no claim of priority for the candidate and does not replace independent specialist adjudication. The 4 September divisor-channel audit verifies the oriented composite-shell bijection and the diagonal-ray density-one argument. It also constructs a separate rational map from $ac - b^2 = -\delta$ to the defect-one Koide surface, with an explicit inverse on stated open sets and exact image denominators; this does not replace the supplied orthogonal map, whose rational image is empty. See Section 8. The quaternionic continuation audit identifies the two proposed Lorentz $4+4+4+1$ slices as the same embedded space, with $w = -\varepsilon\overline{\chi}$, and proves the positive-cone sphere pair (S^{12}, S^8). The retained marking transports the Jordan product to $(MDN + NDM)/2$, $D = \mathrm{diag}(C^{-1}, 3)$, and its closure is the full fifteen-dimensional Hermitian quaternionic algebra. For the supplied scalar mechanics the omitted coordinate $\rho = u^* b$ instead obeys $\ddot{\rho} = -(m^2 + \Gamma s)\rho - \kappa u^* \Phi b$; explicit initial data give $\ddot{\rho} = -2\kappa$. The exact constrained alternative and the gauge-sign correction are in Section 24. Within the unrestricted equations, fixed complex $W \subset u^\perp$ with $b \in W$, $\Phi W \subset W$, and tangent velocities yield global nonzero invariant sectors of configuration dimensions six and eight, without reaction forces. Other initial data have zero normal derivatives through order three but $\rho^{(4)}(0) = -12\kappa$. The fixed-subspace sectors are not exhaustive: an exact periodic moving-kernel family remains in $\rho = 0$ outside all of them. At $(m^2, \Gamma, \kappa) = (1, 1, 6)$ its norm is $s = 1$, energy $7/4$, and conserved moment $-\Omega/2 \neq 0$, so its zero-connection gauge lift fails Gauss' equation. The scalar global-minimum transition at $m^2 = \kappa^2/(27\Gamma)$ and full Hessian multiplicities $1, 8, 5$ are independently verified; no gauge-reduced quantum spectrum follows. The full positive quaternionic carrier admits a split positive observation to $\mathrm{Herm}_2(\mathbb{C})$ whose exact permitted pairs satisfy $G \succeq 0$, $|\rho|^2 \leq \det G$. Every such pair lifts, and explicit positive lifts approach determinant zero with $\rho \neq 0$; no uniform positive gap follows from this cone. For trace-free scalar matrices the separate map $Q \mapsto Q - \lambda_{\min}(Q) I_3$ gives a positive-boundary observation retaining PP^*, not an assumed positive scalar trajectory. The retained rank-six period matrix has complete positive real metric fibres of dimensions twelve over its Hermitian Gram matrix and seventeen over its Lorentz observation. A rank-losing observation is realized by an explicit equivariant metric with diverging relative condition number; bounded relative condition number excludes that rank loss. For the periodic scalar orbit, every homogeneous temporal-gauge lift fails Gauss' equation, even allowing arbitrary spatial matrices. A different interacting radial family solves all gauge and scalar equations exactly. On the positive rank-one boundary, colour-group averaging converts the marked determinant to the gauge invariant $\mathrm{tr}\,\mathrm{adj}(A) = \|b\|^2$, which varies continuously along an explicit family of scalar minima. These are classical observation and field-equation results, not an interacting quantum gap. The exact continuation now supplies a current-free three-real-coordinate sector, nonlinear null waves, and periodic full-gauge motions with arbitrarily small positive frequency and vacuum energy excess at fixed volume. The periodic amplitude stays finite. The full transverse gauge-inertia spectrum and conserved invariant current identify the branch $\mu_{\mathrm{d}}^2 = (2g^2/3)\|b\|^2$ with the colour-averaged Lorentz determinant. The literal earlier rotating orbit has no arbitrary-time real-connection lift and no constant-spatial-field complex lift in its constant-matter frame. A further genuinely complex rotating-pair block class and its constant spatial rotations are excluded. Without that block restriction, Gauss gives $C = \omega T$ with $T = 4(g^2\mathcal{I} + 2\mathcal{K})^{-1}[Z_1, Z_2]$ and the full scalar equation fixes $\omega^2 = (1/4 + \sum_j \|Z_j q\|^2)/\|Tq\|^2$. All remaining matrix/vector equations and both frequency signs are retained in a proved inverse correspondence; its zero set and more general complex time-dependent lifts remain open. The full scalar equations force $s_C^2 < s_{Z_1}^2 + s_{Z_2}^2$ for $s_X = -i\,\mathrm{tr}(HX)$ unless all three vanish; the two spatial equations give an additional positive stress matrix and its exact sum-of-squares discriminant, without a block ansatz.

For each fixed unrestricted spatial pair, we now prove an explicit bounded coupling interval and a nonnegative residual polynomial of degree exactly twenty-eight whenever its domain polynomial is nonzero. Its admissible positive roots are precisely the allowed original g^2: there are at most fourteen, with both frequency signs reconstructed. Three vector tests decide the identically empty Gauss domain. A nonzero, explicitly defined 24×24 Sylvester determinant polynomial further confines the entire spatial solution projection to a closed, nowhere-dense, measure-zero subset of the sixteen-coordinate carrier.

An exact unrestricted witness and rational and finite-field Bezout certificates prove that polynomial is nonzero. The eight-dimensional linear class in which both spatial matrices exchange the matter line with its orthogonal complement lies inside this exceptional set, but a full-equation positivity identity excludes every positive coupling and both frequency signs there. A denominator-free residual identity also gives a strictly positive spatial-error bound on its scalar–Gauss zero set, uniform for bounded spatial norm and coupling bounded away from zero. This scalar–Gauss zero set is now completely parameterized: the classified domain is $\mathcal{D} \times \{1,-1\}^2 \times U(1)^2$, and the literal domain retains the frame phase subject only to $(\zeta, p_+, p_-) \sim (-\zeta, -p_+, -p_-)$. An explicit family attains every original $t = g^2 > 0$, with all scalar and Gauss entries zero. The two spatial residuals cannot vanish simultaneously, and existence on the remaining exceptional set is open.

In the period bridge, Hermitian compatibility makes W a lattice cometric with $h_3 = 2H_3^{-1}$. The native Euclidean choice gives $H_3 = 2I_3$, $G = 2PP^*$ and the exact source/quotient Laplace asymptotics $4\pi^2/y^2$, $8\pi^2/y^2$. Constant observed cometric therefore coexists with expanding periods and vanishing positive Laplace gaps. For the full homogeneous gauge–scalar mechanics, retaining all thirty-eight configuration coordinates and imposing its constant-$SU(3)$ Gauss law, a new direct form estimate proves compact resolvent and a unique invariant quantum ground state. Its first physical excitation gap is strictly positive at each fixed positive volume and coupling. This finite-model theorem does not identify the homogeneous quantum mechanics with an invariant subsector of the full torus field theory: an explicit mixed Fourier-mode quartic term obstructs the automatic fixed-vacuum product construction. No continuum or volume-uniform quantum gap is inferred. For that exact two-coordinate Fourier mechanics, the actual positive quartic fibre ground state gives an isometry $Jf = f\Omega_q$ and $H_2 J - J H_{\mathrm{eff}} = -\hbar^2(2f'\Omega_q' + fQ_q\Omega_q'')/(2\mathcal{V}_3)$. The effective Hamiltonian retains its ground energy and the nonzero Born–Huang potential. Explicit global derivative bounds prove a finite-energy physical-time error of order $\mathcal{V}_3^{-1/2}$, with the other parameters fixed and a stated sufficient volume regime. This is a controlled finite-model correspondence, not a continuum or thermodynamic mass-gap result.

Contents

1 Results and evidence boundary **27**

2 Mathematical results so far **64**

3 Methodology **106**

4 Relation to existing literature and exact scope **107**
- 4.1 Targeted corpus cross-check (29 August 2026) . . . 111
- 4.2 Stable-stem audit of the sphere-valued carriers (3 September 2026) . . . 112
- 4.3 Effective counting for the exact Koide sector . . . 113
- 4.4 Constrained dynamics and symplectic reduction . . . 114

5 Exact integral algebra of the printed monodromy data **117**
- 5.1 The cusp invariant, image, and coinvariant lattices . . . 118
- 5.2 Global coinvariants . . . 118
- 5.3 Finite quotient fibres . . . 120
- 5.4 The terminal presentation and degree-two map . . . 121

6 An exact Erdős–Straus/S6 terminal-matrix crosswalk **123**
- 6.1 The first-half shell and its divisor coordinates . . . 123
- 6.2 The unimodular relation-matrix bijection . . . 124
- 6.3 What the crosswalk proves, and what it does not . . . 126

7 The exact marked Erdős–Straus–S^6 bridge **126**
- 7.1 All objects and conventions . . . 127

7.2 The complete first-half shell selector . . . 128
7.3 The exact selector–terminal-lattice equivalence . . . 130
7.4 The exact remaining arithmetic incidence . . . 133
7.5 Why the labels and support cannot be forgotten . . . 134
7.6 The natural unmarked S^6 point and its complete divisor gate . . . 135
7.7 Exact descendant maps transported through the bridge . . . 136
7.8 Sharp verdict . . . 137

8 Divisor channels, density one, and two distinct arithmetic maps 137
8.1 The complete oriented channel map . . . 137
8.2 The integral presentation, with its exact realization boundary . . . 138
8.3 An exact density-one statement and what it leaves out . . . 138
8.4 The Bézout opening, with the positivity hypothesis retained . . . 139
8.5 The original map's rational nonintersection . . . 140
8.6 A different rational bridge, with exact inverse and denominators . . . 141
8.7 Independent checks and reproducibility . . . 142

9 The fifteen $p = 1201$ selector rays inside $\mathrm{Herm}_3(\mathbb{H})$ 142
9.1 Marked selectors and their prime families . . . 142
9.2 The exact Koide cone map . . . 143
9.3 Embedding and rank obstruction in the quaternionic Jordan algebra . . . 144
9.4 Symmetry and incidence obstructions . . . 145
9.5 Conclusion and next exact test . . . 145

10 The analytic threefold constructed in the candidate 146
10.1 The period family . . . 146
10.2 The toric filling at the cusp . . . 153
10.3 The finite logarithmic fillings and the global gluing . . . 159

11 Flat spectral geometry on the positive radial cusp 163
11.1 The Kummer cover extends across the toric cusp . . . 181
11.2 Cone-compactified sphere carriers for the native cusp quotient . . . 185
11.2.1 Optimal sphere exponents and the exact Euclidean boundary . . . 195
11.3 The full $S^{12} \to S^6$ carrier is null-homotopic . . . 208
11.4 The original cusp coefficient: an explicit framed nullbordism . . . 209
11.5 Fixed-target metric realization and its sharp dimensional scope . . . 214
11.6 The exceptional first-coordinate elliptic quotient . . . 224

12 Classical gauge fluctuation operators on the positive radial cusp 228
12.1 The continuum Yang–Mills Hessian . . . 229
12.2 The finite cubical cochain complex . . . 229
12.3 Exact coordinate-mode eigenvalues and all mesh regimes . . . 232
12.4 Continuum Coulomb reduction and exact spectrum . . . 234
12.5 The physical boundary and the exact no-gap bridge . . . 236

13 Cubical holonomy and finite gauge systems 238
13.1 The transported cubical subdivision and continuum holonomy . . . 238
13.2 An explicit compact-link SU(2) flux . . . 240
13.3 Finite Gaussian quantization with independent cover and mesh . . . 242
13.4 The exponential map and the exact physical boundary . . . 244

14 Integral magnetic sectors and a reducible $SU(2)$ critical point on the radial cusp 245
14.1 The integral coframe . . . 246
14.2 The unit-flux line and the topologically trivial rank-two sum . . . 246
14.3 A non-flat reducible Yang–Mills critical connection . . . 248
14.4 Constant-curvature magnetic Laplacians for arbitrary integral nondegenerate flux . . . 249
14.5 A background-gauge negative mode . . . 252

15 The native full-flux magnetic sector 256
15.1 The invariant class, its cocycle, and both orientations . . . 257
15.2 The exact physical frequencies . . . 258
15.3 Joint Landau levels and their exact multiplicities . . . 259
15.4 The split $SU(2)$ critical connection . . . 262
15.5 The closed Jacobi operator and exact Coulomb reduction . . . 263
15.6 Charge two, all polarizations, and the Morse index . . . 264
15.7 The fixed and varying radial direct integrals . . . 266
15.7.1 Density and self-adjointness on the maximal domain . . . 269
15.7.2 Measurable eigenvectors, including at band crossings . . . 270
15.7.3 Full spectrum and spectral type . . . 270
15.7.4 Exact interpretive boundary . . . 272

16 The native full-flux split-Cartan $SU(r)$ extension 273
16.1 The split bundle, determinant, and Chern number . . . 273
16.2 Every nonzero integral root charge . . . 275
16.3 Closed gauge-fixed and Coulomb operators . . . 275
16.4 Rootwise spectrum, coincidences, index, and nullity . . . 277
16.5 Stability boundary, possible wall bifurcation, and exact scope . . . 279

17 The higher-rung SU(3) Gram cone 279
17.1 Conventions and the Gram morphism . . . 280
17.2 Exact orbit classification . . . 280
17.3 The free locus, its explicit trivialization, and orientation . . . 282
17.4 Lorentz determinant and Plücker norm . . . 283
17.5 The four-coefficient form is additional input . . . 284
17.6 The compact representation quotient . . . 285
17.7 Exact gauge-theoretic boundary . . . 286
17.8 The log-Euclidean metric and Yang–Mills theory are extra choices . . . 287

18 The three-column SU(3) Gram cone and its rank-loss S^8 287
18.1 Conventions . . . 287
18.2 The U(3) orbit space . . . 288
18.3 The SU(3) orbit space . . . 291
18.4 Rank strata, the standard S^8, and determinant winding . . . 293
18.5 The exact rank-two/rank-three distinction . . . 295
18.6 The alternating invariant and the precise nonclaim . . . 295

19 A four-branch toroidal lift and the obstruction to a complex S^8 total space 296
19.1 The exact meaning of the three-way cusp . . . 296
19.2 The rank-three periodic fan . . . 296
19.3 The analytic quotient and the abelian-threefold fibres . . . 297
19.4 Monodromy and the cusp inclusion on fundamental groups . . . 299
19.5 The central fibre, its normalization, and all strata . . . 300

19.6 Two independent obstructions to a complex S^8 globalization . . . 301
19.7 An explicit integral monodromy candidate and its failure . . . 302
19.8 Exact replication requirements for any real singular $T^6 \to S^8 \to S^2$ candidate . . . 303
19.9 The Hermitian determinant boundary is not the four-branch fibre . . . 305
19.10 Exact conclusion and permissible carriers . . . 307

20 An exact toroidal elliptic bridge to the candidate cusp 307
20.1 The candidate lattice and filling data . . . 307
20.2 The rank-three cusp . . . 308
20.3 The exact A_2 quotient of the Kuhn fan . . . 309
20.4 Monodromy and filling compatibility . . . 310
20.5 The actual period correction and the cusp-family morphism . . . 310
20.6 A simultaneous unimodular splitting . . . 312
20.7 The complete finite-order lattice lift . . . 313
20.8 The six toric chambers and the diagonal Gram boundary . . . 314
20.9 The exact Jordan determinant and Lorentz sections . . . 315
20.10 The Gram determinant does not supply this bridge . . . 317
20.11 The remaining global problem . . . 319

21 The rank-six cusp lattice inside the nine-dimensional cone program 319
21.1 The two rank-six groups have different types . . . 319
21.2 The integral S_3-module of the cusp . . . 320
21.3 The six homology generators are not the six chambers . . . 321
21.4 The exact map into the Gram boundary . . . 322

22 The chamber torsor inside the nine-dimensional matrix carrier 323
22.1 The signed quotient and the exact chamber incidence . . . 324
22.2 The real Hermitian representation and its integral defect . . . 326
22.3 The positive six-set is a different equivariant realization . . . 327

23 The Lorentz observation morphism and the full nine-dimensional fibre 327
23.1 The marked angular epimorphism and its exact splitting . . . 327
23.2 A positive observation morphism with all fibres retained . . . 328
23.3 Complete block coordinates and the cubic coupling . . . 329
23.4 The embedded Lorentz group and the meaning of $9 = 4 + 4 + 1$ . . . 330
23.5 The Lorentz cone inside the rank-loss S^8 and its normal opening . . . 332
23.6 The ten-dimensional determinant-phase carrier in the same coordinates . . . 333
23.7 The holomorphic toric seed and its Lorentz-orbit completion . . . 333
23.8 An exact colour-invariant Lorentz momentum and constrained dynamics . . . 334

24 The common marked slice and its dynamics 336
24.1 Conventions, determinant, and the retained marking . . . 336
24.2 The two constructions are the same embedded space . . . 338
24.3 Positive fibres and the embedded sphere pair . . . 339
24.4 The nonunitary Jordan-product correction . . . 340
24.5 The corrected group-action convention . . . 340
24.6 The full carrier still has an exact four-dimensional observation . . . 341
24.7 The actual period-to-observation bridge . . . 342
24.8 The unrestricted and constrained scalar equations . . . 354
24.9 Preserved interacting sectors of the unrestricted equations . . . 356
24.10 Periodic motions outside every fixed-subspace family . . . 358

24.11 The complete scalar vacuum calculation with its kinetic metric . . . 360
24.12 Gauge kinetic sign and precise scope of this audit . . . 363
24.13 The full homogeneous gauge equations and the literal periodic orbit . . . 363
24.14 Constant-matter coordinates and an all-time real-connection obstruction . . . 367
24.15 Constructive homogeneous gauge solutions and their observations . . . 370
24.16 Vacuum kinetics and exact current-free continuations . . . 373
24.17 The gauge-vector spectrum and an invariant vector observable . . . 379
24.18 The retained system and the scope of the result . . . 381
24.19 Universal necessary identities in the unrestricted complex class . . . 383
24.20 A new genuinely complex block exclusion . . . 383
24.21 The precise unresolved part . . . 387
24.22 Exact unrestricted Gauss and frequency elimination . . . 387
24.23 Complete scalar-coordinate identities and an unrestricted signature constraint . . . 389
24.24 A two-by-two stress obstruction before evaluating the full residuals . . . 392
24.25 An exact bounded finite coupling fibre for every unrestricted spatial pair . . . 393
24.26 A proper algebraic exceptional set for the full spatial projection . . . 399
24.27 An eight-dimensional exceptional spatial class with no full solution . . . 401
24.27.1 Literal system and the explicitly selected class . . . 401
24.27.2 Gauss and the complete scalar equations force the temporal form . . . 402
24.27.3 Every gauge entry in the classified frame . . . 404
24.27.4 Exact inclusion of the selected class in the exceptional locus . . . 406
24.27.5 An exact certificate retaining nonzero residuals . . . 408
24.27.6 Scope, provenance, and exact replay . . . 410
24.28 The complete scalar–Gauss domain at every positive coupling . . . 410
24.28.1 Literal data, residuals, and coordinate maps . . . 410
24.28.2 Every scalar and Gauss entry in the classified frame . . . 411
24.28.3 The exact realization of the two Hermitian matrices . . . 412
24.28.4 The complete literal scalar–Gauss domain and its phase quotient . . . 414
24.28.5 A short exact family with every equation checked . . . 419
24.28.6 Exact scope and reproducibility . . . 420
24.29 Smooth literal inverse charts and all positive-coupling fibres . . . 421
24.29.1 Retained objects and the exact parameter domain . . . 421
24.29.2 The phase quotient with every original phase retained . . . 422
24.29.3 Global formulas for literal matrices and their smooth inverses . . . 423
24.29.4 The original coupling is a surjective submersion . . . 425
24.29.5 Exact scope and topology of the positive-coupling level sets . . . 426
24.30 Sources, independent checks, and reproducibility . . . 427

25 Quantum confinement in the homogeneous gauge–scalar model 428
25.1 The retained configuration space and action . . . 429
25.2 The retained cubic and a global scalar bound . . . 431
25.3 Compactness for the entire coupled Hamiltonian . . . 438

26 An exact moving-ground-state comparison for the retained Fourier pair 442
26.1 Exact coefficients, measures, and self-adjoint operators . . . 443
26.2 The actual normalized positive fibre ground state . . . 444
26.3 Explicit global derivative bounds for the true quartic state . . . 446
26.4 The exact projected Hamiltonian and its off-diagonal part . . . 447
26.5 Controlled physical-time and spectral comparisons . . . 448

27 Exact presymplectic reduction of the SU(3)-Gram worldline model **449**
27.1 Objects, constants, and equations . . . 449
27.2 Colour quotient and orbit types . . . 450
27.3 Forms after colour reduction . . . 452
27.4 Positive mass . . . 453
27.5 Massless rank one . . . 454
27.6 The zero-momentum vertex . . . 456
27.7 Kernel dimensions and exact verdict . . . 457
27.8 Deterministic verification companion . . . 457
27.9 Comparison with the extended exact Lorentz packet . . . 458
27.9.1 Claim ledger . . . 459
27.9.2 The full dynamics of the packet's constrained action . . . 459
27.9.3 Relation to the rank-one irregularity . . . 461

28 The Hermitian determinant normal slice as a Lorentzian host **461**
28.1 Fixed matrices and the exact normal metric . . . 461
28.2 Exact Lorentz and affine symmetry . . . 464
28.3 The compact gauge–scalar carrier selected by the standard metrics . . . 465
28.4 The candidate cusp matrix is an exact spacelike vector . . . 467
28.5 Units and the minimal continuum field model . . . 468
28.6 A complete finite regulator and its two limits . . . 470
28.7 An exact classical gauge-theory completion . . . 471
28.8 Exact obstructions and the sufficient spectral bridge . . . 473
28.9 Exact conclusion . . . 474

29 Exact constraints on an S^8 higher-rung analogue **475**
29.1 The literal complex-fourfold proposal . . . 475
29.1.1 Characteristic-class obstruction . . . 475
29.1.2 Independent arithmetic checks . . . 476
29.1.3 Exact scope of the obstruction . . . 476
29.2 Requirements for a real Fabel-style S^8 fibration . . . 476
29.3 Euler constraint for a finite multipole model . . . 477
29.4 Fundamental group and monodromy . . . 477
29.5 Integral homology: the exact gluing tests . . . 478
29.6 The torus-action bound and a maximal real orbit model . . . 479
29.7 A three-column Gram quotient whose quotient boundary is S^8 . . . 479
29.8 Retaining the $T^4/\mathbb{R}^4$ fibre geometry and placing $SU(3)$ on the regular locus . . . 480
29.8.1 The regular locus and its fourth cohomology . . . 480
29.8.2 Exact classification of $SU(3)$ bundles on U . . . 481
29.8.3 Extension across the three special fibres . . . 481
29.8.4 The distinct six-dimensional clutching charge . . . 482
29.8.5 Data required before a gauge-theory conclusion . . . 482
29.9 Consequences for the higher-rung search . . . 482

30 The ternary triality structure and its exact carrier morphisms **483**
30.1 The source construction and the toroidal comparison . . . 483
30.2 Sphere and Euler obstructions . . . 485
30.3 The exact toroidal successor over the cusp . . . 487
30.3.1 The cusp filling map, not merely its boundary lattice . . . 489
30.3.2 Finite peripheral data and the precise globalization gap . . . 491

30.4 The forced S9, S10, and rank-loss S8 . . . 492
30.4.1 A global marked observation on the S9 quotient . . . 496
30.4.2 The exact period-to-S9 and period-to-S8 maps . . . 497
30.5 The 24-dimensional octonionic flag . . . 499
30.5.1 The exact six-chamber comparison . . . 502
30.5.2 The octonionic determinant-boundary sphere . . . 503
30.5.3 The ordered Albert triality tensor and its global descent . . . 505
30.5.4 A coordinate-preserving triality map onto the determinant boundary . . . 512
30.5.5 An explicit equivariant flag cell and its degree-one collapse . . . 514
30.6 What the mod–12 and mod–24 arithmetic actually is . . . 522
30.6.1 The integral peripheral representation . . . 522
30.6.2 The distinct stable mod–12-to-mod–24 injection . . . 524
30.6.3 The original framed cusp coefficient is zero . . . 526
30.6.4 The retained-angle cusp quotient and its nonzero top-cell class . . . 530
30.6.5 The retained cusp class as a three-dimensional bundle inside the triality carrier 538
30.6.6 A nonzero Hopf–cusp morphism to the stable mod–24 group . . . 544
30.6.7 Boundary of the three tested numerical mechanisms . . . 548
30.7 The full Albert orbit quotient and the Cartan family . . . 549
30.7.1 Original space, invariants, and direct diagonalization . . . 549
30.7.2 Exact projectors, smooth orbit inverse, and regular family . . . 550
30.7.3 Retained mean, radius, and angular parameter . . . 551
30.7.4 Exact codimension and global normal framing . . . 552
30.7.5 Cartan equations and spherical principal curvatures . . . 554
30.7.6 Focal projective planes and their endpoint maps . . . 554
30.7.7 Exact map from the equal-diagonal triality slice . . . 555
30.8 The exact spectral map of the common marked lattice . . . 556
30.8.1 Domain and the original matrix . . . 556
30.8.2 Complete determinant and invariant map . . . 556
30.8.3 Exact eigenvalues, projectors, and directional inverse . . . 557
30.8.4 Singular loci and the exact Cartan angle . . . 558
30.8.5 Actual integral witnesses in the orbit family . . . 559
30.9 The Koide cone and the canonical period discriminant . . . 561

31 The modulo-840 lattice, triality, and exact shell maps 563
31.1 The finite multiplicative carrier and its four fibres . . . 563
31.2 The root lattice and its exact discriminant form . . . 564
31.3 The index-72 glue and all glue-class minima . . . 565
31.4 The common D4 root lattice, shortest weights, and triality . . . 568
31.5 The full indicated glue stabilizer and its triality quotient . . . 570
31.6 The exact reduced-shell divisor bijection . . . 573
31.7 A fixed-shell count does not descend to the residue fibre . . . 576
31.8 An actual residue cocycle and a retained-sample Fourier map . . . 577
31.9 The entire lattice automorphism group and its Weyl quotient . . . 578

32 The exact Leech neighbor of the retained index-72 lattice 583
32.1 The Weyl vector and the exact neighbor . . . 583
32.2 A complete rootlessness proof . . . 584
32.3 The exact holy-construction presentation . . . 585
32.4 The indices, discriminant morphism, and cyclic order twelve . . . 586
32.5 An exact Lorentzian quotient and symmetry transport . . . 587

32.6 The exact F_4 Coxeter action of order twelve . . . 588
32.7 An exact cyclic-triality isometry of the entire Niemeier space . . . 588
32.8 The Leech neighbour in the same three octonion coordinates . . . 591
32.9 Exactly which root isometries preserve this glue . . . 592

33 The arithmetic of the retained Albert cubic on both lattices 593
33.1 Original data and the exact polynomial . . . 593
33.2 Finite generators of both original lattices . . . 595
33.3 A complete finite calculation of the cubic value modules . . . 596
33.4 Frame dependence and exact symmetry transport . . . 598

34 The residue cubic action and the common integral Leech correspondence 600
34.1 A necessary and sufficient equality criterion . . . 600
34.2 The exact tetracode obstruction to a fixed neighbor . . . 601
34.3 The canonical orbit and its common integral lattice . . . 601
34.4 The exact discriminant form and its cyclic three planes . . . 603
34.5 The fixed fourth plane and its exact A_2^{12} extension . . . 605

35 The cubic on the common lattice and the fourth Niemeier extension 607
35.1 Integral generators of the common lattice and the fixed extension . . . 607
35.2 Complete finite certificates for the two new value groups . . . 608
35.3 Exact inclusions, quotient maps, and the factor of the Albert determinant . . . 610
35.4 Reproduction and scope of the exact certificate . . . 611

36 A cyclic Wilson Leech lattice in the original triality space 611
36.1 The two retained metrics and the octonion convention . . . 611
36.2 The explicit eight-dimensional lattice and two sublattices . . . 612
36.3 The binary glue code and its exact generators . . . 613
36.4 Cyclic symmetry, the fixed lattice, and exact three-primary glue . . . 614
36.5 Finite lattice symmetries and exact cubic tests . . . 616
36.6 Exact comparison with coordinate lattices and the marked lane . . . 617
36.7 Literal intersections with both retained lattice embeddings . . . 617
36.8 The cubic value group on the literal common lattice . . . 620
36.9 Primary source and exact reproducibility data . . . 620

37 The exact cubic image of the original common lattice 621
37.1 Necessary restrictions in the original coordinates . . . 621
37.2 Three explicit affine lines and their exact values . . . 621
37.3 The global image and a proved reconstruction map . . . 622
37.4 Exact local images with explicit local sections . . . 623
37.5 The exact factorization and quotient metric on the sum-zero slice . . . 624

38 The marked Niemeier lattice, its twelve theta indices, and the umbral shadow 625
38.1 The primary prescription and its complete coefficient matrix . . . 626
38.2 Quadratic refinements and the finite theta transformations . . . 627
38.3 The full Fourier determinant and the retained order-twelve character . . . 629
38.4 The residue Fourier map and what the Weyl quotient retains . . . 630
38.5 The shadow-to-Fricke map with the original scalar seventy-two . . . 633
38.6 Established scope and reproducibility . . . 634

39 Exact arithmetic of the Koide defect-one shell **635**
39.1 Vieta reduction and exact ternary coordinates . . . 635
39.2 The affine quadric and its local densities . . . 639
39.3 The archimedean chamber factor . . . 641
39.4 A genuine norm and the homogeneous counting theorem . . . 642
39.5 The 2-adic orbit sign and Borovoi's coefficient . . . 643
39.6 Calibration and the counting constant . . . 645
39.7 Exact computation and the remaining open problem . . . 646

40 Exact ideal and unit-orbit parametrization of the unchanged sharp Koide count **647**

41 Arithmetic of $\mathbb{Z}[\sqrt{3}]$ **647**

42 Hermite-normal-form parameters for every ideal **648**

43 Complete integral parametrization of the original equation **649**

44 Exact divisor and sharp-height formulas **652**

45 The phase term as exact Hecke-character data **654**

46 An unconditional effective exponent for the exact sector **655**

47 Exact obstruction to the square-root claim **658**

48 Boundary ledger **659**

49 Exact unit-orbit census through height 10^8 **660**
49.1 Complete result and checkpoint table . . . 660
49.2 Certified extrema over every integer . . . 661
49.3 Independent replay and exact algorithm . . . 661

50 Exact quaternion bridge: order, shell, and hyperbolic conjugacy classes **662**
50.1 Quaternion algebra and reduced invariants . . . 663
50.2 The proposed lattice is an order . . . 663
50.3 Trace-zero lattice and exact shell bijection . . . 664
50.4 Division over $\mathbb{Q}$ and splitting over $\mathbb{R}$ . . . 665
50.5 Centralizer, optimal quadratic order, and primitivity . . . 666
50.6 Integral adjoint action and finitely many classes . . . 668
50.7 Exact reformulation of the Koide chamber count . . . 669
50.8 Consequence for the open square-root discrepancy . . . 670
50.9 Missing mathematics before an error theorem can be claimed . . . 671
50.10 Reproducibility and references . . . 671

51 Primary-source theorem audit of effective counting for the fixed Koide ternary quadric sector **672**
51.1 Determination . . . 673
51.2 Exact arithmetic and geometric classification . . . 674
51.2.1 The rational group and stabilizer . . . 674
51.2.2 The exact height really is a norm-sector height . . . 675
51.2.3 Invariant volume with a strong enough secondary estimate . . . 675
51.2.4 The full integral locus is not assumed to be one orbit . . . 676

51.2.5 Exact quaternionic lattice and its congruence status . . . 676
51.3 Result matrix . . . 677
51.4 Benoist–Oh: exact applicability and exact limitation . . . 678
51.4.1 What the theorem assumes . . . 678
51.4.2 Definition 13.1 for the actual B_X . . . 679
51.4.3 Finite-orbit passage and main-term normalization . . . 680
51.4.4 Why the exponent remains unspecified . . . 680
51.5 The sharp hyperbolic frontier . . . 680
51.5.1 Good and Chatzakos–Petridis . . . 680
51.5.2 Current sharp refinements remain at 2/3 . . . 681
51.5.3 Fixed angular harmonics do not give a sharp interval for free . . . 681
51.6 Smooth spectral counting . . . 682
51.7 General effective sector and full-lattice theorems . . . 683
51.7.1 Gorodnik–Nevo's explicit generic exponent . . . 683
51.7.2 Browning–Gorodnik and subsequent full-lattice work . . . 683
51.8 Results in different arithmetic regimes . . . 684
51.8.1 Square discriminant and divergent stabilizer . . . 684
51.8.2 Varying level, isotropic forms, and special rotational quadrics . . . 684
51.9 Spectral-gap, GRH, smoothing, and averaging taxonomy . . . 685
51.10 Citation-ready conclusion . . . 685
51.11 Primary references and exact locators . . . 686
51.12 Audit limitations . . . 688

52 The exact gapless auxiliary operator and the physical boundary 688
52.1 A direct-integral scalar Laplacian with zero essential threshold . . . 688
52.2 Complex data do not select a spectral gap . . . 690
52.3 The exact bridge to a physical Hamiltonian . . . 690

53 Integral topology without the cusp-collapse argument 692
53.1 Integral specialization at the cusp . . . 692
53.2 The global E_2-page . . . 696
53.3 The three transgressions . . . 698
53.4 Based van Kampen and sphere recognition . . . 700

54 The direct algebraic-reduction proof 702

55 Structure-sheaf direct images and uniform non-torsion 705
55.1 The Hodge line and the finite normal characters . . . 705
55.2 Exact product covers, normalization branches, and peripheral kernels of the finite fillings 708
55.2.1 The original lattice, periods, and their complex summands . . . 708
55.2.2 The exact elliptic-product covers of the central tori and surfaces . . . 710
55.2.3 The varying quotient and its full local analytic coordinates . . . 711
55.2.4 Every branch in the normalization of the actual base change . . . 712
55.2.5 Universal covers and the full peripheral inclusion kernel . . . 713
55.2.6 The original punctured overlap and integral cohomology pullback . . . 714
55.2.7 The finite-to-global analytic gluing interface . . . 715
55.2.8 The exact normal character and a smooth trivialization . . . 715
55.3 The cusp structure sheaf . . . 716
55.4 Direct images, base change, and the Picard line . . . 717
55.5 One line bundle non-torsion on every fibre . . . 719
55.6 A local analytic obstruction to a cusp-only Picard argument . . . 720

55.7 The canonical divisor and the exact anticanonical ring . . . 725
55.8 The retained period constant is an effective deformation parameter . . . 726
55.9 Exact global identifications of the retained period parameter . . . 731
55.9.1 The middle family and its complete rational centralizer . . . 731
55.9.2 The exact finite obstruction, including normalization . . . 733
55.9.3 Local maps for every even shift . . . 734
55.9.4 The global additive correction and all overlap identities . . . 735
55.9.5 Intrinsic necessity and the resulting parameter space . . . 737

56 Hodge numbers, cotangent direct images, and the Frölicher page 737
56.1 Vanishing of holomorphic forms . . . 737
56.2 The exact one-parameter Hodge diamond . . . 739
56.3 Generic maximal variation and its elliptic zero . . . 740
56.4 Logarithmic cotangent direct images . . . 741
56.5 Residues and the absolute cotangent direct images . . . 745
56.6 The scalar and the exact Hodge diamond . . . 748
56.7 Bott–Chern and Aeppli dimensions . . . 750

57 Period-coordinate Dolbeault classes and the Bott–Chern algebra 752
57.1 Fixed input and conventions . . . 752
57.2 Exact fibre representatives . . . 753
57.3 Extension of the two Dolbeault inputs . . . 754
57.4 The factor 12 and the Bott–Chern first Chern class . . . 755
57.5 All Bott–Chern generators . . . 756
57.6 Every Bott–Chern product . . . 758

58 The alternate collapse and Mayer–Vietoris route 759
58.1 The abstract cusp-retraction lemma . . . 760
58.2 The candidate-specific equivariant collapse . . . 760
58.3 The integral collapse map . . . 763
58.4 Comparison with analytic nearby cycles . . . 765
58.5 The actual finite-band Mayer–Vietoris maps . . . 766
58.6 The cusp attachment and its primitive blocks . . . 768
58.7 Exact comparison with the Leray transgressions . . . 769
58.8 Complete recognition theorem . . . 771

59 Fibre degree and the cyclic-six centre 773
59.1 Spaces, markings, and the finite-band presentation . . . 773
59.2 The exact fibre-degree row . . . 774
59.3 Normal-line holonomy and the two finite reductions . . . 776
59.4 Coefficient reduction is not the distinguished quotient . . . 777
59.5 The tensor subgroup and the genuine common-centre quotient . . . 777
59.6 Exact boundary of the result . . . 779

60 The non-normal fibre and the CDP obstruction 780
60.1 The exact p. 692 inference . . . 780
60.2 Local algebra of a reduced normal-crossings fibre . . . 781
60.3 A proper counterexample to the printed inference . . . 783
60.4 Cartier-divisor duality without a normality hypothesis . . . 784
60.5 Flatness and top-degree analytic base change . . . 785
60.6 A topology-independent Pic^0 route and three all-Pic routes . . . 786

60.7 An analytic completion of the all-Picard certificate . . . 792
60.8 Abstract conductor theorem for the candidate fibre . . . 803
60.9 A second printed issue: abelianization of split extensions . . . 804
60.10 The corrected theorem that survives both defects . . . 804
60.11 Exact present disposition . . . 807

61 Confirmed local repairs kept separate from the theorem **808**

62 Tool and computational resource disclosure **809**

1 Results and evidence boundary

6 September: explicit spectral and umbral maps, and the original family's parameter identifications. For the literal common lattice $\mathcal{I} = \{\sqrt{2}(a,a,a) : a \in D_4\}$, write $a = a_0 + v$, $t^2 = |v|^2$, $Q = a_0^2 + t^2$ and $P = a_0(a_0^2 - 3t^2)$. Its complete cubic image is

$$\tau(\mathcal{I}) = 2\sqrt{2}\big(18\mathbb{Z} \cup (6\mathbb{Z}+2) \cup (6\mathbb{Z}-2)\big).$$

Three explicit integral affine lines give a section of this value map. The scalar-retaining Albert matrix $X = \mu I + Y(a)$, with cyclic off-diagonal entries $\sqrt{2}a$, has invariants and labelled spectrum

$$(\operatorname{tr} X, \operatorname{tr} X^2, \det X) = (3\mu, 3\mu^2 + 12Q, \mu^3 - 6\mu Q + 4\sqrt{2}P),$$
$$\operatorname{Spec} Y = (2\sqrt{2}a_0, -\sqrt{2}a_0 - \sqrt{6}t, -\sqrt{2}a_0 + \sqrt{6}t).$$

For $t > 0$, the explicit labelled eigenprojector satisfies $(\Pi_1)_{12} = -1/6 - \sqrt{3}(v/t)/6$; thus the frame and spectrum recover a, not only its norm and cubic. The full discriminant is $864t^2(3a_0^2 - t^2)^2$. The full Cartan family retains the scalar μ, radius r, and angle θ, with $r^2 = 12Q$ and $r^3 \cos 3\theta = 24\sqrt{3}P$ on this subfamily.

For the same marked root system $A_5^4 D_4$, the Cheng–Duncan–Harvey prescription gives $S^X = (5I_{12} + J)S_6$, where $J_{rs} = \mathbf{1}_{s=5r\ (12)}$ and $S_{6,r} = \sum_{k\equiv r\ (12)} kq^{k^2/24}$. The coefficient map to the retained Fricke function is

$$T_6 = \frac{\eta(\tau)^5\eta(3\tau)}{\eta(2\tau)\eta(6\tau)^5}, \qquad f^X = -q\frac{d}{dq}\log T_6, \qquad T_6\Big(-\frac{1}{6\tau}\Big) = \frac{72}{T_6(\tau)}.$$

All twelve theta indices, the multiplier determinant, the original glue-group action and its twisted shadows are computed in the proofs. This is an explicit use and comparison of established umbral machinery, not a claim to a new moonshine module.

In the original analytic family $\beta_c = \beta^{\mathrm{part}} + c$, $\operatorname{Im} c < -M$, the actual global identifications are exactly $X_c \simeq X_{c'} \iff c' - c \in 2\mathbb{Z}$. The middle lattices agree for every integer shift; the multiplicity-four filling retains an order-two obstruction to odd shifts. Even shifts have explicit finite-filling and toric cusp maps and a proved global additive gluing correction. The resulting parameter quotient is $w = e^{-\pi i c}$, $0 < |w| < e^{-\pi M}$. This classifies the retained analytic family without using the claimed identification of its total space with S^6.

6 September: cubic arithmetic, integral lattice maps, and the canonical ring. For the retained $A_5^4 \perp D_4$ Niemeier lattice N and its original index-six Leech neighbour Λ, fix the Cayley–Dickson frame $(1, i, j, k, \ell, i\ell, j\ell, k\ell)$ used in the complete cubic proof. Put $\tau(q) = \operatorname{Re}((q_1q_2)q_3)$ and $F = \tau \circ \mathcal{L}$ on the original Niemeier span, and $\mathcal{V}_F(L) = \operatorname{span}_{\mathbb{Z}}\{F(y) : y \in L\}$. The exact additive groups are

$$\mathcal{V}_F(N) = \tfrac{1}{27}\mathbb{Z} \oplus \tfrac{\sqrt{2}}{54}\mathbb{Z} \oplus \tfrac{\sqrt{3}}{18}\mathbb{Z},$$
$$\mathcal{V}_F(\Lambda) = \tfrac{1}{27}\mathbb{Z} \oplus \tfrac{\sqrt{2}}{216}\mathbb{Z} \oplus \tfrac{\sqrt{3}}{54}\mathbb{Z}, \qquad \mathcal{V}_F(\Lambda)/\mathcal{V}_F(N) \cong \mathbb{Z}/12.$$

The Albert determinant is $\lambda^3 - \lambda(y,y) + 2F(y)$. The quotient concerns the displayed value-generated groups in this fixed frame, not an inclusion $N \subset \Lambda$. The actual residue $r = \rho(361)$ cycles three Leech neighbours $\Lambda_j = r^j\Lambda$. Their pairwise intersection is one integral lattice J, with $\det J = 81$ and $[\Lambda_j : J] = 9$. The fixed fourth isotropic plane in $J^\vee/J$ gives M with exactly 72 roots of type A_2^{12}, $M \cap \Lambda_j = J$, and $[M : J] = 9$.

$$\mathcal{V}_F(J) = \mathcal{V}_F(M) = \tfrac{1}{27}\mathbb{Z} \oplus \tfrac{\sqrt{2}}{216}\mathbb{Z} \oplus \tfrac{\sqrt{3}}{18}\mathbb{Z}.$$

The successive value-group inclusions have matrices $\mathrm{diag}(1,4,1)$ and $\mathrm{diag}(1,1,3)$, giving quotients $\mathbb{Z}/4$ and $\mathbb{Z}/3$. Both containments and their reverse generator certificates are calculated from the full cubic polarization, not inferred from lattice indices.

Wilson's construction [81], transported by the explicit isometry $\iota(x) = x/\sqrt{2}$, gives Λ_{cyc} in the original triality metric. Its cyclic fixed Gram matrix is $3G_{E_8}$; its orthogonal complement is explicitly isometric to $A_2 \otimes E_8$, and their sum has index 3^8, with quotient $\mathbb{F}_3^8$. The directions remain $c(q_1,q_2,q_3) = (q_3,q_1,q_2)$ and $c_{\mathrm{W}} = c^{-1}$. Its literal intersections are

$$\Lambda_{\mathrm{cyc}} \cap \mathcal{L}(N) = \Lambda_{\mathrm{cyc}} \cap \mathcal{L}(\Lambda) = \Delta := \{\sqrt{2}(a,a,a) : a \in D_{\mathbb{H}}\},$$

where $D_{\mathbb{H}}$ is the even-coordinate-sum integer lattice in the first four stated Cayley coordinates. Thus Δ has rank 4, Gram matrix $6G_{D_4}$, determinant 5184, minimum squared norm 12, and value-generated subgroup $4\sqrt{2}\mathbb{Z}$ for $\tau|_\Delta$, or $8\sqrt{2}\mathbb{Z}$ for $2\tau|_\Delta$. This is not the actual image: $12\sqrt{2}$ is not attained by τ on Δ. The preceding proofs retain the triality products and spectral boundary graph, the flag's degree-$+1$ collapse, and the retained cusp class's nontrivial three-plane reduction; these exact maps do not supply an interacting quantum mass-gap result.

For the original analytic fibration $f : X \to \mathbb{P}^1$, the full canonical-divisor proof gives $\omega_X \simeq \mathcal{O}_X(-2S_2)$ and the graded anticanonical ring $\bigoplus_{m\geq 0} H^0(X, \omega_X^{-m}) \cong \mathbb{C}[U,V]$, with $\deg U = 1$, $\deg V = 2$. The complete system $|-2K_X|$ recovers f with the specified projective coordinates. This calculation uses the original analytic charts and finite fillings, not an identification of X with S^6, and draws no sphere-topology conclusion.

The retained constant $\beta_c = \beta^{\mathrm{part}} + c$ now gives a proper holomorphic deformation over $\mathrm{Im}\, c < -M$, locally on relatively compact parameter discs. On a fixed regular fibre set $T = \mathrm{Im}\,\tau$, $m = \mathrm{Im}\,\mu$, $D = \mathrm{Im}\,\beta_{c_*} - 6m^2/T < 0$ and $\delta = c - c_*$. In the plus-graph convention its exact Beltrami matrix is

$$\nu_\delta = \frac{\delta}{2iD + \delta}\begin{pmatrix} 0 & 0 \\ -m/T & 1 \end{pmatrix}.$$

The connecting Kodaira–Spencer class is $-[\partial_{\zeta_2} \otimes (d\bar{\zeta}_2 - (m/T)d\bar{\zeta}_1)/(2iD)] \neq 0$. The intrinsic pencil recovers f also over the dual numbers; its three fixed critical values prohibit an infinitesimal projective base change. Consequently this deformation of the whole X_c is nontrivial, not only its marked regular-fibre deformation. The proof uses no sphere recognition or CDP conclusion.

The target is a reconstructed manuscript claiming a compact complex threefold X and holomorphic map $f : X \to \mathbb{P}^1$ with

$$X \cong_{\mathrm{diff}} S^6, \qquad a(X) = 1, \qquad \langle c_3(T_X), [X]\rangle = 2.$$

As of 6 September 2026, the workbench records the following mathematical status. Here **[VERIFIED]** means that a complete proof is written in this workbench; it is not a proxy for a model verdict or for community acceptance.

The 5 September direct-image recheck corrects two written proof steps. Theorem 60.7 now constructs its duality morphism from the analytic trace, not from matching outer terms. The logarithmic gluing is the real translation $a_j(s) = \mathrm{Re}\,\alpha\, v_j + \mathrm{Im}\,\alpha\, J_s v_j$, where $J_s = \Pi(s)^{-1} i \Pi(s)$ and $\alpha = \log s_j/(2\pi i)$. Its full clockwise loop has increment $-v_j$; its mixed sweep integral is $-E(v_j,\lambda)$. The corrected proof of Lemma 60.16 and the based calculation in Proposition 53.6 retain their original signs and coefficients.

The bounded recheck verifies these maps, not every global candidate input. Its before/after sources and primary-source audit are retained in the direct-image evidence folder.

The period continuation now identifies the exact admissible constants: for the cusp-regular $\beta = \beta^{\mathrm{part}} + c_0$, set $M = \max_{B\setminus\{p_0\}} D_{\mathrm{part}}$, an attained finite maximum. The marked real period map is everywhere an isomorphism exactly when $\operatorname{Im} c_0 < -M$ (Proposition 10.3). Omitting the cusp extension permits $\widetilde{\beta} = \beta + i\,t \circ \pi$ with $\widetilde{D} = D + \operatorname{Re} t$ unbounded above. The cusp proof uses charts restricted to Y_ϵ, a bound on a fixed closed subdisc, and the weak inequalities for closed polydiscs (Lemma 10.5). Its exact opposite-edge maps retain the three multipliers $e^{-2\pi i\mu_0}$, $e^{12\pi i\mu_0}$, $e^{-2\pi i(7\mu_0+b_0)}$ with the same-character conventions of equation (20). These bounded period, cusp and finite-filling rechecks add no sphere or CDP theorem.

The new higher-wrap comparison is fully written in Section 30 and in the root draft `s6_higher_rung_24d_preprint.tex`. It gives the octonionic positive-cone compactification (B^{25}, S^{24}), the triality flag $F_4/\operatorname{Spin}(8)$ with integral cohomology ranks $1, 2, 2, 1$, and the typed peripheral and suspension maps involving $\mathbb{Z}/12 \hookrightarrow \mathbb{Z}/24$, $a \mapsto 2a$. The flag facts are attributed to Mare–Willems; the comparison preserves all coordinate, monodromy, and coefficient choices. The particular one-copy cusp homotopy coefficient is now zero: the full original period-corrected derivative frame extends over an explicit collared $D^2 \times T^2$ in one cusp chart times an interval (Theorem 30.49). The retained-angle projection of the same q_2 instead gives $H = p_3 q_2 : X \to S^4$ with $[H] = \eta_4\eta_5 \neq 0$ (Theorem 30.51). Its entire regular inverse image is the original two-torus with the explicitly calculated normal four-frame and Arf invariant one. Thus q_2 is nonnull and detects a quotient $\pi_6(\mathcal{A})/\ker(p_3)_* \cong \mathbb{Z}/2$. Smashing H with the specified Hopf map gives $12\nu_6 \neq 0$ in $\pi_9(S^6) \cong \mathbb{Z}/24$, and the peripheral composite has values $(12, 0, 12)$ with kernel $\chi^{-1}(2\mathbb{Z}/12)$. The full class and exact order of q_2 in its conical target, and its smash powers, are not inferred from this detected quotient. A completed global higher torus wrap remains open. The independent arithmetic continuation in Section 9 proves that the fifteen $p = 1201$ projective selector rays span and Jordan-generate the real three-dimensional symmetric corner, with exact same-residue countercounts.

Section 31 adds the explicit even unimodular lattice $A_5^4 \perp D_4 \subset N$ of index 72, with its original 144 roots, and the full splitting $O(N) = W(A_5^4 D_4) \rtimes L(G)$, $G \cong GL_2(\mathbb{F}_3)$ (Theorem 31.15). This realizes in fixed coordinate/code markings the abstract Weyl quotient already recorded in [74, Table 3]. The residue six-cycle belongs to the Weyl group and centralizes this chosen complement (Theorem 31.16); the D_4 discriminant action has kernel Q_8 in G and image the full triality S_3. The fixed-shell divisor bijection and finite-sample Fourier map retain p, R, P: $A_3(2521) = 0$, $A_3(3361) = 6$, with unordered counts $0, 3$, despite equal residues modulo 840.

The later quaternionic extension is treated separately in Section 24: the two Lorentz thirteen-slices coincide as embedded matrix spaces, but the supplied unconstrained scalar model can leave their trace-free slice. The explicit witness gives $(u^*b)^{\cdot\cdot} = -2\kappa$ at $u^*b = (u^*b)^{\cdot} = 0$. There are also global interacting invariant families within the original equations: fixing $W \subset u^\perp$, retaining $b \in W$ and $\Phi W \subset W$, with tangent velocities, gives configuration dimensions six and eight and identically zero reaction. A separate fourth-order escape example has $\rho^{(j)}(0) = 0$ for $0 \leq j \leq 3$ but $\rho^{(4)}(0) = -12\kappa$. These results distinguish exact invariant families from instantaneous tangency tests. The families do not exhaust the invariant motions: Theorem 24.21 constructs periodic moving-kernel solutions outside every fixed-W family. Their conserved nonzero colour moment gives an explicit Gauss residual for the zero-connection gauge lift, not an obstruction to every possible gauge-field lift. The complete scalar vacuum diagram and full metric Hessian are also proved in Theorems 24.24 and 24.25; their gauge-orbit and quantum counterparts are not inferred from this calculation. The full equations and the different constrained equations are both given; the original candidate's complex-structure claims do not follow from this quaternionic cone calculation.

The full positive fifteen-space also has a split positive observation to $\mathrm{Herm}_2(\mathbb{C})$ with exact image condition $|\rho|^2 \leq \det G$ (Proposition 24.7). Every permitted (G, ρ) occurs; null G forces $\rho = 0$, while $G = \epsilon I_2$, $\rho = \epsilon/2$ permits positive lifts for every $\epsilon > 0$. For trace-free scalar matrices the distinct spectral-floor map (831) gives this positive observation with the original matrix $C = PP^*$ retained. These exact maps do not impose a positive uniform spectral threshold or identify the coordinates as gauge-invariant

physical momenta.

The complete retained-period metric fibres are now proved in Theorem 24.9: they have dimensions twelve over H_3 and seventeen over G. Every smooth positive observed trajectory has an equivariant lift with the original periods unchanged. An explicit $G = \operatorname{diag}(\alpha_{\mathrm{obs}}, \beta_{\mathrm{obs}} y^{-p})$ approaches a nonzero null block; its relative metric condition number is asymptotic to $2\max\{2\alpha_{\mathrm{obs}}/3, 3\lambda\} y^{p+2}/\beta_{\mathrm{obs}}$. Propositions 24.10–24.11 give the exact inverse and bounds. Positivity and period monodromy alone consequently do not select an observed trajectory.

Theorem 24.27 strengthens the rotating-orbit obstruction to every homogeneous spatial connection in temporal gauge. Proposition 24.28 retains the full arbitrary-temporal Gauss equation without inferring the remaining equations from it. Theorem 24.33 constructs a different interacting two-variable family satisfying every field equation. Finally, Proposition 24.35 identifies $\int \det(PgAg^*P^*)\, d\mu = V_\mu \operatorname{tr}\operatorname{adj}(A) = V_\mu \|b\|^2$ on the positive rank-one boundary and exhibits gauge-inequivalent scalar minima on which $\|b\|^2$ ranges continuously. The fixed-mark determinant itself varies within a gauge orbit.

Theorems 24.30 and 24.31 now exclude all homogeneous real connections with arbitrary time dependence, and all complex connections constant in the spatial constant-matter coordinates, respectively, for the literal rotating orbit. Theorem 24.45 additionally excludes the genuinely complex rotating-pair class $C, Z_1 \in \mathfrak{p},\ Z_2 \in \mathfrak{k}$ and its constant $SO(2)$ spatial saturation, where the block splitting is defined by the fixed matter line $\mathbb{C}q$. No unrestricted pair is assumed to admit that splitting. For arbitrary spatial colour matrices in the rotating-pair ansatz, Lemma 24.48 proves $g^2\mathcal{I} + 2\mathcal{K} > 0$ whenever $[Z_1, Z_2] \neq 0$, a condition forced by every solution. Theorem 24.49 gives its exact Gauss/frequency inverse and all four residuals (882). Whether their zero set is empty, and the existence of still more general time-dependent complex lifts, remain open; neither is decided by a scalar contraction alone. The complete scalar equivalence in Proposition 24.50 and the strict stabilizer signature constraint of Theorem 24.51 hold in every complex coordinate, not only in the excluded block class. Proposition 24.52 adds the exact two-by-two spatial stress equation and its sum-of-squares discriminant.

Theorem 24.57 now computes the complete coupling fibre over every unrestricted ordered spatial pair as the admissible zeros of a nonnegative degree-twenty-eight polynomial. There are at most fourteen distinct positive g^2 values, and its explicit inverse retains both signs of ω and every temporal coordinate. Proposition 24.55 gives three exact vector tests for an identically empty denominator domain. Theorem 24.59 additionally places the full spatial projection inside the zero set of a specific nonzero Sylvester determinant polynomial on all sixteen coordinates. This set is closed, nowhere dense and measure zero; an explicit full pair certifies the determinant is nonzero. Proposition 24.62 places the entire eight-dimensional two-cross spatial subspace inside that zero set, and Theorem 24.60 excludes all full solutions there. Proposition 24.63 gives a polynomial residual certificate and a positive uniform lower bound for spatial error on its bounded scalar–Gauss zero locus at coupling bounded away from zero. The full scalar–Gauss zero locus is now completely parameterized by Theorems 24.66 and 24.68:

$$\mathcal{D} = \{(u, v_0, y) : u > 0,\ d = uv_0 - \delta^2 - y^2 > 0,\ c = 4v_0 - 5u > 0\},$$

$$\mathcal{S}_{\mathrm{sg}} \cong \big(\mathcal{D} \times \{1, -1\}^2 \times U(1)^2 \times U(1)\big)/\langle \iota \rangle,$$

$$\iota : (p_+, p_-, \zeta) \mapsto (-p_+, -p_-, -\zeta), \qquad \delta = \sqrt{2/3}.$$

where ι fixes the other coordinates and $t = 72d/c$. The smooth refinement in Corollary 24.75 gives

$$\mathcal{S}_{\mathrm{sg}} \cong \{1, -1\}^2 \times (0, \infty) \times \mathbb{R}^2 \times \mathbb{T}^3.$$

These are four embedded six-dimensional literal components. The original coupling is the displayed positive coordinate and is a surjective submersion; every fixed $t = \tau > 0$ has exactly four embedded five-dimensional components $\mathbb{R}^2 \times \mathbb{T}^3$. For every prescribed original $\tau = g^2 > 0$,

$$\rho = \frac{\tau + \sqrt{\tau^2 + 4608\delta^2}}{96\delta}, \qquad M_\rho = \delta \begin{pmatrix} \rho & 1 \\ 1 & 2\rho \end{pmatrix}$$

gives $t = \tau$ by Corollary 24.70, with every scalar and Gauss entry zero and both original frequency signs retained. The two spatial residuals cannot vanish simultaneously in this class. Existence on the remaining exceptional locus is **[OPEN]**. None of these finite-ansatz results is a theorem about arbitrary time-dependent gauge connections or the interacting continuum quantum spectrum.

The different sector in Proposition 24.38 supplies exact full-gauge solutions. Its periodic family (Theorem 24.39) has $\omega \to 0$ and $E - V_0 = 9h_0^2\omega^2 + O(\omega^4)$ at fixed spatial volume; its amplitude tends to $3h_0$, not zero. Proposition 24.40 proves nonlinear null waves and their finite-volume energy. The entire transverse gauge spectrum is given in Theorem 24.41. The gauge-invariant conserved vector current in Proposition 24.42 detects a branch with $\mu_{\mathrm{d}}^2 = (2g^2/3)\|b\|^2$, exactly proportional to the preceding colour average. Its linear current coefficient vanishes at $b = 0$. These are proved classical equations and observables, not quantum spectral conclusions.

There is now a separate quantum result for the complete homogeneous gauge–scalar mechanics, not inferred from those classical families. Section 25 retains all thirty-eight real configuration coordinates. The exact form bound (974), with its original $\hbar, g^2, \mathcal{V}_3, \Gamma, m^2, \kappa$, proves compact resolvent. Theorems 25.4 and 25.10 impose the spatially constant Gauss constraints and prove a unique invariant ground state; Corollary 25.11 gives $E_1^{\mathrm{phys}} - E_0 > 0$ for fixed positive volume and couplings. This is an exact finite-dimensional interacting quantum result, not a continuum Yang–Mills gap or a uniform infinite-volume estimate. The classical embedding and the failure of the automatic product quantum-subsector map are proved explicitly in Proposition 25.12 and the following Fourier calculation. The bosonic commutator-matrix discreteness mechanism is attributed to Simon's classical work, not claimed as new. Section 26 constructs a further exact correspondence for the retained two-Fourier-coordinate mechanics. Its true positive quartic ground state Ω_q gives $Jf = f\Omega_q$, zero Berry connection and the nonzero Born–Huang term in (988). The full residual (989) prevents this moving subspace from being reducing, but Theorem 26.5 bounds the difference of its physical-time evolutions by the explicit $|t|\mathcal{E}(E)$ in (991). The sufficient regime (979) is proved by min–max and a variational bound, not an assumed spectral gap. No Gaussian is substituted for the quartic ground state. The error is of order $\mathcal{V}_3^{-1/2}$ at fixed $\hbar, \Gamma, L_1, m^2, E, t$; it does not control a simultaneous limit in which $\omega_* \to 0$, nor the full continuum theory.

Theorem 24.12 identifies the original weight W as a lattice cometric. Complex-structure compatibility removes exactly its twelve anti-complex-structure fibre directions and gives $h_3 = 2H_3^{-1}$. The quotient in Theorem 24.14 has $h_2 = 2G^{-1}$ and intertwines the Laplacians on the exact pulled-back character subspace. Proposition 24.15 retains the native Euclidean metric and all inherited period corrections: $H_3 = 2I_3$, $G = 2PP^*$, $\lambda_1^{(3)} \sim 4\pi^2/y^2$ and $\lambda_1^{(2)} \sim 8\pi^2/y^2$. The full marked spectrum recovers the lattice cometric even though the fixed observed matrix does not recover the changing periods.

1. **[VERIFIED]** Let $\Delta \simeq (\mathbb{Z}/3) * (\mathbb{Z}/4)$ act on $\mathfrak{h}_z$, let g_1, g_2 be its elliptic generators with chosen fixed points z_1, z_2, and put $g_0 = (g_1 g_2)^{-1}$. For the normalized period lift $\tau : \mathfrak{h}_z \to \mathfrak{h}$, define

$$\varphi : \Delta \longrightarrow \mathrm{PSL}_2(\mathbb{Z}), \qquad \tau \circ g = \varphi(g) \circ \tau.$$

To avoid confusing the modular transformations with the finite reduced fibres S_j below, write

$$\mathfrak{s}_1(w) = \frac{w-1}{w}, \qquad \mathfrak{s}_2(w) = -\frac{1}{w}, \qquad \mathfrak{p}(w) = w - 1.$$

The finite generators are normalized simultaneously by

$$g_1'(z_1) = e^{-2\pi i/3}, \qquad g_2'(z_2) = -i, \qquad g_0 \text{ parabolic},$$

and the exact modular deck certificate is

$$\varphi(g_1) = \mathfrak{s}_1, \qquad \varphi(g_2) = \mathfrak{s}_2, \qquad \varphi(g_0) = \mathfrak{p}, \qquad \mathfrak{s}_1\mathfrak{s}_2\mathfrak{p} = 1.$$

The sign at z_2 is forced by the simultaneous parabolic normalization; the quadratic ramification calculation alone gives only $(g_2'(z_2))^2 = -1$. The cusp normalization proof now records its covering levels explicitly: it chooses a g_0-stable component $U_0 \subset \pi^{-1}(D_0^*)$ and a component $K \subset j^{-1}(\{|w| > R\})$ containing $\tau(U_0)$. For $P_{\mathrm{cusp}} := \varphi(g_0)$, the auxiliary conjugation

$$\tau' = \gamma_0^{-1}\tau|_{U_0}, \qquad P' = \gamma_0^{-1}P_{\mathrm{cusp}}\gamma_0, \qquad \gamma_0(\infty) = c,$$

where c is the rational boundary cusp of K, proves that $P' \neq 1$ and preserves the selected cusp at infinity, hence is parabolic. The proof then discards γ_0, returns to the original $\mathfrak{s}_1$-normalized frame, performs only the allowed final $\mathfrak{s}_1$-power conjugation, and obtains the displayed modular certificate. Only after that return does it define anew the cusp exponential

$$q_{\mathrm{cusp}} := e^{2\pi i\tau} = t_c u_1(t_c), \qquad u_1(0) \neq 0;$$

this q_{cusp} is distinct from the integral class q below. The resulting nondegenerate period family descends, and the middle quotient

$$J = (\mathcal{T}|_{\mathfrak{h}_z^\circ})/\Delta \longrightarrow B_{\mathrm{mid}} := \mathfrak{h}_z^\circ/\Delta$$

is a proper holomorphic submersion with compact torus fibres. The toric cusp filling, the two finite logarithmic fillings, all three punctured-overlap biholomorphisms, and the global gluing produce a connected compact Hausdorff smooth complex threefold and a proper surjective holomorphic map with connected fibres

$$f : X \longrightarrow B \simeq \mathbb{P}^1.$$

Its full scheme-theoretic cusp fibre $X_{p_0} = W$ is reduced, irreducible, nonnormal, and normal crossings, with local equations

$$z_0, \qquad z_0z_1, \qquad z_0z_1z_2,$$

and normalization $\nu : E_0 \simeq \mathrm{dP}_6 \to W$; it is not a simple-normal-crossings divisor. For the finite fibres, let

$$A_0^{(j)} = (\Lambda \otimes_{\mathbb{Z}} \mathbb{R})/\Lambda \simeq F$$

be the marked central torus on the covering family, and put

$$v_1 = \widehat{\gamma} + 2\widehat{u} - 4\widehat{w}, \qquad v_2 = -(\widehat{\gamma} + 3\widehat{u} - 3\widehat{w}),$$

so that

$$S_j = A_0^{(j)}/\langle x \mapsto A_j x + v_j/m_j\rangle, \qquad f^*(p_j) = m_j S_j, \qquad (m_1, m_2) = (3, 4).$$

The quotient covering and its codomain are, explicitly,

$$\pi_j : A_0^{(j)} \longrightarrow S_j, \qquad x \longmapsto [x].$$

The radial retraction $N_j \to S_j$ identifies finite-fibre specialization exactly with

$$\mathrm{sp}_{p_j}^r = \pi_j^* : H^r(S_j;\mathbb{Z}) \longrightarrow H^r(A_0^{(j)};\mathbb{Z}) \xrightarrow[\text{marking}]{\sim} H^r(F;\mathbb{Z}),$$

with no unrecorded conjugacy or index. This stalk identification is theorem 10.10; see also theorems 10.1, 10.2, 10.6 to 10.9 and 10.11.

2. **[VERIFIED]** The monodromy, cusp invariant lattices, global coinvariants, finite-filling stalk maps, terminal group presentation, and terminal degree-two lattice are proved over $\mathbb{Z}$ in theorems 5.1 to 5.7.

3. **[VERIFIED]** The intrinsic integral nearby-cycle specialization at the cusp is injective with saturated image
$$\operatorname{im}(\mathrm{sp}^q) = H^q(F;\mathbb{Z})^{T_0} \qquad (0 \le q \le 4),$$
where the degree-two invariant class is $q := uw + 6\gamma\delta \in \bigwedge^2 V = H^2(F;\mathbb{Z})$, and the finite stalks and parabolic sheaf calculation give the complete global Leray page. The canonical truncation triangle identifies the explicit Cech discrepancy cocycle with Leray d_2, and multiplicativity then gives
$$d_2(12\gamma) = \omega, \qquad d_2(2q) = \omega[\delta], \qquad d_2(2\gamma uw) = \omega[\gamma\delta].$$
Consequently
$$H^1(X;\mathbb{Z}) = H^2(X;\mathbb{Z}) = H^3(X;\mathbb{Z}) = 0, \qquad H_2(X;\mathbb{Z}) = 0;$$
see theorems 53.1 to 53.5.

4. **[VERIFIED]** The candidate's independent toric-collapse construction is continuous across every stratum and descends equivariantly to a strong deformation retraction $r : N_0' \to W$. Its fixed-fibre restriction $c = r|_F$ has the exact primitive integral sequence
$$0 \longrightarrow \operatorname{im}(\textstyle\bigwedge^q M_0 - I) \longrightarrow \bigwedge^q \Lambda \xrightarrow{c_*} H_q(W;\mathbb{Z}) \longrightarrow 0,$$
and proper-base-change functoriality gives, in every degree,
$$c^* = i_F^* r^* = \mathrm{sp}^q .$$
Thus the former C1–C3 obligations are closed; see theorems 58.5 to 58.7.

5. **[VERIFIED]** With zero-section whiskers and the fixed clockwise meridians, the actual overlap inclusions give
$$\pi_1(X) = \langle c, x, y \mid c \text{ central},\ xy = 1,\ x^3 = c,\ y^4 = c^{-1} \rangle \simeq 1.$$
Together with the preceding integral cohomology this proves
$$X \cong_{\mathrm{diff}} S^6, \qquad \langle c_3(T_X), [X] \rangle = 2;$$
see theorems 53.6 and 53.7.

6. **[VERIFIED]** The very-general fibre has
$$\eta := q = u \wedge w + 6\gamma \wedge \delta, \qquad \mathrm{NS}(F_b) = \mathbb{Z}\eta, \qquad a(F_b) = 0.$$
The square is the oriented wedge identity
$$\eta \wedge \eta = 12\, \gamma \wedge u \wedge w \wedge \delta,$$
so, for the orientation determined by the ordered marking (γ, u, w, δ) and for the complex orientation respectively,
$$\int_{(F_b, \mathrm{or}_{\mathrm{mark}})} \eta \wedge \eta = 12, \qquad \int_{(F_b, \mathrm{or}_{\mathbb{C}})} \eta \wedge \eta = -12.$$
The repaired meromorphic-descent argument gives
$$\mathcal{M}(X) = f^*\mathbb{C}(t), \qquad a(X) = 1,$$
with f the algebraic reduction; see theorems 54.2 and 54.4.

7. **[REFUTED]** The p. 692 sheaf-theoretic inference of Campana–Demailly–Peternell is false in the stated generality. If $\mathscr{K}_S = \ker(\Omega^1_X|_S \to \widetilde{\Omega}^1_S)$, then

$$0 \longrightarrow N^*_{S/X} \longrightarrow \mathscr{K}_S \longrightarrow \operatorname{tors}\Omega^1_S \longrightarrow 0,$$

and the two printed vanishings do not control sections mapping to the conductor-supported quotient. The proper nodal-cubic-times- $\mathbb{P}^1$ family in theorem 60.6 is an explicit counterexample to that inference.

8. **[VERIFIED]** The conductor section, nonnormal Cartier-divisor duality, flatness, and top-degree analytic base change are complete in theorems 60.5 and 60.7 to 60.11. On the cusp model the invariant nowhere-vanishing form

$$\Omega = \frac{dx_1 \wedge dx_2 \wedge dt}{x_1 x_2}$$

first proves intrinsically that $\omega_{N_0} \simeq \mathcal{O}_{N_0}$. After the global gluing, the open embedding $N_0 \hookrightarrow X$ gives the exact restriction chain

$$\omega_X|_{N_0} \simeq \omega_{N_0} \simeq \mathcal{O}_{N_0}.$$

Let $\nu : E_0 \simeq \mathrm{dP}_6 \to W$ be the normalization. Consequently, for every $L \in \mathrm{Pic}^0(X)$,

$$\nu^*((L^\vee \otimes \omega_X)|_W) \simeq \mathcal{O}_{\mathrm{dP}_6},$$

and the conductor–duality–base-change chain gives

$$R^2 f_*(T_X \otimes L) \neq 0$$

without using global topology; see theorems 10.6, 10.11, 60.12 and 60.22.

9. **[VERIFIED]** For arbitrary line bundles the exact trivial-pullback obstruction is

$$r = \nu^* i^* : \mathrm{Pic}(X) \longrightarrow \mathrm{Pic}(\mathrm{dP}_6).$$

The conductor argument applies uniformly to all line bundles exactly when $r = 0$; see theorem 60.13. For the constructed candidate, the Leray edge image of every $c_1(L)$ lies in $E_\infty^{0,2} \subseteq \ker d_2^{0,2}$, while $d_2^{0,2}(2q) = \omega[\delta]$ is an isomorphism. Proper-base-change naturality therefore gives $c_1(L|_W) = 0$, and injectivity of c_1 on $\mathrm{Pic}(\mathrm{dP}_6)$ forces $r = 0$; see theorem 60.14. Combining this with the analytic inputs gives

$$R^2 f_*(T_X \otimes L) \neq 0 \qquad \text{for every } L \in \mathrm{Pic}(X).$$

Thus the topology-free Pic^0 route and the all-Pic route are compatible rather than competing statuses. There are now three all-Pic certificates. The first is the preceding Leray-edge proof. The second uses no global Leray differential: the order-four specialization lattice writes $c_1(L|_F) = 2nq$, and the pair-of-pants crossed defect has boundary values

$$\bar{d}_1 = 4n[\delta], \qquad \bar{d}_2 = -3n[\delta], \qquad \bar{d}_0 = 0.$$

The exact boundary relation forces $n = 0$, and local integral cusp specialization then forces $r(L) = 0$; see theorems 60.15 to 60.18. This second proof is independent of the global transgressions, but it still uses one local cusp bounding chain and the local specialization theorem.

10. **[VERIFIED]** The local cusp Picard map itself is nonzero. The explicit equivariant Cartier divisor

$$\mathcal{D}_Q = \sum_{v \in \mathbb{Z}^2} (v_1^2 - 6v_2^2) E_v$$

descends to $\mathcal{L}_Q \in \mathrm{Pic}(N_0)$ with

$$c_1(\mathcal{L}_Q|_{F_t}) = -2q, \qquad \left(\deg_{C_i} \nu^*(\mathcal{L}_Q|_W)\right)_{i=1}^6 = (-12, 2, 0, -12, 2, 0).$$

Thus neither the cusp geometry nor analytic relative-Picard representability alone can force normalized central restriction to vanish; see theorem 55.15.

11. **[VERIFIED]** After the global coinvariant calculation writes

$$E := c_1(L|_F) = kq \in \mathrm{Alt}^2(\Lambda, \mathbb{Z}) = \bigwedge^2 V,$$

the third all-Pic certificate completes the argument by holomorphic factors and toric support functions. Put $V_{\mathbb{R}} = \mathrm{Hom}(\Lambda, \mathbb{R})$, let $P \subset B^\circ$ be the compact pair-of-pants core with clockwise boundary meridians ρ_1, ρ_2, ρ_0, and let $d : \pi_1(P) \to V_{\mathbb{R}}$ be the factor-of-automorphy crossed homomorphism, put $d_i = d(\rho_i)$, and let a bar denote passage to $V_G \otimes \mathbb{R} = \mathbb{R}[\delta]$. Logarithms of the actual $\mathcal{O}^*$-factors give

$$\bar{d}_1 + \bar{d}_2 + \bar{d}_0 = 0, \qquad \bar{d}_1 = 2k[\delta], \quad \bar{d}_2 = -\tfrac{3}{2}k[\delta], \quad \bar{d}_0 = 0.$$

The finite norm identities now include the fully typed factor-extension repair (5.17-A14c)–(5.17-A15), rather than identifying maps living on different covering levels. Put

$$\widetilde{J}_j = \Delta_j \times \mathbb{C}^2, \qquad r_j : \widetilde{J}_j \longrightarrow J'_j,$$

where r_j is the lattice quotient with deck translations τ_λ. On this vector cover, and only there, define

$$Q_{j,\log} = \widetilde{g}_j^{\,v_j/m_j}, \qquad Q_{j,\mathrm{lin}} = \widetilde{g}_j^{\,0}.$$

Their exact relations are

$$Q_{j,\log}\tau_\lambda Q_{j,\log}^{-1} = \tau_{A_j\lambda}, \quad (Q_{j,\log})^{m_j} = \tau_{v_j}, \qquad Q_{j,\mathrm{lin}}\tau_\lambda Q_{j,\mathrm{lin}}^{-1} = \tau_{A_j\lambda}, \quad (Q_{j,\mathrm{lin}})^{m_j} = 1.$$

Let $q_{j,\log}, q_{j,\mathrm{lin}}$ be their descended order-m_j automorphisms of J'_j, put $J'^*_j = J'_j|_{\Delta_j^*}$, and define

$$p_{j,\log} : J'^*_j \to N_j^* := J'^*_j/\langle q_{j,\log}\rangle, \qquad p_{j,\mathrm{lin}} : J'^*_j \to J^* := J'^*_j/\langle q_{j,\mathrm{lin}}\rangle.$$

The branch-independent fibre translation $h_j : J'^*_j \to J'^*_j$ and its quotient biholomorphism $\bar{h}_j : N_j^* \to J^*$ satisfy

$$h_j q_{j,\log} h_j^{-1} = q_{j,\mathrm{lin}}, \qquad p_{j,\mathrm{lin}} \circ h_j = \bar{h}_j \circ p_{j,\log},$$

and the gluing gives the typed line-bundle comparison

$$p_{j,\log}^*(L|_{N_j^*}) \simeq h_j^* p_{j,\mathrm{lin}}^*(L|_{J^*}).$$

For the filled factor extension return to the vector cover. With

$$P_{j,\log} : J'_j \to N_j, \qquad \pi_j^{\mathrm{vec}} := P_{j,\log} \circ r_j : \widetilde{J}_j \to N_j, \qquad \Gamma_j^{\log} = \langle \tau_\lambda, Q_{j,\log} : \lambda \in \Lambda\rangle,$$

where $P_{j,\log}$ is the full quotient and $P_{j,\log}|_{J'^*_j} = p_{j,\log}$, pull back $L|_{N_j}$ by π_j^{vec}, push its integral factor extension out along $\mathbb{Z} \hookrightarrow \mathbb{R}$, and choose normalized vertical lifts $U_\lambda^{(j)}$. For $\alpha \in \Gamma_j^{\log}$, define $A_\alpha \in \mathrm{Aut}(\Lambda)$ and $T_\alpha \in \mathrm{Aut}(V_{\mathbb{R}})$ by

$$\alpha\tau_\lambda\alpha^{-1} = \tau_{A_\alpha\lambda}, \qquad T_\alpha = (A_\alpha^{-1})^{\mathsf{t}}.$$

For a factor-extension lift $\widetilde{G}_\alpha$, define $\widetilde{d}_j(\alpha) \in V_{\mathbb{R}}$ by

$$\widetilde{G}_\alpha U^{(j)}_{A_\alpha^{-1}\lambda}\widetilde{G}_\alpha^{-1} = \mathbf{c}\Big(\langle \widetilde{d}_j(\alpha), \lambda\rangle\Big)\, U_\lambda^{(j)}.$$

The normalized product gives

$$\widetilde{d}_j(\alpha\beta) = \widetilde{d}_j(\alpha) + T_\alpha \widetilde{d}_j(\beta).$$

Compatible basepoints in the quotient square identify the clockwise boundary lift successively with $q_{j,\mathrm{lin}}$, $q_{j,\log}$, and finally the vector-cover deck transformation $Q_{j,\log}$. The induced extension isomorphism then proves, for some $a_j \in V_{\mathbb{R}}$,

$$\widetilde{d}_j(Q_{j,\log}) = d_j + (T_j - I)a_j, \qquad \widetilde{d}_j(\tau_v) = \iota_v E, \qquad (\iota_v E)(\lambda) = E(v,\lambda).$$

Consequently, with $N_{T_j} = I + T_j + \cdots + T_j^{m_j-1}$,

$$N_{T_j} d_j = N_{T_j} \widetilde{d}_j(Q_{j,\log}) = \widetilde{d}_j((Q_{j,\log})^{m_j}) = \widetilde{d}_j(\tau_{v_j}) = \iota_{v_j} E,$$

because $N_{T_j}(T_j - I) = 0$. This proves the finite boundary values, including their signs, inside the actual filled logarithmic deck group and closes the factor-extension gap; the certificate therefore remains **[VERIFIED]**. Changes of line-bundle frame, central lift, normalized vertical system, or logarithm branch alter only the displayed principal character term and leave $h_j, \bar{h}_j$, the quotient square, and (5.17-A15) unchanged. The cusp equality follows from the argument principle in a Stein chart, not from a Chern evaluation on a bounding chain; hence $k = 0$. For adjacent rays in a maximal cone $\sigma = \{v, w, u\}$, the actual toric-open inclusions are

$$U_{\rho_v} = D(z_w z_u) \subset U_\sigma, \qquad U_{\rho_w} = D(z_v z_u) \subset U_\sigma, \qquad U_{\rho_v} \cap U_{\rho_w} = U_0.$$

The contractible Stein comparison chart

$$S_\sigma = \{(z_v, z_w, z_u) \in \mathbb{C}^3 : |z_v z_w z_u| < \epsilon\}$$

has $\mathrm{Pic}(S_\sigma) = 0$, so its frames give zero horizontal winding along the connected one-skeleton. For the support-function Cartier divisor D_a, the exact deck-translate identity

$$\Psi_\lambda^* D_a - D_a = \mathrm{div}(\chi^{m_\lambda})$$

shows that multiplication by χ^{m_λ} extends from the dense torus to an isomorphism

$$\mu_\lambda^+ : \Psi_\lambda^* \mathcal{O}(D_a) \xrightarrow{\sim} \mathcal{O}(D_a).$$

The map on the negative divisor required by the descent tensor product is the inverse-dual

$$\mu_\lambda^- = ((\mu_\lambda^+)^{-1})^\vee : \Psi_\lambda^* \mathcal{O}(-D_a) \longrightarrow \mathcal{O}(-D_a), \qquad s \longmapsto \chi^{-m_\lambda} s.$$

Here the cusp shear is the typed lattice isomorphism

$$\Lambda_{\mathrm{tor}} := \ker(M_0 - I) = \mathrm{im}(M_0 - I) = \mathbb{Z}\langle \widehat{w}, \widehat{\delta} \rangle, \qquad \overline{\Lambda} := \Lambda / \Lambda_{\mathrm{tor}},$$
$$B_0 : \overline{\Lambda} \xrightarrow{\sim} \Lambda_{\mathrm{tor}}, \qquad B_0 \overline{\widehat{\gamma}} = -\widehat{\delta}, \qquad B_0 \overline{\widehat{u}} = \widehat{w}.$$

Thus its matrix is $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ from the ordered basis $(\overline{\widehat{\gamma}}, \overline{\widehat{u}})$ to $(\widehat{w}, \widehat{\delta})$; in particular B_0 is a map between two named lattices, not an untyped endomorphism, and $\det B_0 = 1$. For $M = L|_{N_0}$, if (d_1, d_2, d_3) are the independent degrees of $\nu^*(M|_W)$, then

$$\left(\deg_{C_i} \nu^*(M|_W)\right)_{i=1}^6 = (d_1, d_2, d_3, d_1, d_2, d_3),$$

and their support function has polarization

$$B \in \mathrm{Sym}^2(\Lambda_{\mathrm{tor}}^*), \qquad [B]_{(\widehat{w},\widehat{\delta})} = \begin{pmatrix} d_2 + d_3 & d_3 \\ d_3 & d_1 + d_3 \end{pmatrix}.$$

For the Appell–Humbert form $E_M := c_1(M|_F) \in \mathrm{Alt}^2(\Lambda, \mathbb{Z})$, one has

$$E_M((\lambda, 0), (0, m)) = -B(B_0 \lambda, m) \quad (\lambda \in \overline{\Lambda},\ m \in \Lambda_{\mathrm{tor}}).$$

The already proved antecedent $k = 0$ gives $E_M = c_1(L|_F) = 0$; only then does the unimodularity of B_0 force $B = 0$, then $d_1 = d_2 = d_3 = 0$, and hence $\nu^* i^* L \simeq \mathcal{O}_{\mathrm{dP}_6}$. Beyond the preceding coinvariant input, this proof uses neither a global integral Leray transgression, local integral nearby-cycle specialization, nor Chern-class evaluation on a cusp bounding chain; see theorems 60.19 to 60.21.

12. **[REFUTED]** The separate split-extension calculation imported from CDP98 replaces monodromy coinvariants by the entire fibre homology. The correct formula is

$$(K \rtimes Q)^{\mathrm{ab}} \simeq Q^{\mathrm{ab}} \oplus (K^{\mathrm{ab}})_Q.$$

13. **[VERIFIED]** A corrected replacement theorem now survives both printed defects. If one $L_0 \in \mathrm{Pic}(X)$ satisfies

$$H^2(X, T_X \otimes L_0) = 0,$$

then generic endpoint vanishing, semicontinuity, and Hirzebruch–Riemann–Roch give

$$H^i(X, T_X \otimes M) = 0 \ (i \neq 1), \qquad \chi(X, T_X \otimes M) = \tfrac{1}{2} c_3(X) \leq 0$$

for general M. It is enough that $R^2 f_*(T_X \otimes L) = 0$ for one L. For reduced full fibres, the invalid p. 692 shortcut is replaced exactly by injectivity of

$$\delta_c : H^0(W_c, \operatorname{tors} \Omega^1_{W_c} \otimes A_c) \longrightarrow H^1(W_c, N^*_{W_c/X} \otimes A_c)$$

together with the two endpoint vanishings. See theorems 60.28 to 60.30. The constructed candidate fails the global and relative conditions for separate reasons. Since $c_3(X) = 2$, the contrapositive of theorem 60.28 gives $H^2(X, T_X \otimes L_0) \neq 0$ for every L_0. Independently, its conductor makes $R^2 f_*(T_X \otimes L) \neq 0$ for every L, so the stronger relative sufficient condition also fails. The original CDP20 theorem therefore remains refuted.

14. **[VERIFIED]** The constructed X satisfies the stated hypotheses of CDP20 Theorem 2.2 but has both $c_3(X) = 2 > 0$ and $a(X) = 1$. Thus it is a counterexample to that theorem and its ensuing S^6 corollary as stated. This theorem-level conclusion uses the complete construction, topology, and algebraic-reduction proofs above; it is not inferred merely from the two printed proof-route defects.

15. **[VERIFIED]** The candidate's Proposition 10.8 conclusion holds. The workbench proves

$$R^1 f_* \mathcal{O}_X \simeq \mathcal{O} \oplus \mathcal{O}(-1), \qquad \mathrm{Pic}(X) \simeq \mathbb{C},$$

and the finite-filling Hodge-line calculation now distinguishes the descended canonical section from the direct-image frame. In the notation of theorem 55.1,

$$\mathrm{div}(\overline{\sigma}_j) = k_j S_j, \qquad f_{j*}\omega_{N_j/D_j} = \mathcal{O}_{D_j} \cdot s_j^{k_j} \vartheta'_j, \qquad k_j = m_j - 1 - a_j.$$

Thus $\overline{\sigma}_2$ has divisor $2S_2$ and is not itself a nowhere-vanishing frame, while the displayed invariant module still has the claimed generator, a unit times $s_j^{-a_j} e$, over the punctured cover. This removes a local wording defect without changing the Hodge line or any subsequent direct-image calculation. The workbench also proves degree-one base change at every scheme-theoretic fibre, and $\mathrm{ord}(\mathcal{O}_X(S_j)|_{S_j}) = m_j$. The smooth-fibre period condition is exactly $c\tau + d\mu - b = 0$; hence every parameter outside $\mathbb{R}$ and a countable set gives one line bundle non-torsion on every full fibre and on both S_j. The cusp extension is proved intrinsically through nearby cycles and does not depend on the separately reconstructed cellular collapse; see theorems 55.11 and 55.13.

16. **[VERIFIED]** The alternate Mayer–Vietoris route is now closed. The finite-band inclusions are exactly $\alpha_q = (\pi_{1*}, -\pi_{2*})$ in named primitive bases; the second cusp attachment has degree-one determinant $-p$ and reduced degree-two block $\mathrm{diag}(-p, 1)$; and the dual obstruction squares commute with the three verified Leray transgressions. For $p = -1$ both global blocks are unimodular, giving an independent integral sphere certificate; see theorems 58.10, 58.11, 58.13 and 58.18.

17. **[VERIFIED]** The Hodge diamond and Frölicher sequence are now exact. Here are the typed logarithmic objects used in the calculation:

$$B=\mathbb{P}^1,\qquad \Sigma=p_0+p_1+p_2,\qquad D=f^{-1}(\Sigma)_{\mathrm{red}}=W+S_1+S_2,$$

$$\mathcal{L}=\Omega^1_B(\log\Sigma)\simeq\mathcal{O}_B(1),\qquad \mathcal{A}=\Omega^1_X(\log D),\qquad \mathcal{E}=\mathcal{A}/f^*\mathcal{L},\qquad \mathcal{G}=R^1f_*\mathcal{O}_X.$$

They occur in the locally free logarithmic conormal sequence

$$0\longrightarrow f^*\mathcal{L}\longrightarrow\mathcal{A}\longrightarrow\mathcal{E}\longrightarrow 0,$$

whose connecting maps have the exact types

$$\kappa_0:f_*\mathcal{E}\longrightarrow\mathcal{L}\otimes\mathcal{G},\qquad \kappa_1:R^1f_*\mathcal{E}\longrightarrow\mathcal{L}\otimes R^2f_*\mathcal{O}_X\simeq\mathcal{O}_B.$$

The logarithmic relative direct images are

$$f_*\mathcal{E}\simeq\mathcal{O}\oplus\mathcal{O}(-1),\qquad R^1f_*\mathcal{E}\simeq\mathcal{O}^{\oplus2}\oplus\mathcal{O}(-1)^{\oplus2},\qquad R^2f_*\mathcal{E}\simeq\mathcal{O}\oplus\mathcal{O}(-1),$$

and are locally free and compatible with base change.

On a finite cover $t=s^m$, with $(m,a)=(3,2)$ or $(4,1)$, choose equivariant relative-cotangent frames (e_0,e_1) and structure-sheaf frames (b_0,b_1) with semilinear characters $(1,\zeta^a)$ and $(1,\zeta^{-a})$, respectively. The actual $\mathbb{C}\{t\}$-bases are

$$\begin{aligned} f_*\mathcal{E}:&\quad e_0,\ s^{m-a}e_1,\\ R^1f_*\mathcal{E}:&\quad e_0b_0,\ s^ae_0b_1,\ s^{m-a}e_1b_0,\ e_1b_1,\\ R^2f_*\mathcal{E}:&\quad s^ae_0(b_0\wedge b_1),\ e_1(b_0\wedge b_1).\end{aligned}$$

Let $\ell=dt/t$, and let $D_{\mathrm{KS}}=(d_{rs})_{0\le r,s\le1}$ be the marked Kodaira–Spencer matrix after its common factor ds is removed. Put

$$A=\frac{s}{m}d_{00},\qquad B_1=\frac{s^{1-a}}{m}d_{01},\qquad C=\frac{s^{1-a}}{m}d_{10},\qquad D_0=\frac{s^{m+1-2a}}{m}d_{11}.$$

In source basis $(e_0,s^{m-a}e_1)$, target basis $(b_0,s^ab_1)\ell$, ordered degree-one source basis $(e_0b_0,s^ae_0b_1,s^{m-a}e_1b_0,e_1b_1)$, and target generator $s^a(b_0\wedge b_1)\ell$, the coarse maps are exactly

$$[\kappa_0]=\begin{pmatrix}A & tB_1\\ C & D_0\end{pmatrix},\qquad [\kappa_1]=(-C,A,-D_0,B_1).$$

The finite residue is the typed map

$$\mathrm{res}_{1,j}:(\mathcal{L}\otimes\mathcal{G})|_{p_j}\longrightarrow H^1(S_j,\mathcal{O}_{S_j}),\qquad \mathrm{res}_{1,j}\big((xb_0+ys^ab_1)\ell\big)=m_jx[b_0]\quad(x,y\in\mathbb{C}),$$

and the derived residue square gives

$$\mathrm{res}_{1,j}\,\kappa_0=0,\qquad \mathrm{im}(\kappa_0\bmod t)=\ker\mathrm{res}_{1,j}=\mathbb{C}\,s^ab_1.$$

At the cusp put

$$\mathcal{Q}_{\mathrm{cusp}}:=\nu_*\mathcal{O}_{\mathrm{dP}_6}/\mathcal{O}_W,$$

and let $\rho:\mathcal{E}|_W\to\mathcal{Q}_{\mathrm{cusp}}$ be the relative normalization–residue quotient map. Retain the connecting isomorphisms

$$\delta_0:H^0(\mathcal{Q}_{\mathrm{cusp}})\xrightarrow{\sim}H^1(W,\mathcal{O}_W),\qquad \delta_1:H^1(\mathcal{Q}_{\mathrm{cusp}})\xrightarrow{\sim}H^2(W,\mathcal{O}_W).$$

For

$$q_1 = (-1, 0, 1), \qquad q_2 = (0, -1, -1), \qquad \eta_i = d\log x_i, \qquad \alpha_i = \delta_0(q_i),$$

let $\theta \in H^1(\mathcal{Q}_{\mathrm{cusp}})$ be the unique class satisfying

$$\delta_1(\theta) = -\alpha_1 \smile \alpha_2.$$

The standard Čech differential $(\delta c)_{i_0 \dots i_{r+1}} = \sum_k (-1)^k c_{i_0 \dots \widehat{i_k} \dots i_{r+1}}$ fixes the sign

$$\delta_1\big(\rho_*(\alpha \otimes \eta_i)\big) = -\alpha \smile \alpha_i.$$

Consequently the two cusp maps, with their domains and codomains, are

$$H^0(\mathcal{E}|_W) \xrightarrow{\sim} H^0(\mathcal{Q}_{\mathrm{cusp}}),$$

$$H^1(\mathcal{E}|_W) = \mathbb{C}\langle \alpha_1\eta_1, \alpha_1\eta_2, \alpha_2\eta_1, \alpha_2\eta_2 \rangle \xrightarrow{(0,1,-1,0)} H^1(\mathcal{Q}_{\mathrm{cusp}}) = \mathbb{C}\theta;$$

the second is surjective.

Finally, the invariant section $\xi = \partial/\partial\zeta_2$, nowhere zero in the locally free relative sheaf $T_{X/B} = \mathcal{E}^\vee$, gives the split evaluation sequence

$$0 \longrightarrow \mathcal{O}(-1) \longrightarrow f_*\mathcal{E} \xrightarrow{\mathrm{ev}_\xi} \mathcal{O} \longrightarrow 0.$$

There is a unique lift $h \in H^0(B, f_*\mathcal{E})$ of $1 \in H^0(B, \mathcal{O})$, because $H^0(B, \mathcal{O}(-1)) = 0$. On the marked cover it has the uniquely determined form

$$h = (a(z), 1)\begin{pmatrix} d\zeta_1 \\ d\zeta_2 \end{pmatrix}, \qquad a(z_2) = \frac{1}{2}.$$

Use the genuine invariant target frame

$$\omega_2 = s\, u_2(s)(\gamma \wedge u)\ell, \qquad u_2(0) = 1.$$

The determinant and final global scalar are

$$\mathrm{div}(\det \kappa_0) = p_1 + p_2, \qquad \kappa_1(h \otimes \gamma)|_{p_2} = -\frac{1}{4}\dot{\mu}(z_2)\, \omega_2|_{p_2} \neq 0 \quad \text{in } (\mathcal{L} \otimes R^2 f_* \mathcal{O}_X)|_{p_2}.$$

This uses $\dot{\tau}(z_2) = 0$ and $K(z_2) = 6\dot{\mu}(z_2)^2 \neq 0$.

Thus, with $\mathcal{K} = \ker \kappa_1$,

$$\mathcal{K} \simeq \mathcal{O} \oplus \mathcal{O}(-1)^{\oplus 2},$$

and the torsion-free absolute cotangent direct images are

$$R^0 f_* \Omega^1_X \simeq \mathcal{O}(-2), \quad R^1 f_* \Omega^1_X \simeq \mathcal{K}, \quad R^2 f_* \Omega^1_X \simeq \mathcal{O} \oplus \mathcal{O}(-1).$$

Therefore

$$h^{1,2} = h^{2,1} = 1, \qquad h^{1,1} = h^{2,2} = 2.$$

In full, with row index q and column index p, the Hodge table is

$q\backslash p$	0	1	2	3
3	0	0	0	1
2	0	1	2	1
1	1	2	1	0
0	1	0	0	0.

The two middle d_1-rows are exact $0 \to \mathbb{C} \to \mathbb{C}^2 \to \mathbb{C} \to 0$, and

$$\dim_{\mathbb{C}} E_1 = 10, \qquad \dim_{\mathbb{C}} E_2 = 2, \qquad E_2 = E_\infty.$$

See theorems 56.5 to 56.7, 56.9 and 56.10.

18. **[REFUTED]** Remark 1.3's assertion that the relative differential sheaves have torsion along all three singular fibres is false already for $\Omega^1_{X/B}$. The $t = xy$ and $t = xyz$ modules inject into the free logarithmic relative module, so $\Omega^1_{X/B}$ is torsion-free but non-locally-free near the cusp. At a point of the smooth finite cover $q_j : J'_j \to N_j$, with $R_j = \mathcal{O}_{J'_j,x} \simeq \mathbb{C}\{s, y_1, y_2\}$, and $t = s^{m_j}$, the exact module is

$$(q_j^*\Omega^1_{N_j/B})_x \simeq \Omega^1_{J'_j/B,x} \simeq R_j\, dy_1 \oplus R_j\, dy_2 \oplus \big(R_j/(s^{m_j-1})\big)ds,$$

and hence

$$(q_j^* \operatorname{tors} \Omega^1_{N_j/B})_x = \big(R_j/(s^{m_j-1})\big)ds.$$

Faithful flatness descends this module and its support. Thus the torsion support is exactly the two multiple fibres and is absent at the cusp; see the local calculation (11.17f).

19. **[VERIFIED]** The Bott–Chern and Aeppli dimensions are now exact as well. McHugh's threefold formula gives

$$h^{1,2}_{BC} = h^{1,2} + \dim E_2^{3,1} - k^{2,0} = 1 + 0 - 0 = 1,$$

because the de Rham image of the pure doubly closed forms surjects onto $K^{2,0}$, so $k^{2,0} \le b_2 = 0$. The complete Bott–Chern rows $q = 3, 2, 1, 0$ are

$$(0,0,1,1), \quad (0,1,2,1), \quad (0,2,1,0), \quad (1,0,0,0),$$

and Aeppli duality gives

$$(0,0,0,1), \quad (0,1,2,0), \quad (1,2,1,0), \quad (1,1,0,0).$$

See theorem 56.11; no dimension remains open.

20. **[VERIFIED]** Global representatives and the complete Bott–Chern product structure are now known. The two Dolbeault inputs must not be conflated. The distinguished class

$$\mathfrak{g} \in H^{0,1}_{\bar\partial}(X) = H^1(X, \mathcal{O}_X)$$

is the unique coherent-Leray class mapping to the invariant section $\gamma^{0,1} \in H^0(B, R^1 f_* \mathcal{O}_X)$. In contrast, the marked integral fibre class is

$$q = \eta = u \wedge w + 6\gamma \wedge \delta \in H^2(F_b; \mathbb{Z}),$$

whose holomorphic extension is a section $\boldsymbol{\eta} \in H^0(B, R^1 f_* \Omega^1_X)$. Choose a coherent-Leray lift

$$\mathfrak{q} \in H^{1,1}_{\bar\partial}(X) \quad \text{with edge value} \quad \boldsymbol{\eta}.$$

This lift is not distinguished: its exact ambiguity is

$$\mathfrak{q} \longmapsto \mathfrak{q} + c\, d_1\mathfrak{g}, \qquad c \in \mathbb{C},$$

because the kernel of the edge map is $H^1(B, f_*\Omega^1_X) = \mathbb{C}\, d_1\mathfrak{g}$.

Choose global $\bar\partial$-closed smooth representatives

$$G \in \mathcal{A}^{0,1}(X), \quad [G]_{\bar\partial} = \mathfrak{g}, \qquad Q \in \mathcal{A}^{1,1}(X), \quad [Q]_{\bar\partial} = \mathfrak{q}.$$

For a marked smooth fibre put

$$y = \operatorname{Im}\tau, \qquad m_\mu = \operatorname{Im}\mu, \qquad r = \operatorname{Im}\beta, \qquad \Delta_F = r - \frac{6m_\mu^2}{y} < 0,$$

and define the translation-invariant forms

$$\Gamma = \frac{i}{2\Delta_F}\left(d\bar{\zeta}_2 - \frac{m_\mu}{y}d\bar{\zeta}_1\right), \qquad U = \frac{i}{2y\Delta_F}\left(r\,d\bar{\zeta}_1 - 6m_\mu\,d\bar{\zeta}_2\right),$$

$$Q_F = -d\zeta_1 \wedge U - 6d\zeta_2 \wedge \Gamma.$$

The restriction statements are equalities of typed Dolbeault classes, not claims about a preferred literal coefficient gauge:

$$i_b^*\mathfrak{g} = [\Gamma]_{\bar\partial} \in H^{0,1}_{\bar\partial}(F_b), \qquad i_b^*\mathfrak{q} = [Q_F]_{\bar\partial} \in H^{1,1}_{\bar\partial}(F_b).$$

After fixing $\mathfrak{g}$ and one $\mathfrak{q}$, changing smooth representatives by $G \mapsto G + \bar\partial\phi$ and $Q \mapsto Q + \bar\partial\Psi$ changes the derived representatives only by $\partial\bar\partial$-exact terms. Thus no preferred literal global coefficient atlas through the singular fillings is asserted or needed. Put

$$\mathsf{A} := [\partial G]_{BC}, \qquad \mathsf{B} := [\bar\partial\bar G]_{BC} \quad \text{in } H^{1,1}_{BC}(X).$$

Let

$$\exp : H^1(X, \mathcal{O}_X) \longrightarrow \mathrm{Pic}(X)$$

be induced by $z \mapsto e^{2\pi i z}$, and orient $\omega \in H^2(B;\mathbb{Z})$ by $\omega = c_1(\mathcal{O}_B(1))$. The integral Picard normalization and Bott–Chern first Chern class are

$$\exp(12\mathfrak{g}) = f^*\mathcal{O}_B(1), \qquad c_{1,BC}(\exp(z\mathfrak{g})) = z\mathsf{A} + \bar z\,\mathsf{B},$$

$$c_{1,BC}(f^*\mathcal{O}_B(1)) = 12(\mathsf{A} + \mathsf{B}).$$

The absolute factor is 12; choosing the opposite primitive orientation replaces $\mathfrak{g}$ by $-\mathfrak{g}$ in the pullback identity. The exact fibre pairings, with the orientation stated, are

$$\int_{(F_b,\mathrm{or}_{\mathrm{mark}})} Q_F^2 = 12, \qquad \int_{(F_b,\mathrm{or}_{\mathbb{C}})} Q_F^2 = -12,$$

$$\int_{(F_b,\mathrm{or}_{\mathbb{C}})} Q_F \wedge \Gamma \wedge d\zeta_2 = -1.$$

They prove the needed Dolbeault nonvanishing, while the constant fibre coefficients give the typed vanishing

$$i_b^* d_1[G \wedge Q]_{\bar\partial} = i_b^*[\partial(G \wedge Q)]_{\bar\partial} = 0 \quad \text{in } H^{2,2}_{\bar\partial}(F_b).$$

Choose $\upsilon \in \mathcal{A}^{3,3}(X)$ with $\int_X \upsilon = 1$. Together with the unit $\mathbf{1} = [1]_{BC} \in H^{0,0}_{BC}(X)$, define

class		bidegree
$\mathsf{A} = [\partial G]_{BC},$	$\mathsf{B} = [\bar\partial\bar G]_{BC}$	$(1,1)$
$\mathsf{C} = [\partial Q]_{BC},$	$\overline{\mathsf{C}} = [\bar\partial\bar Q]_{BC}$	$(2,1),(1,2)$
$\mathsf{D} = [\partial(G\wedge Q)]_{BC},$	$\overline{\mathsf{D}} = [\bar\partial(\bar G\wedge\bar Q)]_{BC}$	$(2,2),(2,2)$
$\mathsf{E} = [\partial(Q^2)]_{BC},$	$\overline{\mathsf{E}} = [\bar\partial(\bar Q^2)]_{BC}$	$(3,2),(2,3)$
$\mathsf{T} = [\upsilon]_{BC},$	$\int_X \mathsf{T} = 1$	$(3,3).$

These are bases of every nonzero positive-degree Bott–Chern group. For their span N_{BC},

$$H^{*,*}_{BC}(X) = \mathbb{C}\mathbf{1} \oplus N_{\mathrm{BC}}, \qquad N_{\mathrm{BC}}^2 = 0.$$

See theorems 57.4 and 57.5. The lift and smooth-gauge ambiguities above change neither the verified dimensions nor the square-zero product algebra.

21. **[VERIFIED]** The positive radial cusp now has an explicitly declared flat quotient metric and an exact scalar eigenvalue calculation. Write

$$\mu_T = \alpha_T + im_T, \qquad h_T = \eta_T + iH_T, \qquad b_T = \xi_T + iB_T, \qquad c_T = \xi_T - \eta_T,$$
$$L_T = T + H_T, \qquad q_T = L_T - B_T, \qquad D_T = L_T q_T + 6m_T^2.$$

Set also

$$m = \operatorname{Im}\mu(0), \qquad H = \operatorname{Im} h(0), \qquad B = \operatorname{Im} b(0).$$

In the ordered lattice basis $(\widehat{\gamma}, \widehat{u}, \widehat{w}, \widehat{\delta})$ and real coordinate order (x_1, y_1, x_2, y_2), the real period matrix, its oriented determinant, and its positive covolume are

$$P_T = \begin{pmatrix} 6\alpha_T & \eta_T & 1 & 0 \\ 6m_T & L_T & 0 & 0 \\ c_T & \alpha_T & 0 & 1 \\ -q_T & m_T & 0 & 0 \end{pmatrix}, \qquad \det P_T = -D_T, \qquad \operatorname{covol}(P_T\mathbb{Z}^4) = D_T > 0, \qquad D_{\mathrm{can},T} = -\frac{D_T}{L_T} < 0.$$

The full inverse-dual certificate is

$$P_T^{-\mathsf{T}} = \begin{pmatrix} 0 & 0 & 1 & 0 \\ \dfrac{m_T}{D_T} & \dfrac{q_T}{D_T} & \dfrac{-6m_T\alpha_T - q_T\eta_T}{D_T} & \dfrac{-m_T c_T - q_T\alpha_T}{D_T} \\ 0 & 0 & 0 & 1 \\ -\dfrac{L_T}{D_T} & \dfrac{6m_T}{D_T} & \dfrac{6(L_T\alpha_T - m_T\eta_T)}{D_T} & \dfrac{L_T c_T - 6m_T\alpha_T}{D_T} \end{pmatrix}.$$

Thus $(P_T\mathbb{Z}^4)^* = P_T^{-\mathsf{T}}\mathbb{Z}^4$, and for $k = (r, s, n_1, n_2)^{\mathsf{t}}$, with

$$R_T(k) = r - 6\alpha_T n_1 - c_T n_2, \qquad S_T(k) = s - \eta_T n_1 - \alpha_T n_2,$$

its member indexed by k is exactly

$$\left(n_1, \frac{m_T R_T(k) + q_T S_T(k)}{D_T}, n_2, \frac{-L_T R_T(k) + 6m_T S_T(k)}{D_T}\right).$$

For the declared Euclidean quotient metric g_T, the first positive scalar eigenvalue is, for all large T,

$$\lambda_1(F_T, g_T) = 4\pi^2 \min\left\{\frac{L_T^2 + m_T^2}{D_T^2}, \frac{q_T^2 + 36m_T^2}{D_T^2}\right\} = \frac{4\pi^2}{T^2}\left(1 + \frac{-2H + 2\min\{B, 0\}}{T} + O(T^{-2})\right).$$

The metric normalization is recorded separately and gives

$$\widehat{g}_T = D_T^{-1/2} g_T, \qquad \operatorname{Vol}(F_T, \widehat{g}_T) = 1,$$
$$\lambda_1(F_T, \widehat{g}_T) = \frac{4\pi^2}{T}\left(1 + \frac{-H - B/2 + 2\min\{B, 0\}}{T} + O(T^{-2})\right),$$
$$\lambda_1(F_T, \widehat{g}_T) \longrightarrow 0.$$

Normalized Haar measure on the fixed angular torus is only the unitary measure convention used below; it is not the metric rescaling $g_T \mapsto \widehat{g}_T$. See theorems 11.1 to 11.3.

22. **[VERIFIED]** The degree- N^2 regular-fibre cover extends across the completed local toric cusp, with the base-change and total degrees kept distinct. Retain

$$B_0 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad Z(s) = sB_0 + C(t), \qquad t = e^{2\pi i s},$$

let $Y = Y_\Sigma$ be the toric threefold of the height-one translation-invariant A_2 fan, and let the target action be

$$\Psi_{\bar\lambda}(x,t) = (e^{2\pi i C(t)\bar\lambda} t^{B_0\bar\lambda} x, t), \qquad f_0 : N_0 := Y/\bar\Lambda \longrightarrow \Delta_t.$$

For every integer $N \geq 1$, after shrinking the disc, put

$$\rho_N(v) = v^N, \qquad C_N(v) = \frac{1}{N} C(v^N), \qquad \Psi^{(N)}_{\bar\lambda}(\xi, v) = (e^{2\pi i C_N(v)\bar\lambda} v^{B_0\bar\lambda}\xi, v).$$

This is a free properly discontinuous action on a copy $Y^{(N)}$ of Y, and

$$f_N : N_0^{\langle N\rangle} := Y^{(N)}/\bar\Lambda \longrightarrow \Delta_v$$

is a proper holomorphic map from a smooth threefold, with reduced irreducible nonnormal central fibre W_N normalized by dP_6. Multiplication by N on the three-dimensional cocharacter lattice gives

$$P_N : Y^{(N)} \longrightarrow Y, \qquad (\xi_1, \xi_2, v) \longmapsto (\xi_1^N, \xi_2^N, v^N),$$

whose maximal-chart ring map is

$$\mathbb{C}\{z_0, z_1, z_2\} \longrightarrow \mathbb{C}\{w_0, w_1, w_2\}, \qquad z_i \longmapsto w_i^N.$$

It is finite flat of degree N^3, is equivariant for the two displayed actions, and descends to a finite flat degree- N^3 map

$$\kappa_N : N_0^{\langle N\rangle} \longrightarrow N_0, \qquad f_0 \circ \kappa_N = \rho_N \circ f_N, \qquad \kappa_N^* W = N W_N, \qquad \mathrm{Ram}(\kappa_N) = (N-1)W_N.$$

For the normal base change $N_0[N] = N_0 \times_{\Delta_t} \Delta_v$, the unique factorization

$$N_0^{\langle N\rangle} \xrightarrow{Q_N} N_0[N] \longrightarrow N_0$$

has generic degree N^2. If $N > 1$, Q_N is quasi-etale and nonflat: its non-etale and nonflat loci are precisely the inverse images of the double and triple strata, where the closed-fibre lengths are respectively $2N - 1 > N$ and $3N^2 - 3N + 1 > N^2$. On reduced central fibres the generic degree is N^2, and the normalization square is

$$\begin{array}{ccc} \mathrm{dP}_6 & \xrightarrow{[N]} & \mathrm{dP}_6 \\ {\scriptstyle \nu_N}\downarrow & & \downarrow{\scriptstyle \nu} \\ W_N & \longrightarrow & W, \end{array}$$

where $[N]$ has degree N^2 and ramification index N on each of the six toric boundary curves. Over $v \neq 0$, Q_N is the cover for

$$\mathbb{Z}\langle \widehat\gamma, \widehat u, N\widehat w, N\widehat\delta\rangle \subset \Lambda, \qquad \mathrm{Deck}(Q_N) \cong (\mathbb{Z}/N\mathbb{Z})^2,$$

and, with $S_N = \mathrm{diag}(1, 1, N, N)$,

$$S_N^{-1} M_0^N S_N = M_0.$$

This is a local completed-cusp morphism, not a degree- N^2 self-map over the unchanged disc, not a normalization, and not a cover over the entire base; see theorems 11.8 to 11.10. Its use in four-dimensional finite-cover unspooling is an ancillary construction: it changes both the fibre lattice and the cusp base. It does not answer the native S^{12}/S^{24} sphere-route problem, whose solution begins from the unaltered quotient fibres Q_T and C_T.

23. **[VERIFIED]** The native cusp quotient and the invariant-circle quotient give distinct exact conical smash-sphere carriers. With

$$K_\Lambda = \mathbb{Z}\langle \widehat{w}, \widehat{\delta}\rangle = \mathrm{im}(M_0 - I), \qquad \overline{\Lambda} = \Lambda/K_\Lambda, \qquad \overline{T}^2 = (\overline{\Lambda}\otimes\mathbb{R})/\overline{\Lambda},$$

there is a fibrewise real-analytic group epimorphism

$$\overline{\pi} : N_0 \setminus W \longrightarrow D_0^* \times \overline{T}^2$$

with connected kernel and trivial target local system. On $F_T = f^{-1}(e^{-2\pi T})$, it is π_T after the isometry

$$\iota_T : \overline{T}^2 \xrightarrow{\sim} Q_T, \qquad [(a,b)] \longmapsto [Y_T(a,b)^{\mathsf{T}}], \qquad Y_T = \begin{pmatrix} 6m_T & L_T \\ -q_T & m_T \end{pmatrix},$$

and, for a fixed constant $C_{\mathrm{sys}} > 0$, $\mathrm{sys}(Q_T) \geq T - C_{\mathrm{sys}}$. For

$$\mathcal{A} = \frac{[0,\epsilon]\times S^1 \times \overline{T}^2}{\{0\}\times S^1\times\overline{T}^2 \sim v_0,\ \{\epsilon\}\times S^1\times\overline{T}^2 \sim v_1} \cong \mathrm{Susp}(T^3),$$

the map

$$q_2 : (X, x_0) \longrightarrow (\mathcal{A}, v_0), \qquad q_2(x) = [r, e^{i\phi}, \overline{\pi}_f(x)]$$

for $t(f(x)) = re^{i\phi}$, with $\overline{\pi}_f = \mathrm{pr}_{\overline{T}^2}\circ\overline{\pi}$, $q_2(W) = v_0$ and $q_2(X\setminus \mathrm{int}\, N_0) = v_1$, is a continuous surjective based quotient. Hence

$$q_2^{\wedge 2} : X\wedge X \cong S^{12} \longrightarrow \mathcal{A}\wedge\mathcal{A}$$

is global, and its synchronized radial restriction is exactly

$$\pi_T\times\pi_T : F_T\times F_T \longrightarrow Q_T\times Q_T, \qquad (Q_T\times Q_T, 0) \xrightarrow[T\to\infty]{\text{pointed}} (\mathbb{R}^4, 0)$$

in pointed local-isometric, pointed smooth, and pointed Gromov–Hausdorff topology.

The cusp-supported one-circle route uses

$$\mathcal{B} = \frac{[0,\epsilon]\times S^1\times S^1}{\{0\}\times S^1\times S^1 \sim b_0,\ \{\epsilon\}\times S^1\times S^1 \sim b_1} \cong \mathrm{Susp}(T^2)$$

and the continuous surjective based quotient

$$q_{1,\mathrm{cusp}} : (X, x_0) \longrightarrow (\mathcal{B}, b_0), \qquad q_{1,\mathrm{cusp}}(x) = [r, e^{i\phi}, \chi_\gamma(x)]$$

on $N_0\setminus W$, with $q_{1,\mathrm{cusp}}(W) = b_0$ and the exterior collapsed to b_1. Its fourth smash power is

$$q_{1,\mathrm{cusp}}^{\wedge 4} : X^{\wedge 4} \cong S^{24} \longrightarrow \mathcal{B}^{\wedge 4},$$

and its synchronized radial restriction is $\chi_{\gamma,T}^{\times 4} : F_T^4 \to C_T^4$.

For the full-base circle route, let

$$P = B \setminus \bigcup_{j=0}^{2} \mathrm{int}\, D_j, \qquad \mathcal{M} = (P\times S^1) \underset{h_1^{S^1}}{\cup} C_1 \underset{h_2^{S^1}}{\cup} C_2, \qquad \partial\mathcal{M} = \partial D_0 \times S^1,$$

and $\mathcal{C} = \mathcal{M}\cup_{\partial\mathcal{M}} \mathrm{Cone}(\partial\mathcal{M})$, with cone vertex c_0. The continuous surjective based quotient

$$q_1 : (X, x_0) \longrightarrow (\mathcal{C}, c_0)$$

equals (f,θ_γ) over P, equivalently $\chi_\gamma|_{f^{-1}(P)}$, and equals the finite-filling maps $N_j \to C_j$ for $j=1,2$, sends $x \in N_0 \setminus W$ to $[r,(e^{i\phi},\theta_\gamma(x))]$, and sends W to c_0. Its fourth smash power is

$$q_1^{\wedge 4} : X^{\wedge 4} \cong S^{24} \longrightarrow \mathcal{C}^{\wedge 4},$$

whose synchronized radial restriction is exactly

$$\begin{aligned} \chi_{\gamma,T}^{\times 4} &: F_T^4 \longrightarrow C_T^4, \\ C_T &= \mathbb{R}/\ell_T\mathbb{Z}, \\ \ell_T &= \frac{D_T}{\sqrt{L_T^2+m_T^2}} = T + H - B + O(T^{-1}), \end{aligned}$$

so

$$(C_T^4, 0) \xrightarrow[T\to\infty]{\text{pointed}} (\mathbb{R}^4, 0)$$

in the same three pointed topologies.

The exponents 2 and 4, and hence the source-sphere dimensions 12 and 24, are optimal for these two native quotient problems with the unscaled quotient of the marked Euclidean fibre metric. Indeed, if a primitive kernel $N_{\ker} \subset \Lambda^{\oplus r}$ defines a connected torus quotient of r cusp fibres whose quotient systole tends to infinity, then every uniformly bounded period must lie in $N_{\ker}$. Thus

$$K_\Lambda^{\oplus r} \subset N_{\ker}, \qquad \operatorname{rank}(\Lambda^{\oplus r}/N_{\ker}) \le 2r.$$

For a quotient over the full regular base, or over any finite cover of that base, monodromy forces the additional sublattice $\mathbb{Z}\langle\widehat{u}\rangle^{\oplus r}$ into $N_{\ker}$: one uses $A_1\widehat{w} = \widehat{u} - \widehat{w}$, and on a finite cover the exact formula

$$(A_2A_1)^m\widehat{w} = \widehat{w} + m(\widehat{u} - \widehat{\delta})$$

together with primitivity of $N_{\ker}$. Consequently

$$\operatorname{rank}(\Lambda^{\oplus r}/N_{\ker}) \le r.$$

A rank-four expanding quotient therefore needs at least two copies cusp-locally and at least four copies globally; the displayed Q_T^2 and C_T^4 constructions attain the two bounds.

After collapsing the coordinate one-skeleton

$$K^{(1)} = ((\mathbb{R}/\mathbb{Z}) \times \{0\}) \cup (\{0\} \times (\mathbb{R}/\mathbb{Z})) \subset \overline{T}^2,$$

the two native routes also give continuous surjective based quotient maps

$$\widehat{q}_3 : X \twoheadrightarrow S^3, \qquad \widehat{q}_\gamma : X \twoheadrightarrow S^2,$$

and hence, using the fixed based choice $X \cong_{\text{diff}} S^6$,

$$\widehat{q}_3^{\wedge 2} : S^{12} \twoheadrightarrow S^6, \qquad \widehat{q}_\gamma^{\wedge 4} : S^{24} \twoheadrightarrow S^8.$$

Both full sphere carriers are based null-homotopic. First let $f = \widehat{q}_\gamma : S^6 \to S^2$ after the chosen based identification. There is an exact factorization

$$S^{12} = S^6 \wedge S^6 \xrightarrow{f\wedge \mathrm{id}_{S^6} = \Sigma^6 f} S^2 \wedge S^6 = S^8 \xrightarrow{\mathrm{id}_{S^2}\wedge f} S^2 \wedge S^2 = S^4.$$

Freudenthal stability gives $\pi_{12}(S^8) \cong \pi_4^{\mathrm{S}} = 0$, so $f \wedge f$ is based null-homotopic. Since $f^{\wedge r} = (f \wedge f) \wedge f^{\wedge(r-2)}$, it follows that

$$[f^{\wedge r}] = 0 \in \pi_{6r}(S^{2r}) \qquad \text{for every } r \ge 2.$$

In particular, the displayed surjective quotient $\widehat{q}_\gamma^{\wedge 4} : S^{24} \twoheadrightarrow S^8$ is itself based null-homotopic.

For the other carrier, let $g : S^6 \to S^3$ be any based map. Its exact smash factorization is

$$g \wedge g = (\Sigma^3 g) \circ (\Sigma^6 g) : S^{12} \longrightarrow S^9 \longrightarrow S^6.$$

Write st for stabilization. The exact low-stem data are

$$\pi_6(S^3) \cong \mathbb{Z}/12, \quad \pi_3^{\mathrm{S}} = \mathbb{Z}/24\langle \nu \rangle, \quad \pi_6^{\mathrm{S}} = \mathbb{Z}/2\langle \nu^2 \rangle,$$

$$\pi_{12}(S^6) = \mathbb{Z}/2\langle \nu_6 \circ \nu_9 \rangle, \qquad \mathrm{st} : \pi_{12}(S^6) \xrightarrow{\cong} \pi_6^{\mathrm{S}}.$$

The full groups are tabulated in [10, p. 339]; the signed smash formula and the unstable stabilization calculation are [13, Propositions 3.1 and 5.11]. Since $12[g] = 0$, one has $\mathrm{st}([g]) = 2a\nu$ for some $a \in \mathbb{Z}/12$. By multiplicativity,

$$\mathrm{st}([g \wedge g]) = \mathrm{st}([g])^2 = 4a^2\nu^2 = 0 \in \pi_6^{\mathrm{S}}.$$

The stabilization isomorphism forces $[g \wedge g] = 0 \in \pi_{12}(S^6)$. Thus $g^{\wedge r}$ is based null-homotopic for every $r \geq 2$. Taking $g = \widehat{q}_3$ proves

$$[\widehat{q}_3^{\wedge 2}] = 0 \in \pi_{12}(S^6), \qquad [\widehat{q}_\gamma^{\wedge 4}] = 0 \in \pi_{24}(S^8).$$

Both displayed maps remain surjective quotient maps. The one-copy classes $[\widehat{q}_3] \in \pi_6(S^3)$ and $[\widehat{q}_\gamma] \in \pi_6(S^2)$ have not been computed. Their synchronized radial restrictions contain based topologically embedded four-spheres, equipped separately with the intrinsic quotient length metrics

$$\mathbb{S}^4_{12,T} = (Q_T / K_T^{(1)})^{\wedge 2} \hookrightarrow S^6, \qquad \mathbb{S}^4_{24,T} = C_T^{\wedge 4} \hookrightarrow S^8,$$

obtained from, respectively,

$$F_T^2 \xrightarrow{\pi_T^{\times 2}} Q_T^2 \xrightarrow{c_T^{\times 2}} (S_T^2)^2 \longrightarrow \mathbb{S}^4_{12,T}, \qquad F_T^4 \xrightarrow{\chi_{\gamma,T}^{\times 4}} C_T^4 \longrightarrow \mathbb{S}^4_{24,T},$$

where the last arrows collapse the fat wedges. The full maps from F_T^2 and F_T^4 retain the nontrivial kernels of their first arrows; only the quotient-stage maps are injective off their explicitly collapsed skeleton/fat-wedge loci. Equip the displayed products with their Euclidean ℓ^2-product metrics and give each four-sphere the induced quotient length metric. If z_T is the image of $[(1/2, 1/2)]$ and

$$\zeta_{12,T} = [z_T, z_T], \qquad \zeta_{24,T} = [\ell_T/2, \ell_T/2, \ell_T/2, \ell_T/2],$$

then every radius-R ball about these points is exactly a Euclidean four-ball once, respectively,

$$T - C_{\mathrm{sys}} > 4R, \qquad \ell_T > 4R.$$

Therefore

$$(\mathbb{S}^4_{12,T}, \zeta_{12,T}) \longrightarrow (\mathbb{R}^4, 0), \qquad (\mathbb{S}^4_{24,T}, \zeta_{24,T}) \longrightarrow (\mathbb{R}^4, 0)$$

in pointed local-isometric and pointed Gromov–Hausdorff topology, with smooth convergence on the eventual exact Euclidean neighborhoods of the displayed base points. Both based inclusion maps

$$i_{12,T} : \mathbb{S}^4_{12,T} \hookrightarrow S^6, \qquad i_{24,T} : \mathbb{S}^4_{24,T} \hookrightarrow S^8$$

are null-homotopic: sliding their fixed interior suspension coordinate to a suspension endpoint contracts them through based maps. Thus $(i_{12,T})_*[S^4] = 0 \in H_4(S^6; \mathbb{Z})$ and $(i_{24,T})_*[S^4] = 0 \in H_4(S^8; \mathbb{Z})$. These slice-inclusion null-homotopies are distinct from the global equalities $[\widehat{q}_3^{\wedge 2}] = 0 \in \pi_{12}(S^6)$ and $[\widehat{q}_\gamma^{\wedge 4}] = 0 \in \pi_{24}(S^8)$.

Deleting the smash basepoint nevertheless gives literal nonclosed topologically embedded copies of $\mathbb{R}^4$:

$$(Q_T \setminus K_T^{(1)})^2 \cong \mathbb{S}^4_{12,T} \setminus \{*\} \cong \mathbb{R}^4 \hookrightarrow S^6, \qquad (C_T \setminus \{0\})^4 \cong \mathbb{S}^4_{24,T} \setminus \{*\} \cong \mathbb{R}^4 \hookrightarrow S^8.$$

They are open only in the embedded four-spheres and are not globally Euclidean-isometric under a fixed compatible compact-target metric.

There are also Hausdorff, second-countable topological five-manifolds with boundary $\mathfrak{E}_{12}$ and $\mathfrak{E}_{24}$ whose positive fibres are $Q^{\times 2}_{1/s}$ and $C^{\times 4}_{1/s}$, respectively, and whose zero fibres are literally $\mathbb{R}^4$. Continuous surjections $\beta_\nu : \mathfrak{E}_\nu \to \mathcal{D}_\nu$ identify every positive fibre with its radial carrier slice and collapse the zero fibre to the cone vertex. They are homeomorphisms off the zero fibre but are nonproper and nonquotient; thus they are comparison/collapse maps, not topological blow-downs or resolutions, and $\mathcal{D}_\nu$ is not the quotient $\mathfrak{E}_\nu/\mathbb{R}^4$ via β_ν.

The metric quantifiers are essential. For every fixed metric compatible with the compact conical target topology, the target slices $j_T(Q^2_T)$, $k^{\mathcal{B}}_T(C^4_T)$, and $k^{\mathcal{C}}_T(C^4_T)$ converge in Hausdorff, hence Gromov–Hausdorff, distance to the smash basepoint. A fixed cusp-adapted metric can realize the intrinsic limits only after a moving-point blow-up: with $r_T = e^{-2\pi T}$, the factors T/r_T and ℓ_T/r_T give the exact induced slice metrics

$$\left(\frac{T}{r_T}\right)^2 j_T^* g_{Z_{12}} = g_{Q_T} \oplus g_{Q_T}, \qquad \left(\frac{\ell_T}{r_T}\right)^2 (k^{\mathcal{D}}_T)^* g_{Z^{\mathcal{D}}_{24}} = g^{\oplus 4}_{C_T}, \quad \mathcal{D} \in \{\mathcal{B}, \mathcal{C}\}.$$

The slice bubbles are $\mathbb{R}^4$, but the corresponding full ambient bubbles are $\mathbb{R}^8$ for the S^{12} route and $\mathbb{R}^{12}$ for either S^{24} route. More generally, if $\lambda_T > 0$ and $\beta_T = \lambda_T r_T$, the slice limit is a point when $\beta_T \to 0$, a square flat four-torus of side L when $\beta_T \to L \in (0, \infty)$, and $\mathbb{R}^4$ when $\beta_T \to \infty$; in the last case the cone vertex is at scaled distance $\beta_T \to \infty$. Hence the $\mathbb{R}^4$ limits are neither vertex tangent cones nor ambient limits of fixed smooth metrics on the source spheres, whose ordinary tangent dimensions are 12 and 24.

The original maps $q_2^{\wedge 2}$, $q_{1,\mathrm{cusp}}^{\wedge 4}$, and $q_1^{\wedge 4}$ have the displayed conically stratified codomains. The repeated-smash null theorems for the sphere-valued composites $\widehat{q}_3$ and $\widehat{q}_\gamma$ do not supply null-homotopies before the final one-skeleton/base-angle quotients. The unstable classes of

$$q_2^{\wedge 2} : S^{12} \to \mathcal{A}^{\wedge 2}, \qquad q_1^{\wedge 4} : S^{24} \to \mathcal{C}^{\wedge 4}$$

remain **[OPEN]**: their zero ordinary Hurewicz images and absence of homotopy sections do not decide nullity. The class of $q_{1,\mathrm{cusp}}^{\wedge 4} : S^{24} \to \mathcal{B}^{\wedge 4}$ has likewise not been computed. The sphere codomains occur only after the additional one-skeleton and base-angle collapses defining $\widehat{q}_3$ and $\widehat{q}_\gamma$. All identifications of the sources with S^{12} and S^{24} use the chosen based diffeomorphism $X \cong_{\mathrm{diff}} S^6$. See theorems 11.11 to 11.14, 11.16, 11.18 to 11.25, 11.27, 11.28, 11.36 to 11.38 and 11.40.

24. **[REFUTED]** The assertion that the literal fixed-target map

$$\chi_\gamma : J \longrightarrow B^\circ \times (\mathbb{R}/\mathbb{Z})$$

extends continuously across W, or that

$$\chi_\gamma^{[4]} : J^{[4]} \longrightarrow B^\circ \times (\mathbb{R}/\mathbb{Z})^4$$

extends with the stated regular-fibre restriction to the compact fourfold fibre product or to X^4, is false: two explicit paths have the same toric limit in W and constant character values $3/4$ and $1/2$. The cone map above changes the codomain by collapsing $\partial D_0 \times S^1$ to c_0, so it does not contradict this nonextension. Moreover

$$X^2 \simeq S^6 \times S^6 \not\simeq S^{12}, \qquad X^4 \simeq (S^6)^4 \not\simeq S^{24};$$

only the smash products satisfy $X^{\wedge 2} \cong S^{12}$ and $X^{\wedge 4} \cong S^{24}$. Neither smash sphere admits an almost-complex structure, and neither route constructs a compact complex sphere; see theorems 11.7 and 11.18.

25. **[VERIFIED]** The exceptional first-coordinate elliptic quotient is now classified on the actual period image, rather than on an arbitrary ambient path. For

$$\mathcal{A}_z = \mathbb{C}^2/\mathbb{Z}\langle (6\mu, \beta), (\tau, \mu), (1,0), (0,1)\rangle,$$

the first coordinate induces a surjective holomorphic homomorphism

$$\mathcal{A}_z \longrightarrow \mathbb{C}/(\mathbb{Z}\tau + \mathbb{Z})$$

if and only if $6\mu = a\tau + b$ for a unique $(a, b) \in \mathbb{Z}^2$. Its connected kernel has period $\tau_f = a\mu - \beta$ and $\operatorname{Im} \tau_f = -D_{\mathrm{can}} > 0$. The exact marked-label action is

$$g_0 : (a, b) \mapsto (a, a + b), \qquad g_2 : (a, b) \mapsto (-b, 6 + a).$$

The union of these loci has finite image in the completed orbifold base and is absent from a punctured cusp neighborhood. Consequently this particular elliptic/Landau sector does not produce the cusp soft modes. This statement does not classify all elliptic quotients, all Néron–Severi classes, or all smooth integral magnetic fluxes. See theorems 11.41 to 11.43.

26. **[VERIFIED]** The fibrewise small eigenvalues are assembled into one specified gapless operator. On

$$\mathscr{H}_{\mathrm{aux}} = L^2([T_0, \infty), dT; L^2_0(\mathbb{R}^4/\mathbb{Z}^4, dy)) \cong \bigoplus_{k \in \mathbb{Z}^4 \setminus \{0\}} L^2([T_0, \infty), dT),$$

put

$$\lambda_k(T) = 4\pi^2 |P_T^{-\mathsf{T}} k|^2, \qquad (\mathscr{L}_{\mathrm{aux}} u)_k(T) = \lambda_k(T) u_k(T),$$

$$D(\mathscr{L}_{\mathrm{aux}}) = \left\{ u : \sum_{k \neq 0} \int_{T_0}^{\infty} \lambda_k(T)^2 |u_k(T)|^2 \, dT < \infty \right\}.$$

This multiplication operator is nonnegative and self-adjoint, $\ker \mathscr{L}_{\mathrm{aux}} = 0$, and the escaping-support Weyl sequence in the $k = (1, 0, 0, 0)$ channel proves

$$0 \in \sigma_{\mathrm{ess}}(\mathscr{L}_{\mathrm{aux}}).$$

This is a spectral-gap theorem for the displayed auxiliary direct integral, not for the complex threefold without a metric and not for a quantum gauge Hamiltonian; see theorem 52.1.

27. **[VERIFIED]** The cover index and the cubical mesh are independent. For integers $M_{\mathrm{cov}} \geq 1$ and $N_{\mathrm{mesh}} \geq 2$, put

$$S_{M_{\mathrm{cov}}} = \operatorname{diag}(1, 1, M_{\mathrm{cov}}, M_{\mathrm{cov}}), \qquad P_{T, M_{\mathrm{cov}}} = P_T S_{M_{\mathrm{cov}}},$$

$$\widetilde{F}_{T, M_{\mathrm{cov}}} = \mathbb{R}^4 / P_{T, M_{\mathrm{cov}}} \mathbb{Z}^4 \longrightarrow F_T, \qquad \mathrm{Deck} \cong (\mathbb{Z}/M_{\mathrm{cov}}\mathbb{Z})^2.$$

This cover has degree M_{cov}^2, Gram matrix and covolume

$$G_{T, M_{\mathrm{cov}}} = P_{T, M_{\mathrm{cov}}}^{\mathsf{T}} P_{T, M_{\mathrm{cov}}}, \qquad \mathcal{D}_{T, M_{\mathrm{cov}}} = M_{\mathrm{cov}}^2 D_T.$$

For the cubical complex write, once and for all,

$$V_{N_{\mathrm{mesh}}} = (\mathbb{Z}/N_{\mathrm{mesh}}\mathbb{Z})^4, \qquad E_{N_{\mathrm{mesh}}} = V_{N_{\mathrm{mesh}}} \times \{1, 2, 3, 4\}.$$

The diffeomorphism

$$\Phi_{T,M_{\mathrm{cov}}} : \mathbb{R}^4/\mathbb{Z}^4 \xrightarrow{\sim} \widetilde{F}_{T,M_{\mathrm{cov}}}, \qquad [y] \longmapsto [P_{T,M_{\mathrm{cov}}} y]$$

transports the whole standard N_{mesh}-cubical complex; its vertex map is precisely

$$(\mathbb{Z}/N_{\mathrm{mesh}}\mathbb{Z})^4 \xrightarrow{\sim} \widetilde{F}_{T,M_{\mathrm{cov}}}[N_{\mathrm{mesh}}], \qquad [n] \longmapsto \left[\frac{P_{T,M_{\mathrm{cov}}} n}{N_{\mathrm{mesh}}}\right],$$

so the finite set is the torsion subgroup, not the entire torus. Its oriented edge is

$$\gamma_{n,j}(t) = \left[\frac{P_{T,M_{\mathrm{cov}}}(n + t\mathbf{e}_j)}{N_{\mathrm{mesh}}}\right], \qquad 0 \le t \le 1.$$

For a compact matrix Lie group G, the parallel-transport convention

$$U'(t) = -A_{\gamma(t)}(\dot\gamma(t))U(t), \qquad U(0) = 1$$

defines the typed map

$$\mathrm{Hol}_{T,M_{\mathrm{cov}},N_{\mathrm{mesh}}} : \Omega^1(\widetilde{F}_{T,M_{\mathrm{cov}}}; \mathfrak{g}) \longrightarrow G^{E_{N_{\mathrm{mesh}}}}, \qquad A \longmapsto (U_A(\gamma_{n,j}))_{n,j},$$

and for $A^g = g^{-1}Ag + g^{-1}dg$ its exact endpoint law is

$$\mathrm{Hol}(A^g)_{n,j} = g(\gamma_{n,j}(1))^{-1}\,\mathrm{Hol}(A)_{n,j}\,g(\gamma_{n,j}(0)).$$

See equation (350) and theorems 13.1 and 13.2.

28. **[VERIFIED]** The compact-link SU(2) flux uses the inverse-transport convention and has an exact metric-weighted action. With

$$H_{\mathrm{link}} = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \qquad \|H_{\mathrm{link}}\|^2 = 1, \qquad \exp(2\pi H_{\mathrm{link}}) = 1, \qquad N = N_{\mathrm{mesh}},$$

set

$$\begin{aligned} U_1(n) &= \exp(-2\pi n_2 H_{\mathrm{link}}/N^2), \\ U_2(n) &= \begin{cases} 1, & 0 \le n_2 \le N-2, \\ \exp(2\pi n_1 H_{\mathrm{link}}/N), & n_2 = N-1, \end{cases} \\ U_3(n) &= U_4(n) = 1. \end{aligned}$$

For

$$U_{ij}(n) = U_i(n)U_j(n+\mathbf{e}_i)U_i(n+\mathbf{e}_j)^{-1}U_j(n)^{-1},$$

including all wrap-around edges,

$$\begin{aligned} U_{12}(n) &= \exp(2\pi H_{\mathrm{link}}/N^2), \\ U_{ij}(n) &= 1 \quad (i<j,\ (i,j) \ne (1,2)). \end{aligned}$$

The principal logarithm is unambiguous because $0 < 2\pi/N^2 \le \pi/2$, and

$$\mathcal{F}_N^{\log}(n) = N^2\,\mathrm{Log}(U_{12}(n))\,dy^1 \wedge dy^2 = 2\pi H_{\mathrm{link}}\,dy^1 \wedge dy^2.$$

With the full inverse Gram metric,

$$S^{\log}_{T,M_{\mathrm{cov}},N}(U) = \frac{\mathcal{D}_{T,M_{\mathrm{cov}}}}{2g_{\mathrm{YM}}^2 N^4} \sum_{n \in V_{N_{\mathrm{mesh}}}} \|\mathcal{F}_N^{\log}(n)\|^2_{G^{-1}_{T,M_{\mathrm{cov}}} \otimes \mathfrak{su}(2)} = \frac{2\pi^2 M_{\mathrm{cov}}^2}{g_{\mathrm{YM}}^2 D_T}.$$

There is no canonical scalar Wilson action for a general nonorthogonal metric obtained by independent scalar plaquette weights. For this configuration the explicitly defined $(1,2)$-sector functional has

$$\kappa^{12}_{T,M_{\mathrm{cov}}} = (G^{-1}_{T,M_{\mathrm{cov}}})_{11}(G^{-1}_{T,M_{\mathrm{cov}}})_{22} - (G^{-1}_{T,M_{\mathrm{cov}}})^2_{12} = D_T^{-2},$$

$$S^{W,12}_{T,M_{\mathrm{cov}},N}(U) = \frac{\mathcal{D}_{T,M_{\mathrm{cov}}}\kappa^{12}_{T,M_{\mathrm{cov}}}}{2g^2_{\mathrm{YM}}} \sum_{n\in V_{N_{\mathrm{mesh}}}} (2 - \operatorname{Re}\operatorname{tr} U_{12}(n)) = \frac{M^2_{\mathrm{cov}}N^4}{2g^2_{\mathrm{YM}}D_T}\left(2 - 2\cos\frac{2\pi}{N^2}\right)$$

Its exact mesh expansion is

$$S^{W,12}_{T,M_{\mathrm{cov}},N}(U) = \frac{M^2_{\mathrm{cov}}}{g^2_{\mathrm{YM}}D_T}\left(2\pi^2 - \frac{2\pi^4}{3N^4} + \frac{4\pi^6}{45N^8} + O(N^{-12})\right),$$

so its $N\to\infty$ limit is $2\pi^2 M^2_{\mathrm{cov}}/(g^2_{\mathrm{YM}}D_T)$. See theorem 13.3.

29. **[VERIFIED]** The finite Gaussian oscillator is defined on the real reduced Coulomb space with the physical inverse-Gram exterior metric. For

$$\vartheta^j_{T,M_{\mathrm{cov}}} = P^{-\mathsf{T}}_{T,M_{\mathrm{cov}}}\mathbf{e}_j, \qquad \mathfrak{d}_{N_{\mathrm{mesh}}}(k)_j = N_{\mathrm{mesh}}(e^{2\pi i k_j/N_{\mathrm{mesh}}} - 1),$$

let

$$C^p_{N_{\mathrm{mesh}}}(T, M_{\mathrm{cov}}) = \operatorname{Map}\left((\mathbb{Z}/N_{\mathrm{mesh}}\mathbb{Z})^4, \bigwedge^p(\mathbb{R}^4)^*\otimes\mathfrak{su}(2)\right).$$

The exterior metric is

$$\left\langle \vartheta^{i_1}\wedge\cdots\wedge\vartheta^{i_p}, \vartheta^{j_1}\wedge\cdots\wedge\vartheta^{j_p}\right\rangle = \det\left((G^{-1}_{T,M_{\mathrm{cov}}})_{i_a j_b}\right)^p_{a,b=1},$$

and the cochain inner product is

$$\langle a, b\rangle_p = \frac{\mathcal{D}_{T,M_{\mathrm{cov}}}}{N^4_{\mathrm{mesh}}} \sum_{n\in V_{N_{\mathrm{mesh}}}} \langle a(n), b(n)\rangle.$$

Writing $\mathsf{T}_j a(n) = a(n+\mathbf{e}_j)$, put

$$D^+_j = N_{\mathrm{mesh}}(\mathsf{T}_j - 1), \qquad D^-_j a(n) = N_{\mathrm{mesh}}(a(n) - a(n-\mathbf{e}_j)),$$

and then

$$d_N = \sum_{j=1}^4 \varepsilon(\vartheta^j)D^+_j, \qquad d^*_N = -\sum_{j=1}^4 \iota_{(\vartheta^j)^\sharp}D^-_j,$$

$$\mathscr{C}_{T,M_{\mathrm{cov}},N_{\mathrm{mesh}}} = \ker(d^*_N : C^1\to C^0)\cap(\ker\Delta^{(1)}_N)^\perp, \qquad \mathcal{L} = g^{-2}_{\mathrm{YM}}d^*_N d_N.$$

The positive eigenvalue on the Fourier fibre $k\in(\mathbb{Z}/N_{\mathrm{mesh}}\mathbb{Z})^4\setminus\{0\}$ is

$$\lambda_{T,M_{\mathrm{cov}},N_{\mathrm{mesh}}}(k) = \frac{1}{g^2_{\mathrm{YM}}}\left\|P_T^{-\mathsf{T}}S^{-1}_{M_{\mathrm{cov}}}\mathfrak{d}_{N_{\mathrm{mesh}}}(k)\right\|^2,$$

with complex transverse multiplicity 9, and

$$\lambda(\mathbf{e}_1) = \frac{4N^2_{\mathrm{mesh}}\sin^2(\pi/N_{\mathrm{mesh}})}{g^2_{\mathrm{YM}}}\frac{L_T^2 + m_T^2}{D_T^2},$$

which is independent of M_{cov}. On $L^2(\mathscr{C}_{T,M_{\mathrm{cov}},N_{\mathrm{mesh}}}, da)$, the Friedrichs realization of

$$\widehat{H}^{\mathrm{Gauss}} = -\frac{1}{2}\Delta_{\mathscr{C}_{T,M_{\mathrm{cov}},N_{\mathrm{mesh}}}} + \frac{1}{2}\langle a, \mathcal{L}a\rangle$$

has a unique Gaussian ground state and

$$\operatorname{gap}(\widehat{H}^{\mathrm{Gauss}}) = \sqrt{\min_{k\neq 0}\lambda(k)} \leq \frac{2N_{\mathrm{mesh}}\sin(\pi/N_{\mathrm{mesh}})}{g_{\mathrm{YM}}}\frac{\sqrt{L_T^2+m_T^2}}{D_T}.$$

For the explicit sequence

$$T_j = j^2, \qquad M_{\mathrm{cov},j} = j, \qquad N_{\mathrm{mesh},j} = j^3,$$

one has simultaneously

$$\operatorname{sys}(\widetilde{F}_{T_j,M_{\mathrm{cov},j}}) \to \infty, \quad \max_r \frac{\|P_{T_j,M_{\mathrm{cov},j}}\mathbf{e}_r\|}{N_{\mathrm{mesh},j}} \to 0, \quad \operatorname{gap}(\widehat{H}^{\mathrm{Gauss}}) \to 0, \quad S^{\log}_{T_j,M_{\mathrm{cov},j},N_{\mathrm{mesh},j}}(U) \to 0.$$

See theorems 13.4 and 13.5.

30. **[VERIFIED]** The componentwise exponential

$$\operatorname{Exp}_{N_{\mathrm{mesh}}} : C^1 \longrightarrow \mathrm{SU}(2)^{E_{N_{\mathrm{mesh}}}}, \qquad a \longmapsto (\exp(a_e/N_{\mathrm{mesh}}))_e$$

is only a local diffeomorphism at zero. Its derivative is

$$D\operatorname{Exp}_{N_{\mathrm{mesh}}}(0)a = N_{\mathrm{mesh}}^{-1}a,$$

and, if a smooth compact-link action S_W has the identity as a critical point, then exactly

$$D^2(S_W \circ \operatorname{Exp}_{N_{\mathrm{mesh}}})_0[a,b] = N_{\mathrm{mesh}}^{-2}D^2(S_W)_{\mathbf{1}}[a,b].$$

It is not globally injective: a cochain supported on one edge with value $2\pi N_{\mathrm{mesh}}H_{\mathrm{link}}$ has the same image as zero. Thus

$$\mathcal{H}^{\mathrm{Gauss}} = L^2(\mathscr{C}_{T,M_{\mathrm{cov}},N_{\mathrm{mesh}}}, da) \quad \text{and} \quad \mathcal{H}^{\mathrm{inv}}_{N_{\mathrm{mesh}}} = L^2(\mathrm{SU}(2)^{E_{N_{\mathrm{mesh}}}}, dU)^{\mathrm{SU}(2)^{V_{N_{\mathrm{mesh}}}}}$$

are not identified by a global coordinate chart or a constructed unitary map; moreover $\delta_U \notin L^2$ for product Haar measure. No nonlinear compact-link Hamiltonian, Gauss-law spectral problem, time direction, reflection-positive transfer matrix, renormalized-scale limit, or Osterwalder–Schrader reconstruction is supplied by these results. See theorem 13.6 and equations (402) and (403).

31. **[VERIFIED]** The reducible direct-sum connection on $\mathscr{E}_T = \mathscr{L}_T \oplus \mathscr{L}_T^{-1}$ is the non-flat Yang–Mills critical point which, in the lifted splitting over $\mathbb{R}^4$, has the connection form

$$\nabla^T = \mathrm{d} + 2\pi u_1\,\mathrm{d}u_2\,H_{\mathrm{mag}}, \qquad H_{\mathrm{mag}} = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix},$$

$$F_{\nabla^T} = \frac{2\pi}{D_T}H_{\mathrm{mag}}\,\mathrm{d}y_1 \wedge \mathrm{d}y_2, \qquad \mathrm{d}^*_{\nabla^T}F_{\nabla^T} = 0.$$

The underlying principal $SU(2)$-bundle is smoothly trivial; after choosing any of the proved global $SU(2)$-trivializations, the displayed lifted-splitting form is carried to its gauge transform on that trivial bundle. It is not asserted to remain literally $\mathrm{d} + 2\pi u_1\,\mathrm{d}u_2\,H_{\mathrm{mag}}$ in an arbitrary global trivialization. For $\mathrm{YM}_T(\nabla) = (2g_{\mathrm{YM}}^2)^{-1}\int_{F_T}|F_\nabla|^2\,\mathrm{d}\,\mathrm{vol}_{F_T}$, its action is

$$\mathrm{YM}_T(\nabla^T) = \frac{2\pi^2}{g_{\mathrm{YM}}^2 D_T} = \frac{2\pi^2}{g_{\mathrm{YM}}^2 T^2}\left(1 + \frac{B-2H}{T} + O(T^{-2})\right).$$

On the background-gauge slice its classical Jacobi operator is

$$(\mathcal{J}_T a)_\nu = -\sum_{\mu=1}^{4} \nabla_{e_\mu}\nabla_{e_\mu} a_\nu - 2\sum_{\mu=1}^{4}[F_{\nu\mu}, a_\mu],$$

and an explicit smooth real unit vector satisfies

$$\mathrm{d}^*_{\nabla^T} a = 0, \qquad \mathcal{J}_T a = -\frac{4\pi}{D_T} a, \qquad \mathrm{YM}''_T\,|_{\nabla^T}(a,a) = -\frac{4\pi}{g^2_{\mathrm{YM}} D_T}.$$

Thus the displayed reducible classical critical point is a saddle for every finite T; this is not a statement about a quantum Yang–Mills Hamiltonian. See theorems 14.3, 14.4 and 14.8.

32. **[VERIFIED]** The monodromy-invariant full-flux class gives an exact split-Cartan $SU(r)$ theorem for every nonzero

$$\mathbf{m} = (m_1, \dots, m_r) \in \mathbb{Z}^r, \qquad \sum_i m_i = 0, \qquad s_2(\mathbf{m}) = \sum_i m_i^2.$$

On the regular native fibre define

$$\mathscr{E}_{\mathbf{m},T} = \bigoplus_{i=1}^{r} \mathscr{L}_{\eta,T}^{\otimes m_i}, \qquad H_{\mathbf{m}} = i\,\mathrm{diag}(m_1, \dots, m_r).$$

The ordered splitting canonically and parallelly trivializes the determinant, and, retaining the two opposite orientations used by the native construction,

$$c_1(\mathscr{E}_{\mathbf{m},T}) = 0, \qquad c_2(\mathscr{E}_{\mathbf{m},T}) = -\frac{s_2(\mathbf{m})}{2}[\eta_T]^2,$$

$$\langle c_2, [F_T]_{\mathrm{mark}}\rangle = -6s_2(\mathbf{m}), \qquad k_{\mathbf{m}} := \langle c_2, [F_T]_{x\text{-complex}}\rangle = 6s_2(\mathbf{m}).$$

Here $s_2(\mathbf{m})$ is even, so this split family realizes precisely a subfamily with $k_{\mathbf{m}} \in 12\mathbb{Z}_{>0}$. The diagonal connection

$$\nabla^{\mathbf{m},T} = \mathrm{d} - 2\pi(6y^1\,\mathrm{d}y^4 + y^2\,\mathrm{d}y^3)H_{\mathbf{m}}, \qquad F_{\mathbf{m},T} = -2\pi H_{\mathbf{m}}\eta_T$$

is a reducible non-flat Yang–Mills critical point and has exact action

$$\mathrm{YM}_T(\nabla^{\mathbf{m},T}) = \frac{2\pi^2 k_{\mathbf{m}}}{g^2_{\mathrm{YM}}} + \frac{\pi^2 s_2(\mathbf{m}) S_T^2}{g^2_{\mathrm{YM}} D_T}.$$

For the positive root $i < j$, put $q_{ij} = |m_i - m_j|$ and

$$\rho_T^* = \frac{\max\{\mathcal{B}_{T,+}, \mathcal{B}_{T,-}\}}{\min\{\mathcal{B}_{T,+}, \mathcal{B}_{T,-}\}}, \qquad N_-(T) = \#\{n \geq 0 : 2n+1 < \rho_T^*\}.$$

Its complete negative normalized Coulomb spectrum is

$$\nu_{ij,n}(T) = q_{ij}\big((2n+1)\mathcal{B}_{T,\min} - \mathcal{B}_{T,\max}\big), \qquad 0 \leq n < N_-(T),$$

with real multiplicity $12q_{ij}^2$ before summing coincident numerical bands. Consequently

$$\mathrm{ind}_{\mathbb{R}}\,\mathcal{J}_{\mathbf{m},T} = 12rN_-(T)s_2(\mathbf{m}),$$

and on the deep cusp tail, where $N_-(T) = 3$, this is $36rs_2(\mathbf{m}) = 6rk_{\mathbf{m}}$. The module also computes the centralizer harmonic kernel and every odd-frequency wall nullity; negative eigenspaces are descent directions, whereas only the zero-mode walls are possible Lyapunov–Schmidt bifurcation parameters. Finally, naturality of c_2 proves that this nonzero-flux bundle cannot extend over a carrier with $H^4(-;\mathbb{Z}) = 0$, in particular not over a space diffeomorphic to S^6, S^{12}, or S^{24}. See theorems 16.1, 16.2 and 16.5 to 16.7.

33. **[VERIFIED]** The Koide relation has an exact ambient quadratic-form interpretation, and it is rigorously separated from the regular canonical cusp family. For

$$Q_K(r) = \frac{2}{3}(r_1+r_2+r_3)^2 - (r_1^2+r_2^2+r_3^2), \qquad \begin{pmatrix} h \\ x \\ y \end{pmatrix} = \begin{pmatrix} 1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} \\ 1/\sqrt{2} & 0 & -1/\sqrt{2} \\ 1/\sqrt{6} & -2/\sqrt{6} & 1/\sqrt{6} \end{pmatrix} \begin{pmatrix} r_1 \\ r_2 \\ r_3 \end{pmatrix},$$

one has $Q_K(r) = h^2 - x^2 - y^2$. The real-linear coefficient-space isomorphism

$$(h,x,y) \longmapsto (t,m,\varrho) = \left(h+y, \frac{x}{\sqrt{6}}, h-y\right)$$

satisfies, with

$$(t,m,\varrho) = (\operatorname{Im}\tau, \operatorname{Im}\mu, \operatorname{Im}\beta),$$

for the canonical realification of $Z = \left(\begin{smallmatrix} 6\mu & \tau \\ \beta & \mu \end{smallmatrix}\right)$,

$$\det P(\tau,\mu,\beta) = t\varrho - 6m^2 = \det \begin{pmatrix} t & \sqrt{6}\,m \\ \sqrt{6}\,m & \varrho \end{pmatrix} = Q_K(r).$$

For $r \in \mathbb{R}^3_{\geq 0} \setminus \{0\}$, the Koide equation $Q_K(r) = 0$ makes this symmetric matrix positive semidefinite of rank one and makes P singular. Its columns therefore do not form a full lattice and do not define a regular compact complex two-torus. In contrast, the actual positive radial cusp has

$$\det P_T = -D_T < 0, \qquad D_T = T^2 + (2H - B)T + O(1), \qquad \det P_T \sim -T^2.$$

Thus the result is an isomorphism of ambient coefficient spaces with their quadratic discriminants, not a map from Koide triples to fibres of the constructed S^6 family; see theorems 30.65 to 30.67.

34. **[VERIFIED]** Independently of that discriminant comparison, the positive defect-one shell has an exact integral parametrization and a power-saving count with its exact constant. Put

$$\Delta(x,y,z) = 4(xy+xz+yz) - (x^2+y^2+z^2)$$

and let $R(X)$ count triples $0 < x \leq y \leq z \leq X$ with $\Delta = -1$ on the terminal Vieta wall $z \geq 4(x+y)$. The integral change of variables

$$n = z - 4(x+y), \qquad q = y - x, \qquad a = x + y - 4n$$

is a bijection onto

$$a^2 - 3q^2 - 18n^2 = -2, \qquad a > 0,\ q, n \geq 0,\ a + 4n > q, \qquad z = 4a + 17n.$$

Write

$$\mathscr{X} = \{(a,q,n) \in \mathbb{A}^3 : a^2 - 3q^2 - 18n^2 = -2\}.$$

The exact local densities are

$$\sigma_2 = \frac{3}{2}, \qquad \sigma_3 = 2, \qquad \sigma_p = \frac{1 - p^{-2}}{1 - (\frac{3}{p})p^{-1}} \qquad (p \geq 5),$$

and the Poincaré-residue measure of the real chamber has leading factor

$$I_\infty = \frac{1}{2\sqrt{3}} \log\left(\frac{3}{2\sqrt{2}}\right).$$

After verifying the Gorodnik–Oh–Shah sector hypotheses and Borovoi's adelic sign coefficient $c_{\mathscr{X}} = 1$, one obtains unconditionally

$$R(X) \sim C_0 X, \qquad C_0 = \frac{9}{2\pi^2} \log(2+\sqrt{3}) \log\Big(\frac{3}{2\sqrt{2}}\Big).$$

The exact ideal parametrization now proves more. For $n \geq 1$, put

$$\begin{aligned} m_n &= 18n^2 - 2 = 2(3n-1)(3n+1), \\ \mathcal{P}(m_n) &= \{(C, D, E) \in \mathbb{Z}_{>0}^2 \times \mathbb{Z} : \\ &\qquad C^2 D = m_n, \quad 0 \leq E < D, \quad E^2 \equiv 3 \pmod{D}\}. \end{aligned}$$

The map

$$(C, D, E) \longmapsto C\big(D\mathbb{Z} + (E + \sqrt{3})\mathbb{Z}\big)$$

is a bijection to the ideals of $\mathbb{Z}[\sqrt{3}]$ of norm m_n. Each ideal has a unique seed $a_0 + q_0\sqrt{3}$ with $a_0 > 2q_0 \geq 0$, and every solution in its orbit is

$$a_k + q_k\sqrt{3} = (a_0 + q_0\sqrt{3})(2+\sqrt{3})^k, \qquad k \geq 0.$$

The inverse is finite and integral: apply $(a, q) \mapsto (2a - 3q, 2q - a)$ while $a \leq 2q$, then recover the unique HNF triple. Consequently every point has multiplicity one and the number of unit orbits at level n is

$$\#\mathcal{P}(m_n) = \rho(m_n) = \sum_{d \mid m_n} \chi_{12}(d) = \rho(3n-1)\rho(3n+1).$$

The omitted $n = 0$ face is separate and exact:

$$a_k + q_k\sqrt{3} = (1+\sqrt{3})(2+\sqrt{3})^k, \qquad k \geq 1,$$

subject to the unchanged strict wall. Its contribution is the floor count $R_0(X)$ of equation (1585), and $R_0(X) = O(\log(2+X))$.

The exact sharp-height floor formula and its Hecke-phase Fourier expansion are written in theorems 43.3 and 44.2. A direct verification of Benoist–Oh effective well-roundedness for the literal five-wall family, followed by their Theorem 13.2 [54, pp. 1931–1932] on each of the finitely many integral spin-group orbits, strengthens the asymptotic to

$$\boxed{R(X) = C_0 X + O(X^{1-\eta})} \qquad \text{for some fixed } \eta > 0.$$

Neither Benoist–Oh's theorem nor its proof supplies $\eta \geq \frac{1}{2}$. The primary-source comparison is exact: Liu–Sarnak give the smooth standard-radial exponent $39/64$ for this congruence arithmetic lattice, while Good and Chatzakos–Petridis give the sharp standard-radial $2/3$ frontier after the relevant small-spectrum terms are displayed. Neither theorem counts the unchanged five-wall region with height $4a + 17n$, and neither proves its pointwise square-root discrepancy; see section 51.1.

The exact census gives $R(10^8) = 3{,}536{,}002$, after an independent six-coordinate replay of all 3,536,002 rows. Its largest $|R(X) - C_0 X|/\sqrt{X}$ on $10^4 \leq X \leq 10^8$ is $0.17140019280881455 97\ldots$, at $X = 867{,}240$; this is finite evidence only. The stronger proposed estimate $R(X) - C_0 X = O_\varepsilon(X^{1/2+\varepsilon})$ is neither asserted nor used in this status item; the proved conclusion is the displayed linear power-saving asymptotic. See theorems 39.4, 39.6, 39.7, 39.11, 39.12 and 46.2 and sections 49 and 51.1.

35. **[VERIFIED]** The same parity-restricted shell has an exact quaternionic realization. In

$$B = (18, -6)_{\mathbb{Q}} = \mathbb{Q}\langle I, J\rangle/(I^2 - 18, J^2 + 6, IJ + JI), \qquad K = IJ,$$

put

$$\mathcal{O} = \mathbb{Z}1 \oplus \mathbb{Z}\frac{I+J}{2} \oplus \mathbb{Z}\frac{I-J}{2} \oplus \mathbb{Z}\frac{K}{2}.$$

The complete multiplication table proves that this is an order. Its reduced discriminant is 108, so it has index 18 in a maximal overorder; the integer 18 is not an Eichler level, since the order is nonmaximal at the ramified primes 2 and 3. The algebra is a division algebra over $\mathbb{Q}$, split over $\mathbb{R}$, and ramified exactly at $\{2, 3\}$. The bijection

$$(a, q, n) \longmapsto \gamma(a, q, n) = 2 + \frac{qI + aJ + nK}{2}$$

identifies $a^2 - 3q^2 - 18n^2 = -2$, with its forced parity $a \equiv q \pmod 2$, with the reduced-trace-4, reduced-norm-1 locus of $\mathcal{O}$. Every such γ is primitive hyperbolic, with

$$\mathcal{O} \cap \mathbb{Q}(\gamma) = \mathbb{Z}[\gamma - 2] \cong \mathbb{Z}[\sqrt{3}], \qquad C_{\mathcal{O}^1}(\gamma) = \{\pm\gamma^k : k \in \mathbb{Z}\},$$

eigenvalues $2 \pm \sqrt{3}$, and translation length $2\log(2+\sqrt{3})$. The full shell is a finite union of $\mathcal{O}^1$-conjugacy classes. For the displayed real splitting, the standard radial Huber parameter is exactly

$$X_i(\gamma)^2 = 1 + \frac{(7a + 15\sqrt{2}\,n)^2}{48},$$

whereas the Koide height is $4a + 17n$. Thus the hyperbolic reduction is exact, but the radial theorem does not transfer a discrepancy estimate to the sharp Koide polyhedron; see sections 50, 50.7 and 50.8.

36. **[VERIFIED]** The fixed-prime Erdős–Straus divisor selector is an exact marked slice of the candidate's terminal integral presentations. For $p \equiv 1 \pmod{12}$ prime, put

$$L = \frac{p-1}{4}, \qquad R_j = 4j + 3, \qquad a_j = \frac{p + R_j}{4}, \qquad 0 \le j < L.$$

If $u \in \mathbb{Z}_{>0}$, $u \mid a_j^2$, $\varepsilon \in \{0, 1\}$, and $R_j \mid 4u + p^\varepsilon$, then

$$T = \frac{4u + p^\varepsilon}{R_j}, \qquad (\ell_0, \ell_1, \ell_2) = \left(\frac{T+1}{4}, u - jT, 1\right)$$

is integral and satisfies

$$12\ell_0 - 4\ell_1 - 3\ell_2 = p^\varepsilon,$$

with the exact relation-matrix identity

$$\begin{pmatrix} 1 & -j \\ 0 & 1 \end{pmatrix} \begin{pmatrix} R_j & -u \\ 4 & -T \end{pmatrix} = \begin{pmatrix} 3 & -\ell_1 \\ 4 & 1 - 4\ell_0 \end{pmatrix}.$$

The inverse is $T = 4\ell_0 - 1$, $u = \ell_1 + jT$. With p, j, the channel ε, positive orientation, and the divisor gate retained, this is a bijection between the complete two-channel selector and the normalized terminal slice. It proves an exact integral-relator crosswalk, not a geometric morphism from the reciprocal cubic to X; see section 6 and theorem 6.3.

37. **[VERIFIED]** The modulo-840 refinement has $[N : R_{\mathrm{ES}}] = 72$, $\det N = 1$, and root system $A_5^4 D_4$. Theorems 31.3, 31.4, 31.15, and 31.16 prove the glue, marked flag-weight comparison, full Weyl quotient and position of the commuting residue six-cycle. For a prime $p \equiv 1 \pmod{24}$ and $R > 0$, $R \equiv 3 \pmod 4$, set $a = (p+R)/4$, $g = \gcd(R, pa)$, $R_0 = R/g$, $N_0 = pa/g$. The exact divisor-to-ordered-pair map is

$$d > 0, \quad d \mid N_0^2, \quad d \equiv -N_0 \pmod{R_0} \quad \longmapsto \quad (b,c) = \left(\frac{N_0 + d}{R_0}, \frac{N_0 + N_0^2/d}{R_0} \right).$$

It is a bijection onto the positive remaining pairs solving $4/p = 1/a + 1/b + 1/c$; the unordered count is $U_R(p) = (A_R(p) + \mathbf{1}_{R_0|2})/2$ (Theorem 31.8). Theorem 31.12 gives $Q_{R,P} = 6W_{R,P} - M_{R,P}u \in A(S_{840}) \subset N$ for the retained finite sample P of primes reducing into $S_{840} = \langle 289 \rangle$. This is not a residue-only existence test.

38. **[VERIFIED]** The two-column $SU(3)$ Gram quotient is the four-dimensional positive Hermitian cone:

$$\mathrm{Mat}_{3\times 2}(\mathbb{C})/SU(3) \cong \mathrm{Herm}_2(\mathbb{C})_{\geq 0}.$$

Its one-point compactified representation sphere satisfies

$$S^{12}_{\mathrm{col}}/SU(3) \cong B^4,$$

with orbit-type boundary S^3. This S^{12}_{col} is the column representation sphere, not the cusp-smash S^{12}; see theorems 17.1 and 17.5.

39. **[VERIFIED]** The three-column orbit calculation produces an exact rank-three carrier, not a conjectural dimensional analogy. For $\mathscr{V}_3 = \mathrm{Mat}_{3\times 3}(\mathbb{C})$,

$$\mathscr{V}_3/U(3) \cong \mathrm{Herm}_3(\mathbb{C})_{\geq 0}$$

by the complete Gram invariant $A^\dagger A$, whereas strict $SU(3)$ retains the alternating invariant:

$$\mathscr{V}_3/SU(3) \cong \{(H,z) : H \geq 0,\ |z|^2 = \det H\} \cong \mathbb{R}^{10}.$$

Thus

$$S^{17}/U(3) \cong B^8, \quad \mathscr{V}_3^+/U(3) \cong B^9, \quad S^{17}/SU(3) \cong S^9, \quad \mathscr{V}_3^+/SU(3) \cong S^{10}.$$

The compactified rank-loss locus is a standard topologically locally flat $S^8 \subset S^{10}$, and its complement is

$$S^{10} \setminus S^8 \cong \mathbb{R}^8 \times \mathbb{C}^* \simeq S^1,$$

whose positive winding generator is $\det A/|\det A|$. The rank-one trace-one stratum inside the rank-loss sphere is $\mathbb{C}P^2$. The rank-three determinant is cubic; this cone is not the quadratic Lorentz cone arising from two columns. See theorems 18.4, 18.5, 18.7, 18.9 and 18.11.

40. **[VERIFIED]** The selectively corrected Jordan determinant/Euler module supplies an exact algebraic refinement, but no new geometric globalization. In the fixed orientation, every Hermitian matrix has a unique form $S + i[a]_\times$, and

$$\det(S + i[a]_\times) = \det S - a^{\mathsf{T}} S a.$$

For every marked positive-definite symmetric form B, the linear section

$$\ell_B(\lambda, a) = \lambda B + i[a]_\times$$

has the exact positive inverse image

$$\ell_B^{-1}(\mathrm{Herm}_3(\mathbb{C})_{\geq 0}) = \{\lambda \geq 0 : a^{\mathsf{T}} B a \leq \lambda^2 \det B\},$$

a future $(3+1)$-Lorentz cone. The explicit order-cone lift $\Theta : \mathrm{Herm}_2(\mathbb{C}) \to \mathbb{R}I_3 \oplus i\mathfrak{so}(3)$ satisfies

$$\det_3 \Theta(H) = \frac{\operatorname{tr} H}{2} \det_2 H$$

and sends positive-semidefinite ranks $2,1,0$ to $3,2,0$. For a rank-one projection P, the matrix $\Theta(P)$ has eigenvalues $0, 1/2, 1$ and is not idempotent, which separately proves that Θ is not a Jordan homomorphism. Neither Θ nor ℓ_B is a toric, period-lattice, or filling map. The claimed index-three translation compression is also rejected: the proved six-to-two operation is the rank-lowering map $P : \mathbb{Z}^3 \twoheadrightarrow \mathbb{Z}^2$, with the three even and three odd Kuhn labels mapping to the two target triangle orbits. See theorems 20.1, 20.7 and 20.8.

41. **[VERIFIED]** The rank-three toroidal successor and its exact quotient to the reconstructed cusp are now constructed. The local analytic family

$$\pi_3 : N_3 \longrightarrow \Delta$$

has smooth fibre E_τ^3, deepest local equation $t = z_0 z_1 z_2 z_3$, cusp monodromy

$$M_3 = \begin{pmatrix} I_3 & I_3 \\ 0 & I_3 \end{pmatrix},$$

and filling homomorphism $a_i \mapsto 0,\ b_i \mapsto e_i$. Its irreducible central fibre has

$$(f_0, f_1, f_2, f_3) = (1, 7, 12, 6), \qquad \chi(W_3) = 6.$$

Its normalization is the permutohedral threefold with orbit vector $(1, 14, 36, 24)$; the positive-codimension strata map with degrees $2, 3, 4$ onto the $7, 12, 6$ quotient strata. Hence a real S^8-total-space fibration retaining W_3 requires its remaining singular fibres to contribute Euler sum -4. For a full-rank integral shear $\mathcal{B} : \bar{L}_3 \hookrightarrow L_3$, the exact generalization is

$$\chi(W_{3,\mathcal{B}}) = 3!\,[L_3 : \mathcal{B}\bar{L}_3] = 3!\,|\det \mathcal{B}|.$$

In the candidate decomposition $\Lambda_F = A_F \oplus B_F$, define

$$P = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix}, \qquad Q = \begin{pmatrix} 0 & -1 & 1 \\ 1 & -1 & 0 \end{pmatrix}.$$

Then

$$\Phi = P \oplus Q : H_1(T^6; \mathbb{Z}) \twoheadrightarrow H_1(T^4; \mathbb{Z})$$

obeys the literal monodromy, filling, and period identities

$$\Phi M_3 = M_F \Phi, \qquad c_F \Phi = Q c_3, \qquad P Z_3(t) = Z_2(t) Q.$$

It extends over the toric strata to a holomorphic map of cusp families with connected elliptic kernel. Moreover an explicit $J \in SL_6(\mathbb{Z})$ gives

$$J M_3 J^{-1} \cong M_F \oplus \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix},$$

and the candidate's free order-three and order-four peripheral actions lift on the added elliptic factor. The latter successful lifts are distinct from the diagonal candidate

$$M_1 = I_3 \otimes S, \qquad M_2 = I_3 \otimes Q, \qquad M_0 = I_3 \otimes T, \qquad M_1 M_2 M_0 = I.$$

Although its finite factors have orders 4 and 6, the maps $I - M_j : T^6 \to T^6$ are isogenies for $j = 1, 2$, so every affine lift of either finite action has a fixed point.

The six source tetrahedron orbits also map S_3-equivariantly to the six marked diagonal A_2 Weyl chambers: with $\nu_i = E_{ii} - I_3/3$, the π-th cell maps to the cone

$$\mathbb{R}_{\geq 0}\nu_{\pi_1} + \mathbb{R}_{\geq 0}(-\nu_{\pi_3}) = \{\lambda_{\pi_1} \geq \lambda_{\pi_2} \geq \lambda_{\pi_3}\}.$$

This is an exact fan/Cartan correspondence, not a geometric identification with the Gram determinant singularity. The global period-and-overlap gluing over both finite-order discs remains **[OPEN]**. See theorems 19.3, 19.5, 19.8 and 20.3 to 20.6.

42. **[VERIFIED]** The $\mathbb{Z}^6$ native to the higher-rung cusp is

$$\Lambda_{\mathrm{tor}} = H_1(T^6; \mathbb{Z}) = L_3 \oplus \overline{L}_3,$$

not the cyclic group $\mathbb{Z}/6$ and not the separate group $H^4(U; \mathbb{Z})$. For $k = (1, 1, 1)$, the exact sequence

$$0 \longrightarrow \mathbb{Z}(k, 0) \oplus \mathbb{Z}(0, k) \longrightarrow \Lambda_{\mathrm{tor}} \xrightarrow{P \oplus Q} A_F \oplus B_F \longrightarrow 0$$

is S_3-equivariant and compatible with monodromy. It splits as an integral monodromy lattice but not as a $\mathbb{Z}[S_3]$-module. Each quotient copy is the A_2 weight lattice; its root lattice has index three. The six standard generators form two S_3-orbits of size three, whereas the six Kuhn/Weyl chambers form the regular S_3-orbit. Thus

$$\operatorname{rk} \Lambda_{\mathrm{tor}} = 6, \qquad |S_3| = 3! = 6, \qquad \chi(W_{3,\mathcal{B}}) = 3! |\det \mathcal{B}|$$

are three precisely related but nonidentical statements. The actual map to the Gram boundary is $\Phi_{\mathrm{G}} \circ \jmath \circ P \circ \mathrm{pr}_{\mathrm{ang}}$: it is an S_3-equivariant, piecewise-linear chamber map on the real cocharacter space, not an additive map, a toric fan morphism, or a map of torus fibres.

The regular chamber orbit itself has an exact rank-nine realization. With $\mathcal{M}_{\mathbb{Z}} = \mathrm{Hom}(\overline{L}_3, L_3)$,

$$\mathcal{M}_{\mathbb{Z}} = \bigoplus_i \mathbb{Z}E_{ii} \oplus \bigoplus_{i \neq j} \mathbb{Z}E_{ij},$$

the second summand is $\mathbb{Z}[S_3]$, and $c_{(\pi_1\pi_2\pi_3)} \mapsto E_{\pi_1,\pi_3}$ identifies its basis with the six chambers. These are precisely the six mixed basis pairs with zero alternating pairing; the three diagonal matrix units encode the excluded pairs with pairing one. The real matrix carrier is S_3-equivariantly isomorphic to $\mathrm{Herm}_3(\mathbb{C})$, but the integral symmetric/skew splitting has index 8, and every off-diagonal matrix unit maps to an indefinite Hermitian matrix. Hence this is an exact representation and incidence theorem inside the nine-dimensional carrier, not a positive-semidefinite or toroidal map. See theorems 21.1 to 21.3, 22.1 and 22.4.

43. **[VERIFIED]** The marked angular epimorphism now gives an exact positive observation morphism from the full rank-nine carrier to Minkowski space, while retaining and classifying every fibre direction. Put

$$P = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix}, \quad u = (1, 1, 1)^{\mathsf{T}}, \quad C = PP^*, \quad R = P^*C^{-1}, \quad \Pi = uu^*/3, \quad X = (R\ u).$$

Then $PR = I_2$, $RP = I_3 - \Pi$, and $\det X = 1$. The maps

$$\mathcal{O}_P(H) = PHP^*, \qquad \jmath_P(G) = RGR^*$$

satisfy

$$\mathcal{O}_P \jmath_P = \mathrm{id}, \qquad \jmath_P \mathcal{O}_P(H) = (I_3 - \Pi) H (I_3 - \Pi),$$

and every $H \in \mathrm{Herm}_3(\mathbb{C})$ has unique coordinates

$$H = X \begin{pmatrix} G & \psi \\ \psi^* & \lambda \end{pmatrix} X^*, \qquad \det H = \lambda \det G - \psi^* \operatorname{adj}(G) \psi.$$

The complete positive fibre over $G \succeq 0$ is

$$\psi \in \operatorname{im} G, \qquad \lambda \geq \psi^* G^+ \psi,$$

and the zero fibre is exactly $\{\lambda u u^* : \lambda \geq 0\}$. For $g \in SL_2(\mathbb{C})$,

$$T_P(g) = RgP + \Pi = X(g \oplus 1)X^{-1}$$

acts by $(G, \psi, \lambda) \mapsto (gGg^*, g\psi, \lambda)$. Thus

$$\mathrm{Herm}_3(\mathbb{C}) \cong \text{Lorentz four-vector} \oplus \text{defining complex spin representation} \oplus \text{real scalar}.$$

The vector action has kernel $\{\pm I_2\}$, but the full action is faithful because $-I_2$ sends ψ to $-\psi$. Moreover

$$e_2(\jmath_P(G)) = \tfrac{1}{3} \det G, \qquad \det(uu^* + \varepsilon \jmath_P(G)) = \varepsilon^2 \det G.$$

Consequently $\jmath_P$ extends properly to an embedding $B^4 \hookrightarrow S^8$; the reverse observation on the rank-loss boundary is nonproper and has no infinity-preserving extension. In the strict $SU(3)$-quotient its determinant phase obeys

$$|\omega|^2 = \lambda \det G - \psi^* \operatorname{adj}(G) \psi.$$

Finally, for $Z \in \mathrm{Mat}_{3\times 2}(\mathbb{C})$ and retained $\alpha, c > 0$, the exact colour-invariant Lorentz momentum and mass are

$$p_\alpha(Z) = \alpha Z^* Z, \qquad m^2_{\alpha,c}(Z) = \frac{\alpha^2}{c^2} \sum_{a<b} |Z_{a1} Z_{b2} - Z_{a2} Z_{b1}|^2.$$

Rank two is equivalent to positive mass, rank at most one to zero mass, and $m^2_{\alpha,c}(rZ) = r^4 m^2_{\alpha,c}(Z)$; hence this unrestricted kinematic carrier is gapless. Its exact presymplectic reduction is now complete. For $m_0 > 0$,

$$(\mathcal{J}_2 \times \mathcal{H}^+_{m_0,c}) / ((x,p) \sim (x + sp, p)) \cong_{\mathrm{symp}} T^* \mathcal{H}^+_{m_0,c},$$

so the rank-two system is exactly the future-timelike spinless particle. For $m_0 = 0$, the intrinsic punctured rank-one reduction is $T^* \mathcal{N}^\times_+$, but the composite multiplier action is irregular:

$$d(\det \circ p_\alpha)_Z = 0 \quad (\mathrm{rank}_{\mathbb{C}}\, Z \leq 1),$$

so $\dot{x} = \lambda p$ does not imply $\lambda = e$; four transverse real rank-opening directions are first-order null without being colour gauge, and at $Z = 0$ every $x(\tau)$ is stationary. On the hard (λ_0, ω_0)-constraint with $\omega_0 \neq 0$, the momentum map has rank four at $w \neq 0$ and rank three at $w = 0$. Thus its own action forces $\dot{x} = 0$ above the sharp threshold and permits $\dot{x} \in \mathbb{R}p$ only at the threshold. Its determinant inequality and multiplication/Fock spectrum are exact, but the latter is not the Hamiltonian of that action and gives a constraint-selected free spectral gap, not a dynamically generated interacting gap. See theorems 23.2 to 23.7, 27.6, 27.9, 27.12 and 27.16.

44. **[VERIFIED]** The determinant normal slice has now been promoted from an exact Lorentzian analogy to a completely typed host for explicit field models, while the additional choices and the spectral limitations remain visible. With

$$\mathcal{N}_P = \{Y \in \mathrm{Herm}_3(\mathbb{C}) : Yu = 0\}, \qquad \jmath_P(G) = RGR^*,$$

the map $\jmath_P : \mathrm{Herm}_2(\mathbb{C}) \to \mathcal{N}_P$ is a real-linear isomorphism with inverse $Y \mapsto PYP^*$, and

$$\det(uu^* + \varepsilon \jmath_P(G)) = \varepsilon^2 \det G.$$

The polarized determinant is exactly the Minkowski form $tt' - xx' - yy' - zz'$, so the affine slice carries a constant metric of signature $(1,3)$, its positive-semidefinite part is the future cone, and the lifted congruence action realizes

$$PSL_2(\mathbb{C}) \cong SO^+(1,3).$$

The standard Hermitian metrics used to define the Moore–Penrose section select

$$W_P = R^*R = C^{-1}, \qquad K_P = \{a \in SL_2(\mathbb{C}) : a^*W_P a = W_P\} \cong SU(2),$$

and the positive invariant form $E_P(G,K) = \mathrm{tr}(GW_PKW_P)$. The real normal representation is $\mathbf{1} \oplus \mathbf{3}$; its centre acts trivially. Promoting K_P to a local gauge symmetry gives the explicit nonnegative action

$$\int_M \left(\frac{|F_{\mathcal{A}}|^2}{2g_{\mathrm{YM}}^2} + \frac{Z_G}{2} |D_{\mathcal{A}} G|^2_{E_P} + \frac{m_G^2}{2} |G|^2_{E_P} + \frac{\lambda_4}{4} (\det G)^2 \right) d\,\mathrm{vol}_g,$$

and the lift $\widehat{\mathcal{A}} = R\mathcal{A}P$ satisfies $D_{\widehat{\mathcal{A}}} \jmath_P(G) = \jmath_P(D_{\mathcal{A}} G)$. This gauging is an additional physical construction; the normal field cannot screen a fundamental centre charge without the centre-nontrivial retained spinor ψ or another added matter field.

The candidate-specific separation is now exact rather than verbal. The retained inverse-dual cusp matrix obeys

$$G_{\eta,T} = \begin{pmatrix} 2L_T & 2m_T \\ 2m_T & -q_T/3 \end{pmatrix}, \qquad \det G_{\eta,T} = -\frac{2}{3} D_T < 0,$$

whereas $p_\alpha(Z) = \alpha Z^* Z$ ranges over the entire positive-semidefinite cone. Positive rescaling and complex congruence cannot change this determinant sign/inertia, so a further morphism is required to place these two matrices in one physical model. On the Lorentzian host, the free scalar Fock Hamiltonian has exact gap m_{KG} when the independently supplied coefficient is positive and is gapless when it is zero. For the massless compact gauge Gaussian, the lowest nonzero transverse frequency is $2\pi/L$, which closes as $L \to \infty$. Entangled vacua and sectors occur in both explicitly gapped and explicitly gapless Hamiltonians. Hence none of the determinant identity, the Gram carrier, or entanglement alone proves a nonlinear four-dimensional mass gap. The full-domain comparison theorem identifies a sufficient exact bridge: if $\|Ru\| \geq c_R \|u\|$ and $\mathfrak{b}[Ru] \leq C_R(\mathfrak{a}_{\mathrm{phys}}[u] - E_{\mathrm{vac}}\|u\|^2)$ throughout the physical excitation form domain and $\mathfrak{b} \geq \gamma$, then

$$\mathrm{gap}(A_{\mathrm{phys}}) \geq \frac{\gamma c_R^2}{C_R}.$$

An exact counterexample proves that an intertwiner defined on only one gapped subspace is insufficient. See theorems 28.1 to 28.3, 28.5 to 28.8, 28.10, 28.12 and 28.16.

45. **[REFUTED]** A literal Fabel–Alpöge-style complex-fourfold construction on a homotopy S^8 cannot exist. If $E = (TS^8, J)$ were complex of rank four, then

$$c_1(E) = c_2(E) = c_3(E) = 0, \qquad c_4(E) = 2u,$$

and therefore $p_2(TS^8) = 2c_4(E) = 4u$. The eight-dimensional signature formula, with $p_1 = 0$ and signature zero, forces $p_2 = 0$, a contradiction. A nonsingular bundle $T^6 \to S^8 \to S^2$ is also impossible: its homotopy sequence would require a surjection $\mathbb{Z} \to \mathbb{Z}^6$, and its Euler characteristic would give $2 = 0$. A merely real singular fibration is not refuted, but it must satisfy $\sum_i \chi(F_i) = 2$ and the exact van Kampen, Wang, and integral Mayer–Vietoris maps proved in the workbench. For regular-piece monodromy ρ, the corrected homological Serre edge sequence is

$$0 \longrightarrow (\textstyle\bigwedge^q \Lambda)_\rho \longrightarrow H_q(E;\mathbb{Z}) \longrightarrow H_1(B^\circ; \textstyle\bigwedge^{q-1} \Lambda) \longrightarrow 0,$$

whereas dually $H^q(E;\mathbb{Z})$ surjects onto the invariant group $(\bigwedge^q \Lambda^\vee)^\rho$. Thus every filling map must respect the coinvariant subobject and the base-one-cycle quotient; the reversed edge maps cannot be used in the Mayer–Vietoris test. Every effective torus action on S^8 has rank at most four, and the sharp coordinate model is

$$S^8/T^4 \cong \mathrm{Susp}(\Delta^3) \cong B^4.$$

It is a real two-pole singular compactification with generic T^4 orbit, not a three-special-fibre T^6-fibration over S^2. See section 29.1 and equations (1098), (1101), (1103), (1105) and (1108) to (1110).

46. **[VERIFIED]** The original reconstructed S^6 carries a separate, exact six-coordinate bundle problem on its regular locus. With

$$D = W \sqcup S_1 \sqcup S_2, \qquad U = X \setminus D,$$

Alexander duality and the computed homology of the three special fibres give

$$H^4(U;\mathbb{Z}) \cong H_1(W;\mathbb{Z}) \oplus H_1(S_1;\mathbb{Z}) \oplus H_1(S_2;\mathbb{Z}) \cong \mathbb{Z}^2 \oplus \mathbb{Z}^2 \oplus \mathbb{Z}^2.$$

Because U has CW dimension at most five, principal $SU(3)$-bundles on U are classified exactly by their complete $c_2 \in H^4(U;\mathbb{Z})$. The connecting map

$$\delta : H^4(U;\mathbb{Z}) \xrightarrow{\sim} H^5(X,U;\mathbb{Z})$$

shows that such a bundle extends across all three deleted fibres if and only if $c_2 = 0$. This is the free lattice $\mathbb{Z}^6$, not the cyclic group $\mathbb{Z}/6\mathbb{Z}$.

47. **[VERIFIED]** The primitive fibre-degree map, its saturated kernel, the finite normal-holonomy quotients, and the common-centre matrix maps now have a complete theorem–proof module in section 59 and theorems 59.1 to 59.4. The result is exact but stops before any gauge-bundle, connection, confinement, or spectral assertion.

In the marked basis

$$(a_0, b_0, a_1, b_1, a_2, b_2) = (\bar\gamma, \bar u, g_1, h_1, g_2, h_2),$$

solving the transpose of the verified one-shot collar matrix gives

$$\mathrm{res}_F\left(\sum_{i=0}^{2}(n_i a_i + r_i b_i)\right) = (12n_0 + 4n_1 - 3n_2)\omega_F, \qquad [\deg_F] = (12, 0, 4, 0, -3, 0),$$

and

$$\ker \deg_F = \mathbb{Z}\langle b_0, b_1, b_2, a_0 - 3a_1, 3a_1 + 4a_2\rangle.$$

The map is primitive, so restriction to the oriented fibre is surjective. Define

$$q_{\mathrm{col}} : L \longrightarrow H_1(X^\circ;\mathbb{Z}), \qquad a_0 \mapsto c, \quad a_1 \mapsto \bar g_1, \quad a_2 \mapsto \bar g_2, \quad b_0, b_1, b_2 \mapsto 0.$$

If $\Phi_1 : H_1(X^\circ;\mathbb{Z}) \to \mathbb{Z}$ has row $(12, 4\ell_1, 3\ell_2)$ in the basis $(c, \bar g_1, \bar g_2)$, then the factorization through the actual collar quotient is

$$\deg_F = \Phi_1 \circ q_{\mathrm{col}}, \qquad \Phi_1(j_*[\widetilde a_0]) = p.$$

Let $\vartheta_1(a_1) = e^{-2\pi i/3}$ and $\vartheta_2(a_2) = e^{-2\pi i/4}$ be the normal-line holonomy characters, trivial on the other marked generators. The candidate twists select

$$\chi_{12}\left(\sum_{i=0}^{2}(n_i a_i + r_i b_i)\right) = \vartheta_1(n_1 a_1 + r_1 b_1)\vartheta_2(n_2 a_2 + r_2 b_2)^{-1} = e^{-2\pi i(4n_1 - 3n_2)/12}.$$

For $\rho_m(k) = e^{-2\pi i k/m}$, the exact typed identities are

$$\boxed{\chi_{12} = \rho_{12} \circ \deg_F, \qquad \chi_6 := \chi_{12}^2 = \rho_6 \circ \deg_F\,.}$$

Consequently

$$H^4(U;\mathbb{Z})/\ker\chi_{12} \cong \mathbb{Z}/12\mathbb{Z}, \qquad H^4(U;\mathbb{Z})/\ker\chi_6 \cong \mathbb{Z}/6\mathbb{Z},$$

whereas ordinary coefficient reduction gives

$$H^4(U;\mathbb{Z}/6) \cong (\mathbb{Z}/6)^6.$$

Thus this six-coordinate coefficient group is not the single distinguished cyclic quotient selected by χ_6. A principal $SU(3)$-bundle $P \to U$ satisfies

$$\chi_6(c_2(P)) = \exp\left(-\frac{2\pi i}{6}\langle c_2(P)|_F, [F]\rangle\right).$$

The order-six quotient is therefore the reduction of a geometrically computed degree and of the combined local normal holonomy, not a torsion subgroup inferred from the rank-six notation.

Tensor product gives an independent group-theoretic match:

$$\iota : SU(2) \times SU(3) \hookrightarrow SU(6), \qquad (A, B) \longmapsto A \otimes B, \qquad \mu_2 \times \mu_3 \xrightarrow{\sim} Z(SU(6)) = \mu_6.$$

Since U has homotopy dimension at most five, c_2 classifies $SU(n)$-bundles for $n \geq 3$. For determinant-trivial bundles of ranks two and three,

$$c_2(E_2 \otimes E_3) = 3c_2(E_2) + 2c_2(E_3).$$

Thus every chosen $SU(6)$-bundle class $h \in H^4(U;\mathbb{Z})$ admits a tensor-subgroup reduction, with all c_2-factorizations

$$(e, f) = (h + 2t, -h - 3t), \qquad t \in H^4(U;\mathbb{Z}).$$

Neither h nor a reduction is preferred. The separate support group is

$$H^5(U;\mathbb{Z}) \cong \ker\{\mathbb{Z}^3 \xrightarrow{(1,1,1)} \mathbb{Z}\} \cong Q(A_2),$$

for the combinatorial form making the three distinguished support basis vectors orthonormal; this is not a cup-product or intersection form. It is not an intrinsic $A_2 \otimes \mathbb{Z}^2$ inside $H^4(U)$, and the rank-five kernel of $\deg_F$ is not thereby an A_5 root lattice.

The finite-group isomorphism $H^4(U)/\ker\chi_6 \cong Z(SU(6))$ supplies neither transition functions nor a connection. The injective tensor map does not factor through $(SU(2) \times SU(3))/\mu_6$, because its central μ_6 maps to nontrivial scalar matrices. The separate common-centre construction is

$$\Gamma_6 = \{((\lambda\mu)^{-1}, \lambda I_2, \mu I_3) : \lambda \in \mu_2,\ \mu \in \mu_3\},$$
$$\frac{U(1) \times SU(2) \times SU(3)}{\Gamma_6} \hookrightarrow U(6),$$
$$[z, A, B] \longmapsto z(A \otimes B),$$

where $\Gamma_6 \cong \mu_6$. Its determinant is z^6, and its image meets $SU(6)$ in exactly $\iota(SU(2) \times SU(3))$; the image is not generally contained in $SU(6)$. No present morphism identifies this degree-four quotient with the degree-two lifting obstruction of a common-centre gauge quotient. Global $SU(3)$-bundles on S^6 are instead classified by $k \in \pi_5(SU(3)) \cong \mathbb{Z}$, with $c_3 = \pm 2ku$, and all restrict trivially to U. See equations (1116), (1118) and (1120) to (1122).

48. **[VERIFIED]** The exact operator-theoretic bridge required for any physical no-gap inference is proved as a typed implication. Let $H \geq 0$ be self-adjoint on $\mathcal{H}_{\mathrm{phys}}$, put

$$P_0 = \mathbf{1}_{\{0\}}(H), \qquad \mathcal{H}_{\mathrm{exc}} = (I - P_0)\mathcal{H}_{\mathrm{phys}}, \qquad \mathfrak{h}[\psi] = \|H^{1/2}\psi\|^2, \qquad \mathcal{Q}(H) = D(H^{1/2}).$$

For a nonnegative quadratic form $\mathfrak{q}_n$ and nonzero $u_n \in D(\mathfrak{q}_n) \subset \mathcal{K}_n$, and for $J_n : \mathrm{span}\{u_n\} \to \mathcal{Q}(H)$, define $\psi_n = (I - P_0)J_n u_n$. If, for some $c > 0$, $C_n \geq 0$, and $\varepsilon_n \geq 0$,

$$\|\psi_n\| \geq c\|u_n\|, \qquad \mathfrak{h}[\psi_n] \leq C_n \mathfrak{q}_n[u_n] + \varepsilon_n \|u_n\|^2,$$

$$C_n \frac{\mathfrak{q}_n[u_n]}{\|u_n\|^2} + \varepsilon_n \longrightarrow 0,$$

then the self-adjoint restriction

$$H_{\mathrm{exc}} = H\big|_{D(H)\cap\mathcal{H}_{\mathrm{exc}}}$$

satisfies

$$\inf \sigma(H_{\mathrm{exc}}) = 0, \qquad 0 \in \sigma_{\mathrm{ess}}(H_{\mathrm{exc}}).$$

The full spectral projection P_0, rather than orthogonality to one chosen vacuum, is essential unless vacuum uniqueness is independently proved; see theorem 52.4.

49. **[OPEN]** The hypotheses needed to apply that bridge to four-dimensional quantum Yang–Mills theory have not been constructed here. In particular, the workbench has no common gauge-invariant positive-metric physical Hilbert space $\mathcal{H}_{\mathrm{phys}}$, no reconstructed quantum Hamiltonian H, and no maps J_n satisfying the displayed norm and energy estimates with control through gauge reduction, reflection-positive reconstruction, renormalization, and the continuum and infinite-volume limits. The exact radial scalar, continuum Hessian, compact-link flux, finite cochain, Gaussian-oscillator, magnetic, and split-Cartan $SU(r)$ Jacobi calculations are **[VERIFIED]** auxiliary geometric, finite-dimensional, or classical statements. They neither prove nor disprove a mass gap for a physical four-dimensional quantum Yang–Mills Hamiltonian; see theorems 12.12 and 52.6.

External validation note. A further context-isolated AI derivation independently recovered the cyclic invariant lattices (11.4), determinant divisor (11.6), finite residue cancellation (11.17), scalar test (11.19)–(11.20), all three absolute cotangent direct images, and $h^{1,2} = 1$, $h^{1,1} = 2$. Independent human specialist replication has not been recorded. This is an external-validation boundary, not a mathematical premise or an unresolved step of the Hodge theorem; see theorem 57.7.

Independent review checkpoint. Reviewer B independently examined the reconstruction existing up to that report's evidentiary cutoff and agreed that the intrinsic nearby-cycle, Leray, and based van Kampen route does not depend on the then-open collapse route. Its disposition was provisionally favourable with focused verification requests. The later Codex proofs recorded here are separate work: they are not retroactively attributed to Reviewer B, and its report remains useful as an independent comparison against the argument it actually saw. Reviewer B's 26 August analytic follow-up also independently proves the uniform top-degree base-change corollary and the conductor–duality nonvanishing from CAND-X, CAND-W, and TOP-H2 [36]; those inputs are not certified by that follow-up itself.

Status language. [VERIFIED] means proved here or reproduced by an explicit certificate. [CONDITIONAL] means a conclusion depending on an explicitly named established or assumed hypothesis not proved in the present argument; it never means doubtful or politely false. A completely proved implication whose antecedents are part of its exact statement is [VERIFIED]. The current candidate-specific ledger has no [CONDITIONAL] row. [OPEN] means that no complete proof or counterexample is recorded. [REFUTED] means an exact assertion has a complete contradiction or counterexample. The status applies to the immediately following assertion, not to an entire paper by contagion.

2 Mathematical results so far

6 September: explicit spectral and umbral maps, and the original family's parameter identifications. For the literal common lattice $\mathcal{I} = \{\sqrt{2}(a,a,a) : a \in D_4\}$, write $a = a_0 + v$, $t^2 = |v|^2$, $Q = a_0^2 + t^2$ and $P = a_0(a_0^2 - 3t^2)$. Its complete cubic image is

$$\tau(\mathcal{I}) = 2\sqrt{2}(18\mathbb{Z} \cup (6\mathbb{Z}+2) \cup (6\mathbb{Z}-2)).$$

Three explicit integral affine lines give a section of this value map. The scalar-retaining Albert matrix $X = \mu I + Y(a)$, with cyclic off-diagonal entries $\sqrt{2}a$, has invariants and labelled spectrum

$$\begin{aligned}(\operatorname{tr} X, \operatorname{tr} X^2, \det X) &= (3\mu, 3\mu^2 + 12Q, \mu^3 - 6\mu Q + 4\sqrt{2}P),\\ \operatorname{Spec} Y &= (2\sqrt{2}a_0, -\sqrt{2}a_0 - \sqrt{6}t, -\sqrt{2}a_0 + \sqrt{6}t).\end{aligned}$$

For $t > 0$, the explicit labelled eigenprojector satisfies $(\Pi_1)_{12} = -1/6 - \sqrt{3}(v/t)/6$; thus the frame and spectrum recover a, not only its norm and cubic. The full discriminant is $864t^2(3a_0^2 - t^2)^2$. The full Cartan family retains the scalar μ, radius r, and angle θ, with $r^2 = 12Q$ and $r^3\cos 3\theta = 24\sqrt{3}P$ on this subfamily.

For the same marked root system $A_5^4D_4$, the Cheng–Duncan–Harvey prescription gives $S^X = (5I_{12} + J)S_6$, where $J_{rs} = \mathbf{1}_{s=5r\ (12)}$ and $S_{6,r} = \sum_{k\equiv r\ (12)} kq^{k^2/24}$. The coefficient map to the retained Fricke function is

$$T_6 = \frac{\eta(\tau)^5\eta(3\tau)}{\eta(2\tau)\eta(6\tau)^5}, \qquad f^X = -q\frac{d}{dq}\log T_6, \qquad T_6\left(-\frac{1}{6\tau}\right) = \frac{72}{T_6(\tau)}.$$

All twelve theta indices, the multiplier determinant, the original glue-group action and its twisted shadows are computed in the proofs. This is an explicit use and comparison of established umbral machinery, not a claim to a new moonshine module.

In the original analytic family $\beta_c = \beta^{\mathrm{part}} + c$, $\operatorname{Im} c < -M$, the actual global identifications are exactly $X_c \simeq X_{c'} \iff c' - c \in 2\mathbb{Z}$. The middle lattices agree for every integer shift; the multiplicity-four filling retains an order-two obstruction to odd shifts. Even shifts have explicit finite-filling and toric cusp maps and a proved global additive gluing correction. The resulting parameter quotient is $w = e^{-\pi i c}$, $0 < |w| < e^{-\pi M}$. This classifies the retained analytic family without using the claimed identification of its total space with S^6.

6 September: cubic arithmetic, integral lattice maps, and the canonical ring. For the retained $A_5^4 \perp D_4$ Niemeier lattice N and its original index-six Leech neighbour Λ, fix the Cayley–Dickson frame $(1, i, j, k, \ell, i\ell, j\ell, k\ell)$ used in the complete cubic proof. Put $\tau(q) = \operatorname{Re}((q_1q_2)q_3)$ and $F = \tau \circ \mathcal{L}$ on the original Niemeier span, and $\mathcal{V}_F(L) = \operatorname{span}_{\mathbb{Z}}\{F(y) : y \in L\}$. The exact additive groups are

$$\begin{aligned}\mathcal{V}_F(N) &= \tfrac{1}{27}\mathbb{Z} \oplus \tfrac{\sqrt{2}}{54}\mathbb{Z} \oplus \tfrac{\sqrt{3}}{18}\mathbb{Z},\\ \mathcal{V}_F(\Lambda) &= \tfrac{1}{27}\mathbb{Z} \oplus \tfrac{\sqrt{2}}{216}\mathbb{Z} \oplus \tfrac{\sqrt{3}}{54}\mathbb{Z}, \qquad \mathcal{V}_F(\Lambda)/\mathcal{V}_F(N) \cong \mathbb{Z}/12.\end{aligned}$$

The Albert determinant is $\lambda^3 - \lambda(y,y) + 2F(y)$. The quotient concerns the displayed value-generated groups in this fixed frame, not an inclusion $N \subset \Lambda$. The actual residue $r = \rho(361)$ cycles three Leech

neighbours $\Lambda_j = r^j\Lambda$. Their pairwise intersection is one integral lattice J, with $\det J = 81$ and $[\Lambda_j : J] = 9$. The fixed fourth isotropic plane in $J^\vee/J$ gives M with exactly 72 roots of type A_2^{12}, $M \cap \Lambda_j = J$, and $[M : J] = 9$.

$$\mathcal{V}_F(J) = \mathcal{V}_F(M) = \tfrac{1}{27}\mathbb{Z} \oplus \tfrac{\sqrt{2}}{216}\mathbb{Z} \oplus \tfrac{\sqrt{3}}{18}\mathbb{Z}.$$

The successive value-group inclusions have matrices $\mathrm{diag}(1,4,1)$ and $\mathrm{diag}(1,1,3)$, giving quotients $\mathbb{Z}/4$ and $\mathbb{Z}/3$. Both containments and their reverse generator certificates are calculated from the full cubic polarization, not inferred from lattice indices.

Wilson's construction [81], transported by the explicit isometry $\iota(x) = x/\sqrt{2}$, gives Λ_{cyc} in the original triality metric. Its cyclic fixed Gram matrix is $3G_{E_8}$; its orthogonal complement is explicitly isometric to $A_2 \otimes E_8$, and their sum has index 3^8, with quotient $\mathbb{F}_3^8$. The directions remain $c(q_1,q_2,q_3) = (q_3,q_1,q_2)$ and $c_{\mathrm{W}} = c^{-1}$. Its literal intersections are

$$\Lambda_{\mathrm{cyc}} \cap \mathcal{L}(N) = \Lambda_{\mathrm{cyc}} \cap \mathcal{L}(\Lambda) = \Delta := \{\sqrt{2}(a,a,a) : a \in D_{\mathbb{H}}\},$$

where $D_{\mathbb{H}}$ is the even-coordinate-sum integer lattice in the first four stated Cayley coordinates. Thus Δ has rank 4, Gram matrix $6G_{D_4}$, determinant 5184, minimum squared norm 12, and value-generated subgroup $4\sqrt{2}\mathbb{Z}$ for $\tau|_\Delta$, or $8\sqrt{2}\mathbb{Z}$ for $2\tau|_\Delta$. This is not the actual image: $12\sqrt{2}$ is not attained by τ on Δ. The preceding proofs retain the triality products and spectral boundary graph, the flag's degree-$+1$ collapse, and the retained cusp class's nontrivial three-plane reduction; these exact maps do not supply an interacting quantum mass-gap result.

For the original analytic fibration $f : X \to \mathbb{P}^1$, the full canonical-divisor proof gives $\omega_X \simeq \mathcal{O}_X(-2S_2)$ and the graded anticanonical ring $\bigoplus_{m\geq 0} H^0(X, \omega_X^{-m}) \cong \mathbb{C}[U,V]$, with $\deg U = 1$, $\deg V = 2$. The complete system $|-2K_X|$ recovers f with the specified projective coordinates. This calculation uses the original analytic charts and finite fillings, not an identification of X with S^6, and draws no sphere-topology conclusion.

The retained constant $\beta_c = \beta^{\mathrm{part}} + c$ now gives a proper holomorphic deformation over $\operatorname{Im} c < -M$, locally on relatively compact parameter discs. On a fixed regular fibre set $T = \operatorname{Im}\tau$, $m = \operatorname{Im}\mu$, $D = \operatorname{Im}\beta_{c_*} - 6m^2/T < 0$ and $\delta = c - c_*$. In the plus-graph convention its exact Beltrami matrix is

$$\nu_\delta = \frac{\delta}{2iD+\delta}\begin{pmatrix} 0 & 0 \\ -m/T & 1 \end{pmatrix}.$$

The connecting Kodaira–Spencer class is $-[\partial_{\zeta_2} \otimes (d\bar{\zeta}_2 - (m/T)d\bar{\zeta}_1)/(2iD)] \neq 0$. The intrinsic pencil recovers f also over the dual numbers; its three fixed critical values prohibit an infinitesimal projective base change. Consequently this deformation of the whole X_c is nontrivial, not only its marked regular-fibre deformation. The proof uses no sphere recognition or CDP conclusion.

5 September: exact corrections to two S6 proof steps. The Cartier duality isomorphism in Theorem 60.7 is induced by the analytic trace with $K_W^\bullet = \omega_W[2]$; outer identities of five-term sequences alone do not define it. For the holomorphic overlap translation by $\alpha(s)\Pi(s)v_j$, the real marking gives $a_j = \operatorname{Re}\alpha\, v_j + \operatorname{Im}\alpha\, J_s v_j$, not multiplication of a real vector by the complex scalar α. The proved covariance $J_{g_j s}A_j = A_j J_s$ and $A_j v_j = v_j$ give $a_j(g_j s) = A_j a_j(s) - v_j/m_j$. On the full clockwise covering circle, $\Delta a_j = -v_j$ and the mixed class of $h_j^*\Omega_E - \Omega_E$ is $-\iota_{v_j}E$. Thus Lemma 60.16 still gives $N_{T_j} d_j = \iota_{v_j}E$, and Proposition 53.6 retains its based meridian relation. The complete proofs now include the base two-form term and its zero restriction to the circle; no coefficient or orientation is suppressed.

5 September: exact period constants and cusp edge maps. For the cusp-regular particular solution define

$$D_{\mathrm{part}} = \operatorname{Im}\beta^{\mathrm{part}} - \frac{6(\operatorname{Im}\mu)^2}{\operatorname{Im}\tau}, \qquad M = \max_{B\setminus\{p_0\}} D_{\mathrm{part}}.$$

Proposition 10.3 proves that M is finite and attained, and that $\beta = \beta^{\mathrm{part}} + c_0$ gives an everywhere invertible marked real period map exactly for $\operatorname{Im} c_0 < -M$. The cusp-free affine solution $\widetilde{\beta} = \beta + i\,t \circ \pi$ has $\widetilde{D} = D + \operatorname{Re} t \to +\infty$ along the positive real base ray. The actual cusp extension excludes it. For $C_v = E_0 \cap E_v$, the cusp identification is $\iota_v = \Psi_{B_0 v}|_{C_v} : C_v \to C_{-v}$. Using the same character $w = \chi^{(a,0)}$ with $av = 0$ on both curves,

$$\iota_v^* w_{-v} = e^{2\pi i\, a C(0) B_0 v} w_v.$$

For $(v, a) = (e_1, (0,1)), (e_2, (1,0)), (e_1 - e_2, (1,1))$, the multipliers are respectively $e^{-2\pi i \mu_0}$, $e^{12\pi i \mu_0}$ and $e^{-2\pi i(7\mu_0 + b_0)}$, with $\mu_0 = \mu(0)$ and $b_0 = b(0)$. Equation (20) retains the full $C(0)$; Lemma 10.5 uses restricted charts and the closed-polydisc condition $\theta_i^{\sigma}(y) \geq 0$ for torus points. These refinements and the bounded finite-filling audit do not supply an additional sphere-identification or CDP theorem.

5 September: the complete two-cross scalar–Gauss domain. Section 24.28 retains the literal matrices, constants, and all three full scalar and Gauss residuals. Put $\delta = \sqrt{2/3}$ and

$$\mathcal{D} = \{(u, v_0, y) \in \mathbb{R}^3 : u > 0,\ d = uv_0 - \delta^2 - y^2 > 0,\ c = 4v_0 - 5u > 0\}.$$

Theorem 24.66 proves a bijection from $\mathcal{D} \times \{1,-1\}^2 \times U(1)^2$ to all classified-frame solutions, with $t = 72d/c$, $e = \epsilon\sqrt{2 + u + v_0}$, $k = \sigma\sqrt{d/(-\det T)}$ and $\omega = e/k$, where

$$M = \begin{pmatrix} u & \delta + iy \\ \delta - iy & v_0 \end{pmatrix}, \quad Q = \operatorname{diag}(1, 2),$$
$$T = \frac{t}{4}\operatorname{diag}(2/3, -5/6) - \frac{1}{2}(QM + MQ).$$

The rank-one matrices $A_\pm = M \pm kT$ have $h_\pm = (A_\pm)_{11} > 0$; the exact vectors and phase inverse are

$$a_\pm = \begin{pmatrix} \sqrt{h_\pm} \\ (A_\pm)_{21}/\sqrt{h_\pm} \end{pmatrix}, \qquad p_\pm = \frac{(v \mp iw)_1}{\sqrt{h_\pm}},$$
$$v = \frac{p_+ a_+ + p_- a_-}{2}, \quad w = \frac{i(p_+ a_+ - p_- a_-)}{2}.$$

Theorem 24.68 proves that the full literal domain is the quotient of these parameters and $\zeta \in U(1)$ by the sole free involution $(p_+, p_-, \zeta) \mapsto (-p_+, -p_-, -\zeta)$. The exact inverse frame recovers every literal temporal and spatial entry. The smooth phase coordinates are $(\eta, \alpha, \beta) = (\zeta^2, p_+/\zeta, p_-/\zeta)$. Theorems 24.73 and 24.74 give

$$\mathcal{S}_{\mathrm{sg}} \cong \{1,-1\}^2 \times (0,\infty) \times \mathbb{R}^2 \times \mathbb{T}^3, \qquad t = \tau = g^2,$$

with coupling coordinates and inverse

$$v_0 = \frac{\sqrt{5}}{2}\sqrt{\delta^2 + y^2} + e^s, \qquad u = \frac{4\tau v_0 + 72(\delta^2 + y^2)}{72 v_0 + 5\tau}, \quad (\tau, y, s) \in (0,\infty) \times \mathbb{R}^2.$$

The literal manifold has four embedded six-dimensional components; each positive-coupling level has four embedded five-dimensional components $\mathbb{R}^2 \times \mathbb{T}^3$ (Corollary 24.75). For each original $\tau = g^2 > 0$ the choice

$$\rho = \frac{\tau + \sqrt{\tau^2 + 4608\delta^2}}{96\delta} > \frac{1}{\sqrt{2}},$$
$$M_\rho = \delta \begin{pmatrix} \rho & 1 \\ 1 & 2\rho \end{pmatrix}, \qquad t = \frac{24\delta(2\rho^2 - 1)}{\rho} = \tau$$

proves nonemptiness with both temporal and frequency signs (Corollary 24.70). Both scalar equations and every Gauss entry vanish, while Theorem 24.60 proves that the two spatial residuals cannot vanish simultaneously on this entire class. The quantitative spatial bound in Proposition 24.63 therefore has an explicitly nonempty domain. Existence on the unrestricted remaining exceptional set stays **[OPEN]**; no quantum gap claim follows.

5 September higher-rung and selector continuation. Section 30 gives the complete higher-wrap portion of the expanded root draft. The explicit octonionic slice has determinant $\lambda^3 - \lambda\sum_i |q_i|^2 + 2\,\mathrm{Re}((q_1q_2)q_3)$ and positive-cone compactification (B^{25}, S^{24}). The flag $F_4/\mathrm{Spin}(8)$ has Poincaré polynomial $1 + 2t^8 + 2t^{16} + t^{24}$ and Euler class $6\beta_1^2\beta_2$. For the retained peripheral character, $(\chi(C_0), \chi(C_1), \chi(C_2')) = (-1, 4, -3)$ in $\mathbb{Z}/12$. The exact suspension is $E^6(a\alpha) = 2a\nu_9$ in $\pi_{12}(S^9) \cong \mathbb{Z}/24$; the particular cusp coefficient is now $a = 0$. Theorem 30.49 constructs the complete framed filling inside one original chart times an interval, retaining the full period functions. Thus $g \simeq *$ and $\Theta_g\chi = 0$. The retained-angle map is different: Theorem 30.51 proves $[p_3q_2] = \eta_4\eta_5 \neq 0$ in $\pi_6(S^4)$. Its full normally four-framed inverse torus has spin form $q(x, y) = x + xy + y$ over $\mathbb{F}_2$ and Arf invariant one, so the original conical quotient q_2 is nonnull. The exact homomorphism $[x] \mapsto [\eta_2 \wedge x]$ sends this value to $12\nu_6 \neq 0$ in $\pi_9(S^6) \cong \mathbb{Z}/24$; its threefold suspension is $12\nu_9$ in the same group receiving $E^6[g] = 0$. Composing with the original χ gives peripheral values $(12, 0, 12)$ and kernel $\chi^{-1}(2\mathbb{Z}/12)$. The full class and exact order of q_2 are not computed by this order-two quotient alone. The arithmetic addition in Section 9 maps the complete fifteen $p = 1201$ selector rays into the real symmetric corner of $\mathrm{Herm}_3(\mathbb{H})$ and proves both linear span and Jordan closure have dimension three. These additions retain the global higher-completion question explicitly.

The modulo-840 lattice, full Weyl quotient, and retained counts. Section 31 proves

$$R_{\mathrm{ES}} = A_5^4 \perp D_4 \subset N, \qquad [N : R_{\mathrm{ES}}] = 72, \qquad \det N = 1, \qquad \#\,\mathrm{Roots}(N) = 144,$$

$$O(N) = W(R_{\mathrm{ES}}) \rtimes L(G), \qquad G \cong GL_2(\mathbb{F}_3) \cong Q_8 \rtimes S_3.$$

The full quotient and residue-position theorems 31.15 and 31.16 give $\rho(S_{840}) \subset W(R_{\mathrm{ES}})$ and $L_g\rho(r) = \rho(r)L_g$ in the original six-coordinate fibre order. The abstract quotient is credited to [74, Table 3]; the explicit code isomorphism, lattice lift and marked D_4 weight comparison are proved here. The shell maps retain the actual prime: $A_3(2521) = 0$, $A_3(3361) = 6$, with unordered counts $0, 3$. For a finite set P of primes reducing into $S_{840} = \langle 289 \rangle$, the count pushforward gives $Q_{R,P} = 6W_{R,P} - M_{R,P}u \in A(S_{840}) \subset N$; Theorem 31.12 gives the exact Fourier inverse for the residue-count function $w_{R,P}$.

This is the short route through the current mathematics. It states what is proved in this workbench and exhibits the load-bearing formulas. Both the intrinsic nearby-cycle/Leray certificate and the independent collapse/Mayer–Vietoris certificate are now complete for the candidate.

1. **[VERIFIED] The analytic threefold exists in the reconstructed coordinate model.** The descended modular periods give

$$\Pi = [Z \mid I_2], \qquad Z = \begin{pmatrix} 6\mu & \tau \\ \beta & \mu \end{pmatrix},$$

with

$$\det_{\mathbb{R}} \Pi = -\det \mathrm{Im}\, Z = \mathrm{Im}\,\tau \left(\mathrm{Im}\,\beta - \frac{6(\mathrm{Im}\,\mu)^2}{\mathrm{Im}\,\tau} \right) < 0.$$

The Cayley coordinates $s_j(z) = (z - z_j)/(z - \bar{z}_j)$, the two elliptic orientations, and the modular deck normalization are fixed simultaneously:

$$g_1'(z_1) = e^{-2\pi i/3}, \qquad g_2'(z_2) = e^{-2\pi i/4} = -i, \qquad s_1 \circ g_1 = e^{-2\pi i/3}s_1, \quad s_2 \circ g_2 = -is_2.$$

The period-deck homomorphism φ is defined by $\tau \circ g = \varphi(g) \circ \tau$. Its cusp image is proved in a fixed covering component, not by an untyped single-valuedness assertion: choose

$$U_0 \subset \pi^{-1}(D_0^*), \qquad g_0U_0 = U_0, \qquad \tau(U_0) \subset K \subset j^{-1}(\{|w| > R\}).$$

For $P = \varphi(g_0)$ one has $P(K) = K$. If c is the boundary cusp of K, choose $\gamma_0(\infty) = c$ and put

$$\tau' = \gamma_0^{-1}\tau|_{U_0}, \qquad P' = \gamma_0^{-1}P\gamma_0, \qquad \tau' \circ g_0 = P' \circ \tau'.$$

If $P' = 1$, then τ' descends to $\bar{\tau}' : D_0^* \to \mathfrak{h}$ and $q' = \exp(2\pi i \bar{\tau}')$ simultaneously has a positive-order zero at the cusp and a single-valued holomorphic logarithm, an impossible winding combination. Thus P' and hence P are nontrivial parabolic.

The auxiliary γ_0-frame is then discarded and the proof returns to $\tau(z_1) = \rho$, $\varphi(g_1) = S_1$. For the trace-+2 lift, with primitive fixed vector $(p, r)^{\mathsf{t}}$, $\gcd(p, r) = 1$, and $k \neq 0$,

$$\widetilde{P} = I + k \begin{pmatrix} -pr & p^2 \\ -r^2 & pr \end{pmatrix},$$

and

$$\widetilde{S}_1 = \begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix}, \qquad \operatorname{tr}(\widetilde{S}_1^{-1} \widetilde{P}^{-1}) = 1 + k(p^2 - pr + r^2) = 0.$$

Thus projective order two of $S_1^{-1} P^{-1}$ gives $-k(p^2 - pr + r^2) = 1$, so $k = -1$ and $p^2 - pr + r^2 = 1$; conjugating the whole deck representation by a power of S_1 leaves S_1 fixed and gives the final normalization below. Rename the resulting period map τ and define anew, in this cusp paragraph,

$$q = e^{2\pi i \tau} = t_c u_1(t_c), \qquad u_1(0) \neq 0.$$

For $S_1(w) = (w - 1)/w$, $S(w) = -1/w$, and $P(w) = w - 1$,

$$\varphi(g_1) = S_1, \qquad \varphi(g_2) = S, \qquad \varphi(g_0) = P, \qquad S_1 S P = 1 \quad \text{in } \mathrm{PSL}_2(\mathbb{Z}).$$

At z_2, comparison of quadratic leading terms gives

$$\big(g_2'(z_2)\big)^2 = S'(i) = -1;$$

the simultaneous parabolic normalization selects $g_2'(z_2) = -i$. The periodic toric cusp action is free and properly discontinuous; its bounded-chart argument uses $|B_t \bar{\lambda}| \geq \frac{1}{2} |\bar{\lambda}|$. Exponentiation identifies the punctured toric quotient with the family of compact two-dimensional complex tori, not with an abelian-surface family. For the finite logarithmic actions the opposite translation signs are retained:

$$v_1 = \varepsilon, \qquad v_2 = -\varepsilon', \qquad \gamma(v_1) = 1, \qquad \gamma(v_2) = -1,$$

and

$$(z, x) \longmapsto (g_j z, A_j x + v_j / m_j).$$

These actions are free by the invariant covector criterion. On an overlap, the compatible logarithm is

$$\ell'(w) = \ell(g_j^{-1} w) - \frac{2\pi i}{m_j}, \qquad h_j \widetilde{g}_j^{\log} h_{\bar{j}}^{-1} = \widetilde{g}_j^{0},$$

so no branch sign is suppressed. Their radial stalk maps are the printed quotient maps, and the three fillings glue to a compact connected smooth complex threefold X with a proper surjective holomorphic map

$$f : X \longrightarrow \mathbb{P}^1.$$

On the cusp cover the toric form

$$\Omega = \frac{dx_1 \wedge dx_2 \wedge dt}{x_1 x_2} = \pm dz_0 \wedge dz_1 \wedge dz_2$$

extends nowhere vanishingly and satisfies $\Psi_{\bar{\lambda}}^* \Omega = \Omega$. This first proves $\omega_{N_0} \simeq \mathcal{O}_{N_0}$; only then does the open embedding $N_0 \hookrightarrow X$ give the typed two-stage chain

$$\omega_X|_{N_0} \simeq \omega_{N_0} \simeq \mathcal{O}_{N_0}.$$

The complete chain is theorems 10.2, 10.6, 10.7 and 10.9 to 10.11.

2. **[VERIFIED] The cusp fibre has the exact geometry needed in the topology and conductor arguments.** Its scheme-theoretic local equations are

$$z_0, \qquad z_0z_1, \qquad z_0z_1z_2.$$

It is reduced, irreducible, nonnormal, and normal crossings (not simple normal crossings). Its normalization is dP_6, with opposite sides of the anticanonical hexagon identified, and its double locus consists of three rational curves through two triple points.

3. **[VERIFIED] The algebraic reduction is computed directly and does not use the topology or CDP20.** For a very general smooth fibre,

$$\mathrm{NS}(F_b) = \mathbb{Z}\eta, \qquad \eta = u \wedge w + 6\gamma \wedge \delta, \qquad \eta \wedge \eta = 12\gamma \wedge u \wedge w \wedge \delta.$$

The inverse-dual matrix G_η associated with the Appell–Humbert form has determinant

$$\det G_\eta = \frac{2}{3} \operatorname{Im}\tau \left(\operatorname{Im}\beta - \frac{6(\operatorname{Im}\mu)^2}{\operatorname{Im}\tau} \right) < 0,$$

and the same inertia as that form, hence signature $(1,1)$ and $a(F_b) = 0$. The repaired meromorphic-descent argument then gives

$$\mathcal{M}(X) = f^*\mathbb{C}(t), \qquad a(X) = 1,$$

and f is the algebraic reduction; see theorems 54.2 and 54.4.

4. **[VERIFIED] The intrinsic nearby-cycle/Leray route proves the integral topology without using the cellular collapse.** For the cusp filling, the integral nearby-cycle map satisfies

$$\mathrm{sp}^r : H^r(W;\mathbb{Z}) \xrightarrow{\sim} H^r(F;\mathbb{Z})^{T_0} \qquad (0 \le r \le 4)$$

as saturated lattices. With $q := uw + 6\gamma\delta \in \bigwedge^2 V = H^2(F;\mathbb{Z})$, the global sheaf calculation gives

$$H^0(R^1) = \mathbb{Z}(12\gamma), \quad H^0(R^2) = \mathbb{Z}(2q), \quad H^0(R^3) = \mathbb{Z}(2\gamma uw).$$

The canonical truncation triangle

$$R^0f_*\mathbb{Z}_X \longrightarrow \tau_{\le 1}Rf_*\mathbb{Z}_X \longrightarrow R^1f_*\mathbb{Z}_X[-1] \xrightarrow{\kappa} R^0f_*\mathbb{Z}_X[1]$$

identifies the explicitly computed Cech discrepancy with Leray d_2. For $(\ell_0,\ell_1,\ell_2) = (0,1,-1)$, hence $p = -1$, orientation of ω may be chosen so that

$$d_2(12\gamma) = \omega, \qquad d_2(2q) = \omega[\delta], \qquad d_2(2\gamma uw) = \omega[\gamma\delta].$$

Therefore

$$H^1(X;\mathbb{Z}) = H^2(X;\mathbb{Z}) = H^3(X;\mathbb{Z}) = 0, \qquad H_2(X;\mathbb{Z}) = 0.$$

5. **[VERIFIED] The based van Kampen maps and sphere recognition close the topology theorem.** Zero-section whiskers fix the common basepoint, the finite logarithmic overlaps impose the primitive relations with one common sign, the cusp inclusion is $\Lambda \to \bar{\Lambda}$, and the cusp meridian bounds the explicit toric zero-section disc with no translation drift. Hence

$$\pi_1(X) = \langle c, x, y \mid c \text{ central},\ xy = 1,\ x^3 = c,\ y^4 = c^{-1} \rangle \simeq 1.$$

Poincaré duality, universal coefficients, Hurewicz, and the smooth homotopy-sphere recognition theorem now give

$$X \cong_{\mathrm{diff}} S^6, \qquad \langle c_3(T_X), [X] \rangle = \chi_{\mathrm{top}}(X) = 2.$$

See theorems 53.1 to 53.7.

6. **[REFUTED] The printed CDP dependency has two exact failures.** For a reduced nonnormal fibre,

$$0 \longrightarrow N^*_{S/X} \longrightarrow \ker(\Omega^1_X|_S \to \widetilde{\Omega}^1_S) \longrightarrow \operatorname{tors}\Omega^1_S \longrightarrow 0,$$

so the vanishings used on p. 692 of [3] do not control the conductor-supported middle term. A proper nodal-cubic-times-$\mathbb{P}^1$ family is a counterexample to that inference. Separately, a split extension satisfies

$$(K \rtimes Q)^{\mathrm{ab}} \simeq Q^{\mathrm{ab}} \oplus (K^{\mathrm{ab}})_Q,$$

not the direct sum with all of K^{ab} unless monodromy acts trivially.

7. **[VERIFIED] The singular-fibre nonvanishing has a topology-free Pic^0 proof and three exact all-Pic certificates.** The invariant toric three-form first trivializes ω_{N_0}, and the open embedding then gives $\omega_X|_{N_0} \simeq \omega_{N_0} \simeq \mathcal{O}_{N_0}$, as recorded above. Let $\nu : \widetilde{W} \simeq \mathrm{dP}_6 \to W$ be the normalization. For $L \in \mathrm{Pic}^0(X)$, naturality of c_1 and injectivity of

$$c_1 : \mathrm{Pic}(\mathrm{dP}_6) \hookrightarrow H^2(\mathrm{dP}_6; \mathbb{Z})$$

give $\nu^*((L^\vee \otimes \omega_X)|_W) \simeq \mathcal{O}_{\mathrm{dP}_6}$. The conductor section, Ramis–Ruget duality on the compact Cartier fibre, and top-degree analytic base change then prove

$$R^2 f_*(T_X \otimes L) \neq 0 \qquad (L \in \mathrm{Pic}^0(X))$$

without any global homology calculation. For arbitrary L, the exact trivial-pullback obstruction is the homomorphism

$$r : \mathrm{Pic}(X) \longrightarrow \mathrm{Pic}(\mathrm{dP}_6), \qquad r([L]) = [\nu^* i^* L].$$

For the candidate, the Leray edge image of $c_1(L)$ lies in $E_\infty^{0,2} \subseteq \ker d_2^{0,2}$, but $d_2^{0,2}(2q) = \omega[\delta]$ is an isomorphism. Hence the edge section is zero; proper base change and naturality give $c_1(L|_W) = 0$, and injectivity of c_1 on dP_6 forces

$$\nu^* i^* L \simeq \mathcal{O}_{\mathrm{dP}_6}, \qquad r([L]) = 0.$$

This proves the conclusion for every $L \in \mathrm{Pic}(X)$ without citing the packaged $H^2(X;\mathbb{Z}) = 0$ theorem, although it reuses the same integral Leray transgression.

The second certificate uses no global Leray differential. The exact order-four filling lattice forces $c_1(L|_F) = 2nq$. Subtracting the invariant vertical form on the pair-of-pants bundle gives a crossed homomorphism with global-coinvariant boundary values. The logarithmic translations and their cyclic norm maps give

$$\bar{d}_1 = 4n[\delta], \qquad \bar{d}_2 = -3n[\delta],$$

while the flat zero-section cusp sweep bounds a literal toric solid torus, so $\bar{d}_0 = 0$. The based relation gives $\bar{d}_1 + \bar{d}_2 + \bar{d}_0 = 0$, hence $n = 0$. Naturality and injectivity of the local integral cusp specialization then give $c_1(L|_W) = 0$ and

$$\nu^* i^* L \simeq \mathcal{O}_{\mathrm{dP}_6}, \qquad r([L]) = 0.$$

This is independent of global d_2, though not of local cusp topology. See theorems 60.12 to 60.14, 60.18, 60.22 and 60.25.

The precise limit of a cusp-only analytic replacement is also computed. For $Q(v) = v_1^2 - 6v_2^2$, the locally finite toric divisor $\mathcal{D}_Q = \sum_v Q(v)E_v$ has an explicit equivariant descent $\mathcal{L}_Q \in \mathrm{Pic}(N_0)$ satisfying

$$c_1(\mathcal{L}_Q|_{F_t}) = -2q, \qquad (\deg_{C_i} \nu^*(\mathcal{L}_Q|_W)) = (-12, 2, 0, -12, 2, 0).$$

Thus $\mathrm{Pic}(N_0) \to \mathrm{Pic}(\mathrm{dP}_6)$ is nonzero, so no cusp-local representability argument can prove the global restriction theorem by itself. The completed third certificate adds the two finite fillings analytically and retains the prior coinvariant input

$$(\textstyle\bigwedge^2 V)^G = \mathbb{Z}q, \qquad q = u \wedge w + 6\gamma \wedge \delta, \qquad V_G \otimes \mathbb{R} = \mathbb{R}[\delta].$$

Thus, for a global line bundle L, $E = c_1(L|_F) = kq$. The finite norm proof keeps every covering level distinct. On the vector cover

$$\widetilde{J}_j = \Delta_j \times \mathbb{C}^2 \xrightarrow{r_j} J'_j$$

define the vector-cover automorphisms

$$Q_{j,\log}, Q_{j,\mathrm{lin}} \in \mathrm{Aut}(\widetilde{J}_j), \qquad Q_{j,\log} = \widetilde{g}_j^{\,v_j/m_j}, \quad Q_{j,\mathrm{lin}} = \widetilde{g}_j^{\,0}.$$

These are not the quotient automorphisms: on the vector cover,

$$\begin{aligned} Q_{j,\log}\tau_\lambda Q_{j,\log}^{-1} &= \tau_{A_j\lambda}, \quad (Q_{j,\log})^{m_j} = \tau_{v_j}, \\ Q_{j,\mathrm{lin}}\tau_\lambda Q_{j,\mathrm{lin}}^{-1} &= \tau_{A_j\lambda}, \quad (Q_{j,\mathrm{lin}})^{m_j} = 1. \end{aligned}$$

The sign in $(Q_{j,\log})^{m_j} = \tau_{v_j}$ is positive because m_j clockwise iterations of $s_j \circ g_j = e^{-2\pi i/m_j} s_j$ produce translation by $+v_j$. Let

$$q_{j,\log}, q_{j,\mathrm{lin}} \in \mathrm{Aut}(J'_j)$$

be their descents. They have order dividing m_j by these relations and exactly m_j because they cover g_j. On $J'^*_j = J'_j|_{\Delta_j^*}$ define

$$\begin{aligned} p_{j,\log} &: J'^*_j \longrightarrow N^*_j = J'^*_j/\langle q_{j,\log}\rangle, \\ p_{j,\mathrm{lin}} &: J'^*_j \longrightarrow J^*_j = J'^*_j/\langle q_{j,\mathrm{lin}}\rangle. \end{aligned}$$

The branch-independent translation and its descended biholomorphism are typed by

$$h_j : J'^*_j \longrightarrow J'^*_j, \qquad \bar{h}_j : N^*_j \longrightarrow J^*_j,$$

and satisfy

$$h_j \circ q_{j,\log} \circ h_j^{-1} = q_{j,\mathrm{lin}}, \qquad p_{j,\mathrm{lin}} \circ h_j = \bar{h}_j \circ p_{j,\log},$$

and the filled line bundle gives the typed pullback comparison

$$p^*_{j,\log}(L|_{N^*_j}) \simeq h_j^* p^*_{j,\mathrm{lin}}(L|_{J^*_j}).$$

For the factor extension return to the filled vector cover:

$$P_{j,\log} : J'_j \longrightarrow N_j, \qquad \pi_j^{\mathrm{vec}} := P_{j,\log} \circ r_j : \widetilde{J}_j \longrightarrow N_j, \qquad P_{j,\log}|_{J'^*_j} = p_{j,\log},$$

$$\Gamma_j^{\log} = \langle \tau_\lambda, Q_{j,\log} : \lambda \in \Lambda \rangle.$$

The pullback $(\pi_j^{\mathrm{vec}})^*(L|_{N_j})$ defines the filled factor extension. Over Δ_j^* the quotient square identifies its commutator form with E; that integer-valued form is locally constant on the connected disc, so it remains E on the filled cover. After trivializing this pullback, choose normalized vertical lifts

$$U_\lambda^{(j)} U_\mu^{(j)} = \mathbf{c}\Big(\frac{1}{2}E(\lambda,\mu)\Big)\, U_{\lambda+\mu}^{(j)},$$

write $\alpha\tau_\lambda\alpha^{-1} = \tau_{A_\alpha\lambda}$ and $T_\alpha = (A_\alpha^{-1})^{\mathsf{t}}$, and define $\widetilde{d}_j(\alpha) \in V_{\mathbb{R}}$ by

$$\widetilde{G}_\alpha U^{(j)}_{A_\alpha^{-1}\lambda} \widetilde{G}_\alpha^{-1} = \mathbf{c}\Big(\langle \widetilde{d}_j(\alpha), \lambda\rangle\Big)\, U_\lambda^{(j)}.$$

It is a crossed homomorphism:

$$\widetilde{d}_j(\alpha\beta) = \widetilde{d}_j(\alpha) + T_\alpha \widetilde{d}_j(\beta).$$

To compare the boundary factors, use the actual common cover and its restricted lattice quotient

$$\widetilde{J}_j^* := \Delta_j^* \times \mathbb{C}^2, \qquad r_j^* := r_j|_{\widetilde{J}_j^*} : \widetilde{J}_j^* \longrightarrow J_j'^*.$$

The two quotient maps from this common domain are

$$\begin{aligned} \Phi_{j,\log} : \widetilde{J}_j^* &\longrightarrow N_j^*, \quad \Phi_{j,\log} := p_{j,\log} \circ r_j^*, \\ \Phi_{j,\mathrm{glob}} : \widetilde{J}_j^* &\longrightarrow J_j^*, \quad \Phi_{j,\mathrm{glob}} := p_{j,\mathrm{lin}} \circ h_j \circ r_j^* = \bar{h}_j \circ \Phi_{j,\log}. \end{aligned}$$

Both maps are $\Gamma_j^{\log}$-invariant. Translation invariance follows from $r_j^* \circ \tau_\lambda = r_j^*$. For every $\widetilde{x} \in \widetilde{J}_j^*$, invariance under the nontrivial generator is the point-valued calculation

$$\begin{aligned} \Phi_{j,\log}(Q_{j,\log}(\widetilde{x})) &= p_{j,\log}(q_{j,\log}(r_j^*(\widetilde{x}))) = \Phi_{j,\log}(\widetilde{x}), \\ \Phi_{j,\mathrm{glob}}(Q_{j,\log}(\widetilde{x})) &= p_{j,\mathrm{lin}}(h_j(q_{j,\log}(r_j^*(\widetilde{x})))) \\ &= p_{j,\mathrm{lin}}(q_{j,\mathrm{lin}}(h_j(r_j^*(\widetilde{x})))) = \Phi_{j,\mathrm{glob}}(\widetilde{x}). \end{aligned}$$

Thus $\widetilde{J}_j^*$ is the same $\Gamma_j^{\log}$-cover of the two biholomorphic quotients N_j^* and J_j^*. Pulling back the gluing isomorphism gives

$$\Theta_j : \Phi_{j,\log}^*(L|_{N_j^*}) \xrightarrow{\sim} \Phi_{j,\mathrm{glob}}^*(L|_{J_j^*}).$$

It intertwines descent and hence identifies the pushed-out filled and global $\Gamma_j^{\log}$-factor extensions while fixing every central $\mathbf{c}(r)$, $r \in \mathbb{R}$.

The loop and its three successive lifts are fixed pointwise as follows. Let

$$\gamma_j^J : [0,1] \longrightarrow J_j^*$$

be the zero-section image of the unwhiskered clockwise boundary meridian, so $\gamma_j^J(1) = \gamma_j^J(0)$. Choose the global zero-section basepoint $\widehat{b} \in J|_P$ and the fixed zero-section whisker

$$\omega_j : [0,1] \longrightarrow J|_P, \qquad \omega_j(0) = \widehat{b}, \qquad \omega_j(1) = \gamma_j^J(0),$$

where the boundary copy of J_j^* is identified with its image in $J|_P$. The retained based total-space loop is the class

$$\widehat{\rho}_j = [\omega_j * \gamma_j^J * \omega_j^{-1}] \in \pi_1(J|_P, \widehat{b}).$$

Choose $x_j \in J_j'^*$ such that $p_{j,\mathrm{lin}}(h_j(x_j)) = \gamma_j^J(0)$, put $x_{j,\mathrm{lin}} = h_j(x_j)$, and let $\widetilde{\gamma}_{j,\mathrm{lin}} : [0,1] \to J_j'^*$ be the $p_{j,\mathrm{lin}}$-lift of γ_j^J beginning at $x_{j,\mathrm{lin}}$. For the chosen clockwise generator,

$$\begin{aligned} p_{j,\mathrm{lin}} \circ \widetilde{\gamma}_{j,\mathrm{lin}} &= \gamma_j^J, \\ \widetilde{\gamma}_{j,\mathrm{lin}}(0) &= x_{j,\mathrm{lin}}, \\ \widetilde{\gamma}_{j,\mathrm{lin}}(1) &= q_{j,\mathrm{lin}}(x_{j,\mathrm{lin}}). \end{aligned}$$

Set

$$\gamma_j^N := \bar{h}_j^{-1} \circ \gamma_j^J : [0,1] \longrightarrow N_j^*, \qquad \widetilde{\gamma}_{j,\log} := h_j^{-1} \circ \widetilde{\gamma}_{j,\mathrm{lin}} : [0,1] \longrightarrow J_j'^*.$$

Then the quotient square gives the point-valued identities

$$\begin{aligned} p_{j,\log} \circ \widetilde{\gamma}_{j,\log} &= \gamma_j^N, \\ \widetilde{\gamma}_{j,\log}(0) &= x_j, \\ \widetilde{\gamma}_{j,\log}(1) &= q_{j,\log}(x_j). \end{aligned}$$

Choose $\widetilde{x}_j \in \widetilde{J}_j^*$ with $r_j^*(\widetilde{x}_j) = x_j$, and let $\widetilde{\gamma}_{j,\mathrm{vec}} : [0,1] \to \widetilde{J}_j^*$ be the r_j^*-lift of $\widetilde{\gamma}_{j,\log}$ beginning at $\widetilde{x}_j$. On adjacent sheets choose compatible logarithm branches

$$\ell'(w) = \ell(g_j^{-1}w) - \frac{2\pi i}{m_j}.$$

The point-valued vector-cover endpoint is

$$\begin{aligned} r_j^* \circ \widetilde{\gamma}_{j,\mathrm{vec}} &= \widetilde{\gamma}_{j,\log}, \\ \widetilde{\gamma}_{j,\mathrm{vec}}(0) &= \widetilde{x}_j, \\ \widetilde{\gamma}_{j,\mathrm{vec}}(1) &= Q_{j,\log}(\widetilde{x}_j). \end{aligned}$$

excluding both $Q_{j,\log}^{-1}(\widetilde{x}_j)$ and every $\tau_\lambda Q_{j,\log}(\widetilde{x}_j)$ with $\lambda \neq 0$. Fibre loops remain the pointwise deck paths from $\widetilde{x}_j$ to $\tau_\lambda(\widetilde{x}_j)$, since h_j is a fibre translation.

The factor of L along the fixed lift of ω_j identifies the target factor over the local boundary segment with the globally based factor. Let $G_{\rho_j}^{(j)}$ be the target lift of the local boundary deck element whose whiskered image is the chosen global lift G_{ρ_j}. It satisfies

$$G_{\rho_j}^{(j)} U_{A_j^{-1}\lambda} (G_{\rho_j}^{(j)})^{-1} = \mathbf{c}(\langle d_j, \lambda \rangle) U_\lambda, \qquad d_j = d(\rho_j).$$

Consequently, for some $a_j \in V_{\mathbb{R}}$ and $b_j \in \mathbb{R}$,

$$\Theta_{j,*}(U_\lambda^{(j)}) = \mathbf{c}(\langle a_j, \lambda \rangle) U_\lambda, \qquad \Theta_{j,*}(\widetilde{G}_{Q_{j,\log}}) = \mathbf{c}(b_j) G_{\rho_j}^{(j)}.$$

The central factor $\mathbf{c}(b_j)$ disappears under conjugation. Therefore

$$\widetilde{d}_j(Q_{j,\log}) = d_j + (T_j - I)a_j, \qquad \widetilde{d}_j(\tau_v) = \iota_v E.$$

Here $A_{Q_{j,\log}} = A_j$ and $T_{Q_{j,\log}} = T_j$. Consequently the exact, sign-sensitive norm chain is

$$\begin{aligned} N_{T_j} d_j &= N_{T_j} \widetilde{d}_j(Q_{j,\log}) = \widetilde{d}_j((Q_{j,\log})^{m_j}) \\ &= \widetilde{d}_j(\tau_{v_j}) = \iota_{v_j} E. \end{aligned}$$

Finally, on a simply connected $U \subset \Delta_j^*$ the logarithm branch ℓ gives the point-typed vector-cover lift

$$H_j^\ell : U \times \mathbb{C}^2 \longrightarrow U \times \mathbb{C}^2, \qquad H_j^\ell(z,\zeta) = \left(z, \zeta + \frac{\ell(z)}{2\pi i} \Pi(z) v_j \right)$$

of h_j. A branch change $\ell \mapsto \ell + 2\pi i n$ gives

$$H_j^{\ell + 2\pi i n} = \tau_{n v_j} \circ H_j^\ell.$$

At factor-extension level this gives

$$\Theta'_{j,*} = \mathrm{Ad}(U_{n v_j}) \circ \Theta_{j,*}.$$

Thus the comparison character changes by $a_j \mapsto a_j + n\iota_{v_j} E$. Since

$$T_j(\iota_{v_j} E) = \iota_{v_j} E,$$

both displayed $\widetilde{d}_j$ and norm identities are unchanged; the descended $h_j, \bar{h}_j$ and quotient square are branch independent. The exact contractions are

$$\iota_{v_1} q = 2\eta_1, \quad \iota_{v_2} q = -3\eta_2, \quad [\eta_1] = 3[\delta], \quad [\eta_2] = 2[\delta],$$

and logarithms of the actual $\mathcal{O}^*$ factors give

$$\bar{d}_1 = 2k[\delta], \qquad \bar{d}_2 = -\tfrac{3}{2}k[\delta], \qquad \bar{d}_0 = 0, \qquad \bar{d}_1 + \bar{d}_2 + \bar{d}_0 = 0,$$

where the cusp equality is an argument-principle calculation in a Stein chart. Hence $k = 0$ and $E = 0$. The toric support-function calculation uses the explicitly typed unimodular map

$$B_0 : \bar{\Lambda} \simeq \mathbb{Z}^2 \longrightarrow \Lambda_{\mathrm{tor}} \simeq \mathbb{Z}^2, \qquad [B_0] = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad \det B_0 = 1.$$

For adjacent rays $\rho_v, \rho_w < \sigma = \{v, w, u\}$ the affine-open directions are, exactly,

$$U_{\rho_v} = D(z_w z_u) \subset U_\sigma, \qquad U_{\rho_w} = D(z_v z_u) \subset U_\sigma, \qquad U_{\rho_v} \cap U_{\rho_w} = D(z_v z_w z_u) = U_0.$$

If $\mu_\lambda^+ : \Psi_\lambda^* \mathcal{O}(D_a) \to \mathcal{O}(D_a)$ is multiplication by χ^{m_λ}, the map needed on the negative divisor is the inverse-dual, not the ordinary dual:

$$\mu_\lambda^- := ((\mu_\lambda^+)^{-1})^\vee : \Psi_\lambda^* \mathcal{O}(-D_a) \longrightarrow \mathcal{O}(-D_a), \qquad s \longmapsto \chi^{-m_\lambda} s.$$

Now set, with its type retained,

$$M = L|_{N_0} \in \mathrm{Pic}(N_0), \qquad E_M = c_1(M|_{F_t}) = E = 0.$$

For $A = \nu^*(M|_W)$ and $(\deg_{C_1} A, \dots, \deg_{C_6} A) = (d_1, d_2, d_3, d_1, d_2, d_3)$, the normalized degrees determine

$$[B] = \begin{pmatrix} d_2 + d_3 & d_3 \\ d_3 & d_1 + d_3 \end{pmatrix}, \qquad E_M((\lambda, 0), (0, m)) = -B(B_0\lambda, m).$$

The antecedent $E_M = 0$ and unimodularity of B_0 force $B = 0$, hence all six degrees vanish and

$$\nu^*(L|_W) = \nu^*(M|_W) \simeq \mathcal{O}_{E_0}; \qquad r([L]) = [\mathcal{O}_{E_0}] = 0 \quad \text{in } \mathrm{Pic}(E_0) \simeq \mathrm{Pic}(\mathrm{dP}_6).$$

This proof uses neither global Leray nor local integral specialization. See theorems 55.15 and 60.19 to 60.21.

8. **[VERIFIED] The constructed threefold is a theorem-level counterexample to CDP20 Theorem 2.2 as stated.** It has

$$H^1(X;\mathbb{Z}) = H^2(X;\mathbb{Z}) = 0, \qquad a(X) = 1, \qquad f : X \to \mathbb{P}^1 \text{ a holomorphic algebraic reduction,}$$

while

$$c_3(X) = 2 > 0, \qquad X \cong_{\mathrm{diff}} S^6.$$

These are precisely the relevant hypotheses and the opposite of the published conclusions $c_3(X) \le 0$ and $a(X) = 0$. This conclusion uses the verified candidate proof, not merely the fact that defects were found in the published proof.

9. **[VERIFIED] The strongest corrected CDP conclusion is now proved, with its exact extra hypothesis.** For any compact complex threefold in the CDP holomorphic-reduction setting, existence of a single L_0 with

$$H^2(X, T_X \otimes L_0) = 0$$

implies, for general M,

$$H^i(X, T_X \otimes M) = 0 \quad (i \neq 1), \qquad \chi(X, T_X \otimes M) = \frac{1}{2} c_3(X) \le 0.$$

The relative condition $R^2 f_*(T_X \otimes L) = 0$ for one L implies this global hypothesis by projection formula, Serre vanishing on the base, and Leray. On a reduced full fibre W_c, after the two endpoint vanishings, the missing ambient-cotangent vanishing is equivalent to injectivity of

$$H^0(W_c, \operatorname{tors} \Omega^1_{W_c} \otimes A_c) \xrightarrow{\delta_c} H^1(W_c, N^*_{W_c/X} \otimes A_c).$$

This repairs the proof route without asserting the false original theorem. Because $c_3(X) = 2$, the contrapositive of the corrected global theorem gives $H^2(X, T_X \otimes L_0) \neq 0$ for every L_0. Independently, the candidate has nonzero $\ker \delta_c$ at its cusp and $R^2 f_*(T_X \otimes L) \neq 0$ for every L. See theorems 60.28 to 60.30.

10. **[VERIFIED] The Proposition 10.8 uniform non-torsion branch is complete.** The reconstructed direct-image calculation gives

$$R^1 f_* \mathcal{O}_X \simeq \mathcal{O} \oplus \mathcal{O}(-1), \qquad \operatorname{Pic}(X) \simeq \mathbb{C},$$

with degree-one base change on all scheme-theoretic fibres. At a finite filling, the invariant canonical section and the direct-image generator are not conflated: if $k_j = m_j - 1 - a_j$, then

$$\operatorname{div}(\overline{\sigma}_j) = k_j S_j, \qquad f_{j*}\omega_{N_j/D_j} = \mathcal{O}_{D_j} \cdot s_j^{k_j} \vartheta'_j, \qquad s_j^{k_j} \vartheta'_j = \frac{e^{\varphi(s_j)}}{m_j} s_j^{-a_j} e \quad \text{on } \Delta_j^*.$$

In particular $k_2 = 2$, so the canonical section is not a frame there; the invariant-module identity is the exact statement that proves the Hodge line used below. The finite character calculation gives $\operatorname{ord}(\mathcal{O}_X(S_j)|_{S_j}) = m_j$. For $\lambda = a\gamma + bu + cw + d\delta$, membership of its Dolbeault image in the global Picard line is exactly

$$c\tau + d\mu - b = 0.$$

Linear independence of $1, \tau, \mu$ makes the rank-two exceptional fibres countable, while rank-one fibres contribute bad parameters only in $\mathbb{R}$. The three full special fibres and the reductions S_j each add only countably many parameters. Thus every $z \in \mathbb{C} \setminus (\mathbb{R} \cup \text{a countable set})$ supplies one L_z non-torsion on every full fibre and on both S_j. The cusp step was first repaired using intrinsic nearby cycles and therefore does not depend on the later cellular-collapse verification.

11. **[VERIFIED] The equivariant cellular collapse and its exact comparison with nearby cycles are now complete.** The positive toric quotient is a compact 3-manifold cobordism between two tori, its central boundary is a strong deformation retract, and the lifted retraction spreads by compact phases and conjugates back to the original twist. On the fixed fibre, the resulting map $c = r|_F$ has

$$0 \longrightarrow \operatorname{im}(\textstyle\bigwedge^q M_0 - I) \longrightarrow \bigwedge^q \Lambda \xrightarrow{c_*} H_q(W; \mathbb{Z}) \longrightarrow 0.$$

In degree two, with the primitive orientations fixed in theorem 58.6,

$$[c_*]_2 = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & -1 & -1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \end{pmatrix},$$

and its kernel is $\mathbb{Z}\langle \widehat{w}\widehat{\delta}, \widehat{\gamma}\widehat{w} + \widehat{u}\widehat{\delta} \rangle$. The neighborhood realization of the nearby-cycle unit then gives, for all q,

$$\mathrm{sp}^q = i_F^* (i_W^*)^{-1} = i_F^* r^* = c^*.$$

Thus C1–C3 are verified integrally, including saturation and the degree-three orientation.

12. **[VERIFIED] The global Mayer–Vietoris realization and its Leray comparison are complete.** The earlier audit conflated the two stages. On $X^\circ = X_1 \cup X_2$, the verified radial markings give the actual maps

$$\alpha_q = (\pi_{1*}, -\pi_{2*}).$$

Their degree-one and degree-two matrices have saturated rank three. On the second attachment $X = N_0' \cup X^\circ$, primitive splittings reduce the global maps to

$$D_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -12 & 0 & -p \end{pmatrix}, \qquad D_2^{\mathrm{red}} = \begin{pmatrix} -p & 0 \\ 0 & 1 \end{pmatrix}.$$

For $p = -1$ they are unimodular. Dualizing the same oriented collar maps gives commuting obstruction squares and

$$d_2(12\gamma) = -p\omega, \quad d_2(2q) = -p\omega[\delta], \quad d_2(2\gamma uw) = -p\omega[\gamma\delta].$$

Thus the alternate route independently gives the integral homology of S^6; see theorems 58.10, 58.11, 58.13 and 58.18.

13. **[VERIFIED] The Hodge diamond and Frölicher sequence are computed completely.** Put

$$\Sigma = p_0 + p_1 + p_2, \qquad D = f^{-1}(\Sigma)_{\mathrm{red}} = W + S_1 + S_2, \qquad \mathcal{L} = \Omega^1_B(\log \Sigma) \simeq \mathcal{O}_B(1).$$

The logarithmic relative bundle $\mathcal{E} = \Omega^1_X(\log D)/f^*\Omega^1_B(\log \Sigma)$ has

$$f_*\mathcal{E} \simeq \mathcal{O} \oplus \mathcal{O}(-1), \qquad R^1 f_*\mathcal{E} \simeq \mathcal{O}^{\oplus 2} \oplus \mathcal{O}(-1)^{\oplus 2}, \qquad R^2 f_*\mathcal{E} \simeq \mathcal{O} \oplus \mathcal{O}(-1).$$

All three are locally free and commute with base change. At a finite point write $t = s^m$ and $\ell = dt/t = m\, ds/s$, with $(m, a) = (3, 2)$ at p_1 and $(m, a) = (4, 1)$ at p_2, and let $D_{\mathrm{KS}} = (d_{rs})$ be the marked Kodaira–Spencer matrix after removing ds. Dots in the raw period determinant mean d/ds, and

$$K = \det(\dot{Z}) = 6\dot{\mu}^{\,2} - \dot{\tau}\dot{\beta}.$$

With $\zeta_m = e^{-2\pi i/m}$, choose relative-cotangent eigenframes (e_0, e_1) with characters $(1, \zeta_m^a)$ and $R^1\mathcal{O}$ eigenframes (b_0, b_1) with characters $(1, \zeta_m^{-a})$. The connecting maps are typed as

$$\kappa_0 : f_*\mathcal{E} \longrightarrow \mathcal{L} \otimes R^1 f_*\mathcal{O}_X, \qquad \kappa_1 : R^1 f_*\mathcal{E} \longrightarrow \mathcal{L} \otimes R^2 f_*\mathcal{O}_X \simeq \mathcal{O}.$$

In the actual invariant source basis $(e_0, s^{m-a}e_1)$ and target basis $(b_0, s^a b_1)\ell$, put

$$A = \frac{s}{m} d_{00}, \quad B = \frac{s^{1-a}}{m} d_{01}, \quad C = \frac{s^{1-a}}{m} d_{10}, \quad D_0 = \frac{s^{m+1-2a}}{m} d_{11}.$$

The two connecting maps then have the exact matrices

$$[\kappa_0] = \begin{pmatrix} A & tB \\ C & D_0 \end{pmatrix}, \qquad [\kappa_1] = (-C, A, -D_0, B),$$

where the ordered source of the row is $(e_0b_0, s^a e_0 b_1, s^{m-a} e_1 b_0, e_1 b_1)$ and its target generator is $s^a(b_0 \wedge b_1)\ell$. Thus

$$\mathrm{div}(\det \kappa_0) = p_1 + p_2, \qquad \mathrm{ord}_{z_1} K = 2, \quad \mathrm{ord}_{z_2} K = 0, \qquad \kappa_1 \text{ is sheaf-surjective.}$$

The residue normalizations are also exact:

$$\mathcal{O}(1) \longrightarrow \mathbb{C}_{p_0} \oplus \mathbb{C}_{p_1} \oplus \mathbb{C}_{p_2}, \qquad \sigma \longmapsto (\sigma(p_0), 3\sigma(p_1), 4\sigma(p_2)),$$

$$\operatorname{res}_{1,j}(xb_0 + ys^a b_1) = m_j x[b_0],$$

and

$$\operatorname{im}(\kappa_0 \bmod t) = \ker \operatorname{res}_{1,j} = \mathbb{C}\, s^a b_1.$$

Hence the two finite cokernel skyscrapers map isomorphically to $H^1(S_1, \mathcal{O}) \oplus H^1(S_2, \mathcal{O})$ and cancel in the absolute residue sequence.

At the cusp, the quotient from ordinary to logarithmic relative differentials is checked on the xy and xyz modules. In the fixed global branch ordering its two triple-point maps are

$$d_P(a) = (a_0 - a_1, a_0 - a_2, a_1 - a_2), \qquad d_Q(a) = (a_2 - a_1, a_0 - a_1, a_0 - a_2),$$

Put $\eta_1 = d\log x_1$ and $\eta_2 = d\log x_2$. These cusp logarithmic forms lie in the logarithmic-form module, whereas the finite-filling covectors denoted η_j above lie in the filling-dual lattice. The exact morphism used below is the branch-difference map ρ from the former module to the branch-difference lattice; no identification of the two source modules is made. Both branch maps give

$$\rho(\eta_1) = q_1 = (-1, 0, 1), \qquad \rho(\eta_2) = q_2 = (0, -1, -1),$$

which is the degree-zero branch-difference isomorphism. With $\alpha_i = \delta_0(q_i)$ and the standard Čech differential, the total-complex sign is

$$\delta_1\big(\rho_*(\alpha \otimes \eta_i)\big) = -\alpha \smile \delta_0(q_i) = -\alpha \smile \alpha_i.$$

Thus, for $\theta = -\delta_1^{-1}(\alpha_1 \smile \alpha_2)$, the induced row is exactly

$$\rho_* = (0, 1, -1, 0) : \mathbb{C}\langle \alpha_1\eta_1, \alpha_1\eta_2, \alpha_2\eta_1, \alpha_2\eta_2\rangle \longrightarrow \mathbb{C}\theta.$$

The same local modules show that $\Omega^1_{X/B}$ is torsion-free but non-locally-free at the cusp; its torsion is supported precisely on the two multiple fibres.

For the remaining scalar, rescale the Cayley coordinate to $s = (z_2 - \bar{z}_2)s_2$, so that $s \circ g_2 = -is$ and $ds/dz(z_2) = 1$; write a dot for d/ds, and put $\ell = dt/t = 4\, ds/s$. In the marked fibre coordinate $\zeta = (\zeta_1, \zeta_2)$, the invariant section $\xi = \partial/\partial\zeta_2$ is nowhere zero in the locally free relative sheaf $T_{X/B} = \mathcal{E}^\vee$ (although its ambient tangent-space value can vanish at a toric fixed point), and the unique lift of its dual unit is $h = (a(z), 1)(d\zeta_1, d\zeta_2)^{\mathsf{t}}$ with $a(z_2) = 1/2$. At the order-four lift the genuine invariant target frame is

$$\omega_2 = s\, u_2(s)(\gamma \wedge u)\ell, \qquad u_2(0) = 1;$$

here b_1 is normalized by $\gamma \wedge b_1 = u_2(s)(\gamma \wedge u)$. $s(\gamma \wedge u)\ell$ alone is not invariant away from $s = 0$. In this frame the signed coarse value is

$$\kappa_1(h \otimes \gamma)(p_2) = -\frac{1}{4}\big(a(z_2)\dot{\tau}(z_2) + \dot{\mu}(z_2)\big).$$

Since $\operatorname{ord}_{z_2}(\tau - i) = 2$, one has $\dot{\tau}(z_2) = 0$ and $K(z_2) = 6\dot{\mu}(z_2)^2 \neq 0$; hence the scalar is exactly $-\frac{1}{4}\dot{\mu}(z_2) \neq 0$. Therefore

$$f_*\Omega^1_X \simeq \mathcal{O}(-2), \quad R^1 f_*\Omega^1_X \simeq \mathcal{O} \oplus \mathcal{O}(-1)^2, \quad R^2 f_*\Omega^1_X \simeq \mathcal{O} \oplus \mathcal{O}(-1),$$

and the exact Hodge table is

$q\backslash p$	0	1	2	3
3	0	0	0	1
2	0	1	2	1
1	1	2	1	0
0	1	0	0	0.

The two middle Frölicher rows are exact copies of $0 \to \mathbb{C} \to \mathbb{C}^2 \to \mathbb{C} \to 0$. Thus the sequence does not degenerate at E_1, while $\dim E_1 = 10$, $E_2 = E_\infty$, and $\dim E_2 = 2$. See theorems 56.5 to 56.7, 56.9 and 56.10. The internal calculation is **[VERIFIED]**, and a further context-isolated AI derivation has independently recovered the invariant lattices, determinant divisor, residue cancellation, scalar test, direct-image splittings, and the same values $h^{1,2} = 1$, $h^{1,1} = 2$. The separate H2 audit record remains **[OPEN]** only because independent human specialist replication is still outstanding; that external replication boundary is not a premise of the Hodge theorem.

14. **[VERIFIED] Every Bott–Chern and Aeppli dimension is fixed.** The auxiliary spaces in McHugh's compact-threefold formulas satisfy $k^{p,q} \le b_{p+q}$: the de Rham image of the pure, doubly closed forms surjects onto $K^{p,q}$. Hence they vanish in all middle degrees for this X; also $E_2^{3,1} = 0$. Thus

$$h^{1,1}_{BC} = 2, \quad h^{1,2}_{BC} = h^{2,1}_{BC} = 1, \quad h^{2,2}_{BC} = 2, \quad h^{2,3}_{BC} = h^{3,2}_{BC} = 1,$$

besides the two endpoint copies of $\mathbb{C}$, and all other Bott–Chern dimensions vanish. Schweitzer duality gives the complete Aeppli diamond. Equivalently,

$$(h^k_{BC}) = (1,0,2,2,2,2,1), \quad (h^k_A) = (1,2,2,2,2,0,1), \quad (\Delta^k) = (0,2,4,4,4,2,0).$$

See theorem 56.11. Coordinate-level representatives and the product structure are certified separately in the next item.

15. **[VERIFIED] The Bott–Chern algebra has explicit generators and square-zero positive ideal.** Choose global $\bar\partial$-closed smooth representatives $G \in \mathcal{A}^{0,1}(X)$ and $Q \in \mathcal{A}^{1,1}(X)$ of the two candidate-specific Dolbeault classes $\mathfrak{g}$ and $\mathfrak{q}$. Here $\mathfrak{g}$ is distinguished, while $\mathfrak{q}$ is a chosen coherent-Leray lift, unique modulo $\operatorname{im} d_1$. Set

$$\mathsf{A} = [\partial G]_{BC}, \qquad \mathsf{B} = [\bar\partial \bar G]_{BC}.$$

On a marked smooth fibre,

$$\sigma_1 = 6\mu\gamma + \tau u + w, \qquad \sigma_2 = \beta\gamma + \mu u + \delta,$$

put

$$y = \operatorname{Im}\tau, \qquad m = \operatorname{Im}\mu, \qquad r = \operatorname{Im}\beta, \qquad \Delta = r - \frac{6m^2}{y} < 0,$$

and define

$$\Gamma = \frac{i}{2\Delta}\left(d\bar\zeta_2 - \frac{m}{y} d\bar\zeta_1\right), \qquad U = \frac{i}{2y\Delta}\left(r\, d\bar\zeta_1 - 6m\, d\bar\zeta_2\right).$$

These represent the fibre images of γ and u, and the global classes satisfy

$$i_b^*\mathfrak{g} = [\Gamma]_{\bar\partial}, \qquad i_b^*\mathfrak{q} = [Q_F]_{\bar\partial}.$$

The Appell–Humbert matrix itself is

$$H_\eta = \frac{1}{y\Delta}\begin{pmatrix} r & -6m \\ -6m & 6y \end{pmatrix}, \qquad H_\eta = 2G_\eta^{-1}, \qquad \det H_\eta = \frac{6}{y\Delta} < 0.$$

The load-bearing wedge identity is

$$\eta = u \wedge w + 6\gamma \wedge \delta = u \wedge \sigma_1 + 6\gamma \wedge \sigma_2 = -\sigma_1 \wedge u - 6\sigma_2 \wedge \gamma,$$

which gives the exact fibre representative

$$Q_F = -d\zeta_1 \wedge U - 6 d\zeta_2 \wedge \Gamma, \qquad \eta \wedge \eta = 12\gamma \wedge u \wedge w \wedge \delta.$$

The marked and complex orientations are opposite because the real determinant of the marked lattice columns is $y\Delta < 0$. Accordingly,

$$\int_{(F_b,\mathrm{or}_{\mathrm{mark}})} Q_F^2 = 12, \qquad \int_{(F_b,\mathrm{or}_{\mathbb{C}})} Q_F^2 = -12.$$

The fibre Dolbeault–Serre pairing is normalized by

$$\int_{F_b} Q_F \wedge \Gamma \wedge d\zeta_2 = -1 \quad \text{for the complex orientation,}$$

so $[\Gamma \wedge Q_F]_{\bar\partial} \neq 0$.

The classes $d_1\mathfrak{g}$ and $\mathfrak{q}$ are independent in $H^{1,1}_{\bar\partial}(X)$: the first restricts to zero on every smooth fibre, whereas the second restricts to the nonzero class of Q_F. Exactness of the first middle Frölicher row therefore gives

$$d_1\mathfrak{q} = [\partial Q]_{\bar\partial} \neq 0,$$

so $\mathsf{C} = [\partial Q]_{BC}$ and its conjugate are nonzero. Moreover, putting

$$v = [G \wedge Q]_{\bar\partial} \in H^{1,2}_{\bar\partial}(X)$$

gives $i_b^* v = [\Gamma \wedge Q_F]_{\bar\partial} \neq 0$. Thus v spans $H^{1,2}_{\bar\partial}(X)$, and injectivity in the second middle row gives

$$d_1 v = [\partial(G \wedge Q)]_{\bar\partial} \neq 0.$$

Consequently $\mathsf{D} = [\partial(G \wedge Q)]_{BC}$ and its conjugate are nonzero.

The global integral normalization is equally explicit. If the base orientation is chosen so that $d_2(12\gamma) = c_1(\mathcal{O}_B(1))$ and $\exp(z) = e^{2\pi i z}$, the holomorphic structure $\bar\partial_z s = -2\pi i z G \otimes s$ on the smooth trivial line has unit-metric Chern connection and curvature

$$\nabla_z = d - 2\pi i(zG + \bar z\,\bar G), \qquad F_{\nabla_z} = -2\pi i(z\,\partial G + \bar z\,\bar\partial \bar G).$$

With the convention $c_{1,BC}(L,h) = [\, iF_{\nabla^h}/(2\pi)\,]_{BC}$, one obtains

$$\exp(12\mathfrak{g}) = f^*\mathcal{O}_B(1), \qquad c_{1,BC}(\exp(z\mathfrak{g})) = z\mathsf{A} + \bar z\,\mathsf{B},$$

and hence

$$c_{1,BC}(f^*\mathcal{O}_B(1)) = 12(\mathsf{A} + \mathsf{B}).$$

With the opposite primitive orientation both occurrences of 12 acquire the common sign -1:

$$\exp(-12\mathfrak{g}) = f^*\mathcal{O}_B(1), \qquad c_{1,BC}(f^*\mathcal{O}_B(1)) = -12(\mathsf{A} + \mathsf{B}).$$

Thus the canonical integral Picard–Leray datum fixes the absolute factor 12; it does not assert that A, B are integral Bott–Chern classes.

Put

$$\mathbf{1} = [1]_{BC}, \qquad \mathsf{T} = [\nu_X]_{BC}, \quad \nu_X \text{ a smooth positive } (3,3)\text{-form with } \int_X \nu_X = 1.$$

Together with the unit, the following nine positive-degree classes form bases of all positive-degree Bott–Chern groups, with T the normalized top class:

$$\mathsf{A} = [\partial G]_{BC},\ \mathsf{B} = [\bar\partial \bar G]_{BC},\ \mathsf{C} = [\partial Q]_{BC},\ \overline{\mathsf{C}},\ \mathsf{D} = [\partial(G \wedge Q)]_{BC},\ \overline{\mathsf{D}},\ \mathsf{E} = [\partial(Q^2)]_{BC},\ \overline{\mathsf{E}},\ \mathsf{T}.$$

If N denotes their span, then

$$H^{*,*}_{BC}(X) = \mathbb{C}\mathbf{1} \oplus N, \qquad N^2 = 0.$$

For the last one-dimensional generator, $[Q^2]_{\bar\partial}$ has nonzero fibre restriction, whereas every class in $\mathrm{im}(d_1 : H^{1,2}_{\bar\partial} \to H^{2,2}_{\bar\partial})$ restricts to zero. Exactness therefore gives

$$d_1[Q^2]_{\bar\partial} = [\partial(Q^2)]_{\bar\partial} \neq 0,$$

which certifies E (and conjugation certifies $\overline{\mathsf{E}}$). Equivalently every product of two positive-degree classes is zero. Put $a = \partial G$, $a^\dagger = \bar\partial \bar G$, and $c = \partial Q$. The exact primitive

$$\partial\bar\partial(G \wedge \bar G) = -a \wedge a^\dagger$$

gives $\mathsf{AB} = 0$. Represent $c_{1,BC}(f^*\mathcal{O}_B(1))$ by the pullback Fubini–Study Chern form; its square vanishes pointwise because B is a curve. Squaring the factor-12 identity and using $\mathsf{AB} = 0$ gives $\mathsf{A}^2 + \mathsf{B}^2 = 0$. Moreover

$$[(a^\dagger)^2]_{\bar\partial} = 0, \qquad [a^2]_{\bar\partial} = d_1[G \wedge a]_{\bar\partial} = 0.$$

Injectivity of $d_1 : H^{1,2}_{\bar\partial} \to H^{2,2}_{\bar\partial}$ then gives a form Φ with

$$G \wedge a = \bar\partial\Phi, \qquad a^2 = \partial\bar\partial\Phi,$$

so $\mathsf{A}^2 = \mathsf{B}^2 = 0$. The last below-top product requiring a pairing is disposed of by

$$\int_X a \wedge c \wedge G = -\int_X G \wedge a \wedge c = -\int_X \bar\partial\Phi \wedge c = 0,$$

together with the perfect Dolbeault–Serre pairing and the isomorphism $H^{3,2}_{BC} \to H^{3,2}_{\bar\partial}$; the Bott–Chern–Aeppli perfect pairing alone would not prove this vanishing. Every top-degree product of two positive-degree generators contains a de Rham exact factor, while

$$H^{3,3}_{BC}(X) \xrightarrow{\int_X} \mathbb{C}$$

is an isomorphism, so those products vanish as well. See theorems 57.4 and 57.5. This asserts global smooth representatives, but not a preferred literal coefficient atlas through the two finite fillings and the cusp: $[i_b^* Q]_{\bar\partial} = [Q_F]_{\bar\partial}$ does not require $i_b^* Q = Q_F$ as forms. The chosen Q cannot be real, although its fibre class is real. After fixing the distinguished $\mathfrak{g}$ and one coherent-Leray lift $\mathfrak{q}$, changing the smooth representatives and horizontal corrections changes G, Q by Dolbeault coboundaries and changes none of the displayed classes, pairings, or products. This is a candidate-specific rank-one line-bundle normalization, not a statement about arbitrary gauge theories.

16. **[VERIFIED] The positive radial cusp has an exact flat period, inverse-dual, and scalar-gap certificate.** Put

$$\mu_T = \alpha_T + i m_T, \qquad h_T = \eta_T + i H_T, \qquad b_T = \xi_T + i B_T, \qquad c_T = \xi_T - \eta_T,$$

$$L_T = T + H_T, \qquad q_T = L_T - B_T, \qquad D_T = L_T q_T + 6m_T^2.$$

Set also

$$m = \mathrm{Im}\,\mu(0), \qquad H = \mathrm{Im}\,h(0), \qquad B = \mathrm{Im}\,b(0).$$

In the ordered marked lattice and interleaved real coordinate bases,

$$P_T = \begin{pmatrix} 6\alpha_T & \eta_T & 1 & 0 \\ 6m_T & L_T & 0 & 0 \\ c_T & \alpha_T & 0 & 1 \\ -q_T & m_T & 0 & 0 \end{pmatrix}, \qquad \det P_T = -D_T, \qquad \mathrm{covol}(P_T\mathbb{Z}^4) = D_T > 0, \qquad D_{\mathrm{can},T} = -\frac{D_T}{L_T} < 0,$$

and the complete inverse transpose is

$$P_T^{-\mathsf{T}} = \begin{pmatrix} 0 & 0 & 1 & 0 \\ \dfrac{m_T}{D_T} & \dfrac{q_T}{D_T} & \dfrac{-6m_T\alpha_T - q_T\eta_T}{D_T} & \dfrac{-m_T c_T - q_T\alpha_T}{D_T} \\ 0 & 0 & 0 & 1 \\ -\dfrac{L_T}{D_T} & \dfrac{6m_T}{D_T} & \dfrac{6(L_T\alpha_T - m_T\eta_T)}{D_T} & \dfrac{L_T c_T - 6m_T\alpha_T}{D_T} \end{pmatrix}.$$

Consequently $(P_T\mathbb{Z}^4)^* = P_T^{-\mathsf{T}}\mathbb{Z}^4$. More explicitly, for $k = (r, s, n_1, n_2)^{\mathsf{t}}$, define

$$R_T(k) = r - 6\alpha_T n_1 - c_T n_2, \qquad S_T(k) = s - \eta_T n_1 - \alpha_T n_2.$$

The corresponding dual vector is

$$\left(n_1, \frac{m_T R_T(k) + q_T S_T(k)}{D_T}, n_2, \frac{-L_T R_T(k) + 6m_T S_T(k)}{D_T}\right),$$

and the scalar eigenvalue is $4\pi^2$ times its squared norm. For every sufficiently large T,

$$\lambda_1(F_T, g_T) = 4\pi^2 \min\left\{\frac{L_T^2 + m_T^2}{D_T^2}, \frac{q_T^2 + 36m_T^2}{D_T^2}\right\} = \frac{4\pi^2}{T^2}\left(1 + \frac{-2H + 2\min\{B, 0\}}{T} + O(T^{-2})\right).$$

This is the gap of the explicitly declared Euclidean quotient metric. Its unit-volume rescaling is a different operator family and obeys

$$\widehat{g}_T = D_T^{-1/2} g_T, \qquad \lambda_1(F_T, \widehat{g}_T) = \frac{4\pi^2}{T}\left(1 + \frac{-H - B/2 + 2\min\{B, 0\}}{T} + O(T^{-2})\right) \longrightarrow 0.$$

The degeneration has an exact fixed-energy density, not merely a lowest-mode estimate. Counting Fourier multiplicities by

$$N_T(E) = \#\{k \in \mathbb{Z}^4 : 4\pi^2 \|P_T^{-\mathsf{T}} k\|^2 \le E\}, \qquad R_E^2 = \frac{E}{4\pi^2},$$

one has, for every fixed $E \ge 0$,

$$N_T(E) = \pi D_T \sum_{n \in \mathbb{Z}^2} (R_E^2 - \|n\|^2)_+ + O_E(T + 1), \qquad (x)_+ = \max\{x, 0\}.$$

The sum is finite. In particular, below the first transverse threshold,

$$0 < E < 4\pi^2 \quad \Longrightarrow \quad N_T(E) = \frac{E}{4\pi} D_T + O_E(T + 1) = \frac{E}{4\pi} T^2 + O_E(T).$$

The proof writes the admissible (r, s)-indices for each fixed transverse $n \in \mathbb{Z}^2$ as the lattice points in a translated ellipse of area $\pi D_T(R_E^2 - \|n\|^2)$ and perimeter $O_E(T + 1)$, then applies the elementary planar area–perimeter lattice bound. Thus the remainder is uniform in the otherwise unrestricted real parts of the periods. See theorem 11.4.

Normalized Haar measure in the angular-coordinate direct integral is a unitary measure convention, not this metric rescaling. See theorems 11.1 to 11.4.

17. **[VERIFIED] The native $\mathbb{T}^2 \times \mathbb{R}^2$ limit and the ancillary covered $\mathbb{R}^4$ limit have different exact constructions.** The real projection $\rho(x_1, y_1, x_2, y_2) = (y_1, y_2)$ gives

$$0 \longrightarrow \mathbb{T}^2_{\mathrm{re}} \longrightarrow F_T \longrightarrow Q_T := \mathbb{R}^2 / Y_T\mathbb{Z}^2 \longrightarrow 0, \qquad Y_T = \begin{pmatrix} 6m_T & L_T \\ -q_T & m_T \end{pmatrix}, \qquad \det Y_T = D_T.$$

The quotient systole is at least $T - C$, whereas the kernel retains its two unit periods; this is the native pointed $\mathbb{T}^2 \times \mathbb{R}^2$ degeneration. Adding

$$\mathsf{S}_M = \operatorname{diag}(1, 1, M, M), \qquad t_c = v^M$$

produces a degree-M^2 relative fibre cover over $v \neq 0$, because

$$\mathsf{S}_M^{-1} M_0^M \mathsf{S}_M = M_0.$$

The cover character indexed by $k \in \mathbb{Z}^4$ has momentum

$$P_T^{-\mathsf{T}} \mathsf{S}_M^{-1} k,$$

and it is pulled back from the original fibre exactly when k_3, k_4 are divisible by M. The covering radius of this momentum lattice is $O(T^{-1} + M^{-1})$; hence $T, M \to \infty$ fills all four momentum directions and gives eventual exact Euclidean balls. This argument by itself proves only a pointed local $\mathbb{R}^4$ limit over the punctured base; the modified-action completion across the singular fibre is the distinct local Kummer construction recorded below, and neither argument alone gives a global cover of the completed threefold or an S12/S24-type globalization.

There is, however, an exact four-copy regular-locus construction. The monodromy-invariant closure of the compact cusp lattice is

$$L_\Lambda = \mathbb{Z}\langle \widehat{u}, \widehat{w}, \widehat{\delta} \rangle, \qquad \Lambda / L_\Lambda \cong \mathbb{Z}\langle \widehat{\gamma} \rangle.$$

The induced real circle quotient has exact circumference

$$\ell_T = \frac{D_T}{\sqrt{L_T^2 + m_T^2}} = T + H - B + O(T^{-1}) \longrightarrow \infty.$$

The fourfold fibre product of $X^\circ = f^{-1}(B^\circ)$ therefore has a real fibrewise-group $\mathbb{T}^4$ quotient over B° whose synchronized radial fibres converge pointed-smoothly to $\mathbb{R}^4$. Two copies are insufficient for this monodromy-invariant-kernel mechanism; four are minimal, and no finite cover of the base repairs the native two-copy quotient. The two finite fillings admit explicit simultaneous-root normalizations over Seifert targets, with degrees 27 and 64 on the reduced central fibres. The toric cusp does not: two paths converging to the same triple point retain different character values $3/4$ and $1/2$, so the literal quotient and its fourfold product have no continuous cusp extension. The Cartesian source is not S^{24}: $X^4 \simeq (S^6)^4$, while that fixed-target fourfold-product map does not factor through the topological smash sphere $X^{\wedge 4} \cong S^{24}$; Borel–Serre also rules out almost-complex structures on S^{12} and S^{24}. Thus this exact regular-locus construction answers the globalization question as far as this mechanism can go and proves its compact-extension obstruction. See theorems 11.6, 11.7 and 11.44.

18. **[VERIFIED] The degree-N^2 regular-fibre cover has an exact completed local toric Kummer model, with direct and relative degrees kept distinct.** Write

$$Z(s) = sB_0 + C(t), \qquad B_0 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad t = e^{2\pi i s},$$

and let the target lattice action on the toric variety $Y = Y_\Sigma$ be

$$\Psi_{\bar{\lambda}}(x, t) = (e^{2\pi i C(t)\bar{\lambda}} t^{B_0 \bar{\lambda}} x, t), \qquad N_0 = Y / \bar{\Lambda}.$$

After the base change $t = v^N$, the correct source factor and action are

$$C_N(v) = \frac{1}{N} C(v^N), \qquad \Psi_{\bar{\lambda}}^{(N)}(\xi, v) = (e^{2\pi i C_N(v)\bar{\lambda}} v^{B_0 \bar{\lambda}} \xi, v).$$

They extend freely and properly discontinuously to a copy $Y^{(N)}$ of the same toric variety. Thus

$$N_0^{\langle N\rangle} = Y^{(N)}/\bar{\Lambda} \longrightarrow \Delta_v$$

is a proper holomorphic map from a smooth threefold, with reduced irreducible nonnormal central fibre W_N, whose normalization is dP_6. The toric power map

$$P_N(\xi_1, \xi_2, v) = (\xi_1^N, \xi_2^N, v^N)$$

is finite flat of degree N^3, is equivariant for the two displayed actions, and descends to

$$\kappa_N : N_0^{\langle N\rangle} \longrightarrow N_0, \qquad \deg \kappa_N = N^3, \qquad \kappa_N^* W = N W_N, \qquad \mathrm{Ram}(\kappa_N) = (N-1)W_N.$$

If $N_0[N] = N_0 \times_{\Delta_t} \Delta_v$, then $N_0[N]$ is normal and

$$N_0^{\langle N\rangle} \xrightarrow{Q_N} N_0[N] \longrightarrow N_0$$

is the unique relative factorization. The generic degree of Q_N is N^2; for $N > 1$ it is quasi-etale but not flat, with non-etale and nonflat loci exactly over the double and triple strata of the base-changed central fibre. It is not a normalization. On reduced central fibres the map has generic degree N^2, and on normalizations it is the toric map

$$[N] : \mathrm{dP}_6 \longrightarrow \mathrm{dP}_6,$$

of degree N^2, ramified with index N along all six toric boundary curves and respecting the paired conductor curves and the two triple-point classes. Over $v \neq 0$, it is exactly the lattice cover

$$\mathbb{Z}\langle \widehat{\gamma}, \widehat{u}, N\widehat{w}, N\widehat{\delta}\rangle \subset \Lambda, \qquad \mathrm{Deck}(Q_N) \cong (\mathbb{Z}/N\mathbb{Z})^2, \qquad S_N^{-1} M_0^N S_N = M_0,$$

where $S_N = \mathrm{diag}(1, 1, N, N)$. The factor $C(v^N)/N$ is essential: reusing the target action would require the generally false equality $e^{2\pi i C(v)\bar{\lambda} N} = e^{2\pi i C(v^N)\bar{\lambda}}$. This is a morphism of completed *local* cusp families. For $N > 1$ the cover lattice is not preserved by the full monodromy, so it gives neither a cover over all of B nor a product- or smash-sphere map. See theorems 11.8 to 11.10.

19. **[VERIFIED] The maps q_2, $q_{1,\mathrm{cusp}}$, and q_1 give exact cone-smash carriers with sources S^{12} and S^{24}, and their intrinsic metrics, ambient metrics, homology, and limitations are all explicit.** Let

$$K_\Lambda = \mathbb{Z}\langle \widehat{w}, \widehat{\delta}\rangle, \qquad \overline{\Lambda} = \Lambda/K_\Lambda, \qquad \overline{T}^2 = (\overline{\Lambda} \otimes \mathbb{R})/\overline{\Lambda}.$$

Fix $C_{\mathrm{sys}} > 0$ so that, for all sufficiently large T,

$$Y_T = T \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} + E_T, \qquad \|E_T\|_{\mathrm{op}} \leq C_{\mathrm{sys}}.$$

Since

$$\mathrm{im}(M_0 - I) = K_\Lambda = \ker(M_0 - I),$$

the punctured cusp has a fibrewise group epimorphism

$$\overline{\pi} : N_0 \setminus W \longrightarrow D_0^* \times \overline{T}^2$$

with connected two-torus kernel and trivial target local system. On the radial fibre, $\overline{T}^2$ is isometric to $Q_T = \mathbb{R}^2/Y_T\mathbb{Z}^2$ and has systole at least $T - C_{\mathrm{sys}}$.

Put

$$\mathcal{A} = \operatorname{Susp}(S^1 \times \overline{T}^2) \cong \operatorname{Susp}(T^3).$$

For $t(f(x)) = re^{i\phi}$, the continuous surjective quotient

$$q_2 : (X, x_0) \longrightarrow (\mathcal{A}, v_0), \qquad q_2(x) = [r, e^{i\phi}, \overline{\pi}_f(x)]$$

on $N_0 \setminus W$, where $\overline{\pi}_f = \operatorname{pr}_{\overline{T}^2} \circ \overline{\pi}$, sends W to v_0 and $X \setminus \operatorname{int} N_0$ to the other suspension vertex v_1. Its smash square

$$q_2^{\wedge 2} : X \wedge X \cong S^{12} \longrightarrow \mathcal{A} \wedge \mathcal{A}$$

restricts on synchronized radial fibres to

$$\pi_T \times \pi_T : F_T^2 \longrightarrow Q_T^2, \qquad (Q_T^2, 0) \longrightarrow (\mathbb{R}^4, 0),$$

where the convergence is pointed local-isometric, pointed smooth, and pointed Gromov–Hausdorff, and every fixed radius ball is eventually exactly Euclidean. The nonzero reduced integral homology groups are

$$\widetilde{H}_k(\mathcal{A}) = \begin{cases} \mathbb{Z}^3, & k = 2, 3, \\ \mathbb{Z}, & k = 4, \end{cases} \qquad \widetilde{H}_k(\mathcal{A}^{\wedge 2}) = \begin{cases} \mathbb{Z}^9, & k = 4, \\ \mathbb{Z}^{18}, & k = 5, \\ \mathbb{Z}^{15}, & k = 6, \\ \mathbb{Z}^6, & k = 7, \\ \mathbb{Z}, & k = 8. \end{cases}$$

More precisely,

$$\mathcal{A}^{\wedge 2} \simeq \bigvee^{9} S^4 \vee \bigvee^{18} S^5 \vee \bigvee^{15} S^6 \vee \bigvee^{6} S^7 \vee S^8.$$

The cusp-supported circle version uses

$$\mathcal{B} = \operatorname{Susp}(S^1 \times S^1) = \operatorname{Susp}(T^2)$$

and the continuous surjective based quotient

$$q_{1,\mathrm{cusp}} : (X, x_0) \longrightarrow (\mathcal{B}, b_0), \qquad q_{1,\mathrm{cusp}}(x) = [r, e^{i\phi}, \chi_\gamma(x)], \qquad q_{1,\mathrm{cusp}}(W) = b_0,$$

with the exterior collapsed to b_1. Its fourth smash power has source

$$q_{1,\mathrm{cusp}}^{\wedge 4} : X^{\wedge 4} \cong S^{24} \longrightarrow \mathcal{B}^{\wedge 4}$$

and restricts to $\chi_{\gamma,T}^{\times 4} : F_T^4 \to C_T^4$. Here

$$\widetilde{H}_k(\mathcal{B}) = \begin{cases} \mathbb{Z}^2, & k = 2, \\ \mathbb{Z}, & k = 3, \end{cases} \qquad \widetilde{H}_k(\mathcal{B}^{\wedge 4}) = \begin{cases} \mathbb{Z}^{16}, & k = 8, \\ \mathbb{Z}^{32}, & k = 9, \\ \mathbb{Z}^{24}, & k = 10, \\ \mathbb{Z}^8, & k = 11, \\ \mathbb{Z}, & k = 12. \end{cases}$$

For the full-base version, glue the regular circle target and the two finite Seifert targets to obtain the compact three-manifold with boundary

$$\mathcal{M} = (P \times S^1) \cup C_1 \cup C_2, \qquad \partial\mathcal{M} = \partial D_0 \times S^1,$$

and cone its boundary:

$$\mathcal{C} = \mathcal{M} \cup_{\partial\mathcal{M}} \mathrm{Cone}(\partial\mathcal{M}).$$

The map $q_1 : X \to \mathcal{C}$ is (f, θ_γ) over P, equivalently $\chi_\gamma|_{f^{-1}(P)}$, is the finite-filling map $N_j \to C_j$ for $j = 1, 2$, and on the cusp is

$$q_1(x) = [r, (e^{i\phi}, \theta_\gamma(x))], \qquad q_1(W) = c_0.$$

It is continuous, surjective, and based. Its fourth smash power is

$$q_1^{\wedge 4} : X^{\wedge 4} \cong S^{24} \longrightarrow \mathcal{C}^{\wedge 4},$$

and its synchronized radial restriction is again $\chi_{\gamma,T}^{\times 4} : F_T^4 \to C_T^4$, with

$$\mathrm{sys}(C_T^4) = \ell_T = T + H - B + O(T^{-1}), \qquad (C_T^4, 0) \longrightarrow (\mathbb{R}^4, 0).$$

The target homology is

$$\widetilde{H}_k(\mathcal{C}) = \begin{cases} \mathbb{Z}, & k = 2, 3, \end{cases} \quad \widetilde{H}_k(\mathcal{C}^{\wedge 4}) = \begin{cases} \mathbb{Z}, & k = 8, 12, \\ \mathbb{Z}^4, & k = 9, 11, \\ \mathbb{Z}^6, & k = 10. \end{cases}$$

In fact,

$$\mathcal{C}^{\wedge 4} \simeq S^8 \vee \bigvee^4 S^9 \vee \bigvee^6 S^{10} \vee \bigvee^4 S^{11} \vee S^{12}.$$

All omitted degrees in these three homology displays are zero. The conical-target maps $q_2^{\wedge 2}$ and $q_1^{\wedge 4}$ induce zero on all positive-degree integral homology and cohomology and have zero ordinary Hurewicz image; neither has a continuous homotopy section. Their unstable homotopy classes remain **[OPEN]**, because those facts do not decide nullity. The class of the cusp-supported conical-target map $q_{1,\mathrm{cusp}}^{\wedge 4} : S^{24} \to \mathcal{B}^{\wedge 4}$ has likewise not been computed.

The two-copy and four-copy exponents are sharp for the unscaled quotient norm induced by the marked Euclidean fibre metric. If a primitive lattice $N_{\mathrm{quot}} \subset \Lambda^{\oplus r}$ defines a connected quotient over the punctured cusp whose radial systole tends to infinity, then every bounded-length class must vanish, so

$$K_\Lambda^{\oplus r} \subset N_{\mathrm{quot}}, \qquad \mathrm{rank}(\Lambda^{\oplus r}/N_{\mathrm{quot}}) \le 2r.$$

Thus a rank-four quotient needs at least two copies, and equality is attained by $(\Lambda/K_\Lambda)^{\oplus 2}$. Over the full regular base, including after a finite base cover, monodromy invariance and primitivity additionally force $\widehat{u} \in N_{\mathrm{quot}}$: indeed $A_1\widehat{w} = \widehat{u} - \widehat{w}$, while a finite-index subgroup contains some power G_*^m satisfying $G_*^m\widehat{w} = \widehat{w} + m(\widehat{u} - \widehat{\delta})$. Consequently

$$\mathrm{rank}(\Lambda^{\oplus r}/N_{\mathrm{quot}}) \le r,$$

so four copies are necessary and sufficient there. This proves the exact minimality of the S^{12} cusp-local and S^{24} full-base smash sources within this native connected torus-quotient class; it asserts no minimality among arbitrary continuous quotient maps or after changing to a T-dependent rescaling. See theorem 11.19.

There are also sphere targets, not merely conical targets. Let

$$K^{(1)} = ((\mathbb{R}/\mathbb{Z}) \times \{0\}) \cup (\{0\} \times (\mathbb{R}/\mathbb{Z})), \qquad c : \overline{T}^2 \twoheadrightarrow \overline{T}^2/K^{(1)} \cong S^2,$$

where c collapses the coordinate one-skeleton. Forgetting the cusp base angle in the reduced suspensions gives continuous surjections

$$\widehat{p}_2 : \mathcal{A} \twoheadrightarrow S^3, \quad [r, e^{i\phi}, u] \longmapsto [r, c(u)], \qquad \widehat{p}_1 : \mathcal{B} \twoheadrightarrow S^2, \quad [r, e^{i\phi}, \theta] \longmapsto [r, \theta].$$

Thus $\widehat{q}_3 = \widehat{p}_2 \circ q_2$ and $\widehat{q}_\gamma = \widehat{p}_1 \circ q_{1,\mathrm{cusp}}$ induce, after the chosen based diffeomorphism $X \cong_{\mathrm{diff}} S^6$, quotient maps

$$\widehat{q}_3^{\wedge 2} : S^{12} \twoheadrightarrow S^6, \qquad \widehat{q}_\gamma^{\wedge 4} : S^{24} \twoheadrightarrow S^8.$$

Both full sphere carriers are based null-homotopic, by two distinct exact calculations. First let $f : S^6 \to S^2$ be any based map. The factorization

$$S^{12} = S^6 \wedge S^6 \xrightarrow{f \wedge \mathrm{id}_{S^6} = \Sigma^6 f} S^2 \wedge S^6 = S^8 \xrightarrow{\mathrm{id}_{S^2} \wedge f} S^2 \wedge S^2 = S^4$$

and $\pi_{12}(S^8) \cong \pi_4^{\mathrm{S}} = 0$ show that $f \wedge f$ is based null-homotopic. Associativity gives $f^{\wedge r} = (f \wedge f) \wedge f^{\wedge(r-2)}$, so

$$[f^{\wedge r}] = 0 \in \pi_{6r}(S^{2r}) \qquad (r \geq 2).$$

Taking $f = \widehat{q}_\gamma$ proves, in particular,

$$[\widehat{q}_\gamma^{\wedge 2}] = 0 \in \pi_{12}(S^4), \qquad [\widehat{q}_\gamma^{\wedge 4}] = 0 \in \pi_{24}(S^8).$$

Thus the full surjective quotient $S^{24} \twoheadrightarrow S^8$ is based null-homotopic.

For the other carrier, let $g : S^6 \to S^3$ be any based map. Its exact smash factorization is

$$g \wedge g = (\Sigma^3 g) \circ (\Sigma^6 g) : S^{12} \longrightarrow S^9 \longrightarrow S^6.$$

Writing st for stabilization, the low-stem data are

$$\pi_6(S^3) \cong \mathbb{Z}/12, \quad \pi_3^{\mathrm{S}} = \mathbb{Z}/24\langle\nu\rangle, \quad \pi_6^{\mathrm{S}} = \mathbb{Z}/2\langle\nu^2\rangle,$$

$$\pi_{12}(S^6) = \mathbb{Z}/2\langle\nu_6 \circ \nu_9\rangle, \qquad \mathrm{st} : \pi_{12}(S^6) \xrightarrow{\cong} \pi_6^{\mathrm{S}}.$$

The full groups are tabulated in [10, p. 339]; the signed smash formula and the unstable stabilization calculation are [13, Propositions 3.1 and 5.11]. Because $12[g] = 0$, its stable image is $\mathrm{st}([g]) = 2a\nu$ for some $a \in \mathbb{Z}/12$. Stable smash multiplication gives

$$\mathrm{st}([g \wedge g]) = \mathrm{st}([g])^2 = 4a^2\nu^2 = 0 \in \pi_6^{\mathrm{S}}.$$

The displayed stabilization isomorphism forces $[g \wedge g] = 0 \in \pi_{12}(S^6)$. Associativity again shows that $g^{\wedge r}$ is based null-homotopic for every $r \geq 2$. Taking $g = \widehat{q}_3$ therefore proves

$$[\widehat{q}_3^{\wedge 2}] = 0 \in \pi_{12}(S^6), \qquad [\widehat{q}_\gamma^{\wedge 4}] = 0 \in \pi_{24}(S^8).$$

Both maps retain their point-set surjectivity and quotient-map status. The one-copy class $[\widehat{q}_3] \in \pi_6(S^3)$ is now zero by Theorem 30.49, while $[\widehat{q}_\gamma] \in \pi_6(S^2)$ remains open. Neither full-map null theorem settles any of the conical-target classes named above; see theorems 11.21, 11.22, 11.25 and 11.27. Give $S_T^2 = \overline{T}^2/K^{(1)}$ the quotient length metric transported from Q_T, and give both smash products the quotient length metrics from the Riemannian ℓ^2-product metrics on Q_T^2 and C_T^4. These moving-slice metrics are assigned intrinsically and are not restrictions of one fixed ambient sphere metric. The targets contain moving topologically embedded four-spheres

$$\mathbb{S}_{12,T}^4 = S_T^2 \wedge S_T^2 \subset S^6, \qquad \mathbb{S}_{24,T}^4 = C_T^{\wedge 4} \subset S^8,$$

and the synchronized radial restrictions are exactly

$$F_T^2 \xrightarrow{\pi_T^{\times 2}} Q_T^2 \xrightarrow{c_T^{\times 2}} (S_T^2)^2 \longrightarrow \mathbb{S}^4_{12,T}, \qquad F_T^4 \xrightarrow{\chi_{\gamma,T}^{\times 4}} C_T^4 \longrightarrow \mathbb{S}^4_{24,T},$$

where the last arrows collapse the corresponding fat wedges. The full maps from F_T^2 and F_T^4 retain the nontrivial kernels of their first arrows; at the quotient stage, precisely the displayed skeleton/fat-wedge loci are collapsed. At the midpoint basepoints, every radius-R ball is exactly a Euclidean four-ball once $T - C_{\mathrm{sys}} > 4R$, respectively $\ell_T > 4R$. Hence both moving four-spheres converge to $\mathbb{R}^4$ in pointed local-isometric and pointed Gromov–Hausdorff topology; on those eventual Euclidean basepoint neighborhoods the convergence is also smooth. Each moving four-sphere is null-homotopic in its ambient sphere by sliding its suspension coordinate to an endpoint, so its fundamental class dies in the target. See theorems 11.20 and 11.24. Removing the smash basepoint gives exact homeomorphisms

$$(Q_T \setminus K_T^{(1)})^2 \cong \mathbb{S}^4_{12,T} \setminus \{*\} \cong \mathbb{R}^4 \hookrightarrow S^6, \qquad (C_T \setminus \{0\})^4 \cong \mathbb{S}^4_{24,T} \setminus \{*\} \cong \mathbb{R}^4 \hookrightarrow S^8.$$

These are nonclosed topological open cells, not globally Euclidean-isometric subspaces for a fixed compatible compact-target metric; see theorem 11.23. Both sphere-to-sphere quotient maps induce zero on every positive-degree integral homology group, and both full maps are based null-homotopic:

$$[\widehat{q}_3^{\wedge 2}] = 0 \in \pi_{12}(S^6), \qquad [\widehat{q}_\gamma^{\wedge 4}] = 0 \in \pi_{24}(S^8).$$

The one-copy map $\widehat{q}_3$ is now null-homotopic by Theorem 30.49; the one-copy class of $\widehat{q}_\gamma$ remains open. These full-map statements are distinct from the two explicit null-homotopies of the moving four-sphere inclusions.

For completeness, the radial target-slice embeddings used below are typed as follows. Put $r_T = e^{-2\pi T}$ and

$$\vartheta_T : \mathbb{R}/\mathbb{Z} \xrightarrow{\cong} C_T, \qquad [a] \longmapsto [\ell_T a].$$

Then

$$j_T : Q_T^2 \hookrightarrow \mathcal{A}^{\wedge 2}, \qquad j_T(y_1, y_2) = \left[[r_T, 1, \iota_T^{-1}(y_1)], [r_T, 1, \iota_T^{-1}(y_2)]\right]_\wedge,$$
$$k_T^{\mathcal{B}} : C_T^4 \hookrightarrow \mathcal{B}^{\wedge 4}, \qquad k_T^{\mathcal{B}}(c_1, \dots, c_4) = \left[[r_T, 1, \vartheta_T^{-1}(c_1)], \dots, [r_T, 1, \vartheta_T^{-1}(c_4)]\right]_\wedge,$$

and, in the cusp-cone piece of $\mathcal{C}$,

$$k_T^{\mathcal{C}} : C_T^4 \hookrightarrow \mathcal{C}^{\wedge 4}, \qquad k_T^{\mathcal{C}}(c_1, \dots, c_4) = \bigwedge_{j=1}^{4} [r_T, (1, \vartheta_T^{-1}(c_j))].$$

These are the embeddings of theorem 11.15.

The intrinsic decompactification can be made into an actual boundary family. Fix T_0 sufficiently large, put $r_0 = e^{-2\pi T_0}$, and define the compact radial carriers

$$\mathcal{D}_{12} = \frac{[0, r_0] \times (\overline{T}^2)^2}{\{0\} \times (\overline{T}^2)^2 \sim d_{12}}, \qquad \mathcal{D}_{24} = \frac{[0, r_0] \times (S^1)^4}{\{0\} \times (S^1)^4 \sim d_{24}}.$$

Both are homeomorphic to $\mathrm{Cone}(\mathbb{T}^4)$. At either cone vertex d_ν,

$$H_k(\mathcal{D}_\nu, \mathcal{D}_\nu \setminus \{d_\nu\}; \mathbb{Z}) \cong \begin{cases} \mathbb{Z}^4, & k = 2, \\ \mathbb{Z}^6, & k = 3, \\ \mathbb{Z}^4, & k = 4, \\ \mathbb{Z}, & k = 5, \\ 0, & \text{otherwise}, \end{cases}$$

so d_ν is not a topological-manifold point, with or without boundary, of any dimension. Set

$$A_T^{(12)} = \operatorname{diag}(Y_T, Y_T), \qquad A_T^{(24)} = \ell_T I_4, \qquad s_0 = T_0^{-1},$$

and, for $\nu \in \{12, 24\}$, quotient $[0, s_0] \times \mathbb{R}^4$ fibrewise by

$$(0, x) \sim_\nu (0, y) \iff x = y, \qquad (s, x) \sim_\nu (s, y) \iff x - y \in A_{1/s}^{(\nu)} \mathbb{Z}^4 \quad (s > 0).$$

The resulting $\mathfrak{E}_\nu$ is a Hausdorff, second-countable topological five-manifold with boundary, with

$$p_{12}^{-1}(s) \cong Q_{1/s}^2, \qquad p_{24}^{-1}(s) \cong C_{1/s}^4 \quad (s > 0), \qquad p_\nu^{-1}(0) \cong \mathbb{R}^4.$$

Its other boundary component, over $s = s_0$, is a compact four-torus. Every fixed Euclidean ball embeds isometrically in all sufficiently small positive fibres. There is a continuous surjective comparison map $\beta_\nu : \mathfrak{E}_\nu \to \mathcal{D}_\nu$ which is a homeomorphism away from the zero fibre, identifies each positive fibre with its radial torus slice, and collapses the entire zero-boundary component $p_\nu^{-1}(0) \cong \mathbb{R}^4$ to the vertex of the compact $\mathcal{D}_\nu \cong \operatorname{Cone}(\mathbb{T}^4)$. Thus the Euclidean space is an exact decompactification boundary, not the whole compact source or target and not a continuous quotient target of a compact source. The map β_ν is neither proper nor a quotient map, so this is not a topological blow-down or a quotient identification $\mathfrak{E}_\nu / \mathbb{R}^4 \cong \mathcal{D}_\nu$. Explicitly, for $a = (1/2, 1/2, 1/2, 1/2)$ and $s_j \downarrow 0$, the points

$$z_j = [s_j, A_{1/s_j}^{(\nu)} a]$$

form a closed subset of $\mathfrak{E}_\nu$. Indeed, in the S^{12} case

$$\inf_{n \in \mathbb{Z}^4} \|A_{1/s_j}^{(12)}(a + n)\| \geq (1/s_j - C_{\mathrm{sys}}) \operatorname{dist}(a, \mathbb{Z}^4) \to \infty,$$

and in the S^{24} case the same infimum equals $\ell_{1/s_j} \operatorname{dist}(a, \mathbb{Z}^4) \to \infty$; hence the sequence has no accumulation in a positive fibre and eventually misses every half-space chart about every zero-fibre point. On the other hand, $\beta_\nu(z_j) = [e^{-2\pi/s_j}, a] \to d_\nu$. Hence the image set is not closed although its inverse image is closed, which proves that β_ν is not quotient; see theorems 11.28 and 11.29.

Unrescaled ambient metrics impose a different conclusion. Put

$$Z_{12} := \mathcal{A}^{\wedge 2}, \qquad Z_{24}^{\mathcal{B}} := \mathcal{B}^{\wedge 4}, \qquad Z_{24}^{\mathcal{C}} := \mathcal{C}^{\wedge 4}.$$

For every fixed compatible metric on these compact targets,

$$\operatorname{diam}_{\mathrm{amb}} j_T(Q_T^2) \to 0, \qquad \operatorname{diam}_{\mathrm{amb}} k_T^{\mathcal{B}}(C_T^4) \to 0, \qquad \operatorname{diam}_{\mathrm{amb}} k_T^{\mathcal{C}}(C_T^4) \to 0,$$

whereas intrinsically

$$\operatorname{diam}(Q_T^2) \geq \tfrac{1}{2}(T - C_{\mathrm{sys}}) \to \infty, \qquad \operatorname{diam}(C_T^4) \geq \tfrac{1}{2}\ell_T \to \infty.$$

There are fixed cusp-adapted compatible length metrics with regular-stratum metric tensors $g_{Z_{12}}$ and $g_{Z_{24}^{\mathcal{Q}}}$, $\mathcal{Q} \in \{\mathcal{B}, \mathcal{C}\}$, for which the exact rescaled first fundamental forms are

$$\left(\frac{T}{r_T}\right)^2 j_T^* g_{Z_{12}} = g_{Q_T} \oplus g_{Q_T}, \qquad \left(\frac{\ell_T}{r_T}\right)^2 (k_T^{\mathcal{Q}})^* g_{Z_{24}^{\mathcal{Q}}} = g_{C_T}^{\oplus 4}, \quad \mathcal{Q} \in \{\mathcal{B}, \mathcal{C}\}.$$

With their induced Riemannian path metrics (not the restricted ambient distance functions), the rescaled slices converge to $\mathbb{R}^4$, but the corresponding full ambient pointed blow-ups converge on their eventual regular-stratum balls to $\mathbb{R}^8$ and $\mathbb{R}^{12}$, respectively. In every rescaling that unrolls a slice to $\mathbb{R}^4$, the cone/fat-wedge basepoint recedes to infinite scaled distance; vertex-based conic tangents

have dimensions 8 and 12 for exact conic target metrics, and unit-volume normalization gives a flat four-torus, not $\mathbb{R}^4$. See theorems 11.36 to 11.40.

Thus coning changes the codomain and does not contradict the proved nonextension of the literal character map to the fixed target $B \times S^1$. Nor do Cartesian powers give spheres:

$$X^2 \simeq S^6 \times S^6 \not\simeq S^{12}, \qquad X^4 \simeq (S^6)^4 \not\simeq S^{24};$$

only the based smash powers are spheres. There is no full-base two-copy four-torus quotient, no fixed T^4 target retaining the four character classes, and no fixed S^4 target extending a degree-one collapse of the radial quotient. No nontrivial finite base cover preserves K_Λ, and the local unspooling cover is not a nontrivial connected unbranched cover of the simply connected compact sources. Finally, S^{12} and S^{24} admit no almost-complex structure, so the smash sources are not compact complex spheres. See theorems 11.12 to 11.14, 11.16, 11.18, 11.26 and 11.27. Every construction in this sphere-route item uses the native quotients of the original, unscaled cusp fibres; none uses the ancillary finite-cover unspooling from the preceding item.

20. **[VERIFIED] The repaired compact-link construction and its finite Gaussian quantization have exact actions, spectra, and a resolved zero-gap diagonal sequence.** Keep the cover index and the mesh index independent:

$$M_{\mathrm{cov}} \in \mathbb{Z}_{\geq 1}, \qquad S_{M_{\mathrm{cov}}} = \mathrm{diag}(1, 1, M_{\mathrm{cov}}, M_{\mathrm{cov}}), \qquad P_{T,M_{\mathrm{cov}}} = P_T S_{M_{\mathrm{cov}}}, \qquad N_{\mathrm{mesh}} \in \mathbb{Z}_{\geq 2}.$$

Then

$$\widetilde{F}_{T,M_{\mathrm{cov}}} = \mathbb{R}^4 / P_{T,M_{\mathrm{cov}}} \mathbb{Z}^4 \longrightarrow F_T$$

has degree M_{cov}^2, Gram matrix and covolume

$$G_{T,M_{\mathrm{cov}}} = P_{T,M_{\mathrm{cov}}}^{\mathsf{T}} P_{T,M_{\mathrm{cov}}}, \qquad \mathcal{D}_{T,M_{\mathrm{cov}}} = \sqrt{\det G_{T,M_{\mathrm{cov}}}} = M_{\mathrm{cov}}^2 D_T.$$

Put

$$V_{N_{\mathrm{mesh}}} = (\mathbb{Z}/N_{\mathrm{mesh}}\mathbb{Z})^4, \qquad E_{N_{\mathrm{mesh}}} = V_{N_{\mathrm{mesh}}} \times \{1, 2, 3, 4\}.$$

Transport of the standard N_{mesh}-cubulation by $[y] \mapsto [P_{T,M_{\mathrm{cov}}} y]$ is an honest cubulation of the whole cover torus; its vertex set is exactly the N_{mesh}-torsion subgroup, via

$$[n] \longmapsto \left[P_{T,M_{\mathrm{cov}}} n / N_{\mathrm{mesh}}\right].$$

Parallel transport along its oriented edges gives the typed gauge-equivariant holonomy map to $\mathrm{SU}(2)^{E_{N_{\mathrm{mesh}}}}$.

Write $M = M_{\mathrm{cov}}$ and $N = N_{\mathrm{mesh}}$ only within the following compact-link formulas. Normalize

$$\langle X, Y \rangle_{\mathfrak{su}(2)} = -\tfrac{1}{2} \operatorname{tr}(XY), \qquad H_{\mathrm{link}} = \mathrm{diag}(i, -i), \qquad \|H_{\mathrm{link}}\|^2 = 1, \qquad \exp(2\pi H_{\mathrm{link}}) = I_2.$$

The periodic inverse-transport links

$$\begin{aligned} U_1(n) &= \exp(-2\pi n_2 H_{\mathrm{link}} / N^2), \\ U_2(n) &= \begin{cases} I_2, & 0 \leq n_2 \leq N-2, \\ \exp(2\pi n_1 H_{\mathrm{link}} / N), & n_2 = N-1, \end{cases} \\ U_3(n) &= U_4(n) = I_2. \end{aligned}$$

have, including every wrap plaquette,

$$U_{12}(n) = \exp(2\pi H_{\mathrm{link}} / N^2), \qquad U_{ij}(n) = I_2 \quad ((i,j) \neq (1,2),\ i < j),$$

The principal-logarithm action and the explicit continuum-matched $(1,2)$-Wilson sector are exactly

$$S^{\log}_{T,M,N}(U) = \frac{2\pi^2 M^2}{g^2_{\mathrm{YM}} D_T},$$

$$S^{W,12}_{T,M,N}(U) = \frac{M^2 N^4}{2 g^2_{\mathrm{YM}} D_T}\left(2 - 2\cos\frac{2\pi}{N^2}\right) = \frac{M^2}{g^2_{\mathrm{YM}} D_T}\left(2\pi^2 - \frac{2\pi^4}{3N^4} + \frac{4\pi^6}{45N^8} + O(N^{-12})\right).$$

The inverse-Gram two-form minor producing these formulas is exactly

$$(G^{-1}_{T,M_{\mathrm{cov}}})_{11}(G^{-1}_{T,M_{\mathrm{cov}}})_{22} - (G^{-1}_{T,M_{\mathrm{cov}}})^2_{12} = D_T^{-2}.$$

For $0 \le p \le 4$, define the real cochain spaces

$$C^p_N = \mathrm{Map}\Big(V_{N_{\mathrm{mesh}}}, \bigwedge\nolimits^p (\mathbb{R}^4)^* \otimes \mathfrak{su}(2)\Big).$$

For $\vartheta^j_{T,M} = P^{-\mathsf{T}}_{T,M}\mathbf{e}_j$, let $\mathsf{T}_j a(n) = a(n+\mathbf{e}_j)$ and set

$$D^+_{N,j} = N(\mathsf{T}_j - I), \qquad D^-_{N,j} a(n) = N(a(n) - a(n-\mathbf{e}_j)),$$

$$d_N a = \sum_{j=1}^4 \varepsilon(\vartheta^j_{T,M}) D^+_{N,j} a, \qquad d^*_N = -\sum_{j=1}^4 \iota_{(\vartheta^j_{T,M})^\sharp} D^-_{N,j},$$

and

$$\Delta^{(1)}_N = d_N d^*_N + d^*_N d_N, \qquad \mathscr{C}_{T,M,N} = \ker(d^*_N : C^1_N \to C^0_N) \cap (\ker \Delta^{(1)}_N)^\perp.$$

This last orthogonal complement removes the zero Fourier harmonic cochains. On this real reduced Coulomb space put

$$\mathcal{L}_{T,M,N} = g^{-2}_{\mathrm{YM}} d^*_N d_N, \qquad \mathfrak{d}_N(k)_j = N(e^{2\pi i k_j/N} - 1).$$

For every $k \neq 0$, its eigenvalue and complex transverse multiplicity are

$$\lambda_{T,M,N}(k) = \frac{1}{g^2_{\mathrm{YM}}}\left\| P_T^{-\mathsf{T}} S^{-1}_{M_{\mathrm{cov}}} \mathfrak{d}_N(k)\right\|^2, \qquad 3\dim \mathfrak{su}(2) = 9.$$

In particular,

$$\lambda_{T,M,N}(\mathbf{e}_1) = \frac{4N^2 \sin^2(\pi/N)}{g^2_{\mathrm{YM}}} \frac{L_T^2 + m_T^2}{D_T^2},$$

independently of M. The Friedrichs realization on $L^2(\mathscr{C}_{T,M,N}, da)$ of

$$\widehat{H}^{\mathrm{Gauss}}_{T,M,N} = -\tfrac{1}{2}\Delta_{\mathscr{C}_{T,M,N}} + \tfrac{1}{2}\langle a, \mathcal{L}_{T,M,N} a\rangle$$

has a unique Gaussian ground state and exact gap

$$\mathrm{gap}(\widehat{H}^{\mathrm{Gauss}}_{T,M,N}) = \sqrt{\lambda_{\min}(\mathcal{L}_{T,M,N})} \le \frac{2N\sin(\pi/N)}{g_{\mathrm{YM}}} \frac{\sqrt{L_T^2 + m_T^2}}{D_T}.$$

For

$$T_j = j^2, \qquad M_j = j, \qquad N_j = j^3,$$

one has simultaneously

$$\mathrm{sys}(\widetilde{F}_{T_j,M_j}) \to \infty, \qquad \max_r \frac{\|P_{T_j,M_j}\mathbf{e}_r\|}{N_j} \to 0, \qquad \mathrm{gap}(\widehat{H}^{\mathrm{Gauss}}_{T_j,M_j,N_j}) \to 0, \qquad S^{\log}_{T_j,M_j,N_j}(U) \to 0.$$

The exponential map $a \mapsto (\exp(a_e/N))_e$ is only a local diffeomorphism at zero, has derivative $N^{-1}\,\mathrm{id}$, is not globally injective, and, if the compact-link action satisfies $DS_W(\mathbf{1}) = 0$, transports its Hessian with the exact factor

$$D^2(S_W \circ \mathrm{Exp}_N)_0 = N^{-2} D^2(S_W)_{\mathbf{1}}.$$

Consequently the displayed Gaussian Hilbert space is not the Haar L^2-space on compact links or its nonlinear gauge-invariant subspace; no unitary identification, nonlinear compact-link Hamiltonian, Gauss-law spectral problem, transfer matrix, renormalized continuum limit, or Osterwalder–Schrader reconstruction has been constructed. This theorem therefore remains strictly on the auxiliary finite Gaussian side of the physical Yang–Mills boundary. See theorems 13.1 to 13.6 and equations (402) and (403).

21. **[VERIFIED] The exceptional first-coordinate elliptic quotient is finite and does not approach the cusp.** For the marked complex torus

$$\mathcal{A}_z = \mathbb{C}^2/\mathbb{Z}\langle(6\mu, \beta), (\tau, \mu), (1, 0), (0, 1)\rangle,$$

the first coordinate descends onto $E_\tau = \mathbb{C}/(\mathbb{Z}\tau + \mathbb{Z})$ exactly on

$$6\mu = a\tau + b, \qquad (a, b) \in \mathbb{Z}^2.$$

The kernel is connected and equals $\mathbb{C}/(\mathbb{Z} + \mathbb{Z}(a\mu - \beta))$, with $\mathrm{Im}(a\mu - \beta) = -D_{\mathrm{can}} > 0$. Under the two generators the labels transform by

$$(a, b) \overset{g_0}{\longmapsto} (a, a + b), \qquad (a, b) \overset{g_2}{\longmapsto} (-b, 6 + a).$$

The marked union is therefore monodromy invariant; its image in the completed orbifold base is finite and avoids a punctured cusp neighborhood. Thus this specific elliptic quotient cannot realize the cusp gap collapse. It remains logically possible for a fibre to have a different elliptic quotient or an integral magnetic class not induced by this projection. See theorems 11.41 to 11.43.

22. **[VERIFIED] A single specified auxiliary operator has zero essential spectral threshold.** Fourier expansion identifies

$$\mathscr{H}_{\mathrm{aux}} = L^2([T_0, \infty), dT; L^2_0(\mathbb{R}^4/\mathbb{Z}^4, dy)) \cong \bigoplus_{k \in \mathbb{Z}^4 \setminus \{0\}} L^2([T_0, \infty), dT).$$

With

$$\lambda_k(T) = 4\pi^2 |P_T^{-\mathsf{T}} k|^2, \qquad (\mathscr{L}_{\mathrm{aux}} u)_k(T) = \lambda_k(T) u_k(T),$$

the exact domain is

$$D(\mathscr{L}_{\mathrm{aux}}) = \left\{ u : \sum_{k \neq 0} \int_{T_0}^{\infty} \lambda_k(T)^2 |u_k(T)|^2 \, dT < \infty \right\}.$$

The operator is nonnegative and self-adjoint, has zero kernel, and the $k = (1, 0, 0, 0)$ singular sequence supported on $[n, n + 1]$ proves

$$0 \in \sigma_{\mathrm{ess}}(\mathscr{L}_{\mathrm{aux}}).$$

The word "gapless" here belongs to this auxiliary Hilbert space and operator; it is not an invariant of the complex threefold alone. See theorems 52.1 and 52.3.

23. **[VERIFIED] The continuum and finite classical gauge fluctuation operators have exact spectra and cusp gaps.** For a compact Lie group with $d_G = \dim_{\mathbb{R}} \mathfrak{g}$, the product-connection Hessian on the reduced Coulomb space is

$$\mathcal{L}_T = g_{\mathrm{YM}}^{-2} \mathrm{d}^* \mathrm{d}, \qquad \lambda_T^{\mathrm{YM}}(k) = \frac{4\pi^2}{g_{\mathrm{YM}}^2} |P_T^{-\mathsf{T}} k|^2$$

on every $k \neq 0$ transverse Fourier fibre, with complex multiplicity $3d_G$. Hence

$$\gamma_T = \frac{4\pi^2}{g_{\mathrm{YM}}^2} \min\left\{ \frac{L_T^2 + m_T^2}{D_T^2}, \frac{q_T^2 + 36m_T^2}{D_T^2} \right\} = \frac{4\pi^2}{g_{\mathrm{YM}}^2 T^2} \left(1 + \frac{-2H + 2\min\{B, 0\}}{T} + O(T^{-2})\right).$$

For the finite cubical complex,

$$\mathfrak{d}_N(k)_j = N(e^{2\pi i k_j/N} - 1), \qquad \zeta_{N,T}(k) = P_T^{-\mathsf{T}} \mathfrak{d}_N(k), \qquad \mathcal{L}_{N,T} = g_{\mathrm{YM}}^{-2} \mathrm{d}_N^* \mathrm{d}_N,$$

and the reduced $k \neq 0$ eigenvalue is exactly

$$\lambda_{N,T}^{\mathrm{YM}}(k) = g_{\mathrm{YM}}^{-2} |P_T^{-\mathsf{T}} \mathfrak{d}_N(k)|^2$$

with the same complex multiplicity. Writing $\varrho_N^{\mathrm{spec}} = 4N^2 \sin^2(\pi/N)$, there is a T_1, independent of $N \geq 2$, such that for every $T \geq T_1$ its gap is

$$\gamma_{N,T} = \frac{\varrho_N^{\mathrm{spec}}}{g_{\mathrm{YM}}^2} \min\left\{ \frac{L_T^2 + m_T^2}{D_T^2}, \frac{q_T^2 + 36m_T^2}{D_T^2} \right\} = \frac{\varrho_N^{\mathrm{spec}}}{g_{\mathrm{YM}}^2 T^2} \left(1 + \frac{-2H + 2\min\{B, 0\}}{T} + O(T^{-2})\right)$$

uniformly for $N \geq 2$. Thus each fixed finite model is positively gapped, but $\gamma_{N(T),T} \to 0$ for every $N(T) \geq 2$; resolving the varying continuum geometry is the stronger condition $N(T)/T \to \infty$. The finite operator is a linearized Lie-algebra-valued cochain model, not a nonlinear compact-link Wilson action. See theorems 12.5, 12.6, 12.8 and 12.10.

24. **[VERIFIED] Every fixed nondegenerate integral flux on the marked four-torus has an exact magnetic-oscillator spectrum whose first spacing tends to zero along the cusp.** In angular coordinates $u \in \mathbb{R}^4/\mathbb{Z}^4$, let

$$G_T = P_T^{\mathsf{T}} P_T, \qquad \sqrt{\det G_T} = D_T, \qquad \mathsf{E} = \sum_{j<k} E_{jk}\, \mathrm{d}u_j \wedge \mathrm{d}u_k \in \bigwedge^2 (\mathbb{Z}^4)^*.$$

The factor of automorphy and connection

$$J_n^{\mathsf{E}}(u) = \exp\left(-2\pi i \sum_{j<k} E_{jk} n_j u_k\right), \qquad \nabla^{\mathsf{E}} = \mathrm{d} + 2\pi i \sum_{j<k} E_{jk} u_j\, \mathrm{d}u_k$$

define a Hermitian line bundle $\mathscr{L}_{\mathsf{E},T}$ with curvature $F_{\nabla^{\mathsf{E}}} = 2\pi i \mathsf{E}$. The nonnegative closed form

$$q_{\mathsf{E},T}[f] = \int_{[0,1]^4} (G_T^{-1})^{jk} \overline{\nabla_j^{\mathsf{E}} f} \nabla_k^{\mathsf{E}} f\, D_T\, \mathrm{d}^4 u$$

has form domain $H^1(F_T; \mathscr{L}_{\mathsf{E},T})$; its Friedrichs operator $\Delta_{\mathsf{E},T} = (\nabla^{\mathsf{E}})^* \nabla^{\mathsf{E}}$ has domain $H^2(F_T; \mathscr{L}_{\mathsf{E},T})$ and compact resolvent. For nondegenerate E, define $b_1, b_2 > 0$ by

$$\mathrm{Spec}\Big(G_T^{-1/2} (2\pi \mathsf{E}) G_T^{-1/2}\Big) = \{ib_1, -ib_1, ib_2, -ib_2\}.$$

The ladder commutators give the complete set of spectral values

$$\mathrm{Spec}(\Delta_{\mathsf{E},T}) = \{(2n_1 + 1)b_1 + (2n_2 + 1)b_2 : n_1, n_2 \in \mathbb{Z}_{\geq 0}\}, \qquad \mathrm{gap}(\Delta_{\mathsf{E},T}) = 2\min\{b_1, b_2\}.$$

Writing

$$d = |\operatorname{Pf}(\mathsf{E})| = |E_{12}E_{34} - E_{13}E_{24} + E_{14}E_{23}| > 0,$$

the determinant identity is

$$b_1 b_2 = \frac{4\pi^2 d}{D_T},$$

and therefore

$$\operatorname{gap}(\Delta_{\mathsf{E},T}) \le 4\pi\sqrt{\frac{d}{D_T}} = \frac{4\pi\sqrt{d}}{T}\left(1 + \frac{B-2H}{2T} + O(T^{-2})\right) \longrightarrow 0.$$

This is a fibrewise scalar magnetic Laplacian on the marked cusp tori. It does not assert that $[\mathsf{E}]$ is fixed by full monodromy or identify this level spacing with a physical Yang–Mills particle mass. See theorems 14.5 and 14.6.

25. **[VERIFIED] A reducible non-flat $SU(2)$ critical connection has exact action and an exact negative Jacobi eigenvalue.** The rank-two sum $\mathscr{E}_T = \mathscr{L}_T \oplus \mathscr{L}_T^{-1}$ has parallel trivial determinant, $c_1 = c_2 = 0$, and an explicitly nullhomotopic $SU(2)$ clutching map. Thus its principal $SU(2)$-bundle is smoothly trivial. In the lifted line-bundle splitting $\mathscr{L}_T \oplus \mathscr{L}_T^{-1}$, the direct-sum connection and its curvature are

$$H_{\mathrm{mag}} = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \qquad \nabla^T = \mathrm{d} + 2\pi u_1\,\mathrm{d}u_2\,H_{\mathrm{mag}}, \qquad F_{\nabla^T} = \frac{2\pi}{D_T} H_{\mathrm{mag}}\,\mathrm{d}y_1 \wedge \mathrm{d}y_2,$$

where the Lie-algebra norm is the same one fixed above, $\langle X, Y\rangle = -\frac{1}{2}\operatorname{tr}(XY)$, so that $\|H_{\mathrm{mag}}\|^2 = 1$. In any chosen global $SU(2)$-trivialization of the smoothly trivial principal bundle this matrix is replaced by its corresponding gauge transform. One has $\mathrm{d}^*_{\nabla^T} F_{\nabla^T} = 0$ and

$$\mathrm{YM}_T(\nabla) = \frac{1}{2g_{\mathrm{YM}}^2}\int_{F_T} |F_\nabla|^2\,\mathrm{d}\,\mathrm{vol}_{F_T},$$

so

$$\mathrm{YM}_T(\nabla^T) = \frac{2\pi^2}{g_{\mathrm{YM}}^2 D_T} = \frac{2\pi^2}{g_{\mathrm{YM}}^2 T^2}\left(1 + \frac{B-2H}{T} + O(T^{-2})\right).$$

Let $(e_\mu)_{\mu=1}^4$ be a constant orthonormal frame for the descended flat metric. The background-gauge Jacobi expression

$$(\mathcal{J}_T a)_\nu = -\sum_{\mu=1}^4 \nabla_{e_\mu}\nabla_{e_\mu} a_\nu - 2\sum_{\mu=1}^4 [F_{\nu\mu}, a_\mu]$$

has an explicit smooth real unit eigenvector satisfying

$$\mathrm{d}^*_{\nabla^T} a = 0, \qquad \mathcal{J}_T a = -\frac{4\pi}{D_T} a, \qquad \mathrm{YM}''_T\,|_{\nabla^T}(a,a) = -\frac{4\pi}{g_{\mathrm{YM}}^2 D_T}.$$

This proves a classical saddle direction at every finite T, not a spectral assertion for a quantum Hamiltonian. See theorems 14.3, 14.4 and 14.8.

26. **[VERIFIED] The native full-flux background extends root by root from split $SU(2)$ to split-Cartan $SU(r)$, with exact topology, action, spectrum, index, and wall nullities.** For every nonzero trace-zero $\mathbf{m} \in \mathbb{Z}^r$, put

$$\mathscr{E}_{\mathbf{m},T} = \bigoplus_i \mathscr{L}_{\eta,T}^{\otimes m_i}, \qquad s_2(\mathbf{m}) = \sum_i m_i^2, \qquad k_{\mathbf{m}} = 6 s_2(\mathbf{m}).$$

Then

$$c_2(\mathscr{E}_{\mathbf{m},T}) = -\frac{s_2(\mathbf{m})}{2}[\eta_T]^2, \quad \langle c_2, [F_T]_{\mathrm{mark}}\rangle = -6 s_2(\mathbf{m}), \quad \langle c_2, [F_T]_{x\text{-complex}}\rangle = k_{\mathbf{m}},$$

and the corresponding reducible Yang–Mills connection satisfies

$$\mathrm{YM}_T = \frac{2\pi^2 k_{\mathbf{m}}}{g_{\mathrm{YM}}^2} + \frac{\pi^2 s_2(\mathbf{m}) S_T^2}{g_{\mathrm{YM}}^2 D_T}.$$

For each positive root $i < j$, the charge is $q_{ij} = |m_i - m_j|$; the complete negative normalized Coulomb bands are

$$q_{ij}\big((2n+1)\mathcal{B}_{T,\min} - \mathcal{B}_{T,\max}\big), \qquad 0 \le n < N_-(T),$$

each contributing real dimension $12q_{ij}^2$, with coincident bands summed. Hence

$$\operatorname{ind}_{\mathbb{R}} \mathcal{J}_{\mathbf{m},T} = 12 r N_-(T) s_2(\mathbf{m}) = 36 r s_2(\mathbf{m})$$

on the deep cusp tail. At $S_T = 0$ the negative spectrum is empty but the root zero modes double, so no strict-stability conclusion is made. These are classical fibrewise theorems, not an interacting quantum spectrum; moreover their nonzero c_2 obstructs extension over every proposed sphere carrier with vanishing fourth integral cohomology. See theorems 16.1, 16.2 and 16.5 to 16.7.

27. **[VERIFIED] The Koide cone is isomorphic only to an ambient period-coefficient discriminant, not to the regular cusp family.** For

$$Q_K(r) = \frac{2}{3}(r_1 + r_2 + r_3)^2 - (r_1^2 + r_2^2 + r_3^2), \qquad \begin{pmatrix} h \\ x \\ y \end{pmatrix} = \begin{pmatrix} 1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} \\ 1/\sqrt{2} & 0 & -1/\sqrt{2} \\ 1/\sqrt{6} & -2/\sqrt{6} & 1/\sqrt{6} \end{pmatrix} \begin{pmatrix} r_1 \\ r_2 \\ r_3 \end{pmatrix},$$

the coordinate change is orthogonal and

$$Q_K(r) = h^2 - x^2 - y^2.$$

The real-linear map

$$(h, x, y) \longmapsto (t, m, \varrho) = \left(h + y, \frac{x}{\sqrt{6}}, h - y\right)$$

is an ambient coefficient-space isomorphism. For the canonical convention

$$Z = \begin{pmatrix} 6\mu & \tau \\ \beta & \mu \end{pmatrix}, \qquad P = \begin{pmatrix} 6\operatorname{Re}\mu & \operatorname{Re}\tau & 1 & 0 \\ 6m & t & 0 & 0 \\ \operatorname{Re}\beta & \operatorname{Re}\mu & 0 & 1 \\ \varrho & m & 0 & 0 \end{pmatrix},$$

one has, independently of the real parts,

$$\det P = t\varrho - 6m^2 = \det \begin{pmatrix} t & \sqrt{6}\, m \\ \sqrt{6}\, m & \varrho \end{pmatrix} = Q_K(r).$$

For $r \in \mathbb{R}^3_{\ge 0} \setminus \{0\}$, the equation $Q_K(r) = 0$ is equivalent to the displayed symmetric matrix being positive semidefinite of rank one. It also forces $\det P = 0$, so the columns of P do not form a full lattice and define no regular compact complex two-torus. Actual radial cusp fibres instead satisfy

$$\det P_T = -D_T < 0, \qquad D_T = T^2 + (2H - B)T + O(1), \qquad \det P_T \sim -T^2.$$

See theorems 30.65 to 30.67.

28. **[VERIFIED] The positive defect-one Koide shell has an exact ideal-and-unit parametrization and an unconditional power-saving counting law.** Define

$$\Delta(x,y,z) = 4(xy+xz+yz) - (x^2+y^2+z^2),$$

and put

$$\mathcal{R} = \{(x,y,z) \in \mathbb{Z}^3_{>0} : x \le y \le z,\ \Delta(x,y,z) = -1,\ z \ge 4(x+y)\},$$
$$R_{\mathrm{K}}(Y) = \#\{(x,y,z) \in \mathcal{R} : z \le Y\}.$$

The terminal Vieta wall is exact: below it, the other root $4(x+y)-z$ is positive and smaller than y. On the wall, the coordinates

$$n = z - 4(x+y), \qquad q = y - x, \qquad a = x + y - 4n$$

give a bijection with the integral chamber

$$a^2 - 3q^2 - 18n^2 = -2, \qquad a > 0, q, n \ge 0, a + 4n > q, \qquad z = 4a + 17n,$$

whose inverse is

$$(x,y,z) = \left(\frac{a+4n-q}{2}, \frac{a+4n+q}{2}, 4a+17n\right).$$

For

$$F_{\mathrm{K}}(a,q,n) = a^2 - 3q^2 - 18n^2, \qquad \mathscr{X} = \{F_{\mathrm{K}} = -2\},$$

direct Hensel counting and the good-prime formula give

$$\sigma_2 = \frac{3}{2}, \qquad \sigma_3 = 2, \qquad \sigma_p = \frac{1-p^{-2}}{1-\left(\frac{3}{p}\right)p^{-1}} \qquad (p \ge 5),$$

and hence

$$\mathfrak{S}(\mathscr{X}) = \frac{9}{2}\frac{\log(2+\sqrt{3})}{\sqrt{3}\,\zeta(2)}.$$

The exact Poincaré-residue integral over the reduced chamber is

$$I_\infty = \frac{1}{2\sqrt{3}} \log\left(\frac{3}{2\sqrt{2}}\right).$$

The Gorodnik–Oh–Shah sector theorem applies to the genuine norm $\max\{|4a+17n|, |q|, |a|\}$. An explicit element of $\mathrm{SO}(F_{\mathrm{K}}, \mathbb{Z}_2)$ with spinor norm -17 exchanges the two Borovoi sign orbits and proves $c_{\mathscr{X}} = 1$. Calibrating both counting theorems against the same Euclidean balls therefore gives

$$C_0 := \frac{9}{2\pi^2}\log(2+\sqrt{3})\log\left(\frac{3}{2\sqrt{2}}\right)$$

and

$$\boxed{R_{\mathrm{K}}(Y) \sim C_0 Y.}$$

For $n \ge 1$, the exact ideal-and-unit parametrization puts

$$m_n = 18n^2 - 2 = 2(3n-1)(3n+1),$$
$$\mathcal{P}(m_n) = \{(C,D,E) \in \mathbb{Z}^2_{>0} \times \mathbb{Z} : C^2 D = m_n, \quad 0 \le E < D, \quad E^2 \equiv 3 \pmod{D}\}.$$

The map

$$(C,D,E) \longmapsto C(D\mathbb{Z} + (E+\sqrt{3})\mathbb{Z})$$

is a bijection to norm-m_n ideals of $\mathbb{Z}[\sqrt{3}]$. Each ideal has exactly one generator $a_0 + q_0\sqrt{3}$ with $a_0 > 2q_0 \geq 0$, and its positive solutions are

$$a_k + q_k\sqrt{3} = (a_0 + q_0\sqrt{3})(2 + \sqrt{3})^k, \qquad k \geq 0.$$

The inverse descent is $(a, q) \mapsto (2a - 3q, 2q - a)$, iterated while $a \leq 2q$, followed by the unique HNF recovery. Hence multiplicity is one and

$$\#\mathcal{P}(m_n) = \rho(m_n) = \sum_{d|m_n} \chi_{12}(d) = \rho(3n-1)\rho(3n+1).$$

The $n = 0$ face is not absorbed into this positive-norm ideal family: it is exactly

$$a_k + q_k\sqrt{3} = (1 + \sqrt{3})(2 + \sqrt{3})^k, \qquad k \geq 1,$$

with the original strict wall, and contributes the explicit floor count $R_0(Y) = O(\log(2 + Y))$ in equation (1585). For every real $Y \geq 0$, the unchanged weak inequality $4a + 17n \leq Y$ is retained in the exact floor formula of theorem 44.2; its discrepancy splits into the radial shifted divisor term and the exact phase discrepancies. If $S_\ell(m)$ denotes the norm-m coefficient of the unitary Hecke character defined in equation (1594), then the angular data obey the exact shifted identity

$$S_\ell(18n^2 - 2) = (-1)^\ell S_\ell(3n-1)S_\ell(3n+1),$$

with the shifted factorization proved in equation (1600); the endpoint half-weights and all five sharp walls are retained in equation (1602). A direct proof of effective well-roundedness for the unchanged sharp family allows Benoist–Oh, Theorem 13.2 [54, pp. 1931–1932], to be applied orbit by orbit and strengthens the count to

$$\boxed{R_{\mathrm{K}}(Y) = C_0 Y + O(Y^{1-\eta})} \qquad \text{for some fixed, presently unspecified } \eta > 0.$$

The primary-source comparison gives no exact-domain square-root theorem: Liu–Sarnak's 39/64 exponent is for a smooth standard-radial count on the exact congruence lattice, and the sharp standard-radial results of Good and Chatzakos–Petridis have the 2/3 frontier after small-spectrum terms. Neither transfers to the five-wall height $4a + 17n$; see section 51.1. The exact finite census gives $R_{\mathrm{K}}(10^8) = 3{,}536{,}002$, with all rows independently replayed; the largest $|R_{\mathrm{K}}(Y) - C_0 Y|/\sqrt{Y}$ for $10^4 \leq Y \leq 10^8$ is $0.17140019280881455 97\ldots$ at $Y = 867{,}240$.

There is also an exact hyperbolic bridge. For

$$B = (18, -6)_{\mathbb{Q}}, \qquad \mathcal{O} = \mathbb{Z}1 \oplus \mathbb{Z}\frac{I+J}{2} \oplus \mathbb{Z}\frac{I-J}{2} \oplus \mathbb{Z}\frac{IJ}{2},$$

the map

$$(a, q, n) \longmapsto 2 + \frac{qI + aJ + nIJ}{2}$$

is a bijection from the parity-restricted ternary shell to the reduced-trace-4, reduced-norm-1 locus. The order has reduced discriminant 108, index 18 in a maximal overorder, and is not an Eichler order; B is ramified exactly at $2, 3$. Each image is primitive hyperbolic with centralizer $\{\pm\gamma^k : k \in \mathbb{Z}\}$, eigenvalues $2 \pm \sqrt{3}$, and length $2\log(2 + \sqrt{3})$. The shell is a finite union of conjugacy classes, but its canonical height $4a + 17n$ differs from the exact radial Huber height $7a + 15\sqrt{2}\,n$, since

$$X_i(\gamma)^2 = 1 + \frac{(7a + 15\sqrt{2}\,n)^2}{48}.$$

This theorem is independent of the existence of the proposed complex structure on S^6. The stronger square-root discrepancy $R_{\mathrm{K}}(Y) - C_0 Y = O_\varepsilon(Y^{1/2+\varepsilon})$ is an explicitly **[OPEN]**problem, not a hypothesis used in the asymptotic; see theorems 39.4, 39.6, 39.7, 39.11, 39.12, 43.3 and 46.2 and sections 49, 50.7 and 51.1.

29. **[VERIFIED] The earlier Erdős–Straus divisor selector and the S^6 terminal presentation are the same marked integral matrix slice.** For $p \equiv 1 \pmod{12}$ prime, define

$$L = \frac{p-1}{4}, \qquad 0 \le j < L, \qquad R_j = 4j+3, \qquad a_j = \frac{p+R_j}{4}.$$

For a selector state

$$u \in \mathbb{Z}_{>0}, \qquad u \mid a_j^2, \qquad R_j \mid 4u + p^{\varepsilon}, \qquad \varepsilon \in \{0,1\},$$

set

$$T = \frac{4u+p^{\varepsilon}}{R_j}, \qquad \ell_0 = \frac{T+1}{4}, \qquad \ell_1 = u - jT, \qquad \ell_2 = 1.$$

Then $T \equiv 3 \pmod 4$,

$$12\ell_0 - 4\ell_1 - 3\ell_2 = p^{\varepsilon},$$

and

$$\boxed{\begin{pmatrix} 1 & -j \\ 0 & 1 \end{pmatrix} \begin{pmatrix} R_j & -u \\ 4 & -T \end{pmatrix} = \begin{pmatrix} 3 & -\ell_1 \\ 4 & 1-4\ell_0 \end{pmatrix}.}$$

The inverse $T = 4\ell_0 - 1$, $u = \ell_1 + jT$ proves bijectivity once the prime, shell label, channel, orientation, and divisor support are retained. The $\varepsilon = 0$ channel lies over terminal determinant parameter 1, and the $\varepsilon = 1$ channel lies over p. The proof also derives the complete selector from the factorization $(R_j y - pa_j)(R_j z - pa_j) = p^2 a_j^2$. This is a structural equality of integer presentations, not a map of the underlying geometric spaces; see section 6.

30. **[VERIFIED] The residue lattice and shell-count maps retain their different domains.** The lattice has the exact index-72 glue and the full coordinate splitting in Theorems 31.3 and 31.15. Its D_4 shortest dual-class sets are the vector, even half-spin and odd half-spin weights of the flag tangent modules, in that marking (Theorem 31.4). For $p \equiv 1 \pmod{24}$ prime and $R > 0$, $R \equiv 3 \pmod 4$, put $a = (p+R)/4$, $g = \gcd(R, pa)$, $R_0 = R/g$, $N_0 = pa/g$. Then

$$\left\{ d > 0 : d \mid N_0^2,\ d \equiv -N_0 \pmod{R_0} \right\} \xrightarrow{\cong} \left\{ (b,c) \in \mathbb{Z}_{>0}^2 : \frac{4}{p} = \frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right\},$$

$$d \longmapsto \left(\frac{N_0 + d}{R_0}, \frac{N_0 + N_0^2/d}{R_0} \right), \qquad U_R(p) = \frac{A_R(p) + \mathbf{1}_{R_0 \mid 2}}{2}.$$

The inverse is $d = R_0 b - N_0$. When $\gcd(R, pa) = 1$ and $p \nmid a$, the exact raw count is $A_R(p) = c_R(a;-1) + 2c_R(a;-p^{-1})$ in the exponent-box notation of Theorem 31.9. Equal prime residues do not determine these counts (Theorem 31.10). The finite-sample map in Theorem 31.12 sends these counts to an integral augmentation vector. Its Fourier inversion recovers the aggregate residue-count function; the centering $Q_{R,P} = 6W_{R,P} - M_{R,P}u$ omits its constant Fourier component. The sample P remains in the domain, and no universal selector nonemptiness is inferred.

31. **[VERIFIED] Two complex columns give the exact four-dimensional $SU(3)$ Gram cone.**

$$\mathrm{Mat}_{3\times 2}(\mathbb{C})/SU(3) \cong \mathrm{Herm}_2(\mathbb{C})_{\ge 0}.$$

For the one-point compactified column representation,

$$S^{12}_{\mathrm{col}}/SU(3) \cong B^4,$$

and the rank-one, zero, and infinity orbit types form its boundary S^3. This representation sphere is not the cusp-smash S^{12}; see theorems 17.1 and 17.5.

32. **[VERIFIED] Three complex columns give an exact cubic rank-three cone with a distinguished rank-loss S^8.** For $\mathscr{V}_3 = \mathrm{Mat}_{3\times 3}(\mathbb{C})$, the complete orbit invariants are

$$\mathscr{V}_3/U(3) \cong \mathrm{Herm}_3(\mathbb{C})_{\geq 0}, \qquad \mathscr{V}_3/SU(3) \cong \{(H,z) : H \geq 0,\ |z|^2 = \det H\}.$$

The second space is $\mathbb{R}^{10}$, so

$$S^{17}/SU(3) \cong S^9, \qquad \mathscr{V}_3^+/SU(3) \cong S^{10}.$$

Inside that S^{10}, the compactified rank-deficient locus is the standard topologically locally flat S^8, with

$$S^{10} \setminus S^8 \cong \mathbb{R}^8 \times \mathbb{C}^* \simeq S^1.$$

The generator is the determinant phase $\det A/|\det A|$, and the trace-one rank-one stratum is $\mathbb{C}P^2$. Quotienting the determinant phase gives $\mathscr{V}_3^+/U(3) \cong B^9$ with boundary S^8. As a whole this is a cubic Jordan/Gram carrier rather than a rank-two quadratic Lorentz cone. The marked angular quotient nevertheless gives an exact split Lorentz observation with every complementary direction retained, as recorded in the next item; see section 18.

33. **[VERIFIED] The marked angular epimorphism gives a split positive Lorentz observation, and its full fibre is retained.** For

$$P = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix}, \qquad u = (1,1,1)^{\mathsf{T}}, \qquad R = P^*(PP^*)^{-1} = \frac{1}{3}\begin{pmatrix} 2 & 1 \\ -1 & 1 \\ -1 & -2 \end{pmatrix},$$

put $\Pi = uu^*/3$, $Q_u = I_3 - \Pi$, and

$$\mathcal{O}_P(H) = PHP^*, \qquad \jmath_P(G) = RGR^*.$$

These positive maps satisfy

$$\mathcal{O}_P\jmath_P = \mathrm{id}, \qquad \jmath_P\mathcal{O}_P(H) = Q_uHQ_u,$$

with

$$\ker \mathcal{O}_P = \{a\,uu^* + uv^* + vu^* : a \in \mathbb{R},\ v \in u^\perp\}, \qquad \mathcal{O}_P^{-1}(0) \cap \mathcal{P}_3 = \{\lambda uu^* : \lambda \geq 0\}.$$

Writing $X = (R\ u)$, every Hermitian matrix is uniquely

$$H = X\begin{pmatrix} G & \psi \\ \psi^* & \lambda \end{pmatrix}X^*, \qquad (G,\psi,\lambda) = \left(PHP^*, \frac{1}{3}PHu, \frac{1}{9}u^*Hu\right),$$

so

$$\mathrm{Herm}_3(\mathbb{C}) \cong \mathrm{Herm}_2(\mathbb{C}) \oplus \mathbb{C}^2 \oplus \mathbb{R}, \qquad 9 = 4+4+1,$$

$$\det H = \lambda \det G - \psi^* \operatorname{adj}(G)\psi, \qquad H \succeq 0 \iff G \succeq 0, \psi \in \operatorname{im} G,\ \lambda \geq \psi^* G^+ \psi.$$

The faithful embedding

$$T_P(g) = RgP + \Pi = X(g \oplus 1)X^{-1}$$

acts by $(G,\psi,\lambda) \mapsto (gGg^*, g\psi, \lambda)$. For $g \in SL_2(\mathbb{C})$, the summands are a Lorentz four-vector, the defining complex spin representation, and a real scalar. The vector action factors through $SO^+(1,3)$, while $-I_2$ negates ψ, so the full carrier retains the spin double cover.

The section is proper and compactifies to

$$\jmath_P^+ : B^4 \cong \mathcal{P}_2^+ \hookrightarrow (\partial\mathcal{P}_3)^+ \cong S^8, \qquad e_2(\jmath_P(G)) = \frac{1}{3}\det G,$$

and

$$\det(uu^* + \varepsilon \jmath_P(G)) = \varepsilon^2 \det G.$$

The reverse boundary observation is nonproper. In the strict $SU(3)$ quotient,

$$|\omega|^2 = \lambda \det G - \psi^* \operatorname{adj}(G)\psi,$$

and $\omega = 0$ is the rank-loss cone compactifying to $S^8 \subset S^{10}$. The four-branch chart supplies the exact toric seed

$$K(z) = \operatorname{diag}(z_0 z_3, z_1 z_2), \qquad \det K = z_0 z_1 z_2 z_3,$$

whose positive $SL_2(\mathbb{C})$-orbit fills the future causal cone. For $Z \in \operatorname{Mat}_{3\times 2}(\mathbb{C})$ and retained $\alpha, c > 0$,

$$p_\alpha(Z) = \alpha Z^* Z, \qquad m^2_{\alpha,c}(Z) = \frac{\alpha^2}{c^2} \sum_{a<b} |Z_{a1} Z_{b2} - Z_{a2} Z_{b1}|^2.$$

This is the complete $SU(3)$-orbit momentum and is Lorentz covariant. Its positive-mass locus is exactly rank two, and $m^2_{\alpha,c}(rZ) = r^4 m^2_{\alpha,c}(Z)$, proving that the unrestricted kinematic carrier has no positive threshold. The exact reduction separates four statements which the weak Z-equation alone does not. First, for $m_0 > 0$ the colour quotient and reparametrization characteristic give

$$T^* \mathcal{H}^+_{m_0,c}$$

with its canonical symplectic form, exactly the future spinless massive particle. Second, the intrinsic nonzero rank-one quotient gives $T^* \mathcal{N}^\times_+$. Third, the full massless multiplier action is not the standard massless variational system: the determinant constraint has zero differential on rank one, its equation is $\dot{x} = \lambda p$ without $\lambda = e$, four transverse rank-opening directions are non-gauge first-order null directions, and $Z = 0$ admits arbitrary $x(\tau)$. Fourth, on the hard $\omega_0 \neq 0$ packet the momentum differential has rank four above threshold and rank three at threshold, so its displayed action has $\dot{x} = 0$ in the strict interior and $\dot{x} \in \mathbb{R}p$ only at the endpoint. The packet's multiplication operator is not the Hamiltonian generating this action; its positive endpoint is a constraint-selected algebraic/free spectral threshold rather than a dynamically generated interacting gap. See theorems 23.2 to 23.7, 27.6, 27.9, 27.12 and 27.16.

34. **[VERIFIED] The determinant normal slice is an exact Lorentzian host for explicit field models, but it does not itself select a mass gap.** The real-linear isomorphism

$$\jmath_P : \operatorname{Herm}_2(\mathbb{C}) \xrightarrow{\sim} \mathcal{N}_P := \{Y \in \operatorname{Herm}_3(\mathbb{C}) : Yu = 0\}, \qquad G \longmapsto RGR^*,$$

has inverse $Y \mapsto PYP^*$, and

$$\det(uu^* + \varepsilon \jmath_P(G)) = \varepsilon^2 \det G.$$

Writing $G(t,x,y,z) = \left(\begin{smallmatrix} t+y & x+iz \\ x-iz & t-y \end{smallmatrix}\right)$, polarization gives

$$\eta_P(\jmath_P(G), \jmath_P(K)) = tt' - xx' - yy' - zz'.$$

The positive-semidefinite matrices are the future cone, the nonzero rank-one matrices its null boundary, and the lifted congruence action induces $PSL_2(\mathbb{C}) \cong SO^+(1,3)$. The future timelike determinant shell, future-null locus, and candidate spacelike determinant shell are respectively

$$SL_2(\mathbb{C})/SU(2), \qquad SL_2(\mathbb{C}) \cdot \operatorname{diag}(1,0), \qquad SL_2(\mathbb{C})/SU(1,1).$$

The standard metrics already retained by the Moore–Penrose construction select

$$W_P = R^* R = C^{-1}, \qquad K_P = \{a : a^* W_P a = W_P,\ \det a = 1\} \cong SU(2),$$

and the positive form $E_P(G,K) = \operatorname{tr}(GW_PKW_P)$. Under K_P,

$$\mathrm{Herm}_2(\mathbb{C}) \cong \mathbf{1} \oplus \mathbf{3},$$

with trivial central action. If this pointwise symmetry is promoted to a gauge symmetry, the action

$$S[\mathcal{A}, G] = \int_M \left(\frac{|F_{\mathcal{A}}|^2}{2g_{\mathrm{YM}}^2} + \frac{Z_G}{2} |D_{\mathcal{A}} G|_{E_P}^2 + \frac{m_G^2}{2} |G|_{E_P}^2 + \frac{\lambda_4}{4} (\det G)^2 \right) d\,\mathrm{vol}_g$$

is nonnegative and gauge invariant, and $\widehat{\mathcal{A}} = R\mathcal{A}P$ intertwines the covariant derivatives. The normal field alone cannot screen a fundamental centre charge; the full (G, ψ, λ) carrier contains the centre-nontrivial spinor ψ, but its kinetic and interaction theory is not inserted by this construction.

The candidate inverse-dual matrix and the positive Gram momentum are also separated by an exact invariant:

$$\det G_{\eta,T} = -\frac{2}{3} D_T < 0, \qquad \operatorname{im} p_\alpha = \mathrm{Herm}_2(\mathbb{C})_{\geq 0}.$$

Neither positive rescaling nor complex congruence can identify their inertia types; a further morphism is required. After a separate length/time dictionary is supplied, the same affine host supports the ordinary Klein–Gordon theory. Its Fock Hamiltonian has excitation gap exactly m_{KG} for $m_{\mathrm{KG}} > 0$ and gap zero for $m_{\mathrm{KG}} = 0$. On a periodic spatial gauge regulator, the massless transverse Gaussian frequency is $2\pi/L \to 0$. Explicit countermodels also show that entanglement occurs in both gapped and gapless systems. A positive auxiliary gap transfers to the physical Hamiltonian only through a map on the entire excitation form domain satisfying

$$\|Ru\| \geq c_R \|u\|, \qquad \mathfrak{b}[Ru] \leq C_R(\mathfrak{a}_{\mathrm{phys}}[u] - E_{\mathrm{vac}} \|u\|^2),$$

in which case $\mathrm{gap}(A_{\mathrm{phys}}) \geq \gamma c_R^2 / C_R$. An intertwiner on a single gapped subspace is not enough. See theorems 28.2, 28.3, 28.5 to 28.10, 28.12 and 28.16.

35. **[VERIFIED] The marked cubic cone contains exact Lorentz sections, but these do not identify it with the toroidal degeneration.** With the oriented cross-product matrix $[a]_\times$,

$$\det(S + i[a]_\times) = \det S - a^{\mathsf{T}} S a.$$

Consequently each marked positive monodromy form B determines the exact linear section

$$\ell_B(\lambda, a) = \lambda B + i[a]_\times, \qquad \ell_B^{-1}(\mathrm{Herm}_3(\mathbb{C})_{\geq 0}) = \{\lambda \geq 0 : a^{\mathsf{T}} B a \leq \lambda^2 \det B\}.$$

This is a future $(3+1)$-Lorentz cone. An explicit real-linear order-cone map Θ from 2×2 Hermitian matrices satisfies

$$\det_3 \Theta(H) = \frac{\operatorname{tr} H}{2} \det_2 H$$

and sends positive-semidefinite ranks $2, 1, 0$ to $3, 2, 0$. It is not a Jordan homomorphism: for a rank-one projection P, the matrix $\Theta(P)$ has eigenvalues $0, 1/2, 1$ and is not idempotent. Neither map carries a toric period lattice, monodromy, or singular-fibre filling. In particular, the toric limit at fixed $B = I_3$ is not the rank-drop operation $\det B \to 0$. The rejected packet's claimed index-three translation compression is likewise not a constructed map: the actual six-to-two bridge is the saturated rank-lowering quotient $P : \mathbb{Z}^3 \twoheadrightarrow \mathbb{Z}^2$, which sends the even and odd Kuhn labels to the two unimodular target-triangle orbits. See theorems 20.1, 20.7 and 20.8.

36. **[VERIFIED] The rank-three toroidal successor maps onto the candidate cusp with a connected elliptic kernel.** The Kuhn fan gives a proper analytic family

$$\pi_3 : N_3 \longrightarrow \Delta, \qquad N_{3,t} \cong E_\tau^3 \quad (t \neq 0),$$

whose deepest local equation is $t = z_0z_1z_2z_3$, whose cusp matrix is $M_3 = \left(\begin{smallmatrix} I_3 & I_3 \\ 0 & I_3 \end{smallmatrix}\right)$, and whose filling map kills the three angular cycles and retains the three deck cycles. The irreducible central fibre has

$$(f_0, f_1, f_2, f_3) = (1, 7, 12, 6), \qquad \chi(W_3) = 6.$$

Its normalization is the permutohedral threefold with orbit vector $(1, 14, 36, 24)$; the positive-codimension strata map with degrees $2, 3, 4$ onto the $7, 12, 6$ quotient strata. Hence any real S^8-total-space fibration retaining W_3 requires the remaining singular fibres to contribute Euler sum -4. For an injective integral shear $\mathcal{B} : \bar{L}_3 \hookrightarrow L_3$, this becomes the exact determinant/index formula

$$\chi(W_{3,\mathcal{B}}) = 3!\,[L_3 : \mathcal{B}\bar{L}_3] = 3!\,|\det \mathcal{B}|.$$

The explicit matrices

$$P = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix}, \qquad Q = \begin{pmatrix} 0 & -1 & 1 \\ 1 & -1 & 0 \end{pmatrix}$$

define a surjection $\Phi = P \oplus Q : H_1(T^6) \twoheadrightarrow H_1(T^4)$ satisfying

$$\Phi M_3 = M_F \Phi, \qquad c_F \Phi = Q c_3, \qquad P Z_3(t) = Z_2(t) Q.$$

It extends over all cusp toric strata to a holomorphic map with connected elliptic kernel. Integrally,

$$M_3 \cong M_F \oplus \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix},$$

and the order-three and order-four free affine peripheral actions have compatible rank-six lifts. The six Kuhn tetrahedron orbits map S_3-equivariantly to the six marked diagonal Weyl chambers, but only as fan/Cartan data; the four-branch toroidal germ and the quadratic Gram-rank germ remain inequivalent. What remains open is the single holomorphic period-and-overlap family across both finite-order discs. See theorems 19.8 and 20.3 to 20.6.

37. **[VERIFIED] The free rank-six cusp lattice, the six chambers, and the Euler factor six are now separated exactly.** The lattice native to the higher-rung construction is

$$\Lambda_{\mathrm{tor}} = H_1(T^6; \mathbb{Z}) = \mathbb{Z}^3_{\mathrm{angular}} \oplus \mathbb{Z}^3_{\mathrm{deck}}.$$

For $k = (1, 1, 1)$, the map $\Phi = P \oplus Q$ gives an exact S_3- and monodromy-equivariant sequence

$$0 \longrightarrow \mathbb{Z}(k, 0) \oplus \mathbb{Z}(0, k) \longrightarrow \Lambda_{\mathrm{tor}} \longrightarrow A_F \oplus B_F \longrightarrow 0.$$

It splits integrally as a monodromy module but not over $\mathbb{Z}[S_3]$. The quotient copies are A_2 weight lattices; their root sublattices have index three. The six standard homology generators are two permutation orbits of size three, whereas the six Kuhn cells and six marked Weyl chambers form one regular S_3-orbit. Consequently

$$\operatorname{rk} \Lambda_{\mathrm{tor}} = 6, \qquad \#\{\text{chambers}\} = 3! = 6, \qquad \chi(W_{3,\mathcal{B}}) = 3!|\det \mathcal{B}|$$

do not identify six lattice directions with six strata. The regular six-element chamber set instead occurs in the rank-nine bilinear carrier

$$\operatorname{Hom}(\overline{L}_3, L_3) = \bigoplus_i \mathbb{Z} E_{ii} \oplus \bigoplus_{i \neq j} \mathbb{Z} E_{ij} :$$

the off-diagonal units are a regular S_3-set, and $c_{(\pi_1\pi_2\pi_3)} \mapsto E_{\pi_1,\pi_3}$ is an equivariant bijection. These are exactly the mixed angular–deck basis pairs with zero alternating pairing; the three excluded

diagonal pairs have pairing one. The ambient real matrix representation is equivariantly isomorphic to $\mathrm{Herm}_3(\mathbb{C})$, but the symmetric/skew integral splitting has index 2^3, and the images of the off-diagonal units are indefinite. This is a representation/incidence bridge, not a positive, Jordan, determinant, or toric morphism. The actual nine-dimensional-boundary map is only the piecewise-linear angular map $\Phi_{\mathrm{G}} \circ \jmath \circ P$; the deck lattice is instead used by the period and filling squares. This map does not descend to a torus fibre. The separate bundle-defect lattice $H^4(U;\mathbb{Z}) \cong (\mathbb{Z}^2)^3$ has a different domain and degree. See section 21 and theorems 21.2, 21.3, 22.1 and 22.4.

The successful lifted candidate actions are distinct from the diagonal local-system candidate

$$M_1 = I_3 \otimes S, \qquad M_2 = I_3 \otimes Q, \qquad M_0 = I_3 \otimes T, \qquad M_1 M_2 M_0 = I.$$

Its finite factors have orders 4 and 6, but $I - M_j : T^6 \to T^6$ is an isogeny for $j = 1, 2$; every affine lift of either finite action therefore has a fixed point.

38. **[REFUTED] There is no literal smooth complex-fourfold analogue on a homotopy S^8.** An almost-complex tangent bundle would have

$$c_1 = c_2 = c_3 = 0, \qquad c_4 = 2u, \qquad p_2 = 2c_4 = 4u,$$

whereas the signature formula forces $p_2 = 0$. A nonsingular T^6-bundle over S^2 with total space S^8 is independently excluded by both the homotopy sequence and Euler characteristic. A real singular T^6-over-S^2 construction remains a legitimate search problem, but its special fibres must obey $\sum_i \chi(F_i) = 2$. For monodromy ρ on the regular piece, its actual filling inclusions must use the corrected Serre filtration

$$0 \longrightarrow (\textstyle\bigwedge^q \Lambda)_\rho \longrightarrow H_q(E;\mathbb{Z}) \longrightarrow H_1(B^\circ; \textstyle\bigwedge^{q-1} \Lambda) \longrightarrow 0,$$

and the dual quotient $H^q(E;\mathbb{Z}) \twoheadrightarrow (\bigwedge^q \Lambda^\vee)^\rho$. Thus the integral Mayer–Vietoris test uses coinvariants as the homological subobject and the base-one-cycle term as quotient, not conversely; see section 29 and equations (1105) and (1108). Every effective torus action on S^8 has rank at most four, and the sharp coordinate model is

$$S^8/T^4 \cong \mathrm{Susp}(\Delta^3) \cong B^4.$$

This is a real two-pole singular compactification with generic T^4 orbit, not a three-special-fibre T^6-fibration over S^2; see equations (1109) and (1110).

39. **[VERIFIED] The regular locus of the reconstructed S^6 has a six-coordinate, nonextendable defect lattice.** For $D = W \sqcup S_1 \sqcup S_2$ and $U = X \setminus D$,

$$\boxed{H^4(U;\mathbb{Z}) \cong H_1(W) \oplus H_1(S_1) \oplus H_1(S_2) \cong \mathbb{Z}^2 \oplus \mathbb{Z}^2 \oplus \mathbb{Z}^2.}$$

Principal $SU(3)$-bundles on U are classified exactly by this c_2-class, and the pair connecting morphism $H^4(U) \xrightarrow{\sim} H^5(X, U)$ proves that extension across all three special fibres occurs exactly for $c_2 = 0$. The displayed group is the free rank-six lattice $\mathbb{Z}^6$, not a cyclic $\mathbb{Z}/6$.

40. **[VERIFIED] The primitive fibre degree has exact order-twelve and order-six normal-holonomy quotients.** Complete proofs, including the one-shot collar matrix, transpose system, saturated kernel, coefficient reduction, tensor-centre map, and common-centre quotient, are given in section 59 and theorems 59.1 to 59.4. Those group and lattice theorems do not themselves define a gauge field or an energy operator.

In the marked basis

$$(a_0, b_0, a_1, b_1, a_2, b_2) = (\bar{\gamma}, \bar{u}, g_1, h_1, g_2, h_2),$$

the transpose of the actual one-shot collar matrix gives

$$\boxed{[\deg_F] = (12, 0, 4, 0, -3, 0)},$$

meaning

$$\operatorname{res}_F\left(\sum_{i=0}^{2}(n_i a_i + r_i b_i)\right) = (12n_0 + 4n_1 - 3n_2)\omega_F.$$

It is primitive, with saturated kernel

$$\boxed{\ker \deg_F = \mathbb{Z}\langle b_0, b_1, b_2, a_0 - 3a_1, 3a_1 + 4a_2\rangle,}$$

Define

$$q_{\mathrm{col}} : L \longrightarrow H_1(X^\circ;\mathbb{Z}), \qquad a_0 \mapsto c, \quad a_1 \mapsto \bar{g}_1, \quad a_2 \mapsto \bar{g}_2, \quad b_0, b_1, b_2 \mapsto 0.$$

If $\Phi_1 : H_1(X^\circ;\mathbb{Z}) \to \mathbb{Z}$ has row $(12, 4\ell_1, 3\ell_2)$ in the basis $(c, \bar{g}_1, \bar{g}_2)$, then the factorization through the same collar functional that kills the final cusp is

$$\deg_F = \Phi_1 \circ q_{\mathrm{col}}, \qquad \Phi_1(j_*[\widetilde{a}_0]) = p.$$

Let $\vartheta_1(a_1) = e^{-2\pi i/3}$ and $\vartheta_2(a_2) = e^{-2\pi i/4}$ be the local normal-line holonomies, trivial on the other marked generators. The actual relative twist gives

$$\chi_{12}\left(\sum_{i=0}^{2}(n_i a_i + r_i b_i)\right) = \vartheta_1(n_1 a_1 + r_1 b_1)\vartheta_2(n_2 a_2 + r_2 b_2)^{-1} = e^{-2\pi i(4n_1 - 3n_2)/12}.$$

If $\rho_m(k) = e^{-2\pi i k/m}$, then

$$\boxed{\chi_{12} = \rho_{12} \circ \deg_F, \qquad \chi_6 := \chi_{12}^2 = \rho_6 \circ \deg_F\,.}$$

Therefore

$$H^4(U;\mathbb{Z})/\ker \chi_{12} \cong \mathbb{Z}/12\mathbb{Z}, \qquad H^4(U;\mathbb{Z})/\ker \chi_6 \cong \mathbb{Z}/6\mathbb{Z}.$$

Ordinary coefficient reduction instead gives

$$H^4(U;\mathbb{Z}/6) \cong (\mathbb{Z}/6)^6;$$

it is not the single distinguished cyclic quotient selected by χ_6. For a principal $SU(3)$-bundle $P \to U$, the order-six character is exactly

$$\chi_6(c_2(P)) = \exp\!\left(-\frac{2\pi i}{6}\langle c_2(P)|_F, [F]\rangle\right).$$

This is a topological congruence for the fibre second-Chern number, not a line-operator or spectral assertion.

The tensor embedding

$$SU(2) \times SU(3) \hookrightarrow SU(6), \qquad (A, B) \longmapsto A \otimes B$$

restricts to an isomorphism $\mu_2 \times \mu_3 \xrightarrow{\sim} Z(SU(6)) = \mu_6$. Since U has homotopy dimension at most five, c_2 classifies $SU(n)$-bundles for $n \geq 3$, while

$$c_2(E_2 \otimes E_3) = 3c_2(E_2) + 2c_2(E_3).$$

Every chosen $SU(6)$-bundle class $h \in H^4(U;\mathbb{Z})$ therefore admits a tensor-subgroup reduction, and all its c_2-factorizations are

$$(e, f) = (h + 2t, -h - 3t), \qquad t \in H^4(U;\mathbb{Z}).$$

No nonzero h or reduction is canonically selected. Separately,

$$H^5(U;\mathbb{Z}) \cong \ker\{\mathbb{Z}^3 \xrightarrow{(1,1,1)} \mathbb{Z}\} \cong Q(A_2),$$

for the combinatorial form making the three distinguished support basis vectors orthonormal. This is not a cup-product or intersection form, and the support-redistribution lattice does not identify the rank-five kernel of $\deg_F$ with A_5.

The finite-group isomorphism $H^4(U)/\ker\chi_6 \cong Z(SU(6))$ supplies no $SU(6)$ transition functions or connection. The injective tensor map does not factor through $(SU(2)\times SU(3))/\mu_6$, because the central μ_6 maps to nontrivial scalar matrices. The distinct common-centre map is

$$\begin{aligned} \Gamma_6 &= \{((\lambda\mu)^{-1}, \lambda I_2, \mu I_3) : \lambda \in \mu_2,\ \mu \in \mu_3\}, \\ \frac{U(1)\times SU(2)\times SU(3)}{\Gamma_6} &\hookrightarrow U(6), \\ [z, A, B] &\longmapsto z(A \otimes B), \end{aligned}$$

where $\Gamma_6 \cong \mu_6$. Its determinant is z^6, and its image meets $SU(6)$ in exactly $\iota(SU(2)\times SU(3))$; the image is not generally contained in $SU(6)$. The missing step is a degree-changing identification with a degree-two common-centre lifting obstruction and its line-operator pairing. The separate global clutching classes on S^6 have $c_3 = \pm 2ku$, $k \in \pi_5(SU(3)) \cong \mathbb{Z}$, and restrict trivially to U; see section 29.8.

41. **[VERIFIED] The exact typed transfer from auxiliary Rayleigh quotients to a physical excitation spectrum is proved.** Let $H \geq 0$ be self-adjoint on $\mathcal{H}_{\mathrm{phys}}$, and set

$$P_0 = \mathbf{1}_{\{0\}}(H), \qquad \mathcal{H}_{\mathrm{exc}} = (I - P_0)\mathcal{H}_{\mathrm{phys}}, \qquad \mathfrak{h}[\psi] = \|H^{1/2}\psi\|^2, \qquad \mathcal{Q}(H) = D(H^{1/2}).$$

For a nonnegative quadratic form $\mathfrak{q}_n$ and nonzero $u_n \in D(\mathfrak{q}_n) \subset \mathcal{K}_n$, suppose

$$J_n : \mathrm{span}\{u_n\} \longrightarrow \mathcal{Q}(H), \qquad \psi_n = (I - P_0)J_n u_n,$$

and, for $c > 0$, $C_n, \varepsilon_n \geq 0$,

$$\|\psi_n\| \geq c\|u_n\|, \qquad \mathfrak{h}[\psi_n] \leq C_n \mathfrak{q}_n[u_n] + \varepsilon_n \|u_n\|^2, \qquad C_n \frac{\mathfrak{q}_n[u_n]}{\|u_n\|^2} + \varepsilon_n \longrightarrow 0.$$

Then the self-adjoint excitation restriction

$$H_{\mathrm{exc}} = H\big|_{D(H)\cap\mathcal{H}_{\mathrm{exc}}}$$

satisfies

$$\inf \sigma(H_{\mathrm{exc}}) = 0, \qquad 0 \in \sigma_{\mathrm{ess}}(H_{\mathrm{exc}}).$$

The projection removes the full kernel of H; orthogonality to a single chosen vacuum is equivalent only after vacuum uniqueness is separately proved. This is a proved implication with fully stated antecedents, not a claim that the present geometry supplies them. See theorem 52.4.

42. **[OPEN] The physical Yang–Mills bridge data are absent.** The workbench has not constructed a common gauge-invariant positive-metric $\mathcal{H}_{\mathrm{phys}}$, a four-dimensional quantum Yang–Mills Hamiltonian H, or maps J_n obeying the norm and energy inequalities through gauge reduction, reflection-positive reconstruction, renormalization, and the continuum and infinite-volume limits. Therefore the verified radial scalar spectra, auxiliary direct integral, classical gauge Hessians, finite cochain operators, compact-link flux configurations, finite Gaussian oscillators, magnetic Laplacians, and reducible split-Cartan $SU(r)$ Jacobi operators do not prove or disprove a mass gap for a physical quantum Yang–Mills Hamiltonian. See theorems 12.12 and 52.6.

Exact-computation certificates. The retained exact-arithmetic harnesses passed all 28 full-manuscript checks, all 38 independently implemented core checks, and all 13 cusp-spectral checks, with no failures; the 13-check harness also passed unchanged under `python -O`. The first two harnesses cover finite and cusp monodromy, invariant and coinvariant lattices in exterior degrees 0 through 4, Smith forms and saturation, finite-quotient norm operators, the relation-group determinant, specialization indices, three twisting parameters, period-form identities, and the corrected existential form of Remark 7.28. The 13-check cusp-spectral harness is scoped to the radial period determinant and full inverse transpose, the restricted dual maps and pairings, the integral coframe wedge identity, the ambient Koide discriminant identity, the finite coordinate eigenvalues, and the radial volume and first-correction expansions. The geometric collapse, finite-band and cusp inclusion maps, Leray comparison, Kummer morphisms, cone carriers, compact-link construction, and Koide counting theorem are proved separately rather than inferred from those executions.

Independent review checkpoint. Reviewer B independently reviewed the reconstruction existing up to its evidentiary cutoff. It withdrew the earlier topology-based rejection, distinguished the intrinsic specialization sp^q from the then-open collapse pullback c^*, and found no load-bearing defect in the non-collapse nearby-cycle/Leray and based van Kampen route. Its provisionally favourable disposition and focused verification requests are useful independent evidence about the argument it saw. The later Codex proofs in this workbench are separate and should not be attributed to that review. In its 26 August analytic follow-up, Reviewer B independently proved the finite-complex/UCT top-degree base-change statement on every scheme-theoretic surface fibre and, from CAND-X, CAND-W, and TOP-H2, confirmed the conductor–duality implication $R^2 f_*(T_X \otimes L) \neq 0$ for every L [36]. Those three inputs remain hypotheses in that review record; its cutoff labels do not replace their later workbench classifications.

Current evidence boundary. No mathematical gap is presently recorded in the construction, either topology certificate, sphere recognition, algebraic dimension, or Hodge and Bott–Chern/Aeppli calculations. Exact marked-fibre period-coordinate representatives, global Bott–Chern generators, and their complete product structure are now included. The canonical data do not single out a literal global coefficient atlas for the chosen smooth representatives through the singular fillings; no such gauge is needed for the class or algebra theorem. The new cusp spectral statements apply to the explicitly declared quotient metrics and auxiliary or classical operators. They now include the complete fixed-energy lattice-point law $N_T(E) = \pi D_T \sum_n (E/(4\pi^2) - \|n\|^2)_+ + O_E(T+1)$ and the full-torus integral magnetic bound $\mathrm{gap}(\Delta_{\mathsf{E},T}) \leq 4\pi\sqrt{|\operatorname{Pf}(\mathsf{E})|/D_T}$. The modified-action toric Kummer map now completes the local degree-N^2 regular-fibre cover, while retaining its direct degree N^3, ramification, nonflat relative map, and strictly local scope. The exact maps $q_2^{\wedge 2}$, $q_{1,\mathrm{cusp}}^{\wedge 4}$, and $q_1^{\wedge 4}$ now provide topological S^{12}- and S^{24}-source carriers only by changing the targets through suspension, collapse, and coning. Two copies are rank-minimal cusp-locally and four are rank-minimal over the full monodromy base. Further quotients $\widehat{q}_3^{\wedge 2} : S^{12} \twoheadrightarrow S^6$ and $\widehat{q}_\gamma^{\wedge 4} : S^{24} \twoheadrightarrow S^8$ contain moving, null-homotopic metric four-spheres with eventual exact Euclidean basepoint balls. Both full maps are themselves based null-homotopic. More generally, $\widehat{q}_\gamma^{\wedge r}$ and $\widehat{q}_3^{\wedge r}$ are based null-homotopic for every $r \geq 2$. The one-copy class of $\widehat{q}_3$ is zero by the explicit framed filling of Theorem 30.49; that of $\widehat{q}_\gamma$ remains open. The conical-target classes of $q_2^{\wedge 2}$, $q_{1,\mathrm{cusp}}^{\wedge 4}$, and $q_1^{\wedge 4}$ remain **[OPEN]**.

The topological five-manifolds $\mathfrak{E}_{12}, \mathfrak{E}_{24}$ realize $\mathbb{R}^4$ as a decompactification boundary fibre. Unrescaled slices collapse in every fixed compatible ambient metric; for the fixed cusp-adapted metrics, the rescaled slice bubbles are $\mathbb{R}^4$ while the full ambient bubbles are $\mathbb{R}^8$ and $\mathbb{R}^{12}$. This leaves intact the nonextension to the fixed product target and supplies no compact complex structure on either smash sphere. The repaired compact-link calculation keeps cover degree and mesh refinement independent and proves the exact finite Gaussian gap and its zero-gap diagonal sequence, but not a nonperturbative compact-link spectrum. The physical Hilbert space, quantum Hamiltonian, and comparison morphisms required by the typed transfer theorem remain **[OPEN]**; accordingly, these calculations neither prove nor disprove a four-dimensional quantum Yang–Mills mass gap. Separately, the exact three-column quotient is now a rank-three cubic carrier with compactified rank-loss pair (S^{10}, S^8), while a literal almost-complex S^8 is refuted. The toroidal angular map gives that cubic carrier a split positive observation onto the ordinary future Lorentz cone, with exact $9 = 4 + 4 + 1$ four-vector–spin-representation–scalar coordinates and a proper $B^4 \hookrightarrow S^8$ section. Its determinant-phase equation, toric seed, and $SU(3)$-invariant momentum are explicit, and $m^2_{\alpha,c}(rZ) = r^4 m^2_{\alpha,c}(Z)$ proves that the unrestricted kinematic carrier has no positive threshold. On the candidate's own regular locus, the complete $SU(3)$-bundle classification is the free defect lattice $H^4(U;\mathbb{Z}) \cong \mathbb{Z}^2 \oplus \mathbb{Z}^2 \oplus \mathbb{Z}^2$, and every nonzero class fails to extend across the three special fibres. The one-shot collar matrix computes the primitive fibre-degree row $(12, 0, 4, 0, -3, 0)$; the candidate's local normal holonomies are exactly its reductions $\chi_{12} = \rho_{12} \circ \deg_F$ and $\chi_6 = \rho_6 \circ \deg_F$. The latter quotient is abstractly the same μ_6 as the centre of the tensor subgroup $SU(2) \times SU(3) \hookrightarrow SU(6)$, but no degree-changing map to a common-centre lifting obstruction, gauge cocycle, connection, or physical line pairing has been constructed. The carrier theorem advances the physical bridge through an exact colour-orbit momentum and a now fully reduced worldline action: its massive stratum is the standard spinless particle, its punctured massless characteristic quotient is standard while its multiplier action and apex are irregular, and the hard-constraint packet propagates only on its threshold stratum. Its normal-slice continuation now supplies a constant affine Minkowski metric, the spin and Poincaré actions, an exact Klein–Gordon/Fock theory and finite regulator, an independently chosen classical compact-group Yang–Mills theory, and the metric-selected $K_P \cong SU(2)$ gauge–scalar model. These constructions show both possible gap values: the free scalar gap is the externally supplied m_{KG}, while the massless scalar and thermodynamic Gaussian gauge limits are gapless. They also isolate the candidate cusp matrix as spacelike and outside the positive Gram-momentum image. Dynamics on the retained ψ, λ, ω directions, a dynamically selected gauge-invariant interacting spectral sector, a continuum quantum reconstruction, and a full-domain comparison map to its physical excitation space remain unconstructed. The labels above mean that complete arguments are present here for audit. Specialist replication and external adjudication are separate validation boundaries, neither hypotheses nor gates on these proofs; they remain welcome. The corrected CDP sufficient theorem is now proved in theorems 60.28 and 60.30. Recovery of the original theorem under its original hypotheses would instead require overturning a verified premise of the constructed counterexample.

3 Methodology

The unit of audit is an exact mathematical assertion, not a paragraph's plausibility. Each assertion is recorded with its domain, codomain, coefficient ring, variance convention, and dependencies. The four outcomes used in this paper are literal:

VERIFIED A complete proof from the stated hypotheses is given or a precisely cited theorem is applied after every hypothesis is checked.

CONDITIONAL The implication is proved, but its conclusion depends on an explicitly named established or assumed hypothesis that is not proved in the present argument. This label is never a politeness substitute for OPEN or REFUTED.

OPEN No complete proof or counterexample is presently recorded. This is not a euphemism for false.

REFUTED A complete contradiction or counterexample is given to the exact assertion. A defect in a proof is not, by itself, a disproof of its theorem.

Repair is an operation, not a fifth status: when a printed assertion is false but a corrected statement is proved, the former is recorded as REFUTED and the latter separately as VERIFIED, with every downstream use recomputed.

Computations are converted into checkable certificates: displayed matrices, integer row or column relations, determinant ideals, primitive kernels, or explicit substitutions. A software pass is corroboration, not a proof. Likewise, an algebraically correct matrix calculation does not identify that matrix with a geometric specialization, attaching, or boundary map; that identification receives its own theorem and complete proof.

A theorem whose antecedents are explicitly part of its statement and whose implication is completely proved is VERIFIED as that exact implication. CONDITIONAL is reserved for a conclusion that imports an explicitly named established or assumed hypothesis not proved in the present argument. The current candidate-specific ledger has no such row.

Relevant human literature is searched proactively, not only when a remembered fact seems doubtful. Results are cited at the point of use with theorem, section, or page locators and with their hypotheses intact. Public working LaTeX editions such as the maintained SGA English source [35] may be used as searchable reading layers, but the published SGA volumes [31, 32] remain canonical. Every reference in the compiled paper is resolvable without access to the editor's computer. The citation and tool-disclosure policy follows the Leiden Declaration [2]; neither a model-generated review nor an internal audit note substitutes for a human mathematical source.

4 Relation to existing literature and exact scope

The theorem-level conflict is with Campana–Demailly–Peternell [3, Theorem 2.2 and Corollary 2.3, p. 680]. Their hypotheses match the now workbench-verified identities $H^1(X;\mathbb{Z}) = H^2(X;\mathbb{Z}) = 0$, algebraic dimension one, and holomorphic algebraic reduction. Their conclusion gives $c_3(X) \leq 0$ and, for a threefold homeomorphic to S^6, algebraic dimension zero. The same constructed threefold instead has $c_3(X) = 2$ and $a(X) = 1$. It is therefore a counterexample to Theorem 2.2 and Corollary 2.3 as stated; this conclusion uses the complete candidate reconstruction, not merely the two printed proof-route defects below, and is not resolved by observing that the candidate proof does not invoke [3].

The first local issue isolated below occurs in the opening reduction of [3, §7, p. 692]. The explicit nodal-cubic family in theorem 60.6 refutes that reduction as a formal sheaf-theoretic inference. A second issue is the replacement of monodromy coinvariants by the full fibre homology in [4, Lemma 3.2, pp. 84–85]; CDP20 later imports the resulting rank argument. Either defect might in principle admit a different repair under stronger hypotheses. The original theorem cannot be repaired under its printed hypotheses because the constructed threefold is a verified counterexample. The strongest intact replacement is now proved in theorem 60.28: one global H^2-vanishing implies the stated generic vanishing and $c_3 \leq 0$. The stronger relative condition in theorem 60.29 is sufficient for such a global vanishing after a sufficiently positive twist from the base. For reduced full fibres, theorem 60.30 replaces the p. 692 shortcut by the exact connecting-map injectivity condition.

For the singular fibre, analytic duality is checked in two independent ways. The proof in theorem 60.7 derives the result from smooth Serre duality [9, Theorem 9.37]; Ramis–Ruget's dualizing-complex theorem provides the singular analytic comparison [5, Theorems 1–2, pp. 81–84, and Proposition 2, p. 91]. Relative duality for a proper analytic morphism is also cited in its precise form as Ramis–Ruget–Verdier, Invent.

Math. 13 (1971), 261–283 [6]. Analytic cohomology and base change is applied in the form of Bănică–Stănăşilă, Chapter III, §3, Theorem 3.4 and Corollary 3.10, together with Grauert's local-freeness/continuity theorem in Chapter III, §4, Theorem 4.12 [7]. These locators cover all degrees used by the structure-sheaf direct-image proof, not only its top-degree specialization. The algebraic top-degree analogue in the Stacks Project, Tag 0EX2 [37], is a useful cross-check but is not substituted for the analytic theorem.

The algebraic nearby-cycle formalism in SGA 7 II, Exposé XIII [32], and the semistable/Raynaud framework in SGA 7 I, Exposé IX [31], do not by themselves cover every step for the present proper holomorphic map, which is not assumed algebraic. The needed analytic formalism is nevertheless available. Dimca's Definition 4.2.1 constructs nearby cycles for complex analytic varieties; Proposition 4.2.2 identifies their stalks with Milnor-fibre cohomology; Proposition 4.2.11 commutes nearby cycles with proper analytic direct image; and Theorem 2.3.26 gives proper base change, all for coefficient rings including $\mathbb{Z}$ [33]. Kashiwara–Schapira [34, §8.6] is an independent formal comparison. Thus the general integral analytic framework is verified. None of these references by itself identifies the candidate's specialization image with the full monodromy-invariant lattice. That candidate-specific conclusion, including saturation, is proved in theorem 53.1 for the reconstructed proper cusp model, using the explicit local equations and a supported Gysin calculation. The direct comparison with the cellular collapse is now proved in theorem 58.7: the neighborhood realization of the nearby-cycle unit is

$$\mathrm{sp}^q = i_F^*(i_W^*)^{-1},$$

and the candidate retraction gives $(i_W^*)^{-1} = r^*$, hence $\mathrm{sp}^q = c^*$ over $\mathbb{Z}$ in every degree.

The topological input to the retraction is Brown's collaring theorem for the boundary of a topological manifold [29], followed by Whitehead and the cofibration argument in Hatcher [10]. Clemens's classical collapse for semistable Kähler degenerations [30] is useful context, but it is not used as a substitute for the candidate-specific proof: the central fibre here is irreducible and nonnormal, so theorems 58.2 to 58.5 constructs and checks the map directly.

For compact-support Poincaré duality with local coefficients and its projection formula we use Bredon [14, Chapter VI], with Hatcher [10, §3.H] as a concise topological overview. This is the precise input behind the multiplicative identification $H^2(B, j_*\mathcal{L}) \simeq M_G$ used in the Leray calculation.

The completed global comparison uses the chain-level Mayer–Vietoris convention of Hatcher [10, §2.2, pp. 149–150]: the inclusion of the intersection complex is $z \mapsto (i_{A*}z, -i_{B*}z)$. This fixes the only global minus sign. The candidate-specific content is not supplied by that citation: the radial markings in the reconstructed finite fillings prove that the two band inclusions retract to π_1 and π_2, while the explicit toric disc and solid tori give the primitive cusp kernels. Those geometric facts produce the matrices and the commuting Leray squares in theorems 58.10, 58.11 and 58.13.

For the Hodge-number question, Ugarte's general restrictions on a hypothetical complex structure on S^6 [51] and Angella's later synthesis [48] provide the correct general comparison class. The candidate-specific calculation is sharper: its reconstructed structure-sheaf direct images and explicit vanishing of holomorphic forms first determine the Hodge diamond up to $m = h^{1,2}$ and force $E_2 = E_\infty$. Neither reference determines m for this family. The workbench now does: the cyclic invariant lattices, logarithmic Kodaira–Spencer divisor, normalized residue sequence, and order-four scalar give $m = 1$, hence $h^{1,1} = 2$. This is a candidate-specific calculation, not a consequence imported from the general literature.

The spectral interpretation has also been audited against the primary physical framework. Jaffe–Witten formulate the four-dimensional Yang–Mills mass gap on $\mathbb{R}^4$ only after construction of the quantum theory: on its physical Hilbert space, the Hamiltonian $H \geq 0$ has the vacuum at energy zero and a mass gap precisely when there is a $\Delta > 0$ such that

$$\sigma(H) \cap (0, \Delta) = \varnothing \qquad (\text{equivalently, } \sigma(H) \subseteq \{0\} \cup [\Delta, \infty)).$$

This is the physical mass-gap condition in [64]. Euclidean correlation data yield such an $\mathcal{H}_{\mathrm{phys}}$, vacuum, and positive time-translation generator only through Osterwalder–Schrader reconstruction; the required

input includes Euclidean covariance and symmetry, distributional regularity/growth, reflection positivity, and clustering, with the precise reconstruction qualifications supplied by the two primary papers [68, 69]. Hodge decompositions, periods, Leray terms, and coherent direct images are not Schwinger functions and supply neither reflection positivity nor a physical Hamiltonian. They also do not determine even a geometric Laplace spectrum: the explicit fixed-complex-torus calculation in theorem 52.3 keeps the complex structure, integral cohomology, Hodge decomposition, coherent direct-image data, and total volume fixed while the Kähler metrics

$$g_R = R^2(dx_1^2 + dy_1^2) + R^{-2}(dx_2^2 + dy_2^2)$$

have $\lambda_1(\Delta_{g_R}) = 4\pi^2/R^2 \to 0$. Thus spectra depend on the chosen metric and operator, and no spectral or physical conclusion follows from the cohomological/direct-image data alone. The auxiliary spectral statements in the workbench are direct operator calculations; none is claimed to be established by these references or to constitute an Osterwalder–Schrader reconstruction of quantum Yang–Mills theory.

For the compact-link appendix, the historical scope has now been made explicit. Wilson supplies the plaquette/character lattice action [65]; Lüscher supplies a selfadjoint strictly positive transfer-matrix construction in the standard Wilson lattice theory [66]; and Menotti–Pelissetto establish Osterwalder–Schrader positivity for the Wilson action in the corresponding reflection setup [67]. These citations identify the standard constructions and the physical reconstruction boundary; they do not prove the appendix's metric-dependent sector normalization, exact flux action, finite Hessian spectrum, or Gaussian gap, all of which remain the displayed candidate-specific calculations. Nor do they turn that finite Gaussian model into an Osterwalder–Schrader reconstruction, whose axiomatic requirements remain those of [68, 69].

The general mechanisms used inside that candidate-specific calculation were also checked against the acquired primary archival PDFs (including Deligne's published erratum); no author- or publisher-supplied primary TeX was found for these historical works. Cartan's analytic quotient theorem identifies the local quotient rings with invariants [39, §1, Theorem 1, pp. 1–2]; Reynolds averaging then gives exactness of finite-group invariants over $\mathbb{C}$. Deligne's logarithmic conormal and normalized-stratum residue formalism covers the $d\log$ bases, relative sequence, and self-crossing residue target [41, Chapter II, §3.2–3.7, pp. 72–78]. Katz–Oda supplies the connecting-map and exterior-product formalism for the Gauss–Manin and Kodaira–Spencer maps [47, §2, equations (5)–(10), pp. 202–204]; Grothendieck and the cited Stacks tags supply the splitting and elementary-modification background on $\mathbb{P}^1$ [46, Theorem 2.1, p. 126]. Finally, Frölicher's original lemmas identify the first and limiting pages with Dolbeault and associated-graded de Rham cohomology [42, p. 643]. None of these sources supplies the printed period derivatives or cusp branch maps; those remain the explicit candidate-specific calculations in the workbench. The subsequent line-by-line audit therefore kept their normalization data explicit: the finite determinant formula is defined up to a nowhere-zero frame unit, and the constant cusp Kodaira–Spencer matrices are special-fibre reductions modulo t. These distinctions leave the divisor, residue, and rank conclusions unchanged.

Once those candidate-specific Hodge and Frölicher data are known, the Bott–Chern and Aeppli dimensions are no longer open. McHugh's general compact-threefold formulas [49, Theorem 6.3, Proposition 6.4, equation (7), and §7.1] express them in terms of Dolbeault dimensions, $E_2^{3,1}$, the de Rham subgroup represented by real $(1,1)$-forms, and the auxiliary spaces $K^{p,q}$. For this candidate $E_2^{3,1} = 0$, $H^2_{\mathrm{dR}}(X) = 0$, and $\dim K^{p,q} \le b_{p+q} = 0$ in every middle degree. The last inequality is not an injection $K^{p,q} \hookrightarrow H^{p+q}_{\mathrm{dR}}$: the image in de Rham cohomology of the pure, ∂- and $\bar\partial$-closed forms surjects onto $K^{p,q}$ because every pure d-exact form lies in McHugh's denominator. Direct substitution therefore yields the Bott–Chern diamond, and Schweitzer's Hodge-star isomorphism [50, §2.c, pp. 9–10] yields the Aeppli diamond, in theorem 56.11. Angella's published synthesis [48, §4] independently confirms the endpoint dimensions, $h^{1,1}_{BC} = 2h^{0,1}$, the relation $h^{1,1}_{BC} + h^{2,2}_{BC} = 2h^{2,1}_{BC} + 2$, and Schweitzer duality. The literature supplies the general identities; the numerical specialization depends on the workbench's candidate-specific proof that $h^{1,2} = 1$ and $E_2 = E_\infty$.

The multiplicative refinement is also now complete. Schweitzer's definitions give the Bott–Chern algebra, its natural map to Dolbeault cohomology, and the Chern-class convention [50, §2 and §8.a], but they do not determine this candidate's generators or products. The workbench constructs global $\bar\partial$-closed

representatives G, Q, proves their exact marked-fibre period-coordinate classes, and uses the two exact Frölicher rows, the factor-12 exponential–Leray comparison, Dolbeault–Serre duality, and top integration to obtain all nine positive-degree Bott–Chern generators and

$$H^{*,*}_{BC}(X) = \mathbb{C}\mathbf{1} \oplus N, \qquad N^2 = 0.$$

See theorems 57.4 and 57.5. This is a candidate-specific calculation, not a consequence of the general dimension identities. It determines the cohomology classes and every product, while making no claim of a preferred literal global coefficient atlas for the noncanonical smooth representatives through the singular fillings.

The algebraic reduction of a compact complex torus is used in the exact form given by Birkenhake–Lange [19, Chapter 2, §6], with the independent statement in Pareschi–Popa–Schnell [22, §8, p. 2437]. The toric chart facts are compared with Fulton [23]. The exact classical-topology formula $\pi_1(X_\Sigma) \cong N/N_\Sigma$ is cited from Cox–Little–Schenck [24, Theorem 12.1.10], with Danilov's original toric treatment [25, §9] as an independent comparison. All assertions specific to the locally finite infinite fan and the candidate's proper-discontinuity argument are proved directly.

Finally, the sphere-recognition endpoint uses Hurewicz and homological Whitehead exactly as stated by Hatcher [10, Theorem 4.32 and Corollary 4.33], Smale's generalized Poincaré theorem [15, Theorem A], and the computation of the smooth endpoint by Smale [16, Theorem 1.1 and Corollary 1.2]. Kervaire–Milnor provide an independent equivalent check: their opening table gives $|\Theta_6| = 1$, while the following Remark identifies Θ_6 with diffeomorphism classes in dimension six [17, pp. 504–505]. Those results close the recognition step after the candidate's fundamental group and integral homology are established in theorems 53.5 and 53.6; they are not used to establish those inputs.

For the normalized cusp-restriction calculation, the public Stacks Project references were checked at the formal spectral-sequence level rather than relying on an unstated edge-map convention. Leray is Tag 01F2; the passage $E_r \mapsto E_{r+1}$ and the stable term are Tags 011N and 011O; functoriality, identification with the associated graded, and convergence are Tags 012O, 012U, and 012W [37]. Together with analytic proper base change [33, Theorem 2.3.26], these results make the evaluation square for $H^2(X;\mathbb{Z}) \to H^0(B, R^2 f_*\mathbb{Z})$ commute with restriction to every fibre. Naturality of the exponential-sequence boundary then sends $c_1(L)$ to $c_1(L|_{X_b})$. The literature supplies this functorial formalism; the injective differential $d_2^{0,2}(2q) = \omega[\delta]$ remains the candidate-specific calculation proved in theorem 53.4.

The independent direct boundary-Picard certificate uses Brown's standard crossed-homomorphism model for cohomology of a group with coefficients in a module [28, Chapter I, §5]. That reference supplies only the identification of H^1 with crossed homomorphisms modulo principal ones. The candidate-specific content is proved directly: the finite-cover norm, the direction of each logarithmic conjugacy, the contraction formula for a translated invariant two-form, the order-four parity lattice, the explicit cusp solid torus, and Chern-class naturality under the local nearby-cycle specialization. These give

$$E = 2nq, \qquad (\bar{d}_1, \bar{d}_2, \bar{d}_0) = (4n, -3n, 0)[\delta], \qquad n = 0.$$

The exact Appell–Humbert source is Birkenhake–Lange, Chapter 1, §2, pp. 5–8, especially Theorem 2.1 on p. 7 [19]; Chapter 2, §6 is instead the separate source for algebraic dimension used above. Schnell's public notes record the precise factor/coboundary and antisymmetrized connecting-cocycle formulas [20, equations (3.3)–(3.10), pp. 3–7], together with the translation contraction $w \mapsto \iota_w E$ [21, equation (6.2), p. 1]. These sources certify the standard fibrewise formalism, not the candidate's chosen lifts, finite norms, signs, or cusp winding; those are proved directly.

The completed third certificate uses only holomorphic $\mathcal{O}^*$ factors, both finite fillings, the cusp argument principle, and the toric support function. Its exact boundary equations are

$$E = kq, \qquad (\bar{d}_1, \bar{d}_2, \bar{d}_0) = (2k, -\tfrac{3}{2}k, 0)[\delta], \qquad k = 0.$$

For normalized central degrees (d_1, d_2, d_3), it then proves

$$B = \begin{pmatrix} d_2 + d_3 & d_3 \\ d_3 & d_1 + d_3 \end{pmatrix}, \qquad E_L((\lambda, 0), (0, m)) = -B(B_0\lambda, m).$$

Fulton's chart, character-divisor, and support-function results [23, §2.1, pp. 28–29; §3.3, pp. 60–63; §3.4, pp. 63–72] and the parallel theorem-numbered statements in Cox–Little–Schenck [24, Example 1.2.21, p. 31; Propositions 4.1.1–4.1.2, p. 171; Theorems 4.2.8, p. 179 and 4.2.12, p. 182] fix the convention: for $D = \sum a_\rho D_\rho$ the standard support function has value $-a_\rho$, while $\operatorname{div}(\chi^m) = \sum \langle m, u_\rho \rangle D_\rho$ has the positive sign used in the proof. Cartan theorem B [40, §2, fundamental theorem, p. 81] and the exponential sequence [9, Proposition 6.28 and equations (6.35)–(6.36), pp. 180–182] trivialize only the residual line bundles whose c_1 has already been proved zero. All infinite-fan local finiteness, descent factors, frame comparisons, and the final cross-block identity remain candidate-specific proofs in section 60.7.

Bingener's analytic representability theorem was checked against the public primary text, not inferred from the algebraic Picard literature. For a proper flat holomorphic map cohomologically flat in dimension zero, the relative Picard functor is represented by a complex space which need not be separated [38, Introduction, p. 318; Satz 6.2, pp. 336–337]. The same introduction states that the direct exponential-sequence quotient works under topological local triviality and cohomological flatness in dimension one, and explains why that quotient argument fails in general. These statements provide representability for the cusp family but do not provide the component-specialization or separatedness assertion required to annihilate the normalized central restriction. The candidate-specific calculation in theorem 55.15 makes this limitation concrete: it constructs a descended local line bundle with generic Chern class $-2q$ and nonzero normalized central degree vector $(-12, 2, 0, -12, 2, 0)$. Thus all three proved global restriction theorems genuinely use data absent from a cusp-local representability argument; no general relative-Picard theorem has silently replaced the candidate-specific finite-filling and common-boundary calculation.

4.1 Targeted corpus cross-check (29 August 2026)

Before extending the validate/repair/disprove chain, the relevant primary texts were reread in the local LaTeX corpus and compared with their public editions. This check adds no unproved hypothesis; it records exactly which formal tools the sources do, and do not, supply.

First, the primary transcription of Campana–Demailly–Peternell (2020), Theorem 2.2 and Corollary 2.3, confirms the printed implication

$$H^1(X, \mathbb{Z}) = H^2(X, \mathbb{Z}) = 0, \quad a(X) = 1 \Longrightarrow c_3(X) \leq 0, \qquad X \simeq S^6 \Longrightarrow a(X) = 0,$$

under the stated compact-threefold and algebraic-reduction hypotheses. Its Proposition 2.4 identifies the obstruction with the vanishing of $R^2 f_*(T_X \otimes L)$, equivalently the fibrewise dual spaces involving $H^0(\Omega^1_X \otimes L^*)$. The workbench's conductor section, absolute-fibre duality, and analytic base-change propositions prove a nonzero singular-fibre contribution for the reconstructed candidate; this is the exact point at which the earlier conditional objection is replaced by a candidate-specific calculation.

The same primary source confirms that Proposition 2.4 is logically separate from the defective attempts to establish its premise. The workbench now proves that premise-to-conclusion implication without importing either defect. If one L has $R^2 f_*(T_X \otimes L) = 0$, projection formula and Serre vanishing on the base produce L_0 with $H^2(X, T_X \otimes L_0) = 0$; generic endpoint vanishing and semicontinuity then give

$$H^i(X, T_X \otimes M) = 0 \quad (i \neq 1), \qquad \chi(X, T_X \otimes M) = \frac{1}{2} c_3(X) \leq 0.$$

This is a corrected sufficient theorem, not a reinstatement of Theorem 2.2: the candidate violates this relative sufficient condition for every L. Separately, $c_3(X) = 2$ and the corrected global theorem imply by contraposition that $H^2(X, T_X \otimes L_0) \neq 0$ for every L_0.

The reread of CDP (1998), Lemma 3.2, confirms a separate printed rank defect: for a split extension with fibre module $H_1(F)$ and monodromy group Q, the abelianization contributes the coinvariants $(H_1(F))_Q$, not the whole $H_1(F)$. The elementary example $\mathbb{Z} \rtimes_{-1} \mathbb{Z}$ has abelianization $\mathbb{Z} \oplus \mathbb{Z}/2$, which is the exact correction used in theorem 5.1.

The SGA 7 II nearby-cycle transcription supplies the specialization–variation sequence and the identity $\mathrm{Var}(\sigma)\alpha = \sigma - 1$; the Stacks/EGA transcriptions supply Leray functoriality and the precise proper/flat base-change hypotheses. Both are algebraic formal references, so neither is silently applied as an analytic existence theorem. The candidate's integral specialization map, saturation, and stalk values remain the explicit local calculations in theorem 53.1.

Finally, the independent Angella and McHugh transcriptions reproduce the known six-sphere Hodge inequalities and the Bott–Chern dimension identities. Substitution of the workbench values

$$(h^{0,1}, h^{0,2}, h^{1,0}, h^{2,0}, h^{1,1}, h^{1,2}) = (1, 0, 0, 0, 2, 1),$$
$$(h^{1,1}_{BC}, h^{1,2}_{BC}, h^{2,2}_{BC}) = (2, 1, 2)$$

agrees with the explicit nine-generator square-zero algebra. This is an external consistency check, not a substitute for the candidate-specific quotient, gluing, and period proofs.

4.2 Stable-stem audit of the sphere-valued carriers (3 September 2026)

Let

$$g = \widehat{q}_3 : (S^6, *) \longrightarrow (S^3, *), \qquad f = \widehat{q}_\gamma : (S^6, *) \longrightarrow (S^2, *)$$

be the based sphere-valued quotients obtained after the chosen based identification of X with S^6.

Hatcher's full low-dimensional homotopy-group table [10, p. 339] gives

$$\pi_6(S^3) = \mathbb{Z}/12, \qquad \pi_{12}(S^6) = \mathbb{Z}/2.$$

In particular $12g = 0$. Toda's Proposition 5.11 [13, Proposition 5.11] identifies the latter group as

$$\pi_{12}(S^6) = \mathbb{Z}/2\{\nu_6 \circ \nu_9\}.$$

The same proposition gives $\pi_{n+6}(S^n) = \mathbb{Z}/2\{\nu_n \circ \nu_{n+3}\}$ for $n \geq 5$. Suspension carries each displayed generator to the next displayed generator. Hence the stabilization homomorphism

$$\mathrm{st} : \pi_{12}(S^6) \longrightarrow \pi_6^{\mathrm{S}}$$

is an isomorphism. Isaksen–Wang–Xu's stable-stem table [12, Table 1, p. 113] identifies

$$\pi_3^{\mathrm{S}} = \mathbb{Z}/24\{\nu\}, \qquad \pi_6^{\mathrm{S}} = \mathbb{Z}/2\{\nu^2\}.$$

Since $12\,\mathrm{st}(g) = 0$ in $\mathbb{Z}/24\{\nu\}$, there is an $a \in \mathbb{Z}/12$ such that

$$\mathrm{st}(g) = 2a\nu.$$

Stabilization is multiplicative for smash products, so

$$\mathrm{st}(g^{\wedge 2}) = \mathrm{st}(g)^2 = 4a^2\nu^2 = 0 \in \pi_6^{\mathrm{S}}.$$

Injectivity of the preceding stabilization isomorphism now gives $[g^{\wedge 2}] = 0 \in \pi_{12}(S^6)$. Thus every based map $S^6 \to S^3$, and in particular $\widehat{q}_3$, has based null-homotopic smash square. Smashing this based null-homotopy with further copies of g also gives

$$[g^{\wedge r}] = 0 \in \pi_{6r}(S^{3r}) \qquad (r \geq 2).$$

For the second quotient, Hatcher's stable-range corollary [10, Corollary 4.24, p. 360] says that $\pi_{n+k}(S^n)$ has reached the stable group π_k^{S} when $n > k+1$. Applying this with $n = 8$ and $k = 4$, and then using the empty $k = 4$ row of Isaksen–Wang–Xu's complete stable-stem table [12, Table 1, p. 113], gives the exact equality

$$\pi_{12}(S^8) \cong \pi_4^{\mathrm{S}} = 0.$$

Consequently the sixfold reduced suspension

$$\Sigma^6 f = f \wedge \mathrm{id}_{S^6} : S^6 \wedge S^6 \longrightarrow S^2 \wedge S^6$$

is based null-homotopic. Under the canonical associativity and symmetry identifications of based sphere smashes, the smash square factors as

$$f^{\wedge 2} = (\mathrm{id}_{S^2} \wedge f) \circ (f \wedge \mathrm{id}_{S^6}) : S^6 \wedge S^6 \longrightarrow S^2 \wedge S^2.$$

Its right-hand factor is $\Sigma^6 f$, so this factorization supplies a based null-homotopy of $f^{\wedge 2}$. For every $r > 2$, associativity gives

$$f^{\wedge r} = f^{\wedge 2} \wedge f^{\wedge (r-2)},$$

and smashing the preceding based null-homotopy with the based map $f^{\wedge(r-2)}$ proves

$$[f^{\wedge r}] = 0 \in \pi_{6r}(S^{2r}) \qquad (r \geq 2).$$

Hence both already constructed continuous surjective quotients

$$\widehat{q}_3^{\wedge 2} : S^{12} \twoheadrightarrow S^6, \qquad \widehat{q}_\gamma^{\wedge 4} : S^{24} \twoheadrightarrow S^8$$

are based null-homotopic. Neither calculation determines the one-copy class $[g] \in \pi_6(S^3)$ or $[f] \in \pi_6(S^2)$. Both arguments use the sphere-valued postcompositions essentially, so they give no null-homotopy of any map into a conically stratified target; those based homotopy classes remain open. Point-set surjectivity, quotient topology, the embedded moving four-spheres, and their intrinsic metric limits are unaffected by nullity of the global sphere-valued maps.

4.3 Effective counting for the exact Koide sector

The effective assertion used in the Koide chapter is matched to its literal domain rather than transferred from a smoother or radial problem. After the five defining walls and the height $H(a, q, n) = 4a + 17n$ are checked directly against Definition 13.1, Benoist–Oh, Theorem 13.2 [54], applies to each of the finitely many integral orbits and proves

$$R(X) = C_0 X + O(X^{1-\eta}) \qquad (\eta > 0 \text{ fixed but unspecified}).$$

Neither the statement nor the proof of that theorem supplies $\eta \geq \frac{1}{2}$.

The hyperbolic literature gives two stronger exponents for different domains. Liu–Sarnak, Proposition 3.2 [57], gives the smooth standard-radial exponent $1/2 + 7/64 = 39/64$. The exact order group is a congruence arithmetic Fuchsian group because, for a maximal overorder $\mathcal{M}$,

$$\mathcal{M}^1(18\mathcal{M}) \subset \mathcal{O}^1 \subset \mathcal{M}^1;$$

Jacquet–Langlands and Kim–Sarnak give $\lambda_1 \geq 975/4096$ for that input [58, 59]. Good's sharp radial theorem and its Chatzakos–Petridis formulation [55, 56] instead have a $2/3$ error after the small-spectrum terms are displayed. The exact radial Huber variable is $7a + 15\sqrt{2}\,n$, whereas the Koide count uses $4a + 17n$ together with five sharp walls. Consequently neither radial theorem transfers its exponent to the target count. A pointwise square-root remainder for the unchanged domain remains open; the complete hypothesis-by-hypothesis audit is section 51.1.

4.4 Constrained dynamics and symplectic reduction

The worldline calculation in section 27 uses three primary sources for its general reduction framework. Dirac's original constrained-Hamiltonian paper [71] supplies the distinction between regular and singular constraint dynamics and motivates retaining the multiplier and all primary constraint equations. Gotay–Nester–Hinds [72] supplies the presymplectic constraint algorithm: one first restricts the closed two-form to the constraint locus, determines its characteristic distribution, and checks whether successive consistency conditions or rank jumps occur. Marsden–Weinstein [73] supplies the regular symplectic quotient by a Hamiltonian symmetry action.

Their role here is structural, not candidate-specific evidence. None of these sources computes the Gram morphism $P_\alpha(Z) = \alpha Z^* Z$, its $SU(3)$-orbits and stabilizers, the rank-one zero of $\mathrm{d}(\det \circ P_\alpha)$, the apex branch, or the hard-constrained packet's threshold rank jump. Those data and the exact cotangent sign are established directly in the cited workbench module. In particular, regular Marsden–Weinstein reduction applies to the massive rank-two stratum and to the intrinsic punctured rank-one stratum after their characteristic quotients have been proved regular; it does not authorize an extension of that quotient through the irregular zero-momentum apex.

Quaternionic linear algebra for the 4 September continuation

The local primary corpus was searched for quaternionic determinant and Jordan-cone sources before promoting the two new Pro packets. The available Alesker paper, *Non-commutative linear algebra and plurisubharmonic functions of quaternionic variables*, arXiv:math/0104209, Theorem 1.1.9, gives the Moore determinant restriction and congruence law. Wei Wang, *The linear algebra in the quaternionic pluripotential theory*, arXiv:1805.07759v1, Propositions 2.1–2.3, supplies the corresponding spectral theorem and complex matrix representation. Its primary LaTeX source was retrieved and inspected. The sign conversion from its representation to the present one is conjugation by $\mathrm{diag}(I, -I)$, stated explicitly in Section 24. These classical results are cited as background. The marked thirteen-slice identity, nonunitary product coefficient, and scalar normal-force witness are calculated explicitly here rather than attributed to those papers.

For the scalar-to-gauge comparison, David Tong's *Lectures on Gauge Theory*, Section 2.2.1, equation (2.27), identifies the A_0 equation that must be retained after choosing temporal gauge: `https://www.damtp.cam.ac.uk/user/tong/gaugetheory/gt.pdf`. The primary PDF was obtained and the surrounding derivation read; the local corpus was first checked for the corresponding moment-map and constraint references. This workbench independently varies its own anti-Hermitian A_0 and derives the matter moment $[\Phi, \dot{\Phi}] + b\dot{b}^* - \dot{b}b^*$, so no coefficient or sign is transferred from Tong's distinct field convention. The exact moving-frame transformation retains the resulting nonzero Gauss residual.

For the vacuum-orbit comparison, the primary LaTeX of Andrea Gambioli, *Eight-dimensional $SU(3)$-manifolds of cohomogeneity one*, arXiv:math/0611796v1 (27 November 2006), was retrieved and its $\mathbb{HP}^2$ example read: `https://arxiv.org/abs/math/0611796`. This is classical cohomogeneity-one geometry, not a new classification claimed here. The displayed family and its invariant $\|b\|^2$ are verified directly in Proposition 24.35; the exact bridge proved there is the Haar average of the marked Lorentz determinant. The source archive and extracted TeX SHA-256 values are retained in the supplement's `GAUGE_METRIC_SOURCE_RECEIPT.md`, together with the separate Tong source used to check retention of Gauss' equation.

4 September: source roles in the integrated kinetic and metric continuation

The canonical quaternionic proof now also uses the classical gauge-invariant vector-operator context in Tong, *Lectures on Gauge Theory*, Section 2.7.3, printed pages 107–108, `https://www.damtp.cam.ac.uk/user/tong/gaugetheory/gt.pdf`. The model-specific eight-eigenvalue inertia calculation, all transverse field equations including Gauss, conserved current, coefficient $-4\|b\|^2$, and relation $\mu_{\mathrm{d}}^2 = (2g^2/3)\|b\|^2$ are

proved directly in Theorem 24.41 and Proposition 24.42; they are not imported from Tong's different scalar example. Gambioli's Section 4 remains background for the orbit types only. The original kinetic norm, nonzero normal acceleration, exact current-free periodic continuation, and null waves are independently proved in Propositions 24.37, 24.38, Theorem 24.39, and Proposition 24.40.

The flat-torus conventions recorded in the integrated metric section are A. Gogolev, *Spectra of flat tori*, Section 2 (27 November 2017), `https://andreygogolev.wordpress.com/2017/11/27/spectra-of-flat-tori/`, and V. Ivrii, *Partial Differential Equations*, Appendix 4.B, `https://www.math.utoronto.ca/courses/apm346h1/20181/PDE-textbook/Chapter4/S4.B.html`. The present integer-valued dual is $\mathcal{N}^{-\mathsf{T}}\mathbb{Z}^6$; Ivrii incorporates 2π in the dual instead. U. Lang, *Lecture Notes on Riemannian Geometry*, 16 June 2020, Definition 4.1 and Proposition 4.4, printed pages 45 and 47, `https://metaphor.ethz.ch/x/2020/fs/401-3532-08L/sc/DG2_16June2020.pdf`, supplies the standard horizontal-isometry definition of a Riemannian submersion. The specialized proofs remain in Theorems 24.12–24.14 and Proposition 24.15: W is a lattice cometric, $h = 2H^{-1}$, native $h = I$ gives $H = 2I$, and the original source/quotient lattice gaps have leading constants $4\pi^2$ and $8\pi^2$ over y^2. These are source-convention pointers to the integrated proofs, not claims of a new independent source retrieval in this metadata pass or of a physical quantum spectral conclusion.

4 September: finite homogeneous quantization. The complete proof in Section 25 uses the classical bosonic-matrix mechanism of B. Simon, *Some quantum operators with discrete spectrum but classically continuous spectrum*, Annals of Physics **146** (1983), 209–220, `https://doi.org/10.1016/0003-4916(83)90057-X`. The original author-hosted scan was read, especially Section 2 and Section 7, equation (3) and Corollary 4. The present proof gives a direct slice-oscillator and finite-cell argument in the original $-\operatorname{tr}$ metric, rather than invoking a matrix-model discreteness theorem. The root-frequency constant is $2\sqrt{6}$, including eigenvalue walls.

The bounded-below Brownian-bridge formula is Theorem 1.1 of B. Simon, *A Feynman–Kac Formula for Unbounded Semigroups*, arXiv:math-ph/9907022v1 (1999), `https://arxiv.org/abs/math-ph/9907022`. Its original LaTeX source was retrieved and read. The proof here retains the anisotropic diffusion matrix and mass-volume measure. D. Tong, *Lectures on Gauge Theory*, Section 2.2.1, equations (2.27)–(2.31), `https://www.damtp.cam.ac.uk/user/tong/gaugetheory/gt.pdf`, supplies the Gauss-after-temporal-gauge convention. All triplet, adjoint and volume factors are derived here, not copied from a different gauge normalization. The corresponding infinite-volume and continuum interacting spectral question remains open.

4 September: moving Fourier-state comparison and retained Gauss law. The original TeX of G. Panati, H. Spohn and S. Teufel, *The time-dependent Born–Oppenheimer approximation*, ESAIM: M2AN **41** (2007), 297–314, `https://arxiv.org/abs/0712.4369`, `https://doi.org/10.1051/m2an:2007023`, was retrieved and read for the projected Hamiltonian, Berry and Born–Huang terms, and the Duhamel/invariance distinction in Sections 1–2.1. In particular compression alone does not justify a dynamical approximation or higher-order expansion. Section 26 therefore proves an explicit residual and fixed-physical-time estimate directly for the retained quartic two-mode operator; no general Born–Oppenheimer theorem is asserted as new or invoked with unchecked hypotheses. The positive ground-state and parameter regularity arguments are included. Tong's cited Section 2.2.1 was also reread: fixing temporal gauge does not remove Gauss. The complex rotating-pair proof above retains it with the original anti-Hermitian conventions.

5 September: exact coupling elimination and spatial projection. The Sylvester map in (906) is the classical polynomial map $(A, B) \mapsto AF + BG$; the primary Stacks Project text, `https://stacks.math.columbia.edu/tag/00UA`, was checked for its matrix description and coprimality criterion. The proof here includes the degree-drop argument, the explicit retained coefficient ring, the nonzero determinant

witness and the elementary measure-zero proof. No resultant theorem is presented as new. The new applications are the exact degree-twenty-eight residual polynomial, its complete coupling-fibre inverse, and the nonzero spatial-projection polynomial for the specified rotating gauge ansatz. All metric, Gauss and scalar conventions are derived in Section 24; they are not inferred from a different model or an assumed quantum spectral comparison.

5 September: two-cross residual certificate. The local primary corpus was checked, including David Tong's *Gauge Theory*, printed p. 40, equations (2.27)–(2.28), `https://www.damtp.cam.ac.uk/user/tong/gaugetheory/gt.pdf`, on retaining the temporal Gauss constraint. The full matter-current equations used here are derived in the retained-system subsection, not imported from the pure-gauge example. The primary Stacks Project Example 10.143.12, Tag 00UA, supplies the classical Sylvester-map context. The degree-drop kernel, block factor, positive residual identity and Cauchy–Schwarz constants in Section 24.27 are proved directly for the literal data. Neither standard linear algebra nor Gauss law is represented as a new discovery or a quantum conclusion.

5 September: the modulo-840 lattice and its full Weyl quotient. Cheng–Duncan–Harvey [74, Table 3, arXiv p. 21] already record the group $GL_2(3)$ for the $A_5^4D_4$ Niemeier lattice; their equation (2.30) defines the umbral group as the automorphism group modulo the Weyl group. This abstract identification is not claimed as new. Section 31 proves the particular residue-indexed index-72 glue, its exact minimum table, the transporter from the earlier glue, and the signed code isomorphism with $GL_2(\mathbb{F}_3)$. Theorems 31.15 and 31.16 exhaust all lattice isometries as wL_g and place the literal residue six-cycle in the Weyl group, centralizing this chosen complement, with every fibre origin retained.

Mare–Willems [75, Definition 2.2, Propositions 2.4 and 2.7, and equations (5.4)–(5.6)] supply the flag $F_4/\operatorname{Spin}(8)$ and its vector/even-half-spin/odd-half-spin tangent modules. The marked dual classes v, s, c are matched to their shortest weight sets by the coordinate proof in Theorem 31.4. The subsequent exact isometry in Theorem 32.4 maps the entire rank-24 real span into the original three octonion sectors, with inverse, glue test and the specified cyclic intertwining equation. The retained residual and Foundation sources [76, 77, 78] provide the earlier shell and glue data. Their exact excerpts and snapshot digests accompany the source package. The raw count and complementary-divisor quotient are separately proved, with $A_3(2521) = 0$, $A_3(3361) = 6$ despite equal prime residues modulo 840. The Fourier map retains the actual finite prime sample and its counts; neither universal selector positivity nor a physical model is asserted by these lattice and counting maps.

6 September: the Leech and full triality comparisons. Conway–Sloane [79, pp. 277–280] give the holy coefficient construction. Borcherds [80, Sections 6–7] proves uniqueness and the uniform construction theorem. Theorem 32.1 now writes that construction for the actual 72 marked glue words: $\rho^2 = 84$, the two index-six inclusions, all shifted-class minima and every norm-two exclusion are proved in the original Euclidean coordinates. The original F_4 Coxeter matrix also records its full D_4 discriminant action, including $Cv_D = c_D + (1, -1, 0, 0)$, not equality of the chosen representatives. Yokota [23, Sections 1.10, 1.14, 1.16; Theorem 2.7.1] fixes the related-triple and Albert conventions. The workbench proves the complete Peirce table, the determinant coefficients, exact Clifford maps, and global associated-bundle descent in those conventions. Mare–Willems [75, Appendix B, p. 519; orientation convention pp. 495–496] supplies the real-flag cell theorem. The explicit finite exponential, inverse chart, orientation sign, quotient continuity and homology maps are all computed here. The Hopf pullback and triality reduction use the previously stated retained H and based source identification; they do not silently recertify the global complex-threefold construction.

5 Exact integral algebra of the printed monodromy data

This section replaces opaque computational status reports by certificates that can be checked by hand. Let

$$V = \mathbb{Z}\langle \gamma, u, w, \delta \rangle, \qquad \Lambda = V^* = \mathbb{Z}\langle \widehat{\gamma}, \widehat{u}, \widehat{w}, \widehat{\delta} \rangle.$$

Vectors are columns in the displayed ordered bases. The printed monodromy on V is

$$T_1 = \begin{pmatrix} 1 & 0 & -6 & 2 \\ 0 & -1 & 1 & 1 \\ 0 & -1 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad T_2 = \begin{pmatrix} 1 & 6 & 0 & -3 \\ 0 & 0 & -1 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \tag{3.1}$$

Put $T_0 = (T_1 T_2)^{-1}$ and let $A_j = (T_j^{-1})^t$, $M_0 = (T_0^{-1})^t$ act on Λ.

Proposition 5.1 (monodromy certificate)**.** ***[VERIFIED]*** *Direct integral multiplication gives*

$$T_1^3 = I, \quad T_2^4 = I, \quad T_0 = \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad (T_0 - I)^2 = 0,$$

and

$$A_1 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 6 & 0 & 1 & 0 \\ -6 & -1 & -1 & 0 \\ -2 & 1 & 0 & 1 \end{pmatrix}, \quad A_2 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ -6 & 1 & 0 & 0 \\ 3 & 0 & 1 & 1 \end{pmatrix}, \quad M_0 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{pmatrix},$$

with $A_1 A_2 M_0 = I$.

Proof. Multiplying the first two matrices in (3.1) gives

$$T_1 T_2 = \begin{pmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Its inverse is the displayed T_0. Thus $T_0 - I$ sends $(a, b, c, d)^t$ to $(d, -c, 0, 0)^t$, and applying it twice gives zero. The complete order certificate is

$$T_1^2 = \begin{pmatrix} 1 & 6 & -6 & -2 \\ 0 & 0 & -1 & 1 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad T_1^2 T_1 = I,$$

$$T_2^2 = \begin{pmatrix} 1 & 6 & -6 & 0 \\ 0 & -1 & 0 & 1 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad T_2^3 = \begin{pmatrix} 1 & 0 & -6 & 3 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad T_2^3 T_2 = I.$$

In particular $T_1 \neq I$ and $T_2^2 \neq I$. Transposing the integral inverses gives the three displayed contragredient matrices; finally $A_1 A_2 = (M_0)^{-1}$ by the definitions, hence $A_1 A_2 M_0 = I$. □

5.1 The cusp invariant, image, and coinvariant lattices

Theorem 5.2 (all exterior-power cusp lattices)**.** ***[VERIFIED]*** *The equality*

$$\ker(M_0 - I) = \operatorname{im}(M_0 - I) = \mathbb{Z}\langle \widehat{w}, \widehat{\delta} \rangle \tag{3.2}$$

holds as an equality of saturated sublattices of Λ*. On* V*,*

$$V^{T_0} = \mathbb{Z}\langle \gamma, u \rangle,$$

$$(\textstyle\bigwedge^2 V)^{T_0} = \mathbb{Z}\langle \gamma u, \gamma\delta, uw, \gamma w - u\delta \rangle, \qquad (\textstyle\bigwedge^3 V)^{T_0} = \mathbb{Z}\langle \gamma uw, \gamma u\delta \rangle. \tag{3.3}$$

Every displayed invariant lattice is saturated. Consequently the invariant ranks in degrees $0, 1, 2, 3, 4$ *are* $(1, 2, 4, 2, 1)$*.*

Proof. From the displayed matrix,

$$(M_0 - I)(a, b, c, d)^t = (0, 0, b, -a)^t.$$

Its kernel is given by $a = b = 0$, and its image consists of every vector $(0, 0, b, -a)^t$; this proves (3.2). The two generators are members of the chosen basis, so the lattice is saturated.

The action of T_0 on basis vectors is

$$T_0\gamma = \gamma, \quad T_0 u = u, \quad T_0 w = w - u, \quad T_0\delta = \delta + \gamma. \tag{3.4}$$

Thus a vector $a\gamma + bu + cw + d\delta$ is fixed exactly when $c = d = 0$. For degree two, write

$$x = a\gamma u + b\gamma w + c\gamma\delta + d\, uw + e\, u\delta + f\, w\delta.$$

Expanding (3.4) and collecting coefficients gives

$$(\textstyle\bigwedge^2 T_0 - I)x = (-b - e + f)\gamma u - f\gamma w - fu\delta.$$

Hence $f = 0$ and $e = -b$, with a, b, c, d arbitrary, which is exactly the middle lattice in (3.3). These are integral equations with a unit coefficient, so their solution lattice is saturated.

In the ordered basis $(\gamma uw, \gamma u\delta, \gamma w\delta, uw\delta)$,

$$(\textstyle\bigwedge^3 T_0 - I)(a, b, c, d)^t = (d, -c, 0, 0)^t.$$

The fixed lattice is therefore generated by the first two basis vectors and is saturated. Degrees zero and four are fixed because $\det T_0 = 1$. □

5.2 Global coinvariants

Let $G = \langle T_1, T_2 \rangle$ and, for a G-lattice L, write $L_G = L/((T_1 - I)L + (T_2 - I)L)$.

Theorem 5.3 (coinvariant certificates)**.** ***[VERIFIED]*** *The common invariant lattices are*

$$V^G = \mathbb{Z}\gamma, \qquad (\textstyle\bigwedge^2 V)^G = \mathbb{Z}q, \qquad (\textstyle\bigwedge^3 V)^G = \mathbb{Z}\gamma uw, \qquad (\textstyle\bigwedge^4 V)^G = \mathbb{Z}\,\mathrm{vol},$$

where $q = uw + 6\gamma\delta$ *and* $\mathrm{vol} = \gamma uw\delta$*. There are torsion-free identifications*

$$V_G = \mathbb{Z}[\delta], \qquad (\textstyle\bigwedge^2 V)_G = \mathbb{Z}[\gamma\delta], \qquad (\textstyle\bigwedge^3 V)_G = \mathbb{Z}[uw\delta], \tag{3.5}$$

and $[uw] = 6[\gamma\delta]$ *in degree two.*

Proof. Solving $(T_1 - I)v = 0$ and $(T_2 - I)v = 0$ in (3.1) gives

$$V^{T_1} = \langle \gamma, 2u + w + 3\delta \rangle, \qquad V^{T_2} = \langle \gamma, u + w + 2\delta \rangle.$$

If a vector belongs to both lattices, comparison of its u- and w-coordinates gives simultaneously $2r = t$ and $r = t$ for the two non-γ coefficients; hence $r = t = 0$. Thus $V^G = \mathbb{Z}\gamma$.

The higher common invariants follow from the same integral nullspace calculation. In the ordered bases

$$(\gamma u, \gamma w, \gamma\delta, uw, u\delta, w\delta), \qquad (\gamma uw, \gamma u\delta, \gamma w\delta, uw\delta),$$

the simultaneous equations $(\bigwedge^k T_1 - I)x = (\bigwedge^k T_2 - I)x = 0$ are respectively equivalent to

$$x_1 = x_2 = x_5 = x_6 = 0, \qquad x_3 = 6x_4,$$

and

$$x_2 = x_3 = x_4 = 0.$$

Thus the integral solution lattices are generated by $uw + 6\gamma\delta = q$ and γuw. Both generators are primitive. Finally $\det T_1 = \det T_2 = 1$, so the common degree-four invariant lattice is $\mathbb{Z}\,\mathrm{vol}$.

In V_G, columns of $T_1 - I$ and $T_2 - I$ give, among others,

$$u + w = 0, \qquad u - 3\gamma = 0, \qquad 2\gamma + u + w = 0, \qquad 2u + w = 0.$$

The first two relations give $w = -u$ and $u = 3\gamma$; the last two then give $2\gamma = 0$ and $3\gamma = 0$. Hence $\gamma = u = w = 0$. Every column has zero δ-coordinate, so no relation involves δ. This proves $V_G = \mathbb{Z}[\delta]$, including absence of torsion.

For degree two use the ordered basis

$$(\gamma u, \gamma w, \gamma\delta, uw, u\delta, w\delta).$$

In this basis the twelve columns of the relation map are

$$B_2 = \left[\begin{array}{cccccc|cccccc} -2 & 1 & 1 & -6 & 2 & -8 & -1 & -1 & 1 & -6 & 6 & -3 \\ -1 & -1 & 1 & -6 & 2 & -6 & 1 & -1 & 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 & 0 & -6 & 0 & 0 & 0 & 0 & 6 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & -2 & 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 0 & 0 & 0 & 0 & -1 & -1 & 0 & 0 & 0 & 0 & 1 & -1 \end{array}\right], \tag{3.6a}$$

where the two blocks are $\bigwedge^2 T_1 - I$ and $\bigwedge^2 T_2 - I$. Its column lattice is

$$\mathrm{im}[\,\textstyle\bigwedge^2 T_1 - I \mid \bigwedge^2 T_2 - I\,] = \langle \gamma u, \gamma w, u\delta, w\delta, uw - 6\gamma\delta \rangle. \tag{3.6}$$

For completeness, the third columns of the two blocks yield γu and $\gamma u + \gamma w$; their difference yields γw. After these are removed, the fifth and sixth columns give $2u\delta + w\delta$ and $u\delta + w\delta$, hence both $u\delta$ and $w\delta$; the remaining nonzero relation is $uw - 6\gamma\delta$. Conversely, substituting the twelve displayed columns of the two exterior powers shows that they all lie in the right side. After ordering the ambient basis as $(\gamma u, \gamma w, u\delta, w\delta, uw, \gamma\delta)$, the five generators have matrix $I_4 \oplus (1, -6)^t$. Its maximal minors have gcd 1, so (3.6) is saturated. The quotient is therefore freely generated by $[\gamma\delta]$, with $[uw] = 6[\gamma\delta]$.

For degree three, let $(a, b, c, d) = (\gamma uw, \gamma u\delta, \gamma w\delta, uw\delta)$. The full relation matrix is

$$B_3 = \left[\begin{array}{cccc|cccc} 0 & 0 & 1 & 2 & 0 & -1 & 0 & -3 \\ 0 & -2 & 1 & -6 & 0 & -1 & -1 & -6 \\ 0 & -1 & -1 & -6 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]. \tag{3.6b}$$

If $c_{i,r}$ denotes column r of the T_i block, then $c_{1,2} - c_{2,3} = -b$, $c_{2,3} = -b - c$, and $c_{1,3} = a + b - c$. Hence a, b, c belong to the column lattice. Conversely every displayed column lies in $\langle a, b, c \rangle$, and the fourth row is zero. The relation lattice is therefore exactly the primitive summand $\langle a, b, c \rangle$, so the quotient is $\mathbb{Z}[uw\delta]$ with no hidden torsion. □

5.3 Finite quotient fibres

Put

$$\eta_1 = 2u + w + 3\delta, \qquad \eta_2 = u + w + 2\delta, \qquad \mathbf{q} = uw + 6\gamma\delta,$$

$m_1 = 3$, $m_2 = 4$, and $v_1 = (1, 2, -4, 0)^t$, $v_2 = -(1, 3, -3, 0)^t$ in Λ.

Proposition 5.4 (finite-fibre pullback indices)**.** ***[VERIFIED]*** *For the free affine actions and finite fibres geometrically realized in theorem 10.9, one has*

$$S_j = A_0^{(j)}/\langle x \mapsto A_j x + v_j/m_j \rangle$$

with $H^*(A_0^{(j)}; \mathbb{Z}) = \bigwedge^* V$, the degree-$m_j$ covering $\pi_j : A_0^{(j)} \to S_j$ has injective pullback with indices, in degrees 0 through 4,

$$(1, 3, 1, 1, 3), \qquad (1, 4, 2, 2, 4). \tag{3.7}$$

Moreover

$$\pi_1^* H^1(S_1) = \langle 3\gamma, \eta_1 \rangle, \qquad \pi_2^* H^1(S_2) = \langle 4\gamma, \eta_2 \rangle,$$
$$\pi_1^* H^2(S_1) = (\textstyle\bigwedge^2 V)^{T_1}, \qquad \pi_2^* H^2(S_2) = \{a\gamma\eta_2 + b\mathbf{q} : a \equiv b \pmod 2\}. \tag{3.8}$$

The identification of these abstract lattices with the actual finite-fibre stalks is theorem 10.10; hence no geometric hypothesis remains open in this proposition.

Proof. The fundamental group is the extension

$$\Gamma_j = \langle \Lambda, g \mid g\lambda g^{-1} = A_j\lambda,\ g^{m_j} = v_j \rangle.$$

After abelianization,

$$H_1(S_j) = (\Lambda_{A_j} \oplus \mathbb{Z}g)/\mathbb{Z}([v_j], -m_j). \tag{3.9}$$

The matrices $A_j - I$ have Smith form $\mathrm{diag}(1, 1, 0, 0)$, so Λ_{A_j} is free of rank two. The relation in (3.9) is primitive: if a prime divides m_j, its image $[v_j]$ modulo that prime is nonzero because $\gamma((A_j - I)\Lambda) = 0$ and $\gamma(v_1) = 1$, $\gamma(v_2) = -1$. Thus $H_1(S_j)$ is free of rank two. The quotient is a closed oriented four-manifold (the linear parts have determinant one) and $e(S_j) = e(A_0^{(j)})/m_j = 0$. Poincaré duality and the universal coefficient theorem therefore give

$$H_k(S_j; \mathbb{Z}) = (\mathbb{Z}, \mathbb{Z}^2, \mathbb{Z}^2, \mathbb{Z}^2, \mathbb{Z})$$

with no torsion. Indeed, Poincaré duality gives $H_3(S_j) \simeq H^1(S_j) \simeq \mathbb{Z}^2$ and $H^3(S_j) \simeq H_1(S_j) \simeq \mathbb{Z}^2$. The universal coefficient sequence for H^3 then forces $\mathrm{Ext}(H_2(S_j), \mathbb{Z}) = 0$, so $H_2(S_j)$ is torsion-free; Euler characteristic zero gives its rank as two.

Transfer gives $\pi_{j*}\pi_j^* = m_j$; hence π_j^* is injective and its image lies in the T_j-invariants. The Lyndon–Hochschild–Serre spectral sequence for $1 \to \Lambda \to \Gamma_j \to \mathbb{Z}/m_j \to 1$ [28, Chapter VII, §6] has

$$E_2^{p,q} = H^p(\mathbb{Z}/m_j; \textstyle\bigwedge^q V).$$

It computes $H^*(S_j; \mathbb{Z})$ because the universal cover of this affine torus quotient is $\mathbb{R}^4$, hence contractible. Its transgression on an invariant $\xi \in V$ evaluates the extension class: $d_2\xi = \xi(v_j) \bmod m_j$. Since $V^{T_j} = \langle \gamma, \eta_j \rangle$ and $\eta_j(v_j) = 0$, the degree-one images and indices in (3.8) follow.

In degree two, the integral invariant lattices are

$$(\textstyle\bigwedge^2 V)^{T_j} = \langle \gamma\eta_j, \mathbf{q} \rangle$$

with no omitted finite-index overlattice. Indeed, writing an invariant as (a,b,c,d,e,f) in the ordered basis $(\gamma u, \gamma w, \gamma\delta, uw, u\delta, w\delta)$, the T_1 equations are $e=f=0$, $a=2b$, $c=3b+6d$, while the T_2 equations are $e=f=0$, $a=b$, $c=2b+6d$. Using $\gamma uw\delta$ as orientation,

$$(\gamma\eta_j)^2 = 0, \quad (\gamma\eta_1)\mathbf{q} = 3, \quad (\gamma\eta_2)\mathbf{q} = 2, \quad \mathbf{q}^2 = 12. \tag{3.10}$$

Let $L_j = \pi_j^* H^2(S_j)$. The intersection lattice of S_j is unimodular, and pullback multiplies its form by m_j. Hence $|\det L_j| = m_j^2$. The two invariant Gram determinants from (3.10) are 9 and 4. If k_j is the index of L_j in the invariant lattice, $k_j^2 |\det(\bigwedge^2 V)^{T_j}| = m_j^2$, so $k_1 = 1$ and $k_2 = 2$.

For $j=2$, multiplicativity of the spectral sequence gives $d_2(\gamma\eta_2) = -[\eta_2] \neq 0$ in $\langle \gamma, \eta_2\rangle / \langle 4\gamma, 2\eta_2\rangle$. Thus $\gamma\eta_2 \notin L_2$. The remaining index-two possibilities are $L_2 = \langle 2\gamma\eta_2, a\gamma\eta_2 + \mathbf{q}\rangle$, $a \in \{0,1\}$. The covering is étale, so $\pi_2^* K_{S_2}$ is trivial and $K_{S_2}^{\otimes 4}$ is trivial. Because $H^2(S_2;\mathbb{Z})$ is torsion-free, $c_1(S_2) = 0$ and $w_2(S_2) = 0$; the intersection form is even. Dividing the pullback pairings by four gives

$$\begin{pmatrix} 0 & 1 \\ 1 & a+3 \end{pmatrix},$$

so $a+3$ is even and $a = 1$. This proves (3.8).

For clarity, the displayed differential contains no suppressed coefficient calculation. For the cyclic group generated by T_2,

$$H^2(\mathbb{Z}/4; V) = V^{T_2}/(I + T_2 + T_2^2 + T_2^3)V = \langle \gamma, \eta_2\rangle / \langle 4\gamma, 2\eta_2\rangle.$$

The extension class gives $d_2\gamma = \gamma(v_2) = -1$ and $d_2\eta_2 = \eta_2(v_2) = 0$; the derivation rule therefore gives $d_2(\gamma\eta_2) = -[\eta_2]$, which is nonzero because $\eta_2 \notin \langle 4\gamma, 2\eta_2\rangle$.

In degree four the index is the covering degree. For degree three, direct exterior calculation gives the primitive invariant bases

$$(\textstyle\bigwedge^3 V)^{T_1} = \langle \gamma uw,\ uw\delta - 4\gamma u\delta - 2\gamma w\delta\rangle,$$
$$(\textstyle\bigwedge^3 V)^{T_2} = \langle \gamma uw,\ uw\delta - 3\gamma u\delta - 3\gamma w\delta\rangle.$$

Against (γ, η_j) their pairing matrices are

$$\begin{pmatrix} 0 & 1 \\ -3 & 0 \end{pmatrix}, \qquad \begin{pmatrix} 0 & 1 \\ -2 & 0 \end{pmatrix},$$

so the absolute determinants are 3 and 2. If the degree-three index is r_j, the degree-one indices already proved are 3 and 4. Pullback multiplies the perfect $H^1(S_j) \times H^3(S_j)$ pairing by m_j, whose rank-two determinant is m_j^2. Hence $3 \cdot 3r_1 = 3^2$ and $2 \cdot 4r_2 = 4^2$; hence $r_1 = 1$, $r_2 = 2$. Together with degrees zero through two this proves (3.7). □

5.4 The terminal presentation and degree-two map

Proposition 5.5 (terminal group, including the nonabelian step)**.** ***[VERIFIED]*** *Consider*

$$G(\ell_0,\ell_1,\ell_2) = \langle c, x, y \mid c \text{ central},\ xy = c^{\ell_0},\ x^3 = c^{\ell_1},\ y^4 = c^{\ell_2}\rangle$$

and put $p = 12\ell_0 - 4\ell_1 - 3\ell_2$. *If* $p \neq 0$, *then* $G \cong \mathbb{Z}/|p|\mathbb{Z}$; *if* $p = 0$, *then* $G \cong \mathbb{Z}$. *For* $(\ell_0,\ell_1,\ell_2) = (0,1,-1)$ *one has* $p = -1$ *and* $G = 1$.

Proof. The relation $xy = c^{\ell_0}$ gives $y = x^{-1}c^{\ell_0}$. Since c is central, x and y commute, so the apparently nonabelian presentation is already abelian. In the ordered additive generators (x, c) the remaining relations form

$$R = \begin{pmatrix} 3 & -\ell_1 \\ 4 & \ell_2 - 4\ell_0 \end{pmatrix}, \qquad \det R = -p.$$

The greatest common divisor of all entries of R is 1 because it divides both 3 and 4. Thus its Smith form is $\mathrm{diag}(1, |p|)$ when $p \neq 0$ and $\mathrm{diag}(1, 0)$ when $p = 0$. The stated groups follow. □

Proposition 5.6 (terminal degree-two lattice)**.** ***[VERIFIED]*** *Let $A : \mathbb{Z}^6 \to \mathbb{Z}^4$ have columns*

$$c_1 = (2,-4,-2,3)^t, \quad c_2 = (1,-2,-2,3)^t,$$
$$c_3 = (3,0,-4,0)^t, \quad c_4 = (0,1,0,-1)^t, \quad c_5 = c_6 = 0.$$

Then $\operatorname{im} A$ *is saturated of rank three,*

$$\operatorname{im} A = \ker(4,2,3,2), \qquad \operatorname{coker} A \cong \mathbb{Z}, \tag{3.11}$$

and $(4,2,3,2)$ *is the primitive left-kernel generator.*

Proof. The row $\Phi = (4,2,3,2)$ annihilates the four nonzero columns:

$$\Phi c_1 = 8-8-6+6 = 0, \quad \Phi c_2 = 4-4-6+6 = 0, \quad \Phi c_3 = 12-12 = 0, \quad \Phi c_4 = 2-2 = 0.$$

It is primitive since the gcd of its entries is 1, so $\ker \Phi \subset \mathbb{Z}^4$ is a saturated rank-three lattice. The 3×3 minor on rows $(1,2,3)$ and columns $(1,2,4)$ is 2, whereas the minor on rows $(1,2,4)$ and the same columns is -3. Consequently the ideal generated by all maximal minors contains $(2,3) = (1)$. This proves that A has rank three and saturated image. We have $\operatorname{im} A \subseteq \ker \Phi$ between rank-three lattices. The finite quotient $\ker \Phi / \operatorname{im} A$ injects into the torsion-free group $\mathbb{Z}^4 / \operatorname{im} A$, so it is zero. Thus equality holds, and the primitive surjection $\Phi : \mathbb{Z}^4 \to \mathbb{Z}$ induces (3.11). □

Proposition 5.7 (all-degree terminal homology package)**.** ***[VERIFIED]*** *Let $C_\bullet$ be the degree-by-degree terminal package printed for the glued six-manifold, and assume only the following algebraic outputs of its displayed maps:*

(i) $H_0(C_\bullet) = H_6(C_\bullet) = \mathbb{Z}$ *and the degree-one presentation is* $G(\ell_0, \ell_1, \ell_2)$ *of theorem 5.5;*

(ii) in degree two there is an exact sequence $0 \to \mathbb{Z} \to H_2^\circ \to \mathbb{Z} \to 0$*, the retained boundary lattice kills the second quotient, and its remaining generator maps to* p *times a primitive generator of the first copy of* $\mathbb{Z}$*;*

(iii) the package obeys integral Poincaré duality and the universal coefficient theorem in formal dimension six, and its Euler characteristic is 2*.*

Then, for $p \neq 0$*,*

$$(H_0(C_\bullet), \dots, H_6(C_\bullet)) \cong (\mathbb{Z}, \mathbb{Z}/|p|, \mathbb{Z}/|p|, \mathbb{Z}/|p|, \mathbb{Z}/|p|, 0, \mathbb{Z}). \tag{3.12}$$

In particular, the printed terminal package has the integral homology of S^6 *when* $p = -1$*.*

Proof. By theorem 5.5, the degree-one group is $\mathbb{Z}/|p|$. In degree two, exactness and the stated retained-boundary map first quotient H_2° down to the primitive first summand and then quotient that summand by p; hence $H_2(C_\bullet) = \mathbb{Z}/|p|$. Both groups are finite, so their Betti numbers vanish. Universal coefficients and duality give

$$H_5 \cong H^1 = 0, \qquad H_4 \cong H^2 \cong \operatorname{Ext}(H_1, \mathbb{Z}) = \mathbb{Z}/|p|.$$

The Euler-characteristic identity is therefore $2 = 1 - b_3 + 1$, so $b_3 = 0$. Finally

$$H_3 \cong H^3 \cong \operatorname{Ext}(H_2, \mathbb{Z}) = \mathbb{Z}/|p|,$$

which proves (3.12). □

Geometric boundary. The calculations in this section deliberately isolate the printed lattices, presentations, and matrices. Their geometric realization is proved separately: the finite-band and cusp maps are identified in theorem 58.10, the based fundamental-group calculation is theorem 58.9, and the comparison with the intrinsic Leray transgressions is theorem 58.13.

6 An exact Erdős–Straus/S6 terminal-matrix crosswalk

This section records an arithmetic relation suggested by comparing the candidate's terminal presentation with the divisor-shell coordinates used in the maintained Erdős–Straus corpus [70]. All maps below are proved over the integers. The conclusion is an exact change of relators; it is not a geometric map from the reciprocal cubic surface to the candidate complex threefold.

6.1 The first-half shell and its divisor coordinates

Proposition 6.1 (first-half denominator)**.** ***[VERIFIED]*** *Let $p \equiv 1 \pmod 4$ be an odd prime. If*

$$\frac{4}{p} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}, \qquad x, y, z \in \mathbb{Z}_{>0}, \qquad x \le y \le z,$$

then

$$\frac{p}{4} < x \le \frac{p}{2}.$$

Proof. The lower bound follows from $1/x < 4/p$. Suppose $x > p/2$. Clearing denominators and reducing modulo p gives

$$4xyz = p(xy + xz + yz), \qquad p \mid xyz.$$

If $p \mid x$, then $x, y, z \ge p$, so the sum of the three reciprocals is at most $3/p$. If $p \mid y$, then $y, z \ge p$, and

$$\frac{1}{x} + \frac{1}{y} + \frac{1}{z} < \frac{2}{p} + \frac{2}{p} = \frac{4}{p}.$$

Both alternatives are impossible. Hence $z = kp$ for an integer $k \ge 1$, and substitution gives

$$(4k - 1)xy = kp(x + y).$$

Neither x nor y is divisible by p, so $p \mid 4k - 1$. Write $4k - 1 = \ell p$. Since $p \equiv 1 \pmod 4$, one has $\ell \equiv 3 \pmod 4$, hence $\ell \ge 3$. Cancelling p gives $\ell xy = k(x + y)$. Since $x \le y$,

$$x \le \frac{2k}{\ell} = \frac{\ell p + 1}{2\ell} = \frac{p}{2} + \frac{1}{2\ell} < \frac{p + 1}{2}.$$

But the integer $x > p/2$ is at least $(p + 1)/2$, a contradiction. □

Fix henceforth a prime $p \equiv 1 \pmod{12}$, and put

$$L = \frac{p - 1}{4}, \qquad R_j = 4j + 3, \qquad a_j = \frac{p + R_j}{4}, \qquad 0 \le j < L.$$

The preceding proposition shows that every solution's smallest denominator lies in the exact candidate set $\{a_j : 0 \le j < L\}$. Notice that

$$4a_j = p + R_j, \qquad \gcd(R_j, p) = \gcd(R_j, a_j) = \gcd(p, a_j) = 1.$$

Proposition 6.2 (complete two-channel divisor selector)**.** ***[VERIFIED]*** *Define*

$$E_j = \{u \in \mathbb{Z}_{>0} : u \mid a_j^2,\ R_j \mid 4u + 1\},$$
$$M_j = \{u \in \mathbb{Z}_{>0} : u \mid a_j^2,\ R_j \mid 4u + p\}.$$

Then the Erdős–Straus equation for p has a positive integral solution if and only if at least one E_j or M_j is nonempty. For $u \in E_j$ set $\varepsilon = 0$, and for $u \in M_j$ set $\varepsilon = 1$. With

$$T = \frac{4u + p^{\varepsilon}}{R_j}, \qquad K = \frac{p^{1-\varepsilon}T + 1}{4},$$

the corresponding solution is

$$(x, y, z) = \left(a_j, \frac{p^{\varepsilon} a_j K}{u}, pK\right),$$

up to interchange of y and z.

Proof. Fix $x = a_j$, and abbreviate $R = R_j$, $a = a_j$, and $N = pa$. The remaining equation is

$$Ryz = N(y + z).$$

It is equivalent to

$$(Ry - N)(Rz - N) = N^2.$$

Both factors are positive. Since $\gcd(R, N) = 1$, positive solutions are in bijection with positive divisors $d \mid N^2$ satisfying

$$d \equiv -N \pmod{R};$$

the complementary divisor $e = N^2/d$ then also satisfies $e \equiv -N \pmod{R}$, and

$$y = \frac{N + d}{R}, \qquad z = \frac{N + e}{R}.$$

Because $\gcd(p, a) = 1$, the p-adic exponent of d is 0, 1, or 2. Complementation $d \leftrightarrow e$ exchanges exponents 0 and 2, so choose the exponent-two representative in that pair. If $d = p^2 u$, then $u \mid a^2$, and

$$R \mid d + N \iff R \mid pu + a \iff R \mid 4u + 1,$$

where $4a \equiv p \pmod{R}$ and p is invertible modulo R. This is the E_j channel. If $d = pu$, then

$$R \mid d + N \iff R \mid u + a \iff R \mid 4u + p,$$

which is the M_j channel. This proves completeness of the two selector sets.

For either channel, $R_j T = 4u + p^{\varepsilon}$. Modulo 4, this says $-T \equiv 1$, so $T \equiv 3 \pmod{4}$; the displayed K is therefore a positive integer. Direct calculation using $4a = p + R_j$ gives

$$R_j K = a + p^{1-\varepsilon} u, \qquad Ta = u + p^{\varepsilon} K.$$

Since $u \mid a^2$, the second identity implies $u \mid p^{\varepsilon} aK$. Finally,

$$\frac{u}{p^{\varepsilon} aK} + \frac{1}{pK} = \frac{p^{1-\varepsilon} u + a}{paK} = \frac{R_j}{pa} = \frac{4}{p} - \frac{1}{a},$$

which proves the asserted formula and its converse. □

6.2 The unimodular relation-matrix bijection

Put

$$\lambda(\ell_0, \ell_1, \ell_2) = 12\ell_0 - 4\ell_1 - 3\ell_2.$$

By theorem 5.5, the candidate's ordered terminal relation matrix in the ordered generator basis (x, c), for $\ell = (\ell_0, \ell_1, \ell_2)$, is

$$\mathsf{R}(\ell) = \begin{pmatrix} 3 & -\ell_1 \\ 4 & \ell_2 - 4\ell_0 \end{pmatrix}, \qquad \det \mathsf{R}(\ell) = -\lambda(\ell).$$

For $\varepsilon \in \{0, 1\}$, define the marked selector space

$$\mathcal{W}_{\varepsilon}(p) = \left\{ (j, u) \in \mathbb{Z} \times \mathbb{Z}_{>0} : \begin{array}{l} 0 \le j < L,\ u \in \mathbb{Z}_{>0},\ u \mid a_j^2, \\ R_j \mid 4u + p^{\varepsilon} \end{array} \right\}$$

and the normalized divisor-supported terminal slice

$$\mathcal{H}_{\varepsilon}(p) = \left\{ (j, \ell_0, \ell_1) \in \mathbb{Z}^3 : \begin{array}{l} 0 \le j < L,\ \lambda(\ell_0, \ell_1, 1) = p^{\varepsilon}, \\ T := 4\ell_0 - 1 > 0, \quad u := \ell_1 + jT > 0, \quad u \mid a_j^2 \end{array} \right\}.$$

Theorem 6.3 (exact marked selector/terminal-slice equivalence)**.** ***[VERIFIED]*** *The map*

$$\Theta_{\varepsilon,p} : \mathcal{W}_\varepsilon(p) \longrightarrow \mathcal{H}_\varepsilon(p), \qquad (j,u) \longmapsto \left(j, \frac{T+1}{4}, u - jT\right), \quad T = \frac{4u+p^\varepsilon}{R_j},$$

is a bijection. Under it, the two relation matrices are related by the explicit unimodular row operation

$$\begin{pmatrix} 1 & -j \\ 0 & 1 \end{pmatrix} \begin{pmatrix} R_j & -u \\ 4 & -T \end{pmatrix} = \begin{pmatrix} 3 & -\ell_1 \\ 4 & 1-4\ell_0 \end{pmatrix} = \mathsf{R}(\ell_0, \ell_1, 1),$$

and both determinants equal $-p^\varepsilon$*.*

Proof. For $(j,u) \in \mathcal{W}_\varepsilon(p)$, T is a positive integer and the congruence computation in theorem 6.2 gives $T \equiv 3 \pmod 4$, so $\ell_0 = (T+1)/4$ is integral. Set $\ell_1 = u - jT$. Since $R_j = 4j+3$,

$$\lambda(\ell_0, \ell_1, 1) = 3(T+1) - 4(u - jT) - 3 = (4j+3)T - 4u = p^\varepsilon.$$

The positivity and divisor conditions in the definition of $\mathcal{H}_\varepsilon(p)$ follow from $u = \ell_1 + jT$. Matrix multiplication gives the displayed identity, and

$$\det \begin{pmatrix} R_j & -u \\ 4 & -T \end{pmatrix} = -R_jT + 4u = -p^\varepsilon.$$

Conversely, from $(j, \ell_0, \ell_1) \in \mathcal{H}_\varepsilon(p)$ recover

$$T = 4\ell_0 - 1, \qquad u = \ell_1 + jT.$$

Expanding $\lambda(\ell_0, \ell_1, 1) = p^\varepsilon$ gives $R_jT = 4u + p^\varepsilon$. Thus $R_j \mid 4u + p^\varepsilon$, and the remaining positivity and divisor hypotheses are already part of $\mathcal{H}_\varepsilon(p)$. These constructions are literal inverses. □

The normalization $\ell_2 = 1$ automatically satisfies the candidate's odd ℓ_2 admissibility condition. It also forces $3 \nmid \ell_1$: if $3 \mid \ell_1$, then $\lambda(\ell_0, \ell_1, 1)$ is divisible by 3, whereas $p^\varepsilon \equiv 1 \pmod 3$. Write $\mathrm{ES}(p)$ for the assertion that $4/p$ is a sum of three positive integral unit fractions. Consequently

$$\mathrm{ES}(p) \Longleftrightarrow \mathcal{H}_0(p) \sqcup \mathcal{H}_1(p) \neq \varnothing.$$

The E-channel is the slice with terminal parameter $p_{\mathrm{S6}} = 1$; the M-channel is the slice with $p_{\mathrm{S6}} = p$.

Here the channel label is part of the datum. The raw selector subsets need not be disjoint: for $p = 13$, $j = 0$, and $u = 2$, one has $u \in E_0 \cap M_0$. Accordingly the occurrence statement above means the tagged disjoint union $\{0\} \times \mathcal{H}_0(p) \sqcup \{1\} \times \mathcal{H}_1(p)$. Moreover the map from tagged selector states to *unordered* unit-fraction solutions is not injective in the M-channel: complementation $u \leftrightarrow a_j^2/u$ interchanges y and z. For example, $p = 13$, $j = 0$, and $u = 2, 8$ give the same unordered solution $\{4, 26, 52\}$. The theorem is a bijection with the marked terminal slice, not a bijection with unordered solutions.

The tags are essential. For every eligible prime q, define the fixed-prime forgetful map

$$\mathfrak{f}_q : \mathcal{H}_0(q) \longrightarrow \mathbb{Z}^3, \qquad \mathfrak{f}_q(j, \ell_0, \ell_1) = (\ell_0, \ell_1, 1).$$

For $p = 13$, the two E-channel selector states satisfy

$$\Theta_{0,13}(0,2) = (0,1,2), \qquad \Theta_{0,13}(1,5) = (1,1,2),$$

so their marked images are distinct, while both are sent by $\mathfrak{f}_{13}$ to $(1,2,1)$. Thus forgetting j destroys injectivity. To compare different prime fibres, define

$$\mathfrak{F} : \bigsqcup_{\substack{q \text{ prime} \\ q \equiv 1 \pmod{12}}} (\{q\} \times \mathcal{H}_0(q)) \longrightarrow \mathbb{Z}^3, \qquad \mathfrak{F}(q; j, \ell_0, \ell_1) = (\ell_0, \ell_1, 1).$$

At $q = 73$, one has $\Theta_{0,73}(1,5) = (1,1,2)$, so this tagged image is also sent by $\mathfrak{F}$ to $(1,2,1)$. Hence forgetting the external prime loses still more information.

6.3 What the crosswalk proves, and what it does not

The theorem identifies the complete fixed-p, normalized, positively oriented, j-marked divisor selector with an arithmetic slice of the candidate's terminal relation matrices. It explains a genuine structural continuity between the earlier reciprocal-equation work and the terminal algebra used here: the common object is a 2×2 integral presentation, and the passage between the two displayed matrices is an element of $\mathrm{SL}_2(\mathbb{Z})$.

The orientation qualification is substantive. The geometrically validated candidate specialization used elsewhere in this workbench has terminal parameter $p_{\mathrm{S6}} = -1$, whereas the E-channel here has $p_{\mathrm{S6}} = +1$. No result in this section identifies those two oriented terminal data. Thus the exact arithmetic equivalence does not, by itself, place an Erdős–Straus selector state on the particular complex threefold reconstructed in the candidate-specific $p = -1$ branch.

It does not prove that the selector is nonempty for every prime, so it does not prove the Erdős–Straus conjecture. It also does not construct a map from the reciprocal cubic surface to X, identify selector divisors with cycles or attaching maps in the candidate geometry, or transfer the candidate's complex structure to the arithmetic surface. Those would be additional geometric assertions, not consequences of unimodular equivalence of relation matrices.

7 The exact marked Erdős–Straus–S^6 bridge

Current lattice and modular continuation

The shared continuation now proves the full cubic image $2\sqrt{2}(18\mathbb{Z} \cup (6\mathbb{Z}+2) \cup (6\mathbb{Z}-2))$ of the literal common Niemeier–Leech lattice, with explicit integral and local sections. The marked $A_5^4 D_4$ root system maps to the Cheng–Duncan–Harvey shadow $(5I_{12} + J)S_6$, where $J_{rs} = \mathbf{1}_{s=5r\ (12)}$. The proof calculates the original glue-group action on twisted shadows, the complete theta determinant character, and the coefficient map to $T_6 = \eta(\tau)^5\eta(3\tau)/(\eta(2\tau)\eta(6\tau)^5)$, including $T_6(-1/(6\tau)) = 72/T_6(\tau)$. See theorem 37.1 and section 38.

Result and scope

For each prime $p \equiv 1 \pmod{12}$, the complete first-half Erdős–Straus divisor selector is in explicit bijection with a normalized, positively oriented, divisor-supported, and shell-marked slice of the terminal parameter lattice occurring in the candidate complex structure on S^6. The coordinate change is integral in both directions. More strongly, the selector relation matrix and the S6 terminal relation matrix have exactly the same row lattice: they differ by a displayed element of $\mathrm{SL}_2(\mathbb{Z})$. Composing the bijection with the shell reconstruction gives positive denominators solving $4/p = 1/x + 1/y + 1/z$, and every positive solution is detected by at least one of the two tagged slices.

The nonlinear condition $u \mid a_j^2$, together with the external labels (p, j, ε), is indispensable. If it is removed, each normalized S6 fibre is an infinite affine line and carries no Erdős–Straus existence information. A particularly natural unmarked S6 point produces precisely the already-solved congruence class $p \equiv 13 \pmod{24}$ and produces no selector state when $p \equiv 1 \pmod{24}$. Thus the bridge is an exact arithmetic equivalence of marked presentations, not a proof of universal selector nonemptiness and not a geometric morphism from the reciprocal cubic to the complex threefold.

The separate continuation in Section 31 constructs the index-72 modulo-840 lattice $A_5^4 \perp D_4 \subset N$ and its exact marked D_4 triality maps. It realizes the known Weyl quotient by the coordinate splitting $O(N) = W(A_5^4 D_4) \rtimes L(G)$, $G \cong GL_2(\mathbb{F}_3)$, and places the residue six-cycle inside the Weyl group, centralizing this chosen complement (Theorems 31.15 and 31.16; [74, Table 3]). Its divisor and Fourier maps retain the prime, shell, and finite sample; in particular the shell count does not descend to the prime residue alone.

7.1 All objects and conventions

Fix throughout a prime

$$p \equiv 1 \pmod{12}, \qquad h = \frac{p-1}{12} \in \mathbb{Z}_{>0},$$

and write

$$L = \frac{p-1}{4} = 3h, \qquad J_p = \{j \in \mathbb{Z} : 0 \le j < L\}.$$

For $j \in J_p$, retain the four integers

$$R_j = 4j+3, \qquad a_j = 3h+j+1 = \frac{p+R_j}{4}, \qquad N_j = pa_j, \qquad 4a_j = p+R_j. \tag{1.1}$$

No label in (p, j) will be suppressed. The inequalities

$$3 \le R_j \le p-2, \qquad \frac{p+3}{4} \le a_j \le \frac{p-1}{2} < p$$

give

$$\gcd(p, a_j) = \gcd(p, R_j) = \gcd(a_j, R_j) = 1. \tag{1.2}$$

Indeed p is prime and $0 < a_j < p$; then $R_j = 4a_j - p$ gives the other two equalities.

Let

$$\mathscr{S}(p) = \left\{(x,y,z) \in \mathbb{Z}^3_{>0} : 4xyz = p(xy+xz+yz)\right\}.$$

The equation is symmetric, but whenever completeness is invoked we choose the ordered representative $x \le y \le z$. Write $\mathrm{ES}(p)$ for the assertion $\mathscr{S}(p) \neq \varnothing$.

For the S6 terminal parameters put

$$\lambda(\ell_0, \ell_1, \ell_2) = 12\ell_0 - 4\ell_1 - 3\ell_2 \tag{1.3}$$

and define the abstract terminal group

$$G(\ell_0, \ell_1, \ell_2) = \left\langle c, x, y \;\middle|\; c \text{ central},\ xy = c^{\ell_0},\ x^3 = c^{\ell_1},\ y^4 = c^{\ell_2} \right\rangle. \tag{1.4}$$

After eliminating y, and using the ordered generator basis (x, c), its ordered row-relation matrix is

$$\mathsf{R}(\ell_0, \ell_1, \ell_2) = \begin{pmatrix} 3 & -\ell_1 \\ 4 & \ell_2 - 4\ell_0 \end{pmatrix}, \qquad \det \mathsf{R} = -\lambda(\ell_0, \ell_1, \ell_2). \tag{1.5}$$

The letter p in this note always denotes the Erdős–Straus prime. The signed terminal value is denoted

$$q_\varepsilon = p^\varepsilon \in \{1, p\}, \qquad \varepsilon \in \{0, 1\}.$$

Thus no identification of the external prime p with the terminal value is made in the $\varepsilon = 0$ channel.

For any integral 2×2 matrix A, define its row-relation lattice and row-presented group by

$$\mathrm{Rel}(A) = \{v^{\mathsf{T}} A : v \in \mathbb{Z}^2\} \subseteq \mathbb{Z}^2, \qquad \mathcal{G}(A) = \mathbb{Z}^2 / \mathrm{Rel}(A) = \mathrm{coker}(A^{\mathsf{T}}). \tag{1.6}$$

This convention makes every matrix direction in the bridge explicit.

7.2 The complete first-half shell selector

Lemma 7.1 (first-half denominator)**.** *Let $p \equiv 1 \pmod 4$ be an odd prime. If $(x, y, z) \in \mathscr{S}(p)$ and $x \le y \le z$, then*

$$\frac{p}{4} < x \le \frac{p}{2}.$$

Proof. Positivity of the two reciprocals other than $1/x$ gives $1/x < 4/p$, hence $x > p/4$. Suppose $x > p/2$. Clearing denominators gives

$$4xyz = p(xy + xz + yz),$$

so $p \mid xyz$. If $p \mid x$, then $x, y, z \ge p$ and the reciprocal sum is at most $3/p$. If $p \mid y$, then $y, z \ge p$, while $1/x < 2/p$, so the sum is strictly less than $4/p$. Both cases are impossible. Therefore $z = kp$ for some $k \in \mathbb{Z}_{>0}$, while neither x nor y is divisible by p. Substitution yields

$$(4k - 1)xy = kp(x + y).$$

Reduction modulo p gives $p \mid 4k - 1$. Write $4k - 1 = rp$. Since $p \equiv 1 \pmod 4$, one has $r \equiv 3 \pmod 4$, hence $r \ge 3$. After cancelling p,

$$rxy = k(x + y).$$

As $x \le y$, division by xy gives

$$x \le \frac{2k}{r} = \frac{rp + 1}{2r} = \frac{p}{2} + \frac{1}{2r} < \frac{p + 1}{2}.$$

The integer $x > p/2$ must be at least $(p + 1)/2$, a contradiction. □

For fixed $j \in J_p$, define the ordered residual shell

$$\mathscr{P}_{p,j} = \{(y, z) \in \mathbb{Z}^2_{>0} : R_j yz = N_j(y + z)\} \tag{2.1}$$

and the oriented divisor shell

$$\mathscr{D}_{p,j} = \{d \in \mathbb{Z}_{>0} : d \mid N_j^2,\ d \equiv -N_j \pmod{R_j}\}. \tag{2.2}$$

Proposition 7.2 (oriented shell bijection)**.** *The maps*

$$\delta_{p,j} : \mathscr{P}_{p,j} \longrightarrow \mathscr{D}_{p,j}, \qquad (y, z) \longmapsto R_j y - N_j,$$

$$\rho_{p,j} : \mathscr{D}_{p,j} \longrightarrow \mathscr{P}_{p,j}, \qquad d \longmapsto \left(\frac{N_j + d}{R_j}, \frac{N_j + N_j^2/d}{R_j}\right)$$

are mutually inverse. The involution

$$\iota_{N_j}(d) = \frac{N_j^2}{d}$$

corresponds exactly to $(y, z) \mapsto (z, y)$.

Proof. Abbreviate $R = R_j$ and $N = N_j$. Equation (2.1) is equivalent to

$$(Ry - N)(Rz - N) = N^2. \tag{2.3}$$

Both factors are positive: for example, division of $Ryz = N(y + z)$ by z gives $Ry = N(1 + y/z) > N$, and similarly for $Rz - N$. Therefore $d = Ry - N$ is a positive divisor of N^2 and $d \equiv -N \pmod R$.

Conversely, let $d \in \mathscr{D}_{p,j}$ and put $e = N^2/d$. By (1.2), d and N are units modulo R; hence

$$e \equiv N^2(-N)^{-1} \equiv -N \pmod R.$$

Thus both components of $\rho_{p,j}(d)$ are positive integers. Their two shifted factors are d and e, so (2.3), and therefore (2.1), holds. The formulas visibly compose to the identity. Replacing d by $e = N^2/d$ exchanges the two reconstructed components. □

For $\varepsilon \in \{0,1\}$, define the fixed-shell selector and the fully marked selector by

$$\mathcal{W}_\varepsilon(p,j) = \{u \in \mathbb{Z}_{>0} : u \mid a_j^2,\ R_j \mid 4u + p^\varepsilon\}, \tag{2.4}$$

$$\mathcal{W}_\varepsilon(p) = \{(j,u) : j \in J_p,\ u \in \mathcal{W}_\varepsilon(p,j)\}. \tag{2.5}$$

The tag $\varepsilon = 0$ is the exterior, or E, channel; $\varepsilon = 1$ is the middle, or M, channel. Whenever a disjoint union of the two channels is written below, it means the tagged coproduct

$$\{0\} \times \mathcal{W}_0(p) \ \sqcup\ \{1\} \times \mathcal{W}_1(p).$$

This convention is necessary because the two raw subsets can intersect.

Proposition 7.3 (exact two-channel decomposition)**.** *The maps*

$$\iota_E : \mathcal{W}_0(p,j) \longrightarrow \mathscr{D}_{p,j}, \qquad u \longmapsto p^2 u,$$
$$\iota_M : \mathcal{W}_1(p,j) \longrightarrow \mathscr{D}_{p,j}, \qquad u \longmapsto pu$$

are bijections onto the subsets on which the p-adic valuations are 2 *and* 1*, respectively. The valuation-zero subset is $\iota_{N_j}(\iota_E(\mathcal{W}_0(p,j)))$. Consequently*

$$\mathscr{D}_{p,j} = \iota_E(\mathcal{W}_0(p,j)) \ \sqcup\ \iota_M(\mathcal{W}_1(p,j)) \ \sqcup\ \iota_{N_j}(\iota_E(\mathcal{W}_0(p,j))), \tag{2.6}$$

and

$$|\mathscr{D}_{p,j}| = 2|\mathcal{W}_0(p,j)| + |\mathcal{W}_1(p,j)|. \tag{2.7}$$

Proof. Because $p \nmid a_j$, every divisor $d \mid p^2 a_j^2$ has $v_p(d) \in \{0,1,2\}$.

If $d = p^2 u$, then $u \mid a_j^2$. The shell congruence is

$$R_j \mid d + N_j = p(pu + a_j).$$

After cancelling the unit p modulo R_j and multiplying by 4, the identity $4a_j = p + R_j$ makes this equivalent to $R_j \mid 4u + 1$. This proves the E-channel assertion in both directions. If $d = pu$, the same calculation gives

$$R_j \mid d + N_j \iff R_j \mid u + a_j \iff R_j \mid 4u + p,$$

which proves the M-channel assertion. Complementation preserves the shell congruence by Proposition 7.2, exchanges valuations 0 and 2, and preserves valuation 1. This proves the disjoint decomposition and its cardinality formula. □

For $u \in \mathcal{W}_\varepsilon(p,j)$, define the two exact quotients

$$T = \frac{4u + p^\varepsilon}{R_j}, \qquad K = \frac{p^{1-\varepsilon}T + 1}{4}. \tag{2.8}$$

Since $R_j \equiv 3 \pmod 4$ and $p^\varepsilon \equiv 1 \pmod 4$, the first equation gives $T \equiv 3 \pmod 4$. Hence $K \in \mathbb{Z}_{>0}$. Direct calculation, without dropping the tag, gives

$$R_j K = a_j + p^{1-\varepsilon} u, \qquad T a_j = u + p^\varepsilon K. \tag{2.9}$$

Indeed, multiplication by 4 changes these identities into

$$4R_jK = p^{1-\varepsilon}(4u + p^\varepsilon) + R_j = 4p^{1-\varepsilon}u + p + R_j,$$
$$4Ta_j = T(p + R_j) = pT + 4u + p^\varepsilon = 4u + 4p^\varepsilon K.$$

Proposition 7.4 (denominator reconstruction)**.** *The map*

$$\Psi_{p,j,\varepsilon} : \mathcal{W}_\varepsilon(p,j) \longrightarrow \mathscr{S}(p), \qquad u \longmapsto \left(a_j,\ pK,\ \frac{p^\varepsilon a_j K}{u}\right) \tag{2.10}$$

is well-defined. Its two oriented shell factors are

$$R_j(pK) - N_j = p^{2-\varepsilon}u, \qquad R_j\left(\frac{p^\varepsilon a_j K}{u}\right) - N_j = p^\varepsilon \frac{a_j^2}{u}. \tag{2.11}$$

Moreover,

$$\mathrm{ES}(p) \quad \Longleftrightarrow \quad (\{0\} \times \mathcal{W}_0(p)) \sqcup (\{1\} \times \mathcal{W}_1(p)) \neq \varnothing. \tag{2.12}$$

Proof. The last coordinate in (2.10) is integral. To see this without assuming it, reduce the second identity in (2.9) modulo u, multiply by a_j, and use $u \mid a_j^2$: this gives $u \mid p^\varepsilon a_j K$. The two identities (2.11) now follow from the first identity in (2.9). Their product is N_j^2, so Proposition 7.2 shows that (2.10) lies in $\mathscr{S}(p)$.

Conversely, suppose $\mathscr{S}(p) \neq \varnothing$, and sort a solution. Lemma 7.1 shows that its smallest denominator is uniquely of the form $x = a_j$ with $j \in J_p$: explicitly

$$R_j = 4x - p \equiv 3 \pmod 4, \qquad j = \frac{4x - p - 3}{4}.$$

Proposition 7.2 supplies a divisor $d \in \mathscr{D}_{p,j}$, and Proposition 7.3 places either d or its complement in the image of ι_E, or places d in the image of ι_M. Hence one of the two marked selector sets is nonempty. The forward implication follows from (2.10). □

7.3 The exact selector–terminal-lattice equivalence

Define the normalized, positively oriented, divisor-supported terminal slice

$$\mathcal{H}_\varepsilon(p) = \left\{ (j,\ell_0,\ell_1) \in \mathbb{Z}^3 : \begin{array}{l} j \in J_p, \quad \lambda(\ell_0,\ell_1,1) = p^\varepsilon, \\ T(\ell_0) := 4\ell_0 - 1 > 0, \\ u(j,\ell_0,\ell_1) := \ell_1 + jT(\ell_0) > 0, \\ u(j,\ell_0,\ell_1) \mid a_j^2 \end{array} \right\}. \tag{3.1}$$

Let $\mathcal{H}_\varepsilon(p,j)$ be the fibre with the displayed marker j.

For a selector state define its relation matrix

$$\mathsf{A}_\varepsilon(p;j,u) = \begin{pmatrix} R_j & -u \\ 4 & -T \end{pmatrix}, \qquad R_jT = 4u + p^\varepsilon. \tag{3.2}$$

For the marker j, retain both inverse unimodular shears

$$\mathsf{U}_j = \begin{pmatrix} 1 & -j \\ 0 & 1 \end{pmatrix}, \qquad \mathsf{L}_j = \mathsf{U}_j^{-1} = \begin{pmatrix} 1 & j \\ 0 & 1 \end{pmatrix}. \tag{3.3}$$

Theorem 7.5 (strong marked bridge theorem)**.** *For every $\varepsilon \in \{0,1\}$, the map*

$$\Theta_{\varepsilon,p} : \mathcal{W}_\varepsilon(p) \longrightarrow \mathcal{H}_\varepsilon(p), \qquad (j,u) \longmapsto \left(j, \frac{T+1}{4}, u - jT\right) \tag{3.4}$$

is a bijection. Its inverse is

$$\Theta_{\varepsilon,p}^{-1}(j,\ell_0,\ell_1) = (j, \ell_1 + j(4\ell_0 - 1)). \tag{3.5}$$

For corresponding points, with $\ell = (\ell_0, \ell_1, 1)$, one has the exact matrix identities

$$\boxed{\mathsf{U}_j \mathsf{A}_\varepsilon(p; j, u) = \mathsf{R}(\ell_0, \ell_1, 1), \qquad \mathsf{A}_\varepsilon(p; j, u) = \mathsf{L}_j \mathsf{R}(\ell_0, \ell_1, 1).} \tag{3.6}$$

In particular,

$$\mathrm{Rel}(\mathsf{A}_\varepsilon(p; j, u)) = \mathrm{Rel}(\mathsf{R}(\ell_0, \ell_1, 1)), \qquad \det \mathsf{A}_\varepsilon = \det \mathsf{R} = -p^\varepsilon. \tag{3.7}$$

Thus the bridge intertwines the entire integral row presentation, not only its determinant.

Proof. Take $(j, u) \in \mathcal{W}_\varepsilon(p)$. Section 2 proves that T is a positive integer satisfying $T \equiv 3 \pmod 4$, so $\ell_0 = (T+1)/4 \in \mathbb{Z}$. Put $\ell_1 = u - jT$. Since $R_j = 4j + 3$,

$$\begin{aligned} \lambda(\ell_0, \ell_1, 1) &= 3(T+1) - 4(u - jT) - 3 \\ &= (4j+3)T - 4u = p^\varepsilon. \end{aligned}$$

The positivity and divisor-support conditions in (3.1) hold because $u = \ell_1 + jT$. Thus (3.4) is well-defined.

Conversely, take $(j, \ell_0, \ell_1) \in \mathcal{H}_\varepsilon(p)$, and set

$$T = 4\ell_0 - 1, \qquad u = \ell_1 + jT.$$

Expanding $\lambda(\ell_0, \ell_1, 1) = p^\varepsilon$ gives

$$(4j+3)T - 4u = p^\varepsilon,$$

so $R_j \mid 4u + p^\varepsilon$. All other selector conditions occur explicitly in (3.1). The two constructions recover T, u, ℓ_0, ℓ_1 literally, which proves bijectivity and (3.5).

Finally, direct multiplication gives

$$\begin{pmatrix} 1 & -j \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 4j+3 & -u \\ 4 & -T \end{pmatrix} = \begin{pmatrix} 3 & -u + jT \\ 4 & -T \end{pmatrix} = \begin{pmatrix} 3 & -\ell_1 \\ 4 & 1 - 4\ell_0 \end{pmatrix}.$$

Both shears have determinant 1, so their integral row operations are invertible and the row lattices are equal. The signed determinant is $-R_j T + 4u = -p^\varepsilon$, proving (3.7). □

Corollary 7.6 (presented groups and the Erdős–Straus equation)**.** *For every corresponding selector/S6 point,*

$$\mathcal{G}(\mathsf{A}_\varepsilon(p; j, u)) = \mathcal{G}(\mathsf{R}(\ell_0, \ell_1, 1)) \cong G(\ell_0, \ell_1, 1) \cong \mathbb{Z}/p^\varepsilon\mathbb{Z}. \tag{3.8}$$

Here $\mathbb{Z}/1\mathbb{Z}$ is the trivial group. Moreover the direct map

$$\widehat{\Psi}_{\varepsilon,p} : \mathcal{H}_\varepsilon(p) \longrightarrow \mathscr{S}(p) \tag{3.9}$$

is obtained by (3.5), (2.8), and

$$(j, \ell_0, \ell_1) \longmapsto \left(a_j,\ pK,\ \frac{p^\varepsilon a_j K}{u}\right), \quad \begin{cases} T = 4\ell_0 - 1, \\ u = \ell_1 + jT, \\ K = (p^{1-\varepsilon}T + 1)/4. \end{cases} \tag{3.10}$$

It satisfies

$$\boxed{\mathrm{ES}(p) \Longleftrightarrow (\{0\} \times \mathcal{H}_0(p)) \sqcup (\{1\} \times \mathcal{H}_1(p)) \neq \varnothing.} \tag{3.11}$$

Proof. The relation $xy = c^{\ell_0}$ in (1.4) gives $y = x^{-1}c^{\ell_0}$; since c is central, x and y commute. The remaining two relations are the rows of (1.5). The greatest common divisor of the four entries of R is 1, because it divides both 3 and 4. Its Smith form is therefore $\mathrm{diag}(1, p^\varepsilon)$, proving (3.8). Formula (3.10) is the composition of Theorem 7.5 with Proposition 7.4, and (3.11) follows from (2.12). □

Remark 7.7 (the selector tag and the solution map). The channel tag is not redundant. For $p = 13$, $j = 0$, and $u = 2$, one has $u \in \mathcal{W}_0(p, j) \cap \mathcal{W}_1(p, j)$, but the two tags give different terminal values and different marked terminal points. Nor is the map to unordered denominator triples injective: in the M-channel at $p = 13, j = 0$, the complementary divisors $u = 2$ and $u = 8 = a_0^2/2$ both give the unordered solution $\{4, 26, 52\}$. The bijection in Theorem 7.5 is a bijection of marked selector states with marked terminal states, not a bijection with unordered solutions.

Corollary 7.8 (fixed-fibre affine coordinates and the intrinsic gate)**.** *Put*

$$C_\varepsilon = \frac{p^\varepsilon + 3}{4}, \qquad D_{\varepsilon,j} = j + C_\varepsilon = \frac{R_j + p^\varepsilon}{4}.$$

Every integral point with $\ell_2 = 1$ and $\lambda(\ell_0, \ell_1, 1) = p^\varepsilon$ is uniquely

$$(\ell_0, \ell_1, \ell_2) = (n, 3n - C_\varepsilon, 1), \qquad n \in \mathbb{Z}, \tag{3.12}$$

and, for each fixed marker $j \in J_p$, define

$$T = 4n - 1, \qquad u = R_j n - D_{\varepsilon,j}, \qquad K = \frac{p^{1-\varepsilon} T + 1}{4}. \tag{3.13}$$

Then $T \equiv 3 \pmod 4$, so $K \in \mathbb{Z}$, and the congruence equation $R_j T = 4u + p^\varepsilon$ holds identically. On the locus $T > 0$ and $u > 0$, one has

$$\gcd(R_j, u) = 1, \qquad u \mid a_j^2 \Longleftrightarrow u \mid K^2. \tag{3.14}$$

Thus the nonlinear divisor gate can be written entirely in the S6 coordinates, but it does not disappear.

Proof. Equation (3.12) is the direct solution of $12\ell_0 - 4\ell_1 - 3 = p^\varepsilon$. Substitution gives the first two formulas in (3.13), and $T \equiv 3 \pmod 4$ proves that the displayed integer K is well defined. The definitions give

$$4u = 4R_j n - (R_j + p^\varepsilon) = R_j T - p^\varepsilon,$$

so the congruence equation holds for every integral point in (3.12). If a positive integer divides both R_j and u, then the congruence equation makes it divide p^ε; (1.2) therefore gives $\gcd(R_j, u) = 1$. Without invoking the selector hypothesis, direct substitution gives

$$4\big(R_j K - p^{1-\varepsilon} u - a_j\big) = p^{1-\varepsilon}(R_j T - 4u) - p = 0.$$

Thus $a_j \equiv R_j K \pmod u$. Since R_j is a unit modulo u, $a_j^2 \equiv R_j^2 K^2 \pmod u$, which proves (3.14) in both directions. □

The normalization automatically lies in the admissible residue class used by the S6 terminal presentation:

$$\ell_2 = 1 \text{ is odd}, \qquad \ell_1 \equiv 2 \pmod 3. \tag{3.15}$$

Indeed reduction of $\lambda(\ell_0, \ell_1, 1) = p^\varepsilon \equiv 1 \pmod 3$ gives $-\ell_1 \equiv 1 \pmod 3$.

Remark 7.9 (orientation and the constructed S6 candidate). The geometrically reconstructed candidate elsewhere in the audited corpus uses the oriented terminal parameter $\lambda(0, 1, -1) = -1$. The two arithmetic channels in this note use the positive terminal values $+1$ and $+p$. No audited map identifies the -1 terminal datum with either positive datum. Consequently the row shear proved here does not, by itself, place any selector state on the particular complex threefold constructed in the -1 branch.

7.4 The exact remaining arithmetic incidence

The bridge can be expressed without an untyped phrase such as "some divisor of a square." The next factor coordinates are canonical and retain their orientation.

Lemma 7.10 (canonical ordered factor coordinates)**.** *For $a, u \in \mathbb{Z}_{>0}$, the condition $u \mid a^2$ is equivalent to the existence of a unique ordered triple $(r, s, t) \in \mathbb{Z}^3_{>0}$ such that*

$$u = r^2 s, \qquad a = rst, \qquad \gcd(r, t) = 1. \tag{4.1}$$

Explicitly, with $g = \gcd(a, u)$,

$$r = \frac{u}{g}, \qquad s = \frac{g^2}{u}, \qquad t = \frac{a}{g}. \tag{4.2}$$

Proof. Write $v_q(a) = \alpha$ and $v_q(u) = \beta \le 2\alpha$ prime by prime. Then

$$v_q(g) = \min(\alpha, \beta), \qquad v_q(g^2/u) = 2\min(\alpha, \beta) - \beta \ge 0.$$

Thus (4.2) consists of positive integers. If $\beta \le \alpha$, then $q \nmid r$; if $\beta \ge \alpha$, then $q \nmid t$. Hence $\gcd(r, t) = 1$, and direct multiplication gives (4.1).

Conversely, (4.1) gives $a^2/u = st^2 \in \mathbb{Z}_{>0}$. It also gives $\gcd(a, u) = rs$, because $\gcd(r, t) = 1$. Formula (4.2) then recovers (r, s, t), proving uniqueness. □

Proposition 7.11 (factor form of both selector channels)**.** *For the unique factorization $u = r^2 s$, $a_j = rst$, $\gcd(r, t) = 1$, one has*

$$u \in \mathcal{W}_0(p, j) \iff R_j \mid pr + t, \tag{4.3}$$

$$u \in \mathcal{W}_1(p, j) \iff R_j \mid r + t. \tag{4.4}$$

Proof. All of r, s, t are units modulo R_j, because their product is a_j and $\gcd(a_j, R_j) = 1$. For the E-channel,

$$t(4u + 1) = 4r^2 st + t = (p + R_j)r + t = (pr + t) + R_j r.$$

Cancelling t modulo R_j proves (4.3). For the M-channel,

$$4u + p \equiv 4(u + a_j) = 4rs(r + t) \pmod{R_j},$$

and cancellation of the unit $4rs$ proves (4.4). □

Theorem 7.12 (necessary-and-sufficient counterexample criterion)**.** *A prime $p \equiv 1 \pmod{12}$ is a counterexample to the Erdős–Straus equation if and only if, for every $j \in J_p$ and every ordered $(r, s, t) \in \mathbb{Z}^3_{>0}$ satisfying*

$$a_j = rst, \qquad \gcd(r, t) = 1,$$

one has simultaneously

$$R_j \nmid pr + t, \qquad R_j \nmid r + t. \tag{4.5}$$

Equivalently, with the tag retained, it is necessary and sufficient that

$$R_j \nmid p^{1-\varepsilon} r + t \tag{4.6}$$

for every $j \in J_p$, every $\varepsilon \in \{0, 1\}$, and every such factor triple.

Proof. Lemma 7.10 is a bijection between all allowed divisors in the selector (2.4) and all displayed ordered factor triples. Proposition 7.11 translates the two selector congruences without loss. The assertion now follows directly from (2.12), or equivalently (3.11), by negating the finite existential statement. □

This theorem isolates the unresolved problem exactly: the S6 row shear has not removed the bounded factor-incidence condition (4.5).

7.5 Why the labels and support cannot be forgotten

Lemma 7.13 (full terminal fibres are affine lattices)**.** *The kernel of* $\lambda : \mathbb{Z}^3 \to \mathbb{Z}$ *is*

$$\ker \lambda = \mathbb{Z}(1,3,0) \oplus \mathbb{Z}(1,0,4). \tag{5.1}$$

If $q \equiv 1 \pmod{12}$ *and* $m_q = (q+11)/12$*, then*

$$\lambda^{-1}(q) = \{(m_q + r + s,\ 2 + 3r,\ 1 + 4s) : r, s \in \mathbb{Z}\}. \tag{5.2}$$

In the normalized slice $\ell_2 = 1$*, this is an infinite affine line.*

Proof. If $12a - 4b - 3c = 0$, reduction modulo 4 gives $c = 4s$. The remaining equation gives $b = 3(a - s)$; putting $r = a - s$ yields $(a, b, c) = r(1,3,0) + s(1,0,4)$, uniquely. Since $\lambda(m_q, 2, 1) = 12m_q - 11 = q$, translation by (5.1) proves (5.2). □

The lemma proves that the mere existence of a terminal parameter with signed value 1 or p is automatic. It cannot imply selector nonemptiness. The following examples show separately why the marker and the divisor-support condition are both part of the theorem's domain.

Example 7.14 (forgetting j and p is noninjective)**.** *For each eligible prime* q*, define the fixed-prime forgetful map*

$$\mathfrak{f}_q : \mathcal{H}_0(q) \longrightarrow \mathbb{Z}^3, \qquad \mathfrak{f}_q(j, \ell_0, \ell_1) = (\ell_0, \ell_1, 1).$$

At $p = 13$*, the two E-states*

$$(j, u, T) = (0, 2, 3), \qquad (1, 5, 3)$$

satisfy

$$\Theta_{0,13}(0, 2) = (0, 1, 2), \qquad \Theta_{0,13}(1, 5) = (1, 1, 2),$$

and both images are sent by $\mathfrak{f}_{13}$ *to* $(1, 2, 1)$*. To compare different prime fibres, form the tagged coproduct and the global forgetful map*

$$\mathfrak{F} : \bigsqcup_{\substack{q \text{ prime} \\ q \equiv 1 \pmod{12}}} (\{q\} \times \mathcal{H}_0(q)) \longrightarrow \mathbb{Z}^3, \qquad \mathfrak{F}(q; j, \ell_0, \ell_1) = (\ell_0, \ell_1, 1).$$

The state $(p, j, u) = (73, 1, 5)$ *has* $\Theta_{0,73}(1, 5) = (1, 1, 2)$*, so its tagged image also maps under* $\mathfrak{F}$ *to* $(1, 2, 1)$*. Therefore neither* j *nor the external prime* p *is recoverable after applying the explicitly displayed forgetful maps.*

Example 7.15 (ambient positivity does not imply divisor support)**.** *For* $p = 13$*,* $j = 2$*, one has* $R_j = 11$ *and* $a_j = 6$*. The positive normalized E-line has*

$$(\ell_0, \ell_1) = (n, 3n - 1), \qquad u = 11n - 3,$$

and the positive normalized M-line has

$$(\ell_0, \ell_1) = (n, 3n - 4), \qquad u = 11n - 6.$$

Both lines are infinite. The only positive $u \le a_j^2 = 36$ *are*

$$E :\ 8, 19, 30, \qquad M :\ 5, 16, 27,$$

and none divides 36*. Hence* $\mathcal{H}_0(13, 2) = \mathcal{H}_1(13, 2) = \varnothing$*, even though both ambient positive S6 lines are nonempty. The prime* 13 *is solved at other markers, so this is a fixed-shell counterexample to forgetting support, not an Erdős–Straus counterexample.*

7.6 The natural unmarked S^6 point and its complete divisor gate

The most direct attempted use of the S6 presentation starts from one point in the signed p-fibre and shears it through every shell. Its exact content can be classified completely.

Write

$$p = 12m - 11, \qquad m = h + 1 \geq 2, \qquad J_p = \{0, \ldots, 3m - 4\}. \tag{6.1}$$

Retain the unmarked S6 point

$$\ell^{\mathrm{aff}} = (m, 2, 1), \qquad \lambda(\ell^{\mathrm{aff}}) = p, \tag{6.2}$$

and for $j \in J_p$ put

$$T_0 = 4m - 1, \quad u_j = 2 + jT_0, \quad a_j = 3m + j - 2, \quad R_j = 4j + 3. \tag{6.3}$$

Then

$$R_j T_0 = 4u_j + p, \qquad R_j m = a_j + u_j, \qquad T_0 a_j - u_j = pm, \tag{6.4}$$

and

$$\mathsf{U}_j \begin{pmatrix} R_j & -u_j \\ 4 & -T_0 \end{pmatrix} = \begin{pmatrix} 3 & -2 \\ 4 & 1 - 4m \end{pmatrix} = \mathsf{R}(m, 2, 1). \tag{6.5}$$

Theorem 7.16 (complete affine-line gate)**.** *For prime $p = 12m - 11$, $m \geq 2$, and $0 \leq j < 3m - 3$,*

$$u_j \mid a_j^2 \iff (j = 0 \text{ and } 2 \mid m) \text{ or } (j = 2 \text{ and } 8 \mid m). \tag{6.6}$$

Consequently the single unmarked point (6.2), with all markers restored, produces an Erdős–Straus witness if and only if $p \equiv 13 \pmod{24}$. In that case $j = 0, u = 2, K = m$ gives

$$\frac{4}{p} = \frac{1}{3m - 2} + \frac{1}{pm} + \frac{1}{pm(3m - 2)/2}. \tag{6.7}$$

For $p \equiv 1 \pmod{24}$, this affine S6 point supplies no supported selector state at any $j \in J_p$.

Proof. The identities (6.4) and (6.5) follow by direct expansion. Assume first that $u_j \mid a_j^2$, and put $g = \gcd(u_j, a_j)$. Write $u_j = gu'$ and $a_j = ga'$ with $\gcd(u', a') = 1$. The assumed divisibility gives $u' \mid g(a')^2$, hence $u' \mid g$, and therefore

$$u_j \mid g^2. \tag{6.8}$$

The last identity in (6.4) gives $g \mid pm$. Since $0 < a_j < p$, the primality of p and $g \mid a_j$ imply $p \nmid g$; therefore $g \mid m$. Reducing $u_j = 2 + j(4m - 1)$ modulo m now gives

$$g \mid j - 2. \tag{6.9}$$

Suppose $j \neq 0, 2$. Then $j \geq 1$, and (6.8)–(6.9) imply $u_j \leq g^2 \leq (j - 2)^2$. But

$$u_j - (j - 2)^2 = j(4m + 3 - j) - 2 > 0 \tag{6.10}$$

For $m \geq 2$ and $1 \leq j < 3m - 3$, indeed $4m + 3 - j \geq m + 7$, so the right-hand side of (6.10) is at least $m + 5 > 0$. This is a contradiction. Thus only $j = 0$ or $j = 2$ can occur.

At $j = 0$, $u_0 = 2$ and $a_0 = 3m - 2$, so $u_0 \mid a_0^2$ exactly when m is even. At $j = 2$, $u_2 = 8m$ and $a_2 = 3m$, so

$$u_2 \mid a_2^2 \iff 8m \mid 9m^2 \iff 8 \mid m.$$

These calculations also prove both converse implications in (6.6).

Now $p = 12m - 11$ is congruent to 13 (mod 24) exactly when m is even. The $j = 0$ state lies in the M-channel, and (2.8) gives $T = T_0$ and $K = (T_0 + 1)/4 = m$. Formula (2.10) is precisely (6.7). If m is odd, (6.6) excludes every marker on this particular S6 point. □

The last sentence is deliberately local to (6.2). For example, $p = 73$ has $m = 7$ odd, so the affine point contributes nothing; nevertheless $(j, u, T; \ell_0, \ell_1) = (1, 1, 11; 3, -10)$ is an M-state. Thus failure of the affine point is not failure of the full selector.

7.7 Exact descendant maps transported through the bridge

The bridge also transports a verified selector witness along multiplicative descendant rays. This is a map from an already nonempty domain, not a nonemptiness theorem.

Fix a supported state at (p, j), retain $R = R_j$, and let $b \in \mathbb{Z}_{>0}$. Define

$$a_b = ba_j, \qquad p_b = 4a_b - R = bp + (b-1)R. \tag{7.1}$$

Whenever p_b is again a prime congruent to $1 \pmod{12}$, the same marker j belongs to J_{p_b}, since $R < p \le p_b$, and the following maps have the displayed domains and codomains:

$$\Phi^E_{b;p} : \mathcal{H}_0(p, j) \longrightarrow \mathcal{H}_0(p_b, j), \qquad (j, \ell_0, \ell_1) \longmapsto (j, \ell_0, \ell_1), \tag{7.2}$$

$$\Phi^M_{b;p} : \mathcal{H}_1(p, j) \longrightarrow \mathcal{H}_1(p_b, j), \qquad (j, \ell_0, \ell_1) \longmapsto (j, b\ell_0, b\ell_1 - j(b-1)). \tag{7.3}$$

Write

$$\theta_{\varepsilon,p,j} : \mathcal{W}_\varepsilon(p, j) \longrightarrow \mathcal{H}_\varepsilon(p, j), \qquad \theta_{\varepsilon,p,j}(u) = \Theta_{\varepsilon,p}(j, u)$$

for the fixed-marker restriction of the marked bridge.

Proposition 7.17 (intertwining of descendant actions)**.** *The selector maps corresponding to (7.2) and (7.3) are the typed maps*

$$\psi^E_{b;p} : \mathcal{W}_0(p, j) \longrightarrow \mathcal{W}_0(p_b, j), \qquad u \longmapsto u,$$

$$\psi^M_{b;p} : \mathcal{W}_1(p, j) \longrightarrow \mathcal{W}_1(p_b, j), \qquad u \longmapsto bu.$$

They intertwine the fixed-marker bridges exactly:

$$\Phi^\bullet_{b;p} \circ \theta_{\varepsilon,p,j} = \theta_{\varepsilon,p_b,j} \circ \psi^\bullet_{b;p}, \qquad (\bullet, \varepsilon) \in \{(E, 0), (M, 1)\}. \tag{7.4}$$

Equivalently,

$$\theta^{-1}_{0,p_b,j} \Phi^E_{b;p} \theta_{0,p,j}(u) = u, \qquad \theta^{-1}_{1,p_b,j} \Phi^M_{b;p} \theta_{1,p,j}(u) = bu.$$

On the accompanying arithmetic metadata these maps replace (a_j, p, u) by (a_b, p_b, u) and (a_b, p_b, bu), respectively. For the latter,

$$T_b = bT + b - 1, \qquad u_b = bu, \qquad K_b = bK, \tag{7.5}$$

and

$$\lambda(b\ell_0, b\ell_1 - j(b-1), 1) = p_b. \tag{7.6}$$

For either tag the maps are injective. For $c \in \mathbb{Z}_{>0}$, put

$$a_{bc} = bc\, a_j, \qquad p_{bc} = 4a_{bc} - R = bc\, p + (bc-1)R.$$

Whenever p_b and p_{bc} are primes congruent to $1 \pmod{12}$, the displayed maps are composable and satisfy

$$\Phi^\bullet_{c;p_b} \circ \Phi^\bullet_{b;p} = \Phi^\bullet_{bc;p} \tag{7.7}$$

Proof. In the E-channel, $u \mid a_j^2$ implies $u \mid a_b^2$, and the congruence $R \mid 4u+1$ and quotient T are unchanged. Hence ℓ_0, ℓ_1 are unchanged by (3.4).

In the M-channel, $bu \mid b^2 a_j^2 = a_b^2$, and

$$4bu + p_b = b(4u + p) + (b-1)R.$$

Thus the descendant is a selector state and its quotient is $T_b = bT + b - 1$. Since $K = (T+1)/4$ in this channel, $K_b = (T_b + 1)/4 = bK$. Applying (3.4) gives

$$\ell_{0,b} = \frac{T_b + 1}{4} = b\ell_0, \qquad \ell_{1,b} = bu - jT_b = b\ell_1 - j(b-1),$$

which proves (7.3)–(7.6), including the intertwining identity (7.4). Each map is visibly injective for fixed b. Substitution into (7.1)–(7.3), with the source prime retained, proves the typed composition law (7.7). □

Reverse descent need not preserve divisor support: the factor b can introduce divisors absent from a proposed parent. Therefore (7.2)–(7.7) propagate witnesses but cannot create the first witness on an empty ray.

7.8 Sharp verdict

The strongest theorem in this selector/terminal section is Theorem 7.5 together with the exhaustive equivalence (3.11): the complete tagged Erdős–Straus selector for a fixed prime $p \equiv 1 \pmod{12}$ is exactly the marked divisor-supported S6 terminal slice, and the two integral relation matrices have the same row lattice. The exact remaining obstruction is (4.5).

Three stronger readings are false or unsupported.

1. The abstract group $G(\ell)$, its determinant, or an unmarked terminal triple does not determine selector support. Every E-state has the same trivial presented group, every M-state has group $\mathbb{Z}/p\mathbb{Z}$, and Examples 7.14 and 7.15 exhibit the information lost by forgetting the marker and divisor gate.

2. The row shear is not a geometric map from the reciprocal cubic $4xyz = p(xy + xz + yz)$ to the candidate complex threefold X. No map of varieties or analytic spaces, no cycle map, no attaching map, and no cohomological correspondence is defined by left multiplication of a relation matrix.

3. The existence of the full S6 affine fibre cannot prove universal Erdős–Straus positivity. Those fibres are always nonempty by (5.2), whereas selector nonemptiness is exactly the additional finite nonlinear incidence in (4.5). The natural affine point is completely exhausted by Theorem 7.16 and contributes nothing to the unresolved $p \equiv 1 \pmod{24}$ half.

Accordingly, the bridge is a genuine theorem and is stronger than a shared determinant or a heuristic analogy. It is also exact enough to prove its own boundary: without a new argument forcing one of the bounded factor congruences in (4.5), it neither proves nor disproves the Erdős–Straus conjecture.

The additional lattice and shell theorems in Section 31 supply further exact morphisms without changing this incidence question. In particular, Theorem 31.10 gives $A_3(2521) = 0$ and $A_3(3361) = 6$ although both primes are 1 modulo 840; Theorem 31.12 retains the actual counts in the integral augmentation vector $Q_{R,P} = 6W_{R,P} - M_{R,P}u \in N$.

8 Divisor channels, density one, and two distinct arithmetic maps

This dated continuation (4 September 2026) checks the supplied Pro-mode *Divisor Channels, Prime Saturation, and Bézout Openings*, version 3, and extends its final arithmetic comparison. The original packet is retained unchanged under `supporting_materials/workbench/evidence/koide_es_v3_audit_20260904/source/`. The claims below concern explicit reciprocal equations, integral matrices, and quadratic surfaces. They do not require a complex structure on a sphere.

8.1 The complete oriented channel map

Fix integers $n \equiv 1 \pmod 4$, $R \equiv 3 \pmod 4$, $0 < R < n$, $\gcd(n, R) = 1$, and put $a = (n + R)/4$, $N = na$. Then $\gcd(n, a) = \gcd(R, a) = 1$. For a positive divisor $\delta \mid n^2$ define

$$\mathcal{C}_\delta(n, R) = \{(u, v) \in \mathbb{Z}^2_{>0} : uv = a^2,\ R \mid 4u + \delta\}.$$

Proposition 8.1. *The following is a bijection onto all ordered positive solutions of* $4/n = 1/a + 1/y + 1/z$*:*

$$(\delta, u, v) \longmapsto (y, z) = \left(\frac{N + (n^2/\delta)u}{R}, \frac{N + \delta v}{R}\right), \qquad (u, v) \in \mathcal{C}_\delta(n, R).$$

Its inverse, retaining this orientation, is

$$d = Ry - N, \quad e = Rz - N, \quad k = \gcd(d, n^2), \quad \delta = n^2/k, \quad u = d/k, \quad v = e/\delta.$$

Exchange of y, z is the fixed-point-free involution $(\delta, u, v) \mapsto (n^2/\delta, v, u)$.

Proof. The reciprocal equation is $Ryz = N(y + z)$, hence $de = N^2$. For any positive solution $Ry = N + Ny/z > N$, so $d > 0$; similarly $e > 0$. Because n and a are coprime, every prime dividing n occurs in de with precisely its exponent in n^2. Thus k extracts the entire n-primary part of d, δ the entire n-primary part of e, and u, v are positive integers with $uv = a^2$. Reducing $N + e$ modulo R gives $0 \equiv na + \delta v \equiv 4a^2 + \delta v = v(4u + \delta)$. Since v is a unit modulo R, this proves the channel condition. Conversely that condition makes $N + e$ divisible by R; also $\delta(N + d) = n(\delta a + nu) \equiv na(\delta + 4u) \equiv 0 \pmod{R}$. The unit δ can be cancelled modulo R, proving integrality of y. The identity $de = N^2$ proves the reciprocal equation. These operations are inverse. A fixed point of the involution would have $\delta = n$, $u = v = a$, so $R \mid 4a + n = 2n + R$, impossible since $R \geq 3$ is odd and coprime to n. □

The package therefore corrects the earlier orientation reversal: the channel order is $n^2/\gcd(Ry - na, n^2)$, not $\gcd(Ry - na, n^2)$. For $n = 9$, $R = 7$, $a = 4$, $(\delta, u, v) = (3, 1, 16)$, the ordered solution is $(y, z) = (9, 12)$, and $\gcd(27, 81) = 27$.

8.2 The integral presentation, with its exact realization boundary

Put $T = (4u + \delta)/R$, $j = (R - 3)/4$, and $\sigma = 1$ for $\delta \equiv 1 \pmod 4$, $\sigma = -1$ otherwise. Define

$$\ell_0 = (T + \sigma)/4, \quad \ell_1 = u - jT = (3T - \delta)/4, \quad \ell_2 = \sigma, \quad \mathsf{R}_\delta = \begin{pmatrix} 3 & -\ell_1 \\ 4 & -T \end{pmatrix}.$$

All coordinates are integral because $T \equiv -\delta \pmod 4$. Direct multiplication and subtraction give

$$\begin{pmatrix} 1 & -j \\ 0 & 1 \end{pmatrix} \begin{pmatrix} R & -u \\ 4 & -T \end{pmatrix} = \mathsf{R}_\delta, \qquad 12\ell_0 - 4\ell_1 - 3\ell_2 = \delta.$$

The entries $3, 4$ are coprime, so $\mathrm{SNF}(\mathsf{R}_\delta) = \mathrm{diag}(1, \delta)$; in particular its cokernel is $\mathbb{Z}/\delta\mathbb{Z}$. Reducing $4\ell_1 = 3T - \delta$ modulo three gives $\ell_1 \equiv -\delta \pmod 3$. Thus the source gluing admissibility $3 \nmid \ell_1$ fails precisely when $3 \mid \delta$. For the example above $(\ell_0, \ell_1, \ell_2) = (0, 0, -1)$. The matrix and its cokernel remain valid, but assigning the source's global manifold homology to this nonadmissible triple is not established by this calculation. This is an explicit admissibility test, not a failure of the reciprocal solution or the integral matrix morphism.

8.3 An exact density-one statement and what it leaves out

Proposition 8.2. *Let $p \equiv 1 \pmod 4$ be prime and $m = (p + 1)/2$. The exterior channel states on $u = a = v$ are indexed exactly by*

$$\{R : R \mid m, \ R \equiv 3 \pmod 4\}.$$

For each such R, set $x = (p + R)/4$ and $k = (p + 1)/R$. Then

$$\frac{4}{p} = \frac{1}{x} + \frac{1}{xk} + \frac{1}{pxk}, \qquad \nu(p) = \frac{1}{2}\left(\tau(m) - \sum_{d \mid m} \chi_4(d)\right),$$

where $\chi_4(d) = 1$ for $d \equiv 1 \pmod 4$ and -1 for $d \equiv 3 \pmod 4$. The count is zero exactly when every prime divisor of m is $1 \pmod 4$.

Proof. For $\delta = 1$ and $u = a$, the condition is $R \mid 4a + 1 = p + R + 1$, equivalently $R \mid p + 1$. Since R is odd this is $R \mid m$. Every such R obeys $3 \le R \le (p+1)/2 < p$, and $k \ge 2$. Adding the final two fractions gives $(p+1)/(pxk) = R/(px)$, and adding $1/x$ gives $(p+R)/(px) = 4/p$. Distinct R give distinct x. Summing the indicator $(1 - \chi_4(d))/2$ over divisors of m proves the count. A divisor of residue three exists exactly when a prime factor of residue three exists: the converse follows because products of residue-one primes and their powers have residue one. □

Proposition 8.3. *For fixed $4 \mid M$, $\gcd(c, M) = 1$, $c \equiv 1 \pmod 4$, the preceding diagonal states cover relative density one of primes $p \equiv c \pmod M$. For a finite set $\mathcal{Q}$ of primes $q \equiv 3 \pmod 4$, $q \nmid M$, the subfamily with some $q \mid p + 1$ has exact relative density*

$$1 - \prod_{q \in \mathcal{Q}} \frac{q-2}{q-1}.$$

Proof. The Chinese remainder theorem gives $\prod(q-1)$ reduced residue classes modulo $M \prod q$ above the specified class modulo M. Avoiding $p \equiv -1 \pmod q$ at every q leaves exactly $\prod(q-2)$ of them. The prime number theorem for each fixed arithmetic progression gives the displayed ratio. For completeness, its modulus-four case and partial summation give

$$\sum_{\substack{q \le Y \\ q \equiv 3 \ (4)}} \frac{1}{q} = \frac{\pi(Y;4,3)}{Y} + \int_2^Y \frac{\pi(t;4,3)}{t^2}\, dt \sim \tfrac{1}{2} \log\log Y.$$

Removing finitely many primes dividing M does not change divergence. Since $\log(1-z) \le -z$ for $0 < z < 1$, the finite products tend to zero. The full family's lower relative density is at least each of these finite union densities, hence is at least one and therefore equals one. No interchange of two unproved asymptotic limits is needed. □

These analytic tools are classical. Elsholtz and Tao, *Counting the number of solutions to the Erdős–Straus equation on unit fractions*, J. Aust. Math. Soc. 94 (2013), 50–105, arXiv:1107.1010v6, Appendix A, Lemma A.3, give a stronger shifted-prime additive-function statement, attributed there to Barban via Elliott. Choosing its prime weight to be the indicator of residue three gives the relevant abundance of factors. The finite-CRT proof above records the precise chosen ray and progression; it is not a claim that almost-all solvability is new.

For $p = 73$, $(p+1)/2 = 37$ has no residue-three prime factor. This prime therefore misses the diagonal ray, although

$$\frac{4}{73} = \frac{1}{20} + \frac{1}{220} + \frac{1}{4015}.$$

Here $R = 7$, $u = 5$, $v = 80$, and the primitive factor ray is $[1:4]$. Thus a prime missed by the selected ray need not be missed by the full selector. Density one does not settle the universal Erdős–Straus claim.

8.4 The Bézout opening, with the positivity hypothesis retained

For coprime positive integers r, t, choose $x, y \in \mathbb{Z}$ with $ry - tx = 1$, set $v = (r,t)^{\mathsf{T}}$, $w = (x,y)^{\mathsf{T}}$, and let

$$B = (v\ w) \in \mathrm{SL}_2(\mathbb{Z}), \qquad A = \delta v v^{\mathsf{T}} - w w^{\mathsf{T}} = B\,\mathrm{diag}(\delta, -1) B^{\mathsf{T}}.$$

Then $\det A = -\delta$ for every real δ. All companions are $w + kv$ for $k \in \mathbb{Z}$: subtraction of two Bézout equations gives $r\Delta y = t\Delta x$, so coprimality gives $(\Delta x, \Delta y) = k(r,t)$. Choosing $|x| \le r/2$ retains an actual companion rather than quotienting away its dependence. For a positive integer channel order,

$$A_{11} = \delta r^2 - x^2 \ge (\delta - 1/4) r^2 > 0.$$

The broader wording "every $\delta > 0$" does not imply that inequality. At $(r, t, \delta) = (2, 1, 1/8)$ all companions have x odd, and $A_{11} = 1/2 - x^2 \le -1/2$. For a specified real order and companion the exact necessary and sufficient condition is $\delta > (x/r)^2$. This correction leaves every positive integral divisor channel intact.

With $A = \left(\begin{smallmatrix} a & b \\ b & c \end{smallmatrix}\right)$ and $a > 0$, the periods $(\tau, \mu, \beta) = (ia, ib/\sqrt{6}, ic)$ give

$$P_A = \begin{pmatrix} 0 & 0 & 1 & 0 \\ \sqrt{6}b & a & 0 & 0 \\ 0 & 0 & 0 & 1 \\ c & b/\sqrt{6} & 0 & 0 \end{pmatrix}, \qquad \det P_A = ac - b^2 = -\delta.$$

This invertible real matrix sends $\mathbb{Z}^4$ to a discrete cocompact lattice in $\mathbb{C}^2$ and therefore defines a complex two-torus. The determinant sign and the factor $\sqrt{6}$ are retained. It does not determine a point on the particular global equivariant period curve of the proposed threefold. That incidence problem remains a distinct map problem, not an obstruction to the torus just constructed.

There is also an explicit integral equivalence of presentations, not merely an equality of determinants. Set

$$P_0 = \begin{pmatrix} -1 & 1 \\ -4 & 3 \end{pmatrix}, \quad Q_0 = \begin{pmatrix} 1 & T - \ell_1 \\ 0 & 1 \end{pmatrix}, \quad J = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Multiplication gives $P_0 \mathsf{R}_\delta Q_0 = \operatorname{diag}(1, -\delta)$ and $(-JB^{-1})A(B^{-\mathsf{T}}J) = \operatorname{diag}(1, -\delta)$. Thus, for $L = P_0^{-1}(-JB^{-1})$, $V = B^{-\mathsf{T}} J Q_0^{-1}$,

$$\mathsf{R}_\delta = LAV.$$

As maps $\mathbb{Z}^2 \to \mathbb{Z}^2$, the commuting square has source isomorphism V^{-1} and target isomorphism L because $\mathsf{R}_\delta V^{-1} = LA$. Hence $[z] \mapsto [Lz]$ is an explicit isomorphism of the two cokernels. It is not by itself a map of tori or of global threefolds.

8.5 The original map's rational nonintersection

Retain the exact form and orthogonal matrix

$$\Delta(q) = 4(q_1q_2 + q_1q_3 + q_2q_3) - q_1^2 - q_2^2 - q_3^2, \quad F = \begin{pmatrix} 1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} \\ 1/\sqrt{2} & 0 & -1/\sqrt{2} \\ 1/\sqrt{6} & -2/\sqrt{6} & 1/\sqrt{6} \end{pmatrix}.$$

For $\det A = -\delta$, $\delta \in \mathbb{Q}_{>0}$, Pro's map is

$$\mathfrak{K}_\delta(A) = F^{-1} \frac{((a + c)/2, b, (a - c)/2)^{\mathsf{T}}}{\sqrt{3\delta}}.$$

It satisfies $\Delta(\mathfrak{K}_\delta(A)) = -1$ but sends no rational A to a rational triple. Indeed, if q were rational, put $S = q_1 + q_2 + q_3$, $U = q_1 - q_3$, $V = q_1 + q_3 - 2q_2$. The rational coordinates $H = (a + c)/2$, $X = b$, $Y = (a - c)/2$ would satisfy

$$H = \sqrt{\delta}\, S, \quad X = \sqrt{3\delta/2}\, U, \quad Y = \sqrt{\delta/2}\, V.$$

Two nonzero coordinates would make one of $\sqrt{3/2}$, $\sqrt{2}$, $\sqrt{3}$ rational. If only H is nonzero the determinant is positive. If only X or only Y is nonzero, the determinant equation forces, respectively, $U^2 = 2/3$ or $V^2 = 2$, impossible over $\mathbb{Q}$. All coordinates zero also contradicts $\delta > 0$. This proves the claim.

8.6 A different rational bridge, with exact inverse and denominators

The preceding nonintersection concerns the specified map, not every map between the quadratic surfaces. The following construction uses projection from the rational point $(1,0,0)$ on $\Delta=-1$; it does not replace F or the physical meaning, if any, assigned to its coordinates.

Lemma 8.4. *The equation $\Delta(q)=0$ has no nonzero rational solution.*

Proof. After clearing denominators use the integral S,U,V just defined. Expansion gives $2\Delta = 2S^2 - 3U^2 - V^2$. Modulo three, $2S^2 = V^2$ forces $3 \mid S,V$. Substitution then forces $3 \mid U$. Repeating gives divisibility of S,U,V by every power of three, hence all vanish. The linear map from q to (S,U,V) has nonzero determinant, so $q=0$. □

Theorem 8.5. *For fixed $\delta \in \mathbb{Q}_{>0}$ write*

$$X_\delta = \{(a,b,c) : ac - b^2 = -\delta\}, \quad Y = \{q : \Delta(q) = -1\},$$

and define

$$D(a,b) = -1 - a^2 - b^2 + 4a + 4b + 4ab, \quad K(a,b) = 1 - 2a - 2b.$$

On $U_\delta = \{(a,b,c) \in X_\delta : aDK \neq 0\}$ the map

$$\mathcal{B}_\delta(a,b,c) = \left(1 + \frac{2K}{D}, \frac{2Ka}{D}, \frac{2Kb}{D}\right)$$

is an isomorphism of open varieties over $\mathbb{Q}$ onto the open part of Y where $q_1 \neq 1$, $q_2 \neq 0$, and $D(q_2/(q_1-1), q_3/(q_1-1)) \neq 0$. Its inverse is

$$a = \frac{q_2}{q_1 - 1}, \qquad b = \frac{q_3}{q_1 - 1}, \qquad c = \frac{b^2 - \delta}{a}.$$

Every integral matrix on X_δ with $a > 0$ belongs to U_δ. For such a matrix, the exact least common denominator of its image is

$$d_{\text{image}} = \frac{|D(a,b)|}{\gcd(|D(a,b)|, |2K(a,b)|)}.$$

In particular its image is integral exactly when $D(a,b) \mid 2K(a,b)$.

Proof. Set $q = (1,0,0) + \lambda(1,a,b)$. Expansion, with no change of scale in Δ, gives

$$\Delta(q) = -1 - 2K\lambda + D\lambda^2.$$

The second intersection with Y is therefore $\lambda = 2K/D$, nonzero on the stated domain. The displayed image and inverse follow. Conversely, for a point of the stated target open set, put $\lambda = q_1 - 1$ and recover a,b. Its equation gives $\lambda D = 2K$, so $K \neq 0$ and the constructions are inverse regular maps on the indicated opens. Recovering c retains the fixed determinant $-\delta$. For integral a,b, K is odd and cannot vanish. Moreover $D = \Delta(1,a,b) \neq 0$ by Lemma 8.4. Finally, $q_1 - 1 = 2K/D$ has reduced denominator $|D|/\gcd(|D|,|2K|)$, and q_2, q_3 are integer multiples of this fraction. This proves both the denominator formula and its exactness. □

For the packet's fixed opening $A_* = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, $\delta = 1$, $D = 9$, $K = -3$, and

$$\mathcal{B}_1(A_*) = (1/3, -2/3, -2/3), \qquad \Delta(1/3, -2/3, -2/3) = -1.$$

This rational image coexists with the irrationality of $\mathfrak{K}_1(A_*)$, because the two maps are different and both have now been specified. The rational bridge does not automatically preserve the integral positive chamber, the integral orbit count, or a dynamics. The denominator condition above is the exact next arithmetic restriction, and the inverse retains δ as a parameter rather than erasing it.

8.7 Independent checks and reproducibility

The separate verifier `supporting_materials/workbench/evidence/koide_es_v3_audit_20260904/verify_independent.py` enumerates the smaller denominator $N/R < y \le 2N/R$, then reconstructs $z = Ny/(Ry - N)$, independently of the packet's divisor enumerator. It agrees with all 1430 oriented states in 665 coprime shells through $n = 161$. It checks the 819 diagonal certificates for 329 primes below 5000, the symbolic rational-map identity and inverse, and 820 exact denominator cases. These finite regressions supplement, rather than replace, the proofs above. The original packet's verifier is replayed separately without modifying its source or its original output. The archive SHA-256 is

```
ab5c21cf00268f9019c95ee24b565d05c4d9337c51e4a576ded9e3cc3f0f235e
```

The correct full hash is recorded byte-for-byte in the evidence manifest; the line breaking here is typographical only.

9 The fifteen $p = 1201$ selector rays inside $\mathrm{Herm}_3(\mathbb{H})$

The common integer 15 in the complete $p = 1201$ selector census and in $\dim_{\mathbb{R}} \mathrm{Herm}_3(\mathbb{H}) = 15$ admits an exact test. The test gives a nontrivial morphism, but it rules out the interpretation of the selector states as fifteen independent Jordan-algebra directions.

9.1 Marked selectors and their prime families

Fix $p_0 = 1201$. A marked state in the complete census is a tuple

$$s = (\chi, R, h, r, t), \qquad \chi \in \{E, M\}, \qquad p_0 = 4rth - R, \qquad \gcd(r, t) = 1,$$

with positive integral R, h, r, t. Its channel gate is

$$\begin{aligned} E: &\quad R \mid 4r^2h + 1, \\ M: &\quad R \mid r + t. \end{aligned} \tag{1}$$

In the exterior channel the first gate is equivalent to $R \mid p_0 r + t$, because

$$p_0 r + t = t(4r^2h + 1) - Rr.$$

Put

$$(u, a, v) = (hr^2, hrt, ht^2), \qquad uv = a^2, \qquad R = 4a - p_0. \tag{2}$$

If $\kappa_E = (p_0 r + t)/R$ and $m = (r + t)/R$, the ordered denominator maps are

$$D_E(s) = (hrt,\ ht\kappa_E,\ p_0 hr\kappa_E), \tag{3}$$

$$D_M(s) = (hrt,\ p_0 hrm,\ p_0 htm). \tag{4}$$

Clearing the full common denominator proves

$$\sum_{d \in D_\chi(s)} \frac{1}{d} = \frac{4}{p_0}.$$

The fifteen states, with orientation and scale retained, are as follows. The integer L is the least prime increment that also preserves $p \equiv 361 \pmod{840}$.

tag	χ	R	h	(r,t)	u	a	v	L	Koide +
E23a	E	23	17	$(1,18)$	17	306	5508	57960	yes
M23a	M	23	3	$(6,17)$	108	306	867	14280	no
M23b	M	23	3	$(17,6)$	867	306	108	14280	no
E23b	E	23	34	$(9,1)$	2754	306	34	57960	no
M31a	M	31	2	$(1,154)$	2	308	47432	9240	yes
E31	E	31	77	$(4,1)$	1232	308	77	52080	no
M31b	M	31	2	$(154,1)$	47432	308	2	9240	no
M39a	M	39	2	$(1,155)$	2	310	48050	26040	yes
E39	E	39	62	$(5,1)$	1550	310	62	10920	no
M39b	M	39	2	$(155,1)$	48050	310	2	26040	no
M47a	M	47	1	$(8,39)$	64	312	1521	43680	no
M47b	M	47	1	$(39,8)$	1521	312	64	43680	no
E71	E	71	53	$(1,6)$	53	318	1908	59640	no
E339	E	339	11	$(35,1)$	13475	385	11	94920	no
E1043	E	1043	17	$(33,1)$	18513	561	17	1376760	no

For an exterior row define

$$h_k = h_0 + Rk, \qquad p_k = p_0 + 4rtRk;$$

for a middle row define

$$h_k = h_0 + k, \qquad p_k = p_0 + 4rtk.$$

The corresponding gate in (1) is preserved. The exterior prime class modulo $4rtR$ and the middle prime class modulo $4rt$ are reduced. Their least intersections with $361 \pmod{840}$ are

$$L_s = \begin{cases} \operatorname{lcm}(4rtR, 840), & \chi = E, \\ \operatorname{lcm}(4rt, 840), & \chi = M. \end{cases} \tag{5}$$

Thus

$$p_n = 1201 + L_s n, \qquad h_n = h_0 + \frac{L_s}{4rt} n \tag{6}$$

retains the same marked ray, and Dirichlet's theorem gives infinitely many prime p_n in every row. The fifteen marked rows give eleven unordered denominator triples at p_0 and ten distinct residue classes $1201 \pmod{L_s}$. These are different quotients: middle reversal produces the first collapse, while E23a and E23b have distinct rays but the same $L_s = 57960$.

9.2 The exact Koide cone map

Define

$$\Gamma(s) = \begin{pmatrix} u & a \\ a & v \end{pmatrix} = h \begin{pmatrix} r \\ t \end{pmatrix} (r \ t). \tag{7}$$

This matrix is positive semidefinite of rank one and has Smith normal form $\operatorname{diag}(h, 0)$. The Lorentz-coordinate map

$$\mathcal{L}(\Gamma) = \left(\frac{u+v}{2}, a, \frac{u-v}{2} \right) = (H, X, Y) \tag{8}$$

satisfies $H^2 - X^2 - Y^2 = uv - a^2 = 0$.

Let

$$F(q_1, q_2, q_3) = \left(\frac{q_1 + q_2 + q_3}{\sqrt{3}}, \frac{q_1 - q_3}{\sqrt{2}}, \frac{q_1 + q_3 - 2q_2}{\sqrt{6}} \right).$$

This is orthogonal. Its inverse at (8) is

$$\begin{aligned} q_1 &= \frac{u+v}{2\sqrt{3}} + \frac{a}{\sqrt{2}} + \frac{u-v}{2\sqrt{6}}, \\ q_2 &= \frac{u+v}{2\sqrt{3}} - \frac{u-v}{\sqrt{6}}, \\ q_3 &= \frac{u+v}{2\sqrt{3}} - \frac{a}{\sqrt{2}} + \frac{u-v}{2\sqrt{6}}. \end{aligned} \tag{9}$$

For

$$Q_K(q) = \frac{2}{3}(q_1+q_2+q_3)^2 - (q_1^2+q_2^2+q_3^2)$$

orthogonality gives $Q_K(q) = H^2 - X^2 - Y^2 = 0$. Hence (9) is an exact coefficient-preserving map over $\mathbb{Q}(\sqrt{2},\sqrt{3})$, not a numerical fit. In the displayed orientation exactly E23a, M31a, and M39a lie in the positive Koide chamber.

9.3 Embedding and rank obstruction in the quaternionic Jordan algebra

Let $J = \mathrm{Herm}_3(\mathbb{H})$ with $A \circ B = (AB+BA)/2$, and put $e = \mathrm{diag}(1,1,0)$. Define

$$\iota : \mathrm{Sym}_2(\mathbb{R}) \longrightarrow J, \qquad \iota\begin{pmatrix} u & a \\ a & v \end{pmatrix} = \begin{pmatrix} u & a & 0 \\ a & v & 0 \\ 0 & 0 & 0 \end{pmatrix}. \tag{10}$$

Block multiplication proves $\iota(A \circ B) = \iota(A) \circ \iota(B)$. Thus ι is an injective real-linear Jordan homomorphism with image

$$C_{\mathbb{R}} := \iota(\mathrm{Sym}_2(\mathbb{R})) = eJe \cap \mathrm{Herm}_3(\mathbb{R}), \qquad \dim_{\mathbb{R}} C_{\mathbb{R}} = 3.$$

The full Peirce corner is $eJe \cong \mathrm{Herm}_2(\mathbb{H})$ of real dimension 6; $C_{\mathbb{R}}$ is its proper real-symmetric Jordan subalgebra. It preserves trace, trace pairing, positivity, and ordinary rank. In the Moore convention

$$\begin{pmatrix} d_1 & q_3 & \bar{q}_2 \\ \bar{q}_3 & d_2 & q_1 \\ q_2 & \bar{q}_1 & d_3 \end{pmatrix},$$

the only nonzero coordinates of $\iota\Gamma(s)$ are $d_1 = u, d_2 = v, q_3 = a \in \mathbb{R}$. Its Moore cubic is zero.

Projectivizing the scale gives the idempotent

$$P_{r,t} = \frac{1}{r^2+t^2}\begin{pmatrix} r^2 & rt & 0 \\ rt & t^2 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad P_{r,t}^2 = P_{r,t}, \quad \mathrm{tr}\, P_{r,t} = 1. \tag{11}$$

It represents the quaternionic line $[r : t : 0]_{\mathbb{H}}$. The fifteen rows have pairwise distinct ratios $[r : t]$, so the composition

$$\Sigma_{1201} \hookrightarrow \mathbb{RP}^1 \hookrightarrow \mathbb{HP}^2, \qquad s \longmapsto [r : t] \longmapsto [r : t : 0]_{\mathbb{H}} \tag{12}$$

is injective.

Theorem 9.1 (Exact span and Jordan closure)**.** *The fifteen matrices $\iota\Gamma(s)$ are fifteen distinct projective rank-one points, but*

$$\mathrm{span}_{\mathbb{R}}\{\iota\Gamma(s) : s \in \Sigma_{1201}\} = \langle \iota\Gamma(\Sigma_{1201})\rangle_{\mathrm{Jordan}} = C_{\mathbb{R}}.$$

In particular, both spaces have real dimension 3*, not* 15*.*

Proof. In coordinates $\left(\begin{smallmatrix} x & y \\ y & z \end{smallmatrix}\right) \leftrightarrow (x,y,z)$, the projective image of (r,t) is (r^2, rt, t^2). The rows E31, E39, and E339 give

$$\begin{pmatrix} 16 & 4 & 1 \\ 25 & 5 & 1 \\ 1225 & 35 & 1 \end{pmatrix}, \qquad \det = -930 \neq 0.$$

Hence the images span the whole three-dimensional real-symmetric corner $C_{\mathbb{R}}$. The space $C_{\mathbb{R}}$ is a Jordan subalgebra by the block-multiplication identity above, so the generated Jordan algebra is contained in $C_{\mathbb{R}}$. A Jordan subalgebra is a real vector space; because it contains the fifteen generators, it contains their span $C_{\mathbb{R}}$. Both containments are equalities. □

The number fifteen is not invariant even after the hard residue class is held fixed. Complete divisor enumeration gives the following exact comparison:

p	1201	4561	8761	9601
$\#E$	7	26	16	17
$\#M$	8	28	20	12
$\#(E \sqcup M)$	15	54	36	29
#{unordered denominator triples}	11	40	26	23

All four displayed primes satisfy $p \equiv 361 \pmod{840}$. Thus neither the hard residue class nor the selector construction singles out a fifteen-element family; the equality $\#\Sigma_{1201} = \dim_{\mathbb{R}} \mathrm{Herm}_3(\mathbb{H}) = 15$ is a feature of the individual prime 1201, not an invariant of the arithmetic family. The deterministic enumeration and primality checks are included in the accompanying verification script.

At fixed p_0 the unprojectivized embedding preserves the markings. From $\iota\Gamma$ one recovers (u, a, v) and then

$$h = \gcd(u, a, v), \quad r = \sqrt{u/h}, \quad t = \sqrt{v/h}, \quad R = 4a - 1201, \quad j = (R - 3)/4.$$

The two channel gates cannot overlap in this census: overlap would imply $R \mid 1200$, and none of $23, 31, 39, 47, 71, 339, 1043$ divides 1200. Projectivization in (11) loses h, and without the fixed prime the residual formula $R = 4a - p$ is unavailable.

9.4 Symmetry and incidence obstructions

Normalized orientation reversal fixes the seven exterior states and swaps the four middle pairs, so its cycle type is $1^7 2^4$. The transvection

$$\tau(x_1, x_2, x_3, x_4) = (x_1 + x_2, x_2, x_3, x_4)$$

on $\mathbb{F}_2^4 \setminus \{0\}$ has the same cycle type. For example, with integers $1, \ldots, 15$ encoding the four binary coordinates from the low bit, the following map is C_2-equivariant:

s	$E23a$	$E23b$	$E31$	$E39$	$E71$	$E339$	$E1043$
$\phi(s)$	1	4	5	8	9	12	13

s	$M23a$	$M23b$	$M31a$	$M31b$
$\phi(s)$	2	3	6	7

s	$M39a$	$M39b$	$M47a$	$M47b$
$\phi(s)$	10	11	14	15

This proves only an equivalence of C_2-sets. It is noncanonical: a set with this cycle type has $7!2^4 4! = 1{,}935{,}360$ automorphisms commuting with the involution.

It cannot be upgraded to the natural projective incidence. The selector points lie on the nonsingular conic $xz - y^2 = 0$ in $\mathbb{P}^2(\mathbb{R})$, so no three distinct selector images are collinear. By contrast, $PG(3, 2)$ has 35 three-point lines. Nor does normalized reversal extend linearly on the canonical real-symmetric corner: it fixes E31, E39, and E339, which span $C_{\mathbb{R}}$, so such a linear map would be the identity; but it must exchange four nonidentical middle pairs.

The other elementary count comparisons do not supply a bridge. Type A_3 has 12 oriented roots, 6 unoriented root lines, and $4 + 6 + 4 = 14$ nonzero minuscule weights. The rank-one idempotents of J form the real eight-dimensional continuum $\mathbb{HP}^2$, while the present images occupy the proper copy $\mathbb{RP}^1$ in (12).

9.5 Conclusion and next exact test

The current structural statement is therefore

$$\boxed{\begin{array}{c}\text{fifteen marked prime-carrying selector rays } \hookrightarrow \mathbb{RP}^1 \hookrightarrow \mathbb{HP}^2, \\ \dim \text{span} = \dim \text{Jordan closure} = 3.\end{array}}$$

The equality between the number of marked rays and $\dim_{\mathbb{R}} J$ is not explained by the defined morphism.

A decisive higher-rung test would construct, from the arithmetic residual and channel markings, an explicit element g_s of a named quaternionic structure group and study

$$g_s \iota(\Gamma(s)) g_s^*.$$

One must then compute the span and Jordan closure over complete multi-prime censuses and prove equivariance with the selector involutions. If all images remain in one common three-dimensional real-symmetric Jordan corner, the obstruction persists; if the markings canonically move them through enough conjugate corners and the closure is all of J, that would be a genuine fifteen-dimensional bridge. No such marking-to-conjugation map is presently defined.

10 The analytic threefold constructed in the candidate

This section audits the construction itself. It does not infer existence from the later topology. The period family, the three local fillings, and the gluing are proved in that order. The notation is the candidate's: $B \simeq \mathbb{P}^1$, $B^\circ = B \setminus \{p_0, p_1, p_2\}$, and the local monodromy orders at p_1, p_2 are $m_1 = 3, m_2 = 4$.

10.1 The period family

Let $\Delta \simeq (\mathbb{Z}/3) * (\mathbb{Z}/4)$ be the orientation-preserving $(3, 4, \infty)$-triangle group acting on a simply connected Riemann surface $\mathfrak{h}_z$, with elliptic generators g_1, g_2 and $g_0 = (g_1 g_2)^{-1}$. Let t be the affine coordinate on B with values $0, 1, \infty$ at p_1, p_2, p_0, respectively. Write

$$\pi : \mathfrak{h}_z \longrightarrow B \setminus \{p_0\}$$

for the orbifold quotient map; its stabilizers at chosen lifts $z_j \in \pi^{-1}(p_j)$ are $\langle g_j \rangle$. Let $\rho_V : \Delta \to \mathrm{GL}(V)$ be the printed representation, $\rho_V(g_j) = T_j$, and put

$$M_g := \rho_V(g)^{\mathsf{t}}, \qquad A_g := M_g^{-1}.$$

Thus $M_{gh} = M_h M_g$, $A_{g_j} = A_j$, and $A_{g_0} = M_0$ in the notation of theorem 5.1.

Lemma 10.1 (simultaneous elliptic-orientation normalization)**.** ***[VERIFIED]*** *The generators may, and henceforth do, satisfy*

$$g_1'(z_1) = e^{-2\pi i/3}, \qquad g_2'(z_2) = e^{-2\pi i/4} = -i, \tag{13}$$

while $g_0 = (g_1 g_2)^{-1}$ remains parabolic. More precisely, the Cayley coordinates

$$s_j(z) = \frac{z - z_j}{z - \bar{z}_j} \tag{14}$$

obey

$$s_1 \circ g_1 = e^{-2\pi i/3} s_1, \qquad s_2 \circ g_2 = -i s_2. \tag{15}$$

The two signs in equation (13) *are a simultaneous orientation choice; they are not obtained by shrinking a coordinate disc.*

Proof. Realize Δ as the orientation-preserving subgroup of the reflection group of a hyperbolic triangle with counterclockwise vertices A, B, C, angles $\pi/3, \pi/4, 0$, and opposite sides a, b, c, respectively. Let $\sigma_a, \sigma_b, \sigma_c$ denote reflection in those sides and put

$$x = \sigma_b \sigma_c, \qquad y = \sigma_c \sigma_a, \qquad z = \sigma_a \sigma_b.$$

The product of reflections in two oriented geodesics meeting at angle θ is the rotation through 2θ. Hence x is the counterclockwise rotation about A through $2\pi/3$, y is the counterclockwise rotation about B through $\pi/2$, and z is parabolic at the ideal vertex C; this is the standard triangle-group construction [27, Chapters 7 and 10]. Moreover

$$xyz = \sigma_b\sigma_c\sigma_c\sigma_a\sigma_a\sigma_b = 1.$$

Set

$$z_1 = A, \qquad z_2 = B, \qquad g_1 = x^{-1}, \qquad g_2 = y^{-1}.$$

Both finite rotations are thereby reversed, so their complex derivatives at their fixed points are exactly those in equation (13). Since $g_2g_1 = y^{-1}x^{-1} = z$, one also has

$$g_1g_2 = g_1zg_1^{-1};$$

thus g_1g_2, and therefore $g_0 = (g_1g_2)^{-1}$, is parabolic.

This simultaneous reversal is forced once the orientation of g_1 and the cusp conjugacy class are fixed. Indeed, in $\langle x\rangle * \langle y\rangle \simeq C_3 * C_4$ every parabolic element is conjugate to a nonzero power of the primitive cusp word $z = y^{-1}x^{-1} = y^3x^2$. The two mixed choices would instead give

$$(x^{-1}y)^{-1} = y^{-1}x = y^3x, \qquad (xy^{-1})^{-1} = yx^{-1} = yx^2.$$

Both words are cyclically reduced of length two. A nontrivial power z^n has cyclically reduced length $2|n|$; hence a mixed word could be parabolic only for $|n| = 1$. But neither y^3x nor yx^2 is a cyclic permutation of y^3x^2 or its inverse xy. Thus neither mixed choice is parabolic. The derivative selected at z_1 consequently fixes the derivative at z_2 as well.

Finally, for an elliptic Möbius transformation g fixing $z_0 \in \mathfrak{h}_z$, the Cayley map $s = (z - z_0)/(z - \bar{z}_0)$ conjugates g to a disc automorphism fixing zero. Such an automorphism is $s \mapsto g'(z_0)s$. Applying this to g_1, g_2 proves equation (15). □

Proposition 10.2 (existence and non-degeneracy of the period matrix)**.** ***[VERIFIED]*** *There are holomorphic functions*

$$\tau : \mathfrak{h}_z \to \mathfrak{h}, \qquad \mu, \beta : \mathfrak{h}_z \to \mathbb{C}$$

satisfying

$$\begin{array}{lll} \tau \circ g_0 = \tau - 1, & \mu \circ g_0 = \mu, & \beta \circ g_0 = \beta + 1, \\ \tau \circ g_2 = -1/\tau, & \mu \circ g_2 = 1 + \mu/\tau, & \beta \circ g_2 = \beta - 3 - 6\mu^2/\tau, \end{array} \tag{2.1}$$

and the corresponding order-three transformation laws obtained from $\tau \circ g_1 = (\tau - 1)/\tau$. They may be chosen so that

$$\operatorname{Im}\tau > 0, \qquad D := \operatorname{Im}\beta - \frac{6(\operatorname{Im}\mu)^2}{\operatorname{Im}\tau} < 0. \tag{2.2}$$

Consequently

$$\Pi = [Z \mid I_2], \qquad Z = \begin{pmatrix} 6\mu & \tau \\ \beta & \mu \end{pmatrix} \tag{2.3}$$

maps the marked rank-four lattice Λ isomorphically onto a lattice in $\mathbb{C}^2$ at every point. The quotients $\mathbb{C}^2/\Pi(z)\Lambda$ form a holomorphic family of compact two-dimensional complex tori over $\mathfrak{h}_z$, equivariant for Δ.

Proof. Normalize the modular function by $j(q) = q^{-1}(1 + O(q))$ and put $J_B = 1728(t \circ \pi)$; the modular facts used here are collected in [26, Chapter VII]. Put

$$\mathfrak{h}_z^\dagger = \mathfrak{h}_z \setminus J_B^{-1}\{0, 1728\}, \qquad \mathfrak{h}_\tau^\dagger = \mathfrak{h} \setminus j^{-1}\{0, 1728\}.$$

Away from the deleted sets, the classical modular map $j : \mathfrak{h}_\tau^\dagger \to \mathbb{C} \setminus \{0, 1728\}$ is an unramified covering with deck group $\mathrm{PSL}_2(\mathbb{Z})$. The deleted set in $\mathfrak{h}_z$ is locally finite. Since $\mathfrak{h}_z$ is simply connected, $\pi_1(\mathfrak{h}_z^\dagger)$

is normally generated by small positive meridians about its points. At a translate of z_1, J_B has local degree three over 0; the corresponding meridian therefore has deck image equal to the third power of the order-three modular generator. At a translate of z_2, J_B has local degree four over 1728; its meridian has deck image equal to the fourth power of the order-two modular generator. Both images are the identity. The covering-space lifting criterion gives a global lift τ with $j(\tau) = J_B$.

Near a deleted point, compose the lift with a biholomorphism $\chi : \mathfrak{h} \to \mathbb{D}$. The bounded map $\chi \circ \tau$ extends by the removable-singularity theorem. Its value at the puncture has modulus strictly less than one: equality would give an interior maximum and force $\chi \circ \tau$ to be constant, although its punctured values lie in $\mathbb{D}$. Thus the extension still lands in $\mathfrak{h}$.

Because J_B is Δ-invariant, $\tau \circ g$ and τ are lifts of the same map through the Galois covering j on the punctured surface. Hence there is a unique deck transformation $\varphi(g) \in \mathrm{PSL}_2(\mathbb{Z})$ such that

$$\tau \circ g = \varphi(g) \circ \tau. \tag{16}$$

The identity extends across the deleted points. Uniqueness and $(\tau \circ g) \circ h = \tau \circ (gh)$ give $\varphi(gh) = \varphi(g)\varphi(h)$, so φ is a homomorphism.

Normalize $\tau(z_1) = \rho = e^{\pi i/3}$. The local cubic expansion of j at ρ, together with the local degree three of J_B, gives a simple zero of $\tau - \rho$ at z_1. At z_2, choose $\gamma \in \mathrm{PSL}_2(\mathbb{Z})$ with $\tau(z_2) = \gamma(i)$. The local quadratic expansion of $j \circ \gamma$ at i, together with the local degree four of J_B, gives a double zero of $\gamma^{-1}\tau - i$ at z_2. At z_1, differentiating equation (16) and using the simple zero gives

$$\varphi(g_1)'(\rho) = g_1'(z_1) = e^{-2\pi i/3}.$$

The stabilizer of ρ is generated by $S_1(\tau) = (\tau - 1)/\tau$, whose derivative at ρ is $e^{-2\pi i/3}$; hence $\varphi(g_1) = S_1$, rather than S_1^{-1}. The order-four generator cannot have trivial modular image: otherwise the invariant local power series $\gamma^{-1}\tau - i$ would have order divisible by four, contrary to its order two. Its modular image therefore has order two. If $P = \varphi(g_0)$, the exact relation $g_1 g_2 g_0 = 1$ gives

$$\varphi(g_2) = S_1^{-1} P^{-1}. \tag{17}$$

It remains to determine the cusp image. Choose a punctured cusp disc $D_0^* \subset B$ and a component

$$U_0 \subset \pi^{-1}(D_0^*)$$

stabilized by $\langle g_0 \rangle$. Choose R and shrink D_0^* so that $\tau(U_0)$ lies in a single component

$$K \subset j^{-1}(\{|w| > R\}).$$

Put $P = \varphi(g_0)$. Since $g_0 U_0 = U_0$,

$$P\tau(U_0) = \tau(g_0 U_0) = \tau(U_0).$$

Hence $P(K) = K$: the two components K and $P(K)$ intersect in the nonempty set $\tau(U_0)$. Let c be the rational boundary cusp of K, choose $\gamma_0 \in \mathrm{PSL}_2(\mathbb{Z})$ with $\gamma_0(\infty) = c$, and set

$$\tau' = \gamma_0^{-1}\tau|_{U_0}, \qquad P' = \gamma_0^{-1} P \gamma_0.$$

Then

$$\tau' \circ g_0 = P' \circ \tau',$$

and P' preserves $K' = \gamma_0^{-1} K$, whose boundary cusp is ∞.

Suppose $P' = 1$. Then τ' is g_0-invariant and descends to a holomorphic map $\bar{\tau}' : D_0^* \to \mathfrak{h}$. Put

$$q' = \exp(2\pi i \bar{\tau}').$$

After increasing R, the q-expansion is valid on K', and $0 < |q'| < 1$. Thus q' extends holomorphically across the puncture. If $|q'(0)| = 1$, the maximum-modulus principle makes q', hence J_B, constant. If $q'(0) \neq 0$, the expansion $j(q') = (q')^{-1}(1 + O(q'))$ makes J_B bounded, contrary to its pole. Therefore $q'(0) = 0$. But a holomorphic zero has positive winding on a small positive circle, while q' has the single-valued holomorphic logarithm $2\pi i\bar{\tau}'$ on D_0^*, so its winding is zero. This contradiction proves $P' \neq 1$. Since P' preserves the selected cusp component K', it is parabolic; hence the original P is a nontrivial parabolic.

We now discard the auxiliary conjugation by γ_0 and return to the original normalization

$$\tau(z_1) = \rho, \qquad \varphi(g_1) = S_1.$$

Choose the trace-+2 lift $\widetilde{P} \in \mathrm{SL}_2(\mathbb{Z})$ of P. For a primitive fixed vector $(p, r)^{\mathrm{t}}$, it has the unique form

$$\widetilde{P} = I + k \begin{pmatrix} -pr & p^2 \\ -r^2 & pr \end{pmatrix}, \qquad p, r \in \mathbb{Z}, \quad \gcd(p, r) = 1, \quad k \neq 0.$$

With

$$\widetilde{S}_1 = \begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix},$$

one has

$$\mathrm{tr}(\widetilde{S}_1^{-1}\widetilde{P}^{-1}) = 1 + k(p^2 - pr + r^2).$$

Since $S_1^{-1}P^{-1} = \varphi(g_2)$ has projective order two, this trace is zero. Hence

$$-k(p^2 - pr + r^2) = 1,$$

so $k = -1$ and $p^2 - pr + r^2 = 1$. Modulo simultaneous sign, the three solutions are cycled transitively by S_1. Conjugating the entire deck representation by a suitable power of S_1, which fixes ρ and leaves S_1 unchanged, therefore gives

$$P(w) = w - 1, \qquad \varphi(g_2)(w) = -\frac{1}{w}.$$

Rename the resulting period map τ. Since z_2 is fixed by the order-four generator, the new normalization has $\tau(z_2) = i$, and the preceding local ramification calculation says exactly that $\tau - i$ has order two at z_2. Let t_c vanish at p_0 and define anew

$$q = e^{2\pi i\tau}.$$

Since $\tau \circ g_0 = \tau - 1$, q descends to D_0^*. The simple pole of $J_B = j(\tau)$ gives

$$q = t_c u_1(t_c), \qquad u_1(0) \neq 0.$$

Writing

$$S_1(w) = \frac{w-1}{w}, \qquad S(w) = -\frac{1}{w}, \qquad P(w) = w - 1,$$

the normalized deck certificate is therefore

$$\varphi(g_1) = S_1, \qquad \varphi(g_2) = S, \qquad \varphi(g_0) = P, \qquad S_1SP = 1 \quad \text{in } \mathrm{PSL}_2(\mathbb{Z}). \tag{18}$$

Differentiation at z_1, where $\tau - \rho$ has order one, gives $g_1'(z_1) = S_1'(\rho) = \rho^{-2} = e^{-2\pi i/3}$. At z_2, comparison of the quadratic leading terms gives only

$$\big(g_2'(z_2)\big)^2 = S'(i) = -1.$$

The choice $g_2'(z_2) = -i$, rather than $+i$, is supplied by the simultaneous parabolic normalization in theorem 10.1; it does not follow from the quadratic ramification calculation alone.

We now solve the two affine cocycle problems, including the orbifold and cusp regularity which the abbreviated argument had suppressed. Define $\widetilde{\jmath}\colon \Delta \times \mathfrak{h}_z \to \mathbb{C}^*$ on the generators by

$$\widetilde{\jmath}(g_1, z) = -\tau(z), \qquad \widetilde{\jmath}(g_2, z) = \tau(z),$$

and extend it by $\widetilde{\jmath}(gh, z) = \widetilde{\jmath}(g, hz)\widetilde{\jmath}(h, z)$. This is well defined: the products around the two finite cyclic subgroups are

$$(-1)^3\tau\frac{\tau-1}{\tau}\frac{-1}{\tau-1} = 1, \qquad \tau\left(-\frac{1}{\tau}\right)\tau\left(-\frac{1}{\tau}\right) = 1.$$

Moreover $\widetilde{\jmath}(g_0, \cdot) = 1$.

For an open $U \subset B$, set $\widetilde{U} = \pi^{-1}(U \setminus \{p_0\})$. Define

$$\mathcal{L}(U) = \left\{\nu \in \mathcal{O}(\widetilde{U}) : \nu \circ g = \nu/\widetilde{\jmath}(g, \cdot) \quad (g \in \Delta)\right\}$$

and let $\mathcal{M}(U)$ be the holomorphic functions on $\widetilde{U}$ satisfying

$$\mu \circ g_1 = \frac{1-\mu}{\tau}, \qquad \mu \circ g_2 = 1 + \frac{\mu}{\tau}. \tag{2.1a}$$

If $p_0 \in U$, both definitions additionally require that the restriction to the distinguished $\langle g_0\rangle$-invariant cusp component, viewed as a function of t_c, extend holomorphically across $t_c = 0$. Because $\widetilde{\jmath}(g_0) = 1$ and $\mu \circ g_0 = \mu$, this requirement is meaningful and is stable under restriction. Thus $\mathcal{L}$ is a line sheaf and $\mathcal{M}$, if locally nonempty, is an affine $\mathcal{L}$-torsor.

Local nonemptiness is explicit. Over an ordinary small disc, prescribe $\mu = 0$ on one sheet and extend using (2.1a). At the cusp prescribe $\mu = 0$ on the distinguished component. At the elliptic points use

$$\mu_1 = \frac{2-\tau}{3} \quad \text{on } \Delta_1, \qquad \mu_2 = \frac{1-\tau}{2} \quad \text{on } \Delta_2. \tag{2.1b}$$

Indeed

$$\mu_1 \circ g_1 = \frac{\tau+1}{3\tau} = \frac{1-\mu_1}{\tau}, \qquad \mu_2 \circ g_2 = \frac{\tau+1}{2\tau} = 1 + \frac{\mu_2}{\tau}.$$

The cyclic products above, or direct iteration of (2.1a), show that these prescriptions close after three and four steps.

To identify $\mathcal{L}$, use that E_6 has a simple zero at the order-two elliptic orbit and no other zero in $\mathfrak{h}$. Since $\tau - i$ has order two at every translate of z_2, the function $E_6 \circ \tau$ has only even-order zeros on the simply connected surface $\mathfrak{h}_z$. Fix a holomorphic square root h, $h^2 = E_6 \circ \tau$, and put

$$\Theta = \frac{(E_4 \circ \tau)^2 h}{\Delta_{\mathrm{mod}} \circ \tau}.$$

For $j = 1, 2$, the quotient $h(g_j z)/(\tau(z)^3 h(z))$ has square one away from its removable zero set, hence is a constant sign. At z_1, $h(z_1) \neq 0$ and $\rho^3 = -1$, so the sign for g_1 is -1. At z_2, choose the linearizing coordinate with $s_2 \circ g_2 = -is_2$. The function h has a simple zero there, and therefore $h(g_2 z)/h(z) \to -i = i^3$; the sign for g_2 is $+1$. The modular transformation formulae

$$E_4(\gamma\tau) = (c\tau + d)^4 E_4(\tau), \quad E_6(\gamma\tau) = (c\tau + d)^6 E_6(\tau), \quad \Delta_{\mathrm{mod}}(\gamma\tau) = (c\tau + d)^{12}\Delta_{\mathrm{mod}}(\tau)$$

now give the exact identities

$$\Theta \circ g_1 = -\Theta/\tau, \qquad \Theta \circ g_2 = \Theta/\tau. \tag{2.1e}$$

These are precisely the automorphy laws defining $\mathcal{L}$. Since E_4 has a simple zero at ρ, whereas E_6 has a simple zero at i, the orders of Θ at z_1, z_2 are respectively two and one. At the distinguished cusp, E_4, E_6 are units and $\Delta_{\mathrm{mod}} = q \cdot (\text{unit})$, with $q = t_c \cdot (\text{unit})$, so $\Theta = t_c^{-1} \cdot (\text{unit})$. Taking the stabilizer orders into

account, division by Θ identifies the ordinary and elliptic stalks with $\mathcal{O}_B$, while the cusp stalk is $\mathfrak{m}_{p_0}\Theta$. Hence

$$\mathcal{L} \simeq \mathcal{O}_B(-p_0) \simeq \mathcal{O}_{\mathbb{P}^1}(-1).$$

Consequently $H^1(B,\mathcal{L}) = H^0(B,\mathcal{L}) = 0$: the locally nonempty torsor $\mathcal{M}$ has a unique global section. Evaluating at the fixed points gives

$$\mu(z_1) = \frac{2-\rho}{3}, \qquad \mu(z_2) = \frac{1-i}{2},$$

and the cusp condition says precisely that its invariant branch extends holomorphically.

For the second affine problem put

$$\phi_1 = 2 - \frac{6(1-\mu)^2}{\tau}, \qquad \phi_2 = -3 - \frac{6\mu^2}{\tau}.$$

Substitution around the finite orbits gives

$$\sum_{k=0}^{2} \phi_1 \circ g_1^k = 0, \qquad \sum_{k=0}^{3} \phi_2 \circ g_2^k = 0. \tag{2.1c}$$

For example, the three summands in the first identity are

$$2 - \frac{6(1-\mu)^2}{\tau}, \quad 2 - \frac{6(\tau-1+\mu)^2}{\tau(\tau-1)}, \quad 2 + \frac{6\mu^2}{\tau-1},$$

Putting them over the common denominator $\tau(\tau-1)$, the three constant terms contribute $6\tau(\tau-1)$, while the remaining terms contribute $-6\tau(\tau-1)$; hence the total numerator is zero. For the order-four identity the bracket after extracting $6/\tau$ is 2τ, cancelling -12.

Define $\mathcal{B}(U)$ to consist of holomorphic β on $\widetilde{U}$ with $\beta \circ g_j - \beta = \phi_j$; at the cusp require $b = \beta + \tau$ to extend holomorphically in t_c. Differences of sections are ordinary holomorphic functions on U, so $\mathcal{B}$ is a pseudo-torsor under $\mathcal{O}_B$. It is locally nonempty: prescribe $\beta = 0$ on one ordinary sheet; prescribe $\beta = -\tau$ on the distinguished cusp component; and at p_j use

$$\beta_j = \frac{1}{m_j} \sum_{k=0}^{m_j-1} k\,(\phi_j \circ g_j^k). \tag{2.1d}$$

Equation (2.1c) makes the last expression telescope to $\beta_j \circ g_j - \beta_j = \phi_j$. Therefore $\mathcal{B}$ is an $\mathcal{O}_B$-torsor. Since $H^1(B,\mathcal{O}_B) = 0$, it has global sections, and any two differ by a constant; by construction $b = \beta + \tau$ is invariant and holomorphic at the cusp.

The real determinant is

$$\det_{\mathbb{R}} \Pi = -\det \operatorname{Im} Z = \operatorname{Im} \tau\, D. \tag{2.4}$$

For completeness, the equivariance can be checked before any quotient is taken. Regard the rows of Π as the two covectors

$$\sigma_1 = 6\mu\gamma + \tau u + w, \qquad \sigma_2 = \beta\gamma + \mu u + \delta \quad \text{in } V \otimes \mathbb{C}. \tag{2.3a}$$

Multiplication by the printed matrices gives

$$\begin{aligned} T_1\sigma_1 &= (6\mu - 6)\gamma + (1-\tau)u - \tau w, \\ T_1\sigma_2 &= (\beta + 2)\gamma + (1-\mu)u + (1-\mu)w + \delta, \\ T_2\sigma_1 &= (6\mu + 6\tau)\gamma - u + \tau w, \\ T_2\sigma_2 &= (\beta + 6\mu - 3)\gamma + u + \mu w + \delta. \end{aligned} \tag{2.3b}$$

Using the four affine transformation laws in (2.1a)–(2.1d), coefficient comparison in (2.3b) yields

$$\begin{aligned} \sigma_1 \circ g_1 &= -\tau^{-1} T_1 \sigma_1, \quad \sigma_2 \circ g_1 = T_1 \sigma_2 + (1-\mu)\tau^{-1} T_1 \sigma_1, \\ \sigma_1 \circ g_2 &= \tau^{-1} T_2 \sigma_1, \qquad \sigma_2 \circ g_2 = T_2 \sigma_2 - \mu \tau^{-1} T_2 \sigma_1. \end{aligned} \tag{2.3c}$$

For instance, dividing the first line of (2.3b) by $-\tau$ gives the coefficients $6(1-\mu)/\tau$, $(\tau-1)/\tau$, and 1, exactly those of $\sigma_1(g_1 z)$; subtracting $(1-\mu)$ times that normalized row from $T_1\sigma_2$ gives $\sigma_2(g_1 z)$. The two g_2 identities follow by the same coefficient check. Thus, with

$$R_{g_1} = \begin{pmatrix} -1/\tau & 0 \\ (1-\mu)/\tau & 1 \end{pmatrix}, \qquad R_{g_2} = \begin{pmatrix} 1/\tau & 0 \\ -\mu/\tau & 1 \end{pmatrix},$$

pairing (2.3c) with $\lambda \in \Lambda$ gives

$$\Pi(gz) = R_g(z)\Pi(z)M_g. \tag{2.3d}$$

Since $M_{gh} = M_h M_g$, uniqueness of the left factor gives $R_{gh}(z) = R_g(hz)R_h(z)$; hence these formulae define an action, not merely generator-wise maps. Taking real determinants in (2.3d), using $\det M_g = 1$ and $\operatorname{Im}\tau(g_j z) = \operatorname{Im}\tau/|\tau|^2$, shows $D(gz) = D(z)$. At the cusp,

$$D = \operatorname{Im} b - \operatorname{Im}\tau - \frac{6(\operatorname{Im}\mu)^2}{\operatorname{Im}\tau} \longrightarrow -\infty.$$

The descended function is bounded above on the complement of a cusp neighbourhood. Replacing β by $\beta + c_0$ adds $\operatorname{Im} c_0$ to D, so a sufficiently negative imaginary constant makes (2.2) hold globally. Equation (2.4) then makes Π a real isomorphism everywhere. The equivariance identity supplies the claimed Δ-action and completes the construction of the period family. □

Proposition 10.3 (exact admissible period constants)**.** *Let $\tau, \mu, \beta^{\mathrm{part}}$ be the cusp-regular functions constructed in theorem 10.2, with all its marked coordinates and transformation laws retained. The descended function*

$$D_{\mathrm{part}} = \operatorname{Im}\beta^{\mathrm{part}} - \frac{6(\operatorname{Im}\mu)^2}{\operatorname{Im}\tau}$$

has an attained finite maximum M on $B \setminus \{p_0\}$. The exact set of constants for which $\beta = \beta^{\mathrm{part}} + c_0$ makes the marked real period map an isomorphism at every point is

$$\{c_0 \in \mathbb{C} : \operatorname{Im} c_0 < -M\}.$$

Proof. The preceding proposition proves continuity on $B\setminus\{p_0\}$, including its elliptic points, and convergence $D_{\mathrm{part}} \to -\infty$ at p_0. Choose a base point b_*. Outside a compact set containing b_*, the function is smaller than $D_{\mathrm{part}}(b_*) - 1$. Its maximum on that compact set is therefore its maximum on the entire base. A constant shift gives $D = D_{\mathrm{part}} + \operatorname{Im} c_0$, so $D < 0$ everywhere exactly for $\operatorname{Im} c_0 < -M$. At equality, $D = 0$ at a maximum point. For $\operatorname{Im} c_0 > -M$, that point has positive D, whereas the cusp still has negative D. A path toward the cusp contains a zero. Since the exact ordered real determinant is $\operatorname{Im}\tau\, D$, these and only these excluded constants make the period matrix singular somewhere. □

Remark 10.4 (the cusp extension is essential). For the same τ, μ, β, the function $\widetilde{\beta} = \beta + i\,t \circ \pi$ is holomorphic on $\mathfrak{h}_z$ and satisfies all the same affine transformation laws, because $t \circ \pi$ is invariant. However, $\widetilde{D} = D + \operatorname{Re} t$. On the actual ray $t = R \to +\infty$, $j(\tau) = 1728R$ gives

$$q = (1728R)^{-1}(1 + O(R^{-1})), \qquad D(R) = -\frac{\log(1728R)}{2\pi} + \operatorname{Im} b(0) + o(1), \quad b = \beta + \tau.$$

Consequently $\widetilde{D}(R) \to +\infty$, and no additive constant makes $\widetilde{D}$ negative everywhere. This function is excluded by the actual construction because $\widetilde{\beta} + \tau = b + it$ has a cusp pole. Thus holomorphic solvability on the open base alone does not supply the required uniform bound; the proved cusp extension supplies it.

For the cusp filling, record the local periods explicitly. Let t_c be the coordinate vanishing at p_0. The simple pole of j gives

$$q = e^{2\pi i\tau} = t_c u_1(t_c), \qquad u_1(0) \neq 0.$$

After choosing a logarithm of u_1, put

$$h = (2\pi i)^{-1} \log u_1, \qquad s = \tau - h, \qquad b = \beta + \tau.$$

Then $e^{2\pi i s} = t_c$, $s \circ g_0 = s - 1$, and μ, b, h are holomorphic functions of t_c at the origin. Consequently

$$B_0 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad Z = sB_0 + C(t_c), \qquad C(t_c) = \begin{pmatrix} 6\mu(t_c) & h(t_c) \\ b(t_c) - h(t_c) & \mu(t_c) \end{pmatrix}, \tag{2.5a}$$

and C is holomorphic on a full cusp disc. In particular, for $\bar\lambda \in \bar\Lambda \simeq \mathbb{Z}^2$,

$$\exp(2\pi i Z\bar\lambda) = t_c^{B_0\bar\lambda} c_{\bar\lambda}(t_c), \qquad c_{\bar\lambda}(t_c) = \exp(2\pi i C(t_c)\bar\lambda). \tag{2.5b}$$

10.2 The toric filling at the cusp

Let $\mathcal{T}_{A_2}$ be the translation-invariant triangulation of $\mathbb{R}^2$ by the two triangles $\{v, v+e_1, v+e_2\}$ and $\{v+e_1, v+e_2, v+e_1+e_2\}$. In $N' = \mathbb{Z}^2 \oplus \mathbb{Z}$, take the cones over all cells at height one, and denote the associated toric threefold by Y. We use the standard orbit–cone and affine-chart constructions of [23, Chapters 1 and 3]; all nonstandard assertions caused by the infinite fan are proved directly below.

For $\bar\lambda \in \bar\Lambda$, define the integral linear map

$$\varphi_{\bar\lambda}(y, y_3) = (y + y_3 B_0\bar\lambda, y_3).$$

On the height-one slice this is translation by $B_0\bar\lambda \in \mathbb{Z}^2$, so it preserves the triangulation and maps the cone over a cell P to the cone over $P + B_0\bar\lambda$. It therefore induces a toric automorphism $\Phi_{\bar\lambda}$ of Y, satisfying

$$\Phi_{\bar\lambda}(x, t) = (t^{B_0\bar\lambda} x, t)$$

on the dense torus. Because t and $C(t)$ are holomorphic on the full cusp model, multiplication by $(c_{\bar\lambda}(t), 1) \in (\mathbb{C}^*)^3$ defines a biholomorphism on all of Y_ϵ. Set

$$\Psi_{\bar\lambda}(p) = (c_{\bar\lambda}(t(p)), 1) \cdot \Phi_{\bar\lambda}(p).$$

The maps $\Phi_{\bar\lambda}$ fix the first two cocharacter directions and preserve t, so they commute with this torus multiplication. The linearity of $C(t)\bar\lambda$ gives $c_{\bar\lambda} c_{\bar\mu} = c_{\bar\lambda+\bar\mu}$; hence $\Psi_{\bar\lambda}\Psi_{\bar\mu} = \Psi_{\bar\lambda+\bar\mu}$. This proves, before taking any quotient, both the holomorphic extension and the action law used below.

For a triangle $\sigma = \{v_0, v_1, v_2\}$, let θ_i^σ be its barycentric coordinates and let $z_i = \chi^{m_i}$ be the coordinates dual to $(v_i, 1)$. Set

$$\sigma^{(\eta)} = \{y : \theta_i^\sigma(y) > -\eta \text{ for } i = 0, 1, 2\}, \quad \mathcal{U}_\sigma(S) = \{|z_i| < S \text{ for all } i\}.$$

We use the restricted charts

$$\mathcal{U}_\sigma(S, \epsilon) = \mathcal{U}_\sigma(S) \cap Y_\epsilon$$

whenever a deck transformation acts on a chart.

Lemma 10.5 (toric chart control)**.** *For a torus point $p = (x, t)$ with $0 < |t| < 1$, put*

$$y(p) = \frac{(\log|x_1|, \log|x_2|)}{\log|t|}.$$

Then

$$p \in \mathcal{U}_\sigma(S) \quad \Longleftrightarrow \quad y(p) \in \sigma^{(\log S/|\log|t||)}. \tag{2.5c}$$

The exact closed-unit-polydisc version is

$$p \in \overline{\mathcal{U}}_\sigma(1) \quad \Longleftrightarrow \quad \theta_i^\sigma(y(p)) \geq 0 \ (i=0,1,2) \quad \Longleftrightarrow \quad y(p) \in \sigma. \tag{19}$$

Every point of $\{|t|<1\} \subset Y$ *belongs to* $\overline{\mathcal{U}}_\sigma(1) = \{|z_i| \leq 1\}$ *for some* σ. *Consequently the sets* $\mathcal{U}_\sigma(2,\epsilon)$ *cover* Y_ϵ, *and every compact subset is contained in the union of finitely many members of this cover.*

Proof. Since $(y,1) = \sum_i \theta_i^\sigma(y)(v_i,1)$, duality gives

$$\log|z_i(p)| = \log|t(p)|\,\theta_i^\sigma(y(p)).$$

Because $\log|t|<0$, this is exactly (2.5c); retaining non-strict inequalities proves the closed-unit-polydisc assertion, including triangle edges and vertices.

For the boundary assertion let p lie in the toric orbit $O_{\widehat{P}}$ of the cone over a cell P. In a chart with $\sigma \supseteq P$, the coordinates corresponding to vertices of P vanish. On the quotient lattice $N'/(N' \cap \operatorname{span}\widehat{P})$, the remaining coordinates are the characters dual to the image cone $\bar{\sigma}$. Hence $p \in \overline{\mathcal{U}}_\sigma(1)$ exactly when $-\operatorname{Log}_{\widehat{P}}(p) \in \bar{\sigma}$. The cones $\bar{\sigma}$, for triangles $\sigma \supseteq P$, cover the quotient real vector space: lift a vector n, choose a relative interior point $p_0 \in P \times \{1\}$, and for sufficiently small $u>0$ place the height-one rescaling of $p_0 + un$ in a triangle of the locally finite triangulation. That triangle contains P, and its quotient cone contains n. This proves the asserted cover. Compactness then supplies a finite subcover. □

Theorem 10.6 (the cusp quotient and its central fibre)**.** ***[VERIFIED]*** *For the period functions and holomorphic cusp matrix* $C(t)$ *constructed above, and sufficiently small* $\epsilon>0$, *the maps*

$$\Psi_{\bar{\lambda}}(x,t) = \big(c_{\bar{\lambda}}^-(t)t^{B_0\bar{\lambda}}x, t\big), \qquad c_{\bar{\lambda}}^-(t) = \exp(2\pi i C(t)\bar{\lambda}), \tag{2.5}$$

extend to a free properly discontinuous action of $\bar{\Lambda} \simeq \mathbb{Z}^2$ *on* $Y_\epsilon = \{|t|<\epsilon\}$. *The quotient*

$$f_0 : N_0 = Y_\epsilon/\bar{\Lambda} \longrightarrow \Delta_\epsilon$$

is a proper holomorphic map from a connected smooth complex threefold. Its nonzero fibres are the tori of theorem 10.2. Its full scheme-theoretic central fibre W *is reduced and has local equations*

$$z_0, \qquad z_0z_1, \qquad z_0z_1z_2. \tag{2.6}$$

The surface W *is irreducible and nonnormal. Its normalization is* $\nu : E_0 \simeq \mathrm{dP}_6 \to W$, *obtained by identifying opposite sides of the toric hexagon; its double locus consists of three rational curves through exactly two triple points. Thus* W *is a normal-crossings surface, but it is not a simple-normal-crossings divisor. Moreover*

$$\omega_{N_0} \simeq \mathcal{O}_{N_0}. \tag{2.6a}$$

Proof. For a bounded cell P, the cone over $P \times \{1\}$ is strongly convex, and two such cones meet in the cone over the common face of the two cells. The locally finite triangulation therefore defines a countable rational fan with support $(\mathbb{R}^2 \times \mathbb{R}_{>0}) \cup \{0\}$. For the determinant calculation, order the height-one generators of the two kinds as

$$((v,1),(v+e_1,1),(v+e_2,1)), \qquad ((v+e_1,1),(v+e_2,1),(v+e_1+e_2,1)).$$

Their determinants are respectively $+1$ and -1, hence have absolute value one. Every cone is therefore unimodular. Here is the pairwise affine-chart separation argument, without assuming finiteness of the fan. For two cones κ, κ', let $\upsilon = \kappa \cap \kappa'$ and $S_\kappa = \kappa^\vee \cap M'$, where $M' = \operatorname{Hom}(N', \mathbb{Z})$. The separation lemma for

cones meeting in a common face gives an integral character m nonnegative on κ, nonpositive on κ', and zero on each exactly along υ [24, Proposition 3.1.3 and its separation lemma, pp. 106–108]. For $a \in S_\upsilon$, choose an integer $k \geq 0$ large enough that $a + km$ is nonnegative on every ray of κ. On the common face this already holds; on each other ray m is positive. Then $a = (a + km) + (-km) \in S_\kappa + S_{\kappa'}$. The reverse inclusion follows by evaluation on υ. Thus the map

$$\mathbb{C}[S_\kappa] \otimes \mathbb{C}[S_{\kappa'}] \longrightarrow \mathbb{C}[S_\upsilon], \qquad \chi^a \otimes \chi^b \longmapsto \chi^{a+b}$$

is surjective. It identifies the gluing graph with a closed algebraic subset of the product of these two affine charts, hence with an analytically closed subset. These chart products cover $Y \times Y$, so its diagonal is closed. Countability of the charts supplies a countable topological basis. Thus Y is a connected Hausdorff smooth complex threefold with dense torus.

If m_0, m_1, m_2 is the basis dual to the height-one generators, then the corresponding affine chart is $\mathbb{C}^3$, with coordinates $z_i = \chi^{m_i}$, and

$$t = z_0 z_1 z_2. \tag{2.7}$$

Since every primitive ray generator has height one, the divisor of t is $\sum_v E_v$, all with multiplicity one. Equation (2.7) proves that the scheme-theoretic fibre $t^{-1}(0)$ is reduced and normal crossings. The star of a height-one vertex has rays $e_1, e_2, e_2 - e_1, -e_1, -e_2, e_1 - e_2$, the complete smooth hexagonal fan; hence every irreducible component E_v is dP_6.

The local canonical trivialization is also intrinsic to this toric model. On the dense torus put

$$\Omega = \frac{dx_1 \wedge dx_2 \wedge dt}{x_1 x_2}.$$

If (z_0, z_1, z_2) is a unimodular toric chart, then

$$\bigwedge_i \frac{dz_i}{z_i} = \pm \frac{dx_1}{x_1} \wedge \frac{dx_2}{x_2} \wedge \frac{dt}{t}, \qquad t = z_0 z_1 z_2,$$

and therefore $\Omega = \pm dz_0 \wedge dz_1 \wedge dz_2$. It extends holomorphically and nowhere vanishingly across every boundary chart. For a deck transformation $\Psi_{\bar\lambda}$, formula (2.5) gives $\Psi_{\bar\lambda}^*(dx_i/x_i) = dx_i/x_i + \alpha_i(t)dt$ and $\Psi_{\bar\lambda}^* dt = dt$; every additional term in the pullback of Ω contains dt twice. Hence Ω is invariant and descends to a nowhere-vanishing three-form on N_0, proving the intrinsic trivialization (2.6a). No global threefold has been invoked at this stage.

Choose a radius $r > 0$ on which the full matrix $C(t)$ is holomorphic, and $0 < r_1 < \min(r, 1)$. Put

$$R(t) = -2\pi \operatorname{Im} C(t), \qquad M = \max_{|t| \leq r_1} \|R(t)\| < \infty.$$

Choose $0 < \epsilon < r_1$ with $|\log \epsilon| > 2M$. At a torus point fixed by a nonzero $\bar\lambda$, taking logarithmic absolute values in (2.5) gives

$$\log|t| B_0 \bar\lambda + R(t)\bar\lambda = 0,$$

impossible because $|B_0 \bar\lambda| = |\bar\lambda|$ and the first term has more than twice the norm of the second. On $t = 0$, a cell is translated by $B_0 \bar\lambda \neq 0$; a bounded cell cannot be fixed by a nonzero translation. Thus the action is free.

For proper discontinuity put

$$B_t = B_0 + \frac{R(t)}{\log|t|}.$$

Then $\|B_t - B_0\| \leq 1/2$, so $|B_t \bar\lambda| \geq |\bar\lambda|/2$. In logarithmic position coordinates, (2.5) is

$$y(\Psi_{\bar\lambda} p) = y(p) + B_t \bar\lambda. \tag{2.8}$$

Put $\eta_0 = \log 2/|\log \epsilon|$. If $\Psi_{\bar\lambda}\mathcal{U}_\sigma(2,\epsilon)$ meets $\mathcal{U}_{\sigma'}(2,\epsilon)$, the intersection is open in Y_ϵ and hence contains a point of the dense invariant torus. By theorem 10.5, for that point p and its image p',

$$y(p) \in \sigma^{(\eta_0)}, \qquad y(p') \in \sigma'^{(\eta_0)}.$$

Both thickenings are bounded, so (2.8) gives

$$\frac{1}{2}|\bar\lambda| \le |B_t\bar\lambda| = |y(p') - y(p)| \le \sup_{a\in\sigma'^{(\eta_0)},\, b\in\sigma^{(\eta_0)}} |a-b|.$$

Thus only finitely many $\bar\lambda$ can make this fixed pair of charts meet. A compact set is contained in finitely many of the charts from theorem 10.5; taking the finite union over all chart pairs proves proper discontinuity. Freeness and proper discontinuity give a Hausdorff complex-manifold quotient and a holomorphic covering map. For completeness the restricted total space is connected: every maximal chart intersects it in

$$\{(z_0,z_1,z_2) \in \mathbb{C}^3 : |z_0z_1z_2| < \epsilon\},$$

a set star shaped about the origin. These connected open sets meet the common connected dense torus $(\mathbb{C}^*)^2 \times \{0 < |t| < \epsilon\}$. Their union Y_ϵ, and hence its quotient, is connected.

To prove properness, fix $\epsilon_1 < \epsilon$. Let $\mathcal{F}_0$ be the finite set of triangles meeting the closed Euclidean ball of radius $\sqrt{2}$, and put

$$K_0 = \bigcup_{\sigma\in\mathcal{F}_0} \left(\overline{\mathcal{U}}_\sigma(1) \cap \{|t| \le \epsilon_1\}\right).$$

This is compact. For a torus point p with $0 < |t(p)| \le \epsilon_1$, put $u = B_{t(p)}^{-1} y(p)$ and choose $\bar\lambda \in \mathbb{Z}^2$ with $|u + \bar\lambda|_\infty \le 1/2$. Since $\|B_t\| \le 3/2$, equation (2.8) gives

$$|y(\Psi_{\bar\lambda}p)| = |B_t(u+\bar\lambda)| \le \frac{3}{2\sqrt{2}} < \sqrt{2}.$$

Choose a triangle containing this logarithmic position. The closed-unit- polydisc identity of theorem 10.5 places the translate in the corresponding $\overline{\mathcal{U}}_\sigma(1) \subset K_0$.

For the central representatives, we prove the needed closed-polydisc cover explicitly. Denote the two ordered triangles at v by

$$L_v = (v, v+e_1, v+e_2), \qquad U_v = (v+e_1, v+e_2, v+e_1+e_2).$$

On the dense orbit of E_0, the characters $(1,0,0)$ and $(0,1,0)$ give the original coordinates x_1, x_2. The coordinate dual to $(0,0,1)$ vanishes there; the other two dual characters have last exponent zero. Pairing with the ordered height-one generators gives the following restrictions of the chart coordinates. Write $a = \log|x_1|$, $b = \log|x_2|$ on this dense orbit.

σ	$(z_0,z_1,z_2)\|_{E_0^{\mathrm{tor}}}$	$\lvert z_i\rvert \le 1$ $(i=0,1,2)$
L_0	$(0, x_1, x_2)$	$a \le 0,\ b \le 0$
L_{-e_1}	$((x_1x_2)^{-1}, 0, x_2)$	$a+b \ge 0,\ b \le 0$
L_{-e_2}	$((x_1x_2)^{-1}, x_1, 0)$	$a+b \ge 0,\ a \le 0$
U_{-e_1}	$(0, x_1^{-1}, x_1x_2)$	$a \ge 0,\ a+b \le 0$
U_{-e_2}	$(x_2^{-1}, 0, x_1x_2)$	$b \ge 0,\ a+b \le 0$
$U_{-e_1-e_2}$	$(x_2^{-1}, x_1^{-1}, 0)$	$a \ge 0,\ b \ge 0.$

The six regions cover $\mathbb{R}^2$: the first and last cover equal signs; for $a \ge 0 \ge b$, the sign of $a+b$ selects the second or fourth; for $b \ge 0 \ge a$, it selects the third or fifth. Equalities are retained throughout. Each corresponding coordinate plane meets $\overline{\mathcal{U}}_\sigma(1)$ in a compact closed bidisc $Q_\sigma \subset E_0$. Their finite union is

compact and hence closed in the already Hausdorff Y. It contains E_0^{tor}, so it contains its closure E_0; the reverse inclusion follows from $Q_\sigma \subset E_0$. Consequently

$$E_0 = \bigcup_{\sigma \ni 0} Q_\sigma.$$

This proves the closed-polydisc cover at every boundary point as well as on the dense orbit, without presupposing compactness of E_0.

Now let p lie in the central orbit belonging to a cell P, choose a vertex $v \in P$, and set $\bar{\lambda} = B_0 v$. The exact map $\Psi_{B_0 v}$ translates the cell to $P - v$, because $B_0^2 v = -v$, and retains the full torus multiplier

$$c_{B_0 v}(0) = \exp\bigl(2\pi i C(0) B_0 v\bigr).$$

This nonzero multiplier preserves the orbit label, so the actual translated point belongs to E_0. The proved cover applies to it regardless of the multiplier's modulus or argument. Each of the six triangles contains zero and therefore belongs to $\mathcal{F}_0$, so the translate lies in K_0. Together with the nonzero-fibre representatives above, this proves that the image of K_0 in the quotient is exactly $f_0^{-1}(\{|t| \le \epsilon_1\})$, which is compact. Every compact subset of the open base disc is closed in one such closed subdisc; its inverse image is therefore closed in that compact Hausdorff inverse image. Thus f_0 is proper.

For $t \neq 0$, choose s with $e^{2\pi i s} = t$. The multiplier in (2.5) is $\exp(2\pi i Z(s)\bar{\lambda})$, so exponentiation identifies the fibre with

$$\mathbb{C}^2/(\mathbb{Z}^2 + Z(s)\mathbb{Z}^2).$$

It is compact because $\operatorname{Im} Z(s)$ is invertible by (2.2).

The quotient map is a local biholomorphism; therefore the local equation (2.7), its multiplicity one, and reducedness descend to W. Every cell has a vertex, and $B_0\bar{\Lambda} = \mathbb{Z}^2$ acts simply transitively on vertices, so every orbit in the central fibre meets E_0. Thus $E_0 \to W$ is surjective. Two interior points cannot be identified; on the boundary, precisely opposite edges and the corresponding hexagon vertices are identified. The fibres have at most three points. Since E_0 is compact, the map is finite; since it is generically injective from a smooth irreducible surface to the reduced surface W, it is the normalization. The two translation-orbits of triangles give the two triple points, and the three unoriented edge directions give the three double curves. Irreducibility follows from the finite surjection by E_0.

The actual opposite-side maps retain the entire matrix $C(0)$. For a neighboring vertex $v \in \{\pm e_1, \pm e_2, \pm(e_1 - e_2)\}$, put

$$C_v = E_0 \cap E_v, \qquad \bar{\lambda}_v = B_0 v, \qquad \iota_v = \Psi_{B_0 v}|_{C_v} : C_v \longrightarrow C_{-v}.$$

Since $B_0^2 v = -v$, this is precisely the translation taking the edge $\{0, v\}$ to $\{0, -v\}$. For a primitive row $a \in \mathbb{Z}^2$ with $av = 0$, use the same character $w = \chi^{(a,0)}$ on both dense edge orbits. Character pullback gives

$$\iota_v^* w_{-v} = \exp\bigl(2\pi i\, aC(0)B_0 v\bigr) w_v, \tag{20}$$

because the exponent of t is $-av = 0$. With $\mu_0 = \mu(0)$ and $b_0 = b(0)$, the three maps are

v	a	$\iota_v^* w_{-v}/w_v$
e_1	$(0,1)$	$e^{-2\pi i \mu_0}$
e_2	$(1,0)$	$e^{12\pi i \mu_0}$
$e_1 - e_2$	$(1,1)$	$e^{-2\pi i(7\mu_0 + b_0)}$.

For the third entry the full exponent coefficient is

$$(1,1)C(0)B_0(e_1 - e_2) = -6\mu_0 - h(0) - (b_0 - h(0)) - \mu_0 = -7\mu_0 - b_0.$$

The cancellation of $h(0)$ occurs in this exact character evaluation; no entry was suppressed. All multipliers are nonzero. The maps extend over both endpoints as restrictions of the ambient biholomorphisms, and ι_{-v} is the inverse map with reciprocal multiplier in the same-character convention. □

Proposition 10.7 (the cusp punctured-overlap biholomorphism)**.** ***[VERIFIED]*** *Let $D_0^* = \{0 < |t_c| < \epsilon\}$, and shrink it if necessary so that its inverse image has a distinguished component U_0 stabilized exactly by $\langle g_0 \rangle$. Exponentiation induces a fibre-preserving biholomorphism*

$$J|_{D_0^*} \xrightarrow{\sim} N_0|_{D_0^*} = N_0 \setminus W. \tag{2.8i}$$

On the covering space it is given by

$$\mathcal{E} : U_0 \times \mathbb{C}^2 \longrightarrow (\mathbb{C}^*)^2 \times D_0^*, \qquad \mathcal{E}(z, \zeta_1, \zeta_2) = \big(e^{2\pi i \zeta_1}, e^{2\pi i \zeta_2}, e^{2\pi i s(z)}\big). \tag{2.8ii}$$

Proof. First identify every deck transformation. From $s \circ g_0 = s - 1$, the equality $t_c \circ \pi = e^{2\pi i s}$, and the fact that t_c is a base coordinate, ds never vanishes on U_0. Thus s is a local biholomorphism to the logarithmic cover of D_0^*. The restriction $U_0 \to D_0^*$ is a connected regular cover with deck group $\langle g_0 \rangle \simeq \mathbb{Z}$. Under the classification of connected covers of the punctured disc, it corresponds to a subgroup of $\pi_1(D_0^*) \simeq \mathbb{Z}$; because its deck group is infinite, that subgroup is zero. Hence U_0 is the universal cover. The logarithmic cover is universal as well, and its deck generator acts by $s \mapsto s - 1$. Therefore s is an equivariant biholomorphism onto that logarithmic cover.

The cusp transformation laws give

$$\Pi(g_0 z) = [Z(s-1) \mid I_2] = [Z(s) - B_0 \mid I_2].$$

In the same block decomposition, the explicit matrix in theorem 5.1 gives

$$M_{g_0} = M_0^{-1} = \begin{pmatrix} I_2 & 0 \\ -B_0 & I_2 \end{pmatrix}.$$

Consequently

$$\Pi(g_0 z) = \Pi(z) M_{g_0}.$$

Comparison with (2.3d) and real invertibility of $\Pi(g_0 z)$ yields $R_{g_0} = I_2$. To make the crossed action explicit, for $\lambda \in \Lambda$ and $g \in \Delta$ put

$$T_\lambda(z, \zeta) = (z, \zeta + \Pi(z)\lambda), \qquad \widetilde{g}(z, \zeta) = (gz, R_g(z)\zeta).$$

Equation (2.3d), together with $A_g = M_g^{-1}$, gives

$$\begin{aligned} \widetilde{g} T_\lambda(z, \zeta) &= (gz, R_g(z)\zeta + R_g(z)\Pi(z)\lambda) \\ &= (gz, R_g(z)\zeta + \Pi(gz) A_g \lambda) = T_{A_g \lambda} \widetilde{g}(z, \zeta). \end{aligned}$$

Thus

$$\widetilde{g} T_\lambda = T_{A_g \lambda} \widetilde{g}. \tag{2.8ii-a}$$

At the cusp $A_{g_0} = M_0$, so

$$\widetilde{g}_0(z, \zeta) = (g_0 z, \zeta), \qquad \widetilde{g}_0 T_\lambda = T_{M_0 \lambda} \widetilde{g}_0. \tag{2.8ii-b}$$

In particular $\mathcal{E}$ is invariant under $\widetilde{g}_0$.

Let $\Lambda_{\mathrm{tor}} = \mathbb{Z}\widehat{w} \oplus \mathbb{Z}\widehat{\delta}$. By theorem 5.2,

$$\Lambda_{\mathrm{tor}} = \ker(M_0 - I) = \mathrm{im}(M_0 - I). \tag{2.8ii-c}$$

Put

$$\mathcal{G}_0 = \langle T_\lambda, \widetilde{g}_0 : \lambda \in \Lambda \rangle \simeq \Lambda \rtimes_{M_0} \langle \widetilde{g}_0 \rangle.$$

Since M_0 fixes Λ_{tor} pointwise, (2.8ii-b) shows that

$$H := \{T_\nu \widetilde{g}_0^n : \nu \in \Lambda_{\mathrm{tor}},\ n \in \mathbb{Z}\} \simeq \Lambda_{\mathrm{tor}} \times \langle \widetilde{g}_0 \rangle. \tag{2.8ii-d}$$

Moreover, for $\lambda \in \Lambda$ and $\nu \in \Lambda_{\mathrm{tor}}$,

$$\widetilde{g}_0 T_\nu \widetilde{g}_0^{-1} = T_\nu, \qquad T_\lambda T_\nu T_\lambda^{-1} = T_\nu, \qquad T_\lambda \widetilde{g}_0 T_\lambda^{-1} = T_{(I-M_0)\lambda}\widetilde{g}_0 \in H. \tag{2.8ii-e}$$

These generator conjugations prove that $H \triangleleft \mathcal{G}_0$. Translation by Λ_{tor} adds an integral vector to ζ, so it lies in the deck group of $\mathcal{E}$. Conversely, if two points have the same image under $\mathcal{E}$, their s-coordinates differ by an integer and their ζ-coordinates by an element of $\mathbb{Z}^2$. Equivariance and injectivity of the logarithmic coordinate then show that they differ by a unique element of H. Thus $\mathcal{E}$ identifies $(U_0 \times \mathbb{C}^2)/H$ with $(\mathbb{C}^*)^2 \times D_0^*$.

It remains to identify the residual action. Write $\bar{\Lambda} = \Lambda/\Lambda_{\mathrm{tor}}$, and let λ be any lift of $\bar{\lambda}$. In the period basis, translation by λ adds $Z(s)\bar{\lambda}$ to ζ, up to an integral vector killed by exponentiation. Therefore (2.5b) gives

$$\mathcal{E}(z, \zeta + \Pi(z)\lambda) = \big(c_{\bar{\lambda}}(t_c) t_c^{B_0\bar{\lambda}} x, t_c\big) = \Psi_{\bar{\lambda}}(\mathcal{E}(z,\zeta)). \tag{2.8iii}$$

This is independent of the chosen lift. Moreover M_0 is the identity on $\bar{\Lambda}$, by (2.8ii-c). The crossed multiplication law is

$$(T_\lambda \widetilde{g}_0^n)(T_\mu \widetilde{g}_0^m) = T_{\lambda + M_0^n \mu} \widetilde{g}_0^{n+m}.$$

It follows that

$$\vartheta : \mathcal{G}_0 \longrightarrow \bar{\Lambda}, \qquad \vartheta(T_\lambda \widetilde{g}_0^n) = \bar{\lambda},$$

is a surjective homomorphism with kernel H. Hence $\mathcal{G}_0/H \simeq \bar{\Lambda}$, and (2.8iii) identifies this residual action with the displayed $\Psi_{\bar{\lambda}}$-action. Dividing by it gives

$$(U_0 \times \mathbb{C}^2)/\mathcal{G}_0 \simeq ((\mathbb{C}^*)^2 \times D_0^*)/\bar{\Lambda}.$$

The left side is $J|_{D_0^*}$, while the right side is $N_0|_{D_0^*}$ because $t^{-1}(\mathbb{C}^*)$ is the dense torus of Y. This proves (2.8i), including holomorphicity of the inverse. □

10.3 The finite logarithmic fillings and the global gluing

Set

$$\varepsilon = \widehat{\gamma} + 2\widehat{u} - 4\widehat{w}, \qquad \varepsilon' = \widehat{\gamma} + 3\widehat{u} - 3\widehat{w}. \tag{2.8a}$$

Direct multiplication by the matrices in theorem 5.1 gives $A_1\varepsilon = \varepsilon$ and $A_2\varepsilon' = \varepsilon'$.

A *linearizing pair* (Δ_j, s_j) consists of a simply connected g_j-invariant neighbourhood of z_j, disjoint from all its translates outside $\langle g_j \rangle$, and an injective coordinate satisfying

$$s_j(z_j) = 0, \qquad s_j \circ g_j = \zeta_j s_j, \qquad \zeta_j = e^{-2\pi i/m_j}. \tag{2.8b}$$

Choose the image of s_j to be a round disc about zero, shrinking the linearizing neighbourhood if necessary. Rotation invariance and translate-disjointness are preserved by this restriction. Put $D_j = \pi(\Delta_j)$, $\Delta_j^* = \Delta_j \setminus \{z_j\}$, and choose the base coordinate t_j by $t_j \circ \pi = s_j^{m_j}$. Let

$$\mathcal{T} = (\mathfrak{h}_z \times \mathbb{C}^2)/\Lambda, \qquad \mathfrak{h}_z^\circ = \mathfrak{h}_z \setminus (\Delta z_1 \cup \Delta z_2), \qquad J = (\mathcal{T}|_{\mathfrak{h}_z^\circ})/\Delta$$

$$J_j' = \mathcal{T}|_{\Delta_j}, \qquad A_0^{(j)} = \mathbb{C}^2/\Pi(z_j)\Lambda. \tag{2.8c}$$

Lemma 10.8 (the middle period quotient is proper)**.** *The induced Δ-action on $\mathcal{T}|_{\mathfrak{h}_z^\circ}$ is free and properly discontinuous. Consequently J is a Hausdorff complex manifold and its natural map*

$$J \longrightarrow B_{\mathrm{mid}} := \mathfrak{h}_z^\circ/\Delta$$

is a proper holomorphic submersion with compact torus fibres.

Proof. The deleted translates contain every elliptic fixed point, so the Δ-action on $\mathfrak{h}_z^\circ$ is free; it is properly discontinuous because it is the restriction of the Fuchsian action. The action on $\mathcal{T}$ projects equivariantly to this base action. Hence an element fixing a point of $\mathcal{T}|_{\mathfrak{h}_z^\circ}$ fixes its base point and is the identity. If K_1, K_2 are compact subsets of the total space and $gK_1 \cap K_2 \neq \varnothing$, then $g\,\mathrm{pr}(K_1) \cap \mathrm{pr}(K_2) \neq \varnothing$. Base proper discontinuity leaves only finitely many such g, proving proper discontinuity upstairs and hence the Hausdorff manifold assertion.

For properness, take a simply connected open set $U \Subset B_{\mathrm{mid}}$ evenly covered by $\mathfrak{h}_z^\circ$, and one lift $\widetilde{U}$. The quotient over U is identified with $\mathcal{T}|_{\widetilde{U}}$. Before quotienting, projection is a holomorphic submersion, and the quotient maps are local biholomorphisms; hence the displayed map is a holomorphic submersion. The flat real-analytic trivialization

$$\widetilde{U} \times (\Lambda \otimes \mathbb{R})/\Lambda \longrightarrow \mathcal{T}|_{\widetilde{U}}, \qquad (z, x + \Lambda) \longmapsto [z, \Pi(z)x]$$

shows this properness explicitly: if $C \subset U$ is compact, its chosen lift $\widetilde{C} \subset \widetilde{U}$ is compact and the inverse image of C is homeomorphic to $\widetilde{C} \times (\Lambda \otimes \mathbb{R})/\Lambda$, hence compact. Thus the map is proper over U. Properness is local on the target, so the displayed map is proper. □

For $b \in \Lambda \otimes \mathbb{R}$, define the varying holomorphic lift

$$\widetilde{g}_j^b(z, \zeta) = (g_j z, R_{g_j}(z)\zeta + \Pi(g_j z)b). \tag{2.8d}$$

Equation (2.3d) says

$$R_{g_j}(z)\Pi(z) = \Pi(g_j z)A_j, \tag{2.8e}$$

so (2.8d) normalizes the lattice action and descends to J_j'. In the flat real-analytic coordinates

$$\Xi(z, x + \Lambda) = [z, \Pi(z)x]$$

one has, with $\Sigma_k = I + A_j + \cdots + A_j^{k-1}$,

$$\Xi^{-1}(\widetilde{g}_j^b)^k \Xi(z, x) = (g_j^k z, A_j^k x + \Sigma_k b). \tag{2.8f}$$

We call $\widetilde{g}_j^0$ the linear lift and, for $v \in \Lambda^{A_j}$, call $\widetilde{g}_j^{\log} := \widetilde{g}_j^{v/m_j}$ the logarithmic lift.

Proposition 10.9 (finite fillings). ***[VERIFIED]*** *Let A_j be the two finite monodromies and choose*

$$v_1 = \varepsilon, \qquad v_2 = -\varepsilon', \qquad \gamma(v_1) = 1, \quad \gamma(v_2) = -1.$$

On the local torus family over a linearizing disc Δ_j, the affine map

$$(z, x) \longmapsto (g_j z, A_j x + v_j/m_j) \tag{2.9}$$

generates a free action of order m_j. The quotient $f_j : N_j \to D_j$ is a smooth complex threefold with

$$f_j^*(p_j) = m_j S_j, \qquad S_j = A_0^{(j)}/\langle x \mapsto A_j x + v_j/m_j \rangle. \tag{2.10}$$

Over the punctured disc it is biholomorphic to the original period family.

Proof. First prove freeness on the actual local family. For a finite-order integral map A, the affine self-map $x \mapsto Ax + b$ of $(\Lambda \otimes \mathbb{R})/\Lambda$ has a fixed point exactly when

$$\varphi(b) \in \mathbb{Z} \quad \text{for every } \varphi \in \mathrm{Hom}(\Lambda, \mathbb{Z}) \text{ with } \varphi \circ A = \varphi. \tag{2.9a}$$

Necessity is immediate. Conversely, the averaging projector $p = m^{-1} \sum_{r=0}^{m-1} A^r$ has image $\ker(A - I)_{\mathbb{R}}$ and kernel $\mathrm{im}(A - I)_{\mathbb{R}}$; the dual lattice of $p(\Lambda)$ is exactly the invariant integral covector lattice. Thus (2.9a) puts pb in $p(\Lambda)$, and $b \in (A - I)(\Lambda \otimes \mathbb{R}) + \Lambda$, which supplies a fixed point.

The exact matrix calculation gives

$$V^{T_1^k} = \langle \gamma, \psi_1 \rangle, \quad \psi_1 = 2u + w + 3\delta \quad (k = 1, 2),$$

and

$$V^{T_2^k} = \langle \gamma, \psi_2 \rangle, \quad \psi_2 = u + w + 2\delta \quad (k = 1, 2, 3). \tag{2.9b}$$

For $v_1 = \varepsilon$ and $v_2 = -\varepsilon'$,

$$\gamma(v_1) = 1, \quad \psi_1(v_1) = 0, \qquad \gamma(v_2) = -1, \quad \psi_2(v_2) = 0. \tag{2.9c}$$

Because $A_j v_j = v_j$, formula (2.8f) shows that the k-th power of the logarithmic lift has translation kv_j/m_j. For $j = 1$, (2.9a) fails for $k = 1, 2$ already on γ, since $k/3 \notin \mathbb{Z}$. For $j = 2$, it fails for $k = 1, 2, 3$, since $-k/4 \notin \mathbb{Z}$. Away from z_j, the base map g_j^k has no fixed point in Δ_j. Hence every nonidentity power is fixed-point free on J_j'. At $k = m_j$, (2.8f) is translation by $v_j \in \Lambda$, hence the identity; the order is exactly m_j because the map covers g_j.

It follows that

$$N_j = J_j'/\langle \widetilde{g}_j^{\log} \rangle$$

is a smooth complex threefold and the quotient map is finite étale of degree m_j. The function $s_j^{m_j} = t_j \circ \pi$ is invariant and descends to $f_j : N_j \to D_j$. The projection $J_j' \to \Delta_j$ is proper: over a compact subset, the image of the compact set obtained by letting x range over a fixed unit parallelotope in $\Lambda \otimes \mathbb{R}$ is the whole inverse image. Together with finiteness of $\Delta_j \to D_j$ and of the quotient, this proves properness of f_j. Pulling the divisor of t_j back through the etale quotient gives the divisor of $s_j^{m_j}$, namely m_j times the reduced central torus. This proves (2.10).

It remains to identify the punctured filling with the original family and to check that no logarithm branch has entered the answer. On a simply connected $U \subset \Delta_j^*$, choose a branch ℓ of $\log s_j$ and put

$$\sigma_j^\ell(z) = \frac{\ell(z)}{2\pi i} \Pi(z) v_j. \tag{2.9d}$$

Changing ℓ by $2\pi i n$ changes (2.9d) by the lattice vector $n\Pi(z)v_j$. The classes therefore glue to a branch-independent holomorphic section σ_j of $J_j'|_{\Delta_j^*}$; translation by that section defines a biholomorphism h_j.

For the conjugacy, use ℓ on U and on $g_j U$ use the branch

$$\ell'(w) = \ell(g_j^{-1} w) - \frac{2\pi i}{m_j}.$$

This is a logarithm of $s_j(w)$ because $s_j \circ g_j = e^{-2\pi i/m_j} s_j$. For $z \in U$, equation (2.8e) and $A_j v_j = v_j$ give

$$\widetilde{g}_j^{\log} h_j^{-1}[z, \zeta] = \Big[g_j z, R_{g_j}(z)\zeta - \Big(\frac{\ell(z)}{2\pi i} - \frac{1}{m_j} \Big) \Pi(g_j z) v_j \Big].$$

But

$$\frac{\ell'(g_j z)}{2\pi i} = \frac{\ell(z)}{2\pi i} - \frac{1}{m_j},$$

so applying h_j cancels the displayed translation and proves the global identity

$$h_j \widetilde{g}_j^{\log} h_j^{-1} = \widetilde{g}_j^0. \tag{2.9e}$$

Finally, by the stabilizer and translate conditions in (2.8b),

$$J|_{D_j^*} = (J_j'|_{\Delta_j^*})/\langle \widetilde{g}_j^0 \rangle.$$

Thus (2.9e) descends to the asserted punctured-disc biholomorphism. □

Lemma 10.10 (radial retraction and the finite-fibre stalk map)**.** ***[VERIFIED]*** *Under theorem 10.9, the finite local model N_j strongly deformation retracts onto its reduced central fibre S_j. After identifying a nearby fibre with the marked real torus $A_s = (\Lambda \otimes \mathbb{R})/\Lambda$ by the flat coordinates Ξ, the specialization map*

$$H^r(S_j;\mathbb{Z}) \longrightarrow H^r(A_s;\mathbb{Z})$$

is the pullback π_j^* of the quotient covering

$$\pi_j : A_0^{(j)} \longrightarrow S_j = A_0^{(j)}/\langle x \mapsto A_j x + v_j/m_j \rangle.$$

Proof. Use the linearizing coordinate to define the radial contraction $r_u : \Delta_j \to \Delta_j$ by

$$s_j(r_u(z)) = (1-u)s_j(z), \qquad 0 \le u \le 1.$$

Because $s_j \circ g_j = \zeta_j s_j$, this contraction commutes with g_j. In the global real-analytic marking

$$\Xi(z, x+\Lambda) = [z, \Pi(z)x],$$

put

$$H_u(\Xi(z, x+\Lambda)) = \Xi(r_u(z), x+\Lambda).$$

Formula (2.8f) shows directly that H_u commutes with the logarithmic lift: it keeps x fixed while both composites send it to $A_j x + v_j/m_j$. Hence H_u descends to a strong deformation retraction of N_j onto S_j.

Fix $t_j \neq 0$ and one root s with $s^{m_j} = t_j$. Every orbit of the logarithmic lift meets the fibre over s exactly once, so that fibre maps isomorphically onto $f_j^{-1}(t_j)$. Following the descended radial retraction sends the marked point $x \in A_s$ to the orbit of the same x in the central torus, namely $\pi_j(x)$. The induced map from central-fibre cohomology to nearby-fibre cohomology is therefore exactly π_j^*, with no unrecorded conjugacy or index. □

Theorem 10.11 (global analytic construction)**.** ***[VERIFIED]*** *Gluing the period family over B° to N_0, N_1, N_2 by the preceding punctured-disc biholomorphisms produces a connected compact Hausdorff complex threefold X and a proper surjective holomorphic map*

$$f : X \longrightarrow B \simeq \mathbb{P}^1$$

with connected fibres. The fibre over p_0 is exactly the reduced nonnormal surface W of theorem 10.6; those over p_1, p_2 have scheme multiplicities three and four. The open embedding $N_0 \hookrightarrow X$ identifies canonical bundles, so

$$\omega_X|_{N_0} \simeq \omega_{N_0} \simeq \mathcal{O}_{N_0}. \tag{21}$$

Proof. Take the disjoint union of the four manifolds and identify the two copies of each punctured-disc restriction by the displayed biholomorphism. The special discs are pairwise disjoint, so every nonsingleton equivalence class has two points and meets each summand at most once. The saturation of an open set is the union of that set with its images under finitely many overlap biholomorphisms, hence is open. Each summand therefore embeds openly in the quotient, and the only transition maps are biholomorphisms. This gives a complex threefold.

Points over distinct base values have disjoint disc preimages. Points over one base value lie in a common one of the four Hausdorff charts, so the quotient is Hausdorff. The images of countable bases of the four open pieces give second countability. The local maps to the base agree on overlaps. Moreover $f^{-1}(D_j)$ is exactly the embedded N_j, since every middle-piece point over D_j^* was identified with its unique representative there; similarly $f^{-1}(B^\circ) = J$. Each of these full restrictions is proper, for the period-family piece by theorem 10.8 and for the three fillings by their local proper constructions. Explicitly, for compact $K \subset B$, choose finitely many open sets V_a covering K, with compact closures inside members U_a of this

base cover. Each $K \cap \overline{V}_a$ is compact in U_a, so its full inverse image is compact. Their finite union is $f^{-1}(K)$; hence the global map is proper. Since the base is compact, $X = f^{-1}(B)$ is compact. All four pieces are connected and every local filling meets the central period-family piece, so X is connected. The fibre statements follow from theorems 10.6 and 10.9; their fibres are connected, and surjectivity follows. Because each summand embeds openly in the quotient, the restriction of the intrinsic canonical bundle ω_X to the open summand N_0 is ω_{N_0}. Equation (2.6a) proves equation (21). □

What has and has not been discharged. The formulas above reconstruct the period functions, both finite fillings, the toric cusp quotient, and all three punctured-overlap biholomorphisms. Thus theorems 10.6 and 10.11 discharge existence, smoothness, properness, the full central-fibre equation, and the normalization assertions formerly labelled (H1)–(H3) in theorem 60.24. This promotion is local to the analytic construction: no collapse, Leray transgression, or Mayer–Vietoris computation is used here. Integral cohomology and sphere recognition remain separate below.

11 Flat spectral geometry on the positive radial cusp

The period family used in this section is the family already constructed and proved nondegenerate in theorem 10.2. In particular, this section introduces no additional period hypothesis. We retain the proved ordered lattices

$$V = \mathbb{Z}\langle \gamma, u, w, \delta \rangle, \qquad \Lambda = V^* = \mathbb{Z}\langle \widehat{\gamma}, \widehat{u}, \widehat{w}, \widehat{\delta} \rangle$$

and the proved period matrix

$$\Pi = [Z \mid I_2], \qquad Z = \begin{pmatrix} 6\mu & \tau \\ \beta & \mu \end{pmatrix}.$$

Thus the columns of Π, in the displayed ordered basis of Λ, are

$$(6\mu, \beta)^{\mathsf{t}}, \quad (\tau, \mu)^{\mathsf{t}}, \quad (1, 0)^{\mathsf{t}}, \quad (0, 1)^{\mathsf{t}}.$$

The canonical cusp calculation has already produced holomorphic functions $\mu(t_c), b(t_c), h(t_c)$ on a full disc about $t_c = 0$, where

$$b = \beta + \tau, \qquad s = \tau - h, \qquad e^{2\pi i s} = t_c,$$

and has proved the exact identity

$$Z = s \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} + \begin{pmatrix} 6\mu & h \\ b - h & \mu \end{pmatrix}. \tag{22}$$

Choose T_0 so that $e^{-2\pi T}$ lies in the proved cusp disc whenever $T \geq T_0$. On the positive radial ray

$$t_c = e^{-2\pi T}, \qquad T \geq T_0,$$

we choose the lift $s = iT$. The other lifts differ by the already accounted for integral deck transformation. Put

$$\begin{aligned} \mu_T &:= \mu(e^{-2\pi T}) = \alpha_T + i m_T, \\ h_T &:= h(e^{-2\pi T}) = \eta_T + i H_T, \\ b_T &:= b(e^{-2\pi T}) = \xi_T + i B_T, \\ c_T &:= \xi_T - \eta_T, \qquad L_T := T + H_T, \qquad q_T := L_T - B_T, \\ D_T &:= L_T q_T + 6 m_T^2. \end{aligned} \tag{23}$$

All letters on the right-hand side of equation (23) are real. We reserve

$$m := \operatorname{Im} \mu(0), \qquad H := \operatorname{Im} h(0), \qquad B := \operatorname{Im} b(0) \tag{24}$$

for their limiting values. Holomorphy at the origin gives, with the same statement for the real parts,

$$m_T = m + O(e^{-2\pi T}), \qquad H_T = H + O(e^{-2\pi T}), \qquad B_T = B + O(e^{-2\pi T}). \tag{25}$$

Identify $\mathbb{C}^2$ with $\mathbb{R}^4$ by the ordered coordinates

$$(z_1, z_2) = (x_1 + iy_1, x_2 + iy_2) \longmapsto (x_1, y_1, x_2, y_2). \tag{26}$$

The metric in this section is, by definition, the quotient of $dx_1^2 + dy_1^2 + dx_2^2 + dy_2^2$. This makes the metric choice explicit; the complex period construction itself neither requires nor singles out this Euclidean metric.

Proposition 11.1 (Exact radial realification and dual lattice)**.** *In the domain basis $(\widehat{\gamma}, \widehat{u}, \widehat{w}, \widehat{\delta})$ and the real coordinate order equation* (26)*, the positive radial period matrix is*

$$P_T = \begin{pmatrix} 6\alpha_T & \eta_T & 1 & 0 \\ 6m_T & L_T & 0 & 0 \\ c_T & \alpha_T & 0 & 1 \\ -q_T & m_T & 0 & 0 \end{pmatrix}. \tag{27}$$

Its oriented determinant, positive covolume, and relation with the determinant function in theorem 10.2 are

$$\det P_T = -D_T, \qquad \operatorname{covol}(P_T\mathbb{Z}^4) = D_T > 0, \qquad D_{\mathrm{can},T} = -\frac{D_T}{L_T} < 0. \tag{28}$$

Here $D_{\mathrm{can},T} = \operatorname{Im}\beta_T - 6(\operatorname{Im}\mu_T)^2/\operatorname{Im}\tau_T$ is the canonical determinant function; in particular, D_T is not a new nondegeneracy assumption.

The complete inverse transpose is

$$P_T^{-\mathsf{T}} = \begin{pmatrix} 0 & 0 & 1 & 0 \\ \dfrac{m_T}{D_T} & \dfrac{q_T}{D_T} & \dfrac{-6m_T\alpha_T - q_T\eta_T}{D_T} & \dfrac{-m_Tc_T - q_T\alpha_T}{D_T} \\ 0 & 0 & 0 & 1 \\ -\dfrac{L_T}{D_T} & \dfrac{6m_T}{D_T} & \dfrac{6(L_T\alpha_T - m_T\eta_T)}{D_T} & \dfrac{L_Tc_T - 6m_T\alpha_T}{D_T} \end{pmatrix}. \tag{29}$$

Equivalently, for $k = (r, s, n_1, n_2)^{\mathsf{t}} \in \mathbb{Z}^4$, define

$$R_T(k) := r - 6\alpha_T n_1 - c_T n_2, \qquad S_T(k) := s - \eta_T n_1 - \alpha_T n_2. \tag{30}$$

Then the Euclidean dual lattice is exactly

$$(P_T\mathbb{Z}^4)^* = \Big\{ \Big(n_1, \frac{m_T R_T(k) + q_T S_T(k)}{D_T}, n_2, \frac{-L_T R_T(k) + 6m_T S_T(k)}{D_T} \Big) : k = (r, s, n_1, n_2) \in \mathbb{Z}^4 \Big\}. \tag{31}$$

Consequently, if

$$F_T := \mathbb{R}^4/P_T\mathbb{Z}^4$$

and Δ_T is the nonnegative self-adjoint scalar Laplacian on $L^2(F_T, \mathrm{d}\operatorname{vol}_{F_T})$, obtained from the closed Dirichlet form

$$\mathfrak{q}_T[f] = \int_{F_T} |\mathrm{d}f|^2 \,\mathrm{d}\operatorname{vol}_{F_T}, \qquad \operatorname{Dom}(\mathfrak{q}_T) = H^1(F_T),$$

with operator domain $\operatorname{Dom}(\Delta_T) = H^2(F_T)$, then, with multiplicity,

$$\operatorname{Spec}(\Delta_T) = \{\lambda_T(k) : k \in \mathbb{Z}^4\}, \tag{32}$$

where

$$\lambda_T(k) = 4\pi^2\Bigg(n_1^2 + n_2^2 + \frac{(m_T R_T(k) + q_T S_T(k))^2 + (-L_T R_T(k) + 6m_T S_T(k))^2}{D_T^2}\Bigg). \tag{33}$$

The eigenfunction belonging to k is the character

$$\chi_{T,k}([x]) = \exp\bigl(2\pi i \langle P_T^{-\mathsf{T}} k, x\rangle\bigr). \tag{34}$$

Proof. Substitution of $s = iT$ into equation (22) gives

$$Z_T = \begin{pmatrix} 6\mu_T & h_T + iT \\ b_T - h_T - iT & \mu_T \end{pmatrix}.$$

Realifying its four columns in the interlaced coordinate order (x_1, y_1, x_2, y_2) gives equation (27); in particular, the two divergent entries have signs $+L_T$ and $-q_T = B_T - L_T$.

Expanding the determinant along the last two columns gives

$$\det P_T = -L_T q_T - 6m_T^2 = -D_T.$$

Moreover,

$$\operatorname{Im}\tau_T = L_T, \qquad \operatorname{Im}\beta_T = B_T - L_T = -q_T,$$

so the canonical identity $\det_{\mathbb{R}} \Pi_T = \operatorname{Im}\tau_T D_{\mathrm{can},T}$ becomes

$$L_T D_{\mathrm{can},T} = L_T\left(-q_T - \frac{6m_T^2}{L_T}\right) = -D_T.$$

The already-proved inequalities $L_T > 0$ and $D_{\mathrm{can},T} < 0$ imply $D_T > 0$.

To compute the inverse transpose without suppressing the real parts, solve $P_T^{\mathsf{T}} x = k$. Its third and fourth equations give

$$x_1 = n_1, \qquad x_3 = n_2.$$

The remaining equations are

$$\begin{pmatrix} 6m_T & -q_T \\ L_T & m_T \end{pmatrix}\begin{pmatrix} x_2 \\ x_4 \end{pmatrix} = \begin{pmatrix} R_T(k) \\ S_T(k) \end{pmatrix}. \tag{35}$$

The determinant of the matrix in equation (35) is D_T, and its inverse is

$$\frac{1}{D_T}\begin{pmatrix} m_T & q_T \\ -L_T & 6m_T \end{pmatrix}.$$

This proves both equation (29) and equation (31).

A character $x \mapsto e^{2\pi i\langle \ell, x\rangle}$ descends to F_T precisely when $\langle \ell, P_T k\rangle \in \mathbb{Z}$ for every $k \in \mathbb{Z}^4$, which is precisely $\ell \in P_T^{-\mathsf{T}}\mathbb{Z}^4$. These characters are the complete Fourier basis of the compact abelian group F_T, and applying the positive Euclidean Laplacian gives $4\pi^2\|\ell\|^2$. Inserting equation (31) proves equations (32) to (34). □

Theorem 11.2 (Shortest cusp modes and their first corrections)**.** *Let*

$$\ell_{\gamma,T} := P_T^{-\mathsf{T}}(1,0,0,0)^{\mathsf{t}}, \qquad \ell_{u,T} := P_T^{-\mathsf{T}}(0,1,0,0)^{\mathsf{t}}. \tag{36}$$

For all sufficiently large T, every shortest nonzero vector of $(P_T\mathbb{Z}^4)^*$ belongs to

$$\{\pm\ell_{\gamma,T},\ \pm\ell_{u,T}\}.$$

Their exact squared lengths are

$$\|\ell_{\gamma,T}\|^2 = \frac{L_T^2 + m_T^2}{D_T^2}, \qquad \|\ell_{u,T}\|^2 = \frac{q_T^2 + 36m_T^2}{D_T^2}. \tag{37}$$

Their exact comparison is

$$D_T^2(\|\ell_{\gamma,T}\|^2 - \|\ell_{u,T}\|^2) = B_T(2L_T - B_T) - 35m_T^2. \tag{38}$$

Thus the shortest Fourier indices are $\pm(0,1,0,0)$ *if the right-hand side of equation* (38) *is positive,* $\pm(1,0,0,0)$ *if it is negative, and both pairs if it is zero.*

In the standard basis e_1, e_2, e_3, e_4 *of the real coordinate space, the vector expansions are*

$$T\ell_{\gamma,T} = -e_4 + \frac{me_2 + (H-B)e_4}{T} + O(T^{-2}), \tag{39}$$

$$T\ell_{u,T} = e_2 + \frac{-He_2 + 6me_4}{T} + O(T^{-2}). \tag{40}$$

Accordingly,

$$4\pi^2\|\ell_{\gamma,T}\|^2 = \frac{4\pi^2}{T^2}\left(1 + \frac{2(B-H)}{T} + O(T^{-2})\right), \tag{41}$$

$$4\pi^2\|\ell_{u,T}\|^2 = \frac{4\pi^2}{T^2}\left(1 - \frac{2H}{T} + O(T^{-2})\right). \tag{42}$$

More generally, every fixed plane index has the first correction

$$\frac{T^2\lambda_T(r,s,0,0)}{4\pi^2} = r^2 + s^2 + \frac{2(B-H)r^2 - 10mrs - 2Hs^2}{T} + O_{r,s}(T^{-2}). \tag{43}$$

In particular, the first positive eigenvalue, including the coefficient after its leading term, is

$$\boxed{\lambda_1(F_T) = \frac{4\pi^2}{T^2}\left(1 + \frac{-2H + 2\min\{B,0\}}{T} + O(T^{-2})\right).} \tag{44}$$

If $B > 0$*, the bottom indices are eventually* $\pm(0,1,0,0)$*; if* $B < 0$*, they are eventually* $\pm(1,0,0,0)$*. If* $B = 0$*, both pairs have the same displayed first correction, and the exact decision is made by equation* (38)*. In particular, if* $B = 0$ *but* $m \neq 0$*, the bottom indices are eventually* $\pm(1,0,0,0)$*. The first positive eigenvalue has complex multiplicity two when exactly one pair minimizes and multiplicity four when both pairs minimize.*

Proof. For $k = (r,s,0,0)$, define

$$K_T := \frac{T}{D_T}\begin{pmatrix} m_T & q_T \\ -L_T & 6m_T \end{pmatrix}.$$

Equations equations (23) and (25) give

$$K_T = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} + O_{\mathrm{op}}(T^{-1}). \tag{45}$$

Hence there is a constant C, independent of $(r,s) \in \mathbb{R}^2$, such that

$$(1 - C/T)\sqrt{r^2+s^2} \le T\|P_T^{-\mathsf{T}}(r,s,0,0)\| \le (1 + C/T)\sqrt{r^2+s^2} \tag{46}$$

for all sufficiently large T. Thus among the vectors with $(n_1,n_2) = (0,0)$, every vector with $r^2 + s^2 \ge 2$ is eventually longer than all four vectors for which $r^2 + s^2 = 1$.

If instead $(n_1, n_2) \neq (0,0)$, equation (31) gives

$$\|P_T^{-\mathsf{T}} k\|^2 \geq n_1^2 + n_2^2 \geq 1,$$

whereas each of the four unit vectors just identified has length $T^{-1}(1 + O(T^{-1}))$. This proves the asserted exhaustion of the shortest vectors.

Taking the first two columns of equation (29) proves equation (37). Since $q_T = L_T - B_T$, their numerator difference is

$$L_T^2 + m_T^2 - q_T^2 - 36m_T^2 = B_T(2L_T - B_T) - 35m_T^2,$$

which proves equation (38) and the exact mode classification.

Finally,

$$D_T = T^2 + (2H - B)T + H(H - B) + 6m^2 + O(Te^{-2\pi T}). \tag{47}$$

Expanding the first two columns of equation (29) with equation (47) gives equations (39) and (40). Taking their squared norms gives equations (41) and (42). The same multiplication for a fixed (r, s) gives equation (43); explicitly, the scaled two-dimensional matrix is

$$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} + \frac{1}{T}\begin{pmatrix} m & -H \\ H - B & 6m \end{pmatrix} + O(T^{-2}),$$

and twice the Euclidean inner product of its leading and first-order terms is $2(B - H)r^2 - 10mrs - 2Hs^2$. The smaller relative correction is

$$\min\{2(B - H), -2H\} = -2H + 2\min\{B, 0\},$$

which proves equation (44). □

Corollary 11.3 (Volume and unit-volume scaling)**.** *The Euclidean volume is exactly*

$$V_T := \mathrm{Vol}(F_T) = D_T = (T + H_T)(T + H_T - B_T) + 6m_T^2, \tag{48}$$

and it has the expansion equation (47)*. Define the unit-volume metric*

$$\widehat{g}_T := D_T^{-1/2} g_T. \tag{49}$$

Let $\widehat{\Delta}_T$ denote the nonnegative self-adjoint scalar Laplacian on $L^2(F_T, \mathrm{d}\,\mathrm{vol}_{\widehat{g}_T})$ obtained from the closed Dirichlet form

$$\widehat{\mathfrak{q}}_T[f] = \int_{F_T} |\mathrm{d}f|^2_{\widehat{g}_T}\, \mathrm{d}\,\mathrm{vol}_{\widehat{g}_T}, \qquad \mathrm{Dom}(\widehat{\mathfrak{q}}_T) = H^1(F_T),$$

with operator domain $\mathrm{Dom}(\widehat{\Delta}_T) = H^2(F_T)$. Then the first positive eigenvalue of $\widehat{\Delta}_T$ is

$$\boxed{\lambda_1(F_T, \widehat{g}_T) = \frac{4\pi^2}{T}\left(1 + \frac{-H - B/2 + 2\min\{B, 0\}}{T} + O(T^{-2})\right).} \tag{50}$$

Thus unit-volume normalization does not prevent the first eigenvalue from tending to zero. In the normalized metric the periods e_1, e_3 have length $D_T^{-1/4} = T^{-1/2}(1 + O(T^{-1}))$, while the two periods with indices $\widehat{\gamma}, \widehat{u}$ have length $T^{1/2}(1 + O(T^{-1}))$.

Proof. The volume of a Euclidean lattice quotient is its covolume, so equation (28) proves equation (48). In real dimension four, replacing g by ag multiplies volume by a^2 and every Laplace eigenvalue by a^{-1}. Therefore equation (49) has volume one and

$$\lambda_1(F_T, \widehat{g}_T) = D_T^{1/2} \lambda_1(F_T, g_T).$$

Equation equation (47) gives

$$D_T^{1/2} = T\left(1 + \frac{H - B/2}{T} + O(T^{-2})\right).$$

Multiplication by equation (44) proves equation (50). The stated period-length estimates follow directly from equation (27): e_1, e_3 have original length one, whereas the $\widehat{\gamma}$- and $\widehat{u}$-columns have original length $T(1 + O(T^{-1}))$; lengths are multiplied by $D_T^{-1/4}$. □

Proposition 11.4 (Exact counting formula and its two cusp regimes)**.** *For $E \geq 0$, count eigenvalues with their Fourier multiplicities by*

$$N_T(E) := \#\{k \in \mathbb{Z}^4 : \lambda_T(k) \leq E\}. \tag{51}$$

Put

$$R_E^2 := \frac{E}{4\pi^2}, \qquad (x)_+ := \max\{x, 0\}.$$

For every fixed E,

$$N_T(E) = \pi D_T \sum_{n \in \mathbb{Z}^2} (R_E^2 - \|n\|^2)_+ + O_E(T + 1). \tag{52}$$

The sum is finite. Equivalently,

$$\lim_{T \to \infty} \frac{N_T(E)}{D_T} = \pi \sum_{n \in \mathbb{Z}^2} (R_E^2 - \|n\|^2)_+. \tag{53}$$

In the first band, $0 < E < 4\pi^2$, this reduces to

$$N_T(E) = \frac{E}{4\pi} D_T + O_E(T + 1). \tag{54}$$

For the unit-volume metric $\widehat{g}_T$, put

$$\widehat{N}_T(E) := \#\{k \in \mathbb{Z}^4 : \lambda_T(k; \widehat{g}_T) \leq E\}. \tag{55}$$

For every fixed $E > 0$,

$$\widehat{N}_T(E) = \frac{E}{4\pi} D_T^{1/2} + O_E(D_T^{1/4} + 1), \qquad \lim_{T \to \infty} \frac{\widehat{N}_T(E)}{D_T^{1/2}} = \frac{E}{4\pi}. \tag{56}$$

There is also a finite counting law at the collapsing spectral scale. Define

$$N_T^{\mathrm{coll}}(E) := \#\left\{k \in \mathbb{Z}^4 : \lambda_T(k) \leq \frac{4\pi^2 E}{T^2}\right\}. \tag{57}$$

If E is not a sum of two integer squares, then, for all sufficiently large T,

$$N_T^{\mathrm{coll}}(E) = \#\{(r, s) \in \mathbb{Z}^2 : r^2 + s^2 \leq E\}. \tag{58}$$

At a threshold $E \in \{r^2+s^2 : r,s \in \mathbb{Z}\}$, the sharp statement available from the zeroth-order limit is

$$\begin{aligned}\#\{(r,s) : r^2+s^2 < E\} &\le \liminf_{T\to\infty} N_T^{\mathrm{coll}}(E) \\ &\le \limsup_{T\to\infty} N_T^{\mathrm{coll}}(E) \le \#\{(r,s) : r^2+s^2 \le E\}.\end{aligned} \tag{59}$$

For a threshold vector (r,s), the coefficient in equation (43) decides its inclusion whenever that coefficient is nonzero. If it vanishes, the exact eigenvalue formula and, if necessary, higher Taylor coefficients decide the inclusion.

Proof. Set

$$\mathcal{A}_T := \begin{pmatrix} 6m_T & -q_T \\ L_T & m_T \end{pmatrix}, \qquad \theta_T(n_1,n_2) := \begin{pmatrix} 6\alpha_T n_1 + c_T n_2 \\ \eta_T n_1 + \alpha_T n_2 \end{pmatrix}. \tag{60}$$

Then $\det \mathcal{A}_T = D_T$, and the exact eigenvalue formula can be rewritten as

$$\frac{\lambda_T(r,s,n_1,n_2)}{4\pi^2} = \|(n_1,n_2)\|^2 + \left\| \mathcal{A}_T^{-1}\left(\begin{pmatrix} r \\ s \end{pmatrix} - \theta_T(n_1,n_2)\right)\right\|^2. \tag{61}$$

Fix $n=(n_1,n_2)$. If $\|n\|^2 > R_E^2$, this index contributes nothing. Otherwise put $\rho_n = (R_E^2 - \|n\|^2)^{1/2}$. The allowed pairs (r,s) are the integer points in the translated ellipse

$$\theta_T(n) + \mathcal{A}_T \overline{B}_{\mathbb{R}^2}(0,\rho_n). \tag{62}$$

Its area is $\pi D_T \rho_n^2$. Its boundary length is $O_E(T+1)$, because $\|\mathcal{A}_T\|_{\mathrm{op}} = O(T+1)$. If $\rho_n = 0$, the set in equation (62) is one point and contributes either zero or one integer point, which is absorbed in the $O_E(1)$ remainder below. We apply the planar area–perimeter estimate only when $\rho_n > 0$.

For completeness, the elementary planar lattice estimate used here is

$$\#(\mathbb{Z}^2 \cap K) = \mathrm{area}(K) + O(\mathrm{length}(\partial K) + 1)$$

for every translated compact convex set K. Indeed, attach to each integer point its centered unit square and let U be the union of the squares centered at points of $K \cap \mathbb{Z}^2$. A point of $U \triangle K$ lies within distance $\sqrt{2}$ of ∂K: this is immediate for $U \setminus K$, while a point of $K \setminus U$ whose distance from the boundary exceeded $\sqrt{2}$ would have its nearest integer point inside K and would belong to the corresponding square. If $P(K)$ is the perimeter, the outer parallel-set formula for a convex polygon gives $\mathrm{area}(K+B_\rho) - \mathrm{area}(K) = \rho P(K) + \pi\rho^2$: its added pieces are the edge rectangles and vertex sectors, whose angles total 2π. Polygonal approximation gives the same bound for every compact convex K. Applying it to the outer and inner boundary strips with $\rho = \sqrt{2}$ gives $\mathrm{area}(U \triangle K) = O(P(K)+1)$. Since the unit squares have disjoint interiors and total area $\#(K \cap \mathbb{Z}^2)$, the lattice estimate follows.

Only finitely many n's occur for fixed E. Applying the estimate to equation (62) and summing proves equation (52). Since $D_T \sim T^2$, division by D_T proves equation (53); if $0 < E < 4\pi^2$, only $n=0$ occurs and equation (54) follows.

For $\widehat{g}_T$, eigenvalues are multiplied by $D_T^{1/2}$, so $\widehat{N}_T(E) = N_T(E/D_T^{1/2})$. For large T, only $n=0$ can occur in equation (61). Its ellipse has radius parameter $\rho = (E/(4\pi^2 D_T^{1/2}))^{1/2}$, hence area $ED_T^{1/2}/(4\pi)$ and perimeter $O_E(TD_T^{-1/4}+1) = O_E(D_T^{1/4}+1)$. The same lattice estimate proves equation (56).

For the collapsing scale, any $(n_1,n_2) \neq (0,0)$ contributes at least $4\pi^2$, so it contributes nothing for all sufficiently large T. For $n_1 = n_2 = 0$, equation (45) says uniformly that

$$\frac{T^2 \lambda_T(r,s,0,0)}{4\pi^2} = \|K_T(r,s)^{\mathsf{t}}\|^2 \longrightarrow r^2+s^2.$$

The lower bound in equation (46) also shows that only a fixed finite set of (r,s)'s can enter a fixed rescaled window. Termwise comparison on that finite set proves equation (58) away from thresholds and the two-sided bound equation (59) at thresholds. □

Theorem 11.5 (Pointed limits and expanding local covers)**.** *Let o_T be the origin of F_T, and set*

$$\mathbb{T}^2_{\mathrm{re}} := (\mathbb{R}e_1 \oplus \mathbb{R}e_3)/(\mathbb{Z}e_1 \oplus \mathbb{Z}e_3). \tag{63}$$

With the product Euclidean metric,

$$(F_T, o_T) \longrightarrow (\mathbb{T}^2_{\mathrm{re}} \times (\mathbb{R}e_2 \oplus \mathbb{R}e_4), (0,0)) \cong (\mathbb{T}^2 \times \mathbb{R}^2, (0,0)) \tag{64}$$

in pointed smooth, and hence pointed Gromov–Hausdorff, topology. More precisely, for every $R < \infty$, the radius-R balls in the two sides of equation (64) *are isometric for all sufficiently large T.*

For $M \in \mathbb{Z}_{\geq 1}$, define the marked local sublattice

$$\Lambda_{T,M} := \mathbb{Z}v_{\gamma,T} \oplus \mathbb{Z}v_{u,T} \oplus \mathbb{Z}(Me_1) \oplus \mathbb{Z}(Me_3), \tag{65}$$

where

$$\begin{aligned} v_{\gamma,T} &= (6\alpha_T, 6m_T, c_T, -q_T), \\ v_{u,T} &= (\eta_T, L_T, \alpha_T, m_T). \end{aligned} \tag{66}$$

Then

$$\widetilde{F}_{T,M} := \mathbb{R}^4/\Lambda_{T,M}, \qquad c_{T,M} : \widetilde{F}_{T,M} \longrightarrow F_T \tag{67}$$

is a regular holomorphic local-isometric cover of degree M^2, with deck group $(\mathbb{Z}/M\mathbb{Z})^2$. For every integer-valued function $M(T) \to \infty$,

$$(\widetilde{F}_{T,M(T)}, \widetilde{o}_{T,M(T)}) \longrightarrow (\mathbb{R}^4, 0) \tag{68}$$

in pointed smooth and pointed Gromov–Hausdorff topology. Again, every fixed radius ball is eventually exactly isometric to its Euclidean counterpart. If the cover is instead equipped with the pullback of the unit-volume metric $\widehat{g}_T$, then it converges in pointed smooth topology, and hence in pointed Gromov–Hausdorff topology, to $\mathbb{R}^4$ if and only if

$$\frac{M(T)}{D_T^{1/4}} \longrightarrow \infty, \quad \textit{equivalently} \quad \frac{M(T)}{\sqrt{T}} \longrightarrow \infty. \tag{69}$$

By contrast, if every covering torus is itself rescaled to volume one, no choice of $M(T)$ gives a pointed smooth limit $\mathbb{R}^4$.

Proof. The period lattice is

$$P_T\mathbb{Z}^4 = \mathbb{Z}v_{\gamma,T} \oplus \mathbb{Z}v_{u,T} \oplus \mathbb{Z}e_1 \oplus \mathbb{Z}e_3. \tag{70}$$

First quotient $\mathbb{R}^4$ by $\mathbb{Z}e_1 \oplus \mathbb{Z}e_3$. This gives the proposed limit

$$X_\infty := \mathbb{T}^2_{\mathrm{re}} \times (\mathbb{R}e_2 \oplus \mathbb{R}e_4).$$

The remaining quotient translations are generated by the images of $v_{\gamma,T}, v_{u,T}$. The projection of their integer combination $av_{\gamma,T} + bv_{u,T}$ to the noncompact (e_2, e_4)-plane is

$$Y_T \begin{pmatrix} a \\ b \end{pmatrix}, \qquad Y_T := \begin{pmatrix} 6m_T & L_T \\ -q_T & m_T \end{pmatrix}. \tag{71}$$

More precisely,

$$Y_T = T \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} + \begin{pmatrix} 6m_T & H_T \\ B_T - H_T & m_T \end{pmatrix}. \tag{72}$$

Holomorphy makes the second matrix uniformly bounded for $T \geq T_0$. Choose C above its operator norm. The first matrix is T times an orthogonal matrix, so the reverse triangle inequality gives

$$\left\| Y_T \begin{pmatrix} a \\ b \end{pmatrix} \right\| \geq (T - C)\sqrt{a^2 + b^2}. \tag{73}$$

The displacement in X_∞ of a translation is at least the norm of its noncompact-plane projection. Hence every nonidentity element of the remaining deck group has displacement at least $T - C$. Together with the fixed lattice vectors e_1, e_3, this also proves

$$\operatorname{sys}(P_T\mathbb{Z}^4) = 1, \qquad \operatorname{inj}(F_T, g_T) = \frac{1}{2} \tag{74}$$

for all sufficiently large T.

Fix R. Once $T - C > 4R$, the quotient map $X_\infty \to F_T$ is injective on the radius-R ball about the origin. It is a local isometry, and every minimizing geodesic of length at most R in the quotient lifts from the origin, so the closed radius-R ball maps onto the closed radius-R ball in F_T. Moreover, a nontrivial translate cannot shorten the distance between two points of the ball: their original distance is at most $2R$, while every translated distance is at least $(T - C) - 2R > 2R$. The balls are therefore isometric. This proves equation (64), including its pointed smooth assertion.

Because the four vectors in equation (70) are a lattice basis, the sublattice equation (65) has index M^2. Thus the identity of $\mathbb{R}^4$ induces the regular holomorphic degree-M^2 local-isometric cover equation (67). The quotient of the two lattices is $(\mathbb{Z}/M\mathbb{Z})^2$, which is its deck group. Consider a nonzero vector of $\Lambda_{T,M}$:

$$av_{\gamma,T} + bv_{u,T} + M(pe_1 + qe_3), \qquad a, b, p, q \in \mathbb{Z}. \tag{75}$$

If $(a, b) \neq (0, 0)$, its (e_2, e_4)-projection and equation (73) show that its norm is at least $T - C$. If $(a, b) = (0, 0)$, its norm is at least M. Therefore

$$\operatorname{sys}(\Lambda_{T,M}) \geq \min\{T - C, M\}. \tag{76}$$

Conversely, Me_1 has length M, while either long generator has length at most $T + C'$ for a fixed C'. Hence

$$\min\{T - C, M\} \leq \operatorname{sys}(\Lambda_{T,M}) \leq \min\{T + C', M\}, \qquad \operatorname{inj}(\widetilde{F}_{T,M}) = \tfrac{1}{2}\operatorname{sys}(\Lambda_{T,M}). \tag{77}$$

For $M = M(T) \to \infty$, this lower bound tends to infinity. Once it is greater than $4R$, the universal covering map $\mathbb{R}^4 \to \widetilde{F}_{T,M(T)}$ identifies the radius-R balls isometrically. This proves equation (68).

For the pullback of $\widehat{g}_T$, all lengths in equation (76) are multiplied by $D_T^{-1/4}$. The long lower bound becomes $(T-C)D_T^{-1/4} \sim \sqrt{T}$, which diverges automatically. The explicit lattice vector Me_1 has normalized length $MD_T^{-1/4}$. Hence the normalized systole tends to infinity exactly under equation (69). Divergence gives Euclidean balls by the preceding covering argument. We now prove necessity from the particular flat lattices rather than from any general injectivity-radius principle. Put

$$\ell_T := M(T)D_T^{-1/4}, \qquad A_T := (T - C)D_T^{-1/4}.$$

Then $A_T \sim \sqrt{T} \to \infty$. Quotienting the normalized Euclidean cover first by $\ell_T\mathbb{Z}e_1 \oplus \ell_T\mathbb{Z}e_3$ gives

$$X_{\ell_T} := (\mathbb{R}/\ell_T\mathbb{Z})^2 \times \mathbb{R}^2.$$

Every remaining nonidentity deck transformation has displacement at least A_T. Consequently, for each fixed R, once $A_T > 4R$, the closed radius-R ball in the normalized covering torus is exactly the closed radius-R ball in X_{ℓ_T}. If $\ell_T \not\to \infty$, pass to a subsequence with $\ell_T \to \ell < \infty$. When $\ell > 0$, these pointed product spaces converge to $(\mathbb{R}/\ell\mathbb{Z})^2 \times \mathbb{R}^2$; when $\ell = 0$, the two circle factors have diameter tending to zero and the pointed Gromov–Hausdorff limit is $\mathbb{R}^2$. Neither limit is $\mathbb{R}^4$. Thus $\ell_T \to \infty$, which is precisely equation (69), is necessary for both the pointed smooth and the pointed Gromov–Hausdorff conclusion.

Finally, the volume of the covering torus with its unscaled pulled-back metric is M^2D_T. If one instead puts

$$g_{T,M}^{\mathrm{cov},1} := (M^2D_T)^{-1/2}g_T \quad \text{on } \widetilde{F}_{T,M}, \tag{78}$$

then its volume is one and its length scale is multiplied by $(M^2 D_T)^{-1/4}$. The explicit periods Me_1 and $v_{\gamma,T}$ consequently have respective lengths

$$\sqrt{\frac{M}{T}}\,(1+o(1)), \qquad \sqrt{\frac{T}{M}}\,(1+o(1)). \tag{79}$$

Their leading factors have product one, so at least one stays bounded. The injectivity radius therefore cannot tend to infinity, which excludes a pointed smooth $\mathbb{R}^4$ limit for the individually unit-volume covers. □

Proposition 11.6 (Exact compact quotient of the native cusp fibre)**.** *Let*

$$\rho : \mathbb{R}^4 \longrightarrow \mathbb{R}^2, \qquad \rho(x_1, x_2, x_3, x_4) = (x_2, x_4), \tag{80}$$

and let Y_T be the matrix in equation (71)*. Then*

$$\det Y_T = D_T > 0. \tag{81}$$

Consequently

$$Q_T := \mathbb{R}^2 / Y_T \mathbb{Z}^2 \tag{82}$$

is a compact real two-torus, and ρ induces an exact sequence of compact real Lie groups

$$0 \longrightarrow \mathbb{T}^2_{\mathrm{re}} \longrightarrow F_T \xrightarrow{\pi_T} Q_T \longrightarrow 0. \tag{83}$$

Moreover,

$$\mathrm{sys}(Q_T) \geq T - C, \tag{84}$$

where C is the constant in equation (73)*. Thus the native $\mathbb{T}^2 \times \mathbb{R}^2$ limit separates exactly into a fixed unit two-torus kernel and an expanding two-torus quotient.*

Proof. Direct calculation gives

$$\det Y_T = (6m_T)(m_T) - L_T(-q_T) = L_T q_T + 6m_T^2 = D_T.$$

The period decomposition equation (70) gives

$$\rho(P_T \mathbb{Z}^4) = Y_T \mathbb{Z}^2,$$

because $\rho(e_1) = \rho(e_3) = 0$ and the projected long generators are the two columns of Y_T. Hence

$$\pi_T([x]_{P_T\mathbb{Z}^4}) := [\rho(x)]_{Y_T\mathbb{Z}^2}$$

is well defined and surjective.

It remains to determine its kernel without suppressing the integral lattices. If $\pi_T([x]) = 0$, there are $a, b \in \mathbb{Z}$ such that

$$\rho(x) = Y_T(a,b)^{\mathsf{t}} = \rho(a v_{\gamma,T} + b v_{u,T}).$$

Thus

$$x - a v_{\gamma,T} - b v_{u,T} \in \mathbb{R}e_1 \oplus \mathbb{R}e_3,$$

so $[x]$ lies in the displayed copy of $\mathbb{T}^2_{\mathrm{re}}$. Conversely that copy is plainly killed by π_T. Finally, if a vector of $P_T\mathbb{Z}^4$ lies in $\mathbb{R}e_1 \oplus \mathbb{R}e_3$, its long coefficients a, b satisfy $Y_T(a,b)^{\mathsf{t}} = 0$. Since $\det Y_T = D_T > 0$, one has $a = b = 0$; the intersection is therefore exactly $\mathbb{Z}e_1 \oplus \mathbb{Z}e_3$. This proves injectivity at the left and exactness. The systolic bound is equation (73). □

Theorem 11.7 (The maximal regular-locus $\mathbb{R}^4$ quotient and its compactification obstruction)**.** ***[VERIFIED]*** *Let*

$$\Lambda = \mathbb{Z}\langle \widehat{\gamma}, \widehat{u}, \widehat{w}, \widehat{\delta}\rangle, \qquad K_\Lambda = \mathbb{Z}\langle \widehat{w}, \widehat{\delta}\rangle,$$

and define

$$\varepsilon_\gamma(a\widehat{\gamma} + b\widehat{u} + c\widehat{w} + d\widehat{\delta}) = a, \qquad L_\Lambda = \ker \varepsilon_\gamma = \mathbb{Z}\langle \widehat{u}, \widehat{w}, \widehat{\delta}\rangle. \tag{G.1}$$

Then the following statements hold.

(i) Regular group quotient. *Over B°, the period family $J = X^\circ = f^{-1}(B^\circ)$ is a holomorphic family of complex Lie groups, not merely an affine torus bundle. The formula*

$$\chi_\gamma([z, \Pi(z)x]) = ([z], \varepsilon_\gamma(x) \bmod \mathbb{Z}) \tag{85}$$

defines a global smooth real-analytic fibrewise group epimorphism

$$\chi_\gamma : J \longrightarrow B^\circ \times (\mathbb{R}/\mathbb{Z}). \tag{86}$$

Write

$$\theta_\gamma := \mathrm{pr}_{\mathbb{R}/\mathbb{Z}} \circ \chi_\gamma : J \longrightarrow \mathbb{R}/\mathbb{Z} \tag{87}$$

for its circle coordinate. Thus $\chi_\gamma = (f, \theta_\gamma)$; the two symbols have different codomains and will not be interchanged below. Its kernel is the connected real three-torus family

$$\mathcal{H} := \frac{\mathfrak{h}^\circ_z \times ((L_\Lambda \otimes \mathbb{R})/L_\Lambda)}{\Delta}, \tag{88}$$

where $\Delta = \pi_1(B^\circ)$ acts diagonally through the deck action on $\mathfrak{h}^\circ_z$ and the restricted monodromy on L_Λ. Thus every fibre of $\mathcal{H}$ is $(L_\Lambda \otimes \mathbb{R})/L_\Lambda$; the family need not be a product. The quotient is real and is not a holomorphic quotient of complex tori.

(ii) Exact radial quotient metric. *On the positive radial cusp, the restriction of equation* (85) *to $F_T = \mathbb{R}^4/P_T\mathbb{Z}^4$ is*

$$\chi_{\gamma,T}([y]) = \left\langle P_T^{-\mathsf{T}} e_\gamma, y \right\rangle \bmod \mathbb{Z}, \qquad e_\gamma = (1,0,0,0)^{\mathsf{T}}. \tag{89}$$

For the Riemannian quotient of the Euclidean metric, the target circle has circumference

$$\ell_T = \frac{1}{\|P_T^{-\mathsf{T}} e_\gamma\|} = \frac{D_T}{\sqrt{L_T^2 + m_T^2}} = T + H - B + O(T^{-1}). \tag{90}$$

In particular, $\ell_T \to \infty$.

(iii) Four copies and minimality. *The source*

$$J^{[4]} = J \times_{B^\circ} J \times_{B^\circ} J \times_{B^\circ} J \tag{91}$$

is the actual fourfold fibre product over all of B°. The product map

$$\chi_\gamma^{[4]} = (\chi_\gamma, \chi_\gamma, \chi_\gamma, \chi_\gamma) : J^{[4]} \longrightarrow B^\circ \times (\mathbb{R}/\mathbb{Z})^4 \tag{92}$$

has kernel $\mathcal{H}^{[4]} = \mathcal{H} \times_{B^\circ} \cdots \times_{B^\circ} \mathcal{H}$, whose fibres are $\mathbb{T}^{12}$. Along the synchronized positive radial cusp, its quotient fibre, with product quotient metric, is

$$G_T = (\mathbb{R}/\ell_T\mathbb{Z})^4, \qquad \mathrm{sys}(G_T) = \ell_T. \tag{93}$$

Consequently

$$(G_T, 0) \longrightarrow (\mathbb{R}^4, 0) \tag{94}$$

in pointed local-isometric, pointed smooth, and pointed Gromov–Hausdorff topology. If $\ell_T > 4R$, the radius-R ball about zero is exactly Euclidean.

Equivalently, the external Cartesian product gives a quotient

$$J^4 \longrightarrow (B^\circ)^4 \times (\mathbb{R}/\mathbb{Z})^4,$$

and $J^{[4]}$ is its pullback along the diagonal $B^\circ \to (B^\circ)^4$. At four independent cusp parameters its target lattice is $\operatorname{diag}(\ell_{T_1}, \dots, \ell_{T_4})\mathbb{Z}^4$*, so the same pointed limit holds whenever* $\min_i \ell_{T_i} \to \infty$.

Among connected torus quotients obtained from r copies by a primitive full-monodromy-invariant kernel and whose quotient systole tends to infinity on the radial cusp, the quotient rank is at most r. Thus four copies are minimal for a four-dimensional all-expanding quotient, and equation (92) *attains the bound.*

The cusp-local two-copy map

$$\pi_T \times \pi_T : F_T \times F_T \longrightarrow Q_T \times Q_T \tag{95}$$

from theorem 11.6 also has pointed limit $\mathbb{R}^4$. It is not a quotient over the full B°, however, and no finite base cover makes it one.

(iv) Finite fillings. *At p_j, use the notation of theorem 10.9 and put*

$$(m_1, v_1, \kappa_1) = (3, \widehat{\gamma} + 2\widehat{u} - 4\widehat{w}, 1),$$

$$(m_2, v_2, \kappa_2) = (4, -\widehat{\gamma} - 3\widehat{u} + 3\widehat{w}, -1), \qquad \kappa_j = \varepsilon_\gamma(v_j). \tag{G.2}$$

In the un-conjugated flat coordinates, the finite filling generator is

$$\Phi_j(s, x) = \left(e^{-2\pi i/m_j} s, A_j x + \frac{v_j}{m_j} \right). \tag{96}$$

The raw character is not invariant:

$$\varepsilon_\gamma \left(A_j x + \frac{v_j}{m_j} \right) = \varepsilon_\gamma(x) + \frac{\kappa_j}{m_j} \pmod{\mathbb{Z}}. \tag{97}$$

It instead descends equivariantly to the Seifert circle family

$$C_j = \frac{\Delta_j \times (\mathbb{R}/\mathbb{Z})}{(s, \theta) \sim (e^{-2\pi i/m_j} s, \theta + \kappa_j/m_j)}. \tag{98}$$

The total space C_j is a smooth solid torus, the map $C_j \to D_j$ has a multiple central circle of order m_j, and the induced map $N_j \to C_j$ is a smooth proper $\mathbb{T}^3$-bundle. The target translation in equation (98) *is exactly compatible with the logarithmic gluing map h_j.*

For four synchronized copies, the smooth simultaneous-root extension is

$$\widetilde{N}_j^{[4]} = \frac{(J_j')^{\times_{\Delta_j} 4}}{\langle (\widetilde{g}_j^{\log})^{\times 4} \rangle}, \tag{99}$$

with target

$$C_j^{[4]} = \frac{\Delta_j \times (\mathbb{R}/\mathbb{Z})^4}{(s, \theta_1, \dots, \theta_4) \sim (e^{-2\pi i/m_j} s, \theta_1 + \kappa_j/m_j, \dots, \theta_4 + \kappa_j/m_j)}. \tag{100}$$

The map $\widetilde{N}_j^{[4]} \to C_j^{[4]}$ is a smooth proper $\mathbb{T}^{12}$-bundle. There is a canonical finite map

$$\nu_j : \widetilde{N}_j^{[4]} \longrightarrow N_j \times_{D_j} N_j \times_{D_j} N_j \times_{D_j} N_j \tag{101}$$

which is an isomorphism over D_j^*, is the normalization of the reduced ordinary filled fibre product, and has degree m_j^3 on the reduced central fibre. These two degrees are 27 *and* 64*. Gluing these two local normalizations to $J^{[4]}$ by $(h_j)^{\times 4}$, and gluing their targets to the product $B^\circ \times (\mathbb{R}/\mathbb{Z})^4$ by $(h_j^{S^1})^{\times 4}$, gives the corresponding normalization and Seifert target over $B \setminus \{p_0\}$.*

(v) Cusp nonextension. *Neither the literal map χ_γ nor $\chi_\gamma^{[4]}$ extends continuously through the toric cusp fibre W. Thus the four-copy quotient is global over B°, and its finite-filling normalization is global over $B \setminus \{p_0\}$, but the displayed map $\chi_\gamma^{[4]}$ has no continuous extension to the compact fourfold fibre product or to X^4 with the stated regular-fibre restriction, and it does not factor through $X^{\wedge 4} \cong S^{24}$. The cohomology class of χ_γ does extend over N_0; this weaker homotopy statement does not extend the displayed quotient map or its metric.*

(vi) Products, smashes, and sphere constructions. *If the recognition $X \cong_{\mathrm{diff}} S^6$ is used, then*

$$X^2 \cong_{\mathrm{diff}} S^6 \times S^6 \not\simeq S^{12}, \qquad H_6(X^2;\mathbb{Z}) \cong \mathbb{Z}^2, \tag{102}$$

and

$$\begin{aligned} X^4 &\cong_{\mathrm{diff}} (S^6)^4 \not\simeq S^{24}, \\ H_6(X^4;\mathbb{Z}) \cong \mathbb{Z}^4, &\qquad H_{12}(X^4;\mathbb{Z}) \cong \mathbb{Z}^6, \quad H_{18}(X^4;\mathbb{Z}) \cong \mathbb{Z}^4. \end{aligned} \tag{103}$$

The synchronized regular fourfold $J^{[4]}$ has real dimension 18*, not* 24*. Choosing a base point gives based homeomorphisms*

$$X \wedge X \cong S^{12}, \qquad X^{\wedge 4} \cong S^{24}, \tag{104}$$

but these smash spheres do not carry either literal fixed-target *quotient above. The later cone constructions theorems 11.12 to 11.14 change the codomain by retaining the radial parameter and collapsing the obstructing boundary cross-section; they therefore do not contradict this nonextension theorem. Moreover, neither S^{12} nor S^{24}, with any differentiable structure, admits an almost-complex structure [63, §15, Proposition 15.1 and the following remark, p. 434]; hence neither can be promoted to a compact complex manifold. This is an obstruction to a proposed complex S^{12}- or S^{24}-globalization, not to the real quotient or its pointed metric limit. Precisely, $\pi_T \times \pi_T$ is a map on F_T^2, and it fails to factor through $F_T \wedge F_T$; $\chi_{\gamma,T}^{[4]}$ is a map on F_T^4, and it fails to factor through $F_T^{\wedge 4}$. Neither map is a map on X^2 or X^4, and no extension with the stated fibre restriction exists.*

There is one exact but different sphere construction. If $C_T = \mathbb{R}/\ell_T\mathbb{Z}$ is based at zero, postcomposition with the target fat-wedge collapse gives

$$F_T^4 \xrightarrow{\chi_{\gamma,T}^{[4]}} C_T^4 \longrightarrow C_T^{\wedge 4} \cong S^4. \tag{105}$$

This map does factor through $F_T^{\wedge 4}$. With the quotient length metric and base point $[\ell_T/2, \ell_T/2, \ell_T/2, \ell_T/2]$, its pointed limit is $\mathbb{R}^4$, with every radius-R ball exactly Euclidean once $\ell_T > 4R$. The metric is singular at the collapsed fat wedge, and the map still does not extend to X^4 or $X^{\wedge 4}$.

Proof. Invariant lattices and the regular group law.

The matrices in theorem 5.1 give

$$(M_0 - I)(a,b,c,d)^{\mathsf{T}} = (0,0,b,-a)^{\mathsf{T}},$$

so

$$K_\Lambda = \ker(M_0 - I) = \mathrm{im}(M_0 - I). \tag{106}$$

They also give

$$\begin{aligned} A_1\widehat{u} &= -\widehat{w} + \widehat{\delta}, \quad A_1\widehat{w} = \widehat{u} - \widehat{w}, \quad A_1\widehat{\delta} = \widehat{\delta}, \\ A_2\widehat{u} &= \widehat{w}, \qquad\quad A_2\widehat{w} = -\widehat{u} + \widehat{\delta}, \quad A_2\widehat{\delta} = \widehat{\delta}. \end{aligned} \tag{107}$$

Thus L_Λ is invariant. It is the smallest invariant lattice containing K_Λ, because $\widehat{u} = A_1\widehat{w} + \widehat{w}$. Both lattices are primitive.

For a row covector (a, b, c, d), invariance under A_1 gives $b = 2c$ and $d = 3c$, whereas invariance under A_2 gives $b = c$ and $d = 2c$. Hence $b = c = d = 0$, and

$$\mathrm{Hom}(\Lambda, \mathbb{Z})^\Delta = \mathbb{Z}\varepsilon_\gamma. \tag{108}$$

The family over the cover is $(\mathfrak{h}_z^\circ \times \mathbb{C}^2)/\Lambda$, and the action used to form J is

$$[z, \zeta] \longmapsto [gz, R_g(z)\zeta].$$

It is complex linear in every fibre and fixes the zero section. Fibrewise addition, inversion, and zero are therefore equivariant and descend to J. In the flat real-analytic marking

$$\Xi(z, x + \Lambda) = [z, \Pi(z)x],$$

equation (2.3d) makes the action

$$(z, x + \Lambda) \longmapsto (gz, A_g x + \Lambda).$$

Equation equation (108) proves that equation (85) is well defined. Primitivity of ε_γ proves surjectivity and connectedness of the kernel. The induced action on $\Lambda/L_\Lambda \simeq \mathbb{Z}$ is trivial, so the target is the product circle bundle. Since the kernel has odd real dimension, this is not a holomorphic quotient.

The quotient metric and the four-copy limit.

The first column of equation (29) is

$$P_T^{-\mathsf{T}} e_\gamma = \left(0, \frac{m_T}{D_T}, 0, -\frac{L_T}{D_T}\right)^{\mathsf{T}}. \tag{109}$$

The quotient metric on $\mathbb{R}/\mathbb{Z}$ is therefore $\ell_T^2 d\theta^2$. Equivalently,

$$P_T(L_\Lambda \otimes \mathbb{R}) = \mathrm{span}_{\mathbb{R}}\{e_1, e_3, v_{u,T}\}$$

has unit normal

$$\nu_T = \frac{(0, m_T, 0, -L_T)}{\sqrt{L_T^2 + m_T^2}},$$

and

$$\mathrm{dist}\big(v_{\gamma,T}, P_T(L_\Lambda \otimes \mathbb{R})\big) = |\langle v_{\gamma,T}, \nu_T\rangle| = \frac{L_T q_T + 6m_T^2}{\sqrt{L_T^2 + m_T^2}} = \ell_T.$$

The expansions in equations (23) and (25) give

$$D_T = T^2 + (2H - B)T + O(1), \qquad \sqrt{L_T^2 + m_T^2} = T + H + O(T^{-1}),$$

which proves equation (90).

After rescaling the circle coordinate by ℓ_T, the product target is $\mathbb{R}^4/\ell_T\mathbb{Z}^4$. If $x, y \in B_R(0)$ and $0 \neq \lambda \in \ell_T\mathbb{Z}^4$, then

$$\|x - y + \lambda\| \geq \|\lambda\| - \|x - y\| > \ell_T - 2R > 2R \geq \|x - y\|$$

when $\ell_T > 4R$. Hence the covering projection is an exact isometry on $B_R(0)$, proving all three convergences in equation (94). The quotient map is global over B°; only the metric limit is evaluated along the radial cusp. The separately chosen Euclidean fibre metric is not claimed to be a canonical global metric away from that cusp.

Minimality and the finite-base-cover obstruction.

Let $N \subset \Lambda^{\oplus r}$ be a primitive invariant kernel whose connected quotient has radial systole tending to infinity. Every element of $K_\Lambda^{\oplus r}$ has uniformly bounded physical length, since

$$P_T \widehat{w} = e_1, \qquad P_T \widehat{\delta} = e_3.$$

If such an element survived in the quotient lattice, its quotient norm would be no greater than its source norm and would give a nonzero uniformly bounded period. Therefore $K_\Lambda^{\oplus r} \subset N$. Invariance and equation (107) then force $L_\Lambda^{\oplus r} \subset N$, and

$$\operatorname{rank}(\Lambda^{\oplus r}/N) \leq 4r - 3r = r.$$

This proves the lower bound $r \geq 4$, while $N = L_\Lambda^{\oplus 4}$ proves sharpness.

For the finite-cover assertion, put

$$H_* = A_2 A_1 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 6 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ -5 & 0 & -1 & 1 \end{pmatrix} = I + N_*. \tag{110}$$

Direct multiplication gives $N_*^2 = 0$, and hence

$$H_*^n \widehat{w} = \widehat{w} + n(\widehat{u} - \widehat{\delta}), \qquad H_*^n \widehat{\delta} = \widehat{\delta}. \tag{111}$$

Thus $H_*^n K_\Lambda \neq K_\Lambda$ for every $n \neq 0$. Every finite-index subgroup of the base fundamental group contains a positive power of the loop with monodromy H_*. Such a subgroup cannot preserve K_Λ, proving that no finite base cover globalizes the native two-torus quotient.

The affine finite fillings and the actual overlap maps.

Equation equation (108) gives $\varepsilon_\gamma A_j = \varepsilon_\gamma$, so equation (97) follows. Define

$$\overline{\Phi}_j(s, \theta) = \left(e^{-2\pi i/m_j} s, \theta + \frac{\kappa_j}{m_j} \right).$$

Then $(s, x) \mapsto (s, \varepsilon_\gamma(x))$ is $(\Phi_j, \overline{\Phi}_j)$-equivariant; denote the descended map by $\overline{\chi}_j : N_j \to C_j$. A nonidentity power which fixes the base must lie over $s = 0$, where it translates the circle by $k\kappa_j/m_j$. Since $\gcd(\kappa_j, m_j) = 1$, this is nonintegral for $0 < k < m_j$. The target action is free; the source action is free by theorem 10.9.

For an explicit solid-torus chart, put $\lambda_j = e^{2\pi i/m_j}$, $u = e^{2\pi i\theta}$, and choose $r_j \in \mathbb{Z}$ with $r_j \kappa_j \equiv 1 \pmod{m_j}$. Then

$$w = s u^{r_j}, \qquad U = u^{m_j}$$

are invariant and identify C_j real-analytically with a solid torus. In these coordinates

$$t_j = s^{m_j} = w^{m_j} U^{-r_j},$$

which proves the central multiplicity. Equivariance and the primitive kernel give the proper $\mathbb{T}^3$-bundle $N_j \to C_j$.

It remains to check the actual logarithmic gluing. Let $\Pi_{\mathbb{R}}(z) : \Lambda \otimes \mathbb{R} \to \mathbb{C}^2$ be the underlying real isomorphism. On a logarithm branch ℓ, define $b_j^\ell(z)$ uniquely by

$$\Pi_{\mathbb{R}}(z) b_j^\ell(z) = \frac{\ell(z)}{2\pi i} \Pi(z) v_j, \qquad \beta_j(z) = \varepsilon_\gamma(b_j^\ell(z)) \bmod \mathbb{Z}. \tag{112}$$

Changing ℓ by $2\pi i n$ adds nv_j to b_j^ℓ, hence adds $n\kappa_j \in \mathbb{Z}$ to its ε_γ-value. Thus β_j is branch independent. In flat coordinates the source conjugacy (2.9e) is translation by b_j^ℓ. Applying ε_γ to that conjugacy gives

$$\beta_j(g_j z) = \beta_j(z) - \frac{\kappa_j}{m_j} \pmod{\mathbb{Z}}. \tag{113}$$

Therefore

$$h_j^{S^1}(z,\theta) = (z, \theta + \beta_j(z))$$

conjugates $\overline{\Phi}_j$ to the linear target action $(z,\theta) \mapsto (g_j z, \theta)$, and the overlap identity is

$$\chi_\gamma \circ h_j = h_j^{S^1} \circ \overline{\chi}_j. \tag{114}$$

The diagonal action on $(J_j')^{\times_{\Delta_j} 4}$ is free, and the same coprimality argument proves that $C_j^{[4]}$ is smooth. Every fibre of the resulting map is a $\mathbb{T}^{12}$-torsor; over D_j^* its structure torus is the fourfold fibre product of $\mathcal{H}$. No group zero is asserted at the affine filling. In local etale coordinates along the central fibre, the ordinary filled fibre product has equations

$$s_i^{m_j} = s_1^{m_j}, \qquad i = 2,3,4. \tag{115}$$

It has m_j^3 smooth branches along the reduced central fibre. The simultaneous-root model separates them, is smooth and hence normal, and is an isomorphism away from the central fibre. The map ν_j is proper: both spaces are proper over D_j, and the inverse image of a compact subset of the target is closed in the inverse image of its compact image in D_j. It is quasi-finite, since its fibres are singletons away from the central fibre and have at most m_j^3 points over the central fibre. Hence ν_j is finite. The universal property of normalization now proves that it is the normalization of the reduced ordinary fibre product. On the central fibre the relative root choices form $(\mathbb{Z}/m_j)^4/\operatorname{diag}(\mathbb{Z}/m_j)$, of order m_j^3, which proves the asserted degrees.

On D_j^*, the source identification is $(h_j)^{\times 4}$, and equation (114) identifies the target overlap with $(h_j^{S^1})^{\times 4}$. These are precisely the fourfolds of the one-copy overlap maps and therefore satisfy the cocycle identities. They glue the two local normalizations and Seifert targets to $J^{[4]}$ and to $B^\circ \times (\mathbb{R}/\mathbb{Z})^4$, respectively. Equivalently, the source gluing follows from uniqueness of normalization.

The fixed product-target trivialization does not extend over X, even if base winding is allowed. Indeed, in the presentation of theorem 53.6, let a homomorphism to $\mathbb{Z}$ restrict to $n\varepsilon_\gamma$ on Λ and take values a, b on $\widehat{\rho}_1, \widehat{\rho}_2$. The three attachment relations give

$$3a = n, \qquad 4b = -n, \qquad a + b = 0.$$

The first two give $a = n/3$, $b = -n/4$, while the third gives $n/12 = 0$. Hence $n = 0$. This is the exact global obstruction behind the fact that the two Seifert targets do not combine into a fixed-circle map on X.

The explicit toric cusp obstruction.

On the positive radial ray, write

$$Z_T = iTB_0 + C(t_c), \qquad B_0 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad t_c = e^{-2\pi T}.$$

For $a = (a_1, a_2)^{\mathsf{T}} \in \mathbb{R}^2$, take the point of the regular fibre represented by $Z_T a$, with zero short-period coordinates, and exponentiate it to the dense torus of the cusp model:

$$x_T(a) = \exp(2\pi i Z_T a) \in (\mathbb{C}^*)^2.$$

Its logarithmic position is

$$y_T(a) = \frac{(\log|x_{T,1}(a)|, \log|x_{T,2}(a)|)}{\log|t_c|} = B_0 a + \frac{\operatorname{Im} C(t_c)a}{T}. \tag{116}$$

The circle character has the constant value

$$\chi_{\gamma,T}(x_T(a)) = a_1 \bmod \mathbb{Z}. \tag{117}$$

Use the affine toric chart of $\sigma = \operatorname{conv}(0, e_1, e_2)$. Its coordinates are

$$z_0 = \frac{t_c}{x_1 x_2}, \qquad z_1 = x_1, \qquad z_2 = x_2. \tag{118}$$

If $B_0 a = (y_1, y_2)$ is in the interior of σ, then $y_1 > 0$, $y_2 > 0$, and $1 - y_1 - y_2 > 0$. For all sufficiently large T, the three absolute values in equation (118) are

$$|t_c|^{1-y_{T,1}-y_{T,2}}, \qquad |t_c|^{y_{T,1}}, \qquad |t_c|^{y_{T,2}},$$

and all tend to zero. Every such path converges to the same toric fixed point and hence to the same triple point of W.

Take

$$a = \left(-\frac{1}{4}, \frac{1}{4}\right), \qquad a' = \left(-\frac{1}{2}, \frac{1}{4}\right).$$

Then

$$B_0 a = \left(\frac{1}{4}, \frac{1}{4}\right), \qquad B_0 a' = \left(\frac{1}{4}, \frac{1}{2}\right)$$

are both in the interior of σ. The paths converge to the same point of W, whereas

$$\chi_{\gamma,T}(x_T(a)) = \frac{3}{4}, \qquad \chi_{\gamma,T}(x_T(a')) = \frac{1}{2}.$$

No continuous extension of the literal map χ_γ exists. Choosing identical marked radial paths in the other three factors and using these two paths in one factor proves the same assertion for $\chi_\gamma^{[4]}$.

The radial family of translations by $\widehat{u}/2$ in $\mathcal{H}$ fails to extend as well. Take two paths whose limiting logarithmic positions are

$$y = \left(\frac{1}{8}, \frac{1}{8}\right), \qquad y' = \left(\frac{3}{8}, \frac{3}{8}\right).$$

Both approach the fixed point of $\operatorname{conv}(0, e_1, e_2)$. Translation by $\widehat{u}/2 \in \mathcal{H}_T$ adds the exact displacement

$$\frac{1}{2} B_0 e_2 + \frac{1}{2T} \operatorname{Im} C(t_c) e_2 \longrightarrow \left(\frac{1}{2}, 0\right)$$

to logarithmic position. The translated limits are

$$\left(\frac{5}{8}, \frac{1}{8}\right) \quad \text{and} \quad \left(\frac{7}{8}, \frac{3}{8}\right).$$

The first lies in the interior of $\operatorname{conv}(0, e_1, e_2)$, while the second lies in the interior of $\operatorname{conv}(e_1, e_2, e_1 + e_2)$. These two triangle types are the two translation orbits in the A_2 triangulation and give the two distinct triple points of W, as proved in theorem 10.6. Thus one regular-fibre translation sends two paths with the same limit to paths with different limits; it has no continuous extension to N_0.

This pointwise obstruction must not be confused with the cohomology class. By theorem 53.6,

$$\pi_1(N_0) = \Lambda / K_\Lambda,$$

and ε_γ kills K_Λ. It therefore defines an integral H^1-class on N_0, hence a map to the Eilenberg–Mac Lane space S^1 whose restriction is homotopic to χ_γ. The explicit paths prove that no such representative agrees with χ_γ on the whole punctured cusp. The phase torus $(K_\Lambda \otimes \mathbb{R})/K_\Lambda$ extends as the compact real torus action of the toric model, generally with boundary stabilizers; the full radial $\widehat{u}/2$-translation and the full quotient do not.

Products and smash quotients.

The integral Kunneth theorem gives

$$H_k(X^n;\mathbb{Z}) \cong \begin{cases} \mathbb{Z}^{\binom{n}{j}}, & k = 6j, \quad 0 \le j \le n, \\ 0, & \text{otherwise}, \end{cases} \qquad \chi(X^n) = 2^n,$$

which proves equations (102) and (103). The synchronized regular n-fold fibre product has dimension

$$6n - 2(n-1) = 4n + 2,$$

which gives 18 for four copies.

A product map factors through a smash quotient if and only if it is constant on the corresponding fat wedge. On $F_T \times \{0\}$, the local two-copy quotient is

$$x \longmapsto (\pi_T(x), 0),$$

which is nonconstant. Thus the literal conclusion is that $\pi_T \times \pi_T$ does not factor through $F_T \wedge F_T$; it would be a type error to call it a map on $X \wedge X$. The restriction of $\chi_{\gamma,T}^{[4]}$ to $F_T \times \{0\}^3$ is similarly

$$x \longmapsto (\chi_{\gamma,T}(x), 0, 0, 0),$$

so it does not factor through $F_T^{\wedge 4}$.

There is no global extension which repairs the domain mismatch: $H^1(X^n;\mathbb{Z}) = 0$, whereas both local quotient maps pull nonzero target H^1-classes back to nonzero classes on the cusp fibre. Hence no map on X^2 or X^4 has the required restriction, and precomposition with the smash collapses in equation (104) cannot produce the local quotient maps.

Identify the metric quotient circle with $C_T = \mathbb{R}/\ell_T\mathbb{Z}$ by $\theta \mapsto \ell_T\theta$. For equation (105), target-smash collapse is constant whenever one source coordinate is the identity, so the modified map does factor through $F_T^{\wedge 4}$. Its chosen base point has distance $\ell_T/2$ from the collapsed target fat wedge. If two points lie within distance R of it, a route through the collapsed point has length at least $\ell_T - 2R$, while their direct Euclidean distance is at most $2R$. When $\ell_T > 4R$, neither the collapse nor circle wrap-around shortens distances in that ball.

Let $u_i \in H^1(F_T^4;\mathbb{Z})$ be the pullback of the generator from the i-th quotient circle. The orientation class of $S^4 = C_T^{\wedge 4}$ pulls back under equation (105) to

$$\pm u_1 \smile u_2 \smile u_3 \smile u_4 \neq 0.$$

But $H^4(X^4;\mathbb{Z}) = 0$. Thus even this modified S^4-target map does not extend to X^4, and a fortiori does not arise from a map on $X^{\wedge 4} \cong S^{24}$.

For completeness, Borel–Serre prove that S^{2n} admits no almost-complex structure for $n \ge 4$. Their argument applies to every differentiable structure on the sphere: on S^{2n}, all lower Chern classes vanish, while their reduced-power calculation forces the top Chern class to vanish modulo a suitable odd prime $p < n$; on the other hand that class is the Euler class, namely $2h$ for a generator $h \in H^{2n}(S^{2n};\mathbb{Z})$, a contradiction. Taking $n = 6$ and $n = 12$ gives the asserted obstructions for S^{12} and S^{24}. This topological obstruction does not itself prohibit the real four-circle target G_T, the target-smash S^4, or their metric limits.

Finally, $X * X \cong S^{13}$ and $X^{*4} \cong S^{27}$, so joins do not produce the proposed S^{12} or S^{24} dimensions. Suspensions can produce spheres of those dimensions only by discarding the torus-fibration maps. □

11.1 The Kummer cover extends across the toric cusp

This section determines exactly what is true in the assertion that the degree-N^2 cover of the regular fibres extends by a toric power map. The answer is positive for the completed *local* cusp family, but the source action has to be changed, and two different degrees must not be conflated.

Retain the cusp data

$$B_0 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad Z(s) = sB_0 + C(t), \qquad t = e^{2\pi i s},$$

where C is holomorphic on a disc Δ_t about zero. Put

$$c_{\bar\lambda}(t) := \exp\bigl(2\pi i C(t)\bar\lambda\bigr) \in (\mathbb{C}^*)^2, \qquad \bar\lambda \in \bar\Lambda \simeq \mathbb{Z}^2.$$

Let Σ be the fan in $N' = \mathbb{Z}^2 \oplus \mathbb{Z}$ consisting of the cones over the cells of the translation-invariant A_2 triangulation at height one, and write $Y = Y_\Sigma$. On the dense torus, with coordinates (x_1, x_2, t), the target action is

$$\Psi_{\bar\lambda}(x,t) = \bigl(c_{\bar\lambda}(t) t^{B_0\bar\lambda} x, t\bigr). \tag{119}$$

Its quotient is the completed cusp family $f_0 \colon N_0 = Y/\bar\Lambda \to \Delta_t$.

Fix an integer $N \geq 1$. Let v be a coordinate on a second disc and make the base change

$$\rho_N \colon \Delta_v \longrightarrow \Delta_t, \qquad \rho_N(v) = v^N, \qquad t = v^N. \tag{120}$$

On a copy $Y^{(N)}$ of the same toric variety, with dense-torus coordinates (ξ_1, ξ_2, v), define

$$C_N(v) := \frac{1}{N} C(v^N), \qquad c^{(N)}_{\bar\lambda}(v) := \exp\bigl(2\pi i C_N(v)\bar\lambda\bigr) \tag{121}$$

and

$$\Psi^{(N)}_{\bar\lambda}(\xi, v) := \bigl(c^{(N)}_{\bar\lambda}(v) v^{B_0\bar\lambda}\xi, v\bigr). \tag{122}$$

Theorem 11.8 (Exact Kummer extension through the cusp)**.** ***[VERIFIED]*** *After shrinking the v-disc, the following statements hold.*

1. *The maps in* (122) *extend to a free, properly discontinuous action on $Y^{(N)}$. The quotient*

$$f_N \colon N_0^{\langle N\rangle} := Y^{(N)}/\bar\Lambda \longrightarrow \Delta_v$$

is a proper holomorphic map from a smooth threefold. Its central fibre W_N is reduced, irreducible, and nonnormal, and its normalization is a toric del Pezzo surface of degree six.

2. *Multiplication by N on the cocharacter lattice defines a finite flat toric morphism*

$$P_N \colon Y^{(N)} \longrightarrow Y, \qquad P_N(\xi_1, \xi_2, v) = (\xi_1^N, \xi_2^N, v^N), \tag{123}$$

of degree N^3. It is equivariant for (122) *and* (119)*, hence descends to a finite flat map*

$$\kappa_N \colon N_0^{\langle N\rangle} \longrightarrow N_0, \qquad f_0 \circ \kappa_N = \rho_N \circ f_N, \tag{124}$$

of degree N^3.

3. *The map κ_N is etale over $v \neq 0$, and*

$$\kappa_N^* W = N W_N, \qquad \mathrm{Ram}(\kappa_N) = (N-1) W_N. \tag{125}$$

In particular, the branch support of the map to the original cusp family is the whole central fibre.

4. *Let*

$$N_0[N] := N_0 \times_{\Delta_t} \Delta_v, \qquad t = v^N. \tag{126}$$

Then $N_0[N]$ is normal and κ_N factors uniquely as

$$N_0^{\langle N\rangle} \xrightarrow{Q_N} N_0[N] \longrightarrow N_0. \tag{127}$$

The map Q_N has generic degree N^2. For $N > 1$ it is quasi-etale but not flat: it is etale over the smooth locus of $N_0[N]$, and its non-etale and nonflat loci are exactly the inverse images of the double and triple strata of the base-changed central fibre. Thus Q_N is not a normalization map.

5. *On the reduced central fibres, the induced map $W_N \to W$ has generic degree N^2. It lifts to a commutative diagram*

$$\begin{array}{ccc} E_N \simeq \mathrm{dP}_6 & \xrightarrow{[N]} & E \simeq \mathrm{dP}_6 \\ {\scriptstyle \nu_N}\big\downarrow & & \big\downarrow{\scriptstyle \nu} \\ W_N & \longrightarrow & W, \end{array} \tag{128}$$

where $[N]$ is toric multiplication by N, of degree N^2, ramified with index N along each of the six toric boundary curves. The diagram respects the paired conductor curves and the two triple-point classes.

6. *Over $v \neq 0$, Q_N is exactly the cover defined by*

$$\mathbb{Z}\langle \widehat{\gamma}, \widehat{u}, N\widehat{w}, N\widehat{\delta}\rangle \subset \mathbb{Z}\langle \widehat{\gamma}, \widehat{u}, \widehat{w}, \widehat{\delta}\rangle.$$

Its deck group is $(\mathbb{Z}/N\mathbb{Z})^2$. The pulled-back cusp monodromy is M_0^N, and in the displayed cover basis it is M_0:

$$S_N^{-1} M_0^N S_N = M_0, \qquad S_N = \mathrm{diag}(1, 1, N, N). \tag{129}$$

Proof. We prove the assertions in the order in which the constructions occur.

The source action. For $\bar{\lambda} \in \bar{\Lambda}$, the integral linear map

$$\varphi_{\bar{\lambda}}(y, y_3) = (y + y_3 B_0 \bar{\lambda}, y_3)$$

translates the height-one slice by the integral vector $B_0\bar{\lambda}$. It preserves the A_2 triangulation and hence induces a toric automorphism $\Phi_{\bar{\lambda}}$ of $Y^{(N)}$. On the dense torus this automorphism is

$$\Phi_{\bar{\lambda}}(\xi, v) = (v^{B_0\bar{\lambda}}\xi, v).$$

Multiplication by the nowhere-zero holomorphic function $c_{\bar{\lambda}}^{(N)}(v)$ extends across every toric chart. Since $C_N(v)\bar{\lambda}$ is additive in $\bar{\lambda}$, the resulting maps satisfy $\Psi_{\bar{\lambda}}^{(N)}\Psi_{\bar{\mu}}^{(N)} = \Psi_{\bar{\lambda}+\bar{\mu}}^{(N)}$.

For completeness, freeness and proper discontinuity can be checked directly. Put

$$R_N(v) := -2\pi \,\mathrm{Im}\, C_N(v).$$

It is bounded at $v = 0$. At a dense-torus point fixed by $\bar{\lambda}$, logarithmic absolute values give

$$\log|v| B_0\bar{\lambda} + R_N(v)\bar{\lambda} = 0.$$

Choose the disc so small that $|\log|v|| > 2\|R_N\|_\infty$. Because B_0 is orthogonal, the last equality is impossible for $\bar{\lambda} \neq 0$. On the central fibre, $\Psi_{\bar{\lambda}}^{(N)}$ translates every bounded cell by the nonzero vector $B_0\bar{\lambda}$, so it fixes no toric orbit.

For a dense-torus point set

$$y(\xi, v) = \frac{(\log|\xi_1|, \log|\xi_2|)}{\log|v|}, \qquad B_{N,v} = B_0 + \frac{R_N(v)}{\log|v|}.$$

Then

$$y(\Psi_{\bar\lambda}^{(N)}(\xi, v)) = y(\xi, v) + B_{N,v}\bar\lambda, \qquad |B_{N,v}\bar\lambda| \geq \tfrac{1}{2}|\bar\lambda|.$$

The logarithmic-position regions associated with any two fixed toric charts are bounded. Hence only finitely many $\bar\lambda$ can translate one of them to meet the other. Local finiteness of the fan reduces the same assertion for compact sets to finitely many chart pairs, proving proper discontinuity. Finally, reducing $B_{N,v}^{-1} y(\xi, v)$ modulo $\mathbb{Z}^2$ places every orbit meeting a closed subdisc in finitely many closed unit toric charts. Their union is compact. This proves properness of f_N. The central-fibre and normalization assertions follow exactly from the height-one fan: locally v is the product of the coordinates belonging to the rays through a cell, every component occurs with multiplicity one, the lattice action is transitive on vertices, and the star of a vertex is the complete smooth hexagonal fan.

The toric map and equivariance. Multiplication by N on N' sends every cone $\sigma \in \Sigma$ onto itself as a real cone. It therefore defines the toric morphism P_N. For a maximal cone, let z_0, z_1, z_2 be the target characters dual to its three primitive height-one ray generators, and let w_0, w_1, w_2 be the corresponding source characters. The map on convergent local rings is

$$\mathbb{C}\{z_0, z_1, z_2\} \longrightarrow \mathbb{C}\{w_0, w_1, w_2\}, \qquad z_i \longmapsto w_i^N. \tag{130}$$

The target-chart preimage is the corresponding source chart, and the right-hand ring is free of rank N^3 over the left-hand ring, with basis

$$\{w_0^{a_0} w_1^{a_1} w_2^{a_2} : 0 \leq a_i < N\}.$$

Thus P_N is finite flat of degree N^3 on every maximal chart, hence globally.

The choice (121) is forced by equivariance. Indeed, writing powers coordinatewise,

$$P_N \Psi_{\bar\lambda}^{(N)}(\xi, v) = \big((c_{\bar\lambda}^{(N)}(v))^N v^{NB_0\bar\lambda} \xi^N, v^N\big) \tag{131}$$

$$= \big(c_{\bar\lambda}(v^N)(v^N)^{B_0\bar\lambda}\xi^N, v^N\big) \tag{132}$$

$$= \Psi_{\bar\lambda} P_N(\xi, v). \tag{133}$$

Consequently P_N descends to (124). To see finiteness and flatness after the infinite-lattice quotient without any implicit descent theorem, take an evenly covered open set in N_0 and one of its lifts $V \subset Y$. Equivariance gives a unique representative of every source orbit above that open set whose image lies in V; two such representatives cannot differ by a nontrivial lattice translate. The descended map is therefore locally identified with $P_N^{-1}(V) \to V$. It is finite flat of degree N^3 by (130).

In the same chart the base characters are

$$t = z_0 z_1 z_2, \qquad v = w_0 w_1 w_2, \qquad P_N^* t = v^N.$$

The Jacobian determinant of (130) is

$$N^3 (w_0 w_1 w_2)^{N-1} = N^3 v^{N-1}.$$

This proves (125); the local quotient maps are biholomorphisms, so the calculation is unchanged downstairs.

The relative degree-N^2 factor. The equality $(f_0 \circ \kappa_N)([\xi, v]) = v^N = (\rho_N \circ f_N)([\xi, v])$ gives the factorization (127). At a triple point the exact local-ring map is

$$\begin{aligned} A_N &:= \frac{\mathbb{C}\{z_0, z_1, z_2, v\}}{(z_0 z_1 z_2 - v^N)}, \qquad B := \mathbb{C}\{w_0, w_1, w_2\}, \\ A_N &\longrightarrow B, \qquad z_i \longmapsto w_i^N, \qquad v \longmapsto w_0 w_1 w_2. \end{aligned} \tag{134}$$

Let

$$G_N = \{(\varepsilon_0, \varepsilon_1, \varepsilon_2) \in \mu_N^3 : \varepsilon_0\varepsilon_1\varepsilon_2 = 1\} \simeq \mu_N^2$$

act diagonally on B. A monomial $w_0^{a_0}w_1^{a_1}w_2^{a_2}$ is G_N-invariant if and only if

$$a_0 \equiv a_1 \equiv a_2 \pmod{N}.$$

Writing their common residue as r, with $0 \le r < N$, expresses that monomial uniquely in the form

$$(w_0w_1w_2)^r(w_0^N)^{b_0}(w_1^N)^{b_1}(w_2^N)^{b_2}.$$

The same coefficientwise argument applies to convergent power series. Hence

$$A_N = B^{G_N}. \tag{135}$$

It follows that A_N is normal, that the map has generic degree $|G_N| = N^2$, and that it is a finite quotient rather than a normalization. The corresponding face-chart calculations prove normality along the double and smooth strata as well, so $N_0[N]$ is normal everywhere.

The stabilizer in G_N is trivial when none or exactly one of the w_i vanishes, is isomorphic to μ_N when exactly two vanish, and is all of G_N when all three vanish. Thus Q_N is etale in codimension one and has no ramification divisor, but it is not etale above the double and triple strata. It is also nonflat at every such point when $N > 1$. At a triple point the closed fibre of (134) has length

$$\dim_{\mathbb{C}} \frac{\mathbb{C}\{w_0, w_1, w_2\}}{(w_0^N, w_1^N, w_2^N, w_0w_1w_2)} = N^3 - (N-1)^3 \tag{136}$$

$$= 3N^2 - 3N + 1, \tag{137}$$

which exceeds the generic rank N^2. At a point of a double stratum, localizing at the nonzero third coordinate and choosing its holomorphic Nth root reduces the ramified factor to

$$\frac{\mathbb{C}\{z_0, z_1, v\}}{(z_0z_1 - v^N)} \longrightarrow \mathbb{C}\{w_0, w_1\}, \qquad z_i \longmapsto w_i^N, \quad v \longmapsto w_0w_1.$$

Its local generic rank is N, whereas its closed fibre has length

$$\dim_{\mathbb{C}} \frac{\mathbb{C}\{w_0, w_1\}}{(w_0^N, w_1^N, w_0w_1)} = 2N - 1 > N. \tag{138}$$

The remaining transverse nonzero torus coordinate is etale locally. Equations (137) and (138), together with etaleness elsewhere, identify the nonflat locus exactly.

The regular fibres and monodromy. Choose a logarithm r with $v = e^{2\pi ir}$. The target period matrix at $t = v^N$ is

$$Z(Nr) = NrB_0 + C(v^N),$$

whereas the source residual action is determined by

$$Z_N(r) = rB_0 + \frac{1}{N}C(v^N) = \frac{1}{N}Z(Nr).$$

The source fibre is therefore

$$\frac{\mathbb{C}^2}{\mathbb{Z}^2 + Z_N(r)\mathbb{Z}^2},$$

and multiplication by N on $\mathbb{C}^2$ maps it to the target fibre. Equivalently, in the unscaled additive coordinate it is the identity map

$$\frac{\mathbb{C}^2}{N\mathbb{Z}^2 + Z(Nr)\mathbb{Z}^2} \longrightarrow \frac{\mathbb{C}^2}{\mathbb{Z}^2 + Z(Nr)\mathbb{Z}^2}. \tag{139}$$

The imaginary part of $Z(Nr)$ is invertible on a regular fibre, so its period lattice meets $\mathbb{Z}^2$ trivially in the required real sense; the index in (139) is N^2. Its deck group is $(\mathbb{Z}/N\mathbb{Z})^2$. In the ordered lattice basis $(\widehat{\gamma}, \widehat{u}, \widehat{w}, \widehat{\delta})$, this is exactly the sublattice obtained by S_N. Since $M_0 = I + R$ with $R^2 = 0$ and $R(a, b, c, d) = (0, 0, b, -a)$, one has

$$M_0^N = I + NR, \qquad S_N^{-1} M_0^N S_N = M_0,$$

proving (129).

The central normalization and conductor. Multiplication by N fixes every ray of Σ. It therefore maps a chosen source component of the toric central fibre to the corresponding target component. The star of that ray is the hexagonal fan, so the restriction is the toric power map $[N]\colon \mathrm{dP}_6 \to \mathrm{dP}_6$. On its dense two-torus it has degree N^2; in each toric boundary chart it is an Nth-power map in the normal coordinate, hence has ramification index N along all six boundary curves. Equation (133) says that the source and target lattice identifications commute with this map. Consequently opposite boundary curves, their conductor identifications, and the two translation-orbits of triangle vertices descend compatibly. This proves (128) and completes the proof. □

Remark 11.9 (Why the unmodified action does not work). If one reuses the target action with v in place of t at the source, then the two sides of equivariance have multiplicative factors

$$c_{\bar{\lambda}}(v)^N v^{NB_0\bar{\lambda}} \quad \text{and} \quad c_{\bar{\lambda}}(v^N) v^{NB_0\bar{\lambda}},$$

respectively. In general $c_{\bar{\lambda}}(v)^N \neq c_{\bar{\lambda}}(v^N)$. Thus the bare formula $(\xi_1, \xi_2, v) \mapsto (\xi_1^N, \xi_2^N, v^N)$ extends on the ambient toric variety but does not descend through the manuscript's quotient unless the source factor (121) is inserted. The explicit logarithm $2\pi i C(v^N)\bar{\lambda}/N$ makes this root holomorphic; no unjustified choice of an Nth root is involved.

Remark 11.10 (Exact scope). The theorem constructs a finite morphism of completed families over the local cusp discs. It is a degree-N^3 map to the original family and a degree-N^2 map only after the target is base changed. It is neither a degree-N^2 self-map over the unchanged disc nor the normalization of the base change.

Nothing here supplies a cover over the entire base B. In fact the displayed sublattice is not preserved by the original full monodromy when $N > 1$: for example, the printed matrix A_1 sends $\widehat{u}$ to a vector whose $\widehat{w}$- and $\widehat{\delta}$-coefficients are -1 and 1. A further global base cover and compatible extensions through both finite fillings would have to be constructed separately. The local Kummer morphism also supplies no map to a product sphere, smash sphere, S^{12}, or S^{24}.

11.2 Cone-compactified sphere carriers for the native cusp quotient

This subsection answers a topological question which is strictly different from the finite-cover construction. No period is enlarged by passing to a cover. Instead, the native expanding quotient is retained on the cusp, the parts on which that quotient is unavailable are collapsed, and the resulting based map is passed through a smash product. The outcome is an exact global map with source S^{12}, and there is an analogous S^{24} map from the full-monodromy-invariant circle quotient. The targets are conically stratified spaces. The Euclidean limits below use the intrinsic metrics inherited from the flat quotient tori on the regular slices; they are not ambient metric limits in one fixed metric on the compact targets.

Fix a closed coordinate disc

$$D_0 = \{t \in \mathbb{C} : |t| \leq \epsilon\} \subset B, \qquad p_0 = \{t = 0\}, \qquad N_0 = f^{-1}(D_0), \qquad W = f^{-1}(0), \tag{140}$$

small enough that it contains no singular value other than p_0. Put

$$K_\Lambda = \mathbb{Z}\langle \widehat{w}, \widehat{\delta} \rangle, \qquad \overline{\Lambda} = \Lambda / K_\Lambda, \qquad \overline{T}^2 = \frac{\overline{\Lambda} \otimes \mathbb{R}}{\overline{\Lambda}}. \tag{141}$$

The ordered basis of $\overline{\Lambda}$ is the image of $(\widehat{\gamma}, \widehat{u})$.

Lemma 11.11 (The native quotient over the whole punctured cusp)**.** *There is a well-defined real-analytic fibrewise group epimorphism*

$$\overline{\pi} : N_0 \setminus W \longrightarrow D_0^* \times \overline{T}^2 \tag{142}$$

whose kernel is the connected two-torus family defined by K_Λ. The target local system is trivial. On the positive radial fibre F_T, $t = e^{-2\pi T}$, the isomorphism

$$\iota_T : \overline{T}^2 \longrightarrow Q_T, \qquad [(a,b)] \longmapsto [Y_T(a,b)^{\mathsf{T}}] \tag{143}$$

satisfies

$$\pi_T = \iota_T \circ \overline{\pi}|_{F_T}. \tag{144}$$

If $\overline{T}^2$ is given the pullback by ι_T of the Euclidean quotient metric on $Q_T = \mathbb{R}^2/Y_T\mathbb{Z}^2$, then ι_T is an isometry and

$$\operatorname{sys}(\overline{T}^2, g_T) = \operatorname{sys}(Q_T) \geq T - C. \tag{145}$$

Proof. On the universal cover of D_0^*, use the real-analytic flat marking

$$(z, x + \Lambda) \longmapsto [z, \Pi(z)x].$$

The deck generator acts by

$$(z, x + \Lambda) \longmapsto (g_0 z, M_0 x + \Lambda).$$

The displayed matrix M_0 gives

$$(M_0 - I)(a,b,c,d)^{\mathsf{T}} = (0,0,b,-a)^{\mathsf{T}}, \qquad \operatorname{im}(M_0 - I) = K_\Lambda. \tag{146}$$

Consequently M_0 preserves K_Λ and induces the identity on $\overline{\Lambda}$. The formula

$$[z, \Pi(z)x] \longmapsto ([z], [x \bmod K_\Lambda])$$

is invariant under the deck action and proves equation (142). The lattice K_Λ is primitive, so the kernel is connected. On the radial fibre, projection to the (e_2, e_4)-coordinates sends $a\widehat{\gamma} + b\widehat{u}$ to $Y_T(a,b)^{\mathsf{T}}$ and kills K_Λ. This proves equations (143) and (144). The metric and systolic assertion is the quotient calculation in theorem 11.6. □

Define

$$\mathcal{A} = \frac{[0,\epsilon] \times S^1 \times \overline{T}^2}{\{0\} \times S^1 \times \overline{T}^2 \sim v_0, \quad \{\epsilon\} \times S^1 \times \overline{T}^2 \sim v_1}. \tag{147}$$

Thus

$$\mathcal{A} \cong \operatorname{Susp}(S^1 \times \overline{T}^2) \cong \operatorname{Susp}(T^3), \tag{148}$$

where Susp denotes the unreduced suspension. The extra circle is the angular coordinate of the punctured base. If one deliberately forgets that angle, the evident map $\mathcal{A} \to \operatorname{Susp}(\overline{T}^2)$ gives a smaller target with the same restriction to every fixed radial fibre. We retain $\mathcal{A}$ because it records the entire punctured-cusp quotient.

Theorem 11.12 (Exact localized S^{12} route)**.** *Choose $x_0 \in W$, and base $\mathcal{A}$ at v_0. There is a continuous surjective based quotient map*

$$q_2 : (X, x_0) \longrightarrow (\mathcal{A}, v_0) \tag{149}$$

with the following exact description. For $x \in N_0 \setminus W$, write

$$t(f(x)) = re^{i\phi}, \qquad 0 < r < \epsilon,$$

and let $\overline{\pi}_f(x) \in \overline{T}^2$ be the fibre component of $\overline{\pi}(x)$. Then

$$q_2(x) = [r, e^{i\phi}, \overline{\pi}_f(x)], \qquad q_2(W) = v_0, \qquad q_2\big(X \setminus \operatorname{int} N_0\big) = v_1. \tag{150}$$

On $f^{-1}(|t| = \epsilon)$, the value is v_1.

The smash square is a global based map

$$q_2^{\wedge 2} : X \wedge X \longrightarrow \mathcal{A} \wedge \mathcal{A}. \tag{151}$$

Using $X \cong_{\mathrm{diff}} S^6$, a based choice of the diffeomorphism identifies its source with

$$X \wedge X \cong S^6 \wedge S^6 \cong S^{12}. \tag{152}$$

For $0 < e^{-2\pi T} < \epsilon$, the canonical product-to-smash quotient maps restrict to embeddings on F_T^2 and on the corresponding interior target-product cross-section. Under those embeddings and ι_T^2, the induced smash map restricts exactly to

$$\pi_T \times \pi_T : F_T \times F_T \longrightarrow Q_T \times Q_T. \tag{153}$$

With the Euclidean ℓ^2-product flat quotient metric,

$$\operatorname{sys}(Q_T \times Q_T) \geq T - C, \qquad (Q_T \times Q_T, (0,0)) \longrightarrow (\mathbb{R}^4, 0) \tag{154}$$

in pointed local-isometric, pointed smooth, and pointed Gromov–Hausdorff topology. Once $T - C > 4R$, the radius-R ball is exactly Euclidean.

The integral homology of the targets is

$$\widetilde{H}_k(\mathcal{A}; \mathbb{Z}) \cong \begin{cases} \mathbb{Z}^3, & k = 2, 3, \\ \mathbb{Z}, & k = 4, \\ 0, & \textit{otherwise}, \end{cases} \tag{155}$$

and

$$\widetilde{H}_k(\mathcal{A} \wedge \mathcal{A}; \mathbb{Z}) \cong \begin{cases} \mathbb{Z}^9, & k = 4, \\ \mathbb{Z}^{18}, & k = 5, \\ \mathbb{Z}^{15}, & k = 6, \\ \mathbb{Z}^6, & k = 7, \\ \mathbb{Z}, & k = 8, \\ 0, & \textit{otherwise}. \end{cases} \tag{156}$$

In particular, the target is not S^4, S^{12}, or a compact four-manifold.

Proof. The equivalence relation in equation (147) is closed: it is the union of the diagonal with the squares of the two closed end cross-sections. A compact Hausdorff space modulo a closed equivalence relation is compact Hausdorff. Rescaling r to the unit interval gives exactly the unreduced suspension in equation (148).

Well-definedness of the interior formula follows from theorem 11.11. A full turn in ϕ changes the marking by M_0, but M_0 is the identity modulo K_Λ.

We check both collapsed strata. If U is an open neighborhood of v_0, its inverse image in the cylinder is open and contains the compact set $\{0\} \times S^1 \times \overline{T}^2$. Hence, for some $\delta > 0$, it contains $[0, \delta) \times S^1 \times \overline{T}^2$. Therefore $q_2^{-1}(U)$ contains $f^{-1}(|t| < \delta)$, a neighborhood of W. The same compactness argument at the other end shows that every neighborhood of v_1 contains the image of $(\epsilon - \delta, \epsilon] \times S^1 \times \overline{T}^2$. Its preimage together with $X \setminus \operatorname{int} N_0$ is a neighborhood of the outer boundary. Continuity on the interior is equation (142); the closed pasting lemma gives continuity on X.

Every interior point has a preimage because $\overline{\pi}$ is fibrewise surjective. The nonempty sets W and $X \setminus \operatorname{int} N_0$ map onto v_0 and v_1, respectively. Thus q_2 is surjective. A continuous surjection from compact X to Hausdorff $\mathcal{A}$ is a quotient map.

The map is based, so functoriality gives equation (151). The standard CW quotient gives $S^m \wedge S^n \cong S^{m+n}$ [10, Chapter 0, Smash Product], proving equation (152).

The base point lies in W, so F_T^2 is disjoint from the source fat wedge. Its image lies in an interior cross-section of $\mathcal{A}^2$, disjoint from the target fat wedge. The source and target fat wedges are closed in the compact Hausdorff products X^2 and $\mathcal{A}^2$, so both smash quotients are Hausdorff. Their restrictions to the relevant compact subspaces are continuous and one-to-one, and are therefore embeddings. Equations equations (144) and (150) give equation (153).

The period lattice of the product target is $\operatorname{diag}(Y_T, Y_T)\mathbb{Z}^4$. Its systole is the systole of $Y_T\mathbb{Z}^2$, proving the first assertion in equation (154). If $x, y \in B_R(0) \subset \mathbb{R}^4$ and $0 \neq \lambda$ is a product period, then

$$\|x - y + \lambda\| \geq \|\lambda\| - \|x - y\| \geq (T - C) - 2R > 2R \geq \|x - y\|$$

when $T - C > 4R$. The quotient projection is an exact isometry on the Euclidean radius-R ball. Conversely, any point of quotient distance less than R has a shortest lattice representative of norm less than R, so this embedded ball is the entire quotient radius-R ball. This proves all three pointed convergences.

Reduced suspension shifts reduced homology by one. Since

$$H_1(T^3) = \mathbb{Z}^3, \qquad H_2(T^3) = \mathbb{Z}^3, \qquad H_3(T^3) = \mathbb{Z},$$

equation (155) follows. All groups are free, so the Kunneth theorem for smash products has no Tor terms. The ranks in degrees $4, 5, 6, 7, 8$ are

$$3 \cdot 3, \quad 2(3 \cdot 3), \quad 2(3 \cdot 1) + 3 \cdot 3, \quad 2(3 \cdot 1), \quad 1,$$

which proves equation (156). □

The localized one-circle construction. The full-monodromy-invariant character also gives a smaller cusp-supported target. Define

$$\mathcal{B} = \frac{[0, \epsilon] \times S^1 \times S^1}{\{0\} \times S^1 \times S^1 \sim b_0, \quad \{\epsilon\} \times S^1 \times S^1 \sim b_1} \cong \operatorname{Susp}(T^2), \tag{157}$$

where the two circle coordinates in an interior cross-section are the base angle and the value of θ_γ.

Proposition 11.13 (Localized S^{24} route with target $\operatorname{Susp}(T^2)^{\wedge 4}$)**.** *There is a continuous surjective based quotient map*

$$q_{1,\mathrm{cusp}} : (X, x_0) \longrightarrow (\mathcal{B}, b_0) \tag{158}$$

given by

$$q_{1,\mathrm{cusp}}(x) = [r, e^{i\phi}, \theta_\gamma(x)] \quad (0 < |t| = r < \epsilon), \qquad q_{1,\mathrm{cusp}}(W) = b_0, \qquad q_{1,\mathrm{cusp}}(X \setminus \operatorname{int} N_0) = b_1. \tag{159}$$

Its fourth smash power has source $X^{\wedge 4} \cong S^{24}$, and on four synchronized radial fibres it is exactly

$$\chi_{\gamma,T}^{\times 4} : F_T^4 \longrightarrow C_T^4. \tag{160}$$

Consequently its intrinsic target slices satisfy the Euclidean limit equation (172)*. The target homology is*

$$\widetilde{H}_k(\mathcal{B}; \mathbb{Z}) \cong \begin{cases} \mathbb{Z}^2, & k = 2, \\ \mathbb{Z}, & k = 3, \\ 0, & \textit{otherwise}, \end{cases} \tag{161}$$

and

$$\widetilde{H}_k(\mathcal{B}^{\wedge 4};\mathbb{Z}) \cong \begin{cases} \mathbb{Z}^{16}, & k=8, \\ \mathbb{Z}^{32}, & k=9, \\ \mathbb{Z}^{24}, & k=10, \\ \mathbb{Z}^{8}, & k=11, \\ \mathbb{Z}, & k=12, \\ 0, & \textit{otherwise.} \end{cases} \tag{162}$$

Proof. The monodromy invariance $\varepsilon_\gamma M_0 = \varepsilon_\gamma$ makes the interior formula single-valued after a turn in the base angle. The two endpoint-continuity arguments are identical to those in the proof of theorem 11.12, with compact cross-section T^2 in place of T^3. Fibrewise surjectivity of θ_γ, together with the nonempty inner fibre and exterior, proves surjectivity. Compact-to-Hausdorff makes the map a quotient map.

Since $x_0 \in W$, the fourth smash power exists. The radial regular fibres and their circle targets avoid the two closed fat wedges. The restrictions of the Hausdorff smash quotients to these compact subspaces are continuous and one-to-one, hence embeddings, and the fibre map is exactly equation (160). The intrinsic metric statement is therefore the four-circle calculation equation (172). Finally, suspension shifts the free homology of T^2, giving equation (161). In the fourth smash power, choosing j of the four factors in degree 3 and the other $4-j$ factors in degree 2 gives degree $8+j$ and rank $\binom{4}{j}2^{4-j}$. This is equation (162). □

The full-base circle target before coning. Choose pairwise disjoint closed coordinate discs D_0, D_1, D_2 about the three singular values and put

$$P = B \setminus \bigcup_{j=0}^{2} \operatorname{int} D_j. \tag{163}$$

For $j = 1, 2$, let C_j be the solid-torus Seifert target in equation (98), and let $h_j^{S^1}$ be the target overlap map in equation (114). Glue

$$\mathcal{M} = (P \times S^1) \underset{h_1^{S^1}}{\cup} C_1 \underset{h_2^{S^1}}{\cup} C_2. \tag{164}$$

The gluing is along boundary tori over $\partial D_1, \partial D_2$. The remaining boundary is

$$\partial\mathcal{M} = \partial D_0 \times S^1 \cong T^2. \tag{165}$$

Define

$$\mathcal{C} = \mathcal{M} \cup_{\partial\mathcal{M}} \operatorname{Cone}(\partial\mathcal{M}), \tag{166}$$

and write c_0 for the cone vertex. This changes the codomain at the cusp: the whole boundary two-torus, including the target circle, is collapsed to c_0. It therefore does not contradict the proved nonextension of the literal map χ_γ to the fixed product target $B \times S^1$.

Theorem 11.14 (Exact cone-compactified S^{24} route)**.** *There is a continuous surjective based quotient map*

$$q_1 : (X, x_0) \longrightarrow (\mathcal{C}, c_0), \qquad x_0 \in W, \tag{167}$$

defined as follows:

1. *over P, it is $(f, \theta_\gamma) : f^{-1}(P) \to P \times S^1$, equivalently the restriction of χ_γ to $f^{-1}(P)$;*
2. *over D_j, $j = 1, 2$, it is the finite-filling map $N_j \to C_j$ from equation* (98)*;*
3. *on $N_0 \setminus W$, writing $t = re^{i\phi}$, it is*

$$x \longmapsto [r, (e^{i\phi}, \theta_\gamma(x))] \in \operatorname{Cone}(\partial D_0 \times S^1), \tag{168}$$

and every point of W maps to c_0.

The fourth smash power is a global based map

$$q_1^{\wedge 4} : X^{\wedge 4} \longrightarrow \mathcal{C}^{\wedge 4}. \tag{169}$$

Using $X \cong_{\mathrm{diff}} S^6$*, its source is*

$$X^{\wedge 4} \cong (S^6)^{\wedge 4} \cong S^{24}. \tag{170}$$

On four synchronized radial fibres, the canonical product-to-smash quotient maps restrict to embeddings on the source fibre product and the interior target-product cross-section. Under those embeddings, equation (169) *restricts exactly to*

$$\chi_{\gamma,T}^{\times 4} : F_T^4 \longrightarrow C_T^4, \qquad C_T = \mathbb{R}/\ell_T\mathbb{Z}. \tag{171}$$

With its Euclidean ℓ^2*-product flat quotient metric,*

$$\mathrm{sys}(C_T^4) = \ell_T = T + H - B + O(T^{-1}), \qquad (C_T^4, 0) \longrightarrow (\mathbb{R}^4, 0) \tag{172}$$

in pointed local-isometric, pointed smooth, and pointed Gromov–Hausdorff topology.

The target homology is

$$\widetilde{H}_k(\mathcal{C};\mathbb{Z}) \cong \begin{cases} \mathbb{Z}, & k = 2, 3, \\ 0, & \textit{otherwise}, \end{cases} \tag{173}$$

and

$$\widetilde{H}_k(\mathcal{C}^{\wedge 4};\mathbb{Z}) \cong \begin{cases} \mathbb{Z}, & k = 8, 12, \\ \mathbb{Z}^4, & k = 9, 11, \\ \mathbb{Z}^6, & k = 10, \\ 0, & \textit{otherwise}. \end{cases} \tag{174}$$

Proof. The compact spaces $P \times S^1, C_1, C_2$ are glued along closed boundary tori by homeomorphisms, producing a compact orientable three-manifold $\mathcal{M}$ with boundary $\partial D_0 \times S^1$. Attaching the cone along this closed boundary gives a compact Hausdorff finite CW space $\mathcal{C}$.

The overlap identity

$$\chi_\gamma \circ h_j = h_j^{S^1} \circ \overline{\chi}_j$$

proves that the first two pieces of q_1 agree. On ∂D_0, the cone base is glued to the remaining boundary of $P \times S^1$, so the first and third pieces agree. If U is a neighborhood of c_0, its inverse image in the cone contains $[0, \delta) \times T^2$ for some $\delta > 0$, by compactness of the cone base. Therefore $q_1^{-1}(U)$ contains $f^{-1}(|t| < \delta)$, a neighborhood of W. This proves continuity at the new stratum; continuity elsewhere follows from the input maps and the closed pasting lemma.

Every regular circle quotient and both finite Seifert targets are surjective, and W maps onto the cone vertex. Thus q_1 is a continuous surjection from compact X to Hausdorff $\mathcal{C}$, hence a quotient map. It is based, so its fourth smash power exists. Iterating $S^m \wedge S^n \cong S^{m+n}$ gives equation (170).

The base point lies in W, while each radial regular fibre is disjoint from W. Hence F_T^4 and its product-circle image avoid the closed source and target fat wedges. The canonical quotient maps from the source product and the indicated interior target-product cross-section to their smash products restrict to embeddings. Under those embeddings, the induced smash map restricts to $\chi_{\gamma,T}^{\times 4}$, which itself retains its displayed nontrivial torus kernel. The quotient-metric calculation in equation (90) gives ℓ_T. The product lattice is $\ell_T\mathbb{Z}^4$, so the radius-R argument, including the shortest-representative proof that the embedded Euclidean ball is the whole quotient ball, in the proof of theorem 11.12 proves equation (172).

For homology, let ρ_0, ρ_1, ρ_2 be the boundary loops of the pair of pants, with $\rho_0 + \rho_1 + \rho_2 = 0$, and let η be the regular target-circle class. With the boundary-loop and circle orientations used in the finite-filling presentation, the meridian relations are

$$3\rho_1 - \eta = 0, \qquad 4\rho_2 + \eta = 0. \tag{175}$$

Reversing the circle orientation reverses both displayed η-signs and changes none of the following groups. Thus

$$H_1(\mathcal{M};\mathbb{Z}) = \frac{\mathbb{Z}\langle \eta, \rho_1, \rho_2\rangle}{\langle 3\rho_1 - \eta,\ 4\rho_2 + \eta\rangle} \cong \mathbb{Z}\langle g\rangle, \tag{176}$$

where

$$\rho_0 = -g, \qquad \rho_1 = 4g, \qquad \rho_2 = -3g, \qquad \eta = 12g. \tag{177}$$

In particular, $H_1(\partial\mathcal{M}) = \mathbb{Z}\langle\rho_0,\eta\rangle \to H_1(\mathcal{M})$ is surjective. The orientable three-manifold $\mathcal{M}$ has one torus boundary, $H_3(\mathcal{M}) = 0$, and $\chi(\mathcal{M}) = \frac{1}{2}\chi(\partial\mathcal{M}) = 0$; hence $H_2(\mathcal{M};\mathbb{Q}) = 0$. To obtain the integral statement without suppressing torsion, Poincare–Lefschetz duality and the universal coefficient theorem give

$$H_2(\mathcal{M};\mathbb{Z}) \cong H^1(\mathcal{M},\partial\mathcal{M};\mathbb{Z}) \cong \mathrm{Hom}(H_1(\mathcal{M},\partial\mathcal{M};\mathbb{Z}),\mathbb{Z}),$$

because $H_0(\mathcal{M},\partial\mathcal{M};\mathbb{Z}) = 0$. Thus $H_2(\mathcal{M};\mathbb{Z})$ is free; its rank is zero, so it vanishes.

The cone attachment is the homotopy cofiber of $\partial\mathcal{M} \hookrightarrow \mathcal{M}$. Since the boundary inclusion is a cofibration,

$$\widetilde{H}_k(\mathcal{C};\mathbb{Z}) \cong H_k(\mathcal{M},\partial\mathcal{M};\mathbb{Z}).$$

Poincare–Lefschetz duality and $H^0(\mathcal{M}) = H^1(\mathcal{M}) = \mathbb{Z}$, $H^2(\mathcal{M}) = H^3(\mathcal{M}) = 0$, give equation (173). The homology is free, so the smash Kunneth theorem says that four factors of degree 2 or 3 produce ranks $\binom{4}{j}$ in degree $8+j$, proving equation (174). □

Definition 11.15 (The three radial target-slice embeddings)**.** Put $r_T = e^{-2\pi T}$ and let

$$\vartheta_T : \mathbb{R}/\mathbb{Z} \longrightarrow C_T = \mathbb{R}/\ell_T\mathbb{Z}, \qquad [a] \longmapsto [\ell_T a].$$

For $y_1, y_2 \in Q_T$, define

$$j_T(y_1,y_2) = \left[[r_T, 1, \iota_T^{-1}(y_1)], [r_T, 1, \iota_T^{-1}(y_2)]\right]_\wedge \in \mathcal{A}^{\wedge 2}.$$

For $c_1,\dots,c_4 \in C_T$, define

$$k_T^{\mathcal{B}}(c_1,\dots,c_4) = \left[[r_T, 1, \vartheta_T^{-1}(c_1)], \dots, [r_T, 1, \vartheta_T^{-1}(c_4)]\right]_\wedge \in \mathcal{B}^{\wedge 4},$$

and, using the cusp-cone piece of $\mathcal{C}$,

$$k_T^{\mathcal{C}}(c_1,\dots,c_4) = \bigwedge_{j=1}^{4} [r_T, (1, \vartheta_T^{-1}(c_j))] \in \mathcal{C}^{\wedge 4}.$$

These are embeddings because every displayed factor lies away from the smash basepoint, so equality in the smash quotient is equality in every factor; compact-to-Hausdorff injectivity then gives a topological embedding.

Proposition 11.16 (Intrinsic expansion versus ambient collapse)**.** *Let D_{12}, $D_{24}^{\mathcal{B}}$, and $D_{24}^{\mathcal{C}}$ be arbitrary fixed metrics compatible with the topologies of $\mathcal{A}^{\wedge 2}$, $\mathcal{B}^{\wedge 4}$, and $\mathcal{C}^{\wedge 4}$, respectively. In their corresponding ambient metrics, the radial target slices satisfy*

$$\mathrm{diam}_{\mathrm{amb}}\big(j_T(Q_T^2)\big) \longrightarrow 0, \qquad \mathrm{diam}_{\mathrm{amb}}\big(k_T^{\mathcal{B}}(C_T^4)\big) \longrightarrow 0, \qquad \mathrm{diam}_{\mathrm{amb}}\big(k_T^{\mathcal{C}}(C_T^4)\big) \longrightarrow 0. \tag{178}$$

By contrast, in the intrinsic quotient metrics,

$$\mathrm{diam}(Q_T^2) \geq \tfrac{1}{2}(T - C) \longrightarrow \infty, \qquad \mathrm{diam}(C_T^4) \geq \tfrac{1}{2}\ell_T \longrightarrow \infty. \tag{179}$$

Thus equations (154) *and* (172) *assert intrinsic, flat pointed limits of varying slices, not Sormani–Wenger intrinsic-flat convergence and not an ambient pointed $\mathbb{R}^4$ limit inside one fixed compact metric target.*

Proof. Every neighborhood of the relevant cone vertex contains every sufficiently small radial cross-section, uniformly in all compact angular and torus coordinates. In a compatible metric, every such cross-section eventually lies in an arbitrarily small ball about the vertex. Products followed by the smash quotient preserve this uniform convergence, proving equation (178).

For a flat torus with systole s, choose a shortest primitive period λ. The class of $\lambda/2$ has distance $s/2$ from zero: for every period μ, the nonzero period $\lambda - 2\mu$ has length at least s. Hence the diameter is at least $s/2$. Equations equations (145) and (172) prove equation (179). □

Proposition 11.17 (Exact almost-complex obstruction on S^{12})**.** *Let M be a smooth manifold homeomorphic to S^{12}. There is no bundle endomorphism $J : TM \to TM$ satisfying $J^2 = -\operatorname{Id}_{TM}$. This conclusion has two independent proofs which use only the homotopy type of M: the original Borel–Serre reduced-power calculation at the single prime 5, and Bott integrality. For the standard embedded sphere there is also an independent stable-tangent Pontryagin-class contradiction.*

These statements concern an unstabilized complex structure on the actual rank-twelve tangent bundle. A stable complex structure on $TS^{12} \oplus \varepsilon^r_{\mathbb{R}}$, a complex structure confined to an open subset or to fibres of a map, or an endomorphism which ceases to be a bundle endomorphism along a singular set is not an almost-complex structure on TS^{12} and therefore lies outside the hypothesis.

Proof. Assume that J exists and put $E = (TM, J)$, a complex vector bundle of rank six. Orient M by the interleaved complex frame

$$(v_1, Jv_1, \dots, v_6, Jv_6),$$

write $[M]_J$ for the resulting fundamental class, and choose $u_J \in H^{12}(M;\mathbb{Z})$ by $\langle u_J, [M]_J \rangle = 1$. Since M has the integral cohomology of S^{12},

$$c_1(E) = c_2(E) = c_3(E) = c_4(E) = c_5(E) = 0. \tag{180}$$

The top-Chern/Euler identity uses exactly the orientation just fixed, so Poincare–Hopf gives

$$c_6(E) = e(E_{\mathbb{R}}) = e(TM) = 2u_J. \tag{181}$$

Thus reversing a previously chosen orientation can change the displayed generator but cannot change the coefficient two.

We first specialize the original calculation of Borel and Serre [63, §12, Proposition 12.4, Lemma 12.5, and Proposition 12.6, p. 430; §15, Proposition 15.1, p. 434]. They write C_{2j} for the universal class now denoted c_j. For $j = 6$, the admissible prime is exactly

$$p = 5 < 6, \qquad 5 \nmid 6,$$

and their identity in $H^*(BU(6);\mathbb{F}_5)$ reads

$$\bar{c}_6 = 6^{-1}\mathcal{P}^1_5(\bar{c}_2) + Q(\bar{c}_1, \dots, \bar{c}_5). \tag{182}$$

Here $6^{-1} = 1$ in $\mathbb{F}_5$. To retain the exact coefficient rather than suppressing the lower-class polynomial, pass to the splitting space, write $x_1, \dots, x_6$ for the degree-two Chern roots, put $s_r = \sum_i x_i^r$, and use $\mathcal{P}^1_5(x_i) = x_i^5$ and the Cartan formula. The resulting universal identity is

$$\begin{aligned}\mathcal{P}^1_5(\bar{c}_2) &= \sum_{i<j}(x_i^5 x_j + x_i x_j^5) = s_5 s_1 - s_6 \\ &= 6\bar{c}_6 + \bar{c}_1^4\bar{c}_2 - \bar{c}_1^3\bar{c}_3 - 4\bar{c}_1^2\bar{c}_2^2 + \bar{c}_1^2\bar{c}_4 + 7\bar{c}_1\bar{c}_2\bar{c}_3 - \bar{c}_1\bar{c}_5 \\ &\quad + 2\bar{c}_2^3 - 6\bar{c}_2\bar{c}_4 - 3\bar{c}_3^2.\end{aligned} \tag{183}$$

Consequently, for the lower-Chern ideal $I = (\bar{c}_1, \dots, \bar{c}_5)$,

$$\mathcal{P}^1_5(\bar{c}_2) \equiv 6\bar{c}_6 = \bar{c}_6 \pmod{I}, \tag{184}$$

which verifies the exact top-class coefficient in equation (182). Pulling back along the classifying map of E, using equation (180) and naturality of $\mathcal{P}_5^1$, gives

$$\overline{c_6(E)} = \mathcal{P}_5^1(\overline{c_2(E)}) = \mathcal{P}_5^1(0) = 0.$$

But equation (181) gives $\overline{c_6(E)} = 2\overline{u}_J \neq 0$ in $H^{12}(M;\mathbb{F}_5)$, the first contradiction.

For the second contradiction, choose an orientation-preserving homotopy equivalence $f : S^{12} \to M$. Bott integrality says that the sixth Chern class of every complex bundle on S^{12} is divisible by $(6-1)! = 120$ [61, Chapter 20, §9, Theorem 9.6 and Corollary 9.8, p. 308]; see also [62, §6]. The coefficient can be checked without normalization ambiguity: Newton's sixth identity and equation (180) give

$$\mathrm{ch}_6(f^*E) = \frac{s_6(f^*E)}{6!} = -\frac{c_6(f^*E)}{5!}.$$

The integral Bott generator therefore forces $c_6(f^*E) = 120k f^*u_J$ for some $k \in \mathbb{Z}$, whereas equation (181) gives $c_6(f^*E) = 2f^*u_J$. Thus $120k = 2$, which has no integral solution.

Finally take $M = S^{12}$ with its standard smooth embedding in $\mathbb{R}^{13}$. There is an explicit stable trivialization

$$TS^{12} \oplus \varepsilon^1_{\mathbb{R}} \xrightarrow{\cong} \varepsilon^{13}_{\mathbb{R}}, \qquad (x; v, a) \longmapsto (x; v + ax), \tag{185}$$

so $p_3(TS^{12}) = 0$. On the other hand,

$$E_{\mathbb{R}} \otimes_{\mathbb{R}} \mathbb{C} \cong E \oplus \overline{E}, \qquad p_3(E_{\mathbb{R}}) = -c_6(E \oplus \overline{E}).$$

Equation equation (180) and $c_6(\overline{E}) = c_6(E)$ give the exact relation

$$p_3(TS^{12}) = -2c_6(E) = -4u_J, \tag{186}$$

contradicting stable triviality in the torsion-free group $H^{12}(S^{12};\mathbb{Z})$. This is the Pontryagin–Euler proof recorded independently in [62, Lemma 4.1 and Theorem 4.2].

Every contradiction above requires a globally defined J on the actual bundle TM. Stabilizing that bundle, deleting points, restricting to regular fibres, or allowing J to fail at a singular locus removes that hypothesis rather than producing an exception to it. □

Theorem 11.18 (Sharp obstructions to stronger interpretations)**.** *The sphere routes above have the following exact limitations.*

1. *Let $r \geq 1$, and let $N \subset \Lambda^{\oplus r}$ be a primitive kernel defining a connected fibrewise torus quotient over the full regular base, or over any finite cover of that base. Equip the quotient with the norm induced by the unrescaled marked Euclidean metric on the original fibres. If its quotient systole tends to infinity on the radial cusp, then*

$$\mathrm{rank}(\Lambda^{\oplus r}/N) \leq r. \tag{187}$$

   Thus two copies cannot give a full-base four-torus quotient. The S^{12} map in theorem 11.12 evades the obstruction only by localizing to the cusp and collapsing the exterior.

2. *No nontrivial finite base cover, including one extending with branching over the original singular values, makes K_Λ a sub-local system over the full regular base. The local cyclic change $t = s^M$ used for the auxiliary degree-M^2 cover cannot extend as a connected cover of $B \cong \mathbb{P}^1$ branched only at p_0.*

3. *There is no map from X^2, $X \wedge X$, X^4, or $X^{\wedge 4}$ to a fixed T^4 whose restriction retains all four target character classes of the displayed product quotient. There is no map to a fixed S^4 whose restriction is a degree-one collapse of Q^2_T, or of the four-circle target, for even one regular fibre. The conical targets evade these obstructions because the fibre character and orientation classes do not extend as global target classes.*

4. *Cartesian products do not produce the named spheres:*

$$X^2 \simeq S^6 \times S^6 \not\simeq S^{12}, \qquad X^4 \simeq (S^6)^4 \not\simeq S^{24}. \tag{188}$$

The two- and four-fold synchronized fibre products have real dimensions 10 *and* 18*, respectively, and generally have singular completed fibres. Joins give*

$$X * X \cong S^{13}, \qquad X^{*4} \cong S^{27}. \tag{189}$$

5. *Since $X \cong S^6$ is simply connected, every connected unbranched finite cover of X, X^r, $X \wedge X \cong S^{12}$, or $X^{\wedge 4} \cong S^{24}$ is trivial. The local degree-M^2 unspooling cover is not the restriction of a nontrivial connected unbranched cover of any of these compact sources. Branched or normalized compactifications are different objects and are not constructed here.*

6. *By theorem 11.17, no smooth manifold homeomorphic to S^{12} admits an almost-complex structure. Nor does S^{24}, with any differentiable structure, by the Borel–Serre theorem [63, §15, Proposition 15.1 and the following remark, p. 434]. Thus the smash sources cannot be promoted to compact complex spheres.*

Proof. Every element of $K_\Lambda^{\oplus r}$ has uniformly bounded physical length on the radial cusp. If one survived in the quotient lattice, it would be a nonzero period of bounded quotient norm. Hence

$$K_\Lambda^{\oplus r} \subset N. \tag{190}$$

For the full monodromy group, $A_1 \widehat{w} = \widehat{u} - \widehat{w}$ and invariance force every copy of $\widehat{u}$ into N. Therefore

$$L_\Lambda^{\oplus r} = \mathbb{Z}\langle \widehat{u}, \widehat{w}, \widehat{\delta} \rangle^{\oplus r} \subset N,$$

which proves equation (187).

The same conclusion holds after a finite base cover. Let H be its finite-index subgroup and put $H_* = A_2 A_1 = I + N_*$, $N_*^2 = 0$. Every finite-index subgroup contains H_*^n for some $n > 0$, while

$$H_*^n \widehat{w} = \widehat{w} + n(\widehat{u} - \widehat{\delta}). \tag{191}$$

Equations equations (190) and (191) give $n\widehat{u} \in N$. Primitivity gives $\widehat{u} \in N$, proving the same rank bound. The orbit also gives $H_*^n K_\Lambda \neq K_\Lambda$ for every $n > 0$, so no finite cover of the regular base preserves the native kernel. A compact cover branched over the singular values restricts to such a cover. Finally, a connected finite cover of $\mathbb{P}^1$ branched only at p_0 would restrict to a connected unbranched finite cover of $\mathbb{P}^1 \setminus \{p_0\} \cong \mathbb{C}$, hence has degree one.

For a fixed T^4 target, $H^1(X^r; \mathbb{Z}) = 0$, as does the first cohomology of the smash spheres. A global map therefore pulls every target character class back to zero. On a regular fibre the quotient homomorphisms inject the target character lattice into the character lattice of the source torus, giving four independent nonzero pullbacks, a contradiction.

For the fixed S^4 obstruction, let $\omega_T \in H^2(Q_T; \mathbb{Z})$ be the orientation class. Since $\pi_T^* : H^1(Q_T) \to H^1(F_T)$ injects two independent classes and torus cohomology is their exterior algebra,

$$(\pi_T \times \pi_T)^*(\omega_T \times \omega_T) \neq 0 \quad \text{in } H^4(F_T^2; \mathbb{Z}). \tag{192}$$

A degree-one collapse $Q_T^2 \to S^4$ pulls the orientation generator back to $\pm(\omega_T \times \omega_T)$. But

$$H^4(X^2; \mathbb{Z}) = H^4(S^{12}; \mathbb{Z}) = 0,$$

so no global extension has this restriction. The four-circle case is the same: its orientation class pulls back to the exterior product of four independent circle characters, while $H^4(X^4) = H^4(S^{24}) = 0$.

The Kunneth theorem gives $H_6(X^2) = \mathbb{Z}^2$, whereas $H_6(S^{12}) = 0$, and gives nonzero groups of ranks $4, 6, 4$ in degrees $6, 12, 18$ for X^4, whereas S^{24} has none. The real dimension of an r-fold synchronized fibre product is $6r - 2(r-1) = 4r + 2$, giving 10 and 18. The join formula $S^m * S^n \cong S^{m+n+1}$ proves equation (189). The remaining assertions follow from simple connectivity, theorem 11.17, and the cited Borel–Serre theorem for S^{24}. □

Canonicity and exact scope. The lattice $K_\Lambda = \ker(M_0 - I) = \mathrm{im}(M_0 - I)$ and the punctured-cusp quotient are intrinsic to the marked cusp monodromy. The compact map q_2 additionally depends on the chosen cusp coordinate, closed cusp disc, and the decision to collapse its outer boundary and the exterior. The identifications $X^{\wedge 2} \cong S^{12}$ and $X^{\wedge 4} \cong S^{24}$ depend on a based choice of $X \cong S^6$. None of these choices is claimed to be holomorphic or canonical in the algebraic category.

The map $q_2^{\wedge 2}$ is the direct two-copy construction from the native $T^2 \times \mathbb{R}^2$ cusp: it is globally defined on the smash sphere, and its radial regular-fibre quotients are $Q_T \times Q_T$, which converge pointed local-isometrically, smoothly, and Gromov–Hausdorff to $\mathbb{R}^4$. It is not a torus quotient over the full regular base. The map $q_{1,\mathrm{cusp}}^{\wedge 4}$ is the parallel cusp-supported construction with target $\mathrm{Susp}(T^2)^{\wedge 4}$. The stronger map $q_1^{\wedge 4}$ uses the quotient over every regular and finite-filled fibre before the cusp cone is attached; four copies are minimal for that full-base all-expanding group quotient. Neither construction gives a compact complex S^{12} or S^{24}, an ambient Euclidean limit in a fixed compact metric target, or a claim about the interacting Yang–Mills spectrum.

The argument does not rule out a separately constructed branched or toroidal compactification of the auxiliary finite covers after additional branching away from p_0. Such a construction would have to specify its global base cover, normalize and resolve every pulled-back singular fibre, track the lifted metric, and determine the topology of the resulting total space. No such smooth compact sphere construction follows from the present data.

11.2.1 Optimal sphere exponents and the exact Euclidean boundary

Within the class of native connected fibrewise torus quotients equipped with the induced, unscaled quotient of the marked Euclidean metric, the two sphere dimensions in the preceding constructions are forced by two different integral quotient problems. Cusp-locally, each copy has an expanding quotient of rank two, so two copies are necessary and sufficient within that class. Over the full regular base, monodromy leaves at most one expanding quotient direction per copy, so four copies are necessary and sufficient within the same class. We now prove this scoped optimality, identify the compact conical carriers, and construct a noncompact comparison manifold in which the limiting $\mathbb{R}^4$ is an actual boundary fibre. There is only a continuous comparison map from this manifold to the compact conical carrier: it set-theoretically collapses the new boundary, is a homeomorphism away from that boundary, and is nonproper and nonquotient.

Throughout this subsubsection,

$$B_R^Z(z) := \{w \in Z : d_Z(z,w) \le R\}$$

denotes the closed radius-R ball; when the ambient space and center are clear, superscripts and centers are suppressed.

Theorem 11.19 (Optimality within the native connected torus-quotient class)**.** *Consider only native connected fibrewise torus quotients of copies of the original marked fibres, and equip every such quotient with the induced, unscaled Euclidean quotient metric.*

1. *For a native connected torus quotient of r copies over the punctured cusp, if the quotient systole tends to infinity on the positive radial ray, then its lattice rank is at most $2r$. A rank-four quotient therefore requires $r \ge 2$. Equality is attained for $r = 2$ by*

$$(\Lambda/K_\Lambda)^{\oplus 2}, \qquad K_\Lambda = \mathbb{Z}\langle \widehat{w}, \widehat{\delta} \rangle,$$

whose physical period lattice is

$$L_T^{(12)} = \mathrm{diag}(Y_T, Y_T)\mathbb{Z}^4.$$

The corresponding smash source is $X^{\wedge 2} \cong S^{12}$.

2. *For a native connected torus quotient of r copies over the full regular base, or over a finite cover of that base, radial divergence of the quotient systole forces its lattice rank to be at most r. A rank-four quotient therefore requires $r \geq 4$. Equality is attained by the four copies of the full-monodromy-invariant character χ_γ; on the radial fibre its period lattice is*

$$L_T^{(24)} = \ell_T \mathbb{Z}^4, \qquad \ell_T = T + H - B + O(T^{-1}),$$

and the corresponding smash source is $X^{\wedge 4} \cong S^{24}$.

Thus, among precisely these native connected torus quotients with their induced unscaled marked Euclidean metrics, S^{12} is the smallest smash sphere produced by the cusp-local rank-two quotient, and S^{24} is the smallest smash sphere produced by a quotient which survives the full regular-base monodromy. No optimality is asserted here among arbitrary continuous quotient maps or after a T-dependent rescaling of the metric.

Proof. For r copies, every element of

$$K_\Lambda^{\oplus r} = \mathbb{Z}\langle \widehat{w}, \widehat{\delta} \rangle^{\oplus r}$$

has uniformly bounded physical length on the radial cusp. Let $N \subset \Lambda^{\oplus r}$ be a primitive sublattice defining a connected torus quotient. If some $v \in K_\Lambda^{\oplus r}$ had nonzero class in $\Lambda^{\oplus r}/N$, that nonzero quotient period would have quotient norm at most the original norm of v, uniformly in T. The quotient systole could not diverge. Hence

$$K_\Lambda^{\oplus r} \subset N,$$

and

$$\operatorname{rank}(\Lambda^{\oplus r}/N) \leq 4r - 2r = 2r.$$

For $r = 2$, the primitive kernel $K_\Lambda^{\oplus 2}$ attains this bound. The cusp monodromy is the identity modulo K_Λ, so the quotient descends over the punctured cusp. Its radial lattice is exactly $\operatorname{diag}(Y_T, Y_T)\mathbb{Z}^4$, and

$$\operatorname{sys}(L_T^{(12)}) = \operatorname{sys}(Y_T \mathbb{Z}^2) \geq T - C \longrightarrow \infty.$$

For a quotient over the full regular base, monodromy invariance gives a stronger bound. Since $\widehat{w} \in N$ and

$$A_1 \widehat{w} = \widehat{u} - \widehat{w},$$

one has $\widehat{u} \in N$. Together with $\widehat{w}, \widehat{\delta} \in N$, this places the rank-three lattice

$$\mathbb{Z}\langle \widehat{u}, \widehat{w}, \widehat{\delta} \rangle^{\oplus r}$$

inside N, and hence

$$\operatorname{rank}(\Lambda^{\oplus r}/N) \leq r.$$

The same conclusion holds after a finite base cover. Indeed, if Γ_{mon} is the corresponding finite-index monodromy subgroup and $G_* = A_2 A_1 = I + N_*$, then $N_*^2 = 0$, and Γ_{mon} contains G_*^m for some $m > 0$. The exact orbit formula

$$G_*^m \widehat{w} = \widehat{w} + m(\widehat{u} - \widehat{\delta})$$

implies $m\widehat{u} \in N$; primitivity of N implies $\widehat{u} \in N$.

The invariant character χ_γ has precisely the complementary rank-one quotient in each copy. The full-base construction, including the two finite Seifert fillings and the cone at the unipotent cusp, proves that four copies attain rank four. Finally, after choosing the based diffeomorphism $X \cong_{\mathrm{diff}} S^6$ already reconstructed in this workbench, $X^{\wedge r} \cong (S^6)^{\wedge r} \cong S^{6r}$, which gives the two claimed sphere dimensions. □

Theorem 11.20 (Exact sphere-to-sphere carriers with moving four-spheres)**.** *There are continuous surjective based quotient maps*

$$\widehat{q}_3 : (X, x_0) \longrightarrow (S^3, *), \qquad \widehat{q}_\gamma : (X, x_0) \longrightarrow (S^2, *),$$

whose smash powers are continuous surjective quotient maps

$$\widehat{q}_3^{\wedge 2} : S^{12} \twoheadrightarrow S^6, \qquad \widehat{q}_\gamma^{\wedge 4} : S^{24} \twoheadrightarrow S^8. \tag{193}$$

Use the marked presentation $\overline{T}^2 = \mathbb{R}^2/\mathbb{Z}^2$ and put

$$K^{(1)} = ((\mathbb{R}/\mathbb{Z}) \times \{0\}) \cup (\{0\} \times (\mathbb{R}/\mathbb{Z})), \qquad c : \overline{T}^2 \longrightarrow \overline{T}^2/K^{(1)} \cong S^2.$$

Here c is the quotient collapsing the coordinate one-skeleton. For every sufficiently large T, put

$$\begin{aligned} r_T &= e^{-2\pi T}, \\ K_T^{(1)} &:= \iota_T(K^{(1)}) \subset Q_T, \\ S_T^2 &:= \overline{T}^2/K^{(1)} \cong Q_T/K_T^{(1)}, \\ c_T &:= c \circ \iota_T^{-1} : Q_T \longrightarrow S_T^2, \end{aligned}$$

where the identification of the two displayed quotients is induced by ι_T. Within this theorem and its proof, c_T denotes the displayed collapse map; the earlier scalar $c_T = \xi_T - \eta_T$ does not enter. At reduced-suspension cross-section height r_T, the first target contains a based topologically embedded four-sphere, equipped with its intrinsic quotient length metric,

$$\mathbb{S}_{12,T}^4 = S_T^2 \wedge S_T^2 \hookrightarrow S^6,$$

and the second target contains, at the same cross-section height, a based topologically embedded four-sphere, equipped with its intrinsic quotient length metric,

$$\mathbb{S}_{24,T}^4 = C_T^{\wedge 4} \hookrightarrow S^8.$$

On synchronized radial fibres, the exact restrictions factor as

$$F_T^2 \xrightarrow{\pi_T \times \pi_T} Q_T^2 \xrightarrow{c_T \times c_T} (S_T^2)^2 \xrightarrow{\rho_{12,T}} S_T^2 \wedge S_T^2 = \mathbb{S}_{12,T}^4, \tag{194}$$

$$F_T^4 \xrightarrow{\chi_{\gamma,T}^{\times 4}} C_T^4 \xrightarrow{\rho_{24,T}} C_T^{\wedge 4} = \mathbb{S}_{24,T}^4, \tag{195}$$

Here c_T collapses $K_T^{(1)}$, and $\rho_{12,T}$ and $\rho_{24,T}$ are the smash quotient maps. Thus the full collapsed locus in Q_T^2 is exactly

$$W_{12,T} = (K_T^{(1)} \times Q_T) \cup (Q_T \times K_T^{(1)}), \tag{196}$$

and the full collapsed locus in C_T^4 is exactly

$$W_{24,T} = \{(x_1, x_2, x_3, x_4) \in C_T^4 : x_j = 0 \pmod{\ell_T} \text{ for some } j\}. \tag{197}$$

The maps $\rho_{12,T} \circ (c_T \times c_T) : Q_T^2 \to S_T^2 \wedge S_T^2$ and $\rho_{24,T} : C_T^4 \to C_T^{\wedge 4}$ send exactly $W_{12,T}$ and $W_{24,T}$, respectively, to the smash basepoint and are injective on their complements. The full composites from F_T^2 and F_T^4 still identify every pair of points identified by $\pi_T \times \pi_T$ and $\chi_{\gamma,T}^{\times 4}$, respectively; in particular, their first arrows retain their nontrivial torus kernels. Their exact smash-basepoint preimages in the full radial sources are

$$(\pi_T^{-1}(K_T^{(1)}) \times F_T) \cup (F_T \times \pi_T^{-1}(K_T^{(1)}))$$

and

$$\{(x_1, x_2, x_3, x_4) \in F_T^4 : \chi_{\gamma,T}(x_j) = 0 \text{ in } C_T \text{ for some } j\},$$

respectively. Outside these basepoint preimages, two source points have the same final image if and only if their images under the corresponding first arrow in equations (194) *and* (195) *are equal.*

Choose the Euclidean Riemannian ℓ^2-product length metrics on Q_T^2 and C_T^4 before taking the smash quotients, and use the successive quotient length metrics specified in the proof. With the specified midpoint basepoints,

$$(\mathbb{S}^4_{12,T}, \zeta_{12,T}) \longrightarrow (\mathbb{R}^4, 0), \qquad (\mathbb{S}^4_{24,T}, \zeta_{24,T}) \longrightarrow (\mathbb{R}^4, 0) \tag{198}$$

in pointed local-isometric and pointed Gromov–Hausdorff topology. Moreover, the convergence is smooth on the eventually Euclidean basepoint neighborhoods. For every $R > 0$, once $T - C > 4R$, respectively once $\ell_T > 4R$, the corresponding quotient map restricts to an isometry from the radius-R ball in Q_T^2, respectively C_T^4, onto *the entire radius-R ball in the indicated smash sphere. The domain ball, and hence the quotient ball, is exactly a Euclidean four-ball.*

Proof. Write $\Sigma_r(Z, z_0)$ for the reduced suspension

$$\frac{[0,\epsilon] \times Z}{(\{0,\epsilon\} \times Z) \cup ([0,\epsilon] \times \{z_0\})},$$

where the displayed union is collapsed to one basepoint.

Use the marked presentation $\overline{T}^2 = \mathbb{R}^2/\mathbb{Z}^2$, let

$$K^{(1)} = ((\mathbb{R}/\mathbb{Z}) \times \{0\}) \cup (\{0\} \times (\mathbb{R}/\mathbb{Z})),$$

and let

$$c : \overline{T}^2 \longrightarrow \overline{T}^2/K^{(1)} \cong S^2$$

be the degree-one quotient obtained by collapsing the one-skeleton. The formula

$$\widehat{p}_2 : \mathcal{A} \longrightarrow \Sigma_r(\overline{T}^2/K^{(1)}, c(K^{(1)})) \cong S^3, \qquad [r, e^{i\phi}, u] \longmapsto [r, c(u)]$$

is well defined: the two suspension ends of $\mathcal{A}$ and the entire cylinder over $K^{(1)}$ map to the reduced-suspension basepoint. It is a continuous surjection. Set $\widehat{q}_3 = \widehat{p}_2 \circ q_2$.

For every sufficiently large T, define the transported one-skeleton and its collapse by

$$\begin{aligned} K_T^{(1)} &:= \iota_T(K^{(1)}) \subset Q_T, \\ S_T^2 &:= \overline{T}^2/K^{(1)} \cong Q_T/K_T^{(1)}, \\ c_T &:= c \circ \iota_T^{-1} : Q_T \longrightarrow S_T^2. \end{aligned} \tag{199}$$

The displayed target identification is the one induced by ι_T, so $c_T^{-1}(*) = K_T^{(1)}$, where $* = c(K^{(1)}) = c_T(K_T^{(1)})$.

Give Q_T its Euclidean quotient length metric and S_T^2 the quotient length metric under c_T. Put

$$y_* = [(1/2, 1/2)] \in \overline{T}^2, \qquad y_T := \iota_T(y_*) \in Q_T, \qquad z_T := c_T(y_T) \in S_T^2.$$

Equivalently, the metric on $\overline{T}^2$ used below is $g_T := \iota_T^* g_{Q_T}$. For every $v \in \mathbb{R}^2$, the exact decomposition

$$Y_T = T \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} + E_T, \qquad \|E_T\|_{\mathrm{op}} \le C,$$

gives

$$\|Y_T v\| \ge (T - C)\|v\|.$$

The standard flat distance from y_* to $K^{(1)}$ is $1/2$; therefore

$$d_T := \mathrm{dist}_{Q_T}(y_T, K_T^{(1)}) = \mathrm{dist}_{g_T}(y_*, K^{(1)}) \geq \frac{T-C}{2}. \tag{200}$$

For a length-space quotient which collapses a closed subset K to one point, the distance between two noncollapsed classes represented by x, y is

$$\min\{d(x,y), d(x,K) + d(y,K)\}.$$

The two displayed terms are upper bounds, obtained respectively from a direct path and from paths from x and y to K. Conversely, a rectifiable quotient path either avoids the collapsed point, in which case its compact image lifts to $X \setminus K$ and has length at least $d(x,y)$, or meets that point, in which case its two pieces have total length at least $d(x,K) + d(y,K)$. Taking infima proves the formula. If $T - C > 4R$, then $d_T > 2R$. The displayed formula and the systolic bound show that

$$c_T : B_R^{Q_T}(y_T) \longrightarrow B_R^{S_T^2}(z_T)$$

is an isometry onto the entire target ball, and both balls are exactly Euclidean two-balls. Indeed, the second term in the minimum cannot realize a distance at most $2R$, and any point of the target ball has a unique noncollapsed representative; the same formula puts that representative in the source ball.

At reduced-suspension coordinate $r_T = e^{-2\pi T}$, S_T^2 is a based embedded subspace of S^3: its point $*$ is the global reduced-suspension basepoint. Hence

$$\mathbb{S}_{12,T}^4 = S_T^2 \wedge S_T^2$$

embeds in $S^3 \wedge S^3 = S^6$. Equip Q_T^2 first with the Euclidean Riemannian ℓ^2-product length metric, equip $(S_T^2)^2$ with the ℓ^2-product of the two quotient length metrics, and let

$$\rho_{12,T} : (S_T^2)^2 \longrightarrow S_T^2 \wedge S_T^2$$

carry the quotient length metric obtained by collapsing the fat wedge

$$\mathcal{W}_{12,T} = (S_T^2 \times \{*\}) \cup (\{*\} \times S_T^2).$$

Since $c_T^{-1}(*) = K_T^{(1)}$, the inverse image of $\mathcal{W}_{12,T}$ under $c_T \times c_T$ is exactly $W_{12,T}$ in equation (196). On $Q_T^2 \setminus W_{12,T}$, the map $c_T \times c_T$ is injective and its image avoids $\mathcal{W}_{12,T}$; the map $\rho_{12,T}$ is injective on the complement of $\mathcal{W}_{12,T}$. Hence their composite identifies all of $W_{12,T}$ to the smash basepoint and makes no other identifications.

The point $\zeta_{12,T} = [z_T, z_T]$ has distance d_T from this fat wedge. The one-factor onto isometry above gives an onto isometry from the radius-R ball about (y_T, y_T) in Q_T^2 to the corresponding ball about (z_T, z_T) in $(S_T^2)^2$. Applying the same collapsed-set distance formula to $\rho_{12,T}$, and using $d_T > 2R$, shows that

$$\rho_{12,T} \circ (c_T \times c_T) : B_R^{Q_T^2}((y_T, y_T)) \xrightarrow{\cong_{\mathrm{isom}}} B_R^{\mathbb{S}_{12,T}^4}(\zeta_{12,T}) \tag{201}$$

is onto as well as isometrically embedded. The systolic bound $\mathrm{sys}(Q_T^2) \geq T - C > 4R$ makes its domain exactly a Euclidean four-ball. On a radial fibre,

$$(\widehat{p}_2 \circ q_2)|_{F_T} = [r_T, c(\iota_T^{-1}(\pi_T(\,\cdot\,)))] = [r_T, c_T(\pi_T(\,\cdot\,))],$$

and taking two factors followed by $\rho_{12,T}$ proves the entire factorization in equation (194).

For the four-copy construction, choose $1 \in S^1$ as character basepoint and define

$$\widehat{p}_1 : \mathcal{B} \longrightarrow \Sigma_r(S^1, 1) \cong S^2, \qquad [r, e^{i\phi}, \theta] \longmapsto [r, \theta].$$

The formula forgets the base angle and collapses the two suspension ends and the entire cylinder with $\theta = 1$ to the reduced-suspension basepoint. It is a continuous surjection. Put $\widehat{q}_\gamma = \widehat{p}_1 \circ q_{1,\mathrm{cusp}}$.

At the interior suspension coordinate $r_T = e^{-2\pi T}$, identify the character circle $C_T = \mathbb{R}/\ell_T\mathbb{Z}$ with $\mathbb{R}/\mathbb{Z} \subset S^2$ by $[x] \mapsto [x/\ell_T]$. It is a based embedded circle, so its fourth smash is an embedded

$$\mathbb{S}^4_{24,T} = C_T^{\wedge 4} \subset (S^2)^{\wedge 4} = S^8.$$

Equip C_T^4 first with the Euclidean Riemannian ℓ^2-product length metric, and let

$$\rho_{24,T} : C_T^4 \longrightarrow C_T^{\wedge 4}$$

carry the quotient length metric obtained by collapsing exactly the fat wedge $W_{24,T}$ of equation (197). Away from that fat wedge the smash quotient is injective, so no further points are identified. The midpoint

$$\zeta_{24,T} = [(\ell_T/2, \ell_T/2, \ell_T/2, \ell_T/2)]$$

has distance $\ell_T/2$ from that fat wedge. If $\ell_T > 4R$, neither a nonzero period nor a path through the collapsed fat wedge can shorten a distance between points in the radius-R ball of C_T^4. Conversely, the collapsed-set distance formula shows that every point of the radius-R quotient ball has its unique noncollapsed representative in that product ball. Consequently

$$\rho_{24,T} : B_R^{C_T^4}((\ell_T/2, \ell_T/2, \ell_T/2, \ell_T/2)) \xrightarrow{\cong_{\mathrm{isom}}} B_R^{\mathbb{S}^4_{24,T}}(\zeta_{24,T}) \tag{202}$$

is onto as well as isometrically embedded, and its domain is exactly a Euclidean four-ball because $\mathrm{sys}(C_T^4) = \ell_T > 4R$. On the radial source the formula discards only the base angle and then performs this smash quotient, giving equation (195).

Both one-copy maps are continuous surjections from compact X to Hausdorff spheres, hence quotient maps, and they are based. Their smash powers are surjective; compact-to-Hausdorff again makes them quotient maps. Using the same chosen based diffeomorphism $X \cong_{\mathrm{diff}} S^6$, the homeomorphisms

$$X^{\wedge 2} \cong S^{12}, \quad (S^3)^{\wedge 2} \cong S^6, \qquad X^{\wedge 4} \cong S^{24}, \quad (S^2)^{\wedge 4} \cong S^8$$

prove equation (193). The exact Euclidean-ball statements prove all three convergences in equation (198). □

Corollary 11.21 (The two-copy rank-two sphere carrier is null-homotopic)**.** *After the based identifications in equation* (193)*, the continuous surjective quotient*

$$\widehat{q}_3^{\wedge 2} : S^{12} \twoheadrightarrow S^6$$

is based null-homotopic. In fact, $g \wedge g : S^{12} \to S^6$ is based null-homotopic for every based map $g : S^6 \to S^3$.

Proof. Apply the full external-smash calculation in theorem 11.30 to $g = \widehat{q}_3$. That theorem proves the stronger assertion that $x \wedge y : S^{12} \to S^6$ is null-homotopic for every pair of based maps $x, y : S^6 \to S^3$. □

Corollary 11.22 (Every repeated smash of the circle-quotient carrier is null)**.** *Put $f = \widehat{q}_\gamma : S^6 \to S^2$ after the chosen based identification $X \cong S^6$. For every $r \geq 2$,*

$$[f^{\wedge r}] = 0 \quad \textit{in} \quad \pi_{6r}(S^{2r}). \tag{203}$$

In particular,

$$[\widehat{q}_\gamma^{\wedge 2}] = 0 \in \pi_{12}(S^4), \qquad [\widehat{q}_\gamma^{\wedge 4}] = 0 \in \pi_{24}(S^8). \tag{204}$$

Thus the displayed surjective quotient $\widehat{q}_\gamma^{\wedge 4} : S^{24} \twoheadrightarrow S^8$ is globally null-homotopic. This does not alter either its point-set surjectivity or the exact intrinsic $\mathbb{R}^4$ limit on the moving four-spheres $\mathbb{S}^4_{24,T}$: those are properties of the specified representative and its radial restrictions, whereas null-homotopy classifies that representative only up to based homotopy.

Proof. Associativity of the smash product gives the exact factorization

$$S^6 \wedge S^6 \xrightarrow{f\wedge \mathrm{id}_{S^6}} S^2 \wedge S^6 \xrightarrow{\mathrm{id}_{S^2}\wedge f} S^2 \wedge S^2. \tag{205}$$

Under the standard based sphere identifications, the first arrow is the sixfold suspension of f, up to the harmless twist homeomorphisms dictated by the left-versus-right suspension convention:

$$f \wedge \mathrm{id}_{S^6} = \Sigma^6 f : S^{12} \longrightarrow S^8. \tag{206}$$

Freudenthal's suspension theorem is already in its isomorphism range here, because $12 \leq 2 \cdot 8 - 2$, and remains in that range after every further suspension. It therefore identifies $\pi_{12}(S^8)$ with the fourth stable stem. The low-stem calculation gives

$$\pi_{12}(S^8) \cong \pi_4^{\mathrm{S}} = 0 \quad \text{[10, Corollary 4.24, p. 360]; [12, Table 1, p. 113].} \tag{207}$$

Hence $\Sigma^6 f$ is based null-homotopic, so the composite in equation (205), namely $f \wedge f$, is based null-homotopic. For $r \geq 2$, the canonical associativity homeomorphisms identify

$$f^{\wedge r} = (f \wedge f) \wedge f^{\wedge(r-2)}, \tag{208}$$

with the convention $f^{\wedge 0} = \mathrm{id}_{S^0}$. Smashing a based null-homotopy with any based map is a based null-homotopy. This proves equations (203) and (204). □

Corollary 11.23 (Literal $\mathbb{R}^4$ open cells in the sphere targets)**.** *Let $*$ denote the smash basepoint and set*

$$U_{12,T} := \mathbb{S}^4_{12,T} \setminus \{*\}, \qquad U_{24,T} := \mathbb{S}^4_{24,T} \setminus \{*\}.$$

The embedded moving four-spheres restrict to topological embeddings

$$U_{12,T} \cong \mathbb{R}^4 \hookrightarrow S^6, \qquad U_{24,T} \cong \mathbb{R}^4 \hookrightarrow S^8.$$

More exactly, the quotient-stage maps restrict to homeomorphisms

$$(Q_T \setminus K_T^{(1)})^2 \xrightarrow{\cong} U_{12,T}, \tag{209}$$

$$(C_T \setminus \{0\})^4 \xrightarrow{\cong} U_{24,T}. \tag{210}$$

These copies of $\mathbb{R}^4$ are nonclosed topological open cells: they are open in their moving four-spheres, not open in the ambient S^6 or S^8, and their closures in those ambient spheres are $\mathbb{S}^4_{12,T}$ and $\mathbb{S}^4_{24,T}$, respectively. They are not globally isometric to Euclidean $\mathbb{R}^4$ for either the assigned compact quotient metrics or the restriction of any fixed compatible metric on the compact sphere targets. The exact midpoint balls in equations (201) *and* (202) *give the precise metric statement: as $T \to \infty$, every fixed Euclidean radius-R ball occurs isometrically and onto the corresponding pointed ball.*

Proof. For a quotient which collapses a closed subset K to a point, restriction to the complement gives a homeomorphism

$$X \setminus K \xrightarrow{\cong} (X/K) \setminus \{*\}.$$

Indeed, the complement of the quotient point is open, its inverse image is the saturated open set $X \setminus K$, and restriction of a quotient map over an open subset of its target is again a quotient map. The restricted map is bijective, and a bijective quotient map is a homeomorphism. Consequently

$$Q_T \setminus K_T^{(1)} \cong S^2_T \setminus \{*\}.$$

The marked coordinate one-skeleton $K^{(1)}$ is the 1-skeleton of the standard one-vertex CW structure on $\overline{T}^2$, with one 0-cell, two 1-cells, and one 2-cell. Its complement is therefore the open 2-cell $(0,1)^2 \cong \mathbb{R}^2$. Transport by ι_T gives

$$Q_T \setminus K_T^{(1)} \cong \mathbb{R}^2.$$

Likewise $C_T \setminus \{0\}$ is an open interval and hence is homeomorphic to $\mathbb{R}$.

The complement of the smash basepoint is the product of the complements of the factor basepoints. Therefore

$$(S_T^2 \wedge S_T^2) \setminus \{*\} \cong (S_T^2 \setminus \{*\})^2 \cong \mathbb{R}^4,$$

and

$$C_T^{\wedge 4} \setminus \{*\} \cong (C_T \setminus \{0\})^4 \cong \mathbb{R}^4.$$

The full moving four-spheres are topologically embedded in their compact sphere targets, so these restrictions are topological embeddings. In each case the omitted smash basepoint lies in the closure of the open cell, which proves the closure assertions. Finally, every compatible metric on a compact target is bounded, as is each compact quotient metric assigned to a moving four-sphere, whereas Euclidean $\mathbb{R}^4$ is unbounded. Thus no global isometry is possible; the exact metric content is precisely the onto-ball calculation already proved. □

Remark 11.24 (The moving four-spheres are null-homotopic in the targets). For a fixed interior reduced-suspension coordinate, the inclusion $S_T^2 \hookrightarrow S^3$ contracts through based maps by sliding that coordinate to a suspension endpoint; throughout the homotopy the collapsed meridian remains at the global basepoint. Hence $S_T^2 \wedge S_T^2 \hookrightarrow S^6$ is null-homotopic. The identical sliding argument contracts $C_T \hookrightarrow S^2$, and therefore also contracts $C_T^{\wedge 4} \hookrightarrow S^8$. Thus the moving metric four-spheres do not contradict the fixed-target orientation obstruction: their fundamental classes die in the ambient targets.

Remark 11.25 (Homotopical status of the two sphere-to-sphere maps). The maps $S^{12} \to S^6$ and $S^{24} \to S^8$ induce zero on all positive-degree integral homology and cohomology groups and have zero ordinary Hurewicz image. They have no classical degree because source and target dimensions differ. The full four-copy map $\widehat{q}_\gamma^{\wedge 4} : S^{24} \to S^8$ is based null-homotopic by theorem 11.22. The distinct two-copy map $\widehat{q}_3^{\wedge 2} : S^{12} \to S^6$ is also based null-homotopic, by theorem 11.21. In both cases the vanishing is a theorem about the full specified sphere-valued map, not merely its Hurewicz image or the inclusion of a moving four-sphere. The one-copy class $[\widehat{q}_3] \in \pi_6(S^3)$ is now zero by Theorem 11.34; the one-copy class $[\widehat{q}_\gamma] \in \pi_6(S^2)$ remains open.

Proposition 11.26 (The compact cone-on-T^4 carriers). *Fix T_0 sufficiently large and put $r_0 = e^{-2\pi T_0} < \epsilon$. Define*

$$\mathcal{D}_{12} = \frac{[0, r_0] \times (\overline{T}^2)^2}{\{0\} \times (\overline{T}^2)^2 \sim d_{12}}, \qquad \mathcal{D}_{24} = \frac{[0, r_0] \times (S^1)^4}{\{0\} \times (S^1)^4 \sim d_{24}}.$$

Both spaces are homeomorphic to $\mathrm{Cone}(T^4)$. *There are topological embeddings*

$$\delta_{12} : \mathcal{D}_{12} \hookrightarrow \mathcal{A} \wedge \mathcal{A}, \qquad [r, u_1, u_2] \longmapsto [\,[r, 1, u_1], [r, 1, u_2],\,], \tag{211}$$

$$\delta_{24} : \mathcal{D}_{24} \hookrightarrow \mathcal{C}^{\wedge 4}, \qquad [r, \theta_1, \dots, \theta_4] \longmapsto \bigwedge_{j=1}^{4} [r, (1, \theta_j)]. \tag{212}$$

Here $1 \in S^1$ fixes the positive radial base angle. For every $T \geq T_0$, put $r_T = e^{-2\pi T}$. The restrictions of the sphere quotient maps to the synchronized radial fibres are exactly the cross-sections at height r_T:

$$F_T^2 \xrightarrow{\pi_T \times \pi_T} Q_T^2 \cong \{r_T\} \times T^4 \subset \mathcal{D}_{12}, \tag{213}$$

$$F_T^4 \xrightarrow{\chi_{\gamma,T}^{\times 4}} C_T^4 \cong \{r_T\} \times T^4 \subset \mathcal{D}_{24}. \tag{214}$$

Every one of these embedded four-torus slices is null-homotopic in its carrier, and hence in the full smash target.

At the cone vertex d_ν, $\nu \in \{12, 24\}$, the local integral homology is

$$H_k(\mathcal{D}_\nu, \mathcal{D}_\nu \setminus \{d_\nu\}; \mathbb{Z}) \cong \begin{cases} \mathbb{Z}^4, & k = 2, \\ \mathbb{Z}^6, & k = 3, \\ \mathbb{Z}^4, & k = 4, \\ \mathbb{Z}, & k = 5, \\ 0, & \textit{otherwise.} \end{cases} \tag{215}$$

In particular, d_ν is not an integral homology-manifold point, with or without boundary, of any dimension. Consequently it is not a topological manifold point, with or without boundary, of any dimension.

Proof. The two displayed domains are cones on $(\overline{T}^2)^2 \cong T^4$ and $(S^1)^4 \cong T^4$. At $r = 0$, every factor on the right of equations (211) and (212) is the relevant smash basepoint, so the formulae descend from the cylinders to the cones. For $0 < r \le r_0$, none of the factors is the smash basepoint. Equality in the smash quotient is then equality in every factor, and the coordinate descriptions of the suspension and cusp cone give equality of r and of all torus coordinates. Thus each δ_ν is injective. Its domain is compact and its target is Hausdorff, so it is an embedding.

The already proved one-factor restriction formulae, with $r_T = e^{-2\pi T}$, are

$$q_2(x) = [r_T, 1, \iota_T^{-1}(\pi_T(x))] \quad (x \in F_T)$$

and, in the cusp-cone piece of $\mathcal{C}$,

$$q_1(x) = [r_T, (1, \vartheta_T^{-1}(\chi_{\gamma,T}(x)))] \quad (x \in F_T).$$

Applying these identities in two and four factors gives equations (213) and (214). Sliding the cone coordinate from r_T to zero contracts each slice to the cone vertex. Finally, excision and the long exact sequence of the pair give, for $L = T^4$,

$$H_k(\mathrm{Cone}(L), \mathrm{Cone}(L) \setminus \{v\}; \mathbb{Z}) \cong \widetilde{H}_{k-1}(L; \mathbb{Z}).$$

The ranks $4, 6, 4, 1$ of H_1, H_2, H_3, H_4 of T^4 prove equation (215). At an interior point of an integral homology n-manifold the local integral homology is $\mathbb{Z}$ in degree n and zero in every other degree, whereas at a boundary point of an integral homology manifold with boundary all these local groups vanish. The simultaneous nonzero groups in degrees $2, 3, 4, 5$, including ranks greater than one, agree with neither pattern for any n. Every topological manifold is locally an integral homology manifold, so the final assertion follows as well. □

Theorem 11.27 (Point-set quotient maps and exact homotopy types)**.** *The smash maps are continuous surjective quotient maps*

$$q_2^{\wedge 2} : S^{12} \twoheadrightarrow \mathcal{A}^{\wedge 2}, \qquad q_1^{\wedge 4} : S^{24} \twoheadrightarrow \mathcal{C}^{\wedge 4}.$$

Their targets have homotopy types

$$\mathcal{A}^{\wedge 2} \simeq \bigvee^{9} S^4 \vee \bigvee^{18} S^5 \vee \bigvee^{15} S^6 \vee \bigvee^{6} S^7 \vee S^8, \tag{216}$$

$$\mathcal{C}^{\wedge 4} \simeq S^8 \vee \bigvee^{4} S^9 \vee \bigvee^{6} S^{10} \vee \bigvee^{4} S^{11} \vee S^{12}. \tag{217}$$

Both maps induce zero on every positive-degree integral homology and cohomology group and have zero ordinary Hurewicz image. Neither map has a continuous homotopy section. These conclusions do not determine whether the unstable homotopy class of either map is zero.

Proof. Let $f : (K, k_0) \to (L, \ell_0)$ be a surjective based map between compact Hausdorff spaces. Every non-basepoint class of $L^{\wedge r}$ has a representative $(y_1, \dots, y_r)$ with each $y_i \neq \ell_0$. Choosing $x_i \in f^{-1}(y_i)$ proves surjectivity of $f^{\wedge r}$; the smash basepoint is also in its image. The source smash is compact. The target fat wedge is closed, hence the target smash is Hausdorff. A continuous surjection from a compact space to a Hausdorff space is a quotient map. This proves the first assertion for q_2 and q_1.

The suspension splitting

$$\Sigma(X_1 \times \cdots \times X_n) \simeq \bigvee_{\varnothing \neq I \subseteq \{1,\dots,n\}} \Sigma\left(\bigwedge_{i \in I} X_i\right)$$

gives

$$\mathcal{A} = \Sigma T^3 \simeq \bigvee^{3} S^2 \vee \bigvee^{3} S^3 \vee S^4.$$

Finite smash products distribute over finite wedges. The ordered degree sums give multiplicities $9, 18, 15, 6, 1$ in dimensions $4, 5, 6, 7, 8$, proving equation (216).

For $\mathcal{C}$, write $x = \rho_1$, $y = \rho_2$, and $h = \eta$. The pair-of-pants product and the two exact Seifert fillings give

$$\pi_1(\mathcal{M}) = \langle x, y, h \mid [x, h] = [y, h] = 1, x^3 h^{-1} = 1,\ y^4 h = 1 \rangle.$$

The remaining boundary torus is generated by h and $\rho_0 = (xy)^{-1}$. Coning this torus kills their normal closure, and therefore

$$\pi_1(\mathcal{C}) \cong \langle x, y \mid x^3 = 1,\ y^4 = 1,\ xy = 1 \rangle = 1,$$

because $y = x^{-1}$ and $\gcd(3, 4) = 1$. The previously computed relative homology gives

$$\widetilde{H}_2(\mathcal{C}; \mathbb{Z}) = \mathbb{Z}, \qquad \widetilde{H}_3(\mathcal{C}; \mathbb{Z}) = \mathbb{Z},$$

and no other reduced homology. Since $\mathcal{C}$ is a simply connected finite three-complex, Hurewicz supplies a sphere representing the H_2-generator, and the Whitehead exact sequence makes $\pi_3(\mathcal{C}) \to H_3(\mathcal{C})$ surjective. A wedge of representing maps $S^2 \vee S^3 \to \mathcal{C}$ is a homology equivalence between simply connected CW complexes, hence a homotopy equivalence. Thus

$$\mathcal{C} \simeq S^2 \vee S^3.$$

Taking the fourth smash power yields equation (217) with the binomial multiplicities $1, 4, 6, 4, 1$.

The first target has no reduced homology above degree eight and the second has none above degree twelve. The source spheres have no positive homology below degrees twelve and twenty-four. Hence all positive-degree homology and cohomology homomorphisms induced by the two maps vanish. Their Hurewicz images lie in $H_{12}(\mathcal{A}^{\wedge 2}) = 0$ and $H_{24}(\mathcal{C}^{\wedge 4}) = 0$, respectively. A section would split a nonzero lower homology group of the target into the zero lower homology of the source sphere, which is impossible. Higher homotopy groups of the displayed wedges need not vanish, so none of these arguments decides the unstable null-homotopy question. □

Theorem 11.28 (Exact decompactification spaces with boundary $\mathbb{R}^4$)**.** *For $\nu \in \{12, 24\}$, set*

$$A_T^{(12)} = \operatorname{diag}(Y_T, Y_T), \qquad A_T^{(24)} = \ell_T I_4, \qquad s_0 = T_0^{-1}.$$

On $[0, s_0] \times \mathbb{R}^4$, define an equivalence relation $\sim_\nu$ fibrewise over s by

$$(0, x) \sim_\nu (0, y) \iff x = y,$$
$$(s, x) \sim_\nu (s, y) \iff x - y \in A_{1/s}^{(\nu)} \mathbb{Z}^4 \quad (0 < s \leq s_0).$$

Let

$$\mathfrak{E}_\nu = ([0, s_0] \times \mathbb{R}^4)/\!\sim_\nu, \qquad p_\nu : \mathfrak{E}_\nu \longrightarrow [0, s_0].$$

Then $\mathfrak{E}_\nu$ is a Hausdorff, second-countable topological five-manifold with boundary, and

$$p_{12}^{-1}(s) \cong Q^2_{1/s} \qquad (s > 0), \qquad p_{12}^{-1}(0) \cong \mathbb{R}^4, \tag{218}$$

$$p_{24}^{-1}(s) \cong C^4_{1/s} \qquad (s > 0), \qquad p_{24}^{-1}(0) \cong \mathbb{R}^4. \tag{219}$$

The other boundary component is the compact four-torus over $s = s_0$.

Give every positive fibre its Euclidean quotient metric and the zero fibre its Euclidean metric. For every $R > 0$, there is $s_R > 0$ such that, for $0 \le s < s_R$, writing $q_{\nu,s}(x) = [s, x]$, the quotient map restricts to an exact isometry

$$q_{\nu,s} : B^{\mathbb{R}^4}_R(0) \xrightarrow{\cong_{\mathrm{isom}}} B^{p_\nu^{-1}(s)}_R([s, 0])$$

onto the entire quotient radius-R ball, not merely onto an embedded subset of that ball. Consequently the positive fibres converge to the zero fibre in pointed local-isometric, pointed smooth, and pointed Gromov–Hausdorff topology.

There are continuous surjective comparison maps which set-theoretically collapse the zero boundary fibres,

$$\mathbb{R}^4 = p_{12}^{-1}(0) \hookrightarrow \mathfrak{E}_{12} \xrightarrow{\beta_{12}} \mathcal{D}_{12} \xhookrightarrow{\delta_{12}} \mathcal{A}^{\wedge 2},$$

and

$$\mathbb{R}^4 = p_{24}^{-1}(0) \hookrightarrow \mathfrak{E}_{24} \xrightarrow{\beta_{24}} \mathcal{D}_{24} \xhookrightarrow{\delta_{24}} \mathcal{C}^{\wedge 4}.$$

For $s > 0$, β_ν is the specified marked homeomorphism from $p_\nu^{-1}(s)$ to the radial torus slice at $r = e^{-2\pi/s}$. Its restriction

$$\beta_\nu : \mathfrak{E}_\nu \setminus p_\nu^{-1}(0) \xrightarrow{\cong} \mathcal{D}_\nu \setminus \{d_\nu\}$$

is a homeomorphism, while

$$\beta_\nu^{-1}(d_\nu) = p_\nu^{-1}(0) \cong \mathbb{R}^4.$$

Neither β_ν is proper or a quotient map. It factors through a continuous bijection

$$\overline{\beta}_\nu : \mathfrak{E}_\nu / p_\nu^{-1}(0) \longrightarrow \mathcal{D}_\nu,$$

but this bijection is not a homeomorphism; equivalently, the topology of $\mathcal{D}_\nu$ is not the quotient topology induced by β_ν. Thus the limiting $\mathbb{R}^4$ is an actual boundary fibre of $\mathfrak{E}_\nu$, whereas the continuous comparison map sends that entire fibre to the cone vertex.

Proof. We give a common proof. Write

$$L_s = A^{(\nu)}_{1/s}\mathbb{Z}^4 \quad (s > 0), \qquad L_0 = \{0\}.$$

The matrices vary continuously for $s > 0$, and

$$\mathrm{sys}(L_s) \longrightarrow \infty \qquad (s \downarrow 0)$$

by $\mathrm{sys}(Y_T\mathbb{Z}^2) \ge T - C$ in the S^{12} case and $\mathrm{sys}(\ell_T\mathbb{Z}^4) = \ell_T$ in the S^{24} case.

The equivalence relation is closed. Indeed, suppose $(s_j, x_j) \sim_\nu (s_j, y_j)$ and $(s_j, x_j, y_j) \to (s, x, y)$. If $s > 0$, write $y_j - x_j = A^{(\nu)}_{1/s_j} n_j$. On a compact neighborhood of s, the inverse matrices are uniformly bounded, so boundedness of $y_j - x_j$ forces n_j to lie in a finite set. Passing to a subsequence makes n_j constant and proves $x \sim_\nu y$. If $s = 0$, any nonzero n_j would give

$$\|y_j - x_j\| \ge \mathrm{sys}(L_{s_j}) \longrightarrow \infty,$$

contrary to convergence. Hence $x = y$.

The quotient projection $q_\nu : [0, s_0] \times \mathbb{R}^4 \to \mathfrak{E}_\nu$ is open. If U is open, its saturation is the union of U with the images, for $n \in \mathbb{Z}^4$, of

$$U \cap ((0, s_0] \times \mathbb{R}^4) \quad \text{under} \quad (s, x) \longmapsto (s, x + A^{(\nu)}_{1/s} n),$$

and every one of these images is open. A closed equivalence relation with an open quotient map has Hausdorff quotient: two inequivalent points admit a product neighborhood disjoint from the relation, and the saturations of the two factor neighborhoods are disjoint open sets. Since the source is second countable and the quotient map is open, images of a countable base form a countable base of the quotient.

Fix $x_0 \in \mathbb{R}^4$. Choose $a > 0$ and $s_1 > 0$ such that $\mathrm{sys}(L_s) > 2a$ for $0 < s < s_1$. The quotient projection is one-to-one on $[0, s_1) \times B^\circ_a(x_0)$, where $B^\circ_a(x_0) = \{x : \|x - x_0\| < a\}$, and its restriction is an open continuous bijection onto an open subset of $\mathfrak{E}_\nu$; it is therefore a homeomorphism. These are half-space charts at $s = 0$. The usual evenly covered small balls give product charts for $s > 0$, and the same argument at $s = s_0$ gives the second boundary component. This proves the manifold assertion and the fibre identifications.

For the metric assertion, choose s_R so that $\mathrm{sys}(L_s) > 4R$ for $s < s_R$. If $x, y \in B_R(0)$ and $0 \neq \lambda \in L_s$, then

$$\|x - y + \lambda\| \geq \|\lambda\| - \|x - y\| > 4R - 2R \geq \|x - y\|.$$

Thus no translated representative shortens the Euclidean distance on that ball, so $q_{\nu,s}$ is an exact isometric embedding there. It is also onto the full quotient ball. Indeed, if $\xi = [s, x]$ satisfies $d([s, 0], \xi) \leq R$, discreteness of L_s makes the infimum in

$$d([s, 0], [s, x]) = \min_{\lambda \in L_s} \|x + \lambda\|$$

attained. For a minimizing λ_0, the representative $x + \lambda_0$ lies in $B^{\mathbb{R}^4}_R(0)$ and maps to ξ. At $s = 0$, $q_{\nu,0}$ is the identity. This proves the asserted onto isometry, and since it holds for every fixed R, it proves all three pointed convergences.

It remains to define these comparison maps, which set-theoretically collapse the zero fibres. For $s > 0$, put $T = 1/s$ and $r = e^{-2\pi T}$. In the S^{12} case, write $x = (x_1, x_2) \in \mathbb{R}^2 \oplus \mathbb{R}^2$ and set

$$\beta_{12}([s, x]) = \big[r, \iota_T^{-1}([x_1]_{Y_T\mathbb{Z}^2}), \iota_T^{-1}([x_2]_{Y_T\mathbb{Z}^2})\big].$$

In the S^{24} case, write $x = (x_1, \ldots, x_4)$ and set

$$\beta_{24}([s, x]) = [r, [x_1/\ell_T], \ldots, [x_4/\ell_T]].$$

These formulae are well defined on the positive fibres and are continuous there. Define $\beta_\nu([0, x]) = d_\nu$ for every $x \in \mathbb{R}^4$. Every neighborhood of d_ν in the cone contains all cross-sections with sufficiently small r. Since $e^{-2\pi/s} \to 0$ as $s \downarrow 0$, continuity at the zero fibre is uniform in x. Surjectivity follows because each positive fibre maps onto its complete cross-section and the zero fibre maps onto the vertex. The restriction away from $p_\nu^{-1}(0)$ is a homeomorphism because $s \mapsto e^{-2\pi/s}$ is a homeomorphism from $(0, s_0]$ to $(0, r_0]$, and the displayed marked-torus identifications are fibrewise homeomorphisms varying continuously with s. The vertex fibre is exactly $p_\nu^{-1}(0)$, so β_ν is not proper.

It is also not a quotient map. Choose $a \in \mathbb{R}^4 \setminus \mathbb{Z}^4$, for example $a = (1/2, 1/2, 1/2, 1/2)$, choose $s_j \downarrow 0$, and put

$$z_j = [s_j, A^{(\nu)}_{1/s_j} a] \in \mathfrak{E}_\nu.$$

For $\nu = 12$,

$$\inf_{n \in \mathbb{Z}^4} \|A^{(12)}_{1/s_j}(a + n)\| \geq (1/s_j - C)\,\mathrm{dist}(a, \mathbb{Z}^4) \longrightarrow \infty,$$

and for $\nu = 24$ the same infimum equals

$$\ell_{1/s_j} \operatorname{dist}(a, \mathbb{Z}^4) \longrightarrow \infty.$$

More explicitly, for every fixed $x_0 \in \mathbb{R}^4$,

$$\inf_{n\in\mathbb{Z}^4} \|A^{(\nu)}_{1/s_j}(a+n) - x_0\| \geq \inf_{n\in\mathbb{Z}^4} \|A^{(\nu)}_{1/s_j}(a+n)\| - \|x_0\| \longrightarrow \infty.$$

Hence $\{z_j : j \geq 1\}$ is closed in $\mathfrak{E}_\nu$: it has no accumulation in a positive fibre and the last estimate shows that it eventually misses every half-space chart about every point of the zero fibre. But

$$\beta_\nu(z_j) = [e^{-2\pi/s_j}, a] \longrightarrow d_\nu,$$

so $\{\beta_\nu(z_j) : j \geq 1\}$ is not closed in $\mathcal{D}_\nu$. Since β_ν is injective off the zero fibre and the latter maps to d_ν, the inverse image of this nonclosed set is the closed set $\{z_j : j \geq 1\}$. Thus β_ν is not a quotient map. This proves the two diagrams and their exact topological scope. Since β_ν is constant exactly on $p_\nu^{-1}(0)$ and otherwise injective, it induces the continuous bijection $\overline{\beta}_\nu$ displayed in the theorem. Were that bijection a homeomorphism, β_ν would be the composite of the quotient projection with a homeomorphism and hence a quotient map, contrary to the counterexample. Therefore $\overline{\beta}_\nu$ is not a homeomorphism. □

Corollary 11.29 (Sharp global meaning of the Euclidean limit)**.** *The native routes constructed above exhibit the underlying space $\mathbb{R}^4$ in the following three precise senses:*

1. *as each of the nonclosed topologically embedded open cells $U_{12,T} \subset S^6$ and $U_{24,T} \subset S^8$;*
2. *as the pointed intrinsic Euclidean limit of the four-torus slices in the global quotient carriers of S^{12} and S^{24}; and*
3. *as the Euclidean decompactification boundary fibre of $\mathfrak{E}_{12}$ and $\mathfrak{E}_{24}$.*

It does not produce $\mathbb{R}^4$ as the whole compact sphere, as the whole compact conical target, or as a continuous quotient target of either compact space. Nor does the unrescaled restriction of any fixed compatible ambient target metric have $\mathbb{R}^4$ as its slice limit, and the open cells in item 1 are not globally Euclidean-isometric in such a metric.

Proof. The three positive assertions are theorems 11.23, 11.26 and 11.28. For the negative assertions, a continuous image of a compact sphere or compact conical target is compact, so it cannot be all of $\mathbb{R}^4$. The local homology calculation equation (215) shows that the carrier vertex is not an integral homology-manifold point, and hence not a manifold point, of any dimension, with or without boundary. Finally, in every fixed compatible metric on a compact cone target, all sufficiently deep radial slices lie uniformly in an arbitrarily small neighborhood of the vertex, so their ambient diameters tend to zero. Their intrinsic diameters tend to infinity because their intrinsic systoles do. Hence without a T-dependent rescaling no fixed ambient metric realizes the intrinsic Euclidean limit on those slices. The fixed cusp-adapted blow-ups are treated separately in theorem 11.37. The comparison diagrams record the unrescaled distinction exactly: they collapse the new $\mathbb{R}^4$ boundary rather than identify it with the compact target. □

Scope. The constructions above use only the original, unscaled fibre metrics and the quotient maps derived from the native S^6 cusp. They do not use the auxiliary finite-cover unspooling. The conical carriers depend on the chosen cusp coordinate, positive radial ray, and based identification $X \cong S^6$. The spaces $\mathfrak{E}_\nu$ additionally remember the marked Euclidean quotient metrics. None of these data defines a compact complex structure on S^{12} or S^{24}, a four-dimensional physical spacetime, or an interacting Yang–Mills Hamiltonian.

11.3 The full $S^{12} \to S^6$ carrier is null-homotopic

This subsection determines the based homotopy class of the full two-copy sphere carrier. It is important that the calculation concerns the full quotient map, not merely the inclusion of one of its moving four-spheres.

Recall the exact maps from theorems 11.12 and 11.20. The cusp quotient is

$$q_2 : (X, x_0) \longrightarrow \mathcal{A} = \mathrm{Susp}(S^1 \times \overline{T}^2),$$

and, for the coordinate one-skeleton

$$K^{(1)} = ((\mathbb{R}/\mathbb{Z}) \times \{0\}) \cup (\{0\} \times (\mathbb{R}/\mathbb{Z})) \subset \overline{T}^2,$$

the degree-one quotient $c : \overline{T}^2 \to \overline{T}^2/K^{(1)} \cong S^2$ defines

$$\widehat{p}_2([r, e^{i\phi}, u]) = [r, c(u)] \in \Sigma_r S^2 \cong S^3.$$

Thus

$$g := \widehat{q}_3 = \widehat{p}_2 \circ q_2 : (X, x_0) \longrightarrow (S^3, *)$$

is the specified continuous surjective based quotient. After the fixed based identification $X \cong_{\mathrm{diff}} S^6$, its smash square is the specified continuous surjective based quotient

$$g^{\wedge 2} : S^{12} = S^6 \wedge S^6 \longrightarrow S^3 \wedge S^3 = S^6.$$

Theorem 11.30 (The full external smash pairing is zero)**.** *For every pair of based maps $x, y : S^6 \to S^3$,*

$$[x \wedge y] = 0 \qquad \textit{in} \qquad \pi_{12}(S^6).$$

In particular, $\widehat{q}_3^{\wedge 2} : S^{12} \twoheadrightarrow S^6$ is simultaneously surjective as a point-set quotient and based null-homotopic.

Proof. Write E for reduced suspension, using the ordered smash coordinates on spheres. The external smash product has the exact factorization

$$x \wedge y = (x \wedge \mathrm{id}_{S^3}) \circ (\mathrm{id}_{S^6} \wedge y) : S^6 \wedge S^6 \longrightarrow S^6 \wedge S^3 \longrightarrow S^3 \wedge S^3. \tag{220}$$

Under the displayed ordered sphere identifications, the two factors, in order of application, represent $E^6 y \in \pi_{12}(S^9)$ and $E^3 x \in \pi_9(S^6)$. Equivalently, Toda's Barratt–Hilton reduced-join formula gives, with $p = q = k = h = 3$,

$$[x \wedge y] = (E^3[x]) \circ (E^6[y]). \tag{221}$$

Here Toda's sign is $(-1)^{(p+k)h} = (-1)^{18} = +1$. Reversing the two displayed factors introduces the separate twist sign $(-1)^{ph} = -1$; no such reversal is used here.

The three exact groups needed below are

$$\pi_6(S^3) \cong \mathbb{Z}/12, \qquad \pi_{12}(S^9) \cong \mathbb{Z}/24, \qquad \pi_{12}(S^6) \cong \mathbb{Z}/2. \tag{222}$$

The first and third are the integral low unstable groups displayed in Hatcher's table extracted from Toda; the middle group is in the stable three-stem [10, opening table, p. 339]; [12, Table 1, p. 113].

Put $\beta = E^6[y] \in \pi_{12}(S^9)$. Since $12[y] = 0$, functoriality of suspension gives $12\beta = 0$. In the cyclic group $\mathbb{Z}/24$, the elements annihilated by 12 are exactly the even elements. Hence there is $\gamma \in \pi_{12}(S^9)$ such that $\beta = 2\gamma$. Postcomposition with $E^3[x] : S^9 \to S^6$ induces a group homomorphism

$$(E^3[x])_* : \pi_{12}(S^9) \longrightarrow \pi_{12}(S^6).$$

The target has exponent two, so equation (221) gives

$$[x \wedge y] = (E^3[x])_*(\beta) = 2(E^3[x])_*(\gamma) = 0.$$

This proves the vanishing of the entire external smash pairing and hence the theorem. □

Corollary 11.31 (All higher smash powers)**.** *For every based $g : S^6 \to S^3$ and every $r \geq 2$,*

$$g^{\wedge r} : S^{6r} \longrightarrow S^{3r}$$

is based null-homotopic.

Proof. With the fixed associativity homeomorphisms for smash products,

$$g^{\wedge r} = g^{\wedge 2} \wedge g^{\wedge (r-2)}.$$

Smashing a based null-homotopy with a based map gives a based null-homotopy. □

Remark 11.32 (The one-copy invariant and its computed value)*.* The proof does not need to determine the one-copy class $[g] \in \pi_6(S^3) \cong \mathbb{Z}/12$. Toda's suspension calculation identifies $\pi_6(S^3)$ with the even subgroup of the stable three-stem; Hatcher gives an integral Serre-spectral-sequence derivation of the same injectivity and identifies twice the stable Hopf–ν generator with a generator of the suspended $\mathbb{Z}/12$-summand [11, Theorem 5.41(b), p. 573, and pp. 577–578]. Thus, after choosing a stable generator ν_9, there is a unique $a \in \mathbb{Z}/12$ such that

$$[E^6 g] = 2a\nu_9 \in \pi_{12}(S^9) \cong \mathbb{Z}/24. \tag{223}$$

Because $a \mapsto 2a$ embeds $\mathbb{Z}/12$ as the even subgroup of $\mathbb{Z}/24$, $E^6 g$ is null-homotopic if and only if g is null-homotopic. Thus the one-copy class and its sixfold suspension are the same integer modulo 12. It is now zero by Theorem 11.34.

The exact geometric invariant which computes a is also identifiable. Choose $0 < r_0 < \epsilon$ and $u_0 \in \overline{T}^2 \setminus K^{(1)}$, and let $y_0 = [r_0, c(u_0)]$ lie in the open three-cell of $\Sigma_r S^2$. From the literal formula for q_2,

$$g^{-1}(y_0) = \{x \in f^{-1}(|t| = r_0) : \overline{\pi}_f(x) = u_0\}. \tag{224}$$

The fibre of $\overline{\pi}$ is the two-torus defined by $K_\Lambda = \mathbb{Z}\langle \widehat{w}, \widehat{\delta} \rangle$, and the cusp monodromy fixes both $\widehat{w}$ and $\widehat{\delta}$. Therefore (224) is the mapping torus of an affine translation of the kernel torus, which is diffeomorphic to $S^1 \times T^2 \cong T^3$: if the translation is $k \mapsto k + v$, choose a real lift $\widetilde{v}$; the map $[\theta, k] \mapsto (e^{2\pi i\theta}, k + \theta\widetilde{v})$ respects $(1, k) \sim (0, k + v)$ and has the inverse obtained by subtracting $\theta\widetilde{v}$. After choosing a smooth chart in the open target cell, the differential of g supplies its normal three-framing. Pontryagin–Thom identifies the stabilized class of g with the framed bordism class of this specifically framed T^3. Since stabilization on $\pi_6(S^3)$ is injective, computing that framing class computes a exactly. The unframed diffeomorphism type T^3, point-set surjectivity, and the zero ordinary homology map do not compute a.

Remark 11.33 (Scope)*.* The theorem changes no point-set or metric statement. A surjective quotient map may be null-homotopic, and the embedded moving four-spheres and their intrinsic pointed $\mathbb{R}^4$-limits remain exactly as constructed. The theorem concerns the sphere-valued postcomposition $\widehat{q}_3^{\wedge 2}$; it does not imply null-homotopy of the earlier conical-target map $q_2^{\wedge 2} : S^{12} \to \mathcal{A}^{\wedge 2}$.

11.4 The original cusp coefficient: an explicit framed nullbordism

Let X be the smooth compact source of the original construction, $f : X \to B$ its map, and t the specified cusp coordinate. Put $W = f^{-1}(0)$, choose the original $x_0 \in W$, and retain the ordered quotient lattice basis induced by $(\widehat{\gamma}, \widehat{u})$, with kernel generated by $(\widehat{w}, \widehat{\delta})$. Thus the quotient torus is $\overline{T}^2 = \mathbb{R}^2/\mathbb{Z}^2$. The original periods are

$$Z(s) = sB_0 + C(t), \qquad e^{2\pi i s} = t, \qquad B_0 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad C(t) = \begin{pmatrix} 6\mu(t) & h(t) \\ b(t) - h(t) & \mu(t) \end{pmatrix}. \tag{225}$$

Every entry of C is holomorphic at $t = 0$. The dense torus of the original toric cusp has coordinates (x_1, x_2, t), with $x = \exp(2\pi i\zeta)$. The residual lattice action is exactly

$$\Psi_\lambda(x, t) = \left(t^{B_0\lambda} \exp(2\pi i C(t)\lambda)x, t\right), \qquad \lambda \in \mathbb{Z}^2. \tag{226}$$

It extends freely and properly discontinuously across the original unimodular toric filling. In particular the quotient projection is a local biholomorphism at every toric fixed point. These are the reconstructed data and proved local properties of the original source (the reconstructed original manuscript, §4, source pp. 24–29; its complete cusp data are displayed here).

Let

$$K^{(1)} = (\mathbb{R}/\mathbb{Z} \times \{0\}) \cup (\{0\} \times \mathbb{R}/\mathbb{Z}), \qquad c : \overline{T}^2 \longrightarrow \overline{T}^2/K^{(1)} \cong S^2$$

be the specified degree-one cell quotient. Its restriction to the open two-cell is taken with the smooth coordinates used in the original regular-fibre description. Both radial endpoints and the complete basepoint cylinder are collapsed in

$$\Sigma_r(S^2, *) = \frac{[0,\epsilon] \times S^2}{(\{0,\epsilon\} \times S^2) \cup ([0,\epsilon] \times \{*\})} \cong S^3.$$

The original based map is

$$g(x) = [|t|, c(\overline{\pi}_f(x))] \quad (0 < |t| < \epsilon), \qquad g(W) = g(X \setminus \operatorname{int} N_0) = *. \tag{227}$$

Theorem 11.34 (The original one-copy cusp map is null-homotopic)**.** *The based map $g : (X, x_0) \to (S^3, *)$ in* (227) *is based null-homotopic. More precisely, a regular inverse image with its actual ordered normal framing induced by (r, c_1, c_2) is framed nullbordant inside one original cusp chart times $[0, 1]$, disjoint from $\{x_0\} \times [0, 1]$.*

Proof. The proof constructs the complete inverse image, compares its actual framing with an explicitly extendible framing, and gives the extension.

1. The exact complete inverse image. Retain the lift

$$u^0 = \begin{pmatrix} -1/4 \\ 1/4 \end{pmatrix}, \qquad \overline{u}^0 = (3/4, 1/4) \pmod{\mathbb{Z}^2}, \qquad B_0 u^0 = \begin{pmatrix} 1/4 \\ 1/4 \end{pmatrix}.$$

Thus $\overline{u}^0 \notin K^{(1)}$, and $B_0 u^0$ lies in the interior of the original triangle $\operatorname{conv}(0, e_1, e_2)$. Use its original unimodular chart

$$z_0 = t/(x_1 x_2), \qquad z_1 = x_1, \qquad z_2 = x_2, \qquad t = z_0 z_1 z_2. \tag{228}$$

Set

$$\ell = \log|t| < 0, \qquad R(t) = -2\pi \operatorname{Im} C(t), \qquad A(t) = \ell B_0 + R(t).$$

For a representative $\zeta = Z(s)u + k$, where $u, k \in \mathbb{R}^2$, $\operatorname{Im} s = -\ell/(2\pi)$ gives the exact identity

$$\begin{pmatrix} \log|x_1| \\ \log|x_2| \end{pmatrix} = A(t)u. \tag{229}$$

The quotient value is $u \bmod \mathbb{Z}^2$. Since B_0 is orthogonal, for all $v \in \mathbb{R}^2$,

$$\|A(t)v\| \geq (|\ell| - \|R(t)\|)\|v\|. \tag{230}$$

The boundedness of R therefore makes $A(t)$ invertible on a sufficiently small punctured disc.

For any point of the fibre quotient over $\overline{u}^0$, subtracting the appropriate original period, equivalently applying $\Psi_{-\lambda}$, gives a representative with the exact lift $u = u^0$. This representative is unique: another residual translate has logarithmic moduli differing by $A(t)\lambda$, which vanishes only for $\lambda = 0$. Conversely every dense-torus point satisfying $\log|x| = A(t)u^0$ has this quotient value, since the remaining real parts of ζ are exactly the kernel coordinates k. For each fixed t, varying $k \bmod \mathbb{Z}^2$ gives all independent phases of x_1, x_2.

The following change of coordinates uses the full original period units:

$$h_j^0(t) = \exp(-2\pi i (C(t)u^0)_j), \qquad j = 1, 2, \tag{231}$$

$$\mathcal{F}(z_0, z_1, z_2) = (w_0, w_1, w_2) = \left(\frac{z_0}{h_1^0(t) h_2^0(t)},\, h_1^0(t) z_1,\, h_2^0(t) z_2 \right). \tag{232}$$

The functions h_j^0 are nowhere-zero holomorphic units, including at $t = 0$. The product $w_0 w_1 w_2$ is exactly t. The inverse of $\mathcal{F}$ is

$$(w_0, w_1, w_2) \longmapsto \left(w_0 h_1^0(t) h_2^0(t),\, \frac{w_1}{h_1^0(t)},\, \frac{w_2}{h_2^0(t)} \right), \qquad t = w_0 w_1 w_2.$$

Thus this is a biholomorphism of the original chart restricted to the cusp disc, not a replacement of $C(t)$.

For $r_0 \in (0, \epsilon)$, the entire inverse image of $y_0 = [r_0, c(\overline{u}^0)]$ becomes exactly

$$T_{r_0} = \{|w_0| = \rho_0,\ |w_1| = \rho_1,\ |w_2| = \rho_2\}, \qquad (\rho_0, \rho_1, \rho_2) = (r_0^{1/2}, r_0^{1/4}, r_0^{1/4}). \tag{233}$$

Indeed, $\log |h^0(t)| = 2\pi \operatorname{Im} C(t) u^0 = -R(t) u^0$, so (229) gives $\log |w_j| = \ell (B_0 u^0)_j$ for $j = 1, 2$; the product identity gives the exponent $1/2$ for w_0. Conversely these moduli recover $u = u^0$ by invertibility of $A(t)$. The angular map is explicitly

$$(\arg t, \arg w_1, \arg w_2) \longleftrightarrow (\arg w_0, \arg w_1, \arg w_2), \qquad \arg w_0 = \arg t - \arg w_1 - \arg w_2 \pmod{2\pi},$$

with inverse $\arg t = \arg w_0 + \arg w_1 + \arg w_2$. Its integral matrix has determinant one. This includes the complete angular mapping torus and all kernel phases, with their original signs.

Let o be the origin of (228). The original free properly discontinuous action supplies a neighborhood V of o on which the quotient projection is injective. To see the local injectivity directly, first choose a relatively compact neighborhood; proper discontinuity leaves only finitely many translates that can intersect it. None of those nonidentity translates fixes o. Shrinking finitely many times separates the neighborhood from all of them. The quotient map is a local biholomorphism, so its restriction to V is a chart of the original source. Choose a polydisc about 0 whose closure lies in $\mathcal{F}(V)$. For sufficiently small r_0, both T_{r_0} and

$$\{|w_0| \le \rho_0,\ |w_1| = \rho_1,\ |w_2| = \rho_2\} \tag{234}$$

lie in this polydisc. This proves that the whole inverse image embeds in one chart and supplies space for the bounding manifold below.

2. The actual normal framing, including its angle dependence. Put $L_j = \log |w_j|$, and use the unique real lift of the quotient coordinate near u^0. From the exact change of coordinates,

$$\begin{pmatrix} L_1 \\ L_2 \end{pmatrix} = \ell B_0 u^0 + A(t)(u - u^0), \qquad L_0 = \ell - L_1 - L_2. \tag{235}$$

Differentiating on $u = u^0$ annihilates the term $(dA(t))(u - u^0)$. Hence no derivative of $C(t)$ has been omitted: the exact differential there is

$$\begin{pmatrix} dL_0 \\ dL_1 \\ dL_2 \end{pmatrix} = J(t) \begin{pmatrix} d\ell \\ du_1 \\ du_2 \end{pmatrix}, \quad J(t) = \begin{pmatrix} 1/2 & -A_{11} - A_{21} & -A_{12} - A_{22} \\ 1/4 & A_{11} & A_{12} \\ 1/4 & A_{21} & A_{22} \end{pmatrix}. \tag{236}$$

Adding the second and third rows to the first gives $(1, 0, 0)$, and expansion in that row proves $\det J(t) = \det A(t)$.

Let $D = Dc_{\overline{u}^0}$ be the actual two-by-two derivative in the ordered open-cell target coordinates (c_1, c_2). It is a fixed invertible matrix; all its entries and signs are retained. Let

$$H = \begin{pmatrix} r_0 & 0 & 0 \\ 0 & D_{11} & D_{12} \\ 0 & D_{21} & D_{22} \end{pmatrix}.$$

Let n_j be the real radial vector w_j in the j-th complex coordinate, with other components zero. Then $dL_i(n_j) = \delta_{ij}$. Writing $N = (n_0, n_1, n_2)$ as the matrix with these vector columns, the actual dg-dual normal framing is, in the normal quotient bundle,

$$\mathfrak{n}(t) = NJ(t)H^{-1}. \tag{237}$$

Indeed, $dg(N) = HJ(t)^{-1}$, so multiplication by $J(t)H^{-1}$ gives the identity. In the Euclidean metric of this chart the radial vectors are perpendicular to the product torus; a different source metric changes their representatives by the canonical projection to its normal bundle and preserves this normal-quotient framing.

Put $\ell_0 = \log r_0$. Decrease r_0 further, if necessary, so that $|\ell_0| > 2\sup_{|t|\le r_0} \|R(t)\|$. For $0 \le \sigma \le 1$, define only a homotopy of frames by

$$A_\sigma(t) = \ell_0 B_0 + \sigma R(t),$$

and let $J_\sigma(t)$ be (236) with A_σ in place of A. For every $v \in \mathbb{R}^2$,

$$\|A_\sigma(t)v\| \ge (|\ell_0| - \|R(t)\|)\|v\| > 0 \quad (v \ne 0).$$

Thus all matrices are invertible. At $\sigma = 0$,

$$J_0 = \begin{pmatrix} 1/2 & \ell_0 & -\ell_0 \\ 1/4 & 0 & \ell_0 \\ 1/4 & -\ell_0 & 0 \end{pmatrix}, \qquad \det J_0 = \ell_0^2 > 0. \tag{238}$$

Continuity also proves $\det J_\sigma(t) > 0$ for every σ, t. The homotopy $NJ_\sigma(t)H^{-1}$, run from $\sigma = 1$ to $\sigma = 0$, compares the actual frame with the exact constant-matrix frame NP, where

$$P = J_0 H^{-1}. \tag{239}$$

The original embedded torus and original functions $C(t)$ have not changed. A framing homotopy is a framed cylinder: choose a smooth parameter function constant near its endpoints and use the displayed frames on $T_{r_0} \times [0,1]$.

3. A collared framed nullbordism with the exact constant matrix. Use a new time coordinate $v \in [0,1]$. Define

$$q(v) = \begin{cases} 0, & 0 \le v \le 1/4, \\ \rho_0^2 \exp(4 - 1/(v - 1/4)), & 1/4 < v \le 1. \end{cases} \tag{240}$$

This function is smooth: all right derivatives of $\exp(-1/(v-1/4))$ at $1/4$ vanish. It is strictly increasing for $v > 1/4$, satisfies $q(1/2) = \rho_0^2$, and $q'(1/2) = 16\rho_0^2 > 0$. In the chart times $[0,1]$, put

$$\mathcal{W} = \{|w_0|^2 + q(v) = \rho_0^2, \quad |w_1|^2 = \rho_1^2, \quad |w_2|^2 = \rho_2^2\}. \tag{241}$$

No point has $v > 1/2$. The three defining differentials are independent: the latter two have nonzero gradients in their separate complex factors; the first has nonzero w_0-gradient whenever $w_0 \ne 0$, and at $w_0 = 0$ it has the nonzero v-derivative $q'(1/2)$. Thus $\mathcal{W}$ is a smooth compact four-manifold. Its only

boundary is $T_{r_0} \times \{0\}$, and on $0 \le v \le 1/4$ it is the exact product collar $T_{r_0} \times [0, 1/4]$. The north locus $w_0 = 0, v = 1/2$ is an interior cap: the implicit function theorem expresses v smoothly in terms of $(\operatorname{Re} w_0, \operatorname{Im} w_0)$ there. The first factor is a disk obtained by rotating its radial profile with this smooth cap and one boundary circle. Therefore $\mathcal{W} \cong D^2 \times T^2$. Its spatial coordinates lie in (234), so the complete manifold lies in the injective original chart.

In the Euclidean chart product metric define

$$V_0 = (w_0, 0, 0, q'(v)/2), \qquad V_1 = (0, w_1, 0, 0), \qquad V_2 = (0, 0, w_2, 0).$$

These are one half of the gradients of the three defining equations in (241). They are mutually orthogonal and never zero. For example, if $F_0 = |w_0|^2 + q(v) - \rho_0^2$, then

$$dF_0(V_0) = 2|w_0|^2 + \frac{1}{2} q'(v)^2 > 0$$

at every point of $\mathcal{W}$. On the entire collar $q' = 0$, so $(V_0, V_1, V_2) = (N, 0)$ exactly. Consequently

$$(V_0, V_1, V_2)P \tag{242}$$

is a framing of the normal three-plane bundle of $\mathcal{W}$, and its restriction to the collar is exactly the framing NP from (239), with every original target-coordinate sign retained. Concatenate the framing cylinder of the preceding paragraph with this nullbordism, rescaling the two interval lengths. Their framings agree exactly on the common product collar, so the concatenation is a smooth framed nullbordism of the actual inverse image.

4. The based map represented by this framed inverse image. Choose the injective chart to exclude x_0 if its center is not x_0. If its center is x_0, the constructed nullbordism still excludes x_0, since $|w_1| = \rho_1 > 0$ and $|w_2| = \rho_2 > 0$ everywhere on it. Compactness therefore gives tubular neighborhoods for the framed cylinder and cap disjoint from $\{x_0\} \times [0, 1]$.

The original continuous quotient map is smooth and a submersion near its entire inverse image of y_0. First make it constant in a neighborhood of x_0, by a based homotopy that changes nothing near that inverse image. Explicitly, take a target chart $\kappa : O \to \mathbb{R}^3$ with $\kappa(*) = 0$, whose image is a ball, and $y_0 \notin O$. By continuity choose a neighborhood U of x_0 with $g(U) \subset O$, and a smooth bump function β supported in U and equal to one near x_0. For $x \in U$, replace $g(x)$ during this homotopy by $\kappa^{-1}((1 - a\beta(x))\kappa(g(x)))$, $0 \le a \le 1$, and leave g fixed off U. Convexity of the target chart image makes the formula well-defined, the vanishing of β near the boundary of its support proves continuity of the pasted formula, and $\kappa(g(x_0)) = 0$ makes the homotopy based. It never introduces an inverse image of y_0. Its endpoint is smooth near x_0 and agrees with g near the entire old inverse image.

Denote the endpoint of the preceding homotopy by g_b. Choose one closed tubular neighborhood U_1 of the inverse image, contained in the smooth submersion region and with the entire inverse image contained in $\operatorname{int} U_1$. The compact set $X \setminus \operatorname{int} U_1$ has image under g_b missing y_0. In a fixed target metric put

$$d_0 = \min_{x \in X \setminus \operatorname{int} U_1} d(g_b(x), y_0) > 0.$$

Relative smooth approximation, fixed on the whole closed set U_1 and on a neighborhood of x_0, gives a smooth based representative g_{sm} whose error on $X \setminus \operatorname{int} U_1$ is less than $d_0/2$. It therefore has the same inverse image and differential on U_1, and $d(g_{\mathrm{sm}}(x), y_0) > d_0/2$ on its complement. Thus there is no uncontrolled intervening region and no additional inverse image. Make the approximation also closer than a fixed target injectivity radius; its short geodesics give the based homotopy.

The relative Pontryagin construction now applies to the explicit framed nullbordism. It produces a based homotopy to a map missing y_0; stereographic contraction of $S^3 \setminus \{y_0\} \cong \mathbb{R}^3$ to the image of $*$ is based and contracts that last map. For completeness, the dependence on the framed inverse image in this

argument is precisely the local product-neighborhood construction and the interpolation lemma of [1, §7, Theorem B and Lemma 4, pp. 43, 48–50]. Their relative versions fix the source basepoint because all tubular modifications are away from it and stereographic interpolation between equal values keeps that value fixed. A framed cylinder realizes a framing homotopy; the empty Pontryagin manifold realizes a map missing the chosen regular value. This proves the claimed based null-homotopy of the exact map g. □

Corollary 11.35. *Under the original fixed based identification $X \cong S^6$, the one-copy coefficient is*

$$[g] = 0 \in \pi_6(S^3) \cong \mathbb{Z}/12, \qquad a = 0 \pmod{12}, \qquad [E^6 g] = 0 \in \pi_{12}(S^9) \cong \mathbb{Z}/24.$$

The previously defined homomorphism $\Theta_g : \mathbb{Z}/12 \to \pi_6(S^3)$, $\Theta_g([n]) = n[g]$, and its composite with the marked peripheral character are zero.

Proof. The theorem states that the actual based map is null-homotopic. Precomposition with the fixed based identification preserves a based null-homotopy. Suspension preserves it as well. Thus the unique coefficient of $[g]$ in any generator of $\mathbb{Z}/12$ is zero. For every $[n]$, $\Theta_g([n]) = n \cdot 0 = 0$, and every composite with this homomorphism is zero. □

Exact scope. The proof computes this sphere-valued postcomposition. It does not compute the homotopy class of the earlier conical-target map $q_2 : X \to \mathrm{Susp}(S^1 \times \overline{T}^2)$, and it does not assert a null-homotopy for that different target. Point-set surjectivity of g and all of its stated metric restrictions remain properties of the original map: a homotopy-class computation does not alter the defining formula (227). The retained-angle companion is proved separately in Theorem 30.51: $p_3 q_2 : X \to S^4$ represents $\eta_4\eta_5 \neq 0$, so q_2 is nonnull. Its Hopf smash represents $12\nu_6 \neq 0$ in $\pi_9(S^6)$. These positive computations use the actual four-framed inverse torus of that different postcomposition; they do not reverse or contradict the preceding null-homotopy through $\widehat{p}_2$.

References

[1] J. W. Milnor, *Topology from the Differentiable Viewpoint*, University Press of Virginia, 1965, §7, Theorem B and Lemma 4, pp. 42–50. `https://www.maths.ed.ac.uk/~aar/papers/milnortop.pdf`.

11.5 Fixed-target metric realization and its sharp dimensional scope

This note concerns the codomains of the three based maps

$$q_2^{\wedge 2} : X^{\wedge 2} \cong S^{12} \longrightarrow Z_{12} := \mathcal{A}^{\wedge 2},$$

$$q_{1,\mathrm{cusp}}^{\wedge 4} : X^{\wedge 4} \cong S^{24} \longrightarrow Z_{24}^{\mathcal{B}} := \mathcal{B}^{\wedge 4}, \qquad q_1^{\wedge 4} : X^{\wedge 4} \cong S^{24} \longrightarrow Z_{24}^{\mathcal{C}} := \mathcal{C}^{\wedge 4}.$$

In particular, S^{12} and S^{24} are the sources, not the spaces in which the metric limits occur. We construct one fixed compatible length metric on each codomain which realizes the native four-dimensional flat limits on the target slices. The realization is sharp: after the same rescaling, the full ambient limits have dimensions 8 and 12, and the collapsed smash base point goes to infinity.

Put

$$r_T = e^{-2\pi T}, \qquad J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad Y_T = TJ + E_T,$$

where E_T is the bounded matrix in the radial cusp expansion. Choose and fix T_0 and $C > 0$ so that $\|E_T\|_{\mathrm{op}} \leq C$ for every $T \geq T_0$, in the fixed Euclidean coordinates used below. This notation avoids confusing the error matrix with the quotient circle C_T used below. On the fixed torus

$$\overline{T}^2 = (\overline{\Lambda} \otimes \mathbb{R})/\overline{\Lambda} \cong \mathbb{R}^2/\mathbb{Z}^2$$

let

$$h_T := \iota_T^* g_{Q_T} = dx^{\mathsf{T}} Y_T^{\mathsf{T}} Y_T dx.$$

Thus

$$T^{-2} h_T \longrightarrow h_\infty := dx^{\mathsf{T}} dx \tag{243}$$

in every C^k norm in the fixed flat coordinates. Indeed, the cusp coefficients are holomorphic in $t = e^{-2\pi T}$, so their T-derivatives are exponentially small, while $T^{-1} Y_T = J + T^{-1} E_T$.

Use normalized angular coordinates

$$u, \theta \in \mathbb{R}/\mathbb{Z};$$

u is the argument of the base coordinate and θ is the value of the invariant character. Increase this same T_0 once more, if necessary, so that all radial cusp formulas used below hold for $T \geq T_0$ and $e^{-2\pi T_0} < \epsilon$, and put

$$r_0 = e^{-2\pi T_0}.$$

For $0 < r \leq r_0$, write

$$T(r) := -\frac{1}{2\pi} \log r.$$

Target-slice notation. We use the embeddings $j_T, k_T^{\mathcal{B}}, k_T^{\mathcal{C}}$ defined in theorem 11.15; the normalized angle $u = 0$ used below is the same positive radial base angle denoted by $1 \in S^1$ there.

Theorem 11.36 (Topology-forced collapse for every fixed compatible metric)**.** *Let $D_{12}, D_{24}^{\mathcal{B}}, D_{24}^{\mathcal{C}}$ be arbitrary fixed metrics compatible with the respective compact target topologies, and let $*$ denote the corresponding smash base point. Then*

$$d_H^{D_{12}}\big(j_T(Q_T^2), \{*\}\big) \longrightarrow 0, \tag{244}$$

$$d_H^{D_{24}^{\mathcal{B}}}\big(k_T^{\mathcal{B}}(C_T^4), \{*\}\big) \longrightarrow 0, \tag{245}$$

$$d_H^{D_{24}^{\mathcal{C}}}\big(k_T^{\mathcal{C}}(C_T^4), \{*\}\big) \longrightarrow 0. \tag{246}$$

Consequently, with the metrics restricted from any one fixed ambient target, all three target slices converge in Gromov–Hausdorff topology to a point, not to $\mathbb{R}^4$.

Proof. We prove the first assertion; the other two are identical. Let

$$G : [0, r_0]^2 \times (T^3)^2 \longrightarrow Z_{12}$$

be the product of the two inner-cone quotient maps followed by the smash quotient. It is continuous and $G(0, 0, z_1, z_2) = *$ for every $(z_1, z_2) \in (T^3)^2$. Since the domain is compact, uniform continuity with respect to D_{12} gives

$$\sup_{(z_1, z_2) \in (T^3)^2} D_{12}\big(G(r, r, z_1, z_2), *\big) \longrightarrow 0.$$

The set $j_T(Q_T^2)$ is contained in such a cross-section, which proves the first Hausdorff convergence. Replacing T^3 by T^2 and taking four factors proves the other two. Hausdorff convergence to a singleton implies Gromov–Hausdorff convergence to that singleton. □

The preceding theorem is unrescaled. It does not prevent a blow-up of one fixed metric. The following construction gives the strongest natural positive statement.

Theorem 11.37 (One fixed cusp-adapted metric realizes the vertical $\mathbb{R}^4$ limits)**.** *There are fixed compatible length metrics*

$$D_{12} \text{ on } Z_{12}, \qquad D_{24}^{\mathcal{B}} \text{ on } Z_{24}^{\mathcal{B}}, \qquad D_{24}^{\mathcal{C}} \text{ on } Z_{24}^{\mathcal{C}}$$

with the following properties.

On the inner cone of $\mathcal{A}$, the metric is induced by

$$g_{\mathcal{A}} = dr^2 + r^2 du^2 + \frac{r^2}{T(r)^2} h_{T(r)}. \tag{247}$$

On the cusp cone of either $\mathcal{B}$ or $\mathcal{C}$, it is induced by

$$g_{\mathcal{D}} = dr^2 + r^2(du^2 + d\theta^2), \qquad \mathcal{D} \in \{\mathcal{B}, \mathcal{C}\}. \tag{248}$$

The metrics on the smash powers are the metric quotients of the product length metrics after the closed fat wedge is collapsed. Write $g_{Z_{12}}$ and $g_{Z_{24}^{\mathcal{D}}}$ for the resulting smooth Riemannian tensors on the open regular strata.

Let

$$p_T^{12} := j_T(0,0), \qquad p_T^{24,\mathcal{D}} := k_T^{\mathcal{D}}(0,0,0,0).$$

Put

$$\Sigma_T^{12} := j_T(Q_T^2), \qquad \Sigma_T^{24,\mathcal{D}} := k_T^{\mathcal{D}}(C_T^4).$$

Let $d_{T,12}^{\mathrm{slice}}$ and $d_{T,24,\mathcal{D}}^{\mathrm{slice}}$ be the intrinsic Riemannian path metrics of the first fundamental forms on these embedded slices. These must be distinguished from the restricted ambient distance functions

$$d_{T,12}^{\mathrm{amb}} := D_{12}|_{\Sigma_T^{12} \times \Sigma_T^{12}}, \qquad d_{T,24,\mathcal{D}}^{\mathrm{amb}} := D_{24}^{\mathcal{D}}|_{\Sigma_T^{24,\mathcal{D}} \times \Sigma_T^{24,\mathcal{D}}}.$$

In general the latter can be smaller, because an ambient path is allowed to leave the slice.

Then the induced first fundamental forms on the target slices satisfy the exact identities

$$\left(\frac{T}{r_T}\right)^2 j_T^* g_{Z_{12}} = g_{Q_T} \oplus g_{Q_T}, \tag{249}$$

and

$$\left(\frac{\ell_T}{r_T}\right)^2 (k_T^{\mathcal{D}})^* g_{Z_{24}^{\mathcal{D}}} = g_{C_T}^{\oplus 4}. \tag{250}$$

Consequently,

$$\left(\Sigma_T^{12}, \frac{T}{r_T} d_{T,12}^{\mathrm{slice}}, p_T^{12}\right) \longrightarrow (\mathbb{R}^4, 0), \tag{251}$$

$$\left(\Sigma_T^{24,\mathcal{D}}, \frac{\ell_T}{r_T} d_{T,24,\mathcal{D}}^{\mathrm{slice}}, p_T^{24,\mathcal{D}}\right) \longrightarrow (\mathbb{R}^4, 0) \tag{252}$$

in pointed local-isometric and pointed Gromov–Hausdorff topology, and smoothly on the exhausting Euclidean slice balls.

The full ambient pointed blow-ups are instead

$$\left(Z_{12}, \frac{T}{r_T} D_{12}, p_T^{12}\right) \longrightarrow (\mathbb{R}^8, 0), \tag{253}$$

$$\left(Z_{24}^{\mathcal{D}}, \frac{\ell_T}{r_T} D_{24}^{\mathcal{D}}, p_T^{24,\mathcal{D}}\right) \longrightarrow (\mathbb{R}^{12}, 0). \tag{254}$$

as pointed metric spaces in pointed Gromov–Hausdorff topology. More precisely, for every fixed radius the corresponding rescaled target ball is eventually contained in an open regular-stratum coordinate neighborhood, and the pulled-back Riemannian tensors converge there in every C^k norm. Thus smooth convergence is asserted on an exhaustion by regular-stratum neighborhoods, not globally on the singular compact targets. Under these limits, the slice $\mathbb{R}^4$ is respectively the coordinate subspace obtained by fixing the two radial and two base-angle coordinates, or by fixing the four radial and four base-angle coordinates.

Proof. Equation (243) shows that the metrics on the compact cross-sections of (247) remain uniformly bounded and converge to a positive definite flat metric. Therefore the cross-section diameter is $O(r)$. Moreover the radial coordinate is 1-Lipschitz and a radial segment from level r to the vertex has length r. The length completion consequently adds exactly one point at $r = 0$, and its topology is the prescribed cone topology. The same assertions are immediate for (248). Interpolation on a closed annulus away from the vertex extends these metrics to fixed compatible length metrics on the compact spaces $\mathcal{A}, \mathcal{B}, \mathcal{C}$.

If a compact metric space (E, d) and a closed subset $W \subset E$ are given, the metric quotient which collapses W to one point is

$$d_{E/W}([x], [y]) = \min\{d(x, y), d(x, W) + d(y, W)\}.$$

Applying this to the fat wedges in the product length spaces gives the claimed fixed compatible metrics on the smash targets.

On the r_T-slice of $\mathcal{A}$, with $u = 0$, the first fundamental form is

$$\frac{r_T^2}{T^2} h_T.$$

The definition $h_T = \iota_T^* g_{Q_T}$ proves (249). On the r_T-slice of $\mathcal{B}$ or $\mathcal{C}$, with $u = 0$, the form is $r_T^2 d\theta^2$. Under $\vartheta_T : [\theta] \mapsto [\ell_T \theta]$, the native quotient-circle metric is $\ell_T^2 d\theta^2$. This proves (250). The native systole estimates

$$\operatorname{sys}(Q_T^2) \geq T - C, \qquad \operatorname{sys}(C_T^4) = \ell_T$$

then prove (252): every prescribed radius ball is eventually exactly a Euclidean four-ball.

It remains to compute the ambient limits. In one $\mathcal{A}$-factor, put

$$\lambda_T = \frac{T}{r_T}, \qquad s = \lambda_T (r - r_T), \qquad \eta = Tu, \qquad z = Y_T x$$

on the universal cover of the torus directions. On every fixed bounded set in (s, η, z), one has

$$\frac{r}{r_T} = 1 + \frac{s}{T}, \qquad T(r) = T + O(T^{-1}), \qquad Y_{T(r)} Y_T^{-1} = I + O(T^{-2}),$$

with the same estimates after any fixed number of derivatives. Thus $\lambda_T^2 g_{\mathcal{A}}$ converges in these coordinates to

$$ds^2 + d\eta^2 + dz_1^2 + dz_2^2.$$

The inward radial boundary is at distance T, the u-period is T, and the z-period lattice is $Y_T \mathbb{Z}^2$, whose systole is at least $T - C$. The outward edge of the fixed cusp chart is also at scaled distance

$$\frac{T}{r_T}(r_0 - r_T) = T\left(\frac{r_0}{r_T} - 1\right) \longrightarrow \infty.$$

Thus the inward vertex, all periods, and the outward edge recede to infinity. A single factor converges on exhausting regular-stratum coordinate neighborhoods to $\mathbb{R}^4$, and two factors converge to $\mathbb{R}^8$.

For one $\mathcal{B}$- or $\mathcal{C}$-factor, put

$$\lambda_T = \frac{\ell_T}{r_T}, \qquad s = \lambda_T (r - r_T), \qquad \eta = \ell_T u, \qquad \zeta = \ell_T \theta.$$

Then

$$\lambda_T^2 g_{\mathcal{D}} = ds^2 + \left(\frac{r}{r_T}\right)^2 (d\eta^2 + d\zeta^2),$$

which converges smoothly on bounded coordinate sets to the Euclidean metric on $\mathbb{R}^3$. The two circle periods and the inward radial boundary are all at scaled distance $\ell_T \to \infty$, while the outward edge is at scaled distance

$$\frac{\ell_T}{r_T}(r_0 - r_T) = \ell_T \left(\frac{r_0}{r_T} - 1 \right) \longrightarrow \infty.$$

Four factors therefore converge on exhausting regular-stratum coordinate neighborhoods to $\mathbb{R}^{12}$.

Finally, a level-r_T factor point has distance exactly r_T from its cone vertex. Indeed, the radial segment has that length. A competing curve contained in the inner chart has length at least r_T, while one which leaves has inner-chart portions of total length at least $(r_0 - r_T) + r_0 > r_T$. The Riemannian-ℓ^2-product distance from a synchronized point to the fat wedge is the minimum of the factor distances, and hence is also r_T.

For the chosen scale put

$$b_T = \lambda_T r_T, \qquad a_T = \lambda_T (r_0 - r_T),$$

and let σ_T be the least scaled period in a factor. Thus $\sigma_T \geq T - C$ in the $\mathcal{A}$ calculation and $\sigma_T = \ell_T$ in the $\mathcal{B}, \mathcal{C}$ calculation. The preceding computations give

$$a_T \longrightarrow \infty, \qquad b_T \longrightarrow \infty, \qquad \sigma_T \longrightarrow \infty.$$

Fix R. A curve of rescaled length at most R from p_T meets neither the fat wedge nor $r = r_0$ for all large T, so the radius-R target ball lies in the open regular stratum. If x, y lie in that ball, a competitor through the fat wedge has length at least

$$(b_T - R) + (b_T - R) > 2R,$$

and a competitor which exits the inner chart and returns has length at least

$$2(a_T - R) > 2R.$$

Every remaining competitor of length at most $2R$ stays in $r/r_T = 1 + o_R(1)$. After lifting it to the angular and torus covers, a nonzero deck translation has length at least $(1 - o_R(1))(\sigma_T - 2R) > 2R$ for all large T. Since the path through p_T has length at most $2R$, none of these three kinds of global shortcut computes the distance between x and y. Thus the metric on every fixed target ball is computed entirely in one lifted regular-stratum chart. The C^∞ convergence of the pulled-back tensors on these exhausting charts implies the pointed Gromov–Hausdorff limits in (254). □

Theorem 11.38 (Complete intrinsic scalar-rescaling trichotomy)**.** *For the fixed metrics of theorem 11.37, let $\lambda_T > 0$ be arbitrary and put*

$$\beta_T := \lambda_T r_T.$$

Choose one of the following triples, retaining the indicated ambient metric:

$$(\Sigma_T, d_T^{\mathrm{slice}}, d_T^{\mathrm{amb}}, p_T; D) = (\Sigma_T^{12}, d_{T,12}^{\mathrm{slice}}, d_{T,12}^{\mathrm{amb}}, p_T^{12}; D_{12}),$$
$$(\Sigma_T, d_T^{\mathrm{slice}}, d_T^{\mathrm{amb}}, p_T; D) = (\Sigma_T^{24,\mathcal{D}}, d_{T,24,\mathcal{D}}^{\mathrm{slice}}, d_{T,24,\mathcal{D}}^{\mathrm{amb}}, p_T^{24,\mathcal{D}}; D_{24}^{\mathcal{D}}), \qquad \mathcal{D} \in \{\mathcal{B}, \mathcal{C}\}.$$

After passing to a subsequence so that β_T has a limit in $[0, \infty]$, the intrinsic pointed slice limit is

$$(\Sigma_T, \lambda_T d_T^{\mathrm{slice}}, p_T) \longrightarrow \begin{cases} \{\mathrm{pt}\}, & \beta_T \longrightarrow 0, \\ ((\mathbb{R}/L\mathbb{Z})^4, 0), & \beta_T \longrightarrow L \in (0, \infty), \\ (\mathbb{R}^4, 0), & \beta_T \longrightarrow \infty. \end{cases} \tag{255}$$

The convergence in (255) *is pointed Gromov–Hausdorff in all three cases. In the finite positive case the flat metrics converge smoothly under the fixed marked-torus identifications; in the unrolling case the*

convergence is pointed local-isometric and smooth on the exhausting Euclidean balls. In the middle line the lattice is orthogonally equivalent to the square lattice $L\mathbb{Z}^4$. At the same time,

$$\lambda_T D(p_T, *) = \beta_T \tag{256}$$

for every one of the displayed choices.

Only in the unrolling regime $\beta_T \to \infty$ do we compare the intrinsic metric with the restricted ambient distance. In that regime, for every $R \in (0, \infty)$,

$$\sup_{x,y \in \overline{B}_{\lambda_T d_T^{\mathrm{slice}}}(p_T, R)} \left| \lambda_T d_T^{\mathrm{amb}}(x, y) - \lambda_T d_T^{\mathrm{slice}}(x, y) \right| \longrightarrow 0. \tag{257}$$

Moreover, there are $\varepsilon_{T,R} \to 0$ such that

$$\overline{B}_{\lambda_T d_T^{\mathrm{slice}}}(p_T, R) \subset \overline{B}_{\lambda_T d_T^{\mathrm{amb}}}(p_T, R) \subset \overline{B}_{\lambda_T d_T^{\mathrm{slice}}}(p_T, (1 + \varepsilon_{T,R})R). \tag{258}$$

Thus the restricted ambient metric has the same pointed $\mathbb{R}^4$ limit on these fixed slice balls. No agreement is asserted in the regimes $\beta_T \to 0$ or $\beta_T \to L \in (0, \infty)$, where radial or fat-wedge shortcuts need not disappear. Every rescaling which unrolls the intrinsic slice sends the cone/fat-wedge base point to infinity, so no such limit is a tangent cone based at that point.

In the same regime $\beta_T \to \infty$, the corresponding full ambient bubbles are

$$(Z_{12}, \lambda_T D_{12}, p_T^{12}) \longrightarrow (\mathbb{R}^8, 0), \tag{259}$$

$$(Z_{24}^{\mathcal{D}}, \lambda_T D_{24}^{\mathcal{D}}, p_T^{24,\mathcal{D}}) \longrightarrow (\mathbb{R}^{12}, 0), \qquad \mathcal{D} \in \{\mathcal{B}, \mathcal{C}\}. \tag{260}$$

These are pointed Gromov–Hausdorff limits of the singular metric targets. The convergence of tensors is C^∞ only under coordinate identifications on exhausting open regular-stratum neighborhoods; no global smooth convergence of the singular targets is claimed.

Proof. For the Q_T^2 route, the rescaled period matrix in each two-dimensional factor is

$$\frac{\lambda_T r_T}{T} Y_T = \beta_T \left(J + \frac{E_T}{T} \right).$$

If $\beta_T \to L \in (0, \infty)$, this lattice converges to $LJ\mathbb{Z}^2$; two factors give the square four-torus in (255). If $\beta_T \to 0$, the lengths of a lattice basis tend to zero, so the diameter tends to zero. If $\beta_T \to \infty$, the reverse triangle inequality gives systole

$$\frac{\beta_T}{T}(T - C) = \beta_T \left(1 - \frac{C}{T} \right) \longrightarrow \infty,$$

so every fixed-radius ball is eventually Euclidean.

For a C_T^4 slice, the four periods after rescaling have the common length

$$\lambda_T r_T = \beta_T,$$

which gives the same three intrinsic cases directly.

Consider first one factor. A curve from a level-r_T point to the cone vertex which remains in the inner chart has length at least r_T, because the dr^2-term makes r 1-Lipschitz there. If it leaves the chart, it has a first crossing of $r = r_0$ and, before ending at the vertex, a last re-entry through $r = r_0$; its two inner-chart portions have total length at least $(r_0 - r_T) + r_0 > r_T$. The radial segment has length r_T, so the factor distance to its based vertex is exactly r_T. The Riemannian-ℓ^2-product distance from every synchronized slice point to the fat wedge is the minimum of these factor distances and hence is also r_T. The metric-quotient formula in the preceding proof therefore gives (256).

It remains to prove the restricted-ambient comparison, so assume $\beta_T \to \infty$ and fix R. Two points in $\overline{B}_{\lambda_T d_T^{\mathrm{slice}}}(p_T, R)$ have intrinsic distance at most $2R/\lambda_T$. The slice path is an ambient competitor, so

$$d_T^{\mathrm{amb}}(x,y) \le d_T^{\mathrm{slice}}(x,y).$$

The quotient shortcut through the fat wedge has rescaled length $2\lambda_T r_T = 2\beta_T$, and hence is longer than $2R$ for all large T. A path which leaves the fixed cusp chart must reach $r = r_0$. Its outward cost is controlled by

$$\lambda_T(r_0 - r_T) = \beta_T\left(\frac{r_0}{r_T} - 1\right) \longrightarrow \infty, \tag{261}$$

so no path of rescaled length at most $2R$ can use the interpolating annulus or the global part of a factor.

Consider now an ambient competitor of length at most $2R/\lambda_T + o(\lambda_T^{-1})$ which stays in the cusp chart. In every factor its radial coordinate satisfies

$$|r - r_T| \le \frac{2R + o(1)}{\lambda_T}, \qquad \frac{r}{r_T} = 1 + O_R(\beta_T^{-1}).$$

On this tube, the horizontal tensors in the directions tangent to the slice are uniformly $(1+\varepsilon_{T,R})$-bilipschitz to their values at $r = r_T$, where $\varepsilon_{T,R} \to 0$. For $\mathcal{B}$ and $\mathcal{C}$ this follows directly from $(r/r_T)^2 = 1 + O_R(\beta_T^{-1})$. For $\mathcal{A}$, the same conclusion follows, without changing the fixed coordinates or the matrix Y_T, from

$$\begin{aligned} T(r) - T &= O_R(\beta_T^{-1}), \\ \frac{T(r)}{T} &= 1 + O_R((\beta_T T)^{-1}), \\ Y_{T(r)}Y_T^{-1} &= I + O_R((\beta_T T)^{-1}). \end{aligned}$$

Projecting the competitor to the torus or character directions, and discarding its radial and base-angle components, produces a path between the same slice endpoints. The bilipschitz estimate therefore gives

$$d_T^{\mathrm{slice}}(x,y) \le (1+\varepsilon_{T,R}) d_T^{\mathrm{amb}}(x,y).$$

Together with the opposite inequality and $\lambda_T d_T^{\mathrm{slice}}(x,y) \le 2R$, this proves (257). The first inclusion in (258) follows from $d_T^{\mathrm{amb}} \le d_T^{\mathrm{slice}}$. For a point x in the middle ball, apply the same fat-wedge and outward exclusions to a near-minimizing ambient path from p_T to x, whose rescaled length is at most $R + o(1)$, and project that path to the slice directions. This gives

$$\lambda_T d_T^{\mathrm{slice}}(p_T, x) \le (1+\varepsilon_{T,R})R$$

after enlarging $\varepsilon_{T,R}$ by a quantity tending to zero, and proves the second inclusion.

For completeness, the full ambient bubbles at an arbitrary unrolling scale are obtained without changing any coordinate or orientation. In one $\mathcal{A}$-factor use

$$s = \lambda_T(r - r_T), \qquad \eta = \beta_T u, \qquad z = \frac{\beta_T}{T} Y_T x.$$

Writing $A_T(r) = Y_{T(r)}Y_T^{-1}$, one obtains the exact expression

$$\lambda_T^2 g_{\mathcal{A}} = ds^2 + \left(\frac{r}{r_T}\right)^2 d\eta^2 + \left(\frac{r}{r_T}\right)^2 \left(\frac{T}{T(r)}\right)^2 dz^{\mathsf{T}} A_T(r)^{\mathsf{T}} A_T(r) dz.$$

On every fixed bounded coordinate set,

$$\frac{r}{r_T} = 1 + O_R(\beta_T^{-1}), \qquad T(r) - T = O_R(\beta_T^{-1}), \qquad A_T(r) = I + O_R((\beta_T T)^{-1}),$$

with the same estimates after every fixed number of derivatives. Hence the tensor converges smoothly to the Euclidean metric on $\mathbb{R}^4$. The η-period is β_T, and the z-period lattice is

$$\frac{\beta_T}{T}Y_T\mathbb{Z}^2, \qquad \operatorname{sys}\left(\frac{\beta_T}{T}Y_T\mathbb{Z}^2\right) \geq \beta_T\left(1-\frac{C}{T}\right) \longrightarrow \infty.$$

In one $\mathcal{B}$- or $\mathcal{C}$-factor use

$$s = \lambda_T(r - r_T), \qquad \eta = \beta_T u, \qquad \zeta = \beta_T \theta.$$

Then

$$\lambda_T^2 g_{\mathcal{D}} = ds^2 + \left(\frac{r}{r_T}\right)^2 (d\eta^2 + d\zeta^2),$$

and both angular periods are $\beta_T \to \infty$.

In all three cases the inward boundary and the fat wedge are at scaled distance β_T, while a lift starting at radius r_T incurs the one-way radial cost in (261) before its first outward crossing of $r = r_0$. Fix R. For large T, all these quantities and all nonzero period lengths exceed $4R$. If x, y lie in the ambient rescaled radius-R ball, their distances from the fat wedge are at least $\beta_T - R$; hence a quotient path through the smash base point has length at least $2(\beta_T - R) > 2R$. Every remaining competitor lifts to the complement of the fat wedge in the product. If such a lift exits the inner product chart, then some factor has a first crossing of $r = r_0$; the radial variation before that crossing is at least $\lambda_T(r_0 - r_T) - R$, and returning to the lift of y incurs the same lower bound. Its length is therefore greater than $2R$. Finally, a path of length at most $2R$ remains in the tube $r/r_T = 1 + O_R(\beta_T^{-1})$; there every nonzero deck translation has length tending to infinity, so such a path cannot wrap around an angular or torus period. The displayed coordinate neighborhoods therefore exhaust the pointed balls without quotient or global shortcuts. Their smooth tensor convergence proves the pointed Gromov–Hausdorff limits in (260), with smooth convergence confined to this regular-stratum exhaustion. □

Corollary 11.39 (Unit-volume normalization does not give $\mathbb{R}^4$)**.** *The four-volume of Q_T^2 is $(\det Y_T)^2 \sim T^4$. Its unit-volume length factor is therefore $(\det Y_T)^{-1/2} \sim T^{-1}$, and its unit-volume limit is a square flat four-torus, not $\mathbb{R}^4$. Likewise, the four-volume of C_T^4 is ℓ_T^4, and its unit-volume normalization is exactly the square unit four-torus.*

Proposition 11.40 (What vertex tangents and compact asymptotic cones can mean)**.** *The compact target topology alone determines no metric tangent cone. For an exact conic length metric built from fixed piecewise-Riemannian metrics on the links, however, the vertex links have topological and Hausdorff dimensions 7 for Z_{12} and 11 for both Z_{24} targets. The resulting vertex cones have topological and Hausdorff dimensions 8 and 12, and are not $\mathbb{R}^4$.*

More precisely, writing $L_3 = T^3$ and $L_2 = T^2$, one has the based topological models

$$Z_{12} \cong \operatorname{Susp}(L_3 * L_3), \qquad Z_{24}^{\mathcal{B}} \cong \operatorname{Susp}(L_2 * L_2 * L_2 * L_2).$$

At the suspension base point their conic links are respectively $L_3 * L_3$ and L_2^{*4}. Also

$$(\mathcal{C}, c_0) \cong (\mathcal{M}/\partial\mathcal{M}, [\partial\mathcal{M}]), \qquad (Z_{24}^{\mathcal{C}}, *) \cong (\mathcal{M}^4/\partial(\mathcal{M}^4), [\partial(\mathcal{M}^4)]),$$

so an exact conic metric at the collapsed boundary has link $\partial(\mathcal{M}^4)$. At the collapsed point of $Z_{24}^{\mathcal{C}}$, the complete integral local-homology calculation is

$$H_i(Z_{24}^{\mathcal{C}}, Z_{24}^{\mathcal{C}} \setminus \{*\}; \mathbb{Z}) \cong \begin{cases} \mathbb{Z}^{\binom{4}{i-1}}, & 2 \leq i \leq 5, \\ 0, & i = 6, 7, \\ \mathbb{Z}^{\binom{4}{i-8}}, & 8 \leq i \leq 12, \\ 0, & \text{otherwise.} \end{cases} \tag{262}$$

In particular, its local homology is not concentrated in top degree. For the first two targets the top local homology occurs in degree 8 *and* 12*, respectively; for the third target* (262) *includes a top* $\mathbb{Z}$ *in degree* 12*. None has the local homology of* $\mathbb{R}^4$*.*

Finally, the standard asymptotic cone obtained by blowing down any fixed compact metric target is a point. Indeed, if $R_n \to \infty$*, then*

$$\operatorname{diam}(Z, R_n^{-1} D) = R_n^{-1} \operatorname{diam}(Z, D) \longrightarrow 0.$$

Thus the limits in theorem 11.37 are moving-base-point blow-ups, often called bubbles. They are neither vertex-based tangent cones nor asymptotic cones of the compact targets.

Proof. For compact spaces V_i with closed subspaces A_i, there is a canonical homeomorphism

$$(V_1/A_1) \wedge \cdots \wedge (V_m/A_m) \cong \frac{V_1 \times \cdots \times V_m}{\bigcup_i V_1 \times \cdots \times A_i \times \cdots \times V_m}.$$

Take $V_i = \operatorname{Cone}(L_i)$ and $A_i = L_i$, its outer boundary. The product of m cones is homeomorphic to the cone on the iterated join $L_1 * \cdots * L_m$, and its displayed union of faces is the outer link. Collapsing that link gives its suspension. This proves the first two models and the link dimensions

$$\dim(T^3 * T^3) = 3 + 3 + 1 = 7, \qquad \dim((T^2)^{*4}) = 4 \cdot 2 + 3 = 11.$$

We next prove the based homeomorphism for $\mathcal{C}$, rather than identifying the two presentations without a map. Put $L = \partial\mathcal{M}$, write the attached cone as

$$\operatorname{Cone}(L) = L \times [0,1]/(L \times \{0\}),$$

with base $L \times \{1\}$, and choose a collar $\kappa : L \times [0,1] \hookrightarrow \mathcal{M}$ satisfying $\kappa(x,0) = x$. There is a homeomorphism

$$\Phi : \mathcal{C} \setminus \{c_0\} \longrightarrow \operatorname{int} \mathcal{M}$$

given on the punctured cone, the inner collar, and the remaining core by

$$\Phi([x,t]) = \kappa(x, t/e) \quad (0 < t \le 1), \qquad \Phi(\kappa(x,s)) = \kappa(x, e^{s-1}) \quad (0 \le s \le 1),$$

$$\Phi(q) = q \quad \text{for} \quad q \in \mathcal{M} \setminus \kappa(L \times [0,1)).$$

The first two formulas agree at the glued points $[x,1] = \kappa(x,0)$, and the second and third agree at $\kappa(x,1)$. Their collar-coordinate images are respectively $L \times (0, e^{-1}]$ and $L \times [e^{-1}, 1]$; hence the displayed piecewise map and its piecewise inverse are continuous.

The compact Hausdorff spaces $\mathcal{C}$ and $\mathcal{M}/L$ are both one-point compactifications of $\operatorname{int}\mathcal{M}$: for $\mathcal{C}$ use Φ, and for $\mathcal{M}/L$ use the quotient map on $\mathcal{M} \setminus L = \operatorname{int}\mathcal{M}$. Uniqueness of the one-point compactification therefore extends Φ to a based homeomorphism

$$(\mathcal{C}, c_0) \xrightarrow{\cong} (\mathcal{M}/\partial\mathcal{M}, [\partial\mathcal{M}]). \tag{263}$$

Only after establishing (263) do we apply the quotient identity at the start of the proof. It gives

$$\mathcal{C}^{\wedge 4} \cong \frac{\mathcal{M}^4}{\bigcup_{a=1}^{4} \mathcal{M}^{a-1} \times \partial\mathcal{M} \times \mathcal{M}^{4-a}} = \mathcal{M}^4/\partial(\mathcal{M}^4),$$

and carries the smash base point to the collapsed boundary.

Put $N = \mathcal{M}^4$. Rounding its product corners preserves the homeomorphism type of the pair $(N, \partial N)$ and makes ∂N an eleven-dimensional closed manifold. A boundary collar descends in $N/\partial N$ to a cone neighborhood of the collapsed point, so the local link is exactly ∂N and

$$H_i(N/\partial N, (N/\partial N) \setminus \{*\}; \mathbb{Z}) \cong \widetilde{H}_{i-1}(\partial N; \mathbb{Z}). \tag{264}$$

The integral homology calculation in the proof of theorem 11.14 for the specific compact orientable three-manifold $\mathcal{M}$ gives

$$H_j(\mathcal{M}; \mathbb{Z}) \cong \begin{cases} \mathbb{Z}, & j = 0, 1, \\ 0, & \text{otherwise.} \end{cases}$$

All groups are free, so the integral Kunneth theorem gives

$$H_j(N; \mathbb{Z}) \cong \begin{cases} \mathbb{Z}^{\binom{4}{j}}, & 0 \le j \le 4, \\ 0, & \text{otherwise.} \end{cases}$$

Poincare–Lefschetz duality, with the product orientation retained, gives

$$H_j(N, \partial N; \mathbb{Z}) \cong H^{12-j}(N; \mathbb{Z}) \cong \begin{cases} \mathbb{Z}^{\binom{4}{j-8}}, & 8 \le j \le 12, \\ 0, & \text{otherwise.} \end{cases}$$

The long exact sequence of $(N, \partial N)$ now has disjoint absolute and relative degree ranges. It gives

$$\widetilde{H}_j(\partial N; \mathbb{Z}) \cong \begin{cases} \mathbb{Z}^{\binom{4}{j}}, & 1 \le j \le 4, \\ 0, & j = 5, 6, \\ \mathbb{Z}^{\binom{4}{j-7}}, & 7 \le j \le 11, \\ 0, & \text{otherwise.} \end{cases}$$

Substitution in (264) proves (262).

An exact metric cone is invariant under every radial blow-up at its vertex, and its Hausdorff dimension is one plus that of its fixed piecewise-Riemannian link. For the first target,

$$H_8\big(\mathrm{Cone}(T^3 * T^3), T^3 * T^3; \mathbb{Z}\big) \cong \mathbb{Z};$$

the analogous top local group for the $\mathcal{B}$ target is $\mathbb{Z}$ in degree 12, while the full $\mathcal{C}$ calculation is (262). Euclidean $\mathbb{R}^4$ has local homology $\mathbb{Z}$ only in degree 4, so none of these conic vertex models is $\mathbb{R}^4$. The diameter calculation proves the final assertion. □

Consequences for the three sphere routes. The exact positive statement is that one fixed metric on each compact codomain contains the native $\mathbb{R}^4$ decompactification as a vertical four-plane in a moving-point Euclidean bubble. The whole bubbles are $\mathbb{R}^8$ for the S^{12} route and $\mathbb{R}^{12}$ for the S^{24} routes. The factor $\beta_T = \lambda_T r_T$ proves that this dimension discrepancy cannot be removed by a different scalar rescaling of the same fixed cusp-adapted metric: the slice unrolls only when the smash base point has already gone to infinite scaled distance.

No metric on a source sphere is supplied by these quotient constructions. If an independent smooth Riemannian metric is placed on S^{12} or S^{24}, its ordinary tangent spaces are $\mathbb{R}^{12}$ or $\mathbb{R}^{24}$. Pulling a codomain metric back through $q_2^{\wedge 2}$ or $q_1^{\wedge 4}$, on the regular strata where the maps are differentiable, gives a degenerate quadratic form along the positive-dimensional quotient kernels, not a Riemannian source metric.

The moving four-sphere carriers

$$\mathbb{S}^4_{12,T} = S^2_T \wedge S^2_T \subset S^6, \qquad \mathbb{S}^4_{24,T} = C_T^{\wedge 4} \subset S^8$$

of theorem 11.20 carry separately assigned metrics. Namely, S^2_T first receives the quotient length metric obtained from $(\overline{T}^2, h_T) \cong (Q_T, g_{Q_T})$ by collapsing $K^{(1)} \subset \overline{T}^2$, equivalently by collapsing $\iota_T(K^{(1)}) \subset Q_T$. Then $\mathbb{S}^4_{12,T}$ receives the quotient length metric of the Riemannian ℓ^2-product $(S^2_T)^2$ after its fat wedge is collapsed. Likewise, $\mathbb{S}^4_{24,T}$ receives the quotient length metric of the Riemannian ℓ^2-product C^4_T after its fat wedge is collapsed. These metrics are assigned directly to the moving carriers; they are unrescaled intrinsic quotient metrics, not restrictions of D_{12} or $D^{\mathcal{B}}_{24}$, whose domains are the conical targets, and no fixed ambient metric on S^6 or S^8 is used to define them. Their Euclidean balls are based at the midpoint classes

$$\begin{aligned} y_* &= [(1/2, 1/2)] \in \overline{T}^2, \qquad z_T = c(y_*), \\ \zeta_{12,T} &= [z_T, z_T], \\ \zeta_{24,T} &= [(\ell_T/2, \ell_T/2, \ell_T/2, \ell_T/2)]. \end{aligned}$$

not at the images of p^{12}_T or $p^{24,\mathcal{B}}_T$, which belong to the subsequently collapsed loci. In particular, the trichotomy in theorem 11.38 is the trichotomy of the uncollapsed intrinsic flat-torus slices; it is not a finite-β_T description of the singular quotient-metric moving four-spheres. Indeed, the restriction of any one compatible metric on the compact sphere S^6 or S^8 has diameter bounded by the diameter of that sphere. By contrast, in the separately assigned moving metrics the midpoint distances to the collapsed loci are at least $(T - C)/2$ and exactly $\ell_T/2$, respectively, and hence tend to infinity. Thus the assigned unrescaled metrics cannot themselves be restrictions of one fixed ambient metric. This does not preclude a separately proved realization after a scalar blow-up of another fixed quotient metric.

Finally, the rank obstruction for a full-base torus quotient must retain its actual metric hypothesis: *quotient systole tends to infinity* means the quotient norm induced by the unrescaled Euclidean fibre metric. Without that qualification the assertion is false, since an arbitrary T-dependent scalar multiplication can make the systole of any fixed nonzero quotient tend to infinity.

11.6 The exceptional first-coordinate elliptic quotient

The native real quotient in equation (83) exists throughout the radial cusp. A holomorphic quotient obtained from the first complex coordinate is a different and much more restrictive construction. We determine its locus without replacing it by an ambient coefficient-space path.

For $z \in \mathfrak{h}_z$, write

$$\begin{aligned} v_{\gamma,z} &= (6\mu(z), \beta(z))^{\mathsf{t}}, \quad v_{u,z} = (\tau(z), \mu(z))^{\mathsf{t}}, \\ v_{w,z} &= (1, 0)^{\mathsf{t}}, \qquad\qquad v_{\delta,z} = (0, 1)^{\mathsf{t}}, \end{aligned} \tag{265}$$

and put

$$\mathcal{A}_z := \mathbb{C}^2/\mathbb{Z}\langle v_{\gamma,z}, v_{u,z}, v_{w,z}, v_{\delta,z}\rangle, \qquad E_{\tau(z)} := \mathbb{C}/(\mathbb{Z}\tau(z) + \mathbb{Z}). \tag{266}$$

These are the marked fibres of theorem 10.2 and the elliptic curves selected by the same period τ.

Proposition 11.41 (Exact first-coordinate descent and connected kernel)**.** *The linear map* $\mathrm{pr}_1 : \mathbb{C}^2 \to \mathbb{C}$ *induces a surjective holomorphic homomorphism*

$$\overline{\mathrm{pr}}_{1,z} : \mathcal{A}_z \longrightarrow E_{\tau(z)} \tag{267}$$

if and only if there is a necessarily unique pair $(a, b) \in \mathbb{Z}^2$ *such that*

$$6\mu(z) = a\tau(z) + b. \tag{268}$$

On this locus define

$$\tau_f(z) := a\mu(z) - \beta(z). \tag{269}$$

Then

$$\operatorname{Im} \tau_f(z) = -D_{\mathrm{can}}(z) > 0, \qquad D_{\mathrm{can}}(z) := \operatorname{Im} \beta(z) - \frac{6(\operatorname{Im} \mu(z))^2}{\operatorname{Im} \tau(z)}, \tag{270}$$

and the kernel in equation (267) *is connected and isomorphic to*

$$E_{\tau_f(z)} = \mathbb{C}/(\mathbb{Z} + \mathbb{Z}\tau_f(z)). \tag{271}$$

Proof. The four first coordinates in equation (265) are $6\mu(z), \tau(z), 1, 0$. Thus the linear projection descends precisely when $6\mu(z) \in \mathbb{Z}\tau(z) + \mathbb{Z}$, which is equation (268). The pair is unique because $\operatorname{Im}\tau(z) > 0$. The periods $v_{u,z}$ and $v_{w,z}$ map onto the two generators of the target lattice, so the induced map is surjective.

Replace the marked lattice basis by the unimodular basis whose first vector is

$$v_{\gamma,z} - av_{u,z} - bv_{w,z} = (0, \beta(z) - a\mu(z))^{\mathsf{t}} = (0, -\tau_f(z))^{\mathsf{t}}. \tag{272}$$

Together with $v_{\delta,z} = (0,1)^{\mathsf{t}}$, these are exactly the vertical periods. Indeed, after subtracting integral multiples of $v_{u,z}$ and $v_{w,z}$, every point in the kernel has first coordinate zero; and an integral combination of $v_{u,z}, v_{w,z}$ has first coordinate zero only when both coefficients vanish. The kernel is therefore the connected one-dimensional complex torus in equation (271).

Taking imaginary parts in equation (268) gives $6\operatorname{Im}\mu = a\operatorname{Im}\tau$. Hence

$$\operatorname{Im}\tau_f = a\operatorname{Im}\mu - \operatorname{Im}\beta = \frac{6(\operatorname{Im}\mu)^2}{\operatorname{Im}\tau} - \operatorname{Im}\beta = -D_{\text{can}} > 0$$

by theorem 10.2. □

For $(a,b) \in \mathbb{Z}^2$, define the marked analytic set

$$\mathcal{D}_{a,b} := \{z \in \mathfrak{h}_z : 6\mu(z) - a\tau(z) - b = 0\}, \qquad \mathcal{D} := \bigcup_{(a,b)\in\mathbb{Z}^2} \mathcal{D}_{a,b}. \tag{273}$$

Theorem 11.42 (Monodromy, cusp exclusion, and finiteness of the descent locus)**.** *The transformation laws of theorem 10.2 act on the integer labels in equation* (273) *by*

$$g_0 : (a,b) \longmapsto (a, a+b), \qquad g_2 : (a,b) \longmapsto (-b, 6+a). \tag{274}$$

Consequently $\mathcal{D}$ is invariant under the triangle group and has a well-defined image in the completed orbifold base $B \simeq \mathbb{P}^1$. That image is finite and is disjoint from some punctured neighborhood of the distinguished cusp p_0.

Proof. Suppose $z \in \mathcal{D}_{a,b}$. From $\tau \circ g_0 = \tau - 1$ and $\mu \circ g_0 = \mu$,

$$6\mu(g_0 z) = a(\tau(g_0 z) + 1) + b = a\tau(g_0 z) + (a+b),$$

so $g_0 z \in \mathcal{D}_{a,a+b}$. Similarly, $\tau \circ g_2 = -1/\tau$ and $\mu \circ g_2 = 1 + \mu/\tau$ give

$$6\mu(g_2 z) = 6 + a + \frac{b}{\tau(z)} = -b\tau(g_2 z) + (6+a),$$

so $g_2 z \in \mathcal{D}_{-b,6+a}$. Both label maps are bijections; and g_0, g_2 generate the triangle group because $g_0 = (g_1 g_2)^{-1}$. This proves invariance.

We next exclude the cusp with the integer labels allowed to vary. If no punctured cusp neighborhood were disjoint from $\mathcal{D}$, there would be points $z_j \in \mathcal{D}_{a_j,b_j}$ in the distinguished cusp component with $t_c(z_j) \to 0$. Put $T_j = \operatorname{Im}\tau(z_j)$ and $m_j = \operatorname{Im}\mu(z_j)$. The cusp construction in theorem 10.2 and the formulas preceding equation (27) give $T_j \to \infty$ while m_j stays bounded. Taking imaginary parts yields

$$a_j = \frac{6m_j}{T_j} \longrightarrow 0. \tag{275}$$

Because a_j is integral, $a_j = 0$ for all sufficiently large j. Then $6\mu(z_j) = b_j$. The cusp extension of μ is bounded, so the integers b_j are bounded; after taking a subsequence, they equal one integer b. The holomorphic function $6\mu - b$ extends across $t_c = 0$ and has zeros accumulating there. It therefore vanishes on the cusp disc and, by the identity theorem, on the connected surface $\mathfrak{h}_z$.

More generally, no identity $6\mu = a\tau + b$ with fixed integers a, b can hold on $\mathfrak{h}_z$. Applying g_0 to such an identity and subtracting it from the original one gives $a = 0$. Then $\mu = b/6$ is constant, whereas the g_2-law would give

$$\frac{b}{6} = 1 + \frac{b}{6\tau}. \tag{276}$$

This is impossible for the nonconstant map τ. The preceding conclusion $6\mu - b \equiv 0$ is therefore a contradiction, proving cusp exclusion.

Choose a closed fundamental polygon for the triangle group and remove the excluded cusp neighborhood. Its closure K is compact in $\mathfrak{h}_z$, including the two elliptic fixed points. The functions τ, μ are holomorphic there and $\inf_K \operatorname{Im}\tau > 0$. At a point of $\mathcal{D}_{a,b} \cap K$,

$$a = \frac{6\operatorname{Im}\mu}{\operatorname{Im}\tau}, \qquad b = 6\mu - a\tau. \tag{277}$$

Thus only finitely many integer pairs (a, b) occur on K. For each of them, $6\mu - a\tau - b$ is a nonzero holomorphic function on a neighborhood of the compact set K, so its zero set meets K in finitely many points. The $(3, 4, \infty)$ triangle group has one cusp orbit. Hence cusp exclusion holds at every lift of the chosen cusp neighborhood. The set $\mathcal{D}$ is triangle-group invariant by the already proved label permutation, and it is disjoint from all those lifted cusp neighborhoods. Consequently every triangle-group orbit contained in $\mathcal{D}$ meets K. Hence the image of $\mathcal{D}$ in the completed orbifold base is finite. □

Remark 11.43 (Exact scope of the elliptic classification). The result classifies only the quotient obtained from the specified first coordinate and the specified target lattice $\mathbb{Z}\tau + \mathbb{Z}$. Its cusp exclusion does not prove that $\mathcal{A}_z$ has no other elliptic quotient or elliptic subtorus, that its Néron–Severi group vanishes, or that it carries no smooth integral magnetic line bundle. Those are different typed questions. In particular, the first-coordinate elliptic construction cannot supply the gap-closing cusp sequence, whereas the native scalar and classical gauge soft modes proved above do not use this exceptional locus.

Theorem 11.44 (Base-changed cover monodromy and exact momentum sectors)**.** *In the ordered integral basis $(\widehat{\gamma}, \widehat{u}, \widehat{w}, \widehat{\delta})$, put*

$$\mathsf{S}_M := \operatorname{diag}(1, 1, M, M), \qquad \Lambda^{(M)} := \mathsf{S}_M \mathbb{Z}^4. \tag{278}$$

Then $P_T \Lambda^{(M)} = \Lambda_{T,M}$, and the cover in equation (67) *is induced by the lattice inclusion*

$$P_T \mathsf{S}_M \mathbb{Z}^4 \subset P_T \mathbb{Z}^4.$$

Writing $M_0 = I + R$, one has

$$R^2 = 0, \qquad \ker R = \operatorname{im} R = \mathbb{Z}\langle \widehat{w}, \widehat{\delta} \rangle, \tag{279}$$

and

$$M_0^M = I + MR, \qquad \mathsf{S}_M^{-1} M_0^M \mathsf{S}_M = M_0. \tag{280}$$

Thus, after the cyclic base change $t_c = v^M$, the degree-M^2 fibre covers form a regular relative cover over $v \neq 0$. This particular statement proves only punctured descent. The distinct completed local extension, with the required modified source action and the separate degree-M^3 and degree-M^2 maps, is proved in theorem 11.8.

Equip $\widetilde{F}_{T,M}$ with the pullback Euclidean metric. Its Riemannian volume is $M^2 D_T$, so normalized Haar measure is exactly

$$\mathrm{d}\widetilde{\nu}_{T,M} := (M^2 D_T)^{-1} \mathrm{d}\operatorname{vol}_{\widetilde{F}_{T,M}}.$$

Let $\widetilde{\Delta}_{T,M}$ be the nonnegative self-adjoint scalar Laplacian on $L^2(\widetilde{F}_{T,M}, \mathrm{d}\widetilde{\nu}_{T,M})$ obtained from the closed Dirichlet form

$$\widetilde{\mathfrak{q}}_{T,M}[f] := \int_{\widetilde{F}_{T,M}} |\mathrm{d}f|^2 \, \mathrm{d}\widetilde{\nu}_{T,M}, \qquad \mathrm{Dom}(\widetilde{\mathfrak{q}}_{T,M}) = H^1(\widetilde{F}_{T,M}),$$

with operator domain $\mathrm{Dom}(\widetilde{\Delta}_{T,M}) = H^2(\widetilde{F}_{T,M})$. Its complete orthonormal character basis is

$$\widetilde{\chi}_{T,M,k}([x]) = \exp\Big(2\pi i \left\langle P_T^{-\mathsf{T}} \mathsf{S}_M^{-1} k, x \right\rangle\Big), \qquad k \in \mathbb{Z}^4, \tag{281}$$

and

$$\mathrm{Spec}(\widetilde{\Delta}_{T,M}) = \left\{ 4\pi^2 \left| P_T^{-\mathsf{T}} \mathsf{S}_M^{-1} k \right|^2 : k \in \mathbb{Z}^4 \right\} \tag{282}$$

with multiplicity. For this covering map one has

$$c_{T,M}^* \chi_{T,\ell} = \widetilde{\chi}_{T,M,\mathsf{S}_M \ell}. \tag{283}$$

Consequently the pullback, or equivalently deck-invariant, sector consists exactly of indices

$$k \in \mathsf{S}_M \mathbb{Z}^4 \quad \Longleftrightarrow \quad k_3 \equiv k_4 \equiv 0 \pmod{M}. \tag{284}$$

There is a constant $C_{\mathrm{mom}} > 0$, independent of sufficiently large T and of $M \geq 1$, such that the Euclidean covering radius of the cover momentum lattice satisfies

$$\sup_{p \in \mathbb{R}^4} \mathrm{dist}\Big(p, P_T^{-\mathsf{T}} \mathsf{S}_M^{-1} \mathbb{Z}^4\Big) \leq C_{\mathrm{mom}} \left(\frac{1}{T} + \frac{1}{M} \right). \tag{285}$$

In particular, if $T_j, M_j \to \infty$, the cover momentum lattices become locally dense in all of $\mathbb{R}^4$, and every $E \geq 0$ is a limit of scalar eigenvalues in equation (282)*.*

Proof. The equality $P_T \Lambda^{(M)} = \Lambda_{T,M}$ follows by multiplying the last two period columns e_1, e_3 by M. The matrix in theorem 5.1 gives

$$R(a,b,c,d)^{\mathsf{t}} = (0,0,b,-a)^{\mathsf{t}}.$$

This proves equation (279) and $R^2 = 0$. The binomial theorem then gives $M_0^M = I + MR$. For every $(a,b,c,d)^{\mathsf{t}}$,

$$\mathsf{S}_M^{-1} M_0^M \mathsf{S}_M (a,b,c,d)^{\mathsf{t}} = (a, b, c+b, d-a)^{\mathsf{t}} = M_0 (a,b,c,d)^{\mathsf{t}},$$

which proves equation (280). In particular $M_0^M \Lambda^{(M)} = \Lambda^{(M)}$, which is precisely the descent condition after $t_c = v^M$.

The dual of $P_T \mathsf{S}_M \mathbb{Z}^4$ is

$$(P_T \mathsf{S}_M)^{-\mathsf{T}} \mathbb{Z}^4 = P_T^{-\mathsf{T}} \mathsf{S}_M^{-1} \mathbb{Z}^4.$$

The standard Fourier theorem for a flat torus proves equations (281) and (282). Substitution of $k = \mathsf{S}_M \ell$ proves equation (283). A character on the cover descends through $c_{T,M}$ exactly when it is invariant under the deck group, and the displayed equality shows that this occurs exactly for $k \in \mathsf{S}_M \mathbb{Z}^4$, proving equation (284).

It remains to prove the quantitative density claim. Let $b_{j,T,M}$ be the four columns of $P_T^{-\mathsf{T}} \mathsf{S}_M^{-1}$. The exact matrix equation (29), the boundedness of $\alpha_T, \eta_T, c_T, m_T$, and

$$L_T, q_T = O(T), \qquad D_T \asymp T^2$$

give, after increasing a uniform constant C_0,

$$|b_{1,T,M}| + |b_{2,T,M}| \leq \frac{C_0}{T}, \qquad |b_{3,T,M}| + |b_{4,T,M}| \leq \frac{C_0}{M}.$$

Given $p \in \mathbb{R}^4$, express it uniquely as $p = \sum_j t_j b_{j,T,M}$, and choose $n_j \in \mathbb{Z}$ with $|t_j - n_j| \le \frac{1}{2}$. Then

$$\left| p - \sum_j n_j b_{j,T,M} \right| \le \frac{1}{2} \sum_j |b_{j,T,M}| \le \frac{C_0}{2} \left(\frac{1}{T} + \frac{1}{M} \right).$$

This proves equation (285). For $E > 0$, apply it to any p with $4\pi^2|p|^2 = E$; continuity gives eigenvalues tending to E. For $E = 0$, use $k = (1, 0, 0, 0)$, whose momentum norm is $O(T^{-1})$. □

Scope and residual data. The conclusions above use the explicitly declared Euclidean quotient metric. The canonical period theorem proves the lattice family but does not itself specify a Riemannian metric, so no assertion about a different metric follows without an explicit comparison.

The constants H and B determine the first relative T^{-1}-correction. When $B = 0$, they do not by themselves determine which of the two unit pairs is strictly shortest at the next order. The canonical functions do determine it through the exact expression $B_T(2L_T - B_T) - 35m_T^2$ in equation (38); if both $B = 0$ and $m = 0$, its sign can depend on higher Taylor coefficients of b and μ. This is an indeterminacy of the two requested zeroth-order constants, not a missing proof obligation. The proved construction permits an additive constant in β and chooses its imaginary part only by the qualitative requirement that the canonical determinant be negative. It records no numerical additive normalization. Accordingly, the formulas retain the resulting value $B = \operatorname{Im} b(0)$ and do not silently impose a sign on it.

Finally, the degree-M^2 covers in equation (67) are local in the marked radial family, exactly as stated. The sublattice $\mathbb{Z}\langle \widehat{\gamma}, \widehat{u}, M\widehat{w}, M\widehat{\delta} \rangle$ is not invariant under the full cusp monodromy when $M > 1$: the proved monodromy sends $\widehat{\gamma}$ to $\widehat{\gamma} - \widehat{\delta}$ and $\widehat{u}$ to $\widehat{u} + \widehat{w}$. Therefore these particular degree-M^2 covers do not, without additional construction, descend around the entire punctured cusp. Its M-th power does preserve the sublattice, because it adds $-M\widehat{\delta}$ to $\widehat{\gamma}$ and $M\widehat{w}$ to $\widehat{u}$. Equivalently, $M_0^M = I + M(M_0 - I)$. Thus, after the degree-M cyclic base change $t_c = v^M$, the cover forms a regular relative degree-M^2 cover over $v \neq 0$. The pointed-limit proof itself uses no central-fibre extension. Nevertheless, theorem 11.8 proves that a corrected source action extends this cover through $v = 0$ as a finite morphism of completed *local* cusp families. It remains true that no cover over the entire original base, and no cover of the compact threefold X, is used or constructed here.

12 Classical gauge fluctuation operators on the positive radial cusp

Retain the positive radial flat tori $F_T = \mathbb{R}^4/P_T\mathbb{Z}^4$, their Euclidean quotient metrics, and the real quantities $D_T, L_T, q_T, m_T, B_T, H, B$ from theorems 11.1 and 11.2. Thus

$$D_T = L_T q_T + 6m_T^2 > 0, \qquad L_T = \operatorname{Im} \tau_T, \qquad q_T = L_T - B_T = -\operatorname{Im} \beta_T,$$

and D_T is the positive covolume. Let

$$\mathbf{e}_\gamma = (1, 0, 0, 0)^{\mathsf{t}}, \qquad \mathbf{e}_u = (0, 1, 0, 0)^{\mathsf{t}};$$

then the already-proved dual covectors are

$$\ell_{\gamma,T} = P_T^{-\mathsf{T}} \mathbf{e}_\gamma, \qquad \ell_{u,T} = P_T^{-\mathsf{T}} \mathbf{e}_u,$$

with exact squared lengths

$$a_{\gamma,T} := \|\ell_{\gamma,T}\|^2 = \frac{L_T^2 + m_T^2}{D_T^2}, \qquad a_{u,T} := \|\ell_{u,T}\|^2 = \frac{q_T^2 + 36m_T^2}{D_T^2}. \tag{286}$$

For all sufficiently large T, the shortest nonzero dual vectors are precisely the minimizing members of $\{\pm\ell_{\gamma,T}, \pm\ell_{u,T}\}$. Put

$$G_T = P_T^{\mathsf{T}} P_T. \tag{287}$$

The matrix G_T is the tangent Gram matrix in the lattice coordinates $y \in \mathbb{R}^4/\mathbb{Z}^4$, where $x = P_T y$.

12.1 The continuum Yang–Mills Hessian

Let G be a compact Lie group with real Lie algebra $\mathfrak{g}$. Fix a positive-definite $\mathrm{Ad}(G)$-invariant inner product on $\mathfrak{g}$, put

$$d_G = \dim_{\mathbb{R}} \mathfrak{g}, \qquad \mathfrak{g}_{\mathbb{C}} = \mathfrak{g} \otimes_{\mathbb{R}} \mathbb{C},$$

and fix a coupling $g_{\mathrm{YM}} > 0$. On the trivial principal bundle

$$\mathcal{P}_T = F_T \times G \longrightarrow F_T$$

let A_0 be the product connection. The Yang–Mills functional is normalized by

$$\mathrm{YM}_T(A) = \frac{1}{2g_{\mathrm{YM}}^2} \int_{F_T} |F_A|^2 \, \mathrm{d}\,\mathrm{vol}_{F_T} . \tag{288}$$

Proposition 12.1 (Hessian at the product connection)**.** *For smooth $a, b \in \Omega^1(F_T; \mathfrak{g})$,*

$$\mathrm{Hess}_{A_0} \mathrm{YM}_T(a, b) = \frac{1}{g_{\mathrm{YM}}^2} \langle \mathrm{d}a, \mathrm{d}b \rangle_{L^2}. \tag{289}$$

Relative to the unscaled L^2 inner product on one-forms, the associated closed Hessian form is

$$\mathfrak{h}_T[a] = g_{\mathrm{YM}}^{-2} \|\mathrm{d}_1 a\|_{L^2}^2, \qquad \mathrm{Dom}(\mathfrak{h}_T) = \mathrm{Dom}(\mathrm{d}_1), \tag{290}$$

on $L^2\Omega^1(F_T; \mathfrak{g})$, where d_1 is the maximal closed exterior derivative

$$\mathrm{Dom}(\mathrm{d}_1) = \{a \in L^2\Omega^1(F_T; \mathfrak{g}) : \mathrm{d}a \in L^2\Omega^2(F_T; \mathfrak{g}) \ \textit{distributionally}\}.$$

The associated nonnegative self-adjoint operator is

$$\mathcal{L}_T = g_{\mathrm{YM}}^{-2} \mathrm{d}_1^* \mathrm{d}_1, \qquad \mathrm{Dom}(\mathcal{L}_T) = \{a \in \mathrm{Dom}(\mathrm{d}_1) : \mathrm{d}_1 a \in \mathrm{Dom}(\mathrm{d}_1^*)\}. \tag{291}$$

Its differential expression on smooth one-forms is $g_{\mathrm{YM}}^{-2} \mathrm{d}^* \mathrm{d}$.

Proof. With the convention $F_{A_0 + a} = \mathrm{d}a + \frac{1}{2}[a \wedge a]$, one has

$$F_{A_0 + sa + tb} = s\,\mathrm{d}a + t\,\mathrm{d}b + O((|s| + |t|)^2).$$

The coefficient of st in the squared norm is $2\langle \mathrm{d}a, \mathrm{d}b \rangle_{L^2}$. Taking the mixed derivative in equation (288) proves equation (289). The maximal distributional exterior derivative is closed and densely defined. Hence its graph-norm square is a closed nonnegative form, and the first representation theorem identifies its operator with $g_{\mathrm{YM}}^{-2} \mathrm{d}_1^* \mathrm{d}_1$ on precisely the domain displayed in equation (291). □

12.2 The finite cubical cochain complex

Fix $N \geq 2$ and put

$$\Lambda_N = (\mathbb{Z}/N\mathbb{Z})^4.$$

The vertex $n \in \Lambda_N$ is realized in F_T as the class of $P_T(n/N)$. Let $E = (\mathbb{R}^4)^*$, with its Euclidean inner product, and let $\mathbf{e}_1, \ldots, \mathbf{e}_4$ be the standard coordinate columns in the lattice-coordinate space, also used as the coefficient columns of its dual coordinate basis. Define the physical coframe

$$\vartheta_T^j = P_T^{-\mathsf{T}} \mathbf{e}_j \in E, \qquad 1 \leq j \leq 4. \tag{292}$$

Thus $\vartheta_T^j = \mathrm{d}y^j$ in physical x-coordinates, and the Gram matrix of this coframe is G_T^{-1}.

For $0 \le p \le 4$, define

$$C_N^p(T;\mathfrak{g}_{\mathbb{C}}) = \operatorname{Map}\left(\Lambda_N, \bigwedge^p E_{\mathbb{C}} \otimes \mathfrak{g}_{\mathbb{C}}\right). \tag{293}$$

These spaces are finite-dimensional. Every finite differential, adjoint, Hodge operator, projection, and reduced Hessian below therefore acts on its entire displayed finite-dimensional source space; no additional operator domain is implicit. The coefficient of $\vartheta_T^{i_1} \wedge \cdots \wedge \vartheta_T^{i_p}$ at n is the coefficient assigned to the positively oriented coordinate p-cell based at n. Let

$$(\mathsf{T}_j a)(n) = a(n + \mathbf{e}_j)$$

and define the forward and backward differences by

$$D_{N,j}^+ = N(\mathsf{T}_j - \mathrm{id}), \qquad (D_{N,j}^- a)(n) = N\big(a(n) - a(n - \mathbf{e}_j)\big). \tag{294}$$

The finite exterior derivative is

$$\mathrm{d}_N a = \sum_{j=1}^{4} \varepsilon(\vartheta_T^j) D_{N,j}^+ a. \tag{295}$$

Lemma 12.2 (Cochain property and induced exterior metric)**.** *The maps in*

$$0 \longrightarrow C_N^0 \xrightarrow{\mathrm{d}_N} C_N^1 \xrightarrow{\mathrm{d}_N} C_N^2 \xrightarrow{\mathrm{d}_N} C_N^3 \xrightarrow{\mathrm{d}_N} C_N^4 \longrightarrow 0 \tag{296}$$

form a cochain complex. Equip C_N^p with

$$\langle a, b\rangle_{N,p,T} = \frac{D_T}{N^4} \sum_{n \in \Lambda_N} \langle a(n), b(n)\rangle_{\wedge^p E \otimes \mathfrak{g}_{\mathbb{C}}}. \tag{297}$$

For increasing multi-indices $I = (i_1, \dots, i_p)$ and $J = (j_1, \dots, j_p)$, the exterior metric is exactly

$$\left\langle \vartheta_T^{i_1} \wedge \cdots \wedge \vartheta_T^{i_p}, \vartheta_T^{j_1} \wedge \cdots \wedge \vartheta_T^{j_p} \right\rangle = \det\left((G_T^{-1})_{i_a j_b} \right)_{a,b=1}^{p}. \tag{298}$$

Proof. The differences $D_{N,j}^+$ commute, whereas the exterior multiplications $\varepsilon(\vartheta_T^j)$ anticommute. Therefore

$$\mathrm{d}_N^2 = \sum_{j,\ell} \varepsilon(\vartheta_T^j)\varepsilon(\vartheta_T^\ell) D_{N,j}^+ D_{N,\ell}^+ = 0,$$

which proves that equation (296) is a complex. The factor D_T/N^4 in equation (297) is the exact Riemann-sum weight: D_T is the covolume of $P_T\mathbb{Z}^4$. The Euclidean inner product of $\vartheta_T^i = P_T^{-\mathsf{T}}\mathbf{e}_i$ and $\vartheta_T^j = P_T^{-\mathsf{T}}\mathbf{e}_j$ is $(G_T^{-1})_{ij}$. The induced inner product on an exterior power is the determinant of the matrix of pairings, proving equation (298). □

Remark 12.3 (Integrated cellular cochains give the same operator). The convention above stores form coefficients. If $c_I(n)$ instead stores the integral over a p-cell, put $a_I(n) = N^p c_I(n)$. The unscaled cellular coboundary $\mathsf{T}_j - \mathrm{id}$, together with the inner product

$$\langle c, c'\rangle_{N,p,T}^{\mathrm{int}} = D_T N^{2p-4} \sum_{n \in \Lambda_N} \langle c(n), c'(n)\rangle_{\wedge^p E \otimes \mathfrak{g}_{\mathbb{C}}},$$

is carried isometrically to equations (295) and (297). Thus the two conventions give identical Hessian symbols and eigenvalues.

Proposition 12.4 (Adjoint and Fourier wedge–contraction symbols)**.** *The adjoint of* $\mathrm{d}_N : C_N^p \to C_N^{p+1}$ *is*

$$\mathrm{d}_N^* = -\sum_{j=1}^{4} \iota_{(\vartheta_T^j)^\sharp} D_{N,j}^-. \tag{299}$$

For $k \in \Lambda_N$*, define*

$$\begin{aligned} \chi_{N,k}(n) &= \exp(2\pi i k \cdot n/N), \\ \mathfrak{d}_N(k)_j &= N\left(e^{2\pi i k_j/N} - 1\right), \end{aligned} \tag{300}$$

$$\zeta_{N,T}(k) = \sum_{j=1}^{4} \mathfrak{d}_N(k)_j \vartheta_T^j = P_T^{-\mathsf{T}} \mathfrak{d}_N(k) \in E_{\mathbb{C}}. \tag{301}$$

For every constant exterior amplitude α*,*

$$\mathrm{d}_N(\chi_{N,k}\alpha) = \chi_{N,k}\varepsilon(\zeta_{N,T}(k))\alpha, \tag{302}$$

$$\mathrm{d}_N^*(\chi_{N,k}\alpha) = \chi_{N,k}\iota_{\overline{\zeta_{N,T}(k)}^\sharp}\alpha, \tag{303}$$

$$(\mathrm{d}_N\mathrm{d}_N^* + \mathrm{d}_N^*\mathrm{d}_N)(\chi_{N,k}\alpha) = |\zeta_{N,T}(k)|^2\chi_{N,k}\alpha. \tag{304}$$

Consequently the harmonic p*-cochains are exactly the constant cochains and have complex dimension* $\binom{4}{p} d_G$*.*

Proof. Translation is unitary for equation (297); hence

$$(D_{N,j}^+)^* = N(\mathsf{T}_j^{-1} - \mathrm{id}) = -D_{N,j}^-.$$

For the induced exterior metric, the adjoint of exterior multiplication by ϑ_T^j is contraction by $(\vartheta_T^j)^\sharp$. This proves equation (299). On $\chi_{N,k}$, the forward difference has eigenvalue $\mathfrak{d}_N(k)_j$, whereas $-D_{N,j}^-$ has eigenvalue $\overline{\mathfrak{d}_N(k)_j}$. Summing proves equations (302) and (303). The Hermitian exterior-algebra identity

$$\iota_{\overline{\zeta}^\sharp}\varepsilon(\zeta) + \varepsilon(\zeta)\iota_{\overline{\zeta}^\sharp} = |\zeta|^2\,\mathrm{id}$$

proves equation (304). Finally, $\mathfrak{d}_N(k) = 0$ exactly when $k = 0$ in Λ_N, and P_T is invertible. Thus the Hodge symbol vanishes exactly on the zero Fourier mode, where the entire constant exterior amplitude is harmonic. □

Define the finite reduced Coulomb slice by

$$\mathscr{C}_{N,T} = \ker(\mathrm{d}_N^* : C_N^1 \to C_N^0) \cap (\ker \Delta_{N,T}^{(1)})^\perp, \qquad \Delta_{N,T}^{(1)} = \mathrm{d}_N\mathrm{d}_N^* + \mathrm{d}_N^*\mathrm{d}_N. \tag{305}$$

Equivalently, one deletes the zero Fourier mode and, for each $k \neq 0$, restricts the amplitude to

$$\mathcal{T}_{N,T,k} = \ker \iota_{\overline{\zeta_{N,T}(k)}^\sharp} \subset E_{\mathbb{C}}. \tag{306}$$

The exact orthogonal projection is

$$\Pi_{N,T,k}\alpha = \alpha - \zeta_{N,T}(k)\frac{\iota_{\overline{\zeta_{N,T}(k)}^\sharp}\alpha}{|\zeta_{N,T}(k)|^2}. \tag{307}$$

Thus the projection uses the physical covector $P_T^{-\mathsf{T}}\mathfrak{d}_N(k)$, including every skew-metric cross term.

The finite quadratic gauge form and its associated operator are

$$Q_{N,T}(a) = \frac{1}{2g_{\mathrm{YM}}^2}\|\mathrm{d}_N a\|_{N,2,T}^2, \qquad \mathcal{L}_{N,T} = g_{\mathrm{YM}}^{-2}\mathrm{d}_N^*\mathrm{d}_N. \tag{308}$$

This is the exact linearized, Lie-algebra-valued cubical discretization of equation (289); it is not a nonlinear compact-link Wilson action.

Corollary 12.5 (Exact finite Coulomb spectrum)**.** *On the $k \neq 0$ Fourier fibre of $\mathscr{C}_{N,T}$, $\mathcal{L}_{N,T}$ has the single eigenvalue*

$$\lambda^{\mathrm{YM}}_{N,T}(k) = \frac{1}{g^2_{\mathrm{YM}}} |P_T^{-\mathsf{T}} \mathfrak{d}_N(k)|^2 = \frac{1}{g^2_{\mathrm{YM}}} \mathfrak{d}_N(k)^* G_T^{-1} \mathfrak{d}_N(k) \tag{309}$$

with complex multiplicity $3d_G$.

Proof. On $\mathcal{T}_{N,T,k}$, the contraction term in

$$\iota_{\overline{\zeta}^\sharp} \varepsilon(\zeta)\alpha = |\zeta|^2 \alpha - \varepsilon(\zeta) \iota_{\overline{\zeta}^\sharp} \alpha$$

vanishes. This proves the eigenvalue. Since $\zeta_{N,T}(k) \neq 0$, the contraction has rank one on $E_{\mathbb{C}}$, and its kernel has dimension three. Tensoring with $\mathfrak{g}_{\mathbb{C}}$ gives the stated multiplicity. □

12.3 Exact coordinate-mode eigenvalues and all mesh regimes

Set

$$\rho_N = 4N^2 \sin^2\left(\frac{\pi}{N}\right). \tag{310}$$

Because $\mathfrak{d}_N(\mathbf{e}_\gamma)$ and $\mathfrak{d}_N(\mathbf{e}_u)$ are the common scalar $N(e^{2\pi i/N} - 1)$ times the corresponding coordinate vector, equations (286) and (309) give the exact eigenvalues

$$\lambda^{\mathrm{YM}}_{N,T}(\mathbf{e}_\gamma) = \frac{\rho_N}{g^2_{\mathrm{YM}}} \frac{L_T^2 + m_T^2}{D_T^2}, \tag{311}$$

$$\lambda^{\mathrm{YM}}_{N,T}(\mathbf{e}_u) = \frac{\rho_N}{g^2_{\mathrm{YM}}} \frac{q_T^2 + 36 m_T^2}{D_T^2}. \tag{312}$$

Each individual complex Fourier fibre contributes multiplicity $3d_G$. For real cochains, the pair $\{\mathbf{e}_j, -\mathbf{e}_j\}$ contributes $6d_G$ dimensions when $N > 2$. When $N = 2$, these two labels coincide, so its real contribution is $3d_G$.

Theorem 12.6 (Finite gap and simultaneous cusp–mesh limits)**.** *There is T_1, independent of $N \geq 2$, such that for $T \geq T_1$ the first eigenvalue of $\mathcal{L}_{N,T}$ on $\mathscr{C}_{N,T}$ is*

$$\gamma_{N,T} = \min\left\{ \lambda^{\mathrm{YM}}_{N,T}(\mathbf{e}_\gamma), \lambda^{\mathrm{YM}}_{N,T}(\mathbf{e}_u) \right\}. \tag{313}$$

Uniformly for $N \geq 2$,

$$\gamma_{N,T} = \frac{\rho_N}{g^2_{\mathrm{YM}} T^2} \left(1 + \frac{-2H + 2\min\{B, 0\}}{T} + O(T^{-2})\right). \tag{314}$$

Let

$$I_{N,T} = \{ j \in \{\gamma, u\} : \lambda^{\mathrm{YM}}_{N,T}(\mathbf{e}_j) = \gamma_{N,T} \}, \qquad \eta_N = \begin{cases} 1, & N = 2, \\ 2, & N > 2. \end{cases}$$

The real multiplicity is exactly

$$3 d_G \eta_N \, \# I_{N,T}. \tag{315}$$

The limiting conclusions are:

(i) For fixed $N \geq 2$,

$$T^2 \gamma_{N,T} \longrightarrow \frac{\rho_N}{g^2_{\mathrm{YM}}}, \qquad \gamma_{N,T} \longrightarrow 0. \tag{316}$$

(ii) For every integer-valued $N(T) \geq 2$, with no growth assumption, there are constants $0 < c < C < \infty$, independent of $N(T)$, such that

$$\frac{c}{T^2} \leq \gamma_{N(T),T} \leq \frac{C}{T^2} \tag{317}$$

for all large T. Hence $\gamma_{N(T),T} \to 0$, although the scaled gaps need not converge when $N(T)$ oscillates.

(iii) If $N(T) \to \infty$, then

$$\gamma_{N(T),T} = \frac{4\pi^2}{g_{\mathrm{YM}}^2 T^2} \left(1 + O(T^{-1}) + O(N(T)^{-2})\right). \tag{318}$$

(iv) The largest physical elementary edge has length

$$\max_{1 \leq j \leq 4} \frac{\|P_T \mathbf{e}_j\|}{N(T)} = \frac{T}{N(T)}(1 + O(T^{-1})). \tag{319}$$

Thus the varying continuum torus is resolved if and only if $N(T)/T \to \infty$. This refinement condition is not needed for the algebraic soft-mode conclusion in (ii).

Proof. Let

$$\sigma_N = 2N \sin(\pi/N), \qquad \rho_N = \sigma_N^2.$$

The least nonzero modulus of a coordinate of $\mathfrak{d}_N(k)$ is σ_N, attained precisely at frequency ± 1, with the two signs equal for $N = 2$. If $\mathfrak{d}_N(k)$ is supported in its first two components but $k \notin \{\pm \mathbf{e}_\gamma, \pm \mathbf{e}_u\}$, then

$$|\mathfrak{d}_N(k)_1|^2 + |\mathfrak{d}_N(k)_2|^2 \geq 2\sigma_N^2. \tag{320}$$

Indeed, either both coordinates are nonzero, or the single nonzero frequency is not ± 1. In the latter case

$$\frac{\sin(2\pi/N)}{\sin(\pi/N)} = 2\cos(\pi/N) \geq \sqrt{2}$$

for $N \geq 4$, and the cases $N = 2, 3$ are immediate.

For $z = (z_1, z_2, 0, 0)^{\mathsf{t}} \in \mathbb{C}^4$, the exact inverse transpose in theorem 11.1 gives

$$P_T^{-\mathsf{T}} z = \left(0, \frac{m_T z_1 + q_T z_2}{D_T}, 0, \frac{-L_T z_1 + 6 m_T z_2}{D_T}\right). \tag{321}$$

The scaled 2×2 matrix

$$\frac{T}{D_T} \begin{pmatrix} m_T & q_T \\ -L_T & 6m_T \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} + O_{\mathrm{op}}(T^{-1})$$

has all singular values $1 + O(T^{-1})$, also on complex vectors. Thus the factor $\sqrt{2}$ in equation (320) separates every other soft-plane frequency from the coordinate frequencies uniformly in N for large T.

For a general $z \in \mathbb{C}^4$, the full inverse transpose in theorem 11.1 has first and third components z_3, z_4. Hence a finite frequency with nonzero third or fourth coordinate satisfies

$$\|P_T^{-\mathsf{T}} \mathfrak{d}_N(k)\|^2 \geq |\mathfrak{d}_N(k)_3|^2 + |\mathfrak{d}_N(k)_4|^2 \geq \sigma_N^2,$$

whereas either coordinate candidate is $O(\sigma_N^2/T^2)$. This proves the exact minimizing list and equation (313); the reality discussion before the theorem proves equation (315).

Multiplying the two expansions in theorem 11.2 by $\rho_N / g_{\mathrm{YM}}^2$ proves equation (314). The elementary sine bounds $2x/\pi \leq \sin x \leq x$ for $0 < x \leq \pi/2$ give

$$16 \leq \rho_N \leq 4\pi^2. \tag{322}$$

Together with equation (314), these bounds prove (i) and (ii). The Taylor expansion $\rho_N = 4\pi^2(1 + O(N^{-2}))$ proves (iii).

The first two columns of P_T, displayed in theorem 11.1, have Euclidean lengths $T(1 + O(T^{-1}))$, while its last two columns have length one. Division by $N(T)$ proves equation (319) and (iv). □

Remark 12.7 (Every fixed reduced model remains positively gapped). For each fixed T, the reduced continuum operator has a positive first eigenvalue. For each fixed pair (T, N), the reduced finite operator also has a positive first eigenvalue. What fails is a positive lower bound uniform as $T \to \infty$. Before reduction, gauge-longitudinal directions and constant harmonic one-forms are separate Hessian-zero directions.

12.4 Continuum Coulomb reduction and exact spectrum

The infinitesimal gauge directions $\mathrm{d}\Omega^0(F_T;\mathfrak{g})$ and the harmonic one-forms are Hessian-zero directions. Coulomb gauge removes the former but not the latter. The reduced Coulomb space is therefore

$$\mathscr{C}_T = \left\{ a \in L^2\Omega^1(F_T;\mathfrak{g}) : \mathrm{d}^* a = 0 \text{ distributionally},\ a \perp \mathcal{H}^1(F_T;\mathfrak{g}) \right\}. \tag{323}$$

The subspace $\mathscr{C}_T$ reduces $\mathcal{L}_T$. On it the closed Hessian form has domain $H^1\Omega^1(F_T;\mathfrak{g}) \cap \mathscr{C}_T$, and the restricted self-adjoint operator has domain $H^2\Omega^1(F_T;\mathfrak{g}) \cap \mathscr{C}_T$. Indeed, on the distributionally coclosed slice the graph norm of d_1, together with the L^2-norm, is the Hodge graph norm

$$\|a\|_{L^2}^2 + \|\mathrm{d}a\|_{L^2}^2 + \|\mathrm{d}^* a\|_{L^2}^2.$$

The Gaffney estimate on the compact flat torus identifies its completion with H^1, and elliptic regularity for the Hodge Laplacian identifies the operator domain with H^2. Orthogonality to the finite-dimensional harmonic space is preserved by the Fourier multiplier.

For $k \in \mathbb{Z}^4$, write

$$\xi_{T,k} = P_T^{-\mathsf{T}} k, \qquad \chi_{T,k}([x]) = \exp\bigl(2\pi i \langle \xi_{T,k}, x \rangle\bigr).$$

For a covector ξ, let $\varepsilon(\xi)$ denote exterior multiplication by ξ, and use the Euclidean metric to define $\xi^\sharp$ and contraction $\iota_{\xi^\sharp}$.

Proposition 12.8 (Continuum Fourier symbol, slice, and multiplicity)**.** *For $\alpha \in (\mathbb{R}^4)^*_{\mathbb{C}} \otimes \mathfrak{g}_{\mathbb{C}}$,*

$$\mathrm{d}(\chi_{T,k}\alpha) = 2\pi i\, \chi_{T,k}\varepsilon(\xi_{T,k})\alpha, \tag{324}$$

$$\mathrm{d}^*(\chi_{T,k}\alpha) = -2\pi i\, \chi_{T,k}\iota_{\xi_{T,k}^\sharp}\alpha, \tag{325}$$

$$\mathrm{d}^*\mathrm{d}(\chi_{T,k}\alpha) = 4\pi^2 \chi_{T,k}\left(|\xi_{T,k}|^2 \alpha - \varepsilon(\xi_{T,k})\iota_{\xi_{T,k}^\sharp}\alpha \right). \tag{326}$$

For $k \neq 0$, the Coulomb polarization space is

$$\mathcal{T}_{T,k} = \ker \iota_{\xi_{T,k}^\sharp} \subset (\mathbb{R}^4)^*_{\mathbb{C}}. \tag{327}$$

It has complex dimension three. On $\mathcal{T}_{T,k} \otimes \mathfrak{g}_{\mathbb{C}}$, the Hessian eigenvalue is

$$\lambda_T^{\mathrm{YM}}(k) = \frac{4\pi^2}{g_{\mathrm{YM}}^2} |P_T^{-\mathsf{T}} k|^2 \tag{328}$$

with complex multiplicity $3d_G$ on that signed Fourier fibre. If

$$K_T(\lambda) = \{ k \in \mathbb{Z}^4 \setminus \{0\} : \lambda_T^{\mathrm{YM}}(k) = \lambda \},$$

the total complex multiplicity, and also the dimension of the real eigenspace, is

$$3d_G \, \# K_T(\lambda). \tag{329}$$

Equivalently, a conjugate pair $\{k, -k\}$ contributes $6d_G$ real dimensions.

Proof. Differentiating the character proves equation (324). On the Euclidean cover,

$$\mathrm{d}^* = -\sum_{j=1}^{4} \iota_{\partial_{x_j}} \partial_{x_j},$$

which proves equation (325). The exterior algebra identity

$$\iota_{\xi^\sharp}\varepsilon(\xi) + \varepsilon(\xi)\iota_{\xi^\sharp} = |\xi|^2 \,\mathrm{id}$$

proves equation (326). For $k \neq 0$, the invertibility of P_T gives $\xi_{T,k} \neq 0$. Contraction by $\xi_{T,k}^\sharp$ has rank one on one-forms, so its kernel has dimension three. On that kernel the second term of equation (326) vanishes, proving the eigenvalue and its per-fibre multiplicity. The zero Fourier mode consists of the constant harmonic one-forms and has already been removed in equation (323). Fourier completeness gives all eigenvectors. Complex conjugation exchanges k with $-k$; the reality condition $\alpha_{-k} = \overline{\alpha_k}$ turns each conjugate pair of complex $3d_G$-dimensional spaces into a real $6d_G$-dimensional space. This proves equation (329). □

Remark 12.9 (The Coulomb projection uses the physical covector metric). In lattice-coordinate one-form coefficients a_y, the mode k obeys

$$k^{\mathsf{T}} G_T^{-1} a_y = 0, \qquad \Pi_{T,k} a_y = a_y - k \frac{k^{\mathsf{T}} G_T^{-1} a_y}{k^{\mathsf{T}} G_T^{-1} k}. \tag{330}$$

After conversion to physical x-coefficients, this is Euclidean projection perpendicular to $P_T^{-\mathsf{T}} k$. It is generally incorrect to impose Euclidean coefficient orthogonality to k, or to delete one coordinate merely because k is a coordinate lattice vector.

Theorem 12.10 (Exact reduced continuum Hessian gap)**.** *For all sufficiently large T, the first eigenvalue of $\mathcal{L}_T$ on $\mathscr{C}_T$ is*

$$\gamma_T = \frac{4\pi^2}{g_{\mathrm{YM}}^2} \min\left\{ \frac{L_T^2 + m_T^2}{D_T^2}, \frac{q_T^2 + 36 m_T^2}{D_T^2} \right\}. \tag{331}$$

It has the radial expansion

$$\gamma_T = \frac{4\pi^2}{g_{\mathrm{YM}}^2 T^2} \left(1 + \frac{-2H + 2\min\{B, 0\}}{T} + O(T^{-2}) \right). \tag{332}$$

Let

$$I_T = \{ j \in \{\gamma, u\} : a_{j,T} = \min\{a_{\gamma,T}, a_{u,T}\} \}.$$

The real multiplicity of γ_T is exactly

$$6 d_G \, \# I_T. \tag{333}$$

Proof. By theorem 12.8, minimizing the reduced Hessian eigenvalue is exactly the shortest-nonzero-dual-vector problem. The shortest-mode theorem theorem 11.2 says that, for all large T, the only possible minimizers are the two signed pairs with exact squared lengths given in equation (286). This proves equation (331). The expansions already proved in theorem 11.2 give relative corrections $2(B-H)$ and $-2H$; their minimum is $-2H + 2\min\{B, 0\}$, proving equation (332). Each minimizing signed pair contributes $6d_G$ real dimensions by theorem 12.8, which proves the multiplicity formula. □

12.5 The physical boundary and the exact no-gap bridge

The operators proved above are a classical Hessian and a finite quadratic cochain approximation on the internal flat manifolds F_T. They do not construct a quantum gauge theory on standard spacetime $\mathbb{R}^4$, and they do not determine the mass spectrum of such a theory. The precise boundaries are as follows.

(1) The unscaled cusp retains two compact geometric directions. The period parameter T has not been identified with a spacetime box size, a lattice spacing, a correlation length, or a renormalization scale.

(2) The operator $g_{\mathrm{YM}}^{-2}\mathrm{d}^*\mathrm{d}$ is the quadratic classical fluctuation operator at the product connection. It contains no nonperturbative interacting measure, dimensional transmutation, or confinement argument.

(3) The spaces $C_N^p(T;\mathfrak{g}_{\mathbb{C}})$ are linear Lie-algebra cochains. They are not compact-group link variables, and equation (308) is not a Wilson plaquette action.

(4) No Euclidean time direction or reflection, reflection-positive measure, transfer matrix, Osterwalder–Schrader reconstruction, Gauss-law physical quotient, local observable algebra, thermodynamic limit, or continuum limit in fixed physical units has been constructed from these operators.

(5) An eigenvalue of a four-dimensional elliptic Hessian has units of squared inverse length. Without a time reconstruction and a scale conversion, it is not an eigenvalue of a physical Hamiltonian. In particular, a dimensionless transfer-matrix gap can behave as $a_t m \to 0$ while the physical mass $m > 0$ remains fixed.

The exact extra statement that would transfer the auxiliary soft sequence to one already constructed physical Hamiltonian is the following. It must use the full zero-energy sector rather than only a chosen vacuum vector.

Proposition 12.11 (Typed state-and-energy no-gap bridge)**.** *Let $\mathcal{H}_{\mathrm{phys}}$ be a Hilbert space and let $H \geq 0$ be self-adjoint on it. Its closed energy form is*

$$\mathfrak{h}[\psi] = \|H^{1/2}\psi\|^2, \qquad \mathcal{D}(\mathfrak{h}) = \mathcal{D}(H^{1/2}). \tag{334}$$

Let

$$P_0 = \mathbf{1}_{\{0\}}(H), \qquad \mathcal{V} = \ker H = P_0\mathcal{H}_{\mathrm{phys}}, \qquad \mathcal{H}_{\mathrm{exc}} = (\mathrm{id} - P_0)\mathcal{H}_{\mathrm{phys}}, \tag{335}$$

and fix a normalized vacuum vector $\Omega \in \mathcal{V}$. Then

$$\mathcal{H}_{\mathrm{phys}} = \mathcal{V} \oplus \mathcal{H}_{\mathrm{exc}} \tag{336}$$

reduces H. Denote its excitation restriction by

$$H_{\mathrm{exc}} = H|_{\mathcal{D}(H)\cap\mathcal{H}_{\mathrm{exc}}}.$$

For every n, let $\mathcal{K}_n$ be an auxiliary Hilbert space with a closed nonnegative quadratic form $\mathfrak{q}_n$. Let $u_n \in \mathcal{D}(\mathfrak{q}_n)$ satisfy

$$\|u_n\|_{\mathcal{K}_n} = 1, \qquad \mathfrak{q}_n[u_n] \longrightarrow 0. \tag{337}$$

Suppose there are linear maps

$$J_n : \mathcal{D}(\mathfrak{q}_n) \longrightarrow \mathcal{D}(\mathfrak{h}), \qquad \psi_n = (\mathrm{id} - P_0)J_n u_n, \tag{338}$$

and numbers $c > 0$, $C_n \geq 0$, and $\varepsilon_n \geq 0$ such that

$$\|\psi_n\| \geq c, \tag{339}$$

$$\mathfrak{h}[\psi_n] \leq C_n\mathfrak{q}_n[u_n] + \varepsilon_n, \qquad C_n\mathfrak{q}_n[u_n] + \varepsilon_n \longrightarrow 0. \tag{340}$$

Then

$$\inf \operatorname{Spec}(H_{\mathrm{exc}}) = 0 \tag{341}$$

and, for every $\epsilon > 0$,

$$\mathbf{1}_{(0,\epsilon)}(H) \neq 0. \tag{342}$$

Moreover,

$$0 \in \operatorname{Spec}(H_{\mathrm{exc}}), \qquad \ker H_{\mathrm{exc}} = \{0\}. \tag{343}$$

Thus zero is a continuous spectral threshold, essential under the usual Weyl convention, and H *has no positive gap above its full zero-energy sector.*

If the vacuum is unique, so that $\mathcal{V} = \mathbb{C}\Omega$*, one may replace the projection in equation* (338) *by the requirement* $J_n u_n \perp \Omega$*. Without uniqueness, orthogonality only to* Ω *is insufficient.*

Proof. The operator P_0 is a spectral projection of H, so it commutes with the resolvent and with the spectral calculus of H. Hence equation (336) is a reducing decomposition and

$$\mathcal{D}(\mathfrak{h}|_{\mathcal{H}_{\mathrm{exc}}}) = \mathcal{D}(H^{1/2}) \cap \mathcal{H}_{\mathrm{exc}}.$$

The normalized vectors

$$v_n = \frac{\psi_n}{\|\psi_n\|}$$

belong to this excitation form domain. By equations (339) and (340),

$$\mathfrak{h}[v_n] \leq c^{-2}\big(C_n \mathfrak{q}_n[u_n] + \varepsilon_n\big) \longrightarrow 0.$$

The form version of the spectral variational identity is

$$\inf \operatorname{Spec}(H_{\mathrm{exc}}) = \inf_{0 \neq v \in \mathcal{D}(H^{1/2}) \cap \mathcal{H}_{\mathrm{exc}}} \frac{\mathfrak{h}[v]}{\|v\|^2}.$$

Nonnegativity and the sequence v_n prove equation (341).

If $\mathbf{1}_{(0,\epsilon)}(H) = 0$ for some $\epsilon > 0$, the spectral measure of every excitation vector would be supported in $[\epsilon, \infty)$, and therefore

$$\mathfrak{h}[v] \geq \epsilon \|v\|^2$$

throughout the excitation form domain. This contradicts $\mathfrak{h}[v_n] \to 0$, proving equation (342). Since the spectrum of a self-adjoint operator is closed, equation (341) puts zero in $\operatorname{Spec}(H_{\mathrm{exc}})$. Its kernel is trivial by the definition of $\mathcal{H}_{\mathrm{exc}}$. A self-adjoint operator has no residual spectrum, and equation (342) makes zero an accumulation point. This proves equation (343) and the essential-threshold assertion.

If $\mathcal{V} = \mathbb{C}\Omega$, orthogonality to Ω is exactly membership in $\mathcal{H}_{\mathrm{exc}}$. If the kernel is larger, a zero-energy vector can be orthogonal to the selected Ω, so that orthogonality alone does not produce an excitation vector. This proves the final assertion. □

Remark 12.12 (The present geometry supplies only the auxiliary side). Choose any $T_n \to \infty$, any $N_n \geq 2$, and a unit real transverse eigenvector $u_n \in \mathscr{C}_{N_n,T_n}$ at a minimizing coordinate mode. With

$$\mathcal{K}_n = \mathscr{C}_{N_n,T_n}, \qquad \mathfrak{q}_n[u] = g_{\mathrm{YM}}^{-2} \|\mathrm{d}_{N_n} u\|^2,$$

theorem 12.6 gives $\mathfrak{q}_n[u_n] = \gamma_{N_n,T_n} \to 0$. Thus the auxiliary soft-state data in equation (337) are present.

What is absent is one already constructed, gauge-invariant, positive-metric physical Hilbert space for four-dimensional quantum Yang–Mills theory, its Hamiltonian H, and maps J_n satisfying equations (339) and (340) with constants controlled in physical units after gauge reduction, reconstruction, renormalization, and all limiting operations. Consequently theorem 12.11 cannot be applied to infer a quantum mass statement from the cusp geometry. Even if such maps were supplied, the proposition would establish only spectrum arbitrarily close to zero for that specified H; it would not by itself identify a one-particle state, establish scattering, or construct the required quantum gauge theory.

13 Cubical holonomy and finite gauge systems

This section records three constructions on the regular cusp fibres while keeping two independent integers separate throughout: M_{cov} is the index used to enlarge two periods of a fibre, whereas N_{mesh} is the number of cubical subdivisions in each coordinate direction. In particular, no mesh refinement is identified with a change of covering space.

Let

$$P_T = \begin{pmatrix} 6\,\mathrm{Re}\,\mu_T & \mathrm{Re}\,\tau_T & 1 & 0 \\ 6\,\mathrm{Im}\,\mu_T & \mathrm{Im}\,\tau_T & 0 & 0 \\ \mathrm{Re}\,\beta_T & \mathrm{Re}\,\mu_T & 0 & 1 \\ \mathrm{Im}\,\beta_T & \mathrm{Im}\,\mu_T & 0 & 0 \end{pmatrix}, \qquad F_T = \mathbb{R}^4/P_T\mathbb{Z}^4. \tag{344}$$

Write

$$L_T = \mathrm{Im}\,\tau_T, \qquad m_T = \mathrm{Im}\,\mu_T, \qquad q_T = -\,\mathrm{Im}\,\beta_T, \qquad D_T = L_T q_T + 6m_T^2 > 0. \tag{345}$$

Expanding the determinant along the last two columns gives

$$\det P_T = L_T\,\mathrm{Im}\,\beta_T - 6m_T^2 = -D_T.$$

Along the positive radial cusp the source data give

$$L_T = T + O(1), \qquad q_T = T + O(1), \qquad m_T = O(1), \tag{346}$$

and all three real parts occurring in (344) are bounded. Equivalently, after ordering physical coordinates as $(x_1, x_3; x_2, x_4)$, the first two columns have block form

$$\begin{pmatrix} A_T \\ B_T \end{pmatrix}, \qquad B_T = \begin{pmatrix} 6m_T & L_T \\ -q_T & m_T \end{pmatrix} = T\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} + C_T, \qquad \sup_T \|C_T\|_{\mathrm{op}} < \infty. \tag{347}$$

For $M_{\mathrm{cov}} \geq 1$, put

$$S_{M_{\mathrm{cov}}} = \mathrm{diag}(1, 1, M_{\mathrm{cov}}, M_{\mathrm{cov}}), \qquad P_{T,M_{\mathrm{cov}}} = P_T S_{M_{\mathrm{cov}}}, \tag{348}$$

and define the degree-M_{cov}^2 cover

$$\widetilde{F}_{T,M_{\mathrm{cov}}} = \mathbb{R}^4/P_{T,M_{\mathrm{cov}}}\mathbb{Z}^4 \longrightarrow F_T. \tag{349}$$

Indeed, $S_{M_{\mathrm{cov}}}\mathbb{Z}^4 \subset \mathbb{Z}^4$ and

$$\mathbb{Z}^4/S_{M_{\mathrm{cov}}}\mathbb{Z}^4 \cong (\mathbb{Z}/M_{\mathrm{cov}}\mathbb{Z})^2,$$

so the identity on $\mathbb{R}^4$ induces the asserted cover and its deck group has order M_{cov}^2. Its Euclidean quotient metric, written on the coordinate torus $\mathbb{R}^4/\mathbb{Z}^4$, has Gram matrix and covolume

$$G_{T,M_{\mathrm{cov}}} = P_{T,M_{\mathrm{cov}}}^{\mathsf{T}} P_{T,M_{\mathrm{cov}}}, \qquad \mathcal{D}_{T,M_{\mathrm{cov}}} = \sqrt{\det G_{T,M_{\mathrm{cov}}}} = M_{\mathrm{cov}}^2 D_T. \tag{350}$$

13.1 The transported cubical subdivision and continuum holonomy

Fix an integer $N_{\mathrm{mesh}} \geq 2$, and let $K_{N_{\mathrm{mesh}}}$ be the standard cubical subdivision of $\mathbb{R}^4/\mathbb{Z}^4$ obtained by dividing every coordinate circle into N_{mesh} equal intervals. Its vertex set and positively oriented coordinate edges are

$$V_{N_{\mathrm{mesh}}} = (\mathbb{Z}/N_{\mathrm{mesh}}\mathbb{Z})^4, \qquad E_{N_{\mathrm{mesh}}} = V_{N_{\mathrm{mesh}}} \times \{1, 2, 3, 4\}. \tag{351}$$

Proposition 13.1 (Exact cubical realization and the torsion vertices)**.** *The linear map $P_{T,M_{\mathrm{cov}}}$ induces a diffeomorphism*

$$\Phi_{T,M_{\mathrm{cov}}} : \mathbb{R}^4/\mathbb{Z}^4 \longrightarrow \widetilde{F}_{T,M_{\mathrm{cov}}}, \qquad [y] \longmapsto [P_{T,M_{\mathrm{cov}}} y]. \tag{352}$$

Transporting $K_{N_{\mathrm{mesh}}}$ by this diffeomorphism gives an honest cubical cell decomposition of the whole torus. On vertices its restriction is the bijection

$$\begin{aligned} \iota_{T,M_{\mathrm{cov}},N_{\mathrm{mesh}}} : \quad & (\mathbb{Z}/N_{\mathrm{mesh}}\mathbb{Z})^4 \xrightarrow{\cong} \widetilde{F}_{T,M_{\mathrm{cov}}}[N_{\mathrm{mesh}}], \\ & [n] \longmapsto \left[\frac{1}{N_{\mathrm{mesh}}} P_{T,M_{\mathrm{cov}}} n\right]. \end{aligned} \tag{353}$$

Here the codomain is the finite N_{mesh}-torsion subgroup, not the entire torus. The coordinate edge (n, j) is the path

$$\gamma_{n,j}(t) = \left[\frac{1}{N_{\mathrm{mesh}}} P_{T,M_{\mathrm{cov}}}(n + t\mathbf{e}_j)\right], \qquad 0 \le t \le 1. \tag{354}$$

More generally, for an ordered subset $I = \{i_1, \dots, i_p\} \subset \{1, 2, 3, 4\}$, the corresponding closed p-cell is parametrized by

$$c_{n,I}(t_1, \dots, t_p) = \left[\frac{1}{N_{\mathrm{mesh}}} P_{T,M_{\mathrm{cov}}} \left(n + \sum_{a=1}^{p} t_a \mathbf{e}_{i_a}\right)\right], \qquad (t_1, \dots, t_p) \in [0, 1]^p. \tag{355}$$

Proof. If $y - y' \in \mathbb{Z}^4$, then $P_{T,M_{\mathrm{cov}}}(y - y')$ belongs to the quotient lattice, so (352) is well defined. The matrix $P_{T,M_{\mathrm{cov}}}$ is invertible, and its inverse induces the inverse map on the two quotients. Hence it is a diffeomorphism and carries every open cell of the standard subdivision diffeomorphically onto an open cell of the target.

Every point displayed in (353) is killed by N_{mesh}. If it is zero, then

$$\frac{P_{T,M_{\mathrm{cov}}} n}{N_{\mathrm{mesh}}} \in P_{T,M_{\mathrm{cov}}} \mathbb{Z}^4.$$

Invertibility gives $n \in N_{\mathrm{mesh}}\mathbb{Z}^4$, proving injectivity. Conversely, if $[x]$ is killed by N_{mesh}, choose a lift $x \in \mathbb{R}^4$. There is $n \in \mathbb{Z}^4$ such that $N_{\mathrm{mesh}} x = P_{T,M_{\mathrm{cov}}} n$, and therefore $[x] = [P_{T,M_{\mathrm{cov}}} n/N_{\mathrm{mesh}}]$. This proves surjectivity. Formula (354) is simply the transport of the corresponding coordinate edge; at a wrap-around endpoint the extra vector $P_{T,M_{\mathrm{cov}}}\mathbf{e}_j$ is a period. □

Let G be a compact matrix Lie group and use the trivial principal bundle over $\widetilde{F}_{T,M_{\mathrm{cov}}}$. A connection is represented by $A \in \Omega^1(\widetilde{F}_{T,M_{\mathrm{cov}}}; \mathfrak{g})$. For an oriented path γ, define $U_A(\gamma)$ by

$$\frac{dU}{dt} = -A_{\gamma(t)}(\dot{\gamma}(t))U(t), \qquad U(0) = 1, \qquad U_A(\gamma) = U(1). \tag{356}$$

Proposition 13.2 (Typed continuum-to-link holonomy map)**.** *The formula*

$$\mathrm{Hol}_{T,M_{\mathrm{cov}},N_{\mathrm{mesh}}}(A)_{n,j} = U_A(\gamma_{n,j}) \tag{357}$$

defines a map

$$\mathrm{Hol}_{T,M_{\mathrm{cov}},N_{\mathrm{mesh}}} : \Omega^1(\widetilde{F}_{T,M_{\mathrm{cov}}}; \mathfrak{g}) \longrightarrow G^{E_{N_{\mathrm{mesh}}}}. \tag{358}$$

For the convention

$$A^g = g^{-1}Ag + g^{-1}dg, \qquad g \in C^\infty(\widetilde{F}_{T,M_{\mathrm{cov}}}, G), \tag{359}$$

it is equivariant for evaluation of g at the vertices:

$$\mathrm{Hol}(A^g)_{n,j} = g(\gamma_{n,j}(1))^{-1} \mathrm{Hol}(A)_{n,j} g(\gamma_{n,j}(0)). \tag{360}$$

Proof. The smooth matrix ODE (356) has a unique solution on every edge, which proves that (358) is defined. Let U solve that ODE for A, set $g_t = g(\gamma_{n,j}(t))$, and put $V(t) = g_t^{-1}U(t)g_0$. Direct differentiation gives

$$V'(t) = -(g_t^{-1}A(\dot\gamma_{n,j})g_t + g_t^{-1}g_t')V(t), \qquad V(0) = 1.$$

This is exactly the parallel-transport equation for A^g. Uniqueness and evaluation at $t = 1$ prove (360). Notice that the endpoint inverse in this formula is forced by the joint conventions (356) and (359). □

13.2 An explicit compact-link SU(2) flux

The ordered plaquette convention used below is most naturally written for the inverse-transport link variable

$$\mathcal{U}(A)_{n,j} = \mathrm{Hol}_{T,M_{\mathrm{cov}},N_{\mathrm{mesh}}}(A)^{-1}_{n,j}. \tag{361}$$

Equation (360) then becomes

$$\mathcal{U}(A^g)_{n,j} = g(\gamma_{n,j}(0))^{-1}\mathcal{U}(A)_{n,j}g(\gamma_{n,j}(1)). \tag{362}$$

Thus the product in (368) transforms by conjugation at its base vertex. The symbols $U_j(n)$ in the remainder of this subsection use this inverse-transport convention.

Use

$$\mathfrak{su}(2) = \{X \in M_2(\mathbb{C}) : X^* = -X,\ \mathrm{tr}\, X = 0\}, \qquad \langle X, Y\rangle_{\mathfrak{su}(2)} = -\frac{1}{2}\,\mathrm{tr}(XY), \tag{363}$$

on real elements, and fix

$$H_{\mathrm{link}} = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}. \qquad \|H_{\mathrm{link}}\|^2 = 1, \qquad \exp(2\pi H_{\mathrm{link}}) = 1. \tag{364}$$

Write $N = N_{\mathrm{mesh}}$ temporarily inside the next displayed link formulas, and represent n_1, n_2 by integers in $\{0, \dots, N-1\}$.

Theorem 13.3 (A periodic compact-link flux and its exact action)**.** *On the oriented edges of K_N, define*

$$U_1(n) = \exp\left(-\frac{2\pi n_2}{N^2}H_{\mathrm{link}}\right), \tag{365}$$

$$U_2(n) = \begin{cases} 1, & 0 \le n_2 \le N-2, \\ \exp(2\pi n_1 H_{\mathrm{link}}/N), & n_2 = N-1, \end{cases} \tag{366}$$

$$U_3(n) = U_4(n) = 1. \tag{367}$$

For

$$U_{ij}(n) = U_i(n)U_j(n+\mathbf{e}_i)U_i(n+\mathbf{e}_j)^{-1}U_j(n)^{-1}, \tag{368}$$

one has

$$\begin{aligned} &U_{12}(n) = \exp(2\pi H_{\mathrm{link}}/N^2) && \text{for every } n, \\ &U_{ij}(n) = 1 && \text{for } 1 \le i < j \le 4, \quad (i,j) \ne (1,2). \end{aligned} \tag{369}$$

Let Log *be the principal logarithm on the displayed plaquette. It is unambiguous because $0 < 2\pi/N^2 \le \pi/2$. Define the logarithmic curvature two-form and its metric-weighted action by*

$$\mathcal{F}_N^{\log}(n) = N^2\,\mathrm{Log}(U_{12}(n))\,dy^1 \wedge dy^2, \tag{370}$$

$$S^{\log}_{T,M_{\mathrm{cov}},N}(U) = \frac{\mathcal{D}_{T,M_{\mathrm{cov}}}}{2g_{\mathrm{YM}}^2N^4}\sum_{n\in V_N}\|\mathcal{F}_N^{\log}(n)\|^2_{G^{-1}_{T,M_{\mathrm{cov}}}\otimes\mathfrak{su}(2)}. \tag{371}$$

Then

$$\mathcal{F}_N^{\log}(n) = 2\pi H_{\mathrm{link}}\, dy^1 \wedge dy^2, \qquad S^{\log}_{T,M_{\mathrm{cov}},N}(U) = \frac{2\pi^2 M_{\mathrm{cov}}^2}{g_{\mathrm{YM}}^2 D_T}. \tag{372}$$

In particular, the native-fibre value $M_{\mathrm{cov}} = 1$ is $2\pi^2/(g_{\mathrm{YM}}^2 D_T)$, independently of the mesh.

There is no canonical scalar Wilson action for a general nonorthogonal metric obtained merely by assigning independent character weights to all coordinate plaquettes. For this configuration, however, the exact continuum-matched $(1,2)$-sector Wilson functional is unambiguous if it is defined explicitly by the following formula. Its ordered plaquette holonomy and real trace are the Wilson lattice-action construction [65]; the metric coefficient and normalization here are the present definition, and their consequences are proved below rather than imported from that source:

$$S^{W,12}_{T,M_{\mathrm{cov}},N}(U) = \frac{\mathcal{D}_{T,M_{\mathrm{cov}}}\kappa^{12}_{T,M_{\mathrm{cov}}}}{2g_{\mathrm{YM}}^2} \sum_{n\in V_N} \left(2 - \operatorname{Re}\operatorname{tr} U_{12}(n)\right), \tag{373}$$

where

$$\kappa^{12}_{T,M_{\mathrm{cov}}} = (G^{-1}_{T,M_{\mathrm{cov}}})_{11}(G^{-1}_{T,M_{\mathrm{cov}}})_{22} - (G^{-1}_{T,M_{\mathrm{cov}}})^2_{12} = D_T^{-2}. \tag{374}$$

Its exact value and continuum-mesh limit are

$$S^{W,12}_{T,M_{\mathrm{cov}},N}(U) = \frac{M_{\mathrm{cov}}^2 N^4}{2g_{\mathrm{YM}}^2 D_T}\left(2 - 2\cos\frac{2\pi}{N^2}\right), \tag{375}$$

$$= \frac{M_{\mathrm{cov}}^2}{g_{\mathrm{YM}}^2 D_T}\left(2\pi^2 - \frac{2\pi^4}{3N^4} + \frac{4\pi^6}{45N^8} + O(N^{-12})\right), \tag{376}$$

$$\lim_{N\to\infty} S^{W,12}_{T,M_{\mathrm{cov}},N}(U) = \frac{2\pi^2 M_{\mathrm{cov}}^2}{g_{\mathrm{YM}}^2 D_T}. \tag{377}$$

Proof. All links lie in the same diagonal circle subgroup, so they commute. If $0 \le n_2 \le N-2$, the two U_2-terms in (368) are trivial and

$$U_{12}(n) = \exp(-2\pi n_2 H_{\mathrm{link}}/N^2)\exp(2\pi(n_2+1)H_{\mathrm{link}}/N^2) = \exp(2\pi H_{\mathrm{link}}/N^2).$$

If $n_2 = N-1$, then $U_1(n+\mathbf{e}_2) = 1$. Moreover,

$$U_2(n+\mathbf{e}_1)U_2(n)^{-1} = \exp(2\pi H_{\mathrm{link}}/N).$$

For $n_1 = N-1$, the exponent computed with representatives differs from $2\pi H_{\mathrm{link}}/N$ by $-2\pi H_{\mathrm{link}}$, whose exponential is one; hence the formula also holds across that wrap boundary. It follows that

$$U_{12}(n) = \exp\left(-\frac{2\pi(N-1)}{N^2}H_{\mathrm{link}}\right)\exp\left(\frac{2\pi}{N}H_{\mathrm{link}}\right) = \exp(2\pi H_{\mathrm{link}}/N^2).$$

The functions U_1, U_2 are independent of n_3, n_4, and $U_3 = U_4 = 1$, proving every remaining assertion in (369). Reversing the orientation gives $U_{21}(n) = U_{12}(n)^{-1}$; it is not an additional trivial plaquette.

The lower-right 2×2 principal block of the cover Gram matrix is $M_{\mathrm{cov}}^2 I_2$. Jacobi's complementary-minor identity therefore gives

$$\kappa^{12}_{T,M_{\mathrm{cov}}} = \frac{\det(M_{\mathrm{cov}}^2 I_2)}{\det G_{T,M_{\mathrm{cov}}}} = \frac{M_{\mathrm{cov}}^4}{M_{\mathrm{cov}}^4 D_T^2} = D_T^{-2},$$

which proves (374). By (364) and the principal-log choice, $N^2 \operatorname{Log} U_{12} = 2\pi H_{\mathrm{link}}$. Hence every one of the N^4 summands in (371) equals $4\pi^2 D_T^{-2}$. Substitution of $\mathcal{D}_{T,M_{\mathrm{cov}}} = M_{\mathrm{cov}}^2 D_T$ proves (372).

Finally, H_{link} has eigenvalues $\pm i$, so

$$\operatorname{Re}\operatorname{tr}\exp(2\pi H_{\mathrm{link}}/N^2) = 2\cos(2\pi/N^2).$$

There are N^4 equal plaquettes. Substitution proves (375). Applying

$$1 - \cos x = \frac{x^2}{2} - \frac{x^4}{24} + \frac{x^6}{720} + O(x^8)$$

with $x = 2\pi/N^2$ proves (376) and (377). □

13.3 Finite Gaussian quantization with independent cover and mesh

Let

$$\vartheta^j_{T,M_{\mathrm{cov}}} = P^{-\mathsf{T}}_{T,M_{\mathrm{cov}}}\mathbf{e}_j \in (\mathbb{R}^4)^*, \qquad 1 \le j \le 4. \tag{378}$$

For $0 \le p \le 4$, define the real finite cochain spaces

$$C^p_{N_{\mathrm{mesh}}}(T, M_{\mathrm{cov}}) = \mathrm{Map}\left(V_{N_{\mathrm{mesh}}}, \bigwedge^p (\mathbb{R}^4)^* \otimes \mathfrak{su}(2)\right). \tag{379}$$

The exterior powers carry the metric induced by the physical coframe (378); explicitly, for increasing multi-indices $I = (i_1, \dots, i_p)$ and $J = (j_1, \dots, j_p)$,

$$\left\langle \vartheta^{i_1} \wedge \cdots \wedge \vartheta^{i_p}, \vartheta^{j_1} \wedge \cdots \wedge \vartheta^{j_p} \right\rangle = \det\left((G^{-1}_{T,M_{\mathrm{cov}}})_{i_a j_b}\right)^p_{a,b=1}. \tag{380}$$

Together with (363), this defines the pointwise tensor-product metric. The cochain inner product is

$$\langle a, b\rangle_p = \frac{\mathcal{D}_{T,M_{\mathrm{cov}}}}{N^4_{\mathrm{mesh}}} \sum_{n \in V_{N_{\mathrm{mesh}}}} \langle a(n), b(n)\rangle. \tag{381}$$

Let $\mathsf{T}_j a(n) = a(n + \mathbf{e}_j)$, set

$$D^+_{N_{\mathrm{mesh}},j} = N_{\mathrm{mesh}}(\mathsf{T}_j - 1), \qquad D^-_{N_{\mathrm{mesh}},j} a(n) = N_{\mathrm{mesh}}(a(n) - a(n - \mathbf{e}_j)), \tag{382}$$

and define

$$d_{N_{\mathrm{mesh}}} a = \sum_{j=1}^4 \varepsilon(\vartheta^j_{T,M_{\mathrm{cov}}}) D^+_{N_{\mathrm{mesh}},j} a. \tag{383}$$

The commuting differences and anticommuting exterior multiplications give $d^2_{N_{\mathrm{mesh}}} = 0$. Its adjoint is

$$d^*_{N_{\mathrm{mesh}}} = -\sum_{j=1}^4 \iota_{(\vartheta^j_{T,M_{\mathrm{cov}}})^\sharp} D^-_{N_{\mathrm{mesh}},j}. \tag{384}$$

Define the real reduced Coulomb space

$$\mathscr{C}_{T,M_{\mathrm{cov}},N_{\mathrm{mesh}}} = \ker(d^*_{N_{\mathrm{mesh}}} : C^1 \to C^0) \cap \left(\ker \Delta^{(1)}_{N_{\mathrm{mesh}}}\right)^\perp, \tag{385}$$

$$\Delta^{(1)}_{N_{\mathrm{mesh}}} = d_{N_{\mathrm{mesh}}} d^*_{N_{\mathrm{mesh}}} + d^*_{N_{\mathrm{mesh}}} d_{N_{\mathrm{mesh}}}. \tag{386}$$

On this finite-dimensional space put

$$\mathcal{L}_{T,M_{\mathrm{cov}},N_{\mathrm{mesh}}} = g^{-2}_{\mathrm{YM}} d^*_{N_{\mathrm{mesh}}} d_{N_{\mathrm{mesh}}}. \tag{387}$$

Proposition 13.4 (Exact finite Hessian spectrum)**.** *For*

$$\mathfrak{d}_{N_{\mathrm{mesh}}}(k)_j = N_{\mathrm{mesh}}\left(e^{2\pi i k_j / N_{\mathrm{mesh}}} - 1\right), \qquad k \in (\mathbb{Z}/N_{\mathrm{mesh}}\mathbb{Z})^4, \tag{388}$$

the positive spectrum of $\mathcal{L}$ is obtained from

$$\lambda_{T,M_{\mathrm{cov}},N_{\mathrm{mesh}}}(k) = \frac{1}{g^2_{\mathrm{YM}}} \left\| P_T^{-\mathsf{T}} S^{-1}_{M_{\mathrm{cov}}} \mathfrak{d}_{N_{\mathrm{mesh}}}(k) \right\|^2, \qquad k \neq 0. \tag{389}$$

Each complex Fourier fibre has transverse multiplicity $3 \dim \mathfrak{su}(2) = 9$. In particular,

$$\lambda_{\min}(\mathcal{L}) = \min_{k \neq 0} \lambda_{T,M_{\mathrm{cov}},N_{\mathrm{mesh}}}(k) > 0. \tag{390}$$

For $\mathbf{e}_1 = (1,0,0,0)^{\mathsf{T}}$, the exact mode value is

$$\lambda_{T,M_{\mathrm{cov}},N_{\mathrm{mesh}}}(\mathbf{e}_1) = \frac{4N^2_{\mathrm{mesh}} \sin^2(\pi/N_{\mathrm{mesh}})}{g^2_{\mathrm{YM}}} \frac{L_T^2 + m_T^2}{D_T^2}; \tag{391}$$

it is independent of M_{cov}.

Proof. Translation is unitary for (381), so $(D_j^+)^* = -D_j^-$. Exterior multiplication by ϑ^j has adjoint contraction by $(\vartheta^j)^\sharp$, proving (384). On the Fourier character $\chi_k(n) = e^{2\pi i k\cdot n/N_{\mathrm{mesh}}}$, the derivative symbol is exterior multiplication by

$$\zeta(k) = P_{T,M_{\mathrm{cov}}}^{-\mathsf{T}}\mathfrak{d}_{N_{\mathrm{mesh}}}(k) = P_T^{-\mathsf{T}}S_{M_{\mathrm{cov}}}^{-1}\mathfrak{d}_{N_{\mathrm{mesh}}}(k).$$

The Hermitian exterior-algebra identity

$$\iota_{\overline{\zeta}^\sharp}\varepsilon(\zeta) + \varepsilon(\zeta)\iota_{\overline{\zeta}^\sharp} = \|\zeta\|^2\,\mathrm{id}$$

shows that on the Coulomb polarization $\ker \iota_{\overline{\zeta}^\sharp} \subset (\mathbb{C}^4)^*$, the operator d^*d is multiplication by $\|\zeta\|^2$. For $k \neq 0$, $\mathfrak{d}_{N_{\mathrm{mesh}}}(k) \neq 0$, so invertibility of $P_{T,M_{\mathrm{cov}}}$ gives $\zeta(k) \neq 0$. The contraction has rank one, and its kernel has dimension three. The zero Fourier fibre consists exactly of constant harmonic cochains and was removed in (385). Fourier completeness now proves (389)–(390).

For $k = \mathbf{e}_1$, the only nonzero component of $\mathfrak{d}_{N_{\mathrm{mesh}}}(k)$ has modulus

$$2N_{\mathrm{mesh}}\sin(\pi/N_{\mathrm{mesh}}),$$

and $S_{M_{\mathrm{cov}}}^{-1}\mathbf{e}_1 = \mathbf{e}_1$. Solving $P_T^{\mathsf{T}}\xi = \mathbf{e}_1$, or directly inverting the lower 2×2 block in (344), gives

$$\|P_T^{-\mathsf{T}}\mathbf{e}_1\|^2 = \frac{L_T^2 + m_T^2}{D_T^2}.$$

Substitution proves (391). □

Let da denote Lebesgue measure on $\mathscr{C}_{T,M_{\mathrm{cov}},N_{\mathrm{mesh}}}$ induced by (381) and define

$$\mathcal{H}^{\mathrm{Gauss}}_{T,M_{\mathrm{cov}},N_{\mathrm{mesh}}} = L^2(\mathscr{C}_{T,M_{\mathrm{cov}},N_{\mathrm{mesh}}}, da). \tag{392}$$

Theorem 13.5 (Finite Gaussian oscillator and a resolved cusp sequence)**.** *The Friedrichs realization on* (392) *of*

$$\widehat{H}^{\mathrm{Gauss}}_{T,M_{\mathrm{cov}},N_{\mathrm{mesh}}} = -\frac{1}{2}\Delta_{\mathscr{C}} + \frac{1}{2}\langle a, \mathcal{L}_{T,M_{\mathrm{cov}},N_{\mathrm{mesh}}}a\rangle \tag{393}$$

has a unique Gaussian ground state and spectral gap

$$\mathrm{gap}(\widehat{H}^{\mathrm{Gauss}}) = \sqrt{\lambda_{\min}(\mathcal{L}_{T,M_{\mathrm{cov}},N_{\mathrm{mesh}}})}. \tag{394}$$

Consequently,

$$\mathrm{gap}(\widehat{H}^{\mathrm{Gauss}}) \le \frac{2N_{\mathrm{mesh}}\sin(\pi/N_{\mathrm{mesh}})}{g_{\mathrm{YM}}}\frac{\sqrt{L_T^2 + m_T^2}}{D_T}. \tag{395}$$

There is an explicit diagonal sequence for which the cover geometry tends locally to $\mathbb{R}^4$, the cubical mesh resolves that geometry, and the Gaussian gap tends to zero. Explicitly, for integers $j \to \infty$, take

$$T_j = j^2, \qquad M_{\mathrm{cov},j} = j, \qquad N_{\mathrm{mesh},j} = j^3. \tag{396}$$

Then

$$\begin{aligned} \mathrm{sys}(\widetilde{F}_{T_j,M_{\mathrm{cov},j}}) &\longrightarrow \infty,\\ \max_r \frac{\|P_{T_j,M_{\mathrm{cov},j}}\mathbf{e}_r\|}{N_{\mathrm{mesh},j}} &\longrightarrow 0,\\ \mathrm{gap}(\widehat{H}^{\mathrm{Gauss}}) &\longrightarrow 0,\\ S^{\log}_{T_j,M_{\mathrm{cov},j},N_{\mathrm{mesh},j}}(U) &\longrightarrow 0. \end{aligned} \tag{397}$$

Thus the pointed cover tori converge locally isometrically to $\mathbb{R}^4$ while their finite linearized Gaussian systems are simultaneously refined.

Proof. The real finite-dimensional operator $\mathcal{L}$ is positive definite on the reduced Coulomb space. Choose an orthonormal eigenbasis with eigenvalues $0 < \lambda_1 \le \cdots \le \lambda_r$. In the resulting coordinates, (393) is the tensor sum

$$\sum_{s=1}^{r} \left(-\frac{1}{2}\frac{\partial^2}{\partial x_s^2} + \frac{1}{2}\lambda_s x_s^2 \right).$$

The normalized Hermite basis has eigenvalues $\sum_s (n_s + \frac{1}{2})\sqrt{\lambda_s}$, with $n_s \in \mathbb{Z}_{\ge 0}$. The all-zero multi-index is the unique ground state, and the smallest possible positive excitation is $\sqrt{\lambda_1}$. This proves (394); applying the minimum to the explicit $\mathbf{e}_1$-mode in (391) proves (395).

For completeness, reorder physical coordinates as in (347). Every vector of the cover lattice has the form

$$\begin{pmatrix} A_T n + M_{\mathrm{cov}} m \\ B_T n \end{pmatrix}, \qquad n, m \in \mathbb{Z}^2.$$

Let $C = \sup_T \|C_T\|_{\mathrm{op}}$. If $n \neq 0$, then $\|B_T n\| \ge (T - C)\|n\|$; if $n = 0$ and the lattice vector is nonzero, its norm is at least M_{cov}. Hence

$$\mathrm{sys}(\widetilde{F}_{T,M_{\mathrm{cov}}}) \ge \min\{T - C, M_{\mathrm{cov}}\}. \tag{398}$$

This tends to infinity for the first two choices in (396). For any fixed $R > 0$, once the systole exceeds $4R$, the quotient map from $\mathbb{R}^4$ is injective and distance-preserving on the radius-R ball: for a nonzero period λ and x, y in that ball,

$$\|x - y + \lambda\| \ge \|\lambda\| - \|x - y\| > 4R - 2R \ge \|x - y\|.$$

This proves the pointed local-isometric $\mathbb{R}^4$ limit.

The first two columns of P_T have norm $O(T)$, while the last two columns of $P_{T,M_{\mathrm{cov}}}$ have norm exactly M_{cov}. Therefore the largest physical edge in the N_{mesh}-subdivision has length $O(\max\{T, M_{\mathrm{cov}}\}/N_{\mathrm{mesh}})$, which is $O(j^{-1})$ for (396). Finally, (346) gives $\sqrt{L_T^2 + m_T^2}/D_T = T^{-1}(1 + O(T^{-1}))$, while $2N \sin(\pi/N) \le 2\pi$. The right-hand side of (395) is therefore $O(j^{-2})$. Moreover, (372) is $O(j^{-2})$, since $M_{\mathrm{cov},j}^2/D_{T_j} = j^2/(j^4 + O(j^2))$. This proves every assertion in (397). □

13.4 The exponential map and the exact physical boundary

Let $C^1 = C^1_{N_{\mathrm{mesh}}}(T, M_{\mathrm{cov}})$, regarded only as a finite-dimensional Lie-algebra cochain space, and define componentwise

$$\mathrm{Exp}_{N_{\mathrm{mesh}}} : C^1 \longrightarrow \mathrm{SU}(2)^{E_{N_{\mathrm{mesh}}}}, \qquad a \longmapsto (\exp(a_e/N_{\mathrm{mesh}}))_e\,. \tag{399}$$

Here, if $a(n) = \sum_{j=1}^{4} \vartheta^j_{T,M_{\mathrm{cov}}} \otimes a_j(n)$, then $a_e = a_j(n)$ for the oriented edge $e = (n, j)$.

Proposition 13.6 (Local chart, not a global embedding)**.** *The map* (399) *is smooth and a local diffeomorphism at zero, but it is not globally injective. Under left trivialization of every tangent space at the identity,*

$$D\,\mathrm{Exp}_{N_{\mathrm{mesh}}}(0)\,a = N_{\mathrm{mesh}}^{-1} a. \tag{400}$$

If S_W is any smooth compact-link action for which the identity configuration is critical, then

$$D^2(S_W \circ \mathrm{Exp}_{N_{\mathrm{mesh}}})_0[a, b] = N_{\mathrm{mesh}}^{-2} D^2(S_W)_{\mathbf{1}}[a, b] \tag{401}$$

under the same tangent identification. Thus, for a prescribed cochain quadratic form Q,

$$D^2(S_W \circ \mathrm{Exp}_{N_{\mathrm{mesh}}})_0 = Q$$

holds exactly when

$$D^2(S_W)_{\mathbf{1}} = N_{\mathrm{mesh}}^2 Q$$

under the coefficient identification. Both this scaling and the tensor structure must be verified for the chosen compact-link action; neither follows from the exponential map alone.

Proof. For the Lie-group exponential, $D\exp_0$ is the identity. Applying the chain rule to $a_e \mapsto a_e/N_{\mathrm{mesh}}$ gives (400), which is invertible, so the inverse function theorem gives a local diffeomorphism. It is not globally injective: a cochain which is zero on every edge except one, where it equals $2\pi N_{\mathrm{mesh}} H_{\mathrm{link}}$, has the same image as the zero cochain by (364). Finally, the second-derivative chain rule contains a term involving $DS_W(\mathbf{1})$, which vanishes by criticality; the two remaining first derivatives of Exp each supply the factor N_{mesh}^{-1}. This proves (401). □

The Gaussian Hilbert space in (392) is not the compact-link Yang–Mills Hilbert space. At this finite lattice, the latter kinematic and gauge-invariant spaces would be

$$\mathcal{H}^{\mathrm{link}}_{N_{\mathrm{mesh}}} = L^2(\mathrm{SU}(2)^{E_{N_{\mathrm{mesh}}}}, dU), \qquad \mathcal{H}^{\mathrm{inv}}_{N_{\mathrm{mesh}}} = \left(\mathcal{H}^{\mathrm{link}}_{N_{\mathrm{mesh}}}\right)^{\mathrm{SU}(2)^{V_{N_{\mathrm{mesh}}}}}, \tag{402}$$

where dU is product normalized Haar measure and the superscript denotes the nonlinear vertex-gauge action. Here "gauge-invariant" refers only to the link-configuration space of this four-dimensional cell complex. Without a chosen time direction or transfer matrix, it is not asserted to be a canonical Hamiltonian physical Hilbert space. Lüscher's construction of a selfadjoint, strictly positive transfer matrix [66] and Menotti–Pelissetto's reflection-positivity theorem for the Wilson action [67] concern the standard Wilson lattice setting after the requisite reflection/time structure has been fixed; they do not supply those omitted choices for the nonorthogonal sector functional (373). The Osterwalder–Schrader reconstruction boundary is likewise the one stated in the primary Euclidean axioms papers [68, 69], not a consequence of a finite Gaussian oscillator calculation. The differences are structural:

1. $\mathcal{H}^{\mathrm{Gauss}}$ is an L^2-space on a noncompact linear Coulomb slice with Lebesgue measure; $\mathcal{H}^{\mathrm{inv}}$ is a gauge-invariant subspace of an L^2-space on compact link variables with Haar measure.

2. The local exponential map is neither a global coordinate chart nor a measure-preserving bijection. No unitary map between the two displayed Hilbert spaces has been constructed.

3. A link configuration U determines the Dirac distribution δ_U, but

$$\delta_U \in \mathcal{D}'(\mathrm{SU}(2)^{E_{N_{\mathrm{mesh}}}}), \qquad \delta_U \notin L^2(\mathrm{SU}(2)^{E_{N_{\mathrm{mesh}}}}, dU). \tag{403}$$

   Indeed, product Haar measure is nonatomic, so a point mass is singular with respect to it and has no square-integrable density.

4. No nonlinear compact-link Hamiltonian, Gauss-law spectral problem, time direction, reflection-positive transfer matrix, renormalized-scale limit, or Osterwalder–Schrader reconstruction is supplied here.

Accordingly, (394) is an exact theorem about the finite quadratic oscillator defined by the Coulomb-restricted operator $g_{\mathrm{YM}}^{-2} d^* d$. It is not an eigenvalue statement about a nonperturbative physical Yang–Mills Hamiltonian, and the small classical action in (372) is not by itself a physical-state Rayleigh quotient.

14 Integral magnetic sectors and a reducible $SU(2)$ critical point on the radial cusp

Retain the positive radial flat tori $F_T = \mathbb{R}^4/P_T\mathbb{Z}^4$, their Euclidean quotient metrics, and the real quantities $L_T, q_T, m_T, D_T, H, B, m$ from equations (23) and (47) and theorem 11.1. In particular,

$$L_T = \operatorname{Im}\tau_T, \qquad q_T = -\operatorname{Im}\beta_T, \qquad D_T = L_T q_T + 6m_T^2 = \mathrm{Vol}(F_T) > 0.$$

The real coordinate order is the already fixed order

$$x = (x_1, y_1, x_2, y_2)^{\mathsf{t}}.$$

14.1 The integral coframe

Let $e_1, \dots, e_4$ be the standard basis of $\mathbb{R}^4$, put $u = P_T^{-1}x$, and define

$$\theta_T^j := \mathrm{d}u_j = (P_T^{-\mathsf{T}} e_j)^{\mathsf{t}} (\mathrm{d}x_1, \mathrm{d}y_1, \mathrm{d}x_2, \mathrm{d}y_2)^{\mathsf{t}}. \tag{404}$$

Proposition 14.1 (Integral coframe and the distinguished unit-flux plane)**.** *The forms θ_T^j are the integral coframe dual to the ordered marked lattice basis:*

$$\int_{[0,1] \ni s \mapsto [x + s P_T e_k]} \theta_T^j = \delta_{jk}.$$

The first two members are exactly

$$\theta_T^1 = \frac{m_T}{D_T}\,\mathrm{d}y_1 - \frac{L_T}{D_T}\,\mathrm{d}y_2, \qquad \theta_T^2 = \frac{q_T}{D_T}\,\mathrm{d}y_1 + \frac{6m_T}{D_T}\,\mathrm{d}y_2, \tag{405}$$

and hence

$$\theta_T^1 \wedge \theta_T^2 = \frac{1}{D_T}\,\mathrm{d}y_1 \wedge \mathrm{d}y_2, \qquad |\theta_T^1 \wedge \theta_T^2|^2 = \frac{1}{D_T^2}. \tag{406}$$

Proof. For $n \in \mathbb{Z}^4$,

$$u(x + P_T n) = u(x) + n,$$

which proves the integral-period assertion. The first two columns of the already-proved matrix $P_T^{-\mathsf{T}}$ in equation (29) give equation (405). Their wedge coefficient is

$$\frac{6m_T^2 + L_T q_T}{D_T^2} = \frac{1}{D_T}.$$

The coordinate forms $\mathrm{d}y_1, \mathrm{d}y_2$ are orthonormal for the fixed quotient metric, proving equation (406). □

14.2 The unit-flux line and the topologically trivial rank-two sum

The following construction is made in the marked coordinates u of (404); it therefore applies to the actual skew lattice $P_T\mathbb{Z}^4$, not to a rectangular replacement.

Proposition 14.2 (Explicit factor of automorphy and connection)**.** *For $n \in \mathbb{Z}^4$, set*

$$j_n(u) = \exp(-2\pi i n_1 u_2). \tag{407}$$

The quotient

$$\mathscr{L}_T = (\mathbb{R}^4 \times \mathbb{C})/\mathbb{Z}^4, \qquad n \cdot (u, z) = (u + n, j_n(u) z),$$

is a Hermitian line bundle over F_T. The connection on its pullback to $\mathbb{R}^4$,

$$\widetilde{\nabla}^{\mathscr{L}_T} = \mathrm{d} + 2\pi i u_1\,\mathrm{d}u_2, \tag{408}$$

descends, and its curvature is

$$F_{\mathscr{L}_T} = 2\pi i\,\mathrm{d}u_1 \wedge \mathrm{d}u_2 = 2\pi i\,\theta_T^1 \wedge \theta_T^2. \tag{409}$$

With the unitary Chern–Weil convention $c_1 = [\,iF/(2\pi)\,]$, one has $c_1(\mathscr{L}_T) = -[\theta_T^1 \wedge \theta_T^2]$.

Proof. For $m, n \in \mathbb{Z}^4$,

$$j_m(u+n)j_n(u) = j_{m+n}(u)\exp(-2\pi i m_1 n_2) = j_{m+n}(u).$$

Thus (407) is a factor of automorphy. Its values are unitary, so the standard fibre metric descends. If $\mathcal{A} = 2\pi i u_1 \,\mathrm{d}u_2$, then

$$\mathcal{A}(u+n) = \mathcal{A}(u) + 2\pi i n_1 \,\mathrm{d}u_2, \qquad j_n^{-1}\mathrm{d}j_n = -2\pi i n_1 \,\mathrm{d}u_2.$$

Hence $\mathcal{A}(u+n)+j_n^{-1}\mathrm{d}j_n = \mathcal{A}(u)$, which is precisely the descent identity for $\mathrm{d}+\mathcal{A}$ under the stated quotient convention. Taking $\mathrm{d}\mathcal{A}$ gives (409). The Chern–Weil statement follows without a sign suppression:

$$\frac{i}{2\pi}F_{\mathscr{L}_T} = -\theta_T^1 \wedge \theta_T^2.$$

□

Let $\mathscr{L}_T^{-1} = \mathscr{L}_T^*$ carry the dual metric and dual connection, and set

$$\mathscr{E}_T = \mathscr{L}_T \oplus \mathscr{L}_T^{-1}. \tag{410}$$

Proposition 14.3 (Determinant, c_2, and explicit $SU(2)$-triviality)**.** *The factors of automorphy of $\mathscr{E}_T$ are*

$$G_n(u) = \begin{pmatrix} j_n(u) & 0 \\ 0 & j_n(u)^{-1} \end{pmatrix} \in SU(2).$$

Evaluation gives a canonical parallel determinant trivialization

$$\det \mathscr{E}_T = \mathscr{L}_T \otimes \mathscr{L}_T^{-1} \xrightarrow{\mathrm{ev}} F_T \times \mathbb{C}.$$

Moreover

$$c_1(\mathscr{E}_T) = 0, \qquad c_2(\mathscr{E}_T) = 0 \quad \textit{in } H^4(F_T;\mathbb{Z}),$$

and the underlying principal $SU(2)$-bundle is smoothly trivial. The last assertion follows from the explicit nullhomotopy below and does not rely only on the vanishing of a de Rham Chern form.

Proof. The determinant assertion follows pointwise from $j_n j_n^{-1} = 1$. The induced determinant connection has connection form

$$\mathrm{tr}\begin{pmatrix} 2\pi i u_1 \,\mathrm{d}u_2 & 0 \\ 0 & -2\pi i u_1 \,\mathrm{d}u_2 \end{pmatrix} = 0,$$

so the evaluation trivialization is parallel.

Let $c = c_1(\mathscr{L}_T)$. The integral Whitney identity gives

$$c(\mathscr{E}_T) = (1+c)(1-c) = 1 - c^2.$$

The factor j_n depends only on (u_1, u_2), so $\mathscr{L}_T$ is the pullback of a line bundle L_0 on $\mathbb{R}^2/\mathbb{Z}^2$ under the marked projection $[u_1, u_2, u_3, u_4] \mapsto [u_1, u_2]$. Hence

$$c^2 = \pi_{12}^*(c_1(L_0)^2) = 0$$

integrally because $H^4(\mathbb{T}^2;\mathbb{Z}) = 0$. This proves the two Chern-class identities on the four-torus.

It remains to trivialize the $SU(2)$-bundle itself. On the fundamental square $0 \le u_1, u_2 \le 1$ for the first two marked coordinates, the only nontrivial seam is

$$(0, t, v) \sim (1, t, \gamma(t)v), \qquad \gamma(t) = \begin{pmatrix} e^{-2\pi i t} & 0 \\ 0 & e^{2\pi i t} \end{pmatrix}.$$

For $0 \le r \le 1$, define

$$a(r,t) = (1-r)e^{-2\pi i t} + r, \qquad s(r) = \sin(\pi r), \qquad N(r,t) = \big(|a(r,t)|^2 + s(r)^2\big)^{1/2},$$

and

$$U(r,t) = \frac{1}{N(r,t)} \begin{pmatrix} a(r,t) & -s(r) \\ s(r) & a(r,t) \end{pmatrix}. \tag{411}$$

If $0 < r < 1$, then $s(r) > 0$; at $r = 0, 1$, one has $|a(r,t)| = 1$. Thus N never vanishes. The two columns in (411) are orthonormal and its determinant is one, so $U(r,t) \in SU(2)$. It is smooth and periodic in t, and

$$U(0,t) = \gamma(t), \qquad U(1,t) = I_2.$$

This is an explicit nullhomotopy of the clutching loop. To see directly that it produces a smooth global frame, choose a smooth $\chi : [0,1] \to [0,1]$ which is zero near 0 and one near 1, and put

$$K(\sigma,t) = U(1-\chi(\sigma),t).$$

Then $K = I_2$ near $\sigma = 0$, $K = \gamma$ near $\sigma = 1$, and $K(\sigma,0) = K(\sigma,1)$. The required trivialization, written in the direction from the clutched bundle to the product bundle, is

$$[\sigma,t,v] \longmapsto \big([\sigma,t], K(\sigma,t)^{-1}v\big). \tag{412}$$

Indeed, the quotient convention gives $[0,t,v] = [1,t,\gamma(t)v]$, and the two representatives have the same image because

$$K(1,t)^{-1}\gamma(t)v = \gamma(t)^{-1}\gamma(t)v = v = K(0,t)^{-1}v.$$

At the other seam, $K(\sigma,0)^{-1} = K(\sigma,1)^{-1}$. The constant collars used in the definition of χ make the local matrix in equation (412) constant after applying the seam transition, so the map and its inverse are smooth across the seam. It is fibrewise unitary and determinant one. Extending it independently of u_3, u_4 and pulling it through the marking gives the required global $SU(2)$-frame of $\mathscr{E}_T$. □

14.3 A non-flat reducible Yang–Mills critical connection

The symbol $H = \operatorname{Im} h(0)$ is already fixed by equation (24). For the requested Lie-algebra element, fix instead the unambiguous notation

$$\langle X, Y\rangle_{\mathfrak{su}(2)} = -\frac{1}{2}\operatorname{Tr}(XY), \qquad H_{\mathrm{mag}} = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}. \tag{413}$$

Theorem 14.4 (Reducible critical point and exact action)**.** *The direct-sum connection on $\mathscr{E}_T$ has, in the lifted splitting,*

$$\nabla^T = \mathrm{d} + 2\pi u_1\,\mathrm{d}u_2\,H_{\mathrm{mag}}, \qquad F_{\nabla^T} = 2\pi H_{\mathrm{mag}}\,\theta_T^1 \wedge \theta_T^2 = \frac{2\pi}{D_T} H_{\mathrm{mag}}\,\mathrm{d}y_1 \wedge \mathrm{d}y_2. \tag{414}$$

After any of the global $SU(2)$-trivializations from theorem 14.3, this is a connection on the trivial principal $SU(2)$-bundle. It is non-flat and satisfies

$$\mathrm{d}^*_{\nabla^T} F_{\nabla^T} = 0.$$

For the normalization

$$\mathrm{YM}_T(\nabla) = \frac{1}{2g_{\mathrm{YM}}^2}\int_{F_T} |F_\nabla|^2\,\mathrm{d}\,\mathrm{vol}_{F_T}, \tag{415}$$

its action is exactly

$$\mathrm{YM}_T(\nabla^T) = \frac{2\pi^2}{g_{\mathrm{YM}}^2 D_T}. \tag{416}$$

Its first inverse-power correction, in the canonical scalar notation $H = \operatorname{Im} h(0)$, $B = \operatorname{Im} b(0)$, is

$$\mathrm{YM}_T(\nabla^T) = \frac{2\pi^2}{g_{\mathrm{YM}}^2 T^2}\left(1 + \frac{B - 2H}{T} + O(T^{-2})\right). \tag{417}$$

Proof. The connection matrix is

$$\begin{pmatrix} 2\pi i u_1 \,\mathrm{d}u_2 & 0 \\ 0 & -2\pi i u_1 \,\mathrm{d}u_2 \end{pmatrix} = 2\pi u_1 \,\mathrm{d}u_2 \, H_{\mathrm{mag}},$$

so equation (414) follows from theorem 14.2 and equation (406). The diagonal endomorphism H_{mag} is a globally defined parallel section of $\operatorname{ad} \mathscr{E}_T$, because every transition G_n commutes with it. The forms θ_T^j are parallel constant forms for the descended flat metric. Thus $\nabla^T F_{\nabla^T} = 0$, which implies $\mathrm{d}^*_{\nabla^T} F_{\nabla^T} = 0$. The curvature is nonzero because $D_T > 0$.

The chosen Lie-algebra normalization gives

$$|H_{\mathrm{mag}}|^2 = -\frac{1}{2}\operatorname{Tr}(H_{\mathrm{mag}}^2) = 1.$$

Equations equations (48) and (406) therefore give

$$\int_{F_T} |F_{\nabla^T}|^2 \,\mathrm{d}\operatorname{vol}_{F_T} = 4\pi^2 D_T \frac{1}{D_T^2} = \frac{4\pi^2}{D_T},$$

which proves equation (416). Finally, equation (47) and the elementary reciprocal expansion

$$\frac{1}{T^2 + aT + O(1)} = \frac{1}{T^2}\left(1 - \frac{a}{T} + O(T^{-2})\right)$$

with $a = 2H - B$ prove equation (417). □

14.4 Constant-curvature magnetic Laplacians for arbitrary integral nondegenerate flux

We next give the Hilbert space, closed form, self-adjoint realization, and oscillator spectrum in a form valid for every nondegenerate integral alternating form on the marked four-torus.

Let

$$\mathsf{E} = \sum_{1 \le j < k \le 4} E_{jk} \,\mathrm{d}u_j \wedge \mathrm{d}u_k \in \bigwedge^2 (\mathbb{Z}^4)^*, \qquad E_{jk} \in \mathbb{Z},$$

and let the same symbol denote its skew matrix. For $n \in \mathbb{Z}^4$, put

$$J_n^{\mathsf{E}}(u) = \exp\left(-2\pi i \sum_{j<k} E_{jk} n_j u_k\right). \tag{418}$$

The integer in

$$\sum_{j<k} E_{jk} m_j n_k$$

shows directly that $J_m^{\mathsf{E}}(u+n) J_n^{\mathsf{E}}(u) = J_{m+n}^{\mathsf{E}}(u)$. Thus (418) defines a Hermitian line bundle $\mathscr{L}_{\mathsf{E},T}$. On its pullback, set

$$\nabla^{\mathsf{E}} = \mathrm{d} + 2\pi i \sum_{j<k} E_{jk} u_j \,\mathrm{d}u_k. \tag{419}$$

Exactly as in theorem 14.2, the connection descends and

$$F_{\nabla^{\mathsf{E}}} = 2\pi i\,\mathsf{E}. \tag{420}$$

In u-coordinates the flat metric matrix is

$$G_T = P_T^{\mathsf{T}} P_T, \qquad \sqrt{\det G_T} = D_T. \tag{421}$$

Define the Hilbert space concretely by

$$\begin{aligned} \mathcal{H}_{\mathsf{E},T} &= \{\, f \in L^2_{\mathrm{loc}}(\mathbb{R}^4) : f(u+n) = J_n^{\mathsf{E}}(u) f(u) \text{ for every } n \in \mathbb{Z}^4 \}, \\ \|f\|^2 &= \int_{[0,1]^4} |f(u)|^2 \sqrt{\det G_T}\, \mathrm{d}^4 u. \end{aligned} \tag{422}$$

Let $\mathcal{C}_{\mathsf{E},T}$ be the smooth functions with the same automorphy. On this dense subspace define

$$q_{\mathsf{E},T}[f] = \int_{[0,1]^4} (G_T^{-1})^{jk} \overline{\nabla_j^{\mathsf{E}} f}\, \nabla_k^{\mathsf{E}} f \sqrt{\det G_T}\, \mathrm{d}^4 u. \tag{423}$$

Theorem 14.5 (Friedrichs realization and oscillator spectrum)**.** *The form* (423) *is nonnegative and closable. Its closed form domain is*

$$\operatorname{Dom} \overline{q}_{\mathsf{E},T} = H^1(F_T; \mathscr{L}_{\mathsf{E},T}).$$

Its Friedrichs operator is the magnetic Laplacian

$$\Delta_{\mathsf{E},T} = (\nabla^{\mathsf{E}})^* \nabla^{\mathsf{E}}, \qquad \operatorname{Dom}(\Delta_{\mathsf{E},T}) = H^2(F_T; \mathscr{L}_{\mathsf{E},T}),$$

and it has compact resolvent.

Assume now that E *is nondegenerate. Let* $b_1, b_2 > 0$ *be defined by the eigenvalues*

$$\operatorname{Spec}\Big(G_T^{-1/2} (2\pi \mathsf{E}) G_T^{-1/2} \Big) = \{ i b_1, -i b_1, i b_2, -i b_2 \}. \tag{424}$$

Then, as a set (with finite multiplicities),

$$\operatorname{Spec}(\Delta_{\mathsf{E},T}) = \{ (2n_1+1) b_1 + (2n_2+1) b_2 : n_1, n_2 \in \mathbb{Z}_{\geq 0} \}. \tag{425}$$

In particular, the ground-to-first-distinct-level spacing is

$$\operatorname{gap}(\Delta_{\mathsf{E},T}) = 2 \min(b_1, b_2). \tag{426}$$

No multiplicity formula is needed for, or asserted in, these identities.

Proof. In local unitary trivializations, the integrand in (423) is the squared norm of a first-order unitary covariant derivative. A finite trivializing cover and a partition of unity show that the form norm $\|f\|^2 + q_{\mathsf{E},T}[f]$ is equivalent to the usual local H^1 norm. Its completion is therefore $H^1(F_T; \mathscr{L}_{\mathsf{E},T})$, proving closability and identifying the closed domain. The representation theorem for closed nonnegative forms gives the Friedrichs operator. Its local principal symbol is $G_T^{-1}(\xi, \xi)$, so it is uniformly elliptic. Standard interior elliptic regularity on the compact manifold without boundary identifies the operator domain with H^2. The compact inclusion $H^1 \hookrightarrow L^2$ implies compact resolvent.

The real skew-normal-form theorem supplies a constant orthonormal frame $v_1, \dots, v_4$ on the Euclidean cover in which the real two-form $2\pi\mathsf{E}$ is

$$b_1 v^1 \wedge v^2 + b_2 v^3 \wedge v^4.$$

Changing the sign of v^2 or v^4, if necessary, makes both frequencies positive. Let ∇_a denote covariant differentiation along v_a. Each ∇_a is formally skew-adjoint on the common invariant core $\mathcal{C}_{\mathsf{E},T}$; the closures used below are the ones determined by the closed magnetic form. On that smooth core one has

$$[\nabla_1, \nabla_2] = ib_1, \qquad [\nabla_3, \nabla_4] = ib_2,$$

and all other commutators vanish. Put

$$C_1 = \nabla_1 - i\nabla_2, \qquad C_2 = \nabla_3 - i\nabla_4, \qquad \Delta_j = -(\nabla_{2j-1}^2 + \nabla_{2j}^2).$$

Direct multiplication, retaining the commutator signs, gives

$$C_j^* C_j = \Delta_j - b_j, \qquad C_j C_j^* = \Delta_j + b_j, \qquad [\Delta_{\mathsf{E},T}, C_j] = -2b_j C_j. \tag{427}$$

Because the resolvent is compact, the lowest eigenvalue has a smooth eigenfunction $\psi_0 \neq 0$. If $C_j\psi_0 \neq 0$, the last identity in (427) would make it an eigenfunction with a strictly smaller eigenvalue. Hence $C_1\psi_0 = C_2\psi_0 = 0$, and the first identity gives

$$\Delta_{\mathsf{E},T}\psi_0 = (b_1 + b_2)\psi_0.$$

Conversely, repeated application of C_1^*, C_2^* produces nonzero eigenfunctions with eigenvalues $(2n_1+1)b_1 + (2n_2+1)b_2$. Nonvanishing follows from $C_jC_j^* = C_j^*C_j + 2b_j$.

It remains to exclude other eigenvalues. For an eigenfunction ψ with eigenvalue $\lambda > b_1 + b_2$, the operator identity

$$(\lambda - b_1 - b_2)\psi = C_1^*C_1\psi + C_2^*C_2\psi$$

expresses ψ as a sum of raising operators applied to the lower eigenfunctions $C_j\psi$. Induction over the finite set of eigenvalues below λ reduces every eigenfunction to raised ground states. This proves (425). Raising once in a least-frequency plane proves (426). □

Corollary 14.6 (Frequency product and cusp spacing bound)**.** *Let*

$$d = |\operatorname{Pf}(\mathsf{E})| = |E_{12}E_{34} - E_{13}E_{24} + E_{14}E_{23}| > 0.$$

Then

$$b_1 b_2 = \frac{4\pi^2 d}{D_T}, \tag{428}$$

and

$$\begin{aligned} \operatorname{gap}(\Delta_{\mathsf{E},T}) &\le 4\pi\sqrt{\frac{d}{D_T}} \\ &= \frac{4\pi\sqrt{d}}{T}\left(1 + \frac{B - 2H}{2T} + O(T^{-2})\right). \end{aligned} \tag{429}$$

Proof. The determinant of the skew matrix in (424) is both $b_1^2 b_2^2$ and

$$\frac{\det(2\pi\mathsf{E})}{\det G_T} = \frac{(2\pi)^4 \operatorname{Pf}(\mathsf{E})^2}{D_T^2}.$$

Taking the positive square root proves (428). Since $\min(b_1, b_2) \le \sqrt{b_1 b_2}$, Theorem 14.5 gives (429). Expanding the square root of equation (47) gives the displayed first correction. □

14.5 A background-gauge negative mode

We finish with the classical background-gauge Jacobi operator at the reducible critical connection. Every sign below uses exactly the inner product and the element H_{mag} fixed in equation (413).

Put

$$\mathscr{H}_T^{\mathrm{ad},1} := L^2\Omega^1(F_T; \operatorname{ad}\mathscr{E}_T).$$

Put also

$$\mathscr{H}_T^{\mathrm{ad},0} := L^2\Omega^0(F_T; \operatorname{ad}\mathscr{E}_T), \qquad D_T^{\max} := \mathrm{d}_{\nabla^T}^{*,\max}.$$

For the resolvent argument these real Hilbert spaces are complexified; all operators commute with the real structure, so their restrictions to real sections are unchanged. Here the maximal distributional adjoint is the closed, densely defined operator whose domain and value are specified without an implicit regularity assumption by

$$\begin{aligned}\operatorname{Dom}(D_T^{\max}) := \{a \in \mathscr{H}_T^{\mathrm{ad},1} :\ &\text{there is a } u \in \mathscr{H}_T^{\mathrm{ad},0} \text{ such that}\\ &\langle a, \mathrm{d}_{\nabla^T}\phi\rangle_{L^2} = \langle u, \phi\rangle_{L^2} \text{ for every } \phi \in C^\infty(F_T; \operatorname{ad}\mathscr{E}_T)\},\\ D_T^{\max} a := u.&\end{aligned} \tag{430}$$

The test space is dense, so u is unique. The definition also proves closedness directly: if $a_n \to a$ and $D_T^{\max} a_n \to u$ in L^2, passage to the limit in the displayed testing identity gives $a \in \operatorname{Dom}(D_T^{\max})$ and $D_T^{\max} a = u$. Consequently the background slice

$$\mathscr{S}_T := \ker D_T^{\max} \subset \mathscr{H}_T^{\mathrm{ad},1}. \tag{431}$$

is a closed subspace.

The scalar covariant Laplacian used below is fixed as follows. On $C^\infty(F_T; \operatorname{ad}\mathscr{E}_T)$ start with the nonnegative form

$$q_T^{(0)}[\phi, \psi] := \langle \mathrm{d}_{\nabla^T}\phi, \mathrm{d}_{\nabla^T}\psi\rangle_{L^2}.$$

Its closure has domain $H^1\Omega^0(F_T; \operatorname{ad}\mathscr{E}_T)$, and its associated Friedrichs operator is

$$\Delta_{\nabla^T}^{(0)} := \mathrm{d}_{\nabla^T}^* \mathrm{d}_{\nabla^T}, \qquad \operatorname{Dom}(\Delta_{\nabla^T}^{(0)}) = H^2\Omega^0(F_T; \operatorname{ad}\mathscr{E}_T). \tag{432}$$

The H^2 domain follows from elliptic regularity on the compact flat torus; in particular, this notation always denotes the nonnegative self-adjoint realization, not merely its differential expression.

For an adjoint-valued one-form a, define the smooth bundle endomorphism

$$(V_T a)_\nu := -2\sum_{\mu=1}^{4} [F_{\nu\mu}, a_\mu].$$

It is symmetric: after interchanging μ, ν, the identities $F_{\mu\nu} = -F_{\nu\mu}$ and skew-adjointness of $\operatorname{ad}(F_{\nu\mu})$ give $\langle V_T a, b\rangle = \langle a, V_T b\rangle$. On the full one-form Hilbert space define

$$\mathfrak{j}_T^{\mathrm{gf}}[a, b] := \langle \nabla^T a, \nabla^T b\rangle_{L^2} + \langle V_T a, b\rangle_{L^2}, \qquad \operatorname{Dom}(\mathfrak{j}_T^{\mathrm{gf}}) = H^1\Omega^1(F_T; \operatorname{ad}\mathscr{E}_T). \tag{433}$$

Because V_T is a bounded symmetric endomorphism on the compact torus, this form is closed and bounded below. Its associated self-adjoint operator $\mathcal{J}_T^{\mathrm{gf}}$ has

$$\operatorname{Dom}(\mathcal{J}_T^{\mathrm{gf}}) = H^2\Omega^1(F_T; \operatorname{ad}\mathscr{E}_T) \tag{434}$$

and differential expression

$$(\mathcal{J}_T^{\mathrm{gf}} a)_\nu = -\sum_{\mu=1}^{4} \nabla_{e_\mu}\nabla_{e_\mu} a_\nu - 2\sum_{\mu=1}^{4} [F_{\nu\mu}, a_\mu].$$

Proposition 14.7 (Intertwining and the reducing Coulomb slice)**.** *At the Yang–Mills connection ∇^T, the following formal identities hold on smooth sections:*

$$\mathcal{J}_T^{\mathrm{gf}}\mathrm{d}_{\nabla^T} = \mathrm{d}_{\nabla^T}\Delta^{(0)}_{\nabla^T}, \qquad \mathrm{d}^*_{\nabla^T}\mathcal{J}_T^{\mathrm{gf}} = \Delta^{(0)}_{\nabla^T}\mathrm{d}^*_{\nabla^T}. \tag{435}$$

For every $z \in \mathbb{C}\setminus\mathbb{R}$, the correctly typed closed-operator resolvent identity is

$$D_T^{\max}(\mathcal{J}_T^{\mathrm{gf}} - z)^{-1} f = (\Delta^{(0)}_{\nabla^T} - z)^{-1} D_T^{\max} f, \qquad f \in \mathrm{Dom}(D_T^{\max}). \tag{436}$$

In particular, $\mathscr{S}_T$ reduces $\mathcal{J}_T^{\mathrm{gf}}$. The part of the full operator in this subspace and its closed form are therefore

$$\mathcal{J}_T := \mathcal{J}_T^{\mathrm{gf}}\big|_{\mathscr{S}_T}, \quad \mathrm{Dom}(\mathcal{J}_T) = H^2\Omega^1(F_T;\mathrm{ad}\,\mathscr{E}_T)\cap\mathscr{S}_T, \quad \mathrm{Dom}(\mathfrak{j}_T) = H^1\Omega^1(F_T;\mathrm{ad}\,\mathscr{E}_T)\cap\mathscr{S}_T \tag{437}$$

with $\mathfrak{j}_T = \mathfrak{j}_T^{\mathrm{gf}}|_{\mathrm{Dom}(\mathfrak{j}_T)}$. The operator $\mathcal{J}_T$ is self-adjoint on $\mathscr{S}_T$, and $\mathfrak{j}_T$ is its closed, lower-bounded form.

Proof. Write $\nabla_\mu = \nabla^T_{e_\mu}$ in the constant orthonormal frame, so $[\nabla_\mu,\nabla_\nu]\phi = [F_{\mu\nu},\phi]$. For a smooth adjoint-valued scalar ϕ, commuting the derivatives without suppressing either curvature contribution gives

$$\begin{aligned}
&\big(\mathcal{J}_T^{\mathrm{gf}}\mathrm{d}_{\nabla^T}\phi - \mathrm{d}_{\nabla^T}\Delta^{(0)}_{\nabla^T}\phi\big)_\nu \\
&\quad = -\nabla_\mu\nabla_\mu\nabla_\nu\phi + \nabla_\nu\nabla_\mu\nabla_\mu\phi - 2[F_{\nu\mu},\nabla_\mu\phi] \\
&\quad = -[\nabla_\mu F_{\mu\nu},\phi] = [(\mathrm{d}^*_{\nabla^T}F_{\nabla^T})_\nu,\phi].
\end{aligned}$$

The last member vanishes by theorem 14.4. This proves the first identity in (435). Since both Laplacians and $\mathcal{J}_T^{\mathrm{gf}}$ are formally symmetric, pairing it against arbitrary smooth zero- and one-forms proves the second identity.

Fix $z \notin \mathbb{R}$, take $f \in \mathrm{Dom}(D_T^{\max})$, and put $u = (\mathcal{J}_T^{\mathrm{gf}} - z)^{-1}f$. The full operator domain gives $u \in H^2$, hence $D_T^{\max}u \in H^1$. Moreover $(\mathcal{J}_T^{\mathrm{gf}} - z)u = f$ and the distributional extension of the second formal identity imply

$$(\Delta^{(0)}_{\nabla^T} - z)D_T^{\max}u = D_T^{\max}f \quad \text{in } \mathcal{D}'(F_T;\mathrm{ad}\,\mathscr{E}_T). \tag{438}$$

The right side is in L^2. Elliptic regularity for the scalar Laplacian upgrades $D_T^{\max}u$ from H^1 to H^2, so it belongs to $\mathrm{Dom}(\Delta^{(0)}_{\nabla^T})$. Applying the bounded scalar resolvent to (438) proves (436); no extension of $D_T^{\max}$ to all of L^2 is asserted or needed.

If $f \in \mathscr{S}_T$, the right side of (436) is zero, hence both resolvents at z and at $\overline{z}$ preserve $\mathscr{S}_T$. For $g \in \mathscr{S}_T^{\perp}$ and $s \in \mathscr{S}_T$, self-adjointness of the full operator gives

$$\langle(\mathcal{J}_T^{\mathrm{gf}} - z)^{-1}g, s\rangle = \langle g, (\mathcal{J}_T^{\mathrm{gf}} - \overline{z})^{-1}s\rangle = 0.$$

Thus the resolvent preserves both $\mathscr{S}_T$ and its orthogonal complement, which is precisely the reducing property. The standard part of a self-adjoint operator on a reducing subspace is self-adjoint, and its domain is the intersection in (437). The reducing orthogonal projection commutes with the spectral calculus of the lower-bounded full operator, including its form square root; this proves the asserted form domain and completes the proof. □

Theorem 14.8 (Exact charged Landau negative mode)**.** *For the self-adjoint Yang–Mills Jacobi operator $\mathcal{J}_T$ on the closed background slice $\mathscr{S}_T$ defined above, the differential expression in a constant orthonormal frame $(e_\mu)_{\mu=1}^4$ is*

$$(\mathcal{J}_T a)_\nu = -\sum_{\mu=1}^{4}\nabla_{e_\mu}\nabla_{e_\mu}a_\nu - 2\sum_{\mu=1}^{4}[F_{\nu\mu},a_\mu]. \tag{439}$$

There exists a nonzero smooth real $\mathfrak{su}(2)$-valued one-form a such that

$$\|a\|_{L^2} = 1, \qquad \mathrm{d}^*_{\nabla^T} a = 0, \qquad \mathcal{J}_T a = -\frac{4\pi}{D_T} a. \tag{440}$$

Equivalently, for the action normalization equation (415)*,*

$$\mathrm{YM}''_T\big|_{\nabla^T}(a,a) = -\frac{4\pi}{g^2_{\mathrm{YM}} D_T}. \tag{441}$$

Thus the reducible non-flat critical connection has a strictly negative classical second-variation direction for every finite T.

Proof. First isolate the exact two-dimensional magnetic problem. Let

$$W = \{x_1 = x_2 = 0\} \subset \mathbb{R}^4,$$

so that (y_1, y_2) are oriented orthonormal coordinates on W. By equation (27), orthogonal projection to W sends the first two marked lattice generators to

$$v_1 = (6m_T, -q_T), \qquad v_2 = (L_T, m_T),$$

and sends the last two to zero. Since $\det(v_1, v_2) = D_T > 0$, projection descends to

$$\varpi_T : F_T \longrightarrow \Sigma_T := W/(\mathbb{Z}v_1 + \mathbb{Z}v_2).$$

The factors and connection of $\mathscr{L}^2_T$ depend only on (u_1, u_2). They therefore define a Hermitian line bundle with unitary connection $(\mathscr{M}_T, \nabla^{\mathscr{M}_T})$ on Σ_T such that

$$(\mathscr{L}^2_T, \nabla^{\mathscr{L}^2_T}) = \varpi_T^*(\mathscr{M}_T, \nabla^{\mathscr{M}_T}). \tag{442}$$

Let $\Delta_+ = (\nabla^{\mathscr{M}_T})^* \nabla^{\mathscr{M}_T}$ be the nonnegative self-adjoint Bochner Laplacian on $L^2(\Sigma_T; \mathscr{M}_T)$, defined by its closed H^1 form and with operator domain $H^2(\Sigma_T; \mathscr{M}_T)$. More precisely, if $c_T > 0$ is the constant Riemannian volume of a fibre of ϖ_T, then

$$U_T : L^2(\Sigma_T; \mathscr{M}_T) \longrightarrow L^2(F_T; \mathscr{L}^2_T), \qquad U_T s = c_T^{-1/2} \varpi_T^* s,$$

is an isometry. Its image

$$\mathscr{B}_T := U_T L^2(\Sigma_T; \mathscr{M}_T)$$

is the closed fibre-basic subspace. The pulled-back realization is, by definition,

$$\Delta_{+,\mathrm{bas}} := U_T \Delta_+ U_T^{-1} \quad \text{on } \mathscr{B}_T, \qquad \mathrm{Dom}(\Delta_{+,\mathrm{bas}}) = U_T H^2(\Sigma_T; \mathscr{M}_T),$$

and its closed form domain is $U_T H^1(\Sigma_T; \mathscr{M}_T)$. Only this operator on $\mathscr{B}_T$ is self-adjoint. In the formulas below we abbreviate $\Delta_{+,\mathrm{bas}}$ again to Δ_+; no self-adjoint operator on the full $L^2(F_T; \mathscr{L}^2_T)$ is thereby defined or asserted. On this root line the curvature is

$$F_{\mathscr{L}^2_T} = 4\pi i\, \theta^1_T \wedge \theta^2_T = i\kappa_T \,\mathrm{d}y_1 \wedge \mathrm{d}y_2, \qquad \kappa_T = \frac{4\pi}{D_T} > 0. \tag{443}$$

We now construct an exact lowest Landau section on this actual projected lattice. In the marked coordinates put

$$\mathcal{D}_1 := \partial_{u_1}, \qquad \mathcal{D}_2 := \partial_{u_2} + 4\pi i u_1,$$

and define

$$\zeta_T := \frac{q_T - 6im_T}{m_T + iL_T}, \qquad v_T := -\operatorname{Im} \zeta_T = \frac{D_T}{m_T^2 + L_T^2} > 0. \tag{444}$$

Indeed,

$$\operatorname{Im}((q_T - 6im_T)(m_T - iL_T)) = -(L_T q_T + 6m_T^2) = -D_T,$$

which also checks the sign $v_T > 0$. The series

$$\Theta_T(u_1, u_2) = \sum_{\ell \in \mathbb{Z}} \exp\Big(-2\pi i \zeta_T (\ell + u_1)^2 + 4\pi i \ell u_2\Big) \tag{445}$$

converges normally with all derivatives, because the absolute value of its ℓ-th summand is $\exp(-2\pi v_T(\ell + u_1)^2)$. Reindexing the series and using $\ell \in \mathbb{Z}$ give

$$\Theta_T(u_1 + 1, u_2) = e^{-4\pi i u_2}\Theta_T(u_1, u_2), \qquad \Theta_T(u_1, u_2 + 1) = \Theta_T(u_1, u_2). \tag{446}$$

Thus Θ_T is a smooth section of $\mathscr{L}_T^2$. Termwise differentiation gives

$$(\mathcal{D}_1 + \zeta_T \mathcal{D}_2)\Theta_T = 0. \tag{447}$$

The exact period block on the (y_1, y_2)-plane is

$$\begin{pmatrix} 6m_T & L_T \\ -q_T & m_T \end{pmatrix}.$$

Its inverse transpose gives the physical covariant derivatives

$$\nabla^+_{y_1} = \frac{m_T \mathcal{D}_1 + q_T \mathcal{D}_2}{D_T}, \qquad \nabla^+_{y_2} = \frac{-L_T \mathcal{D}_1 + 6m_T \mathcal{D}_2}{D_T}. \tag{448}$$

Consequently

$$\nabla^+_{y_1} - i\nabla^+_{y_2} = \frac{m_T + iL_T}{D_T}(\mathcal{D}_1 + \zeta_T \mathcal{D}_2). \tag{449}$$

Orthogonality of the u_2-Fourier modes also gives the nonzero norm

$$\int_0^1 \int_0^1 |\Theta_T|^2 \, \mathrm{d}u_1 \, \mathrm{d}u_2 = \sum_{\ell \in \mathbb{Z}} \int_0^1 e^{-4\pi v_T (\ell + u_1)^2} \, \mathrm{d}u_1 = \frac{1}{2\sqrt{v_T}}.$$

Set

$$\phi_T := \left(\frac{2\sqrt{v_T}}{D_T}\right)^{1/2} \Theta_T$$

and pull it back by ϖ_T. Since the marked-coordinate volume element is $D_T \, \mathrm{d}^4 u$, it has unit $L^2(F_T)$-norm. Its covariant derivatives in the x_1, x_2 directions vanish, and

$$(\nabla^+_{y_1} - i\nabla^+_{y_2})\phi_T = 0, \qquad \Delta_+ \phi_T = \kappa_T \phi_T, \qquad \Delta_+ \phi_T = -((\nabla^+_{y_1})^2 + (\nabla^+_{y_2})^2)\phi_T. \tag{450}$$

Indeed, equation (448) and $[\mathcal{D}_1, \mathcal{D}_2] = 4\pi i$ give

$$[\nabla^+_{y_1}, \nabla^+_{y_2}] = \frac{m_T(6m_T) - q_T(-L_T)}{D_T^2} \, 4\pi i = i\kappa_T.$$

Direct multiplication then gives

$$(\nabla^+_{y_1} - i\nabla^+_{y_2})^*(\nabla^+_{y_1} - i\nabla^+_{y_2}) = \Delta_+ - \kappa_T,$$

and the second equation in (450) follows from the first.

In $\mathfrak{sl}_2(\mathbb{C})$, put

$$X_+ = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad X_- = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

Then

$$[H_{\mathrm{mag}}, X_+] = 2iX_+, \qquad [H_{\mathrm{mag}}, X_-] = -2iX_-.$$

The X_+-root line in $\mathrm{ad}(\mathscr{L}_T \oplus \mathscr{L}_T^{-1}) \otimes \mathbb{C}$ is $\mathscr{L}_T^2$, so

$$a^+ = \phi_T X_+(\mathrm{d}y_1 - i\,\mathrm{d}y_2)$$

is a globally defined complexified adjoint-valued one-form. The first equation in (450) gives

$$\mathrm{d}^*_{\nabla^T} a^+ = -(\nabla^+_{y_1} - i\nabla^+_{y_2})\phi_T\, X_+ = 0.$$

Put $\mathcal{B}_T := 2\pi/D_T$, so that $\kappa_T = 2\mathcal{B}_T$ and $F_{y_1y_2} = \mathcal{B}_T H_{\mathrm{mag}}$. The two nonzero components of a^+ are $a^+_{y_1} = \phi_T X_+$ and $a^+_{y_2} = -i\phi_T X_+$. Substitution into (439), with no suppressed charge factor, gives

$$\begin{aligned}(\mathcal{J}_T a^+)_{y_1} &= \kappa_T\phi_T X_+ - 2[\mathcal{B}_T H_{\mathrm{mag}}, -i\phi_T X_+] = -\kappa_T\phi_T X_+,\\ (\mathcal{J}_T a^+)_{y_2} &= -i\kappa_T\phi_T X_+ - 2[-\mathcal{B}_T H_{\mathrm{mag}}, \phi_T X_+] = i\kappa_T\phi_T X_+.\end{aligned}$$

Thus $\mathcal{J}_T a^+ = -\kappa_T a^+$.

For completeness, formula (439) follows by expanding $F_{\nabla^T + ta} = F_{\nabla^T} + t\,\mathrm{d}_{\nabla^T} a + \frac{1}{2}t^2[a \wedge a]$, integrating the quadratic form by parts on the background slice, and using the flat-base Weitzenbock identity. The Hodge Laplacian contributes $-[F_{\nu\mu}, a_\mu]$, and the curvature term in the second variation contributes the identical term, giving the coefficient -2.

Finally take the anti-Hermitian real form

$$a := \frac{1}{\sqrt{2}}(a^+ - (a^+)^\dagger) = \frac{1}{\sqrt{2}}\left(\phi_T X_+(\mathrm{d}y_1 - i\,\mathrm{d}y_2) - \overline{\phi_T} X_-(\mathrm{d}y_1 + i\,\mathrm{d}y_2)\right). \tag{451}$$

It is nonzero, traceless, and anti-Hermitian. For $X(z) := zX_+ - \overline{z}X_-$, the fixed inner product gives $-\frac{1}{2}\operatorname{Tr}(X(z)^2) = |z|^2$. The two orthonormal one-form components in equation (451) therefore have total pointwise norm $|a|^2 = |\phi_T|^2$, so $\|a\|_{L^2} = 1$. Complex conjugation and Hermitian adjunction preserve both the gauge equation and the real Jacobi operator, so (440) follows from the calculation for a^+. The Hessian identity follows from $\mathrm{YM}''_T(a,a) = g_{\mathrm{YM}}^{-2}\langle a, \mathcal{J}_T a\rangle_{L^2}$ on the background slice. □

15 The native full-flux magnetic sector

This section retains the canonical radial parameter, period matrix, Euclidean quotient metric, marked lattice, and every sign already fixed in section 11. It treats the monodromy-invariant integral class that uses both elementary alternating blocks. It is therefore distinct from the unit-flux plane treated in section 14. No metric rescaling, finite covering, rectangular replacement of the lattice, or orientation suppression occurs below.

For $T > T_0$, retain

$$\begin{aligned} L_T = T + H_T = \operatorname{Im}\tau_T, \qquad q_T = L_T - B_T = -\operatorname{Im}\beta_T,\\ m_T = \operatorname{Im}\mu_T, \qquad D_T = L_T q_T + 6m_T^2 > 0,\end{aligned} \tag{452}$$

and write

$$\eta_T^{\mathrm{re}} := \operatorname{Re} h_T.$$

This scalar is denoted by η_T in section 11; the superscript is retained throughout this section so that it cannot be confused with the native integral two-form introduced below. The complete period matrix is

$$P_T = \begin{pmatrix} 6\alpha_T & \eta_T^{\mathrm{re}} & 1 & 0\\ 6m_T & L_T & 0 & 0\\ c_T & \alpha_T & 0 & 1\\ -q_T & m_T & 0 & 0\end{pmatrix}, \qquad \det P_T = -D_T. \tag{453}$$

Thus $F_T = \mathbb{R}^4/P_T\mathbb{Z}^4$ has the unrescaled Euclidean quotient metric and volume D_T. In the marking coordinates $y \in \mathbb{R}^4$, the map $y \mapsto x = P_T y$ pulls the Euclidean metric back to $G_T = P_T^{\mathsf{T}} P_T$.

The limiting cusp constants remain

$$m = \operatorname{Im}\mu(0), \qquad H = \operatorname{Im} h(0), \qquad B = \operatorname{Im} b(0), \tag{454}$$

so holomorphy at $e^{-2\pi T} = 0$ gives

$$\begin{aligned} m_T &= m + O(e^{-2\pi T}), \quad H_T = H + O(e^{-2\pi T}), \\ B_T &= B + O(e^{-2\pi T}), \quad D_T = T^2 + (2H - B)T \\ &\qquad + (H^2 - BH + 6m^2) + O(Te^{-2\pi T}). \end{aligned} \tag{455}$$

15.1 The invariant class, its cocycle, and both orientations

In the ordered marking $\mathbb{Z}\langle\gamma, u, w, \delta\rangle$, put

$$\eta = u \wedge w + 6\gamma \wedge \delta. \tag{456}$$

The two finite monodromy generators act by

$$\begin{aligned} &T_1\gamma = \gamma, && T_1 u = -u - w, && T_1 w = -6\gamma + u, && T_1\delta = 2\gamma + u + w + \delta, \\ &T_2\gamma = \gamma, && T_2 u = 6\gamma + w, && T_2 w = -u, && T_2\delta = -3\gamma + u + \delta. \end{aligned}$$

Directly, without passing to a normalized representative,

$$\begin{aligned} (\textstyle\bigwedge^2 T_1)\eta &= (-u - w) \wedge (-6\gamma + u) + 6\gamma \wedge (2\gamma + u + w + \delta) \\ &= u \wedge w + 6\gamma \wedge \delta, \\ (\textstyle\bigwedge^2 T_2)\eta &= (6\gamma + w) \wedge (-u) + 6\gamma \wedge (-3\gamma + u + \delta) \\ &= u \wedge w + 6\gamma \wedge \delta. \end{aligned}$$

Hence η is invariant under both generators.

On $Y = \mathbb{R}^4/\mathbb{Z}^4$, with ordered coordinates y^1, y^2, y^3, y^4, its constant representative and alternating matrix are

$$\eta_T = \mathrm{d}y^2 \wedge \mathrm{d}y^3 + 6\,\mathrm{d}y^1 \wedge \mathrm{d}y^4, \qquad E_\eta = \begin{pmatrix} 0 & 0 & 0 & 6 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \\ -6 & 0 & 0 & 0 \end{pmatrix}. \tag{457}$$

In the paired basis $(e_2, e_3; e_1, e_4)$, the alternating elementary divisors are $(1, 6)$; consequently

$$\operatorname{SNF}(E_\eta) = \operatorname{diag}(1, 1, 6, 6), \qquad \operatorname{Pf}(E_\eta) = 6, \qquad \det E_\eta = 36. \tag{458}$$

The actual Hermitian line bundle is the quotient of $\mathbb{R}^4 \times \mathbb{C}$ by

$$\begin{aligned} &(y, z) \longmapsto (y + e_1, e^{12\pi i y^4} z), && (y, z) \longmapsto (y + e_2, e^{2\pi i y^3} z), \\ &(y, z) \longmapsto (y + e_3, z), && (y, z) \longmapsto (y + e_4, z). \end{aligned} \tag{459}$$

The only possibly nontrivial generator commutators acquire the factors $e^{12\pi i} = 1$ and $e^{2\pi i} = 1$, so these maps define a $\mathbb{Z}^4$-action. Denote the quotient by $\mathscr{L}_{\eta,T} \to F_T$, using the marking $[y] \mapsto [P_T y]$. The connection

$$\nabla_{\eta,T} = \mathrm{d} - 2\pi i(6y^1\,\mathrm{d}y^4 + y^2\,\mathrm{d}y^3) \tag{460}$$

descends through the displayed cocycle. For instance translation by e_1 changes its connection form by $-12\pi i\,\mathrm{d}y^4$, which equals the negative logarithmic derivative of $e^{12\pi i y^4}$; the e_2 check is identical. Therefore

$$F_{\nabla_{\eta,T}} = -2\pi i \eta_T, \qquad c_1(\mathscr{L}_{\eta,T}) = \frac{i}{2\pi} F_{\nabla_{\eta,T}} = \eta_T. \tag{461}$$

There are two opposite orientations which must not be conflated. Write

$$\mathrm{or}_{\mathrm{mark}} = \mathrm{d}y^1 \wedge \mathrm{d}y^2 \wedge \mathrm{d}y^3 \wedge \mathrm{d}y^4 \tag{462}$$

and let or_x be the native complex orientation of $(z_1, z_2) = (x_1 + iy_1, x_2 + iy_2)$. Then

$$P_T^* \,\mathrm{or}_x = -D_T \,\mathrm{or}_{\mathrm{mark}}, \tag{463}$$

whereas

$$\frac{1}{2}\eta_T^2 = 6\,\mathrm{or}_{\mathrm{mark}}\,. \tag{464}$$

Consequently

$$\frac{1}{2}\langle \eta_T^2, [F_T]_{\mathrm{mark}} \rangle = 6, \qquad \frac{1}{2}\langle \eta_T^2, [F_T]_{x\text{-complex}} \rangle = -6. \tag{465}$$

15.2 The exact physical frequencies

In the physical Euclidean coordinates the two-form matrix is

$$\Omega_T = P_T^{-\mathsf{T}} E_\eta P_T^{-1} = \frac{1}{D_T} \begin{pmatrix} 0 & -q_T & 0 & -6m_T \\ q_T & 0 & 6m_T & 0 \\ 0 & -6m_T & 0 & 6L_T \\ 6m_T & 0 & -6L_T & 0 \end{pmatrix}. \tag{466}$$

After the exact permutation $(x^1, x^2, x^3, x^4) \mapsto (x^1, x^3; x^2, x^4)$, its off-diagonal block is

$$K_T = \begin{pmatrix} -q_T & -6m_T \\ -6m_T & 6L_T \end{pmatrix}, \qquad \det K_T = -6D_T. \tag{467}$$

Define the magnetic scalars

$$S_T := 6L_T - q_T, \qquad \Xi_T := \sqrt{(6L_T + q_T)^2 + 144 m_T^2}. \tag{468}$$

Here S_T has no argument and is distinct from the affine residual $S_T(k)$ used in the dual-lattice calculation of theorem 11.1. The signed eigenvalues of K_T are

$$\kappa_{T,\pm} = \frac{S_T \pm \Xi_T}{2}. \tag{469}$$

Direct expansion, with no asymptotic replacement, gives

$$\Xi_T^2 - S_T^2 = 24(L_T q_T + 6m_T^2) = 24 D_T > 0. \tag{470}$$

Thus $\kappa_{T,+} > 0 > \kappa_{T,-}$. The two positive charge-one magnetic frequencies are

$$\boxed{\mathcal{B}_{T,+} = \frac{\pi(S_T + \Xi_T)}{D_T}, \qquad \mathcal{B}_{T,-} = \frac{\pi(\Xi_T - S_T)}{D_T}.} \tag{471}$$

They obey the exact identities

$$\mathcal{B}_{T,+}\mathcal{B}_{T,-} = \frac{24\pi^2}{D_T}, \tag{472}$$

$$\mathcal{B}_{T,+} + \mathcal{B}_{T,-} = \frac{2\pi\Xi_T}{D_T}, \qquad \mathcal{B}_{T,+} - \mathcal{B}_{T,-} = \frac{2\pi S_T}{D_T}, \tag{473}$$

$$\mathcal{B}_{T,+} = \frac{24\pi}{\Xi_T - S_T}, \qquad \mathcal{B}_{T,-} = \frac{24\pi}{\Xi_T + S_T}. \tag{474}$$

The last line follows by multiplying by the conjugate factor and using equation (470).

Keeping the three independent limiting coefficients H, B, m, one obtains

$$\mathcal{B}_{T,-} = \frac{2\pi}{T} - \frac{2\pi H}{T^2} + \frac{2\pi}{T^3}\left(H^2 - \frac{6}{7}m^2\right) + O(T^{-4}), \tag{475}$$

$$\mathcal{B}_{T,+} = \frac{12\pi}{T} + \frac{12\pi(B-H)}{T^2} + \frac{12\pi}{T^3}\left((B-H)^2 - \frac{36}{7}m^2\right) + O(T^{-4}), \tag{476}$$

$$\rho_T := \frac{\mathcal{B}_{T,+}}{\mathcal{B}_{T,-}} = 6 + \frac{6B}{T} + \frac{6B^2 - 6BH - 180m^2/7}{T^2} + O(T^{-3}). \tag{477}$$

These follow by substituting equation (455) into equation (474) and expanding the reciprocals; no sign of H, B, m is assumed. In particular, on a sufficiently deep cusp tail,

$$S_T > 0, \qquad \mathcal{B}_{T,+} > \mathcal{B}_{T,-} > 0, \qquad 5 < \rho_T < 7. \tag{478}$$

15.3 Joint Landau levels and their exact multiplicities

Let $\Delta_{\eta,T} = \nabla^*_{\eta,T}\nabla_{\eta,T}$ be the Friedrichs magnetic Laplacian on $L^2(F_T, \mathscr{L}_{\eta,T})$. Its form domain is $H^1(F_T; \mathscr{L}_{\eta,T})$, its operator domain is $H^2(F_T; \mathscr{L}_{\eta,T})$, and compactness of the torus gives compact resolvent. We now prove the multiplicity rather than infer it from the Pfaffian alone.

Define A_T^{pol} on the marked tangent space by

$$G_T(A_T^{\mathrm{pol}} v, w) = 2\pi E_\eta(v, w), \qquad Q_T^{\mathrm{pol}} = \left(-(A_T^{\mathrm{pol}})^2\right)^{1/2}, \qquad J_{\eta,T} = A_T^{\mathrm{pol}}(Q_T^{\mathrm{pol}})^{-1}. \tag{479}$$

The endomorphism A_T^{pol} is G_T-skew-adjoint and invertible, so Q_T^{pol} is positive, commutes with A_T^{pol}, and

$$J_{\eta,T}^2 = -1, \qquad E_\eta(J_{\eta,T}v, J_{\eta,T}w) = E_\eta(v, w), \qquad 2\pi E_\eta(v, J_{\eta,T}v) = G_T(Q_T^{\mathrm{pol}}v, v) > 0. \tag{480}$$

For every signed charge $q \in \mathbb{Z} \setminus \{0\}$, use the convention

$$\mathscr{L}_{\eta,T}^{\otimes q} = \begin{cases} (\mathscr{L}_{\eta,T})^{\otimes q}, & q > 0, \\ (\mathscr{L}_{\eta,T}^{\vee})^{\otimes(-q)}, & q < 0, \end{cases} \qquad A_{q,T}^{\mathrm{pol}} = qA_T^{\mathrm{pol}}, \qquad Q_{q,T}^{\mathrm{pol}} = \left(-(A_{q,T}^{\mathrm{pol}})^2\right)^{1/2} = |q|Q_T^{\mathrm{pol}}, \tag{481}$$

and define

$$J_{q,T} = A_{q,T}^{\mathrm{pol}}(Q_{q,T}^{\mathrm{pol}})^{-1} = \mathrm{sgn}(q)J_{\eta,T}. \tag{482}$$

All these coefficients are constant, hence the Nijenhuis tensor of $J_{q,T}$ vanishes and the marking transports it to a complex structure on F_T. The induced connection $\nabla^{(q)}_{\eta,T}$ has

$$F_{\nabla^{(q)}_{\eta,T}} = -2\pi i q \eta_T, \qquad 2\pi q E_\eta(v, J_{q,T}v) = |q| G_T(Q_T^{\mathrm{pol}}v, v) > 0. \tag{483}$$

Thus its curvature is of type $(1,1)$, and $(\nabla^{(q)}_{\eta,T})^{0,1}$ makes $\mathscr{L}_{\eta,T}^{\otimes q}$ a positive Appell–Humbert line bundle on the auxiliary complex torus $(F_T, J_{q,T})$.

The auxiliary orientation is fixed, rather than repaired by an absolute value. For $q > 0$, compatibility identifies the $J_{q,T}$-complex orientation with the E_η-symplectic orientation; for $q < 0$, replacing J by $-J$ preserves complex orientation in complex dimension two. Hence

$$[F_T]_{J_{q,T}} = [F_T]_{\mathrm{mark}} = -[F_T]_{x\text{-complex}}. \tag{484}$$

This auxiliary positive structure is not the native complex structure inherited from the S^6 construction. Indeed, with $E_\eta = \operatorname{Im} H_{\eta,T}$, the native Appell–Humbert calculation is

$$G_{\eta,T} = \begin{pmatrix} 2L_T & 2m_T \\ 2m_T & -q_T/3 \end{pmatrix}, \qquad \det G_{\eta,T} = -\frac{2}{3} D_T < 0, \qquad H_{\eta,T} = 2G_{\eta,T}^{-1}. \tag{485}$$

Consequently $H_{\eta,T}$, and also $qH_{\eta,T}$ for every nonzero q, has signature $(1,1)$. The native line bundle is holomorphic because $q\eta_T$ is of type $(1,1)$, but it is indefinite and is not the positive bundle just constructed.

The tangent and canonical bundles of the auxiliary complex torus are trivial. Kodaira vanishing and Hirzebruch–Riemann–Roch therefore give

$$\begin{aligned} &\dim_{\mathbb{C}} H^0((F_T, J_{q,T}), \mathscr{L}_{\eta,T}^{\otimes q}) = \int_{[F_T]_{J_{q,T}}} \frac{(q\eta_T)^2}{2} = \int_{[F_T]_{\mathrm{mark}}} \frac{(q\eta_T)^2}{2} = 6q^2, \\ &\dim_{\mathbb{C}} H^0((F_T, J_{\eta,T}), \mathscr{L}_{\eta,T}) = 6, \\ &\dim_{\mathbb{C}} H^0((F_T, J_{2,T}), \mathscr{L}_{\eta,T}^{\otimes 2}) = 24. \end{aligned} \tag{486}$$

No absolute value occurs in this auxiliary Hirzebruch–Riemann–Roch integral. Equivalently, the positive polarization has elementary divisors $(|q|, 6|q|)$, and the Appell–Humbert theta theorem gives their product $6q^2$; see Beauville, Theorems 2.6 (pp. 107–108) and 3.5 (p. 112). On the native complex torus the same signed Hirzebruch–Riemann–Roch calculation instead gives

$$\chi((F_T, J_{\mathrm{native},T}), \mathscr{L}_{\eta,T}^{\otimes q}) = \int_{[F_T]_{x\text{-complex}}} \frac{(q\eta_T)^2}{2} = -6q^2, \tag{487}$$

and positivity vanishing is unavailable there. Thus the positive auxiliary section count and the native signed Euler characteristic concern distinct Dolbeault problems and distinct complex orientations.

Choose a G_T-orthonormal frame $e_+, f_+ = J_{\eta,T}e_+, e_-, f_- = J_{\eta,T}e_-$ which diagonalizes Q_T^{pol}, labelled so its positive eigenvalues are $\mathcal{B}_{T,+}, \mathcal{B}_{T,-}$. Put

$$a_\sigma = \frac{\nabla_{e_\sigma} + i\nabla_{f_\sigma}}{\sqrt{2\mathcal{B}_{T,\sigma}}}, \qquad a_\sigma^* = \frac{-\nabla_{e_\sigma} + i\nabla_{f_\sigma}}{\sqrt{2\mathcal{B}_{T,\sigma}}}, \qquad \sigma \in \{+,-\}. \tag{488}$$

The curvature commutators give

$$[a_\sigma, a_\tau^*] = \delta_{\sigma\tau}, \qquad [a_\sigma, a_\tau] = [a_\sigma^*, a_\tau^*] = 0, \tag{489}$$

and, for $N_\sigma = a_\sigma^* a_\sigma$,

$$\Delta_{\eta,T} = \mathcal{B}_{T,+}(2N_+ + 1) + \mathcal{B}_{T,-}(2N_- + 1). \tag{490}$$

The simultaneous vacuum $\ker a_+ \cap \ker a_- = H^0((F_T, J_{\eta,T}), \mathscr{L}_{\eta,T})$ has complex dimension six by equation (486).

Theorem 15.1 (Complete native joint Landau decomposition)**.** *For every $n_+, n_- \in \mathbb{Z}_{\geq 0}$, the joint eigenspace*

$$\mathcal{H}_{T;n_+,n_-} = \ker(N_+ - n_+) \cap \ker(N_- - n_-) \tag{491}$$

has complex dimension six and energy

$$E_{T;n_+,n_-} = (2n_+ + 1)\mathcal{B}_{T,+} + (2n_- + 1)\mathcal{B}_{T,-}. \tag{492}$$

Moreover

$$L^2(F_T,\mathscr{L}_{\eta,T})=\widehat{\bigoplus}_{n_+,n_-\geq 0}\mathcal{H}_{T;n_+,n_-}, \tag{493}$$

and the map from the vacuum to the joint level is the unitary

$$\frac{(a_+^*)^{n_+}(a_-^*)^{n_-}}{\sqrt{n_+!n_-!}}. \tag{494}$$

For a numerical eigenvalue, the multiplicity is six times the number of joint pairs in equation (492) *producing that number.*

Proof. The elliptic operator has compact resolvent, and the two commuting number operators preserve its finite-dimensional eigenspaces; they can therefore be diagonalized simultaneously. If $N_\sigma\psi=\lambda_\sigma\psi$, then $\lambda_\sigma\geq 0$. Repeated application of a_σ lowers the Laplacian energy by $2\mathcal{B}_{T,\sigma}>0$, so it terminates. If $a_\sigma^r\psi\neq 0$ and $a_\sigma^{r+1}\psi=0$, then

$$0=\|a_\sigma^{r+1}\psi\|^2=(\lambda_\sigma-r)\|a_\sigma^r\psi\|^2,$$

and hence $\lambda_\sigma=r\in\mathbb{Z}_{\geq 0}$. Lowering both indices reaches the simultaneous vacuum.

The commutator identities give, for a vacuum vector v,

$$\|(a_\sigma^*)^n v\|^2=n!\|v\|^2,\qquad a_\sigma^n(a_\sigma^*)^n v=n!v.$$

On the n-th level they also give $(a_\sigma^*)^n a_\sigma^n=n!$. Operators in the two planes commute, so equation (494) is unitary with inverse $a_+^{n_+}a_-^{n_-}/\sqrt{n_+!n_-!}$. This proves the dimension, energy, and completeness assertions. Orthogonal joint eigenspaces with coincident numerical energies must be added, which proves the final sentence. □

The compact-resolvent lowering argument proves that no additional joint eigenspaces occur. Independently, this is exactly the case $m=2$, $k=1$ of Gordon–Kirwin–Schueth–Webb, Proposition 4.5, article pp. 9–10 (volume pp. 175–176), doi:10.1090/pspum/084/1355. Their signed tensor-power statement gives every oscillator tuple multiplicity $|k|^m V_{E_\eta}=|k|^2\,\mathrm{Pf}(E_\eta)=6k^2$, so it also supplies the external completeness check for every $k\neq 0$.

The exact ground energy and first distinct spacing are therefore

$$E_0(T)=\mathcal{B}_{T,+}+\mathcal{B}_{T,-}=\frac{2\pi\Xi_T}{D_T}=\frac{14\pi}{T}+\frac{\pi(12B-14H)}{T^2}+O(T^{-3}), \tag{495}$$

$$g_0(T)=2\min\{\mathcal{B}_{T,+},\mathcal{B}_{T,-}\}=\frac{2\pi(\Xi_T-|S_T|)}{D_T}=\frac{48\pi}{\Xi_T+|S_T|}. \tag{496}$$

On the deep tail equation (478), this is $g_0(T)=2\mathcal{B}_{T,-}$, with expansion

$$g_0(T)=\frac{4\pi}{T}-\frac{4\pi H}{T^2}+\frac{4\pi}{T^3}\left(H^2-\frac{6}{7}m^2\right)+O(T^{-4}). \tag{497}$$

Corollary 15.2 (Heat trace and exact fixed-energy count)**.** *For $t>0$,*

$$\mathrm{Tr}(e^{-t\Delta_{\eta,T}})=\frac{3}{2\sinh(t\mathcal{B}_{T,+})\sinh(t\mathcal{B}_{T,-})}=\frac{D_T}{16\pi^2t^2}(1+O_t(T^{-2})). \tag{498}$$

For $E>0$, let $N_{\eta,T}(E)$ count complex eigenvalues not exceeding E, with multiplicity. Then

$$N_{\eta,T}(E)=\begin{cases}0, & E<E_0(T),\\ 6\displaystyle\sum_{n_+=0}^{\lfloor (E-E_0(T))/(2\mathcal{B}_{T,+})\rfloor}\left(1+\left\lfloor\frac{E-E_0(T)-2\mathcal{B}_{T,+}n_+}{2\mathcal{B}_{T,-}}\right\rfloor\right), & E\geq E_0(T).\end{cases} \tag{499}$$

For fixed $E>0$, as $T\to\infty$,

$$N_{\eta,T}(E)=\frac{D_T}{32\pi^2}(E-E_0(T))^2+O_E(T)=\frac{E^2D_T}{32\pi^2}+O_E(T). \tag{500}$$

Proof. The heat trace is the product of the two geometric series in equation (492), multiplied by six. Expanding both hyperbolic sines and using equation (472) proves its asymptotic. The counting formula enumerates the lattice points

$$(n_+, n_-) \in \mathbb{R}^2_{\geq 0}, \qquad 2\mathcal{B}_{T,+}n_+ + 2\mathcal{B}_{T,-}n_- \leq E - E_0(T).$$

Its triangular area is $(E - E_0(T))^2/(8\mathcal{B}_{T,+}\mathcal{B}_{T,-})$. The difference between the lattice count and the area is bounded by a constant times the perimeter plus one, hence is $O_E(T)$. Multiplying by six and using the exact frequency product gives the first asymptotic; the second follows from $E_0(T) = O(T^{-1})$ and $D_T = O(T^2)$. □

15.4 The split $SU(2)$ critical connection

Put

$$\mathscr{E}_{\eta,T} = \mathscr{L}_{\eta,T} \oplus \mathscr{L}_{\eta,T}^{-1}, \qquad A_{\eta,T} = \nabla_{\eta,T} \oplus \nabla_{\eta,T}^{-1}. \tag{501}$$

The determinant is canonically trivial and

$$c(\mathscr{E}_{\eta,T}) = (1 + \eta_T)(1 - \eta_T) = 1 - \eta_T^2, \qquad c_2(\mathscr{E}_{\eta,T}) = -\eta_T^2. \tag{502}$$

Thus the orientation-sensitive Chern numbers are

$$\langle c_2(\mathscr{E}_{\eta,T}), [F_T]_{\text{mark}} \rangle = -12, \qquad \langle c_2(\mathscr{E}_{\eta,T}), [F_T]_{x\text{-complex}} \rangle = +12. \tag{503}$$

The cohomology class $c_2 = -\eta_T^2$ is nonzero, so this split $SU(2)$-bundle is not smoothly trivial. This is another exact distinction from the unit-flux split bundle of theorem 14.3.

With

$$H_{\text{mag}} = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \qquad \langle X, Y \rangle_{\mathfrak{su}(2)} = -\frac{1}{2}\operatorname{Tr}(XY), \tag{504}$$

one has $|H_{\text{mag}}| = 1$ and

$$F_{A_{\eta,T}} = -2\pi\eta_T H_{\text{mag}}. \tag{505}$$

This curvature is parallel; hence $d^*_{A_{\eta,T}} F_{A_{\eta,T}} = 0$, and $A_{\eta,T}$ is Yang–Mills. For

$$\mathrm{YM}_T(A) = \frac{1}{2g_{\mathrm{YM}}^2} \int_{F_T} |F_A|^2 \, \mathrm{d}\,\mathrm{vol}_{g_T}, \tag{506}$$

diagonalizing equation (467) gives the exact action

$$\mathrm{YM}_T(A_{\eta,T}) = \frac{D_T}{2g_{\mathrm{YM}}^2}(\mathcal{B}_{T,+}^2 + \mathcal{B}_{T,-}^2) \tag{507}$$

$$= \frac{2\pi^2}{g_{\mathrm{YM}}^2 D_T}(q_T^2 + 36L_T^2 + 72m_T^2) \tag{508}$$

$$= \frac{24\pi^2}{g_{\mathrm{YM}}^2} + \frac{2\pi^2 S_T^2}{g_{\mathrm{YM}}^2 D_T}. \tag{509}$$

Indeed $q_T^2 + 36L_T^2 + 72m_T^2 = 12D_T + (6L_T - q_T)^2$. Consequently

$$\mathrm{YM}_T(A_{\eta,T}) = \frac{74\pi^2}{g_{\mathrm{YM}}^2} + \frac{70\pi^2 B}{g_{\mathrm{YM}}^2 T} + O(T^{-2}). \tag{510}$$

15.5 The closed Jacobi operator and exact Coulomb reduction

We next define the operator before computing its spectrum. Let

$$\mathscr{H}^1_T = L^2\Omega^1(F_T; \operatorname{ad}\mathscr{E}_{\eta,T}), \qquad \mathscr{H}^0_T = L^2\Omega^0(F_T; \operatorname{ad}\mathscr{E}_{\eta,T}) \tag{511}$$

be real Hilbert spaces. Their complexifications carry the root decomposition used below, and every operator constructed here commutes with the anti-linear real structure.

Define the maximal distributional codifferential by

$$\begin{aligned}\operatorname{Dom} D^{\max}_T = \{a \in \mathscr{H}^1_T : &\text{ there is } u \in \mathscr{H}^0_T \text{ such that} \\ &\langle a, d_{A_{\eta,T}}\phi\rangle_{L^2} = \langle u, \phi\rangle_{L^2} \text{ for every smooth } \phi\},\end{aligned} \tag{512}$$

$$D^{\max}_T a = u, \qquad \mathscr{S}_T := \ker D^{\max}_T. \tag{513}$$

Uniqueness follows from density of the test space. Passing to the limit in the defining identity proves that $D^{\max}_T$ is closed, hence the Coulomb slice $\mathscr{S}_T$ is a closed subspace.

On adjoint-valued scalars, let $\Delta^{(0)}_{A_{\eta,T}}$ be the nonnegative self-adjoint operator of the closed form

$$\mathfrak{q}^{(0)}_T[\phi,\psi] = \langle d_{A_{\eta,T}}\phi, d_{A_{\eta,T}}\psi\rangle_{L^2}, \qquad \operatorname{Dom}\mathfrak{q}^{(0)}_T = H^1\Omega^0. \tag{514}$$

Elliptic regularity gives

$$\Delta^{(0)}_{A_{\eta,T}} = d^*_{A_{\eta,T}} d_{A_{\eta,T}}, \qquad \operatorname{Dom}\Delta^{(0)}_{A_{\eta,T}} = H^2\Omega^0. \tag{515}$$

In a constant orthonormal frame, define the symmetric zeroth-order endomorphism

$$(V_T a)_\nu = -2\sum_{\mu=1}^{4} [F_{A_{\eta,T},\nu\mu}, a_\mu]. \tag{516}$$

Symmetry follows from invariance of the compact Lie-algebra inner product, antisymmetry of $F_{\nu\mu}$, and an exchange of ν,μ. Consider the gauge-fixed form

$$\mathfrak{j}^{\mathrm{gf}}_T[a,b] = \langle \nabla_{A_{\eta,T}} a, \nabla_{A_{\eta,T}} b\rangle_{L^2} + \langle V_T a, b\rangle_{L^2}, \qquad \operatorname{Dom}\mathfrak{j}^{\mathrm{gf}}_T = H^1\Omega^1. \tag{517}$$

Proposition 15.3 (Closed gauge-fixed realization)**.** *The form equation* (517) *is densely defined, closed, symmetric, and bounded below. Its self-adjoint operator $\mathcal{J}^{\mathrm{gf}}_T$ has compact resolvent, domain $H^2\Omega^1$, and expression*

$$(\mathcal{J}^{\mathrm{gf}}_T a)_\nu = -\sum_{\mu=1}^{4} \nabla_{A_{\eta,T},e_\mu}\nabla_{A_{\eta,T},e_\mu} a_\nu - 2\sum_{\mu=1}^{4}[F_{A_{\eta,T},\nu\mu}, a_\mu]. \tag{518}$$

It is exactly the Hessian operator with the background-gauge term, before the overall factor g^{-2}_{YM}.

Proof. The first summand in equation (517), together with the L^2-norm, is an equivalent H^1-norm. The smooth endomorphism V_T is bounded and symmetric. The bounded perturbation theorem for closed forms gives closedness and a lower bound, and the representation theorem gives a self-adjoint operator. Its principal symbol is $|\xi|^2$ times the identity. Elliptic regularity identifies its maximal L^2-domain with $H^2\Omega^1$; direct substitution supplies the reverse inclusion. The compact embedding $H^1 \hookrightarrow L^2$ gives compact resolvent.

For the final assertion, define $[a\wedge b]_{\mu\nu} = [a_\mu, b_\nu] - [a_\nu, b_\mu]$. The exact two-parameter curvature expansion is

$$F_{A_{\eta,T}+sa+tb} = F_{A_{\eta,T}} + s d_A a + t d_A b + \frac{s^2}{2}[a\wedge a] + st[a\wedge b] + \frac{t^2}{2}[b\wedge b].$$

It yields

$$\mathrm{YM}''_T\big|_{A_{\eta,T}}(a,b) = \frac{1}{g^2_{\mathrm{YM}}}\left(\langle d_A a, d_A b\rangle + \langle F_A, [a\wedge b]\rangle\right). \tag{519}$$

On the flat torus, commuting the covariant derivatives with the retained curvature convention gives

$$\langle d_A a, d_A b\rangle + \langle d_A^* a, d_A^* b\rangle = \langle \nabla_A a, \nabla_A b\rangle - \sum_{\nu,\mu}\langle a_\nu, [F_{\nu\mu}, b_\mu]\rangle,$$
$$\langle F_A, [a\wedge b]\rangle = -\sum_{\nu,\mu}\langle a_\nu, [F_{\nu\mu}, b_\mu]\rangle.$$

The factor $1/2$ in the two-form inner product is included in the second identity. Adding $g_{\mathrm{YM}}^{-2}\langle d_A^* a, d_A^* b\rangle$ to equation (519) gives exactly $g_{\mathrm{YM}}^{-2}\mathfrak{j}_T^{\mathrm{gf}}[a,b]$, including the coefficient -2 in equation (516). □

Proposition 15.4 (Resolvent intertwining and the reducing Coulomb slice)**.** *On smooth sections,*

$$\mathcal{J}_T^{\mathrm{gf}} d_{A_{\eta,T}} = d_{A_{\eta,T}}\Delta^{(0)}_{A_{\eta,T}}, \qquad d^*_{A_{\eta,T}}\mathcal{J}_T^{\mathrm{gf}} = \Delta^{(0)}_{A_{\eta,T}} d^*_{A_{\eta,T}}. \tag{520}$$

For $z\in\mathbb{C}\setminus\mathbb{R}$ and $f\in\operatorname{Dom} D_T^{\max}$,

$$D_T^{\max}(\mathcal{J}_T^{\mathrm{gf}} - z)^{-1} f = (\Delta^{(0)}_{A_{\eta,T}} - z)^{-1} D_T^{\max} f. \tag{521}$$

Thus $\mathscr{S}_T$ reduces $\mathcal{J}_T^{\mathrm{gf}}$, and its part

$$\mathcal{J}_T = \mathcal{J}_T^{\mathrm{gf}}\big|_{\mathscr{S}_T}, \qquad \operatorname{Dom}\mathcal{J}_T = H^2\Omega^1\cap\mathscr{S}_T \tag{522}$$

is self-adjoint. Its closed form is the restriction of $\mathfrak{j}_T^{\mathrm{gf}}$ to $H^1\Omega^1\cap\mathscr{S}_T$, and the Yang–Mills Hessian on this slice is

$$\mathrm{YM}''_T\big|_{A_{\eta,T}} = g_{\mathrm{YM}}^{-2}\mathfrak{j}_T. \tag{523}$$

Proof. For a smooth adjoint-valued scalar ϕ, write $\nabla_\mu = \nabla_{A_{\eta,T},e_\mu}$. Retaining both curvature terms when commuting three derivatives gives

$$\begin{aligned}(\mathcal{J}_T^{\mathrm{gf}} d_A\phi - d_A\Delta_A^{(0)}\phi)_\nu &\\ &= -\nabla_\mu\nabla_\mu\nabla_\nu\phi + \nabla_\nu\nabla_\mu\nabla_\mu\phi - 2[F_{\nu\mu},\nabla_\mu\phi]\\ &= -[\nabla_\mu F_{\mu\nu},\phi] = [(d_A^* F_A)_\nu,\phi] = 0.\end{aligned}$$

This proves the first identity in equation (520); formal adjunction gives the second.

Let $u = (\mathcal{J}_T^{\mathrm{gf}} - z)^{-1} f$. The operator domain gives $u\in H^2$, hence $D_T^{\max}u\in H^1$. Distributionally applying the second intertwining identity to $(\mathcal{J}_T^{\mathrm{gf}} - z)u = f$ yields

$$(\Delta_A^{(0)} - z) D_T^{\max} u = D_T^{\max} f.$$

The right side is in L^2; scalar elliptic regularity upgrades $D_T^{\max}u$ to H^2, and the scalar resolvent proves equation (521). Both resolvents of the self-adjoint gauge-fixed operator preserve $\mathscr{S}_T$; adjunction shows that they also preserve its orthogonal complement. Therefore the slice is reducing and its operator part is self-adjoint with the displayed domain. The reducing projection commutes with the spectral calculus, so the form domain is exactly $H^1\Omega^1\cap\mathscr{S}_T$. The Hessian identity is the restriction of the last assertion of theorem 15.3. □

15.6 Charge two, all polarizations, and the Morse index

Choose root vectors with $[H_{\mathrm{mag}}, E_\pm] = \pm 2iE_\pm$. Then

$$(\operatorname{ad}\mathscr{E}_{\eta,T})_{\mathbb{C}} = \underline{\mathbb{C}}H_{\mathrm{mag}} \oplus \mathscr{L}_{\eta,T}^{\otimes 2}E_+ \oplus \mathscr{L}_{\eta,T}^{\otimes(-2)}E_-. \tag{524}$$

The E_+ root connection has curvature $-4\pi i\eta_T$. Replacing E_η by $2E_\eta$ gives $J_{2,T} = J_{\eta,T}$, doubles both frequencies, and changes the vacuum dimension from six to twenty-four by equation (486). The E_- root has $q = -2$, uses the conjugate auxiliary structure $J_{-2,T} = -J_{\eta,T}$, and has the same positive frequencies and the same dimension $6(-2)^2 = 24$. Repeating the complete ladder proof of theorem 15.1 therefore gives, for either root, the joint charge-two scalar levels

$$E^{(2)}_{T;p,q} = 2(2p+1)\mathcal{B}_{T,+} + 2(2q+1)\mathcal{B}_{T,-}, \qquad p, q \in \mathbb{Z}_{\geq 0}, \tag{525}$$

each of complex dimension

$$|\operatorname{Pf}(2E_\eta)| = 24. \tag{526}$$

Coincident joint pairs must again be summed.

Choose $O_T \in SO(2)$ with $O_T^{\mathsf{T}} K_T O_T = \operatorname{diag}(\kappa_{T,+}, \kappa_{T,-})$, and define the constant orthonormal coframe

$$\begin{pmatrix} \mathrm{d}X_+ \\ \mathrm{d}X_- \end{pmatrix} = O_T^{\mathsf{T}} \begin{pmatrix} \mathrm{d}x^1 \\ \mathrm{d}x^3 \end{pmatrix}, \qquad \begin{pmatrix} \mathrm{d}Y_+ \\ \mathrm{d}Y_- \end{pmatrix} = O_T^{\mathsf{T}} \begin{pmatrix} \mathrm{d}x^2 \\ \mathrm{d}x^4 \end{pmatrix}. \tag{527}$$

Then

$$\eta_T = \frac{\mathcal{B}_{T,+}}{2\pi} \mathrm{d}X_+ \wedge \mathrm{d}Y_+ - \frac{\mathcal{B}_{T,-}}{2\pi} \mathrm{d}X_- \wedge \mathrm{d}Y_-. \tag{528}$$

The odd grouping permutation in equation (527) reverses orientation; it is used only to expose the two blocks and never to infer a Chern sign.

On the E_+ root line, the four constant circular one-forms diagonalize the spin endomorphism. Direct substitution into equation (516), with the charge-two root normalization retained, gives

polarization	V_T-shift	$\mathcal{J}_T^{\mathrm{gf}}$-eigenvalue on (p, q)
$\mathrm{d}X_+ + i\mathrm{d}Y_+$	$-4\mathcal{B}_{T,+}$	$2[(2p-1)\mathcal{B}_{T,+} + (2q+1)\mathcal{B}_{T,-}]$
$\mathrm{d}X_+ - i\mathrm{d}Y_+$	$+4\mathcal{B}_{T,+}$	$2[(2p+3)\mathcal{B}_{T,+} + (2q+1)\mathcal{B}_{T,-}]$
$\mathrm{d}X_- - i\mathrm{d}Y_-$	$-4\mathcal{B}_{T,-}$	$2[(2p+1)\mathcal{B}_{T,+} + (2q-1)\mathcal{B}_{T,-}]$
$\mathrm{d}X_- + i\mathrm{d}Y_-$	$+4\mathcal{B}_{T,-}$	$2[(2p+1)\mathcal{B}_{T,+} + (2q+3)\mathcal{B}_{T,-}]$

The E_- table is its anti-linear conjugate. The Cartan sector is the ordinary nonnegative coefficientwise Laplacian on one-forms.

Theorem 15.5 (Exact negative spectrum, walls, and real Morse index)**.** *Put*

$$\mathcal{B}_{T,\max} = \max\{\mathcal{B}_{T,+}, \mathcal{B}_{T,-}\}, \quad \mathcal{B}_{T,\min} = \min\{\mathcal{B}_{T,+}, \mathcal{B}_{T,-}\}, \quad \rho_T^* = \frac{\mathcal{B}_{T,\max}}{\mathcal{B}_{T,\min}} = \frac{\Xi_T + |S_T|}{\Xi_T - |S_T|}. \tag{529}$$

For every regular T, the negative spectrum of the self-adjoint Coulomb operator is exactly

$$\left\{ \nu_{T,n} = 2[(2n+1)\mathcal{B}_{T,\min} - \mathcal{B}_{T,\max}] = \frac{4\pi}{D_T}[n\Xi_T - (n+1)|S_T|] : 2n+1 < \rho_T^* \right\}. \tag{530}$$

Every indicated band has real multiplicity 48*, and*

$$\operatorname{ind}_{\mathbb{R}}(\mathcal{J}_T) = 48 \left\lceil \frac{\rho_T^* - 1}{2} \right\rceil. \tag{531}$$

At the n-th wall the strict inequality becomes equality. Equivalently,

$$\rho_T^* = 2n+1 \iff n\Xi_T = (n+1)|S_T| \iff 24n^2 D_T = (2n+1)S_T^2. \tag{532}$$

The corresponding branch is then a zero branch and is not counted in the Morse index. If one removes the absolute value by choosing the $\mathcal{B}_{T,+} > \mathcal{B}_{T,-}$ side, the polynomial equation in equation (532) *must be accompanied by $S_T > 0$; otherwise squaring introduces the opposite side as an extraneous solution.*

On the sufficiently deep tail of equation (478)*, the negative branches are exactly* $n = 0, 1, 2$*,*

$$\nu_{T,n} = \frac{4\pi}{D_T}[n\Xi_T - (n+1)S_T], \qquad n = 0, 1, 2, \tag{533}$$

each of real multiplicity 48*. Therefore*

$$\operatorname{ind}_{\mathbb{R}}(\mathcal{J}_T) = 144, \tag{534}$$

and

$$\begin{aligned} \nu_{T,0} &= -\frac{20\pi}{T} + O(T^{-2}), & \nu_{T,1} &= -\frac{12\pi}{T} + O(T^{-2}), \\ \nu_{T,2} &= -\frac{4\pi}{T} + O(T^{-2}), & \nu_{T,3} &= +\frac{4\pi}{T} + O(T^{-2}). \end{aligned} \tag{535}$$

The Hessian operator for equation (506) *has eigenvalues* $\nu_{T,n}/g_{\mathrm{YM}}^2$ *and the same index.*

Proof. Assume first $\mathcal{B}_{T,+} > \mathcal{B}_{T,-}$. In the first row of the polarization table, $p \geq 1$ makes the eigenvalue positive. At $p = 0$ it is

$$2[-\mathcal{B}_{T,+} + (2q+1)\mathcal{B}_{T,-}],$$

which is negative exactly when $2q + 1 < \rho_T^*$. The second and fourth rows are positive, and the third has minimum $2(\mathcal{B}_{T,+} - \mathcal{B}_{T,-}) > 0$. When $\mathcal{B}_{T,-} > \mathcal{B}_{T,+}$, the identical argument with the two planes exchanged shows that only the $q = 0$ part of the third row can be negative and gives the same formula with the ordered frequencies. If the frequencies agree, both candidate lowest branches are zero and none is negative. This proves the exhaustive list for the gauge-fixed operator.

Every negative gauge-fixed eigenform is automatically Coulomb. Elliptic regularity makes it smooth; if $\mathcal{J}_T^{\mathrm{gf}} a = \lambda a$ with $\lambda < 0$, then equation (520) gives

$$\Delta^{(0)}_{A_{\eta,T}}(d^*_{A_{\eta,T}} a) = \lambda d^*_{A_{\eta,T}} a.$$

Nonnegativity of the scalar operator forces $d_A^* a = 0$. Hence the full and Coulomb negative spectral spaces coincide.

One joint charge-two scalar space contributes complex dimension 24 in the E_+ sector. Its conjugate contributes complex dimension 24 in the complexification of the real adjoint problem. A complexified real eigenspace has complex dimension equal to the original real dimension, so the real multiplicity is $24 + 24 = 48$, not 24 or 96.

The count of integers $n \geq 0$ satisfying $2n + 1 < \rho_T^*$ is $\lceil (\rho_T^* - 1)/2 \rceil$, including the correct exclusion at an odd integer wall. Substitution of the ordered frequency formulas gives equation (530). The wall equation is $n\Xi_T = (n+1)|S_T|$; squaring and using $\Xi_T^2 - S_T^2 = 24 D_T$ gives the final polynomial form, and nonnegativity of both sides gives the converse. Finally, equation (478) leaves precisely $n = 0, 1, 2$, and the leading frequency terms give $\nu_{T,n} = (8n - 20)\pi/T + O(T^{-2})$. □

The lowest branch and the action excess are tied by the exact identities

$$\nu_{T,0} = -2|\mathcal{B}_{T,+} - \mathcal{B}_{T,-}| = -\frac{4\pi|S_T|}{D_T}, \qquad \mathrm{YM}_T(A_{\eta,T}) - \frac{24\pi^2}{g_{\mathrm{YM}}^2} = \frac{D_T}{8g_{\mathrm{YM}}^2}\nu_{T,0}^2. \tag{536}$$

15.7 The fixed and varying radial direct integrals

Choose $T_1 > T_0$ so that the complete deep-tail inequalities in equation (478) hold on

$$I := (T_1, \infty).$$

Let $Y = \mathbb{R}^4/\mathbb{Z}^4$. Denote by $\mathscr{L}_\eta \to Y$ the fixed marked Hermitian line bundle defined by the factor of automorphy in equation (459). The marking diffeomorphism and its transported bundle isomorphism are

$$\Phi_T : Y \longrightarrow F_T, \qquad [y] \longmapsto [P_T y], \qquad \widehat{\Phi}_T : \mathscr{L}_\eta \longrightarrow \mathscr{L}_{\eta,T}.$$

They are fixed by requiring

$$\widehat{\Phi}_T^* \nabla_{\eta,T} = \mathrm{d} - 2\pi i(6y^1\,\mathrm{d}y^4 + y^2\,\mathrm{d}y^3),$$

so no gauge, metric, lattice, or orientation normalization is hidden in the identification. Put

$$\mathscr{K}_T := L^2(F_T, \mathscr{L}_{\eta,T}; \mathrm{d}\,\mathrm{vol}_{g_T}).$$

The pullback of the physical volume is

$$\Phi_T^*(\mathrm{d}\,\mathrm{vol}_{g_T}) = \sqrt{\det(P_T^{\mathsf{T}} P_T)}\,\mathrm{d}^4 y = D_T\,\mathrm{d}^4 y. \tag{537}$$

Put

$$\mathscr{H}_* := L^2(Y, \mathscr{L}_\eta; \mathrm{d}^4 y), \qquad \mathscr{D}_* := H^2(Y; \mathscr{L}_\eta),$$

where the Sobolev space is defined using any one fixed smooth reference metric and connection on the fixed compact bundle. Define

$$U_T : \mathscr{K}_T \longrightarrow \mathscr{H}_*, \qquad U_T s := D_T^{1/2} \widehat{\Phi}_T^* s. \tag{538}$$

Then

$$\|U_T s\|^2_{\mathscr{H}_*} = D_T \int_Y |\widehat{\Phi}_T^* s|^2\,\mathrm{d}^4 y = \|s\|^2_{\mathscr{K}_T},$$

and

$$U_T^{-1} v = D_T^{-1/2} (\widehat{\Phi}_T^{-1})^* v.$$

Thus U_T is an onto unitary, not merely an isometric injection.

Transport the fibre operator to the fixed space:

$$A_T := U_T \Delta_{\eta,T} U_T^{-1}, \qquad \mathrm{Dom}(A_T) = \mathscr{D}_*. \tag{539}$$

Its closed quadratic form is

$$\widetilde{\mathfrak{a}}_T[v, w] = \int_Y (G_T^{-1})^{jk} \langle \nabla_{\eta,j} v, \nabla_{\eta,k} w \rangle\,\mathrm{d}^4 y, \qquad G_T = P_T^{\mathsf{T}} P_T. \tag{540}$$

Here $\nabla_{\eta,j}$ is the $\mathrm{d}y^j$-component of the fixed marked connection displayed above. The factor $D_T^{1/2}$ in U_T is constant in y, and it cancels the factor D_T in (537); hence no unrecorded factor appears in (540).

Lemma 15.6 (Common domain and measurable resolvents)**.** *For every compact interval $K \Subset I$, the family*

$$T \longmapsto A_T \in \mathcal{B}(\mathscr{D}_*, \mathscr{H}_*)$$

is norm-continuous, in fact real analytic. For every $z \in \mathbb{C} \setminus \mathbb{R}$, the resolvents $(A_T - z)^{-1}$ are locally norm-continuous as operators on $\mathscr{H}_*$, and hence form a measurable operator field.

Proof. On K, the matrices G_T^{-1} are uniformly positive definite, and their entries are real analytic in T. In fixed local bundle trivializations, (540) therefore gives a uniformly elliptic second-order expression whose coefficients depend real analytically on T. Elliptic regularity identifies every operator domain with the same set $\mathscr{D}_*$, and the standard local coordinate estimate for a differential operator of order two gives norm-continuity in $\mathcal{B}(\mathscr{D}_*, \mathscr{H}_*)$.

Uniform ellipticity on K gives

$$\|v\|_{\mathscr{D}_*} \le C_K(\|A_T v\|_{\mathscr{H}_*} + \|v\|_{\mathscr{H}_*}), \qquad T \in K.$$

For $v = (A_T - z)^{-1} f$, self-adjointness gives $\|v\| \le |\operatorname{Im} z|^{-1}\|f\|$, while $A_T v = f + zv$. Consequently

$$\sup_{T\in K} \|(A_T - z)^{-1}\|_{\mathscr{H}_*\to\mathscr{D}_*} < \infty. \tag{541}$$

The resolvent identity, with its exact sign,

$$(A_T - z)^{-1} - (A_R - z)^{-1} = -(A_T - z)^{-1}(A_T - A_R)(A_R - z)^{-1},$$

together with (541), proves local norm-resolvent continuity. A countable compact exhaustion of I gives measurability on all of I. □

We now define the varying measurable Hilbert field rather than presuppose one. A section $T \mapsto s_T \in \mathscr{K}_T$ is declared measurable precisely when

$$T \longmapsto U_T s_T \in \mathscr{H}_*$$

is strongly measurable. This is the measurable field pulled back from the constant separable field $I \times \mathscr{H}_*$. Its direct integral is

$$\mathscr{H}_{\mathrm{var}} := \int_I^{\oplus} \mathscr{K}_T \,\mathrm{d}T.$$

The fixed radial Hilbert space is

$$\mathscr{H}_{\mathrm{rad}} := L^2(I, \mathrm{d}T; \mathscr{H}_*).$$

By construction,

$$\mathscr{U} := \int_I^{\oplus} U_T \,\mathrm{d}T : \mathscr{H}_{\mathrm{var}} \longrightarrow \mathscr{H}_{\mathrm{rad}}, \qquad (\mathscr{U} s)(T) = U_T s_T, \tag{542}$$

is an isometry. It is onto: if $\Psi \in \mathscr{H}_{\mathrm{rad}}$, then $s_T = U_T^{-1}\Psi(T)$ is measurable by the very definition of the varying field and has the same squared norm integral.

On the fixed direct integral, define the maximal multiplication operator

$$(\widetilde{\mathscr{M}_\eta}\Psi)(T) := A_T \Psi(T) \tag{543}$$

with the exact domain

$$\begin{aligned} \mathrm{Dom}(\widetilde{\mathscr{M}_\eta}) := \{\Psi \in \mathscr{H}_{\mathrm{rad}} : \Psi(T) \in \mathscr{D}_* \text{ for almost every } T,\\ T \mapsto A_T\Psi(T) \text{ is strongly measurable},\\ \int_I \|A_T\Psi(T)\|^2_{\mathscr{H}_*}\,\mathrm{d}T < \infty\}. \end{aligned} \tag{544}$$

On the varying direct integral, define

$$\begin{aligned} \mathrm{Dom}(\mathscr{M}_\eta) := \{s \in \mathscr{H}_{\mathrm{var}} : s_T \in \mathrm{Dom}(\Delta_{\eta,T}) \text{ for almost every } T,\\ T \mapsto U_T\Delta_{\eta,T}s_T \text{ is strongly measurable},\\ \int_I \|\Delta_{\eta,T}s_T\|^2_{\mathscr{K}_T}\,\mathrm{d}T < \infty\}, \end{aligned} \tag{545}$$

and

$$(\mathscr{M}_\eta s)_T := \Delta_{\eta,T} s_T \quad \text{for almost every } T. \tag{546}$$

Every expression in (543) and (546) is now typed on its actual fibre. Equations (539), (542), (544), and (545) give both the domain equality and the exact intertwining identity

$$\mathscr{U}\,\mathrm{Dom}(\mathscr{M}_\eta) = \mathrm{Dom}(\widetilde{\mathscr{M}_\eta}), \qquad \mathscr{U}\,\mathscr{M}_\eta\mathscr{U}^{-1} = \widetilde{\mathscr{M}_\eta}. \tag{547}$$

15.7.1 Density and self-adjointness on the maximal domain

Proposition 15.7 (Maximal-domain realization)**.** *The operators $\widetilde{\mathscr{M}_\eta}$ and $\mathscr{M}_\eta$ are densely defined, nonnegative, and self-adjoint on the domains* (544) *and* (545)*, respectively.*

Proof. For $\chi \in C_c(I)$ and $v \in \mathscr{D}_*$, local continuity of $T \mapsto A_T v$ makes it bounded on $\operatorname{supp}\chi$. Thus $\chi v \in \operatorname{Dom}(\widetilde{\mathscr{M}_\eta})$. Since $\mathscr{D}_*$ is dense in $\mathscr{H}_*$, the algebraic tensor product

$$C_c(I) \odot \mathscr{D}_*$$

is dense in $L^2(I;\mathscr{H}_*)$. Hence the fixed operator is densely defined. Fibrewise self-adjointness and Cauchy–Schwarz give, for Ψ, Θ in its domain,

$$\langle \widetilde{\mathscr{M}_\eta}\Psi, \Theta\rangle = \int_I \langle A_T\Psi(T), \Theta(T)\rangle \,\mathrm{d}T = \langle \Psi, \widetilde{\mathscr{M}_\eta}\Theta\rangle,$$

so it is symmetric.

Fix $z \in \mathbb{C}\setminus\mathbb{R}$. For $F \in \mathscr{H}_{\mathrm{rad}}$, define pointwise

$$(\mathcal{R}_z F)(T) := (A_T - z)^{-1} F(T). \tag{548}$$

The locally norm-continuous resolvent field, approximation of F by simple functions, and

$$\|(A_T - z)^{-1}\|_{\mathscr{H}_*\to\mathscr{H}_*} \le |\operatorname{Im} z|^{-1}$$

show that $\mathcal{R}_z F$ is strongly measurable and

$$\|\mathcal{R}_z F\|_{\mathscr{H}_{\mathrm{rad}}} \le |\operatorname{Im} z|^{-1}\|F\|_{\mathscr{H}_{\mathrm{rad}}}. \tag{549}$$

Pointwise, $(A_T - z)^{-1}F(T) \in \mathscr{D}_*$. The exact identity

$$A_T(A_T - z)^{-1} = I + z(A_T - z)^{-1} \tag{550}$$

gives the global uniform bound

$$\|A_T(A_T - z)^{-1}\| \le 1 + \frac{|z|}{|\operatorname{Im} z|}.$$

The right-hand side of (550), applied to $F(T)$, is strongly measurable and square-integrable. Therefore $\mathcal{R}_z F$ belongs to the full maximal domain (544); this is the domain step that pointwise inversion alone does not prove.

Fibrewise identities now yield

$$(\widetilde{\mathscr{M}_\eta} - z)\mathcal{R}_z = I \quad \text{on } \mathscr{H}_{\mathrm{rad}}, \qquad \mathcal{R}_z(\widetilde{\mathscr{M}_\eta} - z) = I \quad \text{on } \operatorname{Dom}(\widetilde{\mathscr{M}_\eta}).$$

In particular, the ranges of $\widetilde{\mathscr{M}_\eta} - i$ and $\widetilde{\mathscr{M}_\eta} + i$ are both all of $\mathscr{H}_{\mathrm{rad}}$. The range criterion for a densely defined symmetric operator proves self-adjointness. Moreover,

$$\langle \widetilde{\mathscr{M}_\eta}\Psi, \Psi\rangle = \int_I \langle A_T\Psi(T), \Psi(T)\rangle \,\mathrm{d}T \ge 0,$$

so it is nonnegative. The unitary equivalence (547) transfers density, self-adjointness, and nonnegativity to the varying-fibre operator. □

15.7.2 Measurable eigenvectors, including at band crossings

Lemma 15.8 (Explicit measurable eigenvector selection)**.** *For every fixed* $\mathbf{n} = (n_+, n_-) \in \mathbb{Z}^2_{\geq 0}$*, there is a strongly measurable map*

$$I \longrightarrow \mathscr{D}_*, \qquad T \longmapsto \phi_{\mathbf{n}}(T),$$

such that

$$\|\phi_{\mathbf{n}}(T)\|_{\mathscr{H}_*} = 1, \qquad A_T\phi_{\mathbf{n}}(T) = E_{\mathbf{n}}(T)\phi_{\mathbf{n}}(T) \tag{551}$$

for every $T \in I$*. The construction remains valid when* $E_{\mathbf{n}}(T)$ *crosses one or more other bands.*

Proof. Set

$$C_{\mathbf{n},T} := A_T - E_{\mathbf{n}}(T), \qquad \mathrm{Dom}(C_{\mathbf{n},T}) = \mathscr{D}_*.$$

The band $E_{\mathbf{n}}$ is continuous. The proof of Lemma 15.6, applied also to the scalar shift, shows that $(C_{\mathbf{n},T} - z)^{-1}$ is a measurable resolvent field for every nonreal z.

Let

$$\mathsf{P}_{\mathbf{n}}(T) := \mathbf{1}_{\{0\}}(C_{\mathbf{n},T}) = \mathbf{1}_{\{E_{\mathbf{n}}(T)\}}(A_T).$$

This singleton projection is not assumed continuous at a crossing. Instead, the spectral theorem gives the pointwise strong limit

$$\mathsf{P}_{\mathbf{n}}(T) = \operatorname*{s-lim}_{k\to\infty}(I + kC^2_{\mathbf{n},T})^{-1}, \tag{552}$$

and the exact resolvent formula

$$(I + kC^2_{\mathbf{n},T})^{-1} = \frac{1}{2i\sqrt{k}}\left[(C_{\mathbf{n},T} - i/\sqrt{k})^{-1} - (C_{\mathbf{n},T} + i/\sqrt{k})^{-1}\right] \tag{553}$$

shows that every approximating field is strongly measurable. Therefore $T \mapsto \mathsf{P}_{\mathbf{n}}(T)v$ is strongly measurable for every $v \in \mathscr{H}_*$. Proposition 15.1 says that $\mathsf{P}_{\mathbf{n}}(T) \neq 0$ for every T, including crossings.

Choose a countable dense sequence $(v_k)_{k\geq 1}$ in the separable space $\mathscr{H}_*$, and define the measurable partition

$$X_k = \{T : \|\mathsf{P}_{\mathbf{n}}(T)v_k\| > 0\} \cap \bigcap_{\ell<k}\{T : \|\mathsf{P}_{\mathbf{n}}(T)v_\ell\| = 0\}.$$

The sets X_k are pairwise disjoint and cover I: a nonzero bounded projection cannot annihilate a dense set. On X_k, set

$$\phi_{\mathbf{n}}(T) = \frac{\mathsf{P}_{\mathbf{n}}(T)v_k}{\|\mathsf{P}_{\mathbf{n}}(T)v_k\|}.$$

Countable measurable patching gives a strongly measurable unit vector. The range of the singleton spectral projection lies in $\mathrm{Dom}(A_T)$ and satisfies $(A_T - E_{\mathbf{n}}(T))\phi_{\mathbf{n}}(T) = 0$, which proves (551). □

15.7.3 Full spectrum and spectral type

Theorem 15.9 (Purely continuous full half-line spectrum)**.** *For both the fixed-fibre and varying-fibre maximal radial multiplication operators,*

$$\begin{aligned} \sigma(\widetilde{\mathscr{M}_\eta}) &= \sigma_{\mathrm{ess}}(\widetilde{\mathscr{M}_\eta}) = \sigma_{\mathrm{c}}(\widetilde{\mathscr{M}_\eta}) = [0,\infty), \\ \sigma_{\mathrm{p}}(\widetilde{\mathscr{M}_\eta}) &= \varnothing, \qquad \ker \widetilde{\mathscr{M}_\eta} = \{0\}, \end{aligned} \tag{554}$$

and

$$\begin{aligned} \sigma(\mathscr{M}_\eta) &= \sigma_{\mathrm{ess}}(\mathscr{M}_\eta) = \sigma_{\mathrm{c}}(\mathscr{M}_\eta) = [0,\infty), \\ \sigma_{\mathrm{p}}(\mathscr{M}_\eta) &= \varnothing, \qquad \ker \mathscr{M}_\eta = \{0\}. \end{aligned} \tag{555}$$

Here σ_{ess} *is the Weyl essential spectrum.*

Proof. It is enough to prove (554), because (547) is a unitary equivalence. Proposition 15.7 and nonnegativity give

$$\sigma(\widetilde{\mathscr{M}_\eta}) \subset [0,\infty).$$

The zero-energy singular sequence. Let $a_j = T_1 + 2j + 1$. Using the measurable normalized ground eigenvector from Lemma 15.8, define

$$\Psi_j(T) := \mathbf{1}_{[a_j, a_j+1]}(T)\phi_{(0,0)}(T).$$

These vectors have norm one and have pairwise disjoint supports, hence are orthonormal and converge weakly to zero. They belong to the maximal domain because

$$A_T\Psi_j(T) = \mathbf{1}_{[a_j, a_j+1]}(T)E_0(T)\phi_{(0,0)}(T)$$

is strongly measurable and square-integrable. From (495),

$$\|\widetilde{\mathscr{M}_\eta}\Psi_j\|^2 = \int_{a_j}^{a_j+1} E_0(T)^2 \,\mathrm{d}T = \frac{196\pi^2}{a_j^2} + O(a_j^{-3}) \longrightarrow 0.$$

Thus (Ψ_j) is a singular Weyl sequence at zero, and

$$0 \in \sigma_{\mathrm{ess}}(\widetilde{\mathscr{M}_\eta}).$$

Positive energy: escaping crossings before disjoint supports. Fix $\lambda > 0$, and fix once and for all $T_\bullet = T_1 + 1 \in I$. For $n \in \mathbb{Z}_{\geq 0}$, retain the specific canonical band

$$e_n(T) := E_{(0,n)}(T) = \mathcal{B}_{T,+} + (2n+1)\mathcal{B}_{T,-}. \tag{556}$$

Since $\mathcal{B}_{T_\bullet,-} > 0$,

$$e_n(T_\bullet) \longrightarrow \infty \quad (n \to \infty).$$

For each fixed n, equations (475) and (476) give

$$e_n(T) \longrightarrow 0 \quad (T \to \infty).$$

Hence, for every sufficiently large n, the intermediate value theorem gives at least one crossing parameter $r_n > T_\bullet$ satisfying

$$e_n(r_n) = \lambda. \tag{557}$$

We prove explicitly that every such choice of crossing parameters escapes to infinity. Fix $M > T_\bullet$. Positivity and continuity give

$$c_M := \min_{T \in [T_\bullet, M]} \mathcal{B}_{T,-} > 0.$$

For every $T \in [T_\bullet, M]$,

$$e_n(T) \geq (2n+1)c_M.$$

Thus, once $(2n+1)c_M > \lambda$, equation (557) has no solution in $[T_\bullet, M]$, and the chosen r_n must satisfy $r_n > M$. Since M was arbitrary,

$$r_n \longrightarrow \infty. \tag{558}$$

Only now, before choosing any support, pass to a subsequence $n_j \to \infty$ such that, with $r_j := r_{n_j}$,

$$r_{j+1} > r_j + 2. \tag{559}$$

Continuity of e_{n_j} at r_j permits the subsequent choice of $\varepsilon_j \in (0, \frac{1}{4})$ such that

$$J_j := (r_j - \varepsilon_j, r_j + \varepsilon_j) \Subset I, \qquad \sup_{T \in J_j} |e_{n_j}(T) - \lambda| < \frac{1}{j}.$$

The order of construction matters: the separation (559) makes the intervals J_j pairwise disjoint.

Take the measurable normalized eigenvector $\phi_{(0,n_j)}(T)$ from Lemma 15.8; the singleton projection used there remains valid if the selected band crosses another one inside J_j. Define

$$\Theta_j(T) := |J_j|^{-1/2}\mathbf{1}_{J_j}(T)\phi_{(0,n_j)}(T). \tag{560}$$

The sections Θ_j have norm one and are orthogonal by disjointness of their supports. They lie in the maximal domain because

$$A_T\Theta_j(T) = |J_j|^{-1/2}\mathbf{1}_{J_j}(T)e_{n_j}(T)\phi_{(0,n_j)}(T)$$

is strongly measurable and square-integrable. Moreover,

$$\|(\widetilde{\mathscr{M}_\eta} - \lambda)\Theta_j\|^2 = \frac{1}{|J_j|}\int_{J_j} |e_{n_j}(T) - \lambda|^2 \,\mathrm{d}T < \frac{1}{j^2}.$$

Every orthonormal sequence is weakly null, so (Θ_j) is a singular Weyl sequence at λ. Thus every $\lambda > 0$ belongs to the Weyl essential spectrum. Together with the zero sequence and nonnegativity, this proves

$$\sigma(\widetilde{\mathscr{M}_\eta}) = \sigma_{\mathrm{ess}}(\widetilde{\mathscr{M}_\eta}) = [0,\infty). \tag{561}$$

Real analyticity, absence of flat bands, and no eigenvalues. Equation (470) writes

$$\Xi_T = \sqrt{S_T^2 + 24D_T}$$

with a strictly positive real-analytic radicand. The positive square-root function is real analytic on $(0,\infty)$. Hence Ξ_T, $\mathcal{B}_{T,+}$, $\mathcal{B}_{T,-}$, and every $E_{\mathbf{n}}(T)$ are real analytic on the connected interval I. For each fixed $\mathbf{n}$,

$$E_{\mathbf{n}}(T) > 0 \quad (T < \infty), \qquad \lim_{T\to\infty} E_{\mathbf{n}}(T) = 0.$$

It follows that no band is constant: a constant band would have constant value equal to its limit zero, contradicting strict positivity at every finite T. This proves the absence of flat bands, rather than assuming it.

Fix $\lambda \in \mathbb{R}$. For every $\mathbf{n}$, the real-analytic function $E_{\mathbf{n}} - \lambda$ is not identically zero, so its zero set in I is discrete and has Lebesgue measure zero. Since $\mathbb{Z}^2_{\geq 0}$ is countable,

$$Z_\lambda := \{T \in I : \lambda \in \sigma_{\mathrm{p}}(A_T)\} = \bigcup_{\mathbf{n}\in\mathbb{Z}^2_{\geq 0}} \{T : E_{\mathbf{n}}(T) = \lambda\} \tag{562}$$

is a null set.

If $\widetilde{\mathscr{M}_\eta}\Psi = \lambda\Psi$, then $(A_T - \lambda)\Psi(T) = 0$ for almost every T. At every point where $\Psi(T) \neq 0$, Proposition 15.1 forces $T \in Z_\lambda$. Thus the essential support of Ψ is contained in the null set (562), and $\Psi = 0$. A self-adjoint operator has only real eigenvalues, so

$$\sigma_{\mathrm{p}}(\widetilde{\mathscr{M}_\eta}) = \varnothing, \qquad \ker \widetilde{\mathscr{M}_\eta} = \{0\}.$$

Self-adjoint operators have empty residual spectrum. Combining this with (561) proves that the entire spectrum is continuous and gives (554). Unitary equivalence gives (555). □

15.7.4 Exact interpretive boundary

Remark 15.10 (No radial derivative and no interacting Yang–Mills claim). The variable T in (543) and (546) is a direct-integral parameter. Neither operator contains a T-derivative. In particular, there is no term such as $-\partial_T^2$, no radial Sobolev condition, no cusp boundary condition, and no interaction between distinct T-fibres. This is why characteristic functions of radial intervals legitimately occur in the maximal-domain Weyl sequences above.

The theorem is therefore a complete spectral statement for the specified auxiliary decomposable family of native full-flux magnetic Laplacians. It does not construct, identify, or determine the spectrum or mass gap of an interacting quantum Yang–Mills Hamiltonian, and it makes no Osterwalder–Schrader reconstruction claim.

16 The native full-flux split-Cartan $SU(r)$ extension

This section fixes an integer $r \geq 2$ and extends the full-flux construction of section 15 from its already proved split $SU(2)$ case to arbitrary split-Cartan $SU(r)$ backgrounds. Every object remains on the same regular flat four-torus $F_T = \mathbb{R}^4/P_T\mathbb{Z}^4$, with the unrescaled Euclidean quotient metric. In particular, the integral two-form η_T, the line bundle $\mathscr{L}_{\eta,T}$, the two orientations, and the exact quantities

$$\begin{aligned} L_T, \quad q_T, \quad m_T, \quad D_T = L_T q_T + 6m_T^2 > 0, \\ S_T = 6L_T - q_T, \qquad \Xi_T = \sqrt{(6L_T + q_T)^2 + 144m_T^2}. \end{aligned}$$

are precisely those of equations (452) and (468). The charge-one frequencies are

$$\mathcal{B}_{T,+} = \frac{\pi(S_T + \Xi_T)}{D_T}, \qquad \mathcal{B}_{T,-} = \frac{\pi(\Xi_T - S_T)}{D_T}, \tag{563}$$

and we shall use the globally ordered notation

$$\begin{aligned} \mathcal{B}_{T,\max} = \max\{\mathcal{B}_{T,+}, \mathcal{B}_{T,-}\}, \quad \mathcal{B}_{T,\min} = \min\{\mathcal{B}_{T,+}, \mathcal{B}_{T,-}\}, \\ \rho_T^* = \frac{\mathcal{B}_{T,\max}}{\mathcal{B}_{T,\min}}. \end{aligned} \tag{564}$$

Thus every assertion below remains valid on either side of $S_T = 0$.

16.1 The split bundle, determinant, and Chern number

Fix a nonzero trace-zero integral weight vector

$$\mathbf{m} = (m_1, \ldots, m_r) \in \mathbb{Z}^r, \qquad \sum_{j=1}^r m_j = 0, \qquad s_2(\mathbf{m}) := \sum_{j=1}^r m_j^2. \tag{565}$$

Define

$$\mathscr{E}_{\mathbf{m},T} := \bigoplus_{j=1}^r \mathscr{L}_{\eta,T}^{\otimes m_j}, \qquad H_{\mathbf{m}} := i\,\mathrm{diag}(m_1, \ldots, m_r) \in \mathfrak{su}(r). \tag{566}$$

Proposition 16.1 (Parallel determinant trivialization and exact c_2)**.** *The ordered splitting in equation* (566) *has a canonical parallel determinant trivialization. With $c = [\eta_T]$,*

$$c_1(\mathscr{E}_{\mathbf{m},T}) = 0, \qquad c_2(\mathscr{E}_{\mathbf{m},T}) = -\frac{s_2(\mathbf{m})}{2}c^2. \tag{567}$$

The marked and physical-complex Chern numbers are respectively

$$\langle c_2, [F_T]_{\mathrm{mark}} \rangle = -6s_2(\mathbf{m}), \qquad k_{\mathbf{m}} := \langle c_2, [F_T]_{x\text{-complex}} \rangle = 6s_2(\mathbf{m}) > 0. \tag{568}$$

Moreover $s_2(\mathbf{m})$ is even. Hence this direct-sum family realizes only $k_{\mathbf{m}} \in 12\mathbb{Z}_{>0}$, not all topological $SU(r)$ sectors.

Proof. Let $j_n^\eta(y)$ denote the factor of automorphy in equation (459). The transition matrix of the ordered sum is

$$\mathrm{diag}\big((j_n^\eta)^{m_1}, \ldots, (j_n^\eta)^{m_r}\big).$$

Its determinant is $(j_n^\eta)^{\sum_j m_j} = 1$ identically. The tensor product of the ordered unit frames therefore defines a global determinant section, and the determinant connection has connection form proportional to $\sum_j m_j = 0$; the section is parallel.

The splitting principle gives

$$c(\mathscr{E}_{\mathbf{m},T}) = \prod_{j=1}^{r}(1+m_j c), \qquad c_2 = \sum_{i<j} m_i m_j c^2.$$

Since

$$2\sum_{i<j} m_i m_j = \left(\sum_j m_j\right)^2 - \sum_j m_j^2 = -s_2(\mathbf{m}),$$

this proves equation (567). The two evaluations follow from equation (465); retaining its orientation reversal is essential for the sign. Finally

$$s_2(\mathbf{m}) = \sum_j m_j^2 \equiv \sum_j m_j \equiv 0 \pmod 2,$$

because $a^2 \equiv a \pmod 2$. This proves the divisibility statement. □

The diagonal connection induced by the summands is

$$\nabla^{\mathbf{m},T} = \mathrm{d} - 2\pi(6y^1\,\mathrm{d}y^4 + y^2\,\mathrm{d}y^3)H_{\mathbf{m}}, \qquad F_{\mathbf{m},T} = -2\pi H_{\mathbf{m}}\eta_T. \tag{569}$$

We retain the invariant inner product and functional normalization

$$\begin{aligned} \langle X, Y\rangle_{\mathfrak{su}(r)} &= -\frac{1}{2}\operatorname{Tr}(XY), \qquad \|H_{\mathbf{m}}\|^2 = \frac{1}{2}s_2(\mathbf{m}), \\ \mathrm{YM}_T(A) &= \frac{1}{2g_{\mathrm{YM}}^2}\int_{F_T} |F_A|^2\,\mathrm{d}\,\mathrm{vol}_T\,. \end{aligned} \tag{570}$$

Theorem 16.2 (Yang–Mills criticality and exact action)**.** *The connection $\nabla^{\mathbf{m},T}$ is non-flat, reducible, and Yang–Mills. Its action is*

$$\mathrm{YM}_T(\nabla^{\mathbf{m},T}) = \frac{\pi^2 s_2(\mathbf{m})}{g_{\mathrm{YM}}^2 D_T}(q_T^2 + 36L_T^2 + 72m_T^2) \tag{571}$$

$$= \frac{2\pi^2 k_{\mathbf{m}}}{g_{\mathrm{YM}}^2} + \frac{\pi^2 s_2(\mathbf{m}) S_T^2}{g_{\mathrm{YM}}^2 D_T}. \tag{572}$$

On the canonical cusp tail,

$$\mathrm{YM}_T(\nabla^{\mathbf{m},T}) = \frac{\pi^2 s_2(\mathbf{m})}{g_{\mathrm{YM}}^2}\left(37 + \frac{35B}{T} + O(T^{-2})\right). \tag{573}$$

Proof. Both $H_{\mathbf{m}}$ and η_T are parallel for the diagonal connection and flat quotient metric, and their brackets vanish. Hence $d^*_{\nabla^{\mathbf{m},T}} F_{\mathbf{m},T} = 0$. Non-flatness follows from $\mathbf{m} \neq 0$ and $\eta_T \neq 0$.

In the physical orthonormal frame used in equation (528),

$$|\eta_T|^2 = \frac{\mathcal{B}_{T,+}^2 + \mathcal{B}_{T,-}^2}{4\pi^2}.$$

Using equation (570), $\mathrm{vol}(F_T) = D_T$, and the exact frequency square sum from the native module yields equation (571). The identity

$$q_T^2 + 36L_T^2 + 72m_T^2 = 12(L_T q_T + 6m_T^2) + (6L_T - q_T)^2 = 12D_T + S_T^2$$

and $k_{\mathbf{m}} = 6s_2(\mathbf{m})$ give equation (572). Substituting the retained expansions of L_T, q_T, m_T, D_T, without suppressing H, B, m, gives

$$\frac{q_T^2 + 36L_T^2 + 72m_T^2}{D_T} = 37 + \frac{35B}{T} + O(T^{-2}),$$

which proves the last formula. □

16.2 Every nonzero integral root charge

The native module proves the charge-one ladder decomposition and the charge-two specialization used by its $SU(2)$ theorem. The following lemma records, and proves, the exact integer-charge extension needed here.

Lemma 16.3 (Charge-d Landau decomposition)**.** *Let $d \in \mathbb{Z} \setminus \{0\}$, put $q = |d|$, and equip $\mathscr{L}_{\eta,T}^{\otimes d}$ with its tensor-power connection. Its scalar covariant Laplacian has joint levels*

$$q\big((2a+1)\mathcal{B}_{T,+} + (2b+1)\mathcal{B}_{T,-}\big), \qquad a, b \in \mathbb{Z}_{\geq 0}, \tag{574}$$

and every joint space has complex dimension $6q^2$. If different pairs (a, b) give the same numerical value, their orthogonal joint spaces are to be summed.

Proof. The curvature is $-2\pi i d\eta_T$. In the polar construction of theorem 15.1, replace the auxiliary complex structure by $J_{d,T} = \operatorname{sgn}(d) J_{\eta,T}$. Then $d\eta_T$ is positive of type $(1,1)$, so the vacuum is the holomorphic-section space of the positive bundle $\mathscr{L}_{\eta,T}^{\otimes d}$ on that auxiliary complex torus. Its tangent and canonical bundles are trivial. Kodaira vanishing and Hirzebruch–Riemann–Roch give

$$\dim_{\mathbb{C}} H^0 = \frac{1}{2}\int_{[F_T]_{d\eta\text{-symp}}} (d\eta_T)^2 = \frac{1}{2} q^2 \int_{[F_T]_{\mathrm{mark}}} \eta_T^2 = 6q^2.$$

Here changing d to $-d$ does not change the four-dimensional symplectic orientation. Scaling the curvature by d scales both positive oscillator frequencies by q, while the two canonical commutation relations and the lowering argument in theorem 15.1 are unchanged after this rescaling. Thus every joint level is unitarily obtained from the vacuum and has the asserted energy and dimension. Orthogonal joint spaces remain a direct sum when their numerical energies coincide. □

16.3 Closed gauge-fixed and Coulomb operators

Write $A = \nabla^{\mathbf{m},T}$ in this subsection. Let

$$\mathscr{H}^1_{\mathbf{m},T} = L^2\Omega^1(F_T; \operatorname{ad}\mathscr{E}_{\mathbf{m},T}), \qquad \mathscr{H}^0_{\mathbf{m},T} = L^2\Omega^0(F_T; \operatorname{ad}\mathscr{E}_{\mathbf{m},T}) \tag{575}$$

as real Hilbert spaces. Define the maximal distributional codifferential by

$$\begin{aligned} \operatorname{Dom} D^{\max}_{\mathbf{m},T} = \{a \in \mathscr{H}^1_{\mathbf{m},T} : \exists u \in \mathscr{H}^0_{\mathbf{m},T} \text{ such that} \\ \langle a, d_A\phi\rangle_{L^2} = \langle u, \phi\rangle_{L^2} \quad \text{for every} \\ \phi \in C^\infty(F_T; \operatorname{ad}\mathscr{E}_{\mathbf{m},T})\}, \end{aligned} \tag{576}$$

$$D^{\max}_{\mathbf{m},T} a = u, \qquad \mathscr{S}_{\mathbf{m},T} := \ker D^{\max}_{\mathbf{m},T}. \tag{577}$$

Density of smooth test scalars makes u unique. If $a_j \to a$ and $D^{\max}_{\mathbf{m},T} a_j \to u$ in L^2, passage to the limit in the defining identity gives $D^{\max}_{\mathbf{m},T} a = u$. Thus $D^{\max}_{\mathbf{m},T}$ is closed and $\mathscr{S}_{\mathbf{m},T}$ is a closed subspace.

Let $\Delta_A^{(0)}$ be the nonnegative self-adjoint scalar operator of the closed form

$$\mathfrak{q}^{(0)}_{\mathbf{m},T}[\phi, \psi] = \langle d_A\phi, d_A\psi\rangle_{L^2}, \qquad \operatorname{Dom}\mathfrak{q}^{(0)}_{\mathbf{m},T} = H^1\Omega^0. \tag{578}$$

It has domain $H^2\Omega^0$ and equals $d_A^* d_A$. In a constant orthonormal frame set

$$(V_{\mathbf{m},T} a)_\nu = -2\sum_{\mu=1}^{4} [F_{\mathbf{m},T,\nu\mu}, a_\mu]. \tag{579}$$

Invariance of the compact Lie-algebra inner product and $F_{\nu\mu} = -F_{\mu\nu}$ show directly that $V_{\mathbf{m},T}$ is a bounded symmetric endomorphism. Define

$$\mathfrak{j}^{\mathrm{gf}}_{\mathbf{m},T}[a, b] = \langle \nabla_A a, \nabla_A b\rangle_{L^2} + \langle V_{\mathbf{m},T} a, b\rangle_{L^2}, \qquad \operatorname{Dom}\mathfrak{j}^{\mathrm{gf}}_{\mathbf{m},T} = H^1\Omega^1. \tag{580}$$

Proposition 16.4 (Closed realization and exact Coulomb reduction)**.** *The form in equation* (580) *is densely defined, closed, symmetric, and bounded below. Its self-adjoint operator has compact resolvent, domain $H^2\Omega^1$, and expression*

$$(\mathcal{J}^{\mathrm{gf}}_{\mathbf{m},T}a)_\nu = -\sum_{\mu=1}^{4}\nabla_{A,e_\mu}\nabla_{A,e_\mu}a_\nu - 2\sum_{\mu=1}^{4}[F_{\mathbf{m},T,\nu\mu},a_\mu]. \tag{581}$$

For every $z\in\mathbb{C}\setminus\mathbb{R}$ and $f\in\operatorname{Dom}D^{\max}_{\mathbf{m},T}$,

$$D^{\max}_{\mathbf{m},T}(\mathcal{J}^{\mathrm{gf}}_{\mathbf{m},T}-z)^{-1}f = (\Delta^{(0)}_A - z)^{-1}D^{\max}_{\mathbf{m},T}f. \tag{582}$$

Consequently $\mathscr{S}_{\mathbf{m},T}$ reduces $\mathcal{J}^{\mathrm{gf}}_{\mathbf{m},T}$, and

$$\mathcal{J}_{\mathbf{m},T} := \mathcal{J}^{\mathrm{gf}}_{\mathbf{m},T}\big|_{\mathscr{S}_{\mathbf{m},T}}, \qquad \operatorname{Dom}\mathcal{J}_{\mathbf{m},T} = H^2\Omega^1\cap\mathscr{S}_{\mathbf{m},T} \tag{583}$$

is self-adjoint, with compact resolvent and closed form $\mathfrak{j}_{\mathbf{m},T}$ equal to the restriction of $\mathfrak{j}^{\mathrm{gf}}_{\mathbf{m},T}$ to $H^1\Omega^1\cap\mathscr{S}_{\mathbf{m},T}$. The physical Yang–Mills Hessian on this slice is

$$\mathrm{YM}''_T\big|_A = g_{\mathrm{YM}}^{-2}\mathfrak{j}_{\mathbf{m},T}. \tag{584}$$

Thus an eigenvalue ν of $\mathcal{J}_{\mathbf{m},T}$ is the physical Hessian eigenvalue ν/g^2_{YM}, while its sign and Morse-index contribution are unchanged.

Proof. The first term of equation (580), after addition of the L^2-norm, is an equivalent H^1-norm. The second term is a bounded symmetric form. The bounded perturbation theorem for closed forms and the representation theorem therefore give a lower-bounded self-adjoint operator. Its principal symbol is $|\xi|^2$ times the identity, so elliptic regularity gives the H^2 domain. Rellich compactness on the compact torus gives compact resolvent.

The two-parameter curvature expansion and the flat Weitzenböck identity, with the inner product and functional normalization in equation (570), give

$$\mathrm{YM}''_T\big|_A(a,b) + g_{\mathrm{YM}}^{-2}\langle d^*_Aa, d^*_Ab\rangle = g_{\mathrm{YM}}^{-2}\mathfrak{j}^{\mathrm{gf}}_{\mathbf{m},T}[a,b].$$

This proves the Hessian normalization after restriction to Coulomb gauge.

It remains to prove that the restriction is an operator rather than a formal constraint. For a smooth adjoint scalar ϕ, write $\nabla_\mu = \nabla_{A,e_\mu}$. Commuting all three derivatives and using the Yang–Mills equation gives

$$\begin{aligned}&(\mathcal{J}^{\mathrm{gf}}_{\mathbf{m},T}d_A\phi - d_A\Delta^{(0)}_A\phi)_\nu\\ &\quad= -\nabla_\mu\nabla_\mu\nabla_\nu\phi + \nabla_\nu\nabla_\mu\nabla_\mu\phi - 2[F_{\nu\mu},\nabla_\mu\phi]\\ &\quad= -[\nabla_\mu F_{\mu\nu},\phi] = [(d^*_AF_A)_\nu,\phi] = 0.\end{aligned}$$

Formal adjunction yields, on smooth one-forms,

$$\mathcal{J}^{\mathrm{gf}}_{\mathbf{m},T}d_A = d_A\Delta^{(0)}_A, \qquad d^*_A\mathcal{J}^{\mathrm{gf}}_{\mathbf{m},T} = \Delta^{(0)}_Ad^*_A. \tag{585}$$

For $u = (\mathcal{J}^{\mathrm{gf}}_{\mathbf{m},T}-z)^{-1}f$, the operator domain gives $u\in H^2$, hence $D^{\max}_{\mathbf{m},T}u\in H^1$. Applying the second identity distributionally to $(\mathcal{J}^{\mathrm{gf}}_{\mathbf{m},T}-z)u = f$ gives

$$(\Delta^{(0)}_A - z)D^{\max}_{\mathbf{m},T}u = D^{\max}_{\mathbf{m},T}f.$$

Scalar elliptic regularity and the scalar resolvent give equation (582). The two conjugate resolvents preserve $\mathscr{S}_{\mathbf{m},T}$; adjunction then makes its orthogonal complement invariant as well. The slice is reducing, so the self-adjoint part has the operator and form domains displayed above. □

16.4 Rootwise spectrum, coincidences, index, and nullity

For $i \neq j$, the matrix unit E_{ij} obeys

$$[H_{\mathbf{m}}, E_{ij}] = i(m_i - m_j)E_{ij},$$

so its complex root line is

$$\mathrm{Hom}(\mathscr{L}_{\eta,T}^{m_j}, \mathscr{L}_{\eta,T}^{m_i}) = \mathscr{L}_{\eta,T}^{m_i - m_j}, \qquad q_{ij} := |m_i - m_j|. \tag{586}$$

The factor q_{ij} is the complete charge normalization; for the existing $SU(2)$ weights $(1, -1)$, it gives $q_{12} = 2$.

Let the distinct numerical values among $m_1, \ldots, m_r$ have multiplicities $r_1, \ldots, r_\ell$. The neutral centralizer is

$$\mathfrak{z}_{\mathbf{m}} = \{X \in \mathfrak{su}(r) : [H_{\mathbf{m}}, X] = 0\} \cong \mathfrak{s}\left(\bigoplus_{a=1}^{\ell} \mathfrak{u}(r_a)\right), \qquad \dim_{\mathbb{R}} \mathfrak{z}_{\mathbf{m}} = \sum_{a=1}^{\ell} r_a^2 - 1. \tag{587}$$

Theorem 16.5 (Complete negative spectrum and numerical multiplicities)**.** *Put*

$$N_-(T) := \#\{n \in \mathbb{Z}_{\geq 0} : 2n + 1 < \rho_T^*\} = \left\lceil \frac{\rho_T^* - 1}{2} \right\rceil. \tag{588}$$

For each positive root $i < j$ with $q_{ij} > 0$, the complete negative spectrum of the normalized Coulomb operator is

$$\nu_{ij,n}(T) = q_{ij}\big((2n + 1)\mathcal{B}_{T,\min} - \mathcal{B}_{T,\max}\big), \qquad 0 \leq n < N_-(T). \tag{589}$$

The contribution of that positive-root pair to each displayed band has real dimension $12q_{ij}^2$. For an actual numerical eigenvalue $\lambda < 0$, its full real multiplicity is therefore

$$\mathrm{mult}_{\mathbb{R}}(\lambda) = \sum_{\substack{i<j,\ q_{ij}>0,\ 0\leq n<N_-(T) \\ \nu_{ij,n}(T)=\lambda}} 12q_{ij}^2. \tag{590}$$

In particular, coincident root charges or accidental coincidences between different q_{ij}, n add rather than being counted as one band. The real Morse index is

$$\boxed{\mathrm{ind}_{\mathbb{R}}\, \mathcal{J}_{\mathbf{m},T} = 12N_-(T) \sum_{i<j} (m_i - m_j)^2 = 12rN_-(T)s_2(\mathbf{m}).} \tag{591}$$

Every eigenvalue in equation (589) *is divided by g_{YM}^2 for the physical Hessian.*

Proof. On the E_{ij} line, the scalar oscillator is the charge-q_{ij} operator of theorem 16.3. In a skew-normal orthonormal coframe, the four circular one-form polarizations diagonalize equation (579); the two polarizations in a plane of charge-q_{ij} frequency $q_{ij}\mathcal{B}$ shift the scalar energy by $\pm 2q_{ij}\mathcal{B}$. The unfavorable polarizations are positive. For the favorable polarization in the high-frequency plane, high-plane oscillator number zero leaves

$$q_{ij}\big((2n + 1)\mathcal{B}_{T,\min} - \mathcal{B}_{T,\max}\big).$$

Increasing its high-plane oscillator number makes it positive. The favorable low-frequency polarization has bottom $q_{ij}(\mathcal{B}_{T,\max} - \mathcal{B}_{T,\min}) \geq 0$, and increasing either number cannot lower it. This proves that the list is exhaustive.

Every joint scalar space on one complex root line has dimension $6q_{ij}^2$. The opposite root is its anti-linear conjugate. The pair is the complexification of a real root subspace, so its real dimension is $12q_{ij}^2$. Root subspaces and joint oscillator spaces are orthogonal; hence equality of numerical eigenvalues produces precisely the direct sum in equation (590).

The strict inequality in equation (588) counts the integers correctly even when ρ_T^* is an odd integer. Summing dimensions gives the first index formula. Finally,

$$\sum_{i<j}(m_i-m_j)^2 = r\sum_i m_i^2 - \left(\sum_i m_i\right)^2 = rs_2(\mathbf{m}),$$

which gives the second formula.

It remains to justify Coulomb gauge. If $\mathcal{J}^{\mathrm{gf}}_{\mathbf{m},T}a = \lambda a$ with $\lambda < 0$, elliptic regularity and equation (585) give

$$\Delta_A^{(0)} d_A^* a = \lambda d_A^* a.$$

The scalar operator is nonnegative, so $d_A^* a = 0$. Thus the negative gauge-fixed and Coulomb eigenspaces coincide, completing the proof. □

Theorem 16.6 (Exact neutral and wall nullities)**.** *Away from an odd wall, the neutral Coulomb kernel has real dimension*

$$\mathrm{null}_{\mathrm{neutral}}\,\mathcal{J}_{\mathbf{m},T} = 4\left(\sum_{a=1}^{\ell} r_a^2 - 1\right). \tag{592}$$

At a wall $\rho_T^* = 2n+1 > 1$, the nonzero-root part adds

$$12\sum_{i<j} q_{ij}^2 = 12rs_2(\mathbf{m}) \tag{593}$$

real zero modes. At the equality wall $S_T = 0$, equivalently $\rho_T^* = 1$, both favorable plane polarizations vanish and the nonzero root contribution doubles to

$$24\sum_{i<j} q_{ij}^2 = 24rs_2(\mathbf{m}). \tag{594}$$

The total nullity at a wall is the sum of the neutral term and the stated root term.

Proof. On $\mathfrak{z}_{\mathbf{m}}$, both the induced connection commutator and the spin endomorphism vanish. The operator is therefore the ordinary coefficientwise Laplacian on one-forms. Its kernel on a flat four-torus is the space of constant one-forms, all of which are Coulomb. Tensoring the four-dimensional real harmonic-one-form space with $\mathfrak{z}_{\mathbf{m}}$ proves equation (592).

If $\rho_T^* = 2n+1 > 1$, the high-plane favorable polarization at high oscillator number zero and low oscillator number n is the unique zero branch for each nonzero root. Its positive-root-pair real dimension is $12q_{ij}^2$. If $S_T = 0$, the two frequencies agree and the lowest favorable polarization in each of the two planes is zero, so this dimension doubles. These charged zero modes are Coulomb: the intertwining identity shows that their divergences lie in the kernel of the charged scalar Laplacian, but that scalar Laplacian has strictly positive ground energy by equation (574). Summing over positive roots and using $\sum_{i<j} q_{ij}^2 = rs_2(\mathbf{m})$ proves the result. □

For $n \geq 1$, the n-th odd wall is equivalently

$$\rho_T^* = 2n+1 \iff n\Xi_T = (n+1)|S_T| \iff 24n^2 D_T = (2n+1)S_T^2. \tag{595}$$

The final polynomial equation must be accompanied by the unsigned ordering in the middle expression if one needs to distinguish the two sides $S_T > 0$ and $S_T < 0$. The $n = 0$ wall is exactly $S_T = 0$.

On the sufficiently deep cusp tail, $S_T > 0$, $5 < \rho_T^* < 7$, and $N_-(T) = 3$. Hence

$$\mathrm{ind}_{\mathbb{R}}\,\mathcal{J}_{\mathbf{m},T} = 36rs_2(\mathbf{m}) = 6rk_{\mathbf{m}}, \tag{596}$$

and the three negative branches have the retained asymptotics

$$\nu_{ij,n}(T) = \frac{q_{ij}\pi}{T}(4n-10) + O(T^{-2}), \qquad n = 0,1,2. \tag{597}$$

For example, the genuinely $SU(3)$ data are

weights	nonzero q_{ij}	$k_{\mathbf{m}}$	deep index	neutral nullity
$(1,0,-1)$	$1,1,2$	12	216	8
$(1,1,-2)$	$3,3$	36	648	16
$(2,1,-3)$	$1,4,5$	84	1512	8

The middle row exhibits both required edge phenomena: one neutral root from the repeated weight and coincident numerical bands from the two charge-three positive roots. The specialization $r = 2$, $\mathbf{m} = (1,-1)$ is exactly the already proved native $SU(2)$ theorem and is not reproduced here.

16.5 Stability boundary, possible wall bifurcation, and exact scope

Proposition 16.7 (Where the split background is, and is not, unstable)**.** *If $S_T \neq 0$, then $\rho_T^* > 1$, $N_-(T) \geq 1$, and every nonzero positive root contributes a negative Coulomb direction. Since a nonzero trace-zero weight vector has at least one nonzero root, $\nabla^{\mathbf{m},T}$ is then not a local minimum modulo gauge. This applies in particular throughout the sufficiently deep cusp tail.*

If $S_T = 0$, the negative root spectrum is empty. In that case equation (572) *attains the topological value $2\pi^2 k_{\mathbf{m}}/g_{\mathrm{YM}}^2$, while equation* (594) *supplies root zero modes. No assertion of strict stability follows because the Hessian is degenerate.*

Proof. The first paragraph follows immediately from equation (564) and theorem 16.5. For the second, $S_T = 0$ makes the two frequencies equal and hence $\rho_T^* = 1$; the strict negative-band inequality has no solution. The action and nullity statements are exactly equation (572) and theorem 16.6. □

A negative eigenspace is a space of descent directions, not a Lyapunov–Schmidt bifurcation kernel. A local bifurcation analysis must instead be posed at one of the zero-mode parameters in theorem 16.6. It must first quotient or slice the ever-present centralizer-valued harmonic kernel, then identify the remaining kernel and cokernel of the gauge-fixed nonlinear Yang–Mills map, and finally compute the parameter transversality and nonlinear obstruction coefficients. The present linear calculation supplies the exact candidate kernel at each wall; it does not assert existence of a nearby irreducible branch.

The theorem proved here is strictly lower-rung and fibrewise. It constructs explicit reducible split-Cartan $SU(r)$ Yang–Mills critical points on the already existing regular flat four-torus fibres and determines their topology, action, normalized Jacobi spectrum, physical Hessian scaling, Morse index, and nullity. It constructs no new base, spacetime, cone, S^{12}, S^{24}, transfer matrix, or interacting quantum Hamiltonian. The radial multiplication operator already treated in the canonical native module is not repeated here.

There is also an exact obstruction to interpreting the nonzero-flux bundle as the restriction of a bundle on a proposed higher carrier. If $i : F_T \to Y$ is any map and $H^4(Y;\mathbb{Z}) = 0$, then every complex bundle $\mathscr{V} \to Y$ has $c_2(\mathscr{V}) = 0$, so naturality gives

$$c_2(i^*\mathscr{V}) = i^* c_2(\mathscr{V}) = 0.$$

This contradicts equations (567) and (568) for $\mathscr{E}_{\mathbf{m},T}$. Therefore this nonzero-c_2 bundle cannot extend over any carrier with vanishing fourth integral cohomology, including a space diffeomorphic to S^6, S^{12}, or S^{24}. Changing the gauge group changes the root system on the fibre; it does not by itself produce the proposed higher rung or any conclusion about a physical mass gap.

17 The higher-rung SU(3) Gram cone

This section isolates the part of the proposed higher-rung construction that is independent of the candidate complex threefold and proves it from the underlying linear action. The compact object used here is the representation sphere

$$\mathrm{Mat}_{3\times 2}(\mathbb{C})^+,$$

the one-point compactification of a twelve-real-dimensional vector space. It must not be identified with a cusp-induced S^{12} construction unless an additional based identification has first been chosen. The proofs below also distinguish three structures that otherwise look deceptively similar: the Lorentzian determinant on $\mathrm{Herm}_2(\mathbb{C})$, the positive-semidefinite Gram cone inside that vector space, and a separately chosen Euclidean metric on the positive-definite cone.

17.1 Conventions and the Gram morphism

Put

$$\mathscr{V} = \mathrm{Mat}_{3\times 2}(\mathbb{C}), \qquad \Gamma : \mathscr{V} \longrightarrow \mathrm{Herm}_2(\mathbb{C}), \qquad \Gamma(A) = A^{\dagger}A. \tag{598}$$

Write $A = (a\ b)$, with $a, b \in \mathbb{C}^3$, and let $\mathrm{SU}(3)$ act on the left:

$$U \cdot A = UA.$$

For $r = 0, 1, 2$, let

$$\mathscr{V}_{(r)} = \{A \in \mathscr{V} : \operatorname{rank}_{\mathbb{C}} A = r\}.$$

The Hermitian products are linear in the second argument. We orient $\mathscr{V}$ by the complex orientation, in the real coordinate order

$$\begin{aligned}(&\operatorname{Re} A_{11}, \operatorname{Im} A_{11}, \operatorname{Re} A_{12}, \operatorname{Im} A_{12}, \operatorname{Re} A_{21}, \operatorname{Im} A_{21},\\ &\operatorname{Re} A_{22}, \operatorname{Im} A_{22}, \operatorname{Re} A_{31}, \operatorname{Im} A_{31}, \operatorname{Re} A_{32}, \operatorname{Im} A_{32}).\end{aligned} \tag{599}$$

On $\mathrm{Herm}_2(\mathbb{C})$, use the oriented real-linear coordinates

$$\Phi(t, x, y, z) = \begin{pmatrix} t + z & x - iy \\ x + iy & t - z \end{pmatrix}, \qquad \mathrm{or}_{\mathrm{Herm}} = dt \wedge dx \wedge dy \wedge dz. \tag{600}$$

Thus, if $\Gamma(A)_{12} = \langle a, b\rangle$, then

$$\begin{aligned} t &= \frac{1}{2}(\|a\|^2 + \|b\|^2), & z &= \frac{1}{2}(\|a\|^2 - \|b\|^2),\\ x &= \operatorname{Re}\langle a, b\rangle, & y &= -\operatorname{Im}\langle a, b\rangle. \end{aligned} \tag{601}$$

17.2 Exact orbit classification

Theorem 17.1 (The Gram quotient)**.** *The morphism Γ is $\mathrm{SU}(3)$-invariant, has image exactly $\mathrm{Herm}_2(\mathbb{C})_{\geq 0}$, and separates $\mathrm{SU}(3)$-orbits. It induces a proper homeomorphism*

$$\overline{\Gamma} : \mathscr{V}/\mathrm{SU}(3) \xrightarrow{\ \cong\ } \mathrm{Herm}_2(\mathbb{C})_{\geq 0}. \tag{602}$$

Proof. For $U \in \mathrm{SU}(3)$,

$$\Gamma(UA) = A^{\dagger}U^{\dagger}UA = A^{\dagger}A,$$

so Γ is invariant. Every matrix $A^{\dagger}A$ is positive semidefinite. Conversely, for $H \in \mathrm{Herm}_2(\mathbb{C})_{\geq 0}$, let $H^{1/2}$ be its unique positive-semidefinite square root and define

$$s(H) = \begin{pmatrix} H^{1/2} \\ 0 \quad 0 \end{pmatrix} \in \mathrm{Mat}_{3\times 2}(\mathbb{C}). \tag{603}$$

Then $s(H)^{\dagger}s(H) = H$, proving surjectivity and giving a continuous section.

It remains to prove exact orbit separation. Suppose $A^{\dagger}A = B^{\dagger}B$. Define

$$T : \operatorname{im} A \longrightarrow \operatorname{im} B, \qquad T(Au) = Bu.$$

If $Au = 0$, then

$$0 = \|Au\|^2 = u^\dagger A^\dagger A u = u^\dagger B^\dagger B u = \|Bu\|^2,$$

so $Bu = 0$, and T is well defined. For $u, v \in \mathbb{C}^2$,

$$\langle T(Au), T(Av)\rangle = u^\dagger B^\dagger B v = u^\dagger A^\dagger A v = \langle Au, Av\rangle.$$

Thus T is a unitary isomorphism of the image subspaces. Extend it to some $U_0 \in \mathrm{U}(3)$ satisfying $U_0 A = B$. Since $\operatorname{rank} B \le 2$, the orthogonal complement of $\operatorname{im} B$ has positive complex dimension. Choose a unitary map V which fixes $\operatorname{im} B$ pointwise and whose determinant is $(\det U_0)^{-1}$. Then

$$U = V U_0 \in \mathrm{SU}(3), \qquad UA = VB = B.$$

Hence equal Gram matrices are exactly equal $\mathrm{SU}(3)$-orbits.

The action is proper because $\mathrm{SU}(3)$ is compact; hence the orbit space is Hausdorff. The continuous map $H \mapsto [s(H)]$ is inverse to the continuous induced map $\overline{\Gamma}$, so (602) is a homeomorphism. Finally,

$$\|A\|_F^2 = \operatorname{tr}(A^\dagger A) = \operatorname{tr} \Gamma(A). \tag{604}$$

If K is compact in the positive-semidefinite cone, then $\Gamma^{-1}(K)$ is closed and bounded by (604), hence compact. Thus Γ is proper. For compact K, $\overline{\Gamma}^{-1}(K)$ is the continuous image in the orbit space of the compact set $\Gamma^{-1}(K)$, so $\overline{\Gamma}$ is proper as well. □

Proposition 17.2 (Stabilizers and orbit-type strata)**.** *For $A \in \mathscr{V}_{(r)}$,*

$$\operatorname{Stab}_{\mathrm{SU}(3)}(A) \cong \mathrm{SU}(3 - r). \tag{605}$$

The exact dimensions are

r	$\dim_{\mathbb{R}} \mathscr{V}_{(r)}$	$\operatorname{Stab}(A)$	$\dim_{\mathbb{R}}(\mathrm{SU}(3) \cdot A)$	$\dim_{\mathbb{R}}(\mathscr{V}_{(r)}/\mathrm{SU}(3))$
2	12	$\{1\}$	8	4
1	8	$\mathrm{SU}(2)$	5	3
0	0	$\mathrm{SU}(3)$	0	0.

(606)

Consequently the quotient has topological dimension four, but the action is free only on the rank-two stratum.

Proof. The equality $UA = A$ is equivalent to the assertion that U fixes $\operatorname{im} A$ pointwise. Restriction to its orthogonal complement identifies the stabilizer with $\mathrm{SU}(3 - r)$, proving (605). The rank-two locus is open in the twelve-real-dimensional vector space. The complex rank-one locus in the space of 3×2 matrices has complex dimension

$$1(3 + 2 - 1) = 4,$$

hence real dimension eight away from the origin. Subtracting $8 - \dim_{\mathbb{R}} \operatorname{Stab}(A)$ gives the displayed orbit and quotient dimensions. They agree with the target strata: positive-definite Hermitian matrices form a four-dimensional open cone, while every nonzero rank-one positive-semidefinite matrix has a unique expression $\lambda v v^\dagger$ with $\lambda > 0$ and $v \in S^3$, modulo $v \sim e^{i\theta} v$. Hence that target stratum is

$$(0, \infty) \times \mathbb{CP}^1 \cong (0, \infty) \times S^2,$$

of real dimension three. The rank-zero stratum is the origin. □

17.3 The free locus, its explicit trivialization, and orientation

Theorem 17.3 (Determinant-one completion)**.** *There is an* $\mathrm{SU}(3)$*-equivariant diffeomorphism*

$$\mathscr{V}_{(2)} \xrightarrow{\cong} \mathrm{SU}(3) \times \mathrm{Herm}_2(\mathbb{C})_{>0} \tag{607}$$

under which Γ *is projection to the second factor and the group acts by left multiplication on the first. Thus the principal bundle over the free quotient has a global section and is trivial.*

Proof. For $A \in \mathscr{V}_{(2)}$, set

$$H = A^\dagger A > 0, \qquad Q = AH^{-1/2} = (q_1\ q_2).$$

Then $Q^\dagger Q = I_2$, so (q_1, q_2) is an ordered orthonormal two-frame in $\mathbb{C}^3$. With $\varepsilon_{123} = 1$, define

$$(q_3)_i = \sum_{j,k=1}^{3} \varepsilon_{ijk} \overline{(q_1)_j}\, \overline{(q_2)_k}. \tag{608}$$

Antisymmetry gives $q_1^\dagger q_3 = q_2^\dagger q_3 = 0$. The Gram–Plücker identity, proved independently in theorem 17.4, gives $\|q_3\| = 1$. Finally, contraction of the two alternating tensors gives

$$\begin{aligned} \det[q_1\ q_2\ q_3] &= \sum_{i,j,k} \varepsilon_{ijk} (q_1)_i (q_2)_j (q_3)_k \\ &= \|q_1\|^2 \|q_2\|^2 - |q_1^\dagger q_2|^2 = 1. \end{aligned}$$

Thus $U_A = [q_1\ q_2\ q_3] \in \mathrm{SU}(3)$, and

$$A = U_A s(H),$$

where s is (603). The positive square root, inverse square root, and the polynomial-conjugate formula (608) are smooth on the positive-definite locus. Consequently

$$A \longmapsto (U_A, A^\dagger A), \qquad (U, H) \longmapsto U s(H)$$

are smooth inverse maps. Their equivariance is immediate. □

The order of factors in (607) does not by itself fix an orientation. We now fix and calculate it. Give $\mathfrak{su}(3)$ the ordered real basis

$$\begin{gathered} X_{12}, Y_{12}, X_{13}, Y_{13}, X_{23}, Y_{23}, D_1, D_2, \\ X_{jk} = E_{jk} - E_{kj}, \qquad Y_{jk} = i(E_{jk} + E_{kj}), \\ D_1 = i\,\mathrm{diag}(1, -1, 0), \qquad D_2 = i\,\mathrm{diag}(0, 1, -1). \end{gathered} \tag{609}$$

At

$$A_0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix},$$

use the oriented Hermitian basis

$$K_t = I_2, \quad K_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad K_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad K_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \tag{610}$$

and horizontal lifts $B_\alpha = \frac{1}{2} A_0 K_\alpha$. Directly, $d\Gamma_{A_0}(B_\alpha) = K_\alpha$. Relative to (599), the twelve columns

$$(X_{12}A_0, Y_{12}A_0, X_{13}A_0, Y_{13}A_0, X_{23}A_0, Y_{23}A_0, D_1A_0, D_2A_0, B_t, B_x, B_y, B_z)$$

form the exact real coordinate matrix

$$M_{\mathrm{or}} := \begin{pmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & \frac{1}{2} \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 \\
0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 \\
-1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 \\
0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & -\frac{1}{2} \\
0 & 0 & 0 & 0 & 0 & 0 & -1 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0
\end{pmatrix}. \tag{611}$$

Laplace expansion gives

$$\det M_{\mathrm{or}} = \frac{1}{2} > 0. \tag{612}$$

Since both factors in (607) are connected, the sign is constant on the free locus. Therefore the quotient orientation determined by

$$\mathrm{or}_{\mathscr{V}} = \mathrm{or}_{\mathrm{SU}(3)} \wedge \mathrm{or}_{\mathscr{V}_{(2)}/\mathrm{SU}(3)}$$

agrees under Γ with $dt \wedge dx \wedge dy \wedge dz$.

17.4 Lorentz determinant and Plücker norm

For $H = \Phi(t, x, y, z)$, direct expansion gives

$$\operatorname{tr} H = 2t, \qquad \det H = t^2 - x^2 - y^2 - z^2, \tag{613}$$

and the eigenvalues are $t \pm \sqrt{x^2 + y^2 + z^2}$. Hence

$$\begin{aligned}
H > 0 &\iff t > \sqrt{x^2 + y^2 + z^2}, \\
H \geq 0 &\iff t \geq \sqrt{x^2 + y^2 + z^2}, \\
\operatorname{rank} H = 1 &\iff t = \sqrt{x^2 + y^2 + z^2} > 0.
\end{aligned} \tag{614}$$

Thus det has signature $(1, 3)$ when positive directions are listed first, while $-\det$ has signature $(3, 1)$. The quotient image is not all of this Lorentz vector space: it is precisely the closed future causal cone, and its free locus is the open future timelike cone.

Proposition 17.4 (Gram–Plücker identity)**.** *For $A = (a\ b)$,*

$$\det \Gamma(A) = \|a\|^2 \|b\|^2 - |\langle a, b \rangle|^2 = \sum_{1 \leq i < j \leq 3} |a_i b_j - a_j b_i|^2 = \|a \wedge b\|^2. \tag{615}$$

Proof. Expanding gives

$$\begin{aligned}
\|a\|^2 \|b\|^2 - |\langle a, b \rangle|^2 &= \sum_{i,j} |a_i|^2 |b_j|^2 - \sum_{i,j} \overline{a_i} b_i a_j \overline{b_j} \\
&= \sum_{i<j} \left(|a_i|^2 |b_j|^2 + |a_j|^2 |b_i|^2 - \overline{a_i} b_i a_j \overline{b_j} - \overline{a_j} b_j a_i \overline{b_i} \right) \\
&= \sum_{i<j} |a_i b_j - a_j b_i|^2.
\end{aligned}$$

The last expression is the squared Hermitian norm of $a \wedge b$ in the ordered basis $(e_1 \wedge e_2, e_1 \wedge e_3, e_2 \wedge e_3)$. □

The determinant is only one invariant and does not classify orbits. For example,

$$A_1 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad A_2 = \begin{pmatrix} \sqrt{2} & 0 \\ 0 & 1/\sqrt{2} \\ 0 & 0 \end{pmatrix}$$

satisfy $\det \Gamma(A_1) = \det \Gamma(A_2) = 1$, but their Gram matrices are I_2 and $\mathrm{diag}(2, 1/2)$, so theorem 17.1 shows that they lie in distinct orbits.

17.5 The four-coefficient form is additional input

Let $r = (r_1, r_2, r_3, r_4)^{\mathsf{T}} \in \mathbb{R}^4$, and define the orthogonal Hadamard coordinates

$$\begin{pmatrix} h \\ x \\ y \\ z \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix} \begin{pmatrix} r_1 \\ r_2 \\ r_3 \\ r_4 \end{pmatrix}. \tag{616}$$

The proposed coefficient-space form

$$Q_4(r) = \frac{1}{2} \left(\sum_{i=1}^{4} r_i \right)^2 - \sum_{i=1}^{4} r_i^2 \tag{617}$$

satisfies the exact identity

$$Q_4(r) = h^2 - x^2 - y^2 - z^2. \tag{618}$$

Indeed, orthogonality gives $\sum r_i^2 = h^2 + x^2 + y^2 + z^2$ and $\sum r_i = 2h$.

The same Hermitian matrix can be written in the alternative convention

$$\Psi(h, x, y, z) = \begin{pmatrix} h + y & x + iz \\ x - iz & h - y \end{pmatrix}. \tag{619}$$

It agrees exactly with Φ under

$$(t, x, y_\Phi, z_\Phi) = (h, x, -z, y).$$

The transformation $(y, z) \mapsto (-z, y)$ has determinant $+1$, so the two conventions give the same matrix, Lorentz form, and orientation.

Nothing in the SU(3) orbit calculation selects the coefficient $1/2$. More generally,

$$Q_{4,\lambda}(r) = \lambda \left(\sum_i r_i \right)^2 - \sum_i r_i^2 = (4\lambda - 1)h^2 - x^2 - y^2 - z^2. \tag{620}$$

Every $\lambda > 1/4$ therefore has Lorentz signature $(1, 3)$, after a rescaling of h. The value $\lambda = 1/2$ is the unique member for which the fixed orthogonal coordinates in (616) have equal unit coefficients. This is a normalization choice in coefficient space, not a theorem forced by the Gram quotient.

Nor does a nonnegative coefficient tuple automatically determine a point of the positive Gram cone. The tuple $(1, 0, 0, 0)$ has $Q_4 = -1/2$. A coefficient tuple enters the Gram quotient through (619) only when that Hermitian matrix is positive semidefinite, equivalently when $h \geq 0$ and $Q_4 \geq 0$. The hypersurface $z = 0$ is a real-symmetric coefficient slice, but that fact alone does not supply a coordinatewise nonnegative embedding of a three-component mass domain.

17.6 The compact representation quotient

Theorem 17.5 (The S^{12} representation quotient)**.** *Let*

$$S^{12}_{\mathrm{col}} = \mathscr{V}^+ = \mathscr{V} \cup \{\infty\}$$

be the one-point compactification, and extend the $\mathrm{SU}(3)$ *action by fixing* ∞*. Then*

$$S^{12}_{\mathrm{col}}/\mathrm{SU}(3) \cong \left(\mathrm{Herm}_2(\mathbb{C})_{\geq 0}\right)^+ \cong B^4. \tag{621}$$

The orbit-type strata are

source stratum	*quotient stratum*	*stabilizer*
$\operatorname{rank} A = 2$	$\operatorname{int} B^4$	$\{1\}$
$\operatorname{rank} A = 1$	$\partial B^4 \setminus \{p_0, p_\infty\}$	$\mathrm{SU}(2)$
$A = 0$	$\{p_0\}$	$\mathrm{SU}(3)$
$A = \infty$	$\{p_\infty\}$	$\mathrm{SU}(3)$.

(622)

The ball identification is topological on the full stratified quotient. On its smooth interior, it may be chosen to carry the quotient orientation to the orientation fixed in (600)*.*

Proof. As a real vector space, $\mathscr{V} \cong \mathbb{R}^{12}$, so its one-point compactification is S^{12}. On $\mathscr{V} \cong \mathbb{C}^3 \oplus \mathbb{C}^3$, the action of $U \in \mathrm{SU}(3)$ has complex determinant $(\det U)^2 = 1$, hence real determinant one; it preserves the standard orientation before and after one-point compactification.

The properness proved from (604) implies that Γ extends continuously to

$$\Gamma^+ : \mathscr{V}^+ \longrightarrow \left(\mathrm{Herm}_2(\mathbb{C})_{\geq 0}\right)^+, \qquad \Gamma^+(\infty) = \infty.$$

It separates the finite orbits by theorem 17.1, and the added point is a distinct fixed orbit. Hence it induces a continuous bijection

$$\mathscr{V}^+/\mathrm{SU}(3) \longrightarrow \left(\mathrm{Herm}_2(\mathbb{C})_{\geq 0}\right)^+.$$

The source is compact Hausdorff because a compact group acts on the compact Hausdorff space $\mathscr{V}^+$. The target is Hausdorff because the positive-semidefinite cone is locally compact Hausdorff. The continuous bijection is therefore a homeomorphism.

By (614), the map

$$(t, \mathbf{x}) \longmapsto (t - \|\mathbf{x}\|, \mathbf{x})$$

is a homeomorphism from the closed future cone to the closed half-space $\mathbb{R}_{\geq 0} \times \mathbb{R}^3$. For $u = (s, \mathbf{x})$ in that half-space, inverse stereographic projection is

$$u \longmapsto \frac{1}{1 + \|u\|^2}(2s, 2\mathbf{x}, \|u\|^2 - 1) \in S^4 \subset \mathbb{R}^5. \tag{623}$$

Its image is the closed hemisphere with first coordinate nonnegative, minus the north pole. The point at infinity supplies that pole. A closed hemisphere is a closed four-ball, proving (621).

Positive-definite matrices correspond to the open half-space and hence the ball interior. Nonzero rank-one matrices correspond to $\{0\} \times (\mathbb{R}^3 \setminus \{0\})$. The zero matrix and the point at infinity give two distinct endpoints in the compactification of that three-dimensional boundary. Combining this with theorem 17.2 proves (622). The orientation statement on the connected interior is the calculation following (612). □

The notation S^{12}_{col} records the construction. There is a canonical pointed homeomorphism of representation spheres

$$(\mathbb{C}^3)^+ \wedge (\mathbb{C}^3)^+ \cong (\mathbb{C}^3 \oplus \mathbb{C}^3)^+ = S^{12}_{\mathrm{col}}.$$

If a different based six-sphere X is used, a based homeomorphism $X \cong (\mathbb{C}^3)^+$ transports the action to $X \wedge X$. The transported action depends on that choice. Without an independently proved equivariance statement, it is not induced by a torus fibration, cusp monodromy, period matrix, or fibrewise flux on X.

17.7 Exact gauge-theoretic boundary

Theorem 17.6 (No native nonzero S^4 sector from the Gram quotient)**.** *The construction above supplies no native principal* SU(3) *bundle of nonzero second Chern number on* S^4*. More precisely:*

1. *the orbit projection over the rank-two locus is the product bundle in* (607)*, so its second Chern class is zero;*
2. *on the rank-one boundary the orbit projection is not a principal* SU(3)*-bundle, because its stabilizer is* SU(2)*;*
3. *the full compact quotient is* B^4*, not* S^4*; and*
4. *replacing* B^4 *by the one-point compactification of its interior is a different construction whose clutching class is additional data.*

Proof. The first two assertions are theorems 17.2 and 17.3; the third is theorem 17.5. For the fourth, the matrix logarithm is a diffeomorphism

$$\log : \mathrm{Herm}_2(\mathbb{C})_{>0} \xrightarrow{\cong} \mathrm{Herm}_2(\mathbb{C}) \cong \mathbb{R}^4.$$

Its one-point compactification is therefore S^4, but this operation collapses the entire finite rank-one boundary of the full Gram quotient into the new point at infinity. Principal SU(3)-bundles on this new S^4 are obtained by clutching two trivial bundles over hemispheres with a map

$$\kappa : S^3 \longrightarrow \mathrm{SU}(3).$$

The fibration $\mathrm{SU}(2) \to \mathrm{SU}(3) \to S^5$, together with $\pi_4(S^5) = \pi_3(S^5) = 0$, shows that

$$\pi_3(\mathrm{SU}(3)) \cong \pi_3(\mathrm{SU}(2)) = \pi_3(S^3) \cong \mathbb{Z}.$$

Choose the generator so that its associated bundle has second Chern number $+1$. A clutching class $k \in \mathbb{Z}$ then has second Chern number k. Extending the explicit product framing in theorem 17.3 by the same framing gives the constant clutching map and $k = 0$. Neither $A^\dagger A$ nor $\|a \wedge b\|^2$ selects a nonconstant κ. Thus $k \neq 0$ is valid additional gauge data, but it is not produced by the Gram construction. □

There are two further cohomological checks. If a candidate total space is diffeomorphic to S^6, then

$$H^2(S^6; \mathbb{Z}) = H^4(S^6; \mathbb{Z}) = 0.$$

Consequently no nonzero fibrewise degree-two class can be the restriction of a degree-two class on that total space, and no c_2 on the total space can restrict to a nonzero degree-four class on a fibre. This does not forbid independently chosen higher characteristic data; it proves exactly that the degree-two and degree-four classes in question do not extend.

Likewise, for every continuous map $f : S^{12} \to S^4$ and generator $u_4 \in H^4(S^4; \mathbb{Z})$,

$$f^* u_4 = 0$$

because $H^4(S^{12}; \mathbb{Z}) = 0$. Therefore the pullback along any such map of an SU(3) bundle with nonzero $c_2 = k u_4$ has $c_2 = 0$. This statement does not assert that every intrinsic SU(3) bundle on S^{12} is trivial; it says that an S^4 second-Chern charge is not inherited by pullback to the representation sphere.

17.8 The log-Euclidean metric and Yang–Mills theory are extra choices

The logarithm identifies the open positive cone with a real vector space. One may therefore declare

$$X = \log H, \qquad g_{\log} = \frac{1}{2}\operatorname{tr}(dX\, dX), \tag{624}$$

which is a complete flat Euclidean metric in the X-coordinates. This metric is not induced by the Frobenius metric on $\mathscr{V}$. To see the difference without discarding any scale factor, set

$$H(s) = sI_2, \qquad A(s) = \sqrt{s}\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad s > 0.$$

For $\xi \in \mathfrak{su}(3)$,

$$\operatorname{Re}\operatorname{tr}(\dot{A}^{\dagger}\xi A) = \frac{\dot{s}}{2}\operatorname{Re}(\xi_{11} + \xi_{22}) = 0,$$

so $\dot{A}$ is horizontal for the Frobenius quotient. Hence

$$\|\dot{H}\|^2_{\text{quot}} = \|\dot{A}\|^2_F = \frac{\dot{s}^2}{2s}. \tag{625}$$

By contrast, $X(s) = (\log s)I_2$, and

$$\|\dot{H}\|^2_{\log} = \frac{1}{2}\operatorname{tr}(\dot{X}^2) = \frac{\dot{s}^2}{s^2}. \tag{626}$$

The ratio of (625) and (626) depends on s, so no global constant rescaling identifies the metrics. The Lorentz form (613) is a third object and is indefinite.

After choosing (624), one may define classical Yang–Mills theory on the trivial product bundle supplied by theorem 17.3. The additional choice of the zero connection has $F_0 = 0$, so it solves the nonlinear classical equation $d_0^* F_0 = 0$; its Coulomb Hessian is the transverse free Laplacian. These are the standard zero-charge vacuum statements on a chosen Euclidean $\mathbb{R}^4$. The Gram morphism itself supplies no nonzero clutching class, no nonflat connection, no interacting Hamiltonian, no quantum Hilbert-space identification, and no mass-gap estimate. Thus the exact logical implication is

four-dimensional Gram/Lorentz quotient: proved;
$S^{12}_{\text{col}}/\mathrm{SU}(3) \cong B^4$: proved;
nonzero native SU(3) sector on S^4 : not produced;
interacting or quantum Yang–Mills inheritance: not produced.

18 The three-column SU(3) Gram cone and its rank-loss S^8

This section proves the topology forced by the left actions of U(3) and SU(3) on three complex columns. The calculation produces a canonical topological S^8, but it does not equip that sphere with a complex structure, a fibration, a gauge bundle, or a physical interpretation.

18.1 Conventions

Put

$$\mathscr{V}_3 = \mathrm{Mat}_{3\times 3}(\mathbb{C}), \qquad \|A\|^2_F = \operatorname{tr}(A^{\dagger}A). \tag{627}$$

This is a real vector space of dimension 18. Both U(3) and SU(3) act on the left:

$$U \cdot A = UA. \tag{628}$$

Let

$$\mathcal{P}_3 = \{H \in \mathrm{Herm}_3(\mathbb{C}) : H \succeq 0\}, \qquad \mathcal{K}_3 = \{H \in \mathcal{P}_3 : \operatorname{tr} H = 1\}. \tag{629}$$

Write $S_F^{17} \subset \mathscr{V}_3$ for the Frobenius unit sphere and $\mathscr{V}_3^+ = \mathscr{V}_3 \cup \{\infty\} \cong S^{18}$ for the one-point compactification, with ∞ fixed by each group.

18.2 The U(3) orbit space

Theorem 18.1 (Exact Gram classification)**.** *The map*

$$\Gamma_U : \mathscr{V}_3 \longrightarrow \mathcal{P}_3, \qquad \Gamma_U(A) = A^\dagger A \tag{630}$$

is U(3)*-invariant, is surjective, and separates* U(3)*-orbits. It induces a homeomorphism*

$$\overline{\Gamma}_U : \mathscr{V}_3/\mathrm{U}(3) \xrightarrow{\sim} \mathcal{P}_3. \tag{631}$$

Proof. Invariance is the identity

$$(UA)^\dagger(UA) = A^\dagger U^\dagger U A = A^\dagger A.$$

For $H \succeq 0$, its positive-semidefinite square root satisfies

$$(H^{1/2})^\dagger H^{1/2} = H,$$

so the map is surjective.

Suppose $A^\dagger A = B^\dagger B$. Define

$$T : \operatorname{im} A \longrightarrow \operatorname{im} B, \qquad T(Ax) = Bx.$$

If $Ax = 0$, then

$$\|Bx\|^2 = x^\dagger B^\dagger B x = x^\dagger A^\dagger A x = \|Ax\|^2 = 0,$$

so T is well-defined. For $x, y \in \mathbb{C}^3$,

$$\langle T(Ax), T(Ay)\rangle = x^\dagger B^\dagger B y = x^\dagger A^\dagger A y = \langle Ax, Ay\rangle.$$

Thus T is a complex-linear isometry. Extend it to a unitary $U \in \mathrm{U}(3)$. Then $UA = B$, so the Gram matrix separates orbits.

Finally,

$$\|A\|_F^2 = \operatorname{tr}(A^\dagger A).$$

Hence Γ_U is proper. The induced map $\overline{\Gamma}_U$ is a proper continuous bijection between locally compact Hausdorff spaces and is therefore a homeomorphism. □

Lemma 18.2 (Explicit ball chart on the trace-one Gram body)**.** *Let*

$$E = \{X \in \mathrm{Herm}_3(\mathbb{C}) : \operatorname{tr} X = 0\} \tag{632}$$

with Frobenius inner product. Then $\dim_{\mathbb{R}} E = 8$*, and there is an explicit homeomorphism*

$$\chi : \mathcal{K}_3 \xrightarrow{\sim} B(E) \cong B^8 \tag{633}$$

which sends the singular locus in $\mathcal{K}_3$ *onto* $S(E)$*.*

Proof. Put $c = I_3/3$. If $u \in S(E)$, then $u \neq 0$ and $\operatorname{tr} u = 0$, so $\lambda_{\min}(u) < 0$. Define

$$R(u) = -\frac{1}{3\lambda_{\min}(u)} > 0. \tag{634}$$

The eigenvalues of $c + su$ are $1/3 + s\lambda_j(u)$. Therefore, for $s \geq 0$,

$$c + su \succeq 0 \quad \Longleftrightarrow \quad 0 \leq s \leq R(u), \tag{635}$$

and $c + R(u)u$ is singular.

Every $H \in \mathcal{K}_3 \setminus \{c\}$ has the unique expression

$$H = c + su, \qquad s = \|H - c\|_F, \qquad u = \frac{H - c}{\|H - c\|_F}.$$

Define

$$\chi(c) = 0, \qquad \chi(c + su) = \frac{s}{R(u)}u, \tag{636}$$

and

$$\chi^{-1}(0) = c, \qquad \chi^{-1}(ru) = c + rR(u)u \quad (0 < r \leq 1). \tag{637}$$

The minimum-eigenvalue function is continuous. Since it is strictly negative on the compact sphere $S(E)$, R is continuous, finite, positive, and bounded above and below. Equations (636) and (637) are therefore continuous also at the origin and are inverse homeomorphisms. Equation (635) shows that

$$\chi\big(\{H \in \mathcal{K}_3 : \operatorname{rank} H \leq 2\}\big) = S(E).$$

□

Corollary 18.3 (Unit-sphere U(3) quotient)**.** *There is a homeomorphism of pairs*

$$\Big(S_F^{17}/\mathrm{U}(3), \{[A] : \operatorname{rank} A \leq 2\}\Big) \cong (B^8, S^7). \tag{638}$$

Proof. On S_F^{17},

$$\operatorname{tr}(A^\dagger A) = \|A\|_F^2 = 1.$$

Apply Theorem 18.1 and Lemma 18.2. □

Theorem 18.4 (Full U(3) cone and compactification)**.** *There are homeomorphisms*

$$\mathscr{V}_3/\mathrm{U}(3) \cong \mathcal{P}_3 \cong \mathbb{R}^8 \times \mathbb{R}_{\geq 0} \tag{639}$$

and

$$\mathscr{V}_3^+/\mathrm{U}(3) \cong \mathcal{P}_3^+ \cong B^9. \tag{640}$$

Under the second homeomorphism,

$$\partial B^9 = \{[A] : \operatorname{rank} A \leq 2\} \cup \{[\infty]\} \cong S^8. \tag{641}$$

Proof. For $H \neq 0$, write

$$H = r\rho, \qquad r = \operatorname{tr} H > 0, \qquad \rho = \frac{H}{\operatorname{tr} H} \in \mathcal{K}_3.$$

Thus $\mathcal{P}_3$ is the infinite cone on $\mathcal{K}_3 \cong B^8$.

We give the compactification map explicitly. Put

$$t = \frac{r - 1}{r + 1} \in (-1, 1)$$

and define

$$\mathcal{D}^9 = \{(y,t) \in E \oplus \mathbb{R} : \|y\|_F + |t| \le 1\}. \tag{642}$$

Define

$$\begin{aligned} \Xi_U(r\rho) &= \big((1-|t|)\chi(\rho), t\big), \\ \Xi_U(0) &= (0,-1), \\ \Xi_U(\infty) &= (0,1). \end{aligned} \tag{643}$$

For $|t| < 1$, the inverse is

$$x = \frac{y}{1-|t|}, \qquad \rho = \chi^{-1}(x), \qquad r = \frac{1+t}{1-t}, \qquad H = r\rho. \tag{644}$$

As $r \downarrow 0$ or $r \to \infty$, the factor $1-|t|$ tends uniformly to zero. A subset of $\mathcal{P}_3$ is unbounded if and only if its trace is unbounded. Thus Ξ_U is a homeomorphism $\mathcal{P}_3^+ \to \mathcal{D}^9$.

For $p = (y,t) \neq 0$, put

$$\nu(p) = \sqrt{\|y\|_F^2 + t^2}, \qquad \gamma(p) = \|y\|_F + |t|$$

and define

$$\Theta(p) = \frac{\gamma(p)}{\nu(p)} p, \qquad \Theta(0) = 0. \tag{645}$$

Then $\|\Theta(p)\|_2 = \gamma(p) \le 1$. For nonzero $q \in B(E \oplus \mathbb{R})$, the inverse is

$$\Theta^{-1}(q) = \frac{\|q\|_2}{\|q_E\|_F + |q_{\mathbb{R}}|}\, q. \tag{646}$$

Hence $\Theta : \mathcal{D}^9 \to B^9$ is a homeomorphism and takes $\partial\mathcal{D}^9$ onto S^8.

For finite $H = r\rho$, Equation (643) lies on $\partial\mathcal{D}^9$ exactly when $\|\chi(\rho)\|_F = 1$, equivalently when H is singular. The points 0 and ∞ are the two boundary tips. This proves (641).

The canonical map

$$\mathscr{V}_3^+/\mathrm{U}(3) \longrightarrow (\mathscr{V}_3/\mathrm{U}(3))^+$$

is a continuous bijection: the finite orbits agree and the added orbit is $\{\infty\}$. Its source is compact and its target is Hausdorff, hence it is a homeomorphism. This proves (640).

Finally, $\Theta \circ \Xi_U$ sends ∞ to $N = (0,\dots,0,1) \in \partial B^9$. The Cayley map

$$(x', x_9) \longmapsto \left(\frac{2x'}{\|x'\|^2 + (1-x_9)^2}, \frac{1-\|x\|^2}{\|x'\|^2 + (1-x_9)^2} \right) \tag{647}$$

is a homeomorphism

$$B^9 \setminus \{N\} \longrightarrow \mathbb{R}^8 \times \mathbb{R}_{\ge 0}.$$

Its inverse is

$$(v,s) \longmapsto \left(\frac{2v}{\|v\|^2 + (s+1)^2}, \frac{\|v\|^2 + s^2 - 1}{\|v\|^2 + (s+1)^2} \right). \tag{648}$$

This proves (639). □

18.3 The $\mathrm{SU}(3)$ orbit space

Theorem 18.5 (Gram matrix plus alternating determinant)**.** *Define*

$$\Gamma_{SU} : \mathscr{V}_3 \longrightarrow \mathrm{Herm}_3(\mathbb{C}) \times \mathbb{C}, \qquad \Gamma_{SU}(A) = (A^\dagger A, \det A). \tag{649}$$

Its image is exactly

$$\mathcal{Y}_3 = \{(H, z) \in \mathcal{P}_3 \times \mathbb{C} : |z|^2 = \det H\}. \tag{650}$$

It separates $\mathrm{SU}(3)$*-orbits and induces a homeomorphism*

$$\mathscr{V}_3/\mathrm{SU}(3) \xrightarrow{\sim} \mathcal{Y}_3. \tag{651}$$

Proof. The map is $\mathrm{SU}(3)$-invariant, and

$$\det(A^\dagger A) = \overline{\det A} \det A = |\det A|^2.$$

Thus its image is contained in $\mathcal{Y}_3$.

Conversely, take $(H, z) \in \mathcal{Y}_3$. If H is singular, then $z = 0$, and $A = H^{1/2}$ realizes the pair. If $H \succ 0$, put

$$\omega = \frac{z}{\sqrt{\det H}} \in S^1, \qquad Q_\omega = \mathrm{diag}(\omega, 1, 1) \in \mathrm{U}(3), \qquad A = Q_\omega H^{1/2}.$$

Then $A^\dagger A = H$ and $\det A = z$. This proves surjectivity.

Suppose

$$A^\dagger A = B^\dagger B, \qquad \det A = \det B.$$

Theorem 18.1 gives $U_0 \in \mathrm{U}(3)$ with $U_0 A = B$. If A is invertible, then

$$\det U_0 = \frac{\det B}{\det A} = 1,$$

so $U_0 \in \mathrm{SU}(3)$.

If $\operatorname{rank} A \le 2$, choose a unitary V which is the identity on $\operatorname{im} B$ and has determinant $(\det U_0)^{-1}$ on $(\operatorname{im} B)^\perp$. This is possible because the orthogonal complement has positive complex dimension. Then

$$VB = B, \qquad \det(VU_0) = 1, \qquad (VU_0)A = B.$$

Thus the invariant separates $\mathrm{SU}(3)$-orbits.

Since $\operatorname{tr} H = \|A\|_F^2$, the invariant map is proper. The induced continuous bijection onto the closed Hausdorff set $\mathcal{Y}_3$ is therefore a homeomorphism. □

Theorem 18.6 (The unit-sphere $\mathrm{SU}(3)$ quotient)**.** *Let*

$$\mathcal{Y}_{3,1} = \{(H, z) \in \mathcal{Y}_3 : \operatorname{tr} H = 1\}. \tag{652}$$

There is an explicit homeomorphism

$$S_F^{17}/\mathrm{SU}(3) \cong \mathcal{Y}_{3,1} \xrightarrow{\sim} S(E \oplus \mathbb{C}) \cong S^9. \tag{653}$$

Proof. The first homeomorphism follows from Theorem 18.5. To construct the second, let

$$x = \chi(H) \in B(E), \qquad r = \|x\|_F.$$

If $r < 1$, then $H \succ 0$ and $z \neq 0$. Define

$$\Omega(H, z) = \left(x, \sqrt{1 - r^2}\,\frac{z}{|z|}\right) \quad (r < 1), \qquad \Omega(H, 0) = (x, 0) \quad (r = 1). \tag{654}$$

Its norm is exactly one:

$$\|x\|_F^2 + \left|\sqrt{1-r^2}\,\frac{z}{|z|}\right|^2 = r^2 + (1-r^2) = 1.$$

The map is continuous on the full-rank locus. As H approaches the singular locus, $r \to 1$, so its complex coordinate tends to zero regardless of phase. Hence it is continuous everywhere.

Given $(x, w) \in S(E \oplus \mathbb{C})$, set $H = \chi^{-1}(x)$. If $\|x\|_F < 1$, then $w \neq 0$, and the unique inverse value is

$$z = \sqrt{\det H}\,\frac{w}{|w|}. \tag{655}$$

If $\|x\|_F = 1$, then $w = 0$, H is singular, and $z = 0$. Thus Ω is a continuous bijection from a compact space to a Hausdorff space and hence is a homeomorphism. □

Theorem 18.7 (The weighted cone and its compactification)**.** *There is a homeomorphism*

$$\Phi_{SU} : \mathcal{Y}_3 \xrightarrow{\sim} E \oplus \mathbb{C} \cong \mathbb{R}^{10} \tag{656}$$

and therefore

$$\mathscr{V}_3^+/\mathrm{SU}(3) \cong S^{10}. \tag{657}$$

Proof. For $(H, z) \neq (0, 0)$, set

$$s = \sqrt{\operatorname{tr} H}, \qquad h = \frac{H}{s^2}, \qquad \zeta = \frac{z}{s^3}. \tag{658}$$

The exponents are forced by scalar multiplication in the original source: for every real $\lambda > 0$,

$$(\lambda A)^\dagger(\lambda A) = \lambda^2 A^\dagger A, \qquad \det(\lambda A) = \lambda^3 \det A, \qquad s(\lambda A) = \lambda s(A). \tag{659}$$

Moreover,

$$\operatorname{tr} h = 1, \qquad |\zeta|^2 = \frac{|z|^2}{s^6} = \frac{\det H}{s^6} = \det h.$$

Thus $(h, \zeta) \in \mathcal{Y}_{3,1}$. Define

$$\Phi_{SU}(0, 0) = 0, \qquad \Phi_{SU}(H, z) = s\,\Omega(h, \zeta) \quad (s > 0). \tag{660}$$

Since $\Omega(h, \zeta)$ has norm one,

$$\|\Phi_{SU}(H, z)\| = s.$$

For nonzero $v \in E \oplus \mathbb{C}$, the inverse is obtained from

$$s = \|v\|, \qquad (h, \zeta) = \Omega^{-1}\left(\frac{v}{\|v\|}\right), \qquad H = s^2 h, \qquad z = s^3\zeta.$$

The norm identity proves continuity at the origin in both directions, so Φ_{SU} is a homeomorphism.

As in the proof of Theorem 18.4, compactness of SU(3) gives

$$\mathscr{V}_3^+/\mathrm{SU}(3) \cong (\mathscr{V}_3/\mathrm{SU}(3))^+.$$

The one-point compactification of $\mathbb{R}^{10}$ is S^{10}. □

18.4 Rank strata, the standard S^8, and determinant winding

Proposition 18.8 (Stabilizers and dimensions)**.** *Let*

$$\mathscr{V}_{3,(r)} = \{A \in \mathscr{V}_3 : \operatorname{rank}_{\mathbb{C}} A = r\}.$$

For $A \in \mathscr{V}_{3,(r)}$,

$$\operatorname{Stab}_{\mathrm{U}(3)}(A) \cong \mathrm{U}(3-r), \qquad \operatorname{Stab}_{\mathrm{SU}(3)}(A) \cong \mathrm{SU}(3-r), \tag{661}$$

with the rank-three and SU(1) *stabilizers understood to be trivial. The quotient-stratum dimensions are*

r	$\dim_{\mathbb{R}} \mathscr{V}_{3,(r)}$	$\dim U(3)\cdot A$	$\dim \mathscr{V}_{3,(r)}/U(3)$	$\dim SU(3)\cdot A$	$\dim \mathscr{V}_{3,(r)}/SU(3)$
3	*18*	*9*	*9*	*8*	*10*
2	*16*	*8*	*8*	*8*	*8*
1	*10*	*5*	*5*	*5*	*5*
0	*0*	*0*	*0*	*0*	*0*

On the unit sphere, the nonzero quotient-stratum dimensions are

r	3	2	1
$U(3)$	8	7	4
$SU(3)$	9	7	4.

The trace-one rank-one locus is $\mathbb{C}P^2$.

Proof. A left stabilizer fixes $\operatorname{im} A$ pointwise. Its unrestricted unitary action on $(\operatorname{im} A)^{\perp}$ gives $\mathrm{U}(3-r)$. Imposing determinant one gives $\mathrm{SU}(3-r)$. The real dimension of the complex rank-r matrix stratum is

$$2r(6-r).$$

Subtracting the orbit dimensions gives the table; intersecting a nonzero rank stratum with the unit sphere subtracts one more dimension.

A trace-one positive-semidefinite matrix has rank one exactly when it has the form $vv^{\dagger}$ with $\|v\| = 1$. The phase of v does not change $vv^{\dagger}$, so this locus is

$$S^5/S^1 \cong \mathbb{C}P^2.$$

□

Theorem 18.9 (The exact rank-loss pair and complement)**.** *Let*

$$\mathcal{Y}_{3,\mathrm{sing}} = \{(H,0) \in \mathcal{Y}_3 : \operatorname{rank} H \le 2\}. \tag{662}$$

The homeomorphism Φ_{SU} *gives homeomorphisms of pairs*

$$(\mathcal{Y}_3, \mathcal{Y}_{3,\mathrm{sing}}) \cong (E \oplus \mathbb{C}, E \oplus \{0\}) \tag{663}$$

and

$$(\mathcal{Y}_3^+, \mathcal{Y}_{3,\mathrm{sing}}^+) \cong (S^{10}, S^8_{\mathrm{lin}}). \tag{664}$$

Consequently the compactified rank-loss locus is a standard, topologically locally flat codimension-two $S^8 \subset S^{10}$*, and*

$$S^{10} \setminus S^8 \cong E \times (\mathbb{C} \setminus \{0\}) \cong \mathbb{R}^8 \times (\mathbb{C} \setminus \{0\}) \simeq S^1. \tag{665}$$

The map

$$\vartheta(H,z) = \frac{z}{|z|} = \frac{\det A}{|\det A|} \tag{666}$$

is the angular deformation-retraction coordinate and represents a generator of

$$H^1(S^{10} \setminus S^8; \mathbb{Z}) \cong \pi_1(S^{10} \setminus S^8) \cong \mathbb{Z}$$

after choosing the standard positive orientation on S^1.

Proof. In Equation (654), H is singular exactly when $\|\chi(H)\|_F = 1$, in which case the complex coordinate is zero. Conversely, a zero complex coordinate on the unit S^9 has its E-coordinate on $S(E)$, and its inverse Gram matrix is singular. Radial extension through Equation (660) therefore gives exactly (663).

The homeomorphism Φ_{SU} is proper and extends to one-point compactifications, proving (664). The second member is the one-point compactification of the linear pair

$$(E \oplus \mathbb{C}, E \oplus \{0\}).$$

It is the standard topological pair $(S^{10}, S^8_{\text{lin}})$; therefore the S^8 is topologically locally flat at every point.

This local-flatness statement is not a smooth-stratification assertion. The unit rank-deficient S^7 contains the rank-one orbit stratum $\mathbb{C}P^2$, with rank-two complement $S^7 \setminus \mathbb{C}P^2$. The homeomorphism χ uses the minimum eigenvalue and need not be smooth across eigenvalue-collision strata.

Removing the linear subspace from the affine pair gives (665). Under Φ_{SU}, the phase of its complex coordinate is $z/|z|$, so ϑ is the usual angular projection

$$E \times (\mathbb{C} \setminus \{0\}) \longrightarrow S^1.$$

This is a deformation retraction. The explicit quotient loop

$$A_\theta = \frac{1}{\sqrt{3}} \operatorname{diag}(e^{i\theta}, 1, 1), \qquad 0 \le \theta \le 2\pi, \tag{667}$$

satisfies

$$A_\theta^\dagger A_\theta = \frac{1}{3} I_3, \qquad \det A_\theta = \frac{e^{i\theta}}{3\sqrt{3}}.$$

Thus $\vartheta([A_\theta]) = e^{i\theta}$ winds once and is a generator. □

Corollary 18.10 (Residual determinant-phase circle)**.** *The residual group*

$$\mathrm{U}(3)/\mathrm{SU}(3) \cong S^1$$

acts on $E \oplus \mathbb{C}$ *by*

$$e^{i\theta} \cdot (x, w) = (x, e^{i\theta} w). \tag{668}$$

Its fixed set after compactification is the standard $S^8 \subset S^{10}$*, and*

$$S^{10}/S^1 \cong B^9, \qquad \operatorname{Fix}(S^1) = S^8 = \partial B^9. \tag{669}$$

Proof. If $U \in \mathrm{U}(3)$, left multiplication fixes $H = A^\dagger A$ and sends $z = \det A$ to $(\det U)z$. Every phase occurs as a determinant. Equation (654) therefore converts the residual action into (668). On the unit sphere,

$$S(E \oplus \mathbb{C})/S^1 \cong \{(x, |w|) : \|x\|_F^2 + |w|^2 = 1,\ |w| \ge 0\} \cong B(E).$$

On the affine cone,

$$(E \oplus \mathbb{C})/S^1 \cong E \times \mathbb{R}_{\ge 0}.$$

One-point compactification gives B^9, and the fixed set is $(E \oplus \{0\})^+ \cong S^8$, which maps to the boundary. □

18.5 The exact rank-two/rank-three distinction

Theorem 18.11 (Quadratic Lorentz cone versus cubic rank-three cone)**.** *For* $\mathscr{V}_2 = \mathrm{Mat}_{3\times 2}(\mathbb{C})$*, the left* SU(3)*- and* U(3)*-orbit relations coincide, and*

$$\mathscr{V}_2/\mathrm{SU}(3) \cong \mathrm{Herm}_2(\mathbb{C})^{\succeq 0}. \tag{670}$$

In the coordinates

$$H_2(t, x_1, x_2, x_3) = \begin{pmatrix} t + x_3 & x_1 - ix_2 \\ x_1 + ix_2 & t - x_3 \end{pmatrix}, \tag{671}$$

$$\operatorname{tr} H_2 = 2t, \qquad \det H_2 = t^2 - x_1^2 - x_2^2 - x_3^2, \tag{672}$$

and

$$H_2 \succeq 0 \quad \Longleftrightarrow \quad t \geq 0, \qquad t^2 \geq x_1^2 + x_2^2 + x_3^2. \tag{673}$$

Thus the two-column quotient is exactly the closed future $3+1$*-dimensional Lorentz cone; its one-point compactification is* B^4*, with rank-one-plus-infinity boundary* S^3*.*

By contrast, the three-column U(3) *quotient is the nine-real-dimensional, Jordan-rank-three cone* $\mathrm{Herm}_3(\mathbb{C})^{\succeq 0}$*. Its determinant is cubic, its trace-one section is* B^8*, and its compactified rank-deficient-plus-infinity boundary is* S^8*. It is not a Lorentz cone.*

Proof. Every 3×2 matrix has rank at most two, so the determinant-adjustment step in the proof of Theorem 18.5 converts every left U(3) equivalence into a left SU(3) equivalence. The Gram proof of Theorem 18.1, with two columns, gives (670).

Direct expansion of (671) gives (672). A 2×2 Hermitian matrix is positive semidefinite exactly when its trace and determinant are nonnegative, which gives (673). Its trace-one section is

$$t = \frac{1}{2}, \qquad x_1^2 + x_2^2 + x_3^2 \leq \frac{1}{4},$$

a closed 3-ball. The compactification argument of Theorem 18.4, with E of dimension three, gives B^4 with boundary S^3.

For three columns, Theorem 18.4 gives the B^9/S^8 statements. The determinant of a 3×3 Hermitian matrix is homogeneous of degree three, so this rank-three cone is not the rank-two quadratic Lorentz cone. □

18.6 The alternating invariant and the precise nonclaim

The extra invariant retained by the square-matrix SU(3) quotient is

$$z = \det A = \varepsilon_{ijk} A_{i1} A_{j2} A_{k3}. \tag{674}$$

It is exactly the complex alternating volume of the three columns:

$$\det(UA) = \det(U) \det(A).$$

It is invariant for $U \in \mathrm{SU}(3)$, its phase is rotated by the residual $\mathrm{U}(3)/\mathrm{SU}(3)$, and it vanishes exactly when the three columns are linearly dependent.

This invariant theory supplies exact carrier and orbit data for a physical translation, but does not by itself complete that translation. The next construction must specify a quark field, colour-charge and baryon observables, a principal SU(3) bundle with connection and characteristic classes, and an interacting Yang–Mills sector, together with maps relating each object to Equation (674). Likewise, Theorem 18.4 supplies a distinguished S^8, while an almost-complex or integrable-complex structure, a three-fibre structure, and

analogues of the candidate S^6 special-fibre attaching maps remain exact construction problems rather than consequences already proved by the compactification theorem.

The exact conclusion is the topological carrier theorem

$$\boxed{\mathscr{V}_3^+/\mathrm{U}(3) \cong B^9, \quad \partial B^9 \cong S^8, \quad \mathscr{V}_3^+/\mathrm{SU}(3) \cong S^{10}, \quad S^{10} \setminus S^8 \simeq S^1.} \tag{675}$$

Any gauge-theoretic or geometric use of this carrier requires additional explicit descent data.

19 A four-branch toroidal lift and the obstruction to a complex S^8 total space

19.1 The exact meaning of the three-way cusp

The cusp in the audited threefold is controlled by the $\mathbb{Z}^2$-periodic A_2-triangulation of $\mathbb{R}^2$. Its quotient has one vertex, three edges, and two triangles. The central fibre is globally irreducible; its normalization is the degree-six del Pezzo surface, whose six toric boundary curves are paired into three double curves. At either of the two deepest points the smooth total threefold maps to the disc by

$$(z_0, z_1, z_2) \longmapsto t = z_0 z_1 z_2. \tag{676}$$

Thus the three relevant statements are distinct: there are three local branches at a deepest point, three global double curves, and only one global irreducible component. The generalized dual intersection complex is

$$\mathcal{T}_{A_2}/\mathbb{Z}^2 \cong \mathbb{R}^2/\mathbb{Z}^2 = T^2. \tag{677}$$

These baseline data are transcribed from the audited canonical-source snapshot of 620,782 bytes and 6,315 lines, with SHA–256 digest

`0696B6B7FAB83804DFEC947D45607601E5D7F57D64C6148487969FCCB6C67D2D.`

line 320 records the periodic A_2 triangulation and its $(1, 3, 2)$ quotient cell counts; lines 372 and 377 record the normalization, three double curves, and two triple points; lines 1431–1461 construct the fan and its unimodular toric charts; lines 1647–1678 give the quotient local models and orbit strata; and line 3966 gives the deepest central-fibre equation $\{z_0 z_1 z_2 = 0\}$. The line references and digest fix the exact source edition without introducing a machine-local dependency.

The higher-dimensional construction below raises the rank of this periodic triangulation from two to three. It does not replace three line bundles by four line bundles.

19.2 The rank-three periodic fan

Let

$$L = \mathbb{Z}^3, \qquad L_{\mathbb{R}} = \mathbb{R}^3, \qquad \mathbf{1} = e_1 + e_2 + e_3.$$

For $m \in L$ and $\pi = (\pi_1, \pi_2, \pi_3) \in S_3$, put

$$\sigma_{m,\pi} = \mathrm{conv}\left\{m,\ m + e_{\pi_1},\ m + e_{\pi_1} + e_{\pi_2},\ m + \mathbf{1}\right\}. \tag{678}$$

Inside the cube $m + [0,1]^3$, this is the region in which the coordinates, in the order π_1, π_2, π_3, are weakly decreasing. The six possible orders partition the cube, and the induced partitions agree on common faces. Their integral translates therefore give a locally finite L-periodic triangulation $\mathcal{T}_3$ of $\mathbb{R}^3$.

Put

$$N' = L \oplus \mathbb{Z}e_4$$

and, for every cell P of $\mathcal{T}_3$, define

$$\widehat{P} = \mathrm{cone}\{(v,1) : v \in P\} \subset N'_{\mathbb{R}}. \tag{679}$$

Because all cells are bounded, $\widehat{P} \cap \widehat{Q} = \widehat{P \cap Q}$. Hence the cones $\widehat{P}$, together with the zero cone, form a fan Σ_3.

Lemma 19.1 (Unimodularity)**.** *Every maximal cone of Σ_3 is unimodular.*

Proof. For the four lifted vertices of $\sigma_{m,\pi}$, subtract the first column $(m,1)$ from the other three columns. Up to the orientation sign, the determinant is

$$\det\left(e_{\pi_1},\ e_{\pi_1} + e_{\pi_2},\ \mathbf{1}\right) = \pm 1.$$

The four primitive ray generators therefore form a basis of N'. Every face is unimodular as well. □

Let $Y_3 = X_{\Sigma_3}$ be the associated, generally non-quasi-compact, toric fourfold. It is smooth by theorem 19.1. Let

$$M' = \mathrm{Hom}(N', \mathbb{Z}), \qquad h = (0,0,0,1) \in M',$$

and define the toric character

$$t = \chi^h : Y_3 \longrightarrow \mathbb{C}. \tag{680}$$

The character h has value one on every primitive ray generator $(v,1)$. Consequently

$$\mathrm{div}(t) = \sum_{v \in L} E_v \tag{681}$$

with multiplicity one on every toric divisor E_v.

Proposition 19.2 (Four-branch local equation)**.** *If a tetrahedron σ has vertices v_0, v_1, v_2, v_3, its toric chart is $\mathbb{C}^4$, and in its dual-character coordinates (z_0, z_1, z_2, z_3) one has*

$$t = z_0 z_1 z_2 z_3. \tag{682}$$

In particular, four distinct components of the divisor upstairs meet transversely at the torus fixed point of $\widehat{\sigma}$.

Proof. Let m_0, m_1, m_2, m_3 be the basis of M' dual to $(v_0,1), \ldots, (v_3,1)$, and set $z_i = \chi^{m_i}$. Unimodularity identifies the affine chart with $\mathbb{C}^4$. Since h takes value one on every ray generator,

$$h = m_0 + m_1 + m_2 + m_3.$$

Taking characters gives (682). □

19.3 The analytic quotient and the abelian-threefold fibres

For $\lambda \in L$, define the integral shear

$$\varphi_\lambda : N' \longrightarrow N', \qquad \varphi_\lambda(y,r) = (y + r\lambda, r). \tag{683}$$

It sends $\widehat{P}$ to $\widehat{P + \lambda}$, so it induces a toric automorphism Φ_λ of Y_3. These automorphisms preserve t and satisfy $\Phi_\lambda \Phi_\mu = \Phi_{\lambda+\mu}$. On the dense torus and over $t \neq 0$, their formula is

$$\Phi_\lambda(x_1, x_2, x_3, t) = (t^{\lambda_1} x_1, t^{\lambda_2} x_2, t^{\lambda_3} x_3, t). \tag{684}$$

The lattice map (683), rather than the expression with possibly negative powers in (684), proves that the action extends over the toric boundary.

Fix $0 < \epsilon < 1$ and set

$$Y_{3,\epsilon} = \{p \in Y_3 : |t(p)| < \epsilon\}.$$

Define

$$N_3 = Y_{3,\epsilon}/L, \qquad \pi_3 : N_3 \longrightarrow \Delta_\epsilon. \tag{685}$$

Theorem 19.3 (The rank-three toroidal cusp)**.** *The L-action on $Y_{3,\epsilon}$ is free and properly discontinuous. The quotient N_3 is a smooth complex fourfold and π_3 is proper. For $q \neq 0$, its fibre is the abelian threefold*

$$N_{3,q} \cong \mathbb{C}^3/(\mathbb{Z}^3 + \tau\mathbb{Z}^3) \cong E_\tau^3, \qquad q = e^{2\pi i\tau}, \quad \tau \in \mathbb{H}.$$

Proof. A boundary point belongs to a unique torus orbit $O_{\widehat{P}}$. If Φ_λ fixes it, then $O_{\widehat{P+\lambda}} = O_{\widehat{P}}$, whence $P + \lambda = P$. Since P is bounded, $\lambda = 0$. On a fibre $t = q \neq 0$, (684) is free because $|q| < 1$ and the equalities $q^{\lambda_i} = 1$ for all i force $\lambda = 0$.

For proper discontinuity and properness, use on the dense fibre the logarithmic coordinate

$$\operatorname{trop}_q(x) = \left(\frac{\log|x_1|}{\log|q|}, \frac{\log|x_2|}{\log|q|}, \frac{\log|x_3|}{\log|q|}\right). \tag{686}$$

Equation (684) translates this coordinate by λ. Every orbit can therefore be moved into $[0,1]^3$. If

$$r = \sum_{i=0}^{3} \alpha_i v_i$$

belongs to $\sigma = \operatorname{conv}(v_0, v_1, v_2, v_3)$, then $\alpha_i \geq 0$, $\sum_i \alpha_i = 1$, and the dual-character coordinates of theorem 19.2 obey

$$\frac{\log|z_i|}{\log|q|} = \alpha_i, \qquad |z_i| = |q|^{\alpha_i} \leq 1.$$

For $\epsilon' < \epsilon$, let $F_{\epsilon'}$ be the closure of the points with $0 < |q| \leq \epsilon'$ and $\operatorname{trop}_q(x) \in [0,1]^3$. For each of the six tetrahedra in that cube, the preceding identities put these points in the closed unit polydisc $\{|z_0|, \dots, |z_3| \leq 1\}$ of its $\mathbb{C}^4$-chart. The finite union of those polydiscs is compact and closed in the Hausdorff toric variety; $F_{\epsilon'}$, being closed in that union, is compact. Its translates cover the dense locus, and the same holds on $t = 0$ because every cell has an L-translate meeting the fundamental cube.

It remains to verify proper discontinuity at the boundary, where trop_q itself is undefined. Let $K \subset Y_{3,\epsilon}$ be compact. Local compactness and the toric affine cover give finitely many maximal-cone charts U_{σ_j} and compact subsets $K_j \subset U_{\sigma_j}$ whose interiors cover K. On the dense part of one such compact set, write

$$r = \sum_{i=0}^{3} \alpha_i v_i, \qquad \alpha_i = \frac{\log|z_i|}{\log|t|}, \qquad \sum_i \alpha_i = 1.$$

The coordinate moduli are uniformly bounded above. When $|t|$ is small, this gives a uniform lower bound for every α_i, and the identity $\sum_i \alpha_i = 1$ gives a uniform upper bound. When $|t|$ is bounded away from zero, the product $t = z_0 z_1 z_2 z_3$ and the upper bounds give uniform positive lower bounds for the $|z_i|$, hence again uniform bounds for every α_i. Since only finitely many vertex sets occur, all dense points of K consequently have tropical coordinates in one bounded set $C_K \subset \mathbb{R}^3$.

If $p, \Phi_\lambda(p) \in K$ are dense, then $\lambda \in (C_K - C_K) \cap L$, a finite set. If p lies on the boundary, its orbit cell and that of $\Phi_\lambda(p)$ lie among the finitely many faces of the σ_j, and they satisfy $P + \lambda = Q$; again only finitely many λ occur. Thus a compact set meets only finitely many of its translates, proving proper discontinuity. Finally, the image of $F_{\epsilon'}$ is the entire inverse image of $\{|q| \leq \epsilon'\}$, which proves that π_3 is proper. Freeness on the smooth $Y_{3,\epsilon}$ makes N_3 smooth.

For $q \neq 0$, all non-dense toric orbits lie in $t = 0$; hence $Y_{3,q} = (\mathbb{C}^*)^3$, and

$$N_{3,q} = (\mathbb{C}^*)^3/q^{\mathbb{Z}^3}.$$

The map

$$[w_1, w_2, w_3] \longmapsto [e^{2\pi i w_1}, e^{2\pi i w_2}, e^{2\pi i w_3}]$$

induces the asserted isomorphism from $\mathbb{C}^3/(\mathbb{Z}^3 + \tau\mathbb{Z}^3)$. □

The preceding construction is the untwisted $B_0 = I$, $C(t) = 0$ instance of the Tate–Mumford construction. The general analytic construction is due to Mumford, *An analytic construction of degenerating abelian varieties over complete rings*, Compositio Math. **24** (1972), 239–272. No appeal to that generality is needed for the displayed quotient.

19.4 Monodromy and the cusp inclusion on fundamental groups

Use the integral basis

$$(a_1, a_2, a_3, b_1, b_2, b_3)$$

of $H_1(N_{3,q}; \mathbb{Z})$, where a_i is the unit period and b_i is the τ-period in the i-th factor. A positive loop about $q = 0$ sends τ to $\tau + 1$.

Proposition 19.4 (Cusp monodromy and filling map)**.** *The monodromy is*

$$M_0 = \begin{pmatrix} I_3 & I_3 \\ 0 & I_3 \end{pmatrix}, \qquad R = M_0 - I = \begin{pmatrix} 0 & I_3 \\ 0 & 0 \end{pmatrix}. \tag{687}$$

It satisfies

$$R^2 = 0, \qquad \operatorname{rank} R = 3, \qquad \ker R = \operatorname{im} R = \mathbb{Z}\langle a_1, a_2, a_3 \rangle.$$

Moreover

$$\pi_1(N_3) \cong L \cong \mathbb{Z}^3,$$

and the inclusion of a smooth fibre induces

$$a_i \longmapsto 0, \qquad b_i \longmapsto e_i \in L. \tag{688}$$

Proof. The period change $b_i \mapsto b_i + a_i$, $a_i \mapsto a_i$, gives (687); the assertions about R follow by direct block multiplication.

The primitive rays

$$(0, 1), \quad (e_1, 1), \quad (e_2, 1), \quad (e_3, 1)$$

generate N'. We record the toric fundamental-group calculation needed here. If Σ_0 is a finite subfan of this smooth fan and N_{Σ_0} is the sublattice generated by its primitive ray vectors, then

$$\pi_1(X_{\Sigma_0}) \cong N'/N_{\Sigma_0}.$$

Indeed, the dense torus has fundamental group N'. Adjoining a neighbourhood of the generic point of the divisor belonging to a primitive ray v fills precisely the circle v, so van Kampen quotients by $\langle v \rangle$. The remaining orbit strata have complex codimension at least two; general position for loops and disc homotopies in the smooth real manifold shows that adding them changes no fundamental group. Applying van Kampen over all rays gives the displayed quotient.

Every loop in Y_3 has compact image and is therefore contained in the union of finitely many toric affine charts. Enlarge their finite subfan by the four rays displayed above. Those rays generate N', so the finite subfan formula makes this enlarged toric open set simply connected. The loop bounds there, and hence $\pi_1(Y_3) = 0$. For $0 < u \leq 1$, the real one-parameter torus element $\vartheta_u = (1, 1, 1, u)$ satisfies

$$t(\vartheta_u p) = u\, t(p).$$

Any loop in $Y_{3,\epsilon}$ bounds a disc in Y_3. Choosing u small moves that entire disc into $Y_{3,\epsilon}$, while $\vartheta_{u'}$, $u \leq u' \leq 1$, gives a homotopy back to the original loop. Hence $\pi_1(Y_{3,\epsilon}) = 0$. The quotient map in (685) is therefore the universal cover with deck group L, so $\pi_1(N_3) = L$.

The loop a_i is the angular loop in the i-th $\mathbb{C}^*$-coordinate and is killed by the corresponding toric divisor. The path representing b_i closes precisely by the deck transformation Φ_{e_i}. This proves (688). □

19.5 The central fibre, its normalization, and all strata

Let

$$W_3 = \pi_3^{-1}(0) = \left(\bigcup_{v \in L} E_v \right) / L. \tag{689}$$

The toric orbit–cone correspondence identifies its open strata with L-orbits of cells of $\mathcal{T}_3$. Its generalized dual intersection complex is therefore

$$\mathcal{T}_3 / L \cong \mathbb{R}^3 / L = T^3. \tag{690}$$

Theorem 19.5 (Forced stratum counts)**.** *The quotient cell structure and the open torus stratification of W_3 are*

cell	*number*	*open stratum*	*local branches*
vertex	1	$(\mathbb{C}^*)^3$	1
edge	7	$(\mathbb{C}^*)^2$	2
triangle	12	$\mathbb{C}^*$	3
tetrahedron	6	pt	4.

In particular,

$$\chi(W_3) = 6. \tag{691}$$

Proof. There is one vertex orbit. A fundamental parallelepiped of L has normalized three-volume $3! = 6$, whereas every unimodular tetrahedron has normalized volume one. Hence there are six tetrahedron orbits. Every triangular face is incident to two tetrahedra, so

$$f_2 = \frac{4 f_3}{2} = 12.$$

Since the quotient is T^3,

$$0 = \chi(T^3) = f_0 - f_1 + f_2 - f_3 = 1 - f_1 + 12 - 6,$$

and therefore $f_1 = 7$. Explicit edge representatives are

$$[0, e_1],\ [0, e_2],\ [0, e_3],\ [0, e_1 + e_2],\ [0, e_1 + e_3],\ [0, e_2 + e_3],\ [0, \mathbf{1}].$$

A k-cell gives a torus orbit of complex dimension $3 - k$ and an intersection of $k + 1$ local branches. Finally every positive-dimensional $(\mathbb{C}^*)^j$ has Euler characteristic zero. Additivity over the finite stratification leaves the six point strata and proves (691). □

Theorem 19.6 (Normalization)**.** *W_3 is globally irreducible. The quotient map induces its normalization*

$$\nu : E_0 \longrightarrow W_3.$$

The variety E_0 is the smooth three-dimensional permutohedral toric variety. It has fourteen toric boundary surfaces, thirty-six invariant curves, and twenty-four fixed points. Under ν, these map respectively two-to-one, three-to-one, and four-to-one onto seven double surfaces, twelve triple curves, and six quadruple points.

Proof. The lattice L acts simply transitively on the vertices and therefore transitively on the components E_v. The quotient W_3 is irreducible. Every cell has a vertex and can be translated to one containing zero, so the map $E_0 \to W_3$ is surjective. Only finitely many cells contain zero, hence the map is finite, and it is injective on the dense torus. The star fan at zero is complete and unimodular, so E_0 is smooth, compact, and normal. The finite birational map is consequently the normalization.

The rays of the star fan are

$$\pm\sum_{i\in I} e_i, \qquad \varnothing \neq I \subseteq \{1,2,3\}.$$

These are the fourteen rays of the A_3 braid fan. Its maximal cones are indexed by the twenty-four permutations of four symbols; hence the fan is the permutohedral fan. There are 24 maximal cones. Each has three two-dimensional faces and each such face is shared by two maximal cones, so there are $24 \cdot 3/2 = 36$ two-dimensional cones. Together with the fourteen rays this gives the asserted orbit counts.

An edge has two vertices, a triangle has three, and a tetrahedron has four. The normalization preimage of the corresponding generic quotient stratum therefore has respectively two, three, or four points. Division of the normalization counts by these degrees gives

$$14/2 = 7, \qquad 36/3 = 12, \qquad 24/4 = 6.$$

□

Remark 19.7 (Local branches are not global components). At each quadruple point the four local branches are four germs of the same global irreducible central fibre. The conductor in the normalization is the union of fourteen toric boundary surfaces, paired into seven quotient surfaces. No canonical four-term decomposition of a vector bundle or line bundle follows from this local branch number.

More generally, let $\bar{L} \cong \mathbb{Z}^3$ and let $B_0 : \bar{L} \hookrightarrow L$ have finite index d. Replacing λ by $B_0\lambda$ in (683) gives d vertex orbits and hence d global central components. The stratum counts are

$$(f_0, f_1, f_2, f_3) = d(1, 7, 12, 6). \tag{692}$$

Corollary 19.8 (Euler characteristic as the determinant of the shear)**.** *Choose oriented integral bases of $\bar{L}$ and L, and let $[B_0] \in \mathrm{Mat}_{3\times 3}(\mathbb{Z})$ be the matrix of the injective shear. The associated central fibre satisfies*

$$\chi(W_{3,B_0}) = 3!\,[L : B_0\bar{L}] = 3!\,|\det[B_0]|. \tag{693}$$

The right side is unchanged by independent unimodular changes of the two marked bases.

Proof. The index of the image of a full-rank integral matrix is $[L : B_0\bar{L}] = |\det[B_0]|$. By (692), the quotient has $6d$ tetrahedron orbits. Their zero-dimensional torus strata are the only strata with nonzero Euler characteristic, so $\chi(W_{3,B_0}) = 6d = 3!d$. Replacing the source and target bases multiplies the determinant by units in $\mathbb{Z}$, and hence does not change its absolute value. □

If a polarization additionally identifies $[B_0]$ with a symmetric positive-definite matrix, then its determinant is the cubic Jordan norm on that real symmetric subcone of $\mathrm{Herm}_3(\mathbb{C})_{\geq 0}$. Without those additional hypotheses $[B_0]$ is only an integral lattice map, not a positive Hermitian matrix. In either case the scalar in (693) does not determine the fan, the period correction, the singular filling, or the total space.

Thus four global components occur only after the independent choice $d = 4$; four local branches already occur for $d = 1$. A nontrivial Fable-style twist would additionally require holomorphic translations $c_\lambda(t)$ satisfying the exact group cocycle, extension, freeness, and properness identities.

19.6 Two independent obstructions to a complex S^8 globalization

Theorem 19.9 (Euler obstruction to the literal three-special-fibre pattern)**.** *Let $g : M^8 \to S^2$ be a proper triangulable map which is a locally trivial T^6-bundle away from finitely many critical values. If one singular fibre is W_3 and every other singular fibre is a finite free quotient of T^6, then*

$$\chi(M) = 6.$$

In particular, M is not homeomorphic to S^8.

Proof. Choose a triangulation for g in which every critical value is a vertex. Additivity of compactly supported Euler characteristic over the open regular stratum and the critical singleton strata, proved cell by cell in this finite triangulation, gives

$$\chi(M) = \chi_c(S^2 \setminus \Sigma)\chi(T^6) + \sum_{p\in\Sigma} \chi(g^{-1}(p)).$$

Here multiplicativity on the regular stratum follows by subdividing its locally trivial bundle over base cells. Since M is compact, $\chi_c(M) = \chi(M)$, and since $\chi(T^6) = 0$, the first term vanishes. By (691), the cusp contributes 6. A finite free quotient by a group G has Euler characteristic $\chi(T^6)/|G| = 0$. Therefore $\chi(M) = 6$, whereas $\chi(S^8) = 2$. □

Consequently, retaining the cusp W_3 in a real S^8 candidate would require additional genuinely singular fibres whose Euler characteristics sum to -4. Smooth multiple torus fibres cannot supply them.

Theorem 19.10 (No smooth complex S^8 total space)**.** *A smooth manifold homotopy equivalent to S^8 admits no almost-complex structure. Hence a complex structure on the regular locus of a putative T^6-fibration on S^8 cannot extend through all its singular fillings.*

Proof. Suppose $E = (TM, J)$ is a complex rank-four bundle whose underlying real bundle is TM. Orient M by J, and let $u \in H^8(M;\mathbb{Z})$ evaluate to one. Since

$$H^2(M;\mathbb{Z}) = H^4(M;\mathbb{Z}) = H^6(M;\mathbb{Z}) = 0,$$

one has

$$c_1(E) = c_2(E) = c_3(E) = 0, \qquad c_4(E) = e(TM) = \chi(M)u = 2u.$$

For the underlying real bundle,

$$p_2(TM) = c_2(E)^2 - 2c_1(E)c_3(E) + 2c_4(E) = 4u. \tag{694}$$

On the other hand $H^4(M) = 0$ gives $p_1(TM) = 0$, and the middle cohomology vanishes, so $\operatorname{sign}(M) = 0$. Hirzebruch's eight-dimensional signature formula gives

$$0 = \left\langle \frac{7p_2(TM) - p_1(TM)^2}{45}, [M] \right\rangle,$$

which forces $p_2(TM) = 0$. This contradicts (694). For the standard sphere the same vanishing follows immediately from $TS^8 \oplus \varepsilon^1_{\mathbb{R}} \cong \varepsilon^9_{\mathbb{R}}$. □

If every local filling were a smooth complex toroidal fourfold and every overlap map were biholomorphic, these local complex tangent bundles would glue to an almost-complex structure on all of S^8, contradicting theorem 19.10. At least one filling, transition, or total-space singularity must therefore leave the smooth complex category.

19.7 An explicit integral monodromy candidate and its failure

Put

$$S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad Q = S^{-1}T^{-1} = \begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix}.$$

Then

$$S^4 = I, \qquad Q^6 = I, \qquad SQ = T^{-1}. \tag{695}$$

On $\Lambda = \mathbb{Z}^3 \otimes \mathbb{Z}^2 \cong \mathbb{Z}^6$, define

$$M_1 = I_3 \otimes S, \qquad M_2 = I_3 \otimes Q, \qquad M_0 = I_3 \otimes T. \tag{696}$$

They are integral and symplectic, and

$$M_1 M_2 M_0 = I.$$

After reordering the tensor-product basis as $(a_1, a_2, a_3, b_1, b_2, b_3)$, M_0 is exactly the cusp matrix (687). Thus (696) is an explicit local system over a thrice-punctured sphere with finite orders $4, 6$ and a rank-three unipotent cusp.

Proposition 19.11 (Fixed-point obstruction)**.** *No translation makes either finite linear action M_1 or M_2 free on T^6.*

Proof. Neither S nor Q has eigenvalue one. More explicitly,

$$\det(I - S) = 2, \qquad \det(I - Q) = 1.$$

Consequently $I - M_j : T^6 \to T^6$ is a surjective isogeny for $j = 1, 2$. For any translation parameter $c \in T^6$, the fixed-point equation for

$$x \longmapsto M_j x + c$$

is

$$(I - M_j)x = c,$$

and it always has a solution. Therefore every affine lift has a fixed point. □

This is an integral symplectic homology local system compatible with a diagonal rank-three period ansatz. The three matrices alone do not define a polarized variation of Hodge structure or a holomorphic family E^3_τ; those require an equivariant period map in addition. Any isotrivial affine quotient filling built from either displayed finite generator is singular by the fixed-point calculation. A successful alternative would need finite monodromies with invariant torus directions that admit fixed-point-free affine lifts, while retaining the product relation and the rank-three cusp. It would still have to overcome both obstructions in section 19.6.

19.8 Exact replication requirements for any real singular $T^6 \to S^8 \to S^2$ candidate

The preceding results exclude the direct smooth holomorphic construction, not an arbitrary real stratified map. Let

$$\Sigma = \{p_0, \dots, p_s\} \subset S^2, \qquad B^\circ = S^2 \setminus \Sigma.$$

A candidate must first specify an affine integral monodromy

$$\widetilde{\rho} : \pi_1(B^\circ) \longrightarrow \mathrm{GL}(6, \mathbb{Z}) \ltimes T^6 \tag{697}$$

with based peripheral matrices satisfying

$$M_0 M_1 \cdots M_s = I.$$

The corresponding affine peripheral transformations $\widetilde{M}_i = (M_i, c_i)$ must satisfy the stronger identity

$$\widetilde{M}_0 \widetilde{M}_1 \cdots \widetilde{M}_s = (I, 0), \qquad c_0 + M_0 c_1 + \cdots + M_0 \cdots M_{s-1} c_s = 0 \quad \text{in } T^6. \tag{698}$$

Choose pairwise disjoint small closed discs D_i around the p_i, and put

$$B = S^2 \setminus \bigcup_i \operatorname{int} D_i, \qquad J = g^{-1}(B).$$

It must then give each compact filling $N_i = g^{-1}(D_i)$, its boundary mapping torus

$$A_i = J \cap N_i,$$

and the actual based inclusion maps

$$A_i \longrightarrow J, \qquad A_i \longrightarrow N_i.$$

Here one fixes a base point in J, a base point in each A_i, and a whisker in J joining them; changing a whisker only conjugates the corresponding homomorphism and does not change the pushout below.

Proposition 19.12 (Fundamental-group replication map)**.** *The fundamental group of the glued total space is the iterated pushout*

$$\pi_1(M) \cong \pi_1(J) *_{\pi_1(A_0)} \pi_1(N_0) *_{\pi_1(A_1)} \cdots *_{\pi_1(A_s)} \pi_1(N_s), \tag{699}$$

where both homomorphisms out of each $\pi_1(A_i)$ are the literal based inclusion homomorphisms. If, in addition, every $\pi_1(A_i) \to \pi_1(N_i)$ is surjective, then

$$\pi_1(M) \cong \pi_1(J) \Big/ \Big\langle\!\Big\langle (j_{J,i})_* \ker\{\pi_1(A_i) \to \pi_1(N_i)\} : 0 \le i \le s \Big\rangle\!\Big\rangle. \tag{700}$$

Consequently a proposed S^8 construction must either compute the full pushout in (699)*, or prove the stated surjectivities and prove that the displayed normal generators kill $\pi_1(J)$.*

Proof. Apply van Kampen successively to

$$M = J \cup N_0 \cup \cdots \cup N_s$$

with connected collar intersections A_i. Each attachment quotients by the relations identifying the two images of every element of $\pi_1(A_i)$, which is precisely the iterated amalgamated product (699). If the map to $\pi_1(N_i)$ is surjective, then every generator coming from $\pi_1(N_i)$ can be represented by an element of $\pi_1(A_i)$. Eliminating those generators leaves exactly the normal closure of the image in $\pi_1(J)$ of the kernel of the map to $\pi_1(N_i)$. Iterating proves (700). □

For homology, define the literal Mayer–Vietoris map

$$\alpha_q = \bigoplus_i ((j_{J,i})_*, -(j_{N_i})_*) : \bigoplus_i H_q(A_i;\mathbb{Z}) \longrightarrow H_q(J;\mathbb{Z}) \oplus \bigoplus_i H_q(N_i;\mathbb{Z}). \tag{701}$$

Theorem 19.13 (Integral homology-sphere criterion)**.** *Assume that J and the N_i are compact, connected, collar-compatible eight-manifolds with nonempty boundary, so their top homology vanishes. The glued space has the integral homology of S^8 if and only if*

q	*condition on* α_q
0	α_0 *is injective and* $\operatorname{coker}\alpha_0 \cong \mathbb{Z}$,
$1 \le q \le 6$	α_q *is an isomorphism,*
7	α_7 *is surjective and* $\ker\alpha_7 \cong \mathbb{Z}$.

Proof. Insert (701) into the Mayer–Vietoris long exact sequence. For every $1 \le q \le 7$, exactness gives the short exact sequence

$$0 \longrightarrow \operatorname{coker}\alpha_q \longrightarrow H_q(M;\mathbb{Z}) \longrightarrow \ker\alpha_{q-1} \longrightarrow 0.$$

Thus simultaneous vanishing in degrees $1,\dots,7$ is equivalent to α_0 being injective, α_q being an isomorphism for $1 \le q \le 6$, and α_7 being surjective. In degree zero, $H_0(M) \cong \operatorname{coker}\alpha_0$, which gives the remaining degree-zero condition. Since the pieces have zero H_8, exactness at

$$0 \longrightarrow H_8(M) \longrightarrow \bigoplus_i H_7(A_i) \xrightarrow{\alpha_7} H_7(J) \oplus \bigoplus_i H_7(N_i) \longrightarrow H_7(M) \longrightarrow 0$$

identifies $H_8(M)$ with $\ker\alpha_7$. Hence $H_8(M) \cong \mathbb{Z}$ is exactly the last kernel condition. □

If A_i is the mapping torus of a torus automorphism with integral monodromy M_i, its integral homology is fixed by the Wang exact sequence:

$$0 \longrightarrow \operatorname{coker}\left(I - \bigwedge^{q} M_i\right) \longrightarrow H_q(A_i;\mathbb{Z}) \longrightarrow \ker\left(I - \bigwedge^{q-1} M_i\right) \longrightarrow 0. \tag{702}$$

The right-hand group is free, so this sequence splits as a sequence of abelian groups, though not canonically. Equations (697), (699), (701), and (702), together with the actual tangent clutching maps, are the minimum replication data. For an S^8 total space those clutching maps must give

$$p_1 = p_2 = 0, \qquad e = 2u,$$

while preventing the regular-locus complex structure from extending, as required by theorem 19.10.

19.9 The Hermitian determinant boundary is not the four-branch fibre

Let

$$\mathcal{C}_3 = \operatorname{Herm}_3(\mathbb{C})_{\geq 0}.$$

The real dimension of the positive-semidefinite rank-r stratum is

$$2 \cdot 3r - r^2.$$

Indeed, write $H = BB^*$ with B a full-rank complex $3 \times r$ matrix; the entire redundancy is the right action of $U(r)$. Thus the rank $3, 2, 1, 0$ strata have real dimensions

$$9, \quad 8, \quad 5, \quad 0. \tag{703}$$

The rank-two stratum is smooth because the derivative of the determinant is the trace pairing with $\operatorname{adj}(H)$, which is nonzero exactly in rank two. The singular locus is the rank-at-most-one locus.

Proposition 19.14 (The real Hermitian boundary compactifies to S^8)**.** *The determinant-zero boundary*

$$\partial\mathcal{C}_3 = \{H \in \operatorname{Herm}_3(\mathbb{C}) : H \geq 0,\ \det H = 0\}$$

is homeomorphic to $\mathbb{R}^8$. Its one-point compactification is therefore S^8, with the rank strata in (703) *giving a non-smooth stratification.*

Proof. Intersect $\mathcal{C}_3$ with the affine hyperplane $\operatorname{tr} H = 1$:

$$K = \{H \geq 0 : \operatorname{tr} H = 1\}.$$

This is a compact convex body with nonempty interior in the eight-dimensional affine space $\{H = H^* : \operatorname{tr} H = 1\}$. Radial projection from the interior point $I_3/3$ gives a homeomorphism from K to the closed eight-ball. Explicitly, after choosing a Euclidean norm on the trace-zero translation space, every ray

$$I_3/3 + sU, \qquad \|U\| = 1, \quad s \geq 0,$$

meets ∂K once, at a radius $r(U) > 0$; compactness and convexity make r continuous, and $(\rho, U) \mapsto I_3/3 + \rho r(U)U$, $0 \leq \rho \leq 1$, is the required radial homeomorphism after collapsing $\{0\} \times S^7$ to the centre. Thus $\partial K \cong S^7$.

Every nonzero $H \in \partial\mathcal{C}_3$ has the unique expression

$$H = rP, \qquad r = \operatorname{tr} H > 0, \qquad P \in \partial K.$$

Adding $H = 0$ identifies

$$\partial\mathcal{C}_3 \cong ([0,\infty) \times S^7)/(\{0\} \times S^7) \cong \mathbb{R}^8.$$

The Alexandroff one-point compactification of $\mathbb{R}^8$ is S^8. □

Theorem 19.15 (Non-SNC determinant boundary)**.** *At a rank-one point, the semialgebraic boundary of $\mathcal{C}_3$ has a quadratic-cone normal slice rather than a union of smooth crossing faces. The ambient real determinant hypersurface, and its complexification, are not normal-crossings hypersurfaces there. Neither that rank-one germ nor the germ at the zero matrix is the four-branch germ in* (682)*.*

Proof. Unitary conjugation and positive rescaling reduce any rank-one point to

$$H_0 = \operatorname{diag}(1, 0, 0).$$

Near H_0, write

$$H = \begin{pmatrix} a & b^* \\ b & C \end{pmatrix}, \qquad a \in \mathbb{R}, \quad b \in \mathbb{C}^2, \quad C \in \operatorname{Herm}_2(\mathbb{C}).$$

On $a > 0$, the real-analytic coordinate change

$$(a, b, C) \longmapsto (a, b, S), \qquad S = C - a^{-1} b b^*$$

is a local diffeomorphism. The Schur-complement identities are exact:

$$\det H = a \det S, \qquad H \geq 0 \quad \Longleftrightarrow \quad S \geq 0. \tag{704}$$

Writing

$$S = \begin{pmatrix} u & x + iy \\ x - iy & v \end{pmatrix}$$

gives the exact semialgebraic normal boundary

$$uv - x^2 - y^2 = 0, \qquad u \geq 0, \quad v \geq 0. \tag{705}$$

The defining quadratic form has rank four. A product of two real linear forms has quadratic rank at most two, so the ambient real-algebraic hypersurface is not a union of smooth real hyperplane branches; imposing the displayed positive-semidefinite inequalities cannot turn its quadratic cone into transverse smooth faces.

After complexification, the normal slice is

$$s_{11} s_{22} - s_{12} s_{21} = 0.$$

All four first derivatives vanish only at the origin. A product of two linear forms would be singular along their codimension-two intersection, so the quadratic is irreducible and is not a coordinate crossing. At the zero matrix the 3×3 determinant is homogeneous of degree three and therefore has multiplicity three. The germ $z_0 z_1 z_2 z_3 = 0$ has multiplicity four. Neither germ is the other. □

There is also a dimension obstruction. The toroidal central fibre W_3 has complex dimension three and real dimension six, while the dense rank-two boundary of $\mathcal{C}_3$ has real dimension eight. The one-point compactification in theorem 19.14 is a stratified S^8, but it is not the central divisor of the complex fourfold N_3.

The two occurrences of the number three are independent: $\det_{\mathrm{Herm}_3}$ has three eigenvalue factors, whereas a semistable degeneration of complex three-tori has $3 + 1 = 4$ local branches at a deepest point because one includes the base parameter. Moving to Herm_4 makes the determinant quartic but changes the real ambient dimension to 16 and the natural compactified boundary dimension to 15; it does not produce an S^8.

19.10 Exact conclusion and permissible carriers

The local higher-rung toroidal object is exact:

$$\boxed{\text{rank-three periodic fan} \quad \Longrightarrow \quad t = z_0 z_1 z_2 z_3, \quad \mathcal{D}(W_3) \cong T^3, \quad (f_0, f_1, f_2, f_3) = (1, 7, 12, 6).}$$

It lives on the smooth noncompact complex fourfold $N_3 \to \Delta$, not on a complex S^8. A compact complex globalization must have different topology. A real S^8 may retain the complex-torus structure only on its regular locus and requires explicitly non-complex or singular fillings satisfying every map in section 19.8. The Hermitian Gram cone provides a separate real stratified S^8 boundary, not an SNC toroidal fibre. Stable complex data are also permissible because

$$TS^8 \oplus \varepsilon^2_{\mathbb{R}} \cong \varepsilon^{10}_{\mathbb{R}},$$

but this stable structure cannot destabilize to an almost-complex structure on TS^8.

Finally, the six tetrahedron orbits are naturally indexed by the six permutations in S_3; neither this integer six nor the rank-six lattice $H_1(T^6; \mathbb{Z})$ is by itself a cyclic group $\mathbb{Z}/6\mathbb{Z}$. Any cyclic-order-six or gauge-centre interpretation requires an additional exact homomorphism.

20 An exact toroidal elliptic bridge to the candidate cusp

This section asks for the smallest typed morphism between the rank-three toroidal construction and the reconstructed Fabel–Alpöge threefold. It also tests whether that morphism is supplied by the nine-dimensional Hermitian Gram cone. The conclusions are distinct:

1. a quotient of integral lattices extends to a morphism of toric cusp families and commutes with monodromy and the cusp filling maps;
2. the quotient has an explicit unimodular splitting as the candidate cusp plus one elliptic cusp;
3. the order-three and order-four affine peripheral actions admit exact rank-six lattice lifts;
4. none of these maps is induced by the determinant-rank stratification or the rank-one $\mathbb{CP}^2$ normal cones of the Gram construction.

The last assertion is an obstruction to a structure-preserving identification. It is not the false assertion that no continuous map between the underlying spheres exists.

20.1 The candidate lattice and filling data

Retain the candidate map

$$f : X \longrightarrow B \simeq \mathbb{P}^1, \qquad B^\circ = B \setminus \{p_0, p_1, p_2\}.$$

Its smooth fibre has marked first homology

$$\Lambda_F = \mathbb{Z}\langle \widehat{\gamma}, \widehat{u}, \widehat{w}, \widehat{\delta} \rangle.$$

Put

$$A_F = \mathbb{Z}\langle \widehat{w}, \widehat{\delta} \rangle, \qquad B_F = \mathbb{Z}\langle \widehat{\gamma}, \widehat{u} \rangle.$$

Thus $\Lambda_F = A_F \oplus B_F$, with the decomposed order $(\widehat{w}, \widehat{\delta}; \widehat{\gamma}, \widehat{u})$. Define

$$B_0 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} : B_F \longrightarrow A_F. \tag{TEB.1}$$

The cusp monodromy is

$$M_F(a,b) = (a + B_0 b, b), \tag{TEB.2}$$

or, in the manuscript order $(\widehat{\gamma}, \widehat{u}, \widehat{w}, \widehat{\delta})$,

$$[M_F] = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{pmatrix}. \tag{TEB.3}$$

In particular,

$$(M_F - I)^2 = 0, \qquad \ker(M_F - I) = \operatorname{im}(M_F - I) = A_F. \tag{TEB.4}$$

The toric cusp neighbourhood retracts onto its central fibre W. In the marking above, the induced filling homomorphism is

$$c_F : H_1(T^4;\mathbb{Z}) = A_F \oplus B_F \longrightarrow H_1(W;\mathbb{Z}) = B_F, \qquad c_F(a,b) = b. \tag{TEB.5}$$

Thus the two toric angular cycles are killed and the two deck cycles survive. The exact homology of the central fibre is

$$H_*(W;\mathbb{Z}) = (\mathbb{Z}, \mathbb{Z}^2, \mathbb{Z}^4, \mathbb{Z}^2, \mathbb{Z}). \tag{TEB.6}$$

At the other two special points, the finite monodromies are

$$A_1 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 6 & 0 & 1 & 0 \\ -6 & -1 & -1 & 0 \\ -2 & 1 & 0 & 1 \end{pmatrix}, \quad A_2 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ -6 & 1 & 0 & 0 \\ 3 & 0 & 1 & 1 \end{pmatrix}, \tag{TEB.7}$$

in the manuscript order. They satisfy

$$A_1^3 = I, \qquad A_2^4 = I, \qquad A_1 A_2 M_F = I. \tag{TEB.8}$$

The invariant primitive vectors

$$v_1 = \widehat{\gamma} + 2\widehat{u} - 4\widehat{w}, \qquad v_2 = -\widehat{\gamma} - 3\widehat{u} + 3\widehat{w} \tag{TEB.9}$$

define free affine actions

$$x \longmapsto A_j x + v_j/m_j, \qquad (m_1, m_2) = (3,4). \tag{TEB.10}$$

20.2 The rank-three cusp

Let

$$L_3 = \mathbb{Z}\langle e_1, e_2, e_3 \rangle.$$

For $m \in L_3$ and $\pi = (\pi_1, \pi_2, \pi_3) \in S_3$, put

$$\sigma_{m,\pi} = \operatorname{conv}\{m, m + e_{\pi_1}, m + e_{\pi_1} + e_{\pi_2}, m + e_1 + e_2 + e_3\}. \tag{TEB.11}$$

These tetrahedra give the periodic Kuhn triangulation $\mathcal{T}_3$ of $L_{3,\mathbb{R}}$. Coning all cells at height one gives a smooth toric fourfold Y_3, because every maximal cone is unimodular. The height character gives

$$t : Y_3 \longrightarrow \mathbb{C},$$

and a maximal affine chart has the exact equation

$$t = z_0 z_1 z_2 z_3. \tag{TEB.12}$$

For the untwisted quotient, a noncentral fibre is

$$\mathbb{C}^3/(s\mathbb{Z}^3+\mathbb{Z}^3), \qquad t=e^{2\pi i s},$$

and its monodromy, in angular–deck order, is

$$M_3=\begin{pmatrix} I_3 & I_3 \\ 0 & I_3 \end{pmatrix}, \qquad M_3(a,b)=(a+b,b). \tag{TEB.13}$$

The toric filling map is

$$c_3 : H_1(T^6;\mathbb{Z}) = L_3 \oplus \overline{L}_3 \longrightarrow \overline{L}_3, \qquad c_3(a,b)=b. \tag{TEB.14}$$

Indeed, each angular generator bounds a toric disc, while a deck generator closes only by the corresponding lattice transformation and survives in the filling group.

20.3 The exact A_2 quotient of the Kuhn fan

In A_F define

$$n_1=\widehat{w}, \qquad n_2=\widehat{\delta}-\widehat{w}, \qquad n_3=-\widehat{\delta}. \tag{TEB.15}$$

They are primitive and satisfy $n_1+n_2+n_3=0$. Define

$$P : L_3 \longrightarrow A_F, \qquad P(e_i)=n_i. \tag{TEB.16}$$

In the bases (e_1,e_2,e_3) and $(\widehat{w},\widehat{\delta})$,

$$P=\begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix}, \qquad \ker P=\mathbb{Z}(1,1,1). \tag{TEB.17}$$

Define the deck-lattice quotient

$$Q=B_0^{-1}P=\begin{pmatrix} 0 & -1 & 1 \\ 1 & -1 & 0 \end{pmatrix} : \overline{L}_3 \longrightarrow B_F. \tag{TEB.18}$$

Then

$$B_0Q=P, \qquad \ker Q=\mathbb{Z}(1,1,1). \tag{TEB.19}$$

Proposition 20.1 (The Kuhn-to-A_2 fan morphism)**.** *The map P sends every tetrahedron of $\mathcal{T}_3$ onto a triangle of the candidate's A_2 triangulation. Consequently*

$$\widehat{P} : L_3 \oplus \mathbb{Z} \longrightarrow A_F \oplus \mathbb{Z}, \qquad \widehat{P}(x,r)=(Px,r), \tag{TEB.20}$$

is a map of height-one fans and induces a toric morphism

$$F : Y_3 \longrightarrow Y_2, \qquad t \circ F = t. \tag{TEB.21}$$

Proof. The four vertices of $\sigma_{m,\pi}$ map to

$$Pm, \quad Pm+n_{\pi_1}, \quad Pm+n_{\pi_1}+n_{\pi_2}=Pm-n_{\pi_3}, \quad Pm. \tag{TEB.22}$$

Write $(n_1,n_2,n_3)=(e_1,e_2-e_1,-e_2)$ in the target basis. For the six permutations, the nonrepeated images are

$$\begin{array}{c|c} (1,2,3) & \{v,\, v+e_1,\, v+e_2\} \\ (2,1,3) & \{v,\, v-e_1+e_2,\, v+e_2\} \\ (1,3,2) & \{v,\, v+e_1,\, v+e_1-e_2\} \\ (3,1,2) & \{v,\, v-e_2,\, v+e_1-e_2\} \\ (2,3,1) & \{v,\, v-e_1+e_2,\, v-e_1\} \\ (3,2,1) & \{v,\, v-e_2,\, v-e_1\}. \end{array} \tag{TEB.23}$$

These are precisely the six triangles incident to $v = Pm$ in the periodic triangulation by

$$\mathrm{conv}\{u, u+e_1, u+e_2\}, \quad \mathrm{conv}\{u+e_1, u+e_2, u+e_1+e_2\}.$$

Thus every source cone maps into a target cone. The toric fan criterion gives F; the last coordinate in (TEB.20) is unchanged, so the height character is unchanged. □

The kernel $\mathbb{Z}(1,1,1)$ is invariant under permutations of the three coordinates. Hence the quotient lattice is natural up to the Weyl action on the A_2 target. The ordered basis and orientation used in (TEB.1) choose one representative of that Weyl orbit.

20.4 Monodromy and filling compatibility

Theorem 20.2 (Exact local lattice bridge)**.** *The homomorphism*

$$\Phi = P \oplus Q : L_3 \oplus \overline{L}_3 \longrightarrow A_F \oplus B_F, \qquad \Phi(a,b) = (Pa, Qb), \tag{TEB.24}$$

is surjective and obeys

$$\Phi M_3 = M_F \Phi, \qquad c_F \Phi = Q c_3. \tag{TEB.25}$$

Thus the diagram

$$\begin{array}{ccc} H_1(T^6;\mathbb{Z}) & \xrightarrow{\Phi} & H_1(T^4;\mathbb{Z}) \\ {\scriptstyle c_3}\downarrow & & \downarrow{\scriptstyle c_F} \\ \overline{L}_3 & \xrightarrow{Q} & B_F \end{array} \tag{TEB.26}$$

commutes.

Proof. Both P and Q are surjective. Using (TEB.13),

$$\Phi M_3(a,b) = (P(a+b), Qb).$$

Using (TEB.2) and $B_0 Q = P$,

$$M_F \Phi(a,b) = (Pa + B_0 Qb, Qb) = (Pa + Pb, Qb).$$

This proves monodromy equivariance. The filling identity is the literal calculation

$$c_F \Phi(a,b) = Qb = Q c_3(a,b).$$

□

20.5 The actual period correction and the cusp-family morphism

The candidate period matrix at the cusp has the exact form

$$Z_2(t) = sB_0 + C(t), \qquad t = e^{2\pi i s}, \tag{TEB.27}$$

where $C(t)$ is holomorphic and bounded at $t = 0$. Fix the integral right inverse

$$S = \begin{pmatrix} 1 & 1 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad PS = I_2, \tag{TEB.28}$$

and define

$$C_3(t) = SC(t)Q, \qquad Z_3(t) = sI_3 + C_3(t). \tag{TEB.29}$$

Theorem 20.3 (Holomorphic toroidal elliptic bridge)**.** *After shrinking the cusp disc,* $\operatorname{Im} Z_3(t)$ *is invertible and*

$$PZ_3(t) = Z_2(t)Q. \tag{TEB.30}$$

Consequently $P : \mathbb{C}^3 \to \mathbb{C}^2$ *descends on every punctured fibre to a surjective holomorphic homomorphism*

$$\frac{\mathbb{C}^3}{Z_3(t)\mathbb{Z}^3 + \mathbb{Z}^3} \longrightarrow \frac{\mathbb{C}^2}{Z_2(t)\mathbb{Z}^2 + \mathbb{Z}^2}, \tag{TEB.31}$$

with connected elliptic kernel and integral homology map Φ*.*

Define the source and target toric actions by

$$\Psi^{(3)}_\lambda(x,t) = \big(\exp(2\pi i C_3(t)\lambda)t^\lambda x, t\big), \qquad \lambda \in \mathbb{Z}^3, \tag{TEB.32}$$

$$\Psi^{(2)}_\mu(y,t) = \big(\exp(2\pi i C(t)\mu)t^{B_0\mu} y, t\big), \qquad \mu \in \mathbb{Z}^2. \tag{TEB.33}$$

Then

$$F\Psi^{(3)}_\lambda = \Psi^{(2)}_{Q\lambda} F. \tag{TEB.34}$$

The actions are free and properly discontinuous on sufficiently small toric cusp neighbourhoods, and (TEB.21) descends to a holomorphic map

$$\overline{F} : N_3^C \longrightarrow N_0 \tag{TEB.35}$$

over the cusp disc. Its smooth-fibre and filling maps are (TEB.24) and (TEB.26).

Proof. Equations (TEB.19), (TEB.28), and (TEB.29) give

$$PZ_3 = sP + PSCQ = sB_0Q + CQ = Z_2Q.$$

Because $\operatorname{Im} s \to +\infty$ at the cusp while $C_3(t)$ is bounded, the determinant of $\operatorname{Im} Z_3(t)$ has leading term $(\operatorname{Im} s)^3$; it is nonzero on a sufficiently small punctured disc. Hence (TEB.31) is a well-defined homomorphism of complex tori.

Both kernel lattices in (TEB.17) and (TEB.19) are generated by $k = (1,1,1)$, and $C_3(t)k = 0$. The kernel is therefore

$$\mathbb{C}k/(\mathbb{Z}k + s\mathbb{Z}k),$$

an elliptic curve. Saturation of the primitive kernel proves connectedness.

On the dense torus, (TEB.30) gives

$$\begin{aligned} F\Psi^{(3)}_\lambda(x,t) &= \big(\exp(2\pi i P C_3(t)\lambda)t^{P\lambda}F(x), t\big) \\ &= \big(\exp(2\pi i C(t)Q\lambda)t^{B_0Q\lambda}F(x), t\big), \end{aligned}$$

which is (TEB.34). Proposition 20.1 extends the first map over every toric stratum.

For completeness, freeness at a boundary point follows from orbit labels. A nonzero λ translates the bounded cell labelling its toric orbit to that cell plus λ; equality would force $\lambda = 0$. On a smooth fibre a fixed point would give $Z_3(t)\lambda \in \mathbb{Z}^3$, and taking imaginary parts gives $\operatorname{Im} Z_3(t)\lambda = 0$, again forcing $\lambda = 0$. Here is the required uniform estimate for proper discontinuity and properness. Put $y = \operatorname{Im} s = -\log|t|/(2\pi)$, and, on a punctured fibre, use

$$\operatorname{trop}_t(x) = \frac{1}{\log|t|}\big(\log|x_1|, \log|x_2|, \log|x_3|\big).$$

The exact transformation law is

$$\operatorname{trop}_t(\Psi^{(3)}_\lambda x) = \operatorname{trop}_t(x) + A(t)\lambda, \qquad A(t) = I_3 + \frac{\operatorname{Im} C_3(t)}{y}. \tag{TEB.35a}$$

Let

$$K = \sup_{|t|\le\epsilon} \|\operatorname{Im} C_3(t)\|_{\mathrm{op}} < \infty.$$

After shrinking ϵ, $K/y \le 1/2$, and hence

$$\frac{1}{2}\|\lambda\| \le \|A(t)\lambda\| \le \frac{3}{2}\|\lambda\| \tag{TEB.35b}$$

uniformly. If a compact set meets its λ-translate in the dense locus, its tropical coordinates lie in one bounded set C, so $\|A(t)\lambda\| \le \operatorname{diam}(C)$ and (TEB.35b) leaves only finitely many integral λ. On the toric boundary the cell label is translated by λ, because the nonzero factor $\exp(2\pi i C_3(t)\lambda)$ does not change an orbit label; finitely many cells again allow only finitely many λ.

For properness, given a tropical point r, choose $\lambda \in \mathbb{Z}^3$ by rounding the three coordinates of $A(t)^{-1}r$. Then $r - A(t)\lambda$ lies in $A(t)[0,1]^3$, a uniformly bounded set by (TEB.35b). Over a smaller closed disc, the closures of these representatives lie in the finite union of the closed toric polydiscs belonging to the six Kuhn tetrahedra, exactly as in the untwisted rank-three quotient. That union is compact. Thus the action is properly discontinuous, the inverse image of every smaller closed disc has compact quotient, and the quotient and descended map in (TEB.35) exist. The homology assertions follow from Theorem 20.2. □

20.6 A simultaneous unimodular splitting

Proposition 20.4 (Candidate cusp plus one elliptic cusp)**.** *Let*

$$r = \begin{pmatrix} 0 & 0 & 1 \end{pmatrix},$$

$$A = \begin{pmatrix} P \\ r \end{pmatrix} = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} Q \\ r \end{pmatrix} = \begin{pmatrix} 0 & -1 & 1 \\ 1 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \tag{TEB.36}$$

and

$$D = B_0 \oplus [1] = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \tag{TEB.37}$$

Then

$$\det A = \det B = 1, \qquad A = DB. \tag{TEB.38}$$

For $J = \operatorname{diag}(A, B) \in SL(6, \mathbb{Z})$,

$$JM_3J^{-1} = \begin{pmatrix} I_3 & D \\ 0 & I_3 \end{pmatrix}. \tag{TEB.39}$$

In the order $(A_F, k_A; B_F, k_B)$, this is the literal matrix

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

In the coordinates

$$(A_F, k_A; B_F, k_B) = (Pa, ra; Qb, rb),$$

and after reordering them as

$$(A_F, B_F; k_A, k_B),$$

the right side of (TEB.39) is

$$M_F \oplus C_0, \qquad C_0 = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}. \tag{TEB.40}$$

Explicitly, in the reordered basis it is

$$\begin{pmatrix} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} M_F & 0 \\ 0 & C_0 \end{pmatrix},$$

where M_F on the right is in the decomposed order $(A_F; B_F)$. Projection to the candidate block is exactly $\Phi = P \oplus Q$.

Proof. The matrix A is upper triangular with unit diagonal. Expanding B along its last row gives

$$\det B = \det \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix} = 1.$$

Direct multiplication gives $DB = A$. Hence

$$JM_3J^{-1} = \begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix} \begin{pmatrix} I & I \\ 0 & I \end{pmatrix} \begin{pmatrix} A^{-1} & 0 \\ 0 & B^{-1} \end{pmatrix} = \begin{pmatrix} I & AB^{-1} \\ 0 & I \end{pmatrix} = \begin{pmatrix} I & D \\ 0 & I \end{pmatrix}.$$

The first two rows of A and B are P and Q; their last row records the primitive diagonal kernel coordinate. Since $D = B_0 \oplus 1$, the asserted permutation separates the candidate and kernel blocks exactly. □

This proposition proves integrally, rather than only rationally, that the rank-three cusp module is the candidate cusp module plus one standard rank-one cusp. It also identifies both kernel cycles with the primitive diagonal vector $(1, 1, 1)$.

20.7 The complete finite-order lattice lift

Define

$$C_1 = \begin{pmatrix} -1 & -1 \\ 1 & 0 \end{pmatrix}, \qquad C_2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad C_0 = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}. \tag{TEB.41}$$

Direct multiplication gives

$$C_1^3 = I, \qquad C_2^4 = I, \qquad C_1C_2 = C_0^{-1}, \qquad (C_0 - I)^2 = 0, \quad \operatorname{rank}(C_0 - I) = 1. \tag{TEB.42}$$

Proposition 20.5 (Rank-six lift of the $(3, 4, \infty)$ peripheral actions)**.** *On $\widetilde{\Lambda} = \Lambda_F \oplus \mathbb{Z}^2$, put*

$$\widetilde{A}_1 = A_1 \oplus C_1, \qquad \widetilde{A}_2 = A_2 \oplus C_2, \qquad \widetilde{M}_0 = M_F \oplus C_0. \tag{TEB.43}$$

Then

$$\widetilde{A}_1^3 = I, \qquad \widetilde{A}_2^4 = I, \qquad \widetilde{A}_1\widetilde{A}_2\widetilde{M}_0 = I, \tag{TEB.44}$$

and $\widetilde{M}_0 - I$ has square zero, rank three, and equal kernel and image of rank three. Projection

$$\mathrm{pr} : \widetilde{\Lambda} \longrightarrow \Lambda_F$$

intertwines all three monodromies.

The affine maps

$$\widetilde{g}_j(x, y) = (A_j x + v_j/m_j, C_j y) \quad \text{on } T^4 \times T^2 \tag{TEB.45}$$

have orders $m_1 = 3, m_2 = 4$ and are free. Their quotient maps project to the candidate finite-fibre quotient models.

Proof. Equations (TEB.8) and (TEB.42) prove (TEB.44) blockwise. The nilpotent part of $\widetilde{M}_0$ is the direct sum of the candidate rank-two nilpotent and the rank-one nilpotent $C_0 - I$, which proves all cusp assertions. Block projection gives equivariance.

The vectors $(v_j, 0)$ are invariant under $\widetilde{A}_j$, so the m_j-th iterate of (TEB.45) is translation by the integral vector $(v_j, 0)$, hence the identity on the torus. If a nonidentity iterate had a fixed point, its projection to T^4 would be a fixed point of the corresponding nonidentity iterate of (TEB.10). The latter actions are free. Thus (TEB.45) is free and has exact order m_j. Projection descends because it is equivariant. □

Proposition 20.5 proves that the orders $3, 4$, affine freeness, and a rank-three cusp are mutually compatible at the integral topological level. It does not yet identify these block lifts with one holomorphic period family on the full triangle-group cover. That is part of the remaining problem below.

20.8 The six toric chambers and the diagonal Gram boundary

The numerical agreement between the six deepest toric strata and the six A_2 Weyl chambers has the following exact, but purely fan-theoretic, form. Let

$$\mathfrak{a} = \{\mathrm{diag}(\lambda_1, \lambda_2, \lambda_3) : \lambda_1 + \lambda_2 + \lambda_3 = 0\} \subset \mathrm{Herm}_3^0(\mathbb{C})$$

and put

$$\nu_i = E_{ii} - \frac{1}{3} I_3, \qquad \nu_1 + \nu_2 + \nu_3 = 0. \tag{TEB.45a}$$

Because $n_1 + n_2 + n_3 = 0$, there is a unique real-linear isomorphism

$$\jmath : A_{F,\mathbb{R}} \xrightarrow{\sim} \mathfrak{a}, \qquad \jmath(n_i) = \nu_i. \tag{TEB.45b}$$

For $H \in \mathrm{Herm}_3^0(\mathbb{C})$, let

$$\Phi_{\mathrm{G}}(H) = H - \lambda_{\min}(H) I_3.$$

This is the $SU(3)$-equivariant homeomorphism onto the singular positive-semidefinite Gram cone used in the preceding higher-rung analysis.

Proposition 20.6 (The Kuhn strata are the marked A_2 chambers)**.** *For every $\pi = (\pi_1, \pi_2, \pi_3) \in S_3$,*

$$\jmath(P(\sigma_{0,\pi})) = \mathrm{conv}\{0, \nu_{\pi_1}, -\nu_{\pi_3}\}, \tag{TEB.45c}$$

and the cone on this triangle is the closed Weyl chamber

$$\boxed{\mathbb{R}_{\geq 0}\nu_{\pi_1} + \mathbb{R}_{\geq 0}(-\nu_{\pi_3}) = \{\lambda_{\pi_1} \geq \lambda_{\pi_2} \geq \lambda_{\pi_3}\}.} \tag{TEB.45d}$$

Consequently the six tetrahedron orbits, hence the six quadruple-point strata of the untwisted toroidal central fibre, are identified S_3-equivariantly with the six marked diagonal Weyl chambers.

On this chamber the Gram-boundary homeomorphism is the explicit piecewise-linear map

$$\Phi_{\mathrm{G}}(H) = H - \lambda_{\pi_3}(H) I_3. \tag{TEB.45e}$$

In particular,

$$\Phi_{\mathrm{G}}(\nu_{\pi_1}) = E_{\pi_1\pi_1}, \qquad \Phi_{\mathrm{G}}(-\nu_{\pi_3}) = I_3 - E_{\pi_3\pi_3}. \tag{TEB.45f}$$

Proof. Equation (TEB.22) with $m = 0$ gives the three distinct image vertices $0, n_{\pi_1}, -n_{\pi_3}$, proving (TEB.45c). For $a, b \geq 0$, the diagonal entries of $a\nu_{\pi_1} - b\nu_{\pi_3}$ satisfy

$$\lambda_{\pi_1} - \lambda_{\pi_2} = a, \qquad \lambda_{\pi_2} - \lambda_{\pi_3} = b.$$

Conversely these two nonnegative differences recover a, b, since the trace is zero. This proves (TEB.45d). Both the permutation action on the Kuhn labels and the Weyl action on diagonal entries are the same action of S_3, so the correspondence is equivariant. On the displayed chamber the least eigenvalue is λ_{π_3}, which proves (TEB.45e). Substitution of the two extreme rays gives (TEB.45f). □

This is stronger than the bare equality $6 = |S_3|$, but it is not an identification of singular spaces. After quotienting the target fan by A_F, the six triangles incident to a marked vertex form two global triangle orbits. Moreover $\Phi_{\mathrm{G}} \circ \jmath$ is a map on the real cocharacter space, not a holomorphic map of toroidal total spaces. It neither supplies the period correction nor changes the analytic factorization comparison below.

20.9 The exact Jordan determinant and Lorentz sections

The determinant–index identity in Corollary 19.8 has a precise extension inside the nine-dimensional Hermitian cone. It is an algebraic extension of the scalar invariant; it is not a map from the toroidal degeneration to the Hermitian cone.

Write

$$\mathcal{J}_3 = \mathrm{Herm}_3(\mathbb{C}), \qquad \mathcal{J}_3^{\succeq 0} = \{H \in \mathcal{J}_3 : H \succeq 0\}, \qquad \Delta(H) = \det H.$$

On the ambient vector spaces, the inclusion

$$\iota : \mathrm{Sym}_3(\mathbb{R}) \hookrightarrow \mathcal{J}_3, \qquad \iota(S) = S + i0 \tag{TEB.45g}$$

is an injective unital homomorphism of real Jordan algebras for the product $A \circ B = (AB + BA)/2$. Its restriction sends the positive-definite cone into the positive-definite Hermitian cone. The ambient formulation matters: a positive cone is not itself a vector space or a Jordan algebra. The map also uses the retained marking of the integral lattice; it is not a coordinate-free map from an unmarked congruence class of polarizations to a fixed matrix space.

For $a = (a_1, a_2, a_3)^{\mathsf{T}} \in \mathbb{R}^3$, fix the oriented cross-product matrix

$$[a]_\times = \begin{pmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{pmatrix}. \tag{TEB.45h}$$

Every $H \in \mathcal{J}_3$ has one and only one decomposition

$$H = S + i[a]_\times, \qquad S \in \mathrm{Sym}_3(\mathbb{R}), \quad a \in \mathbb{R}^3. \tag{TEB.45i}$$

Theorem 20.7 (Hermitian determinant splitting)**.** *For every real symmetric S and every $a \in \mathbb{R}^3$,*

$$\boxed{\det(S + i[a]_\times) = \det S - a^{\mathsf{T}} S a.} \tag{TEB.45j}$$

If S is positive definite, then

$$S + i[a]_\times \succeq 0 \quad \Longleftrightarrow \quad a^{\mathsf{T}} S a \le \det S, \tag{TEB.45k}$$

and positive semidefiniteness and the weak scalar inequality may both be replaced by strict positivity and a strict scalar inequality.

Proof. Direct expansion in the fixed coordinates (TEB.45h) gives

$$\det(S + i[a]_\times) = \det S - a^{\mathsf{T}} S a.$$

For the order statement let $M = S^{-1/2}$, the positive-definite symmetric inverse square root. The identity

$$M[a]_\times M^{\mathsf{T}} = \det(M)[M^{-\mathsf{T}} a]_\times \tag{TEB.45l}$$

follows by applying both sides to two vectors and evaluating against a third with the fixed oriented volume form. Hence

$$S^{-1/2}(S + i[a]_\times)S^{-1/2} = I_3 + i[b]_\times, \qquad b = \det(S)^{-1/2} S^{1/2} a,$$

and

$$\|b\|^2 = \frac{a^{\mathsf{T}} S a}{\det S}.$$

The three eigenvalues of $I_3 + i[b]_\times$ are $1 - \|b\|, 1, 1 + \|b\|$. Congruence preserves inertia, so (TEB.45k) and its strict version follow. □

For a fixed marked positive-definite symmetric monodromy form B, define the injective real-linear map

$$\ell_B : \mathbb{R} \oplus \mathbb{R}^3 \longrightarrow \mathcal{J}_3, \qquad \ell_B(\lambda, a) = \lambda B + i[a]_\times. \tag{TEB.45m}$$

Theorem 20.7, applied first to $\lambda > 0$ and then directly at $\lambda = 0$, gives

$$\det \ell_B(\lambda, a) = \lambda(\lambda^2 \det B - a^{\mathsf{T}} B a), \tag{TEB.45n}$$

and the exact inverse image

$$\boxed{\ell_B^{-1}(\mathcal{J}_3^{\geq 0}) = \{(\lambda, a) : \lambda \geq 0,\ a^{\mathsf{T}} B a \leq \lambda^2 \det B\}.} \tag{TEB.45o}$$

Indeed, if $\lambda < 0$, then $\operatorname{tr} \ell_B(\lambda, a) = \lambda \operatorname{tr} B < 0$, which is impossible for a positive-semidefinite matrix. Thus every marked positive monodromy ray determines a literal future $(3+1)$-Lorentz cone with quadratic form

$$q_B(\lambda, a) = \lambda^2 \det B - a^{\mathsf{T}} B a. \tag{TEB.45p}$$

At $B = I_3$ this is $\lambda^2 - a_1^2 - a_2^2 - a_3^2$. The equality

$$\chi(W_{3,I_3}) = 6 = 6 q_{I_3}(1, 0) \tag{TEB.45q}$$

is therefore exact, while remaining an equality of scalar invariants rather than a morphism of degenerations.

There is also an exact order-cone lift of the rank-two Lorentz cone from Theorem 18.11. In its coordinates

$$H_2(t, x_1, x_2, x_3) = \begin{pmatrix} t + x_3 & x_1 - i x_2 \\ x_1 + i x_2 & t - x_3 \end{pmatrix},$$

put

$$\Theta(H_2(t, x_1, x_2, x_3)) = \begin{pmatrix} t & -i x_3 & -i x_2 \\ i x_3 & t & -i x_1 \\ i x_2 & i x_1 & t \end{pmatrix}. \tag{TEB.45r}$$

Proposition 20.8 (Order-cone lift and its complete rank behaviour)**.** *The map Θ is a real-linear bijection from* $\mathrm{Herm}_2(\mathbb{C})$ *onto* $\mathbb{R} I_3 \oplus i\mathfrak{so}(3)$*, and*

$$H_2 \succeq 0 \quad \Longleftrightarrow \quad \Theta(H_2) \succeq 0. \tag{TEB.45s}$$

Moreover

$$\boxed{\det_3 \Theta(H_2) = \frac{\operatorname{tr} H_2}{2} \det_2 H_2.} \tag{TEB.45t}$$

On the positive-semidefinite cone, Θ sends ranks $2, 1, 0$ respectively to ranks $3, 2, 0$.

Proof. Put $r = (x_1^2 + x_2^2 + x_3^2)^{1/2}$. The eigenvalues of H_2 are $t - r, t + r$, while those of $\Theta(H_2)$ are $t - r, t, t + r$. Both matrices are positive semidefinite exactly when $t \geq r$. Multiplication of the displayed eigenvalues gives

$$\det_3 \Theta(H_2) = t(t^2 - r^2) = \frac{\operatorname{tr} H_2}{2} \det_2 H_2.$$

The same eigenvalues give all three rank assertions. The displayed matrix also makes real-linearity, injectivity, and the stated image immediate. □

If P is a rank-one projection in $\mathrm{Herm}_2(\mathbb{C})$, then its Lorentz coordinates satisfy $t = r = 1/2$, so $\Theta(P)$ has eigenvalues $0, 1/2, 1$ and is not an idempotent. Thus Θ is not a Jordan-algebra homomorphism. Globally, the pullback of the cubic determinant hypersurface is

$$\Theta^{-1}\{\det_3 = 0\} = \{\operatorname{tr} H_2 = 0\} \cup \{\det_2 H_2 = 0\}; \tag{TEB.45u}$$

only on the positive cone does the first set add no points beyond the origin. Neither ℓ_B nor Θ is a holomorphic toric map, a map of integral period lattices, or a singular-fibre filling map. For the actual one-parameter toric degeneration the monodromy form is fixed at $B = I_3$ while the local parameter in $z_0z_1z_2z_3 = t$ tends to zero. Consequently the induced Jordan datum is the constant point I_3, not a path to $\det B = 0$. Degeneration at fixed monodromy and rank loss of the monodromy form are different operations.

Finally, the cyclic group $C_3 \subset S_3$ does have two orbits on the six Kuhn labels. Under P, the even labels $(123), (231), (312)$ give translates of the lower target triangle $\mathrm{conv}\{u, u+e_1, u+e_2\}$, while the odd labels $(132), (213), (321)$ give translates of the upper target triangle $\mathrm{conv}\{u+e_1, u+e_2, u+e_1+e_2\}$. This action is by coordinate permutations and the exact bridge lowers lattice rank through P; it is not translation by an index-three superlattice. No such translation-lattice realization is constructed by the packet supplying (TEB.45j)–(TEB.45t). More decisively, for any rank-three lattice Λ', a fundamental domain has normalized volume $3! = 6$, so every Λ'-unimodular triangulation of that domain has six maximal tetrahedra. The canonical six-to-two construction is instead the rank-lowering fan map $P : L_3 \to A_F$ in (TEB.16)–(TEB.23). It must not be replaced by an unproved semistable translation quotient.

Audit record 20.9 (Completed Jordan–toric packet, 3 September 2026)**.** The source archive has SHA-256 8BA0BD7FD7BADEAAF2C44E2F2B4EFD42C25DA9119DA3180017FC488CA364E96F. Its supplied symbolic identities were replayed independently, and separate Euler/stratification, morphism/type, fan/Weyl, and theorem-map audits were performed. Equations (TEB.45j)–(TEB.45t) are the surviving exact content. The positive-cone Jordan typing, the normalization stratum attribution, and the index-three compression were corrected rather than copied. The audit reports and exact replay script are retained under agent_logs/ and agent_scratch/jordan_toric_symbolic_replay_20260903/ in the local research record.

20.10 The Gram determinant does not supply this bridge

Let

$$V = \mathrm{Mat}_3(\mathbb{C}), \qquad \mathcal{G} = V^+/SU(3), \qquad \mathcal{D} = \{\det = 0\}^+/SU(3).$$

Here $V^+ = V \cup \{\infty\}$ is the one-point compactification and $\{\det = 0\}^+ = \{\det = 0\} \cup \{\infty\}$. The exact Gram classification gives

$$(\mathcal{G}, \mathcal{D}) \cong (S^{10}, S^8), \qquad \mathcal{G} \setminus \mathcal{D} \cong \mathbb{R}^9 \times S^1. \tag{TEB.46}$$

The real fixed form gives

$$\Big(\mathrm{Mat}_3(\mathbb{R})^+/SO(3), \{\det = 0\}^+/SO(3)\Big) \cong (S^6, S^5). \tag{TEB.47}$$

Theorem 20.10 (No determinant-stratified identification)**.** *There is no homeomorphism of pairs from the real Gram pair (TEB.47) to*

$$\Big(X, f^{-1}\{p_0, p_1, p_2\}\Big). \tag{TEB.48}$$

There is no homotopy equivalence from the complex Gram complement in (TEB.46) to the candidate regular part $f^{-1}(B^\circ)$. The rank-one trace-one Gram stratum $\mathbb{CP}^2$ is not homotopy equivalent to any of the three candidate special fibres. Finally, no equivalence of function germs identifies its normal determinant with any of the three candidate local filling maps.

Proof. The determinant-zero set in (TEB.47) is a connected codimension-one S^5, and its complement has two components. The set in (TEB.48) is the disjoint union of three connected real four-dimensional fibres, hence has codimension two and three components. Its complement is connected because it is a torus bundle over the connected pair of pants B°. This proves the first assertion.

The complex Gram complement has fundamental group $\mathbb{Z}$. The homotopy sequence of the candidate regular torus bundle gives

$$1 \longrightarrow \mathbb{Z}^4 \longrightarrow \pi_1\big(f^{-1}(B^\circ)\big) \longrightarrow \pi_1(B^\circ) = F_2 \longrightarrow 1. \tag{TEB.49}$$

Its fundamental group surjects onto the nonabelian group F_2, so it is not $\mathbb{Z}$.

The rank-one trace-one stratum has

$$H_*(\mathbb{CP}^2;\mathbb{Z}) = (\mathbb{Z},0,\mathbb{Z},0,\mathbb{Z}), \qquad \chi(\mathbb{CP}^2) = 3.$$

The cusp fibre has (TEB.6), so $b_1 = 2, b_2 = 4, \chi = 2$. Each finite reduced fibre is a free finite quotient of T^4, hence has $\chi = 0$, and its audited first homology has free rank two. These invariants rule out all three homotopy equivalences.

At a rank-one Gram matrix, Schur-complement coordinates put the normal determinant in the exact form

$$uv - x^2 - y^2, \tag{TEB.50}$$

a nondegenerate quadratic form of signature $(1,3)$. Its complexification is the irreducible quadratic determinant

$$s_{11}s_{22} - s_{12}s_{21}. \tag{TEB.51}$$

At a deepest candidate cusp point, the map is instead

$$(z_0,z_1,z_2) \longmapsto z_0z_1z_2, \tag{TEB.52}$$

with three smooth complex branches. The source dimensions, multiplicities, and number of irreducible branches in (TEB.50)–(TEB.52) disagree. The two finite transverse maps have multiplicities three and four, while the Gram determinant has multiplicity two along the rank-one stratum and multiplicity three at the zero matrix. It has no order-four stratum. Orders of vanishing and analytic factorization are invariant under equivalence of function germs, which proves the final assertion. □

There are two further typed obstructions. First, the determinant phase map

$$\arg\det : \mathcal{G} \setminus \mathcal{D} \longrightarrow S^1$$

has trivial page monodromy after the $SU(3)$ quotient. Before quotienting, its page is diffeomorphic to $\mathbb{R}_{>0} \times SL(3,\mathbb{C})$, hence is simply connected, and its geometric monodromy is the scalar central element of order three. It supplies no $\mathbb{Z}^4$- or $\mathbb{Z}^6$-valued first-homology local system and no order-four peripheral loop.

Second, the trace-one rank-one locus is the homogeneous space

$$\mathbb{CP}^2 = SU(3)/S(U(1)\times U(2)).$$

Any $SU(3)$-invariant map from it to the candidate base, on which no $SU(3)$-action is part of the construction, is constant. Choosing a line $\ell \in \mathbb{CP}^2$ does define the exact positive compression

$$\mathrm{Herm}_3(\mathbb{C})_{\geq 0} \longrightarrow \mathrm{Herm}(\ell^\perp)_{\geq 0}, \qquad H \longmapsto P_{\ell^\perp} H|_{\ell^\perp}, \tag{TEB.53}$$

but its target is a four-real-dimensional Lorentz cone, not $\mathbb{P}^1$, and it contains no integral torus lattice. An integral quotient such as P requires the additional choice of maximal-torus and weight-lattice data. Irreducibility of the fundamental $SU(3)$ representation forbids an $SU(3)$-equivariant rank-two linear quotient of $\mathbb{C}^3$.

In particular, the homomorphism (TEB.14) belongs to the toroidal degeneration, not to the Gram cone. A generic conjugacy orbit in the trace-normalized Hermitian cone is $SU(3)/T^2$, which is simply connected, and the determinant page is simply connected as well. Neither has first homology $\mathbb{Z}^6$. Calling any map from it a T^6 filling map before constructing a torus bundle would be a type error.

20.11 The remaining global problem

The local bridge and the finite lattice lift isolate the remaining question. One must construct a proper complex, or explicitly stratified-real, map

$$\widetilde{f} : \widetilde{X}^8 \longrightarrow \mathbb{P}^1$$

and a map over the same base

$$\Pi : \widetilde{X} \longrightarrow X \tag{TEB.54}$$

with all of the following literal properties:

1. on every regular fibre, Π is the elliptic extension (TEB.31) and its marked integral map is Φ;
2. over the cusp disc, Π is the toric morphism (TEB.35), including the central strata and the filling square (TEB.26);
3. over each finite-order disc, the affine monodromy and translation are those in (TEB.43)–(TEB.45), and its punctured overlap agrees with the regular period family;
4. the actual based van Kampen maps and integral Mayer–Vietoris maps are computed from these inclusions, rather than inferred from the displayed monodromy matrices.

Equations (TEB.24)–(TEB.45) prove that there is no integral monodromy or affine-freeness obstruction to these requirements. They do not supply the global period and overlap construction in item 3.

There is also an unavoidable topological conclusion. The rank-three toroidal central fibre has Euler characteristic six. The two finite fibres in (TEB.45) are free finite quotients of T^6, so each has Euler characteristic zero. If these are the only three special fibres, Euler integration gives

$$\chi(\widetilde{X}) = 6. \tag{TEB.55}$$

Therefore this direct complex globalization cannot be S^8, whose Euler characteristic is two. Independently, S^8 admits no almost-complex structure. A real S^8 construction would have to add non-complex singular filling data with total Euler contribution -4, then pass the full fundamental-group, integral-homology, and tangent-clutching tests.

The nine-dimensional Gram cone remains a valid independent representation-theoretic carrier. Relating it to (TEB.54) requires a new, explicit morphism from determinant-rank data to period, lattice, and filling data. Neither the determinant, the Cartan cubic, nor the $\mathbb{CP}^2$-family of normal Lorentz cones supplies such a morphism.

21 The rank-six cusp lattice inside the nine-dimensional cone program

The rank-three toroidal construction contains a free rank-six lattice before any finite quotient is taken. This section identifies that lattice, its integral symmetric-group action, its monodromy, and the exact part of it seen by the Hermitian Gram boundary. It also separates it from the different rank-six group which classifies bundle defects on the regular locus of the reconstructed S^6.

21.1 The two rank-six groups have different types

Retain the notation of section 20.2. The lattice native to the rank-three toroidal cusp is

$$\Lambda_{\mathrm{tor}} := H_1(T^6;\mathbb{Z}) = L_3 \oplus \overline{L}_3 = \mathbb{Z}\langle e_1, e_2, e_3\rangle \oplus \mathbb{Z}\langle \bar{e}_1, \bar{e}_2, \bar{e}_3\rangle. \tag{706}$$

Its first summand consists of angular cycles and its second summand consists of deck cycles. The cusp monodromy and filling map are

$$M_3(a,b) = (a+b,b), \qquad c_3(a,b) = b. \tag{707}$$

The bundle-defect lattice on the original threefold has a different source and a different cohomological degree. With $D = W \sqcup S_1 \sqcup S_2$ and $U = X \setminus D$, it is

$$L_{\mathrm{def}} := H^4(U;\mathbb{Z}) \cong H_1(W;\mathbb{Z}) \oplus H_1(S_1;\mathbb{Z}) \oplus H_1(S_2;\mathbb{Z}) \cong (\mathbb{Z}^2)^3. \tag{708}$$

The map $\Phi = P \oplus Q$ below is defined on Λ_{tor}, whereas the fibre-degree character of theorem 59.1 is defined on L_{def}. Equality of their ranks supplies no map between them. In particular, neither group is the cyclic group $\mathbb{Z}/6\mathbb{Z}$.

There is also no component-permuting symmetric-group action on L_{def} supplied by the marked candidate fibration. The three deleted fibres have, respectively, unipotent, order-three, and order-four local monodromy. An automorphism of the marked fibration which permutes the punctures must preserve these conjugacy-order types and hence fixes all three punctures. One may impose an abstract permutation of the three displayed summands in (708) after choosing three identifications with $\mathbb{Z}^2$, but that is a re-marking, not an automorphism furnished by the geometry. This argument does not classify automorphisms fixing the base pointwise or internal actions on the individual summands; in particular it does not assert that the abstract group admits no S_3-action.

21.2 The integral S_3-module of the cusp

Let R_σ permute the three standard basis vectors by $R_\sigma e_i = e_{\sigma(i)}$, and let S_3 act diagonally on $\Lambda_{\mathrm{tor}} = L_3 \oplus \overline{L}_3$. Put

$$k = e_1 + e_2 + e_3, \qquad K = \mathbb{Z}(k,0) \oplus \mathbb{Z}(0,\bar{k}), \quad \bar{k} = \bar{e}_1 + \bar{e}_2 + \bar{e}_3. \tag{709}$$

The group S_3 fixes K pointwise, commutes with M_3, and makes c_3 equivariant for the permutation action on $\overline{L}_3$: $c_3(R_\sigma a, R_\sigma b) = R_\sigma c_3(a,b)$.

Recall

$$P = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix}, \qquad Q = \begin{pmatrix} 0 & -1 & 1 \\ 1 & -1 & 0 \end{pmatrix}, \qquad \Phi = P \oplus Q. \tag{710}$$

For $\sigma \in S_3$, define integral actions on A_F and B_F by

$$\rho_A(\sigma)(Px) = P(R_\sigma x), \qquad \rho_B(\sigma)(Qx) = Q(R_\sigma x). \tag{711}$$

These definitions are well posed because $\ker P = \ker Q = \mathbb{Z}k$.

Theorem 21.1 (The exact rank-six S_3-module sequence)**.** *The map Φ fits into an exact sequence of integral S_3-modules*

$$0 \longrightarrow K \longrightarrow \Lambda_{\mathrm{tor}} \xrightarrow{\Phi} A_F \oplus B_F \longrightarrow 0. \tag{712}$$

It is simultaneously equivariant for the cusp monodromies:

$$\Phi M_3 = M_F \Phi, \qquad M_3|_K = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \quad \text{in the basis } ((k,0),(0,\bar{k})). \tag{713}$$

As a sequence of free abelian groups with monodromy it splits integrally, as shown by theorem 20.4. As a sequence of integral S_3-modules it does not split. After tensoring with $\mathbb{Q}$,

$$\Lambda_{\mathrm{tor},\mathbb{Q}} \cong \mathbf{21} \oplus 2V_{\mathrm{std}}, \qquad (A_F \oplus B_F)_{\mathbb{Q}} \cong 2V_{\mathrm{std}}, \qquad K_{\mathbb{Q}} \cong \mathbf{21}. \tag{714}$$

Proof. The kernels of P and Q are both the primitive diagonal lattice $\mathbb{Z}k$, and both maps are surjective. This proves exactness. Equations (711) make P, Q, and hence Φ, equivariant. Since $R_\sigma(a+b) = R_\sigma a + R_\sigma b$, the source action commutes with M_3. Moreover

$$M_3(xk, y\bar{k}) = ((x+y)k, y\bar{k}),$$

which gives (713). The target equivariance with M_F is equivalently the already verified identity $\Phi M_3 = M_F \Phi$.

It remains to prove the integral non-splitting. Any S_3-equivariant homomorphism $r : \mathbb{Z}^3 \to \mathbb{Z}$, where the target is trivial, has the same value $a \in \mathbb{Z}$ on all three standard basis vectors. Thus $r(k) = 3a$, so no such map retracts the inclusion $\mathbb{Z}k \hookrightarrow \mathbb{Z}^3$. For two copies, every entry of the matrix obtained by restricting an equivariant map $(\mathbb{Z}^3)^2 \to \mathbb{Z}^2$ to $(\mathbb{Z}k)^2$ is divisible by three. That restriction cannot be the identity matrix. Hence (712) does not split over $\mathbb{Z}[S_3]$.

Over $\mathbb{Q}$, the invariant diagonal line has the equivariant complement $\{x_1 + x_2 + x_3 = 0\}$, the standard representation. Applying this to both copies gives (714). □

For reference, if s_{12} and s_{23} are the adjacent transpositions, the induced target matrices are

$$\rho_A(s_{12}) = \begin{pmatrix} -1 & 0 \\ 1 & 1 \end{pmatrix}, \quad \rho_A(s_{23}) = \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}, \tag{715}$$

and

$$\rho_B(s_{12}) = \begin{pmatrix} 1 & -1 \\ 0 & -1 \end{pmatrix}, \quad \rho_B(s_{23}) = \begin{pmatrix} -1 & 0 \\ -1 & 1 \end{pmatrix}. \tag{716}$$

Direct multiplication gives

$$\rho_A(\sigma)P = PR_\sigma, \qquad \rho_B(\sigma)Q = QR_\sigma, \qquad \rho_A(\sigma)B_0 = B_0\rho_B(\sigma). \tag{717}$$

The quotient $\mathbb{Z}^3/\mathbb{Z}k$ is the A_2 weight lattice, not the A_2 root lattice. Indeed, the augmentation-zero lattice is generated by $e_1 - e_2$ and $e_2 - e_3$, whose images under P are the columns of

$$\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}. \tag{718}$$

Its determinant is three. Thus the root lattice is the index-three sublattice of A_F generated by the differences of the three weights n_1, n_2, n_3. The integral non-splitting in theorem 21.1 is the same index-three phenomenon viewed before passing to the quotient.

21.3 The six homology generators are not the six chambers

There are three related but distinct appearances of the integer six.

Theorem 21.2 (The three sixes)**.** *For the rank-three toroidal cusp and every injective integral shear $\mathcal{B} : \overline{L}_3 \hookrightarrow L_3$,*

$$\operatorname{rk} \Lambda_{\mathrm{tor}} = 2 \operatorname{rk} L_3 = 6, \tag{719}$$

$$\#\{\textit{Kuhn maximal cells modulo } L_3\} = |S_3| = 3! = 6, \tag{720}$$

$$\chi(W_{3,\mathcal{B}}) = |S_3|[L_3 : \mathcal{B}\overline{L}_3] = 3! \,|\det \mathcal{B}|. \tag{721}$$

The equalities (720) *and* (721) *have the same toric source: the maximal cells are indexed by orderings of three angular coordinates. Equation* (719) *has a different source: first homology contains one angular and one deck copy of the rank-three lattice. For a general shear $\mathcal{B}$, this cell count does not assert that the diagonal S_3-action descends to the sheared quotient; that requires the additional commutation equations $R_\sigma \mathcal{B} = \mathcal{B} R_\sigma$.*

In particular, the six standard basis vectors of Λ_{tor} are not S_3-equivariantly identifiable with the six Kuhn cells. The basis vectors form two orbits of size three; the maximal cells form one free orbit of size six.

Proof. The rank formula follows from (706). A Kuhn cell in one lattice cube is labelled by the order in which the three coordinate increments are taken, hence by a unique element of S_3. Left multiplication is free and transitive on these labels, proving (720). The normalization and orbit calculation in theorem 19.8 gives (721).

Under the diagonal permutation action, the angular basis $\{e_1, e_2, e_3\}$ and the deck basis $\{\bar{e}_1, \bar{e}_2, \bar{e}_3\}$ are two distinct transitive three-element sets. The cell labels form the regular six-element S_3-set. No equivariant bijection exists. Equivalently, over $\mathbb{Q}$ the character of the basis permutation module on the conjugacy classes $(1), (12), (123)$ is

$$(6, 2, 0),$$

whereas the regular representation has character $(6, 0, 0)$. The representations are not isomorphic even over $\mathbb{Q}$. □

This distinction also locates the chamber construction: the Kuhn fan uses the angular summand L_3, while the deck summand is needed for periods, monodromy, and filling. The map P produces the chamber fan; the different map Q makes the filling square commute. Replacing both by a single untyped "six-direction map" loses precisely the information that makes the cusp morphism valid.

21.4 The exact map into the Gram boundary

Let

$$\mathfrak{a} = \{\mathrm{diag}(\lambda_1, \lambda_2, \lambda_3) : \lambda_1 + \lambda_2 + \lambda_3 = 0\} \subset \mathrm{Herm}_3^0(\mathbb{C}),$$

let $\jmath : A_{F,\mathbb{R}} \to \mathfrak{a}$ be the marked isomorphism $\jmath(n_i) = E_{ii} - I_3/3$, and let

$$\Phi_{\mathrm{G}}(H) = H - \lambda_{\min}(H) I_3.$$

Write $\mathrm{pr}_{\mathrm{ang}}(a, b) = a$. The maps which actually exist are

$$\Psi = \jmath \circ P \circ \mathrm{pr}_{\mathrm{ang}} : \Lambda_{\mathrm{tor},\mathbb{R}} \longrightarrow \mathfrak{a}, \tag{722}$$

and

$$\Gamma = \Phi_{\mathrm{G}} \circ \Psi : \Lambda_{\mathrm{tor},\mathbb{R}} \longrightarrow \{H \in \mathrm{Herm}_3(\mathbb{C}) : H \succeq 0,\ \det H = 0\}. \tag{723}$$

Proposition 21.3 (Exact scope of the Gram-boundary map)**.** *The map Ψ is linear and S_3-equivariant, with*

$$\ker \Psi = \mathbb{R}(k, 0) \oplus (0, \overline{L}_{3,\mathbb{R}}). \tag{724}$$

The map Γ is continuous, positively homogeneous, S_3-equivariant, and piecewise linear. On the six projected Kuhn cones it is exactly the six diagonal Weyl-chamber map of theorem 20.6.

The map Γ is not additive, is not proper, ignores the complete deck summand, and does not descend to a map from a smooth torus fibre. Thus it is an exact continuous piecewise-linear chamber map on a real cocharacter space into the diagonal part of the nine-dimensional Gram carrier's eight-dimensional boundary, not a toric fan morphism and not a morphism from the toroidal total space to the Hermitian Gram cone.

Proof. Linearity and equivariance of Ψ follow from (717) and the Weyl permutation of diagonal entries. Since $\jmath$ is injective and $\ker P = \mathbb{R}k$, its kernel is exactly (724). The least-eigenvalue function is continuous, positively homogeneous, and linear on each closed Weyl chamber, which proves the corresponding assertions for Γ. The chamber formula is theorem 20.6.

The function $H \mapsto H - \lambda_{\min}(H) I_3$ is not additive across different Weyl chambers. The kernel in (724) has positive dimension, so Γ is not proper. It ignores the deck summand by definition. Finally, translation by the angular lattice does not leave it invariant: for example $\Gamma(e_1, 0) \neq \Gamma(0, 0)$. Hence it cannot descend through the quotient of the universal cover by Λ_{tor}. □

There is one further integral check on what the elliptic quotient does not preserve. On Λ_{tor}, define the standard alternating form

$$E_3((a,b),(a',b')) = a^{\mathsf{T}}b' - b^{\mathsf{T}}a'. \tag{725}$$

It is invariant under S_3 and M_3, but on the kernel it satisfies

$$E_3((k,0),(0,\bar{k})) = 3. \tag{726}$$

Every alternating form pulled back along Φ vanishes whenever one argument lies in K. Therefore E_3 is not the pullback of an alternating form on $A_F \oplus B_F$. In particular, the holomorphic elliptic quotient in theorem 20.3 is not a morphism preserving this chosen principal lattice form. This is an integral statement about (725); it does not assert that either nonalgebraic torus family carries a polarization.

Audit record 21.4 (Rank-six conclusion)**.** The native higher-rung object is the monodromy lattice $\Lambda_{\mathrm{tor}} = H_1(T^6;\mathbb{Z})$, not a cyclic group. Its exact relation to the nine-dimensional Hermitian construction factors through the angular quotient to the A_2 weight lattice and then through the marked Cartan/Weyl fan. The deck half is indispensable for the period and filling squares but invisible to the Gram-boundary chamber map. The separate six-coordinate group $H^4(U;\mathbb{Z})$ remains the correct classifier for bundle defects on the original S^6 regular locus; no identification of these two rank-six groups is used.

22 The chamber torsor inside the nine-dimensional matrix carrier

The preceding section separates the two three-element basis orbits of Λ_{tor} from the regular six-element set of Kuhn chambers. There is nevertheless a natural place where the regular six-set occurs. It is not the lattice Λ_{tor} itself, but the off-diagonal part of the rank-nine bilinear carrier

$$\mathcal{M}_{\mathbb{Z}} := \mathrm{Hom}(\overline{L}_3, L_3) \cong \mathrm{Mat}_{3\times 3}(\mathbb{Z}).$$

This is the integral real form underlying the nine-real-dimensional Hermitian carrier. The distinction between the full carrier and its positive-semidefinite cone will be essential.

Retain the marked bases e_1, e_2, e_3 of L_3 and $\bar{e}_1, \bar{e}_2, \bar{e}_3$ of $\overline{L}_3$. Let E_{ij} denote the homomorphism which sends $\bar{e}_j$ to e_i and annihilates the other two basis vectors. The diagonal S_3-action on the two rank-three lattices induces

$$\sigma \cdot X = R_\sigma X R_\sigma^{-1}, \qquad \sigma \cdot E_{ij} = E_{\sigma(i),\sigma(j)}. \tag{727}$$

Put

$$\Delta = \bigoplus_{i=1}^{3} \mathbb{Z}E_{ii}, \qquad \mathcal{O} = \bigoplus_{i\neq j} \mathbb{Z}E_{ij}. \tag{728}$$

Theorem 22.1 (The regular chamber torsor in rank nine)**.** *There is an integral S_3-module decomposition*

$$\mathcal{M}_{\mathbb{Z}} = \Delta \oplus \mathcal{O}, \qquad \Delta \cong \mathbb{Z}[S_3/S_2], \qquad \mathcal{O} \cong \mathbb{Z}[S_3]. \tag{729}$$

If a chamber is indexed in one-line notation by $\pi = (\pi_1\pi_2\pi_3) \in S_3$, then

$$\vartheta(c_\pi) = E_{\pi_1,\pi_3} \tag{730}$$

is an S_3-equivariant bijection from the six Kuhn chambers at a marked vertex to the six-element basis of $\mathcal{O}$. Equivalently, these are the six ordered mixed pairs

$$\big((e_i,0),(0,\bar{e}_j)\big), \qquad i \neq j. \tag{731}$$

They are exactly the mixed basis pairs which are orthogonal for the integral alternating form E_3, since

$$E_3\big((e_i,0),(0,\bar{e}_j)\big) = \delta_{ij}. \tag{732}$$

The three excluded diagonal pairs $i = j$ comprise the basis of Δ and are the symplectically dual mixed pairs.

Proof. Equation (728) partitions the nine matrix units and is preserved by (727), proving the direct-sum assertion. The action is transitive on the three diagonal units, and the stabilizer of E_{11} is the subgroup permuting 2 and 3; hence $\Delta \cong \mathbb{Z}[S_3/S_2]$.

The action on ordered pairs (i,j) with $i \neq j$ is free: a permutation fixing two distinct letters fixes the remaining letter and is therefore the identity. It is transitive because any ordered pair of distinct letters can be carried to any other. Thus $\mathcal{O}$ is the regular permutation module. For each ordered pair (i,j), there is a unique remaining letter k, and hence a unique one-line permutation (i,k,j). This proves that (730) is a bijection. Moreover,

$$\vartheta(c_{\sigma\pi}) = E_{\sigma(\pi_1),\sigma(\pi_3)} = \sigma \cdot E_{\pi_1,\pi_3},$$

so it is equivariant. Finally, (732) is immediate from the defining formula $E_3((a,b),(a',b')) = a^{\mathsf{T}}b' - b^{\mathsf{T}}a'$. □

22.1 The signed quotient and the exact chamber incidence

The repeated indices on the angular and deck bases determine a marking $\iota : \overline{L}_3 \to L_3$, $\iota(\bar{e}_i) = e_i$. Define

$$s : A_F \oplus B_F \longrightarrow A_F, \qquad s(x,y) = x - B_0 y, \qquad D = s \circ (P \oplus Q). \tag{733}$$

Since $B_0 Q = P$, this means, in the retained bases,

$$D(a,b) = Pa - B_0 Q b = P(a - \iota b), \qquad [D] = [P \mid -P]. \tag{734}$$

The formula is typed: Q is applied to the deck lattice and B_0 then carries its value from B_F to A_F.

Theorem 22.2 (Unimodular pair–Kuhn–Weyl incidence)**.** *The map D is surjective, with saturated kernel*

$$\ker D = \mathbb{Z}\langle (e_1,\bar{e}_1), (e_2,\bar{e}_2), (e_3,\bar{e}_3), (k,0)\rangle, \qquad k = e_1 + e_2 + e_3. \tag{735}$$

For $i \neq j$, put

$$U_{ij} = \mathbb{Z}(e_i,0) \oplus \mathbb{Z}(0,\bar{e}_j).$$

Then U_{ij} is primitive and isotropic for E_3, and

$$D|_{U_{ij}} : U_{ij} \xrightarrow{\sim} A_F, \qquad (e_i,0) \longmapsto n_i, \quad (0,\bar{e}_j) \longmapsto -n_j \tag{736}$$

is an integral isomorphism.

Let k_{ij} be the remaining index and $\pi_{ij} = (i,k_{ij},j)$. Under the Cartan marking $\jmath(n_i) = \nu_i = E_{ii} - I_3/3$, the positive quadrant in $U_{ij,\mathbb{R}}$ maps isomorphically to the Weyl chamber

$$\mathcal{C}_{ij} = \mathbb{R}_{\geq 0}\nu_i + \mathbb{R}_{\geq 0}(-\nu_j) = \{\mathrm{diag}(\lambda) : \lambda_i \geq \lambda_{k_{ij}} \geq \lambda_j,\ \textstyle\sum_r \lambda_r = 0\}. \tag{737}$$

The assignments

$$((e_i,0),(0,\bar{e}_j)) \longmapsto \pi_{ij} \longmapsto [\sigma_{0,\pi_{ij}}] \longmapsto \mathcal{C}_{ij} \tag{738}$$

are equivariant bijections of S_3-torsors, and

$$P(\sigma_{0,\pi_{ij}}) = \mathrm{conv}\{0, n_i, -n_j\}. \tag{739}$$

The vector

$$D(e_i,\bar{e}_j) = n_i - n_j \tag{740}$$

is the interior A_2-root bisector of this chamber. With the source and target orientations fixed by the displayed bases,

$$\det[e_i \mid e_i + e_{k_{ij}} \mid k] = \mathrm{sgn}(\pi_{ij}) = \det[n_i \mid -n_j]. \tag{741}$$

Proof. Since P is surjective, so is D. Equation (734) gives $D(a,b)=0$ precisely when $a-\iota b=tk$ for a unique $t\in\mathbb{Z}$. Hence

$$(a,b)=\sum_{r=1}^{3} b_r(e_r,\bar{e}_r)+t(k,0),$$

and uniqueness gives (735). Surjectivity implies that the kernel is saturated.

The plane U_{ij} is spanned by two members of a basis of Λ_{tor}, hence is primitive. Equation (732) makes it isotropic exactly when $i\neq j$. In its displayed basis the restriction of D has columns $[n_i \mid -n_j]$. Directly from

$$n_1=\begin{pmatrix}1\\0\end{pmatrix},\qquad n_2=\begin{pmatrix}-1\\1\end{pmatrix},\qquad n_3=\begin{pmatrix}0\\-1\end{pmatrix},$$

one obtains $\det[n_i \mid -n_j]=\pm 1$ for $i\neq j$, proving (736).

For $x,y\geq 0$, the diagonal matrix $H=x\nu_i-y\nu_j$ satisfies

$$\lambda_i-\lambda_{k_{ij}}=x,\qquad \lambda_{k_{ij}}-\lambda_j=y.$$

Conversely these two differences, together with trace zero, recover H, which proves (737) and the cone isomorphism. The four vertices of $\sigma_{0,(i,k_{ij},j)}$ have P-images

$$0,\quad n_i,\quad n_i+n_{k_{ij}}=-n_j,\quad n_1+n_2+n_3=0,$$

proving (739). Ordered distinct pairs, one-line permutations, Kuhn-cell translation classes, and Weyl chambers are all free transitive S_3-sets, and the displayed assignments commute with the coordinate action. At $x=y=1$, $\jmath(n_i-n_j)=E_{ii}-E_{jj}$, whose ordered diagonal entries in positions (i,k_{ij},j) are $(1,0,-1)$; this is the interior bisector.

Finally, relative to the ordered basis $(e_i,e_{k_{ij}},e_j)$, the columns $(e_i,e_i+e_{k_{ij}},k)$ form an upper unitriangular matrix. Their determinant in the fixed source basis is therefore $\operatorname{sgn}(\pi_{ij})$. For $(i,k_{ij},j)=(1,2,3)$, $[n_i \mid -n_j]=I_2$; equivariance and $\det\rho_A(\sigma)=\operatorname{sgn}(\sigma)$ prove the second equality in (741). □

Proposition 22.3 (Exact marking and monodromy boundary)**.** *There is no endomorphism $T:A_F\to A_F$ satisfying*

$$DM_3=TD. \tag{742}$$

Moreover, for every $c\in\mathbb{Z}$, the shear

$$S_c(a,b)=(a+c\iota b,b) \tag{743}$$

is an integral automorphism which fixes the angular lattice and its Kuhn fan, preserves c_3 and E_3, and commutes with M_3 and the diagonal S_3-action, but

$$DS_c(a,b)=D(a,b)+cP(\iota b). \tag{744}$$

Thus the pair–chamber correspondence is canonical relative to the retained angular/deck bases, their common indexing, P,Q,B_0, and the sign in (733)*; it is not an invariant of the unmarked data (M_3,E_3,c_3) together with the angular fan.*

Proof. The vector $(e_i,\bar{e}_i)$ lies in $\ker D$, whereas

$$M_3(e_i,\bar{e}_i)=(2e_i,\bar{e}_i),\qquad D(2e_i,\bar{e}_i)=n_i\neq 0.$$

Thus $\ker D$ is not M_3-invariant, which is necessary for (742).

The inverse of S_c is S_{-c}. Direct substitution gives

$$S_cM_3(a,b)=(a+(c+1)\iota b,b)=M_3S_c(a,b),$$

and proves preservation of c_3 and commutation with the diagonal coordinate action. Symmetry of the dot product gives

$$E_3(S_c(a,b),S_c(a',b')) = (a+c\iota b)^{\mathsf{T}}b' - b^{\mathsf{T}}(a'+c\iota b') = E_3((a,b),(a',b')).$$

Equation (744) follows from (734). For $c \neq 0$, the deck complement is moved, so the retained marked pair set and D are changed. □

The parity and opposition operations also retain exact information. The even and odd chamber triples are

$$\vartheta\{c_{123},c_{231},c_{312}\} = \{E_{13},E_{21},E_{32}\}, \tag{745}$$

$$\vartheta\{c_{132},c_{213},c_{321}\} = \{E_{12},E_{23},E_{31}\}. \tag{746}$$

Thus they are the two orientations of the directed three-cycle. Reversal of one-line notation, $(i,k,j) \mapsto (j,k,i)$, is carried by ϑ to matrix transposition $E_{ij} \mapsto E_{ji}$. This is the exact chamber opposition involution; it exchanges the two parity triples.

22.2 The real Hermitian representation and its integral defect

Let $\mathrm{Herm}_3(\mathbb{C})$ carry the S_3-action by conjugation with the real permutation matrices. Define

$$\Xi : \mathrm{Mat}_{3\times 3}(\mathbb{R}) \longrightarrow \mathrm{Herm}_3(\mathbb{C}), \qquad \Xi(X) = \frac{X+X^{\mathsf{T}}}{2} + \frac{i(X-X^{\mathsf{T}})}{2}. \tag{747}$$

Proposition 22.4 (Exact representation bridge and the positivity boundary)**.** *The map Ξ is an S_3-equivariant real-linear isomorphism. Its inverse sends a Hermitian matrix $H = S + iA$, with S real symmetric and A real skew-symmetric, to $S + A$. Consequently*

$$\mathrm{Herm}_3(\mathbb{C}) \cong_{\mathbb{R}[S_3]} 2\cdot\mathbf{1} \oplus 3\cdot V_{\mathrm{std}} \oplus \mathbf{sgn}. \tag{748}$$

For each unordered pair $\{i,j\}$, the symmetric and skew generators $E_{ij}+E_{ji}$ and $E_{ij}-E_{ji}$ span an index-two sublattice of $\mathbb{Z}E_{ij} \oplus \mathbb{Z}E_{ji}$. Hence the symmetric–skew off-diagonal splitting has total integral index

$$2^3 = 8. \tag{749}$$

In particular, Ξ is a real representation isomorphism, not an identification of the two preferred integral lattices.

Moreover, for $i \neq j$, $\Xi(E_{ij})$ has eigenvalues $2^{-1/2}, -2^{-1/2}, 0$. Thus the regular chamber torsor is located in the nine-dimensional Hermitian representation, but its matrix-unit image is not contained in the positive-semidefinite Gram cone or its boundary.

Proof. Every real matrix has the unique decomposition $X = S + A$, where $S = (X+X^{\mathsf{T}})/2$ is symmetric and $A = (X-X^{\mathsf{T}})/2$ is skew-symmetric. Equation (747) sends this pair to $S + iA$, which is Hermitian, and the displayed inverse proves bijectivity. Permutation conjugation commutes with transpose and with these two projections, proving equivariance.

The real permutation representation on three letters is $\mathbf{1} \oplus V_{\mathrm{std}}$. Therefore

$$\mathrm{Mat}_{3\times 3}(\mathbb{R}) \cong (\mathbf{1} \oplus V_{\mathrm{std}})^{\otimes 2} \cong 2\cdot\mathbf{1} \oplus 3\cdot V_{\mathrm{std}} \oplus \mathbf{sgn},$$

where $V_{\mathrm{std}} \otimes V_{\mathrm{std}} \cong \mathbf{1} \oplus V_{\mathrm{std}} \oplus \mathbf{sgn}$. For one unordered pair, the change-of-basis matrix from (E_{ij},E_{ji}) to $(E_{ij}+E_{ji}, E_{ij}-E_{ji})$ has determinant -2. Multiplication over the three unordered pairs proves (749).

Finally, on the i,j coordinate plane, $\Xi(E_{ij})$ has block

$$\begin{pmatrix} 0 & (1+i)/2 \\ (1-i)/2 & 0 \end{pmatrix},$$

whose eigenvalues are $\pm|1+i|/2 = \pm 2^{-1/2}$; the remaining coordinate has eigenvalue zero. Hence it is indefinite. □

22.3 The positive six-set is a different equivariant realization

Although the matrix units themselves are indefinite, the spectral-floor map of the preceding section gives a second realization of the same chamber torsor in the rank-two positive-semidefinite boundary. If $\pi = (i, k, j)$, then

$$G_\pi = 2E_{ii} + E_{kk} \tag{750}$$

has diagonal entries equal to a permutation of $(2, 1, 0)$. The assignment

$$E_{ij} \longmapsto 2E_{ii} + E_{kk}, \qquad \{i, j, k\} = \{1, 2, 3\}, \tag{751}$$

is therefore an equivariant bijection of the two marked six-element sets. It extends uniquely by linearity to a homomorphism $\mathcal{O} \to \Delta$, but that extension is neither the map Ξ nor the nonlinear spectral-floor map on all of $\mathrm{Herm}_3^0(\mathbb{C})$. Thus (751) records exact chamber incidence; it does not manufacture a positive monodromy-compatible morphism from the toroidal family.

Audit record 22.5 (What the native $\mathbb{Z}^6$ result becomes)**.** The six Kuhn chambers are not the six generators of $H_1(T^6; \mathbb{Z})$. They are canonically, relative to the retained marking, the six symplectically orthogonal ordered angular–deck pairs and hence the six off-diagonal basis elements in the associated rank-nine bilinear carrier. This places the regular S_3-torsor exactly inside the nine-dimensional program while preserving the different roles of angular cycles, deck cycles, monodromy, and the positive-semidefinite Gram boundary. The construction is a representation and incidence theorem; positivity and global toroidal descent remain separate questions and are not inferred from the equality of dimensions or orbit cardinalities.

23 The Lorentz observation morphism and the full nine-dimensional fibre

The rank-three Hermitian carrier and the four-real-dimensional Lorentz cone are not identical objects. The required next step is therefore to compute their exact relation. This section constructs a split positive observation morphism, determines every one of its fibres, identifies the five retained directions beyond the observed four-vector, and proves the complete cubic coupling among the resulting Lorentz-vector, spinor, and scalar coordinates. No direction is removed from the source calculation.

23.1 The marked angular epimorphism and its exact splitting

Retain the angular lattice matrix and its primitive kernel vector:

$$P = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix}, \qquad u = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \qquad Pu = 0. \tag{752}$$

The standard Hermitian products in these displayed marked bases give

$$C := PP^* = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \qquad C^{-1} = \frac{1}{3}\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \qquad \det C = 3. \tag{753}$$

Define the corresponding orthogonal right inverse and the projection onto the displayed kernel line by

$$R := P^* C^{-1} = \frac{1}{3}\begin{pmatrix} 2 & 1 \\ -1 & 1 \\ -1 & -2 \end{pmatrix}, \qquad \Pi := \frac{uu^*}{3}, \qquad Q_u := I_3 - \Pi. \tag{754}$$

The choice of R uses the displayed marking and its standard Hermitian products. All formulas below retain that choice explicitly.

Lemma 23.1 (Exact splitting identities)**.** *The matrices in* (752)–(754) *satisfy*

$$PR = I_2, \qquad u^*R = 0, \qquad R^*R = C^{-1}, \qquad RP = Q_u, \qquad \Pi R = 0, \qquad P\Pi = 0. \tag{755}$$

Moreover, with

$$X = (R\ u) = \begin{pmatrix} 2/3 & 1/3 & 1 \\ -1/3 & 1/3 & 1 \\ -1/3 & -2/3 & 1 \end{pmatrix}, \tag{756}$$

one has

$$\det X = 1, \qquad X^{-1} = \begin{pmatrix} P \\ u^*/3 \end{pmatrix}. \tag{757}$$

Proof. Direct multiplication, with every denominator retained, gives

$$PR = \frac{1}{3}\begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix}\begin{pmatrix} 2 & 1 \\ -1 & 1 \\ -1 & -2 \end{pmatrix} = I_2$$

and

$$RP = \frac{1}{3}\begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix} = I_3 - \frac{1}{3}\begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} = Q_u.$$

The identities involving u, R^*R, and Π follow from the same displayed matrices and from $R = P^*(PP^*)^{-1}$. Finally,

$$\det(3X) = \det\begin{pmatrix} 2 & 1 & 3 \\ -1 & 1 & 3 \\ -1 & -2 & 3 \end{pmatrix} = 27,$$

so $\det X = 27/3^3 = 1$. Both products

$$\begin{pmatrix} P \\ u^*/3 \end{pmatrix}(R\ u) \quad \text{and} \quad (R\ u)\begin{pmatrix} P \\ u^*/3 \end{pmatrix}$$

are the identity by (755); this proves (757). □

23.2 A positive observation morphism with all fibres retained

Put

$$\mathcal{J}_3 = \mathrm{Herm}_3(\mathbb{C}), \qquad \mathcal{J}_2 = \mathrm{Herm}_2(\mathbb{C}), \qquad \mathcal{P}_r = \{H \in \mathcal{J}_r : H \succeq 0\}.$$

Define real-linear maps

$$\mathcal{O}_P : \mathcal{J}_3 \longrightarrow \mathcal{J}_2, \qquad \mathcal{O}_P(H) = PHP^*, \tag{758}$$

and

$$\jmath_P : \mathcal{J}_2 \longrightarrow \mathcal{J}_3, \qquad \jmath_P(G) = RGR^*. \tag{759}$$

Theorem 23.2 (Split observation and the exact five-dimensional kernel)**.** *The observation morphism is surjective and positive, and its displayed section is a positive linear embedding:*

$$\mathcal{O}_P\jmath_P = \mathrm{id}_{\mathcal{J}_2}, \qquad \jmath_P\mathcal{O}_P(H) = Q_uHQ_u. \tag{760}$$

Consequently

$$\mathcal{J}_3 = \operatorname{im}\jmath_P \oplus \ker\mathcal{O}_P, \qquad \dim_{\mathbb{R}}\operatorname{im}\jmath_P = 4, \qquad \dim_{\mathbb{R}}\ker\mathcal{O}_P = 5. \tag{761}$$

The kernel is explicitly

$$\ker \mathcal{O}_P = \{a\,uu^* + uv^* + vu^* : a \in \mathbb{R},\ v \in u^\perp\}. \tag{762}$$

It is exactly the tangent space at $H_0 = uu^*$ to the nonzero rank-one positive-semidefinite stratum

$$\mathcal{R}_1 = \{ww^* : w \in \mathbb{C}^3 \setminus \{0\}\} \subset \mathcal{P}_3. \tag{763}$$

On the positive cone, the complete zero fibre is the retained ray

$$\mathcal{O}_P^{-1}(0) \cap \mathcal{P}_3 = \{\lambda uu^* : \lambda \geq 0\}. \tag{764}$$

Proof. Equations (755) give

$$\mathcal{O}_P(\jmath_P(G)) = PRGR^*P^* = G$$

and

$$\jmath_P(\mathcal{O}_P(H)) = RPHP^*R^* = Q_uHQ_u.$$

Both maps preserve positive semidefiniteness because they are congruences. The first identity proves surjectivity, injectivity of $\jmath_P$, and the direct sum in (761). Since $\dim_{\mathbb{R}} \mathcal{J}_3 = 9$ and $\dim_{\mathbb{R}} \mathcal{J}_2 = 4$, the dimensions follow.

The condition $PHP^* = 0$ is equivalent to $Q_uHQ_u = 0$. Relative to $\mathbb{C}^3 = u^\perp \oplus \mathbb{C}u$, such a Hermitian operator has a zero $u^\perp$-to-$u^\perp$ block and therefore has the unique form (762). The differential at u of $F(w) = ww^*$ is

$$dF_u(\dot{w}) = u\dot{w}^* + \dot{w}u^*.$$

Writing $\dot{w} = (a/2)u + v + ibu$, with $a, b \in \mathbb{R}$ and $v \in u^\perp$, shows that its image is exactly (762); the imaginary radial direction ibu is the one-real-dimensional kernel of dF_u. Thus the image has dimension five and is the stated tangent space.

Finally let $H \succeq 0$ and $PHP^* = 0$. For every $\xi \in \mathbb{C}^2$,

$$0 = \xi^*PHP^*\xi = \|H^{1/2}P^*\xi\|^2.$$

Hence $H^{1/2}$ annihilates $\operatorname{im} P^* = u^\perp$. The range of H is contained in $\mathbb{C}u$, and Hermitian positivity then gives $H = \lambda uu^*$ for one $\lambda \geq 0$. The converse follows from $Pu = 0$. □

The observation morphism is not a dimension count. It factors through the already proved two-column Gram carrier. For $H \in \mathcal{P}_3$, put

$$\mathcal{Z}_P(H) = H^{1/2}P^* \in \mathrm{Mat}_{3\times 2}(\mathbb{C}). \tag{765}$$

Then, with no omitted scalar,

$$\mathcal{Z}_P(H)^*\mathcal{Z}_P(H) = PHP^* = \mathcal{O}_P(H). \tag{766}$$

23.3 Complete block coordinates and the cubic coupling

Theorem 23.3 (Lorentz-vector–spinor–scalar coordinates)**.** *Every $H \in \mathcal{J}_3$ has a unique expression*

$$H = X\begin{pmatrix} G & \psi \\ \psi^* & \lambda \end{pmatrix}X^* = RGR^* + R\psi u^* + u\psi^*R^* + \lambda uu^*, \tag{767}$$

where

$$G = PHP^*, \qquad \psi = \frac{1}{3}PHu, \qquad \lambda = \frac{1}{9}u^*Hu, \tag{768}$$

with $G \in \mathcal{J}_2$, $\psi \in \mathbb{C}^2$, and $\lambda \in \mathbb{R}$. Thus the displayed marking gives the exact real-vector-space decomposition

$$\mathcal{J}_3 \xrightarrow{\sim} \mathcal{J}_2 \oplus \mathbb{C}^2 \oplus \mathbb{R}, \qquad 9 = 4 + 4 + 1. \tag{769}$$

For $G = \left(\begin{smallmatrix} a & b \\ \bar{b} & d \end{smallmatrix}\right)$, *retain*

$$\operatorname{adj}(G) = \begin{pmatrix} d & -b \\ -\bar{b} & a \end{pmatrix}. \tag{770}$$

Then the complete determinant is

$$\boxed{\det H = \lambda \det G - \psi^* \operatorname{adj}(G)\psi.} \tag{771}$$

For $G \succeq 0$, *let* G^+ *be its Moore–Penrose inverse on* $\operatorname{im} G$, *extended by zero on* $\ker G$. *The complete positive fibre is*

$$\mathcal{O}_P^{-1}(G) \cap \mathcal{P}_3 = \left\{ X \begin{pmatrix} G & \psi \\ \psi^* & \lambda \end{pmatrix} X^* : \psi \in \operatorname{im} G, \; \lambda \geq \psi^* G^+ \psi \right\}. \tag{772}$$

In particular,

$$H \succ 0 \quad \Longleftrightarrow \quad G \succ 0 \;\; \textit{and} \;\; \lambda > \psi^* G^{-1} \psi. \tag{773}$$

If G *is singular and* $H \succeq 0$, *then* H *is singular. If* $G \succ 0$, *the boundary of its fibre is the graph* $\lambda = \psi^* G^{-1} \psi$.

Proof. Equation (757) gives

$$X^{-1} H X^{-*} = \begin{pmatrix} PHP^* & PHu/3 \\ u^*HP^*/3 & u^*Hu/9 \end{pmatrix},$$

which proves uniqueness, (767), and (768). Because $\det X = 1$, direct expansion of the 3×3 block determinant gives

$$\det H = \lambda(ad - |b|^2) - \begin{pmatrix} \bar{\psi}_1 & \bar{\psi}_2 \end{pmatrix} \begin{pmatrix} d & -b \\ -\bar{b} & a \end{pmatrix} \begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix},$$

which is (771).

Invertible congruence by X preserves positivity. Suppose the block matrix is positive semidefinite. Its upper-left principal block gives $G \succeq 0$. If $v \in \ker G$, positivity of the restriction to $\mathbb{C}(v, 0) \oplus \mathbb{C}(0, 1)$ forces $v^*\psi = 0$; hence $\psi \perp \ker G$, or equivalently $\psi \in \operatorname{im} G$. Write $\psi = G^{1/2} w$ with $w \in \operatorname{im} G$. Completing the square gives

$$\begin{pmatrix} x \\ s \end{pmatrix}^* \begin{pmatrix} G & \psi \\ \psi^* & \lambda \end{pmatrix} \begin{pmatrix} x \\ s \end{pmatrix} = \|G^{1/2}x + sw\|^2 + |s|^2(\lambda - \|w\|^2).$$

Here $\|w\|^2 = \psi^* G^+ \psi$. This proves necessity and sufficiency of (772). The strict case is the ordinary Schur-complement criterion (773).

If G is singular, then $\psi \in \operatorname{im} G$ and $\operatorname{adj}(G)$ has image in $\ker G$. Thus $\psi^* \operatorname{adj}(G)\psi = 0 = \det G$, so (771) gives $\det H = 0$. If $G \succ 0$, the same formula becomes

$$\det H = \det G \, (\lambda - \psi^* G^{-1} \psi),$$

which proves the last assertion. □

23.4 The embedded Lorentz group and the meaning of $9 = 4 + 4 + 1$

For $g \in GL_2(\mathbb{C})$, define

$$T_P(g) = RgP + \Pi = X \begin{pmatrix} g & 0 \\ 0 & 1 \end{pmatrix} X^{-1}. \tag{774}$$

Theorem 23.4 (Exact Lorentz-vector, Weyl-spinor, and scalar action)**.** *The map $T_P : GL_2(\mathbb{C}) \to GL_3(\mathbb{C})$ is an injective group homomorphism satisfying*

$$T_P(g)u = u, \qquad PT_P(g) = gP, \qquad T_P(g)R = Rg, \qquad \det T_P(g) = \det g. \tag{775}$$

Under congruence by $T_P(g)$, the block coordinates transform exactly as

$$(G, \psi, \lambda) \longmapsto (gGg^*, g\psi, \lambda). \tag{776}$$

Consequently

$$\mathcal{O}_P(T_P(g)HT_P(g)^*) = g\mathcal{O}_P(H)g^*, \qquad T_P(g)\jmath_P(G)T_P(g)^* = \jmath_P(gGg^*). \tag{777}$$

For $g \in SL_2(\mathbb{C})$, congruence on $\mathcal{J}_2$ is the standard double-cover action on Minkowski space. In the retained coordinates

$$G(t, x_1, x_2, x_3) = \begin{pmatrix} t + x_3 & x_1 - ix_2 \\ x_1 + ix_2 & t - x_3 \end{pmatrix}, \tag{778}$$

one has

$$\det G = t^2 - x_1^2 - x_2^2 - x_3^2, \qquad G \succeq 0 \Longleftrightarrow t \geq 0, \quad t^2 \geq x_1^2 + x_2^2 + x_3^2. \tag{779}$$

The induced Lorentz-vector homomorphism

$$\rho_{\mathrm{vec}} : SL_2(\mathbb{C}) \longrightarrow GL(\mathcal{J}_2), \qquad \rho_{\mathrm{vec}}(g)(G) = gGg^*, \tag{780}$$

has kernel $\{I_2, -I_2\}$ and image $SO^+(1,3)$. The embedding T_P, and hence the action on the full $4+4+1$ carrier, is faithful. Indeed, on the full nine-dimensional carrier $-I_2$ fixes G and λ but sends $\psi \mapsto -\psi$. Thus the full carrier retains the spin double cover, and (769) *is exactly the real representation sum*

$$\boxed{\textit{Lorentz four-vector} \ \oplus \ \textit{defining complex spin representation } \mathbb{C}^2 \ \oplus \ \textit{real scalar.}} \tag{781}$$

With the convention that g acts on undotted spinor indices, the middle summand is called a left Weyl spinor; reversing that convention reverses the chirality name but not the displayed action $\psi \mapsto g\psi$.

Proof. The second expression in (774) and (757) prove the homomorphism, determinant, and injectivity assertions. Equations (755) give the other identities directly. Moreover,

$$T_P(g)X = X(g \oplus 1),$$

so congruence of (767) gives (776). The two equivariance identities follow.

For $g \in SL_2(\mathbb{C})$,

$$\det(gGg^*) = |\det g|^2 \det G = \det G,$$

and positive semidefiniteness is preserved. Hence congruence defines a homomorphism to the identity-component (proper orthochronous) Lorentz group of (779). If $gGg^* = G$ for every Hermitian G, then $gg^* = I_2$ by taking $G = I_2$; the remaining equations say that g commutes with every Hermitian matrix. Thus g is scalar, and $\det g = 1$ gives $g = \pm I_2$.

The differential has zero kernel: if $AG + GA^* = 0$ for all Hermitian G, then $A + A^* = 0$ at $G = I_2$, and the remaining equation is $[A, G] = 0$ for every Hermitian G. Hence A is scalar; $\operatorname{tr} A = 0$ gives $A = 0$. Both $SL_2(\mathbb{C})$ and $SO^+(1,3)$ have real dimension six, so the image is an open subgroup of the connected group $SO^+(1,3)$, hence the whole group. Finally, (776) gives the action of the central element and identifies the three standard real representations. □

The cubic in (771) is invariant under the full embedded $SL_2(\mathbb{C})$ action. Indeed,

$$\operatorname{adj}(gGg^*) = g^{-*} \operatorname{adj}(G) g^{-1} \qquad (g \in SL_2(\mathbb{C})), \tag{782}$$

so $(g\psi)^* \operatorname{adj}(gGg^*)(g\psi) = \psi^* \operatorname{adj}(G)\psi$. For the larger $GL_2(\mathbb{C})$ action it is instead a relative invariant:

$$\det(T_P(g)HT_P(g)^*) = |\det g|^2 \det H. \tag{783}$$

Thus no GL_2-invariance is being inferred from the SL_2 formula.

23.5 The Lorentz cone inside the rank-loss S^8 and its normal opening

Let

$$e_2(K) = \frac{1}{2}\big((\operatorname{tr} K)^2 - \operatorname{tr}(K^2)\big) \qquad (K \in \mathcal{J}_3). \tag{784}$$

Theorem 23.5 (Proper Lorentz-cone section and determinant opening)**.** *For every $G \in \mathcal{J}_2$,*

$$\operatorname{rank} \jmath_P(G) = \operatorname{rank} G, \qquad \det \jmath_P(G) = 0, \qquad e_2(\jmath_P(G)) = \frac{1}{3}\det G. \tag{785}$$

Hence $\jmath_P$ maps the entire future Lorentz cone properly and injectively into

$$\partial\mathcal{P}_3 = \{H \succeq 0 : \det H = 0\} = \mathcal{P}_{3,\le 2}. \tag{786}$$

It extends to an embedding of one-point compactifications

$$\jmath_P^+ : \mathcal{P}_2^+ \cong B^4 \hookrightarrow (\partial\mathcal{P}_3)^+ \cong S^8. \tag{787}$$

The timelike, nonzero null, and zero strata map respectively to rank two, rank one, and rank zero.

For every real ε, the affine slice through $H_0 = uu^*$ satisfies the exact identity

$$\boxed{\det\big(uu^* + \varepsilon \jmath_P(G)\big) = \varepsilon^2 \det G.} \tag{788}$$

For $\varepsilon > 0$ and $G \succeq 0$, timelike G maps to the interior of $\mathcal{P}_3$, nonzero null G maps to its rank-two boundary, and $G = 0$ maps to H_0. The restricted determinant Hessian is

$$D^2(\det)_{H_0}\big(\jmath_P(G), \jmath_P(K)\big) = \operatorname{tr} G \operatorname{tr} K - \operatorname{tr}(GK), \tag{789}$$

so

$$\frac{1}{2}D^2(\det)_{H_0}\big(\jmath_P(G), \jmath_P(G)\big) = \det G = t^2 - x_1^2 - x_2^2 - x_3^2. \tag{790}$$

The surjection $\mathcal{O}_P : \partial\mathcal{P}_3 \to \mathcal{P}_2$ is not proper and therefore does not extend by sending infinity to infinity. Thus (787) *is an actual compactified embedding, whereas the reverse compactified arrow is not supplied by $\mathcal{O}_P$.*

Proof. Because R has complex rank two, congruence by R preserves the rank of a 2×2 Hermitian matrix. Its image has rank at most two, so its 3×3 determinant vanishes. When G is invertible, the product of the two nonzero eigenvalues of RGR^* equals

$$\det(GR^*R) = \det G \det(C^{-1}) = \frac{1}{3}\det G.$$

That product is $e_2(RGR^*)$. Both sides are polynomial in the four real entries of G, so the identity extends to singular G, proving (785). The map is a linear embedding between finite-dimensional spaces, so its restriction to the closed cone is proper. The compactification statement now follows from the previously proved identifications $\mathcal{P}_2^+ \cong B^4$ and $(\partial\mathcal{P}_3)^+ \cong S^8$.

Using (756),

$$uu^* + \varepsilon \jmath_P(G) = X \begin{pmatrix} \varepsilon G & 0 \\ 0 & 1 \end{pmatrix} X^*.$$

Since $\det X = 1$, taking determinants proves (788). The rank and positivity statements follow from this same block matrix. Polarizing $\det(sG + tK)$ gives

$$\det(sG + tK) = s^2 \det G + st\big(\operatorname{tr} G \operatorname{tr} K - \operatorname{tr}(GK)\big) + t^2 \det K,$$

which proves (789) and (790).

For nonproperness choose $G_0 = \operatorname{diag}(1,0)$ and

$$H_n = \jmath_P(G_0) + nuu^*, \qquad n \in \mathbb{N}.$$

The two summands have orthogonal ranges, so $H_n \succeq 0$ has rank two and lies in $\partial\mathcal{P}_3$. Its norm tends to infinity, while $\mathcal{O}_P(H_n) = G_0$ for every n. A continuous extension taking the source point at infinity to the target point at infinity would therefore send this convergent compactified sequence both to G_0 and to infinity, which is impossible. □

23.6 The ten-dimensional determinant-phase carrier in the same coordinates

The square-matrix $SU(3)$-orbit theorem already identifies its affine quotient with

$$\mathcal{Y}_3 = \{(H,\omega) : H \in \mathcal{P}_3,\ |\omega|^2 = \det H\}. \tag{791}$$

Substitution of (767) gives an exact reformulation.

Corollary 23.6 (Lorentz-spinor-scalar-phase form of the $SU(3)$ quotient)**.** *There is a homeomorphism*

$$\mathcal{Y}_3 \cong \left\{(G,\psi,\lambda,\omega) : \begin{pmatrix} G & \psi \\ \psi^* & \lambda \end{pmatrix} \succeq 0, \quad |\omega|^2 = \lambda \det G - \psi^* \operatorname{adj}(G)\psi \right\}. \tag{792}$$

The $SL_2(\mathbb{C})$-action is

$$(G,\psi,\lambda,\omega) \longmapsto (gGg^*, g\psi, \lambda, \omega). \tag{793}$$

The complex determinant coordinate is invariant, and its zero locus is exactly the rank-deficient cone:

$$\omega = 0 \quad \Longleftrightarrow \quad \det H = 0. \tag{794}$$

After the proved one-point compactifications, this is the distinguished S^8 inside the ten-real-dimensional S^{10} carrier. The ray $\{(G,\psi,\omega) = (0,0,0), \lambda \geq 0\}$ is retained inside that zero locus.

Proof. The block-coordinate map is an invertible real-linear congruence, so it preserves positivity and is a homeomorphism. Equation (771) gives the displayed scalar relation. Equations (776) and (782) prove invariance. The equivalence (794) follows directly from $|\omega|^2 = \det H$. The compactification and the identification of its zero-determinant locus are the previously proved $SU(3)$-orbit and rank-loss compactification theorems. □

This corollary makes the relation among the dimensions explicit. The nine-real-dimensional Hermitian carrier has the $4+4+1$ decomposition in (769). Adding the complex determinant coordinate and retaining its one-real-modulus equation gives $9 + 2 - 1 = 10$ real dimensions. The determinant-zero locus removes the phase and imposes the cubic boundary equation, leaving the eight-real-dimensional rank-loss cone whose one-point compactification is S^8.

23.7 The holomorphic toric seed and its Lorentz-orbit completion

The local four-branch chart has the already typed monomial map

$$q(z_0,z_1,z_2,z_3) = (z_0z_3, z_1, z_2).$$

Define, without identifying angular and filling lattices,

$$\kappa(w_0,w_1,w_2) = \operatorname{diag}(w_0, w_1w_2), \qquad K = \kappa q. \tag{795}$$

Then

$$K(z) = \operatorname{diag}(z_0z_3, z_1z_2), \qquad \boxed{\det K(z) = z_0z_1z_2z_3.} \tag{796}$$

This is a holomorphic toric morphism into the diagonal complex matrices. On the retained positive real locus put

$$a = z_0 z_3, \qquad b = z_1 z_2, \qquad p_0 = \frac{a+b}{2}, \qquad p_3 = \frac{a-b}{2}, \qquad p_1 = p_2 = 0.$$

Then $K(z) = G(p) \succeq 0$ and

$$p_0^2 - p_3^2 = ab = z_0 z_1 z_2 z_3.$$

Thus the toric products are exactly the two light-cone coordinates $p_0 + p_3$ and $p_0 - p_3$.

For a retained $\mu > 0$, define the affine section

$$\Psi_\mu(G) = \mu u u^* + \jmath_P(G). \tag{797}$$

Equation (788) gives

$$\det \Psi_\mu(G) = \mu \det G, \qquad \Psi_\mu(G) \succeq 0 \Longleftrightarrow G \succeq 0, \qquad \operatorname{rank} \Psi_\mu(G) = 1 + \operatorname{rank} G. \tag{798}$$

Every $G \succeq 0$ has a spectral decomposition $G = v \operatorname{diag}(a, b) v^*$ with $a, b \geq 0$ and $v \in SU(2)$: a scalar phase changes the determinant of a chosen unitary diagonalizer without changing its congruence action. Hence

$$\Theta_\mu : SL_2(\mathbb{C}) \times \mathbb{R}^4_{\geq 0} \longrightarrow \Psi_\mu(\mathcal{P}_2), \qquad \Theta_\mu(g, z) = \Psi_\mu(g K(z) g^*) \tag{799}$$

is surjective. It obeys

$$\det \Theta_\mu(g, z) = \mu z_0 z_1 z_2 z_3. \tag{800}$$

On $(\mathbb{C}^*)^4$, the complete auxiliary fibre of K is

$$(\rho, \sigma) \cdot (z_0, z_1, z_2, z_3) = (\rho z_0, \sigma z_1, \sigma^{-1} z_2, \rho^{-1} z_3), \qquad (\rho, \sigma) \in (\mathbb{C}^*)^2. \tag{801}$$

Thus the local toric chart supplies a diagonal $1 + 1$ seed, its $SL_2(\mathbb{C})$-orbit supplies all future causal directions, and the two-complex-dimensional fibre in (801) is retained as the exact redundancy of this parametrization.

23.8 An exact colour-invariant Lorentz momentum and constrained dynamics

The preceding results supply a pointwise carrier with commuting colour and spin actions and a complete Lorentz-covariant orbit coordinate. Let

$$\mathscr{V} = \operatorname{Mat}_{3\times 2}(\mathbb{C}), \qquad U \cdot Z = UZ \quad (U \in SU(3)), \qquad g \triangleright Z = Z g^* \quad (g \in SL_2(\mathbb{C})). \tag{802}$$

These two left actions commute. Retain positive conversion constants $\alpha > 0$ and $c > 0$, with c carrying speed dimension, and define

$$p_\alpha(Z) = \alpha Z^* Z \in \mathcal{P}_2, \qquad m^2_{\alpha,c}(Z) = \frac{\det p_\alpha(Z)}{c^2}. \tag{803}$$

If p_α is assigned momentum dimension, then $[\alpha] = [p]/[Z]^2$ and $[m^2_{\alpha,c}] = [p]^2/[c]^2 = [\text{mass}]^2$; no value of α or c is selected or suppressed. Equivalently, $\det p_\alpha = m^2_{\alpha,c} c^2$, as required when the time component of the Hermitian momentum matrix is E/c.

Theorem 23.7 (Complete colour-orbit momentum and its mass invariant)**.** *The map p_α is $SU(3)$-invariant, is covariant for the spin action, and separates $SU(3)$-orbits:*

$$p_\alpha(UZ) = p_\alpha(Z), \qquad p_\alpha(g \triangleright Z) = g p_\alpha(Z) g^*. \tag{804}$$

It induces a homeomorphism

$$\mathscr{V}/SU(3) \xrightarrow{\sim} \mathcal{P}_2, \qquad [Z] \longmapsto p_\alpha(Z). \tag{805}$$

Writing the rows of Z as three two-component covectors, one has

$$\boxed{m^2_{\alpha,c}(Z) = \frac{\alpha^2}{c^2} \sum_{1\le a<b\le 3} |Z_{a1}Z_{b2} - Z_{a2}Z_{b1}|^2.} \tag{806}$$

It is invariant under both $SU(3)$ and $SL_2(\mathbb{C})$, and

$$m^2_{\alpha,c}(Z) > 0 \Longleftrightarrow \operatorname{rank}_{\mathbb{C}} Z = 2, \qquad m^2_{\alpha,c}(Z) = 0 \Longleftrightarrow \operatorname{rank}_{\mathbb{C}} Z \le 1. \tag{807}$$

For every rank-two Z_0,

$$m^2_{\alpha,c}(rZ_0) = r^4 m^2_{\alpha,c}(Z_0) \qquad (r > 0), \tag{808}$$

so the unrestricted kinematic carrier contains positive masses arbitrarily close to zero.

Proof. The covariance equations follow directly from (802). The orbit-separation and homeomorphism statement is the exact two-column $SU(3)$ Gram quotient, with multiplication by the retained positive scalar α. The 2×2 Gram determinant is the Cauchy–Binet identity

$$\det(Z^* Z) = \sum_{a<b} |Z_{a1}Z_{b2} - Z_{a2}Z_{b1}|^2,$$

which proves (806). Congruence by an $SL_2(\mathbb{C})$ matrix preserves the determinant, while left $SU(3)$ preserves Z^*Z. The sum vanishes exactly when all 2×2 minors vanish, which is equivalent to rank at most one. Finally $p_\alpha(rZ) = r^2 p_\alpha(Z)$, and taking the 2×2 determinant gives (808). □

For a retained mass $m_0 \ge 0$, the constraint

$$\mathscr{C}_{m_0,\alpha,c} = \{Z \in \mathscr{V} : \det p_\alpha(Z) = m_0^2 c^2\} \tag{809}$$

is invariant under both actions, and for $m_0 > 0$ its $SU(3)$-orbit space is exactly the future mass hyperboloid

$$\{p \in \mathcal{P}_2 : \det p = m_0^2 c^2\}. \tag{810}$$

Retain the Lorentz bilinear pairing

$$\langle A, B\rangle_L = \frac{1}{2}\big(\operatorname{tr} A \operatorname{tr} B - \operatorname{tr}(AB)\big), \qquad \langle A, A\rangle_L = \det A. \tag{811}$$

For paths $x : I \to \mathcal{J}_2$, $Z : I \to \mathscr{V}$, and a real multiplier $e : I \to \mathbb{R}$, define the first-order constrained action

$$\mathcal{S}_{m_0,\alpha,c}[x, Z, e] = \int_I \left(\langle p_\alpha(Z), \dot{x}\rangle_L - \frac{e}{2}\big(\det p_\alpha(Z) - m_0^2 c^2\big) \right) d\tau. \tag{812}$$

It is invariant under arbitrary τ-dependent left $SU(3)$ rotations of Z and under the constant $SL_2(\mathbb{C})$ action

$$x \longmapsto g x g^*, \qquad Z \longmapsto Z g^*. \tag{813}$$

If x has length dimension and τ retains its chosen parameter dimension, then $[e] = [x]/([\tau][p])$, so both summands in the integrand have dimension $[p][x]/[\tau]$ and the action has dimension $[p][x]$. Variation of e gives (809); compactly supported variation of x gives

$$\frac{d}{d\tau} p_\alpha(Z) = 0. \tag{814}$$

For a variation $\delta Z : I \to \mathscr{V}$, retain the exact differential

$$\delta p_\alpha = \alpha(\delta Z^* Z + Z^* \delta Z). \tag{815}$$

The Z-Euler–Lagrange equation is the weak identity

$$\langle \alpha(\delta Z^* Z + Z^* \delta Z), \dot{x} - e p_\alpha(Z) \rangle_L = 0 \quad \text{for every } \delta Z \in \mathscr{V}. \tag{816}$$

On the rank-two stratum this is equivalent to

$$\dot{x} = e p_\alpha(Z). \tag{817}$$

Indeed, for arbitrary $A = A^* \in \mathcal{J}_2$, the variation

$$\delta Z_A = \frac{1}{2\alpha} Z (Z^* Z)^{-1} A \tag{818}$$

satisfies $\delta p_\alpha = A$. Hence the differential of the Gram map is surjective there, and nondegeneracy of $\langle\ ,\ \rangle_L$ proves (817). On the rank-at-most-one stratum the differential need not be surjective, so the complete equation is (816); it is not replaced by the rank-two formula. The invariance of (811) follows by polarizing $\det(gAg^*) = \det A$. The multiplier and position equations follow by direct variation and integration by parts, while (816) follows from $D(\det)_p[A] = 2\langle p, A \rangle_L$.

Equation (812) is a complete constrained kinematic dynamics with an independently retained mass parameter. It does not dynamically select $m_0 > 0$: choosing $m_0 = 0$ is equally compatible, and (808) proves that the unrestrained carrier has no positive kinematic threshold. The next physical construction is thereby specified: introduce a field-theoretic kinetic operator and interaction on the full $(G, \psi, \lambda, \omega)$ carrier, derive rather than insert the selected spectral sector, and compare its gauge-invariant spectrum with the measured regime. The exact morphisms and representations proved above are the data that such a construction must preserve.

24 The common marked slice and its dynamics

Two Pro-mode packets dated 4 September 2026 describe a thirteen-dimensional continuation of the marked nine-dimensional Hermitian carrier. One uses the condition $u^* b = 0$ in a complex-plus-triplet decomposition; the other requires the observed quaternionic block PHP^* to be complex. We prove that these are the same embedded space, not merely isomorphic quadratic models. We retain the original nonunitary marking throughout. We then compute the omitted-coordinate force in the packet's scalar mechanics, give a constrained alternative, and correct two action conventions. The results below concern these specified matrices and dynamics; they do not certify every theorem in either supplied packet.

The subsequent two-cross continuation computes the entire scalar–Gauss domain in Section 24.28. Its classified parameters form $\mathcal{D} \times \{1, -1\}^2 \times U(1)^2$; retaining the frame phase ζ gives the literal domain after the sole identification $(\zeta, p_+, p_-) \sim (-\zeta, -p_+, -p_-)$. Corollary 24.70 constructs tuples at every original $t = g^2 > 0$. The two spatial residuals cannot vanish simultaneously on this class by Theorem 24.60; existence on the remaining unrestricted exceptional set is open.

Section 24.29 proves smooth embeddedness and the exact product $\mathcal{S}_{\mathrm{sg}} \cong \{1, -1\}^2 \times (0, \infty) \times \mathbb{R}^2 \times \mathbb{T}^3$, with $t = g^2$ the displayed positive coordinate. The four literal components have dimension six; every fixed positive coupling has four embedded five-dimensional components $\mathbb{R}^2 \times \mathbb{T}^3$. The inverse charts retain all phases and both signs, without a physical gauge quotient or an additional spatial equation.

24.1 Conventions, determinant, and the retained marking

Write $q = a + b\mathbf{j}$, with $a, b \in \mathbb{C}$, $\mathbf{j}^2 = -1$, $\mathbf{j}a = \bar{a}\mathbf{j}$. Thus

$$(a + b\mathbf{j})(c + d\mathbf{j}) = ac - b\bar{d} + (ad + b\bar{c})\mathbf{j}, \qquad \overline{a + b\mathbf{j}} = \bar{a} - b\mathbf{j}.$$

For complex matrices of the appropriate sizes this gives

$$(A + B\mathbf{j})(C + D\mathbf{j}) = AC - B\bar{D} + (AD + B\bar{C})\mathbf{j}, \qquad (A + B\mathbf{j})^* = A^* - B^{\mathsf{T}}\mathbf{j}.$$

In particular a quaternionic Hermitian matrix is uniquely $A + B\mathbf{j}$ with $A = A^{*}$ and $B = -B^{\mathsf{T}}$. For $b \in \mathbb{C}^3$ define

$$B(b) = \begin{pmatrix} 0 & \bar{b}_3 & -\bar{b}_2 \\ -\bar{b}_3 & 0 & \bar{b}_1 \\ \bar{b}_2 & -\bar{b}_1 & 0 \end{pmatrix}, \qquad \Theta(A, b) = A + B(b)\mathbf{j}.$$

The map $\Theta : \mathrm{Herm}_3(\mathbb{C}) \oplus \mathbb{C}^3 \to \mathrm{Herm}_3(\mathbb{H})$ is consequently a real-linear bijection.

We use the Moore determinant with value one on I_3. Its Hermitian rank-three formula is

$$N \begin{pmatrix} a & q_3 & \bar{q}_2 \\ \bar{q}_3 & d & q_1 \\ q_2 & \bar{q}_1 & f \end{pmatrix} = adf - a|q_1|^2 - d|q_2|^2 - f|q_3|^2 + 2\,\mathrm{Re}(q_1 q_2 q_3). \tag{819}$$

The quaternionic spectral theorem and determinant congruence law used below are recorded in S. Alesker, *Non-commutative linear algebra and plurisubharmonic functions of quaternionic variables*, Theorem 1.1.9, arXiv:math/0104209; see also W. Wang, *The linear algebra in the quaternionic pluripotential theory*, Propositions 2.1–2.3, arXiv:1805.07759. In our signs the faithful complex representation is

$$\mathcal{R}(A + B\mathbf{j}) = \begin{pmatrix} A & B \\ -\bar{B} & \bar{A} \end{pmatrix}.$$

Wang's representation has the opposite signs on both off-diagonal blocks; the exact conversion is conjugation by $\mathrm{diag}(I, -I)$.

Lemma 24.1 (The full cubic)**.** *For every $A = A^{*} \in \mathrm{Mat}_3(\mathbb{C})$ and $b \in \mathbb{C}^3$,*

$$N(\Theta(A, b)) = \det A - b^{*} A b. \tag{820}$$

Proof. Write the cyclic off-diagonal entries of A as $A_{23} = z_1$, $A_{31} = z_2$, $A_{12} = z_3$. Then $q_i = z_i + \bar{b}_i\mathbf{j}$ in (819), and $|q_i|^2 = |z_i|^2 + |b_i|^2$. The complex part of their product is

$$z_1 z_2 z_3 - \bar{b}_1 b_2 z_3 - z_1 \bar{b}_2 b_3 - \bar{b}_1 \bar{z}_2 b_3.$$

The three additional cross terms in twice its real part are the negatives of the off-diagonal terms of $b^{*}Ab$. The additional diagonal terms in (819) are $-a|b_1|^2 - d|b_2|^2 - f|b_3|^2$. All remaining terms are precisely the ordinary determinant of A. This proves the identity with all signs. □

Retain the marked matrices

$$\begin{gathered} P = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix}, \quad u = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \quad C = PP^{*} = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \\ R = P^{*}C^{-1} = \frac{1}{3}\begin{pmatrix} 2 & 1 \\ -1 & 1 \\ -1 & -2 \end{pmatrix}, \quad X = (R\ u), \quad X^{-1} = \begin{pmatrix} P \\ u^{*}/3 \end{pmatrix}, \quad \det X = 1. \end{gathered} \tag{821}$$

Direct multiplication gives $PR = I_2$, $u^{*}R = 0$, $RP = I_3 - uu^{*}/3$. In particular the metric retained by this marking is

$$D = X^{*}X = \mathrm{diag}(C^{-1}, 3), \qquad C^{-1} = \frac{1}{3}\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}. \tag{822}$$

It is not the identity matrix.

24.2 The two constructions are the same embedded space

Define, in the original quaternionic matrix space,

$$\mathcal{J}_P = \{H \in \mathrm{Herm}_3(\mathbb{H}) : PHP^* \in \mathrm{Herm}_2(\mathbb{C})\}, \qquad \mathcal{L}_u = \{\Theta(A,b) : u^*b = 0\}.$$

For any (A, b) put

$$X^{-1}AX^{-*} = \begin{pmatrix} G & z \\ z^* & \lambda \end{pmatrix}, \qquad X^*b = \begin{pmatrix} \chi \\ \rho \end{pmatrix}, \qquad \varepsilon = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}. \tag{823}$$

Here G is complex Hermitian, $z, \chi \in \mathbb{C}^2$, $\lambda \in \mathbb{R}$, $\rho \in \mathbb{C}$, and $\rho = u^*b$.

Theorem 24.2 (Embedded reconciliation)**.** *The two subspaces satisfy $\mathcal{J}_P = \mathcal{L}_u$. Their complete coordinate relation, with no change of the ambient matrix H, is*

$$H = X \begin{pmatrix} G & z + w\mathbf{j} \\ (z + w\mathbf{j})^* & \lambda \end{pmatrix} X^*, \qquad w = -\varepsilon\bar{\chi}, \qquad \chi = \varepsilon\bar{w}. \tag{824}$$

The inverse in the original complex-plus-triplet coordinates is

$$A = X \begin{pmatrix} G & z \\ z^* & \lambda \end{pmatrix} X^*, \qquad b = X^{-*} \begin{pmatrix} \varepsilon\bar{w} \\ 0 \end{pmatrix}.$$

Thus both thirteen-dimensional constructions have the same positive cone, the same restricted determinant, and the same embedded old complex space, which is $w = 0$, equivalently $\chi = 0$.

Proof. The skew matrices obey the following two identities in the fixed basis:

$$PB(b)P^{\mathsf{T}} = \overline{u^*b}\,\varepsilon, \qquad X^{-1}B(b)X^{-\mathsf{T}} = B(X^*b). \tag{825}$$

For the first identity the upper-right entry is $\bar{b}_3 + \bar{b}_2\mathbf{j} + \bar{b}_1\mathbf{j}$; skew symmetry gives the other entries. For the second, write $B(b)_{ij} = \sum_k \epsilon_{ijk}\bar{b}_k$. For any real invertible T, expansion of the alternating determinant gives

$$\sum_{a,b} T_{ia}T_{jb}\epsilon_{abk} = \det(T) \sum_l \epsilon_{ijl}(T^{-1})_{kl}.$$

Multiplication by $\bar{b}_k$ therefore gives $TB(b)T^{\mathsf{T}} = \det(T)B(T^{-\mathsf{T}}b)$. Set $T = X^{-1}$ and use $\det X = 1$. This proves (825). Its first identity proves the asserted equality of subspaces. At $\rho = 0$ the second identity is the explicit matrix

$$B\begin{pmatrix} \chi \\ 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & -\bar{\chi}_2 \\ 0 & 0 & \bar{\chi}_1 \\ \bar{\chi}_2 & -\bar{\chi}_1 & 0 \end{pmatrix}.$$

Its upper-right column is $-\varepsilon\bar{\chi}$. This is exactly (824); $\varepsilon^2 = -I_2$ gives the inverse. Every claim about the embedded matrix and its positive cone follows from this identity, rather than from matching dimensions or determinant levels. □

Proposition 24.3 (The restricted and unrestricted cubics)**.** *In* (823) *the unrestricted cubic is*

$$\begin{aligned} N_{15} = \lambda \det G - z^* \operatorname{adj}(G) z - \chi^* G \chi \\ - 2\,\mathrm{Re}(\rho\chi^* z) - \lambda|\rho|^2. \end{aligned} \tag{826}$$

On the common thirteen-space this becomes

$$N_{13} = \lambda \det G - z^* \operatorname{adj}(G) z - w^* \operatorname{adj}(G) w. \tag{827}$$

Proof. Since $\det X = 1$, the ordinary block determinant gives $\det A = \lambda \det G - z^* \operatorname{adj}(G) z$ first for invertible G and then, as a polynomial identity, for all G. Also

$$b^* A b = (\chi^*, \bar{\rho}) \begin{pmatrix} G & z \\ z^* & \lambda \end{pmatrix} \begin{pmatrix} \chi \\ \rho \end{pmatrix}.$$

Expansion and Lemma 24.1 prove (826). For $G = \left(\begin{smallmatrix} a & c \\ \bar{c} & d \end{smallmatrix}\right)$, direct multiplication gives $\varepsilon^* \operatorname{adj}(G) \varepsilon = G^{\mathsf{T}}$. Substituting $w = -\varepsilon \bar{\chi}$ shows $w^* \operatorname{adj}(G) w = \chi^* G \chi$, proving (827). In the second packet's notation $w = \varepsilon \bar{\psi}_R$, the exact dictionary is consequently $\chi = -\psi_R$. □

24.3 Positive fibres and the embedded sphere pair

Proposition 24.4 (All positive fibres)**.** *The matrix in* (824) *is positive semidefinite exactly when*

$$G \succeq 0, \qquad z, w \in \operatorname{Ran}_{\mathbb{C}} G, \qquad s := \lambda - z^* G^{\dagger} z - w^* G^{\dagger} w \geq 0.$$

Its quaternionic rank is $\operatorname{rank}_{\mathbb{C}} G + \mathbf{1}_{s>0}$*. Here* $G^{\dagger}$ *is the inverse on* $\operatorname{Ran} G$*, extended by zero on* $\ker G$*. In particular the fibre over* $G = 0$ *is exactly* $\{\lambda u u^* : \lambda \geq 0\}$*.*

Proof. Congruence by the invertible real matrix X preserves positivity and rank. Write $\psi = z + w\mathbf{j}$. For any complex Hermitian T, quaternionic multiplication gives

$$\psi^* T \psi = z^* T z + w^* T w;$$

the $\mathbf{j}$ coefficient is $z^* T w - w^{\mathsf{T}} \bar{T} \bar{z} = 0$, while the scalar w term is the conjugate of the real number $w^* T w$. Testing the full Hermitian form on $(v, 0)$ gives $G \succeq 0$. If $v \in \ker G$ and $v^* \psi \neq 0$, testing on (tv, q) with q chosen to make $\operatorname{Re}(v^* \psi q) \neq 0$ makes the form negative for one sufficiently large signed real t. Thus ψ is in the quaternionic range of G, exactly the two complex range conditions above. With $a = G^{\dagger} \psi$, direct multiplication gives

$$\begin{pmatrix} I & 0 \\ -a^* & 1 \end{pmatrix} \begin{pmatrix} G & \psi \\ \psi^* & \lambda \end{pmatrix} \begin{pmatrix} I & -a \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} G & 0 \\ 0 & s \end{pmatrix}.$$

Both triangular factors are invertible, proving positivity and rank in both directions. At $G = 0$ the two spinors vanish, and return through X gives the asserted ray. □

Proposition 24.5 (The old boundary inside the new boundary)**.** *Let* $\mathcal{C} = \mathcal{J}_P \cap \{H \succeq 0\}$*. Its determinant-zero locus is its boundary, and its one-point compactification satisfies*

$$(\mathcal{C}^+, \partial \mathcal{C}^+) \cong (B^{13}, S^{12}), \qquad ((\partial \mathcal{C})^+, (\partial \operatorname{Herm}_3(\mathbb{C})_{\succeq 0})^+) \cong (S^{12}, S^8).$$

The sphere pair is the standard linear pair and $S^{12} \setminus S^8 \cong S^3 \times \mathbb{R}^9$*.*

Proof. The identity belongs to $\mathcal{J}_P$, since $P I_3 P^* = C$ is complex. Set $\mathcal{J}_P^0 = \{K \in \mathcal{J}_P : \operatorname{tr} K = 0\}$. If $\lambda_{\min}(K)$ is the least quaternionic Hermitian eigenvalue, the mutually inverse maps are

$$K \longmapsto K - \lambda_{\min}(K) I_3, \qquad H \longmapsto H - \frac{\operatorname{tr} H}{3} I_3.$$

They identify $\mathcal{J}_P^0$ with the positive singular locus. Indeed, the first produces least eigenvalue zero; for singular positive H, the least eigenvalue after the second map is $-\operatorname{tr}(H)/3$. Eigenvalue continuity proves both maps continuous. More generally

$$(K, t) \longmapsto K + (t - \lambda_{\min}(K)) I_3, \quad (K, t) \in \mathcal{J}_P^0 \times [0, \infty)$$

is a homeomorphism onto $\mathcal{C}$, taking $t = 0$ to its boundary. This also proves that no other boundary points occur. Its inverse sends H to $(H - (\operatorname{tr} H/3) I_3, \lambda_{\min}(H))$. The one-point compactification of a closed

half-space is a closed ball, with the boundary compactifying to a sphere; for example inverse stereographic projection from a point on the equator gives this pair. On complex matrices the quaternionic eigenvalues are the same ordinary Hermitian eigenvalues. The trace-free pair is therefore the linear pair $(\mathbb{R}^{12}, \mathbb{R}^8)$ under the displayed homeomorphisms. Both compactifications have their infinity point in the smaller sphere, so its complement is $\mathbb{R}^8 \times (\mathbb{R}^4 \setminus \{0\}) \cong \mathbb{R}^8 \times S^3 \times \mathbb{R}$. This proves the full pair assertion, not just the two dimensions. □

24.4 The nonunitary Jordan-product correction

Theorem 24.6 (Completion with the retained metric)**.** *The Jordan subalgebra generated by $\mathcal{J}_P$ under $H \circ K = (HK + KH)/2$ is all of* $\mathrm{Herm}_3(\mathbb{H})$*. In the marked block coordinates the transported product is*

$$M \circ_D N = \frac{1}{2}(MDN + NDM), \qquad D = \mathrm{diag}(C^{-1}, 3), \tag{828}$$

not the ordinary block product.

Proof. For $F_X(M) = XMX^*$, multiplication gives $F_X(M) \circ F_X(N) = F_X((MDN + NDM)/2)$ exactly. Take

$$M = \begin{pmatrix} 0 & \psi \\ \psi^* & 0 \end{pmatrix}, \quad N = \begin{pmatrix} 0 & \phi \\ \phi^* & 0 \end{pmatrix}, \quad \psi = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \phi = \begin{pmatrix} 0 \\ \mathbf{j} \end{pmatrix}.$$

The upper-left block of $M \circ_D N$ is

$$\frac{3}{2}\begin{pmatrix} 0 & -\mathbf{j} \\ \mathbf{j} & 0 \end{pmatrix}.$$

The bottom scalar is zero: its two terms are $\mathbf{j}/3$ and $-\mathbf{j}/3$. The other blocks vanish. Replacing $\mathbf{j}$ by $\mathbf{k} = \mathbf{ij}$ gives the independent second missing direction. The common slice already contains every other coordinate in $\mathrm{Herm}_2(\mathbb{H}) \oplus \mathbb{H}^2 \oplus \mathbb{R}$. Hence the linear span of the slice and these two products is the whole fifteen-dimensional algebra. Any Jordan subalgebra containing the slice contains that span. This proves the closure. The 1/2 product printed in the quaternionic packet is correct for the ordinary product in the block frame, but the coefficient for the stated embedded slice under the original product is 3/2. □

24.5 The corrected group-action convention

Put $Z = (z\ w)$. The ordinary direct product has the following left action:

$$(g, k) \cdot (G, Z, \lambda) = (gGg^*, gZk^{-1}, \lambda), \quad g \in SL_2(\mathbb{C}), \quad k \in SU(2). \tag{829}$$

Composition gives $g_1 g_2 Z k_2^{-1} k_1^{-1} = (g_1 g_2) Z (k_1 k_2)^{-1}$, as required. The packet's gZk describes commuting left and right actions, or a left action of $SL_2(\mathbb{C}) \times SU(2)^{\mathrm{op}}$; it is not a left action of the usual direct product with those parameters. For instance $k_1 = \varepsilon$, $k_2 = \mathrm{diag}(i, -i)$ do not commute.

Both conventions give the same transformations after $k \mapsto k^{-1}$. To prove invariance, use

$$\mathrm{adj}(gGg^*) = (g^*)^{-1}\,\mathrm{adj}(G)\,g^{-1}, \qquad \det g = 1,$$

first on invertible G and then polynomially on all G. The two spinor terms in (827) are $\mathrm{tr}(Z^*\,\mathrm{adj}(G)Z)$; conjugation by k does not change this trace. Positivity under the left factor follows by quaternionic congruence with $\mathrm{diag}(g, 1)$ in the block frame. For the right factor, a unit quaternion $a + b\mathbf{j}$ acts on Z by

$$Z\begin{pmatrix} a & b \\ -\bar{b} & \bar{a} \end{pmatrix}.$$

This follows by multiplying $(z + w\mathbf{j})(a + b\mathbf{j})$ explicitly; it preserves $\psi^* G^\dagger \psi$, a real scalar, and the range conditions in Proposition 24.4. Finally, if an element fixes every G, then g is unitary and commutes with every complex Hermitian matrix; hence $g = \pm I_2$. Fixing every Z then forces $k = g$. The effective group is therefore $(SL_2(\mathbb{C}) \times SU(2))/\{(I, I), (-I, -I)\}$.

24.6 The full carrier still has an exact four-dimensional observation

Requiring the observed quaternionic block to be complex defines the thirteen-slice above. A different operation retains every coordinate of the full fifteen-dimensional carrier and applies a positive observation map. These two operations have different domains, with an exact relation.

Proposition 24.7 (Full positive observation and the hidden-coordinate bound)**.** *Fix the same complex subfield $\mathbb{C} = \mathbb{R} + \mathbb{R}\mathbf{i}$ and real matrix P. For a quaternionic Hermitian 2×2 matrix A put*

$$\pi_{\mathbb{C}}(A) = \tfrac{1}{2}(A + (\mathbf{i}I_2)^* A(\mathbf{i}I_2)).$$

Then the real-linear map

$$\mathscr{O} : \mathrm{Herm}_3(\mathbb{H}) \longrightarrow \mathrm{Herm}_2(\mathbb{C}), \qquad \mathscr{O}(H) = \pi_{\mathbb{C}}(PHP^*)$$

is positive, onto, and has positive section $G \mapsto RGR^*$. Its real kernel has dimension eleven. On writing $H = \Theta(A_0, b)$, its observed value is $G = PA_0P^*$ and $\rho = u^*b$ obeys

$$H \succeq 0 \quad \Longrightarrow \quad G \succeq 0, \qquad |\rho|^2 \le \det_{\mathbb{C}} G. \tag{830}$$

Every pair (G, ρ) satisfying these inequalities occurs. Thus the possible hidden ρ at a fixed G are exactly the closed complex disc of radius $\sqrt{\det G}$. At a null observed block, $\det G = 0$ *forces $\rho = 0$.*

Proof. For $q = a + b\mathbf{j}$, $a, b \in \mathbb{C}$, direct multiplication gives $(q - \mathbf{i}q\mathbf{i})/2 = a$. Thus $\pi_{\mathbb{C}}$ extracts the complex matrix part. It is positive as the average of the identity congruence and unitary congruence by $\mathbf{i}I_2$. Compression by P is positive as well. Since $PR = I_2$ and G is complex, $\mathscr{O}(RGR^*) = G$; positivity of this section is again congruence. Rank–nullity gives $15 - 4 = 11$.

The original skew identity (825) gives

$$PHP^* = G + \bar{\rho}\varepsilon\mathbf{j} = \begin{pmatrix} g_{11} & g_{12} + \bar{\rho}\mathbf{j} \\ \bar{g}_{12} - \bar{\rho}\mathbf{j} & g_{22} \end{pmatrix}.$$

Its Moore determinant is exactly $g_{11}g_{22} - |g_{12}|^2 - |\rho|^2 = \det_{\mathbb{C}} G - |\rho|^2$. A Hermitian quaternionic 2×2 matrix with real diagonal a, d and upper entry q is positive semidefinite precisely when $a, d \ge 0$ and $ad \ge |q|^2$: for $a > 0$ its quadratic form is $a|x + a^{-1}qy|^2 + (d - |q|^2/a)|y|^2$; for $a = 0$, positivity forces $q = 0$ by varying x, and then requires $d \ge 0$. Applying this criterion proves (830) and its converse for the displayed compressed block.

To realize every permitted pair in the original coordinates, take any $\lambda \ge 0$ and set

$$A_0 = RGR^* + \lambda uu^*, \qquad b = (\rho/3)u, \qquad H = \Theta(A_0, b).$$

The identities for $X = (R\ u)$ give, with $\chi = R^*b = 0$,

$$X^{-1}HX^{-*} = \begin{pmatrix} G + \bar{\rho}\varepsilon\mathbf{j} & 0 \\ 0 & \lambda \end{pmatrix}.$$

The preceding criterion and congruence prove positivity of H, while its observation and contraction are exactly G and ρ. This proves the claimed complete image and every disc fibre for (G, ρ). It does not discard the further z, w, λ coordinates in a general H. □

In explicit Lorentz coordinates, retain

$$G = \begin{pmatrix} p_0 + p_3 & p_1 - ip_2 \\ p_1 + ip_2 & p_0 - p_3 \end{pmatrix}, \qquad \det G = p_0^2 - p_1^2 - p_2^2 - p_3^2.$$

The bound is therefore an exact inequality between this Lorentz quadratic form and the omitted complex coordinate. It does not supply a positive uniform gap: taking $G = \epsilon I_2$, $\rho = \epsilon/2$, $\lambda = \epsilon$, $\epsilon > 0$, in the realization above gives positive definite H, nonzero ρ, and $\det G = \epsilon^2 \to 0$. An interpretation as physical mass or as a sector with a nonzero threshold requires a state-space, units, dynamics, and sector dictionary. The formula identifies the exact quantity that such a dictionary must retain. It also avoids imposing $\rho = 0$ on unrestricted motion merely to obtain a four-dimensional observation. Positivity of a chosen dynamical trajectory itself must still be proved. In fact a positive semidefinite trace-free Hermitian matrix has all three eigenvalues nonnegative with sum zero, so it is zero. Thus the nonzero scalar configurations below cannot themselves be positive matrices.

There is an exact bridge that retains this domain distinction. Put $\mathcal{J}_{14} = \{Q \in \mathrm{Herm}_3(\mathbb{H}) : \operatorname{tr} Q = 0\}$. The restriction $\mathscr{O}|_{\mathcal{J}_{14}}$ is onto all of $\mathrm{Herm}_2(\mathbb{C})$, not only its positive cone: its section is

$$j_0(G) = RGR^* - \frac{\operatorname{tr}(C^{-1}G)}{3} uu^*.$$

Indeed $R^*R = C^{-1}$, $\operatorname{tr}(uu^*) = 3$, and $Pu = 0$, so $\operatorname{tr} j_0(G) = 0$ and $\mathscr{O}(j_0(G)) = G$. Its kernel therefore has dimension ten.

For a positive boundary observation of every scalar configuration retain the separate, nonlinear map

$$\mathcal{F}(Q) = Q - \lambda_{\min}(Q) I_3, \qquad \widehat{G}(Q) = \mathscr{O}(\mathcal{F}(Q)) = P\Phi P^* - \lambda_{\min}(Q) C, \quad Q = \Theta(\Phi, b). \tag{831}$$

It maps $\mathcal{J}_{14}$ bijectively and continuously to the positive singular boundary of $\mathrm{Herm}_3(\mathbb{H})_{\succeq 0}$. The inverse is $H \mapsto H - (\operatorname{tr} H/3) I_3$: the least eigenvalue of this trace-free matrix is $-\operatorname{tr} H/3$. Eigenvalue continuity proves continuity in both directions. This is the full-space version of the earlier sphere-pair construction. The map leaves b, and hence ρ, unchanged, and therefore

$$\widehat{G}(Q) \succeq 0, \qquad \det \widehat{G}(Q) \geq |u^* b|^2.$$

The retained matrix $C = \left(\begin{smallmatrix} 2 & -1 \\ -1 & 2 \end{smallmatrix}\right)$ is essential in (831). This exact observation of a scalar trajectory is not an assertion that the trajectory preserves a positive linear slice; at repeated least eigenvalues the observation need not even be differentiable. No Hamiltonian intertwining is asserted.

The fixed marked observation is not automatically a gauge-invariant four-momentum. There is an exact covariance rule: under $H' = UHU^*$, $U \in SU(3)$, transform the observer as $P' = PU^*$; then $P'H'P'^* = PHP^*$ and hence its complex observation agrees. Keeping P fixed gives a chosen observer; specifying how that observer and the integral angular lattice participate in a physical model is additional data, not supplied by an equality of dimensions.

24.7 The actual period-to-observation bridge

The common marking also occurs in the rank-three toroidal extension of the native rank-two cusp. To specify an actual map, retain its data

$$B_0 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad Q_{\mathrm{latt}} = B_0^{-1} P = \begin{pmatrix} 0 & -1 & 1 \\ 1 & -1 & 0 \end{pmatrix}, \quad S = \begin{pmatrix} 1 & 1 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

Here $PS = I_2$. Let $C_2(t)$ be the retained bounded holomorphic period correction at $t = 0$. On the logarithmic cover $t = e^{2\pi i s}$ put

$$Z_2 = sB_0 + C_2(t), \quad C_3(t) = SC_2(t)Q_{\mathrm{latt}}, \quad Z_3 = sI_3 + C_3(t), \quad \Phi_{\mathrm{latt}} = P \oplus Q_{\mathrm{latt}}.$$

These are the local corrected periods of the toroidal elliptic bridge; the underlying toric and compact-fibre morphisms are recorded in the workbench's *Higher-rung toroidal elliptic bridge*, equations (TEB.16)–(TEB.35). The new calculation below only needs the displayed matrices and retains the entire correction $C_2(t)$.

Proposition 24.8 (Metric-dependent period observation)**.** *Set* $\Omega_3 = (I_3\ Z_3)$ *and* $\Omega_2 = (I_2\ Z_2)$*. For every real symmetric positive-definite* 6×6 *matrix* W_3*,*

$$H_3 = \Omega_3 W_3 \Omega_3^*, \quad W_2 = \Phi_{\mathrm{latt}} W_3 \Phi_{\mathrm{latt}}^{\mathsf{T}} \quad \Longrightarrow \quad P H_3 P^* = \Omega_2 W_2 \Omega_2^*.$$

Both sides are positive definite. No fixed positive W_3 *makes* H_3 *periodic under* $s \mapsto s+1$*. Nevertheless every retained positive matrix* W_* *gives an explicit descending choice*

$$M(x) = \begin{pmatrix} I_3 & xI_3 \\ 0 & I_3 \end{pmatrix}, \quad x = \operatorname{Re} s, \quad W_3(s) = M(-x) W_* M(-x)^{\mathsf{T}}.$$

For this choice, with $y = \operatorname{Im} s$*, the descended matrix is*

$$H_3(t) = (I_3\ iyI_3 + C_3(t)) W_* (I_3\ iyI_3 + C_3(t))^*.$$

If $W_* = \left(\begin{smallmatrix} E & F \\ F^{\mathsf{T}} & K \end{smallmatrix}\right)$*, then* $G(t) = P H_3(t) P^*$ *has leading term* $y^2 PKP^*$*, with* $PKP^* \succ 0$*, and* $\det G(t) = y^4 \det(PKP^*) + O(y^3)$ *at the cusp.*

Proof. Direct multiplication gives $PZ_3 = Z_2 Q_{\mathrm{latt}}$, hence $P\Omega_3 = \Omega_2 \Phi_{\mathrm{latt}}$ and the claimed Gram identity. Both P and Q_{latt} have rank two, so Φ_{latt} is surjective and W_2 is positive definite. Each period matrix has an identity block, proving positivity of both Gram matrices.

Write a fixed W_3 in $3+3$ blocks. Since $Z_3(s+n) = Z_3(s) + nI_3$, the coefficient of n^2 in $H_3(s+n)$ is its positive-definite lower diagonal block. Therefore $H_3(s+n)$ cannot be constant in $n \in \mathbb{Z}$. In particular the period-derived observation cannot be made single-valued by silently retaining a constant lattice metric.

Put $M_3 = M(1)$. The proposed varying metric is positive by congruence and satisfies $W_3(s+1) = M_3^{-1} W_3(s) M_3^{-\mathsf{T}}$. Since $\Omega_3(s+1) = \Omega_3(s) M_3$, its Gram matrix is invariant. Also $\Omega_3(s) M(-x) = (I_3\ iyI_3 + C_3(t))$, which proves the exact descended formula. The complete expansion, without removing its lower terms, is

$$\begin{aligned} H_3 = y^2 K + iy(F^{\mathsf{T}} - F + KC_3^* - C_3 K) \\ + E + FC_3^* + C_3 F^{\mathsf{T}} + C_3 K C_3^*. \end{aligned}$$

Boundedness of C_3 gives the asserted leading term and determinant estimate. The principal block K is positive definite and P^* is injective, so PKP^* is positive definite. This proves every assertion. □

For the explicitly specified example $W_* = I_6$ the exact expression is $H_3 = y^2 I_3 + iy(C_3^* - C_3) + I_3 + C_3 C_3^*$ and $\det G = 3y^4 + O(y^3)$, retaining $C = PP^*$ and $\det C = 3$. Thus this period-derived observation grows in a timelike, full-rank direction, not toward a null determinant. It is an actual positive matrix-valued map on the punctured cusp, not an inverse recovery of periods or of individual torus points. Its extension through the successor's finite-order overlaps and special fillings, and a dynamical interpretation of the chosen lattice metric, are further constructions.

All positive period-metric fibres and the cost of rank loss

The following calculation keeps the periods in Proposition 24.8 fixed. In particular $C_3(t) = SC_2(t)Q_{\mathrm{latt}}$ still uses the displayed S, not the different right inverse $R = P^* C^{-1}$. Choose a punctured cusp on which the retained correction is bounded, and fix a finite constant $K \geq 0$ such that $\|C_3(t)\| \leq K$ there. Such a constant exists because $\|C_3(t)\| \leq \|S\| \|C_2(t)\| \|Q_{\mathrm{latt}}\|$. All norms below are the Euclidean operator norms. Write

$$t = e^{2\pi i s}, \qquad s = x + iy, \qquad y = -\frac{\log|t|}{2\pi},$$

and restrict to the nonempty smaller cusp $y > \max\{K, 1\}$. This changes no period or coefficient. It identifies a domain where the inverse used below is proved to exist.

Theorem 24.9 (The complete positive metric fibre)**.** *Retain*

$$A_{\mathrm{per}}(s) = xI_3 + \operatorname{Re} C_3(t), \qquad Y_{\mathrm{per}}(t) = yI_3 + \operatorname{Im} C_3(t),$$

and define the real period map and its inverse by

$$\mathcal{N}(s) = \begin{pmatrix} I_3 & A_{\mathrm{per}}(s) \\ 0 & Y_{\mathrm{per}}(t) \end{pmatrix}, \qquad \mathcal{N}(s)^{-1} = \begin{pmatrix} I_3 & -A_{\mathrm{per}}(s)Y_{\mathrm{per}}(t)^{-1} \\ 0 & Y_{\mathrm{per}}(t)^{-1} \end{pmatrix}. \tag{MF.1}$$

For every prescribed Hermitian matrix $H = E + iF \succ 0$, where $E = E^{\mathsf{T}}$ and $F = -F^{\mathsf{T}}$ are real, all real symmetric positive-definite lattice matrices giving the unchanged value $\Omega_3 W_3 \Omega_3^* = H$ are exactly

$$W_3 = \mathcal{N}^{-1} \left[\frac{1}{2} \begin{pmatrix} E & -F \\ F & E \end{pmatrix} + \begin{pmatrix} U & V \\ V & -U \end{pmatrix} \right] \mathcal{N}^{-\mathsf{T}}, \tag{MF.2}$$

where U, V are real symmetric 3×3 matrices and the bracket is positive definite. This is a nonempty open convex twelve-real-dimensional fibre. An explicit description of its entire parameter domain is

$$E_+ = E/2 + U \succ 0, \qquad E/2 - U - (-F/2 + V)^{\mathsf{T}} E_+^{-1} (-F/2 + V) \succ 0. \tag{MF.3}$$

In particular $U = V = 0$ always gives a positive lift.

Every smooth map $H_3(t) \succ 0$ on the smaller punctured cusp has a smooth positive lift (MF.2) with

$$W_3(s+1) = M_3^{-1} W_3(s) M_3^{-\mathsf{T}}. \tag{MF.4}$$

The full set of positive observations $G(t) = PH_3(t)P^*$ obtainable from such metrics is exactly the set of all smooth $\mathrm{Herm}_2(\mathbb{C})_{\succ 0}$-valued maps on that cusp. For fixed $G \succ 0$, all the intermediate positive H are

$$H = X \begin{pmatrix} G & z \\ z^* & z^* G^{-1} z + \tau \end{pmatrix} X^*, \qquad z \in \mathbb{C}^2, \quad \tau > 0. \tag{MF.5}$$

The positive real lattice-metric fibre of G has real dimension seventeen.

Proof. For every real vector v,

$$\|Y_{\mathrm{per}} v\| \geq y\|v\| - \|\operatorname{Im} C_3\, v\| \geq (y - K)\|v\|.$$

The inequality $\|\operatorname{Im} C_3\| \leq \|C_3\|$ follows from $\operatorname{Im} C_3 = (C_3 - \bar{C}_3)/(2i)$ and $\|\bar{C}_3\| = \|C_3\|$. Since $y > K$, Y_{per} has zero kernel and is an invertible square real matrix. Block multiplication now proves both inverse identities in (MF.1). This is the original real period map: if $a, b \in \mathbb{R}^3$, then

$$a + Z_3 b = (a + A_{\mathrm{per}} b) + i Y_{\mathrm{per}} b.$$

Thus the inverse takes $v + iw$ to $(v - A_{\mathrm{per}} Y_{\mathrm{per}}^{-1} w,\ Y_{\mathrm{per}}^{-1} w)$. No correction in C_3 has been removed.

Put $\mathcal{J} = (I_3 \ \ iI_3)$. The exact factorization is $\Omega_3 = \mathcal{J}\mathcal{N}$. Congruence gives a bijection between real positive W_3 and real positive $T = \mathcal{N} W_3 \mathcal{N}^{\mathsf{T}}$. For

$$T = \begin{pmatrix} E_0 & F_0 \\ F_0^{\mathsf{T}} & K_0 \end{pmatrix}$$

one obtains by multiplication

$$\mathcal{J} T \mathcal{J}^* = E_0 + K_0 + i(F_0^{\mathsf{T}} - F_0). \tag{MF.6}$$

Consequently this equals $E + iF$ precisely when $E_0 = E/2 + U$, $K_0 = E/2 - U$, and $F_0 = -F/2 + V$ with U, V symmetric. Indeed, the first equality in (MF.6) fixes $K_0 = E - E_0$, and $F_0^{\mathsf{T}} - F_0 = F$ is equivalent to symmetry of $F_0 + F/2$. These recover U, V uniquely, proving the full fibre formula.

For real v, w, with $q = v + iw$,

$$\begin{pmatrix} v \\ w \end{pmatrix}^{\mathsf{T}} \begin{pmatrix} E & -F \\ F & E \end{pmatrix} \begin{pmatrix} v \\ w \end{pmatrix} = q^* H q.$$

The right side is positive for every nonzero (v, w), so the first bracket term in (MF.2) is positive definite. This proves nonemptiness. The kernel parameters have dimension $6 + 6 = 12$. Positive definiteness is an open convex condition on real symmetric matrices, so its intersection with this affine twelve-dimensional fibre is open and convex and has dimension twelve. Finally, for $B_+ = -F/2 + V$,

$$\begin{pmatrix} E_+ & B_+ \\ B_+^{\mathsf{T}} & E/2 - U \end{pmatrix} = \begin{pmatrix} I & 0 \\ B_+^{\mathsf{T}} E_+^{-1} & I \end{pmatrix} \begin{pmatrix} E_+ & 0 \\ 0 & E/2 - U - B_+^{\mathsf{T}} E_+^{-1} B_+ \end{pmatrix} \begin{pmatrix} I & E_+^{-1} B_+ \\ 0 & I \end{pmatrix}.$$

The two triangular factors are transposes of each other and are invertible; the identity proves the necessity and sufficiency of (MF.3), including every positivity restriction on the kernel parameters.

The original correction is single-valued in t, so

$$\mathcal{N}(s+1) = \mathcal{N}(s) M_3, \qquad \mathcal{N}(s+1)^{-1} = M_3^{-1} \mathcal{N}(s)^{-1}. \tag{MF.7}$$

For any smooth prescribed $H_3(t)$, its real and imaginary parts are smooth and periodic on the cover. The inverse of Y_{per} is smooth, since its entries are its adjugate divided by its nonzero determinant. Formula (MF.2) with $U = V = 0$ is therefore smooth and satisfies (MF.4). More generally, smooth periodic U, V satisfying (MF.3) give all such smooth lifts: conversely (MF.7) and (MF.4) make $\mathcal{N} W_3 \mathcal{N}^{\mathsf{T}}$ periodic, and its blocks recover periodic U, V by (MF.6). No holomorphic condition is being imposed on the real metric or on the prescribed Hermitian map.

If $W_3 \succ 0$ is real, it is also positive on nonzero complex columns: for $p + iq$, its quadratic form is $p^{\mathsf{T}} W_3 p + q^{\mathsf{T}} W_3 q$. Since Ω_3^* is injective by the identity block of Ω_3, one has $H_3 \succ 0$. Since P^* is injective, also $G = P H_3 P^* \succ 0$. This proves necessity of strict positivity; no singular G occurs at an interior point for a positive metric.

For the converse, the marked inverse $X^{-1} = (P;\, u^*/3)$ makes the upper-left block of $X^{-1} H X^{-*}$ equal to PHP^*. With that block fixed at $G \succ 0$, the complete positive block matrices are

$$\begin{pmatrix} G & z \\ z^* & \ell \end{pmatrix}, \qquad \ell - z^* G^{-1} z = \tau > 0.$$

To prove this exact positivity statement, multiply

$$\begin{pmatrix} I_2 & 0 \\ z^* G^{-1} & 1 \end{pmatrix} \begin{pmatrix} G & 0 \\ 0 & \tau \end{pmatrix} \begin{pmatrix} I_2 & G^{-1} z \\ 0 & 1 \end{pmatrix}.$$

The result is the displayed block with $\ell = z^* G^{-1} z + \tau$. The invertible triangular factors prove both directions, and congruence by X gives (MF.5). Taking $z = 0$ and any fixed $\tau = \lambda > 0$ already gives the smooth lift $H_3(t) = R G(t) R^* + \lambda u u^*$. Applying (MF.2) proves every asserted positive observation is realized. The intermediate H fibre has four real coordinates in z and one in τ. Equivalently, the real-linear metric-to-observation map has rank four: the constructions above give a right inverse on all Hermitian matrices before imposing positivity. Its real symmetric domain has dimension $6 \cdot 7/2 = 21$, so its affine kernel has dimension $21 - 4 = 17$. The positive part is nonempty and relatively open, proving the stated seventeen-dimensional fibre without a dimension-only surjectivity argument. □

The freedom above does not replace the retained periods by an identity period matrix. It computes the metric which compensates their full real period map. The correction itself has not lost information in passing from rank two to rank three: the exact inverse on its prescribed image is

$$C_2 = P C_3 S B_0,$$

because $PS = I_2$ and $Q_{\mathrm{latt}} S = B_0^{-1}$. To identify exactly what an unrestricted metric observation cannot recover, take two retained corrections with real period maps $\mathcal{N}$ and $\widetilde{\mathcal{N}}$ on a common sufficiently small cusp. The invertible real transformation

$$T_{\mathrm{per}} = \widetilde{\mathcal{N}}^{-1}\mathcal{N}, \qquad \widetilde{W}_3 = T_{\mathrm{per}} W_3 T_{\mathrm{per}}^{\mathsf{T}} \tag{MF.8}$$

preserves positivity and gives $\widetilde{\Omega}_3 \widetilde{W}_3 \widetilde{\Omega}_3^* = \Omega_3 W_3 \Omega_3^*$ by the two exact factorizations $\Omega_3 = \mathcal{J}\mathcal{N}$. Furthermore $T_{\mathrm{per}}(s+1) = M_3^{-1} T_{\mathrm{per}}(s) M_3$, so (MF.8) preserves the prescribed metric monodromy. This is an explicit correspondence between period–metric pairs with the same H_3 and G. Thus these observations alone cannot recover the period correction when the metric is allowed this freedom. The constants remain in the metric formula; they have not been set to zero.

Proposition 24.10 (Relative-metric bounds obstruct projective rank loss)**.** *For an equivariant positive metric put*

$$W_*(t) = M(x) W_3(s) M(x)^{\mathsf{T}}, \qquad \widetilde{\Omega}_3(t) = (I_3 \;\; iyI_3 + C_3(t)).$$

These are single-valued on the punctured cusp, and $H_3 = \widetilde{\Omega}_3 W_* \widetilde{\Omega}_3^*$. If fixed constants $0 < a \le b < \infty$ obey $aI_6 \preceq W_*(t) \preceq bI_6$, then

$$a\big(1 + (y-K)^2\big) C \preceq G(t) \preceq b\big(1 + (y+K)^2\big) C. \tag{MC.1}$$

In particular the unchanged eigenvalues satisfy

$$\lambda_{\min}(G) \ge a\big(1 + (y-K)^2\big)\lambda_{\min}(C), \qquad \lambda_{\max}(G) \le b\big(1 + (y+K)^2\big)\lambda_{\max}(C). \tag{MC.2}$$

With the original C, $\lambda_{\min}(C) = 1$ and $\lambda_{\max}(C) = 3$. More generally, writing $\kappa_*(t) = \lambda_{\max}(W_*)/\lambda_{\min}(W_*)$, one always has

$$\frac{\lambda_{\min}(G)}{\lambda_{\max}(G)} \ge \frac{1 + (y-K)^2}{3\kappa_*(t)\big(1 + (y+K)^2\big)}. \tag{MC.3}$$

Thus if the observed eigenvalue ratio tends to zero as $t \to 0$, the relative metric condition number tends to infinity. Bounded relative condition number excludes this projective rank loss even when the common scale of the metric is not bounded above or below.

Proof. The identity $M(x+1) = M(x)M_3$, together with (MF.4), proves that W_* is periodic. Multiplication gives $\Omega_3 M(-x) = \widetilde{\Omega}_3$, which proves the stated Gram formula. For every complex vector v,

$$(y-K)\|v\| \le \|(iyI_3 + C_3)^* v\| \le (y+K)\|v\|.$$

Both inequalities are the triangle inequality, its reverse, and $\|C_3^*\| \le K$. Since $\widetilde{\Omega}_3 \widetilde{\Omega}_3^* = I_3 + (iyI_3 + C_3)(iyI_3 + C_3)^*$, they give

$$\big(1 + (y-K)^2\big) I_3 \preceq \widetilde{\Omega}_3 \widetilde{\Omega}_3^* \preceq \big(1 + (y+K)^2\big) I_3.$$

Congruencing $aI_6 \preceq W_* \preceq bI_6$ by $\widetilde{\Omega}_3$, and then by P, proves (MC.1). The minimum and maximum Rayleigh quotients prove (MC.2). The characteristic polynomial of the original C is $(z-1)(z-3)$, retaining the factors 1 and 3.

At each point take $a(t) = \lambda_{\min}(W_*)$ and $b(t) = \lambda_{\max}(W_*)$ in the same inequalities. Dividing the resulting positive scalar bounds gives (MC.3); this division is a comparison of eigenvalues, not a rescaling or replacement of G or W_*. Its period-dependent factor tends to 1. For fixed a, b, the lower bound tends to $a/(3b) > 0$; for example it is at least $a/(27b)$ once $y \ge 2K$. If the ratio on the left tends to zero, (MC.3) implies

$$\kappa_*(t) \ge \frac{1 + (y-K)^2}{3\big(1 + (y+K)^2\big)} \frac{\lambda_{\max}(G)}{\lambda_{\min}(G)} \longrightarrow +\infty.$$

This proves the full asserted obstruction, including metrics whose overall scale varies. □

Proposition 24.11 (A retained-period rank-loss family and its exact cost)**.** *Fix, without rescaling any matrix, four real constants*

$$\alpha_{\mathrm{obs}} > 0, \qquad \beta_{\mathrm{obs}} > 0, \qquad \lambda > 0, \qquad p > 0,$$

and set

$$\delta(t) = \beta_{\mathrm{obs}} y^{-p}, \qquad G(t) = \begin{pmatrix} \alpha_{\mathrm{obs}} & 0 \\ 0 & \delta(t) \end{pmatrix}, \qquad H_3(t) = RG(t)R^* + \lambda uu^*. \tag{RL.1}$$

Take the lift (MF.2) with $U = V = 0$. It is a smooth equivariant positive real lattice metric at every point of the smaller punctured cusp, with the original periods and correction unchanged. Its observation has

$$\det G(t) = \alpha_{\mathrm{obs}}\beta_{\mathrm{obs}} y^{-p} = \alpha_{\mathrm{obs}}\beta_{\mathrm{obs}} \left(-\frac{\log|t|}{2\pi}\right)^{-p} \longrightarrow 0, \qquad G(t) \longrightarrow \mathrm{diag}(\alpha_{\mathrm{obs}}, 0). \tag{RL.2}$$

Put $D_2(t) = \operatorname{Im} C_2(t)$ and $Y = Y_{\mathrm{per}}(t) = yI_3 + \operatorname{Im} C_3(t)$. The relative metric has the exact determinant

$$\det W_*(t) = \frac{\alpha_{\mathrm{obs}}^2 \lambda^2 \beta_{\mathrm{obs}}^2\, y^{-2p}}{64\, \det(Y)^2}, \qquad \det(Y) = y\left[y^2 + y\,\operatorname{tr}(D_2 B_0^{-1}) + \det D_2\right]. \tag{RL.3}$$

In particular

$$\det W_*(t) \sim \frac{\alpha_{\mathrm{obs}}^2 \lambda^2 \beta_{\mathrm{obs}}^2}{64} y^{-2p-6}. \tag{RL.4}$$

If $L_{\max} = \max\{2\alpha_{\mathrm{obs}}/3,\, 3\lambda\}$, then

$$\lambda_{\min}(W_*) \sim \frac{\beta_{\mathrm{obs}}}{4}\, y^{-p-2}, \qquad \lambda_{\max}(W_*) \longrightarrow \frac{L_{\max}}{2}, \qquad \kappa_*(t) \sim \frac{2L_{\max}}{\beta_{\mathrm{obs}}}\, y^{p+2}. \tag{RL.5}$$

Every constant and the complete correction in the exact formulas (RL.1)–(RL.3) is retained.

Proof. Formula (MF.5) with $z = 0$ and $\tau = \lambda$ gives $H_3 \succ 0$ and $PH_3P^* = G$. Formula (MF.2) proves the metric, smoothness, and monodromy assertions. Equation (RL.2) is the direct determinant of the displayed matrix and uses the original logarithmic-cover factor 2π. The ratio of its least and greatest eigenvalues is eventually $\delta/\alpha_{\mathrm{obs}}$, so this is projective rank loss in the precise sense used in Proposition 24.10.

Write $A_c = \operatorname{Re} C_3$ and

$$\widetilde{\mathcal{N}} = \begin{pmatrix} I_3 & A_c \\ 0 & Y \end{pmatrix}, \qquad \mathcal{N} = \widetilde{\mathcal{N}} M(x).$$

Here H_3 is real symmetric. The relative metric is exactly

$$W_* = \frac{1}{2}\widetilde{\mathcal{N}}^{-1} \begin{pmatrix} H_3 & 0 \\ 0 & H_3 \end{pmatrix} \widetilde{\mathcal{N}}^{-\mathsf{T}} = \frac{1}{2} \begin{pmatrix} H_3 + A_c Y^{-1} H_3 Y^{-\mathsf{T}} A_c^{\mathsf{T}} & -A_c Y^{-1} H_3 Y^{-\mathsf{T}} \\ -Y^{-1} H_3 Y^{-\mathsf{T}} A_c^{\mathsf{T}} & Y^{-1} H_3 Y^{-\mathsf{T}} \end{pmatrix}. \tag{RL.6}$$

Since $H_3 = X\,\mathrm{diag}(G, \lambda)X^*$ and $\det X = 1$, $\det H_3 = \lambda\alpha_{\mathrm{obs}}\delta$. Taking the real 6×6 determinant in (RL.6) gives $\det W_* = (\det H_3)^2/(2^6 \det Y^2)$, proving the first identity in (RL.3).

The second identity retains the correction rather than replacing Y by yI_3. Indeed $Y = yI_3 + SD_2Q_{\mathrm{latt}}$ and $Q_{\mathrm{latt}}S = B_0^{-1}$. For rectangular U, V, the identity $\det(I + UV) = \det(I + VU)$ follows by eliminating either off-diagonal block of $\left(\begin{smallmatrix} I & U \\ -V & I \end{smallmatrix}\right)$ with triangular determinant-one row operations. Applying it here gives

$$\det Y = y^3 \det\left(I_2 + y^{-1} D_2 B_0^{-1}\right) = y\left[y^2 + y\,\operatorname{tr}(D_2 B_0^{-1}) + \det D_2\right],$$

because $\det B_0 = 1$. If $D_2 = \left(\begin{smallmatrix} d_{11} & d_{12} \\ d_{21} & d_{22} \end{smallmatrix}\right)$, this is the complete polynomial

$$\det Y = y\left[y^2 + y(d_{12} - d_{21}) + d_{11}d_{22} - d_{12}d_{21}\right].$$

The entries of D_2 remain the prescribed bounded functions of t. Thus $\det Y/y^3 \to 1$, proving (RL.4) with the displayed coefficient.

For the eigenvalue assertions, $H_3 u = 3\lambda u$ because $R^* u = 0$. On the two-dimensional invariant plane $u^\perp$, its trace is

$$\operatorname{tr}(C^{-1}G) = \frac{2}{3}(\alpha_{\mathrm{obs}} + \delta)$$

and the product of the two eigenvalues is $\det H_3/(3\lambda) = \alpha_{\mathrm{obs}}\delta/3$. The invariance of the plane follows also from Hermiticity and the displayed u eigenvector. Its two eigenvalues are therefore exactly

$$h_\pm = \frac{\alpha_{\mathrm{obs}} + \delta \pm \sqrt{\alpha_{\mathrm{obs}}^2 - \alpha_{\mathrm{obs}}\delta + \delta^2}}{3}. \tag{RL.7}$$

They are positive: their sum and product are positive. Rationalizing the numerator of h_- gives

$$\frac{h_-}{\delta} = \frac{\alpha_{\mathrm{obs}}}{\alpha_{\mathrm{obs}} + \delta + \sqrt{\alpha_{\mathrm{obs}}^2 - \alpha_{\mathrm{obs}}\delta + \delta^2}} \longrightarrow \frac{1}{2}.$$

Also $h_+ \to 2\alpha_{\mathrm{obs}}/3$. Consequently $\lambda_{\min}(H_3) \sim \delta/2$ and $\lambda_{\max}(H_3) \to L_{\max}$, with H_3 bounded.

The explicit blocks (RL.6), the bound $\|Y^{-1}\| \le (y-K)^{-1}$, and bounded A_c, H_3 imply

$$\left\| W_* - \tfrac{1}{2}\operatorname{diag}(H_3, 0) \right\| = O(y^{-2}).$$

For real symmetric matrices the maximum eigenvalue changes by at most the operator norm of a perturbation: apply the Rayleigh-quotient maximum to each matrix and subtract. Hence $\lambda_{\max}(W_*) \to L_{\max}/2$.

To control the least eigenvalue without assuming a bounded condition number of H_3, use the exact inverse

$$W_*^{-1} = 2\widetilde{\mathcal{N}}^{\mathsf{T}} \operatorname{diag}(H_3^{-1}, H_3^{-1})\widetilde{\mathcal{N}}.$$

For $y > 1$, the decomposition $\widetilde{\mathcal{N}} = \operatorname{diag}(I_3, yI_3) + \left(\begin{smallmatrix} 0 & A_c \\ 0 & \operatorname{Im} C_3 \end{smallmatrix}\right)$ gives $\|\widetilde{\mathcal{N}}\| \le y + \sqrt{2}K$. Thus

$$\lambda_{\max}(W_*^{-1}) \le \frac{2(y + \sqrt{2}K)^2}{\lambda_{\min}(H_3)}.$$

Its lower-right principal block includes the positive term $2Y^{\mathsf{T}} H_3^{-1} Y$. If v is a unit eigenvector of H_3^{-1} for its greatest eigenvalue, testing that block on $Y^{-1}v/\|Y^{-1}v\|$ gives

$$\lambda_{\max}(W_*^{-1}) \ge \frac{2(y-K)^2}{\lambda_{\min}(H_3)}.$$

All denominators are positive. Taking reciprocals yields

$$\frac{\lambda_{\min}(H_3)}{2(y+\sqrt{2}K)^2} \le \lambda_{\min}(W_*) \le \frac{\lambda_{\min}(H_3)}{2(y-K)^2}. \tag{RL.8}$$

Both bounds are asymptotic to $\delta/(4y^2)$, proving the first assertion of (RL.5). Combining it with the maximum-eigenvalue limit proves the condition-number asymptotic with its full coefficient. □

This is metric-choice freedom, not a selection of a physical momentum or mass. For the same unchanged periods the allowed choice $W_* = I_6$ instead gives exactly

$$H_3 = I_3 + (iyI_3 + C_3)(iyI_3 + C_3)^*, \qquad G = y^2 C + O(y), \qquad \det G = 3y^4 + O(y^3),$$

whereas (RL.1)–(RL.2) give a determinant tending to zero and retain the same C_2, C_3. These two explicit admissible choices, the complete twelve-dimensional metric fibres, and the seventeen-dimensional observed fibres prove that the specified period data and positivity alone do not select an observed trajectory. The eigenvalue ratios above only diagnose rank loss; no field, period, metric, or parameter was replaced by a normalized one. Units, a metric-selection law, dynamics, and an interpretation of the marked observation would be additional specified data. No extension across other finite-order overlaps or special fillings, and no physical or quantum gap conclusion, is asserted here.

Hermitian flat metrics, the original period lattice, and its dual spectrum

Here a flat metric means a translation-invariant metric on each fixed torus fibre. Its coefficients may depend smoothly on the cusp parameter t, but not on the point of that fibre. The word *physical* below distinguishes the original real coordinates of $\mathbb{C}^3$ from integral lattice coordinates; it does not assert a physical field theory. Retain, without changing any basis or period,

$$B_0 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad P = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix}, \quad Q_{\mathrm{latt}} = B_0^{-1} P, \quad S = \begin{pmatrix} 1 & 1 \\ 0 & 1 \\ 0 & 0 \end{pmatrix},$$

$$C_3 = SC_2Q_{\mathrm{latt}}, \qquad Z_3 = sI_3 + C_3, \qquad Z_2 = sB_0 + C_2, \qquad s = x + iy, \quad t = e^{2\pi i s}, \quad y = -\frac{\log|t|}{2\pi}.$$

Choose finite bounds $K_2, K \geq 0$ on $\|C_2(t)\|, \|C_3(t)\|$, respectively, and restrict to $y > \max\{K_2, K\} + 1$. Norms are the original Euclidean operator norms. Write

$$A = xI_3 + \operatorname{Re} C_3, \quad Y = yI_3 + \operatorname{Im} C_3, \quad \mathcal{N} = \begin{pmatrix} I_3 & A \\ 0 & Y \end{pmatrix}, \quad \mathcal{J} = (I_3 \;\; iI_3), \quad J_0 = \begin{pmatrix} 0 & -I_3 \\ I_3 & 0 \end{pmatrix}.$$

Thus $\Omega_3 = (I_3 \;\; Z_3) = \mathcal{J}\mathcal{N}$ and $\mathcal{N}^{-1} = \left(\begin{smallmatrix} I_3 & -AY^{-1} \\ 0 & Y^{-1} \end{smallmatrix}\right)$. Indeed $\|Yv\| \geq (y - K)\|v\|$ proves invertibility. The same argument, using $\|B_0 v\| = \|v\|$, proves invertibility of $Y_2 = \operatorname{Im} Z_2 = yB_0 + \operatorname{Im} C_2$. For $r = 2, 3$ put

$$\Lambda_r = \Omega_r \mathbb{Z}^{2r}, \qquad T_r = \mathbb{C}^r / \Lambda_r, \qquad \mathcal{N}_2 = \begin{pmatrix} I_2 & \operatorname{Re} Z_2 \\ 0 & Y_2 \end{pmatrix}.$$

The real coordinates of $z \in \mathbb{C}^3$ are $q = (\operatorname{Re} z, \operatorname{Im} z)^{\mathsf{T}} = \mathcal{N}\xi$, where $\xi \in \mathbb{R}^6/\mathbb{Z}^6$. This is an exact change of coordinates, not a replacement of the lattice. For any complex matrix D define its realification by

$$\mathfrak{r}(D) = \begin{pmatrix} \operatorname{Re} D & -\operatorname{Im} D \\ \operatorname{Im} D & \operatorname{Re} D \end{pmatrix}.$$

Block multiplication gives $\mathfrak{r}(D_1 D_2) = \mathfrak{r}(D_1)\mathfrak{r}(D_2)$ and $\mathfrak{r}(D^*) = \mathfrak{r}(D)^{\mathsf{T}}$.

Theorem 24.12 (The precise Hermitian metric in a period fibre)**.** *Let $g_{\mathrm{phys}} \succ 0$ be a real symmetric metric matrix and put*

$$T = g_{\mathrm{phys}}^{-1}, \qquad g_{\mathrm{latt}} = \mathcal{N}^{\mathsf{T}} g_{\mathrm{phys}} \mathcal{N}, \qquad W = g_{\mathrm{latt}}^{-1} = \mathcal{N}^{-1} T \mathcal{N}^{-\mathsf{T}}. \tag{HS.1}$$

Then the earlier period weight W has the exact geometric meaning of a lattice cometric*; the covariant lattice metric is its inverse. The observation is $H = \Omega_3 W \Omega_3^* = \mathcal{J} T \mathcal{J}^*$. For a fixed positive Hermitian $H = E + iF$, its full real cometric fibre is*

$$T = \frac{1}{2}\mathfrak{r}(H) + \mathcal{A}(U, V), \qquad \mathcal{A}(U, V) = \begin{pmatrix} U & V \\ V & -U \end{pmatrix}, \qquad U = U^{\mathsf{T}}, \quad V = V^{\mathsf{T}}, \quad T \succ 0. \tag{HS.2}$$

The real metric is compatible with the given complex structure, meaning $J_0^{\mathsf{T}} g_{\mathrm{phys}} J_0 = g_{\mathrm{phys}}$, if and only if $U = V = 0$. With the explicit Hermitian metric convention

$$g_{\mathrm{phys}}(v, w) = \mathrm{Re}(v^* h_3 w), \qquad v, w \in \mathbb{C}^3,$$

the unique compatible member over H is

$$h_3 = 2H^{-1}, \qquad h_3^{-1} = H/2, \qquad g_{\mathrm{phys}} = 2\mathfrak{r}(H^{-1}), \qquad T = \tfrac{1}{2}\mathfrak{r}(H), \tag{HS.3}$$

$$g_{\mathrm{latt}} = 2\mathcal{N}^{\mathsf{T}}\mathfrak{r}(H^{-1})\mathcal{N}, \qquad W = \tfrac{1}{2}\mathcal{N}^{-1}\mathfrak{r}(H)\mathcal{N}^{-\mathsf{T}}. \tag{HS.4}$$

For an arbitrary positive real cometric, its observed Hermitian part is precisely the positive J_0-invariant average

$$\frac{T + J_0 T J_0^{\mathsf{T}}}{2} = \frac{1}{2}\mathfrak{r}(H). \tag{HS.5}$$

Proof. Substitution of $q = \mathcal{N}\xi$ into the covariant quadratic form gives the middle formula in (HS.1); multiplication verifies its inverse. The observation is positive: $\mathcal{J}^*$ is injective, and a real positive matrix T is also positive on complex vectors, since for a, b real, $(a+ib)^* T (a+ib) = a^{\mathsf{T}} T a + b^{\mathsf{T}} T b$. For a real symmetric block matrix $T = \left(\begin{smallmatrix} D & B \\ B^{\mathsf{T}} & L \end{smallmatrix}\right)$,

$$\mathcal{J} T \mathcal{J}^* = D + L + i(B^{\mathsf{T}} - B).$$

Thus equality to $E + iF$ is equivalent to $D = E/2 + U$, $L = E/2 - U$, $B = -F/2 + V$ with U, V symmetric. This proves both directions and uniqueness of the parameters in (HS.2).

Because $J_0^{\mathsf{T}} = -J_0$ and $J_0^2 = -I_6$, metric compatibility is equivalent to $g_{\mathrm{phys}} J_0 = J_0 g_{\mathrm{phys}}$, which is equivalent, by multiplying by the inverse on both sides, to $T J_0 = J_0 T$. Direct multiplication shows that $\mathfrak{r}(H)$ commutes with J_0, whereas $\mathcal{A} J_0 = -J_0 \mathcal{A}$. Hence compatibility is equivalent to $J_0 \mathcal{A} = 0$, and the invertibility of J_0 makes this equivalent to $\mathcal{A} = 0$, or $U = V = 0$.

For $z = v + iw$ one has $(v, w)^{\mathsf{T}} \mathfrak{r}(H)(v, w) = z^* H z > 0$ unless $z = 0$. Thus $\mathfrak{r}(H)$ is positive. Realification respects products, so inversion of $T = \mathfrak{r}(H)/2$ gives (HS.3), and (HS.1) gives (HS.4). Finally $J_0 \mathfrak{r}(H) J_0^{\mathsf{T}} = \mathfrak{r}(H)$ and $J_0 \mathcal{A} J_0^{\mathsf{T}} = -\mathcal{A}$, proving (HS.5). This average is an average of cometrics, not an average of covariant metrics. The metric associated to it is the inverse of the average. □

Thus the twelve real fibre coordinates in (HS.2) describe genuine non-Hermitian translation-invariant real metrics, and compatibility removes exactly those coordinates. For example, $H = 2I_3$ and $U = \varepsilon\,\mathrm{diag}(1, 0, 0)$, $V = 0$ give

$$T_\varepsilon = \mathrm{diag}(1 + \varepsilon, 1, 1, 1 - \varepsilon, 1, 1), \qquad -1 < \varepsilon < 1.$$

All have the same H; only $\varepsilon = 0$ is Hermitian-compatible. They have different marked cometrics, and the next theorem proves that some marked character eigenvalue therefore changes.

Theorem 24.13 (The lattice cometric is the marked character spectrum)**.** *For any $T \succ 0$ as in (HS.1), define the nonnegative Laplacian by $\Delta = -\operatorname{div}\operatorname{grad}$. Its complete character eigenbasis and eigenvalues on T_3 are*

$$\chi_k([\xi]) = e^{2\pi i k^{\mathsf{T}} \xi}, \qquad k \in \mathbb{Z}^6, \qquad \lambda_k = 4\pi^2 k^{\mathsf{T}} W k = 4\pi^2 p_k^{\mathsf{T}} T p_k, \quad p_k = \mathcal{N}^{-\mathsf{T}} k. \tag{HS.6}$$

Here p_k is in the integer-valued dual of $\mathcal{N}\mathbb{Z}^6$: its pairing with $\mathcal{N}n$ is $k^{\mathsf{T}} n \in \mathbb{Z}$. Conversely every integer-valued dual covector is of this form. The eigenspace multiplicity is exactly the number of integer k with the specified value of (HS.6). The volume, in the original scales, is

$$\mathrm{Vol}(T_3, g_{\mathrm{phys}}) = |\det Y| \sqrt{\det g_{\mathrm{phys}}} = |\det Y| \det h_3 = \frac{8|\det Y|}{\det H} \tag{HS.7}$$

in the compatible case. Without compatibility the first equality still holds.

The marked spectrum recovers every entry of W:

$$W_{aa} = \frac{\lambda_{e_a}}{4\pi^2}, \qquad W_{ab} = \frac{\lambda_{e_a+e_b} - \lambda_{e_a} - \lambda_{e_b}}{8\pi^2} \quad (a \neq b). \tag{HS.8}$$

If $H(t)$ is single-valued and (HS.4) is used, all these metrics descend with the original monodromy

$$\mathcal{N}(s+1) = \mathcal{N}(s)M_3, \quad M_3 = \begin{pmatrix} I_3 & I_3 \\ 0 & I_3 \end{pmatrix}, \quad W(s+1) = M_3^{-1}W(s)M_3^{-\mathsf{T}}. \tag{HS.9}$$

The same character has labels k and $k' = M_3^{\mathsf{T}}k$ in the two markings, and $\lambda'_{k'} = \lambda_k$.

Proof. The covariant metric and its determinant are constant in the fibre coordinate ξ, so the defining divergence formula becomes $\Delta = -\sum_{a,b=1}^{6} W_{ab}\partial_{\xi_a}\partial_{\xi_b}$. Two derivatives of χ_k prove (HS.6). The dual-lattice assertion follows from $p^{\mathsf{T}}\mathcal{N}n \in \mathbb{Z}$ for all integer n: taking the six unit vectors is equivalent to $\mathcal{N}^{\mathsf{T}}p \in \mathbb{Z}^6$.

For completeness, no unlisted eigenfunctions are needed. Integration over $[0,1]^6$ gives zero for $\chi_k\overline{\chi_l}$ when $k \neq l$; the common squared norm in Riemannian measure is the volume. To prove density, the one-dimensional Fejer kernels

$$F_M(\theta) = \frac{1}{M}\left|\sum_{r=0}^{M-1} e^{2\pi i r\theta}\right|^2 = \sum_{|j|<M}\left(1 - \frac{|j|}{M}\right)e^{2\pi i j\theta}$$

are nonnegative and have integral one. For $\operatorname{dist}(\theta,\mathbb{Z}) \geq \eta > 0$ they are bounded by $1/(M\sin^2(\pi\eta))$, using the finite geometric sum. The product of six such kernels also has integral one; its mass outside a coordinate cube of radius η tends to zero by the union bound over its six coordinates. Uniform continuity then proves that convolution of any continuous periodic function with these product kernels converges uniformly to that function. Each convolution is a finite linear combination of characters. Continuous functions are dense in $L^2([0,1]^6)$: step functions on finite unions of boxes are dense by regular approximation in Lebesgue measure, and a box indicator is approximated in L^2 by a continuous function that differs only in a boundary strip of arbitrarily small measure. Multiplication by the constant Riemannian density preserves this argument.

Consequently the characters, divided by the square root of the volume, are a complete orthonormal basis. On Fourier coefficients the closure of Δ is the real diagonal operator with domain $\{(a_k) : \sum_k \lambda_k^2|a_k|^2 < \infty\}$; its adjoint has that same domain, as testing against each coordinate vector forces the displayed weighted square summability. It is therefore self-adjoint. Finite Fourier sums are a core by truncation of both the ordinary and weighted square sums. For a smooth function, integration by parts identifies the Fourier coefficients of its smooth Laplacian with $\lambda_k a_k$; their square sum is finite. Thus the operator on all smooth functions lies in this diagonal operator and has the same closure as its restriction to finite Fourier sums. Since $\lambda_k \geq 4\pi^2\lambda_{\min}(W)\|k\|^2$, bounded eigenvalue intervals contain only finitely many integer indices. This proves the completeness, discreteness and multiplicity assertions, including a positive smallest eigenvalue away from $k = 0$.

The parallelepiped $\mathcal{N}[0,1]^6$ has coordinate volume $|\det\mathcal{N}| = |\det Y|$, proving the first equality in (HS.7). For Hermitian h_3, the determinant identity $\det\mathfrak{r}(h_3) = |\det h_3|^2 = (\det h_3)^2$ follows by conjugating the realification over $\mathbb{C}$ to $\operatorname{diag}(h_3, \overline{h_3})$ with the invertible change $(v,w) \mapsto (v+iw, v-iw)$. Positive h_3 has positive determinant. Now $\det(2H^{-1}) = 8/\det H$, proving (HS.7). Expansion of the quadratic form at $e_a + e_b$ proves (HS.8).

The correction $C_3(t)$ is unchanged by $s \mapsto s+1$, giving the first formula in (HS.9). The cometric formula gives the last. Since $q = \mathcal{N}\xi = \mathcal{N}M_3\xi'$, one has $\xi = M_3\xi'$. Hence $k' = M_3^{\mathsf{T}}k$ and substitution in (HS.6) proves the final assertion. All matrix inversions depend smoothly on the entries where their determinants are nonzero, so a smooth $H(t)$ gives the asserted smooth descending family. □

Theorem 24.14 (The original torus quotient is a metric and spectral map)**.** *Let $H_3 \succ 0$, $G = PH_3P^*$, and equip T_3 with $h_3 = 2H_3^{-1}$. Retain the original integral map*

$$L = P \oplus Q_{\mathrm{latt}} : \mathbb{Z}^6 \longrightarrow \mathbb{Z}^4, \qquad P_{\mathbb{R}} = \mathrm{diag}(P, P).$$

Then $f_P : T_3 \to T_2$, $[z] \mapsto [Pz]$, is a surjective holomorphic homomorphism with connected kernel

$$\mathbb{C}u/(\mathbb{Z}u + s\mathbb{Z}u), \qquad u = (1,1,1)^{\mathsf{T}}. \tag{HS.10}$$

There is a unique target translation-invariant metric making this map a Riemannian submersion. Its Hermitian metric, complex cometric, real cometric and lattice cometric are, respectively,

$$h_2 = 2G^{-1}, \qquad h_2^{-1} = G/2 = Ph_3^{-1}P^*, \qquad \mathsf{T}_2 = \tfrac{1}{2}\mathfrak{r}(G) = P_{\mathbb{R}}\mathsf{T}_3P_{\mathbb{R}}^{\mathsf{T}}, \tag{HS.11}$$

$$W_2 = \mathcal{N}_2^{-1}\mathsf{T}_2\mathcal{N}_2^{-\mathsf{T}} = LW_3L^{\mathsf{T}}. \tag{HS.12}$$

Here $\mathsf{T}_3 = \mathfrak{r}(H_3)/2$ is the real source cometric, distinguished from the torus T_3. The horizontal lift is the exact complex map

$$\eta : \mathbb{C}^2 \longrightarrow \mathbb{C}^3, \qquad \eta = H_3P^*G^{-1}. \tag{HS.13}$$

For every smooth function f on the target,

$$\Delta_3(f \circ f_P) = (\Delta_2 f) \circ f_P. \tag{HS.14}$$

For a target character $\ell \in \mathbb{Z}^4$, its pullback is the source character labelled by $L^{\mathsf{T}}\ell$, and its eigenvalue is unchanged. These are exactly the source characters constant on the fibres. Thus the target spectrum, with each character's multiplicity counted, is a subcollection of the source character spectrum, not an identification with all source modes.

Proof. Direct multiplication gives $PS = I_2$, $B_0Q_{\mathrm{latt}} = P$ and $PZ_3 = Z_2Q_{\mathrm{latt}}$. Thus

$$P\Omega_3 = \Omega_2 L, \qquad P_{\mathbb{R}}\mathcal{N} = \mathcal{N}_2 L. \tag{HS.15}$$

Moreover $\mathrm{diag}(S, SB_0)$ is an integral right inverse of L, since $Q_{\mathrm{latt}}SB_0 = I_2$. Therefore $P\Lambda_3 = \Lambda_2$, proving that the torus homomorphism is well-defined and surjective.

If $Pz \in \Lambda_2$, surjectivity on lattices supplies $\gamma \in \Lambda_3$ with $P\gamma = Pz$, so $z - \gamma \in \ker_{\mathbb{C}} P = \mathbb{C}u$. To find its periods, if $\Omega_3(a,b)$ is in $\mathbb{C}u$, then $\Omega_2(Pa, Q_{\mathrm{latt}}b) = 0$. The invertibility of $\mathcal{N}_2$ forces $Pa = Q_{\mathrm{latt}}b = 0$. Solving the two displayed difference equations gives $a = ru$, $b = vu$ with $r, v \in \mathbb{Z}$. Since $C_3u = 0$, $\Omega_3(a,b) = (r + sv)u$. This proves (HS.10), including connectedness. Its lattice has real rank two because $y > 0$.

Put $K_3 = h_3^{-1} = H_3/2$, $K_2 = PK_3P^* = G/2$. Surjectivity of P makes P^* injective, so $K_2 \succ 0$. The lift in (HS.13) is $K_3P^*K_2^{-1}$ and satisfies

$$P\eta = I_2, \qquad \eta^*h_3v = K_2^{-1}Pv = 0 \quad (v \in \ker P), \qquad \eta^*h_3\eta = K_2^{-1} = h_2. \tag{HS.16}$$

Every $w \in \mathbb{C}^3$ has the orthogonal decomposition $w = \eta(Pw) + (w - \eta(Pw))$, whose second term lies in $\ker P$. Hence P restricts to an isometry on the orthogonal complement of its kernel; this is precisely the definition of a Riemannian submersion. It also proves uniqueness of h_2, since each target vector has a unique horizontal lift. Taking realifications proves (HS.11), and multiplying (HS.15) by the inverses on both sides proves (HS.12).

In lattice coordinates $f_P([\xi]) = [L\xi]$. The chain rule gives $\partial_{\xi_a}\partial_{\xi_b} f(L\xi) = \sum_{i,j} L_{ia}L_{jb}(\partial_i\partial_j f)(L\xi)$. Contraction with $-W_{3,ab}$ and (HS.12) proves (HS.14). Also $e^{2\pi i\ell^{\mathsf{T}}L\xi} = \chi_{L^{\mathsf{T}}\ell}(\xi)$; inserting this index in (HS.6) proves equality of the eigenvalues.

The lattice kernel is $\mathbb{Z}u \oplus \mathbb{Z}u$. The characters constant on the connected kernel annihilate its real tangent space, so their indices (m,n) obey $u^{\mathsf{T}}m = u^{\mathsf{T}}n = 0$. Conversely these equalities make the character constant there. Every integral zero-sum triple $m = (m_1, m_2, m_3)$ is $P^{\mathsf{T}}(m_1, -m_3)^{\mathsf{T}}$, and $\mathrm{im}_{\mathbb{Z}}\, Q_{\mathrm{latt}}^{\mathsf{T}} = \mathrm{im}_{\mathbb{Z}}\, P^{\mathsf{T}}$ because B_0 is unimodular. Thus the stated indices are exactly $L^{\mathsf{T}}\mathbb{Z}^4$. Injectivity of L^{T} preserves the number of target characters at each eigenvalue. □

The compatibility restriction still leaves genuine source-metric freedom over a fixed target metric. Indeed, using the original $R = P^*C^{-1}$, $C = PP^*$ and $X = (R \;\; u)$, all positive H_3 with $PH_3P^* = G$ are

$$H_3 = X \begin{pmatrix} G & z \\ z^* & z^*G^{-1}z + \tau \end{pmatrix} X^*, \qquad z \in \mathbb{C}^2, \quad \tau > 0. \tag{HS.17}$$

To prove the assertion, $PX = (I_2 \;\; 0)$ and $X^{-1} = \begin{pmatrix} P \\ u^*/3 \end{pmatrix}$ show that the top-left block of $X^{-1}H_3X^{-*}$ is G. For any remaining blocks z, d, its quadratic form is

$$(v + G^{-1}zw)^*G(v + G^{-1}zw) + (d - z^*G^{-1}z)|w|^2.$$

It is positive exactly when $d = z^*G^{-1}z + \tau$ with $\tau > 0$. This proves (HS.17) in both directions and its five real parameters. The injectivity of inversion makes distinct H_3 distinct compatible source metrics; (HS.11)–(HS.14) prove that the quotient metric and quotient spectrum stay fixed. This specifies exactly what the remaining observation fibre means geometrically.

Proposition 24.15 (Native Euclidean observation and expanding dual spectrum)**.** *Equip the source with its native Euclidean metric in the retained coordinates: $h_3 = I_3$, $g_{\mathrm{phys}} = I_6$. Then*

$$W_3 = \mathcal{N}^{-1}\mathcal{N}^{-\mathsf{T}}, \qquad H_3 = 2I_3, \qquad G = 2C, \qquad C = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}. \tag{HS.18}$$

The induced target metric is $h_2 = C^{-1}$, not I_2. The source eigenvalues, with no correction suppressed, are

$$\lambda_{m,n} = 4\pi^2 \left(\|m\|^2 + \|Y^{-\mathsf{T}}(n - A^{\mathsf{T}}m)\|^2 \right), \qquad m, n \in \mathbb{Z}^3. \tag{HS.19}$$

For $y > K + 1$, its smallest positive eigenvalue is exactly

$$\lambda_1^{(3)} = 4\pi^2 \min_{0 \neq n \in \mathbb{Z}^3} \|Y^{-\mathsf{T}}n\|^2, \qquad \frac{4\pi^2}{(y+K)^2} \leq \lambda_1^{(3)} \leq \frac{4\pi^2}{(y-K)^2}. \tag{HS.20}$$

For the induced target metric, its smallest positive eigenvalue is

$$\lambda_1^{(2)} = 4\pi^2 \min_{\substack{0 \neq n \in \mathbb{Z}^3 \\ u^{\mathsf{T}}n = 0}} \|Y^{-\mathsf{T}}n\|^2, \qquad \frac{8\pi^2}{(y+K)^2} \leq \lambda_1^{(2)} \leq \frac{8\pi^2}{(y-K)^2}. \tag{HS.21}$$

In particular the exact leading constants are

$$\lambda_1^{(3)} \sim \frac{4\pi^2}{y^2} = \frac{16\pi^4}{(\log|t|)^2}, \qquad \lambda_1^{(2)} \sim \frac{8\pi^2}{y^2} = \frac{32\pi^4}{(\log|t|)^2}. \tag{HS.22}$$

The source volume is the retained polynomial

$$\mathrm{Vol}(T_3, I) = \det Y = y\left[y^2 + y\,\mathrm{tr}\big((\mathrm{Im}\,C_2)B_0^{-1}\big) + \det(\mathrm{Im}\,C_2)\right] \sim y^3. \tag{HS.23}$$

Proof. In (HS.1) the physical cometric is $T = I_6$, so $H_3 = \mathcal{J}\mathcal{J}^* = 2I_3$. Multiplication by P, P^* and (HS.11) prove all of (HS.18). The exact inverse transpose is

$$\mathcal{N}^{-\mathsf{T}} = \begin{pmatrix} I_3 & 0 \\ -Y^{-\mathsf{T}}A^{\mathsf{T}} & Y^{-\mathsf{T}} \end{pmatrix},$$

which gives (HS.19). The singular values of Y lie between $y - K$ and $y + K$, since $(y - K)\|v\| \leq \|Yv\| \leq (y + K)\|v\|$. The same singular-value bounds for the inverse transpose give

$$\frac{\|n\|}{y+K} \leq \|Y^{-\mathsf{T}}n\| \leq \frac{\|n\|}{y-K}. \tag{HS.24}$$

For source modes with $m \neq 0$, (HS.19) is at least $4\pi^2$. The mode $m = 0, n = e_1$ has eigenvalue at most $4\pi^2/(y-K)^2 < 4\pi^2$. Thus a minimizer has $m = 0$. The minimum exists because the inverse-transpose quadratic form is positive definite and only finitely many integer vectors have bounded value. Nonzero integer vectors have norm at least one, and a unit vector exists, giving (HS.20).

Target modes have $m = P^{\mathsf{T}}a$, $n = Q_{\mathrm{latt}}^{\mathsf{T}}b$ for $a, b \in \mathbb{Z}^2$. A nonzero integer zero-sum triple has squared norm at least two: it has at least one positive and at least one negative integer coordinate. The value two is attained by $(1, -1, 0)^{\mathsf{T}}$. Therefore $a \neq 0$ costs at least $8\pi^2$, whereas a mode with $a = 0$ and a shortest nonzero zero-sum n has eigenvalue at most $8\pi^2/(y-K)^2 < 8\pi^2$. The complete index characterization in the preceding theorem now proves (HS.21). Multiplying the bounds by y^2 and taking the limit proves (HS.22), with the original definition of y.

Finally $D_2 = \operatorname{Im} C_2$ is real and $Y = yI_3 + SD_2Q_{\mathrm{latt}}$. Since $Q_{\mathrm{latt}}S = B_0^{-1}$, the identity $\det(I + UV) = \det(I + VU)$ gives

$$\det Y = y^3 \det(I_2 + y^{-1}D_2B_0^{-1}).$$

For completeness that determinant identity follows by taking the determinant of $\left(\begin{smallmatrix} I & U \\ -V & I \end{smallmatrix}\right)$ and eliminating first the upper-right block or first the lower-left block using block triangular matrices with determinant one. Expansion of the two-by-two determinant and $\det B_0 = 1$ give the displayed polynomial in (HS.23). Its sign is positive: $yI_3 + rSD_2Q_{\mathrm{latt}}$ is invertible for $0 \leq r \leq 1$ by the same bound $y - rK > 0$, so its determinant has the positive sign of y^3. Formula (HS.7) and boundedness of D_2 finish the proof. □

Equations (HS.18)–(HS.23) exhibit the exact relationship rather than a nonidentity alone: the constant observation is twice the native physical cometric and twice its quotient cometric, while the unchanged period map enters the dual lattice in (HS.19). The periods expand, the volume grows, and both displayed positive spectral gaps tend to zero. No period coefficient was replaced and no metric scale was divided out. At fixed periods, H_3 uniquely specifies the Hermitian-compatible source metric; at fixed G the five-parameter family (HS.17) specifies its precise residual freedom. When periods are also allowed to vary, the constant $H_3 = 2I_3$ alone cannot recover their marked cometric; (HS.8) proves that the full marked spectrum does recover it. These are geometric and spectral assertions, not a selection of a preferred dynamical metric or a field-theoretic mass.

Primary reference conventions. The nonnegative Laplacian and integer-valued dual-lattice convention agree with A. Gogolev, *Spectra of flat tori*, Section 2 (27 November 2017), available at `https://andreygogolev.wordpress.com/2017/11/27/spectra-of-flat-tori/`. V. Ivrii, *Partial Differential Equations*, Appendix 4.B, `https://www.math.utoronto.ca/courses/apm346h1/20181/PDE-textbook/Chapter4/S4.B.html`, instead places 2π in the definition of the dual lattice; his dual is exactly $2\pi\mathcal{N}^{-\mathsf{T}}\mathbb{Z}^6$ here. The horizontal-isometry definition and uniqueness of quotient metrics are those in U. Lang, *Lecture Notes on Riemannian Geometry*, 16 June 2020, Definition 4.1 and Proposition 4.4, printed pages 45 and 47, `https://metaphor.ethz.ch/x/2020/fs/401-3532-08L/sc/DG2_16June2020.pdf`. All specialized matrix identities and period/spectral asymptotics used here have been proved above in the retained coordinates.

24.8 The unrestricted and constrained scalar equations

Equation (826) gives the exact normal covector at $\rho = 0$:

$$d_\rho N_{15}(\delta\rho) = -2\operatorname{Re}(\delta\rho\,\chi^* z). \tag{832}$$

It need not vanish. For $G = I_2$, $z = \chi = (1, 0)^{\mathsf{T}}$, $\lambda = 3$, $\rho = t \in \mathbb{R}$,

$$N_{15}(t) = 1 - 2t - 3t^2.$$

At $t=0$ this is a positive-definite quaternionic matrix: $w=(0,1)^{\mathsf T}$ and the Schur excess is $3-1-1=1$. Thus the nonzero normal covector occurs even in the interior of the positive cone. This alone is a statement about the cubic; to determine motion one must also retain the specified kinetic form.

Theorem 24.16 (Normal force in the supplied scalar model)**.** *Use the triality packet's fourteen-real-dimensional scalar variables and potential*

$$\begin{aligned}\Phi=\Phi^{*},\quad \operatorname{tr}\Phi=0,\quad b\in\mathbb{C}^3,\quad s=\operatorname{tr}\Phi^2+2b^{*}b,\\ V(\Phi,b)=\frac{m^2}{2}s+\frac{\Gamma}{4}s^2-\kappa(\det\Phi-b^{*}\Phi b),\\ L=\tfrac12\operatorname{tr}(\dot\Phi^2)+\dot b^{*}\dot b-V,\quad m^2\ge0,\quad \Gamma>0,\quad \kappa\in\mathbb{R}.\end{aligned}\tag{833}$$

This is the explicitly specified spatially constant scalar mechanics with gauge fields omitted, not a reduction claimed for arbitrary gauge fields. For its solutions, with $\rho=u^{*}b$,

$$\ddot b=-[(m^2+\Gamma s)I_3+\kappa\Phi]b,\qquad \ddot\rho=-(m^2+\Gamma s)\rho-\kappa u^{*}\Phi b.\tag{834}$$

The phase-space constraint $\rho=\dot\rho=0$ is not invariant for $\kappa\neq0$. At initial data

$$\Phi_0=\operatorname{diag}(1,-1,0),\quad b_0=(1,-1,0)^{\mathsf T},\quad \dot\Phi_0=0,\quad \dot b_0=0,$$

one has $\ddot\rho(0)=-2\kappa$.

Proof. For a variation δb,

$$\delta_b V=2\operatorname{Re}\big(\delta b^{*}[(m^2+\Gamma s)I_3+\kappa\Phi]b\big).$$

The kinetic variation has the same factor two, giving the first equation in (834); multiply by u^{*} for the second. At the displayed initial data $u^{*}b_0=0$ but $\Phi_0b_0=(1,1,0)^{\mathsf T}$, so the acceleration is -2κ. The equations have polynomial right-hand sides and hence a unique local solution for these initial data. Taylor's theorem gives $\rho(t)=-\kappa t^2+o(t^2)$, which leaves the constraint. For comparison with the marked cubic, if $X^{-1}\Phi X^{-*}=\left(\begin{smallmatrix}G & z\\ z^{*} & \lambda\end{smallmatrix}\right)$, then

$$\operatorname{tr}(C^{-1}G)+3\lambda=0,\qquad u^{*}\Phi b=3(z^{*}\chi+\lambda\rho).$$

Thus at $\rho=0$ the normal acceleration is $-3\kappa z^{*}\chi$. This factor three is fixed by the marking and the kinetic form; it is not removed by rewriting the cubic. □

Proposition 24.17 (Exact constrained alternative)**.** *Let $\Pi_u=uu^{*}/3$, $Q_u=I_3-\Pi_u$. Restrict the Lagrangian* (833) *to $b\in u^{\perp}$ and velocities in $u^{\perp}$. Its Euler–Lagrange equations are*

$$\begin{aligned}\ddot b&=-(m^2+\Gamma s)b-\kappa Q_u\Phi b,\\ \ddot\Phi&=-(m^2+\Gamma s)\Phi+\kappa\big(\operatorname{adj}\Phi-bb^{*}\big)_0,\qquad T_0=T-\tfrac13(\operatorname{tr}T)I_3.\end{aligned}\tag{835}$$

The reaction force added to the unrestricted b equation is exactly $\kappa\Pi_u\Phi b$. The constrained solution is also an unrestricted solution exactly when this force vanishes throughout the solution (or automatically when $\kappa=0$).

Proof. Allowed b variations are projected by the orthogonal matrix Q_u, so projecting (834) gives the first equation. In particular $u^{*}\ddot b=0$, and initial data tangent to the constraint remain there. For the Φ equation, variations are trace-free Hermitian and

$$\delta_\Phi V=\operatorname{tr}\big([(m^2+\Gamma s)\Phi-\kappa\operatorname{adj}\Phi+\kappa bb^{*}]\delta\Phi\big).$$

Projection onto trace-free matrices gives the second equation. The unrestricted Φ equation is unchanged. Subtracting the two b equations gives the stated reaction, proving the last equivalence. □

The unrestricted model retains all six real coordinates of b; the constrained model retains four and supplies the displayed reaction force. Neither is excluded by the failure of the other identification. The old complex sector $b = \dot{b} = 0$ *is* invariant by (834). By contrast, a nonzero fixed complex subspace of $\mathbb{C}^3$ invariant under all possible Hermitian Φ must be the whole space: traceless Hermitian matrices together with I_3 complex-linearly span every matrix unit, and the matrix units applied to a nonzero vector span $\mathbb{C}^3$. Thus a four-real-dimensional complex-linear transverse sector cannot retain all these unrestricted couplings without the explicit constraint or an additional restriction on Φ.

24.9 Preserved interacting sectors of the unrestricted equations

The failure of the whole slice to be invariant does not exclude invariant motions within it. We now construct such motions using the original equations, not by adding the reaction in Proposition 24.17. Write $a(t) = m^2 + \Gamma s(t)$, and retain the trace-free Hermitian variable Φ, the standard metric on $b \in \mathbb{C}^3$, and $u^* u = 3$.

Lemma 24.18 (Global existence in the scalar mechanics)**.** *Every initial position and velocity in* (833) *has a unique solution for all real time. Its conserved energy is*

$$E = \tfrac{1}{2} \operatorname{tr} \dot{\Phi}^2 + \dot{b}^* \dot{b} + V(\Phi, b).$$

Proof. The unrestricted equations for both variables are

$$\ddot{\Phi} = -a\Phi + \kappa(\operatorname{adj} \Phi - bb^*)_0, \qquad \ddot{b} = -ab - \kappa \Phi b. \tag{836}$$

They form a polynomial first-order system on positions and velocities, so local existence and uniqueness hold. Put $t_\Phi = \operatorname{tr} \Phi^2$ and $n_b = b^* b$. Direct differentiation of the kinetic energy gives

$$-a \operatorname{tr}(\dot{\Phi}\Phi) + \kappa \operatorname{tr}(\dot{\Phi} \operatorname{adj} \Phi) - \kappa b^* \dot{\Phi} b - 2a \operatorname{Re}(\dot{b}^* b) - 2\kappa \operatorname{Re}(\dot{b}^* \Phi b).$$

The trace projection drops out because $\operatorname{tr} \dot{\Phi} = 0$. Differentiating the displayed potential gives precisely the negative of this expression, including the two terms from differentiating $b^* \Phi b$. Thus E is constant.

For completeness, a bound in the unchanged scalar coordinates suffices to rule out escape. The three real eigenvalues of Φ give $|\det \Phi| \le (t_\Phi/3)^{3/2}$ by the arithmetic–geometric mean inequality applied to their squares. Cauchy–Schwarz and the Frobenius norm give $|b^* \Phi b| \le n_b \sqrt{t_\Phi}$. Since $s = t_\Phi + 2n_b$, these are the explicit inequalities

$$|\det \Phi - b^* \Phi b| \le (t_\Phi/3)^{3/2} + n_b \sqrt{t_\Phi} \le \left(\frac{1}{3\sqrt{3}} + \frac{1}{2} \right) s^{3/2}.$$

This bound is not asserted to be sharp and does not change V. Consequently

$$V \ge \frac{\Gamma}{4} s^2 + \frac{m^2}{2} s - |\kappa| \left(\frac{1}{3\sqrt{3}} + \frac{1}{2} \right) s^{3/2} \longrightarrow +\infty \quad (s \longrightarrow \infty).$$

The energy identity bounds positions; the global lower bound for V then bounds velocities. A polynomial vector field and its derivative are bounded on this compact phase-space set. A finite endpoint therefore has a limiting initial state to which local existence applies, extending the solution. Neither a finite forward nor a finite backward endpoint occurs. □

Theorem 24.19 (Fixed-subspace invariant sectors)**.** *Fix a complex subspace $W \subset u^\perp$ of dimension $r = 1$ or 2, and let E_W be its orthogonal projection in the original metric. Define*

$$\mathcal{T}_W = \{(\Phi, b) : \Phi = \Phi^*,\ \operatorname{tr} \Phi = 0,\ [\Phi, E_W] = 0,\ b \in W\}.$$

With tangent initial velocities, its tangent bundle $T\mathcal{T}_W$ is invariant under the full equations (836) *for all real time. For each fixed W, the configuration dimensions are respectively* 6 *and* 8*, and the phase-space dimensions are* 12 *and* 16*. Every such solution satisfies*

$$\rho = 0, \qquad u^*\Phi b = 0, \qquad \kappa(uu^*/3)\Phi b = 0$$

identically, without any constraint force.

Proof. For Hermitian Φ, preservation of W also preserves $W^\perp$: $\langle \Phi x, w\rangle = \langle x, \Phi w\rangle = 0$ for $x \in W^\perp$, $w \in W$. This is equivalent to $[\Phi, E_W] = 0$. The trace-free adjugate identity is

$$\operatorname{adj}\Phi = \Phi^2 - \tfrac{1}{2}(\operatorname{tr}\Phi^2)I_3. \tag{837}$$

Indeed the characteristic polynomial is $z^3 - \frac{1}{2}(\operatorname{tr}\Phi^2)z - \det\Phi$; Cayley–Hamilton and the adjugate identity prove (837) when Φ is invertible, and polynomial continuity proves it at singular Φ as well. Thus the adjugate commutes with E_W. So does bb^*: it takes W to W and vanishes on $W^\perp$. Scalar trace subtraction preserves this property. Both accelerations in (836) consequently remain in the specified linear spaces. The position derivatives do so when the velocities satisfy $[\dot\Phi, E_W] = 0$, $\dot b \in W$. The restricted polynomial vector field has a local solution; its inclusion is a solution of the full system. Uniqueness and Lemma 24.18 give the claimed global invariance.

The Hermitian blocks on W and $W^\perp$ have real dimensions r^2 and $(3-r)^2$. The trace imposes one real equation, and b contributes $2r$ real coordinates. Hence $\dim_{\mathbb{R}} \mathcal{T}_W = r^2 + (3-r)^2 - 1 + 2r$. Velocities double this dimension. Finally $b, \Phi b \in W \subset u^\perp$ proves each asserted zero contraction. The parameter W is fixed, not an additional moving coordinate: for $r = 1$ the choices form $\mathbb{P}(u^\perp) \cong \mathbb{CP}^1$, and for $r = 2$ the only choice is $W = u^\perp$. □

Here is an explicit interacting member that also retains a nonzero marked cross-coordinate z. Keep

$$v = (1,-1,0)^{\mathsf T}, \quad w_0 = (1,1,-2)^{\mathsf T}, \quad E = vv^*/2, \quad F = w_0w_0^*/6,$$

and take

$$W = \mathbb{C}v, \quad \Phi_0 = E - F + w_0u^* + uw_0^*, \quad b_0 = v, \quad \dot\Phi_0 = 0, \quad \dot b_0 = 0. \tag{838}$$

The vectors u, v, w_0 are orthogonal with squared lengths $3, 2, 6$. Thus $\operatorname{tr}\Phi_0 = 0$, $\Phi_0 v = v$, $\operatorname{tr}\Phi_0^2 = 38$, $s_0 = 42$, and

$$b_0^*\Phi_0 b_0 = 2, \qquad \ddot b_0 = -(m^2 + 42\Gamma + \kappa)v.$$

For example $\kappa > 0$ gives nonstationary interacting motion. In the original marked coordinates, not a newly chosen basis,

$$\chi_0 = R^*b_0 = (1,0)^{\mathsf T}, \qquad z_0 = P\Phi_0 u/3 = (0,3)^{\mathsf T}, \qquad \lambda_0 = 0.$$

Hence $z_0 \neq 0$ while $z_0^*\chi_0 = 0$. Preservation of the scalar slice does not require $z = 0$ for every solution. The theorem constructs exact families; it does not assert they exhaust all invariant initial data.

Proposition 24.20 (Tangency through third order is insufficient)**.** *For any unrestricted solution put $c = u^*\Phi b$ and $h = u^*\dot\Phi\dot b$. The exact identities are*

$$\begin{aligned} \ddot c &= -2ac - \frac{\kappa s}{3}\rho + 2h, \\ \rho^{(4)} &= \left(-\ddot a + a^2 + \frac{\kappa^2 s}{3}\right)\rho - 2\dot a\dot\rho + 3a\kappa c - 2\kappa h. \end{aligned} \tag{839}$$

There are initial data with $\rho^{(j)}(0) = 0$ for $0 \le j \le 3$ but $\rho^{(4)}(0) = -12\kappa$.

Proof. Equation (837) gives, with $t_\Phi = \operatorname{tr}\Phi^2$ and $n_b = b^*b$,

$$(\operatorname{adj}\Phi - bb^*)_0 = \Phi^2 - bb^* + \frac{n_b - t_\Phi}{3} I_3, \qquad (\operatorname{adj}\Phi - bb^*)_0 b = \Phi^2 b - \frac{s}{3} b.$$

Differentiate c twice and substitute (836):

$$\ddot{c} = u^*(\ddot{\Phi}b + 2\dot{\Phi}\dot{b} + \Phi\ddot{b}) = -2ac - \frac{\kappa s}{3}\rho + 2h.$$

The two terms $\kappa u^*\Phi^2 b$ cancel exactly. Twice differentiating $\ddot{\rho} = -a\rho - \kappa c$ and substituting this identity proves the second formula. In particular, at $\rho = \dot{\rho} = 0$, vanishing second and third derivatives requires $c = \dot{c} = 0$ when $\kappa \neq 0$; then the fourth derivative is $-2\kappa h$.

For the promised initial data use the same u, v, w_0 as above, but now set

$$\Phi_0 = 0, \quad b_0 = v, \quad \dot{b}_0 = w_0, \quad \dot{\Phi}_0 = (uw_0^* + w_0u^*)/3.$$

This matrix is trace-free Hermitian, $\dot{\Phi}_0 v = 0$ and $\dot{\Phi}_0 w_0 = 2u$. Consequently $\rho_0 = \dot{\rho}_0 = c_0 = \dot{c}_0 = 0$, whereas $h_0 = 6$. Formula (839) gives $\rho^{(4)}(0) = -12\kappa$, and Taylor's theorem gives $\rho(t) = -(\kappa/2)t^4 + o(t^4)$. For $\kappa \neq 0$ this solution leaves the slice despite the lower-order tangencies. For $\kappa = 0$, by contrast, the full locus $\rho = \dot{\rho} = 0$ is invariant because its equation is the homogeneous linear equation $\ddot{\rho} = -a(t)\rho$ with zero initial data. □

24.10 Periodic motions outside every fixed-subspace family

The families in Theorem 24.19 do not exhaust the motions inside $\rho = 0$. The following exact relative equilibria rotate the instantaneous kernel of Φ while retaining all the original couplings.

Theorem 24.21 (A moving-kernel family)**.** *Choose $\Gamma > 0$, $\kappa \neq 0$, and real x, a satisfying*

$$0 < \frac{x}{\kappa} \leq \frac{1}{6\Gamma}, \qquad a > \frac{\kappa x}{3}.$$

Choose $B, d > 0$ with

$$B^2 + d^2 = \frac{3ax}{\kappa} - x^2, \qquad m^2 = a\left(1 - \frac{6\Gamma x}{\kappa}\right). \tag{840}$$

The right-hand sides give a positive sum and $m^2 \geq 0$. Retain u, v, w_0 from (838)*, and put*

$$J_0 = \begin{pmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ -1 & 1 & 0 \end{pmatrix}, \quad \Pi = uu^*/3, \quad Q_u = I_3 - \Pi, \quad \Omega = \sqrt{a/3}\, J_0,$$

$$O(t) = \Pi + \cos(\sqrt{a}\, t)Q_u + \frac{\sin(\sqrt{a}\, t)}{\sqrt{3}} J_0. \tag{841}$$

Define the original-coordinate initial data and their motion by

$$\begin{aligned} b_0 &= \frac{B}{\sqrt{2}} v, \qquad \Phi_0 = -\frac{x}{6} w_0 w_0^* + \frac{x}{3} uu^* + \frac{d}{\sqrt{18}}(w_0u^* + uw_0^*), \\ b(t) &= O(t)b_0, \qquad \Phi(t) = O(t)\Phi_0 O(t)^*. \end{aligned} \tag{842}$$

These are global periodic solutions of (836)*. They satisfy*

$$s = \frac{6ax}{\kappa}, \qquad m^2 + \Gamma s = a, \qquad \Phi(t)b(t) = 0, \qquad \rho(t) = 0.$$

No fixed complex $W \subset u^\perp$ contains this trajectory in $T\mathcal{T}_W$.

Proof. Direct multiplication gives $J_0^* = -J_0$, $J_0 u = 0$, $J_0^2 = -3Q_u$, and $\Pi J_0 = J_0 \Pi = 0$. Consequently $O(t)^* O(t) = I_3$, $O(t)u = u$, and $\dot{O} = \Omega O = O\Omega$. Its determinant is one by continuity from $O(0) = I_3$. This proves that the displayed rotation is in the original Euclidean metric, with its frequency $\sqrt{a}$ unchanged.

For the following verification only, use the explicit orthogonal matrix

$$K = \left(\frac{v}{\sqrt{2}}, \frac{w_0}{\sqrt{6}}, \frac{u}{\sqrt{3}} \right), \quad K^* K = I_3,$$

and retain its inverse K^* and all its factors. It gives

$$K^* \Phi_0 K = H = \begin{pmatrix} 0 & 0 & 0 \\ 0 & -x & d \\ 0 & d & x \end{pmatrix}, \quad K^* b_0 = B e_1, \quad K^* \Omega K = A = \begin{pmatrix} 0 & -\sqrt{a} & 0 \\ \sqrt{a} & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Thus $s = 2(x^2 + d^2 + B^2) = 6ax/\kappa$, as asserted. In these exact coordinates $HBe_1 = 0$ and

$$(\operatorname{adj} H - B^2 e_1 e_1^*)_0 = \operatorname{diag}(-s/3, s/6, s/6).$$

The required accelerations are explicitly

$$[A,[A,H]] = \begin{pmatrix} -2ax & 0 & 0 \\ 0 & 2ax & -ad \\ 0 & -ad & 0 \end{pmatrix} = -aH + \kappa \operatorname{diag}(-s/3, s/6, s/6), \qquad A^2 B e_1 = -aBe_1.$$

Conjugation by K therefore verifies both equations at $t = 0$. The full force commutes with orthogonal conjugation on Φ and the same orthogonal action on b: the adjugate, trace subtraction, norms, and product Φb all transform in this way. Since O commutes with Ω, conjugation by $O(t)$ verifies both equations at every time, not merely to a finite Taylor order. This also proves the stated kernel and normal-coordinate identities.

At times 0 and $\pi/(2\sqrt{a})$, the two values of b are nonzero multiples of v and w_0. They span $u^\perp$ over $\mathbb{C}$. Any fixed W containing the whole trajectory would consequently equal $u^\perp$. But $u^* \Phi_0 w_0 = \sqrt{18}\, d \neq 0$, so Φ_0 does not preserve this plane. This rules out every fixed-W family in the earlier theorem. □

A particularly small exact instance is

$$\Gamma = m^2 = 1, \quad \kappa = 6, \quad x = \tfrac{1}{2}, \quad a = 2, \quad B = d = \frac{1}{2\sqrt{2}},$$
$$b_0 = \frac{1}{4} \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}, \quad \Phi_0 = \frac{1}{4} \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & -2 \end{pmatrix}. \tag{843}$$

Here $s = 1$ and the period of b is $2\pi/\sqrt{2}$. Its original marked coordinates at zero are $\chi_0 = (1/4, 0)^{\mathsf{T}}$, $z_0 = (0, 1/4)^{\mathsf{T}}$, and $\lambda_0 = 1/6$. Although $\det \Phi - b^* \Phi b = 0$ along this solution, its cubic force does not vanish: at zero it is $6(\operatorname{adj} \Phi_0 - b_0 b_0^*)_0 \neq 0$. It is the coupled equations, not removal of the cubic, that sustain the rotating-kernel solution.

Proposition 24.22 (Conserved charge and the precise gauge comparison)**.** *In the scalar mechanics the anti-Hermitian matrix*

$$\mathcal{J} = [\Phi, \dot{\Phi}] + b\dot{b}^* - \dot{b} b^* \tag{844}$$

is constant. Its trace-free part is the $SU(3)$ moment matrix, using the pairing $\operatorname{tr}(\mathcal{J}\xi)$ with $\xi \in \mathfrak{su}(3)$. On (842)*,*

$$\mathcal{J} = -\frac{s}{2}\Omega, \qquad T = \frac{as}{2}, \qquad E = m^2 s + \frac{3\Gamma}{4} s^2.$$

In particular (843) *has $\mathcal{J} = -\Omega/2 \neq 0$, $T = 1$, $V = 3/4$, and $E = 7/4$. This scalar solution is not a source-free homogeneous gauge solution with all $\mathcal{A}_\mu = 0$: it fails the $\mathcal{A}_0$ field equation.*

Proof. Using (836), polynomial commutation of Φ with its adjugate, and commutation with the scalar trace correction, gives

$$\frac{d}{dt}[\Phi,\dot\Phi] = -\kappa[\Phi, bb^*], \qquad \frac{d}{dt}(b\dot b^* - \dot b b^*) = \kappa[\Phi, bb^*].$$

Their sum vanishes. Under the infinitesimal simultaneous symmetry $\delta\Phi = [\xi,\Phi]$, $\delta b = \xi b$, the kinetic momentum paired with this variation is

$$\operatorname{tr}(\dot\Phi[\xi,\Phi]) + 2\operatorname{Re}(\dot b^*\xi b) = \operatorname{tr}(\mathcal{J}\xi).$$

This proves the pairing, including its coefficient and sign. In the explicit frame K, compute

$$[H,[A,H]] + Be_1(ABe_1)^* - (ABe_1)(Be_1)^* = -(x^2+d^2+B^2)A = -\frac{s}{2}A.$$

It commutes with A, so conjugation by $O(t)$ retains the same constant matrix. Similarly $\frac{1}{2}\operatorname{tr}[A,H]^2 + |ABe_1|^2 = as/2$. The determinant and $b^*\Phi b$ both vanish on the trajectory; substitution of $a = m^2 + \Gamma s$ gives the displayed energy.

For the gauge comparison retain the packet's covariant derivatives $D_0\Phi = \dot\Phi + [\mathcal{A}_0,\Phi]$, $D_0 b = \dot b + \mathcal{A}_0 b$. Varying $\mathcal{A}_0$ by $\xi \in \mathfrak{su}(3)$ in the scalar kinetic density gives $\operatorname{tr}(\mathcal{J}\xi)$ at $\mathcal{A}_0 = 0$. With homogeneous fields and every $\mathcal{A}_i = 0$, the electric field vanishes and the gauge kinetic variation contributes zero. Thus its Euler–Lagrange equation requires the trace-free part of $\mathcal{J}$ to vanish. The exhibited matrix is already trace-free and nonzero, so this particular zero-connection lift fails. This is the Gauss equation obtained by variation before imposing temporal gauge; compare D. Tong, *Lectures on Gauge Theory*, Section 2.2.1, equation (2.27), `https://www.damtp.cam.ac.uk/user/tong/gaugetheory/gt.pdf`. Our matter contribution has been derived here in the retained anti-Hermitian convention, not imported from a different convention. □

There is an exact moving-frame map, not an assertion that these theories are unrelated. With $U(t) = O(t)^*$ the gauge transformation rule is

$$\Phi' = U\Phi U^{-1} = \Phi_0, \quad b' = Ub = b_0, \quad \mathcal{A}_0' = U\mathcal{A}_0U^{-1} - \dot U U^{-1} = \Omega.$$

The covariant velocities remain $D_0'\Phi_0 = [\Omega,\Phi_0]$ and $D_0' b_0 = \Omega b_0$. The same nonzero charge and Gauss residual therefore remain; making the ordinary coordinates static does not remove them. A gauge-field lift would have to supply the electric contribution to Gauss' equation and satisfy the remaining coupled equations. Neither its existence nor its nonexistence follows from failure of the zero-connection lift, and no quantum spectral-gap conclusion is inferred from this periodic motion.

24.11 The complete scalar vacuum calculation with its kinetic metric

This subsection validates the homogeneous scalar vacuum theorem in the triality packet. It does not assert a new discovery of that phase diagram, and does not include the packet's subsequent gauge-orbit claims. Set $Q = \Theta(\Phi, b)$ with $\operatorname{tr} Q = 0$, and retain

$$g(Y,Z) = \operatorname{Re}\operatorname{tr}(YZ), \qquad s(Q) = g(Q,Q) = \operatorname{tr}\Phi^2 + 2b^*b, \qquad n(Q) = N(Q) = \det\Phi - b^*\Phi b.$$

The kinetic term is $g(\dot Q,\dot Q)/2$, not the unweighted sum of squares of the independent entries. Explicitly, $g(Y,Y) = \sum_i Y_{ii}^2 + 2\sum_{i<j}|Y_{ij}|^2$ for Hermitian Y.

Lemma 24.23 (Sharp bound and all equality cases)**.** *For every trace-free quaternionic Hermitian Q,*

$$54\,n(Q)^2 \le s(Q)^3.$$

For $Q \neq 0$, equality holds exactly at $Q = x(3P - I_3)$, $x \in \mathbb{R}\setminus\{0\}$, where $P = P^* = P^2$ has quaternionic rank one. There $s = 6x^2$ and $n = 2x^3$.

Proof. Here are the spectral and determinant details in the stated conventions. The real-valued function v^*Qv on the unit sphere of $\mathbb{H}^3$ attains a maximum. Its Lagrange-multiplier equation gives $\operatorname{Re}(w^*(Qv - v\lambda)) = 0$ for every w, with real $\lambda = v^*Qv$. Taking $w = Qv - v\lambda$ proves $Qv = v\lambda$. The quaternionic orthogonal complement of v is invariant because $v^*Qw = \lambda v^*w$. Iteration gives a unitary eigenbasis with real eigenvalues $\lambda_1, \lambda_2, \lambda_3$.

For quaternionic matrices, the real trace is cyclic by $\operatorname{Re}(pq) = \operatorname{Re}(qp)$, entry by entry. Moreover, the original cubic (819) obeys

$$6n(Q) = (\operatorname{tr} Q)^3 - 3\operatorname{tr}(Q)\operatorname{tr}(Q^2) + 2\operatorname{tr}(Q^3). \tag{845}$$

To check it without changing the quaternionic order, in the ordered entries of (819) the cubic trace is the sum of the diagonal cubes, plus $3(d+f)|q_1|^2 + 3(a+f)|q_2|^2 + 3(a+d)|q_3|^2$, plus $6\operatorname{Re}(q_1q_2q_3)$. Substitution gives each term of six times (819). Unitary conjugation therefore preserves s, n, which are respectively $\sum_i \lambda_i^2$ and $\lambda_1\lambda_2\lambda_3$ in the eigenbasis. Their sum is zero. The exact polynomial identity

$$s^3 - 54n^2 = 2(\lambda_1 - \lambda_2)^2(\lambda_1 - \lambda_3)^2(\lambda_2 - \lambda_3)^2$$

follows by substituting $\lambda_3 = -\lambda_1 - \lambda_2$ and expanding. It proves the bound. For nonzero Q, equality means a repeated eigenvalue, hence spectrum $(2x, -x, -x)$ for some $x \neq 0$. Letting P project onto the simple eigenline gives $Q = x(3P - I_3)$. The converse follows from this same spectrum. □

Theorem 24.24 (All scalar global minima)**.** *For $\Gamma > 0$, $m^2 \geq 0$, and $\kappa \neq 0$, put $m_{\mathrm{c}}^2 = \kappa^2/(27\Gamma)$. The global minima of $V = m^2 s/2 + \Gamma s^2/4 - \kappa n$ are exactly:*

- *$Q = 0$ if $m^2 > m_{\mathrm{c}}^2$;*
- *$Q = 0$ and $Q = \frac{\kappa}{9\Gamma}(3P - I_3)$ for every rank-one quaternionic Hermitian projector P, if $m^2 = m_{\mathrm{c}}^2$;*
- *$Q = x_*(3P - I_3)$ for every such P, if $0 \leq m^2 < m_{\mathrm{c}}^2$, where*

$$x_* = \frac{\kappa + \operatorname{sgn}(\kappa)\sqrt{\kappa^2 - 24\Gamma m^2}}{12\Gamma}.$$

Each nonzero component is diffeomorphic to $\mathbb{HP}^2$. For $\kappa = 0$, the origin is the unique global minimum.

Proof. For $\kappa = 0$, the positive quartic proves the assertion, including $m^2 = 0$. Otherwise at each fixed original norm $r = \sqrt{s}$ the sharp bound is attained with $x = \operatorname{sgn}(\kappa) r/\sqrt{6}$ and $Q = x(3P - I_3)$. For $r > 0$, these are exactly its equality and sign cases. Thus the full minimization is equivalent, with all fibres of the norm map accounted for, to minimizing

$$f(r) = \frac{m^2}{2} r^2 + \frac{\Gamma}{4} r^4 - \frac{|\kappa|}{3\sqrt{6}} r^3, \qquad r \geq 0.$$

Set $r_{\mathrm{c}} = 2|\kappa|/(3\sqrt{6}\Gamma)$. Exact factorization gives

$$f(r) = \frac{\Gamma}{4} r^2 (r - r_{\mathrm{c}})^2 + \frac{m^2 - m_{\mathrm{c}}^2}{2} r^2.$$

This proves the first two cases, including every zero. Below the transition, $f(r_{\mathrm{c}}) < 0$, so a global minimum is nonzero. Put $a_0 = |\kappa|/\sqrt{6}$. The derivative is $f'(r) = r(\Gamma r^2 - a_0 r + m^2)$, with roots

$$r_\pm = \frac{a_0 \pm \sqrt{a_0^2 - 4\Gamma m^2}}{2\Gamma}.$$

Below the transition its discriminant is strictly positive. If $m^2 > 0$, f' has signs $+, -, +$ across these two positive roots, so r_+ is the only positive local minimum. If $m^2 = 0$, $r_- = 0$, and the signs on $(0, r_+)$ and

(r_+, ∞) are $-, +$. Coercivity and the negative value exclude the origin in both cases. Converting r_+ to x proves the third case and the completeness of its matrix equality locus.

For the manifold statement, a right quaternionic line represented by $v \neq 0$ has orthogonal projector $P_v = v(v^*v)^{-1}v^*$. This is unchanged under $v \mapsto vh$, $h \in \mathbb{H}^\times$, and every rank-one Hermitian projector is the unique projection onto its image line. The formula and its inverse line charts are smooth. For a fixed nonzero x, the maps $P \mapsto x(3P - I_3)$ and $Q \mapsto (Q/x + I_3)/3$ are mutually inverse. This proves the claimed $\mathbb{HP}^2$ identification without any gauge quotient. □

Theorem 24.25 (Full metric Hessian)**.** *At every nonzero minimum $Q_* = x(3P - I_3)$, the Hessian relative to g has eigenvalues*

eigenvalue	*real multiplicity*
$x(12\Gamma x - \kappa)$	1
0	8
$3\kappa x$	5

Both nonzero eigenvalues are positive. The entire kernel is the tangent space of the nonzero minimum manifold. At the origin the metric Hessian is $m^2 I$ in all fourteen directions.

Proof. The preceding spectral proof and cubic trace identity show that quaternionic unitary conjugation preserves both g and V. Transport the Hessian isometrically to $Q_* = xD$, $D = \operatorname{diag}(2, -1, -1)$. Every trace-free Hermitian variation has the unique, g-orthogonal decomposition

$$Y = yD + \begin{pmatrix} 0 & v^* \\ v & 0_2 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & B_0 \end{pmatrix}, \qquad y \in \mathbb{R}, \quad v \in \mathbb{H}^2, \quad B_0 = B_0^*, \quad \operatorname{tr} B_0 = 0.$$

Write $B_0 = \left(\begin{smallmatrix} \beta & \omega \\ \bar{\omega} & -\beta \end{smallmatrix}\right)$, $\beta \in \mathbb{R}$, $\omega \in \mathbb{H}$. The respective dimensions are $1, 8, 5$, and the retained norm is

$$g(Y, Y) = 6y^2 + 2v^*v + \|B_0\|^2, \qquad \|B_0\|^2 = 2\beta^2 + 2|\omega|^2.$$

For $h = x + \varepsilon y$, the full cubic, including every mixed direction, is

$$n(Q_* + \varepsilon Y) = 2h^3 + \varepsilon^2 h(v^*v - \|B_0\|^2) + \varepsilon^3 v^* B_0 v. \tag{846}$$

Indeed the diagonal and lower off-diagonal contributions are $2h(h^2 - \varepsilon^2\beta^2) - 2h\varepsilon^2|\omega|^2$. The upper entries contribute $h\varepsilon^2(|v_1|^2 + |v_2|^2) + \varepsilon^3\beta(|v_1|^2 - |v_2|^2)$, and the triangular term is $2\varepsilon^3 \operatorname{Re}(\bar{v}_1 \omega v_2)$. These sum to (846), preserving the quaternionic order. Also

$$s(Q_* + \varepsilon Y) = 6x^2 + 12\varepsilon xy + \varepsilon^2(6y^2 + 2v^*v + \|B_0\|^2).$$

The second derivative of the cubic is therefore $12xy^2 + 2xv^*v - 2x\|B_0\|^2$. Differentiating the potential gives the complete quadratic form before using stationarity:

$$\begin{aligned} D^2 V_{xD}[Y, Y] = {} & 6(m^2 + 18\Gamma x^2 - 2\kappa x)y^2 \\ & + 2(m^2 + 6\Gamma x^2 - \kappa x)v^*v \\ & + (m^2 + 6\Gamma x^2 + 2\kappa x)\|B_0\|^2. \end{aligned}$$

Every nonzero minimum satisfies $m^2 - \kappa x + 6\Gamma x^2 = 0$. Thus this becomes

$$6x(12\Gamma x - \kappa)y^2 + 3\kappa x\|B_0\|^2.$$

There are no mixed quadratic terms; polarization therefore gives no mixed bilinear blocks. Comparison with $g(Y, Y)$ proves every listed eigenvalue and multiplicity, with no rescaling of the kinetic metric.

At coexistence the positive values are $\kappa^2/(27\Gamma)$ and $\kappa^2/(3\Gamma)$. Below it, $12\Gamma x-\kappa=\operatorname{sgn}(\kappa)\sqrt{\kappa^2-24\Gamma m^2}$ and x has the same sign as κ, proving positivity. In the line chart

$$P(z)=\frac{1}{1+z^*z}\begin{pmatrix}1 & z^*\\ z & zz^*\end{pmatrix},\qquad DP_0[\zeta]=\begin{pmatrix}0 & \zeta^*\\ \zeta & 0_2\end{pmatrix},$$

the derivative of $x(3P(z)-I_3)$ spans exactly the eight-dimensional zero block. Isometric conjugation transports this to every P. At the origin, the cubic and quartic contribute zero Hessian, giving $m^2 g$ directly. In particular, at $m^2=\kappa=0$ the origin is still the unique minimum, although its Hessian vanishes in all directions. □

These are scalar Hessian values of a specified finite-dimensional model. Their exact multiplicities do not assert a gauge-reduced physical spectrum or an interacting quantum mass gap. The current calculation includes the full tangent and normal blocks, not solely the radial restriction.

24.12 Gauge kinetic sign and precise scope of this audit

In the packet's gauge lift, $\mathcal{A}_\mu\in\mathfrak{su}(3)$ and $D_\mu b=\partial_\mu b+\mathcal{A}_\mu b$ fix anti-Hermitian gauge generators. Retain Minkowski signature $(+---)$, consistent with the positive time-derivative scalar term in that lift. Write $\|T\|^2=-\operatorname{tr}(T^2)$ for anti-Hermitian T, and $E_i=\mathcal{F}_{0i}$, $B_i=\frac{1}{2}\epsilon_{ijk}\mathcal{F}_{jk}$. Raising the time-space index pair changes its sign and summing both orders of each antisymmetric pair gives

$$\operatorname{tr}(\mathcal{F}_{\mu\nu}\mathcal{F}^{\mu\nu})=2\sum_i(\|E_i\|^2-\|B_i\|^2).$$

Consequently the packet's coefficient $-1/(2g_c^2)$ gives negative electric kinetic energy in these conventions. Keeping its anti-Hermitian fields and scalar kinetic terms, the positive-energy gauge density is

$$\mathcal{L}_{\text{gauge}}=+\frac{1}{2g_c^2}\operatorname{tr}(\mathcal{F}_{\mu\nu}\mathcal{F}^{\mu\nu})=\frac{1}{g_c^2}\sum_i(\|E_i\|^2-\|B_i\|^2).$$

For an immediate check take $\mathcal{F}_{01}=\operatorname{diag}(i,-i,0)$ and all other independent components zero: the trace contraction is 4, so the two signs give $-2/g_c^2$ and $+2/g_c^2$, respectively. This sign correction changes neither the scalar potential nor its derivative calculated above. It does matter before interpreting the full gauge lift as a positive-energy physical model.

The equal-diagonal thirteen-space in the triality packet remains a different embedded slice: its intersection with $\mathrm{Herm}_3(\mathbb{C})$ has dimension seven, whereas the common Lorentz slice contains all nine complex-Hermitian coordinates. Their precise common ambient space is $\mathrm{Herm}_3(\mathbb{H})$; Theorem 24.2 identifies the *Lorentz* constructions, not the equal-diagonal one. No result here identifies these cone-boundary spheres with the claimed integrable complex structure on S^6, assigns new spinor periods and toroidal fillings, or proves a continuum interacting Yang–Mills spectral gap. The exact next data are the transition maps and lattice extension for the additional spinor, followed by a physical state space and Hamiltonian comparison retaining the couplings above.

24.13 The full homogeneous gauge equations and the literal periodic orbit

Retain signature $(+---)$, $\mathcal{A}_\mu\in\mathfrak{su}(3)$, $\Phi=\Phi^*$, $\operatorname{tr}\Phi=0$, and $b\in\mathbb{C}^3$. Write $g^2>0$ for the gauge coupling in this subsection and keep

$$s=\operatorname{tr}\Phi^2+2b^*b,\qquad a=m^2+\Gamma s,\qquad V=\frac{m^2}{2}s+\frac{\Gamma}{4}s^2-\kappa(\det\Phi-b^*\Phi b).$$

Thus $m^2\geq 0$, $\Gamma>0$, and $\kappa\in\mathbb{R}$ are unchanged. For a matrix Z write $Z_{\text{tf}}=Z-\operatorname{tr}(Z)I_3/3$. The full density is

$$\mathcal{L}=\frac{1}{2g^2}\operatorname{tr}(\mathcal{F}_{\mu\nu}\mathcal{F}^{\mu\nu})+\frac{1}{2}\operatorname{tr}(D_\mu\Phi D^\mu\Phi)+(D_\mu b)^*D^\mu b-V,\tag{847}$$

where $\mathcal{F}_{\mu\nu} = \partial_\mu \mathcal{A}_\nu - \partial_\nu \mathcal{A}_\mu + [\mathcal{A}_\mu, \mathcal{A}_\nu]$, $D_\mu \Phi = \partial_\mu \Phi + [\mathcal{A}_\mu, \Phi]$, and $D_\mu b = \partial_\mu b + \mathcal{A}_\mu b$. We consider spatially homogeneous fields on a finite-volume flat $\mathbb{T}^3$, or locally on $\mathbb{R}^3$. A nonzero constant energy density is not asserted to have finite total energy on $\mathbb{R}^3$.

Proposition 24.26 (Every homogeneous field equation)**.** *Set*

$$E_i = \dot{\mathcal{A}}_i + [\mathcal{A}_0, \mathcal{A}_i], \qquad F_{ij} = [\mathcal{A}_i, \mathcal{A}_j], \qquad X = \dot{\Phi} + [\mathcal{A}_0, \Phi], \qquad v = \dot{b} + \mathcal{A}_0 b,$$

and define the lower-index matter currents

$$J_0 = [\Phi, X] + b v^* - v b^*, \qquad J_i = [\Phi, [\mathcal{A}_i, \Phi]] - b b^* \mathcal{A}_i - \mathcal{A}_i b b^*.$$

The complete homogeneous Euler–Lagrange system is

$$D_0 X - \sum_i [\mathcal{A}_i, [\mathcal{A}_i, \Phi]] + a\Phi - \kappa(\operatorname{adj} \Phi - b b^*)_{\mathrm{tf}} = 0, \tag{848}$$

$$D_0 v - \sum_i \mathcal{A}_i^2 b + (a I_3 + \kappa \Phi) b = 0, \tag{849}$$

$$\frac{2}{g^2} \sum_i [\mathcal{A}_i, E_i] - (J_0)_{\mathrm{tf}} = 0, \tag{850}$$

$$D_0 E_i - \sum_j [\mathcal{A}_j, [\mathcal{A}_j, \mathcal{A}_i]] - \frac{g^2}{2} (J_i)_{\mathrm{tf}} = 0 \quad (i = 1, 2, 3). \tag{851}$$

Here $D_0 X = \dot{X} + [\mathcal{A}_0, X]$, $D_0 v = \dot{v} + \mathcal{A}_0 v$, and similarly for E_i.

Proof. First retain the homogeneous density without deleting $\mathcal{A}_0$:

$$\mathcal{L}_{\mathrm{hom}} = -\frac{1}{g^2} \sum_i \operatorname{tr} E_i^2 + \frac{1}{g^2} \sum_{i<j} \operatorname{tr} F_{ij}^2 + \frac{1}{2} \operatorname{tr} X^2 + v^* v - \frac{1}{2} \sum_i \operatorname{tr} [\mathcal{A}_i, \Phi]^2 - \sum_i |\mathcal{A}_i b|^2 - V.$$

Both signs in the gauge terms follow by raising indices in $(+---)$ and counting the two orders of each antisymmetric pair. For trace-free Hermitian $\delta\Phi$ and arbitrary δb,

$$\delta V = \operatorname{tr}\big[(a\Phi - \kappa(\operatorname{adj} \Phi - b b^*)_{\mathrm{tf}}) \delta\Phi \big] + 2 \operatorname{Re}\big(\delta b^* (a I_3 + \kappa \Phi) b \big).$$

Covariant integration by parts therefore gives $D_\mu D^\mu \Phi + a\Phi - \kappa(\operatorname{adj} \Phi - b b^*)_{\mathrm{tf}} = 0$ and $D_\mu D^\mu b + (a I_3 + \kappa\Phi) b = 0$. Homogeneity gives the first two displayed equations, including the minus signs in their spatial sums.

For the gauge variation, antisymmetry gives

$$\delta \mathcal{L}_{\mathrm{gauge}} = \frac{2}{g^2} \operatorname{tr}((D_\mu \delta \mathcal{A}_\nu) \mathcal{F}^{\mu\nu}).$$

After integration by parts its coefficient is $-2 D_\mu \mathcal{F}^{\mu\nu} / g^2$. The scalar coefficient of $\delta \mathcal{A}_\nu$ is

$$J^\nu = [\Phi, D^\nu \Phi] + b (D^\nu b)^* - (D^\nu b) b^*.$$

Indeed cyclicity and anti-Hermiticity respectively give

$$\operatorname{tr}(D^\nu \Phi [\delta \mathcal{A}_\nu, \Phi]) = \operatorname{tr}([\Phi, D^\nu \Phi] \delta \mathcal{A}_\nu),$$
$$2 \operatorname{Re}((D^\nu b)^* \delta \mathcal{A}_\nu b) = \operatorname{tr}\big((b (D^\nu b)^* - (D^\nu b) b^*) \delta \mathcal{A}_\nu \big).$$

Variation over $\mathfrak{su}(3)$ consequently yields

$$D_\mu \mathcal{F}^{\mu\nu} = \frac{g^2}{2} (J^\nu)_{\mathrm{tf}}.$$

For $\nu = 0$, $\mathcal{F}^{i0} = E_i$ and $J^0 = J_0$. For $\nu = i$, $\mathcal{F}^{0i} = -E_i$, $\mathcal{F}^{ji} = [\mathcal{A}_j, \mathcal{A}_i]$, and $J^i = -J_i$. These substitutions give exactly (850) and (851). In particular the $\mathcal{A}_0$ equation survives after choosing temporal gauge. This is the same variational principle emphasized by D. Tong, *Lectures on Gauge Theory*, Section 2.2.1, equation (2.27); the matter currents, signs, and coefficient $g^2/2$ above have been derived in the present conventions. □

Theorem 24.27 (No temporal-gauge lift retaining the literal orbit)**.** *Keep exactly the moving-kernel scalar functions in Theorem 24.21, with $B, d > 0$ and $x \neq 0$. There are no homogeneous spatial fields $\mathcal{A}_i(t)$ for which these same scalar functions, together with $\mathcal{A}_0 = 0$, solve the full system* (848)–(851)*.*

Proof. The scalar functions already solve the unrestricted scalar equations. Subtracting them from (848)–(849), with $\mathcal{A}_0 = 0$, gives

$$\sum_i [\mathcal{A}_i, [\mathcal{A}_i, \Phi]] = 0, \qquad \sum_i \mathcal{A}_i^2 b = 0.$$

Taking the trace pairing with Φ and the Hermitian pairing with b gives respectively

$$-\sum_i \operatorname{tr}[\mathcal{A}_i, \Phi]^2 = 0, \qquad -\sum_i |\mathcal{A}_i b|^2 = 0.$$

Each $[\mathcal{A}_i, \Phi]$ is Hermitian, so both are sums of nonnegative squared norms with the displayed common minus sign. Thus $[\mathcal{A}_i, \Phi] = 0$ and $\mathcal{A}_i b = 0$ separately for every i.

To classify this stabilizer use the retained orthogonal matrix $K = (v/\sqrt{2}, w_0/\sqrt{6}, u/\sqrt{3})$ from the moving-family proof, and put $r^2 = x^2 + d^2 > 0$. At time zero the data are

$$H = K^* \Phi_0 K = \begin{pmatrix} 0 & 0 & 0 \\ 0 & -x & d \\ 0 & d & x \end{pmatrix}, \qquad K^* b_0 = B e_1, \qquad A = K^* \Omega K = \begin{pmatrix} 0 & -\sqrt{a} & 0 \\ \sqrt{a} & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

An anti-Hermitian traceless matrix killing Be_1 has zero first row and column and a lower $\mathfrak{su}(2)$ block. Commutation with $\left(\begin{smallmatrix} -x & d \\ d & x \end{smallmatrix}\right)$, whose distinct eigenvalues are $\pm r$, forces that block to be its real multiple times i. Therefore the stabilizer Lie algebra is $\mathbb{R}\, i\Phi(t)$. The full stabilizer group is the circle

$$\left\{ P_b + \cos\theta (I_3 - P_b) + i \frac{\sin\theta}{r} \Phi(t) : \theta \in \mathbb{R} \right\}, \qquad P_b = \frac{b(t) b(t)^*}{B^2}.$$

Indeed it fixes b, has eigenvalues $1, e^{i\theta}, e^{-i\theta}$, and exhausts the unitary commutant fixing the zero eigenspace pointwise. Consequently

$$\mathcal{A}_i = i c_i(t) \Phi(t), \qquad c_i(t) \in \mathbb{R}.$$

The original orbit has

$$[\Phi, \dot{\Phi}] = -r^2 \Omega, \qquad b\dot{b}^* - \dot{b} b^* = -B^2 \Omega, \qquad J_0 = -(r^2 + B^2)\Omega.$$

For verification, $H^2 = r^2 \operatorname{diag}(0, 1, 1)$, $HAH = 0$, and hence $[H, [A, H]] = -r^2 A$; the vector term is $-B^2 A$. Conjugation back by $O(t)K$, with the rotation commuting with Ω, gives the stated original-coordinate identities. Now

$$F_{ij} = 0, \qquad E_i = i\dot{c}_i \Phi + i c_i \dot{\Phi}, \qquad \sum_i [\mathcal{A}_i, E_i] = r^2 \left(\sum_i c_i^2 \right) \Omega.$$

Thus the full Gauss residual, with precisely the convention in (850), is

$$\boxed{\left(\frac{2r^2}{g^2} \sum_i c_i^2 + r^2 + B^2 \right) \Omega.}$$

Its coefficient is strictly positive and $\Omega \neq 0$, proving the obstruction even before the spatial gauge equations are used.

For completeness those equations force all c_i to vanish as well: here $J_i = F_{ij} = 0$, so they require $i(\ddot{c}_i\Phi + 2\dot{c}_i\dot{\Phi} + c_i\ddot{\Phi}) = 0$. At a fixed time conjugate by the retained moving frame. The $(1,1)$ entries of H and $[A,H]$ vanish, while that of $[A,[A,H]]$ is $-2ax \neq 0$. Hence $c_i = 0$. The remaining Gauss residual is then $(r^2 + B^2)\Omega$, still nonzero. □

Proposition 24.28 (Arbitrary temporal connection and exact Gauss inertia)**.** *For arbitrary homogeneous scalar and spatial-gauge positions and velocities, define*

$$J_{\mathrm{sc}} = [\Phi, \dot{\Phi}] + b\dot{b}^* - \dot{b}b^*, \qquad \mathscr{I}_{\Phi,b}Z = \big([\Phi,[\Phi,Z]] + bb^*Z + Zbb^*\big)_{\mathrm{tf}},$$

$$\mathscr{K}_{\mathcal{A}}Z = -\sum_i [\mathcal{A}_i,[\mathcal{A}_i,Z]].$$

Then Gauss' equation is exactly

$$\left(\mathscr{I}_{\Phi,b} + \frac{2}{g^2}\mathscr{K}_{\mathcal{A}}\right)\mathcal{A}_0 = (J_{\mathrm{sc}})_{\mathrm{tf}} - \frac{2}{g^2}\sum_i [\mathcal{A}_i, \dot{\mathcal{A}}_i]. \tag{852}$$

The operator on the left is self-adjoint and positive semidefinite for $\langle Z,W\rangle = -\operatorname{tr}(ZW)$ on $\mathfrak{su}(3)$. Its kernel is exactly

$$\{Z : [Z,\Phi] = 0,\ Zb = 0,\ [Z,\mathcal{A}_i] = 0 \text{ for every } i\}.$$

The right-hand side always annihilates that kernel. Thus Gauss' equation has a pointwise solution for $\mathcal{A}_0$, unique modulo this kernel. This pointwise statement asserts neither a smooth selection across kernel-rank changes nor a solution of the remaining field equations.

Proof. Expansion gives $J_0 = J_{\mathrm{sc}} - \mathscr{I}_{\Phi,b}\mathcal{A}_0$ after trace-free projection, and

$$\sum_i [\mathcal{A}_i, E_i] = \sum_i [\mathcal{A}_i, \dot{\mathcal{A}}_i] + \mathscr{K}_{\mathcal{A}}\mathcal{A}_0.$$

Substitution proves (852). Cyclicity gives

$$\langle Z, \mathscr{I}_{\Phi,b}Z\rangle = \operatorname{tr}[Z,\Phi]^2 + 2|Zb|^2, \qquad \langle Z, \mathscr{K}_{\mathcal{A}}Z\rangle = \sum_i \|[\mathcal{A}_i, Z]\|^2_{\mathfrak{su}},$$

where $\|T\|^2_{\mathfrak{su}} = -\operatorname{tr} T^2$. More explicitly, cyclicity gives the bilinear identities

$$\langle Z, \mathscr{I}_{\Phi,b}W\rangle = \operatorname{tr}([Z,\Phi][W,\Phi]) + 2\operatorname{Re}((Zb)^*Wb), \qquad \langle Z, \mathscr{K}_{\mathcal{A}}W\rangle = \sum_i \langle [\mathcal{A}_i,Z],[\mathcal{A}_i,W]\rangle.$$

These are symmetric in Z, W and prove self-adjointness; the sums of squares prove the stated kernel. If Z is in the kernel, the matter moment paired with Z vanishes because $[Z,\Phi] = Zb = 0$, and $\operatorname{tr}(Z[\mathcal{A}_i,\dot{\mathcal{A}}_i]) = \operatorname{tr}([Z,\mathcal{A}_i]\dot{\mathcal{A}}_i) = 0$. Thus the right-hand side is orthogonal to the kernel. In a finite dimensional real inner-product space a self-adjoint operator has image equal to the orthogonal complement of its kernel: this follows by diagonalizing its real symmetric matrix. It gives existence and the claimed exact ambiguity. □

For the literal moving orbit, subtracting the original scalar equations while retaining arbitrary $\mathcal{A}_0$ gives the exact additional equations

$$[\dot{\mathcal{A}}_0,\Phi] + 2[\mathcal{A}_0,\dot{\Phi}] + [\mathcal{A}_0,[\mathcal{A}_0,\Phi]] - \sum_i [\mathcal{A}_i,[\mathcal{A}_i,\Phi]] = 0,$$

$$\dot{\mathcal{A}}_0 b + 2\mathcal{A}_0\dot{b} + \mathcal{A}_0^2 b - \sum_i \mathcal{A}_i^2 b = 0.$$

Their norm consequences are

$$\sum_i \operatorname{tr}[\mathcal{A}_i,\Phi]^2 = \operatorname{tr} X^2 - \operatorname{tr}\dot\Phi^2, \qquad \sum_i |\mathcal{A}_i b|^2 = |v|^2 - |\dot b|^2.$$

For example the trace pairing of the first equation uses $\operatorname{tr}(\Phi[\dot{\mathcal{A}}_0,\Phi]) = 0$ and $\operatorname{tr}(\Phi[\mathcal{A}_0,\dot\Phi]) = -\operatorname{tr}([\mathcal{A}_0,\Phi]\dot\Phi)$; expansion of X^2 then gives the first identity. The real pairing of the second equation with b gives the second by anti-Hermiticity. These right-hand sides need not be zero; temporal gauge cannot be imposed while silently keeping the same literal scalar functions.

Here is an exact counterexample to extending the temporal-gauge stabilizer deduction to nonzero $\mathcal{A}_0$. For real constants $\alpha,\beta_1,\beta_2,\beta_3$ satisfying $\alpha^2 - \sum_i \beta_i^2 = 1$, keep the same scalar orbit and set

$$\mathcal{A}_0 = (\alpha-1)\Omega, \qquad \mathcal{A}_i = \beta_i\Omega.$$

All curvatures vanish, while $D_0\Phi = \alpha[\Omega,\Phi]$, $D_0 b = \alpha\Omega b$, $D_i\Phi = \beta_i[\Omega,\Phi]$, and $D_i b = \beta_i\Omega b$. Their covariant wave operators therefore equal the original second derivatives, proving both scalar equations exactly. Taking $\alpha = 5/3$, $\beta_1 = 4/3$, $\beta_2 = \beta_3 = 0$ gives $\mathcal{A}_1 b \neq 0$: the spatial field does not stabilize the scalars. Nevertheless $J_0 = \alpha J_{\rm sc} \neq 0$, since $|\alpha| \geq 1$, and the zero electric field cannot satisfy Gauss' equation. These are scalar-compatible flat connections, not full gauge solutions.

There is also no lift of the moving orbit with all spatial $\mathcal{A}_i = 0$, even when $\mathcal{A}_0$ is arbitrary. Indeed $J_{\rm sc} = -\mathscr{I}_{\Phi,b}\Omega$, so (852) gives $\mathscr{I}_{\Phi,b}(\mathcal{A}_0+\Omega) = 0$. Its kernel description implies $D_0\Phi = D_0 b = 0$. Equation (849) would then require $(aI_3 + \kappa\Phi)b = ab = 0$, impossible because $a > 0$ and $B > 0$. This argument, and Theorem 24.27, do not exclude the general case with both nonzero $\mathcal{A}_0$ and nonzero spatial fields satisfying all coupled equations.

24.14 Constant-matter coordinates and an all-time real-connection obstruction

Retain exactly the parameters and literal scalar functions of Theorem 24.21. In particular $a > 0$, $B, d > 0$, and $x \neq 0$. Keep the same original-coordinate $O(t)$, Ω, and the exact orthogonal map and inverse

$$K = \left(v/\sqrt{2}, w_0/\sqrt{6}, u/\sqrt{3}\right), \qquad K^{-1} = K^*.$$

The squared lengths of v, w_0, u are $2, 6, 3$, their pairwise inner products vanish, and $\det(v, w_0, u) = 6$; hence $K \in SO(3)$. Put

$$H = K^*\Phi_0 K = \begin{pmatrix} 0 & 0 & 0 \\ 0 & -x & d \\ 0 & d & x \end{pmatrix}, \qquad q = K^* b_0 = Be_1, \qquad A = K^*\Omega K = \begin{pmatrix} 0 & -\sqrt{a} & 0 \\ \sqrt{a} & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Thus $Hq = 0$. Write $r^2 = x^2 + d^2$ and $\mu = r^2 + B^2$. The retained force, including the cubic term, is

$$F_H = -aH + \kappa(\operatorname{adj} H - qq^*)_{\rm tf} = \begin{pmatrix} -2ax & 0 & 0 \\ 0 & 2ax & -ad \\ 0 & -ad & 0 \end{pmatrix} = [A,[A,H]], \qquad A^2 q = -aq.$$

Indeed $s = 2(r^2+B^2) = 6ax/\kappa$, $\operatorname{adj} H = \operatorname{diag}(-r^2, 0, 0)$, and $(\operatorname{adj} H - qq^*)_{\rm tf} = \operatorname{diag}(-s/3, s/6, s/6)$, which give these identities by direct multiplication.

Proposition 24.29 (An exact constant-matter reformulation)**.** *Set*

$$C = K^*O^*\mathcal{A}_0 OK + A, \qquad Y_i = K^*O^*\mathcal{A}_i OK, \qquad \mathcal{E}_i = \dot Y_i + [C, Y_i],$$

$$\mathscr{I}_{H,q} Z = ([H,[H,Z]] + qq^*Z + Zqq^*)_{\rm tf}, \qquad \mathscr{K}_Y Z = -\sum_i [Y_i,[Y_i,Z]].$$

The prescribed literal scalar functions have a homogeneous gauge lift if and only if the following complete system holds:

$$[\dot{C}, H] + [C, [C, H]] - \sum_i [Y_i, [Y_i, H]] = F_H, \tag{853}$$

$$(\dot{C} + C^2 - \sum_i Y_i^2)q = -aq, \tag{854}$$

$$(\mathscr{I}_{H,q} + 2\mathscr{K}_Y/g^2)C = -\frac{2}{g^2}\sum_i [Y_i, \dot{Y}_i], \tag{855}$$

$$\dot{\mathcal{E}}_i + [C, \mathcal{E}_i] - \sum_j [Y_j, [Y_j, Y_i]] + \frac{g^2}{2}\mathscr{I}_{H,q} Y_i = 0 \quad (i = 1, 2, 3). \tag{856}$$

The inverse connection maps are exactly

$$\mathcal{A}_0 = OKCK^*O^* - \Omega, \qquad \mathcal{A}_i = OKY_iK^*O^*.$$

Neither the literal functions nor their time scale has been changed.

Proof. For a spatially constant $U(t) \in SU(3)$, the transformations are $\Phi' = U\Phi U^*$, $b' = Ub$, $\mathcal{A}_0' = U\mathcal{A}_0U^* - \dot{U}U^*$, and $\mathcal{A}_i' = U\mathcal{A}_iU^*$. The product rule gives $D_\nu'(Ub) = UD_\nu b$ and $D_\nu'(U\Phi U^*) = U(D_\nu\Phi)U^*$. Their commutators give $\mathcal{F}_{\nu\rho}' = U\mathcal{F}_{\nu\rho}U^*$. Traces, trace projection, and the potential are invariant under this unitary action; adjugate covariance follows from the inverse formula on invertible matrices and polynomial continuity. Thus all the equations of Proposition 24.26, including Gauss' equation, transform covariantly.

Taking $U = K^*O^*$ gives the constant matter fields H, q and the displayed connection maps, because $-\dot{U}U^* = K^*\Omega K = A$. The scalar covariant derivatives are $[C, H]$ and Cq; their second derivatives are $[\dot{C}, H] + [C, [C, H]]$ and $(\dot{C} + C^2)q$. The trace-free currents are $(J_0)_{\mathrm{tf}} = -\mathscr{I}_{H,q}C$ and $(J_i)_{\mathrm{tf}} = -\mathscr{I}_{H,q}Y_i$. Finally $\sum_i [Y_i, \mathcal{E}_i] = \sum_i [Y_i, \dot{Y}_i] + \mathscr{K}_Y C$. Substitution gives every displayed equation with its stated sign and factor. The explicitly invertible maps prove both directions. □

Theorem 24.30 (No real-connection lift with any spatial time dependence)**.** *The literal moving-kernel scalar functions admit no homogeneous lift with $\mathcal{A}_\nu(t) \in \mathfrak{so}(3)$ for all four indices, on any nonempty open time interval. The temporal connection is arbitrary; the spatial connections may be arbitrary twice differentiable functions, including rotating, multifrequency, and nonperiodic functions. No commutativity or spatial-stationarity assumption is imposed.*

Proof. Since K, O, Ω are real, the exact frame maps preserve real skew-symmetry in both directions. Define

$$L_1 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \quad L_2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}, \quad L_3 = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad L(z) = \sum_j z_j L_j \quad (z \in \mathbb{R}^3).$$

Direct multiplication gives

$$L(z)w = z \times w, \quad L(z)L(w) = wz^{\mathsf{T}} - (z \cdot w)I_3, \quad [L(z), L(w)] = L(z \times w), \quad -\operatorname{tr}(L(z)L(w)) = 2z \cdot w.$$

These formulas specify the commutator orientation and every norm factor. Write

$$C = L(c), \quad Y_i = L(y_i), \quad h = \dot{c}, \quad P = \sum_i y_i y_i^{\mathsf{T}}, \quad Q = cc^{\mathsf{T}} - P = (q_{jk})_{j,k=1}^3.$$

Here the scalar entries q_{jk} do not replace the earlier fixed matter vector $q = Be_1$. Since $A = \sqrt{a}L_3$, the residual in (854), divided by B, is

$$R_b = \begin{pmatrix} a - q_{22} - q_{33} \\ h_3 + q_{12} \\ -h_2 + q_{13} \end{pmatrix}.$$

The residual in (853) is a symmetric matrix R_H whose six independent entries are

$$\begin{aligned}
(R_H)_{11} &= 2ax - 4dq_{23} + 2xq_{22} - 2xq_{33},\\
(R_H)_{12} &= dh_2 + 3dq_{13} + xh_3 - 3xq_{12},\\
(R_H)_{13} &= -dh_3 + 3dq_{12} + xh_2 + 3xq_{13},\\
(R_H)_{22} &= -2ax - 2dh_1 + 2dq_{23} + 4xq_{11} + 2xq_{33},\\
(R_H)_{23} &= ad - 4dq_{11} - dq_{22} - dq_{33} - 2xh_1,\\
(R_H)_{33} &= 2dh_1 + 2dq_{23} - 4xq_{11} - 2xq_{22}.
\end{aligned}$$

These entries follow by multiplying the explicitly given matrices in

$$R_H = [L(h), H] + \sum_{j,k} q_{jk}[L_j, [L_k, H]] - a[L_3, [L_3, H]], \qquad R_b = L(h)e_1 + \sum_{j,k} q_{jk}L_jL_ke_1 + ae_1.$$

Thus no spatial time-dependence ansatz has entered this calculation.

The equation $R_b = 0$ gives $q_{22} + q_{33} = a$, $h_3 = -q_{12}$, $h_2 = q_{13}$. The two equations $(R_H)_{12} = (R_H)_{13} = 0$ then give $dq_{13} - xq_{12} = 0$ and $dq_{12} + xq_{13} = 0$. Since $x^2 + d^2 > 0$, they force $q_{12} = q_{13} = h_2 = h_3 = 0$. Next $(R_H)_{11} = 0$ gives $xq_{22} = dq_{23}$. After these substitutions, $(R_H)_{23} = (R_H)_{33} = 0$ give $xh_1 = -2dq_{11}$ and $dh_1 = 2xq_{11}$. They imply $2(x^2 + d^2)q_{11} = 0$, and hence $q_{11} = h_1 = 0$. Conversely the following values make every residual entry zero:

$$\dot c = 0, \qquad Q = \begin{pmatrix} 0 & 0 & 0 \\ 0 & d\tau/x & \tau \\ 0 & \tau & a - d\tau/x \end{pmatrix}, \qquad \tau = q_{23}(t) \in \mathbb{R}. \tag{857}$$

This is the exact necessary-and-sufficient scalar equation within the real algebra. In particular $Qe_1 = 0$ and $\operatorname{tr} Q = a > 0$.

The covariance $P = \sum_i y_i y_i^{\mathsf T}$ is positive semidefinite. Suppose $c_1 \neq 0$ at any time. For $v \in e_1^{\perp}$ define $z = v - (c \cdot v)e_1/c_1$. Then $c \cdot z = 0$ and $Qe_1 = 0$, so

$$v^{\mathsf T}Qv = z^{\mathsf T}Qz = (c \cdot z)^2 - z^{\mathsf T}Pz = -\sum_i (y_i \cdot z)^2 \leq 0.$$

The lower 2×2 block of Q would therefore be negative semidefinite, contradicting its positive trace a. Equivalently, $P_{11} = c_1^2$ and the Schur complement of P_{11} in $P = cc^{\mathsf T} - Q$ is exactly minus that lower block of Q. Thus $c_1 = 0$ at every time. Since $Q_{11} = c_1^2 - \sum_i y_{i1}^2 = 0$, every $y_{i1} = 0$ as well. Differentiation on the open time interval now gives

$$c, y_i, \dot y_i \in e_1^{\perp}, \qquad \sum_i y_i \times \dot y_i \in \mathbb{R}e_1.$$

The exact inertia on this real algebra is

$$\mathscr{I}_{H,q}L_1 = 4r^2L_1, \qquad \mathscr{I}_{H,q}L_2 = \mu L_2, \qquad \mathscr{I}_{H,q}L_3 = \mu L_3.$$

To prove these formulas, $H^2 = r^2 \operatorname{diag}(0,1,1)$ and $HL_1H = -r^2L_1$ give $[H,[H,L_1]] = 4r^2L_1$, while the qq^* terms vanish. For $j = 2,3$, multiplication gives $HL_jH = 0$, $H^2L_j + L_jH^2 = r^2L_j$, and $qq^*L_j + L_jqq^* = B^2L_j$. Every resulting real-skew matrix is already traceless. This proves the three identities and gives $\mathscr{I}_{H,q}C = \mu C$.

For $Z, W \in \mathfrak{su}(3)$ set $\langle Z, W\rangle = -\operatorname{tr}(ZW)$ and $\|Z\|^2 = \operatorname{tr}(Z^*Z) = -\operatorname{tr} Z^2$. Cyclicity gives $\langle Z, [Y,W]\rangle = -\langle [Y,Z], W\rangle$, so

$$\langle C, \mathscr{K}_Y C\rangle = \sum_i \|[Y_i, C]\|^2 = 2\sum_i |y_i \times c|^2.$$

Pair the full Gauss equation (855) with C. Its right side vanishes because $c \perp \sum_i y_i \times \dot{y}_i$. The left side, with the exact factor 2 from the trace inner product, is

$$2\mu|c|^2 + \frac{4}{g^2}\sum_i |y_i \times c|^2 = 0.$$

Since $\mu > 0$ and $g^2 > 0$, this forces $c = 0$. Then $Q = -P$ has $\operatorname{tr} Q = -\sum_i |y_i|^2 \le 0$, contradicting $\operatorname{tr} Q = a > 0$. Every full solution must obey the two scalar equations and Gauss' equation used here. Their contradiction already excludes a full lift, without imposing or deleting any spatial gauge equation. □

Corollary 24.31 (Constant spatial fields with arbitrary complex color)**.** *There is no lift with all Y_i constant in the constant-matter frame, even when $C(t)$ is arbitrary and the fields take values in the full complex algebra $\mathfrak{su}(3)$.*

Proof. Cyclicity and anti-Hermiticity give

$$\langle Z, \mathscr{I}_{H,q} Z\rangle = \operatorname{tr}[Z,H]^2 + 2|Zq|^2, \qquad \langle Z, \mathscr{K}_Y Z\rangle = \sum_i \|[Y_i, Z]\|^2.$$

For completeness, trace projection pairs trivially with traceless Z; cyclicity gives $-\operatorname{tr}(Z[H,[H,Z]]) = \operatorname{tr}[Z,H]^2$, and $-\operatorname{tr}(Z(qq^*Z + Zqq^*)) = -2q^*Z^2q = 2|Zq|^2$. The second formula follows from the same commutator trace identity used in the theorem. The matrix $[Z,H]$ is Hermitian, so all displayed terms are nonnegative.

When $\dot{Y}_i = 0$, Gauss' equation is $(\mathscr{I}_{H,q} + 2\mathscr{K}_Y/g^2)C = 0$. Its pairing with C therefore forces $Cq = 0$, $[C,H] = 0$, and $[C,Y_i] = 0$ for every i at every time. Since q is constant, $Cq = 0$ implies $\dot{C}q = 0$ and $C^2q = 0$. The vector equation becomes $(-\sum_i Y_i^2 + aI_3)q = 0$. Pairing with q gives $\sum_i |Y_iq|^2 + aB^2 = 0$, which contradicts $a > 0$, $B > 0$. □

Corollary 24.32 (Matter-stabilizer gauge saturation)**.** *There is no lift for which a differentiable real $\theta(t)$ makes*

$$C' = SCS^* - i\dot{\theta}H, \qquad Y_i' = SY_iS^*, \qquad S(t) = \exp(i\theta(t)H)$$

all real skew-symmetric. Nor is there a lift when such a transformation makes all Y_i' constant, without a reality restriction.

Proof. H is Hermitian and traceless, so $S \in SU(3)$. It commutes with H, and $Hq = 0$ gives $Sq = q$. Differentiation gives $\dot{S}S^* = i\dot{\theta}H$. Thus these are exactly the gauge transformations proved in Proposition 24.29, retaining the same constant matter fields. They would transform any claimed lift into one excluded by Theorem 24.30 or Corollary 24.31, respectively. □

These are exact exclusions for the stated classes, not an assertion that every $\mathfrak{su}(3)$ connection is real or belongs to their matter-stabilizer gauge saturation. Arbitrary time-dependent complex $\mathfrak{su}(3)$ lifts outside those classes remain open here.

24.15 Constructive homogeneous gauge solutions and their observations

Theorem 24.33 (A real radial family of full gauge solutions)**.** *Let $D = \operatorname{diag}(-2,1,1)$ and retain*

$$\Phi(t) = x(t)D, \qquad b(t) = \beta(t)e_1, \qquad x(t), \beta(t) \in \mathbb{R}, \qquad \mathcal{A}_\mu = 0.$$

This is a full homogeneous gauge–scalar solution exactly when

$$\ddot{x} = -ax - \kappa x^2 + \frac{\kappa}{3}\beta^2,$$
$$\ddot{\beta} = -(a - 2\kappa x)\beta, \qquad a = m^2 + \Gamma(6x^2 + 2\beta^2).$$

For every real initial position and velocity these equations have a global solution. They give finite-energy homogeneous solutions on a finite-volume flat $\mathbb{T}^3$, and local solutions on $\mathbb{R}^3$, without asserting finite total energy on $\mathbb{R}^3$.

Proof. Here $[\Phi, \dot{\Phi}] = 0$ and $b\dot{b}^* - \dot{b}b^* = 0$ because x, β are real. Thus every J_μ vanishes identically and so do all curvatures. The gauge equations hold exactly, not after discarding Gauss' equation. Directly

$$(\mathrm{adj}(xD) - \beta^2 e_1 e_1^*)_{\mathrm{tf}} = \left(-x^2 + \frac{\beta^2}{3}\right) D, \qquad (aI_3 + \kappa x D)\beta e_1 = (a - 2\kappa x)\beta e_1.$$

The two scalar equations are consequently the displayed real system. Its conserved density is

$$\mathcal{E} = 3\dot{x}^2 + \dot{\beta}^2 + V(xD, \beta e_1), \qquad V = \frac{m^2}{2} s + \frac{\Gamma}{4} s^2 - 2\kappa x(\beta^2 - x^2), \quad s = 6x^2 + 2\beta^2.$$

The stated conservation follows by differentiating and substituting the two real equations; the x kinetic coefficient is 3, not 1/2. Moreover $|x| \le \sqrt{s/6}$ and $\beta^2 + x^2 \le s/2$, so $|2x(\beta^2 - x^2)| \le s^{3/2}/\sqrt{6}$ and

$$V \ge \frac{\Gamma}{4} s^2 - \frac{|\kappa|}{\sqrt{6}} s^{3/2}.$$

This is coercive and globally bounded below. Conservation bounds the positions and velocities. The polynomial first-order vector field is locally Lipschitz and bounded with bounded derivative on their compact range; hence no finite-time escape is possible, proving global existence. Multiplying the finite density by the finite volume of $\mathbb{T}^3$ gives finite total energy there. □

Proposition 24.34 (A full-gauge vacuum and its marked Lorentz observation)**.** *At $\Gamma = m^2 = 1$, $\kappa = 6$, let*

$$x_0 = \frac{3 + \sqrt{3}}{12}, \qquad \Phi = x_0 \,\mathrm{diag}(-2, 1, 1), \qquad b = 3x_0 e_1.$$

Together with $\mathcal{A}_\mu = 0$ this is an equilibrium full gauge solution at a global minimum of the scalar potential. For $Q = \Theta(\Phi, b)$ its spectrum is $(4x_0, -2x_0, -2x_0)$. Its spectral-floor matrix $H = Q + 2x_0 I_3$ has rank one and is positive. The complex observation and hidden coordinate are

$$G = 3x_0 \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}, \qquad \rho = 3x_0, \qquad \det G = 9x_0^2 = |\rho|^2.$$

More generally its $SU(3)$ orbit, parameterized by unit vectors $e = ge_1$, has

$$\Phi = x_0(I_3 - 3ee^*), \quad b = 3x_0 e, \quad G = 3x_0 P(I_3 - ee^*)P^*, \quad \det G = 9x_0^2 |u^* e|^2 = |\rho|^2.$$

For fixed P, this one gauge orbit contains both null and timelike observations. The fixed-P determinant is therefore not a gauge-invariant mass observable.

Proof. For a nonzero real equilibrium with $\beta = 3x > 0$, the radial equations reduce to $24\Gamma x^2 - 2\kappa x + m^2 = 0$. The displayed x_0 is the larger root at $(1, 1, 6)$. Its quaternionic matrix is

$$Q = \begin{pmatrix} -2x_0 & 0 & 0 \\ 0 & x_0 & 3x_0 \mathbf{j} \\ 0 & -3x_0 \mathbf{j} & x_0 \end{pmatrix}.$$

The lower block has eigenvalues $x_0 \pm 3x_0$, proving the spectrum. For an independent global-minimum check, every trace-free quaternionic Hermitian matrix satisfies $54N(Q)^2 \le s(Q)^3$: for eigenvalues summing to zero, the exact identity is $s^3 - 54N^2 = 2\prod_{i<j}(\lambda_i - \lambda_j)^2$. Thus, with $r = \sqrt{s} \ge 0$, $V \ge r^2/2 + r^4/4 - 2r^3/\sqrt{6}$,

with equality at eigenvalues $(2y, -y, -y)$, $y \geq 0$, $r = \sqrt{6}y$. On this equality family $V = 3y^2 + 9y^4 - 12y^3$ and $dV/dy = 6y(1 - 6y + 6y^2)$. The larger positive root is $y = (3 + \sqrt{3})/6 = 2x_0$; the derivative is positive, negative, then positive on the three successive intervals, and this minimum has negative value, since at $y = 2/3$ the potential is $-4/9$. It is therefore the global minimum, attained by the displayed Q.

The complex part of H is $3x_0 \operatorname{diag}(0, 1, 1)$, and compression by the retained P gives the displayed G. The $SU(3)$ action on pairs is $(\Phi, b) \mapsto (g\Phi g^*, gb)$; it is exactly quaternionic congruence under Θ, since $B(gb) = gB(b)g^{\mathsf{T}}$ for $g \in SU(3)$. Consequently the spectrum and spectral-floor shift are unchanged. Writing $v = Pe$ and using the determinant lemma in dimension two gives

$$\det(C - vv^*) = \det C\,(1 - v^*C^{-1}v) = 3\left(1 - e^*\left(I_3 - \frac{uu^*}{3}\right)e\right) = |u^*e|^2.$$

This proves the general observation formula. A unit $e \in u^\perp$ gives determinant zero, while $e = e_1$ gives a positive determinant. These vectors are related by $SU(3)$: extend either to a unitary basis and adjust the phase of one of the remaining columns to make determinant one. Thus the examples are genuinely on one gauge orbit.

The exact covariance map retains the observer as part of the data: under $H' = gHg^*$, $P' = Pg^*$ one has $P'H'P'^* = PHP^*$. Complex-component projection therefore has the same value. Simultaneously $u' = gu$, $b' = gb$ gives $u'^*b' = u^*b$. This is covariance of the marked observer, not invariance of a fixed observer under an active transformation of the state. □

Proposition 24.35 (Gauge averaging and distinct minimum orbits)**.** *Let $d\mu$ be Haar measure on $SU(3)$ with its retained positive total mass V_μ, not set to one. For every complex Hermitian A,*

$$\det(PAP^*) = u^* \operatorname{adj}(A)u, \qquad \int_{SU(3)} \det(PgAg^*P^*)\,d\mu(g) = V_\mu \operatorname{tr}\operatorname{adj}(A).$$

For a quaternionic positive rank-one matrix $H = \Theta(A, b)$, $\operatorname{adj}(A) = bb^*$. Hence its averaged observation is $V_\mu\|b\|^2$.

Put $\chi = 2x_0 > 0$ for the preceding minimum and, for $0 \leq \theta \leq \pi/4$, define

$$v_\theta = (\cos\theta,\ \sin\theta\,\mathbf{j},\ 0)^{\mathsf{T}}, \qquad Q_\theta = \chi(3v_\theta v_\theta^* - I_3).$$

Every Q_θ is a scalar global minimum, and its complex-plus-triplet coordinate satisfies

$$b_\theta = (0, 0, -3\chi\cos\theta\sin\theta)^{\mathsf{T}}, \qquad \|b_\theta\|^2 = \frac{9\chi^2}{4}\sin^2(2\theta).$$

Consequently this minimum family contains pairwise $SU(3)$-inequivalent states with invariant $\|b\|^2$ ranging from 0 to $9\chi^2/4$. The coefficient agrees with the supplied packet's invariant formula; this calculation validates that coefficient rather than claiming it as a newly discovered one.

Proof. The signed 2-by-2 minors of P form precisely $u = (1, 1, 1)^{\mathsf{T}}$. The exterior-square form of the Cauchy–Binet expansion therefore gives $\det(PAP^*) = u^* \operatorname{adj}(A)u$; equivalently this follows by expanding the six entries of PAP^* and collecting the cofactors. The identity $\operatorname{adj}(gAg^*) = g\operatorname{adj}(A)g^*$ holds for invertible A and then for every A by polynomial continuity. The Haar average of these conjugates commutes with every element of $SU(3)$. Commutation with diagonal determinant-one phase matrices kills its off-diagonal entries, and commutation with determinant-one cyclic permutations makes its diagonal entries equal. Its trace fixes the average at $V_\mu \operatorname{tr}(\operatorname{adj} A)I_3/3$. Since $u^*u = 3$, the stated integral follows with all factors retained.

Write $r = b^*b$. The skew convention gives directly $B(b)\overline{B(b)} = bb^* - rI_3$. The complex part of H^2 is therefore $A^2 - bb^* + rI_3$, and $e_2(H) = e_2(A) - r$. For positive rank-one H, its spectral identity is $H^2 = (\operatorname{tr} H)H$ and $e_2(H) = 0$. Taking complex parts and using $\operatorname{tr} H = \operatorname{tr} A$ gives $A^2 - (\operatorname{tr} A)A + e_2(A)I_3 = bb^*$. The left side is $\operatorname{adj}(A)$, proving the assertion.

Finally $v_\theta^* v_\theta = 1$, so Q_θ has fixed spectrum $(2\chi, -\chi, -\chi)$ and hence the same minimum value. Its $(1,2)$ quaternionic entry is $-3\chi\cos\theta\sin\theta\,\mathbf{j}$, with the other quaternionic off-diagonal coefficients zero. The retained definition of $B(b)$ therefore gives the displayed b_θ with its minus sign. Taking its norm proves the formula. It is strictly increasing on the given interval; since $b \mapsto gb$ preserves its norm, distinct parameters there cannot lie on one $SU(3)$ orbit. The positive spectral-floor matrix is $3\chi v_\theta v_\theta^*$, so the averaging identity applies to the same triplet coordinate. □

The classical $SU(3)$ cohomogeneity-one geometry of $\mathbb{HP}^2$ is discussed by A. Gambioli, *Eight-dimensional $SU(3)$-manifolds of cohomogeneity one*, arXiv:math/0611796, Section 4, the $\mathbb{HP}^2 = M(\mathbb{CP}^2, S^5)$ example, `https://arxiv.org/abs/math/0611796`. The explicit family and its separating invariant above are proved directly; no new classification of this classical orbit space is claimed. The averaging formula supplies the exact relation of that invariant to the marked Lorentz determinant. A continuum of classical potential minima is not, by itself, a calculation of a quantum Hamiltonian spectrum.

24.16 Vacuum kinetics and exact current-free continuations

Retain the real scalar state space

$$\mathcal{E} = \mathrm{Herm}_3(\mathbb{C})_0 \oplus \mathbb{C}^3, \qquad \langle(\phi, c), (\psi, d)\rangle = \mathrm{tr}(\phi\psi) + 2\,\mathrm{Re}(c^* d).$$

The subscript 0 means trace zero, not a quotient. The scalar kinetic density is exactly one half of this metric:

$$\mathcal{L}_{\mathrm{sc}} = \tfrac{1}{2}\,\mathrm{tr}(D_\mu\Phi D^\mu\Phi) + (D_\mu b)^* D^\mu b - V, \qquad V = \tfrac{m^2}{2}s + \tfrac{\Gamma}{4}s^2 - \kappa(\det\Phi - b^*\Phi b), \quad s = \mathrm{tr}\,\Phi^2 + 2b^* b.$$

The spacetime metric is $(+---)$, $m^2 \geq 0$, $\Gamma > 0$; the nonzero vacuum results below use $\kappa > 0$. The gauge action is $(\Phi, b) \mapsto (g\Phi g^*, gb)$, $g \in SU(3)$. Its anti-Hermitian connection is $\mathcal{A}_\mu$, and the retained gauge density is $\mathrm{tr}(\mathcal{F}_{\mu\nu}\mathcal{F}^{\mu\nu})/(2g^2)$, with $g^2 > 0$. All constructed connections below are identically zero. We make no claim about connections not considered here.

For $Z_{\mathrm{tf}} = Z - \mathrm{tr}(Z)I_3/3$, variation gives, at $\mathcal{A}_\mu = 0$,

$$\begin{aligned} \Box\Phi &= -a\Phi + \kappa(\mathrm{adj}\,\Phi - bb^*)_{\mathrm{tf}}, & a &= m^2 + \Gamma s, \\ \Box b &= -(aI_3 + \kappa\Phi)b, & J_\mu &= [\Phi, \partial_\mu\Phi] + b(\partial_\mu b)^* - (\partial_\mu b)b^*. \end{aligned} \tag{858}$$

Indeed

$$dV(\delta\Phi, \delta b) = \mathrm{tr}([a\Phi - \kappa(\mathrm{adj}\,\Phi - bb^*)_{\mathrm{tf}}]\delta\Phi) + 2\,\mathrm{Re}[((aI_3 + \kappa\Phi)b)^*\delta b]\,.$$

This proves both scalar equations with their original metric factors. Variation in $\mathcal{A}_\mu$ pairs the matter variation with $\mathrm{tr}(J^\mu\delta\mathcal{A}_\mu)$. Consequently $J_\mu = 0$ and $\mathcal{F}_{\mu\nu} = 0$ satisfy every gauge equation, including Gauss, before any restriction of variations.

We retain the quaternionic convention

$$\Theta(\Phi, b) = \Phi + B(b)\mathbf{j}, \qquad B(b) = \begin{pmatrix} 0 & \bar{b}_3 & -\bar{b}_2 \\ -\bar{b}_3 & 0 & \bar{b}_1 \\ \bar{b}_2 & -\bar{b}_1 & 0 \end{pmatrix}, \qquad \mathbf{j}z = \bar{z}\mathbf{j}.$$

Its real trace metric is exactly the displayed metric on $\mathcal{E}$: the diagonal sum of $B(b)\overline{B(c)}$ has real part $-2\,\mathrm{Re}(b^* c)$. Its cubic is $N(\Theta(\Phi, b)) = \det\Phi - b^*\Phi b$. To verify the latter identity, write $\Phi_{23} = z_1, \Phi_{31} = z_2, \Phi_{12} = z_3$ and use the Hermitian Moore formula

$$N\begin{pmatrix} a & q_3 & \bar{q}_2 \\ \bar{q}_3 & d & q_1 \\ q_2 & \bar{q}_1 & f \end{pmatrix} = adf - a|q_1|^2 - d|q_2|^2 - f|q_3|^2 + 2\,\mathrm{Re}(q_1 q_2 q_3).$$

Here $q_i = z_i + \bar{b}_i\mathbf{j}$. The complex part of the triple product is $z_1 z_2 z_3 - \bar{b}_1 b_2 z_3 - z_1\bar{b}_2 b_3 - \bar{b}_1\bar{z}_2 b_3$. Its extra real terms and the three extra diagonal terms are precisely $-b^*\Phi b$.

Lemma 24.36 (Retained nonzero minimum and its curve)**.** *Suppose* $0 \le m^2 \le \kappa^2/(27\Gamma)$*,* $\kappa > 0$*, and put*

$$\chi = \frac{\kappa + \sqrt{\kappa^2 - 24\Gamma m^2}}{12\Gamma} > 0.$$

For a unit $v \in \mathbb{H}^3$*,* $Q = \chi(3vv^* - I_3)$ *is a global minimum of* V *on the trace-free quaternionic Hermitian state space. The origin is an additional minimum only at* $m^2 = \kappa^2/(27\Gamma)$*. For* $0 \le \theta \le \pi/4$*, let*

$$v_\theta = (\cos\theta, \sin\theta\,\mathbf{j}, 0)^{\mathsf{T}}, \qquad Q_\theta = \chi(3v_\theta v_\theta^* - I_3).$$

Its pair coordinates and potential value are

$$\begin{aligned} \Phi_\theta &= \chi\,\mathrm{diag}(3\cos^2\theta - 1,\ 3\sin^2\theta - 1,\ -1), \\ b_\theta &= -3\chi\cos\theta\sin\theta\, e_3, \\ V_0 &= 3m^2\chi^2 + 9\Gamma\chi^4 - 2\kappa\chi^3 = \kappa\chi^3 - 9\Gamma\chi^4. \end{aligned} \tag{859}$$

Proof. The quaternionic spectral theorem gives real eigenvalues $\lambda_1 + \lambda_2 + \lambda_3 = 0$ and $s = \sum \lambda_i^2$. On each sphere $s > 0$, extrema of the product have at most two distinct eigenvalues: subtracting two Lagrange equations for the product gives $(\lambda_i - \lambda_j)(-\lambda_k - 2\beta) = 0$. If all three were distinct they would all equal -2β, a contradiction. The two-eigenvalue trace-free possibilities are $(2t, -t, -t)$ and their negatives and permutations, where $t = \sqrt{s/6} > 0$. Thus $N \le 2t^3$, sharply, and

$$V \ge F(t) = 3m^2t^2 + 9\Gamma t^4 - 2\kappa t^3, \qquad F'(t) = 6t(m^2 - \kappa t + 6\Gamma t^2).$$

For $m^2 > 0$ the larger positive root is χ, the only positive local minimum; the smaller root is a local maximum. For $m^2 = 0$, $F'(t) < 0$ on $0 < t < \chi$ and > 0 on $t > \chi$. Using $m^2 = \kappa\chi - 6\Gamma\chi^2$ yields $F(\chi) = \chi^3(\kappa - 9\Gamma\chi)$, negative below the displayed threshold and zero at it. Comparing with $F(0) = 0$ proves the global claim and all equality cases used here. The spectrum of Q_θ is $(2\chi, -\chi, -\chi)$. Its $(1,2)$ entry is $-3\chi\cos\theta\sin\theta\,\mathbf{j}$, giving the stated sign of b_θ. Direct substitution gives the potential value. □

Proposition 24.37 (Horizontal metric and normal acceleration)**.** *Write* $q_\theta = (\Phi_\theta, b_\theta)$*,* $T = \partial_\theta q_\theta$*,* $N = \partial_\theta^2 q_\theta$*. Then*

$$\langle T, T\rangle = 18\chi^2, \qquad \langle T, N\rangle = 0, \qquad \langle N, N\rangle = 72\chi^2.$$

The curve is $SU(3)$*-horizontal and its induced kinetic density is* $9\chi^2\dot\theta^2$*. Every homogeneous unconstrained solution confined to this fixed-*χ *curve has constant* θ*. The vector* N *is normal to the entire fixed-*χ *minimum manifold. Its radial projection is* $-3q_\theta$*, and* $\|N + 3q_\theta\|^2 = 18\chi^2$*; hence there is a nonzero normal component in addition to the overall radial one.*

Proof. With $D = \mathrm{diag}(-1, 1, 0)$, differentiation gives

$$T_\Phi = 3\chi\sin(2\theta)D, \quad T_b = -3\chi\cos(2\theta)e_3, \quad N_\Phi = 6\chi\cos(2\theta)D, \quad N_b = 6\chi\sin(2\theta)e_3.$$

Since $\mathrm{tr}\, D^2 = 2$, the three metric identities follow without changing the coefficient of the triplet metric. All commutators of the diagonal matrices vanish, and $b_\theta T_b^* - T_b b_\theta^* = 0$ because both coefficients are real. For every $\xi \in \mathfrak{su}(3)$,

$$\langle T, ([\xi, \Phi_\theta], \xi b_\theta)\rangle = \mathrm{tr}\big(([\Phi_\theta, T_\Phi] + b_\theta T_b^* - T_b b_\theta^*)\xi\big) = 0.$$

Thus this nonzero tangent is horizontal and cannot itself be a nonzero vertical tangent in this positive metric. At a vacuum the potential gradient vanishes, so the homogeneous equation is $T\ddot\theta + N\dot\theta^2 = 0$. Pairing with T first gives $\ddot\theta = 0$, and the nonzero norm of N then gives $\dot\theta = 0$. The exact vector that would have to be supplied as an extra force for a constrained constant-speed motion is $N\dot\theta^2$. Its force covector is

$$\mathrm{tr}(N_\Phi\dot\theta^2\,\delta\Phi) + 2\,\mathrm{Re}((N_b\dot\theta^2)^*\delta b).$$

For the assertion about the whole minimum manifold, $v_\theta^* v'_\theta = 0$, $v_\theta'^* v'_\theta = 1$, and $v''_\theta = -v_\theta$. Therefore

$$Q''_\theta = 6\chi(v'_\theta v_\theta'^* - v_\theta v_\theta^*).$$

Every tangent to the rank-one projector manifold has the form $3\chi(z v_\theta^* + v_\theta z^*)$, $v_\theta^* z = 0$. The real trace pairing with both projectors in Q''_θ is zero, by $v_\theta^* v'_\theta = 0$ and cyclicity of real trace. This proves normality. Finally $\|q_\theta\|^2 = 6\chi^2$ and $\langle N, q_\theta\rangle = -18\chi^2$; subtracting the radial projection gives the asserted remaining norm. Thus the constant-speed curve is an intrinsic geodesic of the minimum manifold but not an unconstrained homogeneous solution. □

Proposition 24.38 (Exact three-real-coordinate full-gauge sector)**.** *Let h, k, β be real fields and set*

$$\Phi = \operatorname{diag}(h+k, h-k, -2h), \qquad b = \beta e_3, \qquad \mathcal{A}_\mu = 0.$$

This ansatz satisfies all full field equations if and only if

$$\begin{aligned} \Box h &= -ah - \kappa h^2 + \tfrac{\kappa}{3}(k^2+\beta^2), \\ \Box k &= -(a - 2\kappa h)k, \qquad \Box\beta = -(a-2\kappa h)\beta, \\ a &= m^2 + \Gamma(6h^2 + 2k^2 + 2\beta^2). \end{aligned} \tag{860}$$

In particular it is an unconstrained invariant sector for homogeneous initial data with tangent velocities. Its homogeneous density is

$$\mathcal{L}_{\mathrm{hom}} = 3\dot h^2 + \dot k^2 + \dot\beta^2 - V(h,k,\beta), \quad V = \tfrac{m^2}{2}s + \tfrac{\Gamma}{4}s^2 - 2\kappa h(k^2+\beta^2-h^2), \quad s = 6h^2+2k^2+2\beta^2.$$

Every homogeneous initial condition in this sector has a global solution for $m^2 \geq 0, \Gamma > 0, \kappa \in \mathbb{R}$.

Proof. Direct multiplication gives

$$\det\Phi - b^*\Phi b = 2h(k^2+\beta^2-h^2), \qquad \operatorname{adj}\Phi - bb^* = \operatorname{diag}(-2h(h-k), -2h(h+k), h^2-k^2-\beta^2).$$

Its trace is $-3h^2 - k^2 - \beta^2$. Removing one third of this trace and substituting in (858) gives exactly (860), including the factor $1/3$ in the h equation. The last component of the triplet force is $-(a-2\kappa h)\beta$; all other entries remain zero. The currents vanish for each spacetime index: the matrices and their derivatives are diagonal, while b and its derivative are real multiples of e_3. The zero connection therefore satisfies all gauge equations. This also proves consistency under unrestricted variations, not merely stationary points of a restricted Lagrangian. As a check, that Lagrangian has Euler–Lagrange weights $6, 2, 2$ on h, k, β, respectively.

For homogeneous fields the smooth polynomial vector field is tangent to the displayed real linear sector, so local uniqueness proves invariance. Its conserved energy density is

$$E = 3\dot h^2 + \dot k^2 + \dot\beta^2 + V.$$

If $r^2 = k^2 + \beta^2$, then

$$|2h(r^2-h^2)| \leq 2|h|(h^2+r^2) \leq \frac{s^{3/2}}{\sqrt{6}}, \qquad V \geq \frac{\Gamma}{4}s^2 - \frac{|\kappa|}{\sqrt{6}}s^{3/2}.$$

Here $|h| \leq \sqrt{s/6}$ and $h^2 + r^2 \leq s/2$. This lower bound is coercive, so a fixed energy bounds all positions and velocities. The polynomial first-order system remains in a compact set on every finite interval and extends globally. This is an ODE existence proof, not a general nonlinear PDE global-existence claim. □

On $r > 0$, introduce $k = r\cos(2\theta)$, $\beta = -r\sin(2\theta)$. The original kinetic density and homogeneous equations become

$$\begin{aligned} \mathcal{L}_{\mathrm{kin}} &= 3\dot{h}^2 + \dot{r}^2 + 4r^2\dot{\theta}^2, \\ \ddot{h} &= -ah - \kappa h^2 + \kappa r^2/3, \qquad \ddot{r} = 4r\dot{\theta}^2 - (a - 2\kappa h)r, \\ (r^2\dot{\theta})^{\cdot} &= 0, \qquad a = m^2 + \Gamma(6h^2 + 2r^2). \end{aligned} \tag{861}$$

These follow by differentiating k, β, resolving the two equations into radial and angular components, and using $k\dot{\beta} - \beta\dot{k} = -2r^2\dot{\theta}$. The Cartesian sector includes $r = 0$; the polar chart does not. Even permitting a varying positive amplitude $Q = \chi(t)(3v_\theta v_\theta^* - I_3)$ gives $r = 3h > 0$. Then the h equation implies $3\ddot{h} = -(a - 2\kappa h)r$, whereas the r equation has the extra term $4r\dot{\theta}^2$. Since $\ddot{r} = 3\ddot{h}$, one obtains $4r\dot{\theta}^2 = 0$. Thus changing this single amplitude does not by itself make angular homogeneous motion consistent.

Theorem 24.39 (Exact gauge-invariant periodic continuation)**.** *Suppose* $0 < m^2 < \kappa^2/(27\Gamma)$, $\Gamma > 0, \kappa > 0$. *Define*

$$d_0 = \sqrt{\kappa^2 - 24\Gamma m^2}, \quad h_0 = \frac{\kappa + d_0}{24\Gamma} = \frac{\chi}{2}, \quad h_0 < h < \frac{\kappa}{6\Gamma}, \quad \Delta(h) = 1 - \frac{6\Gamma h}{\kappa}.$$

Put

$$a(h) = \frac{m^2 + 12\Gamma h^2}{\Delta(h)}, \qquad R^2(h) = \frac{3a(h)h}{\kappa} + 3h^2, \qquad \omega^2(h) = a(h) - 2\kappa h = \frac{m^2 - 2\kappa h + 24\Gamma h^2}{\Delta(h)}.$$

Take the positive roots R, ω. *Then the sector fields*

$$h(t) = h, \quad k(t) = R\cos(\omega t), \quad \beta(t) = R\sin(\omega t), \quad \mathcal{A}_\mu = 0$$

are nonconstant, spatially homogeneous, exact full gauge–scalar solutions. The gauge-invariant observable $\|b(t)\|^2 = R^2\sin^2(\omega t)$ *is nonconstant. Their conserved energy density is exactly*

$$E(h) = R^2\omega^2 + m^2(3h^2 + R^2) + \Gamma(3h^2 + R^2)^2 - 2\kappa h(R^2 - h^2). \tag{862}$$

Let

$$\Delta_0 = 1 - \frac{6\Gamma h_0}{\kappa} = \frac{3\kappa - d_0}{4\kappa} > 0, \qquad V_0 = 12m^2h_0^2 + 144\Gamma h_0^4 - 16\kappa h_0^3.$$

As $h \downarrow h_0$,

$$\begin{aligned} R &\longrightarrow 3h_0, \qquad \omega \longrightarrow 0, \\ \omega^2 &= \frac{2d_0}{\Delta_0}(h - h_0) + O((h - h_0)^2), \\ E(h) - V_0 &= \frac{18h_0^2 d_0}{\Delta_0}(h - h_0) + O((h - h_0)^2) = 9h_0^2\omega^2 + O(\omega^4). \end{aligned} \tag{863}$$

These are finite-volume classical statements. On a flat torus with retained volume $\mathcal{V}_3 > 0$, *the total energy and its excess are* $\mathcal{V}_3 E(h)$ *and* $\mathcal{V}_3(E(h) - V_0)$. *The amplitude tends to the nonzero value* $3h_0$*: the family approaches the minimum manifold while traversing a finite range of its invariant* $\|b\|^2$, *rather than becoming a small oscillation about one fixed vacuum. Its period tends to infinity.*

Proof. The denominator is positive on the stated interval; $a > 0$ and $R^2 > 0$. The numerator of ω^2 has roots $(\kappa - d_0)/(24\Gamma)$ and h_0; with its positive leading coefficient it is positive for $h > h_0$. Moreover $d_0 < \kappa$, so $h_0 < \kappa/(12\Gamma) < \kappa/(6\Gamma)$; the interval is nonempty. The identities

$$a = m^2 + \Gamma(6h^2 + 2R^2), \quad -ah - \kappa h^2 + \kappa R^2/3 = 0, \quad \omega^2 = a - 2\kappa h$$

follow by substitution. They are precisely the constant-h and the two oscillatory equations in (860). The previous proposition proves all gauge equations. The triplet norm is preserved by the full $SU(3)$ action, so its nonconstant value rules out a purely gauge motion.

Substitution in the conserved energy gives (862). At $h = h_0$, $m^2 = 2\kappa h_0 - 24\Gamma h_0^2$, so $a = 2\kappa h_0$, $R^2 = 9h_0^2$ and $\omega^2 = 0$. The limiting field at any phase is a global minimum: the 2-by-2 quaternionic block has eigenvalues $h_0 \pm 3h_0$, and the last eigenvalue is $-2h_0$; hence the full spectrum is $(4h_0, -2h_0, -2h_0)$. Writing the frequency numerator as $n(h)$, $n(h_0) = 0$ and $n'(h_0) = 2d_0$. Taylor expansion proves the frequency estimate. The curve $h \mapsto (h, R(h), 0)$ is smooth near h_0, and its endpoint is a critical point of V. Consequently its potential differs from V_0 by $O((h - h_0)^2)$. Its kinetic density is

$$R^2\omega^2 = \Big(9h_0^2 + O(h - h_0)\Big)\left(\frac{2d_0}{\Delta_0}(h - h_0) + O((h - h_0)^2)\right).$$

This proves both energy expansions. Since the leading coefficient of ω^2 is strictly positive, replacing $h - h_0$ by its smooth inverse as a function of ω^2 gives the last equality. For every $h > h_0$, $E(h) > V_0$ because the kinetic term is positive and the potential is bounded below by V_0. The torus assertion is integration of a constant spatial density. Nonzero homogeneous energy density on all of $\mathbb{R}^3$ is not asserted to have finite total energy. □

Proposition 24.40 (Linearized physical tangent and exact null waves)**.** *At each fixed vacuum q_{θ_0}, the perturbation*

$$\delta\Phi = T_\Phi(\theta_0)\eta, \qquad \delta b = T_b(\theta_0)\eta, \qquad \delta\mathcal{A}_\mu = 0$$

solves the full linearized equations precisely when $\Box\eta = 0$. Its quadratic density is

$$9\chi^2((\partial_t\eta)^2 - |\nabla\eta|^2).$$

For a homogeneous mode on a torus of volume $\mathcal{V}_3$, its momentum and quadratic Hamiltonian are

$$\pi_\eta = 18\chi^2\mathcal{V}_3\dot\eta, \qquad H_\eta = \frac{\pi_\eta^2}{36\chi^2\mathcal{V}_3}.$$

There is also an exact nonlinear inhomogeneous continuation at fixed χ: for any smooth real f,

$$\theta(t, x) = f(t - x_1), \qquad q(t, x) = q_{\theta(t,x)}, \qquad \mathcal{A}_\mu = 0$$

solves the full nonlinear field equations wherever these fields are defined. On the rectangular flat torus with sides L_1, L_2, L_3, choose f L_1-periodic. Its excess over the vacuum energy is

$$E_{\text{wave}} - \mathcal{V}_3V_0 = 18\chi^2L_2L_3\int_0^{L_1}(f'(u))^2\,du, \qquad \mathcal{V}_3 = L_1L_2L_3.$$

Proof. Differentiate $\nabla V(q_\theta) = 0$ in θ. The metric Hessian annihilates T. The linearized currents are

$$([\Phi_{\theta_0}, T_\Phi] + b_{\theta_0}T_b^* - T_b b_{\theta_0}^*)\partial_\mu\eta = 0.$$

Thus the linearized scalar equation is $T\Box\eta = 0$, and $T \neq 0$. The gauge equations have zero current and zero field strength. The kinetic norm $18\chi^2$ proves the quadratic density and its Legendre transform with the stated volume.

For the nonlinear assertion, the chain rule in signature $(+---)$ gives

$$\Box q_\theta = T(\theta)\Box\theta + N(\theta)\partial_\mu\theta\,\partial^\mu\theta.$$

Both coefficients vanish for $\theta = f(t - x_1)$. Every value remains a critical point of V; the trigonometric formulas define it for all real θ, or one can restrict the range of f to the displayed minimum section. The exact currents vanish for all spacetime derivatives as in the sector proposition. This verifies every nonlinear equation. The scalar energy density is the sum of one temporal and one spatial term, each $9\chi^2(f')^2$, plus V_0. Periodicity makes its integral independent of time and gives the formula. For $f(u) = \theta_0 + \varepsilon\cos(2\pi u/L_1)$ with range inside the section, the excess is $9\chi^2\mathcal{V}_3\varepsilon^2(2\pi/L_1)^2$. □

Gauge-invariant coordinate and endpoint scope. The invariant

$$I = \|b_\theta\|^2 = \frac{9\chi^2}{4}\sin^2(2\theta), \qquad \frac{dI}{d\theta} = \frac{9\chi^2}{2}\sin(4\theta)$$

has nonzero derivative on $0 < \theta < \pi/4$. Therefore the displayed linearized mode is visible through the genuine gauge-invariant linear observable $\delta I = (9\chi^2/2)\sin(4\theta_0)\eta$ in the interior. With $M = 9\chi^2/4$, its interior coordinate expression is

$$9\chi^2(\partial\theta)^2 = \frac{9\chi^2}{16I(M-I)}(\partial I)^2.$$

The denominator vanishes at the endpoints because I ceases to be a regular linear coordinate there, not because the norm of T diverges.

For completeness the stabilizers on this section can be checked without importing an orbit classification. In the interior the three diagonal entries of Φ_θ are distinct, so a stabilizer must be diagonal. Fixing the nonzero b_3 and its determinant then gives $\mathrm{diag}(e^{it}, e^{-it}, 1)$, of dimension one. At $\theta = 0$, $b = 0$ and the eigenvalues are $2\chi, -\chi, -\chi$, giving $S(U(1) \times U(2))$, of dimension four. At $\theta = \pi/4$, fixing $b_3 \neq 0$ and the two-dimensional first eigenspace gives $\mathrm{diag}(U, 1)$, $U \in SU(2)$, of dimension three. The gauge-orbit dimensions are consequently $7, 4, 5$, respectively. The fixed-χ minimum manifold has dimension eight: the unit sphere in $\mathbb{H}^3$ has dimension eleven and the free unit-quaternion action $v \mapsto va$ has dimension three, with vv^* precisely its quotient. Thus the horizontal tangent dimensions within that manifold are $1, 4, 3$, respectively. In particular the interior has exactly one horizontal vacuum direction. At the endpoints the invariant linear function I vanishes to first order; no assertion of a smooth one-dimensional linear quotient there is made. This agrees with the classical cohomogeneity-one description in A. Gambioli, *Eight-dimensional $SU(3)$-manifolds of cohomogeneity one*, arXiv:math/0611796, Section 4, Examples: $\mathbb{HP}^2$ has singular orbit types $\mathbb{CP}^2$ and S^5. The kinetic and current statements above are independently proved here, not imported from that classification.

Fixed observation and classical scope. Retain

$$P = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix}, \qquad u = (1, 1, 1)^{\mathsf{T}}.$$

The positive spectral-floor matrix $H_\theta = Q_\theta + \chi I_3$ has complex part $A_\theta = 3\chi\,\mathrm{diag}(\cos^2\theta, \sin^2\theta, 0)$. Consequently

$$G_\theta = PA_\theta P^* = 3\chi \begin{pmatrix} 1 & -\sin^2\theta \\ -\sin^2\theta & \sin^2\theta \end{pmatrix}, \qquad \rho = u^* b_\theta = -3\chi\cos\theta\sin\theta, \qquad \det G_\theta = |\rho|^2 = I.$$

These are direct matrix products with the original P, u, χ. They do not assert gauge invariance of a fixed-P observation away from this chosen section: under a general gauge transformation its complex part is $gA_\theta g^*$, so the observation is $PgA_\theta g^* P^*$, not generally $PA_\theta P^*$.

On a rectangular torus the linearized wave equation has classical Fourier frequencies $\omega_{\mathbf{n}}^2 = 4\pi^2 \sum_i n_i^2/L_i^2$; the spatial zero mode has zero linearized frequency. The exact homogeneous periodic family additionally has arbitrarily small positive classical frequency and energy excess at fixed volume. The null waves give exact nonlinear inhomogeneous solutions with arbitrarily small amplitude. None of these statements quantizes the model, identifies a Hilbert-space spectrum, proves a quantum mass gap or its absence, or assigns physical units that the retained parameters have not been supplied. No finite-total-energy assertion on all of $\mathbb{R}^3$ is inferred from a finite-volume solution.

24.17 The gauge-vector spectrum and an invariant vector observable

We now compute masses in the linearized classical field equations of (847), not by assigning a physical meaning to a determinant in isolation. Retain a positive nonzero minimum amplitude χ from Theorem 24.24; thus $\kappa > 0$, $0 \le m^2 \le \kappa^2/(27\Gamma)$ and

$$\chi = \frac{\kappa + \sqrt{\kappa^2 - 24\Gamma m^2}}{12\Gamma}.$$

The nonzero branch is meant at coexistence. Keep $0 \le \theta \le \pi/4$, and introduce the notation $q = \cos(2\theta)$, not a change in the field or its kinetic metric. The same minimum representatives are

$$\Phi_\theta = \chi \operatorname{diag}\left(\frac{1+3q}{2}, \frac{1-3q}{2}, -1\right), \qquad b_\theta = -\frac{3\chi}{2}\sin(2\theta)e_3, \qquad \beta^2 := b_\theta^* b_\theta = \frac{9\chi^2}{4}(1-q^2).$$

Every constant representative together with $\mathcal{A}_\mu = 0$ solves the full classical field equations.

Theorem 24.41 (All eight gauge-inertia eigenvalues)**.** *Use the positive inner product $\langle Z, W\rangle = -\operatorname{tr}(ZW)$ on $\mathfrak{su}(3)$. At the displayed minimum, the exact spectrum of $\mathscr{I}_{\Phi_\theta, b_\theta}$ is*

generator space	*inertia eigenvalue*	*real multiplicity*
$i\operatorname{diag}(1,-1,0)$	0	1
$i\operatorname{diag}(1,1,-2)$	$3\chi^2(1-q^2)$	1
$\langle E_{12}-E_{21},\, i(E_{12}+E_{21})\rangle$	$9\chi^2 q^2$	2
$\langle E_{13}-E_{31},\, i(E_{13}+E_{31})\rangle$	$\frac{9}{2}\chi^2(1+q)$	2
$\langle E_{23}-E_{32},\, i(E_{23}+E_{32})\rangle$	$\frac{9}{2}\chi^2(1-q)$	2

(GS.1)

Here E_{ij} are the actual matrix units. The unscaled generators in the row order with the six off-diagonal generators first have kinetic Gram matrix $\operatorname{diag}(2,2,2,2,2,2,2,6)$*. The squared masses of transverse solutions of the full linearized field equations are exactly $g^2/2$ times the eigenvalues in (GS.1). In particular the diagonal nonzero branch has*

$$\mu_{\mathrm{d}}^2 = \frac{3}{2}g^2\chi^2(1-q^2) = \frac{2g^2}{3}\|b_\theta\|^2. \tag{GS.2}$$

There is one zero eigenvalue for $0 < q < 1$, four at $q = 1$, and three at $q = 0$. These are the Lie-algebra dimensions of the stabilizers $U(1)_{1,-1,0}$, $S(U(1)\times U(2))$, and $SU(2)$, respectively.

Proof. For a diagonal Hermitian $\Phi = \operatorname{diag}(d_1, d_2, d_3)$, the operator $[\Phi, [\Phi, Z]]$ multiplies the ij off-diagonal entry by $(d_i - d_j)^2$. Here $bb^* = \operatorname{diag}(0, 0, \beta^2)$. Consequently $\{bb^*, Z\}$ contributes zero on the 12 root space and β^2 on each of the 13 and 23 root spaces; these entries have zero trace, so no trace-free correction changes them. Retaining the displayed differences gives

$$(d_1-d_2)^2 = 9\chi^2q^2, \quad (d_1-d_3)^2+\beta^2 = \frac{9}{2}\chi^2(1+q), \quad (d_2-d_3)^2+\beta^2 = \frac{9}{2}\chi^2(1-q).$$

For $Z = i\operatorname{diag}(z_1, z_2, z_3)$, $\sum z_i = 0$, the diagonal contribution is

$$\mathscr{I}Z = 2i\beta^2 z_3\left(E_{33} - \frac{1}{3}I_3\right).$$

It annihilates $i\operatorname{diag}(1,-1,0)$ and multiplies $i\operatorname{diag}(1,1,-2)$ by $4\beta^2/3 = 3\chi^2(1-q^2)$. The eight displayed generators are linearly independent and span $\mathfrak{su}(3)$; multiplication and trace give their stated diagonal Gram matrix. This proves the entire generalized Hessian spectrum without replacing that metric by an identity matrix.

For the dynamical assertion, perturb only the connection at first order: $\mathcal{A}_\nu = \varepsilon Z a_\nu$, with both scalar perturbations zero. At the constant scalar background the first-order scalar equations reduce

to $[Z,\Phi_\theta]\partial_\nu a^\nu = 0$ and $Zb_\theta\,\partial_\nu a^\nu = 0$. The first-order current is $(J^\nu)_{\rm tf} = -\varepsilon\mathscr{I}(Z)a^\nu$. For an inertia eigenvector with eigenvalue λ, the full gauge equation is therefore

$$(\Box + g^2\lambda/2)a^\nu - \partial^\nu(\partial_\mu a^\mu) = 0. \tag{GS.3}$$

Choose $a_0 = 0$ and $a_j = e_j\cos(k\cdot x - \omega t)$ with $k\cdot e = 0$. Then $\partial_\mu a^\mu = 0$ and all scalar equations hold. The temporal gauge equation is $\sum_j \partial_j \dot{a}_j = 0$, so Gauss is satisfied as well, not omitted. The remaining equations give $\omega^2 = |k|^2 + g^2\lambda/2$. At nonzero frequency or spatial momentum, the corresponding field strength is nonzero. Thus these are actual transverse solutions, not pure-gauge variations. For a specified rectangular flat spatial torus, $k_j = 2\pi n_j/L_j$, with every length L_j retained. Their positive kinetic sign follows from $\langle Z,Z\rangle/g^2 > 0$ in the original gauge density.

For $0 < q < 1$, every displayed eigenvalue except the first is positive. At $q = 1$ the second and the 23 pair additionally vanish; at $q = 0$ the 12 pair additionally vanishes. A stabilizer satisfies $[Z,\Phi_\theta] = 0$ and $Zb_\theta = 0$ by the proved inertia kernel. For $0 < q < 1$ distinct diagonal entries and nonzero b_θ give $\mathrm{diag}(e^{i\alpha}, e^{-i\alpha}, 1)$. At $q = 1$, $b_\theta = 0$ and the last two entries of Φ_θ coincide, giving the stated $U(2)$ subgroup. At $q = 0$ the first two entries coincide and $b_\theta \neq 0$ fixes e_3, giving the upper-left $SU(2)$. This proves the full group statements, including the endpoints. □

Proposition 24.42 (An exact invariant current and its mass branch)**.** *The full nonlinear gauge–scalar theory has the real, gauge-invariant conserved vector current*

$$j_\nu = -i\,(b^* D_\nu b - (D_\nu b)^* b)\,, \qquad \partial_\nu j^\nu = 0. \tag{GS.4}$$

At any displayed vacuum with $\beta^2 > 0$, the transverse perturbation $\mathcal{A}_\nu = \varepsilon i\,\mathrm{diag}(1,1,-2)a_\nu$, with zero scalar perturbations, has

$$j_\nu = -4\varepsilon\beta^2 a_\nu + O(\varepsilon^2). \tag{GS.5}$$

It therefore detects the branch (GS.2) in the exact linearized system. For the positive rank-one spectral-floor matrix $H_\theta = Q_\theta + \chi I_3 = \Theta(A_\theta, b_\theta)$,

$$\boxed{\mu_{\rm d}^2 = \frac{2g^2}{3V_\mu}\int_{SU(3)} \det(PgA_\theta g^* P^*)\,d\mu(g).} \tag{GS.6}$$

The Haar volume $V_\mu > 0$ and the original marking P are retained. At $\beta = 0$ the linear coefficient in (GS.5) vanishes; that endpoint does not supply a nonzero linear interpolator through this current.

Proof. Under $b \mapsto Ub$ and $D_\nu b \mapsto U D_\nu b$, each scalar product in (GS.4) is unchanged. Their difference is purely imaginary, so j_ν is real. It is the Noether current for the constant phase action $b \mapsto e^{i\alpha}b$: allowing α to depend on spacetime produces $j^\nu\partial_\nu\alpha$ in the variation of the kinetic term. The potential is phase invariant. Directly, the covariant product rule gives

$$\partial_\nu j^\nu = -i\,(b^* D_\nu D^\nu b - (D_\nu D^\nu b)^* b)\,.$$

The exact equation is $D_\nu D^\nu b = -(aI_3 + \kappa\Phi)b$; the matrix in parentheses is Hermitian. The two scalar products are therefore equal, proving conservation in the full interacting theory.

At the specified constant background, $b_\theta^* \mathcal{A}_\nu b_\theta = -2i\varepsilon\beta^2 a_\nu$. Subtracting its conjugate and multiplying by $-i$ gives $-4\varepsilon\beta^2 a_\nu$, with no rescaling of the Cartan generator. Every off-diagonal generator, and $i\,\mathrm{diag}(1,-1,0)$, has zero 33 entry, so none contributes to this linear current when scalar perturbations are zero. For an arbitrary scalar perturbation δb, its additional linear term is $2\,\mathrm{Im}(b_\theta^*\partial_\nu\delta b)$, a pure gradient at the constant background; the transverse sector used in (GS.3) sets it to zero. Thus the nonzero transverse mode is seen by a gauge-invariant observable. Finally Proposition 24.35 proves the integral in (GS.6) equals $V_\mu\beta^2$. Substitution into (GS.2) proves the exact relation with all coefficients and domains fixed. □

This follows the classical Higgs-phase use of gauge-invariant vector operators discussed by D. Tong, *Lectures on Gauge Theory*, Section 2.7.3, especially the fundamental-scalar example on printed pages 107–108, `https://www.damtp.cam.ac.uk/user/tong/gaugetheory/gt.pdf`. The eight-eigenvalue calculation, current coefficient, and marked determinant identity here are derived for this specified adjoint-plus- fundamental model; they are not transferred from that example. Equation (GS.6) is a relation to a linearized gauge-vector mass in this model, not an identification of all entries of G with spacetime momentum, and not a spectral theorem for pure interacting quantum Yang–Mills.

Corollary 24.43 (Variation of the massive branches along the minimum family)**.** *For $0 < q < 1$, the smallest strictly positive inertia eigenvalue is*

$$\lambda_{\mathrm{pos}}(q) = \begin{cases} 9\chi^2 q^2, & 0 < q \le 1/2, \\ \frac{9}{2}\chi^2(1-q), & 1/2 \le q < 1. \end{cases} \tag{GS.7}$$

It tends to zero at both ends and has unique maximum $9\chi^2/4$ at $q = 1/2$. At that point five eigenvalues coincide at $9\chi^2/4$, two equal $27\chi^2/4$, and one is zero. This positive degeneracy does not enlarge the stabilizer. The retained spectral traces are

$$\operatorname{tr}\mathscr{I} = \chi^2(36 - 15\delta), \qquad \operatorname{tr}(\mathscr{I}^2) = \chi^4(324 - 405\delta + 171\delta^2), \qquad \delta = 1 - q^2. \tag{GS.8}$$

Proof. The 13 eigenvalue is at least the 23 eigenvalue. Subtracting the 12 eigenvalue from the diagonal and 23 eigenvalues gives $3\chi^2(1-4q^2)$ and $\frac{9}{2}\chi^2(1-q-2q^2)$, respectively; both are nonnegative exactly for $0 \le q \le 1/2$ on the stated interval. For $q \ge 1/2$, the diagonal minus the 23 eigenvalue is $\frac{3}{2}\chi^2(1-q)(2q-1) \ge 0$. These comparisons prove (GS.7). The first expression increases strictly and the second decreases strictly, proving the maximum and endpoint limits. Substitution of $q = 1/2$ in (GS.1) gives the multiplicities, while its kernel remains one-dimensional. Adding the eight eigenvalues and their squares gives $\chi^2(21 + 15q^2)$ and $\chi^4(90 + 63q^2 + 171q^4)$, respectively. Replacing the notation q^2 by $1 - \delta$ proves (GS.8). □

There is always a zero transverse branch in this linearized system. The strictly positive branches alone also have no positive lower bound uniform over the minimum family. These statements concern the proved quadratic spectrum and its exact parameters. They do not determine radiative selection of a minimum, a renormalized continuum theory, or the interacting Hamiltonian spectrum.

24.18 The retained system and the scope of the result

The literal constant-matter data are

$$H = \begin{pmatrix} 0 & 0 & 0 \\ 0 & -1/2 & 1/(2\sqrt{2}) \\ 0 & 1/(2\sqrt{2}) & 1/2 \end{pmatrix}, \qquad q = \frac{e_1}{2\sqrt{2}}, \qquad A = \sqrt{2}\begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

The original scalar couplings are $m^2 = \Gamma = 1$, $\kappa = 6$; $s = \operatorname{tr} H^2 + 2q^*q = 1$ and $a = m^2 + \Gamma s = 2$. Write $g^2 > 0$ for the gauge coupling and keep $\omega \neq 0$ real. Every unknown C, Z_1, Z_2 belongs to $\mathfrak{su}(3)$, the real vector space of traceless anti-Hermitian complex 3×3 matrices. For any matrix M, set $M_{\mathrm{tf}} = M - (\operatorname{tr} M)I_3/3$, and define

$$\mathcal{I}X = ([H,[H,X]] + qq^*X + Xqq^*)_{\mathrm{tf}}, \qquad \mathcal{K}X = -[Z_1,[Z_1,X]] - [Z_2,[Z_2,X]],$$

$$F = [A,[A,H]] = -2H + \operatorname{diag}(-2,1,1).$$

The last identity follows from $Hq = 0$, $H^2 = (3/8)\,\mathrm{diag}(0,1,1)$ and $6(\mathrm{adj}\,H - qq^*)_{\mathrm{tf}} = \mathrm{diag}(-2,1,1)$. The five equations to be retained are exactly

$$[C,[C,H]] - [Z_1,[Z_1,H]] - [Z_2,[Z_2,H]] = F, \tag{864}$$

$$(C^2 - Z_1^2 - Z_2^2)q = -2q, \tag{865}$$

$$(\mathcal{I} + 2\mathcal{K}/g^2)C = (4\omega/g^2)[Z_1,Z_2], \tag{866}$$

$$-\omega^2 Z_1 - 2\omega[C,Z_2] + [C,[C,Z_1]] + \mathcal{K}Z_1 + (g^2/2)\mathcal{I}Z_1 = 0, \tag{867}$$

$$-\omega^2 Z_2 + 2\omega[C,Z_1] + [C,[C,Z_2]] + \mathcal{K}Z_2 + (g^2/2)\mathcal{I}Z_2 = 0. \tag{868}$$

This section does not solve or exclude all triples in $\mathfrak{su}(3)^3$. It proves necessary identities for every such triple and a no-solution theorem for the precisely stated complex block class below. No remaining general equation is assumed true, and no numerical candidate is a result.

Exact map to the original literal fields

For clarity, keep

$$u = (1,1,1)^{\mathsf{T}}, \quad v_0 = (1,-1,0)^{\mathsf{T}}, \quad w_0 = (1,1,-2)^{\mathsf{T}}, \quad K = \left(v_0/\sqrt{2}, w_0/\sqrt{6}, u/\sqrt{3}\right),$$

$$\Omega = \sqrt{2/3}\begin{pmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ -1 & 1 & 0 \end{pmatrix}, \quad O(t) = e^{t\Omega}, \quad \Phi_0 = \frac{1}{4}\begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & -2 \end{pmatrix}, \quad b_0 = \frac{1}{4}(1,-1,0)^{\mathsf{T}}.$$

The dot products and determinant give $K \in SO(3)$, and multiplication gives $K^*\Phi_0 K = H$, $K^* b_0 = q$, $K^*\Omega K = A$. The prescribed scalar fields remain $\Phi(t) = O(t)\Phi_0 O(t)^*$, $b(t) = O(t)b_0$; their frequency is $\sqrt{2}$. For a solution of (864)–(868), set

$$Y_1 = \cos(\omega t)Z_1 - \sin(\omega t)Z_2, \quad Y_2 = \sin(\omega t)Z_1 + \cos(\omega t)Z_2, \quad Y_3 = 0,$$

$$\mathcal{A}_0 = OKCK^*O^* - \Omega, \qquad \mathcal{A}_i = OKY_iK^*O^*. \tag{869}$$

The inverse is $C = K^*O^*\mathcal{A}_0 OK + A$ and $Y_i = K^*O^*\mathcal{A}_i OK$. These are gauge transformations by K^*O^*, including the inhomogeneous temporal term.

Here is a direct verification of the complete finite reduction. In the constant-matter frame put $E_i = \dot{Y}_i + [C,Y_i]$. The two scalar equations are $[C,[C,H]] - \sum_i [Y_i,[Y_i,H]] = F$ and $(C^2 - \sum_i Y_i^2)q = -2q$. The projected currents are $J_0 = -\mathcal{I}C$ and $J_i = -\mathcal{I}Y_i$. Gauss is $(2/g^2)\sum_i [Y_i,E_i] + \mathcal{I}C = 0$, and the spatial equations are

$$\dot{E}_i + [C,E_i] - \sum_j [Y_j,[Y_j,Y_i]] + (g^2/2)\mathcal{I}Y_i = 0.$$

These follow by homogeneous substitution in the covariant equations with $D_\mu = \partial_\mu + \mathcal{A}_\mu$, signature $(+---)$ and gauge density $\mathrm{tr}(\mathcal{F}_{\mu\nu}\mathcal{F}^{\mu\nu})/(2g^2)$. Specifically the adjoint derivative is a commutator, and $J_\mu = [\Phi, D_\mu\Phi] + b(D_\mu b)^* - (D_\mu b)b^*$; at constant matter it is minus the displayed inertia. Thus no temporal constraint has been removed. The rotation preserves $\sum_i Y_i^2$ and every quadratic commutator sum, and $\sum_i [Y_i,\dot{Y}_i] = -2\omega[Z_1,Z_2]$. Applying $(\partial_t + \mathrm{ad}_C)^2$ to the rotating pair gives the terms $-\omega^2 Z_i$, the two displayed terms with coefficient 2ω, and $[C,[C,Z_i]]$. The third spatial equation is identically zero. This proves equivalence with all five equations above at every time. The product rule proves covariance under (869), so this equivalence also holds in the original coordinates.

24.19 Universal necessary identities in the unrestricted complex class

Use $\langle X, Y\rangle = -\operatorname{tr}(XY)$ and $\|X\|^2 = -\operatorname{tr} X^2$ on $\mathfrak{su}(3)$. This is a real positive inner product because $X^* = -X$. Define the nonnegative real quantities

$$c = \langle C, \mathcal{I}C\rangle, \quad z = \sum_{j=1}^{2} \langle Z_j, \mathcal{I}Z_j\rangle, \quad k = \sum_{j=1}^{2} \|[C, Z_j]\|^2, \quad d = \|[Z_1, Z_2]\|^2, \quad z_0 = \|Z_1\|^2 + \|Z_2\|^2,$$

and the signed real number $b = \langle C, [Z_1, Z_2]\rangle$.

Proposition 24.44. *Every solution of* (864)–(868) *satisfies*

$$c - z = 2, \qquad (g^2/2)c + k = 2\omega b, \qquad \omega^2 z_0 = k + 2d + g^2 c + (g^2/2)z. \tag{870}$$

In particular c, k, d, z_0 are strictly positive, $\omega b > 0$, and

$$8kd + 2g^2 kz \geq g^4 c^2. \tag{871}$$

Proof. Cyclicity and anti-Hermiticity give

$$\langle X, \mathcal{I}X\rangle = \operatorname{tr}[X, H]^2 + 2|Xq|^2, \qquad \langle X, \mathcal{K}X\rangle = \sum_j \|[Z_j, X]\|^2.$$

Trace projection pairs trivially with traceless X. The first commutator term follows by cyclically moving H twice, and the vector term is $-2q^*X^2q = 2|Xq|^2$. The second identity follows by applying $\langle X, [Z, Y]\rangle = -\langle [Z, X], Y\rangle$ twice. Because $[X, H]$ is Hermitian, all terms are nonnegative.

Pair (864) with H using ordinary trace and pair (865) with $2q^*$, taking the real part. Their left sides add to $-c + z$. Their right sides add to $\operatorname{tr}(HF) - 4q^*q = -2\operatorname{tr} H^2 - 4q^*q = -2$, proving the first identity. Pair (866) with C to obtain the second. Pair (867) with Z_1 and (868) with Z_2. The gyroscopic terms sum to $4\omega b$, the double C terms sum to $-k$, and the spatial double commutators sum to $2d$. The result is $-\omega^2 z_0 + 4\omega b - k + 2d + (g^2/2)z = 0$. Substitute the second identity to obtain the third.

The first identity gives $c \geq 2$. The second gives $\omega b > 0$. Thus $b \neq 0$, which forces $d > 0$, $z_0 > 0$, and $k > 0$, since $b = \langle [C, Z_1], Z_2\rangle$. Apply Cauchy–Schwarz to the ordered pairs $([C, Z_1], [C, Z_2])$ and $(Z_2, -Z_1)$ to obtain $4b^2 \leq kz_0$. The second identity gives $\omega^2 z_0 \geq (k + g^2 c/2)^2/k$. Subtract this lower bound from the third identity and multiply by $4k$; the result is exactly (871). □

These identities rule out any attempted solution with zero spatial commutator or zero electric coupling to C. They do not rule out a general noncommuting complex triple.

24.20 A new genuinely complex block exclusion

Let $P = qq^*/(q^*q) = \operatorname{diag}(1, 0, 0)$ and put

$$\mathfrak{p} = \{X \in \mathfrak{su}(3) : PXP = (I - P)X(I - P) = 0\}, \qquad \mathfrak{k} = \{X \in \mathfrak{su}(3) : [X, P] = 0\}.$$

Thus $\mathfrak{p}$ consists of arbitrary complex matrices exchanging the q-line with its orthogonal complement; $\mathfrak{k}$ consists of the block-diagonal matrices preserving that line. Neither definition imposes reality of the entries.

Theorem 24.45 (Cross–cross–block exclusion)**.** *There is no solution of all five equations with $C, Z_1 \in \mathfrak{p}$ and $Z_2 \in \mathfrak{k}$. There is likewise no solution for which a constant real rotation of the ordered pair (Z_1, Z_2) puts it in that class while $C \in \mathfrak{p}$. The assertion allows all complex phases and every $g^2 > 0$, $\omega \neq 0$.*

Exact coordinates used only for the proof

Set $x = 1/2$, $d_0 = 1/(2\sqrt{2})$, $r = \sqrt{x^2 + d_0^2} = \sqrt{6}/4$, and

$$S = \frac{1}{\sqrt{2r(r+x)}} \begin{pmatrix} d_0 & -(r+x) \\ r+x & d_0 \end{pmatrix}, \qquad U = \operatorname{diag}(1, S).$$

Multiplication gives $S^{\mathsf{T}} S = I_2$, $\det S = 1$, $U^* H U = \operatorname{diag}(0, r, -r)$, and $U^* q = q$. All connections and A are conjugated by U^*, with inverse conjugation by U. Brackets, trace projection, the force, and the inertia transform by these exact invertible maps; hence all five equations are equivalent. The subspaces $\mathfrak{p}, \mathfrak{k}$ are preserved because U commutes with P. No field, period, kinetic coefficient, or coupling is divided out.

For $v \in \mathbb{C}^2$ define

$$X(v) = \begin{pmatrix} 0 & -v^* \\ v & 0_{2\times 2} \end{pmatrix}.$$

In either of the proof frames below, H has zero first row and column, and its lower block squares to $r^2 I_2$. Since $q^* q = 1/8$,

$$\mathcal{I} X(v) = (r^2 + 1/8) X(v) = \tfrac{1}{2} X(v). \tag{872}$$

Indeed $H X(v) H = 0$, $H^2 X(v) + X(v) H^2 = r^2 X(v)$, and the two qq^* terms sum to $X(v)/8$; all these matrices are traceless.

Lemma 24.46 (Scalar classification of the block generator)**.** *Under the hypotheses of Theorem 24.45, the two scalar equations put the block generator into exactly one of the following two forms by an explicitly invertible constant unitary map fixing q:*

$$\text{(I)} \quad H = \operatorname{diag}(0, r, -r), \qquad Z_2 = i \operatorname{diag}(0, \sigma, -\sigma), \quad \sigma \in \mathbb{R};$$

$$\text{(II)} \quad H = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & r \\ 0 & r & 0 \end{pmatrix}, \qquad Z_2 = i \operatorname{diag}(e, 0, -e), \quad e \neq 0.$$

In both cases write $C = X(v)$, $Z_1 = X(w)$ and $U_v = vv^* - ww^* = (u_{jk})$. In case (I) scalar compatibility is exactly

$$u_{11} = 1 - \delta > 0, \quad u_{22} = 1 + \delta > 0, \qquad \delta = \frac{1}{2r} = \sqrt{2/3}.$$

In case (II) it is exactly

$$\operatorname{tr} U_v = 2 + e^2, \qquad \operatorname{Re} u_{12} = -\delta \neq 0.$$

Proof. Start in the displayed diagonal-H frame. An arbitrary element of $\mathfrak{k}$ is $Z_2 = i \operatorname{diag}(\eta, B)$, where B is Hermitian, $\operatorname{tr} B = -\eta$, and η is real. Write it without deleting any coefficient as

$$B = -\frac{\eta}{2} I_2 + \begin{pmatrix} z & x_1 - i x_2 \\ x_1 + i x_2 & -z \end{pmatrix}, \qquad x_1, x_2, z \in \mathbb{R}, \qquad D = \operatorname{diag}(r, -r).$$

The lower block of the adjoint scalar equation is

$$-\{U_v, D\} + [B, [B, D]] = I_2 - 2D.$$

Matrix multiplication gives

$$[B, [B, D]] = 4r \begin{pmatrix} x_1^2 + x_2^2 & -z(x_1 - i x_2) \\ -z(x_1 + i x_2) & -(x_1^2 + x_2^2) \end{pmatrix}.$$

Its off-diagonal equation is $z(x_1 - ix_2) = 0$. The two diagonal equations give $u_{11} - u_{22} = -1/r$ and $\operatorname{tr} U_v = 2 + 4(x_1^2 + x_2^2)$. The vector equation gives $\operatorname{tr} U_v = 2 + \eta^2$. Thus $\eta^2 = 4(x_1^2 + x_2^2)$.

If $x_1 = x_2 = 0$, then $\eta = 0$ and $B = \operatorname{diag}(z, -z)$. This is case (I) with $\sigma = z$, and its diagonal scalar equations give the two values stated. They are positive because $\delta^2 = 2/3 < 1$.

Otherwise $z = 0$ and $e := \eta \neq 0$. Let $\gamma = 2(x_1 - ix_2)/\eta$, so $|\gamma| = 1$. Choose either complex square root ζ of $-\bar{\gamma}$ and set

$$W = \frac{\zeta}{\sqrt{2}} \begin{pmatrix} \gamma & \gamma \\ 1 & -1 \end{pmatrix}.$$

Then $|\zeta| = 1$, $W^*W = I_2$, $\det W = 1$, $W^*BW = \operatorname{diag}(0, -e)$, and $W^*DW = r\,(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix})$. These identities follow by multiplying the two columns, which are the zero and $-e$ eigenvectors of B, with the stated phases. Conjugation by $\operatorname{diag}(1, W)^*$, with inverse its adjoint, therefore gives case (II), fixes q, and preserves both block spaces.

With $\sigma_x = (\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix})$, in this frame the lower scalar equation is $-\{U_v, r\sigma_x\} + e^2 r\sigma_x = I_2 - 2r\sigma_x$. Its diagonal entries give $\operatorname{Re} u_{12} = -1/(2r)$ and its off-diagonal entries give $\operatorname{tr} U_v = 2 + e^2$. The vector equation gives the same trace. Conversely direct substitution sets every scalar residual entry to zero in each of the two cases. □

All gauge equations in the two frames

In either case write

$$Z_2 = i \operatorname{diag}(\eta, d_1, d_2), \quad n_j = \eta - d_j, \quad V_j = |v_j|^2, \quad W_j = |w_j|^2,$$
$$V = V_1 + V_2, \quad W_0 = W_1 + W_2, \quad N = V + W_0,$$
$$p = v^*w = \alpha + ib, \quad b_j = \operatorname{Im}(\bar{v}_j w_j),$$
$$A_j = \frac{g^2}{4} + n_j^2 + W_0 > 0, \quad P_j = 3b + 2\omega n_j, \quad B_j = A_j - (\omega^2 + N).$$

Here b is only the imaginary inner product in this section, not the triplet or the signed pairing in Proposition 24.44. Direct block multiplication gives

$$-[X(w), [X(w), X(v)]] = X\big(W_0 v + w(\bar{p} - 2p)\big), \qquad [X(v), Z_2] = X\big(i \operatorname{diag}(n_1, n_2)v\big). \tag{873}$$

Using (872), the lower first-column equations of Gauss and the first spatial equation are exactly

$$A_j v_j = (\alpha + iP_j)w_j, \qquad B_j w_j = -(\alpha - iP_j)v_j \quad (j = 1, 2). \tag{874}$$

There are no other nonzero blocks in these two equations.

The second spatial equation has only a block-diagonal residual. Write $\mathcal{I}Z_2 = i \operatorname{diag}(\ell_0, \ell_1, \ell_2)$; it is diagonal in both frames. Its lower diagonal entries are

$$-\omega^2 d_j + 2n_j(V_j - W_j) + 4\omega b_j + (g^2/2)\ell_j = 0. \tag{875}$$

Its upper diagonal entry is

$$-\omega^2 \eta - 2\sum_j n_j(V_j - W_j) - 4\omega b + (g^2/2)\ell_0 = 0. \tag{876}$$

Its lower off-diagonal entry is

$$i(n_1 + n_2)(v_1\bar{v}_2 - w_1\bar{w}_2) + 2\omega(-v_1\bar{w}_2 + w_1\bar{v}_2) = 0. \tag{877}$$

For verification, the lower block of $[C, Z_1]$ is $-vw^* + wv^*$; the lower block contributed by $[C, [C, Z_2]] + \mathcal{K}Z_2$ has entry $i(n_j + n_k)(v_j\bar{v}_k - w_j\bar{w}_k)$. Its upper entry is minus its lower trace. These multiplications give (875)–(877) with every sign and coefficient. They retain the entire second spatial equation.

Lemma 24.47. *Every solution of these gauge equations has $\alpha = 0$ and*

$$\omega^2 W_0 = \frac{g^2}{4} N + \sum_j n_j^2 (V_j + W_j), \tag{878}$$

$$V_j - W_j = \frac{V_j + W_j}{\omega^2 + N} \left(\omega^2 + V - W_0 - \frac{g^2}{2} - 2n_j^2 \right). \tag{879}$$

Proof. Conjugate the first equation in (874) and multiply by w_j. Summing gives $p = \sum_j (\alpha - iP_j) W_j / A_j$. Taking real parts gives $\alpha = \alpha \sum_j W_j / A_j$. If $W_0 = 0$ the sum is zero. Otherwise $A_j > W_0$ gives $\sum_j W_j / A_j < \sum_j W_j / W_0 = 1$. Hence $\alpha = 0$ in all cases. The two equations are now $A_j v_j = iP_j w_j$ and $B_j w_j = iP_j v_j$. Their real pairings with v_j and w_j give $A_j V_j = -P_j b_j$ and $B_j W_j = P_j b_j$. Adding them gives $A_j (V_j + W_j) = (\omega^2 + N) W_j$. Summing and inserting A_j proves (878). Subtracting twice this last formula divided by $\omega^2 + N$ from $V_j + W_j$ proves (879). The denominator is positive because $\omega \neq 0$ and $N \geq 0$. □

Proof of Theorem 24.45. Use Lemma 24.46 and retain every gauge equation as just derived.

In case (I), $\eta = 0$, $n_1 = -\sigma$, $n_2 = \sigma$, $\mathcal{I} Z_2 = 0$, and $V_j - W_j = 1 \mp \delta > 0$. If $\sigma = 0$, then $Z_2 = 0$ and Gauss paired with C gives $\langle C, \mathcal{I} C \rangle + (2/g^2) \langle C, \mathcal{K} C \rangle = 0$. Both terms are nonnegative and (872) forces $C = 0$, contradicting the two positive scalar differences. Thus $\sigma \neq 0$. Both v_j are nonzero. Gauss then forces $w_j \neq 0$ and $P_j \neq 0$. Since $\alpha = 0$,

$$w_j = i t_j v_j, \qquad t_j = -A_j / P_j \in \mathbb{R} \setminus \{0\}.$$

Here $A_1 = A_2 = g^2/4 + \sigma^2 + W_0$. Because $n_1 + n_2 = 0$, (877) becomes $2i\omega(t_1 + t_2) v_1 \bar{v}_2 = 0$. Thus $t_1 + t_2 = 0$, which implies $P_1 + P_2 = 0$, or $6b = 0$. But (876) gives

$$b = \frac{\sigma((1-\delta) - (1+\delta))}{2\omega} = -\frac{\sigma \delta}{\omega} \neq 0.$$

This contradiction excludes case (I).

In case (II), $\eta = e$, $d_1 = 0$, $d_2 = -e$, $n_1 = e$, $n_2 = 2e$, and scalar compatibility gives $V - W_0 = 2 + e^2$ and $\operatorname{Re}(v_1 \bar{v}_2 - w_1 \bar{w}_2) = -\delta \neq 0$. In particular $V_j + W_j > 0$ for both j. The transformations $(Z_1, Z_2) \mapsto (-Z_1, -Z_2)$ and $(C, \omega) \mapsto (-C, -\omega)$ each preserve all five equations, as substitution verifies. They allow $e > 0$ and $\omega > 0$ without changing the given matter fields or their original frequency.

Direct multiplication in this frame gives

$$\mathcal{I} Z_2 = ie \operatorname{diag}(1/6, 2/3, -5/6).$$

The adjoint and trace-projected vector parts, respectively, are

$$ie \operatorname{diag}(0, 2r^2, -2r^2), \qquad ie \operatorname{diag}(1/6, -1/12, -1/12).$$

Their sum proves the formula with $r^2 = 3/8$. The right side of (878) is positive, so $W_0 > 0$. Since $N = 2W_0 + 2 + e^2 > 2W_0$ and $n_j^2 \geq e^2$, that identity gives

$$\omega^2 W_0 \geq (g^2/4 + e^2) N > (g^2/2 + 2e^2) W_0, \qquad \omega^2 > g^2/2 + 2e^2.$$

Formula (879) consequently implies

$$V_1 - W_1 = \frac{V_1 + W_1}{\omega^2 + N} (\omega^2 + 2 - e^2 - g^2/2) > 0.$$

The $j = 1$ equation in (875) is exactly

$$2e(V_1 - W_1) + 4\omega b_1 + e g^2 / 3 = 0,$$

so $b_1 < 0$. Gauss gives $b_1 = -P_1 W_1/A_1$. Since $V_1 > 0$, it also forces $W_1 > 0$ and hence $P_1 > 0$. Now $P_2 = P_1 + 2e\omega > 0$; because $V_2 + W_2 > 0$, Gauss forces both $V_2, W_2 > 0$ as well.

Taking imaginary parts in the proof of Lemma 24.47 gives

$$b\left(1 + 3\sum_j \frac{W_j}{A_j}\right) = -2\omega \sum_j \frac{n_j W_j}{A_j} < 0.$$

Therefore $b < 0$ and $0 < P_j = 3b + 2\omega n_j < 2\omega n_j$. Put $\rho_j = A_j/P_j > 0$. Since $A_j > n_j^2$, one has the strict lower bound

$$\rho_1 + \rho_2 > \frac{n_1 + n_2}{2\omega} = \frac{3e}{2\omega}. \tag{880}$$

Gauss also gives $w_j = -i\rho_j v_j$. Substitution into the full off-diagonal spatial equation (877), followed by division by the nonzero number $iv_1\bar{v}_2$, gives

$$3e(1 - \rho_1\rho_2) - 2\omega(\rho_1 + \rho_2) = 0.$$

As $\rho_1\rho_2 > 0$, this proves the contradictory strict upper bound

$$\rho_1 + \rho_2 = \frac{3e}{2\omega}(1 - \rho_1\rho_2) < \frac{3e}{2\omega}.$$

This excludes case (II) with arbitrary complex phases of v, w.

Finally a constant $SO(2)$ rotation of the ordered pair (Z_1, Z_2) preserves their sum of squares, every quadratic commutator sum, and $[Z_1, Z_2]$. It commutes with the spatial rotation generator $\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, so the two spatial equations transform by the same invertible rotation. It leaves C, H, q unchanged. Thus a solution that could be put in the excluded class by such a rotation would produce a solution already excluded, proving the final assertion. □

24.21 The precise unresolved part

Theorem 24.45 is a complex restriction, not the real-algebra obstruction restated. In case (II) one has $q^* Z_2 q = ie/8 \neq 0$. A real-skew matrix paired with the real vector q has $q^* Z q = 0$, and any unitary transformation fixing q preserves this pairing. Thus these nonzero block generators cannot be turned into real-skew generators by a matter-stabilizer transformation.

Nevertheless the theorem does not put arbitrary C, Z_1, Z_2 into its block class. In particular it assumes $C \in \mathfrak{p}$ and the stated decomposition of the spatial pair; neither follows here from all five unrestricted equations. The general finite system remains unresolved. The universal identities in Proposition 24.44 continue to apply outside this class. There is no asserted full-complex lift and no asserted full-complex no-lift theorem in this section.

The companion `check_complex_rotation.py` verifies the constant-frame matrix reductions with arbitrary complex block coordinates, the two branch inertia formulas, and the ratio identity by exact arithmetic. It also checks the universal pairing formulas on one exact full-algebra test triple. The universal proofs, branch inequalities, symmetry arguments, and original time-dependent gauge-map equivalence are proved in the text. The checker does not solve the unrestricted finite system.

24.22 Exact unrestricted Gauss and frequency elimination

The word unrestricted in this subsection refers to all complex colour coordinates in the specified constant-C, two-spatial-rotation ansatz, not to all time-dependent homogeneous connections. Retain the literal H, q, F, the positive coupling g^2, and all five equations (864)–(868). No block-class restriction from Theorem 24.45 is imposed here. For arbitrary $Z_1, Z_2 \in \mathfrak{su}(3)$ put

$$\mathscr{B} = [Z_1, Z_2], \qquad L = g^2\mathcal{I} + 2\mathcal{K}.$$

Lemma 24.48 (Positive Gauss inverse on every possible solution)**.** *If $\mathscr{B} \neq 0$, then L is a positive-definite self-adjoint operator on the original real eight-dimensional metric space $(\mathfrak{su}(3), -\operatorname{tr})$. Every solution of the five equations has $\mathscr{B} \neq 0$. Thus its Gauss equation is exactly*

$$C = \omega T, \qquad T = 4L^{-1}\mathscr{B}.$$

Proof. Cyclicity gives the full bilinear identities

$$\begin{aligned}\langle X, \mathcal{I}Y\rangle &= \operatorname{tr}([X,H][Y,H]) + 2\operatorname{Re}((Xq)^* Yq),\\ \langle X, \mathcal{K}Y\rangle &= \sum_{j=1}^{2}\langle [Z_j, X], [Z_j, Y]\rangle.\end{aligned}$$

They are real and symmetric. Their values at $Y = X$ are sums of nonnegative squared norms, since $[X, H]$ is Hermitian. Consequently

$$\ker L = \{X : [X,H] = 0,\ Xq = 0,\ [X, Z_1] = [X, Z_2] = 0\}.$$

For the converse direction in this equality, each displayed vanishing condition makes the corresponding double-commutator or vector term zero directly. For the forward direction, vanishing of their positive weighted sum forces every one of the norms to vanish.

The eigenvalues of H are the three distinct real numbers $0, r, -r$, $r = \sqrt{6}/4$, and q spans its zero eigenline. Conjugate by the exact matrix U already displayed in the block proof. A matrix commuting with this diagonal H has no off-diagonal entries. Anti-Hermiticity makes its diagonal entries imaginary; annihilation of q sets the first entry to zero; trace zero makes the last two opposite. Therefore

$$\{X : [X,H] = 0,\ Xq = 0\} = \mathbb{R}\, iH.$$

If a nonzero element of this line also commuted with both Z_j, both Z_j would commute with H. The same distinct-eigenvalue argument would make them diagonal in its eigenbasis, hence mutually commuting. This contradicts $\mathscr{B} \neq 0$. Thus $\ker L = 0$, and positivity and finite dimension give the asserted inverse.

For any solution, Proposition 24.44 proves $\|[Z_1, Z_2]\|^2 > 0$. Equivalently, if that bracket were zero, Gauss paired with C would give $c = 0$, whereas the scalar contraction gives $c - z = 2$ and $z \geq 0$. Finally multiplying Gauss by g^2 gives $LC = 4\omega\mathscr{B}$, exactly the stated inverse relation. No temporal equation was omitted. □

Theorem 24.49 (Full residual zero set and a two-sign inverse)**.** *On the explicit open domain*

$$\mathcal{D} = \{(Z_1, Z_2, g^2) : Z_j \in \mathfrak{su}(3),\ g^2 > 0,\ [Z_1, Z_2] \neq 0,\ Tq \neq 0\},$$

retain the just-defined T, and set

$$\tau = \frac{1/4 + \|Z_1 q\|^2 + \|Z_2 q\|^2}{\|Tq\|^2} > 0. \tag{881}$$

Define the four full residuals, with their original codomains,

$$\begin{aligned}R_H &= \tau[T,[T,H]] - \sum_{j=1}^{2}[Z_j,[Z_j,H]] - F \in \operatorname{Herm}_3(\mathbb{C})_0,\\ R_q &= (\tau T^2 - Z_1^2 - Z_2^2)q + 2q \in \mathbb{C}^3,\\ R_1 &= \tau(-Z_1 - 2[T,Z_2] + [T,[T,Z_1]]) + \mathcal{K}Z_1 + (g^2/2)\mathcal{I}Z_1 \in \mathfrak{su}(3),\\ R_2 &= \tau(-Z_2 + 2[T,Z_1] + [T,[T,Z_2]]) + \mathcal{K}Z_2 + (g^2/2)\mathcal{I}Z_2 \in \mathfrak{su}(3).\end{aligned} \tag{882}$$

Let $\mathcal{Z} \subset \mathcal{D}$ be their common zero set. The map

$$(Z_1, Z_2, g^2, \varepsilon) \longmapsto \left(C = \varepsilon\sqrt{\tau}\,T,\ Z_1, Z_2, \omega = \varepsilon\sqrt{\tau},\ g^2\right), \qquad \varepsilon \in \{+1, -1\}, \tag{883}$$

is a bijection from $\mathcal{Z} \times \{+1, -1\}$ onto all solutions of the five retained equations. Its inverse forgets C, ω and retains $\varepsilon = \operatorname{sign}(\omega)$.

Proof. For a solution, the preceding lemma gives $C = \omega T$ and a nonzero spatial bracket. Pair its full vector equation with q^*. Anti-Hermiticity of the three matrices gives exactly

$$\omega^2\|Tq\|^2 = 2q^*q + \|Z_1q\|^2 + \|Z_2q\|^2 = \tfrac{1}{4} + \|Z_1q\|^2 + \|Z_2q\|^2 > 0.$$

Thus $Tq \neq 0$ and $\omega^2 = \tau$. The constant $1/4$ is the original $2q^*q$; no scalar or frequency has been scaled to one. This contraction determines the frequency but does not replace the vector equation. Substitution of $C = \omega T$ into all the scalar and spatial equations gives every entry of (882). Hence the original solution determines a point of $\mathcal{Z}$ and its unique sign.

Conversely, start at any point of $\mathcal{Z}$ and choose either sign in (883). By definition $LT = 4\mathscr{B}$, so $LC = 4\omega\mathscr{B}$ and Gauss holds. The substitutions $C = \omega T$, $\omega^2 = \tau$ make each of the other four original residuals identical to its displayed R. They therefore vanish as full matrices or vectors, not only after contraction. The constructions plainly recover all original coordinates and signs, proving the bijection. □

This is an exact seventeen-real-variable formulation (sixteen spatial matrix coordinates and the retained positive g^2), not a proof that its zero set is nonempty or empty. The inverse returns all original fields through (869). In particular the nonzero denominator and bracket domain must be retained in any polynomial elimination. The scalar contraction alone is insufficient: the accompanying exact test point has that contraction zero while its full R_q is nonzero. No such point is reported as a solution.

24.23 Complete scalar-coordinate identities and an unrestricted signature constraint

The following calculation retains all eight real coordinates of each of C, Z_1, Z_2. It is not a cross–cross–block restriction. Let $x = 1/2$, $d_0 = 1/(2\sqrt{2})$, $r = \sqrt{6}/4$, and

$$S = \frac{1}{\sqrt{2r(r+x)}}\begin{pmatrix} d_0 & -(r+x) \\ r+x & d_0 \end{pmatrix}, \qquad U = \operatorname{diag}(1, S), \qquad \delta = \frac{1}{2r} = \sqrt{2/3}.$$

Both columns of the numerator of S have squared length $2r(r+x)$, their inner product is zero, and its determinant is $2r(r+x) > 0$. Multiplication by the lower block of the literal H shows that the columns are eigenvectors for $r, -r$ in that order. Thus

$$U \in SO(3), \qquad U^*HU = H_d = \operatorname{diag}(0, r, -r), \qquad U^*q = q.$$

Conjugate every connection and force by U^*, with inverse conjugation by U. Products, commutators, the adjoint operation, trace projection, and the five equations all transform by these exact inverse maps. In this frame $F_d = \operatorname{diag}(-2, 1-2r, 1+2r)$.

For each original connection X define its eight real coordinates by

$$\widetilde{X} = U^*XU = \begin{pmatrix} ia_X & -\overline{v_{X1}} & -\overline{v_{X2}} \\ v_{X1} & ib_X & z_X \\ v_{X2} & -\overline{z_X} & -i(a_X + b_X) \end{pmatrix}, \qquad a_X, b_X \in \mathbb{R}, \quad v_{X1}, v_{X2}, z_X \in \mathbb{C}, \tag{884}$$

and put $h_X = a_X + 2b_X$. This representation and $X = U\widetilde{X}U^*$ are mutually inverse. In particular no entry or phase has been removed. The stabilizer coefficient has the original-coordinate formula

$$h_X = \frac{\langle iH, X\rangle}{r} = \frac{-i\,\operatorname{tr}(HX)}{r}. \tag{885}$$

Indeed $\operatorname{tr}(H_d\widetilde{X}) = ir(a_X + 2b_X)$.

For a scalar expression $f(X)$ define its signed sum

$$\mathscr{S}(f) = f(C) - f(Z_1) - f(Z_2).$$

This notation retains the temporal plus sign and both spatial minus signs; it is not a positive covariance. Define

$$D_a = \mathscr{S}(a^2), \qquad D_z = \mathscr{S}(|z|^2), \qquad D_j = \mathscr{S}(|v_j|^2) \quad (j = 1, 2),$$
$$m_j = \mathscr{S}(hv_j), \quad m_z = \mathscr{S}(hz), \qquad n_1 = \mathscr{S}(zv_2) + \frac{i}{2}\mathscr{S}(av_1), \quad n_2 = \mathscr{S}(\bar{z}v_1) - \frac{i}{2}\mathscr{S}(av_2).$$

Proposition 24.50 (All scalar equations in unrestricted coordinates)**.** *For arbitrary $C, Z_1, Z_2 \in \mathfrak{su}(3)$, the two scalar equations* (864)–(865) *hold if and only if the following three real and five complex identities hold:*

$$D_a = 4D_z, \qquad D_1 + 2D_z = 1 - \delta, \qquad D_2 + 2D_z = 1 + \delta, \tag{886}$$
$$m_1 = m_2 = m_z = n_1 = n_2 = 0. \tag{887}$$

Proof. For a diagonal Hermitian matrix with entries λ_j,

$$[X, [X, H_d]]_{jk} = \sum_{\ell} X_{j\ell} X_{\ell k} (\lambda_j + \lambda_k - 2\lambda_\ell).$$

Use this identity with $(\lambda_1, \lambda_2, \lambda_3) = (0, r, -r)$ and the full matrix (884). Let Q_H and Q_q denote respectively the left sides minus the right sides of the two scalar equations in this diagonal frame. Their diagonal and matter-line entries are

$$\begin{aligned}
(Q_H)_{11} &= 2r(D_1 - D_2) + 2, \\
(Q_H)_{22} &= -2rD_1 - 4rD_z - (1 - 2r), \\
(Q_H)_{33} &= 2rD_2 + 4rD_z - (1 + 2r), \\
(Q_q)_1 &= q_1(2 - D_a - D_1 - D_2), \qquad q_1 = 1/(2\sqrt{2}).
\end{aligned}$$

The first of these three matrix entries is minus the sum of the next two, since the full residual is traceless. The latter two vanish exactly when $D_1 + 2D_z = 1 - \delta$ and $D_2 + 2D_z = 1 + \delta$. Their sum gives $D_1 + D_2 = 2 - 4D_z$; the vector's first entry vanishes exactly when $D_a = 4D_z$.

For clarity the individual lower matter-column products are

$$(\widetilde{X}^2)_{21} = i(a_X + b_X)v_{X1} + z_X v_{X2}, \qquad (\widetilde{X}^2)_{31} = -ib_X v_{X2} - \bar{z}_X v_{X1},$$

and the three independent lower/upper matrix products are

$$\begin{aligned}
[\widetilde{X}, [\widetilde{X}, H_d]]_{21} &= r\big(i(a_X - b_X)v_{X1} + 3z_X v_{X2}\big), \\
[\widetilde{X}, [\widetilde{X}, H_d]]_{31} &= r\big(-i(2a_X + b_X)v_{X2} + 3\bar{z}_X v_{X1}\big), \\
[\widetilde{X}, [\widetilde{X}, H_d]]_{23} &= -2irh_X z_X.
\end{aligned}$$

Since $b_X = (h_X - a_X)/2$, the complete remaining residual entries are

$$\begin{aligned}
(Q_q)_2 &= q_1(n_1 + im_1/2), & (Q_H)_{21} &= r(3n_1 - im_1/2), \\
(Q_q)_3 &= q_1(-n_2 - im_2/2), & (Q_H)_{31} &= r(3n_2 - im_2/2), \\
& & (Q_H)_{23} &= -2irm_z.
\end{aligned}$$

The remaining off-diagonal entries of Q_H are the conjugates, since it is Hermitian. Both q_1 and r are nonzero. For either j, the two relevant equations first give $n_j = -im_j/2$ and then $3n_j - im_j/2 = -2im_j = 0$. Thus $m_j = n_j = 0$; the last matrix entry gives $m_z = 0$. Conversely these five complex equalities make every displayed off-diagonal and vector entry vanish. Together with the three real relations already proved, they set the entire scalar residuals to zero. This proves both directions without a rank or phase restriction. □

Theorem 24.51 (Quantitative spacelike stabilizer constraint)**.** *Every scalar-compatible unrestricted triple satisfies the following inequalities. Put*

$$H_s = h_{Z_1}^2 + h_{Z_2}^2, \qquad S_j = \sum_{\ell=1}^{2} (|v_{Z_\ell j}|^2 + 2|z_{Z_\ell}|^2), \qquad c_1 = 1-\delta > 0, \quad c_2 = 1+\delta > 0.$$

Then

$$h_C^2 (S_j + c_j) \le H_s S_j \quad (j = 1, 2). \tag{888}$$

In particular exactly one of the following possibilities holds:

$$h_C = h_{Z_1} = h_{Z_2} = 0, \qquad \text{or} \qquad h_C^2 < h_{Z_1}^2 + h_{Z_2}^2.$$

Equivalently, the three original-coordinate numbers $\langle iH, C\rangle, \langle iH, Z_1\rangle, \langle iH, Z_2\rangle$ are all zero or have strictly negative squared norm for the retained signature $(+--)$. Every zero of the unrestricted reduced residuals consequently satisfies

$$\tau h_T^2 \le H_s \min\left\{ \frac{S_1}{S_1 + 1 - \delta}, \frac{S_2}{S_2 + 1 + \delta} \right\}. \tag{889}$$

All denominators in this formula are strictly positive.

Proof. For each $j = 1, 2$ and each $X \in \{C, Z_1, Z_2\}$ put $V_{Xj} = (v_{Xj}, \sqrt{2} z_X) \in \mathbb{C}^2$. The identities $m_j = m_z = 0$ give

$$h_C V_{Cj} = h_{Z_1} V_{Z_1 j} + h_{Z_2} V_{Z_2 j}.$$

Their squared norms and Cauchy–Schwarz on the ordered pair in $\mathbb{C}^2 \oplus \mathbb{C}^2$ imply

$$h_C^2 \|V_{Cj}\|^2 \le (h_{Z_1}^2 + h_{Z_2}^2)(\|V_{Z_1 j}\|^2 + \|V_{Z_2 j}\|^2) = H_s S_j.$$

The diagonal scalar identities give $\|V_{Cj}\|^2 - S_j = D_j + 2D_z = c_j$, proving (888). Since $0 < \delta < 1$, each c_j is strictly positive. If $H_s = 0$, the inequality forces $h_C = 0$. If $H_s > 0$, division by $S_j + c_j > 0$ gives $h_C^2 \le H_s S_j/(S_j + c_j) < H_s$. This covers $S_j = 0$ as well. The original-coordinate version follows by multiplication by $r^2 > 0$ in (885). Finally $C = \varepsilon\sqrt{\tau}\, T$ gives $h_C^2 = \tau h_T^2$, so both inequalities give the last formula, independently of the frequency sign. □

The quantitative inequalities are attained by scalar-compatible triples; equality in them does not certify the gauge equations. For an exact witness in the original coordinates retain A, H, q and set

$$C = \frac{5}{3} A + \frac{4}{3} iH, \qquad Z_1 = 0, \qquad Z_2 = \frac{4}{3} A + \frac{5}{3} iH.$$

The equality $(5/3)^2 - (4/3)^2 = 1$ makes the signed quadratic expressions equal to those of $(A, 0, iH)$: both cross terms cancel in $C^2 - Z_2^2$ and in $[C,[C,H]] - [Z_2,[Z_2,H]]$. Since $Hq = 0$ and $[iH, H] = 0$, these are exactly the original two scalar equations. The A matrix has only matter-cross entries in the diagonal frame. Their squared moduli are $1-\delta, 1+\delta$, either by the displayed U or by applying the scalar diagonal identities to $(A, 0, 0)$. Also $h_A = 0$ and $h_{iH} = 2r$ by (885). Thus

$$h_C = 8r/3, \qquad h_{Z_2} = 10r/3, \qquad S_j = \frac{16}{9} c_j,$$

and both sides of (888) are equal. This witness is not a full gauge solution: $[Z_1, Z_2] = 0$ contradicts the necessary noncommutation proved in Lemma 24.48.

There is a second unrestricted test using both spatial equations, not only their trace pairing. We give all its coefficients and its exact nonnegative discriminant below.

24.24 A two-by-two stress obstruction before evaluating the full residuals

In the coordinates (884), direct multiplication gives the full inertia matrix

$$U^*(\mathcal{I}X)U = \begin{pmatrix} ia_X/6 & -\overline{v_{X1}}/2 & -\overline{v_{X2}}/2 \\ v_{X1}/2 & -ia_X/12 & 3z_X/2 \\ v_{X2}/2 & -3\bar{z}_X/2 & -ia_X/12 \end{pmatrix}. \tag{890}$$

For off-diagonal entries the double commutator multiplies by the square of the eigenvalue difference; these are $r^2 = 3/8$ and $4r^2 = 3/2$. The qq^* terms add $1/8$ to the two matter-cross entries. For the diagonal entries those terms are $i\,\mathrm{diag}(a_X/4, 0, 0)$, whose trace-free part is $i\,\mathrm{diag}(a_X/6, -a_X/12, -a_X/12)$. This proves every entry and also

$$\langle X, \mathcal{I}X\rangle = a_X^2/4 + |v_{X1}|^2 + |v_{X2}|^2 + 3|z_X|^2.$$

The eigenvalues of $\mathcal{I}$ are $0, 1/6, 1/2, 3/2$ with respective real multiplicities $1, 1, 4, 2$. Indeed the h diagonal direction, the a diagonal direction with $b = -a/2$, the four cross-entry directions, and the two lower off-diagonal directions are mutually orthogonal and are multiplied by precisely these scalars in the displayed formula.

For an arbitrary point of $\mathcal{D}$ from Theorem 24.49, define the real symmetric two-by-two matrices and real scalars

$$G_{jk} = \langle Z_j, Z_k\rangle, \qquad D_{jk} = \langle [T, Z_j], [T, Z_k]\rangle, \qquad M_{jk} = \langle Z_j, \mathcal{I}Z_k\rangle,$$
$$d = \|[Z_1, Z_2]\|^2 > 0, \qquad b_T = \langle T, [Z_1, Z_2]\rangle, \qquad N = G + D - 2b_T I_2, \qquad B = dI_2 + (g^2/2)M.$$

All these objects are functions of the unrestricted spatial pair and g^2, with no frequency sign selected. M is positive semidefinite: for $x \in \mathbb{R}^2$, $x^T M x = \langle \sum_j x_j Z_j, \mathcal{I}\sum_j x_j Z_j\rangle \geq 0$. Hence B is positive definite.

Proposition 24.52 (Exact stress matrix and nonnegative discriminant)**.** *Every zero of the four full reduced residuals satisfies*

$$\tau N = B. \tag{891}$$

In particular N must be positive definite. On the open domain $N_{11} > 0$, $\det N > 0$, define

$$\Delta = (B_{11}N_{22} + B_{22}N_{11} - 2B_{12}N_{12})^2 - 4\det B \det N. \tag{892}$$

Then $\Delta \geq 0$, and the stress equation is equivalent to the two scalar conditions

$$\Delta = 0, \qquad \tau = B_{11}/N_{11}. \tag{893}$$

These are necessary tests on the unrestricted reduced zero set, not replacements for R_H, R_q, R_1, R_2.

Proof. For arbitrary C, Z_1, Z_2, ω, g^2, let Q_1, Q_2 be the left sides of the two original spatial equations. Write $\beta = \langle C, [Z_1, Z_2]\rangle$ and $D^C_{jk} = \langle [C, Z_j], [C, Z_k]\rangle$. Cyclicity gives the exact matrix of pairings

$$\langle Z_j, Q_k\rangle = -\omega^2 G_{jk} - D^C_{jk} + (g^2/2)M_{jk} + (d + 2\omega\beta)\delta_{jk}. \tag{894}$$

For the double C commutator the term is $-D^C_{jk}$ by invariance of the trace inner product. The spatial double commutators have diagonal pairing d and off-diagonal pairing zero: expand $\langle Z_j, \mathcal{K}Z_k\rangle = \sum_\ell \langle [Z_\ell, Z_j], [Z_\ell, Z_k]\rangle$. The gyroscopic diagonal pairing in either equation is $2\omega\beta$; the off-diagonal pairings vanish because $\langle X, [C, X]\rangle = 0$. These facts prove every entry of (894), including its symmetry. On a reduced zero, $C = \omega T$, $\omega^2 = \tau$, $D^C = \tau D$, and $\omega\beta = \tau b_T$. Setting the pairing matrix to zero gives $\tau N = B$. Since B is positive definite and $\tau > 0$, N must be positive definite.

Put $n = N_{11} > 0$, $b = N_{12}$, $c = N_{22}$, and $h = nc - b^2 > 0$. Expansion of the following sum of squares is an exact polynomial identity:

$$\begin{aligned} n^2\Delta = {} & \big(B_{11}(nc - 2b^2) + 2bnB_{12} - n^2B_{22}\big)^2 \\ & + 4h(nB_{12} - bB_{11})^2. \end{aligned} \tag{895}$$

Indeed the coefficients of $B_{11}^2, B_{22}^2, B_{12}^2, B_{11}B_{22}, B_{11}B_{12}, B_{22}B_{12}$ on either side are respectively $n^2c^2, n^4, 4n^3c, -2n^3c + 4n^2b^2, -4n^2bc, -4n^3b$. Thus $\Delta \geq 0$. Equality holds exactly when $nB_{12} = bB_{11}$ and $B_{11}(nc - 2b^2) + 2bnB_{12} - n^2B_{22} = 0$. Substitution of the first equality in the second gives $n(cB_{11} - nB_{22}) = 0$. Since $n > 0$, the two equalities are exactly $B = (B_{11}/n)N$. Conversely this proportionality makes the sum of squares zero. Finally $B_{11} > 0$, so this proportionality factor is positive, and $\tau N = B$ holds exactly when $\tau = B_{11}/n$. This proves the claimed equivalence and every denominator condition. □

The scalar-coordinate equivalence, the quantitative stabilizer inequality, and the stress discriminant apply to the full complex finite system. They neither force a cross–cross–block ansatz nor establish a zero or emptiness of the unrestricted reduced zero set. Their exact inverse coordinate maps and the reconstruction (883) preserve all original fields, couplings, signs, and both frequency branches.

24.25 An exact bounded finite coupling fibre for every unrestricted spatial pair

Retain the literal matrices H, q, F, all eight real coordinates of each $Z_j \in \mathfrak{su}(3)$, and the original positive coupling g^2. The inner product remains $\langle X, Y\rangle = -\operatorname{tr}(XY)$. The letter t below is a polynomial indeterminate whose evaluated value is the original g^2; no physical constant or amplitude is set to one. This subsection constructs the complete fibre over a fixed ordered pair (Z_1, Z_2) of the four-residual zero set (882). The ordered pair is not constrained to a real subalgebra, a block class, or a prescribed phase relation.

Proposition 24.53 (An explicit upper bound for the original coupling)**.** *For an arbitrary fixed ordered pair with $[Z_1, Z_2] \neq 0$, define*

$$z_0 = \|Z_1\|^2 + \|Z_2\|^2, \qquad z = \sum_{j=1}^{2} \langle Z_j, \mathcal{I}Z_j\rangle, \qquad d = \|[Z_1, Z_2]\|^2,$$

$$c_* = 2 + z, \qquad k_* = 2z_0(z_0 + 4), \qquad t_* = \frac{k_* z + \sqrt{k_*^2 z^2 + 8k_* d c_*^2}}{c_*^2}.$$

All these numbers are real, with $z \geq 0$ and $z_0, d, c_*, k_*, t_* > 0$. Every full solution over this ordered pair satisfies

$$\|C\|^2 \leq z_0 + 4, \qquad 0 < g^2 \leq t_*, \qquad 0 < \omega^2 \leq \frac{k_* + 2d + t_*(c_* + z/2)}{z_0}. \tag{896}$$

The same bounds apply to both signs of the original frequency.

Proof. We include the signed norm calculation so that the amplitude estimate has no unproved input from a proposed coordinate restriction. In the full coordinates (884), put $h_X = a_X + 2b_X$. Multiplication of the diagonal and off-diagonal entries gives

$$\|X\|^2 = \tfrac{3}{2}a_X^2 + \tfrac{1}{2}h_X^2 + 2(|v_{X1}|^2 + |v_{X2}|^2 + |z_X|^2).$$

The scalar identities (886) give $D_a = 4D_z$ and $D_1 + D_2 = 2 - 4D_z$. Consequently

$$\begin{aligned} \|C\|^2 - z_0 &= \tfrac{3}{2}D_a + \tfrac{1}{2}(h_C^2 - h_{Z_1}^2 - h_{Z_2}^2) + 2(D_1 + D_2 + D_z) \\ &= 4 + \tfrac{1}{2}(h_C^2 - h_{Z_1}^2 - h_{Z_2}^2). \end{aligned}$$

For completeness, the sign of the last correction follows directly from the full scalar cross identities. Write $V_X = (v_{X1}, \sqrt{2}z_X)$, $S_1 = \|V_{Z_1}\|^2 + \|V_{Z_2}\|^2$, and $H_s = h_{Z_1}^2 + h_{Z_2}^2$. Those identities give $h_C V_C = h_{Z_1}V_{Z_1} + h_{Z_2}V_{Z_2}$, while the diagonal identities give $\|V_C\|^2 = S_1 + 1 - \sqrt{2/3}$. Cauchy–Schwarz therefore proves

$$h_C^2(S_1 + 1 - \sqrt{2/3}) \le H_s S_1.$$

Since $1 - \sqrt{2/3} > 0$, this implies $h_C^2 \le H_s$, also when $S_1 = 0$ or $H_s = 0$. This proves $\|C\|^2 \le z_0 + 4$.

For arbitrary anti-Hermitian matrices X, Y, diagonalize X unitarily, writing its eigenvalues as $i\lambda_a$ with real λ_a. In that same basis,

$$\|[X,Y]\|^2 = \sum_{a,b}(\lambda_a - \lambda_b)^2|Y_{ab}|^2 \le 2\Big(\sum_a \lambda_a^2\Big)\Big(\sum_{a,b}|Y_{ab}|^2\Big) = 2\|X\|^2\|Y\|^2.$$

Here $(\lambda_a - \lambda_b)^2 \le 2(\lambda_a^2 + \lambda_b^2) \le 2\sum_a \lambda_a^2$ proves the inequality term by term. Thus the retained $k = \sum_j \|[C, Z_j]\|^2$ satisfies $k \le k_*$. The universal identities (870) give $c = c_*$ and

$$(g^2)^2 c_*^2 \le 8kd + 2g^2 kz \le 8k_* d + 2g^2 k_* z.$$

The quadratic on moving the right side to the left has positive leading coefficient and roots $\big(k_* z \pm \sqrt{k_*^2 z^2 + 8k_* d c_*^2}\big)/c_*^2$. Its negative root is strictly negative, so its nonpositive region at $g^2 > 0$ is exactly $g^2 \le t_*$. Finally the universal spatial trace identity is

$$\omega^2 z_0 = k + 2d + g^2(c_* + z/2).$$

Substitution of the two upper bounds proves the last assertion. Nonzero bracket implies $z_0, d > 0$; positivity of $\mathcal{I}$ and the definitions prove all remaining strict and weak sign assertions. □

We now eliminate the temporal stabilizer component exactly before clearing any frequency denominator. This preserves the original stabilizer vector $J = iH$, whose squared norm is $3/4$. Set

$$W = \{X \in \mathfrak{su}(3) : \langle J, X\rangle = 0\}, \qquad \Pi X = X - \frac{\langle J, X\rangle}{3/4} J, \qquad \mathscr{B} = [Z_1, Z_2].$$

For the fixed spatial pair define

$$\eta = \langle J, \mathcal{K}J\rangle, \qquad w = \Pi\mathcal{K}J, \qquad \beta = \langle J, \mathscr{B}\rangle,$$

and operators and a vector on the original metric subspace W by

$$\mathcal{I}_7 = \mathcal{I}|_W, \qquad \mathcal{S}_7 = \Pi\mathcal{K}|_W - \frac{w \otimes w}{\eta}, \qquad f_7 = \Pi\mathscr{B} - \frac{\beta}{\eta} w.$$

Here $(w \otimes w)x = w\langle w, x\rangle$; no choice of basis alters any of these maps.

Lemma 24.54 (Full stabilizer elimination and polynomial Gauss numerator)**.** *If $\mathscr{B} \neq 0$, then $\eta > 0$, $\mathcal{S}_7 \succeq 0$, and $\mathcal{I}_7 \succ 0$. Define the polynomial endomorphism and its determinant and adjugate by*

$$\mathscr{M}(t) = t\mathcal{I}_7 + 2\mathcal{S}_7, \qquad D_7(t) = \det_W \mathscr{M}(t), \qquad V_7(t) = 4\,\mathrm{adj}_W(\mathscr{M}(t))f_7,$$

$$N_7(t) = V_7(t) + \frac{2\beta D_7(t) - \langle w, V_7(t)\rangle}{\eta} J. \tag{897}$$

Then $D_7(t) > 0$ for $t > 0$, D_7 has degree exactly seven and leading coefficient $3/128$, $\deg V_7 \le 6$, and $\deg N_7 \le 7$. For every original $g^2 > 0$,

$$T = 4(g^2\mathcal{I} + 2\mathcal{K})^{-1}\mathscr{B} = \frac{N_7(g^2)}{D_7(g^2)}, \qquad Tq = \frac{V_7(g^2)q}{D_7(g^2)}. \tag{898}$$

In particular all temporal coordinates, including the entire stabilizer coefficient, are reconstructed by the displayed formula.

Proof. The positive form of $\mathcal{K}$ gives $\eta = \sum_j \|[Z_j, J]\|^2 \geq 0$. If it were zero, both Z_j would commute with H. The three distinct eigenvalues $0, \sqrt{6}/4, -\sqrt{6}/4$ of H would make both matrices diagonal in its eigenbasis; hence $[Z_1, Z_2] = 0$, a contradiction. Thus $\eta > 0$. The kernel calculation in Lemma 24.48 gives $\ker \mathcal{I} = \mathbb{R}J$. Self-adjointness makes W invariant, and positivity then makes $\mathcal{I}_7$ positive definite.

For $x \in W$ and $s \in \mathbb{R}$, use $\langle J, \mathcal{K}x\rangle = \langle w, x\rangle$ to complete the square:

$$\langle x + sJ, \mathcal{K}(x + sJ)\rangle = \langle x, \mathcal{S}_7 x\rangle + \eta \left(s + \frac{\langle w, x\rangle}{\eta}\right)^2.$$

Taking $s = -\langle w, x\rangle/\eta$ and using positivity of $\mathcal{K}$ proves $\mathcal{S}_7 \succeq 0$. Thus $\mathscr{M}(t)$ is positive definite for every $t > 0$.

Write the full Gauss unknown as $T = x + sJ$ with $x \in W$, with no restriction on s. Pair $(t\mathcal{I}+2\mathcal{K})T = 4\mathscr{B}$ with J and then apply Π. These operations give respectively

$$s = \frac{2\beta - \langle w, x\rangle}{\eta}, \qquad (t\mathcal{I}_7 + 2\mathcal{S}_7)x = 4f_7.$$

Conversely these two equations recover the J component and the W projection of Gauss; the orthogonal decomposition $\mathfrak{su}(3) = W \oplus \mathbb{R}J$ therefore recovers the entire equation. Inverting the positive operator on W yields $x = V_7(t)/D_7(t)$, and the formula for s yields (897). Since $Jq = 0$, it also yields the second formula in (898).

Determinants and adjugates may be calculated in any real basis of W; similarity preserves their resulting scalar and endomorphism. The inertia computation (890) gives the seven eigenvalues $1/6, 1/2, 1/2, 1/2, 1/2, 3/2, 3/2$ on W. Thus $\det_W \mathcal{I}_7 = (1/6)(1/2)^4(3/2)^2 = 3/128$. The determinant consequently has degree seven with this leading coefficient; its positivity for $t > 0$ follows from positive definiteness. The adjugate of a seven-by-seven matrix with affine entries has entries of degree at most six, and (897) proves the remaining degree bound. □

Put

$$\alpha = \frac{1}{4} + \|Z_1 q\|^2 + \|Z_2 q\|^2 > 0, \qquad Q_7(t) = \|V_7(t)q\|^2.$$

The real polynomial Q_7 is nonnegative at every real t and has degree at most twelve. Its identically-zero case has the following finite test, which does not require computing a polynomial determinant.

Proposition 24.55 (Three exact vector tests for an empty Gauss domain)**.** *For every fixed noncommuting spatial pair, Q_7 is identically zero if and only if the following three vector equalities all hold:*

$$f_7 q = 0, \qquad (\mathcal{S}_7 f_7)q = 0, \qquad (\mathcal{S}_7^2 f_7)q = 0. \tag{899}$$

When they hold, no positive original coupling over that spatial pair belongs to the original four-residual domain.

Proof. Put $\mathcal{N} = \{X \in W : Xq = 0\}$. In the complete coordinates (884), membership of W sets $h_X = 0$; the first column condition $Xq = 0$ sets $a_X = v_{X1} = v_{X2} = 0$. The only remaining entry is the arbitrary complex z_X. Thus $\dim_{\mathbb{R}} \mathcal{N} = 2$, and the full inertia formula gives $\mathcal{I}_7|_{\mathcal{N}} = (3/2)\,\mathrm{id}_{\mathcal{N}}$. Self-adjointness also makes $\mathcal{N}^{\perp} \cap W$ invariant. Consequently $\mathcal{I}_7^{-1} y \in \mathcal{N}$ if and only if $y \in \mathcal{N}$, and on this subspace the inverse is multiplication by $2/3$.

Suppose first that Q_7 is identically zero. For every $t > 0$ the vector $x(t) = 4(t\mathcal{I}_7 + 2\mathcal{S}_7)^{-1} f_7$ then belongs to $\mathcal{N}$. Set $B_7 = \mathcal{I}_7^{-1}$. Factoring the pencil and using the exact finite inverse identity $(I + A)^{-1} = I - A + A^2 - A^3(I + A)^{-1}$ gives

$$x(t) = \frac{4}{t} B_7 f_7 - \frac{8}{t^2} B_7 \mathcal{S}_7 B_7 f_7 + \frac{16}{t^3} B_7 \mathcal{S}_7 B_7 \mathcal{S}_7 B_7 f_7 + O(t^{-4}).$$

Here the remainder has bounded norm after multiplication by t^4: for $A = 2B_7\mathcal{S}_7/t$ and $t > 4\|B_7\mathcal{S}_7\|$, the inverse has norm at most two, since $\|(I+A)y\| \geq (1-\|A\|)\|y\| \geq \|y\|/2$. The finite-dimensional subspace $\mathcal{N}$ is closed. Multiplying the expansion by t and taking the limit shows $B_7 f_7 \in \mathcal{N}$, hence $f_7 \in \mathcal{N}$. Subtracting the first term, multiplying by t^2 and taking the limit shows $B_7\mathcal{S}_7 B_7 f_7 \in \mathcal{N}$, hence $\mathcal{S}_7 f_7 \in \mathcal{N}$. Subtracting the next term and multiplying by t^3 similarly gives $\mathcal{S}_7^2 f_7 \in \mathcal{N}$. In the latter two steps every occurrence of B_7 on the already established vectors in $\mathcal{N}$ is exactly multiplication by $2/3$. These are precisely the three displayed tests.

Conversely suppose the three tests hold. If $f_7 = 0$, then $x(t) = 0$ for all $t > 0$ and the conclusion follows. If $f_7 \neq 0$ and $\mathcal{S}_7 f_7$ is dependent on f_7, their one-dimensional span is invariant under $\mathcal{S}_7$. Otherwise those two vectors span all of $\mathcal{N}$, and the third test makes that two-dimensional span invariant as well. In either case let V be the resulting invariant span. It is contained in $\mathcal{N}$, so it is invariant under $\mathcal{I}_7$ and therefore under the positive pencil $t\mathcal{I}_7 + 2\mathcal{S}_7$. Its restriction to V is injective and, in finite dimension, onto V; the full inverse thus carries $f_7 \in V$ back into V. Hence $x(t)q = 0$ for all $t > 0$, and the polynomial Q_7 vanishes on the entire positive half-line. A polynomial with infinitely many roots is identically zero. Formula (898) then proves the asserted exclusion from the original domain for every $g^2 > 0$. □

Define four polynomial residuals with the original matrix and vector codomains:

$$\begin{aligned}
P_H(t) &= \alpha[N_7,[N_7,H]] - Q_7\left(\sum_j [Z_j,[Z_j,H]] + F\right),\\
P_q(t) &= \alpha N_7^2 q + Q_7(2I_3 - Z_1^2 - Z_2^2)q,\\
P_1(t) &= \alpha\left(-D_7^2 Z_1 - 2D_7[N_7,Z_2] + [N_7,[N_7,Z_1]]\right) + Q_7\left(\mathcal{K}Z_1 + \frac{t}{2}\mathcal{I}Z_1\right),\\
P_2(t) &= \alpha\left(-D_7^2 Z_2 + 2D_7[N_7,Z_1] + [N_7,[N_7,Z_2]]\right) + Q_7\left(\mathcal{K}Z_2 + \frac{t}{2}\mathcal{I}Z_2\right).
\end{aligned} \tag{900}$$

All polynomials on the right are evaluated at the same t. Their codomains are respectively $\mathrm{Herm}_3(\mathbb{C})_0$, $\mathbb{C}^3$, $\mathfrak{su}(3)$, and $\mathfrak{su}(3)$ for real t. Define the single real polynomial

$$p_Z(t) = \mathrm{tr}(P_H(t)^2) + P_q(t)^* P_q(t) - \mathrm{tr}(P_1(t)^2) - \mathrm{tr}(P_2(t)^2). \tag{901}$$

Proposition 24.56 (The exact degree and positive leading coefficient)**.** *Retain an arbitrary fixed ordered pair $Z_1, Z_2 \in \mathfrak{su}(3)$ with $\mathscr{B} = [Z_1, Z_2] \neq 0$, and all original definitions above: $J, \eta, \beta, \mathcal{I}_7$, D_7, V_7, N_7, Q_7, α, and P_H, P_q, P_1, P_2, p_Z. Put*

$$\delta_7 = \det_W \mathcal{I}_7 = \frac{3}{128}, \qquad a_J = \frac{2\beta}{\eta}, \qquad A_J X = [J, X],$$

and define an endomorphism of the original real product metric space $\mathfrak{su}(3) \oplus \mathfrak{su}(3)$ by

$$\mathcal{L}_{a_J}(X_1, X_2) = (X_1 + a_J A_J X_2,\ X_2 - a_J A_J X_1).$$

Here a_J is an explicitly defined scalar belonging to the fixed spatial pair; it is not a new coupling or a rescaling of the original fields. The coefficient of t^{28} in the original polynomial p_Z is

$$[t^{28}]p_Z = \alpha^2 \delta_7^4 \left\|\mathcal{L}_{a_J}^2(Z_1, Z_2)\right\|^2. \tag{902}$$

If Q_7 is not identically zero, this coefficient is strictly positive. Consequently $\deg p_Z = 28$ on every such spatial pair.

Proof. First write $N_7 = V_7 + s_7 J$. The previously proved bounds are $\deg V_7 \leq 6$, $\deg s_7 \leq 7$, and $\deg Q_7 \leq 12$. Since $[J, H] = 0$ and $Jq = 0$, the exact expressions are

$$[N_7,[N_7,H]] = [V_7,[V_7,H]] + s_7[J,[V_7,H]], \qquad N_7^2 q = V_7^2 q + s_7 J V_7 q.$$

Hence $\deg P_H, \deg P_q \le 13$. The full spatial formulas give $\deg P_1, \deg P_2 \le 14$, including the $tQ_7\mathcal{I}/2$ terms. Taking their displayed squared norms yields $\deg p_Z \le 28$. The established degree statements give

$$D_7(t) = \delta_7 t^7 + \text{terms of degree at most six}, \qquad V_7(t) = \text{terms of degree at most six}.$$

The defining full stabilizer numerator is

$$N_7(t) = V_7(t) + \frac{2\beta D_7(t) - \langle w, V_7(t)\rangle}{\eta} J.$$

Its coefficient of t^7 is therefore exactly $a_J\delta_7 J$, also when $\beta = 0$ and that coefficient is zero. Since $\deg Q_7 \le 12$, its contributions $Q_7(\mathcal{K}Z_j + t\mathcal{I}Z_j/2)$ have degree at most thirteen. It follows directly from the full spatial numerator formulas that

$$[t^{14}]P_1 = \alpha\delta_7^2(-Z_1 - 2a_J A_J Z_2 + a_J^2 A_J^2 Z_1),$$
$$[t^{14}]P_2 = \alpha\delta_7^2(-Z_2 + 2a_J A_J Z_1 + a_J^2 A_J^2 Z_2).$$

Direct composition, with both signs retained, gives

$$\mathcal{L}_{a_J}^2(Z_1, Z_2) = (Z_1 + 2a_J A_J Z_2 - a_J^2 A_J^2 Z_1, Z_2 - 2a_J A_J Z_1 - a_J^2 A_J^2 Z_2).$$

The pair of leading spatial coefficients is thus $-\alpha\delta_7^2\mathcal{L}_{a_J}^2(Z_1, Z_2)$. The scalar numerator degrees satisfy $\deg P_H, \deg P_q \le 13$, so their squared norms contribute no term of degree twenty-eight. The two spatial squared norms therefore give exactly (902).

It remains to prove strict positivity under the stated domain-polynomial hypothesis. Invariance of $\langle X, Y\rangle = -\operatorname{tr}(XY)$ gives

$$\langle X, A_J Y\rangle = -\langle A_J X, Y\rangle.$$

For arbitrary $X = (X_1, X_2)$ and $Y = (Y_1, Y_2)$, this implies

$$\begin{aligned}\langle X, \mathcal{L}_{a_J} Y\rangle &= \langle X_1, Y_1\rangle + \langle X_2, Y_2\rangle + a_J\langle X_1, A_J Y_2\rangle - a_J\langle X_2, A_J Y_1\rangle \\ &= \langle X_1 + a_J A_J X_2, Y_1\rangle + \langle X_2 - a_J A_J X_1, Y_2\rangle = \langle \mathcal{L}_{a_J} X, Y\rangle.\end{aligned}$$

Thus $\mathcal{L}_{a_J}$ is self-adjoint on this positive metric space. If the coefficient in (902) were zero, then $\mathcal{L}_{a_J}^2 Z = 0$ for $Z = (Z_1, Z_2)$. Consequently

$$\|\mathcal{L}_{a_J} Z\|^2 = \langle Z, \mathcal{L}_{a_J}^2 Z\rangle = 0,$$

so $\mathcal{L}_{a_J} Z = 0$. If $a_J = 0$, this gives $Z_1 = Z_2 = 0$, contradicting $\mathscr{B} \neq 0$. Hence $a_J \neq 0$, and the two component equations are

$$a_J[J, Z_1] = Z_2, \qquad a_J[J, Z_2] = -Z_1.$$

The full spatial operator now satisfies the exact identity

$$\begin{aligned}a_J\mathcal{K}J &= -a_J[Z_1, [Z_1, J]] - a_J[Z_2, [Z_2, J]] \\ &= a_J[Z_1, [J, Z_1]] + a_J[Z_2, [J, Z_2]] \\ &= [Z_1, Z_2] + [Z_2, -Z_1] = 2\mathscr{B}.\end{aligned}$$

Apply the original orthogonal projection Π to this equality and use $w = \Pi\mathcal{K}J$ and $a_J/2 = \beta/\eta$. It yields

$$f_7 = \Pi\mathscr{B} - \frac{\beta}{\eta} w = \frac{a_J}{2} w - \frac{a_J}{2} w = 0.$$

The exact definition $V_7(t) = 4\operatorname{adj}_W(\mathscr{M}(t))f_7$ then makes V_7 identically zero, and hence makes $Q_7(t) = \|V_7(t)q\|^2$ identically zero. This contradicts the hypothesis. The coefficient is nonnegative by its squared-norm formula, so it must be strictly positive. Since the global degree bound is twenty-eight, the degree is exactly twenty-eight. □

Theorem 24.57 (Complete finite coupling fibre and its exact inverse)**.** *For every ordered pair $Z_1, Z_2 \in \mathfrak{su}(3)$ the complete fibre of the four-residual zero set is as follows. If $[Z_1, Z_2] = 0$, it is empty. Otherwise form the explicitly defined polynomials above. If Q_7 is identically zero, the fibre is empty. If Q_7 is not identically zero, then p_Z is not identically zero,* $\deg p_Z = 28$*, and the complete set of admissible couplings is exactly*

$$\{t \in (0, t_*] : Q_7(t) > 0,\ p_Z(t) = 0\}\,. \tag{903}$$

This set has at most fourteen elements. Equivalently, the projection $\mathcal{Z} \longrightarrow \mathfrak{su}(3)^2$ that forgets g^2 has fibres of cardinality at most fourteen, including empty fibres.

For each root in (903) *and each $\varepsilon \in \{+1, -1\}$, the complete original finite-system solution is*

$$g^2 = t, \qquad \omega = \varepsilon \frac{\sqrt{\alpha}\, D_7(t)}{\sqrt{Q_7(t)}}, \qquad C = \varepsilon \frac{\sqrt{\alpha}\, N_7(t)}{\sqrt{Q_7(t)}}, \qquad (Z_1, Z_2) = (Z_1, Z_2). \tag{904}$$

Thus every fibre of the five-equation solution set over the fixed ordered spatial pair has at most twenty-eight elements, counting both frequency signs. No root with $Q_7(t) = 0$ is included.

Proof. A zero spatial bracket is impossible by Lemma 24.48. For a nonzero bracket, (898) gives $\|Tq\|^2 = Q_7(t)/D_7(t)^2$ for every $t > 0$. Thus the exact original domain condition $Tq \neq 0$ is precisely $Q_7(t) > 0$. If Q_7 is identically zero, the domain has no point over this pair. On the remaining domain the unchanged frequency is

$$\tau = \frac{\alpha D_7(t)^2}{Q_7(t)}.$$

Substitution in all four full residuals gives, entry by entry,

$$(R_H, R_q, R_1, R_2) = \frac{1}{Q_7(t)}(P_H(t), P_q(t), P_1(t), P_2(t)). \tag{905}$$

Every denominator here is strictly positive on the asserted domain. For real t, each summand in (901) is a squared norm in its stated real inner-product space. The Hermitian and anti-Hermitian assertions follow directly from the commutators in (900); they do not require a residual to vanish. Therefore $p_Z(t) \geq 0$ for every real t, and its vanishing is equivalent to vanishing of every full polynomial residual. This proves the exact correspondence of roots and full residual zeros, before the bound on the number of roots is established.

To prove the degree bounds, write $N_7 = V_7 + s_7 J$, with $\deg V_7 \leq 6$ and $\deg s_7 \leq 7$. Since $[J, H] = 0$ and $Jq = 0$,

$$[N_7, [N_7, H]] = [V_7, [V_7, H]] + s_7[J, [V_7, H]], \qquad N_7^2 q = V_7^2 q + s_7 J V_7 q.$$

Their degrees are at most thirteen. Hence $\deg P_H, \deg P_q \leq 13$. In the two spatial residuals the terms $D_7^2 Z_j$, $D_7[N_7, Z_k]$, and $[N_7, [N_7, Z_j]]$ have degree at most fourteen; the terms involving $Q_7\mathcal{K}$ and $tQ_7\mathcal{I}$ have degree at most twelve and thirteen. Thus $\deg P_1, \deg P_2 \leq 14$ and $\deg p_Z \leq 28$.

Suppose now that Q_7 is not identically zero but p_Z is identically zero. A nonzero univariate polynomial has only finitely many real roots, so there exists $t > t_*$ with $Q_7(t) \neq 0$. Nonnegativity of Q_7 gives $Q_7(t) > 0$. The established exact residual correspondence then makes this t a full zero; the two-sign inverse of Theorem 24.49 reconstructs a solution of all five original equations. This contradicts Proposition 24.53. Consequently p_Z is a nonzero polynomial. Every real root of a real polynomial that is nonnegative on all of $\mathbb{R}$ has even multiplicity: at a root t_0, factor it as $(t - t_0)^m u(t)$ with $u(t_0) \neq 0$; continuity makes u have a constant nonzero sign near t_0, and an odd m would give opposite signs on the two sides. Thus a nonzero polynomial of degree at most twenty-eight of this type has at most fourteen distinct real roots. Imposing the explicit interval and denominator conditions can only decrease that number.

The stronger equality $\deg p_Z = 28$ follows from the independent leading-coefficient proof in Proposition 24.56.

Finally $D_7(t) > 0$ identifies the sign of the displayed ω with ε and gives $C = \omega T$ exactly. The frequency square is the unchanged τ. Formula (905) proves that all remaining equations vanish, while the Schur inverse proves the full Gauss equation. Conversely every original solution determines its fixed pair, its root $t = g^2$, and its unique sign, and these formulas recover C, ω. This proves the stated bijection and both cardinality bounds. □

The original spacetime reconstruction is retained literally: with the already specified $O(t_{\mathrm{phys}})$, K, and Ω, put

$$Y_1(t_{\mathrm{phys}}) = \cos(\omega t_{\mathrm{phys}})Z_1 - \sin(\omega t_{\mathrm{phys}})Z_2,$$
$$Y_2(t_{\mathrm{phys}}) = \sin(\omega t_{\mathrm{phys}})Z_1 + \cos(\omega t_{\mathrm{phys}})Z_2, \qquad Y_3 = 0.$$
$$\mathcal{A}_0 = OKCK^*O^* - \Omega, \qquad \mathcal{A}_i = OKY_iK^*O^*.$$

The prescribed scalar fields remain $\Phi(t_{\mathrm{phys}}) = O\Phi_0O^*$ and $b(t_{\mathrm{phys}}) = Ob_0$, with their original frequency $\sqrt{2}$. The polynomial indeterminate $t = g^2$ is distinct from the physical time t_{phys}.

This theorem solves the coupling fibre exactly as a bounded finite root set while keeping every unrestricted spatial coordinate. It does not establish that any ordered spatial pair has a nonempty fibre, or that all such fibres are empty. Existence or emptiness of the complete sixteen-coordinate spatial projection remains unresolved by this result.

24.26 A proper algebraic exceptional set for the full spatial projection

We now retain all sixteen spatial coordinates simultaneously. The new statement is about the entire spatial projection, not only a selected block class or one value of the coupling. The five original equations and their prescribed rotating ansatz remain unchanged.

Use the exact fixed matrix U of (884). For each $j = 1, 2$, write $\widetilde{Z}_j = U^*Z_jU$ using the eight real coordinates $(a_j, h_j, \operatorname{Re} v_{j1}, \operatorname{Im} v_{j1}, \operatorname{Re} v_{j2}, \operatorname{Im} v_{j2}, \operatorname{Re} z_j, \operatorname{Im} z_j)$, with $b_j = (h_j - a_j)/2$. This is an invertible linear coordinate map on the original spatial carrier. Denote the resulting sixteen-coordinate vector by ξ. No coordinate is held at zero in the theorem below.

Lemma 24.58 (Polynomial diagonal residuals with retained denominators)**.** *Let $\eta = \langle J, \mathcal{K}J\rangle$ be the original nonnegative polynomial in ξ. On $\eta > 0$ define*

$$F_\xi(t) = \eta^{16}(U^*P_H(t)U)_{11}, \qquad G_\xi(t) = \eta^{16}(U^*P_H(t)U)_{22}.$$

Both extend as polynomials in (ξ, t) with real algebraic coefficients, and their degree in t is at most twelve. On the stated domain the inverse formulas are exactly $(U^*P_HU)_{11} = F_\xi/\eta^{16}$ and $(U^*P_HU)_{22} = G_\xi/\eta^{16}$.

Proof. All real coordinate components of $\mathcal{K}, w, \eta, \beta, \Pi\mathscr{B}$ are polynomials in ξ with fixed real algebraic coefficients; the projection denominator $\langle J, J\rangle = 3/4$ is a fixed nonzero constant. Choose once and for all a fixed basis of W with real algebraic coordinates in the displayed diagonal-frame carrier. The entries of $\mathscr{M} = t\mathcal{I}_7 + 2\mathcal{S}_7$ have denominator dividing η, and those of f_7 have the same property. The determinant expansion shows that η^7D_7 is polynomial. Each adjugate entry is a six-by-six minor, so η^7V_7 is polynomial as well. The full formula (897) therefore makes η^8N_7 polynomial. Complex conjugation here conjugates fixed coefficients and the displayed complex coordinate combinations, while every component of ξ and t is real. Thus $\eta^{14}Q_7$ is polynomial. The full formula for P_H now proves polynomial extension after the displayed multiplication by η^{16}. No value at $\eta = 0$ is treated as an inverse Gauss solution.

To prove the sharper degree bound, work through the exact conjugation by U^*, so $H_d = \operatorname{diag}(0, r, -r)$ and $J_d = iH_d$ are both diagonal. Write $N_7 = V_7 + s_7J$ as before. Since $[J, H] = 0$,

$$[N_7, [N_7, H]] = [V_7, [V_7, H]] + s_7[J, [V_7, H]].$$

For every matrix Y, $([J_d, Y])_{jj} = 0$ follows by subtracting $(J_d)_{jj}Y_{jj}$ from itself. Consequently the second term contributes zero to each diagonal entry after conjugation. The first term has degree at most twelve

since $\deg_t V_7 \le 6$. The other term in P_H is Q_7 times a matrix independent of t, and $\deg_t Q_7 \le 12$. Multiplication by η^{16} does not change the degree in t. This proves every assertion. □

Define a fixed-size polynomial matrix $\mathscr{S}(\xi)$ by the linear map

$$\mathbb{R}[t]_{<12} \oplus \mathbb{R}[t]_{<12} \longrightarrow \mathbb{R}[t]_{<24}, \qquad (A, B) \longmapsto AF_\xi + BG_\xi, \qquad \mathscr{R}(\xi) = \det \mathscr{S}(\xi). \tag{906}$$

Use the increasing-power bases $1, t, \dots$ and put the A summand before the B summand. This specifies the determinant sign and its twenty-four columns, including when either polynomial loses degree. This is the classical Sylvester resultant map; see the Stacks Project, Tag 00UA. The argument below proves the properties needed here directly, without replacing either polynomial by a monic one.

Theorem 24.59 (Generic exclusion in all sixteen spatial coordinates)**.** *The explicitly defined polynomial $\mathscr{R}$ is not identically zero. The spatial projection of every full solution of* (864)–(868) *is contained in*

$$\{\xi \in \mathbb{R}^{16} : \mathscr{R}(\xi) = 0\}.$$

This exceptional set is closed, has empty interior, and has sixteen-dimensional Lebesgue measure zero. Thus the complement is an open dense set of unrestricted spatial pairs, each admitting no positive coupling and no frequency sign solving the five retained equations. This theorem does not decide whether any point of the exceptional set admits a solution.

Proof. Any full solution has $[Z_1, Z_2] \neq 0$, hence $\eta > 0$ by Lemma 24.54. It also has $Q_7(t) > 0$ at the original $t = g^2$, and the exact residual identity (905) makes the entire $P_H(t)$ zero. Therefore $F_\xi(t) = G_\xi(t) = 0$. Evaluation at that real t is a nonzero linear functional on $\mathbb{R}[t]_{<24}$ (it sends 1 to 1), and it annihilates the image of (906). That image cannot be all of the codomain. The determinant is zero. This proof includes every degree-drop case and does not use a converse resultant criterion outside the original denominator domain.

Here is an exact witness that the determinant polynomial is nonzero. Write

$$M(a, h, v_1, v_2, z) = \begin{pmatrix} ia & -\bar{v}_1 & -\bar{v}_2 \\ v_1 & i(h-a)/2 & z \\ v_2 & -\bar{z} & -i(a+h)/2 \end{pmatrix}$$

and set, in the displayed diagonal coordinates,

$$\widetilde{Z}_1 = M(1, 2, 1+i, -1+2i, 2-i), \qquad \widetilde{Z}_2 = M(-2, 1, 2+3i, 1-i, -1+2i).$$

The original pair is $Z_j = U\widetilde{Z}_j U^*$, not a replacement of the prescribed H, q, F. Every one of its sixteen displayed real spatial coordinates is nonzero. Substitution in the original definitions of D_7, V_7, N_7, Q_7, P_H gives two diagonal polynomials of degree twelve in $\mathbb{Q}(\sqrt{6})[t]$. Their coprimality can be verified by the following small certificate, with all coefficient lists in descending powers. Reduce their coefficients by the explicit map $\sqrt{6} \mapsto 39$ to $\mathbb{F}_{101}$, where $39^2 = 6$. Each rational coefficient denominator is nonzero modulo 101; the exact coefficient file and replay test check them individually. The two reduced polynomials f, g and the Bezout coefficients a, b are

$$\begin{aligned} f &: (39, 85, 79, 93, 67, 20, 38, 98, 26, 56, 8, 25, 28), \\ g &: (1, 87, 96, 7, 76, 88, 75, 93, 53, 73, 64, 41, 16), \\ a &: (4, 60, 31, 59, 74, 31, 34, 18, 59, 97, 95, 96), \\ b &: (46, 85, 47, 5, 14, 36, 38, 21, 35, 48, 34, 53). \end{aligned}$$

Multiplication gives $af + bg = 1$ in $\mathbb{F}_{101}[t]$. Both f and g have degree exactly twelve. If a pair (a_0, b_0) of degrees less than twelve were in the kernel of their Sylvester map, coprimality would imply that

f divides b_0 and g divides a_0. The degree bounds force both to be zero. Thus that finite-field map, between spaces of equal dimension twenty-four, has nonzero determinant. Reduction commutes with the determinant polynomial. It follows that the Sylvester determinant of the two original diagonal polynomials is nonzero in $\mathbb{Q}(\sqrt{6})$ as well. This uses the localized coefficient ring with all its denominators inverted, not a homomorphism from the whole field $\mathbb{Q}$ to a finite field.

For explicit verification at the witness, multiplication gives

$$[\widetilde{Z}_1, \widetilde{Z}_2]_{21} = 9 - \tfrac{7}{2}i \neq 0, \qquad \eta = \tfrac{3}{4}(7+15) + 3(5+5) = \tfrac{93}{2} > 0.$$

Multiplying both diagonal polynomials by the same nonzero scalar η^{16} multiplies each of the twenty-four columns by η^{16}, and hence multiplies their determinant by $\eta^{384} \neq 0$. Consequently $\mathscr{R}$ is nonzero at this exact witness and cannot be the zero polynomial. The finite-field calculation is solely a certificate of this real algebraic statement; it does not change the physical carrier or any coupling.

As a separate certificate, write $j_0 = \langle J, J\rangle = 3/4$. In $W \oplus \mathbb{R}J$, without changing the length of J, the full Gauss endomorphism has block matrix

$$L = \begin{pmatrix} t\mathcal{I}_7 + 2\Pi\mathcal{K}|_W & 2w \\ 2\langle w, \cdot\rangle/j_0 & 2\eta/j_0 \end{pmatrix}.$$

Subtracting $j_0 w/\eta$ times its last row from the upper block rows preserves the determinant and makes the upper-right block zero; its upper-left block becomes $\mathscr{M}(t)$. Thus $\det L = (2\eta/j_0)D_7$ exactly. The independent replay reconstructs both original diagonal polynomials from that matrix modulo 101. It obtains Sylvester determinant 40; after the retained η^{16} multiplication it obtains $40\eta^{384} = 60 \neq 0$ in $\mathbb{F}_{101}$. These residues refer to the exact basis order in (906).

For completeness, a nonzero real polynomial on $\mathbb{R}^n$ has a measure-zero zero set by induction on n. For $n = 1$ it has finitely many roots. For $n > 1$, write it as a polynomial in the last variable whose coefficients are polynomials in the first $n-1$ variables. Choose one nonzero coefficient. The set where all coefficients vanish is contained in the measure-zero zero set of that coefficient, by induction. At every other base point the last-variable polynomial has only finitely many roots. Applying Fubini on each bounded box shows that the zero set in that box has measure zero; countably many boxes cover $\mathbb{R}^n$. Its complement is open because the polynomial is continuous. Every nonempty open ball has positive measure, so no ball is contained in the zero set, and the complement is dense. This proves the assertions at $n = 16$. □

The exact large-coefficient identity over $\mathbb{Q}(\sqrt{6})$ and the small finite-field identity are retained separately in `PAIR_BEZOUT_CERTIFICATE.json` and `PAIR_MODULAR_CERTIFICATE.json`. The replay `check_pair_bezout.py` constructs the original full pair and every Gauss and scalar numerator, then checks the identities; the integer-only certificate replay checks the displayed finite-field products separately. Neither a finite failed search nor two failed test couplings proves this theorem: the proof uses an explicit nonzero polynomial vanishing on every possible spatial projection, with its nonzeroness certified at the literal witness.

24.27 An eight-dimensional exceptional spatial class with no full solution

24.27.1 Literal system and the explicitly selected class

All vector spaces in this note retain their real structures. In particular, $\mathfrak{su}(3)$ is the eight-real-dimensional space of traceless anti-Hermitian complex 3×3 matrices, with positive inner product $\langle X, Y\rangle = -\operatorname{tr}(XY)$. Retain the original matrices

$$H = \begin{pmatrix} 0 & 0 & 0 \\ 0 & -1/2 & 1/(2\sqrt{2}) \\ 0 & 1/(2\sqrt{2}) & 1/2 \end{pmatrix}, \quad q = \frac{e_1}{2\sqrt{2}}, \quad A = \sqrt{2}\begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

and the original scalar constants $m^2 = \Gamma = 1$, $\kappa = 6$. For $Y_{\mathrm{tf}} = Y - \mathrm{tr}(Y)I_3/3$, define

$$\mathcal{I}X = ([H,[H,X]] + qq^*X + Xqq^*)_{\mathrm{tf}}, \qquad \mathcal{K}X = -[Z_1,[Z_1,X]] - [Z_2,[Z_2,X]],$$

$$F = [A,[A,H]] = -2H + \mathrm{diag}(-2,1,1).$$

The last identity is verified by the two displayed matrix products; also $H^2 = (3/8)\,\mathrm{diag}(0,1,1)$ and $q^*q = 1/8$. The indeterminate/value $t = g^2$ is the original positive gauge coupling, not physical time. The original frequency satisfies $\omega \in \mathbb{R} \setminus \{0\}$. The five retained equations are

$$[C,[C,H]] - [Z_1,[Z_1,H]] - [Z_2,[Z_2,H]] = F, \tag{907}$$

$$(C^2 - Z_1^2 - Z_2^2)q = -2q, \tag{908}$$

$$(t\mathcal{I} + 2\mathcal{K})C = 4\omega[Z_1,Z_2], \tag{909}$$

$$-\omega^2 Z_1 - 2\omega[C,Z_2] + [C,[C,Z_1]] + \mathcal{K}Z_1 + (t/2)\mathcal{I}Z_1 = 0, \tag{910}$$

$$-\omega^2 Z_2 + 2\omega[C,Z_1] + [C,[C,Z_2]] + \mathcal{K}Z_2 + (t/2)\mathcal{I}Z_2 = 0. \tag{911}$$

Multiplying the original Gauss equation by its retained positive t gives (909); division by that same t is its exact inverse.

Put $P = qq^*/(q^*q) = \mathrm{diag}(1,0,0)$ and define

$$\mathfrak{p} = \{X \in \mathfrak{su}(3) : PXP = (I_3 - P)X(I_3 - P) = 0\}, \qquad \mathfrak{k} = \{X \in \mathfrak{su}(3) : [X,P] = 0\}.$$

Thus $\mathfrak{p}$ exchanges the matter line and its orthogonal complement, while $\mathfrak{k}$ preserves them. Every $X \in \mathfrak{p}$ has the unique form

$$X(v) = \begin{pmatrix} 0 & -v^* \\ v & 0_{2\times 2} \end{pmatrix}, \qquad v \in \mathbb{C}^2,$$

whose inverse takes the lower first column. This is a real linear isomorphism $\mathbb{C}^2 \to \mathfrak{p}$, so $\mathfrak{p} \times \mathfrak{p}$ has real dimension eight. Block decomposition gives the orthogonal direct sum $\mathfrak{su}(3) = \mathfrak{k} \oplus \mathfrak{p}$, with both summands of real dimension four.

Theorem 24.60 (Two-cross spatial exclusion)**.** *There is no solution of* (907)–(911) *with both* $Z_1, Z_2 \in \mathfrak{p}$. *No hypothesis on* C *is imposed in this statement. Every complex coordinate of each of the two cross vectors is allowed, and both signs of* $\omega \neq 0$ *and every original* $t > 0$ *are retained.*

This is an explicitly restricted spatial class, not a claim that arbitrary spatial pairs can be put into this class. The proof below also shows that the entire class lies in the previously defined real algebraic exceptional zero set. Consequently the result excludes an eight-dimensional linear subspace inside that zero set, but does not decide existence in its remaining part.

24.27.2 Gauss and the complete scalar equations force the temporal form

Block multiplication gives $[\mathfrak{p},\mathfrak{p}] \subset \mathfrak{k}$, $[\mathfrak{k},\mathfrak{p}] \subset \mathfrak{p}$. Because H and qq^* are block diagonal, $\mathcal{I}$ preserves both summands. Direct multiplication using H^2 and q^*q gives

$$\mathcal{I}X(v) = \left(\frac{3}{8} + \frac{1}{8}\right) X(v) = \frac{1}{2}X(v). \tag{912}$$

In this computation $HX(v)H = 0$, the two H^2 products sum to $(3/8)X(v)$, the two qq^* products sum to $X(v)/8$, and every term is traceless. If $Z_1, Z_2 \in \mathfrak{p}$, their double commutators therefore make $\mathcal{K}$ preserve both summands as well. Cyclicity gives, for every $Y \in \mathfrak{su}(3)$,

$$\langle Y, \mathcal{K}Y \rangle = \sum_{j=1}^{2} \|[Z_j, Y]\|^2 \geq 0.$$

Project (909) onto $\mathfrak{p}$. Its right side lies in $\mathfrak{k}$, hence

$$\left(\frac{t}{2}\,\mathrm{id}_{\mathfrak{p}} + 2\mathcal{K}|_{\mathfrak{p}}\right) C_{\mathfrak{p}} = 0.$$

Pairing with $C_{\mathfrak{p}}$ gives a sum whose first term is $(t/2)\|C_{\mathfrak{p}}\|^2$ and whose other terms are nonnegative. Since $t > 0$, $C_{\mathfrak{p}} = 0$. Thus Gauss forces $C \in \mathfrak{k}$; this was not an additional restriction on the temporal unknown.

Here are all frame changes, together with their inverses. Put

$$x = \frac{1}{2}, \quad d_0 = \frac{1}{2\sqrt{2}}, \quad r = \frac{\sqrt{6}}{4}, \quad \delta = \frac{1}{2r} = \sqrt{\frac{2}{3}},$$

$$S = \frac{1}{\sqrt{2r(r+x)}} \begin{pmatrix} d_0 & -(r+x) \\ r+x & d_0 \end{pmatrix}, \qquad U = \mathrm{diag}(1, S).$$

The relation $r^2 = x^2 + d_0^2$ shows that the numerator columns have squared length $2r(r+x)$, are orthogonal, and have determinant $2r(r+x) > 0$. Multiplication by the lower block of H gives their eigenvalues $r, -r$ in that order. Hence

$$U \in SO(3), \qquad U^* H U = \mathrm{diag}(0, r, -r), \qquad U^* q = q.$$

For every matrix Y the change is $Y \mapsto U^* Y U$, with inverse $Y' \mapsto U Y' U^*$. It is applied to H, F, C, Z_1, Z_2 simultaneously. It commutes with products, brackets, adjoints and trace projection, and fixes q. Thus it intertwines $\mathcal{I}, \mathcal{K}$ and each of the five full residuals exactly. It preserves $\mathfrak{p}, \mathfrak{k}$ because it commutes with P. No physical constant is rescaled.

In this first frame write the full temporal block matrix as

$$C_d = \begin{pmatrix} ia & 0 & 0 \\ 0 & i(h-a)/2 & z \\ 0 & -\bar{z} & -i(a+h)/2 \end{pmatrix}, \qquad a, h \in \mathbb{R}, \quad z \in \mathbb{C}.$$

Write the spatial matrices as $X(v), X(w)$ and put $M = vv^* + ww^*$ in this frame only for the following calculation. The vector scalar residual is exactly

$$(C_d^2 - X(v)^2 - X(w)^2)q + 2q = (-a^2 + \mathrm{tr}\, M + 2)q.$$

The complete nonzero block of the adjoint scalar residual, including its upper entry, is

$$\begin{pmatrix} 2 - 2r(M_{11} - M_{22}) & 0 & 0 \\ 0 & -4r|z|^2 + 2rM_{11} - 1 + 2r & -2irhz \\ 0 & 2irh\bar{z} & 4r|z|^2 - 2rM_{22} - 1 - 2r \end{pmatrix}.$$

For example, if the lower Hermitian block $B = -i(C_d)_{\mathrm{lower}}$ and $D = \mathrm{diag}(r, -r)$ are used, its lower block is $-[B,[B,D]] + \{M, D\} - (I_2 - 2D)$; multiplying these two-by-two matrices gives every entry just displayed. The two scalar equations are therefore equivalent to

$$a^2 = 2 + \mathrm{tr}\, M, \qquad 2|z|^2 - M_{11} = 1 - \delta, \qquad 2|z|^2 - M_{22} = 1 + \delta, \qquad hz = 0. \tag{913}$$

The upper entry is minus the sum of the two lower diagonal entries, so no scalar entry is left unchecked. Adding the middle two equations and comparing with the first gives $a^2 = 4|z|^2$. Since $M_{22} \geq 0$ and $1 + \delta > 0$, $z \neq 0$. Consequently

$$h = 0, \qquad a \neq 0, \qquad |z| = |a|/2. \tag{914}$$

Set $e = a$, retaining either sign, and $\gamma = -2iz/e$, so $|\gamma| = 1$. Choose either square root ζ of $-\bar{\gamma}$ and put

$$W = \frac{\zeta}{\sqrt{2}} \begin{pmatrix} \gamma & \gamma \\ 1 & -1 \end{pmatrix}, \qquad V = \mathrm{diag}(1, W).$$

Then $|\zeta| = 1$, $W^*W = I_2$ and $\det W = -\zeta^2\gamma = \bar\gamma\gamma = 1$. The lower Hermitian block of C_d is

$$\frac{e}{2}\begin{pmatrix} -1 & \gamma \\ \bar\gamma & -1 \end{pmatrix}.$$

Its products with $(\gamma, 1)^{\mathsf T}$ and $(\gamma, -1)^{\mathsf T}$ have eigenvalues $0, -e$, respectively. Those same columns give $W^* \operatorname{diag}(r, -r) W = r\,(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix})$. Thus the second exact change $Y_d \mapsto V^* Y_d V$ gives

$$C' = ie\operatorname{diag}(1, 0, -1), \quad H' = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & r \\ 0 & r & 0 \end{pmatrix}, \quad q' = q, \quad F' = -2H' + \operatorname{diag}(-2, 1, 1). \tag{915}$$

The inverse for every original matrix is, explicitly,

$$Y = UVY'V^*U^*. \tag{916}$$

For spatial vectors the forward change is the corresponding lower unitary action $v \mapsto W^*v$ and its inverse is $v' \mapsto Wv'$. The unitary parameters are retained; the assertion is not that H can be changed while the other data are left fixed. As with U, this simultaneous conjugation intertwines all five equations. In the remainder of the proof primes are omitted only to ease typography; all matrices are the exact objects in (915) and have inverse (916).

24.27.3 Every gauge entry in the classified frame

Write $Z_1 = X(v)$, $Z_2 = X(w)$ with arbitrary $v, w \in \mathbb{C}^2$, and define, afresh in the second frame,

$$V_0 = v^*v, \quad W_0 = w^*w, \quad p_0 = v^*w = \alpha_0 + ib, \quad M = vv^* + ww^*, \quad N = i(vw^* - wv^*).$$

Both M and N are Hermitian; $M \succeq 0$. Put

$$u = M_{11}, \quad v_0 = M_{22}, \quad m = M_{12}, \quad d_M = \det M = |v_1 w_2 - v_2 w_1|^2 \geq 0.$$

The determinant identity follows by expanding the four products in $M_{11}M_{22} - M_{12}M_{21}$ and collecting the squared modulus. The symbol v_0 here is a real diagonal entry, not the vector v. All eight real components of the two vectors remain independent.

Multiplication gives the full scalar residuals in this frame:

$$Q_H = \begin{pmatrix} 2 - 4r\operatorname{Re} m & 0 & 0 \\ 0 & 2r\operatorname{Re} m - 1 & r(\operatorname{tr} M - e^2 + 2) \\ 0 & r(\operatorname{tr} M - e^2 + 2) & 2r\operatorname{Re} m - 1 \end{pmatrix}, \quad Q_q = (\operatorname{tr} M - e^2 + 2)q.$$

Consequently scalar compatibility is exactly

$$e^2 = 2 + \operatorname{tr} M, \qquad \operatorname{Re} m = \delta > 0. \tag{917}$$

In particular $m \neq 0$. No phase of m, v, w has been set to a chosen value.

Set

$$Q = \operatorname{diag}(1, 2), \qquad E = \operatorname{diag}(2/3, -5/6).$$

Direct multiplication, keeping $r^2 = 3/8$ and $q^*q = 1/8$, gives

$$\mathcal{I}C = ie\operatorname{diag}(1/6, 2/3, -5/6), \qquad \mathcal{I}X(v) = \tfrac{1}{2}X(v). \tag{918}$$

For the first formula the two contributions, respectively from H and from the trace-free q term, are

$$ie\operatorname{diag}(0, 3/4, -3/4), \qquad ie\operatorname{diag}(1/6, -1/12, -1/12).$$

The second formula is (912), or follows from $(H')^2 = (3/8)\,\mathrm{diag}(0,1,1)$ directly. The entire spatial bracket and the entire $\mathcal{K}C$ are

$$[Z_1, Z_2] = \begin{pmatrix} -2ib & 0 \\ 0 & iN \end{pmatrix}, \qquad \mathcal{K}C = \begin{pmatrix} 2ie\,\mathrm{tr}(QM) & 0 \\ 0 & -ie(QM+MQ) \end{pmatrix}.$$

For example $[C, X(v)] = X(-ieQv)$, and the two double cross products give the displayed anticommutator. Thus the full Gauss equation is equivalent to the upper entry and lower Hermitian block

$$\frac{et}{6} + 4e\,\mathrm{tr}(QM) + 8\omega b = 0, \qquad etE - 2e(QM+MQ) - 4\omega N = 0. \tag{919}$$

The upper entry is the negative lower trace, since $\mathrm{tr}\,N = 2b$. No off-block entries occur. The entries needed below are therefore exact consequences of the full Gauss equation:

$$b = -\frac{e}{2\omega}\left(u + 2v_0 + \frac{t}{24}\right), \qquad N_{12} = -\frac{3e}{2\omega}m, \qquad N_{22} = -\frac{e}{\omega}\left(2v_0 + \frac{5t}{24}\right). \tag{920}$$

All divisions use the original $\omega \neq 0$.

Put

$$D = \left(\frac{t}{4} - \omega^2\right) I_2 - e^2 Q^2.$$

The complete spatial residuals are $X(s_1), X(s_2)$, respectively, where

$$s_1 = (D + W_0 I_2)v + (-\alpha_0 I_2 + i(2\omega eQ - 3bI_2))\,w, \tag{921}$$

$$s_2 = (D + V_0 I_2)w + (-\alpha_0 I_2 - i(2\omega eQ - 3bI_2))\,v. \tag{922}$$

To verify every term, $[C, X(v)] = X(-ieQv)$ gives the two gyroscopic terms and the term $-e^2Q^2$. The inertia gives $tI_2/4$. For the remaining double commutators, block multiplication gives

$$\mathcal{K}X(v) = X\big(W_0 v + (\bar{p}_0 - 2p_0)w\big), \qquad \mathcal{K}X(w) = X\big(V_0 w + (p_0 - 2\bar{p}_0)v\big).$$

Since $\bar{p}_0 - 2p_0 = -\alpha_0 - 3ib$ and $p_0 - 2\bar{p}_0 = -\alpha_0 + 3ib$, these are exactly (921)–(922). Their upper rows are the negative adjoints of their lower first columns, and all other entries are zero. Thus $s_1 = s_2 = 0$ is equivalent to the two full spatial equations.

We now use a consequence of those equations; no converse is needed. Multiplying (921) by v^*, (922) by w^* and adding gives the exact matrix identity

$$s_1 v^* + s_2 w^* = DM + d_M I_2 + (4bI_2 - 2\omega eQ)N. \tag{923}$$

Indeed $\mathrm{tr}(M)M - M^2 = d_M I_2$ follows by multiplying the two-by-two matrix M. Its off-diagonal entries are zero, and its diagonal entries are both $M_{11}M_{22} - M_{12}M_{21}$. The cubic terms on the left of (923) are

$$W_0 v v^* + V_0 w w^* - \alpha_0(v w^* + w v^*) + 3ib(v w^* - w v^*).$$

Those in $\mathrm{tr}(M)M - M^2$ have $-ib$ instead of $3ib$; their difference is $4bN$. The gyroscopic terms sum to $-2\omega eQN$. These expansions prove (923) entry by entry, without a reality or orthogonality restriction on v, w.

Proof of Theorem 24.60. Gauss and the scalar equations have already forced the exact temporal form and (917). In particular $e \neq 0$, $m \neq 0$. For a full solution, (923) is zero. Set

$$\chi = \frac{3e}{2\omega}, \qquad B_2 = 4b - 4\omega e, \quad D_2 = \frac{t}{4} - \omega^2 - 4e^2.$$

Its $(2,1)$ entry is $D_2\bar{m} + B_2 N_{21} = \bar{m}(D_2 - \chi B_2) = 0$ by Gauss. Since $m \neq 0$, it follows that $D_2 = \chi B_2$. Its $(2,2)$ entry is therefore

$$0 = d_M + B_2(\chi v_0 + N_{22}).$$

Using exactly (920) gives

$$\chi v_0 + N_{22} = -\frac{e}{2\omega}\left(v_0 + \frac{5t}{12}\right), \qquad B_2 = -\frac{2e}{\omega}\left(u + 2v_0 + \frac{t}{24} + 2\omega^2\right).$$

Consequently every proposed full solution in this class would satisfy

$$0 = d_M + \frac{e^2}{\omega^2}\left(u + 2v_0 + \frac{t}{24} + 2\omega^2\right)\left(v_0 + \frac{5t}{12}\right). \tag{924}$$

But $u, v_0 \geq 0$, $d_M \geq 0$, $t > 0$, $\omega^2 > 0$ and $e^2 > 0$. Both parentheses are strictly positive. The second term on the right is strictly positive for either sign of e and either sign of ω, contradicting (924). This proves the theorem without selecting a frequency or deleting any cross-vector phase. □

Corollary 24.61 (A further necessary condition on every full solution)**.** *Every solution of the unrestricted five equations satisfies*

$$\sum_{j=1}^{2}\left(\|PZ_jP\|_{\mathrm{HS}}^2 + \|(I_3 - P)Z_j(I_3 - P)\|_{\mathrm{HS}}^2\right) > 0.$$

Proof. Each summand is a squared Hilbert–Schmidt norm and is nonnegative. The sum vanishes if and only if both spatial matrices lie in $\mathfrak{p}$, which is excluded by Theorem 24.60. □

The scalar-compatible part of the class is nonempty even with a nonzero spatial bracket. This is not a full solution: in the frame (915), choose the exact values

$$v = \sqrt{\delta}(1,1)^{\mathsf{T}}, \qquad w = (i,0)^{\mathsf{T}}, \qquad e = \sqrt{3 + 2\delta}.$$

Then $\operatorname{tr} M = 1 + 2\delta$, $M_{12} = \delta$, and $[X(v), X(w)]_{11} = -2i\sqrt{\delta} \neq 0$. Thus both full scalar equations hold by their displayed residuals. To embed this example into the literal frame, use the fixed $W = i\left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)/\sqrt{2}$ and (916). The choice of the component i is a specified example, not a normalization of the general class. The theorem proves that no retained $t > 0, \omega \neq 0$ can complete this example to all five equations.

24.27.4 Exact inclusion of the selected class in the exceptional locus

We give the determinant map explicitly to specify the comparison with the full sixteen-coordinate projection, including its degree conventions. In this section the symbols H, q, F again denote their literal original matrices in (907)–(911), identical to the earlier retained data of (864)–(868). For any such matrix write $\widetilde{Y} = U^*YU$ when using its first, H-diagonal-frame value. The inverse remains $Y = U\widetilde{Y}U^*$. Put

$$J = iH, \quad j_0 = \langle J, J\rangle = \frac{3}{4}, \quad \mathscr{W} = J^{\perp}, \quad \Pi X = X - \frac{\langle J, X\rangle}{j_0}J, \quad \mathscr{B} = [Z_1, Z_2],$$

$$\eta = \langle J, \mathcal{K}J\rangle, \quad w_J = \Pi\mathcal{K}J, \quad \beta = \langle J, \mathscr{B}\rangle, \quad \mathcal{I}_7 = \mathcal{I}|_{\mathscr{W}},$$

$$\mathcal{S}_7 = \Pi\mathcal{K}|_{\mathscr{W}} - \frac{w_J \otimes w_J}{\eta}, \qquad f_7 = \Pi\mathscr{B} - \frac{\beta}{\eta}w_J.$$

Here $(w_J \otimes w_J)X = w_J\langle w_J, X\rangle$ and the original length of J is retained. On $\eta > 0$ define

$$\mathscr{M}_7(t) = t\mathcal{I}_7 + 2\mathcal{S}_7, \qquad D_7 = \det_{\mathscr{W}}\mathscr{M}_7, \qquad V_7 = 4\,\mathrm{adj}_{\mathscr{W}}(\mathscr{M}_7)f_7,$$

$$N_7 = V_7 + \frac{2\beta D_7 - \langle w_J, V_7\rangle}{\eta} J, \qquad Q_7 = \|V_7 q\|^2, \quad \alpha = \frac{1}{4} + \|Z_1 q\|^2 + \|Z_2 q\|^2,$$

$$P_H(t) = \alpha[N_7, [N_7, H]] - Q_7 \left(\sum_{j=1}^{2} [Z_j, [Z_j, H]] + F \right).$$

Let $F_\xi = \eta^{16}(\widetilde{P}_H)_{11}$ and $G_\xi = \eta^{16}(\widetilde{P}_H)_{22}$ denote the exact polynomials used for the original sixteen-coordinate exceptional set, with inverse division by η^{16} on $\eta > 0$. The matrix P_H in its definition is computed with the literal original H, q, F; $\widetilde{P}_H = U^* P_H U$ is then its exact conjugate, not a second change of already-conjugated data. They extend polynomially in the sixteen real spatial coordinates ξ. For clarity, their polynomial extension follows from the definitions: $\eta^7 D_7$, $\eta^7 V_7$, $\eta^8 N_7$ and $\eta^{14} Q_7$ are polynomial, by determinant and minor expansions. Hence $\eta^{16} P_H$ is polynomial. All other denominators are fixed nonzero constants. Their two selected diagonal entries are real because P_H is Hermitian for real t. The universal degree bounds are $\deg V_7 \le 6$ and $N_7 = V_7 + s_7 J$ with $\deg s_7 \le 7$. Since $[\widetilde{J}, \widetilde{H}] = 0$ and every diagonal entry of $[\widetilde{J}, Y]$ is zero, the diagonal entries of $[\widetilde{N}_7, [\widetilde{N}_7, \widetilde{H}]]$ have degree at most twelve; also $\deg Q_7 \le 12$. Thus F_ξ, G_ξ have degree at most twelve in t, including every specialization where their degrees drop.

The exceptional polynomial is the determinant $\mathscr{R}(\xi)$ of the fixed map

$$\mathbb{R}[t]_{<12} \oplus \mathbb{R}[t]_{<12} \longrightarrow \mathbb{R}[t]_{<24}, \qquad (A_0, B_0) \longmapsto A_0 F_\xi + B_0 G_\xi, \tag{925}$$

in increasing-power bases, with the first summand before the second. Theorem 24.59 proves this polynomial is nonzero on the ambient sixteen-dimensional carrier and that all full solutions project to its zero set. The following inclusion is proved directly, independently of whether any point of that zero set admits a solution.

Proposition 24.62. *The exact linear embedding $\mathfrak{p} \times \mathfrak{p} \hookrightarrow \mathfrak{su}(3)^2$ has image inside $\{\mathscr{R} = 0\}$. On every nonzero pair in this class the two diagonal Gauss-eliminated polynomials have the common factor $D_4(t)^2$, where*

$$D_4(t) = \det_{\mathfrak{p}} \left(\frac{t}{2} \mathrm{id}_{\mathfrak{p}} + 2\mathcal{K}|_{\mathfrak{p}} \right)$$

has degree exactly four and leading coefficient $1/16$.

Proof. The first-frame $\widetilde{J} = i\,\mathrm{diag}(0, r, -r)$ satisfies $[\widetilde{J}, X(v)] = X(i\,\mathrm{diag}(r, -r)v)$. The change by U preserves $\mathfrak{p}$ and the inner product. Thus, for the selected class,

$$\eta = \sum_j \|[Z_j, J]\|^2 = r^2(\|Z_1\|^2 + \|Z_2\|^2).$$

It is positive except at the zero pair. This also shows why this class is compatible with the stabilizer decomposition: $-\mathrm{ad}_J^2$ is exactly $r^2\,\mathrm{id}$ on $\mathfrak{p}$.

Put $\mathfrak{k}_0 = \mathfrak{k} \cap J^\perp$. The orthogonal decomposition is $\mathscr{W} = \mathfrak{k}_0 \oplus \mathfrak{p}$, with dimensions three and four. The already-proved block preservation of $\mathcal{I}, \mathcal{K}$, and $J \in \mathfrak{k}$, give $w_J, f_7 \in \mathfrak{k}_0$. Consequently $\mathcal{S}_7$ preserves both summands and equals $\mathcal{K}$ on $\mathfrak{p}$. Thus

$$\mathscr{M}_7 = \begin{pmatrix} \mathscr{M}_3 & 0 \\ 0 & \mathscr{M}_4 \end{pmatrix}, \qquad \mathscr{M}_4 = \frac{t}{2}\,\mathrm{id}_{\mathfrak{p}} + 2\mathcal{K}|_{\mathfrak{p}}, \quad D_7 = D_3 D_4.$$

Its four-dimensional determinant has degree four and leading coefficient $(1/2)^4 = 1/16$. Moreover $\mathscr{M}_4$ is positive for $t > 0$. For completeness, $\mathcal{I}$ has kernel $\mathbb{R}J$: its quadratic form is $\mathrm{tr}[X, H]^2 + 2\|Xq\|^2$; a zero commutes with the three-distinct-eigenvalue H and annihilates q, so has diagonal entries $0, is, -is$. The same formula proves positivity on $\mathscr{W}$. Completing the square in $\langle X + sJ, \mathcal{K}(X + sJ)\rangle$ gives

$$\langle X, \mathcal{S}_7 X\rangle + \eta \left(s + \frac{\langle w_J, X\rangle}{\eta} \right)^2 \ge 0,$$

so choosing the displayed square to be zero proves $\mathcal{S}_7 \succeq 0$. Thus $\mathscr{M}_7$ and its three-dimensional block are positive for every $t > 0$.

Define $V_3 = 4\,\mathrm{adj}(\mathscr{M}_3) f_7 \in \mathfrak{k}_0$. The block inverse for $t > 0$ gives

$$V_7 = D_4 V_3, \qquad N_7 = D_4 N_3, \qquad N_3 = V_3 + \frac{2\beta D_3 - \langle w_J, V_3 \rangle}{\eta} J.$$

These are polynomial identities in t: the two sides agree for every positive t, while a nonzero polynomial has only finitely many roots, as follows by repeated division by its linear root factors. Therefore $Q_7 = D_4^2 Q_3$, where $Q_3 = \|V_3 q\|^2$, and

$$P_H = D_4^2 P_{H,3}, \qquad P_{H,3} = \alpha[N_3, [N_3, H]] - Q_3 \left(\sum_j [Z_j, [Z_j, H]] + F \right). \tag{926}$$

The three-dimensional minors give $\deg V_3 \le 2$, $\deg D_3 \le 3$. Writing $N_3 = V_3 + s_3 J$, one has $[J, H] = 0$. In the diagonal frame, $[J, Y]_{jj} = 0$ for every matrix Y, by subtracting the identical diagonal products. Hence the selected diagonal entries of $[N_3, [N_3, H]] = [V_3, [V_3, H]] + s_3 [J, [V_3, H]]$ have degree at most four. Also $\deg Q_3 \le 4$. Thus at each fixed pair with $\eta > 0$,

$$F_\xi = D_4^2 F_3, \qquad G_\xi = D_4^2 G_3, \qquad \deg F_3, \deg G_3 \le 4,$$

where F_3, G_3 retain the scalar multiplier η^{16}. If they are not both zero, $(A_0, B_0) = (G_3, -F_3)$ is a nonzero kernel element of (925), with both degrees less than twelve. If both are zero, $(1, 0)$ is such a kernel element instead. The determinant therefore vanishes for every nonzero pair in the class, including every degree-drop case. At the zero pair it vanishes by continuity of the original polynomial $\mathscr{R}$ along this linear subspace. This proves the entire asserted inclusion without using a converse resultant test or interpreting $\eta = 0$ as an inverse solution. □

24.27.5 An exact certificate retaining nonzero residuals

The exclusion has a quantitative extension. It also gives a polynomial identity before the Gauss or spatial residuals are set to zero. Use the entire second-frame variables above, with every entry of v, w retained. More precisely, return here to the matrices and inverse in (915)–(916), not the literal-frame H used in the preceding determinant comparison. All of $M, N, m, u, v_0, b, s_1, s_2$ in this subsection are computed in that same second frame as in (921)–(922). Define the full spatial residual matrix and the Hermitian Gauss components

$$\mathcal{A} = s_1 v^* + s_2 w^*, \qquad \gamma_0 = et/6 + 4e(u + 2v_0) + 8\omega b, \qquad \mathcal{G} = etE - 2e(QM + MQ) - 4\omega N.$$

Here $\mathcal{A}$ is not the original generator A: its calligraphic symbol denotes exactly this two-by-two residual matrix. The full Gauss residual is $i\,\mathrm{diag}(\gamma_0, \mathcal{G})$, with the second block of size two. Set $B_2 = 4b - 4\omega e$ and

$$\mathcal{P} = d_M + \frac{e^2}{\omega^2}(u + 2v_0 + t/24 + 2\omega^2)(v_0 + 5t/12).$$

This expression is defined for the original nonzero ω; the identity below has no such denominator after multiplication.

Proposition 24.63 (Polynomial residual certificate and strict error bound)**.** *For every $v, w \in \mathbb{C}^2$ and real e, t, ω in the displayed second-frame matrices, the following polynomial identity holds:*

$$\begin{aligned} &4\omega^2(\bar{m}\mathcal{A}_{22} - v_0 \mathcal{A}_{21}) + \omega B_2(\bar{m}\mathcal{G}_{22} - v_0 \mathcal{G}_{21}) \\ &\qquad + e(v_0 + 5t/12)\bar{m}\gamma_0 \\ &= 4\bar{m}\left[\omega^2 d_M + e^2(u + 2v_0 + t/24 + 2\omega^2)(v_0 + 5t/12)\right]. \end{aligned} \tag{927}$$

No scalar, Gauss or spatial equation is assumed in this identity.

Now restrict to the exactly specified set of original tuples with $Z_1, Z_2 \in \mathfrak{p}$, $t > 0$, $\omega \neq 0$, for which both scalar equations and Gauss hold. Let $R_1, R_2 \in \mathfrak{su}(3)$ be their two full spatial residuals. Then

$$(\|R_1\|_{\mathrm{HS}}^2 + \|R_2\|_{\mathrm{HS}}^2)^{1/2} \geq \frac{\sqrt{2}\,\mathcal{P}}{\sqrt{\operatorname{tr} M\,(1 + v_0^2/|m|^2)}} > \frac{5\sqrt{2}\,e^2 t}{6\sqrt{\operatorname{tr} M\,(1 + v_0^2/|m|^2)}}. \tag{928}$$

The full matrix norms are preserved by the two exact unitary frame maps. In particular, on any such set with $\operatorname{tr} M \leq L$ and $t \geq t_0 > 0$, where $L > 0$, the strictly positive uniform bound is

$$(\|R_1\|_{\mathrm{HS}}^2 + \|R_2\|_{\mathrm{HS}}^2)^{1/2} > \frac{5\sqrt{2}\,t_0}{3\sqrt{L(1 + L^2/\delta^2)}}, \qquad \delta = 1/(2r). \tag{929}$$

The complete scalar–Gauss solution set is parameterized in Theorems 24.66 and 24.68, and Corollary 24.70 proves it is nonempty at every original $t = g^2 > 0$. No sequence in its displayed bounded portion can make both spatial residuals tend to zero.

Proof. The complete residual formulas above give

$$\mathcal{A}_{21} = D_2\bar{m} + B_2 N_{21}, \qquad \mathcal{A}_{22} = D_2 v_0 + d_M + B_2 N_{22}, \quad D_2 = t/4 - \omega^2 - 4e^2,$$

and, without any equation imposed,

$$\mathcal{G}_{21} = -6e\bar{m} - 4\omega N_{21}, \qquad \mathcal{G}_{22} = -5et/6 - 8ev_0 - 4\omega N_{22}.$$

Subtracting the two indicated spatial entries cancels D_2 exactly:

$$\bar{m}\mathcal{A}_{22} - v_0\mathcal{A}_{21} = \bar{m}d_M + B_2(\bar{m}N_{22} - v_0 N_{21}).$$

Multiplication of the indicated Gauss entries gives

$$\bar{m}\mathcal{G}_{22} - v_0\mathcal{G}_{21} = -e(2v_0 + 5t/6)\bar{m} - 4\omega(\bar{m}N_{22} - v_0 N_{21}).$$

These cancel the two N entries on the left of (927). The remaining factor is

$$\gamma_0 - 2\omega B_2 = 4e(u + 2v_0 + t/24 + 2\omega^2).$$

It yields exactly its stated right side. Each displayed step is a polynomial identity, including at $m = 0$ or $\omega = 0$; no inverse or solution is assigned at those values.

On the stated scalar–Gauss zero set the preceding proof forces the second-frame form, $e^2 = 2 + \operatorname{tr} M$ and $\operatorname{Re} m = \delta > 0$. Hence $m \neq 0$ and $\operatorname{tr} M > 0$. Setting $\gamma_0 = \mathcal{G} = 0$ in the polynomial identity and dividing by its retained nonzero factors gives

$$\mathcal{A}_{22} - \frac{v_0}{\bar{m}}\mathcal{A}_{21} = \mathcal{P} > 0.$$

Cauchy–Schwarz on these two entries implies

$$\mathcal{P} \leq \sqrt{1 + v_0^2/|m|^2}\,\|\mathcal{A}\|_{\mathrm{HS}}.$$

Let $S = (s_1\ s_2)$ and $V = (v\ w)$ be the two column matrices. Then $\mathcal{A} = SV^*$. Applying Cauchy–Schwarz to each entry and summing proves $\|SV^*\|_{\mathrm{HS}}^2 \leq \|S\|_{\mathrm{HS}}^2 \|V\|_{\mathrm{HS}}^2$. The right factors are $\|s_1\|^2 + \|s_2\|^2$ and $\operatorname{tr} M$, respectively. Directly from the two off-diagonal blocks of $X(s)$, $\|X(s)\|_{\mathrm{HS}}^2 = 2\|s\|^2$. Thus $R_j = X(s_j)$ proves the first inequality in (928).

The exact expansion

$$\begin{aligned}&(u + 2v_0 + t/24 + 2\omega^2)(v_0 + 5t/12)\\ &= 5t\omega^2/6 + v_0(u + 2v_0 + t/24 + 2\omega^2) + (5t/12)(u + 2v_0 + t/24)\end{aligned}$$

has nonnegative remainder terms, the last strictly positive for $t > 0$. Together with $d_M \geq 0$ and $e^2 > 0$ it gives $\mathcal{P} > 5e^2 t/6$, proving the strict second inequality. Finally $e^2 = 2 + \operatorname{tr} M \geq 2$, $0 \leq v_0 \leq \operatorname{tr} M \leq L$, and $|m| \geq \delta$ give (929). Since unitary conjugation preserves the Hilbert–Schmidt norm, the same inequalities hold for the complete spatial residuals in the original literal matter frame. □

24.27.6 Scope, provenance, and exact replay

Theorem 24.60 and Proposition 24.62 together prove an empty admissible fibre at every point of the explicit eight-dimensional two-cross linear subspace inside the full exceptional set. The result does not prove global emptiness or produce a full solution. It does not assert a result for arbitrary time-dependent gauge connections, and it makes no quantum mass-gap assertion. The earlier cross–cross–block exclusion in Theorem 24.45 instead assumes $C, Z_1 \in \mathfrak{p}$, $Z_2 \in \mathfrak{k}$; here only $Z_1, Z_2 \in \mathfrak{p}$ were assumed and $C \in \mathfrak{k}$ was proved.

The retained literal system is (864)–(868); its complete scalar coordinates are given in (884) and Proposition 24.50. The ambient projection statement and determinant convention are Theorem 24.59 and (906). The complete source proof, exact checker, and replay receipt for this additional exclusion are retained under `supporting_materials/workbench/evidence/two_cross_exceptional_20260905/`. All matrix identities needed for the theorem are derived above; no external classification theorem, numerical root search, or unproved literature identity is used.

The companion `check_two_cross_exceptional.py` checks 26 exact groups: the original force and constants, both frame maps (the second over the entire coordinate ring $\gamma\bar{\gamma} = 1$), every scalar residual before and after temporal classification, both inertia maps, the full bracket and Gauss residual, both complete spatial residuals, the nonnegative determinant identity, the full outer-product identity and the final positivity formula. Every real component of v, w is an independent symbol. It uses no floating point acceptance criterion and writes no files. One serial run passed all 26 groups in 0.416 seconds of measured checker time. The determinant block-factor proof and all positivity/domain arguments are analytic proofs in the preceding sections, not claims inferred from sampled determinant values.

The additional denominator-free residual identity and the full spatial error bound are proved in Proposition 24.63. The companion `check_residual_certificate.py` verifies its polynomial identity and exact positive remainder. The Cauchy–Schwarz estimates and their scalar–Gauss domain are proved in the text, not inferred from tests.

24.28 The complete scalar–Gauss domain at every positive coupling

24.28.1 Literal data, residuals, and coordinate maps

All complex vector spaces below retain their underlying real structures. The adjoint is denoted by $*$, and $[X, Y] = XY - YX$. Keep exactly

$$H_0 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & -1/2 & 1/(2\sqrt{2}) \\ 0 & 1/(2\sqrt{2}) & 1/2 \end{pmatrix}, \qquad q = \frac{1}{2\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \qquad A_0 = \sqrt{2} \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

The original constants are $m^2 = \Gamma = 1$ and $\kappa = 6$. For $Y_{\mathrm{tf}} = Y - (\operatorname{tr} Y) I_3/3$, define

$$\mathcal{I}_H Y = ([H, [H, Y]] + qq^* Y + Y qq^*)_{\mathrm{tf}}, \qquad \mathcal{K}_{Z_1, Z_2} Y = -\sum_{j=1}^{2} [Z_j, [Z_j, Y]].$$

Direct multiplication gives

$$F_0 = [A_0, [A_0, H_0]] = -2H_0 + \operatorname{diag}(-2, 1, 1), \quad H_0^2 = \frac{3}{8} \operatorname{diag}(0, 1, 1), \quad q^* q = \frac{1}{8}.$$

The scalar–Gauss domain in question consists of

$$(t, \omega, C_0, Z_{1,0}, Z_{2,0}), \quad t = g^2 > 0, \quad \omega \in \mathbb{R} \setminus \{0\}, \quad C_0, Z_{j,0} \in \mathfrak{su}(3),$$

where $P = qq^*/(q^* q) = \operatorname{diag}(1, 0, 0)$ and

$$P Z_{j,0} P = (I_3 - P) Z_{j,0} (I_3 - P) = 0,$$

satisfying exactly the following three full residual equations:

$$[C_0,[C_0,H_0]]-\sum_j[Z_{j,0},[Z_{j,0},H_0]]-F_0=0, \tag{930}$$

$$(C_0^2-Z_{1,0}^2-Z_{2,0}^2)q+2q=0, \tag{931}$$

$$(t\mathcal{I}_{H_0}+2\mathcal{K}_{Z_{1,0},Z_{2,0}})C_0-4\omega[Z_{1,0},Z_{2,0}]=0. \tag{932}$$

Equation (932) is the original Gauss equation multiplied by its retained positive t; division by the same t is its inverse. Neither spatial equation is imposed anywhere in the construction.

The exact maps to and from the classified frame are as follows. Retain

$$r=\frac{\sqrt{6}}{4},\quad \delta=\frac{1}{2r}=\sqrt{\frac{2}{3}},\quad x=\frac{1}{2},\quad d_0=\frac{1}{2\sqrt{2}},$$
$$S=\frac{1}{\sqrt{2r(r+x)}}\begin{pmatrix}d_0 & -(r+x)\\ r+x & d_0\end{pmatrix},\quad U=\operatorname{diag}(1,S).$$

The identity $r^2=x^2+d_0^2$ gives $S^*S=I_2$, $\det S=1$; multiplication of H_0 by its two displayed lower columns gives

$$U^*H_0U=\operatorname{diag}(0,r,-r),\qquad U^*q=q.$$

Choose any $\zeta\in\mathbb{C}$ with $|\zeta|=1$, and set

$$\gamma=-\zeta^{-2},\qquad W=\frac{\zeta}{\sqrt{2}}\begin{pmatrix}\gamma & \gamma\\ 1 & -1\end{pmatrix}=\frac{1}{\sqrt{2}}\begin{pmatrix}-\zeta^{-1} & -\zeta^{-1}\\ \zeta & -\zeta\end{pmatrix},\qquad V=\operatorname{diag}(1,W),\quad B=UV.$$

Here $|\gamma|=1$ and $\zeta^2=-\overline{\gamma}$, retaining exactly the original phase relation. Every unit γ occurs, with its two possible square roots ζ. The displayed entries give

$$W^*W=I_2,\quad \det W=1,\quad W^*\operatorname{diag}(r,-r)W=\begin{pmatrix}0 & r\\ r & 0\end{pmatrix}.$$

Thus, simultaneously for all data,

$$Y'=B^*Y_0B,\qquad Y_0=BY'B^*,\qquad H'=\begin{pmatrix}0&0&0\\0&0&r\\0&r&0\end{pmatrix},\quad q'=q,\quad F'=-2H'+\operatorname{diag}(-2,1,1). \tag{933}$$

These are inverse maps, with no constant or unknown rescaled. In fact $B^*(XY)B=(B^*XB)(B^*YB)$, the same equality holds for brackets, trace and adjoints, and $Bq=q$. Therefore $\mathcal{I}$ and $\mathcal{K}$ and every full residual in (930)–(932) are conjugated exactly (the vector residual is multiplied by B^*). This proves that zero residuals in the classified frame give zero residuals in the literal frame, with the displayed inverse recovering every datum. The block-diagonal B commutes with P, preserving the two-cross class.

24.28.2 Every scalar and Gauss entry in the classified frame

For $v,w\in\mathbb{C}^2$ write

$$X(v)=\begin{pmatrix}0 & -v^*\\ v & 0_{2\times2}\end{pmatrix},\qquad Z_1=X(v),\quad Z_2=X(w),\quad C=ie\operatorname{diag}(1,0,-1),\quad e\in\mathbb{R}\setminus\{0\}.$$

The inverse of $v\mapsto X(v)$ is the lower first column; every $X(v)$ is traceless anti-Hermitian. In this section all unprimed matrices are these classified-frame matrices, with literal inverse (933). Define

$$M=vv^*+ww^*=\begin{pmatrix}u & m\\ \bar{m} & v_0\end{pmatrix},\quad N=i(vw^*-wv^*),$$
$$b=\operatorname{Im}(v^*w),\quad Q=\operatorname{diag}(1,2),\quad E=\operatorname{diag}(2/3,-5/6).$$

The subscript on the real scalar v_0 distinguishes it from the vector v.

Here are the complete residuals, rather than only contractions of them:

$$Q_H = \begin{pmatrix} 2 - 4r\,\mathrm{Re}\,m & 0 & 0 \\ 0 & 2r\,\mathrm{Re}\,m - 1 & r(\mathrm{tr}\,M - e^2 + 2) \\ 0 & r(\mathrm{tr}\,M - e^2 + 2) & 2r\,\mathrm{Re}\,m - 1 \end{pmatrix}, \tag{934}$$

$$Q_q = (\mathrm{tr}\,M - e^2 + 2)q, \tag{935}$$

$$Q_G = i\,\mathrm{diag}(\gamma_0, \mathcal{G}), \tag{936}$$

$$\gamma_0 = et/6 + 4e(u + 2v_0) + 8\omega b, \qquad \mathcal{G} = etE - 2e(QM + MQ) - 4\omega N. \tag{937}$$

In (936) the second diagonal block has size two. To derive every displayed entry, multiplication gives

$$X(v)^2 = \mathrm{diag}(-v^*v, -vv^*), \quad [C, X(v)] = X(-ieQv), \quad [X(v), X(w)] = \mathrm{diag}(-2ib, iN).$$

The first two identities and $H'_{23} = H'_{32} = r$ give (934) and (935); all their omitted entries are zero. The two contributions to $\mathcal{I}_{H'}C$ are

$$ie\,\mathrm{diag}(0, 3/4, -3/4), \qquad ie\,\mathrm{diag}(1/6, -1/12, -1/12),$$

so $\mathcal{I}_{H'}C = ie\,\mathrm{diag}(1/6, 2/3, -5/6)$. Also

$$\mathcal{K}C = \mathrm{diag}\big(2ie\,\mathrm{tr}(QM), -ie(QM + MQ)\big).$$

Substitution of these full blocks gives (936). Finally $\mathrm{tr}\,N = 2b$, since $\mathrm{tr}(vw^*) = w^*v = \overline{v^*w}$. Thus $\mathrm{tr}\,\mathcal{G} = -\gamma_0$; the upper Gauss entry vanishes whenever the full lower block does, and this implication is an identity, not an omitted equation. The two scalar equations are equivalent to

$$e^2 = 2 + u + v_0, \qquad \mathrm{Re}\,m = \delta. \tag{938}$$

24.28.3 The exact realization of the two Hermitian matrices

Lemma 24.64. *For every pair v, w, let $D = (v\ w)$ and $J = \left(\begin{smallmatrix} 0 & i \\ -i & 0 \end{smallmatrix}\right)$. Then $M = DD^*$, $N = DJD^*$ and*

$$M + N = (v - iw)(v - iw)^*, \qquad M - N = (v + iw)(v + iw)^*.$$

If M is positive definite, then

$$\mathrm{tr}(M^{-1}N) = 0, \qquad \det N = -\det M. \tag{939}$$

Proof. Multiplication of the displayed two-by-two blocks gives all three factorizations. If $M > 0$, D is invertible because $z^*Mz = \|D^*z\|^2$ has zero kernel. Consequently $M^{-1}N = D^{-*}JD^*$ is similar to J, whose trace is zero and determinant is -1. Taking the trace and determinant proves (939), with every factor retained. □

Lemma 24.65. *Every classified scalar–Gauss solution has $M > 0$.*

Proof. The scalar equation gives $m = \delta + iy \neq 0$, so $u, v_0 > 0$ and $M \neq 0$. It is already positive semidefinite as DD^*. If M had rank one, the range of $N = DJD^*$ would be contained in its one-dimensional range. A Hermitian matrix with that range and its orthogonal complement in its kernel is a real scalar multiple of M, so $N = \nu M$ for some real ν. Set $k = e/\omega \neq 0$. The full Gauss equation reads

$$N = kT, \qquad T = \frac{t}{4}E - \frac{1}{2}(QM + MQ) = \begin{pmatrix} t/6 - u & -3m/2 \\ -3\bar{m}/2 & -5t/24 - 2v_0 \end{pmatrix}. \tag{940}$$

Its off-diagonal entry gives $\nu = -3k/2$ because $m \neq 0$. The $(1,1)$ entry then gives $t/6 - u = -3u/2$, or $t/6 = -u/2$, which contradicts $t > 0$ and $u > 0$. Thus M has rank two and is positive definite. □

Theorem 24.66 (Complete classified-frame parameterization)**.** *Define the open real parameter set*

$$\mathcal{D} = \{(u, v_0, y) \in \mathbb{R}^3 : u > 0, \quad d = uv_0 - \delta^2 - y^2 > 0, \quad c = 4v_0 - 5u > 0\}.$$

For $(u, v_0, y) \in \mathcal{D}$, retain

$$m = \delta + iy, \qquad M = \begin{pmatrix} u & m \\ \bar{m} & v_0 \end{pmatrix}, \qquad t = \frac{72d}{c}, \tag{941}$$

$$T = \frac{t}{4}E - \frac{1}{2}(QM + MQ), \qquad k = \sigma\sqrt{\frac{d}{-\det T}}, \qquad e = \epsilon\sqrt{2 + u + v_0}, \tag{942}$$

$$\omega = e/k, \qquad N = kT, \qquad \epsilon, \sigma \in \{1, -1\}. \tag{943}$$

The denominator $-\det T$ is strictly positive. The two matrices $A_+ = M + N$ and $A_- = M - N$ are positive semidefinite of rank one, and their $(1,1)$ entries h_+, h_- are strictly positive. Define

$$a_\pm = \begin{pmatrix} \sqrt{h_\pm} \\ (A_\pm)_{21}/\sqrt{h_\pm} \end{pmatrix}, \quad p_\pm \in \mathbb{C}, \quad |p_\pm| = 1, \quad v = \frac{p_+a_+ + p_-a_-}{2}, \quad w = \frac{i(p_+a_+ - p_-a_-)}{2}. \tag{944}$$

These formulas give exactly all classified-frame solutions of (934)–(937)*. In fact they are a bijection from $\mathcal{D} \times \{1, -1\}^2 \times U(1)^2$ to that solution set with coordinates (t, ω, e, v, w).*

Proof. First consider any solution. Lemma 24.65 gives $M > 0$ and $d > 0$. Inverting its two-by-two matrix gives

$$M^{-1} = \frac{1}{d}\begin{pmatrix} v_0 & -m \\ -\bar{m} & u \end{pmatrix}.$$

The trace condition in (939), divided by the retained nonzero k, now gives exactly

$$\begin{aligned} 0 = d\,\mathrm{tr}(M^{-1}T) &= v_0(t/6 - u) + u(-5t/24 - 2v_0) + 3|m|^2 \\ &= \frac{t(4v_0 - 5u)}{24} - 3d. \end{aligned} \tag{945}$$

Hence $c > 0$ and $t = 72d/c$. The determinant condition gives $k^2 \det T = -d$, so $\det T < 0$ and k is precisely one of the two choices in (942). The scalar conditions give precisely the two signs of e stated there. The original $\omega = e/k$ is unchanged, with sign $\epsilon\sigma$.

Conversely, start with parameters in $\mathcal{D}$. Positivity of M follows from the exact factorization

$$M = LL^*, \qquad L = \begin{pmatrix} \sqrt{u} & 0 \\ \bar{m}/\sqrt{u} & \sqrt{d}/\sqrt{u} \end{pmatrix}, \qquad \det L = \sqrt{d} > 0.$$

The specified positive t makes (945) zero. Put $R = L^{-1}TL^{-*}$. This is Hermitian and $\mathrm{tr}\,R = \mathrm{tr}(M^{-1}T) = 0$, so its full form is

$$R = \begin{pmatrix} a & z \\ \bar{z} & -a \end{pmatrix}, \qquad a \in \mathbb{R}.$$

The nonzero m makes $T_{12} = -3m/2 \neq 0$, hence $R \neq 0$. Thus $\det R = -a^2 - |z|^2 < 0$ and $\det T = d \det R < 0$. This proves that both signs of k are well defined. Let $J_0 = kR$. It is Hermitian, has trace zero and determinant -1, and multiplying the displayed matrix shows $J_0^2 = I_2$. For either sign and every column z_0, one has

$$z_0^*(I_2 \pm J_0)z_0 = \frac{1}{2}\|(I_2 \pm J_0)z_0\|^2 \geq 0.$$

Also $\det(I_2 \pm J_0) = 0$ and $\mathrm{tr}(I_2 \pm J_0) = 2$, so these matrices have rank one. Therefore

$$A_\pm = M \pm N = L(I_2 \pm J_0)L^*$$

are positive semidefinite of rank one, as claimed.

Their first diagonal entries are nonnegative. Suppose $h_s = 0$ for a sign $s \in \{1, -1\}$, where $A_s = M + sN$. Its determinant is zero, so $|(A_s)_{12}|^2 = 0$. But

$$(A_s)_{12} = m(1 - 3sk/2), \quad h_s = u + sk(t/6 - u).$$

Since $m \neq 0$, this forces $sk = 2/3$ and then $h_s = u/3 + t/9 > 0$, a contradiction. Thus $h_\pm > 0$. Using $\det A_\pm = 0$ gives $(A_\pm)_{22} = |(A_\pm)_{21}|^2/h_\pm$; consequently $A_\pm = a_\pm a_\pm^*$ with exactly the vectors in (944).

For those vectors, $v - iw = p_+ a_+$ and $v + iw = p_- a_-$. Their two outer products are A_+, A_- because the phases have unit modulus. Adding and subtracting the identities in Lemma 24.64 proves

$$vv^* + ww^* = M, \qquad i(vw^* - wv^*) = N.$$

Equation (938) holds by construction, so both full scalar residuals vanish. Substitution of $N = kT$ and $\omega k = e$ gives, with all coefficients retained,

$$\mathcal{G} = etE - 2e(QM + MQ) - 4e\left(\frac{t}{4}E - \frac{1}{2}(QM + MQ)\right) = 0.$$

The exact identity $\gamma_0 = -\operatorname{tr}\mathcal{G}$ then checks the remaining upper Gauss entry. Thus every entry of all three original residuals vanishes after applying (933).

For bijectivity, u, v_0, y are recovered from M, ϵ is the sign of e, and σ the sign of e/ω. The two phase factors are recovered without any zero denominator by

$$p_+ = \frac{(v - iw)_1}{\sqrt{h_+}}, \qquad p_- = \frac{(v + iw)_1}{\sqrt{h_-}}.$$

Indeed a column c with $cc^* = a_\pm a_\pm^*$ and $h_\pm > 0$ has $|c_1| = \sqrt{h_\pm}$, and the $(2,1)$ entry forces its second component to be the same unit multiple of $(a_\pm)_2$. These formulas prove both surjectivity and the exact inverse, without setting any spatial phase to a selected value. □

24.28.4 The complete literal scalar–Gauss domain and its phase quotient

The classified-frame bijection of Theorem 24.66 has an exact extension to the entire literal two-cross domain. We give the forced temporal classification, the full parameter map, and the complete fibres of that map. All spaces retain their underlying real structures, $[X, Y] = XY - YX$, and $*$ denotes the adjoint.

The literal domain and the retained constants Retain exactly

$$H_0 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & -1/2 & 1/(2\sqrt{2}) \\ 0 & 1/(2\sqrt{2}) & 1/2 \end{pmatrix}, \qquad q = \frac{1}{2\sqrt{2}}\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \qquad A_0 = \sqrt{2}\begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

$$m^2 = \Gamma = 1, \qquad \kappa = 6, \qquad F_0 = [A_0, [A_0, H_0]] = -2H_0 + \operatorname{diag}(-2, 1, 1).$$

Here $\mathfrak{su}(3)$ consists of the traceless anti-Hermitian complex 3×3 matrices, with positive real inner product $\langle X, Y\rangle = -\operatorname{tr}(XY)$. For $Y_{\mathrm{tf}} = Y - (\operatorname{tr} Y)I_3/3$, define

$$\mathcal{I}_{H_0} Y = ([H_0, [H_0, Y]] + qq^* Y + Y qq^*)_{\mathrm{tf}}, \qquad \mathcal{K}_{Z_1, Z_2} Y = -\sum_{j=1}^{2} [Z_j, [Z_j, Y]].$$

Multiplication of the retained matrices gives $H_0 q = 0$, $H_0^2 = (3/8)\operatorname{diag}(0, 1, 1)$ and $q^* q = 1/8$. Consequently $\operatorname{tr} H_0^2 + 2q^* q = 1$, so the original scalar coefficient $m^2 + \Gamma(\operatorname{tr} H_0^2 + 2q^* q)$ is 2. Also $\operatorname{adj} H_0 = \operatorname{diag}(-3/8, 0, 0)$ gives $6(\operatorname{adj} H_0 - qq^*)_{\mathrm{tf}} = \operatorname{diag}(-2, 1, 1)$, agreeing with the displayed force.

Put

$$P = \frac{qq^*}{q^*q} = \operatorname{diag}(1,0,0), \qquad \mathfrak{p} = \{X \in \mathfrak{su}(3) : PXP = (I_3 - P)X(I_3 - P) = 0\}.$$

Let $\mathcal{S}_{\mathrm{sg}}$ be exactly the set of tuples

$$(t, \omega, C_0, Z_{1,0}, Z_{2,0}), \qquad t = g^2 > 0, \quad \omega \in \mathbb{R} \setminus \{0\}, \quad C_0 \in \mathfrak{su}(3), \quad Z_{j,0} \in \mathfrak{p},$$

for which the following three full residuals vanish:

$$[C_0, [C_0, H_0]] - \sum_j [Z_{j,0}, [Z_{j,0}, H_0]] - F_0 = 0, \tag{946}$$

$$(C_0^2 - Z_{1,0}^2 - Z_{2,0}^2)q + 2q = 0, \tag{947}$$

$$(t\mathcal{I}_{H_0} + 2\mathcal{K}_{Z_{1,0},Z_{2,0}})C_0 - 4\omega[Z_{1,0}, Z_{2,0}] = 0. \tag{948}$$

Their codomains are, respectively, $\mathrm{Herm}_3(\mathbb{C})_0$, $\mathbb{C}^3$, and $\mathfrak{su}(3)$. The original Gauss residual is

$$(\mathcal{I}_{H_0} + 2\mathcal{K}_{Z_{1,0},Z_{2,0}}/g^2)C_0 - (4\omega/g^2)[Z_{1,0}, Z_{2,0}].$$

Multiplication by the same retained positive $t = g^2$ gives (948), and division by t is its inverse. No spatial residual is required to vanish in the definition of $\mathcal{S}_{\mathrm{sg}}$.

Temporal classification and all inverse frame maps For $v \in \mathbb{C}^2$, define

$$X(v) = \begin{pmatrix} 0 & -v^* \\ v & 0_2 \end{pmatrix}, \qquad \mathfrak{k} = \{X \in \mathfrak{su}(3) : [X, P] = 0\}.$$

Every element of $\mathfrak{p}$ has exactly this form, whose inverse is the lower first column. Define the fixed constants and matrix

$$r = \frac{\sqrt{6}}{4}, \qquad \delta = \frac{1}{2r} = \sqrt{\frac{2}{3}}, \qquad x = \frac{1}{2}, \qquad d_0 = \frac{1}{2\sqrt{2}},$$

$$S = \frac{1}{\sqrt{2r(r+x)}} \begin{pmatrix} d_0 & -(r+x) \\ r+x & d_0 \end{pmatrix}, \qquad U = \operatorname{diag}(1, S).$$

The identity $r^2 = x^2 + d_0^2$ shows that the numerator columns are orthogonal, have squared length $2r(r+x)$, and have determinant $2r(r+x) > 0$. Multiplication by the lower block of H_0 gives their respective eigenvalues $r, -r$. Thus

$$U^*U = I_3, \qquad \det U = 1, \qquad U^*H_0U = \operatorname{diag}(0, r, -r), \qquad Uq = q.$$

Lemma 24.67 (The temporal form is forced by scalars and Gauss)**.** *Every tuple in $\mathcal{S}_{\mathrm{sg}}$ has $C_0 \in \mathfrak{k}$. In the fixed first frame $C_d = U^*C_0U$, there are unique $e \in \mathbb{R} \setminus \{0\}$ and $z \in \mathbb{C}$ with $|z| = |e|/2$ such that*

$$C_d = \begin{pmatrix} ie & 0 & 0 \\ 0 & -ie/2 & z \\ 0 & -\bar{z} & -ie/2 \end{pmatrix}.$$

Set $\gamma = -2iz/e$. Then $|\gamma| = 1$. For either square root $\zeta^2 = -\bar{\gamma}$, define

$$W(\zeta) = \frac{\zeta}{\sqrt{2}} \begin{pmatrix} \gamma & \gamma \\ 1 & -1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} -\zeta^{-1} & -\zeta^{-1} \\ \zeta & -\zeta \end{pmatrix}, \qquad B(\zeta) = U \operatorname{diag}(1, W(\zeta)).$$

The simultaneous change and its inverse are

$$Y' = B(\zeta)^* Y_0 B(\zeta), \qquad Y_0 = B(\zeta) Y' B(\zeta)^*. \tag{949}$$

They give

$$C' = ie\,\mathrm{diag}(1,0,-1), \qquad H' = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & r \\ 0 & r & 0 \end{pmatrix}, \qquad q' = q, \qquad F' = -2H' + \mathrm{diag}(-2,1,1).$$

These maps preserve the spatial class and intertwine all three full residuals in (946)–(948).

Proof. Block multiplication gives the orthogonal direct sum $\mathfrak{su}(3) = \mathfrak{k} \oplus \mathfrak{p}$, together with $[\mathfrak{p},\mathfrak{p}] \subset \mathfrak{k}$ and $[\mathfrak{k},\mathfrak{p}] \subset \mathfrak{p}$. Since H_0, qq^* are block diagonal, $\mathcal{I}_{H_0}$ preserves both summands. More explicitly, $H_0 X(v) H_0 = 0$; the two H_0^2 products sum to $(3/8)X(v)$, and the two qq^* products sum to $X(v)/8$. Each term is traceless. Thus

$$\mathcal{I}_{H_0}|_{\mathfrak{p}} = \tfrac{1}{2}\,\mathrm{id}_{\mathfrak{p}}\,.$$

The two bracket inclusions make $\mathcal{K} = \mathcal{K}_{Z_{1,0},Z_{2,0}}$ preserve both summands as well. Cyclicity gives, for every $Y \in \mathfrak{su}(3)$,

$$\langle Y, \mathcal{K}Y\rangle = -\sum_j \mathrm{tr}([Z_{j,0},Y]^2) = \sum_j \|[Z_{j,0},Y]\|^2 \geq 0.$$

The $\mathfrak{p}$ projection of Gauss is therefore

$$(\tfrac{t}{2}\,\mathrm{id}_{\mathfrak{p}} + 2\mathcal{K}|_{\mathfrak{p}})(C_0)_{\mathfrak{p}} = 0.$$

Pairing with $(C_0)_{\mathfrak{p}}$ gives nonnegative summands, including $(t/2)\|(C_0)_{\mathfrak{p}}\|^2$. Since $t > 0$, $(C_0)_{\mathfrak{p}} = 0$, proving $C_0 \in \mathfrak{k}$.

The arbitrary first-frame temporal block consequently has the form

$$C_d = \begin{pmatrix} ia & 0 & 0 \\ 0 & i(h-a)/2 & z \\ 0 & -\bar{z} & -i(a+h)/2 \end{pmatrix}, \qquad a,h \in \mathbb{R}, \quad z \in \mathbb{C}.$$

Write the spatial matrices in this first frame as $X(\widetilde{v}), X(\widetilde{w})$, and define $\widetilde{M} = \widetilde{v}\widetilde{v}^* + \widetilde{w}\widetilde{w}^*$. The whole vector residual is

$$(-a^2 + \mathrm{tr}\,\widetilde{M} + 2)q,$$

and the whole adjoint residual is

$$\begin{pmatrix} 2 - 2r(\widetilde{M}_{11} - \widetilde{M}_{22}) & 0 & 0 \\ 0 & -4r|z|^2 + 2r\widetilde{M}_{11} - 1 + 2r & -2irhz \\ 0 & 2irh\bar{z} & 4r|z|^2 - 2r\widetilde{M}_{22} - 1 - 2r \end{pmatrix}.$$

For completeness, if $D = \mathrm{diag}(r,-r)$, the lower spatial contribution is $\widetilde{M}D + D\widetilde{M}$, whose diagonal entries are $2r\widetilde{M}_{11}, -2r\widetilde{M}_{22}$ and whose off-diagonal entries vanish. The temporal double commutator has diagonal entries $-4r|z|^2, 4r|z|^2$ and off-diagonal entries $-2irhz, 2irh\bar{z}$. Subtracting the lower force $I_2 - 2D$ gives the displayed lower block. The upper spatial contribution and force give $2 - 2r(\widetilde{M}_{11} - \widetilde{M}_{22})$. Finally $X(v)^2 = \mathrm{diag}(-v^*v, -vv^*)$ gives the displayed vector residual, with all its other components zero.

Thus both scalar equations are equivalent to

$$a^2 = 2 + \mathrm{tr}\,\widetilde{M}, \qquad 2|z|^2 - \widetilde{M}_{11} = 1 - \delta, \qquad 2|z|^2 - \widetilde{M}_{22} = 1 + \delta, \qquad hz = 0.$$

The upper adjoint entry is minus the lower trace. Adding the middle equations gives $4|z|^2 = 2 + \mathrm{tr}\,\widetilde{M} = a^2$. Since $\widetilde{M}_{22} \geq 0$ and $1 + \delta > 0$, $z \neq 0$. Hence $h = 0$ and $a \neq 0$. This proves the claimed temporal block with $e = a$, where its entries recover e, z uniquely.

Now $|\gamma| = 1$ and each root $\zeta^2 = -\bar{\gamma}$ has $|\zeta| = 1$, with $\gamma = -\zeta^{-2}$. The displayed entries give $W^*W = I_2$ and $\det W = -\zeta^2\gamma = 1$. The lower Hermitian block of $-iC_d$ is

$$\frac{e}{2}\begin{pmatrix} -1 & \gamma \\ \bar{\gamma} & -1 \end{pmatrix}.$$

Its products with $(\gamma, 1)^{\mathsf{T}}$ and $(\gamma, -1)^{\mathsf{T}}$ have eigenvalues $0, -e$, respectively. The same columns give $W^*DW = \left(\begin{smallmatrix} 0 & r \\ r & 0 \end{smallmatrix}\right)$. This proves the stated classified matrices.

The block-diagonal unitary $B(\zeta)$ fixes q and commutes with P. Its conjugation therefore preserves $\mathfrak{p}$ and $\mathfrak{k}$. Unitarity gives $B^*(XY)B = (B^*XB)(B^*YB)$; adjoints, commutators and trace projection satisfy the corresponding equalities. It follows that $\mathcal{I}, \mathcal{K}$ and every full matrix residual are intertwined by conjugation. The vector residual is multiplied by B^*. Equation (949) gives the exact inverse on every datum, with no rescaling. □

The full parameter map and the quotient bijection We restate the classified parameters so that the literal map is explicit. Put

$$Q = \operatorname{diag}(1, 2), \qquad E = \operatorname{diag}(2/3, -5/6),$$
$$\mathcal{D} = \{(u, v_0, y) \in \mathbb{R}^3 : u > 0,\ d = uv_0 - \delta^2 - y^2 > 0,\ c = 4v_0 - 5u > 0\}.$$

For $(u, v_0, y) \in \mathcal{D}$, independent $\epsilon, \sigma \in \{1, -1\}$, and $p_+, p_- \in U(1)$, define

$$\begin{aligned} &m = \delta + iy, && M = \begin{pmatrix} u & m \\ \bar{m} & v_0 \end{pmatrix}, && t = \frac{72d}{c}, \\ &T = \frac{t}{4}E - \frac{1}{2}(QM + MQ), && k = \sigma\sqrt{\frac{d}{-\det T}}, && e = \epsilon\sqrt{2 + u + v_0}, \\ &\omega = e/k, && N = kT, && A_\pm = M \pm N. \end{aligned} \tag{950}$$

Theorem 24.66 proves that $\det T < 0$, that $A_\pm$ are positive semidefinite of rank one, and that $h_\pm = (A_\pm)_{11} > 0$. Thus the following vectors are defined on the whole displayed domain:

$$a_\pm = \begin{pmatrix} \sqrt{h_\pm} \\ (A_\pm)_{21}/\sqrt{h_\pm} \end{pmatrix}, \qquad v = \frac{p_+a_+ + p_-a_-}{2}, \qquad w = \frac{i(p_+a_+ - p_-a_-)}{2}. \tag{951}$$

In particular $vv^* + ww^* = M$ and $i(vw^* - wv^*) = N$. The cited theorem proves that these parameters are in bijection with all solutions of the two full scalar equations and full Gauss in the frame $C' = ie\operatorname{diag}(1, 0, -1)$, $H'_{23} = H'_{32} = r$, $q' = q$, and $F' = -2H' + \operatorname{diag}(-2, 1, 1)$. It also proves the exact inverse phases

$$p_+ = \frac{(v - iw)_1}{\sqrt{h_+}}, \qquad p_- = \frac{(v + iw)_1}{\sqrt{h_-}}. \tag{952}$$

Define the full raw parameter space

$$\mathcal{P}_{\text{lit}} = \mathcal{D} \times \{1, -1\}^2 \times U(1)^2 \times U(1)$$

with coordinates $(u, v_0, y, \epsilon, \sigma, p_+, p_-, \zeta)$. For the last unit phase set $\gamma = -\zeta^{-2}$ and use the matrix $B(\zeta)$ of Lemma 24.67. Define the map $\mathfrak{F} : \mathcal{P}_{\text{lit}} \to \mathcal{S}_{\text{sg}}$ by (950)–(951) and the following entire literal tuple:

$$\begin{aligned} \mathfrak{F}(u, v_0, y, \epsilon, \sigma, p_+, p_-, \zeta) &= (t, \omega, C_0, Z_{1,0}, Z_{2,0}), \\ C_0 &= B(\zeta)\big(ie\operatorname{diag}(1, 0, -1)\big)B(\zeta)^*, \\ Z_{1,0} &= B(\zeta)X(v)B(\zeta)^*, \\ Z_{2,0} &= B(\zeta)X(w)B(\zeta)^*. \end{aligned} \tag{953}$$

The codomain assertion follows from the classified theorem and the exact residual intertwining in Lemma 24.67.

Theorem 24.68 (Exact quotient parameterization of the entire literal domain)**.** *The map $\mathfrak{F}$ is onto $\mathcal{S}_{\mathrm{sg}}$. The map*

$$\iota(u, v_0, y, \epsilon, \sigma, p_+, p_-, \zeta) = (u, v_0, y, \epsilon, \sigma, -p_+, -p_-, -\zeta) \tag{954}$$

is a free involution of $\mathcal{P}_{\mathrm{lit}}$. Each fibre of $\mathfrak{F}$ is exactly one two-element ι-orbit. Consequently (953) *induces a bijection of sets*

$$\overline{\mathfrak{F}} : \mathcal{P}_{\mathrm{lit}}/\langle\iota\rangle \longrightarrow \mathcal{S}_{\mathrm{sg}}. \tag{955}$$

The inverse of this bijection is recovered from the literal matrices by the formulas in the proof, with both square-root representatives retained.

Proof. We first construct a preimage of every literal tuple, recording all inverse coordinates. Given $(t, \omega, C_0, Z_{1,0}, Z_{2,0}) \in \mathcal{S}_{\mathrm{sg}}$, compute

$$C_d = U^* C_0 U, \qquad e = -i(C_d)_{11}, \qquad z = (C_d)_{23}, \qquad \gamma = -2iz/e.$$

Lemma 24.67 proves $e \in \mathbb{R} \setminus \{0\}$ and $|\gamma| = 1$. Choose either square root $\zeta^2 = -\bar{\gamma}$. Define

$$Z'_j = B(\zeta)^* Z_{j,0} B(\zeta), \qquad v = \begin{pmatrix} (Z'_1)_{21} \\ (Z'_1)_{31} \end{pmatrix}, \qquad w = \begin{pmatrix} (Z'_2)_{21} \\ (Z'_2)_{31} \end{pmatrix}.$$

The lemma proves that these are classified solutions with the original t, ω, e. Compute

$$M = vv^* + ww^*, \qquad N = i(vw^* - wv^*), \qquad u = M_{11}, \quad v_0 = M_{22}, \quad y = \operatorname{Im} M_{12},$$

$$\epsilon = \operatorname{sign} e, \qquad \sigma = \operatorname{sign}(e/\omega).$$

Theorem 24.66 proves $(u, v_0, y) \in \mathcal{D}$, $M_{12} = \delta + iy$, and all equations in (950) with this unchanged t, ω, e. It proves $h_\pm = (M \pm N)_{11} > 0$, so (952) recovers unique unit p_+, p_-. The vector formulas recover the displayed v, w and the inverse frame map then recovers all three literal matrices. Thus (953) sends the recovered raw parameter to the original tuple. This proves surjectivity.

Next we prove that (954) preserves each fibre. It preserves the parameter domain, and applying it twice is the identity. It has no fixed point because $\zeta = -\zeta$ would imply $\zeta = 0$, contrary to $|\zeta| = 1$. Replacing ζ by $-\zeta$ preserves $\gamma = -\zeta^{-2}$ and gives $W(-\zeta) = -W(\zeta)$. Put $D_0 = \operatorname{diag}(1, -I_2)$. Then

$$\begin{aligned} B(-\zeta) &= B(\zeta) D_0, \\ D_0 \big(ie \operatorname{diag}(1, 0, -1)\big) D_0 &= ie \operatorname{diag}(1, 0, -1), \\ D_0 X(v) D_0 &= X(-v). \end{aligned}$$

Replacing (p_+, p_-) by $(-p_+, -p_-)$ replaces (v, w) by $(-v, -w)$ in (951), while leaving M, N, t, e, k, ω fixed. The preceding matrix equalities therefore give $\mathfrak{F} \circ \iota = \mathfrak{F}$ on the entire raw parameter domain. This proves that $\overline{\mathfrak{F}}$ is well defined.

To determine every fibre, suppose two raw parameters have the same literal image. The common temporal matrix C_d recovers the same e, z, γ by the inverse formulas above. Their frame roots therefore satisfy either $\zeta_2 = \zeta_1$ or $\zeta_2 = -\zeta_1$, since they square to the same nonzero complex number. If $\zeta_2 = \zeta_1$, the classified temporal and spatial matrices obtained by that common inverse frame map coincide. The classified bijection of Theorem 24.66 then forces equality of all other raw parameters.

If $\zeta_2 = -\zeta_1$, the identity $B(-\zeta) = B(\zeta) D_0$ implies

$$Z'_{j,2} = D_0 Z'_{j,1} D_0, \qquad v_2 = -v_1, \qquad w_2 = -w_1.$$

Hence $M_2 = M_1$ and $N_2 = N_1$, so the inverse coordinates u, v_0, y are equal. The original e, ω are equal, which also gives equal ϵ, σ. Consequently $h_\pm$ agree, and (952) gives

$$p_{+,2} = -p_{+,1}, \qquad p_{-,2} = -p_{-,1}.$$

Thus the two raw parameters differ by exactly ι. No further case is possible. Each fibre is therefore precisely one two-element orbit. This proves injectivity of $\overline{\mathfrak{F}}$ as well as its already-proved surjectivity.

Finally the inverse formulas starting from a literal tuple give the same orbit for either root choice, by the second fibre case. Starting from a raw parameter, applying $\mathfrak{F}$ and then those inverse formulas gives its orbit, by the two fibre cases. Starting from a literal tuple, applying the inverse formulas and then $\overline{\mathfrak{F}}$ gives that literal tuple by the first paragraph. These prove both compositions are identities on their stated domains, completing the quotient bijection. □

The construction retains $\operatorname{sign} e = \epsilon$ and $\operatorname{sign}\omega = \epsilon\sigma$. The involution changes neither sign. The two independent vector phases and the frame phase remain coordinates, subject only to the proved simultaneous-sign identification.

For explicit scope, the additional full spatial residuals would be

$$\begin{aligned} R_1 &= -\omega^2 Z_{1,0} - 2\omega[C_0, Z_{2,0}] + [C_0,[C_0, Z_{1,0}]] + \mathcal{K}_{Z_{1,0},Z_{2,0}} Z_{1,0} + (t/2)\mathcal{I}_{H_0} Z_{1,0},\\ R_2 &= -\omega^2 Z_{2,0} + 2\omega[C_0, Z_{1,0}] + [C_0,[C_0, Z_{2,0}]] + \mathcal{K}_{Z_{1,0},Z_{2,0}} Z_{2,0} + (t/2)\mathcal{I}_{H_0} Z_{2,0}. \end{aligned}$$

Neither $R_1 = 0$ nor $R_2 = 0$ is part of $\mathcal{S}_{\mathrm{sg}}$ or used in either proof above. The bijection therefore classifies exactly the two scalar equations and Gauss; it does not assert a solution of these two additional spatial equations.

24.28.5 A short exact family with every equation checked

Proposition 24.69. *Choose arbitrary $p_+, p_- \in U(1)$ and independent $\epsilon, \sigma \in \{1, -1\}$. Put*

$$M = \delta \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}, \quad t = 24\delta, \quad e = \epsilon\sqrt{2 + 3\delta}, \quad k = \frac{2\sigma}{3\sqrt{13}}, \quad \omega = \frac{3\sigma\sqrt{13}}{2} e,$$

$$N = \frac{\sigma\delta}{\sqrt{13}} \begin{pmatrix} 2 & -1 \\ -1 & -6 \end{pmatrix}, \qquad a_\pm = \sqrt{\delta} \begin{pmatrix} \sqrt{1 \pm 2\sigma/\sqrt{13}} \\ (1 \mp \sigma/\sqrt{13})/\sqrt{1 \pm 2\sigma/\sqrt{13}} \end{pmatrix}.$$

Define v, w by (944)*. For any retained unit frame phase ζ, these give original tuples by*

$$C_0 = B(ie\,\mathrm{diag}(1, 0, -1))B^*, \qquad Z_{1,0} = BX(v)B^*, \qquad Z_{2,0} = BX(w)B^*.$$

Every such tuple satisfies (930)–(932) *at the positive original coupling $g^2 = 24\delta$.*

Proof. The first diagonal factors are positive because $2/\sqrt{13} < 1$. For a sign s, the exact identity

$$(1 - s\sigma/\sqrt{13})^2 = (1 + 2s\sigma/\sqrt{13})(2 - 6s\sigma/\sqrt{13})$$

follows by multiplying out and using $\sigma^2 = 1$. Hence $a_\pm a_\pm^* = M \pm N$. The construction therefore has exactly the displayed M, N for every pair of phases. In particular

$$u = \delta, \quad v_0 = 2\delta, \quad m = \delta, \quad \operatorname{tr} M = 3\delta, \quad d = \delta^2, \quad b = \tfrac{1}{2}\operatorname{tr} N = -\frac{2\sigma\delta}{\sqrt{13}}.$$

The scalar identities are $e^2 = 2 + 3\delta$ and $2r\delta = 1$; substitution in (934) and (935) makes every entry zero. For the full Gauss block, direct multiplication gives

$$QM + MQ = \delta \begin{pmatrix} 2 & 3 \\ 3 & 8 \end{pmatrix}, \qquad tE - 2(QM + MQ) = \delta \begin{pmatrix} 12 & -6 \\ -6 & -36 \end{pmatrix}.$$

The same matrix is $(4\omega/e)N$, since $\omega/e = 3\sigma\sqrt{13}/2$. Thus $\mathcal{G} = 0$ entry by entry, and the upper entry is explicitly

$$\gamma_0 = 4e\delta + 20e\delta + 8\left(\frac{3\sigma\sqrt{13}}{2}e\right)\left(-\frac{2\sigma\delta}{\sqrt{13}}\right) = 0.$$

Finally (933) proves all three literal equations exactly. As an additional check on the construction, its determinant data are

$$T = \frac{3\delta}{2}\begin{pmatrix} 2 & -1 \\ -1 & -6 \end{pmatrix}, \quad \det T = -\frac{117}{4}\delta^2, \quad k^2 = \frac{4}{117},$$

agreeing with the general parameterization. Both signs of the original frequency are present because $\operatorname{sign}\omega = \epsilon\sigma$. □

Corollary 24.70 (Every positive original coupling occurs)**.** *For every specified original $\tau = g^2 > 0$, the literal two-cross scalar–Gauss domain contains tuples at $t = \tau$, with either original frequency sign and both temporal signs. It contains continuous phase families at each such coupling.*

Proof. For $\rho > 1/\sqrt{2}$ choose

$$M_\rho = \delta\begin{pmatrix} \rho & 1 \\ 1 & 2\rho \end{pmatrix}.$$

Its parameters obey $u = \delta\rho > 0$, $d = \delta^2(2\rho^2 - 1) > 0$, $c = 3\delta\rho > 0$ and $m = \delta$. Thus Theorem 24.66 gives the complete solution family at

$$t = \frac{72\delta^2(2\rho^2 - 1)}{3\delta\rho} = \frac{24\delta(2\rho^2 - 1)}{\rho}.$$

For a given $\tau > 0$, set

$$\rho = \frac{\tau + \sqrt{\tau^2 + 4608\delta^2}}{96\delta}.$$

Because $\sqrt{4608}\delta = 48\sqrt{2}\delta$, this is strictly greater than $1/\sqrt{2}$. Squaring the displayed defining equality gives $48\delta\rho^2 - \tau\rho - 24\delta = 0$, which is precisely $t = \tau$. No limiting passage or change of coupling is used. The independent choices of $\epsilon, \sigma, p_+, p_-$ supply the stated signs and phases; the inverse in Theorem 24.66 ensures different phase pairs yield different classified tuples. □

24.28.6 Exact scope and reproducibility

The results concern the simultaneous zero set of both scalar equations and Gauss. The construction never imposes either spatial equation and does not claim a full rotating-ansatz solution. It proves that the previously specified scalar–Gauss domain of the two-cross quantitative residual certificate is nonempty, in fact at every original positive coupling. This is an algebraic statement about the retained classical residuals and makes no quantum assertion.

The companion `check_scalar_gauss_domain.py` is a finite exact symbolic replay. It checks the full scalar and Gauss matrix formulas with eight independent real spatial coordinates, the determinant/trace identities, the explicit factors and both signs, and the original unitary frame maps. The rank, positivity, parameter domains, completeness and inverse-map proofs are given in this document; they are not inferred from sampled values or from a numerical residual tolerance.

The exact full literal parameter map and its inverse are proved in Theorem 24.68; its only identification changes the signs of the frame phase and both vector phases together. Combining Corollary 24.70 with Proposition 24.63 makes the stated spatial-error bound non-vacuous at every positive original coupling. This does not alter the full-solution exclusion in Theorem 24.60.

The complete source versions, the independently checked literal-domain addendum and the exact symbolic replay are retained under `supporting_materials/workbench/evidence/scalar_gauss_domain_20260905/`. The retained equations are those derived from the parent model in (864)–(868); this proof neither changes that model nor replaces its two spatial equations by the Gauss equation.

24.29 Smooth literal inverse charts and all positive-coupling fibres

The exact algebraic bijections of Theorems 24.66 and 24.68 admit a smooth refinement. We prove the parameter domain's convexity, give the phase quotient its explicit smooth coordinates, and construct smooth inverse maps from the literal matrix entries. We then give a global product description in which the first continuous coordinate is the unchanged original coupling $t = g^2$. The topology of every fixed-coupling fibre will follow from explicit inverse formulas.

24.29.1 Retained objects and the exact parameter domain

Keep the literal matrices and constants

$$H_0 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & -1/2 & 1/(2\sqrt{2}) \\ 0 & 1/(2\sqrt{2}) & 1/2 \end{pmatrix}, \quad q = \frac{1}{2\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad A_0 = \sqrt{2} \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

$$m^2 = \Gamma = 1, \qquad \kappa = 6, \qquad F_0 = [A_0, [A_0, H_0]] = -2H_0 + \operatorname{diag}(-2, 1, 1),$$

and set

$$r = \frac{\sqrt{6}}{4}, \quad \delta = \frac{1}{2r} = \sqrt{\frac{2}{3}}, \quad x = \frac{1}{2}, \quad d_0 = \frac{1}{2\sqrt{2}}, \quad U = \operatorname{diag}\left(1, \frac{1}{\sqrt{2r(r+x)}} \begin{pmatrix} d_0 & -(r+x) \\ r+x & d_0 \end{pmatrix}\right).$$

Thus $U^*U = I_3$, $\det U = 1$, $Uq = q$, and $U^*H_0U = \operatorname{diag}(0, r, -r)$, as proved in Lemma 24.67. Write

$$P = \operatorname{diag}(1, 0, 0), \qquad X(v) = \begin{pmatrix} 0 & -v^* \\ v & 0_2 \end{pmatrix}, \qquad \mathfrak{p} = X(\mathbb{C}^2) \subset \mathfrak{su}(3).$$

The original real ambient manifold for this selected spatial class is

$$\mathcal{E} = (0, \infty) \times (\mathbb{R} \setminus \{0\}) \times \mathfrak{su}(3) \times \mathfrak{p} \times \mathfrak{p}.$$

It has real dimension $1 + 1 + 8 + 4 + 4 = 18$. For $Y_{\mathrm{tf}} = Y - (\operatorname{tr} Y)I_3/3$, put

$$\mathcal{I}_{H_0} Y = ([H_0, [H_0, Y]] + qq^*Y + Yqq^*)_{\mathrm{tf}}, \qquad \mathcal{K}_{Z_1, Z_2} Y = -\sum_{j=1}^{2} [Z_j, [Z_j, Y]].$$

The subset $\mathcal{S}_{\mathrm{sg}} \subset \mathcal{E}$ is exactly the simultaneous zero set

$$\begin{aligned} [C_0, [C_0, H_0]] - \sum_j [Z_{j,0}, [Z_{j,0}, H_0]] - F_0 &= 0, \\ (C_0^2 - Z_{1,0}^2 - Z_{2,0}^2)q + 2q &= 0, \\ (t\mathcal{I}_{H_0} + 2\mathcal{K}_{Z_{1,0}, Z_{2,0}})C_0 - 4\omega[Z_{1,0}, Z_{2,0}] &= 0. \end{aligned}$$

Neither additional spatial residual is included in this definition. The topology of $\mathcal{S}_{\mathrm{sg}}$ initially means its subspace topology in $\mathcal{E}$; its embedded smooth structure will be proved.

For real coordinates (u, v_0, y) define

$$h = \delta^2 + y^2, \qquad m = \delta + iy, \qquad M = \begin{pmatrix} u & m \\ \bar{m} & v_0 \end{pmatrix}, \qquad d = uv_0 - h, \qquad c = 4v_0 - 5u,$$

$$\mathcal{D} = \{(u, v_0, y) \in \mathbb{R}^3 : u > 0,\ d > 0,\ c > 0\}.$$

Here h is a scalar, distinct from the two factor entries $h_\pm$ used below.

Lemma 24.71 (Convexity of the entire continuous parameter domain)**.** *The set $\mathcal{D}$ is a nonempty open convex subset of $\mathbb{R}^3$. In particular it is connected and has real dimension three.*

Proof. The inequalities are strict inequalities for continuous functions, so $\mathcal{D}$ is open. For $u > 0$ and $(a, b) \in \mathbb{C}^2$, the exact quadratic-form identity is

$$\begin{pmatrix} \bar{a} & \bar{b} \end{pmatrix} M \begin{pmatrix} a \\ b \end{pmatrix} = u \left| a + \frac{m}{u} b \right|^2 + \frac{d}{u} |b|^2.$$

It proves $M > 0$ when $u > 0, d > 0$. Conversely, $M > 0$ gives $u > 0$ by the first coordinate vector, and evaluation at $(a, b) = (-m/u, 1)$ gives $d/u > 0$. Thus the conditions $u > 0, d > 0$ are exactly positive definiteness of the Hermitian matrix in the affine slice $\operatorname{Re} M_{12} = \delta$.

If M_0, M_1 are positive definite and $0 \le \lambda \le 1$, then for every nonzero $z \in \mathbb{C}^2$,

$$z^*((1-\lambda)M_0 + \lambda M_1)z = (1-\lambda)z^* M_0 z + \lambda z^* M_1 z > 0.$$

The affine slice is preserved by this combination, and the additional linear inequality $4v_0 - 5u > 0$ is preserved as well. The coordinate map $(u, v_0, y) \mapsto M$ is affine and injective, so these facts prove convexity in the original real coordinates. Finally $(u, v_0, y) = (\delta, 2\delta, 0)$ has $u = \delta > 0$, $d = \delta^2 > 0$, and $c = 3\delta > 0$. It supplies a point and, by openness, a three-dimensional neighbourhood. Straight segments inside a convex set join any two points, proving connectedness. □

24.29.2 The phase quotient with every original phase retained

Write $\mathbb{T} = U(1) = \{z \in \mathbb{C} : |z| = 1\}$. On the original three phases, in the order (ζ, p_+, p_-), define

$$\Psi : \mathbb{T}^3 \longrightarrow \mathbb{T}^3, \qquad \Psi(\zeta, p_+, p_-) = \left(\zeta^2, \frac{p_+}{\zeta}, \frac{p_-}{\zeta}\right) = (\eta, \alpha, \beta). \tag{956}$$

Lemma 24.72 (Exact smooth phase quotient)**.** *The map Ψ is a surjective smooth group homomorphism with kernel $\{(1,1,1), (-1,-1,-1)\}$. It induces a diffeomorphism*

$$\mathbb{T}^3/\{(1,1,1), (-1,-1,-1)\} \;\longrightarrow\; \mathbb{T}^3.$$

The quotient smooth charts have explicit local inverses retaining both original square-root representatives.

Proof. Products, inverses on $\mathbb{T}$, and the square map are smooth; the displayed formulas commute with coordinatewise multiplication. An element is in the kernel precisely when $\zeta^2 = 1$ and $p_+ = p_- = \zeta$, giving exactly the two listed elements. Given any $(\eta, \alpha, \beta) \in \mathbb{T}^3$, choose either root $\zeta^2 = \eta$ and set $p_+ = \zeta\alpha$, $p_- = \zeta\beta$. This gives all preimages, so Ψ is onto and each fibre is exactly one simultaneous-sign pair.

Here are the smooth local charts, rather than only a set quotient. Fix $\eta_0 \in \mathbb{T}$ and one root $\zeta_0^2 = \eta_0$. On the open arc $\eta = \eta_0 e^{i\theta}$ with $-\pi < \theta < \pi$, the two lifts are exactly

$$\zeta_\pm = \pm\zeta_0 e^{i\theta/2}, \qquad p_{+,\pm} = \zeta_\pm \alpha, \qquad p_{-,\pm} = \zeta_\pm \beta.$$

Each formula is smooth. The two root arcs are disjoint and open in the preimage, and each maps diffeomorphically to the displayed arc times $\mathbb{T}^2$, using (956) as its inverse. Their simultaneous-sign interchange identifies exactly these two charts. The quotient topology agrees with these charts: the quotient projection is open because the inverse image of the image of any open set is that set united with its simultaneous-sign translate. On overlapping arcs the quotient coordinates are the same (η, α, β), so their transitions are the ordinary smooth torus transitions. These charts prove the asserted diffeomorphism. They do not select a global square root or discard either representative. □

24.29.3 Global formulas for literal matrices and their smooth inverses

We record all the algebraic parameters used by the smooth map. For $(u, v_0, y) \in \mathcal{D}$, set

$$Q = \operatorname{diag}(1, 2), \qquad E = \operatorname{diag}(2/3, -5/6), \qquad t = \frac{72d}{c}, \qquad T = \frac{t}{4}E - \frac{1}{2}(QM + MQ).$$

For independent signs $\epsilon, \sigma \in \{1, -1\}$ put

$$e = \epsilon\sqrt{2 + u + v_0}, \qquad k = \sigma\sqrt{\frac{d}{-\det T}}, \qquad \omega = e/k, \qquad N = kT, \qquad A_\pm = M \pm N.$$

Theorem 24.66 proves on this entire domain that $-\det T > 0$, that $A_\pm$ are positive semidefinite of rank one, and that $h_\pm = (A_\pm)_{11} > 0$. Consequently

$$a_\pm = \begin{pmatrix} \sqrt{h_\pm} \\ (A_\pm)_{21}/\sqrt{h_\pm} \end{pmatrix}$$

are defined everywhere in $\mathcal{D}$ for both signs.

For $(\eta, \alpha, \beta) \in \mathbb{T}^3$ put

$$v^\flat = \frac{\alpha a_+ + \beta a_-}{2}, \qquad w^\flat = \frac{i(\alpha a_+ - \beta a_-)}{2}.$$

$$L(\eta) = \frac{1}{\sqrt{2}}\begin{pmatrix} -1 & -1 \\ \eta & -\eta \end{pmatrix}, \qquad B^\flat(\eta) = U \operatorname{diag}(1, L(\eta)).$$

The superscript $\flat$ names these exact intermediate coordinates; it does not denote a metric dual or a change of magnitude. Multiplication gives

$$L(\eta)^* L(\eta) = I_2, \qquad \det L(\eta) = \eta, \qquad \det B^\flat(\eta) = \eta.$$

In particular these auxiliary matrices are unitary with their displayed determinant; they are not asserted to have determinant one. The precise relation to the original determinant-one frame is, for either root $\zeta^2 = \eta$,

$$\gamma = -\zeta^{-2}, \qquad W(\zeta) = \frac{1}{\sqrt{2}}\begin{pmatrix} -\zeta^{-1} & -\zeta^{-1} \\ \zeta & -\zeta \end{pmatrix}, \qquad L(\eta) = \zeta W(\zeta).$$

If $p_+ = \zeta\alpha$, $p_- = \zeta\beta$, the original vectors satisfy $v = \zeta v^\flat$ and $w = \zeta w^\flat$. Thus every phase and every determinant factor is retained.

Let

$$\mathcal{X} = \{1, -1\}^2 \times \mathcal{D} \times \mathbb{T}^3$$

have coordinates $(\epsilon, \sigma, u, v_0, y, \eta, \alpha, \beta)$. Define $\mathcal{F}^\flat : \mathcal{X} \to \mathcal{E}$ by

$$\begin{aligned} \mathcal{F}^\flat(\epsilon, \sigma, u, v_0, y, \eta, \alpha, \beta) &= (t, \omega, C_0, Z_{1,0}, Z_{2,0}), \\ C_0 &= B^\flat(\eta)\big(ie \operatorname{diag}(1, 0, -1)\big)B^\flat(\eta)^*, \\ Z_{1,0} &= B^\flat(\eta)X(v^\flat)B^\flat(\eta)^*, \\ Z_{2,0} &= B^\flat(\eta)X(w^\flat)B^\flat(\eta)^*. \end{aligned} \tag{957}$$

Theorem 24.73 (Four smooth connected components in literal coordinates)**.** *The image of $\mathcal{F}^\flat$ is exactly $\mathcal{S}_{\mathrm{sg}}$, and $\mathcal{F}^\flat$ is a diffeomorphism onto an embedded six-real-dimensional submanifold of $\mathcal{E}$. The submanifold has exactly four connected components, indexed by*

$$\epsilon = \operatorname{sign} e, \qquad \sigma = \operatorname{sign}(e/\omega).$$

Each component is diffeomorphic to $\mathcal{D} \times \mathbb{T}^3$. The smooth structure and inverse charts refer to the literal matrix subset; no quotient of literal configurations by a physical gauge action is being taken.

Proof. First verify exactly which literal matrices the displayed formulas give. Let $D_\zeta = \operatorname{diag}(1, \zeta I_2)$ and let $B(\zeta) = U \operatorname{diag}(1, W(\zeta))$ be the original frame. Then

$$B^\flat(\eta) = B(\zeta) D_\zeta,$$
$$D_\zeta (ie \operatorname{diag}(1,0,-1)) D_\zeta^* = ie \operatorname{diag}(1,0,-1).$$

$$D_\zeta X(v^\flat) D_\zeta^* = X(\zeta v^\flat) = X(v), \qquad D_\zeta X(w^\flat) D_\zeta^* = X(w).$$

Thus (957) gives exactly the same literal tuple as the original map for each of its two phase lifts. Lemma 24.72 and the already-proved literal quotient bijection, Theorem 24.68, therefore show that its image is $\mathcal{S}_{\mathrm{sg}}$ and that $\mathcal{F}^\flat$ is bijective.

Here is its inverse directly in literal matrix entries. Given a tuple in $\mathcal{S}_{\mathrm{sg}}$, compute

$$C_d = U^* C_0 U, \qquad e = -i(C_d)_{11}, \qquad z = (C_d)_{23}, \qquad \eta = -\frac{2i\bar{z}}{e}. \tag{958}$$

The temporal classification proves that e is real and nonzero, $|z| = |e|/2$, and hence $|\eta| = 1$. The sign is checked by multiplying the lower temporal block in (957):

$$(C_d)_{\mathrm{lower}} = -\frac{ie}{2} \begin{pmatrix} 1 & \bar{\eta} \\ \eta & 1 \end{pmatrix}.$$

In particular $z = -ie\bar{\eta}/2$, so $-2i\bar{z}/e = \eta$ exactly.

Define the two matrices and their entire lower first columns by

$$Z_j^\flat = B^\flat(\eta)^* Z_{j,0} B^\flat(\eta), \qquad v^\flat = \begin{pmatrix} (Z_1^\flat)_{21} \\ (Z_1^\flat)_{31} \end{pmatrix}, \qquad w^\flat = \begin{pmatrix} (Z_2^\flat)_{21} \\ (Z_2^\flat)_{31} \end{pmatrix}.$$

Then recover

$$M = v^\flat (v^\flat)^* + w^\flat (w^\flat)^*, \qquad N = i(v^\flat (w^\flat)^* - w^\flat (v^\flat)^*),$$
$$u = M_{11}, \quad v_0 = M_{22}, \quad y = \operatorname{Im} M_{12}, \qquad \epsilon = \operatorname{sign} e, \quad \sigma = \operatorname{sign}(e/\omega),$$
$$h_\pm = (M \pm N)_{11}, \qquad \alpha = \frac{(v^\flat - i w^\flat)_1}{\sqrt{h_+}}, \qquad \beta = \frac{(v^\flat + i w^\flat)_1}{\sqrt{h_-}}. \tag{959}$$

Indeed multiplication by the common unit ζ preserves both M and N. The original phase inverse therefore gives $\alpha = p_+/\zeta$, $\beta = p_-/\zeta$ in these formulas, with $h_\pm > 0$, $|\alpha| = |\beta| = 1$. The algebraic inverse proves that all recovered coordinates lie in their exact domains. The displayed formulas recover the original parameter from (957), and the forward map recovers the original literal tuple from them.

All forward functions are real smooth on each sign sector: $c > 0$, $d > 0$, $-\det T > 0$, $2 + u + v_0 > 0$, and $h_\pm > 0$ are precisely the denominators and square-root arguments appearing above, while $k \neq 0$. Matrix products and conjugates are real smooth. In the inverse, $e \neq 0$ and $h_\pm > 0$ ensure smoothness of every non-discrete formula. The signs are locally constant. The inverse formulas in particular give a continuous inverse for the subspace topology on $\mathcal{S}_{\mathrm{sg}}$.

We check embeddedness, rather than just transferring an abstract smooth structure through a bijection. At any one image point choose open torus arcs around η, α, β, with their real angle coordinates. The inverse formulas extend as real smooth functions to an open neighbourhood in $\mathcal{E}$ as follows. Keep $e \neq 0$ and the computed $h_\pm > 0$ by taking a sufficiently small neighbourhood. The same matrix formulas define the complex quantities η, α, β there, even when the tuple is off the zero set. For a unit base point η_0, on the complex neighbourhood $\operatorname{Re}(\eta/\eta_0) > 0$, its local angle is the smooth real function

$$\arctan\left(\frac{\operatorname{Im}(\eta/\eta_0)}{\operatorname{Re}(\eta/\eta_0)}\right).$$

Use the corresponding expressions about α_0, β_0. On the zero set these are exactly its three local torus coordinates. Together with u, v_0, y, they give a real smooth six-coordinate map from an ambient neighbourhood whose composition with $\mathcal{F}^\flat$ on the parameter chart is the identity. The derivative of that composition is the identity, so the derivative of $\mathcal{F}^\flat$ has rank six at every point.

For clarity, the real inverse function theorem [82, §1, pp. 4–5] now applies with its full-rank hypothesis verified: choose six ambient linear coordinate projections whose composite with this rank-six parametrization has invertible derivative. Locally the image is a smooth graph over those six coordinates. The continuous inverse already proved prevents other parameter patches from contributing additional image points arbitrarily near this point outside its chart: the inverse image of the chart is open in the subspace topology. Thus the graph is a neighbourhood of the point in the whole subset $\mathcal{S}_{\mathrm{sg}}$. This proves it is an embedded six-dimensional submanifold and that the displayed inverse is smooth for its embedded structure.

It remains to count components exactly. By Lemma 24.71, $\mathcal{D}$ is path connected. Each circle is path connected: for chosen real arguments of two unit endpoints, interpolating those arguments gives an explicit path on the circle. Products of these paths prove $\mathbb{T}^3$ is path connected. Hence each of the four sign sectors in $\mathcal{X}$ is nonempty and connected. On the literal domain the two sign functions in (959) are continuous with values in $\{1, -1\}$ because e, ω never vanish. They cannot change on a connected subset, so distinct sectors cannot belong to the same component. This proves there are exactly four connected components.

Finally the only quotient used to construct the coordinates was the two-element kernel of the parameter map. Its two representatives give identical literal matrices, as the first displayed equalities proved. Distinct literal tuples remain distinct under the bijection. No further gauge equivalence or spatial-pair equivalence has been imposed. □

24.29.4 The original coupling is a surjective submersion

On $\mathcal{D}$ the original coupling function is

$$t(u, v_0, y) = \frac{72(uv_0 - \delta^2 - y^2)}{4v_0 - 5u} = \frac{72d}{c}.$$

Theorem 24.74 (Exact differential and global coupling coordinates)**.** *The function $t : \mathcal{D} \to (0, \infty)$ is a smooth surjective submersion. Its exact partial derivatives are*

$$\begin{aligned} \frac{\partial t}{\partial u} &= \frac{72(4v_0^2 - 5h)}{c^2}, \qquad \frac{\partial t}{\partial v_0} = \frac{72(4h - 5u^2)}{c^2}, \\ \frac{\partial t}{\partial y} &= -\frac{144y}{c}, \qquad h = \delta^2 + y^2. \end{aligned} \tag{960}$$

In fact $\partial t/\partial u > 0$ throughout $\mathcal{D}$.

Define the fixed open set

$$\mathcal{B} = \{(v_0, y) \in \mathbb{R}^2 : v_0 > \tfrac{\sqrt{5}}{2}\sqrt{\delta^2 + y^2}\}.$$

The map

$$\Theta : \mathcal{D} \longrightarrow (0, \infty) \times \mathcal{B}, \qquad \Theta(u, v_0, y) = (t(u, v_0, y), v_0, y) \tag{961}$$

is a diffeomorphism, with exact inverse

$$u = \frac{4\tau v_0 + 72(\delta^2 + y^2)}{72v_0 + 5\tau}, \qquad (\tau, v_0, y) \in (0, \infty) \times \mathcal{B}. \tag{962}$$

Moreover $\mathcal{B}$ is diffeomorphic to $\mathbb{R}^2$ by the explicit inverse pair

$$\begin{aligned} (y, s) &\longmapsto \left(\tfrac{\sqrt{5}}{2}\sqrt{\delta^2 + y^2} + e^s,\ y\right), \\ (v_0, y) &\longmapsto \left(y, \log\left(v_0 - \tfrac{\sqrt{5}}{2}\sqrt{\delta^2 + y^2}\right)\right). \end{aligned} \tag{963}$$

Here $\tau = t = g^2$ is the unchanged positive original coupling.

Proof. Since $c > 0$, the quotient defining t is smooth. The derivatives $d_u = v_0$, $d_{v_0} = u$, $d_y = -2y$, $c_u = -5$, $c_{v_0} = 4$, $c_y = 0$ give

$$\frac{\partial t}{\partial u} = \frac{72(v_0 c + 5d)}{c^2} = \frac{72(4v_0^2 - 5h)}{c^2}, \qquad \frac{\partial t}{\partial v_0} = \frac{72(uc - 4d)}{c^2} = \frac{72(4h - 5u^2)}{c^2},$$

and the last derivative in (960) is $72(-2y)/c$. On $\mathcal{D}$, $v_0 = (d + h)/u > 0$, $c > 0$, and $d > 0$. Hence $v_0 c + 5d > 0$, proving $\partial t/\partial u > 0$. The differential therefore has rank one everywhere, proving the submersion assertion independently of any fibre-topology argument. In particular its three partial derivatives cannot simultaneously vanish. The proposed algebraic simultaneous-zero test also agrees: the first two zero equations would give $4v_0^2 = 5h$ and $4h = 5u^2$, so positivity of u, v_0 would imply $v_0 = 5u/4$, and then both $c = 0$ and $d = 0$, excluded from $\mathcal{D}$.

We now prove the exact global inverse and surjectivity. For a point of $\mathcal{D}$, the identity

$$4v_0^2 - 5h = v_0 c + 5d > 0$$

and $v_0 > 0$ imply $(v_0, y) \in \mathcal{B}$. Writing $\tau = 72d/c > 0$, multiply by the retained c to obtain

$$\tau(4v_0 - 5u) = 72(uv_0 - h), \qquad (72v_0 + 5\tau)u = 4\tau v_0 + 72h.$$

Its denominator $72v_0 + 5\tau$ is strictly positive, giving (962).

Conversely fix any $\tau > 0$ and $(v_0, y) \in \mathcal{B}$, and use (962). Here $h > 0$, $v_0 > 0$, and the numerator and denominator of u are positive. Exact multiplication, with no sign or factor removed, gives

$$\begin{aligned} c = 4v_0 - 5u &= \frac{72(4v_0^2 - 5h)}{72v_0 + 5\tau} > 0, \\ d = uv_0 - h &= \frac{\tau(4v_0^2 - 5h)}{72v_0 + 5\tau} > 0. \end{aligned} \tag{964}$$

Thus the recovered point is in $\mathcal{D}$, and $72d/c = \tau$ exactly. These calculations prove both compositions of (961) and (962) are identities. Both maps are smooth on their stated open domains.

The set $\mathcal{B}$ is nonempty; for example $(v_0, y) = (\sqrt{5}\delta/2 + 1, 0)$ belongs to it. Thus every $\tau > 0$ has a preimage, proving surjectivity. Finally $\delta > 0$ makes $\sqrt{\delta^2 + y^2}$ smooth for all real y, and the defining inequality of $\mathcal{B}$ makes the logarithm argument in (963) strictly positive. Exponential and logarithm are inverse smooth functions on their stated real domains. Substitution into (963) proves both compositions are identities, establishing the diffeomorphism with $\mathbb{R}^2$. □

24.29.5 Exact scope and topology of the positive-coupling level sets

Corollary 24.75 (All positive-coupling levels and their literal inverse charts)**.** *In the literal embedded manifold $\mathcal{S}_{\mathrm{sg}}$, the original coupling projection*

$$\pi : \mathcal{S}_{\mathrm{sg}} \longrightarrow (0, \infty), \qquad (t, \omega, C_0, Z_{1,0}, Z_{2,0}) \longmapsto t$$

is a smooth surjective submersion. There is an explicit diffeomorphism

$$\mathcal{S}_{\mathrm{sg}} \;\cong\; \{1, -1\}^2 \times (0, \infty) \times \mathbb{R}^2 \times \mathbb{T}^3 \tag{965}$$

under which π is projection onto the displayed unchanged $(0, \infty)$ coordinate. For every specified original $\tau > 0$, the complete fixed-coupling level $\pi^{-1}(\tau)$ is an embedded five-real-dimensional submanifold, with exactly four connected components. Each component is diffeomorphic to $\mathbb{R}^2 \times \mathbb{T}^3$. These statements concern precisely the scalar–Gauss domain with both original signs and all three phase coordinates retained.

Proof. Compose the literal diffeomorphism of Theorem 24.73 with the diffeomorphisms of Theorem 24.74. More explicitly, given

$$(\epsilon, \sigma, \tau, y, s, \eta, \alpha, \beta) \in \{1, -1\}^2 \times (0, \infty) \times \mathbb{R}^2 \times \mathbb{T}^3,$$

first put

$$v_0 = \frac{\sqrt{5}}{2}\sqrt{\delta^2 + y^2} + e^s, \qquad u = \frac{4\tau v_0 + 72(\delta^2 + y^2)}{72v_0 + 5\tau}.$$

Then apply (957) with these u, v_0, y and the retained signs and phases. Equation (964) proves that its coupling is exactly τ and every denominator is positive. The inverse begins with the literal matrix formulas (958) and (959), keeps $\tau = t$ from the original tuple, and recovers

$$s = \log\left(v_0 - \frac{\sqrt{5}}{2}\sqrt{\delta^2 + y^2}\right).$$

The preceding theorems prove that these maps are smooth, mutually inverse, and defined on every point of their stated domains. This proves (965).

In these exact product coordinates π is the projection onto τ. Its derivative is onto the one-dimensional tangent space of $(0, \infty)$, and every $\tau > 0$ occurs. The fixed-coordinate slice is a smooth embedded codimension-one submanifold. Transport through the diffeomorphism makes $\pi^{-1}(\tau)$ an embedded five-dimensional submanifold of $\mathcal{S}_{\mathrm{sg}}$, and hence of the original ambient manifold $\mathcal{E}$. Its inclusion into the fixed-coupling ambient hyperplane has the same smooth structure, since that coordinate is identically τ.

The formula identifies the entire fibre, without a missing parameter portion, with $\{1, -1\}^2 \times \mathbb{R}^2 \times \mathbb{T}^3$. The real plane is path connected by line segments; the three circles are path connected by the explicit angle paths used earlier. Their product is therefore connected. The two continuous nonvanishing sign invariants prevent any connection between distinct sign sectors. This proves both the exact count of four components and their stated diffeomorphism type. The fibre topology follows from these explicit inverse maps, not merely from the fact that π is a submersion. □

These manifolds are subsets of the space of ordered literal tuples. The only phase identification was equality of two parameter representatives producing the same tuple. No spatial equation has been imposed, and no physical gauge quotient, frequency-sign identification, or exchange of $Z_{1,0}$ and $Z_{2,0}$ has been made. The construction and all level-set statements use the open domain $t > 0$; they assign no boundary fibre at $t = 0$.

24.30 Sources, independent checks, and reproducibility

The evidence is under `supporting_materials/workbench/evidence/`, in subdirectory `quaternionic_4441_audit_20260904/`. The supplied files are preserved unchanged in its `source/` directory. The source claims used here are the complex-plus-triplet and Lorentz-slice sections of *Triality 4-4-4-1 Dynamic Completion*, and the coordinates, symmetry, positive fibres, and Jordan-completion sections of *Quaternionic 4441 Continuation*, both dated 4 September 2026. Their source TeX and original verification results remain distinguishable from this audit and from the independent checker `verify_exact.py` in that evidence directory. It implements quaternionic matrix multiplication independently, expands the full fifteen-real-variable determinant identity, checks the marked skew transport and charge conjugation, and verifies the positive-cone counterexample, retained Jordan coefficient, group composition, actual scalar acceleration, and electric sign. These checks supplement the proofs above. The separate `verify_invariant_sectors.py` expands the trace-free adjugate and the full complex tangency identity, and checks the retained nonzero-sector coordinates and fourth-order escape coefficient. The invariant-family theorem is proved by tangency and uniqueness, not inferred from finite tests. These checks do not substitute for checking other packet claims such as the gauge-orbit part of the vacuum theorem or the finite-lattice quantization theorem. The separate moving-sector and vacuum verifiers test the exact periodic equations, conserved charge,

fourteen-coordinate Hessian cubic, and metric eigenvalues. The scalar global-minimum theorem is proved above, independently of those gauge and quantum claims. The portable `verify_gauge_lift.py` checks all fourteen potential derivatives symbolically and all forty-six real homogeneous variational components at a retained exact two-jet, as well as the orbit, radial, and observation identities. This finite variational test supplements, and does not replace, the full variational proof above. The portable `exact_period_metric_checker.py` verifies the unchanged periods, inverse real period map, full affine fibres, monodromy, rank-loss constants, and a nonreal rational positive-definite example. The smoothness and operator inequalities are proved above rather than inferred from finite tests. The further portable checks `exact_hermitian_metric_spectrum_checker.py`, `verify_vacuum_kinetics.py`, `verify_mass_spectrum.py`, and `verify_arbitrary_gauge_lift.py` test the retained metric/cometric factors, integral quotient and spectral identities, all current-free continuation equations, gauge-generator Gram matrix and eigenvalues, and the complete real-connection residual elimination. Their exact assertion counts and replay logs are in `CLASSICAL_SPECTRUM_EXACT_REPLAYS.json`. Full proofs of existence, spectral completeness, limits and exclusions are given above; finite symbolic tests do not replace those arguments. The further portable `check_complex_rotation.py` verifies the constant-frame blocks, retained original-coordinate map, weighted Gauss inverse, and both sign reconstructions at an exact unrestricted test pair. `check_complex_constraints.py` expands the full unrestricted scalar entries, inertia, spatial stress and polynomial discriminant. Neither checker certifies a zero of the remaining system. The strict inequalities, equivalences, domains and exclusion proofs are the analytic arguments above.

The 5 September finite-fibre and projection evidence is in `supporting_materials/workbench/evidence/finite_coupling_20260905/`. The two original symbolic checkers retain the full eight-dimensional Gauss equation and all residual codomains. The distinct `check_projection_independent.py` uses only standard-library rational-pair and finite-field arithmetic, independently reconstructs the full eight-dimensional Gauss matrix, and reproduces the nonzero Sylvester determinant. Its 101 evaluations verify degree-twelve polynomial identities, not a sampled real coupling exclusion. Complete definitions and proofs are in Theorems 24.57 and 24.59; no existence inside the exceptional algebraic set, or general complex time-dependent lift, is inferred. The original scripts and certificates are retained in the evidence directory's `checkers/` subdirectories.

The separately dated scalar–Gauss continuation and independent review are retained in `supporting_materials/workbench/evidence/scalar_gauss_domain_20260905/`. The maintained proof is `scalar_gauss_domain_shared.tex`; it includes the full temporal-classification and literal-phase-quotient addendum. The portable `check_scalar_gauss_domain.py` replay passes all 25 exact groups, including every scalar and Gauss entry, both sign choices, the continuous phases, inverse frames and the prescribed-coupling formula. Positivity, rank exclusion, completeness and the two-element fibres are proved analytically in Theorems 24.66 and 24.68. This continuation supersedes the earlier absence of a scalar–Gauss nonemptiness assertion; it does not alter the historical proof snapshot: the two spatial residuals still cannot vanish simultaneously. The unrestricted exceptional-set existence problem remains open, and no quantum spectral conclusion follows from these classical residuals.

The subsequent shared source `scalar_gauss_smooth_shared.tex` proves the smooth phase quotient, embedded literal inverse charts, and the global trivialization over the unchanged positive coupling axis (Theorems 24.73 and 24.74, Corollary 24.75). These analytic proofs establish the component and fibre topology beyond the earlier set bijections; the earlier exact replay receipt retains its original scope.

25 Quantum confinement in the homogeneous gauge–scalar model

The object quantized here is a precisely specified interacting matrix mechanics: three anti-Hermitian traceless gauge matrices, one Hermitian traceless adjoint scalar, and one complex triplet. It is obtained by spatially constant substitution in the gauge–scalar action used in the quaternionic continuation. All of these fields remain variables. In particular the triplet is not constrained to a four-real-dimensional slice or

to a scalar minimum curve.

The conclusions are a compact-resolvent Hamiltonian, a unique spatially constant $SU(3)$-invariant ground state, and a strictly positive first excitation gap at fixed positive volume and couplings. The explicit form inequality (974) is the quantitative step. It combines the exact $SU(3)$ root frequencies with the full quartic scalar potential. Bosonic commutator-matrix discreteness is a classical result; the primary attribution and the complete proof in the retained metric are given below. The present use is to answer a concrete quantization question for the already specified adjoint-plus-triplet action, not to claim a new general discreteness mechanism.

For the classical comparison the spacetime signature is $(+---)$ and the original action density is

$$\mathcal{L} = \frac{1}{2g^2}\operatorname{tr} F_{\mu\nu}F^{\mu\nu} + \frac{1}{2}\operatorname{tr}(D_\mu\Phi D^\mu\Phi) + (D_\mu b)^* D^\mu b - V(\Phi, b),$$

where

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu], \quad D_\mu\Phi = \partial_\mu\Phi + [A_\mu, \Phi], \quad D_\mu b = \partial_\mu b + A_\mu b.$$

The potential is written in full below. The finite-model theorem does not require the construction of a compact complex structure on S^6, and it neither verifies nor refutes that separate global construction. The precise classical map to the torus field theory, its noninjective map on gauge orbits, and an explicit obstruction to one proposed quantum-subsector map are proved at the end.

25.1 The retained configuration space and action

Fix a flat spatial torus of volume $\mathcal{V}_3 > 0$, a gauge coupling $g^2 > 0$, and $\hbar > 0$. Retain $m^2 \geq 0$, $\Gamma > 0$ and $\kappa \in \mathbb{R}$. We quantize the finite-dimensional homogeneous mechanics, not a claimed invariant quantum sector of the full field theory. The configuration space, with no quotient already performed, is

$$\mathcal{Q} = \mathfrak{su}(3)^3 \oplus \operatorname{Herm}_3(\mathbb{C})_0 \oplus \mathbb{C}^3, \qquad \dim_{\mathbb{R}} \mathcal{Q} = 24 + 8 + 6 = 38.$$

Its elements are $q = (A_1, A_2, A_3, \Phi, b)$. On $\mathfrak{su}(3)$ use $(X, Y)_{\mathfrak{g}} = -\operatorname{tr}(XY)$; the scalar metric is exactly $\operatorname{tr}(d\Phi^2) + 2\,db^* db$. The trace-free condition on Φ is part of the linear state space.

For a homogeneous temporal connection A_0, set

$$E_i = \dot{A}_i + [A_0, A_i], \quad X = \dot{\Phi} + [A_0, \Phi], \quad v = \dot{b} + A_0 b, \quad s = \operatorname{tr}\Phi^2 + 2b^* b.$$

The spatial integral of the retained $(+---)$, anti-Hermitian density is

$$\begin{aligned} L_{\mathrm{hom}} &= \mathcal{V}_3\left[\frac{1}{g^2}\sum_i \|E_i\|_{\mathfrak{g}}^2 + \frac{1}{2}\operatorname{tr} X^2 + v^* v - U(q)\right], \\ U(q) &= \frac{1}{g^2}\sum_{i<j}\|[A_i, A_j]\|_{\mathfrak{g}}^2 + \frac{1}{2}\sum_i \operatorname{tr}[A_i, \Phi]^2 + \sum_i \|A_i b\|^2 + V(\Phi, b), \\ V(\Phi, b) &= \frac{m^2}{2}s + \frac{\Gamma}{4}s^2 - \kappa(\det\Phi - b^*\Phi b). \end{aligned} \tag{966}$$

Here $[A_i, \Phi]$ is Hermitian, so its displayed squared trace is nonnegative. There is no velocity for A_0.

Proposition 25.1 (Canonical momenta, Hamiltonian and exact Gauss map)**.** *Use the canonical one-form*

$$\vartheta = \sum_i -\operatorname{tr}(\Pi_i\, dA_i) + \operatorname{tr}(\Pi_\Phi\, d\Phi) + 2\operatorname{Re}(\pi_b^* db).$$

The momenta and Hamiltonian are

$$\begin{aligned} \Pi_i &= \frac{2\mathcal{V}_3}{g^2}E_i, \qquad \Pi_\Phi = \mathcal{V}_3 X, \qquad \pi_b = \mathcal{V}_3 v, \\ H_{A_0} &= H_0 - \mathcal{M}_{A_0}, \\ H_0 &= \frac{g^2}{4\mathcal{V}_3}\sum_i \|\Pi_i\|_{\mathfrak{g}}^2 + \frac{1}{2\mathcal{V}_3}\operatorname{tr}\Pi_\Phi^2 + \frac{1}{\mathcal{V}_3}\pi_b^*\pi_b + \mathcal{V}_3 U(q). \end{aligned} \tag{967}$$

For $\xi \in \mathfrak{su}(3)$, write

$$K_\xi(q) = ([\xi, A_i]_{i=1}^3, [\xi, \Phi], \xi b), \qquad \mathcal{M}_\xi = \vartheta(K_\xi) = (\mathcal{G}, \xi)_{\mathfrak{g}}.$$

The exact matrix-valued momentum map is

$$\mathcal{G} = \sum_i [A_i, \Pi_i] - [\Phi, \Pi_\Phi] - (b\pi_b^* - \pi_b b^*)_{\mathrm{tf}} = \mathcal{V}_3\left(\frac{2}{g^2}\sum_i [A_i, E_i] - (J_0)_{\mathrm{tf}}\right), \tag{968}$$

where $J_0 = [\Phi, X] + bv^* - vb^*$. Thus $\mathcal{G} = 0$ is precisely the full homogeneous Gauss equation.

Proof. Differentiate (966) using the stated duality pairings. In particular, differentiation of v^*v gives $2\mathcal{V}_3 \operatorname{Re}(v^*\delta\dot{b})$, explaining $\pi_b = \mathcal{V}_3 v$ rather than $2\mathcal{V}_3 v$ in this complex duality convention. The velocity is the covariant velocity minus K_{A_0}; its Legendre transform is consequently $H_0 - \vartheta(K_{A_0})$. The ordinary quadratic Legendre transform gives every coefficient of (967). Cyclicity gives

$$-\operatorname{tr}(\Pi_i[\xi, A_i]) = -\operatorname{tr}([A_i, \Pi_i]\xi), \quad \operatorname{tr}(\Pi_\Phi[\xi, \Phi]) = \operatorname{tr}([\Phi, \Pi_\Phi]\xi),$$

while anti-Hermiticity gives

$$2\operatorname{Re}(\pi_b^*\xi b) = \operatorname{tr}((b\pi_b^* - \pi_b b^*)\xi).$$

These identities prove the first expression for $\mathcal{G}$; substitution of the momenta proves the second. Variation of the independent A_0 multiplier imposes $\mathcal{G} = 0$, even after a temporal gauge is selected for the evolution. The primary momentum conjugate to A_0 is zero; A_0 is not an additional coordinate of $\mathcal{Q}$. Since $\mathcal{M}_\xi = p \cdot K_\xi$, canonical Poisson brackets give $\{\mathcal{M}_\xi, \mathcal{M}_\eta\} = \mathcal{M}_{[\xi,\eta]}$. Gauge invariance of every trace and norm in H_0 also gives $\{H_0, \mathcal{M}_\xi\} = 0$. □

Coordinates and measure without changing the metrics. Choose any fixed orthonormal basis T_a, $1 \le a \le 8$, for $-\operatorname{tr}$ on $\mathfrak{su}(3)$, and put $S_a = iT_a$. Then $\operatorname{tr}(S_a S_b) = \delta_{ab}$. Write

$$A_i = \sum_a a_{ia} T_a, \quad \Phi = \sum_a \phi_a S_a, \quad b_r = x_r + iy_r.$$

These are coordinates on the original spaces, not a rescaling of their metrics. The mass metric read from (966) is

$$M = \operatorname{diag}\left(\frac{2\mathcal{V}_3}{g^2} I_{24},\ \mathcal{V}_3 I_8,\ 2\mathcal{V}_3 I_6\right).$$

We use its constant volume measure

$$d\mu_M = C_M\, d^{24}a\, d^8\phi\, d^3x\, d^3y, \qquad C_M = \sqrt{\det M} = \frac{2^{15}\mathcal{V}_3^{19}}{(g^2)^{12}},$$

retaining this constant rather than setting it to one. The auxiliary Hilbert space is $\mathscr{H} = L^2(\mathcal{Q}, d\mu_M; \mathbb{C})$. In these coordinates canonical momentum is $-i\hbar\partial$. Thus the exact kinetic operator is

$$T = -\frac{\hbar^2 g^2}{4\mathcal{V}_3}\Delta_a - \frac{\hbar^2}{2\mathcal{V}_3}\Delta_\phi - \frac{\hbar^2}{4\mathcal{V}_3}(\Delta_x + \Delta_y). \tag{969}$$

Equivalently the complex momentum is $\widehat{\pi}_{b_r} = -i\hbar\partial_{\bar{b}_r} = -i\hbar(\partial_{x_r} + i\partial_{y_r})/2$; its adjoint product is $-\hbar^2(\partial_{x_r}^2 + \partial_{y_r}^2)/4$. The Schrödinger equation is $i\hbar\partial_t\psi = H\psi$.

25.2 The retained cubic and a global scalar bound

Lemma 25.2 (Sharp cubic bound in the original complex coordinates)**.** *For every traceless Hermitian* $\Phi \in \mathrm{Herm}_3(\mathbb{C})$ *and* $b \in \mathbb{C}^3$*, put*

$$s = \operatorname{tr} \Phi^2 + 2b^* b, \qquad N = \det \Phi - b^* \Phi b.$$

Then

$$|N| \le \frac{s^{3/2}}{3\sqrt{6}}. \tag{970}$$

Consequently, with all original coefficients retained,

$$V(\Phi, b) = \frac{m^2 s}{2} + \frac{\Gamma s^2}{4} - \kappa N \;\ge\; \frac{\Gamma s^2}{8} - C_\kappa, \qquad C_\kappa = \frac{\kappa^4}{54\Gamma^3}, \tag{971}$$

whenever $m^2 \ge 0$*,* $\Gamma > 0$*, and* $\kappa \in \mathbb{R}$*. This is an inequality for the full potential, not its replacement by the right-hand side.*

Proof. Choose a unitary diagonalization of Φ and apply the same unitary to b. The inverse conjugation restores both original fields; it preserves s, N and introduces no restriction on either field. Write the three real eigenvalues as $\lambda_1, \lambda_2, \lambda_3$, with sum zero, and let $z_i = |b_i|^2$, $u = \sum_i z_i$. If $u > 0$, the original cubic is the convex combination

$$N = \sum_{i=1}^{3} \frac{z_i}{u} (\lambda_1 \lambda_2 \lambda_3 - u\lambda_i).$$

For one index i and the other two indices j, k, put $d_i = \lambda_j - \lambda_k$. Trace zero gives

$$\lambda_j \lambda_k = (\lambda_i^2 - d_i^2)/4, \qquad s = (3\lambda_i^2 + d_i^2)/2 + 2u.$$

Hence each term in parentheses is exactly

$$\lambda_i(\lambda_j \lambda_k - u) = \lambda_i^3 - \frac{s\lambda_i}{2}, \qquad |\lambda_i| \le \sqrt{2s/3}.$$

If $u = 0$, the same identity equals N for each i. If $s = 0$, both fields vanish and the assertion is immediate. For $s > 0$, the critical points of $x^3 - sx/2$ on $[-\sqrt{2s/3}, \sqrt{2s/3}]$ are $x = \pm\sqrt{s/6}$. At these two points and the two endpoints, its absolute value is $s^{3/2}/(3\sqrt{6})$. The elementary extreme-value theorem proves the bound for every term and hence for their convex combination. Equality is attained, for example, at $\Phi = \operatorname{diag}(2r, -r, -r)$, $b = 0$, $r \ne 0$, since then $s = 6r^2$ and $N = 2r^3$.

Put $\alpha = |\kappa|/(3\sqrt{6})$ and $t = \sqrt{s} \ge 0$. Keep $m^2 s/2$ as a nonnegative term when bounding the original potential. The remaining estimate is

$$V \ge \frac{\Gamma t^4}{8} + \left(\frac{\Gamma t^4}{8} - \alpha t^3 \right).$$

For $\alpha > 0$ the expression in parentheses has its global minimum at $t = 6\alpha/\Gamma$, with value $-54\alpha^4/\Gamma^3 = -\kappa^4/(54\Gamma^3)$. For $\alpha = 0$ it is nonnegative. This proves (971) including both signs of κ. □

Theorem 25.3 (Friedrichs realization with an explicit domain)**.** *Let* $W(q) = \mathcal{V}_3 U(q)$ *and*

$$D = \operatorname{diag}\left(\frac{\hbar^2 g^2}{4\mathcal{V}_3} I_{24},\ \frac{\hbar^2}{2\mathcal{V}_3} I_8,\ \frac{\hbar^2}{4\mathcal{V}_3} I_6 \right), \qquad C = \frac{\mathcal{V}_3 \kappa^4}{54\Gamma^3}.$$

The differential expression $T+W=-\nabla\cdot D\nabla+W$ on $C_c^\infty(\mathcal{Q})$ has the following well-defined semibounded Friedrichs realization H. Its closed form and form domain are

$$\begin{aligned}\mathfrak{h}[\psi]&=\int\Big((\nabla\psi)^*D\nabla\psi+W|\psi|^2\Big)\,d\mu_M,\\ \mathscr{D}&=H^1(\mathcal{Q},d\mu_M)\cap L^2(\mathcal{Q},(W+C)d\mu_M),\\ D(H)&=\{u\in\mathscr{D}:-\nabla\cdot D\nabla u+Wu\in L^2(\mathcal{Q},d\mu_M)\ \textit{in distributions}\}.\end{aligned}\tag{972}$$

The operator is self-adjoint, bounded below by $-C$, and C_c^∞ is a form core. No compactness assertion is needed for this construction.

Proof. Lemma 25.2 proves directly in the original complex coordinates that $V\ge\Gamma s^2/8-\kappa^4/(54\Gamma^3)$. All the other terms of U are nonnegative. In particular $W+C$ is a continuous nonnegative polynomial, locally bounded everywhere.

The norm

$$\|u\|_{\mathscr{D}}^2=\int\Big((\nabla u)^*D\nabla u+(W+C+1)|u|^2\Big)\,d\mu_M$$

is complete. Indeed a Cauchy sequence converges in H^1 and in the weighted L^2 space, and their limits agree on every compact set. This proves closedness. Smooth radial cutoffs converging to one approximate any $u\in\mathscr{D}$ in this norm: the potential and gradient tails tend to zero, and the cutoff-gradient term is bounded by a constant times $R^{-2}\|u\|^2$. On a fixed compact set ordinary mollification converges in H^1, and the bounded weight there gives weighted convergence. Thus C_c^∞ is a form core and the form is densely defined.

For clarity one may construct its self-adjoint operator directly. The positive form $\mathfrak{h}+(C+1)\langle\cdot,\cdot\rangle$ is an inner product on $\mathscr{D}$. Riesz representation assigns to every $f\in\mathscr{H}$ a unique $u=Rf\in\mathscr{D}$ such that

$$\mathfrak{h}(u,v)+(C+1)\langle u,v\rangle=\langle f,v\rangle\quad(v\in\mathscr{D}).$$

It follows that R, viewed on $\mathscr{H}$, is bounded, self-adjoint, positive and injective; its range is dense because its kernel is zero. The spectral theorem for bounded self-adjoint operators gives the self-adjoint inverse R^{-1} on $\operatorname{Ran}R$. Set $H=R^{-1}-(C+1)$. This is precisely the Friedrichs operator of the original form core. Integration by parts on compact test functions identifies its distributional expression. Conversely, for $u\in\mathscr{D}$ whose displayed distribution is $f\in L^2$, that integration-by-parts identity extends from the form core to all of $\mathscr{D}$; the Riesz characterization gives $u\in D(H)$, $Hu=f$. This proves exactly the asserted domain. It does not replace the domain by an unjustified separate requirement $Wu,\Delta u\in L^2$. □

Theorem 25.4 (Spatially constant Gauss constraints and averaging)**.** *For $g\in SU(3)$, let*

$$R_gq=(gA_ig^{-1},g\Phi g^*,gb),\qquad(\mathcal{U}(g)\psi)(q)=\psi(R_{g^{-1}}q).$$

This is a strongly continuous unitary representation on $\mathscr{H}$. For each $\xi\in\mathfrak{su}(3)$, the operator

$$\widehat{\mathcal{M}}_\xi=-i\hbar K_\xi\cdot\nabla\quad\textit{on } C_c^\infty(\mathcal{Q})$$

is essentially self-adjoint. Its closure has domain

$$D(\widehat{\mathcal{M}}_\xi)=\{\psi\in\mathscr{H}:K_\xi\cdot\nabla\psi\in\mathscr{H}\ \textit{distributionally}\},\qquad\mathcal{U}(e^{t\xi})=e^{-it\widehat{\mathcal{M}}_\xi/\hbar}.$$

The spatially constant physical Hilbert space and projection are

$$\begin{aligned}\mathscr{H}_{\text{phys}}&=\{\psi:\mathcal{U}(g)\psi=\psi\ \forall g\in SU(3)\}=\bigcap_{a=1}^{8}\ker\widehat{\mathcal{M}}_{T_a},\\ \mathsf{P}\psi&=\frac{1}{\mathcal{V}_\mu}\int_{SU(3)}\mathcal{U}(g)\psi\,d\mu(g),\qquad\mathcal{V}_\mu=\int_{SU(3)}d\mu(g)>0.\end{aligned}\tag{973}$$

Here the Haar mass is retained explicitly. The space is infinite dimensional. The set $C_c^\infty(\mathcal{Q})^{SU(3)} = \mathsf{P}C_c^\infty(\mathcal{Q})$ *is dense in it and is a form core for its physical Hamiltonian*

$$H_{\rm phys} = H|_{\mathscr{H}_{\rm phys}}, \qquad D(H_{\rm phys}) = D(H) \cap \mathscr{H}_{\rm phys}.$$

Thus the physical Hamiltonian is self-adjoint and is exactly the Friedrichs operator of the restricted form; no quotient-coordinate measure or singular-orbit boundary condition has been inserted.

Proof. Each R_g preserves the three original inner products, hence the mass metric, its measure and the kinetic form. It preserves U by conjugation invariance of the trace/determinant and unitarity of the triplet action. Continuity holds first on compact smooth functions and then on L^2 by density and unitarity. The vector field K_ξ is linear with zero divergence: its flow is an orthogonal linear map for each block. Its action on compact smooth functions is therefore skew-adjoint under integration, so $-i\hbar K_\xi \cdot \nabla$ is symmetric. Differentiating the displayed unitary action gives the stated generator sign. Stone's theorem gives a self-adjoint generator. Its domain is the displayed distributional domain: one direction follows by differentiation against test functions; for the converse, distributional integration along the flow gives

$$\mathcal{U}(e^{t\xi})\psi - \psi = -\int_0^t \mathcal{U}(e^{s\xi})(K_\xi \cdot \nabla\psi)\, ds$$

as an L^2 identity, and strong continuity gives differentiability.

To verify that the original compact smooth domain is a core, use smooth cutoffs depending on the blockwise invariant Euclidean radius. They commute with K_ξ. Convolution with a radial smooth kernel also commutes with its orthogonal flow. Cutoff followed by convolution therefore approximates both ψ and $K_\xi \cdot \nabla\psi$ in L^2. This proves graph-core density and essential self-adjointness. On that common smooth core,

$$[K_\xi \cdot \nabla, K_\eta \cdot \nabla] = -K_{[\xi,\eta]} \cdot \nabla, \qquad [\widehat{\mathcal{M}}_\xi, \widehat{\mathcal{M}}_\eta] = i\hbar\widehat{\mathcal{M}}_{[\xi,\eta]}.$$

There is no divergence correction or ordering anomaly.

An invariant vector lies in every generator domain with derivative zero. Conversely the eight vanishing distributional derivatives give the vanishing derivative for every ξ, hence invariance under every one-parameter subgroup. These generate connected $SU(3)$. For example each unitary determinant-one matrix is unitarily diagonalizable with eigenphases whose sum can be chosen to be zero, so it is the exponential of a trace-free anti-Hermitian matrix. This proves the kernel characterization with its domain included.

The Bochner integral in (973) is bounded in norm by one. Haar invariance and inversion imply $\mathsf{P}^2 = \mathsf{P} = \mathsf{P}^*$, with range precisely the invariant vectors. Averaging a compact smooth function remains smooth and has compact support, because the union of its supports under a compact group is compact. Applying P to any compact-smooth approximation proves L^2 density. Nonzero smooth functions supported in pairwise disjoint annuli of the invariant squared radius are invariant and orthogonal. This proves infinite dimensionality by explicit test states.

The form and its domain are $\mathcal{U}(g)$-invariant. The Riesz characterization above therefore gives $\mathcal{U}(g)R = R\mathcal{U}(g)$; averaging gives $\mathsf{P}R = R\mathsf{P}$. Functional calculus proves that P reduces H, so its restriction has the claimed self-adjoint domain. Jensen's inequality for the positive form $\mathfrak{h} + (C+1)\|\cdot\|^2$ makes P a contraction in the form norm. Averaging a form-core approximation of an invariant vector proves the final form-core statement. □

An explicit quantum bound for the three retained gauge matrices

Let $\mathfrak{g} = \mathfrak{su}(3)$ as a real eight-dimensional vector space, with the retained metric

$$\langle X, Y\rangle = -\operatorname{tr}(XY), \qquad \|X\|^2 = -\operatorname{tr}(X^2).$$

For anti-Hermitian X, Y, the trace of XY is real, by taking its conjugate and using cyclicity. The metric is positive definite because $-\operatorname{tr}(X^2) = \operatorname{tr}(X^*X)$. Lebesgue measure and all Laplacians below are those of

this metric, not of the negative Killing form. Write $A = (A_1, A_2, A_3) \in \mathfrak{g}^3$. For two arbitrary retained numbers $a_g, b_g > 0$, consider on $L^2(\mathfrak{g}^3, dA)$ the nonnegative quadratic form

$$q_{a_g,b_g}[\psi] = a_g \sum_{i=1}^{3} \int \|\nabla_{A_i}\psi\|^2 \, dA + b_g \sum_{1\le i<j\le 3} \int \|[A_i, A_j]\|^2 |\psi|^2 \, dA, \tag{MQ.1}$$

initially on $C_c^\infty(\mathfrak{g}^3; \mathbb{C})$. The coefficients a_g, b_g are independent retained positive numbers; their subscripts distinguish them from the scalar data. No assertion about the scalar coupling or a Gauss-reduced operator is included in the theorem below.

Lemma 25.5 (The exact adjoint frequency sum, including root walls)**.** *Let $X \in \mathfrak{g}$, and let $\theta_1 \le \theta_2 \le \theta_3$ be the real eigenvalues of $-iX$, with $\sum_p \theta_p = 0$. The positive semidefinite real operator $K_X = -\operatorname{ad}_X^2$ has its eight eigenvalues, with real multiplicity, equal to*

$$0, 0, \quad (\theta_1-\theta_2)^2, (\theta_1-\theta_2)^2, \quad (\theta_1-\theta_3)^2, (\theta_1-\theta_3)^2, \quad (\theta_2-\theta_3)^2, (\theta_2-\theta_3)^2. \tag{MQ.2}$$

In particular, zeros in this list remain zeros at repeated eigenvalues. Define the frequency sum with real multiplicities by $\mathcal{S}(X) = \operatorname{tr}_{\mathbb{R}\mathfrak{g}} \sqrt{K_X}$. Then

$$\mathcal{S}(X) = 2\sum_{p<q} |\theta_p - \theta_q| = 4(\theta_3 - \theta_1), \qquad 2\sqrt{6}\,\|X\| \le \mathcal{S}(X) \le 4\sqrt{2}\,\|X\|. \tag{MQ.3}$$

Both constants are sharp. The lower equality at nonzero X occurs exactly when two eigenvalues coincide; the upper equality occurs exactly when the eigenvalues are $(-r, 0, r)$ for some $r \ge 0$. If each positive root is counted only once instead of using real multiplicity, the sum is $\mathcal{S}(X)/2$ and its sharp lower constant is $\sqrt{6}$.

Proof. Cyclicity gives $\langle [X,Y], Z\rangle = -\langle Y, [X,Z]\rangle$. Thus ad_X is skew-adjoint and $\langle Y, K_X Y\rangle = \|[X,Y]\|^2$. Unitary diagonalization of X is an isometry for the retained metric. Multiplication of that unitary by a scalar phase adjusts its determinant to one without changing its conjugation, so this diagonalization can also be performed by $SU(3)$.

For diagonal $X = i\operatorname{diag}(\theta_1, \theta_2, \theta_3)$, use the orthonormal Cartan vectors

$$D_1 = \frac{i}{\sqrt{2}} \operatorname{diag}(1, -1, 0), \qquad D_2 = \frac{i}{\sqrt{6}} \operatorname{diag}(1, 1, -2),$$

and, for $p < q$, the orthonormal pair

$$U_{pq} = \frac{E_{pq} - E_{qp}}{\sqrt{2}}, \qquad V_{pq} = \frac{i(E_{pq} + E_{qp})}{\sqrt{2}}.$$

Direct multiplication gives $[X, U_{pq}] = (\theta_p - \theta_q) V_{pq}$ and $[X, V_{pq}] = -(\theta_p - \theta_q) U_{pq}$, while D_1, D_2 commute with X. These eight orthonormal vectors span $\mathfrak{g}$, proving (MQ.2) and the exact sum in (MQ.3).

Put $r = \theta_2 - \theta_1 \ge 0$ and $s = \theta_3 - \theta_2 \ge 0$. The original eigenvalues are exactly

$$\theta_1 = -(2r+s)/3, \quad \theta_2 = (r-s)/3, \quad \theta_3 = (r+2s)/3, \quad \|X\|^2 = \frac{2}{3}(r^2 + rs + s^2), \quad \mathcal{S}(X) = 4(r+s).$$

Consequently

$$\mathcal{S}(X)^2 - 24\|X\|^2 = 16rs, \qquad 32\|X\|^2 - \mathcal{S}(X)^2 = \frac{16}{3}(r-s)^2.$$

Their nonnegativity proves the bounds and their vanishing proves every equality case. In particular the lower bound does not fail at a repeated eigenvalue: at a nonzero wall exactly four real frequencies remain positive. At $X = 0$ both sides vanish. □

Lemma 25.6 (Slice oscillators with all coefficients)**.** *For fixed $X \in \mathfrak{g}$ and $\alpha, \beta > 0$,*

$$-\alpha\Delta_Y + \beta\|[X,Y]\|^2 \;\geq\; \sqrt{\alpha\beta}\,\mathcal{S}(X) \quad \text{on } L^2(\mathfrak{g}, dY) \tag{MQ.4}$$

as a quadratic-form inequality. No positive frequency is assigned to the commuting coordinates.

Proof. For $\omega \geq 0$ and $f \in C_c^\infty(\mathbb{R};\mathbb{C})$, integration by parts proves the exact identity

$$\int |\sqrt{\alpha} f' + \sqrt{\beta}\,\omega y f|^2\,dy = \alpha \int |f'|^2\,dy + \beta\omega^2 \int y^2|f|^2\,dy - \sqrt{\alpha\beta}\,\omega \int |f|^2\,dy.$$

Indeed $2\,\mathrm{Re}(f'\bar{f}) = (|f|^2)'$ and the boundary term vanishes. Thus each real oscillator contributes $\sqrt{\alpha\beta}\,\omega$, including the correct zero contribution when $\omega = 0$.

Diagonalize the real symmetric K_X in an orthonormal basis. This orthogonal coordinate change preserves measure and Δ_Y, and the potential becomes the sum $\sum_{\nu=1}^{8} \omega_\nu^2 y_\nu^2$ with the frequencies in (MQ.2). Apply the preceding identity in each coordinate, integrate the other coordinates, and sum to obtain (MQ.4). This proof is for each fixed X. It requires no differentiable choice of the diagonalizing basis as X varies, and introduces no derivatives in X. □

Theorem 25.7 (An explicit coercive form bound and compact resolvent)**.** *The closure of (MQ.1) has domain*

$$\mathcal{D}(q_{a_g,b_g}) = \left\{ \psi \in H^1(\mathfrak{g}^3) : \sum_{i<j} \int \|[A_i, A_j]\|^2 |\psi|^2\,dA < \infty \right\}. \tag{MQ.5}$$

Let H_A be its nonnegative Friedrichs operator. Its formal expression on compactly supported smooth functions is exactly

$$H_A = -a_g \sum_i \Delta_{A_i} + b_g \sum_{i<j} \|[A_i, A_j]\|^2. \tag{MQ.6}$$

For every $0 \leq \vartheta < 1$ the following form bound holds:

$$q_{a_g,b_g}[\psi] \geq \vartheta a_g \int \|\nabla\psi\|^2\,dA + 2\sqrt{6(1-\vartheta)a_g b_g} \int \left(\sum_i \|A_i\| \right) |\psi|^2\,dA. \tag{MQ.7}$$

In particular, retaining half of the original kinetic energy gives

$$H_A \geq -\frac{a_g}{2}\Delta_{\mathfrak{g}^3} + 2\sqrt{3a_g b_g} \sum_i \|A_i\|. \tag{MQ.8}$$

The form-domain inclusion into $L^2(\mathfrak{g}^3)$ is compact. Consequently $(H_A + \mu I)^{-1}$ is compact for every $\mu > 0$, and H_A has only isolated eigenvalues of finite multiplicity, with no finite accumulation point.

Proof. The potential in (MQ.6) is a continuous nonnegative polynomial. The right side of (MQ.5), equipped with $\|\psi\|_2^2 + q_{a_g,b_g}[\psi]$, is complete: an associated Cauchy sequence converges in H^1 and also after multiplication by the square root of the potential; an almost-everywhere convergent subsequence identifies the second limit as that product. It is densely defined since it contains C_c^∞. Smooth cutoffs equal to one on a ball of radius R, zero outside the ball of radius $2R$, with derivative bounded by a constant times R^{-1}, approximate each member in this norm as $R \to \infty$. The derivative error tends to zero by the H^1 norm and the R^{-1} bound; the weighted error tends to zero by integrability. For each cutoff, convolution with a compactly supported smooth approximate identity converges in H^1, and also in weighted L^2 because the potential is bounded on the common compact support. This proves the stated core and closure.

For the lower bound start on this core and split the quadratic form exactly as

$$q_{a_g,b_g}[\psi] = \vartheta a_g \|\nabla\psi\|_2^2 + \sum_{\substack{i,j\in\{1,2,3\}\\ i\neq j}} \left\{ \frac{(1-\vartheta)a_g}{2} \|\nabla_{A_j}\psi\|_2^2 + \frac{b_g}{2} \int \|[A_i, A_j]\|^2 |\psi|^2\,dA \right\}. \tag{MQ.9}$$

Every derivative appears in two directed pairs, and every unordered potential pair also appears twice. Hence no kinetic or potential term has been counted more than its original coefficient. Apply (MQ.4) inside each directed summand with $X = A_i$, $Y = A_j$, $\alpha = (1-\vartheta)a_g/2$, $\beta = b_g/2$. Fubini integration in A_i and in the third matrix gives the lower bound

$$\frac{\sqrt{(1-\vartheta)a_g b_g}}{2} \int \mathcal{S}(A_i)|\psi|^2 \, dA$$

for that summand. There are two partners j for each fixed i. Their sum and (MQ.3) prove (MQ.7), and $\vartheta = 1/2$ gives (MQ.8). Extension to (MQ.5) follows from core convergence and Fatou's lemma for the nonnegative position weight.

Here is a direct compactness proof. Fix $0 < \vartheta < 1$ and put $c_\vartheta = 2\sqrt{6(1-\vartheta)a_g b_g} > 0$. In any one retained orthonormal basis of $\mathfrak{g}^3 = \mathbb{R}^{24}$, let $Q_R = [-R, R]^{24}$. Outside this cube, $\sum_i \|A_i\| \geq (\sum_i \|A_i\|^2)^{1/2} \geq R$. Thus

$$\int_{\mathbb{R}^{24}\setminus Q_R} |\psi|^2 \leq \frac{q_{a_g,b_g}[\psi]}{c_\vartheta R}, \qquad \int |\nabla\psi|^2 \leq \frac{q_{a_g,b_g}[\psi]}{\vartheta a_g}. \tag{MQ.10}$$

Partition Q_R into finitely many cubes of side $h = 2R/N$, and let $\Pi_{R,N}\psi$ be its mean on each small cube and zero outside Q_R. This has finite-dimensional range. On any cube Q of side h,

$$\int_Q |\psi - \psi_Q|^2 \leq h^2 \int_Q |\nabla\psi|^2. \tag{MQ.11}$$

For completeness, in one dimension the variance identity gives $\int_0^h |f - f_{\mathrm{av}}|^2 = (2h)^{-1} \int_0^h \int_0^h |f(x) - f(y)|^2 \, dx \, dy$. Cauchy–Schwarz in the fundamental theorem of calculus bounds this by $h^2 \int_0^h |f'|^2$. In a product cube, let P_j be averaging in the jth coordinate. The commuting orthogonal projections give the orthogonal decomposition

$$I - \prod_{j=1}^{24} P_j = \sum_{j=1}^{24} \left(\prod_{k<j} P_k \right) (I - P_j).$$

Applying the one-dimensional estimate to each summand and using that prior-coordinate averaging contracts the derivative norm proves (MQ.11), first for smooth functions and then by H^1 density.

Equations (MQ.10)–(MQ.11) give the explicit approximation bound

$$\|\psi - \Pi_{R,N}\psi\|_2^2 \leq q_{a_g,b_g}[\psi] \left(\frac{h^2}{\vartheta a_g} + \frac{1}{c_\vartheta R} \right). \tag{MQ.12}$$

On a bounded form-domain set choose first R large and then N large. The error is uniformly as small as prescribed, while $\Pi_{R,N}$ maps that set to a bounded subset of a fixed finite-dimensional space. Such a set has a finite net at any positive radius, by subdividing a bounding coordinate box. Thus the original set is totally bounded in L^2, proving compactness of the form-domain inclusion without a matrix-model discreteness theorem.

The closed form defines its Friedrichs operator by the usual Riesz construction: for each $f \in L^2$ and $\mu > 0$ there is a unique u with $q_{a_g,b_g}(u, v) + \mu\langle u, v\rangle = \langle f, v\rangle$ for every form-domain v. This follows by representing the bounded functional on the Hilbert space with that form inner product; coercivity is supplied by $\mu\|v\|_2^2$. The solution is $(H_A + \mu I)^{-1} f$ and obeys $q_{a_g,b_g}[u] + \mu\|u\|_2^2 \leq \|f\|_2^2/\mu$. Hence this resolvent factors through a bounded set in the form domain and its compact inclusion, so it is compact. The spectral theorem for the resulting compact self-adjoint resolvent gives finite multiplicities and eigenvalues tending to zero for the resolvent, equivalently eigenvalues of H_A tending to infinity. Since the resolvent is injective and its range is dense in an infinite-dimensional space, it has infinitely many such eigenvalues. This proves all assertions. □

The inequality is a quadratic-form inequality, not a pointwise lower bound on the quartic potential. That potential still vanishes on every commuting triple. The exact mechanism is (MQ.9): localizing the other matrices transversely to a large fixed matrix costs the nonzero oscillator ground energies in (MQ.4); their sum remains coercive even at a root wall.

Proposition 25.8 (An additional explicit lower bound on the energy origin)**.** *With precisely the energy origin in (MQ.6),*

$$H_A \geq 9(3/2)^{1/3} a_g^{2/3} b_g^{1/3}\, I. \tag{MQ.13}$$

This is a lower bound on the bottom of the operator, not a claimed lower bound of the same size on the difference between its first two energy levels.

Proof. First, for any $\alpha, \gamma > 0$,

$$-\alpha \frac{d^2}{dx^2} + \gamma|x| \geq \frac{3}{2^{8/3}} (\alpha\gamma^2)^{1/3}. \tag{MQ.14}$$

To prove it, take $f \in C_c^\infty(\mathbb{R})$ with $\|f\|_2 = 1$, and write $t = \|f'\|_2^2$ and $M = \int |x||f(x)|^2\, dx > 0$. Integration of $(|f|^2)'$ from both ends of the line gives $\|f\|_\infty^2 \leq \|f\|_2 \|f'\|_2 = \sqrt{t}$. Markov's inequality puts at least half of the mass inside $[-2M, 2M]$, so $1/2 \leq 4M\|f\|_\infty^2 \leq 4M\sqrt{t}$. Thus $tM^2 \geq 1/64$. Minimizing $\alpha/(64M^2) + \gamma M$ over $M > 0$, whose unique minimum occurs at $M^3 = \alpha/(32\gamma)$, proves (MQ.14). Homogeneity in f removes the unit-norm restriction.

Write $x_{i\nu}$ for the original metric-orthonormal real coordinates, $1 \leq i \leq 3$, $1 \leq \nu \leq 8$. Cauchy–Schwarz gives $\|A_i\| \geq 8^{-1/2} \sum_{\nu=1}^{8} |x_{i\nu}|$. By (MQ.8),

$$H_A \geq \sum_{i,\nu} \left\{ -\frac{a_g}{2} \partial_{x_{i\nu}}^2 + \sqrt{\frac{3a_g b_g}{2}}\, |x_{i\nu}| \right\}.$$

Apply (MQ.14) in each of these twenty-four coordinates and integrate the others. Its total constant is exactly

$$24 \frac{3}{2^{8/3}} \left(\frac{a_g}{2} \frac{3a_g b_g}{2} \right)^{1/3} = 9(3/2)^{1/3} a_g^{2/3} b_g^{1/3}.$$

Core closure gives the operator-form bound stated. □

Exact relation to the retained homogeneous gauge density. The canonical density uses signature $(+---)$ and anti-Hermitian gauge fields, and its pure gauge part is

$$\mathcal{L}_{\text{gauge,hom}} = \frac{1}{g^2} \sum_i \|E_i\|^2 - \frac{1}{g^2} \sum_{i<j} \|[A_i, A_j]\|^2, \qquad E_i = \dot{A}_i + [A_0, A_i].$$

The full scalar potential retained there is $V = m^2 s/2 + \Gamma s^2/4 - \kappa(\det \Phi - b^* \Phi b)$, with $s = \operatorname{tr} \Phi^2 + 2b^* b$. It is not part of the pure matrix operator (MQ.6). For the pure gauge homogeneous kinetic form on a finite torus of retained volume $\mathcal{V}_3 > 0$, fixing temporal gauge at the unreduced coordinate level gives $L = \mathcal{V}_3 \sum_i \|\dot{A}_i\|^2/g^2 - \mathcal{V}_3 \sum_{i<j} \|[A_i, A_j]\|^2/g^2$. Its canonical momenta are exactly $p_i = 2\mathcal{V}_3 \dot{A}_i/g^2$; the Legendre transform is

$$H_{\text{cl}} = \frac{g^2}{4\mathcal{V}_3} \sum_i \|p_i\|^2 + \frac{\mathcal{V}_3}{g^2} \sum_{i<j} \|[A_i, A_j]\|^2.$$

The specified canonical substitution $p_i = -i\hbar \nabla_{A_i}$ with $\hbar > 0$ therefore gives the exact coefficient map

$$a_g = \frac{\hbar^2 g^2}{4\mathcal{V}_3}, \qquad b_g = \frac{\mathcal{V}_3}{g^2}, \qquad a_g b_g = \frac{\hbar^2}{4}. \tag{MQ.15}$$

No coupling or volume is set to one. Under this map, (MQ.8) reads

$$H_A \geq -\frac{\hbar^2 g^2}{8\mathcal{V}_3}\Delta + \sqrt{3}\,\hbar \sum_i \|A_i\|,$$

and (MQ.13) reads $H_A \geq 9(3\hbar^4 g^2/(32\mathcal{V}_3))^{1/3} I$. These coefficient identities concern the displayed unreduced homogeneous quantum model. They impose neither the Gauss constraint nor the omitted scalar coupling, and assert no continuum or infinite-volume field-theoretic gap.

Primary literature and scope. The discreteness phenomenon for bosonic commutator matrix potentials is classical. B. Simon, *Some quantum operators with discrete spectrum but classically continuous spectrum*, Annals of Physics **146** (1983), 209–220, DOI `https://doi.org/10.1016/0003-4916(83)90057-X`, treats a compact semisimple Lie-algebra matrix operator in equation (3) and Section 7, Corollary 4. Section 2 gives the elementary zero-point-oscillator argument for the two-variable model. The present proof combines a direct slice argument with the explicit $\mathfrak{su}(3)$ root sum in the metric $-\operatorname{tr}(XY)$ and proves compactness directly by (MQ.12). It does not use the Fefferman–Phong discreteness theorem employed in that paper's Section 7, nor claim the underlying discreteness result as new.

The related finite-volume Yang–Mills analysis is M. Lüscher, *Some analytic results concerning the mass spectrum of Yang–Mills gauge theories on a torus*, Nuclear Physics B **219** (1983), 233–261, DOI `https://doi.org/10.1016/0550-3213(83)90436-4`. That field-theoretic small-volume setting is not substituted for the precisely specified matrix operator here; none of its perturbative conclusions or coefficients is assumed in the foregoing proof.

25.3 Compactness for the entire coupled Hamiltonian

Theorem 25.9 (All thirty-eight coordinates are quantum confined)**.** *Retain the configuration metric*

$$\gamma_E = \operatorname{tr}(d\Phi^2) + 2\,db^* db, \qquad E = \operatorname{Herm}_3(\mathbb{C})_0 \oplus \mathbb{C}^3,$$

and the metric $-\operatorname{tr}$ *on each of the three* $\mathfrak{su}(3)$ *factors. Let* $T_A = -\sum_i \Delta_{A_i}$ *and* $T_E = -\Delta_{\gamma_E}$*; these are nonnegative Laplacians. For every* $0 < \eta < 1$*, the full Hamiltonian* $H_{\mathrm{hom}} = H$ *obeys the quadratic-form inequality*

$$\begin{aligned} H_{\mathrm{hom}} \;\geq\; & \frac{\eta\hbar^2 g^2}{4\mathcal{V}_3} T_A + \frac{\hbar^2}{2\mathcal{V}_3} T_E + \hbar\sqrt{6(1-\eta)} \sum_{i=1}^{3} \|A_i\| \\ & + \frac{\mathcal{V}_3 \Gamma}{8} s^2 - \mathcal{V}_3 C_\kappa. \end{aligned} \tag{974}$$

Its Friedrichs realization on the entire thirty-eight-dimensional configuration space has compact resolvent. No commuting or scalar directions have been deleted in this assertion.

Proof. Apply the matrix estimate with $a_g = \hbar^2 g^2/(4\mathcal{V}_3)$ and $b_g = \mathcal{V}_3/g^2$. Their product is exactly $\hbar^2/4$. The scalar-gradient and gauge–scalar potentials

$$\frac{\mathcal{V}_3}{2} \sum_i \operatorname{tr}[A_i, \Phi]^2, \qquad \mathcal{V}_3 \sum_i \|A_i b\|^2$$

are nonnegative: $[A_i, \Phi]$ is Hermitian. Retain these terms in the definition of the operator and use their signs only in this lower bound. Lemma 25.2 now proves (974) on compactly supported smooth functions, and core closure extends it to the form domain.

Write q for the closed form and $\widetilde{q}[\psi] = q[\psi] + \mathcal{V}_3 C_\kappa \|\psi\|_2^2$. The preceding bound makes $\widetilde{q}$ nonnegative. In the original metric the squared radial coordinate is $r^2 = \sum_i \|A_i\|^2 + s$. The retained positive weight

$$W_\eta = \hbar\sqrt{6(1-\eta)} \sum_i \|A_i\| + \mathcal{V}_3 \Gamma s^2/8$$

tends to infinity as $r \to \infty$. More explicitly, outside the ball $r < R$ either $\sum_i \|A_i\|^2 \geq R^2/2$ or $s \geq R^2/2$, so

$$W_\eta \geq w_R := \min\left\{\hbar\sqrt{3(1-\eta)}R, \mathcal{V}_3\Gamma R^4/32\right\}.$$

Both coefficients are strictly positive for the stated parameters. Thus

$$\int_{r\geq R} |\psi|^2 \leq \widetilde{q}[\psi]/w_R.$$

The derivative bound from (974) is

$$c_0 \int |\nabla\psi|^2 \leq \widetilde{q}[\psi], \qquad c_0 = \min\{\eta\hbar^2 g^2/(4\mathcal{V}_3), \hbar^2/(2\mathcal{V}_3)\} > 0,$$

where the gradient uses the product metric just stated. Metric-orthonormal coordinates are an isometric representation, not a change to any coupling, original field, or volume. On the cube $[-R, R]^{38}$, replace a function by its averages on finitely many cubes of side $h = 2R/N$ and by zero outside the large cube. The coordinate-averaging argument in the matrix proof works in thirty-eight coordinates without changing its constant: the squared local error is at most $h^2 \int |\nabla\psi|^2$. The cube's complement lies outside $r < R$, so the total error is at most

$$\widetilde{q}[\psi]\left(h^2/c_0 + 1/w_R\right).$$

On every form-norm bounded set this tends uniformly to zero by first increasing R and then increasing N. The approximations have finite-dimensional range, and their norms are bounded by $\|\psi\|_2$. Therefore that set is totally bounded in L^2. The complete Hilbert form domain has compact inclusion in L^2. The Riesz construction with the positive form $\widetilde{q} + \|\cdot\|_2^2$ maps bounded L^2 sets to bounded form-domain sets. Its composition with this inclusion is the resolvent, proving compactness for the full operator. □

The original classical potential remains zero in its pure-gauge part for every commuting triple. Inequality (974) uses the derivative cost in directions transverse to such triples. The strictly positive coefficient in front of $\sum_i \|A_i\|$ does not assert a pointwise classical potential that was absent from the original density.

Theorem 25.10 (Strict positivity and the bottom spectral projection)**.** *For the actual polynomial Hamiltonian above and every $\tau > 0$, $e^{-\tau H}$ is positivity improving: if $0 \neq f \geq 0$, then $e^{-\tau H} f > 0$ almost everywhere. The same holds on the invariant Hilbert space. Moreover*

$$\inf \sigma(H_{\mathrm{phys}}) = \inf \sigma(H) =: E_0, \qquad \operatorname{rank} \mathbf{1}_{\{E_0\}}(H) \leq 1.$$

Every nonzero vector in that bottom eigenspace is a scalar multiple of one strictly positive normalized, $SU(3)$-invariant real function. These conclusions hold without assuming compactness.

Proof. We give the heat-kernel argument with its constants. In the above real coordinates the free kernel relative to Lebesgue measure is

$$p_\tau(q, r) = (4\pi\tau)^{-19} (\det D)^{-1/2} \exp\left(-\frac{(q-r)^{\mathsf{T}} D^{-1}(q-r)}{4\tau}\right).$$

Let B_s, $0 \leq s \leq \tau$, be a standard 38-dimensional Brownian motion and form the bridge

$$Z_s = (1 - s/\tau)q + (s/\tau)r + \sqrt{2D}\,(B_s - (s/\tau)B_\tau).$$

The Friedrichs kernel, relative to the retained measure $d\mu_M$, is

$$K_\tau(q, r) = \frac{p_\tau(q, r)}{C_M}\, \mathbb{E} \exp\left(-\int_0^\tau W(Z_s)\, ds\right). \tag{975}$$

Here τ is the semigroup parameter with inverse-energy units; Euclidean physical time is $\hbar\tau$. No value of $\hbar$ has been set to one in D.

For completeness, the formula can be justified within the present bounded-below polynomial setting. Put $W_n = \min(W + C, n)$. For bounded W_n, the Duhamel iteration of the free semigroup converges in norm, since its j-th ordered term is bounded by $(n\tau)^j/j!$. The Markov Gaussian densities of the bridge identify exactly these ordered integrals with the power-series expansion of $\mathbb{E}\exp(-\int_0^\tau W_n(Z_s)\,ds)$. Thus the kernel formula holds for the bounded potentials. The forms with potentials W_n increase to the form with potential $W+C$. Their resolvents converge strongly: for $u_n = (T+W_n+\lambda)^{-1}f$, $\lambda > 0$, the variational identities bound u_n in H^1; testing each fixed truncated form and passing weakly shows that any weak limit belongs to the limiting form domain. Compactly supported tests identify its unique limiting resolvent equation. The identities

$$\mathfrak{h}_n[u_n] + \lambda\|u_n\|^2 = \mathrm{Re}\langle f, u_n\rangle$$

and lower semicontinuity then force convergence of the L^2 norms, hence strong convergence. The spectral functional calculus gives strong convergence of their semigroups. Every bridge path is continuous on a compact interval and has bounded image. The continuous polynomial W is therefore bounded on that path, and monotone convergence of its truncated integrals and dominated convergence of their exponentials yield (975), restoring the factor $e^{C\tau}$. This is the bounded-below Brownian-bridge Feynman–Kac formula, also stated in B. Simon, *A Feynman–Kac Formula for Unbounded Semigroups*, arXiv:math-ph/9907022, Theorem 1.1, `https://arxiv.org/abs/math-ph/9907022`.

In particular the exponential in (975) is strictly positive almost surely and bounded above by $e^{C\tau}$. Since $p_\tau(q,r) > 0$, the kernel is strictly positive for all q, r. Its Gaussian upper bound makes it a valid L^2 kernel action, and integration against any nonzero nonnegative function proves positivity improvement. The invariant subspace is reducing and the action is the same positive kernel, giving its assertion as well.

To prove equality of the two spectral infima without presupposing an eigenfunction, take any $f \in \mathscr{D}$ and define

$$F(q) = \left(\frac{1}{\mathcal{V}_\mu}\int_{SU(3)} |f(R_{g^{-1}}q)|^2\,d\mu(g)\right)^{1/2}.$$

This is an invariant nonnegative function with $\|F\| = \|f\|$. Apply the norm inequality to the Hilbert-valued Sobolev function $q \mapsto (g \mapsto f(R_{g^{-1}}q))$. More explicitly, differentiating $(\int |f(R_{g^{-1}}q)|^2 + \varepsilon^2)^{1/2}$, using Cauchy–Schwarz, and then taking $\varepsilon \downarrow 0$, gives

$$\int (\nabla F)^* D \nabla F\,d\mu_M \le \frac{1}{\mathcal{V}_\mu}\int_{SU(3)}\int (\nabla \mathcal{U}(g)f)^* D \nabla \mathcal{U}(g) f\,d\mu_M\,d\mu(g).$$

One first argues locally for smooth functions and then uses form approximation and lower semicontinuity. Invariance of W gives equality of the potential integrals. Thus $F \in \mathscr{D}$ and $\mathfrak{h}[F] \le \mathfrak{h}[f]$. The variational characterization of the spectral infimum proves $\inf\sigma(H_{\mathrm{phys}}) \le \inf\sigma(H)$; the reverse inequality follows from restriction to a subspace.

Finally suppose the bottom spectral projection is nonzero, and take a real nonzero bottom eigenfunction u; real and imaginary parts are eigenfunctions because H is real. The weak-gradient inequality $|\nabla|u|| \le |\nabla u|$ gives $\mathfrak{h}[|u|] \le E_0\|u\|^2$. The lower spectral bound forces equality, and the spectral theorem then gives $|u| \in \ker(H - E_0)$. Positivity improvement makes $|u| > 0$ almost everywhere. A sign-changing real bottom eigenfunction is impossible: its positive and negative parts are both sent to strictly positive functions, so

$$|e^{-\tau H}u| < e^{-\tau H}|u|$$

almost everywhere, contradicting the eigenfunction identities. Two independent real bottom eigenfunctions would yield, by orthogonalization against the positive one, such a sign-changing eigenfunction. Thus the complex eigenspace has dimension one. Every $\mathcal{U}(g)|u|$ is another positive normalized bottom eigenfunction, so its one-dimensional proportionality constant is positive and of modulus one, hence one. This proves gauge invariance and all assertions of the theorem. □

Corollary 25.11 (Unique physical ground state and a finite-model gap)**.** *The actual homogeneous quantum Hamiltonian has one normalized positive ground state, unique up to phase and invariant under spatially constant $SU(3)$. The physical eigenvalues, counted with multiplicity, satisfy*

$$E_0 = E_0^{\mathrm{phys}} < E_1^{\mathrm{phys}} \le E_2^{\mathrm{phys}} \le \cdots, \qquad E_n^{\mathrm{phys}} \longrightarrow +\infty.$$

In particular $E_1^{\mathrm{phys}} - E_0 > 0$, without a numerical value being claimed.

Proof. The positive compact resolvent $(H + C + 1)^{-1}$ proved above has a nonzero top eigenvalue, whose inverse supplies an eigenfunction at the bottom of H. The strict-positivity theorem makes it unique and invariant. The physical space is infinite dimensional, and its compact resolvent has finite-multiplicity eigenvalues accumulating only at zero. Applying the inverse affine spectral relation proves the displayed sequence and its strict first inequality. □

This is a gap of the specified finite-dimensional quantum mechanics. No numerical value follows from positivity, and classical slow solutions do not compute its level spacings. The proof retains fixed positive $\hbar, g^2, \mathcal{V}_3, \Gamma$; it asserts no uniform gap in limits of these parameters or in a continuum limit.

Proposition 25.12 (Exact classical embedding and its quotient limitations)**.** *Let ι send each homogeneous configuration and velocity to the corresponding spatially constant fields on the trivial $SU(3)$ bundle over $\mathbb{T}^3$. The full action pulls back to* (966)*. Its homogeneous Euler–Lagrange and Gauss solutions are exact full classical solutions. On phase space use constant momentum densities $\Pi_i/\mathcal{V}_3, \Pi_\Phi/\mathcal{V}_3, \pi_b/\mathcal{V}_3$; then the full canonical one-form pulls back to ϑ exactly. The map is equivariant for spatially constant $SU(3)$. The induced map from its constant-group orbit space to the full torus gauge orbit space is not injective.*

Proof. Spatial derivatives of the retained coefficients vanish. Substitution in the density gives (966), and the full local Euler–Lagrange residuals are then constant in space. Vanishing of their homogeneous variations for every time-dependent matrix variation therefore makes those residuals zero pointwise, including the A_0 residual. They consequently vanish against arbitrary, not necessarily homogeneous, spacetime test variations. This proves the exact classical implication without a symmetry reduction heuristic. Integrating constant momentum densities against constant configuration variations multiplies them by $\mathcal{V}_3$, proving the asserted one-form pullback and thus its symplectic pullback. Constant-group equivariance is direct.

For the limitation, let L_1 be the first torus period and set

$$g(x_1) = \mathrm{diag}(e^{2\pi i x_1/L_1}, e^{-2\pi i x_1/L_1}, 1).$$

With $D = \partial + A$, gauge transformation is

$$A_i' = gA_ig^{-1} - (\partial_i g)g^{-1}, \quad \Phi' = g\Phi g^*, \quad b' = gb.$$

The homogeneous state $A_i = 0, \Phi = b = 0$ is thereby equivalent in the full torus theory to the homogeneous state

$$A_1' = -\frac{2\pi i}{L_1}\,\mathrm{diag}(1, -1, 0), \qquad A_2' = A_3' = 0, \qquad \Phi' = b' = 0.$$

No constant conjugation sends the zero A_1 to this nonzero matrix, proving noninjectivity of the orbit-space map. On the other hand the same spatial gauge function sends the constant $\Phi = E_{12} + E_{21}$ to

$$e^{4\pi i x_1/L_1}E_{12} + e^{-4\pi i x_1/L_1}E_{21},$$

which is not homogeneous. Thus spatial gauge maps do not act on the whole homogeneous ansatz. More generally, a gauge map preserving every homogeneous triplet $b = e_j$, $j = 1, 2, 3$, must have all three columns constant, and hence must itself be spatially constant. This argument applies in particular to any genuinely large transformation not homotopic to a constant. The explicit diagonal example above is not being mislabeled as topologically large in $SU(3)$; its periodicity alone already proves the required quotient obstruction. □

Why the classical embedding is not a quantum-sector theorem. There is a direct interaction obstruction already in a finite Fourier cutoff. Choose the trace-free Hermitian $S = \mathrm{diag}(1, -1, 0)/\sqrt{2}$, so $\mathrm{tr}\, S^2 = 1$, $\det S = 0$, and put

$$\Phi(x) = (q + y\cos(2\pi x_1/L_1))S, \qquad A_i = b = 0.$$

The quartic part of the spatially integrated potential is exactly

$$\frac{\Gamma\mathcal{V}_3}{4}\left(q^4 + 3q^2y^2 + \frac{3}{8}y^4\right).$$

Indeed the averages of the cosine, its square, its cube and its fourth power are $0, 1/2, 0, 3/8$. The mass and gradient terms separate in q, y, the cubic determinant vanishes, but the mixed term $3\Gamma\mathcal{V}_3q^2y^2/4$ remains. The kinetic density is $\mathcal{V}_3\dot{q}^2/2 + \mathcal{V}_3\dot{y}^2/4$; thus this example also retains the unequal Fourier-mode masses.

A putative product sector consisting of all q-wavefunctions times one fixed nonzero normalizable y-wavefunction Ω is not invariant under this interacting cutoff Hamiltonian. Here invariance means a reducing quantum subspace, equivalently that its orthogonal projection commutes with the unitary evolution. This includes the necessary domain issue: if that projection commuted with the Hamiltonian, applying it to $f(q)v(y) \in C_c^\infty$, with $\langle\Omega, v\rangle \neq 0$, would put $f(q)\Omega(y)$ in the operator domain for every $f \in C_c^\infty(\mathbb{R})$. Its distributional Hamiltonian is the separated q-term plus $f(q)H_y\Omega + cq^2f(q)y^2\Omega$, $c = 3\Gamma\mathcal{V}_3/4$. Integrating against two compact q-test functions with distinct averaged q^2 values isolates $H_y\Omega$ and $y^2\Omega$ as L^2 functions. Thus the following projections are legitimate even if no regularity of Ω was assumed initially. Projecting its action orthogonally to Ω, for arbitrary compact smooth q-wavefunctions, would require

$$(H_y\Omega)_\perp + \frac{3\Gamma\mathcal{V}_3}{4}q^2(y^2\Omega)_\perp = 0 \quad \text{for almost every } q.$$

Its two coefficients must vanish, implying $y^2\Omega = \lambda_\Omega\Omega$. Multiplication by y^2 has no nonzero $L^2(\mathbb{R})$ eigenfunction: its level sets have Lebesgue measure zero. Thus no fixed discarded-mode vacuum yields that product invariant sector. The distribution $\delta(y)$, which would impose a literally vanishing discarded coordinate, is not an L^2 state either. This does not rule out every possible correspondence with a full quantum theory; it disproves the specific automatic-subsector inference and states exactly which finite model has been quantized. No continuum quantum Hilbert space, full local-gauge quotient, renormalized Hamiltonian, or continuum mass-gap value is constructed by the finite-dimensional argument.

Gauss-law reference. D. Tong, *Lectures on Gauge Theory*, Section 2.2.1, equations (2.27)–(2.31), `https://www.damtp.cam.ac.uk/user/tong/gaugetheory.html`, retains the Gauss equation after temporal gauge and imposes it on physical states. His Hermitian-field normalization is not substituted for the retained anti-Hermitian normalization here; the exact factors and the sign of (968) were derived above from the given density. For this compact spatial torus, spatially constant $SU(3)$ is the finite gauge subgroup being imposed, rather than an asymptotic charge group at spatial infinity.

26 An exact moving-ground-state comparison for the retained Fourier pair

The result below concerns precisely the two-coordinate variational truncation

$$\Phi(x) = (q + y\cos(kx_1))S, \qquad S = \mathrm{diag}(1, -1, 0)/\sqrt{2}, \qquad k = 2\pi/L_1, \qquad A_i = b = 0.$$

It does not assume that this two-dimensional plane is an invariant subspace of the full classical field equations or a reducing subspace of an independently constructed full quantum field theory. Throughout, retain

$$\hbar > 0, \quad \mathcal{V}_3 > 0, \quad \Gamma > 0, \quad L_1 > 0, \qquad m^2 \geq 0,$$

$$\omega_* := \sqrt{m^2 + k^2} > 0, \qquad \omega(q) := \sqrt{\omega_*^2 + 3\Gamma q^2}.$$

All bounds will refer to these original parameters and coordinates. The quartic discarded-mode interaction is never removed from the operator.

26.1 Exact coefficients, measures, and self-adjoint operators

Since $\operatorname{tr} S^2 = 1$ and $\det S = 0$, the pullback of the cubic determinant is zero. The averages of $\cos(kx_1)$ and its first four powers are $0, 1/2, 0, 3/8$ for powers one through four. The exact integrated Lagrangian is therefore

$$L_2 = \frac{\mathcal{V}_3}{2}\dot{q}^2 + \frac{\mathcal{V}_3}{4}\dot{y}^2 - V_q(q) - \frac{\mathcal{V}_3\omega(q)^2}{4}y^2 - cy^4, \tag{976}$$
$$V_q(q) := \mathcal{V}_3\left(\frac{m^2q^2}{2} + \frac{\Gamma q^4}{4}\right), \qquad c := \frac{3\Gamma\mathcal{V}_3}{32} > 0.$$

In particular the gradient term is $\mathcal{V}_3 k^2 y^2/4$ and the mixed quartic term is $3\Gamma\mathcal{V}_3 q^2 y^2/4$. The momenta are $p_q = \mathcal{V}_3\dot{q}$ and $p_y = \mathcal{V}_3\dot{y}/2$. Thus the mass metric and its measures are

$$M_2 = \operatorname{diag}(\mathcal{V}_3, \mathcal{V}_3/2), \quad d\mu_q = \sqrt{\mathcal{V}_3}\,dq, \quad d\mu_y = \sqrt{\mathcal{V}_3/2}\,dy, \quad d\mu_2 = \frac{\mathcal{V}_3}{\sqrt{2}}\,dq\,dy = d\mu_q d\mu_y.$$

We retain these constants. Put

$$\mathscr{K} = L^2(\mathbb{R}, d\mu_y), \qquad \mathscr{L} = L^2(\mathbb{R}, d\mu_q), \qquad \mathscr{H}_2 = L^2(\mathbb{R}, d\mu_q; \mathscr{K}).$$
$$a := \frac{\hbar^2}{2\mathcal{V}_3}, \quad a_y := \frac{\hbar^2}{\mathcal{V}_3}, \quad b_q := \frac{\mathcal{V}_3\omega(q)^2}{4}.$$

The Hamiltonian is exactly

$$H_2 = -a\partial_q^2 + V_q(q) + h(q), \qquad h(q) = -a_y\partial_y^2 + b_q y^2 + cy^4. \tag{977}$$

The symbols a, a_y, b_q, c denote these retained coefficients, not a rescaling of either variable.

Every displayed nonnegative Schrödinger expression is realized by its Friedrichs form. For example

$$\mathcal{D}_y = H^1(\mathbb{R}, d\mu_y) \cap \{u : y^2u \in \mathscr{K}\}, \qquad h_q[u] = a_y\|u'\|^2 + b_q\|yu\|^2 + c\|y^2u\|^2.$$

Its form domain is independent of q, since $b_q, c > 0$. The full form domain consists of $\Psi \in H^1(\mathbb{R}^2, d\mu_2)$ for which the original nonnegative multiplication potential in (977) has finite integral against $|\Psi|^2$. Completeness follows by convergence in H^1 and after multiplication by the square root of that potential; an almost-everywhere subsequence identifies the limits. Smooth cutoffs and then compactly supported mollification prove that C_c^∞ is a form core. The operator domain is the form-domain vectors whose displayed differential expression belongs to L^2 distributionally, as follows by testing against the core and using the Riesz representation of the shifted form. Since $b_q y^2 + cy^4 \to \infty$ in y, the fibre form-domain inclusion is compact: the potential controls the L^2 tail and the derivative controls finite-cell averaging errors on bounded intervals. The same argument proves compactness for H_2, using $V_q(q) + cy^4 \to \infty$ as $q^2 + y^2 \to \infty$.

We shall also use the following elementary operator-core fact. For a smooth nonnegative real potential on $\mathbb{R}^n$ and a constant positive kinetic matrix, C_c^∞ is an operator core for the Friedrichs operator. Indeed an L^2 distributional solution of $(H^* + 1)u = 0$ is locally H^2 by local elliptic regularity. Testing against $\chi_R^2 u$, with a smooth compact cutoff whose gradient is $O(R^{-1})$, gives

$$\int (|\nabla(\chi_R u)|_D^2 + (V+1)|\chi_R u|^2) = \int |\nabla\chi_R|_D^2|u|^2 = O(R^{-2})\|u\|^2.$$

Letting $R \to \infty$ proves $u = 0$. Thus the range of $H + 1$ on the minimal domain is dense; its lower bound makes the range of its closure closed. That closure is surjective at -1 and symmetric, hence self-adjoint. It is therefore the Friedrichs realization and the stated domain is an operator core. This argument applies to the fibres, the full operator, and the effective operator below.

26.2 The actual normalized positive fibre ground state

Lemma 26.1 (Fibre energies and a proved quantitative separation)**.** *For every q, $h(q)$ has a simple lowest eigenvalue $e(q)$ and a unique real strictly positive eigenfunction Ω_q with $\|\Omega_q\|_{\mathscr{K}} = 1$. Write $e_1(q)$ for its second eigenvalue, counting multiplicity. Then*

$$\frac{\hbar\omega(q)}{2} \le e(q) \le \frac{\hbar\omega(q)}{2} + \frac{9\Gamma\hbar^2}{32\mathcal{V}_3\omega(q)^2}, \qquad e_1(q) \ge \frac{3\hbar\omega(q)}{2}. \tag{978}$$

In the explicit parameter regime

$$\frac{9\Gamma\hbar}{32\mathcal{V}_3\omega_*^3} \le \frac{1}{2}, \quad \textit{equivalently} \quad \mathcal{V}_3 \ge \frac{9\Gamma\hbar}{16\omega_*^3}, \tag{979}$$

these imply, simultaneously for every q,

$$e(q) \le \hbar\omega(q), \qquad e_1(q) - e(q) \ge \lambda(q) := \frac{\hbar\omega(q)}{2} > 0. \tag{980}$$

Proof. Compactness gives existence of a bottom eigenfunction. The scalar kinetic form does not increase on taking its modulus, so a nonnegative real eigenfunction can be selected. Alternatively, the continuous nonnegative potential has the Brownian-bridge kernel

$$\frac{\exp[-(y-z)^2/(4a_y\tau)]}{\sqrt{\mathcal{V}_3/2}\sqrt{4\pi a_y\tau}}\mathbb{E}\exp\left[-\int_0^\tau (b_q Z_s^2 + cZ_s^4)\,ds\right] > 0 \qquad (\tau > 0).$$

Here Z is the bridge from y to z with diffusion covariance $2a_y$. Each path is continuous with bounded image, so its exponential is strictly positive and bounded by one. The formula follows first for bounded truncations by the convergent Duhamel series and then by increasing-form resolvent convergence and dominated convergence of the bridge exponential. Consequently the semigroup is positivity improving. The modulus of any bottom eigenfunction is again a bottom eigenfunction; if a real bottom eigenfunction changed sign, positivity improvement on both its positive and negative parts would imply $|e^{-\tau h(q)}u| < e^{-\tau h(q)}|u|$, contradicting its eigenvalue equations. Orthogonalization against a positive bottom eigenfunction therefore rules out a second independent real one. Reality gives complex dimension one. The eigenfunction is smooth by the one-dimensional eigenvalue equation. A zero of a nonnegative smooth solution would have derivative zero, and uniqueness for that second-order ODE would force the solution to vanish. Thus the selected eigenfunction is strictly positive everywhere.

Put $\beta(q) = \mathcal{V}_3\omega(q)/(2\hbar)$ and use the exactly normalized harmonic comparison vector

$$\chi_q(y) = \big(\mathcal{V}_3/2\big)^{-1/4}\big(\beta(q)/\pi\big)^{1/4}\exp[-\beta(q)y^2/2], \qquad \|\chi_q\|_{\mathscr{K}} = 1.$$

The harmonic part $h_{\mathrm{osc}}(q) = -a_y\partial_y^2 + b_q y^2$ factorizes as

$$h_{\mathrm{osc}}(q) = a_y(-\partial_y + \beta y)(\partial_y + \beta y) + \hbar\omega/2.$$

Its ground state is χ_q and its first excitation has energy $3\hbar\omega/2$. To check the latter without presuming a missing spectral level, the harmonic operator has compact resolvent; its lowering operator $\partial_y + \beta y$ maps any eigenfunction above the ground to an L^2 distributional eigenfunction with energy lowered by $\hbar\omega$. Its norm is nonzero by the factorization. No energy strictly between $\hbar\omega/2$ and $3\hbar\omega/2$ is possible. The function $y\chi_q$ attains the latter energy. The operator-core fact above identifies the distributional eigenfunctions with the self-adjoint operator domain.

Since $cy^4 \ge 0$, the variational principle and its two-dimensional min–max version give the lower bounds for e and e_1. Gaussian integration gives, in the retained measure,

$$\langle\chi_q, y^2\chi_q\rangle = \frac{\hbar}{\mathcal{V}_3\omega}, \qquad \langle\chi_q, y^4\chi_q\rangle = \frac{3\hbar^2}{\mathcal{V}_3^2\omega^2}.$$

Its Rayleigh quotient is the upper bound in (978). This is only a trial-vector bound: $h(q)\chi_q = (\hbar\omega/2+cy^4)\chi_q$ is not proportional to χ_q because $c > 0$. In particular the true state Ω_q is not replaced by this Gaussian. Finally $\omega(q) \geq \omega_*$ and (979) make the upper correction at most $\hbar\omega(q)/2$, proving (980). □

Lemma 26.2 (An exact smooth eigenline, with its phase fixed)**.** *The functions $q \mapsto e(q)$ and $q \mapsto \Omega_q \in \mathscr{K}$ are locally real analytic, the latter also in the common local operator graph norm. Thus they define a global normalized positive ground-state line over the entire original q-axis. In this phase,*

$$\langle \Omega_q, \partial_q \Omega_q \rangle = 0, \qquad \mathcal{A}(q) := i\langle \Omega_q, \partial_q \Omega_q \rangle = 0.$$

Put $P_q = |\Omega_q\rangle\langle\Omega_q|$, $Q_q = I - P_q$, and

$$R_q = (h(q) - e(q))^{-1} Q_q, \qquad d := \frac{3\Gamma\mathcal{V}_3}{2}.$$

The inverse is on the orthogonal complement of the actual ground state. The exact derivative identities are

$$\begin{aligned} e'(q) &= dq\, \langle \Omega_q, y^2 \Omega_q \rangle, \\ \Omega_q' &= -dq\, R_q Q_q y^2 \Omega_q, \\ Q_q \Omega_q'' &= -R_q Q_q \big[dy^2 \Omega_q + 2(dqy^2 - e'(q))\Omega_q' \big], \\ \langle \Omega_q, \Omega_q'' \rangle &= -\|\Omega_q'\|^2. \end{aligned} \tag{981}$$

These identities hold without imposing (979)*.*

Proof. For u in the operator domain of a fixed fibre $h(q_0)$,

$$c\|y^2 u\|^2 \leq \langle u, h(q_0)u \rangle \leq \|u\|\, \|h(q_0)u\|.$$

The arithmetic–geometric mean inequality implies that multiplication by y^2 is $h(q_0)$-bounded with arbitrarily small relative bound. Since $h(q) - h(q_0) = (b_q - b_{q_0})y^2$, nearby real fibres have the same operator domain. One can see the corresponding resolvent analyticity directly: at a fixed resolvent point z,

$$(h(q) - z)^{-1} = (h(q_0) - z)^{-1} \big[I + (b_q - b_{q_0}) y^2 (h(q_0) - z)^{-1} \big]^{-1},$$

where the last inverse is a locally convergent Neumann series. For real q, the closed symmetric perturbation is self-adjoint: the same formula gives surjectivity at two conjugate nonreal points chosen far enough from the real axis. It agrees with the Friedrichs realization on the compact smooth operator core. The displayed formula is analytic also as a map into the original operator domain equipped with its graph norm.

A small contour around the isolated simple eigenvalue $e(q_0)$ defines the spectral projection by the resolvent integral. It is locally analytic in that graph norm, remains of rank one because close projections have isomorphic ranges, and gives the locally analytic eigenvalue. For a fully fixed positive reference vector use $\chi_*(y) = \chi_0(y)$, the preceding Gaussian at ω_*. Then the actual state is exactly

$$\Omega_q = \frac{P_q \chi_*}{\|P_q \chi_*\|_{\mathscr{K}}}. \tag{982}$$

The denominator is nonzero because the ground state and χ_* are both strictly positive. This formula selects the same real positive vector on overlaps of the local contour constructions, so the local analytic sections give a global section. It can equivalently be obtained by the convergent normalized heat iteration $e^{-\tau h(q)}\chi_*/\|e^{-\tau h(q)}\chi_*\|$ as $\tau \to \infty$, by the discrete spectral expansion and the nonzero ground-state overlap.

The real normalization identity gives $\langle \Omega_q, \Omega_q' \rangle = 0$. Differentiating the eigenvalue equation in its common graph domain gives $(h - e)\Omega' = -(dqy^2 - e')\Omega$. Taking its ground-state component proves the first identity, and taking its complementary component proves the second. A second differentiation gives $(h - e)\Omega'' = -(dy^2 - e'')\Omega - 2(dqy^2 - e')\Omega'$. Its complementary part gives the asserted reduced-resolvent formula; differentiating the normalization twice gives its ground-state component. All products in these identities are in $\mathscr{K}$, since y^2 is bounded in the local graph norm. □

26.3 Explicit global derivative bounds for the true quartic state

Lemma 26.3 (Moment and reduced-resolvent estimates)**.** *Assume the proved parameter regime* (979)*. Then, globally in q,*

$$\|\Omega_q'\| \le M_1 := \frac{\sqrt{15\Gamma}}{\omega_*}, \tag{983}$$

$$\|Q_q\Omega_q''\| \le M_2 := \frac{18\sqrt{5}\,\Gamma}{\omega_*^2} + \frac{192}{5^{3/4}}\sqrt{\frac{\Gamma\mathcal{V}_3}{\hbar\omega_*}}. \tag{984}$$

Proof. In this proof all fibre inner products use the retained measure $d\mu_y$. Let $\mu_j = \langle \Omega_q, y^j\Omega_q\rangle$ for even j. The energy identity gives $b_q\mu_2 \le e(q)$. To get the necessary sharper fourth moment, take the bounded Lipschitz multiplier $F_R(y) = \max(-R, \min(y, R))$. Testing the weak eigenvalue equation with $F_R^2\Omega_q$ gives the exact identity

$$a_y\|(F_R\Omega_q)'\|^2 + \int (b_q y^2 + cy^4)F_R^2|\Omega_q|^2\,d\mu_y = e(q)\|F_R\Omega_q\|^2 + a_y\int |F_R'|^2|\Omega_q|^2\,d\mu_y.$$

The test lies in the form domain because F_R and its derivative are bounded. Since $|F_R| \le |y|$, $|F_R'| \le 1$, dropping nonnegative terms and letting $R \to \infty$ by Fatou yields

$$b_q\mu_4 \le e(q)\mu_2 + a_y, \qquad \mu_4 \le \frac{e(q)^2}{b_q^2} + \frac{a_y}{b_q}.$$

By (980), this gives the explicit estimates

$$\mu_2 \le \frac{4\hbar}{\mathcal{V}_3\omega(q)}, \qquad \|y^2\Omega_q\| = \sqrt{\mu_4} \le \frac{2\sqrt{5}\,\hbar}{\mathcal{V}_3\omega(q)}. \tag{985}$$

No Gaussian moment has been assigned to the quartic ground state.

From the proved gap, $\|R_q\| \le 1/\lambda(q)$. If $v \in \mathscr{K}$ and $w = R_qQ_qv$, the exact equation $(h-e)w = Q_qv$ and $w \perp \Omega_q$ give

$$c\|y^2w\|^2 \le \langle w, h(q)w\rangle = e(q)\|w\|^2 + \mathrm{Re}\langle w, v\rangle \le \frac{e(q)+\lambda(q)}{\lambda(q)^2}\|v\|^2.$$

Thus (981) implies the pointwise estimates

$$\begin{aligned} \|\Omega_q'\| &\le L(q) := \frac{6\sqrt{5}\,\Gamma|q|}{\omega(q)^2}, \\ \|y^2\Omega_q'\| &\le L(q)\sqrt{\frac{e(q)+\lambda(q)}{c}} \le 4L(q)\sqrt{\frac{\hbar\omega(q)}{\Gamma\mathcal{V}_3}}, \\ |e'(q)| &\le \frac{6\Gamma\hbar|q|}{\omega(q)}. \end{aligned} \tag{986}$$

Substitution in the exact second-derivative identity gives

$$\|Q_q\Omega_q''\| \le \frac{6\sqrt{5}\,\Gamma}{\omega(q)^2} + 144\sqrt{5}\,\Gamma^{3/2}\sqrt{\frac{\mathcal{V}_3}{\hbar}}\frac{q^2}{\omega(q)^{5/2}} + 144\sqrt{5}\,\Gamma^2\frac{q^2}{\omega(q)^4}. \tag{987}$$

For completeness the exact scalar maxima used to make these bounds global are

$$\begin{aligned} \sup_q \frac{|q|}{\omega_*^2 + 3\Gamma q^2} &= \frac{1}{2\sqrt{3\Gamma}\,\omega_*}, \\ \sup_q \frac{q^2}{(\omega_*^2 + 3\Gamma q^2)^2} &= \frac{1}{12\Gamma\omega_*^2}, \\ \sup_q \frac{q^2}{(\omega_*^2 + 3\Gamma q^2)^{5/4}} &= \frac{4}{3\Gamma\,5^{5/4}\sqrt{\omega_*}}. \end{aligned}$$

Differentiation in $|q|$ or q^2 shows that the maxima occur, respectively, at $q^2 = \omega_*^2/(3\Gamma)$, $q^2 = \omega_*^2/(3\Gamma)$, and $q^2 = 4\omega_*^2/(3\Gamma)$; each function decreases to zero at infinity. Applying these maxima to (986) and (987) gives (983) and (984), including the constants $18\sqrt{5}$ and $192/5^{3/4}$. □

26.4 The exact projected Hamiltonian and its off-diagonal part

Theorem 26.4 (Exact isometric correspondence, not an invariant product)**.** *For all of the retained positive parameters, without the extra restriction* (979)*, define the isometry*

$$J : \mathscr{L} \longrightarrow \mathscr{H}_2, \qquad (Jf)(q,y) = f(q)\Omega_q(y), \qquad (J^*\Psi)(q) = \langle \Omega_q, \Psi(q,\cdot)\rangle_{\mathscr{K}}.$$

Its range projection is the fibrewise projection $(\mathsf{P}\Psi)(q) = P_q\Psi(q)$*, and* $\mathsf{Q} = I - \mathsf{P}$*. The exact pullback of the full quadratic form is the Friedrichs form of*

$$H_{\mathrm{eff}} = -\frac{\hbar^2}{2\mathcal{V}_3}\partial_q^2 + V_q(q) + e(q) + B(q), \qquad B(q) = \frac{\hbar^2}{2\mathcal{V}_3}\|\Omega_q'\|_{\mathscr{K}}^2. \tag{988}$$

Its domain is precisely

$$\mathcal{D}_{\mathrm{eff}} = \left\{ f \in H^1(\mathbb{R}, d\mu_q) : \int (V_q + e + B)|f|^2 \, d\mu_q < \infty \right\},$$

and $J^{-1}(\mathcal{D}(H_2^{1/2}) \cap \operatorname{Ran} J) = \mathcal{D}_{\mathrm{eff}}$*. The Berry connection is zero in the fixed positive real phase, but* $B(0) = 0$ *and* $B(q) > 0$ *for every* $q \neq 0$*.*

For every $f \in C_c^\infty(\mathbb{R})$*,* $Jf \in D(H_2)$ *and the exact operator identity is*

$$\begin{aligned} H_2 Jf &= J H_{\mathrm{eff}} f + \mathcal{R}f, \\ (\mathcal{R}f)(q) &= -a\big(2f'(q)\Omega_q' + f(q)Q_q\Omega_q''\big), \qquad \mathsf{P}\mathcal{R}f = 0. \end{aligned} \tag{989}$$

Thus $J^*H_2J = H_{\mathrm{eff}}$ *on this core, while* $\mathsf{Q}H_2J = \mathcal{R}$ *is the retained off-diagonal residual. The range of* J *is not a reducing subspace.*

Under (979)*, these identities extend to all* $f \in D(H_{\mathrm{eff}})$*, and*

$$\|\mathcal{R}f\| \le 2\sqrt{a}\, M_1 \|H_{\mathrm{eff}}^{1/2} f\| + aM_2\|f\|, \qquad 0 \le B(q) \le \frac{15\Gamma\hbar^2}{2\mathcal{V}_3\omega_*^2}. \tag{990}$$

Proof. Normalization of the actual fibre state and Fubini give $\|Jf\| = \|f\|$ and the displayed adjoint and projection. For compact smooth f,

$$\partial_q(Jf) = f'\Omega_q + f\Omega_q', \qquad \|\partial_q(Jf)(q)\|^2 = |f'|^2 + |f|^2\|\Omega_q'\|^2,$$

because $\langle \Omega_q, \Omega_q'\rangle = 0$. The fibre energy is exactly $e(q)|f(q)|^2$, not a harmonic replacement. These identities prove equality of the pulled-back forms on the core. Core approximation gives the forward domain inclusion. Conversely, if Jf lies in the full form domain, local Hilbert-valued Sobolev differentiation gives $f' = \langle \Omega_q, \partial_q(Jf)\rangle$ distributionally; the term with Ω_q' vanishes by normalization. Hence $f \in H^1$, and the displayed nonnegative derivative and fibre-energy identities give every required integral in $\mathcal{D}_{\mathrm{eff}}$. This proves equality of the domains and closed forms. The positive smooth effective potential defines its self-adjoint Friedrichs operator with the operator core proved above. It has compact resolvent because it is bounded below by $V_q(q)$, which tends to infinity, and its kinetic coefficient a is strictly positive.

By local graph regularity of Ω_q, Jf is in the full form domain and its distributional Hamiltonian is in $\mathscr{H}_2$. It therefore lies in $D(H_2)$. Expanding $\partial_q^2(f\Omega_q)$ and using $\langle \Omega_q, \Omega_q''\rangle = -\|\Omega_q'\|^2$ proves (989) and the orthogonality of its residual.

At $q = 0$, (981) gives $\Omega_0' = 0$. If $q \neq 0$ and $\Omega_q' = 0$, that identity and injectivity of the reduced inverse on $Q_q\mathscr{K}$ would give $Q_q y^2 \Omega_q = 0$. It would follow that $y^2\Omega_q$ is a scalar multiple of Ω_q. Multiplication by

y^2 has no nonzero L^2 eigenfunction, because every level set has measure zero. Thus $\Omega_q' \neq 0$ for $q \neq 0$ and $B(q) > 0$ there. If the range of J were reducing, its residual would vanish for every compact smooth f. Since its coefficients are continuous in q, choose at any $q_0 \neq 0$ a test function with $f(q_0) = 0$, $f'(q_0) = 1$. The residual at q_0 is then $-2a\Omega_{q_0}' \neq 0$, a contradiction.

Under (979), the global derivative estimates and the triangle inequality give

$$\|\mathcal{R}f\| \le a(2M_1\|f'\| + M_2\|f\|).$$

All terms in the effective potential are nonnegative, so $\sqrt{a}\|f'\| \le \|H_{\mathrm{eff}}^{1/2} f\|$. This proves (990) and defines $\mathcal{R}$ as a bounded map from the effective form domain with its form norm to $\mathscr{H}_2$. For $f \in D(H_{\mathrm{eff}})$, choose an operator-core sequence. It converges also in form norm by the spectral inequality $\|H_{\mathrm{eff}}^{1/2}u\|^2 \le \|u\|\|H_{\mathrm{eff}}u\|$. The right side of (989) then converges in $\mathscr{H}_2$. Closedness of H_2 shows that $Jf \in D(H_2)$ and proves the identity on the full effective operator domain. The bound on B follows directly from aM_1^2. □

26.5 Controlled physical-time and spectral comparisons

Theorem 26.5 (Explicit finite-energy error in a proved parameter regime)**.** *Assume* (979)*. Let $f \in \mathcal{D}_{\mathrm{eff}}$ have $\|f\| = 1$ and effective energy expectation $\|H_{\mathrm{eff}}^{1/2} f\|^2 \le E$. For every physical time $t \in \mathbb{R}$,*

$$\left\| e^{-itH_2/\hbar} Jf - Je^{-itH_{\mathrm{eff}}/\hbar} f \right\| \le |t|\, \mathcal{E}(E), \tag{991}$$

$$\mathcal{E}(E) := \frac{\sqrt{30\Gamma E}}{\omega_* \sqrt{\mathcal{V}_3}} + \frac{9\sqrt{5}\,\Gamma\hbar}{\mathcal{V}_3 \omega_*^2} + \frac{96}{5^{3/4}} \sqrt{\frac{\Gamma\hbar}{\mathcal{V}_3 \omega_*}}.$$

The same bound holds for the leakage norm $\|\mathsf{Q} e^{-itH_2/\hbar} Jf\|$. In particular, for any prescribed $T > 0$, $0 < \rho < 1$, the entirely explicit sufficient volume regime

$$\mathcal{V}_3 \ge \max\left\{ \frac{9\Gamma\hbar}{16\omega_*^3},\ \frac{270\Gamma E T^2}{\rho^2 \omega_*^2},\ \frac{27\sqrt{5}\,\Gamma\hbar T}{\rho\omega_*^2},\ \frac{288^2 \Gamma\hbar T^2}{5^{3/2}\rho^2\omega_*} \right\} \tag{992}$$

makes both norms at most ρ simultaneously for $|t| \le T$. All compared Hamiltonians still contain the original quartic interaction, and no coupling has been set to one.

Proof. First take $f \in D(H_{\mathrm{eff}})$. The established operator-domain map permits differentiation of $e^{-i(t-s)H_2/\hbar} Je^{-isH_{\mathrm{eff}}/\hbar} f$. Integrating its derivative gives the exact Duhamel identity

$$e^{-itH_2/\hbar} Jf - Je^{-itH_{\mathrm{eff}}/\hbar} f = -\frac{i}{\hbar} \int_0^t e^{-i(t-s)H_2/\hbar} \mathcal{R} e^{-isH_{\mathrm{eff}}/\hbar} f\, ds.$$

Unitarity and preservation of the effective form norm give the bound $|t|(2\sqrt{a}M_1\sqrt{E} + aM_2)/\hbar$. Substitution of the exact a, M_1, M_2 yields (991). For a general form-domain vector, effective spectral cutoffs converge both in norm and form norm and give the same estimate by continuity. The comparison vector on the right lies in $\operatorname{Ran} J$, so applying the contraction Q proves the leakage bound. The last three lower bounds on volume in (992) make respectively the three terms of $T\mathcal{E}(E)$ at most $\rho/3$; the first is the separately proved fibre-gap regime. Squaring the first and third nonnegative inequalities and directly solving the second gives exactly their displayed constants. □

Corollary 26.6 (An explicit approximate-eigenstate relation)**.** *In* (979)*, every normalized effective eigenvector $H_{\mathrm{eff}} f = \Lambda f$ satisfies*

$$\|(H_2 - \Lambda)Jf\| \le 2\sqrt{a}M_1\sqrt{\Lambda} + aM_2 = \hbar\mathcal{E}(\Lambda), \qquad \operatorname{dist}(\Lambda, \sigma(H_2)) \le \hbar\mathcal{E}(\Lambda).$$

Proof. The effective operator has compact resolvent as proved above, so its eigenvectors are in its operator domain, and $\Lambda > 0$ because $e(q) \geq \hbar\omega_*/2$. The exact intertwining identity and residual estimate prove the norm bound. The spectral theorem for H_2 gives $\|(H_2 - \Lambda)u\| \geq \operatorname{dist}(\Lambda, \sigma(H_2))\|u\|$ for every operator-domain vector. Apply it to the normalized Jf. □

The preceding comparison is an actual isometry, an exact compressed Hamiltonian, and an explicit finite-time error estimate for this specific interacting two-coordinate mechanics. At fixed $\hbar, \Gamma, L_1, m^2, E, T$, its stated error tends to zero as $\mathcal{V}_3 \to \infty$, for any family of initial states satisfying the stated energy bound. This is not a volume-uniform mass-gap claim and not a continuum limit. The estimate is conservative; in particular the second-derivative bound deliberately uses the positive quartic energy to control $y^2 R_q$, which is sufficient for the proved $\mathcal{V}_3^{-1/2}$ error without an unproved adiabatic expansion or omission of y^4.

Context and attribution. Moving spectral subspaces, the Berry connection, diagonal Born–Huang energy, and errors in time-dependent Born–Oppenheimer comparisons are established concepts; see G. Panati, H. Spohn, and S. Teufel, *The time-dependent Born–Oppenheimer approximation*, ESAIM: M2AN **41** (2007), 297–314, `https://arxiv.org/abs/0712.4369`, `https://doi.org/10.1051/m2an:2007023`. No new general approximation theorem is claimed here. The spectral bundle, every retained coefficient, and the conservative residual and time estimates used above were proved directly for (977). The bounded-below bridge formula used in the positivity argument is also stated in B. Simon, *A Feynman–Kac Formula for Unbounded Semigroups*, Theorem 1.1, `https://arxiv.org/abs/math-ph/9907022`; its general unbounded-below extension is not needed here.

27 Exact presymplectic reduction of the $SU(3)$-Gram worldline model

The constrained Hamiltonian framework used below goes back to Dirac's analysis of singular Lagrangians and constrained Hamiltonian systems [71]. The separation of constraint surface, characteristic distribution, and reduced phase space is the finite-dimensional presymplectic constraint problem treated systematically by Gotay–Nester–Hinds [72]; the regular quotient step is the symplectic reduction mechanism of Marsden–Weinstein [73]. These references supply the general framework only. Every constraint rank, orbit, stabilizer, kernel, one-form, two-form, quotient map, sign, equation of motion, and singular counterexample used for this particular (SU(3))-Gram model is computed explicitly below.

27.1 Objects, constants, and equations

Let

$$\mathcal{J}_2 = \{A \in \mathrm{Mat}_{2\times 2}(\mathbb{C}) : A = A^*\}, \qquad \langle A, B\rangle_L = \frac{1}{2}\big(\operatorname{tr} A \operatorname{tr} B - \operatorname{tr}(AB)\big).$$

Then $\langle A, A\rangle_L = \det A$, and the pairing is nondegenerate. Write

$$\mathcal{P}_2 = \{p \in \mathcal{J}_2 : p \succeq 0\}, \quad \mathcal{P}_2^{++} = \{p \in \mathcal{J}_2 : p \succ 0\}, \quad \mathcal{N}_+^{\times} = \{p \in \mathcal{P}_2 : \operatorname{rank}_{\mathbb{C}} p = 1\}.$$

Retain

$$\mathscr{V} = \mathrm{Mat}_{3\times 2}(\mathbb{C}), \quad \alpha > 0, \quad c > 0, \quad m_0 \geq 0,$$

and define

$$P_\alpha(Z) = \alpha Z^* Z, \tag{993}$$

$$DP_{\alpha,Z}(\delta Z) = \alpha(\delta Z^* Z + Z^* \delta Z), \tag{994}$$

$$\Phi_{m_0,\alpha,c}(Z) = \frac{1}{2}\left(\det P_\alpha(Z) - m_0^2 c^2\right). \tag{995}$$

The history configuration space on a retained parameter interval I is

$$\mathfrak{C}_I = C^\infty(I, \mathcal{J}_2) \times C^\infty(I, \mathscr{V}) \times C^\infty(I, \mathbb{R}),$$

with coordinates (x, Z, e). Because the Lagrangian is first order in x, contains no $\dot{Z}$, and contains no $\dot{e}$, the corresponding instantaneous presymplectic phase carrier is

$$\mathcal{E} = \mathcal{J}_{2x} \times \mathscr{V}_Z \times \mathbb{R}_e.$$

Define

$$\theta = \langle P_\alpha(Z), \mathrm{d}x \rangle_L, \qquad \Omega = -\mathrm{d}\theta, \qquad \mathcal{H} = e\Phi_{m_0,\alpha,c}. \tag{996}$$

For tangent vectors $v = (v_x, v_Z, v_e)$ and $w = (w_x, w_Z, w_e)$,

$$\Omega(v, w) = \langle v_x, DP_{\alpha,Z}(w_Z) \rangle_L - \langle w_x, DP_{\alpha,Z}(v_Z) \rangle_L. \tag{997}$$

Proposition 27.1 (Exact presymplectic Euler–Lagrange equation)**.** *A path (x, Z, e) is stationary under compactly supported variations if and only if*

$$\det P_\alpha(Z) = m_0^2 c^2, \tag{998}$$

$$\frac{\mathrm{d}}{\mathrm{d}\tau} P_\alpha(Z) = 0, \tag{999}$$

$$\langle DP_{\alpha,Z}(\delta Z), \dot{x} - eP_\alpha(Z) \rangle_L = 0 \quad \text{for every } \delta Z \in \mathscr{V}. \tag{1000}$$

Equivalently, $\iota_{\dot{\gamma}} \Omega = \mathrm{d}\mathcal{H}$.

Proof. Since

$$D(\det)_p(A) = 2\langle p, A \rangle_L,$$

one has

$$\mathrm{d}\mathcal{H}(w) = \Phi_{m_0,\alpha,c} w_e + e\langle P_\alpha(Z), DP_{\alpha,Z}(w_Z) \rangle_L.$$

Insert (997) into $\iota_{\dot{\gamma}} \Omega = \mathrm{d}\mathcal{H}$ and test successively w_x, w_e, w_Z. This gives (999), (998), and (1000). □

27.2 Colour quotient and orbit types

The left action is $U \cdot (x, Z, e) = (x, UZ, e)$.

Lemma 27.2 (Local colour is a zero-moment null symmetry)**.** *For $\xi \in \mathfrak{su}(3)$, let $\xi^\# = (0, \xi Z, 0)$. Then*

$$DP_{\alpha,Z}(\xi Z) = 0, \qquad \iota_{\xi^\#} \theta = 0, \qquad \iota_{\xi^\#} \Omega = 0.$$

Thus the zero map to $\mathfrak{su}(3)^*$ is a presymplectic moment map. The action is invariant under every τ-dependent $U : I \to SU(3)$, with no connection term.

Proof. Because $\xi^* = -\xi$,

$$DP_{\alpha,Z}(\xi Z) = \alpha(Z^* \xi^* Z + Z^* \xi Z) = 0.$$

The form identities follow from (996) and (997). Pointwise, $(UZ)^*(UZ) = Z^* Z$, even when U depends on τ. □

Theorem 27.3 (Gram quotient, fibres, and stabilizers)**.** *The map P_α induces a homeomorphism*

$$\mathscr{V}/SU(3) \xrightarrow{\sim} \mathcal{P}_2, \qquad [Z] \longmapsto P_\alpha(Z).$$

Its orbit types are

$\operatorname{rank}_{\mathbb{C}} Z$	*fibre*	$\operatorname{Stab}_{SU(3)}(Z)$
2	$SU(3)$	$\{I_3\}$
1	$SU(3)/SU(2) \cong S^5$	$SU(2)$
0	*a point*	$SU(3)$

The rank-two quotient is smooth and free. Rank one is a smooth nonprincipal orbit-type stratum with nontrivial stabilizer; the zero vertex is the singular apex of the total stratified quotient.

Proof. If $Z_1^* Z_1 = Z_2^* Z_2$, the map $Z_1 v \mapsto Z_2 v$ is an isometry between their images and extends to a unitary map of $\mathbb{C}^3$. Its action on the orthogonal complement can be adjusted to make its determinant one. In rank two the complement is one-dimensional and the adjustment is unique, hence the stabilizer is trivial. In rank one the stabilizer is $SU(2)$ on the two-dimensional complement. At zero it is all of $SU(3)$. Surjectivity has the retained section

$$Z_p = \alpha^{-1/2} \begin{pmatrix} p^{1/2} & \\ 0 & 0 \end{pmatrix}.$$

Compactness of $SU(3)$ and continuity of the positive square root give the quotient homeomorphism. On $\mathcal{P}_2^{++}$ the section is smooth. On $\mathcal{N}_+^{\times}$, the rank-one identity

$$p^{1/2} = \frac{p}{\sqrt{\operatorname{tr} p}}$$

shows directly that the same section is smooth; together with the constant orbit type $SU(3)/SU(2)$, this gives the stated smooth nonprincipal stratum. □

Lemma 27.4 (Exact ranks of the Gram differential)**.** (i) *If* $\operatorname{rank}_{\mathbb{C}} Z = 2$*, then*

$$\operatorname{im} DP_{\alpha,Z} = \mathcal{J}_2, \qquad \ker DP_{\alpha,Z} = T_Z(SU(3) \cdot Z),$$

of real dimensions 4 *and* 8*.*

(ii) *If* $\operatorname{rank}_{\mathbb{C}} Z = 1$ *and* $p = P_\alpha(Z)$*, then*

$$\operatorname{im} DP_{\alpha,Z} = p^{\perp_L} = T_p \mathcal{N}_+^{\times}$$

has real dimension 3*. The full ambient kernel has dimension* 9*, but*

$$\ker(DP_{\alpha,Z}|_{T_Z \mathscr{V}_1}) = T_Z(SU(3) \cdot Z)$$

has dimension 5*, where* $\mathscr{V}_1 = \{Z : \operatorname{rank}_{\mathbb{C}} Z = 1\}$*. The remaining four real kernel directions are*

$$\operatorname{Hom}_{\mathbb{C}}(\ker Z, (\operatorname{im} Z)^{\perp});$$

they raise rank and are not colour gauge.

(iii) *At* $Z = 0$*,* $DP_{\alpha,0} = 0$*.*

Proof. For rank two and arbitrary $A = A^*$, retain

$$\delta Z_A = \frac{1}{2\alpha} Z (Z^* Z)^{-1} A.$$

Substitution into (994) gives $DP_{\alpha,Z}(\delta Z_A) = A$. Hence the kernel has dimension $12 - 4 = 8$, equal to the free colour-orbit tangent.

For rank one write, without fixing either norm,

$$Z = ab^*, \qquad a \in \mathbb{C}^3 \setminus \{0\}, \quad b \in \mathbb{C}^2 \setminus \{0\}, \quad r = a^* a > 0.$$

Then $p = \alpha r b b^*$. If $w = \delta Z^* a$, every $w \in \mathbb{C}^2$ occurs by taking $\delta Z = a w^*/r$, and

$$\operatorname{im} DP_{\alpha,Z} = \{\alpha(wb^* + bw^*) : w \in \mathbb{C}^2\}.$$

The kernel of $w \mapsto wb^* + bw^*$ is $\{itb : t \in \mathbb{R}\}$, so the image has dimension three. It is contained in $p^{\perp_L}$, and equality follows by dimension.

Use the exact orthogonal splittings

$$\mathbb{C}^2 = (\ker Z)^\perp \oplus \ker Z, \qquad \mathbb{C}^3 = \operatorname{im} Z \oplus (\operatorname{im} Z)^\perp.$$

Retaining the nonzero singular value $\sigma > 0$, adapted unit vectors give

$$Z = \begin{pmatrix} \sigma & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}, \qquad \delta Z = \begin{pmatrix} a_0 & b_0 \\ c_1 & d_1 \\ c_2 & d_2 \end{pmatrix},$$

and

$$\alpha^{-1} DP_{\alpha,Z}(\delta Z) = \begin{pmatrix} 2\sigma \operatorname{Re} a_0 & \sigma b_0 \\ \sigma \overline{b_0} & 0 \end{pmatrix}.$$

The ambient kernel has $\operatorname{Re} a_0 = b_0 = 0$ and dimension nine. The linearized rank-one equations also impose $d_1 = d_2 = 0$. The five remaining directions are the $SU(3)/SU(2)$-orbit tangent; the d_1, d_2 block is the displayed complex Hom-space. The zero case follows from (994). □

27.3 Forms after colour reduction

Define

$$\overline{\mathcal{E}} = \mathcal{J}_{2x} \times \mathcal{P}_{2p} \times \mathbb{R}_e \cong \mathcal{E}/SU(3), \qquad q(x, Z, e) = (x, P_\alpha(Z), e).$$

Proposition 27.5 (Reduced one-form and two-form)**.** *The forms are basic for the colour action and satisfy*

$$\theta = q^* \overline{\theta}, \qquad \Omega = q^* \overline{\Omega},$$

where

$$\overline{\theta} = \langle p, \mathrm{d}x \rangle_L, \tag{1001}$$

$$\overline{\Omega} = -\mathrm{d}\overline{\theta}, \tag{1002}$$

$$\overline{\Omega}((v_x, v_p), (w_x, w_p)) = \langle v_x, w_p \rangle_L - \langle w_x, v_p \rangle_L. \tag{1003}$$

After omitting the multiplier coordinate, this is symplectic on $\mathcal{J}_2 \times \mathcal{P}_2^{++}$. On $\mathcal{J}_2 \times \mathcal{N}_+^\times$, its kernel is exactly

$$\ker \overline{\Omega}_{(x,p)} = \operatorname{span}_{\mathbb{R}}\{(p, 0)\}.$$

On $\mathcal{J}_2 \times \{0\}$, the stratum-restricted two-form is zero.

Proof. Basicness follows from Lemma 27.2; the formulas follow by differentiation. The rank-two assertion uses that $\mathcal{P}_2^{++}$ is open in $\mathcal{J}_2$. At nonzero rank-one p, one has

$$T_p \mathcal{N}_+^\times = p^{\perp_L}.$$

If (v_x, v_p) annihilates (1003), testing arbitrary w_x gives $v_p = 0$, and testing $w_p \in p^{\perp_L}$ gives $v_x \in (p^{\perp_L})^{\perp_L} = \operatorname{span}_{\mathbb{R}}\{p\}$. At $p = 0$, the zero stratum has no momentum tangent direction. □

27.4 Positive mass

For $m_0 > 0$, define the retained future shell

$$\mathcal{H}^+_{m_0,c} = \{p \in \mathcal{P}_2^{++} : \det p = m_0^2 c^2\}.$$

The parameter α remains present in the exact quotient map $p = \alpha Z^* Z$; the displayed subset of momentum space itself depends on m_0, c.

Theorem 27.6 (Exact massive reduction)**.** *Let $m_0 > 0$. Then:*

(a) *The constraint surface*

$$\Sigma_{m_0,\alpha,c} = \mathcal{J}_2 \times \{Z : \det P_\alpha(Z) = m_0^2 c^2\} \times \mathbb{R}_e$$

is smooth and the colour action on it is free.

(b) *Its colour quotient is*

$$\overline{\Sigma}_{m_0,c} = \mathcal{J}_2 \times \mathcal{H}^+_{m_0,c} \times \mathbb{R}_e.$$

Before colour reduction the kernel of the restricted two-form is the direct sum of the eight-dimensional colour-orbit tangent, $\mathrm{span}_{\mathbb{R}}\{(p,0)\}$*, and* $\mathrm{span}_{\mathbb{R}}\{\partial_e\}$*. After colour reduction and omission of e, only* $\mathrm{span}_{\mathbb{R}}\{(p,0)\}$ *remains.*

(c) *The equations are*

$$\dot{p} = 0, \qquad \dot{x} = e\,p, \qquad \det p = m_0^2 c^2,$$

while $\dot{e}$ and motion in the colour fibre are arbitrary.

(d) *The physical phase space*

$$\mathcal{R}_{m_0,c} = \frac{\mathcal{J}_2 \times \mathcal{H}^+_{m_0,c}}{(x,p) \sim (x + sp, p),\ s \in \mathbb{R}}$$

is canonically symplectomorphic to $T^*\mathcal{H}^+_{m_0,c}$.

(e) *The colour-reduced positive-mass model is exactly the standard future-directed spinless first-order relativistic particle of mass m_0 and speed constant c.*

Proof. The constraint implies $\mathrm{rank}_{\mathbb{C}}\, Z = 2$. By Lemma 27.4, P_α is a submersion there, and

$$D(\det)_p = 2\langle p, \cdot\rangle_L \neq 0.$$

This proves smoothness. Theorem 27.3 gives freeness and the colour quotient. Since

$$T_p\mathcal{H}^+_{m_0,c} = p^{\perp_L},$$

the proof of Proposition 27.5 gives the characteristic line. Pullback through the Gram submersion adds exactly the colour-orbit kernel, and the multiplier adds ∂_e.

Surjectivity of $DP_{\alpha,Z}$ turns (1000) into $\dot{x} = ep$; the other equations are (998) and (999). A path with constant p lies in one free Gram fibre and differs locally from a constant lift by a τ-dependent colour transformation.

For the physical identification, if $A \in T_p\mathcal{H}^+_{m_0,c}$, define

$$\xi_{[x],p}(A) = -\langle x, A\rangle_L.$$

This is unchanged by $x \mapsto x + sp$. Conversely, the annihilator of $p^{\perp_L}$ is $\mathrm{span}_{\mathbb{R}}\{p\}$, so every cotangent covector has one and only one class $[x]$. Hence

$$([x], p) \longmapsto (p, \xi_{[x],p})$$

is a fibrewise linear bijection to $T^*\mathcal{H}^+_{m_0,c}$. The characteristic $\mathbb{R}$-action is free because every shell momentum is nonzero. It is proper: choose any Euclidean norm on $\mathcal{J}_2$. For compact sets $K_1, K_2 \subset \mathcal{J}_2 \times \mathcal{H}^+_{m_0,c}$, the momenta occurring in them have a positive minimum Euclidean norm δ; if $(x,p) \in K_1$ and $(x+sp,p) \in K_2$, boundedness of the two position projections gives $|s|\delta \le C(K_1,K_2)$. Thus the orbit space is smooth. The smooth vector-bundle map

$$(x,p) \longmapsto \big(p,\ A \mapsto -\langle x, A\rangle_L\big)$$

is fibrewise onto with kernel $\operatorname{span}_{\mathbb{R}}\{p\}$, and therefore descends with a smooth inverse. The displayed bijection is consequently a diffeomorphism.

The basic one-form on the shell is

$$\theta_{\mathrm{basic}} = -\langle x, \mathrm{d}p\rangle_L. \tag{1004}$$

It is the pullback of the tautological cotangent one-form, and

$$\overline{\theta} = \mathrm{d}\langle p, x\rangle_L + \theta_{\mathrm{basic}}. \tag{1005}$$

Therefore $\overline{\Omega} = -\mathrm{d}\theta_{\mathrm{basic}}$, the canonical cotangent symplectic form under the convention $\omega_{\mathrm{can}} = -\mathrm{d}\lambda_{\mathrm{can}}$. The original $\overline{\theta}$ is not horizontal on the massive characteristic because $\overline{\theta}(p,0) = m_0^2c^2$; the exact boundary term in (1005) produces the reduced one-form.

Finally the reduced action is

$$\int_I \left(\langle p, \dot{x}\rangle_L - \frac{e}{2}(\det p - m_0^2c^2) \right) \mathrm{d}\tau.$$

Because $\mathcal{P}_2^{++}$ is open in $\mathcal{J}_2$, p has every infinitesimal Hermitian variation. This is the standard covariant first-order spinless particle action on the future-timelike sector, with no residual internal symplectic factor. □

Proposition 27.7 (Exact Hamiltonian gauge quasi-symmetry)**.** *For a smooth real $\varepsilon(\tau)$, set*

$$\delta_\varepsilon x = \varepsilon P_\alpha(Z), \qquad \delta_\varepsilon Z = 0, \qquad \delta_\varepsilon e = \dot{\varepsilon}.$$

Then

$$\delta_\varepsilon L = \frac{\mathrm{d}}{\mathrm{d}\tau}\left[\frac{\varepsilon}{2}(\det P_\alpha(Z) + m_0^2c^2) \right]. \tag{1006}$$

For compactly supported ε this is an exact gauge quasi-symmetry, whose phase-space part generates the shell characteristic where that characteristic is regular.

Proof. Put $p = P_\alpha(Z)$ and $q = \det p = \langle p, p\rangle_L$. Since $\dot{q} = 2\langle p, \dot{p}\rangle_L$,

$$\begin{aligned} \delta_\varepsilon L &= \langle p, \dot{\varepsilon} p + \varepsilon \dot{p}\rangle_L - \frac{\dot{\varepsilon}}{2}(q - m_0^2c^2) \\ &= \frac{\dot{\varepsilon}}{2}(q + m_0^2c^2) + \frac{\varepsilon}{2}\dot{q}, \end{aligned}$$

which is (1006). □

27.5 Massless rank one

Set

$$\mathcal{C}_{0,\alpha,c} = \{Z : \det P_\alpha(Z) = 0\} = \{Z : \operatorname{rank}_{\mathbb{C}} Z \le 1\}.$$

The constant $c > 0$ is retained in the model, although it cancels from the set-theoretic $m_0 = 0$ equation.

Lemma 27.8 (The pulled-back massless constraint is irregular)**.** *Writing the rows of Z as $Z_{a\bullet}$,*

$$\det P_\alpha(Z) = \alpha^2 \sum_{1\le a<b\le 3} |Z_{a1}Z_{b2} - Z_{a2}Z_{b1}|^2. \tag{1007}$$

Consequently,

$$\mathrm{d}(\det P_\alpha)_Z = 0 \quad \textit{for every } Z \in \mathcal{C}_{0,\alpha,c}.$$

The single real multiplier equation describes the rank-at-most-one set, but its differential supplies none of the four real normal equations on the smooth rank-one stratum.

Proof. Equation (1007) is Cauchy–Binet with the factor α^2 retained. On its zero set every complex minor vanishes, so differentiation of the sum of squared absolute values gives zero. Equivalently, at rank one,

$$\mathrm{d}(\det P_\alpha)_Z(\delta Z) = 2\langle p, DP_{\alpha,Z}(\delta Z)\rangle_L = 0$$

by Lemma 27.4; at $Z = 0$ use $DP_{\alpha,0} = 0$. □

Theorem 27.9 (Nonzero massless equations and physical quotient)**.** *Suppose $m_0 = 0$ and $\operatorname{rank}_{\mathbb{C}} Z = 1$, and put $p = P_\alpha(Z) \in \mathcal{N}_+^\times$. Then:*

(a) *The colour fibre is $SU(3)/SU(2)$, with stabilizer $SU(2)$.*

(b) *The complete equations are*

$$\dot p = 0, \qquad \dot x = \lambda(\tau)p \quad \textit{for some real } \lambda(\tau), \qquad e(\tau) \textit{ arbitrary.} \tag{1008}$$

No equation identifies λ with the written multiplier e.

(c) *On the intrinsic smooth rank-one shell, the kernel before colour reduction is the direct sum of*

$$T_Z(SU(3)\cdot Z), \qquad \operatorname{span}_{\mathbb{R}}\{(p,0)\}, \qquad \operatorname{span}_{\mathbb{R}}\{\partial_e\}.$$

After colour reduction and omission of e, only the characteristic $\operatorname{span}_{\mathbb{R}}\{(p,0)\}$ remains.

(d) *The physical phase space*

$$\mathcal{R}_0^\times = \frac{\mathcal{J}_2 \times \mathcal{N}_+^\times}{(x,p) \sim (x + sp, p)}$$

is canonically symplectomorphic to $T^*\mathcal{N}_+^\times$, with reduced one-form $-\langle x, \mathrm{d}p\rangle_L$ and its canonical two-form. No quotient by positive rescaling of p occurs.

(e) *Thus the unparameterized spacetime curves and physical symplectic quotient agree with the standard future-directed nonzero spinless massless particle. The original triples (x, Z, e), their off-shell neighbourhood, and their multiplier equations are not exactly the standard massless system.*

Proof. The fibre statement is Theorem 27.3. By Lemma 27.4, equation (1000) says

$$\dot x - ep \in (p^{\perp_L})^{\perp_L} = \operatorname{span}_{\mathbb{R}}\{p\}.$$

Since ep lies in this line, this is equivalent to $\dot x = \lambda p$, with no relation between λ and e. Equation (999) gives $\dot p = 0$. A path in a fixed rank-one Gram fibre differs locally from a constant representative by a τ-dependent $SU(3)$ transformation; the $SU(2)$ stabilizer makes this gauge description reducible.

On the intrinsic rank-one manifold, Lemma 27.4 identifies the vertical kernel with the five colour directions. Proposition 27.5 adds the one spacetime characteristic, and e adds ∂_e.

Since $T_p\mathcal{N}_+^\times = p^{\perp_L}$, the massive cotangent construction applies without division by $\det p$:

$$([x], p) \longmapsto (p,\ A \longmapsto -\langle x, A\rangle_L)\,.$$

Every $p \in \mathcal{N}_+^\times$ is nonzero. The freeness, properness, and smooth vector-bundle argument in the massive proof applies to compact subsets of $\mathcal{J}_2 \times \mathcal{N}_+^\times$ verbatim, and proves that this map is a diffeomorphism to $T^*\mathcal{N}_+^\times$. Moreover, (1005) identifies the canonical form. Reparametrizing the worldline changes x along p; it does not identify distinct positive magnitudes of p. Lemma 27.8 and the absence of $\lambda = e$ establish the final inequivalence statement. □

Counterexample 27.10 (Rank-one multiplier mismatch)**.** *Retain arbitrary $s > 0$, and set*

$$Z_s = \begin{pmatrix} s & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}, \qquad p_s = \alpha \begin{pmatrix} s^2 & 0 \\ 0 & 0 \end{pmatrix}.$$

The path

$$Z(\tau) = Z_s, \qquad x(\tau) = \tau p_s, \qquad e(\tau) = 0$$

satisfies all equations in Proposition 27.1 for $m_0 = 0$. The standard action with momentum independently varied in $\mathcal{J}_2$ and the same $e = 0$ would require $\dot{x} = ep_s = 0$.

Under Proposition 27.7, a solution $\dot{x} = \lambda p$, $\dot{p} = 0$, transforms by

$$\lambda \longmapsto \lambda + \dot{\varepsilon}, \qquad e \longmapsto e + \dot{\varepsilon}.$$

Thus $\lambda - e$ is invariant under the stated Hamiltonian gauge; the discrepancy remains when the original multiplier is retained.

Counterexample 27.11 (Four non-colour first-order null directions)**.** *Retain arbitrary $s > 0$ and $t \in \mathbb{C} \setminus \{0\}$, and put*

$$Z_s = \begin{pmatrix} s & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}, \qquad \Delta_t = \begin{pmatrix} 0 & 0 \\ 0 & t \\ 0 & 0 \end{pmatrix}.$$

Then

$$DP_{\alpha,Z_s}(\Delta_t) = 0, \qquad \det P_\alpha(Z_s + \varepsilon\Delta_t) = \alpha^2 s^2 |t|^2 \varepsilon^2 \quad (\varepsilon \in \mathbb{R}). \tag{1009}$$

The vector Δ_t is not tangent to a colour orbit, since every ξZ_s has zero second column. It is not tangent to the rank-one variety, because the perturbed matrix has rank two for $\varepsilon \neq 0$. Allowing both complex coordinates in $(\operatorname{im} Z_s)^\perp$ gives the four real transverse directions of Lemma 27.4. They are second-order shell-opening directions, not gauge.

27.6 The zero-momentum vertex

Theorem 27.12 (Additional singular branch)**.** *Let $m_0 = 0$ and $Z = 0$. Then $p = 0$, the stabilizer is all of $SU(3)$, and both θ and Ω vanish on $\mathcal{J}_2 \times \{0\} \times \mathbb{R}_e$. Every pair of paths*

$$x : I \to \mathcal{J}_2, \qquad e : I \to \mathbb{R},$$

with $Z(\tau) \equiv 0$, satisfies the Euler–Lagrange equations.

The phase part of the Hamiltonian gauge vanishes at $p = 0$, while the presymplectic kernel contains every x-direction. Quotienting by the actual colour and Hamiltonian-gauge orbits therefore leaves the whole $\mathcal{J}_2$ branch, whereas quotienting by all characteristic leaves would collapse it to one point. This is an exact failure of regular Dirac reduction at the vertex.

Proof. At $Z = 0$, one has $p = 0$ and $DP_{\alpha,0} = 0$. Every equation in Proposition 27.1 is identically satisfied, independent of $\dot{x}, e, \dot{e}$. The form and gauge statements follow from (996) and $\delta_\varepsilon x = \varepsilon p = 0$. □

Counterexample 27.13 (Stationary zero momentum with nonconstant position)**.** *The retained paths*

$$x(\tau) = \begin{pmatrix} \tau^2 & 0 \\ 0 & 0 \end{pmatrix}, \qquad Z(\tau) = 0, \qquad e(\tau) = \sin\tau$$

are stationary for the present $m_0 = 0$ action. The standard independent-momentum action gives $\dot{x} = ep = 0$ at $p = 0$, and therefore excludes this path. Removing $p = 0$, as is conventional for physical massless phase space, removes this vertex counterexample but does not regularize the off-shell rank-one constraint.

27.7 Kernel dimensions and exact verdict

Proposition 27.14 (Every kernel jump)**.** *Omit the always-null multiplier direction for the table. On the ambient* 16*-real-dimensional $\mathcal{J}_{2x} \times \mathcal{V}_Z$,*

$\operatorname{rank}_{\mathbb{C}} Z$	$\operatorname{rank}_{\mathbb{R}} DP_{\alpha,Z}$	$\operatorname{rank}_{\mathbb{R}} \Omega$	$\dim_{\mathbb{R}} \ker \Omega$
2	4	8	8
1	3	6	10
0	0	0	16

On the 12*-real-dimensional intrinsic rank-one stratum $\mathcal{J}_{2x} \times \mathcal{V}_1$, the pulled-back form has rank* 6 *and nullity* 6*: five colour directions and one spacetime characteristic.*

Proof. Using the Lorentz pairing to identify $\mathcal{J}_2 \cong \mathcal{J}_2^*$, the matrix of Ω is

$$\begin{pmatrix} 0 & G_L DP_{\alpha,Z} \\ -(DP_{\alpha,Z})^{\mathsf{T}} G_L & 0 \end{pmatrix}.$$

Its rank is twice the rank of $DP_{\alpha,Z}$. Apply Lemma 27.4 in the ambient space and on $T_Z\mathcal{V}_1$. □

Theorem 27.15 (Final comparison)**.** *The strongest correct result is:*

(1) *for $m_0 > 0$, colour reduction gives exactly the standard future-timelike first-order spinless particle, including its equations, constraint characteristic, and physical cotangent symplectic space;*

(2) *for $m_0 = 0$ and $p \neq 0$, colour and characteristic reduction gives exactly the standard future-null physical phase space and unparameterized null trajectories;*

(3) *the full $m_0 = 0$ pulled-back multiplier action is not the standard massless variational system: its constraint has zero differential, its rank-one equation is $\dot{x} = \lambda p$ rather than $\dot{x} = ep$, its ambient kernel contains four non-colour transverse directions, and its vertex admits arbitrary $x(\tau)$.*

The exact relating morphism is $P_\alpha(Z) = \alpha Z^* Z$; the exact failure locus is the singular differential of $\det \circ P_\alpha$ *on the rank-at-most-one variety.*

Proof. Combine Theorem 27.6, Theorem 27.9, Lemma 27.8, and Counterexamples 27.10–27.13. □

27.8 Deterministic verification companion

The companion program `verify_worldline_reduction.py` checks the determinant polarization, the rank-two Gram right inverse, all Gram and presymplectic ranks, all colour-orbit ranks, the intrinsic rank-one tangent calculation, the zero gradient and quadratic transverse opening of the massless constraint, the cotangent-form sign, and (1006), using exact symbolic arithmetic.

27.9 Comparison with the extended exact Lorentz packet

This subsection records a bounded line-specific audit of source lines 875–1654. The audited source file is

`exact_lorentz_observation_bridge.tex`.

It is stored beneath `supporting_materials/workbench/evidence` in the directory

`worldline_presymplectic_reduction_20260904`.

Its file SHA-256 is

`2BE16E1DC71B2E5DC04238F0FBE440138258D0825A4841AB2F29662587681599`.

That packet adds a third column $\phi \in \mathbb{C}^3$, writes $B = [Z\ \phi]$, and uses

$$G = Z^* Z, \quad z = Z^* \phi, \quad \lambda = \phi^* \phi, \quad \omega = \det B.$$

Its mass convention is $m^2_{\mathrm{packet}} = \det p$, with physical mass $M_{\mathrm{phys}} = \sqrt{\det p}/c$. The convention in the audited S^6 module is $m^2_{\alpha,c} = \det p / c^2$. The exact dictionary is therefore

$$m^2_{\mathrm{packet}} = c^2 m^2_{\alpha,c}, \qquad M_{\mathrm{phys}} = m_{\alpha,c};$$

there is no mathematical disagreement.

27.9.1 Claim ledger

Source lines	Verdict	Exact scope
875–950	validated	$p = \alpha Z^* Z$ is a proper surjection, its fibres are the $SU(3)$-orbits, and $\det p = \alpha^2 \sum_{a<b} \lvert\Delta_{ab}(Z)\rvert^2$ has range $[0, \infty)$. The zero fibre is rank at most one.
952–1027	validated as a multiplication-operator model	With the stated pushforward measure, the observer-energy and mass-squared multiplication operators have essential ranges $[0, \infty)$, no point spectrum, and no positive gap after a vacuum is adjoined. This is a kinematic/free spectral statement, not a statement about an interacting field Hamiltonian.
1029–1476	validated	For $\lambda_0 > 0$, $\omega_0 \neq 0$, the constrained quotient, associated-ball description, momentum fibres, and sharp lower bound $\det p \geq \alpha^2 \lvert\omega_0\rvert^2 / \lambda_0$ are algebraic and differential-topological statements. Since $\omega_0 \neq 0$ forces B invertible and Z rank two, the massless rank-one defect proved above does not invalidate these claims.
1482–1531	valid invariance and constraints, incomplete dynamics	The displayed action is invariant and its multiplier and x variations give the stated constraints and $\dot{p} = 0$. Its omitted (Z, ϕ) equations materially change the dynamical interpretation; Theorem 27.16 below computes them intrinsically.
1533–1603	validated with scope qualification	The multiplication spectra and the direct-sum/Fock vacuum gap equal the displayed threshold. The threshold is inserted by fixing $\lambda = \lambda_0$ and $\omega = \omega_0 \neq 0$; it is not a dynamically generated interacting mass gap and is not a Yang–Mills mass-gap theorem.
1605–1633	validated and scope-sharpening	The two explicit degenerating sequences prove that controlling only one of λ and ω gives no positive threshold. Thus the positive lower endpoint is exactly constraint-selected.
1635–1653	validated	The diagonal boost excludes a positive-definite quadratic form on the spinor carrier invariant under all of $SL_2(\mathbb{C})$.

The independent replay and line-by-line claim ledger for this packet are retained under the project-relative path `supporting_materials/workbench/evidence/exact_lorentz_observation_bridge_audit_20260904`. The packet checker passes 33/33 assertions and the separately written checker passes 30/30. The audit-report SHA–256 is `c8d01af28836340798c2fa7e27f8e097d67e0b8d7c13f75271af473958d391df`; the promotion manifest, which verifies all retained audit artifacts, has SHA–256 `52f53746558fa219260b224abdb71c86d2cd6c13192774d0d2fb1a1ed71bcf35`.

27.9.2 The full dynamics of the packet's constrained action

Retain the packet's constants

$$\lambda_0 > 0, \qquad \omega_0 \in \mathbb{C}^*, \qquad \alpha > 0$$

and its smooth colour quotient

$$\mathcal{Q}_{\lambda_0,\omega_0} \cong \mathrm{Herm}_2^0(\mathbb{C}) \times \{w \in \mathbb{C}^2 : \|w\|^2 < \lambda_0\}.$$

Put

$$\mu(w) = \frac{|\omega_0|}{\sqrt{\lambda_0 - \|w\|^2}}, \qquad G(A,w) = \mu(w)e^A, \qquad p(A,w) = \alpha G(A,w). \tag{1010}$$

Eliminating the three regular multiplier constraints from source lines 1484–1495 gives the intrinsic action

$$S_{\mathrm{packet}}[x,A,w] = \int_I \langle p(A,w), \dot{x}\rangle_L \,\mathrm{d}\tau \tag{1011}$$

on $\mathcal{J}_{2x} \times \mathcal{Q}_{\lambda_0,\omega_0}$. The use of the intrinsic constrained action loses no Euler–Lagrange equation: the constraint map is a submersion on $\omega_0 \neq 0$, so the Lagrange-multiplier theorem identifies ambient stationarity with stationarity against every tangent vector of the constrained manifold. This is a statement about classical stationary paths; it does not assert equality of quantum measures or path integrals after formally integrating out the multipliers.

Theorem 27.16 (Complete reduced dynamics of the constrained packet)**.** *For the momentum projection in* (1010)*:*

(i) *If $w \neq 0$, then $Dp_{(A,w)}$ has real rank* 4*, and its kernel has dimension* 3*, tangent to the S^3 momentum fibre. The Euler–Lagrange equations of* (1011) *are*

$$\dot{p} = 0, \qquad \dot{x} = 0,$$

with motion along the fixed S^3 fibre presymplectically null.

(ii) *If $w = 0$, then*

$$\det p = \frac{\alpha^2|\omega_0|^2}{\lambda_0},$$

$Dp_{(A,0)}$ has rank 3*, its image is $p^{\perp_L}$, and its kernel consists of all four real w-directions. The pointwise equations are*

$$\dot{p} = 0, \qquad \dot{x} = \lambda(\tau)p.$$

Every actual solution remaining on the threshold has constant (A,w), because the momentum fibre over a threshold momentum is a single point.

(iii) *Consequently, the source action is not a family of ordinary massive particle actions over all masses above the lower endpoint. Only its threshold stratum has the ordinary worldline characteristic; strict-interior configurations have $\dot{x} = 0$. The observer multiplication operator of source lines 1533–1603 is a separate positive-energy representation and is not the Hamiltonian generating τ-evolution in* (1011)*.*

Proof. For fixed w, varying A varies p through the three-dimensional fixed-determinant future hyperboloid, so the image of the A-differential is $p^{\perp_L}$. Direct differentiation without suppressing any constant gives

$$D\mu_w(\delta w) = \frac{|\omega_0|\,\mathrm{Re}\langle w, \delta w\rangle}{(\lambda_0 - \|w\|^2)^{3/2}}. \tag{1012}$$

When $w \neq 0$, a radial δw makes (1012) nonzero and changes $\det p = \alpha^2\mu(w)^2$. This supplies a fourth direction transverse to $p^{\perp_L}$. The remaining w-directions satisfy $\mathrm{Re}\langle w, \delta w\rangle = 0$, form a three-dimensional space, and are exactly tangent to the S^3 momentum fibre.

At $w = 0$, equation (1012) vanishes in every direction, so the four w-directions lie in $\ker Dp$ and the image is only $p^{\perp_L}$. These directions open the mass above its minimum at second order:

$$\det p(A, \varepsilon v) = \frac{\alpha^2 |\omega_0|^2}{\lambda_0 - \varepsilon^2 \|v\|^2}.$$

For an action $\int \langle p(q), \dot{x} \rangle_L \mathrm{d}\tau$, variation of x gives $\dot{p} = 0$, while variation of q gives

$$\langle Dp_q(\delta q), \dot{x} \rangle_L = 0 \quad \text{for all } \delta q.$$

If $w \neq 0$, the image is all of $\mathcal{J}_2$, hence $\dot{x} = 0$. If $w = 0$, the image is $p^{\perp_L}$, hence $\dot{x} \in \mathrm{span}_{\mathbb{R}}\{p\}$. The packet's own momentum-fibre theorem states that the fibre over a threshold momentum is a singleton, which completes the solution statement. □

27.9.3 Relation to the rank-one irregularity

The packet's fixed-$\omega_0 \neq 0$ constrained manifold never meets $\mathrm{rank}_{\mathbb{C}} Z \leq 1$, so the zero differential of $\det \circ P_\alpha$ does not invalidate its topology or positive lower bound. The exact relation is instead:

1. on the unrestricted carrier, the rank-one and zero strata give the massless irregularities proved in Theorems 27.9 and 27.12;
2. on the packet's constrained carrier, Z remains rank two, but the momentum projection develops a different four-real-dimensional critical kernel at its positive-mass threshold $w = 0$;
3. the algebraic inequality is a sharp, externally constraint-selected threshold; a dynamically generated gap would require a specified interacting Hamiltonian or field operator whose spectrum produces the scale rather than receiving λ_0, ω_0 as fixed input.

28 The Hermitian determinant normal slice as a Lorentzian host

This independently extractable section restates one already verified finite-dimensional identity and constructs exact physical models from it. When it is placed after the nine-dimensional observation construction, the notation agrees by $C_{A_2} = C$, $R_P = R$, $H_0 = uu^*$, $Q_\perp = Q_u$, and the same map $\jmath_P(G) = R_P G R_P^*$. The positive conclusion is not merely that the identity resembles Minkowski space: the affine normal slice has a canonically polarized Lorentz metric, the displayed $SL_2(\mathbb{C})$ congruence subgroup induces the spin double cover of the proper orthochronous Lorentz group, and after one explicit choice of length scale it supports a completely specified Klein–Gordon theory and finite Gaussian regulators. The same host also supports classical Yang–Mills theory for any separately chosen compact gauge group. In addition, the standard metrics entering the Moore–Penrose section select one exact compact $SU(2)$-type representation internal to the normal data; promoting its pointwise action to a local gauge symmetry gives the gauge–scalar model below and is an additional construction. The proofs identify exactly which structures come from the determinant and which are additional physical choices.

28.1 Fixed matrices and the exact normal metric

Put

$$u = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \qquad P = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix}, \qquad C_{A_2} = PP^* = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \tag{1013}$$

and use the standard Hermitian products in these displayed bases. The standard-metric Moore–Penrose right inverse is

$$R_P = P^*(PP^*)^{-1} = \frac{1}{3}\begin{pmatrix} 2 & 1 \\ -1 & 1 \\ -1 & -2 \end{pmatrix}. \tag{1014}$$

The denominators in (1014) are essential: this is a metric-selected complex-linear section, not a homomorphism splitting the displayed integral lattices. Set

$$H_0 = uu^*, \qquad \mathcal{N}_P = \{X \in \mathrm{Herm}_3(\mathbb{C}) : Xu = 0\}, \qquad \jmath_P(G) = R_P G R_P^*. \tag{1015}$$

Proposition 28.1 (Exact identification of the normal slice)**.** *The following identities hold without a change of normalization:*

$$PR_P = I_2, \qquad u^*R_P = 0, \qquad R_P^*R_P = C_{A_2}^{-1}, \qquad R_P P = I_3 - \frac{1}{3}uu^*. \tag{1016}$$

The real-linear map $\jmath_P : \mathrm{Herm}_2(\mathbb{C}) \to \mathcal{N}_P$ is an isomorphism. Its inverse is

$$\jmath_P^{-1}(X) = PXP^*. \tag{1017}$$

For all $G \in \mathrm{Herm}_2(\mathbb{C})$ and $\varepsilon \in \mathbb{R}$,

$$\det(H_0 + \varepsilon \jmath_P(G)) = \varepsilon^2 \det G. \tag{1018}$$

Proof. The first three equalities in (1016) follow by direct multiplication. The last member is the orthogonal projection onto the row space of P, which is $u^\perp$; direct multiplication gives the displayed matrix as well. Hence $P\jmath_P(G)P^* = G$. Conversely, if $X = X^*$ and $Xu = 0$, then also $u^*X = 0$, and therefore

$$R_P(PXP^*)R_P^* = (R_P P)X(R_P P) = X.$$

This proves the isomorphism and its inverse.

For the determinant, put $v_0 = u/\sqrt{3}$, choose an orthonormal complement $U_\perp$, and let $U = (v_0\ U_\perp)$ and $L = U_\perp^* R_P$. Then

$$U^*H_0U = \mathrm{diag}(3, 0, 0), \qquad U^*\jmath_P(G)U = 0 \oplus LGL^*,$$

while

$$|\det L|^2 = \det(L^*L) = \det(R_P^*R_P) = \det(C_{A_2}^{-1}) = \frac{1}{3}.$$

Taking the determinant in this orthonormal frame gives

$$\det(H_0 + \varepsilon \jmath_P(G)) = 3\varepsilon^2 |\det L|^2 \det G = \varepsilon^2 \det G.$$

□

Put $X_P = (R_P\ u)$. The same block computation also records the exact position of this module inside the full nine-real-dimensional carrier:

$$H_0 + \varepsilon \jmath_P(G) = X_P \begin{pmatrix} \varepsilon G & 0 \\ 0 & 1 \end{pmatrix} X_P^*.$$

Thus, in the full (G, ψ, λ) block coordinates, this section uses the strict sublocus $(\psi, \lambda) = (0, 1)$. Its field models live on the observed four-vector section; they do not yet define dynamics for the retained spinor ψ and scalar λ directions.

For $G, K \in \mathrm{Herm}_2(\mathbb{C})$, define

$$\eta_2(G, K) = \frac{1}{2}\big(\det(G + K) - \det G - \det K\big). \tag{1019}$$

Transport this form to $\mathcal{N}_P$ by

$$\eta_P(\jmath_P(G), \jmath_P(K)) = \eta_2(G, K). \tag{1020}$$

Theorem 28.2 (The determinant Hessian is exactly Minkowski)**.** *For tangent vectors $X = \jmath_P(G)$ and $Y = \jmath_P(K)$,*

$$\eta_P(X, Y) = \frac{1}{2} D^2(\det)_{H_0}[X, Y]. \tag{1021}$$

Writing

$$G(t, x, y, z) = \begin{pmatrix} t + y & x + iz \\ x - iz & t - y \end{pmatrix}, \tag{1022}$$

Relative to the convention $\left(\begin{smallmatrix} t+x_3 & x_1-ix_2 \\ x_1+ix_2 & t-x_3 \end{smallmatrix}\right)$ used in the observation construction, the exact coordinate dictionary is $(x_1, x_2, x_3) = (x, -z, y)$. Thus the two displayed Pauli conventions differ only by this stated spatial relabelling and sign, not by a change of Lorentz form. one has

$$\eta_2(G(t, x, y, z), G(t', x', y', z')) = tt' - xx' - yy' - zz'. \tag{1023}$$

Thus $(\mathcal{N}_P, \eta_P)$ has signature $(1, 3)$. Its distinguished future cone is

$$\mathcal{C}_P^+ = \jmath_P(\mathrm{Herm}_2(\mathbb{C})_{\geq 0}) = \{\jmath_P(G(t, \mathbf{x})) : t \geq |\mathbf{x}|\}. \tag{1024}$$

Its nonzero null boundary consists exactly of the images of rank-one positive semidefinite matrices.

Proof. Proposition 28.1, applied to $sG + rK$, gives

$$\det(H_0 + s\jmath_P(G) + r\jmath_P(K)) = \det(sG + rK).$$

The coefficient of sr on the right is $2\eta_2(G, K)$. Taking one derivative in each variable proves (1021). Direct expansion of (1022) gives

$$\det G(t, x, y, z) = t^2 - x^2 - y^2 - z^2,$$

and polarization gives (1023). A two-by-two Hermitian matrix is positive semidefinite precisely when its trace is nonnegative and its determinant is nonnegative, which here is $t \geq 0$ and $t^2 \geq |\mathbf{x}|^2$. On the nonzero boundary its determinant vanishes and its rank is one. □

The corresponding tetrad inside $\mathrm{Herm}_3(\mathbb{C})$ is completely explicit. If $E_0 = \jmath_P(G(1, 0, 0, 0))$, $E_1 = \jmath_P(G(0, 1, 0, 0))$, $E_2 = \jmath_P(G(0, 0, 1, 0))$, and $E_3 = \jmath_P(G(0, 0, 0, 1))$, then

$$E_0 = \frac{1}{9}\begin{pmatrix} 5 & -1 & -4 \\ -1 & 2 & -1 \\ -4 & -1 & 5 \end{pmatrix}, \qquad E_1 = \frac{1}{9}\begin{pmatrix} 4 & 1 & -5 \\ 1 & -2 & 1 \\ -5 & 1 & 4 \end{pmatrix}, \tag{1025}$$

$$E_2 = \frac{1}{3}\begin{pmatrix} 1 & -1 & 0 \\ -1 & 0 & 1 \\ 0 & 1 & -1 \end{pmatrix}, \qquad E_3 = \frac{1}{3}\begin{pmatrix} 0 & i & -i \\ -i & 0 & i \\ i & -i & 0 \end{pmatrix}. \tag{1026}$$

Every row and column of each E_μ sums to zero, and

$$\eta_P(E_\mu, E_\nu) = \mathrm{diag}(1, -1, -1, -1)_{\mu\nu}. \tag{1027}$$

These identities follow by multiplying the matrices in (1014) and then applying (1023); no orthonormalization of the marked lattice is being suppressed.

The metric is a genuine normal Hessian. It is not a positive energy form: it takes both signs. Energy appears only after choosing a time function and passing to a Hamiltonian formulation below.

Because $\det(H_0 + \jmath_P(G)) = \det G$ on the entire affine slice, not just to second order, its restriction there is quadratic. Hence the polarized Hessian η_P is constant at every point of the affine slice; affine translations really are isometries of this metric.

28.2 Exact Lorentz and affine symmetry

For $A \in SL_2(\mathbb{C})$, define

$$\widehat{A}_P = \frac{1}{3}uu^* + R_P A P \in SL_3(\mathbb{C}). \tag{1028}$$

Theorem 28.3 (The spin action inside the normal slice)**.** *The matrix $\widehat{A}_P$ is invertible, with*

$$\widehat{A}_P^{-1} = \frac{1}{3}uu^* + R_P A^{-1} P, \qquad \det \widehat{A}_P = \det A = 1. \tag{1029}$$

It fixes u and satisfies

$$\widehat{A}_P H_0 \widehat{A}_P^* = H_0, \qquad \widehat{A}_P \jmath_P(G) \widehat{A}_P^* = \jmath_P(AGA^*). \tag{1030}$$

Consequently $SL_2(\mathbb{C})$ acts on $\mathcal{N}_P$ by future-cone preserving η_P-isometries. The kernel is $\{I_2, -I_2\}$, and the induced map

$$PSL_2(\mathbb{C}) \xrightarrow{\sim} SO^+(1,3) \tag{1031}$$

is an isomorphism of real Lie groups.

The kernel $\{\pm I_2\}$ is a statement about the normal-vector action. On the full (G, ψ, λ) carrier the same embedded $SL_2(\mathbb{C})$-action is faithful, because $-I_2$ fixes G and λ but sends $\psi \mapsto -\psi$.

Proof. Let $Q_0 = uu^*/3$ and $Q_\perp = R_P P = I_3 - Q_0$. The relations $Q_0 R_P = 0$, $P Q_0 = 0$, and $P R_P = I_2$ show that the two matrices in (1029) multiply to $Q_0 + Q_\perp = I_3$. In the ordered decomposition $\mathbb{C}u \oplus R_P\mathbb{C}^2$, the map is $1 \oplus A$, proving its determinant. The same relations prove (1030). Since $\det(AGA^*) = |\det A|^2 \det G = \det G$, the action preserves η_P; congruence preserves positive semidefiniteness, hence the chosen future cone.

If $AGA^* = G$ for every Hermitian G, taking $G = I_2$ makes A unitary, after which A commutes with every Hermitian matrix. Hence A is scalar, and determinant one gives $A = \pm I_2$. The differential of the induced map has zero kernel: if $XG + GX^* = 0$ for every Hermitian G, then $G = I_2$ makes X skew-Hermitian and the remaining equations make it scalar; trace zero then gives $X = 0$. Domain and target Lie algebras both have real dimension six, so the image contains a neighborhood of the identity. It is therefore an open subgroup of the connected group $SO^+(1,3)$, hence the whole group. □

Translations are also intrinsic to the affine slice. If $B \in \mathrm{Herm}_2(\mathbb{C})$, define

$$(B, A) \cdot (H_0 + \jmath_P(G)) = H_0 + \jmath_P(B + AGA^*). \tag{1032}$$

With multiplication $(B, A)(C, D) = (B + ACA^*, AD)$, this is the spin double cover of the proper orthochronous Poincaré action. Its kernel is $\{(0, I_2), (0, -I_2)\}$. Under the dimensional chart (1054) introduced below, a physical translation by $b \in \mathbb{R}^{1,3}$ is the element $B = G(b/\ell)$.

Proposition 28.4 (Ambient $SU(3)$ is not an internal color action)**.** *The subgroup of $SU(3)$ preserving H_0 is*

$$\mathrm{Stab}_{SU(3)}(H_0) = \{g \in SU(3) : gu = e^{i\theta}u\} \cong S(U(1) \times U(2)). \tag{1033}$$

It preserves $\mathcal{N}_P$. A general element of $SU(3)$ moves H_0 and hence moves the chosen affine normal slice to a different one. Furthermore, $\widehat{A}_P$ in (1028) *is unitary exactly when*

$$A^* C_{A_2}^{-1} A = C_{A_2}^{-1}; \tag{1034}$$

this compact subgroup is conjugate to $SU(2)$, whereas generic Lorentz boosts are not unitary. Thus the Lorentz spin action supplied by the normal slice is not the ambient $SU(3)$ orbit action and does not itself define an internal $SU(3)$ gauge symmetry.

Proof. The equality $gH_0g^* = H_0$ is equivalent to equality of the rank-one lines and norms, hence to $gu = e^{i\theta}u$. The orthogonal complement is then invariant, giving the displayed block stabilizer and preservation of Hermitian operators supported on $u^\perp$. On $R_P\mathbb{C}^2 = u^\perp$, the standard norm pulls back to the form $R_P^*R_P = C_{A_2}^{-1}$. Therefore $\widehat{A}_P$ is unitary precisely when (1034) holds. Positive-definite congruence identifies this group with $SU(2)$. □

The proposition concerns the congruence action of the ambient $SU(3)$ on $\mathrm{Herm}_3(\mathbb{C})$. It is compatible with, but distinct from, the left $SU(3)$-action on $\mathrm{Mat}_{3\times2}(\mathbb{C})$ whose invariant Gram map is $Z \mapsto Z^*Z$: the source spaces and actions are different, and no identification of them is being made here.

28.3 The compact gauge–scalar carrier selected by the standard metrics

The Lorentz congruence group is noncompact, so it cannot preserve a positive field-space norm. The standard metrics already used in (1014) do, however, select the positive matrix and compact subgroup

$$W_P := R_P^*R_P = C_{A_2}^{-1} = \frac{1}{3}\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \qquad K_P := \{a \in SL_2(\mathbb{C}) : a^*W_Pa = W_P\}. \tag{1035}$$

Because $W_P = C_{A_2}^{-1}$, inversion of the defining equality gives

$$a^*W_Pa = W_P \quad \Longleftrightarrow \quad aC_{A_2}a^* = C_{A_2}.$$

Accordingly, K_P is exactly the Lorentz little group of the future timelike vector C_{A_2}, equivalently the observer stabilizer selected by the standard metrics. Its promotion to a local internal gauge symmetry below is mathematically valid but is an additional physical interpretation.

Proposition 28.5 (Exact compact representation on the normal slice)**.** *Conjugation $a \mapsto W_P^{1/2}aW_P^{-1/2}$ is an isomorphism $K_P \cong SU(2)$. The real bilinear form*

$$E_P(G,K) := \mathrm{tr}\big(\jmath_P(G)\jmath_P(K)\big) = \mathrm{tr}(GW_PKW_P) \tag{1036}$$

is a positive-definite K_P-invariant inner product on $\mathrm{Herm}_2(\mathbb{C})$*. For $a \in K_P$, the lift*

$$\widehat{a}_P = \frac{1}{3}uu^* + R_PaP \tag{1037}$$

lies in $SU(3)$, fixes H_0, and obeys

$$\widehat{a}_P\jmath_P(G)\widehat{a}_P^* = \jmath_P(aGa^*). \tag{1038}$$

After whitening, $\widetilde{G} = W_P^{1/2}GW_P^{1/2}$, the action is ordinary unitary congruence. Therefore, as a real K_P-module,

$$\mathrm{Herm}_2(\mathbb{C}) = \mathbb{R}W_P^{-1} \oplus \{G \in \mathrm{Herm}_2(\mathbb{C}) : \mathrm{tr}(W_PG) = 0\} \cong \mathbf{1} \oplus \mathbf{3}. \tag{1039}$$

The centre $\{\pm I_2\}$ acts trivially. Thus the metric-selected normal field is a singlet plus an adjoint triplet; it is neither a fundamental weak doublet nor an $SU(3)$ color triplet.

Proof. The defining identity in (1035) is equivalent to $(W_P^{1/2}aW_P^{-1/2})^*(W_P^{1/2}aW_P^{-1/2}) = I_2$; determinants are one, so this identifies K_P with $SU(2)$. Cyclicity of the trace and $R_P^*R_P = W_P$ give the second equality in (1036). Moreover $E_P(G,G) = \mathrm{tr}(\jmath_P(G)^2)$, which is positive for nonzero Hermitian $\jmath_P(G)$. Equation (1030) and unitarity of $\widehat{a}_P$ prove invariance. Proposition 28.4 gives $\widehat{a}_P \in U(3)$, and the block action on $u^\perp \oplus \mathbb{C}u$ gives $\det\widehat{a}_P = \det a = 1$. The whitened formula is direct multiplication. Unitary congruence fixes the scalar line and acts through the adjoint rotation representation on traceless Hermitian matrices; the central elements act by $G \mapsto (\pm I_2)G(\pm I_2)^* = G$. □

Theorem 28.6 (A fully typed positive gauge–scalar realization)**.** *Let (M, g) be an oriented Riemannian d-manifold and use the trivial principal K_P-bundle. Take*

$$\mathcal{A} \in \Omega^1(M; \mathfrak{k}_P), \qquad G \in C^\infty(M; \mathrm{Herm}_2(\mathbb{C})) \tag{1040}$$

as the connection and normal scalar. For $a \in C^\infty(M, K_P)$, define

$$G^a = aGa^*, \qquad \mathcal{A}^a = a\mathcal{A}a^{-1} - (da)a^{-1}, \tag{1041}$$

$$D_{\mathcal{A}}G = dG + \mathcal{A}G + G\mathcal{A}^*, \qquad F_{\mathcal{A}} = d\mathcal{A} + \mathcal{A} \wedge \mathcal{A}. \tag{1042}$$

Choose on $\mathfrak{k}_P$ the positive $\mathrm{Ad}(K_P)$-invariant form transported from $-\operatorname{Re}\operatorname{tr}(XY)$ on $\mathfrak{su}(2)$. For $g_{\mathrm{YM}}, Z_G > 0$ and $m_G, \lambda_4 \geq 0$, the Euclidean action

$$\begin{aligned} S^{\mathrm{cpt}}_{g_{\mathrm{YM}}, Z_G, m_G, \lambda_4}[\mathcal{A}, G] = \int_M \Big(& \frac{1}{2g_{\mathrm{YM}}^2} |F_{\mathcal{A}}|^2 + \frac{Z_G}{2} |D_{\mathcal{A}}G|^2_{E_P} + \frac{m_G^2}{2} |G|^2_{E_P} \\ & + \frac{\lambda_4}{4} (\det G)^2 \Big) \, d\,\mathrm{vol}_g \end{aligned} \tag{1043}$$

is nonnegative and gauge invariant. The associated normal-matrix field

$$H_\varepsilon = H_0 + \varepsilon \jmath_P(G) \tag{1044}$$

is equivariant under the lifted action $H^a_\varepsilon = \widehat{a}_P H_\varepsilon \widehat{a}^*_P$. If $\widehat{\mathcal{A}} = R_P \mathcal{A} P$, then

$$D_{\widehat{\mathcal{A}}} \jmath_P(G) = \jmath_P(D_{\mathcal{A}}G), \qquad D_{\widehat{\mathcal{A}}} H_0 = 0. \tag{1045}$$

On $M = \mathbb{R}^d$, the scalar Hessian at $(\mathcal{A}, G) = (0, 0)$ is the nonnegative self-adjoint operator

$$\begin{aligned} \mathcal{K}_{Z_G, m_G} &= (-Z_G \Delta + m_G^2) \otimes I_{\mathrm{Herm}_2}, \\ \mathrm{Dom}(\mathcal{K}_{Z_G, m_G}) &= H^2(\mathbb{R}^d; \mathrm{Herm}_2(\mathbb{C})), \\ \mathrm{Spec}(\mathcal{K}_{Z_G, m_G}) &= [m_G^2, \infty). \end{aligned} \tag{1046}$$

Thus the fixed normal data plus the standard metrics select the compact representation $K_P \curvearrowright \mathrm{Herm}_2(\mathbb{C})$ and its positive fibre metric. After the additional choice to gauge that pointwise action, the formulas above give an exact compact gauge–scalar model, but they still do not select $m_G, Z_G, g_{\mathrm{YM}}, \lambda_4$, a quantum state, or an interacting gap.

Proof. Direct substitution gives $D_{\mathcal{A}^a} G^a = a(D_{\mathcal{A}}G)a^*$ and $F_{\mathcal{A}^a} = aF_{\mathcal{A}}a^{-1}$. Invariance of the two positive inner products and $\det(aGa^*) = |\det a|^2 \det G = \det G$ prove gauge invariance; every term in (1043) is nonnegative. Equation (1038) gives equivariance of (1044). Using $PR_P = I_2$, $Pu = 0$, and $u^* R_P = 0$, direct multiplication proves (1045). Fourier transformation gives the self-adjoint operator, its full domain, and its spectrum in (1046). Finally, $(\det G)^2$ is homogeneous of degree four, so it has zero Hessian at $G = 0$. □

Corollary 28.7 (The selected normal field cannot screen a fundamental centre charge)**.** *Every tensor power and every polynomial composite of the $\mathbf{1} \oplus \mathbf{3}$ normal representation has trivial action by $-I_2 \in K_P$. The fundamental two-dimensional representation has action $-I_2$. Therefore the field G in Theorem 28.6 cannot by itself screen a fundamental centre charge; such a model requires an additional centre-nontrivial matter field and its independently specified kinetic, mass, and interaction terms. This non-screening statement is confined to the G-only normal section. The full (G, ψ, λ) carrier already contains the centre-nontrivial spinor ψ; what remains missing is a specified kinetic and interaction theory for that retained field.*

Proof. The centre acts trivially on G, hence trivially on every tensor power and subquotient generated from it. It acts as minus the identity in the fundamental representation, so that representation cannot occur in this tensor category. □

28.4 The candidate cusp matrix is an exact spacelike vector

The preceding construction applies to every Hermitian two-by-two matrix. For the particular inverse-dual matrix retained in the cusp calculation, let

$$G_{\eta,T} = \begin{pmatrix} 2L_T & 2m_T \\ 2m_T & -q_T/3 \end{pmatrix}, \qquad D_T = L_T q_T + 6m_T^2 > 0. \tag{1047}$$

Proposition 28.8 (Exact candidate-specific Lorentz coordinates)**.** *In the coordinates* (1022)*,*

$$(t_T, x_T, y_T, z_T) = \left(L_T - \frac{q_T}{6},\ 2m_T,\ L_T + \frac{q_T}{6},\ 0 \right), \tag{1048}$$

and

$$\eta_2(G_{\eta,T}, G_{\eta,T}) = \det G_{\eta,T} = -\frac{2}{3} D_T < 0. \tag{1049}$$

If $L_T/T \to 1$, $q_T/T \to 1$, and $m_T/T \to 0$, then

$$\frac{1}{T}(t_T, x_T, y_T, z_T) \longrightarrow \left(\frac{5}{6}, 0, \frac{7}{6}, 0 \right). \tag{1050}$$

Thus this particular candidate vector is spacelike for every T. It is not a positive-mass four-momentum. This statement concerns the indefinite inverse-dual matrix $G_{\eta,T}$; it does not concern the different positive Gram momentum $p_\alpha(Z) = \alpha Z^* Z$ used in the retained-c worldline construction. The latter is positive semidefinite by definition, and its image is all of $\mathrm{Herm}_2(\mathbb{C})_{\geq 0}$*: for every $G \succeq 0$, the block matrix $Z = \alpha^{-1/2} \left(\begin{smallmatrix} G^{1/2} \\ 0_{1\times 2} \end{smallmatrix} \right) \in \mathrm{Mat}_{3\times 2}(\mathbb{C})$ satisfies $p_\alpha(Z) = G$. Hence*

$$G_{\eta,T} \notin \operatorname{im} p_\alpha \qquad (T > 0,\ \alpha > 0).$$

Positive rescaling preserves the determinant sign and $GL_2(\mathbb{C})$ congruence preserves inertia, so neither operation repairs this mismatch. The Gram momentum lies on $\det p_\alpha = m_0^2 c^2$ *only after that mass-shell constraint and the parameters α, c, m_0 are supplied. Interpreting $G_{\eta,T}$ as a field value, a normal deformation, or a spacelike wave-vector requires an additional map; the determinant calculation alone does not choose among those interpretations.*

Proof. Solving $t + y = 2L_T$, $t - y = -q_T/3$, and $x + iz = 2m_T$ gives (1048). Substitution gives

$$t_T^2 - x_T^2 - y_T^2 - z_T^2 = (2L_T)(-q_T/3) - 4m_T^2 = -\frac{2}{3} D_T.$$

The limit is componentwise. □

Theorem 28.9 (Exact determinant-level orbit geometry)**.** *Fix $\varepsilon > 0$ and write*

$$\mathbb{M}_\varepsilon := H_0 + \varepsilon \jmath_P(\mathrm{Herm}_2(\mathbb{C})).$$

For every $r > 0$, the future timelike determinant shell

$$\mathscr{P}^+_{r,\varepsilon} := \{ H_0 + \varepsilon \jmath_P(G) : G \succ 0,\ \det G = r^2 \} \cong SL_2(\mathbb{C})/SU(2) \tag{1051}$$

is one orbit of the lifted congruence action. It equals

$$\{ H \in \mathbb{M}_\varepsilon : H \succ 0,\ \det H = \varepsilon^2 r^2 \}.$$

The nonzero future-null locus

$$\mathscr{N}^+_\varepsilon := \{ H_0 + \varepsilon \jmath_P(G) : G \succeq 0,\ \operatorname{rank} G = 1 \} \tag{1052}$$

is also one $SL_2(\mathbb{C})$-orbit; each of its matrices is positive semidefinite of ambient rank two. Finally, with $r_T = \sqrt{2D_T/3}$, the candidate-specific spacelike shell is

$$\mathscr{S}_{T,\varepsilon} := \{H_0 + \varepsilon \jmath_P(G) : \det G = -r_T^2\} \cong SL_2(\mathbb{C})/SU(1,1), \tag{1053}$$

and every matrix in it has inertia $(2,1)$. Consequently the normal slice contains every timelike mass shell and the future null cone after a unit dictionary is supplied, but it selects neither r nor a physical mass.

Proof. If $G \succ 0$ and $\det G = r^2$, then $A = (G/r)^{1/2}$ lies in $SL_2(\mathbb{C})$ and $A(rI_2)A^* = G$. The stabilizer of rI_2 is $SU(2)$, proving (1051); positivity and the ambient determinant follow from the block decomposition in the proof of Proposition 28.1. Every nonzero positive rank-one G is vv^*. Choose $w \in \mathbb{C}^2$ with $\det(v\ w) = 1$; then $A = (v\ w) \in SL_2(\mathbb{C})$ and $A\,\mathrm{diag}(1,0)A^* = vv^*$. This proves null transitivity, and the orthogonal block decomposition gives ambient rank $1 + 1 = 2$.

Let $J = \mathrm{diag}(1,-1)$. A Hermitian G with $\det G = -r_T^2$ has inertia $(1,1)$, so Sylvester congruence gives $G = B(r_T J)B^*$ for some $B \in GL_2(\mathbb{C})$. Determinants give $|\det B| = 1$. If $\det B = e^{i\theta}$, then $D = \mathrm{diag}(e^{-i\theta}, 1)$ stabilizes J, and $A = BD$ has determinant one. Thus the action is transitive, and the stabilizer is

$$\{A \in SL_2(\mathbb{C}) : AJA^* = J\} = SU(1,1).$$

Equation (1049) puts the retained candidate on this shell. The block decomposition gives one positive eigenvalue from H_0 and one of each sign from G, proving the inertia claim. □

28.5 Units and the minimal continuum field model

All entries in (1013)–(1015) are dimensionless. A physical spacetime requires a scale. For $\ell > 0$, set

$$\kappa_\ell(x^0, x^1, x^2, x^3) = H_0 + \jmath_P\left(G\left(\frac{x^0}{\ell}, \frac{x^1}{\ell}, \frac{x^2}{\ell}, \frac{x^3}{\ell}\right)\right). \tag{1054}$$

Equip the affine slice with the constant metric $g_{P,\ell} = \ell^2 \eta_P$. Then

$$\kappa_\ell^* g_{P,\ell} = (dx^0)^2 - (dx^1)^2 - (dx^2)^2 - (dx^3)^2, \qquad \det \kappa_\ell(x) = \ell^{-2} \eta_{\mu\nu} x^\mu x^\nu. \tag{1055}$$

For the scaled normal displacement $H_0 + \varepsilon \jmath_P(G(x/\ell))$, the determinant is multiplied by ε^2. For $\varepsilon \neq 0$, the corresponding isometric metric rescaling is $(\ell/|\varepsilon|)^2 \eta_P$. Hence ε fixes no absolute energy or length by itself.

The exact unit dictionary is:

symbol	role	status from the determinant data
$P, u, R_P, \jmath_P$	marked linear algebra	fixed and dimensionless
ε	normal-displacement amplitude	fixed only when supplied
ℓ	conversion from matrix coordinates to length	independent choice
$x^0 = ct$	time coordinate	requires the independent constant c
m_{KG}	inverse Compton length in the scalar model	independent choice
$m_{\mathrm{phys}} = \hbar m_{\mathrm{KG}}/c$	physical scalar mass	requires $\hbar, c$
$L = Na$	infrared box size and ultraviolet mesh	regulator choices
K, g_{YM}	compact gauge group and coupling	K arbitrary; $K_P \cong SU(2)$ is the metric-selected little group and its gauging is optional

The retained worldline mass m_0, the Klein–Gordon inverse length m_{KG}, and the dimensionless determinant-shell radius r are independent inputs. Only after declaring that the worldline and field theory describe the same physical species may one impose the unit conversion

$$m_{\mathrm{KG}} = \frac{m_0 c}{\hbar}, \qquad \Delta E_{\mathrm{free}} = \hbar c\, m_{\mathrm{KG}} = m_0 c^2.$$

No determinant identity derives this identification.

Use natural units $c = \hbar = 1$ in the remaining formulas. The minimal field placed on the Lorentzian host is a real scalar $\phi \in C_c^\infty(\mathbb{R}^{1,3}; \mathbb{R})$ with action

$$S_{m_{\mathrm{KG}}}[\phi] = \frac{1}{2} \int_{\mathbb{R}^{1,3}} \left(\partial_\mu \phi\, \partial^\mu \phi - m_{\mathrm{KG}}^2 \phi^2 \right) d^4x, \qquad m_{\mathrm{KG}} \geq 0. \tag{1056}$$

Its Euler–Lagrange operator is $\Box + m_{\mathrm{KG}}^2$. The Poincaré action (1032) acts on fields by pullback. The action, the stress-energy tensor

$$T_{\mu\nu} = \partial_\mu \phi \partial_\nu \phi - \frac{1}{2} \eta_{\mu\nu} (\partial_\alpha \phi \partial^\alpha \phi - m_{\mathrm{KG}}^2 \phi^2), \tag{1057}$$

and the mass shell $p_\mu p^\mu = m_{\mathrm{KG}}^2$ are invariant. Smeared fields $\phi(f) = \int \phi f\, d^4x$ are the basic linear observables.

On $x^0 = 0$, smooth compactly supported initial data (ϕ_0, π_0) carry the symplectic form

$$\sigma((\phi, \pi), (\psi, \rho)) = \int_{\mathbb{R}^3} (\phi \rho - \pi \psi)\, d^3x \tag{1058}$$

and energy

$$E_{m_{\mathrm{KG}}}(\phi, \pi) = \frac{1}{2} \int_{\mathbb{R}^3} (|\pi|^2 + |\nabla \phi|^2 + m_{\mathrm{KG}}^2 |\phi|^2)\, d^3x. \tag{1059}$$

For $m_{\mathrm{KG}} > 0$, the energy completion is $H^1(\mathbb{R}^3) \oplus L^2(\mathbb{R}^3)$; for $m_{\mathrm{KG}} = 0$, it is $\dot{H}^1(\mathbb{R}^3) \oplus L^2(\mathbb{R}^3)$. The symplectic pairing (1058) is exact on the dense smooth, compactly supported data. In the massless homogeneous completion it need not extend continuously to every pair without an additional dual-space choice.

Theorem 28.10 (Exact free-field spectral alternative)**.** *Let*

$$K_{m_{\mathrm{KG}}} = -\Delta + m_{\mathrm{KG}}^2, \quad \mathrm{Dom}(K_{m_{\mathrm{KG}}}) = H^2(\mathbb{R}^3), \quad \omega_{m_{\mathrm{KG}}} = K_{m_{\mathrm{KG}}}^{1/2} \tag{1060}$$

on $L^2(\mathbb{R}^3)$. On the bosonic Fock space $\mathscr{F}_s(L^2(\mathbb{R}^3))$, define the vacuum-subtracted Hamiltonian

$$\mathbb{H}_{m_{\mathrm{KG}}} = d\Gamma(\omega_{m_{\mathrm{KG}}}). \tag{1061}$$

Explicitly, on the orthogonal decomposition $\mathscr{F}_s = \bigoplus_{n=0}^\infty L^2(\mathbb{R}^3)^{\otimes_s n}$, its domain consists of the vectors $\Psi = (\Psi_n)$ for which each Ψ_n belongs to the domain of $\sum_{j=1}^n I \otimes \cdots \otimes \omega_{m_{\mathrm{KG}}} \otimes \cdots \otimes I$ and the squared norms of those images are summable. For every $m_{\mathrm{KG}} \geq 0$, the Fock vacuum is the unique zero-energy vector. Its excitation gap is exactly

$$\inf \mathrm{Spec}(\mathbb{H}_{m_{\mathrm{KG}}}|_{\Omega^\perp}) = m_{\mathrm{KG}}. \tag{1062}$$

Thus the same determinant normal geometry admits a positive gap of any chosen value $m_{\mathrm{KG}} > 0$ and a gapless model at $m_{\mathrm{KG}} = 0$.

Proof. Fourier transform identifies $K_{m_{\mathrm{KG}}}$ with multiplication by $|p|^2 + m_{\mathrm{KG}}^2$ and $\omega_{m_{\mathrm{KG}}}$ with multiplication by $\sqrt{|p|^2 + m_{\mathrm{KG}}^2}$. The latter has zero kernel even when $m_{\mathrm{KG}} = 0$, since the singleton $p = 0$ has Lebesgue measure zero. Hence the only Fock vector annihilated by $d\Gamma(\omega_{m_{\mathrm{KG}}})$ is the vacuum. On the n-particle sector, the Hamiltonian is the sum of n copies of $\omega_{m_{\mathrm{KG}}}$, so the nonvacuum spectral bottom is at least m_{KG}. The one-particle sector is reducing and has spectral bottom m_{KG}, proving equality. For $m_{\mathrm{KG}} = 0$, normalized Fourier-space functions supported in balls $|p| < 1/n$ have energy expectation at most $1/n$, which makes the absence of a positive gap explicit. □

There is no contradiction between the absence of a positive kinematic threshold on the unrestricted Gram carrier and the free Fock gap m_{KG}: the former allows the rescaling $Z \mapsto rZ$, whereas the latter first fixes an external mass coefficient and quantizes the chosen free Hamiltonian.

28.6 A complete finite regulator and its two limits

Let $L > 0$, $N \geq 2$, $a = L/N$, and $V_N = (\mathbb{Z}/N\mathbb{Z})^3$. Give $Q_{L,N} = \mathbb{R}^{V_N}$ the inner product

$$\langle q, r\rangle_a = a^3 \sum_{v \in V_N} q(v) r(v). \tag{1063}$$

The positive periodic lattice Laplacian is

$$(-\Delta_a q)(v) = a^{-2} \sum_{j=1}^{3} (2q(v) - q(v + e_j) - q(v - e_j)). \tag{1064}$$

Let $d\lambda_a$ be Lebesgue measure associated with the Euclidean space $(Q_{L,N}, \langle\cdot,\cdot\rangle_a)$, and let $\Delta_{Q_{L,N}}$ denote its coordinate Laplacian, so that $-\Delta_{Q_{L,N}}$ is nonnegative. For $m_{\mathrm{KG}} > 0$, define on $L^2(Q_{L,N}, d\lambda_a)$

$$\mathbb{H}_{m_{\mathrm{KG}},L,N} = \frac{1}{2}\Big(-\Delta_{Q_{L,N}} + \langle q, (-\Delta_a + m_{\mathrm{KG}}^2) q\rangle_a\Big) - E_{0,m_{\mathrm{KG}},L,N}, \tag{1065}$$

where $E_{0,m_{\mathrm{KG}},L,N}$ is its finite oscillator ground energy. More precisely, before subtracting $E_{0,m_{\mathrm{KG}},L,N}$, this is the self-adjoint operator associated with the closed form

$$\mathfrak{h}_{m_{\mathrm{KG}},L,N}[\Psi] = \frac{1}{2} \int_{Q_{L,N}} \Big(|\nabla \Psi(q)|_a^2 + \langle q, (-\Delta_a + m_{\mathrm{KG}}^2) q\rangle_a |\Psi(q)|^2 \Big)\, d\lambda_a(q), \tag{1066}$$

whose domain is

$$H^1(Q_{L,N}) \cap L^2\Big(Q_{L,N}, \langle q, (-\Delta_a + m_{\mathrm{KG}}^2) q\rangle_a \, d\lambda_a(q)\Big). \tag{1067}$$

Theorem 28.11 (Exact regulated spectrum and continuum limits)**.** *For $n = (n_1, n_2, n_3) \in V_N$, the squared normal-mode frequencies are*

$$\omega_{m_{\mathrm{KG}},L,N}(n)^2 = m_{\mathrm{KG}}^2 + \frac{4}{a^2} \sum_{j=1}^{3} \sin^2\!\left(\frac{\pi n_j}{N}\right). \tag{1068}$$

The operator $\mathbb{H}_{m_{\mathrm{KG}},L,N}$ is self-adjoint on the tensor-product oscillator domain, has a unique vacuum, and has exact gap m_{KG}. The subtracted constant is

$$E_{0,m_{\mathrm{KG}},L,N} = \frac{1}{2} \sum_{n \in V_N} \omega_{m_{\mathrm{KG}},L,N}(n). \tag{1069}$$

For $m_{\mathrm{KG}} = 0$, the constant mode is a free particle and the unreduced operator has no normalized ground state. On the mean-zero configuration space

$$Q_{L,N}^0 = \{q \in Q_{L,N} : \textstyle\sum_v q(v) = 0\}, \tag{1070}$$

the vacuum-subtracted oscillator has a unique ground state and gap

$$\Delta_{L,N}^0 = \frac{2}{a} \sin\!\left(\frac{\pi}{N}\right). \tag{1071}$$

At fixed L, for every fixed $r \in \mathbb{Z}^3$,

$$\omega_{m_{\mathrm{KG}},L,N}(r)^2 \longrightarrow m_{\mathrm{KG}}^2 + \frac{4\pi^2 |r|^2}{L^2} \qquad (N \to \infty), \tag{1072}$$

and

$$\Delta^0_{L,N} \longrightarrow \frac{2\pi}{L}. \tag{1073}$$

After the thermodynamic limit $L \to \infty$, the massive gap remains m_{KG}, whereas the massless positive-mode gap tends to zero.

Proof. The characters $v \mapsto L^{-3/2} e^{2\pi i n\cdot v/N}$ diagonalize $-\Delta_a$. Substitution gives (1068). Orthogonal diagonalization of the real configuration space turns (1065) into a tensor sum of one-dimensional harmonic oscillators with frequencies $\omega_{m_{\mathrm{KG}},L,N}(n)$. For $m_{\mathrm{KG}} > 0$, their minimum is attained at $n = 0$ and equals m_{KG}, proving the ground-state and gap claims.

When $m_{\mathrm{KG}} = 0$, the constant coordinate has zero restoring force, hence gives the free-particle operator and no L^2 ground state. Removing that coordinate leaves the smallest character with one component congruent to ± 1 and the others zero, giving (1071). With $a = L/N$, the elementary limit $N \sin(\pi r_j/N) \to \pi r_j$ proves (1072) and (1073). The last claims follow by sending L to infinity. □

This regulator supplies a literal configuration space, state Hilbert space, quadratic Hamiltonian, observables generated by the coordinates and momenta, and exact fixed-mode ultraviolet and lowest-frequency infrared limits. The displayed limits do not assert strong- or norm-resolvent convergence, convergence of Gaussian measures, or convergence of Fock representations. The mass parameter is not hidden in P, R_P, ε, or the determinant: it is the independent coefficient in (1056).

28.7 An exact classical gauge-theory completion

Let K be any compact matrix Lie group with Lie algebra $\mathfrak{k}$ and an $\mathrm{Ad}\, K$-invariant positive inner product. On the trivial bundle over the Lorentzian host, take a smooth compactly supported gauge potential $A \in \Omega^1_c(\mathbb{R}^{1,3}; \mathfrak{k})$, with curvature

$$F_A = dA + A \wedge A. \tag{1074}$$

The gauge group of smooth maps $\mathbb{R}^{1,3} \to K$ that equal the identity off a compact set acts by

$$A^g = gAg^{-1} - (dg)g^{-1}, \qquad F_{A^g} = gF_Ag^{-1}. \tag{1075}$$

For an independently chosen coupling $g_{\mathrm{YM}} > 0$, the classical action is

$$S_{K,g_{\mathrm{YM}}}[A] = -\frac{1}{4g^2_{\mathrm{YM}}} \int \langle F_{\mu\nu}, F^{\mu\nu} \rangle \, d^4x. \tag{1076}$$

It is invariant under (1075) and the Poincaré action. For a closed curve γ and a finite-dimensional representation ρ of K,

$$W_{\gamma,\rho}(A) = \mathrm{tr}\, \rho(\mathrm{Hol}_A(\gamma)) \tag{1077}$$

is gauge invariant.

On the spatial regulator $\Sigma_L = (\mathbb{R}/L\mathbb{Z})^3$, the smooth Hamiltonian phase space before reduction consists of $(A, E) \in \Omega^1(\Sigma_L; \mathfrak{k})^2$, with gauge action

$$(A, E)^g = (gAg^{-1} - (dg)g^{-1},\, gEg^{-1}). \tag{1078}$$

Its Gauss constraint and energy are

$$d^*_A E = 0, \qquad \mathcal{E}_{K,g}(A, E) = \frac{1}{2g^2_{\mathrm{YM}}} \int_{\Sigma_L} (|E|^2 + |F_A|^2) \, d^3x, \tag{1079}$$

and the reduced classical state space is the constraint locus modulo $C^\infty(\Sigma_L, K)$. At this smooth level it is a set-theoretic orbit space, generally stratified by stabilizer type; making it a smooth symplectic space requires Sobolev completions and hypotheses on the action. Equations (1075) and (1079) show

directly that the constraint locus, energy, and Wilson observables descend to this quotient: in particular, $d^*_{A^g} E^g = g(d^*_A E)g^{-1}$, because the Hodge star acts only on the base and the adjoint action is orthogonal. In temporal gauge the canonical momentum is $\Pi_E = g_{\mathrm{YM}}^{-2} E$. Equivalently, the symplectic form on tangent vectors is

$$\Omega_{(A,E)}\big((a,e),(a',e')\big) = \frac{1}{g_{\mathrm{YM}}^2}\int_{\Sigma_L} \big(\langle e, a'\rangle - \langle e', a\rangle\big)\, d\,\mathrm{vol}\,. \tag{1080}$$

Proposition 28.12 (The exact nonzero-mode Gaussian gauge gap)**.** *At $(A,E) = (0,0)$, impose the Coulomb conditions $d^*A = d^*E = 0$ and remove harmonic one-forms. Relative to the unweighted L^2 configuration metric, the energy Hessian is $g_{\mathrm{YM}}^{-2}\Delta_1$, where the generalized squared-frequency operator is*

$$\Delta_1 = d^*d, \quad \mathrm{Dom}(\Delta_1) = H^2\Omega^1(\Sigma_L;\mathfrak{k}) \cap \ker d^* \cap (\ker \Delta_1)^{\perp}. \tag{1081}$$

on the closed Hilbert space

$$\mathscr{K}_L^{\mathrm{tr}} = \big(L^2\Omega^1(\Sigma_L;\mathfrak{k}) \cap \ker d^*\big) \cap (\ker \Delta_1)^{\perp}. \tag{1082}$$

Its spectrum is

$$\left\{ \frac{4\pi^2|n|^2}{L^2} : n \in \mathbb{Z}^3 \setminus \{0\} \right\}, \tag{1083}$$

with transverse-polarization and $\dim \mathfrak{k}$ *multiplicities. On the unreduced transverse Gaussian Fock space after this explicit harmonic-mode truncation, the gap is $2\pi/L$, which tends to zero as $L \to \infty$. Constant gauge transformations remain after Coulomb gauge, so this statement does not assert that its one-particle level survives restriction to a gauge-invariant Fock subspace.*

Proof. The curvature linearization at zero is $A \mapsto dA$. On the Coulomb slice, the configuration part of the energy has quadratic form $g_{\mathrm{YM}}^{-2}\|dA\|^2/2$, whose Hessian relative to unweighted L^2 is $g_{\mathrm{YM}}^{-2} d^*d$ with the displayed elliptic domain. Dividing by the kinetic metric $g_{\mathrm{YM}}^{-2} I$ gives the squared-frequency operator $\Delta_1 = d^*d$. Fourier characters on the three-torus give (1083); for each nonzero wave-vector, the condition $d^*A = 0$ leaves its two orthogonal polarizations. Gaussian quantization replaces each Hessian eigenvalue λ by the oscillator frequency $\sqrt{\lambda}$: in canonical variables a single mode has Hamiltonian $\frac{1}{2}(g_{\mathrm{YM}}^2 P_Q^2 + \lambda g_{\mathrm{YM}}^{-2} Q^2)$, whose frequency is $\sqrt{\lambda}$. The smallest nonzero wave-vector has length $2\pi/L$, proving the result. □

Thus the determinant slice is an exact four-dimensional Lorentzian host for classical gauge theory, not merely an analogy. It does not select K: the same construction works for $U(1)$, $SU(2)$, $SU(3)$, or any other compact matrix group. In particular, Proposition 28.4 prevents the ambient $SU(3)$ orbit label from being silently reused as a local color gauge group. Nor does the free Hessian establish the nonlinear four-dimensional Yang–Mills mass gap; that problem concerns the physical gauge-invariant quantum Hamiltonian after a continuum construction [64].

A cubical mesh gives the exact holonomy regulator of Wilson [65]. To type the state dependence, let V be the vertices, choose one orientation E^+ of the edges, and put

$$\mathcal{C}_E = K^{E^+}, \qquad \mathcal{G}_V = K^V, \qquad (h\cdot U)_e = h_{s(e)} U_e h_{t(e)}^{-1}. \tag{1084}$$

Reverse orientation means $U_{\bar{e}} = U_e^{-1}$. For a closed edge path $\gamma = e_1 \cdots e_r$ and a nontrivial irreducible representation ρ, use the normalized Wilson observable

$$w_{\gamma,\rho}(U) = \frac{1}{\dim \rho}\,\mathrm{tr}\,\rho(U_{e_1}\cdots U_{e_r}). \tag{1085}$$

Proposition 28.13 (Link kinematics selects no confinement state)**.** *There are two $\mathcal{G}_V$-invariant probability measures on the same configuration space $\mathcal{C}_E$ such that*

$$\mathbb{E}_{\mathrm{pg}}[w_{\gamma,\rho}] = 1 \quad \textit{for every closed } \gamma, \qquad \mathbb{E}_{\mathrm{Haar}}[w_{\gamma,\rho}] = 0 \tag{1086}$$

whenever γ contains an unoriented edge exactly once in the second formula. Their convex mixtures give every expectation between zero and one. Thus the link configuration and gauge action do not determine an area law or string tension.

Proof. For the first measure, give K^V product Haar measure and push it forward under $h \mapsto U(h)$, where $U(h)_e = h_{s(e)} h_{t(e)}^{-1}$. Left invariance of vertex Haar measure makes the push-forward gauge invariant, and every closed holonomy telescopes to the identity, proving the first value. For the second, give the independent variables U_e, $e \in E^+$, product Haar measure. Bi-invariance makes this measure gauge invariant. Integrating first over an edge that occurs exactly once inserts $\int_K \rho(k)\, dk = 0$, because a nontrivial irreducible representation has no invariant vector. This proves the second value. Convexity proves the remaining assertion. □

The link system is a valid ultraviolet regulator only after a time direction, Euclidean rotation, coupling normalization, action, and probability measure have been chosen. Proposition 28.13 shows why the normal-slice geometry and link space alone do not supply confinement.

28.8 Exact obstructions and the sufficient spectral bridge

Proposition 28.14 (No Lorentz-invariant positive quadratic energy)**.** *There is no positive-definite quadratic form on $\mathcal{N}_P$ invariant under the full $PSL_2(\mathbb{C})$ action of Theorem 28.3. Consequently the determinant Hessian cannot itself be a nonnegative Hamiltonian.*

Proof. Suppose a positive-definite norm were invariant. In the coordinates (1022), Lorentz boosts send $(1,0,0,0)$ to $(\cosh s, \sinh s, 0, 0)$. Invariance would keep the positive-definite norm of this orbit constant. Every positive-definite norm on a finite-dimensional space is equivalent to the Euclidean norm, whereas the Euclidean norms of these vectors diverge with $|s|$. This is a contradiction. □

The next two models make the role of entanglement exact. They use bipartite tensor-product Hilbert kinematics but have opposite gap behavior.

Proposition 28.15 (Entanglement determines neither positivity nor magnitude of a gap)**.** *Let $\Omega = (e_0 \otimes e_0 + e_1 \otimes e_1)/\sqrt{2}$ in $\mathbb{C}^2 \otimes \mathbb{C}^2$. Then*

$$H_{\mathrm{gap}} = I - |\Omega\rangle\langle\Omega| \tag{1087}$$

is bounded, nonnegative, and self-adjoint, has the entangled vector Ω as its unique vacuum, and has gap one.

On $\ell^2(\mathbb{N}_0) \otimes \ell^2(\mathbb{N}_0)$, put

$$\Omega_0 = \frac{e_0 \otimes e_0 + e_1 \otimes e_1}{\sqrt{2}}, \qquad \psi_n = \frac{e_{2n} \otimes e_{2n} + e_{2n+1} \otimes e_{2n+1}}{\sqrt{2}} \quad (n \geq 1). \tag{1088}$$

Define a bounded self-adjoint operator by

$$H_{\mathrm{ent}}\Omega_0 = 0, \qquad H_{\mathrm{ent}}\psi_n = n^{-1}\psi_n, \qquad H_{\mathrm{ent}} = I \text{ on } \overline{\mathrm{span}}\{\Omega_0, \psi_n : n \geq 1\}^{\perp}. \tag{1089}$$

Put

$$\mathcal{K}_{\mathrm{ent}} = \overline{\mathrm{span}}\{\psi_n : n \geq 1\}.$$

Every nonzero vector in $\mathcal{K}_{\mathrm{ent}}$ is entangled, $\mathcal{K}_{\mathrm{ent}}$ reduces H_{ent}, and

$$\inf \mathrm{Spec}(H_{\mathrm{ent}}|_{\mathcal{K}_{\mathrm{ent}}}) = 0.$$

Thus even a closed, completely entangled reducing sector need not have a positive energy floor.

Proof. Equation (1087) is an orthogonal projection with kernel $\mathbb{C}\Omega$. The vectors in (1088) are mutually orthonormal and each has Schmidt rank two. Equation (1089) is a bounded real diagonal operator in the indicated orthogonal decomposition. Its kernel is $\mathbb{C}\Omega_0$, while $1/n \to 0$. Under the standard coefficient-operator identification of a bipartite tensor, a nonzero vector $\sum_{n \geq 1} c_n \psi_n$ has, for every n with $c_n \neq 0$, two equal nonzero diagonal entries on the coordinates $2n, 2n+1$. Its Schmidt rank is therefore at least two. The displayed diagonal action also proves reduction of $\mathcal{K}_{\mathrm{ent}}$ and spectral bottom zero. □

Finally, let A_{phys} be a lower-semibounded self-adjoint physical Hamiltonian on $\mathcal{H}_{\mathrm{phys}}$, put $E_{\mathrm{vac}} = \inf \mathrm{Spec}(A_{\mathrm{phys}})$, and assume that E_{vac} is an eigenvalue. Let $P_0 = \mathbf{1}_{\{E_{\mathrm{vac}}\}}(A_{\mathrm{phys}})$, let $\mathfrak{a}_{\mathrm{phys}}$ be its closed quadratic form, and put $\mathcal{H}_{\mathrm{exc}} = (I - P_0)\mathcal{H}_{\mathrm{phys}}$. This is a reducing subspace. Let $\mathcal{H}_{\mathrm{aux}}$ be another Hilbert space and let $B \geq 0$ be self-adjoint on it, with closed form $\mathfrak{b}$ satisfying

$$\mathfrak{b}[v] \geq \gamma \|v\|_{\mathrm{aux}}^2 \quad (v \in \mathrm{Dom}\, \mathfrak{b}) \tag{1090}$$

for some $\gamma > 0$.

Theorem 28.16 (Full-domain coercive transfer)**.** *Suppose $\mathcal{H}_{\mathrm{exc}} \neq \{0\}$ and a linear map*

$$R : \mathrm{Dom}(\mathfrak{a}_{\mathrm{phys}}) \cap \mathcal{H}_{\mathrm{exc}} \longrightarrow \mathrm{Dom}(\mathfrak{b}) \tag{1091}$$

is defined on the entire physical excitation form domain and, for fixed $c_R, C_R > 0$, satisfies

$$\|Ru\|_{\mathrm{aux}} \geq c_R \|u\|_{\mathrm{phys}}, \qquad \mathfrak{b}[Ru] \leq C_R(\mathfrak{a}_{\mathrm{phys}}[u] - E_{\mathrm{vac}}\|u\|^2). \tag{1092}$$

Then

$$\inf \mathrm{Spec}(A_{\mathrm{phys}}|_{\mathcal{H}_{\mathrm{exc}}}) - E_{\mathrm{vac}} \geq \frac{\gamma c_R^2}{C_R}. \tag{1093}$$

An intertwiner from only one gapped subspace cannot replace (1091)*.*

Proof. For every vector in the domain of (1091),

$$C_R(\mathfrak{a}_{\mathrm{phys}}[u] - E_{\mathrm{vac}}\|u\|^2) \geq \mathfrak{b}[Ru] \geq \gamma \|Ru\|_{\mathrm{aux}}^2 \geq \gamma c_R^2 \|u\|_{\mathrm{phys}}^2.$$

The variational characterization of the spectral bottom gives (1093). For the final assertion, take $B = I$ on $\mathbb{C}$ and

$$A = 0 \oplus I \oplus M_x \quad \text{on} \quad \mathbb{C}\Omega \oplus \mathbb{C} \oplus L^2((0,1), dx),$$

where $(M_x f)(x) = x f(x)$. The isometry $Jz = 0 \oplus z \oplus 0$ obeys $AJ = JB$, yet A has unique vacuum Ω and excitation spectral bottom zero because of the multiplication summand. □

28.9 Exact conclusion

The determinant identity therefore supplies all of the following, exactly: a canonical dimensionless affine Minkowski space, its Lorentz metric and future cone, the spin and Poincaré actions, a natural time orientation, and a dimensionless normal coordinate system. After the independent choice of one length/time dictionary this carrier becomes a spacetime model supporting a standard continuum Klein–Gordon model, an exact finite periodic regulator, and any chosen classical compact-group gauge theory. The standard metrics also select the compact $K_P \cong SU(2)$ singlet–adjoint representation; Theorem 28.6 gives one exact gauging of that representation. Those are substantive physical constructions with typed configuration spaces, states, actions, observables, spectra, and limits.

The determinant normal-slice data alone do not select a preferred inertial observer, an internal gauge interpretation, a coupling, a mass, a probability measure, a quantum reconstruction, or a nonlinear spectral gap. A separate two-column Gram construction supplies a left $SU(3)$-action and the positive momentum $p_\alpha(Z) = \alpha Z^* Z$, but it retains the independent parameters α, c, m_0 and does not dynamically select $m_0 > 0$. These limitations are exact rather than semantic: the inverse-dual candidate cusp vector (1048) is spacelike; no positive quadratic energy is Lorentz invariant; the same host has the gapped and gapless scalar realizations of Theorem 28.10; the massless gauge Gaussian gap closes as $L \to \infty$; and entangled states occur in both gapped and gapless Hamiltonians. A claim about a physical mass gap must therefore be proved from the physical spectrum or from a global comparison controlling its entire excitation domain; Theorem 28.16 is one sufficient such criterion.

29 Exact constraints on an S^8 higher-rung analogue

The phrase "an S^8 analogue" can denote three mathematically different proposals, which must not be conflated. A literal smooth complex-fourfold analogue is impossible because a homotopy 8-sphere is not almost complex. A merely real map to S^2 with generic fibre T^6 is not excluded if it has genuine singular fibres, but its local fillings must satisfy the exact Euler, fundamental-group, Wang, and Mayer–Vietoris tests below. Finally, explicit real stratified quotient models with boundary or total space S^8 do exist; they do not thereby acquire a complex-fourfold structure.

There is also a different, well-typed alternative: retain the four-real- dimensional torus fibres of the reconstructed S^6 candidate and place $SU(3)$ bundle data on its regular locus. Subsection 29.8 computes the exact extension obstruction. No gauge-theoretic or physical conclusion is asserted without a specified bundle, connection, trace convention, boundary condition, and analytic operator statement.

29.1 The literal complex-fourfold proposal

29.1.1 Characteristic-class obstruction

Let M be a smooth manifold homotopy equivalent to S^8, and suppose that $J\colon TM \to TM$ satisfies $J^2 = -1$. Then

$$E = (TM, J)$$

is a complex vector bundle of rank four whose underlying oriented real bundle is TM. Since

$$H^2(M;\mathbb{Z}) = H^4(M;\mathbb{Z}) = H^6(M;\mathbb{Z}) = 0,$$

one has

$$c_1(E) = c_2(E) = c_3(E) = 0.$$

Orient M by J, and choose $u \in H^8(M;\mathbb{Z})$ with $\langle u, [M]\rangle = 1$. The top-Chern–Euler identity and the Poincaré–Hopf theorem give

$$c_4(E) = e(TM) = \chi(M)u = 2u. \tag{1094}$$

For a complex bundle, the second Pontryagin class of the underlying real bundle is

$$p_2(E_{\mathbb{R}}) = c_2(E)^2 - 2c_1(E)c_3(E) + 2c_4(E).$$

Consequently,

$$p_2(TM) = 4u. \tag{1095}$$

On the other hand, $H^4(M;\mathbb{Z}) = 0$ gives $p_1(TM) = 0$, and the middle cohomology vanishes, so $\operatorname{sign}(M) = 0$. Hirzebruch's eight-dimensional signature formula is

$$\operatorname{sign}(M) = \left\langle \frac{7p_2(TM) - p_1(TM)^2}{45}, [M] \right\rangle.$$

It forces $\langle p_2(TM), [M]\rangle = 0$, contradicting (1095). For the standard embedded sphere, the signature step may instead be replaced by the explicit stable trivialisation

$$TS^8 \oplus \varepsilon^1_{\mathbb{R}} \cong \varepsilon^9_{\mathbb{R}},$$

which gives $p_2(TS^8) = 0$ directly. Hence

$$\boxed{S^8 \text{ admits no almost-complex structure.}} \tag{1096}$$

Integrability is stronger than being almost complex, so a complex-fourfold structure is excluded a fortiori.

29.1.2 Independent arithmetic checks

Borel and Serre prove that S^{2n} admits no almost-complex structure for $n \geq 4$, including arbitrary differentiable structures on the sphere. In the S^8 specialization, their mod-3 reduced-power identity is

$$\mathcal{P}^1(c_2) = c_1^2 c_2 - 2c_2^2 - c_1 c_3 + 4c_4.$$

All lower Chern classes vanish on S^8, so it would follow that $c_4 = 0 \pmod 3$, whereas (1094) gives $c_4 = \pm 2u \neq 0 \pmod 3$.

Bott integrality gives a second independent check. For a complex bundle over S^{2n}, the top Chern number is divisible by $(n-1)!$. At $n = 4$,

$$6 \mid \langle c_4(E), [S^8] \rangle,$$

which is incompatible with the Euler number ± 2.

29.1.3 Exact scope of the obstruction

The obstruction rules out a smooth complex fourfold diffeomorphic to S^8; a proper holomorphic family of complex three-tori over $\mathbb{P}^1$ whose smooth total space is diffeomorphic to S^8; and an almost-complex structure on all of TS^8, whether or not integrable. It also rules out a symplectic or Kähler structure: if ω were symplectic, then $H^2(S^8; \mathbb{R}) = 0$ would give $\omega = d\alpha$, while

$$\int_{S^8} \omega^4 = \int_{S^8} d(\alpha \wedge \omega^3) = 0,$$

contradicting symplectic volume.

It does not rule out a complex structure on an open subset, a stable complex structure, a singular complex analytic space whose tangent sheaf is not a vector bundle, or a real stratified quotient whose underlying boundary is S^8. The standard sphere is stably complex because

$$TS^8 \oplus \varepsilon_{\mathbb{R}}^2 \cong \varepsilon_{\mathbb{R}}^{10},$$

but this is not a complex structure on the actual rank-eight tangent bundle. Moreover, the usual link of an isolated complex singularity of complex dimension d has real dimension $2d - 1$. Thus an S^8 obtained as a quotient boundary is not, merely for that reason, an ordinary complex-analytic link.

29.2 Requirements for a real Fabel-style S^8 fibration

The reconstructed S^6 candidate has a map

$$f \colon X^6 \longrightarrow B \cong \mathbb{P}^1$$

whose regular fibre is a complex two-torus, hence a real T^4, and whose three special fibres comprise one non-normal toric degeneration and two multiple bielliptic fibres. The direct dimension analogue on a real eight-manifold would be a map

$$g \colon M^8 \longrightarrow S^2$$

with regular fibre T^6. If M is required to be a smooth complex fourfold and the regular fibres are complex three-tori, the obstruction in Subsection 29.1 ends the proposal. If g is only a real smooth or stratified map, the proposal remains topologically meaningful.

The adjective “singular” is essential. There is no fibre bundle

$$T^6 \longrightarrow S^8 \longrightarrow S^2.$$

Indeed, its long exact homotopy sequence would contain

$$\pi_2(S^8) = 0 \longrightarrow \pi_2(S^2) = \mathbb{Z} \xrightarrow{\partial} \pi_1(T^6) = \mathbb{Z}^6 \longrightarrow \pi_1(S^8) = 0.$$

Exactness would require $\partial\colon \mathbb{Z} \to \mathbb{Z}^6$ to be surjective, which is impossible. The Euler identity for a fibre bundle gives the same conclusion:

$$\chi(S^8) = 2 \neq \chi(S^2)\chi(T^6) = 2 \cdot 0 = 0.$$

29.3 Euler constraint for a finite multipole model

Let $g\colon M^8 \to S^2$ be a proper triangulable map with finite critical-value set $\Sigma = \{p_1, \dots, p_s\}$, and suppose that every fibre over $S^2 \setminus \Sigma$ is T^6. Euler integration for a constructible map gives

$$\begin{aligned} \chi(M) &= \chi(S^2 \setminus \Sigma)\chi(T^6) + \sum_{i=1}^{s} \chi(g^{-1}(p_i)) \\ &= \sum_{i=1}^{s} \chi(F_i). \end{aligned} \tag{1097}$$

Equivalently, one may cut out small discs, use multiplicativity on the regular bundle, and note that each boundary mapping torus has Euler characteristic zero. For $M \cong S^8$ one therefore needs

$$\boxed{\sum_{i=1}^{s} \chi(F_i) = 2.} \tag{1098}$$

Multiple fibres that are finite free quotients of tori have Euler characteristic zero and cannot account for S^8. At least one special fibre must contain a genuine collapse or fixed stratum with nonzero Euler characteristic. The numerical pattern consisting of two Euler-zero multiple fibres and one fibre of Euler characteristic two is compatible with (1098), but that numerical compatibility is not a construction.

29.4 Fundamental group and monodromy

Put $B^\circ = S^2 \setminus \Sigma$, $E = g^{-1}(B^\circ)$, and $\Lambda = H_1(T^6; \mathbb{Z}) \cong \mathbb{Z}^6$. The regular bundle has monodromy

$$\rho\colon \pi_1(B^\circ) \longrightarrow \mathrm{GL}(\Lambda).$$

If the fibres and total space are oriented compatibly, its image lies in $\mathrm{SL}(6, \mathbb{Z})$. For oriented peripheral loops $\gamma_1, \dots, \gamma_s$ satisfying $\gamma_1 \cdots \gamma_s = 1$, the matrices $M_i = \rho(\gamma_i)$ satisfy

$$M_1 M_2 \cdots M_s = I. \tag{1099}$$

When $s \geq 1$, B° has $\pi_2(B^\circ) = 0$. The homotopy sequence of the regular bundle therefore gives the exact extension

$$1 \longrightarrow \Lambda \longrightarrow \pi_1(E) \longrightarrow \pi_1(B^\circ) \longrightarrow 1. \tag{1100}$$

After choosing a section and a compatible basepoint system, (1100) becomes an affine semidirect presentation; without those choices it remains the displayed exact extension.

Let $N_i = g^{-1}(D_i)$ be the filling over a small closed disc D_i about p_i, and let $A_i = E \cap N_i$, a T^6 mapping torus. Van Kampen gives

$$\pi_1(M) \cong \pi_1(E) \Big/ \Big\langle\!\Big\langle \ker\bigl(\pi_1(A_i) \to \pi_1(N_i)\bigr),\ 1 \leq i \leq s \Big\rangle\!\Big\rangle. \tag{1101}$$

Thus an S^8 construction must derive the actual local inclusion maps and prove that their kernels normally generate all of (1100). Monodromy matrices or Smith forms alone do not prove that (1101) is trivial.

29.5 Integral homology: the exact gluing tests

Replace B° by a compact sphere with s open discs removed and replace E, N_i by compact collar-compatible pieces. Define the actual Mayer–Vietoris maps

$$\alpha_q = \bigoplus_{i=1}^{s}((j_{E,i})_*, -(j_{N,i})_*)\colon \bigoplus_i H_q(A_i;\mathbb{Z}) \longrightarrow H_q(E;\mathbb{Z}) \oplus \bigoplus_i H_q(N_i;\mathbb{Z}). \tag{1102}$$

Because the pieces have nonempty boundary, their top homology vanishes. The Mayer–Vietoris sequence and $H_q(S^8;\mathbb{Z}) = 0$ for $1 \le q \le 7$ imply the following necessary and sufficient conditions:

q	condition on α_q
0	α_0 is injective and $\operatorname{coker}\alpha_0 \cong \mathbb{Z}$,
$1 \le q \le 6$	α_q is an isomorphism,
7	α_7 is surjective and $\ker\alpha_7 \cong \mathbb{Z}$.

(1103)

The generator of $\ker\alpha_7$ is the boundary of the fundamental class of the closed eight-manifold obtained by gluing the pieces. Equations (1102)–(1103), using the actual inclusion maps, are the integral replication test for a proposed S^8 multipole construction.

Each boundary group is constrained by monodromy. If A_i is the mapping torus of $\phi_i\colon T^6 \to T^6$ and $M_i = (\phi_i)_*|_{H_1(T^6;\mathbb{Z})}$, the Wang sequence gives

$$0 \longrightarrow \operatorname{coker}\bigl(I - \textstyle\bigwedge^q M_i\bigr) \longrightarrow H_q(A_i;\mathbb{Z}) \longrightarrow \ker\bigl(I - \textstyle\bigwedge^{q-1} M_i\bigr) \longrightarrow 0. \tag{1104}$$

The right group is free, so (1104) splits as an abelian group, although not canonically. Since the regular base retracts to a wedge of $s-1$ circles, the homological Serre spectral sequence gives the filtration

$$0 \longrightarrow H_0\bigl(B^\circ;\textstyle\bigwedge^q \Lambda\bigr) \xrightarrow{\iota_q} H_q(E;\mathbb{Z}) \xrightarrow{\pi_q} H_1\Bigl(B^\circ;\textstyle\bigwedge^{q-1}\Lambda\Bigr) \longrightarrow 0. \tag{1105}$$

To type both edge terms explicitly, put $V_q = \bigwedge^q \Lambda$, choose oriented free generators $\gamma_1, \dots, \gamma_{s-1}$ of $\pi_1(B^\circ)$, and write $M_j = \rho(\gamma_j)$. In the Wang sign convention the cellular differential is

$$d_q\colon \bigoplus_{j=1}^{s-1} V_q \longrightarrow V_q, \qquad (v_j)_j \longmapsto \sum_{j=1}^{s-1} (I - \textstyle\bigwedge^q M_j)\, v_j, \tag{1106}$$

so the two homological edge groups are

$$H_0(B^\circ;V_q) = \operatorname{coker} d_q = (V_q)_\rho = \frac{V_q}{\sum_{j=1}^{s-1}(I - \bigwedge^q M_j)V_q}, \qquad H_1(B^\circ;V_{q-1}) = \ker d_{q-1}. \tag{1107}$$

Thus the coinvariants are the Serre filtration subobject and the base-one-cycle group is the quotient. Since $\ker d_{q-1}$ is a subgroup of a free abelian group, it is free, and (1105) splits noncanonically as an abstract abelian group.

Dually, set $V_q^\vee = \bigwedge^q \Lambda^\vee$; its monodromy along γ_j is $\bigwedge^q(M_j^{-\mathsf{T}})$. The cohomological filtration is

$$0 \longrightarrow H^1(B^\circ;V_{q-1}^\vee) \longrightarrow H^q(E;\mathbb{Z}) \longrightarrow H^0(B^\circ;V_q^\vee) \longrightarrow 0, \qquad \begin{aligned} H^0(B^\circ;V_q^\vee) &= (V_q^\vee)^\rho \\ &= \bigcap_{j=1}^{s-1} \ker\Bigl(I - \textstyle\bigwedge^q(M_j^{-\mathsf{T}})\Bigr). \end{aligned} \tag{1108}$$

Accordingly, each boundary inclusion in α_q must send its Wang coinvariant filtration term through ι_q, and its base-one-cycle term must be tested after π_q; neither exact sequence supplies a canonical splitting. The corrected sequence identifies $H_q(E;\mathbb{Z})$ noncanonically with the direct sum of its two edge groups, so a rank or torsion count made from that direct sum survives. The natural edge maps do not: every map-level Mayer–Vietoris test must use (1105) in the displayed order. Equations (1102)–(1108) retain every lattice, monodromy, kernel, coinvariant, invariant, and inclusion that a correct proof must compute.

29.6 The torus-action bound and a maximal real orbit model

For a smooth torus action on a compact manifold,

$$\chi(M) = \chi(M^T).$$

Since $\chi(S^8) = 2$, every torus action on S^8 has a fixed point. Equip S^8 with a T-invariant Riemannian metric. At a fixed point the isotropy representation is a sum of four real two-dimensional weight planes and possibly trivial summands. If the global action is effective, this tangent representation is faithful: an isometry fixing a point with identity derivative is the identity on the connected manifold. Four weights span a character lattice of rank at most four. Hence

$$\boxed{\operatorname{rank} T \le 4.} \tag{1109}$$

In particular, a generic T^6 fibration of S^8 cannot be the orbit map of a global effective torus action. This does not rule out a monodromic T^6-bundle over the regular part of S^2, because such a bundle need not supply a global T^6 action.

The bound is sharp. Write

$$S^8 = \left\{ (z_1, z_2, z_3, z_4, y) \in \mathbb{C}^4 \oplus \mathbb{R} : \sum_{i=1}^{4} |z_i|^2 + y^2 = 1 \right\},$$

and let T^4 rotate the four complex coordinates. The orbit map is

$$q(z_1, z_2, z_3, z_4, y) = (|z_1|, |z_2|, |z_3|, |z_4|, y),$$

with image

$$Q = \left\{ (r_1, r_2, r_3, r_4, y) : r_i \ge 0, \sum r_i^2 + y^2 = 1 \right\}.$$

Two points have the same image exactly when they lie in the same orbit, so compact–Hausdorff descent gives $S^8/T^4 \cong Q$. Away from the two poles, write $r_i = \sqrt{1 - y^2}\, u_i$. Then

$$(u_1, u_2, u_3, u_4) \in \{ u_i \ge 0,\ \sum u_i^2 = 1 \} \cong \Delta^3,$$

and the values $y = \pm 1$ collapse the two ends. Therefore

$$\boxed{S^8/T^4 \cong \operatorname{Susp}(\Delta^3) \cong B^4.} \tag{1110}$$

The locus on which all $r_i > 0$ has free T^4 orbit. Exactly j zero complex coordinates give isotropy T^j, and the two poles are fixed by the whole torus. This is a rigorous two-pole singular torus compactification of S^8, but it has four-dimensional quotient and generic T^4 orbit. It is not a three-special-fibre map to S^2.

29.7 A three-column Gram quotient whose quotient boundary is S^8

Let

$$V = \operatorname{Mat}_{3\times 3}(\mathbb{C}), \qquad \dim_{\mathbb{R}} V = 18,$$

and let $U(3)$ act on the left. Define

$$\Gamma \colon V \longrightarrow \operatorname{Herm}_3(\mathbb{C})_{\ge 0}, \qquad \Gamma(A) = A^*A. \tag{1111}$$

The map is surjective because every $Q \ge 0$ is the Gram matrix of $Q^{1/2}$. It separates $U(3)$-orbits. Indeed, if $A^*A = B^*B$, the rule $Ax \mapsto Bx$ is a well-defined isometry from $\operatorname{im} A$ to $\operatorname{im} B$: equality of the Gram

matrices implies both equality of kernels and preservation of the Hermitian inner product. Extend this isometry to a unitary U of $\mathbb{C}^3$; then $B = UA$. Hence

$$V/U(3) \cong \mathrm{Herm}_3(\mathbb{C})_{\geq 0}. \tag{1112}$$

For the Frobenius unit ball $\mathbb{B}(V) = \{A : \mathrm{tr}(A^*A) \leq 1\}$, (1111) gives

$$\mathbb{B}(V)/U(3) \cong K := \{Q \in \mathrm{Herm}_3(\mathbb{C}) : Q \geq 0, \mathrm{tr}\, Q \leq 1\}. \tag{1113}$$

The real vector space of Hermitian 3×3 matrices has dimension nine. The set K is compact, convex, and has nonempty interior $\{Q > 0, \mathrm{tr}\, Q < 1\}$. Radial projection from an interior point gives a homeomorphism of pairs $(K, \partial K) \cong (B^9, S^8)$, and therefore

$$\boxed{\partial(\mathbb{B}^{18}/U(3)) \cong S^8.} \tag{1114}$$

Two qualifications are essential. First, the quotient of the source sphere is not the boundary in (1114):

$$S^{17}/U(3) \cong \{Q \geq 0 : \mathrm{tr}\, Q = 1\}.$$

The space on the right is an eight-dimensional compact convex body and hence is homeomorphic to B^8. The rest of ∂K consists of rank-deficient Gram matrices with $\mathrm{tr}\, Q < 1$. Thus taking the boundary and taking the orbit quotient do not commute in this construction.

Second, strict $SU(3)$ is different. For invertible matrices, A^*A does not record the phase of $\det A$, while left multiplication by $SU(3)$ preserves $\det A$. The strict quotient is described by

$$(Q, d) = (A^*A, \det A), \qquad Q \geq 0, \qquad |d|^2 = \det Q. \tag{1115}$$

It has generic real dimension ten, not nine. The construction in (1114) therefore uses

$$U(3) \cong (SU(3) \times U(1))/\mu_3,$$

not colour $SU(3)$ alone. It is a precise S^8 real quotient boundary, but no morphism has been produced from its Gram or rank strata to the period matrix, torus bundle, logarithmic twists, or three special fibres of the Fabel–Alpöge construction.

29.8 Retaining the $T^4/\mathbb{R}^4$ fibre geometry and placing $SU(3)$ on the regular locus

This subsection classifies the required topological bundle data and its obstruction to crossing the singular fibres. It makes no physical identification.

29.8.1 The regular locus and its fourth cohomology

Let $X \cong_{\mathrm{diff}} S^6$ be the reconstructed candidate and put

$$D = W \sqcup S_1 \sqcup S_2, \qquad U = X \setminus D = f^{-1}(B^\circ), \qquad B^\circ = \mathbb{P}^1 \setminus \{p_0, p_1, p_2\}.$$

Here W is the non-normal central fibre and S_1, S_2 are the underlying smooth bielliptic surfaces of the two multiple fibres. The reconstructed integral calculations give

$$H_1(W; \mathbb{Z}) \cong \mathbb{Z}^2, \qquad H_1(S_1; \mathbb{Z}) \cong H_1(S_2; \mathbb{Z}) \cong \mathbb{Z}^2.$$

The compact analytic set D is triangulable, so Alexander duality in $X \cong S^6$ applies:

$$\widetilde{H}^i(U; \mathbb{Z}) \cong \widetilde{H}_{5-i}(D; \mathbb{Z}).$$

At $i = 4$, it gives

$$\boxed{H^4(U;\mathbb{Z}) \cong H_1(D;\mathbb{Z}) \cong \mathbb{Z}^6.} \tag{1116}$$

There is an independent fibrewise description. The pair of pants B° retracts to a wedge of two circles, and the T^4 bundle retracts to its restriction over that graph. Hence U has the homotopy type of a CW complex of dimension at most five. If $\mathcal{V}^q = R^q f_* \mathbb{Z}$ is the monodromy local system, the Serre spectral sequence has only columns zero and one and yields

$$0 \longrightarrow H^1(B^\circ;\mathcal{V}^3) \longrightarrow H^4(U;\mathbb{Z}) \xrightarrow{\mathrm{res}_F} H^0(B^\circ;\mathcal{V}^4) \longrightarrow 0. \tag{1117}$$

Holomorphic fibre monodromy preserves the real orientation, so $H^0(B^\circ;\mathcal{V}^4) \cong \mathbb{Z}$. Thus one coordinate of (1116) is the constant evaluation on a smooth T^4 fibre; the remaining rank-five kernel in (1117) consists of mixed base–fibre classes.

29.8.2 Exact classification of $SU(3)$ bundles on U

Let $P \to U$ be a principal $SU(3)$ bundle and $E_P = P \times_{SU(3)} \mathbb{C}^3$ its determinant-trivial rank-three complex bundle. Through degree five,

$$\pi_i(BSU(3)) = 0 \quad (i = 1, 2, 3, 5), \qquad \pi_4(BSU(3)) \cong \mathbb{Z},$$

with universal generator c_2. Because U has CW dimension at most five, obstruction theory gives a bijection

$$\boxed{\{\text{principal } SU(3) \text{ bundles on } U\}/\cong \xrightarrow{\ c_2\ } H^4(U;\mathbb{Z}) \cong \mathbb{Z}^6.} \tag{1118}$$

The fibre integer

$$k_F = \langle c_2(E_P)|_{F_b}, [F_b] \rangle, \qquad b \in B^\circ, \tag{1119}$$

is independent of b. A nonzero k_F is only one possible obstruction: a class with $k_F = 0$ may still have a nonzero mixed component in the kernel of (1117).

29.8.3 Extension across the three special fibres

The cohomology sequence of the pair (X, U) contains

$$H^4(X;\mathbb{Z}) \longrightarrow H^4(U;\mathbb{Z}) \xrightarrow{\delta} H^5(X,U;\mathbb{Z}) \longrightarrow H^5(X;\mathbb{Z}).$$

Both outer groups vanish because $X \cong S^6$. Therefore

$$\boxed{\delta\colon H^4(U;\mathbb{Z}) \xrightarrow{\ \cong\ } H^5(X,U;\mathbb{Z}).} \tag{1120}$$

Every $SU(3)$ bundle on all of X has $c_2 = 0$, so its restriction to U is trivial by (1118). Conversely, the trivial bundle on U extends as the trivial bundle on X. Thus, for unframed topological bundles,

$$\boxed{P \to U \text{ extends across } D \iff c_2(P) = 0 \iff \delta c_2(P) = 0.} \tag{1121}$$

Under Alexander duality, (1120) identifies the obstruction with a six-integer class supported on the one-cycles of $W \sqcup S_1 \sqcup S_2$. Because the three components are disjoint and $H_1(D)$ is their direct sum, nonzero components do not vanish merely by declaring that charges at distinct special fibres cancel. They cancel as an extension obstruction only when the complete class in $\mathbb{Z}^6$ is zero.

This is a statement about the bundle. Even when $c_2 = 0$, a specified connection on the trivial bundle may fail to extend smoothly because of its holonomy or growth near D; this is an analytic boundary-value problem not decided by (1121).

29.8.4 The distinct six-dimensional clutching charge

Principal $SU(3)$ bundles on the complete sphere are classified by

$$[S^6, BSU(3)] \cong \pi_5(SU(3)) \cong \mathbb{Z}.$$

If k is the clutching integer, Bott integrality fixes the top Chern class as

$$c_3(E_P) = \pm 2k\, u \in H^6(S^6; \mathbb{Z}). \tag{1122}$$

Every such global bundle restricts to the trivial bundle on U, because its restricted c_2 vanishes and (1118) is complete. Thus a global six-dimensional $SU(3)$ clutching charge can be concentrated in the gluing across the deleted fibres, but it is a $c_3/\pi_5(SU(3))$ class, not a four-dimensional c_2 instanton class on the regular fibres.

Alexander duality also gives $H^5(U; \mathbb{Z}) \cong \widetilde{H}_0(D; \mathbb{Z}) \cong \mathbb{Z}^2$. The pair sequence therefore yields

$$0 \longrightarrow \mathbb{Z}^2 \longrightarrow H^6(X, U; \mathbb{Z}) \longrightarrow \mathbb{Z} \longrightarrow 0,$$

and hence

$$H^6(X, U; \mathbb{Z}) \cong \mathbb{Z}^3. \tag{1123}$$

After choosing the complex orientations of the three fibre components, the three relative top classes map to the global fundamental class by the augmentation

$$(n_0, n_1, n_2) \longmapsto n_0 + n_1 + n_2. \tag{1124}$$

The rank-two kernel records redistribution among the three supports with the same global total. To promote (1123)–(1124) from exact cohomological data to a relative $SU(3)$ bundle theorem with prescribed framings, one must give the three clutching maps and compute their homotopy classes; arbitrary relative cohomology integers are not silently declared realised.

29.8.5 Data required before a gauge-theory conclusion

A mathematical colour or gauge proposal on U must specify:

1. a principal $SU(3)$ bundle $P \to U$, equivalently its complete class $c_2(P) \in \mathbb{Z}^6$;
2. a connection A and a fixed trace convention for its curvature Chern–Weil form;
3. tubular boundary conditions or framings at each of W, S_1, S_2;
4. whether the invariant under study is the fibre number (1119), the extension defect (1120), or the six-dimensional clutching number (1122); and
5. an explicit analytic statement about a Yang–Mills operator, action, or spectrum.

The topology alone proves no mass gap, confinement statement, particle identification, or Yang–Mills dynamics. It supplies a carrier and an exact obstruction ledger on which such a calculation can be posed.

29.9 Consequences for the higher-rung search

Replacing S^6 by S^8 does not permit a literal smooth complex-fourfold version of the Fabel–Alpöge construction. A real T^6-over-S^2 singular fibration could still exist, but it cannot arise as a global effective torus action and must pass the map-level tests (1098), (1101), and (1103). The standard T^4 action gives an exact and economical real S^8 multipole model, but its base is B^4, not S^2, and its singular locus is a face stratification rather than three isolated fibres.

The 3×3 $U(3)$ Gram construction gives an exact quotient boundary S^8 and is the strongest precise realisation presently obtained for the "one rung higher" intuition. Strict $SU(3)$ retains the determinant phase and does not give the same quotient. Keeping the original T^4 fibres and placing $SU(3)$ on $X \setminus D$ is instead mathematically well typed: its unframed bundle classes form $\mathbb{Z}^6$, every nonzero c_2 is an exact nonextendable divisor defect, and global $SU(3)$ bundles on S^6 carry the even-valued c_3 charge in (1122) while restricting trivially to the regular locus.

None of these statements identifies a lepton, quark, or four-dimensional Yang–Mills theory.

Sources. The almost-complex obstruction is cross-checked against A. Borel and J.-P. Serre, *Groupes de Lie et puissances réduites de Steenrod*, Amer. J. Math. **75** (1953), 409–448, Proposition 15.1, `doi:10.2307/2372495`, and P. Konstantis and M. Parton, *Almost complex structures on spheres*, Differential Geom. Appl. **57** (2018), 10–22, `doi:10.1016/j.difgeo.2017.10.010`. The fixed-point identity used in (1109) is from S. Kobayashi, *Fixed points of isometries*, Nagoya Math. J. **13** (1958), 63–68, `doi:10.1017/S0027763000023497`. For singular torus fibrations and the coordinate sphere action, see respectively Y. Matsumoto, *Torus fibrations over the 2-sphere with the simplest singular fibers*, J. Math. Soc. Japan **37** (1985), 605–636, `doi:10.2969/jmsj/03740605`, and H. Ishida, Y. Fukukawa, and M. Masuda, *Topological toric manifolds*, Moscow Math. J. **13** (2013), 57–98, `doi:10.17323/1609-4514-2013-13-1-57-98`. The theorem-level calculations above are given in full and do not rely on these citations as substitutes for omitted morphisms.

30 The ternary triality structure and its exact carrier morphisms

The complete proof source is also available in the root draft `s6_higher_rung_24d_preprint.tex`. The higher target is a three-directional structure encoded by the ordered triality modules and their invariant trilinear tensor. The proofs retain the Peirce coefficient $1/2$, determinant coefficient 2, spectral graph, flag-cell inverse and degree-$+1$ collapse, and the full retained cusp-angle calculation. The complex-toroidal calculations concern an auxiliary comparison and do not define the higher target.

The single shared source, beginning at Section 31, retains the original modulo-840 lattice, full Weyl quotient, exact index-six Leech neighbour and cyclic-triality isometry. Its continuation now computes the fixed-Cayley-frame value groups for $F = \tau \circ \mathcal{L}$, with $\tau(q) = \mathrm{Re}((q_1 q_2) q_3)$, whose inclusion has quotient $\mathbb{Z}/12$; the Albert determinant contains $2F$, not F. It also gives the common Leech intersection J of determinant 81 and index 9, and its fixed fourth A_2^{12} extension with 72 roots. Wilson's construction [81] supplies the cyclic fixed Gram matrix $3G_{E_8}$, the $A_2 \otimes E_8$ complement, the quotient $\mathbb{F}_3^8$, and the exact rank-four literal intersections with the two marked lattice images. All metrics, cycle directions and the distinction between the cubic image and its value-generated subgroup remain explicit in the shared proofs.

30.1 The source construction and the toroidal comparison

The higher-rung investigation asks how a structure with three interacting directions can extend the source geometry. The present algebraic datum is the vector and two half-spin sectors of triality, together with their invariant trilinear tensor. The complex-torus extension studied first below is one specified comparison with the source construction. Its complex structure is a hypothesis of those toroidal theorems; it is not a requirement on the higher-rung target. The retained cusp-angle map and the marked $A_5^4 D_4$ lattice are retained as inputs to the subsequent comparison maps.

We use the following exact input from the reconstructed candidate construction [2]. Let

$$B = \mathbb{P}^1, \qquad \Sigma = \{p_0, p_1, p_2\}, \qquad B^\circ = B \setminus \Sigma,$$

where p_0 is a cusp and p_1, p_2 have orbifold orders $3, 4$. There is a claimed compact connected complex threefold and proper holomorphic map

$$f : X \longrightarrow B, \qquad X \cong_{\mathrm{diff}} S^6,$$

whose regular fibres are complex two-tori. In the basis (γ, u, w, δ) of $V = \mathbb{Z}^4$ the integral monodromies are generated by

$$T_1 = \begin{pmatrix} 1 & 0 & -6 & 2 \\ 0 & -1 & 1 & 1 \\ 0 & -1 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad T_2 = \begin{pmatrix} 1 & 6 & 0 & -3 \\ 0 & 0 & -1 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \tag{1125}$$

with $T_1^3 = T_2^4 = I$ and $T_0 = (T_1 T_2)^{-1} = I + N$, $N^2 = 0$. On the dual lattice $\Lambda = V^*$ the monodromies are $A_j = (T_j^{-1})^{\mathsf{T}}$. The period matrix is

$$\Pi(z) = \begin{pmatrix} 6\mu(z) & \tau(z) & 1 & 0 \\ \beta(z) & \mu(z) & 0 & 1 \end{pmatrix}, \qquad F_z = \mathbb{C}^2 / \Pi(z)\Lambda. \tag{1126}$$

At p_0 the filling is a Mumford toric quotient with reduced irreducible normal-crossing central fibre W: its normalization is the degree-six del Pezzo surface, opposite sides of its anticanonical hexagon are identified, and $\chi(W) = 2$. At p_j, $j = 1, 2$, the filling is a logarithmic transform with

$$f^* p_1 = 3S_1, \qquad f^* p_2 = 4S_2,$$

where the S_j are smooth bielliptic surfaces and hence have Euler characteristic zero. The primitive affine translations have

$$v_1 = \widehat{\gamma} + 2\widehat{u} - 4\widehat{w}, \qquad v_2 = -\widehat{\gamma} - 3\widehat{u} + 3\widehat{w}, \qquad (\ell_0, \ell_1, \ell_2) = (0, 1, -1). \tag{1127}$$

They give $\pi_1(X) = \mathbb{Z}/|12\ell_0 - 4\ell_1 - 3\ell_2| = 0$ and the integral homology of S^6.

The later marked observation uses the integral difference matrix and its fixed right inverse

$$P = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix}, \quad u = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \quad C = PP^* = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \quad R = P^* C^{-1} = \frac{1}{3} \begin{pmatrix} 2 & 1 \\ -1 & 1 \\ -1 & -2 \end{pmatrix}. \tag{1128}$$

Thus $PR = I_2$, $Pu = 0$, $u^* R = 0$, and for $X_P = (R\ u)$,

$$X_P^{-1} = \begin{pmatrix} P \\ u^*/3 \end{pmatrix}, \qquad \det X_P = 1, \qquad X_P^* X_P = \operatorname{diag}(C^{-1}, 3). \tag{1129}$$

Definition 30.1 (Data for the complex-toroidal comparison)**.** A *full higher toroidal wrap* consists of all the following data and proofs. Neither its dimension nor its homeomorphism type is prescribed.

(F1) A compact connected smooth carrier obtained from the stated pieces, with its smooth and homotopy type computed from the attaching maps.

(F2) A geometric tangent structure on the whole carrier. For a literal toroidal continuation this is an integrable complex structure; a different category must state its replacement rather than inherit the word "complex".

(F3) A proper holomorphic map to the same one-dimensional base, with a complex torus of the dimension forced by total dimension as regular fibre.

(F4) Specified local spaces over the cusp and the two finite points, including their multiplicities, reduced fibres, attaching maps, and Euler characteristics.

(F5) An integral period lattice, monodromy representation, affine translations, and actual specialization maps at every filling.

(F6) The van Kampen and integral Mayer–Vietoris maps proving the claimed smooth sphere topology.

(F7) A marked observation whose domain, codomain, lattice map, and compatibility with periods and fillings are specified.

(F8) Rigidity: after all markings above are fixed, no auxiliary metric, phase section, fan identification, or gluing choice is left unrecorded, and existence and uniqueness up to the stated notion of isomorphism are proved.

This definition specifies the complex-toroidal comparison studied in the following sections. The three-sector triality construction is developed through its own explicitly defined products, bundles, boundary, and lattice maps below; the list above does not constrain it to be another complex structure.

For a rank-three successor over the regular base, require a complex-linear surjection $P_z : \mathbb{C}^3 \to \mathbb{C}^2$ with $K_{\mathbb{C},z} = \ker P_z$, period embeddings $\widetilde{\Pi}_z : \widetilde{\Lambda} \hookrightarrow \mathbb{C}^3$ and $\Pi_z : \Lambda \hookrightarrow \mathbb{C}^2$, and

$$P_z\widetilde{\Pi}_z = \Pi_z\Phi, \qquad K_{\mathbb{C},z} \cap \widetilde{\Pi}_z(\widetilde{\Lambda}) = \widetilde{\Pi}_z\kappa(K_{\mathbb{Z}}).$$

Put

$$E_z = K_{\mathbb{C},z}/\widetilde{\Pi}_z\kappa(K_{\mathbb{Z}}), \qquad \widetilde{F}_z = \mathbb{C}^3/\widetilde{\Pi}_z(\widetilde{\Lambda}), \qquad F_z = \mathbb{C}^2/\Pi_z(\Lambda).$$

The desired portion of the map diagram is then

$$\begin{array}{ccccccccc}
0 & \longrightarrow & K_{\mathbb{Z}} & \xrightarrow{\kappa} & \widetilde{\Lambda} & \xrightarrow{\Phi} & \Lambda & \longrightarrow & 0 \\
 & & {\scriptstyle \widetilde{\Pi}_z\kappa}\downarrow & & {\scriptstyle \widetilde{\Pi}_z}\downarrow & & {\scriptstyle \Pi_z}\downarrow & & \\
0 & \longrightarrow & K_{\mathbb{C},z} & \longrightarrow & \mathbb{C}^3 & \xrightarrow{P_z} & \mathbb{C}^2 & \longrightarrow & 0 \\
 & & {\scriptstyle q_K}\downarrow & & {\scriptstyle q_3}\downarrow & & {\scriptstyle q_2}\downarrow & & \\
0 & \longrightarrow & E_z & \longrightarrow & \widetilde{F}_z & \xrightarrow{\overline{P}_z} & F_z & \longrightarrow & 0.
\end{array} \tag{1130}$$

All three rows are exact. The maps $\overline{P}_z$ must assemble to a map $\widetilde{X}^\circ \to X^\circ$ over B°, together with commuting specialization squares at all three deleted points. Sections 30.3 and 30.4 prove exactly which parts of this diagram exist.

30.2 Sphere and Euler obstructions

Theorem 30.2 (The S^8 tangent obstruction with all hypotheses)**.** *Let M be a closed smooth manifold homotopy equivalent to S^8. There is no bundle endomorphism $J : TM \to TM$ satisfying $J^2 = -\mathrm{id}$. Consequently no smooth manifold in that homotopy type is a complex fourfold, and in particular no proper holomorphic complex-three-torus fibration over $\mathbb{P}^1$ can have such a total space.*

Proof. Suppose J exists and let $E = (TM, J)$, a complex rank-four bundle. Orient M by J and choose $a \in H^8(M;\mathbb{Z})$ with $\langle a, [M]\rangle = 1$. Since the cohomology below the top degree vanishes,

$$c_1(E) = c_2(E) = c_3(E) = 0.$$

The top Chern class of a complex bundle is the Euler class of its underlying oriented real bundle. Poincaré–Hopf and $\chi(M) = 2$ therefore give

$$c_4(E) = e(TM) = 2a.$$

For the underlying real bundle,

$$p_2(E_{\mathbb{R}}) = c_2(E)^2 - 2c_1(E)c_3(E) + 2c_4(E) = 4a,$$

whereas $p_1(TM) = 0$ because $H^4(M;\mathbb{Z}) = 0$. The middle cohomology of M vanishes, so $\operatorname{sign}(M) = 0$. Hirzebruch's degree-eight signature formula now gives

$$0 = \operatorname{sign}(M) = \left\langle \frac{7p_2(TM) - p_1(TM)^2}{45}, [M] \right\rangle = \frac{28}{45},$$

a contradiction. The last displayed rational number is not being asserted to be a possible signature: its nonintegrality is precisely the contradiction forced by the incompatible characteristic classes. □

Theorem 30.3 (Parity and the complete higher-sphere exclusion)**.** *Let M be a smooth homotopy m-sphere with $m > 6$. Then M admits no almost-complex structure and hence no integrable complex structure. In particular S^9 cannot be a complex carrier, and there is no different higher sphere dimension to which the literal complex part of the candidate can be moved.*

Proof. If m is odd, an endomorphism J with $J^2 = -I$ would make each real tangent space a complex vector space, whose real dimension is even. Hence m must be even. Write $m = 2n$; then $m > 6$ gives $n \geq 4$.

If $E = (TM, J)$ existed, $c_i(E) = 0$ for $i < n$ and, after using the complex orientation, $c_n(E) = 2a$ for the integral top generator a. Pull E back along a homotopy equivalence $S^{2n} \to M$. Bott integrality says that the top Chern class of every complex bundle on S^{2n} is divisible by $(n-1)!$ [7, Chapter 17, Section 7, Theorem 7.2, pp. 274–275]. Since $(n-1)! \geq 6$ does not divide 2, this contradicts $c_n(E) = 2a$. Borel and Serre prove the same conclusion for every differentiable structure on S^{2n} in [4, Section 15, Proposition 15.1 and the following remark, p. 434]; their universal reduced-power input is [4, Section 12, Propositions 12.4 and 12.6 and Lemma 12.5, p. 430]. □

Proposition 30.4 (The direct rank-three Euler mismatch)**.** *Consider the direct rank-three toroidal local model obtained by coning the periodic Kuhn triangulation of $\mathbb{R}^3$ and dividing by $\mathbb{Z}^3$. Its central fibre W_3 has*

$$\chi(W_3) = 6.$$

If a proper triangulable map $\widetilde{f} : \widetilde{X}^8 \to S^2$ has this as one special fibre, has precisely two other special fibres that are free finite quotients of T^6, and is a T^6-bundle elsewhere, then $\chi(\widetilde{X}) = 6$. Therefore the direct three-fibre pattern cannot have total space S^8 even after forgetting complex structures.

Proof. A fundamental cube of the Kuhn triangulation is divided into the six tetrahedra indexed by permutations of its three coordinate directions. The central toric fibre is stratified by algebraic tori; every positive-dimensional torus stratum has Euler characteristic zero, while the zero-dimensional strata are indexed by the maximal tetrahedra modulo $\mathbb{Z}^3$. There are exactly six, so additivity over the finite torus stratification gives $\chi(W_3) = 6$.

Let Σ_f be the three critical values. Euler integration, or a cut into the regular bundle and three filling pieces, gives

$$\chi(\widetilde{X}) = \chi(S^2 \setminus \Sigma_f)\chi(T^6) + \sum_{p \in \Sigma_f} \chi(\widetilde{f}^{-1}(p)).$$

The regular term vanishes since $\chi(T^6) = 0$. A free finite quotient of a torus has Euler characteristic zero: the covering formula multiplies its Euler characteristic by the positive degree and the covering torus has Euler characteristic zero. Only W_3 remains, giving six. Since $\chi(S^8) = 2$, the total spaces cannot be homeomorphic. □

Corollary 30.5 (The exact singular-fibre repair equation)**.** *Let $g : M \to S^2$ be a proper triangulable map which is locally trivial with fibre T^6 off a finite set Δ, and suppose the direct Kuhn cusp fibre W_3 occurs over $p_0 \in \Delta$. Then*

$$\chi(M) = 6 + \sum_{p \in \Delta \setminus \{p_0\}} \chi(g^{-1}(p)). \tag{1131}$$

Consequently an S^8 carrier requires, exactly at the Euler-characteristic level,

$$\sum_{p\in\Delta\setminus\{p_0\}} \chi(g^{-1}(p)) = -4. \tag{1132}$$

If there are only the original two finite values, their replacement fibres must have Euler characteristics summing to -4. If those two fibres remain free finite quotients of T^6, they still contribute zero, so the cusp fibre itself must instead be replaced by a filling with Euler characteristic 2. Without either change, topology is an output and every closed carrier of the unchanged wrap necessarily has Euler characteristic 6.

Proof. Euler integration gives $\chi(M) = \chi(S^2 \setminus \Delta)\chi(T^6) + \sum_{p\in\Delta} \chi(g^{-1}(p))$. The first term is zero and $\chi(W_3) = 6$ by proposition 30.4, giving (1131). Substitution of $\chi(S^8) = 2$ gives (1132). The remaining assertions are its two specializations, using the covering calculation in the proof of proposition 30.4. □

Remark 30.6 (What remains possible topologically). A merely real singular map $S^8 \to S^2$ with regular fibre T^6 is not excluded by theorem 30.2; it must have special-fibre Euler sum 2. It cannot be a fibre bundle: the homotopy sequence would require the connecting map $\pi_2(S^2) = \mathbb{Z} \to \pi_1(T^6) = \mathbb{Z}^6$ to be surjective. A real $S^9 \to S^2$ model with regular fibre T^7 would likewise require genuine singularities and special-fibre Euler sum 0, but neither its carrier nor its odd-dimensional regular fibre can be complex in the required sense.

30.3 The exact toroidal successor over the cusp

We now retain, rather than discard, every exact map that survives the obstructions. Put

$$A_F = \mathbb{Z}\langle \widehat{w}, \widehat{\delta}\rangle, \qquad B_F = \mathbb{Z}\langle \widehat{\gamma}, \widehat{u}\rangle, \qquad B_0 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} : B_F \to A_F.$$

In angular–deck order the candidate cusp monodromy and filling map are

$$M_F(a,b) = (a + B_0 b, b), \qquad c_F : A_F \oplus B_F \longrightarrow B_F, \quad c_F(a,b) = b. \tag{1133}$$

Let $L_3 = \mathbb{Z}^3$, let $k = (1,1,1)^{\mathsf{T}}$, and put

$$P = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix}, \qquad Q = B_0^{-1} P = \begin{pmatrix} 0 & -1 & 1 \\ 1 & -1 & 0 \end{pmatrix}. \tag{1134}$$

Both maps are onto, $\ker P = \ker Q = \mathbb{Z}k$, and $B_0 Q = P$. The rank-three cusp module and its specialization are

$$M_3(a,b) = (a + b, b), \qquad c_3 : L_3 \oplus L_3 \longrightarrow L_3, \quad c_3(a,b) = b. \tag{1135}$$

Theorem 30.7 (Primitive lattice, monodromy, and filling quotient)**.** *The homomorphism*

$$\Phi = P \oplus Q : L_3 \oplus L_3 \longrightarrow A_F \oplus B_F, \qquad \Phi(a,b) = (Pa, Qb),$$

fits into the exact sequence

$$0 \longrightarrow \mathbb{Z}^2 \xrightarrow{\kappa} \mathbb{Z}^6 \xrightarrow{\Phi} \mathbb{Z}^4 \longrightarrow 0, \qquad \kappa(m,n) = (mk, nk), \tag{1136}$$

and satisfies

$$\Phi M_3 = M_F \Phi, \qquad c_F \Phi = Q c_3. \tag{1137}$$

The kernel in (1136) *is primitive.*

Proof. The integer matrix

$$S = \begin{pmatrix} 1 & 1 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$$

satisfies $PS = I_2$ and $QSB_0 = I_2$. Thus P and Q, and hence Φ, are integrally surjective. Their displayed kernels give $\ker \Phi = \mathbb{Z}(k,0) \oplus \mathbb{Z}(0,k)$, proving exactness. A kernel of a surjection between free abelian groups is primitive.

Directly,

$$\Phi M_3(a,b) = (P(a+b), Qb),$$

while

$$M_F \Phi(a,b) = (Pa + B_0 Qb, Qb) = (Pa + Pb, Qb).$$

The filling identity is equally literal: $c_F \Phi(a,b) = Qb = Qc_3(a,b)$. □

Let $t = e^{2\pi i s}$ on a logarithmic cusp cover. Retain the complete bounded holomorphic correction $C_2(t)$ of the candidate and define

$$Z_2 = sB_0 + C_2(t), \qquad C_3(t) = SC_2(t)Q, \qquad Z_3 = sI_3 + C_3(t). \tag{1138}$$

We use the angular–deck period convention $\Omega_r = (I_r \; Z_r)$, and write $\Omega_r(t)$ when the dependence through s and $C_r(t)$ must be displayed. The manuscript convention $(Z_r \; I_r)$ is related by the displayed integral block permutation $\left(\begin{smallmatrix} 0 & I_r \\ I_r & 0 \end{smallmatrix}\right)$; no period, sign, or lattice coordinate is silently changed.

Theorem 30.8 (Holomorphic elliptic extension and its exact fibre)**.** *After shrinking the cusp disc,* $\operatorname{Im} Z_3$ *is invertible and*

$$PZ_3 = Z_2 Q, \qquad P\Omega_3 = \Omega_2 \Phi. \tag{1139}$$

Consequently $P : \mathbb{C}^3 \to \mathbb{C}^2$ *descends to a short exact sequence of complex tori*

$$0 \longrightarrow E_s \longrightarrow \frac{\mathbb{C}^3}{\mathbb{Z}^3 + Z_3\mathbb{Z}^3} \xrightarrow{\overline{P}_s} \frac{\mathbb{C}^2}{\mathbb{Z}^2 + Z_2\mathbb{Z}^2} \longrightarrow 0, \qquad E_s \cong \frac{\mathbb{C}}{\mathbb{Z} + s\mathbb{Z}}. \tag{1140}$$

The correction C_2 *is recoverable from its prescribed lift by the exact left inverse*

$$C_2 = PC_3SB_0. \tag{1141}$$

Thus the extension neither deletes the original periods nor replaces them by an identity period matrix.

Proof. Since C_3 is bounded while $\operatorname{Im} s \to +\infty$, $\operatorname{Im} Z_3 = (\operatorname{Im} s) I_3 + \operatorname{Im} C_3$ is invertible on a sufficiently small cusp. Using $PS = I_2$ and $B_0 Q = P$,

$$PZ_3 = sP + PSC_2Q = sB_0Q + C_2Q = Z_2Q,$$

which is exactly (1139). It makes the complex linear map P a homomorphism of the displayed quotient tori. It is surjective on vector spaces and its lattice image is the whole target lattice because Φ is onto; therefore its kernel is connected.

The vector-space kernel is $\mathbb{C}k$. Since $Qk = 0$, one has $C_3 k = 0$ and $Z_3 k = sk$. If a source period lies in $\mathbb{C}k$, its image under P is the zero target period. The real period map $\Omega_2 : \mathbb{R}^4 \to \mathbb{C}^2$ is an isomorphism, so (1139) and $\ker \Phi = \mathbb{Z}(k,0) \oplus \mathbb{Z}(0,k)$ show that all such periods are precisely $\mathbb{Z}k + s\mathbb{Z}k$. This proves the fibre formula.

Finally $QS = B_0^{-1}$, whence

$$PC_3SB_0 = PSC_2QSB_0 = C_2.$$

□

30.3.1 The cusp filling map, not merely its boundary lattice

For $m \in \mathbb{Z}^3$ and a permutation $\pi = (\pi_1, \pi_2, \pi_3) \in S_3$, let

$$\sigma_{m,\pi} = \operatorname{conv}\{m, m + e_{\pi_1}, m + e_{\pi_1} + e_{\pi_2}, m + e_1 + e_2 + e_3\}. \tag{1142}$$

These are the tetrahedra of the periodic Kuhn triangulation; denote that triangulation by $\mathcal{T}_3$. Let $\mathcal{T}_2$ be the periodic triangulation of $\mathbb{R}^2$ whose triangles are

$$\operatorname{conv}\{u, u + e_1, u + e_2\}, \qquad \operatorname{conv}\{u + e_1, u + e_2, u + e_1 + e_2\}, \qquad u \in \mathbb{Z}^2.$$

For $r = 2, 3$, let $\mathfrak{S}_r$ be the fan in $\mathbb{Z}^r \oplus \mathbb{Z}$ consisting of all faces of the cones over $\tau \times \{1\}$ for $\tau \in \mathcal{T}_r$, and put $Y_r = X_{\mathfrak{S}_r}$. In the target lattice write

$$n_1 = \widehat{w}, \qquad n_2 = \widehat{\delta} - \widehat{w}, \qquad n_3 = -\widehat{\delta}; \qquad n_1 + n_2 + n_3 = 0.$$

The map P in (1134) is exactly $P(e_i) = n_i$.

Proposition 30.9 (Exact fan and quotient-filling morphism)**.** *Coning the Kuhn triangulation at height one gives a smooth toric fourfold Y_3 with height map locally equal to*

$$t = z_0 z_1 z_2 z_3.$$

The integral map

$$\widehat{P} : \mathbb{Z}^3 \oplus \mathbb{Z} \longrightarrow A_F \oplus \mathbb{Z}, \qquad \widehat{P}(x, r) = (Px, r),$$

is a map from this fan to the candidate's periodic A_2 fan and hence induces a toric morphism $F : Y_3 \to Y_2$ over the t-line.

On the dense torus define

$$\Psi^{(3)}_\lambda(x, t) = \big(\exp(2\pi i C_3(t)\lambda) t^\lambda x, t\big), \qquad \lambda \in \mathbb{Z}^3, \tag{1143}$$

$$\Psi^{(2)}_\mu(y, t) = \big(\exp(2\pi i C_2(t)\mu) t^{B_0\mu} y, t\big), \qquad \mu \in \mathbb{Z}^2. \tag{1144}$$

Then

$$F \Psi^{(3)}_\lambda = \Psi^{(2)}_{Q\lambda} F. \tag{1145}$$

For a sufficiently small open cusp disc $\Delta_\epsilon = \{t \in \mathbb{C} : |t| < \epsilon\}$, set

$$Y_{r,\epsilon} = t^{-1}(\Delta_\epsilon), \qquad N_3^C = Y_{3,\epsilon} / \{\Psi^{(3)}_\lambda : \lambda \in \mathbb{Z}^3\}, \qquad N_0 = Y_{2,\epsilon} / \{\Psi^{(2)}_\mu : \mu \in \mathbb{Z}^2\}. \tag{1146}$$

After shrinking Δ_ϵ, both actions are free and properly discontinuous, and F descends to a holomorphic filling morphism

$$\overline{F} : N_3^C \longrightarrow N_0 \tag{1147}$$

whose map on smooth-fibre H_1 is Φ and whose monodromy compatibility is (1137)*. Under $H_1(T^6) = L_3 \oplus L_3$ and $H_1(T^4) = A_F \oplus B_F$, the angular generators bound the toric coordinate discs while the deck generators survive. Hence the inclusion-induced specialization maps are exactly $c_3(a, b) = b$ and $c_F(a, b) = b$, and they form the commutative square*

$$\begin{array}{ccc} H_1(T^6; \mathbb{Z}) & \xrightarrow{\Phi} & H_1(T^4; \mathbb{Z}) \\ {\scriptstyle c_3}\downarrow & & \downarrow{\scriptstyle c_F} \\ \mathbb{Z}^3 & \xrightarrow{Q} & \mathbb{Z}^2. \end{array} \tag{1148}$$

Proof. The four vertices of (1142) map to

$$Pm, \quad Pm + n_{\pi_1}, \quad Pm - n_{\pi_3}, \quad Pm.$$

In target coordinates $(n_1, n_2, n_3) = (e_1, e_2 - e_1, -e_2)$. For the six permutations the three distinct images are the six A_2 triangles incident to Pm:

$$\begin{array}{c|c|c|c} (1,2,3) & \{v, v+e_1, v+e_2\} & (2,1,3) & \{v, v-e_1+e_2, v+e_2\} \\ (1,3,2) & \{v, v+e_1, v+e_1-e_2\} & (3,1,2) & \{v, v-e_2, v+e_1-e_2\} \\ (2,3,1) & \{v, v-e_1+e_2, v-e_1\} & (3,2,1) & \{v, v-e_2, v-e_1\}. \end{array}$$

Thus every source cone maps into a target cone. The toric fan criterion gives F, and preservation of the height coordinate gives $t \circ F = t$. The maximal Kuhn cones are unimodular, giving the displayed local monomial equation and smoothness of Y_3.

On the dense torus, $PC_3 = C_2 Q$ and $P = B_0 Q$ give

$$F\Psi_\lambda^{(3)}(x,t) = \big(\exp(2\pi i C_2 Q\lambda) t^{B_0 Q\lambda} F(x), t\big),$$

which proves (1145); the fan map extends it over every toric orbit.

For the analytic quotient, put $y = -\log|t|/(2\pi)$ and

$$\operatorname{trop}_t(x) = \frac{1}{\log|t|}(\log|x_1|, \log|x_2|, \log|x_3|).$$

The exact transformation law is

$$\operatorname{trop}_t(\Psi_\lambda^{(3)} x) = \operatorname{trop}_t(x) + A(t)\lambda, \qquad A(t) = I_3 + \frac{\operatorname{Im} C_3(t)}{y}.$$

Boundedness of C_3 permits the cusp to be shrunk until

$$\tfrac{1}{2}\|\lambda\| \le \|A(t)\lambda\| \le \tfrac{3}{2}\|\lambda\|.$$

For the target use

$$\operatorname{trop}_t^{(2)}(y) = \frac{1}{\log|t|}(\log|y_1|, \log|y_2|).$$

Its exact transformation law is

$$\operatorname{trop}_t^{(2)}(\Psi_\mu^{(2)} y) = \operatorname{trop}_t^{(2)}(y) + A_2(t)\mu, \qquad A_2(t) = B_0 + \frac{\operatorname{Im} C_2(t)}{y}.$$

Since B_0 is orthogonal, after one further shrinking the same bounds hold with $A_2(t), \mu$ in place of $A(t), \lambda$. Therefore a compact set can meet only finitely many of its translates for either action. On a boundary orbit the bounded cell label is translated by λ, so the same conclusion holds there; the target argument uses the invertible lattice map B_0 in exactly the same way. A fixed point in a smooth fibre would give $\operatorname{Im} Z_3\lambda = 0$, hence $\lambda = 0$; equality of a boundary cell with its translate likewise forces $\lambda = 0$. The target equalities give $\operatorname{Im} Z_2\mu = 0$ in the smooth fibre and $B_0\mu = 0$ on the boundary, hence $\mu = 0$. Rounding the coordinates of $A(t)^{-1}\operatorname{trop}_t(x)$ gives a uniformly bounded set of representatives. The closures of these representatives lie in the finite union of closed toric polydiscs belonging to the six Kuhn tetrahedra; that finite union is compact. This proves proper discontinuity. The same construction with $A_2(t)$ supplies target representatives. Over every smaller closed disc $\overline{\Delta}_{\epsilon'}$, $0 < \epsilon' < \epsilon$, the displayed finite unions are compact, proving properness there. The equivariant F therefore descends, and (1148) is exactly the second identity in (1137). □

30.3.2 Finite peripheral data and the precise globalization gap

In the manuscript lattice order the candidate matrices are

$$A_1 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 6 & 0 & 1 & 0 \\ -6 & -1 & -1 & 0 \\ -2 & 1 & 0 & 1 \end{pmatrix}, \qquad A_2 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ -6 & 1 & 0 & 0 \\ 3 & 0 & 1 & 1 \end{pmatrix}. \tag{1149}$$

The matrices A_1, A_2 are written in the manuscript order $(\widehat{\gamma}, \widehat{u}, \widehat{w}, \widehat{\delta}) = B_F \oplus A_F$, whereas M_F in (1133) is written in the angular–deck order $A_F \oplus B_F$. With the explicit block swap

$$J = \begin{pmatrix} 0 & I_2 \\ I_2 & 0 \end{pmatrix}, \qquad M_F^{\mathrm{man}} = J M_F J = \begin{pmatrix} I_2 & 0 \\ B_0 & I_2 \end{pmatrix}, \tag{1150}$$

one has $A_1^3 = A_2^4 = I$ and $A_1 A_2 M_F^{\mathrm{man}} = I$. Define

$$C_1 = \begin{pmatrix} -1 & -1 \\ 1 & 0 \end{pmatrix}, \quad C_2' = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad C_0 = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}. \tag{1151}$$

For uniform notation below set $D_1 = C_1$ and $D_2 = C_2'$. Then $(C_1)^3 = (C_2')^4 = I$, $C_1 C_2' = C_0^{-1}$, and $(C_0 - I)^2 = 0$ with rank one.

The kernel block in this presentation is not an unrelated direct summand of the rank-three cusp lattice. Put

$$\begin{gathered} r = \begin{pmatrix} 0 & 0 & 1 \end{pmatrix}, \qquad A = \begin{pmatrix} P \\ r \end{pmatrix} = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} Q \\ r \end{pmatrix} = \begin{pmatrix} 0 & -1 & 1 \\ 1 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \\ D = B_0 \oplus [1] = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad U_6 = \mathrm{diag}(A, B) \in SL(6, \mathbb{Z}). \end{gathered} \tag{1152}$$

Lemma 30.10 (The identical cusp lattice in quotient–kernel coordinates)**.** *Let*

$$\rho(Pa, ra; Qb, rb) = (Pa, Qb; ra, rb), \qquad \rho_{\mathrm{man}} = (J \oplus I_2)\rho.$$

Then

$$U_6 M_3 U_6^{-1} = \begin{pmatrix} I_3 & D \\ 0 & I_3 \end{pmatrix}, \tag{1153}$$

$$\rho U_6 M_3 U_6^{-1} \rho^{-1} = M_F \oplus C_0, \tag{1154}$$

$$\rho_{\mathrm{man}} U_6 M_3 U_6^{-1} \rho_{\mathrm{man}}^{-1} = M_F^{\mathrm{man}} \oplus C_0. \tag{1155}$$

Projection onto the candidate block is exactly $\mathrm{pr}_{1,2}\, \rho U_6(a, b) = (Pa, Qb) = \Phi(a, b)$ *in angular–deck coordinates and* $J\Phi(a, b)$ *in manuscript coordinates.*

Proof. Both A and B have determinant one, and direct multiplication gives $A = DB$, so $AB^{-1} = D$. Conjugating $M_3 = \left(\begin{smallmatrix} I_3 & I_3 \\ 0 & I_3 \end{smallmatrix}\right)$ by U_6 proves (1153). The first two rows of A, B are exactly P, Q, while $r(1, 1, 1)^{\mathsf{T}} = 1$; regrouping the first two and last coordinates therefore gives (1154). Applying the already displayed block swap J proves (1155) and both projection formulas. □

Proposition 30.11 (Exact finite-order affine lift)**.** *On* $\widetilde{\Lambda} = \Lambda \oplus \mathbb{Z}^2$*, identified with the original* $\mathbb{Z}^3 \oplus \mathbb{Z}^3$ *by lemma 30.10, set*

$$\widetilde{A}_1 = A_1 \oplus C_1, \qquad \widetilde{A}_2 = A_2 \oplus C_2', \qquad \widetilde{M}_0 = M_F^{\mathrm{man}} \oplus C_0.$$

Then

$$\widetilde{A}_1^3 = I, \qquad \widetilde{A}_2^4 = I, \qquad \widetilde{A}_1\widetilde{A}_2\widetilde{M}_0 = I.$$

The affine transformations

$$\widetilde{g}_j(x,y) = (A_j x + v_j/m_j, D_j y), \qquad (m_1,m_2) = (3,4), \tag{1156}$$

of $T^4 \times T^2$ have exact orders m_j, act freely, and project to the candidate finite-fibre actions.

Proof. All three matrix identities follow blockwise from the identities just displayed and the candidate identities. The vector $(v_j, 0)$ is fixed by $\widetilde{A}_j$. The m_j-th power of (1156) is translation by the integral vector $(v_j, 0)$ and hence is the identity on the torus. If a smaller positive power fixed a point, its projection to T^4 would fix a point under the corresponding smaller power of the candidate affine transformation. The logarithmic-transform action in the candidate is free, so this is impossible. Projection to the first block is equivariant by construction. □

Remark 30.12 (Exact endpoint). Theorems 30.7–30.8 and Propositions 30.9–30.11 prove the regular cusp, the entire toric cusp filling, and compatible finite *integral affine* data. They do not construct one holomorphic rank-three period family on the full $(3,4,\infty)$ cover whose two finite analytic fillings glue to the cusp family. Even if that analytic gluing is constructed, proposition 30.4 proves that the unchanged three-fibre pattern has Euler characteristic six and hence is not S^8. This is a proved morphism with a proved endpoint, not an assertion that the rank-three and rank-two objects are unrelated.

30.4 The forced S^9, S^{10}, and rank-loss S^8

Put

$$\mathscr{V} = \mathrm{Mat}_{3\times 3}(\mathbb{C}), \qquad \mathcal{P}_3 = \{H \in \mathrm{Herm}_3(\mathbb{C}) : H \succeq 0\}, \qquad \mathcal{K}_3 = \{H \in \mathcal{P}_3 : \operatorname{tr} H = 1\}.$$

The superscript + denotes one-point compactification.

Theorem 30.13 (Complete $U(3)$ and $SU(3)$ orbit invariants)**.** *The maps*

$$\Gamma_U(A) = A^*A, \tag{1157}$$

$$\Gamma_{SU}(A) = (A^*A, \det A) \tag{1158}$$

induce homeomorphisms

$$\mathscr{V}/U(3) \xrightarrow{\sim} \mathcal{P}_3, \tag{1159}$$

$$\mathscr{V}/SU(3) \xrightarrow{\sim} \mathcal{Y}_3 := \{(H,z) \in \mathcal{P}_3 \times \mathbb{C} : |z|^2 = \det H\}. \tag{1160}$$

For a rank-r matrix A, the two orbit fibres and stabilizers are

$$U(3)\cdot A \cong U(3)/U(3-r), \qquad SU(3)\cdot A \cong SU(3)/SU(3-r),$$

with $SU(1)$ interpreted as the trivial group.

Proof. Both maps are invariant under the indicated left actions, and $|\det A|^2 = \det(A^*A)$. Every $H \succeq 0$ is the Gram matrix of $H^{1/2}$. If $A^*A = B^*B$, the rule $Ax \mapsto Bx$ is well-defined because the two matrices have the same kernel, and is isometric because

$$\langle Bx, By\rangle = x^*B^*By = x^*A^*Ay = \langle Ax, Ay\rangle.$$

Extending it orthogonally gives $U \in U(3)$ with $UA = B$. Thus (1157) separates $U(3)$-orbits.

Given $(H,z) \in \mathcal{Y}_3$, if H is singular then $z = 0$ and $H^{1/2}$ realizes it. If H is positive definite, put

$$\omega = z/\sqrt{\det H}, \qquad A = \mathrm{diag}(\omega, 1, 1)H^{1/2}.$$

Then $(A^*A, \det A) = (H, z)$. If two matrices have both invariants equal, the unitary U above has determinant one when they are invertible. In rank at most two, multiply U on the left by a unitary that fixes $\operatorname{im} B$ pointwise and corrects the determinant on its nonzero orthogonal complement. Hence (1158) separates $SU(3)$-orbits.

Both invariant maps are proper because $\operatorname{tr}(A^*A) = \|A\|_F^2$. The induced continuous bijections to the displayed closed Hausdorff sets are therefore homeomorphisms. A left stabilizer fixes $\operatorname{im} A$ pointwise and is arbitrary unitary, respectively special unitary, on $(\operatorname{im} A)^\perp$, proving the fibre statements. □

Let

$$E = \{X \in \mathrm{Herm}_3(\mathbb{C}) : \operatorname{tr} X = 0\}, \qquad \dim_{\mathbb{R}} E = 8, \qquad c = I_3/3.$$

Equip E with the Frobenius inner product $\langle X, Y\rangle_F = \operatorname{tr}(XY)$ and its norm; $B(E)$ and $S(E)$ below denote its closed unit ball and unit sphere. For $v \in S(E)$ define

$$R_*(v) = -\frac{1}{3\lambda_{\min}(v)}. \tag{1161}$$

Since a nonzero traceless Hermitian matrix has a negative eigenvalue, $R_*(v)$ is finite and positive.

Lemma 30.14 (Explicit eight-ball chart)**.** *The maps*

$$\chi(c) = 0, \qquad \chi(c + sv) = \frac{s}{R_*(v)} v, \quad 0 < s \le R_*(v), \tag{1162}$$

$$\chi^{-1}(rv) = c + rR_*(v)v, \quad 0 < r \le 1 \tag{1163}$$

are inverse homeomorphisms $\mathcal{K}_3 \cong B(E) \cong B^8$. They send the rank-deficient locus onto $S(E)$.

Proof. The eigenvalues of $c + sv$ are $1/3 + s\lambda_j(v)$. Hence this matrix is positive semidefinite exactly for $0 \le s \le R_*(v)$ and is singular exactly at the upper endpoint. Every $H \neq c$ in $\mathcal{K}_3$ lies on one unique such ray. Continuity of the least eigenvalue makes $v \mapsto -\lambda_{\min}(v)$ a positive continuous function on the compact space $S(E)$, hence bounded above and bounded away from zero. Thus $R_*(v)$ is continuous and uniformly bounded above and away from zero. The formulas (1162)–(1163) are consequently continuous, including at c and at the origin, and prove all assertions. □

Theorem 30.15 (The forced sphere ladder and explicit inverses)**.** *Let $S_F^{17} = \{A \in \mathscr{V} : \|A\|_F = 1\}$. Then*

$$S_F^{17}/SU(3) \cong \mathcal{Y}_{3,1} := \{(H, z) \in \mathcal{Y}_3 : \operatorname{tr} H = 1\} \cong S^9. \tag{1164}$$

An explicit second homeomorphism is

$$\Omega(H, z) = \begin{cases} \left(\chi(H), \sqrt{1 - \|\chi(H)\|^2}\, z/|z|\right), & H \succ 0, \\ (\chi(H), 0), & H \text{ singular}, \end{cases} \tag{1165}$$

from $\mathcal{Y}_{3,1}$ to $S(E \oplus \mathbb{C})$.

Furthermore the weighted radial map

$$\Phi_{SU}(H, z) = \begin{cases} 0, & (H, z) = (0, 0), \\ s\,\Omega(H/s^2, z/s^3), & s = \sqrt{\operatorname{tr} H} > 0 \end{cases} \tag{1166}$$

is a homeomorphism $\mathcal{Y}_3 \cong E \oplus \mathbb{C} \cong \mathbb{R}^{10}$. It extends to one-point compactifications and gives a homeomorphism of pairs

$$\left(\mathscr{V}^+/SU(3), \{\det A = 0\}^+/SU(3)\right) \cong \left(S^{10}, S^8_{\mathrm{lin}}\right). \tag{1167}$$

The complement is

$$S^{10} \setminus S^8 \cong E \times (\mathbb{C} \setminus \{0\}) \simeq S^1, \tag{1168}$$

and $\arg\det A$ *is the angular deformation-retraction coordinate.*

Proof. The first identification in (1164) follows from theorem 30.13. If $H \succ 0$, then $z \neq 0$; if H is singular, then $z = 0$ and lemma 30.14 gives $\|\chi(H)\| = 1$. Thus (1165) lands on the unit sphere and is continuous even at rank loss because the coefficient of $z/|z|$ tends to zero. Given $(x, w) \in S(E \oplus \mathbb{C})$, put $H = \chi^{-1}(x)$. If $\|x\| < 1$, then $w \neq 0$ and the unique inverse determinant is

$$z = \sqrt{\det H}\, w/|w|. \tag{1169}$$

If $\|x\| = 1$, then $w = 0$, H is singular, and $z = 0$. This proves the S^9 homeomorphism with its inverse.

For $(H, z) \neq 0$, the normalized pair $(H/s^2, z/s^3)$ belongs to $\mathcal{Y}_{3,1}$ because $|z/s^3|^2 = \det(H/s^2)$. Conversely, for $v \neq 0$ in $E \oplus \mathbb{C}$, set

$$s = \|v\|, \qquad (h, \zeta) = \Omega^{-1}(v/\|v\|), \qquad H = s^2 h, \qquad z = s^3\zeta.$$

This is the inverse of (1166). The norm identity $\|\Phi_{SU}(H, z)\| = s$ proves continuity of both maps at zero and properness, so the map extends to compactifications.

Rank loss is exactly $z = 0$. Under Ω, the normalized rank-loss set maps to $S(E \oplus 0)$, and radialization maps the entire affine rank-loss set to $E \oplus 0$. Its one-point compactification is the standard linear $S^8 \subset S^{10}$. Removing it gives $E \times (\mathbb{C} \setminus 0)$; projection of the second factor to its phase is a deformation retraction. By construction this phase is $z/|z| = \det A/|\det A|$. □

Put

$$\Sigma_3 = \{H \in \mathrm{Herm}_3(\mathbb{C}) : H \succeq 0,\ \det H = 0\}.$$

Proposition 30.16 (The $U(3)$ ball and why its boundary is the S^8)**.** *The map*

$$E \times \mathbb{R}_{\geq 0} \longrightarrow \mathcal{P}_3, \qquad (K, t) \longmapsto K + (t - \lambda_{\min}(K))I_3 \tag{1170}$$

is a homeomorphism with inverse

$$H \longmapsto \left(H - \frac{\operatorname{tr} H}{3} I_3, \lambda_{\min}(H)\right).$$

Consequently

$$\mathscr{V}^+/U(3) \cong \mathcal{P}_3^+ \cong B^9, \qquad \partial B^9 = \Sigma_3^+ \cong S^8. \tag{1171}$$

By contrast, the quotient of the source unit sphere is

$$S_F^{17}/U(3) \cong \mathcal{K}_3 \cong B^8, \tag{1172}$$

whose boundary is the trace-one rank-loss S^7. Thus "take the source sphere quotient" and "take the boundary of the compactified full quotient" are different operations.

Proof. The least eigenvalue of the image of (1170) is t. Conversely, subtracting one third of the trace from H produces a traceless matrix K whose least eigenvalue is $\lambda_{\min}(H) - \operatorname{tr}(H)/3$; substitution recovers H. Eigenvalue continuity proves that the formulas are inverse homeomorphisms. The one-point compactification of a closed half-space is a closed ball, and its boundary is the one-point compactification of the boundary hyperplane. Now use theorem 30.13. Equation (1172) follows from $\|A\|_F^2 = \operatorname{tr}(A^*A)$ and lemma 30.14. □

Corollary 30.17 (Residual determinant circle and its fibres)**.** *The residual group $U(3)/SU(3) \cong S^1$ acts on the pair in* (1167) *by*

$$e^{i\theta} \cdot (x, w) = (x, e^{i\theta} w).$$

Its fixed set is precisely the rank-loss S^8, and the orbit map has

$$S^{10}/S^1 \cong B^9, \qquad q^{-1}(q(p)) \cong \begin{cases} S^1, & p \notin S^8, \\ \{p\}, & p \in S^8. \end{cases} \tag{1173}$$

On the normalized section, $S^9/S^1 \cong B^8$ and the fixed set is the rank-deficient $S^7 = \partial B^8$.

Proof. Left multiplication by $U \in U(3)$ fixes $H = A^*A$ and sends $z = \det A$ to $(\det U)z$. Every phase is a determinant. Equations (1165) and (1166) therefore turn the residual action into scalar rotation of the complex coordinate. Its orbit space on $E \oplus \mathbb{C}$ is $E \times \mathbb{R}_{\geq 0}$, whose one-point compactification is a closed 9-ball; the zero complex coordinate is its boundary. The stabilizer is trivial for $w \neq 0$ and all of S^1 for $w = 0$, giving the fibres. On the unit sphere, replacing w by $|w|$ gives the closed unit ball in E and the last statement. □

The S^8 in (1167) can also be defined without first forming an orbit quotient.

Proposition 30.18 (The exact residual-circle degeneration and join)**.** *Under* (1164)*, forgetting determinant phase is exactly*

$$q_1 : S(E \oplus \mathbb{C}) = S^9 \longrightarrow B(E) = B^8, \qquad q_1(x, w) = x. \tag{1174}$$

Its fibre is S^1 for $\|x\| < 1$ and a singleton for $\|x\| = 1$. More precisely,

$$S^9 = S(E \oplus \mathbb{C}) \cong S(E) * S(\mathbb{C}) \cong S^7 * S^1, \tag{1175}$$

and q_1 collapses the S^1 join factor at the S^7 endpoint. On the full compactification the analogous map is

$$q_+ : S^{10} \longrightarrow B^9, \tag{1176}$$

with circle fibre off the rank-loss S^8 and point fibre on that S^8. The integral class of the circle on either complement is generated by $\arg\det A$.

Proof. For fixed $H \succ 0$ of trace one, the equation $|z|^2 = \det H > 0$ leaves exactly its phase, an S^1; for singular H it forces $z = 0$. Formula (1165) converts H into $x = \chi(H)$ and the determinant phase into the phase of w, so the orbit map is literally (1174). Every unit pair has the unique form

$$(x, w) = (\cos\alpha\, u, \sin\alpha\, v), \qquad u \in S(E),\ v \in S(\mathbb{C}),\ 0 \leq \alpha \leq \pi/2,$$

away from the endpoints, with the usual endpoint identifications. This is the defining quotient of the join and proves (1175). The full statement is corollary 30.17 and proposition 30.16. The generator claim follows from the loop

$$A_\theta = 3^{-1/2}\,\mathrm{diag}(e^{i\theta}, 1, 1), \qquad 0 \leq \theta \leq 2\pi,$$

whose Gram matrix is $I_3/3$ and whose determinant phase winds once. □

Proposition 30.19 (Spectral-floor chart and the marked fibres)**.** *The maps*

$$\mathcal{F} : E \longrightarrow \Sigma_3, \qquad \mathcal{F}(K) = K - \lambda_{\min}(K) I_3, \qquad \mathcal{F}^{-1}(H) = H - \frac{\operatorname{tr} H}{3} I_3 \tag{1177}$$

are inverse homeomorphisms. Hence $\Sigma_3^+ \cong S^8$, and it is exactly the S^8 of (1167)*.*

The fixed marked compression

$$\mathscr{O} : \Sigma_3 \longrightarrow \mathrm{Herm}_2(\mathbb{C})_{\succeq 0}, \qquad \mathscr{O}(H) = PHP^* \tag{1178}$$

is onto. For a fixed $G \succeq 0$, put

$$X_P^{-1} H X_P^{-*} = \begin{pmatrix} G & a \\ a^* & \lambda \end{pmatrix}.$$

Its complete fibre in Σ_3 is

$$\mathscr{O}^{-1}(G) = \left\{ X_P \begin{pmatrix} G & a \\ a^* & a^* G^\dagger a + \tau \end{pmatrix} X_P^* : a \in \operatorname{Ran} G,\ \tau \geq 0,\ \left(\operatorname{rank} G < 2 \text{ or } \tau = 0\right) \right\}. \tag{1179}$$

Consequently the fibres are homeomorphic to

$$\begin{cases} \mathbb{C}^2, & G \succ 0, \\ \mathbb{C} \times \mathbb{R}_{\geq 0}, & \operatorname{rank} G = 1, \\ \mathbb{R}_{\geq 0}, & G = 0. \end{cases} \tag{1180}$$

The positive section is $G \mapsto RGR^*$. The affine map (1178) *is not proper, so these formulas do not assert an unproved extension through the one-point compactification point.*

Proof. For $K \in E$, the matrix $\mathcal{F}(K)$ is positive semidefinite and has least eigenvalue zero. If $H \in \Sigma_3$, then $K = H - (\operatorname{tr} H/3)I_3$ is traceless and has least eigenvalue $-\operatorname{tr} H/3$; therefore the two formulas in (1177) are inverse. Continuity of the least eigenvalue proves the homeomorphism. Its one-point compactification is $E^+ \cong S^8$. The proof of theorem 30.15 identifies the same rank-loss affine space with $E \oplus 0$, so the two S^8 descriptions agree.

Equations (1128)–(1129) show that the upper-left block of $X_P^{-1} H X_P^{-*}$ is $PHP^* = G$. Congruence by the invertible X_P reduces positivity to positivity of the displayed block matrix. The exact generalized Schur-complement criterion is

$$G \succeq 0, \qquad a \in \operatorname{Ran} G, \qquad \tau := \lambda - a^* G^\dagger a \geq 0.$$

Indeed, multiplying on both sides by the invertible triangular matrices with off-diagonal entry $-G^\dagger a$ reduces it to $\operatorname{diag}(G, \tau)$; if a had a component on $\ker G$, testing the quadratic form on a large signed multiple of a kernel vector would violate positivity. The same congruence gives

$$\operatorname{rank} H = \operatorname{rank} G + \mathbf{1}_{\tau > 0}.$$

Thus H is singular precisely when $\operatorname{rank} G < 2$ or $\tau = 0$, proving (1179). The three fibre types follow by using $\dim_{\mathbb{C}} \operatorname{Ran} G = 2, 1, 0$. Setting $a = 0$ and $\tau = 0$ gives $H = RGR^*$ and proves surjectivity and the section. Finally, for fixed $G \succ 0$ the parameter $a \in \mathbb{C}^2$ is unbounded, so the inverse image of the compact singleton $\{G\}$ is not compact; $\mathscr{O}$ is not proper. □

30.4.1 A global marked observation on the S^9 quotient

Although compression on the affine rank-loss S^8 is nonproper, the trace-one S^9 has a global compact observation with exactly computable fibres. Define

$$\mathscr{O}_9 : \mathcal{Y}_{3,1} \longrightarrow \mathcal{B}_P := \{G \succeq 0 : \operatorname{tr}(C^{-1}G) \leq 1\}, \qquad \mathscr{O}_9(H, z) = PHP^*. \tag{1181}$$

The set $\mathcal{B}_P$ is a compact convex body with nonempty interior in $\operatorname{Herm}_2(\mathbb{C})$, hence is homeomorphic to B^4 by radial projection from an interior point.

Theorem 30.20 (Every fibre of the S^9 marked observation)**.** *The map* (1181) *is a continuous surjection. For $G \in \mathcal{B}_P$ set*

$$L(G) = \frac{1 - \operatorname{tr}(C^{-1}G)}{3}. \tag{1182}$$

Its fibre is exactly the set of pairs

$$a \in \operatorname{Ran} G, \qquad a^* G^\dagger a \leq L(G), \qquad z \in \mathbb{C}, \qquad |z|^2 = \det G\big(L(G) - a^* G^\dagger a\big), \tag{1183}$$

where (H, z) is recovered uniquely from

$$H = X_P \begin{pmatrix} G & a \\ a^* & L(G) \end{pmatrix} X_P^*. \tag{1184}$$

The fibre is S^5 when $G \succ 0$ and $L(G) > 0$; it is a point when $G \succ 0$ and $L(G) = 0$; it is a closed two-disc when $\operatorname{rank} G = 1$ and $L(G) > 0$, degenerating to a point when $L(G) = 0$; and it is a point at $G = 0$. These are orbit-space fibres, not torus fibres.

Proof. Since $\operatorname{tr} H = \operatorname{tr}(X_P^* X_P K)$ for $K = X_P^{-1} H X_P^{-*}$, (1129) gives

$$1 = \operatorname{tr} H = \operatorname{tr}(C^{-1}G) + 3\lambda.$$

Thus $\lambda = L(G)$. The Schur-complement argument in proposition 30.19 gives exactly the first two conditions in (1183). Since $\det X_P = 1$,

$$\det H = \det G\big(L(G) - a^* G^\dagger a\big)$$

when G is invertible, and the same identity follows polynomially, or from rank, when G is singular. The defining equation $|z|^2 = \det H$ of $\mathcal{Y}_{3,1}$ gives the last condition. Conversely these conditions construct a unique point by (1184); taking $a = 0$ proves surjectivity for every $G \in \mathcal{B}_P$.

If $G \succ 0$, put $b = G^{-1/2}a$ and $w = z/\sqrt{\det G}$. Equation (1183) becomes $\|b\|^2 + |w|^2 = L(G)$ in $\mathbb{C}^2 \oplus \mathbb{C}$. It is S^5 for $L(G) > 0$ and a point for $L(G) = 0$. If $\operatorname{rank} G = 1$, then $z = 0$ and the allowed a form the closed Hermitian norm disc in the one-dimensional complex space $\operatorname{Ran} G$. If $G = 0$, then $a = z = 0$ and $H = uu^*/3$ is the unique point. □

30.4.2 The exact period-to-S^9 and period-to-S^8 maps

For t in the punctured cusp and $G_6 \in \operatorname{Sym}_6^+(\mathbb{R})$, retain both arguments and put

$$G_4 = \Phi G_6 \Phi^{\mathsf{T}}, \qquad H_3(t; G_6) = \Omega_3(t) G_6 \Omega_3(t)^*, \qquad H_2(t; G_6) = \Omega_2(t) G_4 \Omega_2(t)^*. \tag{1185}$$

Proposition 30.21 (Commuting marked Gram square)**.** *The matrices $G_4, H_3(t; G_6)$, and $H_2(t; G_6)$ are positive definite, and*

$$P H_3(t; G_6) P^* = H_2(t; G_6). \tag{1186}$$

Equivalently, for every fixed t the square

$$\begin{array}{ccc} \operatorname{Sym}_6^+(\mathbb{R}) & \xrightarrow{G \mapsto \Omega_3(t) G \Omega_3(t)^*} & \operatorname{Herm}_3(\mathbb{C})_{\succ 0} \\ {\scriptstyle G \mapsto \Phi G \Phi^{\mathsf{T}}}\big\downarrow & & \big\downarrow{\scriptstyle H \mapsto PHP^*} \\ \operatorname{Sym}_4^+(\mathbb{R}) & \xrightarrow{G \mapsto \Omega_2(t) G \Omega_2(t)^*} & \operatorname{Herm}_2(\mathbb{C})_{\succ 0} \end{array} \tag{1187}$$

commutes. Normalize without suppressing either argument:

$$h_3(t; G_6) = \frac{H_3(t; G_6)}{\operatorname{tr} H_3(t; G_6)}, \qquad z_3(t; G_6) = +\sqrt{\det h_3(t; G_6)}.$$

Then the explicit point

$$\jmath(t; G_6) = \big(h_3(t; G_6), z_3(t; G_6)\big) \in \mathcal{Y}_{3,1} \cong S^9, \qquad \mathscr{O}_9\big(\jmath(t; G_6)\big) = \frac{H_2(t; G_6)}{\operatorname{tr} H_3(t; G_6)}. \tag{1188}$$

The positive square root is a specified phase-zero section; it does not erase the existence of the other determinant phases in S^9.

The spectral-floor map gives a second exact, generally noncompact-chart map

$$\begin{aligned} \sigma(t; G_6) &= H_3(t; G_6) - \lambda_{\min}\big(H_3(t; G_6)\big) I_3 \in \Sigma_3 \hookrightarrow \Sigma_3^+ \cong S^8, \\ \mathscr{O}\big(\sigma(t; G_6)\big) &= H_2(t; G_6) - \lambda_{\min}\big(H_3(t; G_6)\big) C. \end{aligned} \tag{1189}$$

Proof. Surjectivity of Φ makes Φ^{T} injective; hence $G_4 = \Phi G_6 \Phi^{\mathsf{T}}$ is positive definite. The identity blocks in $\Omega_r(t)$ make $\Omega_r(t)^*$ injective, so both $H_r(t; G_6)$ are positive definite. Using $P\Omega_3(t) = \Omega_2(t)\Phi$ from (1139),

$$P H_3(t; G_6) P^* = P \Omega_3(t) G_6 \Omega_3(t)^* P^* = \Omega_2(t) \Phi G_6 \Phi^{\mathsf{T}} \Omega_2(t)^* = H_2(t; G_6).$$

This proves the square and (1188). Subtracting the least eigenvalue makes a positive matrix singular without altering its eigenspaces; applying P and $P I_3 P^* = C$ proves (1189). □

There is one useful cusp endpoint for a fixed, explicitly chosen metric. Take $G_6 = I_6$ in the real period frame, choose the standard logarithmic strip $0 \le x < 1$ for $s = x + iy$, and abbreviate $H_{3,I}(t) = H_3(t; I_6)$. The exact formula is

$$\begin{aligned} H_{3,I}(t) &= I_3 + Z_3(t)Z_3(t)^* \\ &= (1 + |s|^2)I_3 + sC_3(t)^* + \overline{s}C_3(t) + C_3(t)C_3(t)^*. \end{aligned} \tag{1190}$$

Put $h_{3,I}(t) = H_{3,I}(t)/\operatorname{tr} H_{3,I}(t)$, $z_{3,I}(t) = +\sqrt{\det h_{3,I}(t)}$, and $\jmath_I(t) = \jmath(t; I_6)$.

Corollary 30.22 (A proved cusp limit in the S^9 carrier)**.** *For* (1190)*, boundedness of* $C_3(t)$ *gives*

$$h_{3,I}(t) \longrightarrow I_3/3, \qquad z_{3,I}(t) \longrightarrow 1/(3\sqrt{3}), \qquad \mathscr{O}_9(\jmath_I(t)) \longrightarrow C/3 \quad (t \to 0). \tag{1191}$$

Thus the period metric has an exact continuous marked limit in the interior of the S^9 *orbit carrier. This is a map of period–metric data, not a compactification of every point of the torus family and not a fourth singular filling.*

Proof. Equation (1190) and boundedness of C_3 give $y^{-2}H_{3,I} \to I_3$ as $y \to \infty$: indeed $0 \le x < 1$ gives $|s|^2/y^2 \to 1$, while the two terms linear in s are $O(y)$ and the remaining correction is $O(1)$. Taking traces and determinants yields $h_{3,I} \to I_3/3$ and the positive square-root limit for $z_{3,I}$. Compressing by P and using (1188) gives the last limit. □

Theorem 30.23 (Which Gram carrier is strongest)**.** *Within the specified left-action invariant theory, the information order is strict:*

$$\mathcal{Y}_3 \xrightarrow{\textit{forget } \arg z} \mathcal{P}_3, \qquad \mathcal{Y}_{3,1} \xrightarrow{\textit{forget } \arg z} \mathcal{K}_3.$$

The maps have circle fibres on full-rank matrices and point fibres at rank loss. Hence:

(i) $S^9 = S^{17}_F/SU(3)$ *is the strongest* normalized single-sphere *carrier: it retains the complete Gram matrix, determinant phase, its integral winding, the exact circle degeneration* (1174)*, and the global marked observation* (1181)*.*

(ii) (S^{10}, S^8) *is the strongest* unnormalized compact pair*: it additionally retains the radial scale, and its* S^8 *is exactly the fixed rank-loss locus.*

(iii) *The standalone* $S^8 = \partial(\mathscr{V}^+/U(3))$ *is the strongest rank-loss sphere, but it has forgotten determinant phase and does not carry the global compact extension of the affine compression* (1178)*.*

None of this ordering supplies a complex tangent structure, a T^6 *regular fibre, the base* $\mathbb{P}^1$*, the three special values, or their integral specialization maps.*

Proof. The orbit-separation theorem shows that (H, z) is the complete $SU(3)$ invariant and H the complete $U(3)$ invariant. For $H \succ 0$, changing the phase of z changes the $SU(3)$ orbit but not the $U(3)$ orbit; at rank loss $z = 0$, so no phase remains. This proves strict information loss and the fibre claim. Propositions 30.16 and 30.18, Theorem 30.20, and Equation (1167) prove (i)–(iii). The absent structures cannot be inferred from the quotient map: its base has dimensions eight or nine, its generic fibre is S^1, its singular values form the full boundary S^7 or S^8, and its complement has fundamental group $\mathbb{Z}$. In contrast, since $f|_{X^\circ} : X^\circ \to B^\circ$ is a T^4-bundle and $\pi_2(B^\circ) = 0$, its homotopy exact sequence gives

$$1 \longrightarrow \mathbb{Z}^4 \longrightarrow \pi_1(X^\circ) \longrightarrow \pi_1(B^\circ) = F_2 \longrightarrow 1,$$

which surjects onto the nonabelian free group F_2. Thus even the regular complements cannot be homotopy equivalent. The tangent obstructions in theorems 30.2 and 30.3 finish the claimed scope. □

30.5 The 24-dimensional octonionic flag

We now construct the exceptional Jordan object instead of selecting it by dimension. Let

$$\mathfrak{h}_3(\mathbb{O}) = \left\{ \begin{pmatrix} x_1 & p & q \\ \overline{p} & x_2 & r \\ \overline{q} & \overline{r} & x_3 \end{pmatrix} : x_1, x_2, x_3 \in \mathbb{R},\ p, q, r \in \mathbb{O} \right\}, \qquad a \circ b = \frac{1}{2}(ab + ba).$$

Define

$$\mathbb{OP}^2 = \{a \in \mathfrak{h}_3(\mathbb{O}) : a^2 = a,\ \operatorname{tr} a = 1\}$$

and, without replacing the incidence equation by a mnemonic,

$$\mathrm{Fl}(\mathbb{O}) = \{(a, b) \in \mathbb{OP}^2 \times \mathbb{OP}^2 : \operatorname{Re}\operatorname{tr}(ab) = 0\}. \tag{1192}$$

Let

$$F_4 = \operatorname{Aut}_{\mathbb{R}}(\mathfrak{h}_3(\mathbb{O}), \circ), \qquad d_1 = \operatorname{diag}(1, 0, 0), \qquad d_2 = \operatorname{diag}(0, 1, 0), \qquad x_{\mathbb{O},0} = (d_1, d_2).$$

Here F_4 denotes the compact real form. These definitions and all claims in the next theorem are recorded in the primary calculation of Mare and Willems [10, Definition 2.2, p. 488; Proposition 2.4, p. 489, and Proposition 2.7, p. 491, and the root-space discussion on pp. 491–492].

Theorem 30.24 (The exact triality tangent module)**.** *The diagonal F_4-action on $\mathrm{Fl}(\mathbb{O})$ is transitive, the stabilizer of $x_{\mathbb{O},0}$ is $\mathrm{Spin}(8)$, and hence*

$$\mathrm{Fl}(\mathbb{O}) \cong F_4/\operatorname{Spin}(8).$$

Fix pairwise distinct real numbers $\lambda_1, \lambda_2, \lambda_3$ with $\lambda_1 + \lambda_2 + \lambda_3 = 0$, and set $d_3 = I_3 - d_1 - d_2$ and $d_\lambda = \operatorname{diag}(\lambda_1, \lambda_2, \lambda_3)$. The exact comparison with the diagonal-orbit model is the equivariant diffeomorphism

$$\begin{aligned} \Xi_\lambda : \mathrm{Fl}(\mathbb{O}) &\longrightarrow F_4 d_\lambda \subset \mathfrak{h}_3^0(\mathbb{O}), \\ (a, b) &\longmapsto \lambda_1 a + \lambda_2 b + \lambda_3(I_3 - a - b). \end{aligned} \tag{1193}$$

For $z \in F_4 d_\lambda$, its inverse is $(p_1(z), p_2(z))$, where

$$p_i(z) = \frac{(z - \lambda_j I_3) \circ (z - \lambda_k I_3)}{(\lambda_i - \lambda_j)(\lambda_i - \lambda_k)}, \qquad \{i, j, k\} = \{1, 2, 3\}. \tag{1194}$$

Define the three coordinate subspaces of $\mathfrak{h}_3^0(\mathbb{O})$ by

$$\begin{aligned} \mathfrak{h}_{\gamma_1} &= \left\{ \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & r \\ 0 & \overline{r} & 0 \end{pmatrix} : r \in \mathbb{O} \right\}, \\ \mathfrak{h}_{\gamma_2} &= \left\{ \begin{pmatrix} 0 & 0 & q \\ 0 & 0 & 0 \\ \overline{q} & 0 & 0 \end{pmatrix} : q \in \mathbb{O} \right\}, \\ \mathfrak{h}_{\gamma_3} &= \left\{ \begin{pmatrix} 0 & p & 0 \\ \overline{p} & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} : p \in \mathbb{O} \right\}. \end{aligned} \tag{1195}$$

As real $\mathrm{Spin}(8)$-modules,

$$T_{x_{\mathbb{O},0}} \mathrm{Fl}(\mathbb{O}) \xrightarrow[\cong]{d\Xi_\lambda} T_{d_\lambda}(F_4 d_\lambda) = \mathfrak{h}_{\gamma_1} \oplus \mathfrak{h}_{\gamma_2} \oplus \mathfrak{h}_{\gamma_3} \cong V_8 \oplus S_8^+ \oplus S_8^-. \tag{1196}$$

In particular,

$$\dim_{\mathbb{R}} \mathrm{Fl}(\mathbb{O}) = 8 + 8 + 8 = 24.$$

Proof. Mare and Willems prove that the stabilizer of (d_1, d_2) consists exactly of the automorphisms fixing the diagonal Jordan subalgebra pointwise, which is Spin(8), and that the action is transitive. Their restricted-root decomposition of $\mathfrak{h}_3^0(\mathbb{O})$ has the three spaces in (1195); the tangent space to the orbit is their direct sum. To verify the model comparison, write $(a, b) = (hd_1, hd_2)$ with $h \in F_4$. Since $hI_3 = I_3$, $I_3 - a - b = hd_3$, so $\Xi_\lambda(a, b) = hd_\lambda$. The three orthogonal diagonal idempotents satisfy the polynomial identities $p_i(d_\lambda) = d_i$. Jordan automorphisms preserve these identities, whence $p_i(hd_\lambda) = hd_i$. This proves both inverse compositions in (1193)– (1194). All formulas are polynomial on the ambient real vector spaces with nonzero constant denominators, so the maps are smooth. Their differentials are inverse, which gives the displayed tangent identification at the correctly typed basepoints. Under the stabilizer, the three summands are respectively the vector representation and the two real half-spin representations [10, pp. 491–492, especially the discussion around Equation (2.10)]. Each is the displayed copy of the eight-dimensional real vector space $\mathbb{O}$, which proves the dimension calculation without suppressing any summand. □

For $k = 1, 2, 3$, let

$$\mathcal{E}_k = F_4 \times_{\mathrm{Spin}(8)} \mathfrak{h}_{\gamma_k}.$$

These are rank-eight real subbundles, and

$$T\,\mathrm{Fl}(\mathbb{O}) = \mathcal{E}_1 \oplus \mathcal{E}_2 \oplus \mathcal{E}_3. \tag{1197}$$

They are associated bundles; the notation does not assert that any $\mathcal{E}_k$ is trivial.

Theorem 30.25 (Integral ring, cells, and Euler class)**.** *There are orientations and a basis $\beta_1, \beta_2 \in H^8(\mathrm{Fl}(\mathbb{O}); \mathbb{Z})$ for which*

$$\begin{aligned} e_1 &:= e(\mathcal{E}_1) = 2\beta_1 - \beta_2, \\ e_2 &:= e(\mathcal{E}_2) = -\beta_1 + 2\beta_2, \\ e_3 &:= e(\mathcal{E}_3) = e_1 + e_2 = \beta_1 + \beta_2, \end{aligned} \tag{1198}$$

and

$$H^*(\mathrm{Fl}(\mathbb{O}); \mathbb{Z}) \cong \frac{\mathbb{Z}[\beta_1, \beta_2]}{(\beta_1^2 - \beta_1\beta_2 + \beta_2^2,\ \beta_1\beta_2(\beta_1 - \beta_2))}, \qquad |\beta_1| = |\beta_2| = 8. \tag{1199}$$

Writing

$$u_{\mathbb{O}} = \beta_1^2\beta_2 = \beta_1\beta_2^2 \in H^{24}(\mathrm{Fl}(\mathbb{O}); \mathbb{Z})$$

with the induced orientation, $u_{\mathbb{O}}$ is a primitive top generator and

$$e\big(T\,\mathrm{Fl}(\mathbb{O})\big) = e_1e_2e_3 = 6u_{\mathbb{O}}, \qquad \langle e(T\,\mathrm{Fl}(\mathbb{O})), [\mathrm{Fl}(\mathbb{O})]\rangle = 6. \tag{1200}$$

The integral cohomology is free and has Poincaré polynomial

$$1 + 2t^8 + 2t^{16} + t^{24}. \tag{1201}$$

More precisely, on $\mathfrak{d}^0 = \{\mathrm{diag}(x_1, x_2, x_3) : x_1 + x_2 + x_3 = 0\}$, put

$$\gamma_1(x) = x_3 - x_2, \qquad \gamma_2(x) = x_1 - x_3, \qquad \gamma_3(x) = x_1 - x_2 = \gamma_1(x) + \gamma_2(x).$$

In the symmetric-pair normalization the actual restricted roots are $\{\pm\gamma_1/2, \pm\gamma_2/2, \pm\gamma_3/2\}$, of type A_2. The doubled labels γ_i name the same root spaces and reflection hyperplanes; this factor of two is the one stated in the source's root-normalization footnote [10, footnote 3, p. 491]. There is a cell decomposition

$$\mathrm{Fl}(\mathbb{O}) = \bigsqcup_{\pi \in S_3} C_\pi, \qquad C_\pi \cong \mathbb{R}^{8\ell(\pi)}. \tag{1202}$$

For the stabilizer action,

$$\mathrm{Fl}(\mathbb{O})^{\mathrm{Spin}(8)} = \{\pi x_{\mathbb{O},0} : \pi \in S_3\}. \tag{1203}$$

Thus both the cell calculation and the Euler-class calculation give $\chi(\mathrm{Fl}(\mathbb{O})) = 6$.

Proof. Mare and Willems' integral theorem gives a free ring generated by β_1, β_2, with relations

$$S_i(\beta_1, \beta_2 - \beta_1, -\beta_2) = 0, \qquad i = 2, 3,$$

and gives (1198) [10, Theorem 1.1, p. 484, and Equation (3.2), p. 496]. Expanding without changing the generators,

$$S_2 = -\beta_1^2 + \beta_1\beta_2 - \beta_2^2, \qquad S_3 = \beta_1\beta_2(\beta_1 - \beta_2),$$

which is exactly (1199). The second relation gives $\beta_1^2\beta_2 = \beta_1\beta_2^2$. Multiplying the first relation by β_1 and by β_2 then gives $\beta_1^3 = \beta_2^3 = 0$. Consequently

$$(2\beta_1 - \beta_2)(-\beta_1 + 2\beta_2)(\beta_1 + \beta_2) = 3\beta_1^2\beta_2 + 3\beta_1\beta_2^2 = 6u_{\mathbb{O}}.$$

The primary calculation states that one sixth of this Euler product is a basis of the top integral cohomology [10, Equation (3.3) and the following paragraph, p. 497]; hence $u_{\mathbb{O}}$ is primitive and (1200) follows.

The same source constructs the S_3-indexed cells, with one eight-dimensional real root space for each inversion of π, and identifies the fixed-point set [10, Equations (2.11)–(2.12) and Proposition 2.10, pp. 493–494]. Their real dimensions are

$$0, \quad 8, 8, \quad 16, 16, \quad 24.$$

All cells are even-dimensional. The cellular chain groups therefore have zero differentials, yielding (1201) and a second calculation of the Euler characteristic as the number of cells, namely six. Equivalently, the cellwise complex coordinates have dimensions $0, 4, 4, 8, 8, 12$, since $C_\pi \cong \mathbb{C}^{4\ell(\pi)}$ as real cells. This cell notation supplies no global complex tangent structure. □

Proposition 30.26 (The exact relation between the two Weyl data)**.** *The ordinary Coxeter number of the ambient root system F_4 is*

$$h(F_4) = 12.$$

The restricted root system that governs (1202)*, however, is A_2; its Weyl group is S_3 and its Coxeter number is* 3*. For a common maximal torus of* Spin(8) *and F_4, there is a split exact sequence*

$$1 \longrightarrow W(D_4) \longrightarrow W(F_4) \longrightarrow S_3 \longrightarrow 1, \qquad [W(F_4) : W(D_4)] = 1152/192 = 6.$$

The quotient permutes the three triality representations. It gives an exact Weyl-group relation behind the number six; the scalar $h(F_4) = 12$ alone does not determine the cells or Euler characteristic.

Proof. For a finite root system the invariant degrees satisfy $d_i = e_i + 1$, and the largest degree is the Coxeter number. The authoritative F_4 table gives

$$(d_1, d_2, d_3, d_4) = (2, 6, 8, 12), \qquad (e_1, e_2, e_3, e_4) = (1, 5, 7, 11),$$

so $h(F_4) = 12$ [9, Section 10.1, p. 213, and the table on p. 216]. By contrast, the roots $\gamma_1, \gamma_2, \gamma_1 + \gamma_2$ in the proof of theorem 30.25 form A_2. Its two simple reflections act as the adjacent transpositions of three coordinates, and their product is a three-cycle; hence its Coxeter element has order 3. Mare and Willems explicitly identify this restricted Weyl group with S_3 [10, pp. 492–493].

For completeness the extension can be checked in root coordinates. Take an orthonormal basis $\varepsilon_1, \dots, \varepsilon_4$. The long roots of F_4 are $\{\pm\varepsilon_i \pm \varepsilon_j : i < j\}$, the D_4 roots. Partition the short roots into

$$\mathcal{V} = \{\pm\varepsilon_i\}, \qquad \mathcal{S}_\pm = \left\{ \tfrac{1}{2}\sum_{i=1}^{4} \delta_i\varepsilon_i : \delta_i \in \{1, -1\},\ \prod_i \delta_i = \pm 1 \right\}.$$

These are exactly the three connected components of the graph joining orthogonal short roots: distinct components have inner products $\pm 1/2$, and within each component every non-antipodal distinct pair is

orthogonal. Thus $W(F_4)$ permutes the three sets. An element in the kernel preserves $\mathcal{V}$, so is a signed permutation matrix; preserving both half-spin sets forces an even number of sign changes. These are exactly $W(D_4)$, of order $2^3 \cdot 4! = 192$. The reflections in ε_1 and $h = (\varepsilon_1 + \varepsilon_2 + \varepsilon_3 + \varepsilon_4)/2$ induce two distinct transpositions: the first exchanges $\mathcal{S}_+, \mathcal{S}_-$, and the second exchanges $\mathcal{V}, \mathcal{S}_-$. Their product has order three, since $\langle \varepsilon_1, h \rangle = 1/2$. They generate a copy of S_3 mapping isomorphically onto the permutation quotient. This proves surjectivity and splitting, and gives $|W(F_4)| = 6 \cdot 192 = 1152$. The three displayed short-root sets are the weights of the three triality modules, so this is the asserted permutation of those modules. □

30.5.1 The exact six-chamber comparison

Let $\mathscr{K}_3^{\max}/\mathbb{Z}^3$ be the translation classes of the maximal Kuhn tetrahedra $\sigma_{m,\pi}$ in (1142). Translation by $-m$ gives the exact identification

$$\mathscr{K}_3^{\max}/\mathbb{Z}^3 = \{[\sigma_{0,\pi}] : \pi \in S_3\}.$$

The zero-dimensional torus strata of W_3 are indexed by this set; call the stratum corresponding to $[\sigma_{0,\pi}]$ by z_π.

Theorem 30.27 (The exact marked map of finite indexing sets)**.** *After fixing the coordinate order (e_1, e_2, e_3) and the base flag $x_{\mathbb{O},0}$, the formula*

$$\Theta : W_3^{(0)} \longrightarrow \mathrm{Fl}(\mathbb{O})^{\mathrm{Spin}(8)}, \qquad \Theta(z_\pi) = \pi x_{\mathbb{O},0} \tag{1204}$$

is an S_3-equivariant bijection. Here $W_3^{(0)} = \{z_\pi : \pi \in S_3\}$ is a finite set of torus zero-strata, and S_3 acts on the source by permuting coordinate directions and on the target as the restricted Weyl group. It identifies the six summands contributing to the two Euler characteristics, but it does not extend to an asserted identification of the spaces.

Proof. Every translation class of a maximal Kuhn tetrahedron has one and only one representative $\sigma_{0,\pi}$, so the source of Θ has exactly the six displayed elements. Equation (1203) gives exactly the six displayed target elements. Thus Θ is bijective. For $\tau \in S_3$, permutation of coordinate directions gives

$$\tau \cdot [\sigma_{0,\pi}] = [\sigma_{0,\tau\pi}],$$

while the restricted Weyl action gives $\tau \cdot (\pi x_{\mathbb{O},0}) = (\tau\pi) x_{\mathbb{O},0}$. Hence Θ is equivariant.

For W_3, every positive-dimensional algebraic-torus stratum has Euler characteristic zero, so its six zero-strata produce $\chi(W_3) = 6$. For the flag, the six cells in (1202) are all even-dimensional, so each contributes $+1$. The bijection matches the index sets; the two stratifications and their positive-dimensional pieces are not identified by Θ. More formally, let $I_{\mathrm{cell}} = \{\mathrm{label}(C_\pi) : \pi \in S_3\}$ be a set of six labels and put $\tau\,\mathrm{label}(C_\pi) = \mathrm{label}(C_{\tau\pi})$. Then $z_\pi \mapsto \pi x_{\mathbb{O},0} \mapsto \mathrm{label}(C_\pi)$ is a chain of equivariant bijections of finite sets. The last action is defined on labels only: cells of different dimensions occur, so no action carrying each cell homeomorphically to the cell with its translated label is asserted. □

Proposition 30.28 (A genuine embedded lower flag, and the degree boundary)**.** *Choose $\iota \in \mathbb{O}$ with $\iota^2 = -1$ and put $\mathbb{C}_\iota = \mathrm{span}_{\mathbb{R}}\{1, \iota\}$. Entrywise inclusion in $\mathfrak{h}_3(\mathbb{O})$ gives a topological embedding*

$$\jmath_\iota : \mathrm{Fl}_3(\mathbb{C}_\iota) \hookrightarrow \mathrm{Fl}(\mathbb{O}) \tag{1205}$$

which sends the six coordinate flags to the six points in (1203)*.*

There is also an ungraded-ring isomorphism

$$\rho : H^*(\mathrm{Fl}_3(\mathbb{C}); \mathbb{Z}) \xrightarrow{\cong} H^*(\mathrm{Fl}(\mathbb{O}); \mathbb{Z}), \qquad x_i \longmapsto \beta_i, \quad i = 1, 2, \tag{1206}$$

where the source has the presentation in (1199) *with $|x_i| = 2$, while $|\beta_i| = 8$. No continuous map in either direction can induce this degree regrading.*

Proof. The subalgebra $\mathbb{C}_\iota$ is associative and isomorphic to $\mathbb{C}$. Rank-one Hermitian projectors in $\mathfrak{h}_3(\mathbb{C}_\iota)$ remain idempotent trace-one elements after entrywise inclusion, and orthogonality is the unchanged equation $\operatorname{Re}\operatorname{tr}(ab) = 0$. The resulting map is injective. Its domain is compact and its target Hausdorff, so it is a topological embedding. The diagonal projectors are unchanged, which proves the fixed-point assertion.

The standard complete-flag presentation is

$$H^*(\mathrm{Fl}_3(\mathbb{C}); \mathbb{Z}) \cong \frac{\mathbb{Z}[x_1, x_2]}{(x_1^2 - x_1x_2 + x_2^2,\ x_1x_2(x_1 - x_2))}, \qquad |x_i| = 2;$$

Mare and Willems derive the octonionic presentation precisely by comparison with this A_2 flag ring [10, Theorem 1.1, p. 484, and the proof on pp. 496–497]. Thus ρ is an isomorphism after forgetting degrees. If $f : \mathrm{Fl}(\mathbb{O}) \to \mathrm{Fl}_3(\mathbb{C})$ were to induce ρ, then $f^*(x_i) \in H^2(\mathrm{Fl}(\mathbb{O}); \mathbb{Z}) = 0$. If a map in the other direction were to induce ρ^{-1}, then the image of each β_i would lie in $H^8(\mathrm{Fl}_3(\mathbb{C}); \mathbb{Z}) = 0$. Either conclusion contradicts the nonzero generators in (1206). □

30.5.2 The octonionic determinant-boundary sphere

For $\lambda \in \mathbb{R}$ and $q_1, q_2, q_3 \in \mathbb{O}$, retain the ordered coordinates

$$Q_{\mathbb{O}}(\lambda; q_1, q_2, q_3) = \begin{pmatrix} \lambda & q_3 & \overline{q}_2 \\ \overline{q}_3 & \lambda & q_1 \\ q_2 & \overline{q}_1 & \lambda \end{pmatrix} \in \mathfrak{h}_3(\mathbb{O}). \tag{1207}$$

The equal-diagonal slice and its positive cone are

$$\mathcal{T}(\mathbb{O}) = \{Q_{\mathbb{O}}(\lambda; q_1, q_2, q_3)\} \cong \mathbb{R} \oplus \mathbb{O}^3, \qquad \mathcal{C}(\mathbb{O}) = \mathcal{T}(\mathbb{O}) \cap \mathfrak{h}_3(\mathbb{O})_{\succeq 0}. \tag{1208}$$

Thus $\dim_{\mathbb{R}} \mathcal{T}(\mathbb{O}) = 1 + 3 \cdot 8 = 25$. The Jordan determinant on this slice is the exact cubic

$$\det_{\mathbb{O}} Q_{\mathbb{O}} = \lambda^3 - \lambda(|q_1|^2 + |q_2|^2 + |q_3|^2) + 2\operatorname{Re}\big((q_1q_2)q_3\big). \tag{1209}$$

The real part is unchanged if $q_1(q_2q_3)$ is used, because the octonionic associator has zero real part.

Theorem 30.29 (The exact octonionic S^{24} determinant carrier)**.** *The relative boundary of $\mathcal{C}(\mathbb{O})$ inside $\mathcal{T}(\mathbb{O})$ is*

$$\partial\mathcal{C}(\mathbb{O}) = \{Q \in \mathcal{C}(\mathbb{O}) : \det_{\mathbb{O}} Q = 0\}.$$

There are homeomorphisms of pairs

$$(\mathcal{C}(\mathbb{O})^+, (\partial\mathcal{C}(\mathbb{O}))^+) \cong (B^{25}, S^{24}). \tag{1210}$$

In particular

$$S^{24}_{\det,\mathbb{O}} := (\partial\mathcal{C}(\mathbb{O}))^+ \cong S^{24}$$

is an exact determinant-boundary sphere. It is not the 24*-dimensional flag* $F_4/\operatorname{Spin}(8)$*.*

Proof. The nonnegative cone of the Euclidean Jordan algebra $\mathfrak{h}_3(\mathbb{O})$ is closed and convex. Its intersection

$$K_{\mathbb{O}} = \{(q_1, q_2, q_3) \in \mathbb{O}^3 : Q_{\mathbb{O}}(1; q_1, q_2, q_3) \succeq 0\}$$

with the affine hyperplane $\lambda = 1$ is therefore closed and convex. Every positive semidefinite 2×2 principal corner gives $|q_i| \le 1$, so $K_{\mathbb{O}}$ is bounded and hence compact. The origin of $\mathbb{O}^3$ corresponds to I_3, which is positive definite, so it is an interior point of $K_{\mathbb{O}}$.

For $v \in S(\mathbb{O}^3)$, let

$$R_{\mathbb{O}}(v) = \max\{r \geq 0 : rv \in K_{\mathbb{O}}\}.$$

Convexity, compactness, and the interior-point property make $R_{\mathbb{O}} : S(\mathbb{O}^3) \to \mathbb{R}_{>0}$ continuous. Indeed, choose $0 < \epsilon < M$ with $\epsilon B^{24} \subset K_{\mathbb{O}} \subset MB^{24}$. The Minkowski functional

$$p_K(x) = \inf\{t > 0 : x \in tK_{\mathbb{O}}\}$$

is positive for $x \neq 0$, positively homogeneous, and subadditive: if $x \in aK_{\mathbb{O}}$ and $y \in bK_{\mathbb{O}}$, convexity gives $x + y \in (a + b)K_{\mathbb{O}}$. Taking infima proves subadditivity. As $p_K(w) \leq \|w\|/\epsilon$, subadditivity in both directions gives $|p_K(x) - p_K(y)| \leq \|x - y\|/\epsilon$. Hence $R_{\mathbb{O}}(v) = 1/p_K(v)$ is continuous. Radial rescaling

$$0 \longmapsto 0, \qquad rv \longmapsto \frac{r}{R_{\mathbb{O}}(v)}v$$

is consequently a homeomorphism $K_{\mathbb{O}} \cong B^{24}$ carrying $\partial K_{\mathbb{O}}$ onto S^{23}. Denote it by F_K; its inverse sends sv, $0 < s \leq 1$, to $sR_{\mathbb{O}}(v)v$, and sends zero to zero. Both maps are continuous at zero because the radial function is bounded above and below by positive constants.

A nonzero element of $\mathcal{C}(\mathbb{O})$ has $\lambda > 0$: if $\lambda = 0$, the same principal-corner inequalities force $q_1 = q_2 = q_3 = 0$. Positive scaling therefore reconstructs the whole cone as

$$\mathcal{C}(\mathbb{O}) = \{0\} \cup \{\lambda Q_{\mathbb{O}}(1; q_1, q_2, q_3) : \lambda > 0,\ (q_1, q_2, q_3) \in K_{\mathbb{O}}\}.$$

Here are explicit maps through the Lorentz cone, with $q = (q_1, q_2, q_3)$:

$$\begin{aligned}
\mathcal{C}(\mathbb{O}) &\longrightarrow L_{25} := \{(s, x) \in \mathbb{R} \oplus \mathbb{O}^3 : s \geq \|x\|\},\\
Q_{\mathbb{O}}(\lambda; q) &\longmapsto (\lambda, \lambda F_K(q/\lambda)) \quad (\lambda > 0),\\
0 &\longmapsto (0, 0),\\
L_{25} &\longrightarrow [0, \infty) \times \mathbb{O}^3,\\
(s, x) &\longmapsto (s - \|x\|, x).
\end{aligned}$$

The inverses are $(s, x) \mapsto Q_{\mathbb{O}}(s; sF_K^{-1}(x/s))$ for $s > 0$, with zero sent to zero, and $(t, x) \mapsto (t + \|x\|, x)$, respectively. The boundedness of F_K and F_K^{-1} proves continuity at the cone vertices. The boundary maps to $s = \|x\|$, then to $t = 0$, proving an explicit pair homeomorphism with a closed half-space and its boundary.

To identify this boundary with determinant zero in the relative topology, use the spectral decomposition in the Euclidean Jordan algebra. A positive definite element has three positive eigenvalues and is an interior point even in the full algebra, hence in the slice. Conversely, if a positive semidefinite Q has a zero eigenvalue, then $Q - \delta I_3 \in \mathcal{T}(\mathbb{O})$, for every $\delta > 0$, has a negative eigenvalue. Such points lie outside $\mathcal{C}(\mathbb{O})$ and converge to Q; thus Q is not a relative interior point. Since the cone is closed, these are exactly its relative boundary points. The determinant is the product of the three nonnegative eigenvalues, so its zero set in the cone is exactly that boundary.

Fixing an orthonormal identification $\mathbb{O}^3 \cong \mathbb{R}^{24}$, the map

$$(t, x) \longmapsto \left(\frac{2x}{D}, \frac{t^2 + \|x\|^2 - 1}{D}\right) \in B^{25}, \qquad D = 1 + t^2 + \|x\|^2,$$

extends at infinity to $(0, 1) \in \partial B^{25}$. It is inverse stereographic projection to the closed hemisphere in S^{25} with first coordinate nonnegative, followed by deletion of that first coordinate. The hemisphere projection is bijective, with inverse adjoining $\sqrt{1 - \|y\|^2}$; hence the displayed map is a homeomorphism after one-point compactification. At $t = 0$ it is

$$x \longmapsto \left(\frac{2x}{1 + \|x\|^2}, \frac{\|x\|^2 - 1}{1 + \|x\|^2}\right) \in S^{24},$$

with infinity sent to $(0, 1)$. This proves (1210), including its specified boundary. □

30.5.3 The ordered Albert triality tensor and its global descent

Original coordinates and octonion identities Let $\mathbb{O}$ be the real octonion division algebra, with its original conjugation, norm $N(a) = a\bar{a} = \bar{a}a$, and real inner product $h(a,b) = \mathrm{Re}(a\bar{b})$. Put $J = \mathfrak{h}_3(\mathbb{O})$, with its Albert Jordan product $X \circ Y = (XY + YX)/2$. Matrix multiplication here involves only products of two octonions in each entry; no product of three octonionic matrices is left unparenthesized. Set $e_i = \mathrm{diag}(\delta_{1i}, \delta_{2i}, \delta_{3i})$, $I = e_1 + e_2 + e_3$, and

$$E_1(a) = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & a \\ 0 & \bar{a} & 0 \end{pmatrix}, \quad E_2(b) = \begin{pmatrix} 0 & 0 & \bar{b} \\ 0 & 0 & 0 \\ b & 0 & 0 \end{pmatrix}, \quad E_3(c) = \begin{pmatrix} 0 & c & 0 \\ \bar{c} & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

The specified equal-diagonal object remains exactly

$$Q_{\mathbb{O}}(\lambda; q_1, q_2, q_3) = \lambda I + E_1(q_1) + E_2(q_2) + E_3(q_3) = \begin{pmatrix} \lambda & q_3 & \bar{q}_2 \\ \bar{q}_3 & \lambda & q_1 \\ q_2 & \bar{q}_1 & \lambda \end{pmatrix}. \tag{1211}$$

Let V_i be three labelled copies of $\mathbb{O}$ and $J_i = E_i(V_i)$. Labelling does not identify their group actions. The original trace metric on J is

$$\begin{aligned} G(X,Y) &= \mathrm{tr}(X \circ Y), \\ G\left(\sum_i \xi_i e_i + \sum_i E_i(a_i), \sum_i \eta_i e_i + \sum_i E_i(b_i)\right) &= \sum_i \xi_i \eta_i + 2\sum_i h(a_i, b_i). \end{aligned} \tag{1212}$$

In particular $E_i^* G = 2h$, not h.

We use alternativity, $N(ab) = N(a)N(b)$, and conjugation reversing products. The parenthesis changes used below follow as follows. Because $\bar{a} = 2\,\mathrm{Re}(a) - a$, alternativity gives

$$\bar{a}(ab) = N(a)b, \quad a(\bar{a}b) = N(a)b, \quad (ba)\bar{a} = N(a)b, \quad (b\bar{a})a = N(a)b. \tag{1213}$$

Polarizing norm composition gives $h(ab, ac) = N(a)h(b,c)$ and $h(ba, ca) = N(a)h(b,c)$. For $a \neq 0$, these equations and (1213) imply $L_a^* = L_{\bar{a}}$ and $R_a^* = R_{\bar{a}}$; for $a = 0$ the identities are immediate. Consequently

$$\begin{aligned} h(ab, c) &= h(b, \bar{a}c) = h(a, c\bar{b}), \\ \mathrm{Re}((ab)c) &= \mathrm{Re}(a(bc)) = \mathrm{Re}((bc)a). \end{aligned} \tag{1214}$$

For the first real-part equality, both expressions equal $h(c, \bar{b}\bar{a})$ by the adjoint identities. For the second, use $\mathrm{Re}(xy) = h(x, \bar{y}) = h(y, \bar{x}) = \mathrm{Re}(yx)$. These are identities of real parts; they do not equate the octonions $(ab)c$ and $a(bc)$.

The complete Peirce product table A cyclic triple means exactly $(i,j,k) = (1,2,3), (2,3,1), (3,1,2)$.

Proposition 30.30. *The decomposition $J = \bigoplus_i \mathbb{R}e_i \oplus \bigoplus_i J_i$ has the complete product table*

$$\begin{aligned} e_i \circ e_j &= \delta_{ij} e_i, \\ e_i \circ E_i(a) &= 0, & e_j \circ E_i(a) &= e_k \circ E_i(a) = \tfrac{1}{2} E_i(a), \\ E_i(a) \circ E_i(b) &= h(a,b)(e_j + e_k), \\ E_i(a) \circ E_j(b) &= \tfrac{1}{2} E_k(\overline{ab}). \end{aligned} \tag{1215}$$

Reversing the Jordan arguments uses commutativity of $\circ$, without reversing the octonion order in the cyclic formula.

Proof. The first three formulas follow entry by entry. For example the two nonzero diagonal entries of $E_1(a)\circ E_1(b)$ are $(a\bar{b}+b\bar{a})/2$ and $(\bar{a}b+\bar{b}a)/2$, both $h(a,b)$. The mixed products have precisely the following nonzero entries:

product	first entry	second entry
$E_1(a)\circ E_2(b)$	$(1,2)=\bar{b}\bar{a}/2$	$(2,1)=ab/2$
$E_2(b)\circ E_3(c)$	$(2,3)=\bar{c}\bar{b}/2$	$(3,2)=bc/2$
$E_3(c)\circ E_1(a)$	$(3,1)=\bar{a}\bar{c}/2$	$(1,3)=ca/2$

These are respectively $E_3(\overline{ab})/2$, $E_1(\overline{bc})/2$, and $E_2(\overline{ca})/2$, by the original definitions of E_1,E_2,E_3. □

Three typed products, adjoints, and inverses Define the trilinear form and all three bilinear maps by

$$\begin{aligned} \tau &: V_1\times V_2\times V_3\longrightarrow\mathbb{R}, & \tau(a,b,c)&=\mathrm{Re}((ab)c), \\ \mu_{12} &: V_1\times V_2\longrightarrow V_3, & \mu_{12}(a,b)&=\overline{ab}=\bar{b}\bar{a}, \\ \mu_{23} &: V_2\times V_3\longrightarrow V_1, & \mu_{23}(b,c)&=\overline{bc}=\bar{c}\bar{b}, \\ \mu_{31} &: V_3\times V_1\longrightarrow V_2, & \mu_{31}(c,a)&=\overline{ca}=\bar{a}\bar{c}. \end{aligned} \tag{1216}$$

Every V_i carries h, with its label retained. Equation (1214) and conjugation give

$$\tau(a,b,c)=h(\mu_{12}(a,b),c)=h(\mu_{23}(b,c),a)=h(\mu_{31}(c,a),b). \tag{1217}$$

For example $h(\overline{ab},c)=\mathrm{Re}(\overline{ab}\bar{c})=\mathrm{Re}(\overline{c(ab)})=\mathrm{Re}((ab)c)$. Thus these are exactly the three Riesz-curried maps for h. In the original matrix metric,

$$\tau(a,b,c)=G(E_1(a)\circ E_2(b),E_3(c))=\mathrm{tr}((E_1(a)\circ E_2(b))\circ E_3(c)). \tag{1218}$$

Proposition 30.31. *For cyclic (i,j,k), $a\in V_i$, and $b\in V_j$,*

$$N(\mu_{ij}(a,b))=N(a)N(b), \tag{1219}$$

$$\mu_{ki}(\mu_{ij}(a,b),a)=N(a)b, \tag{1220}$$

$$\mu_{jk}(b,\mu_{ij}(a,b))=N(b)a. \tag{1221}$$

For fixed $a\neq 0$, the real-linear isomorphism $M_a:V_j\to V_k$, $M_a(b)=\mu_{ij}(a,b)$, has

$$M_a^*(c)=\mu_{ki}(c,a),\quad M_a^*M_a=N(a)\,\mathrm{id}_{V_j},\quad M_aM_a^*=N(a)\,\mathrm{id}_{V_k},\quad M_a^{-1}(c)=\frac{\mu_{ki}(c,a)}{N(a)}.$$

For fixed $b\neq 0$, the inverse of $a\mapsto\mu_{ij}(a,b)$ is $c\mapsto\mu_{jk}(b,c)/N(b)$, and its numerator is also the adjoint.

Proof. Norm composition proves the first formula. The inverse numerators are, with their parentheses preserved,

$$\overline{\overline{ab}\,a}=\bar{a}(ab)=N(a)b,\qquad \overline{b\,\overline{ab}}=(ab)\bar{b}=N(b)a.$$

Their final equalities are (1213). Relabelling by the three stated cyclic triples proves all six inverse identities. The adjoint statements are (1217). Both adjoint-composition identities follow from the two inverse identities, including with the other argument nonzero. Division by $N(a)$ or $N(b)$ gives the inverses. No unit-length hypothesis has been imposed. □

Retaining G gives the following different numerical identities. For $X=E_i(a)\in J_i, Y=E_j(b)\in J_j$ with $i\neq j$, put $B_{ij}(X,Y)=2X\circ Y=E_k(\mu_{ij}(a,b))$, using cyclic argument order. Then

$$G(B_{ij}(X,Y),B_{ij}(X,Y))=\tfrac{1}{2}G(X,X)G(Y,Y),\qquad G(X\circ Y,X\circ Y)=\tfrac{1}{8}G(X,X)G(Y,Y). \tag{1222}$$

Indeed the left sides are $2N(a)N(b)$ and $N(a)N(b)/2$, whereas $G(X,X)G(Y,Y)=4N(a)N(b)$. The coefficient-one composition identity uses the explicitly stated h, not the trace metric G.

The determinant and the factor six in polarization

Proposition 30.32. *For arbitrary real ξ_i and the original octonionic coordinates q_i, let $X = \sum_i \xi_i e_i + \sum_i E_i(q_i)$. Its Jordan determinant is*

$$d(X) = \xi_1\xi_2\xi_3 - \sum_{i=1}^{3} \xi_i N(q_i) + 2\,\mathrm{Re}((q_1q_2)q_3). \tag{1223}$$

In particular,

$$d(Q_{\mathbb{O}}(\lambda; q_1, q_2, q_3)) = \lambda^3 - \lambda\big(N(q_1) + N(q_2) + N(q_3)\big) + 2\,\mathrm{Re}((q_1q_2)q_3).$$

Proof. The Albert cubic norm is given by the Jordan trace formula

$$d(X) = \tfrac{1}{3}\,\mathrm{tr}(X \circ (X \circ X)) - \tfrac{1}{2}\,\mathrm{tr}(X)\,\mathrm{tr}(X \circ X) + \tfrac{1}{6}\,\mathrm{tr}(X)^3.$$

The product table computes the diagonal entries and three coordinates of X^2:

$$(X^2)_{ii} = \xi_i^2 + N(q_j) + N(q_k), \qquad (X^2)_i = (\xi_j + \xi_k)q_i + \overline{q_jq_k}, \quad (i,j,k) \text{ cyclic.}$$

With $s = \xi_1 + \xi_2 + \xi_3$, it follows that

$$\begin{aligned} \mathrm{tr}(X) &= s, \\ \mathrm{tr}(X^2) &= \sum_i \xi_i^2 + 2\sum_i N(q_i), \\ \mathrm{tr}(X \circ X^2) &= \sum_i \xi_i^3 + 3\sum_i (\xi_j + \xi_k) N(q_i) + 6\,\mathrm{Re}((q_1q_2)q_3). \end{aligned}$$

Each scalar product of q_i with $\overline{q_jq_k}$ equals $\tau(q_1, q_2, q_3)$ by (1217); the trace metric supplies the factor 2. Substitution into the cubic gives coefficient $(\xi_j + \xi_k) - s = -\xi_i$ for $N(q_i)$ and coefficient $6/3 = 2$ for τ. Direct expansion gives the scalar part

$$\tfrac{1}{3}\sum_i \xi_i^3 - \tfrac{1}{2}s\sum_i \xi_i^2 + \tfrac{1}{6}s^3 = \xi_1\xi_2\xi_3.$$

This proves (1223). Setting all three $\xi_i = \lambda$ retains exactly the original equal-diagonal slice. □

Define the symmetric polarization $\mathcal{D}$ by $\mathcal{D}(X, X, X) = d(X)$, explicitly

$$\begin{aligned} 6\mathcal{D}(X,Y,Z) = {} & d(X+Y+Z) - d(X+Y) - d(X+Z) - d(Y+Z) \\ & + d(X) + d(Y) + d(Z). \end{aligned} \tag{1224}$$

Expansion of homogeneous cubic monomials proves symmetry and trilinearity: inclusion–exclusion removes every term missing an argument, and degree three forces each remaining argument to occur once. On the diagonal the multiplier is $27 - 3 \cdot 8 + 3 = 6$. The same expansion proves

$$\frac{\partial^3}{\partial r\partial s\partial t} d(rX + sY + tZ)\bigg|_{r=s=t=0} = 6\mathcal{D}(X,Y,Z).$$

For the three distinct sectors one obtains

$$\begin{aligned} \partial_r\partial_s\partial_t d(rE_1(a) + sE_2(b) + tE_3(c))|_0 &= 2\tau(a,b,c), \\ \mathcal{D}(E_1(a), E_2(b), E_3(c)) &= \tfrac{1}{3}\tau(a,b,c). \end{aligned} \tag{1225}$$

Indeed that determinant is $2rst\tau(a,b,c)$. Relative to the original G, the three Riesz representatives for fixed $E_1(a), E_2(b)$ are

scalar form on $J_1 \times J_2 \times J_3$	representative in J_3
τ	$E_3(\overline{ab})/2 = E_1(a) \circ E_2(b)$
$\partial_r\partial_s\partial_t d$	$E_3(\overline{ab}) = 2E_1(a) \circ E_2(b)$
$\mathcal{D}$	$E_3(\overline{ab})/6 = (E_1(a) \circ E_2(b))/3$

Thus the determinant coefficient 2 and polarization factor 6 are separate, fully specified operations.

The exact stabilizer and its three actions Let $\kappa(a) = \bar{a}$ and define

$$H_\tau = \{(g_1, g_2, g_3) \in O(V_1, h) \times O(V_2, h) \times O(V_3, h) : \\ \tau(g_1 a, g_2 b, g_3 c) = \tau(a, b, c) \text{ for every } a, b, c\}. \tag{1226}$$

By (1217), orthogonality and nondegeneracy of h, this is equivalent to each of the equations

$$\begin{aligned} \mu_{12}(g_1 a, g_2 b) &= g_3 \mu_{12}(a, b), \\ \mu_{23}(g_2 b, g_3 c) &= g_1 \mu_{23}(b, c), \\ \mu_{31}(g_3 c, g_1 a) &= g_2 \mu_{31}(c, a). \end{aligned} \tag{1227}$$

With every conjugation retained, the first equation reads

$$(g_1 a)(g_2 b) = \kappa g_3 \kappa(ab). \tag{1228}$$

Consequently $(g_1, g_2, g_3) \mapsto (g_1, g_2, \kappa g_3 \kappa)$ is a group isomorphism onto the ordinary product-related triples $\mathcal{R} = \{(A, B, C) \in O(8)^3 : (Ax)(By) = C(xy) \text{ for all } x, y\}$.

Theorem 30.33. *Every element of H_τ has $g_i \in SO(8)$, and H_τ is isomorphic to the compact simply connected* Spin(8)*. The map*

$$\Phi : H_\tau \longrightarrow \{g \in \mathrm{Aut}(J, \circ) : g(e_i) = e_i \ (i = 1, 2, 3)\},$$
$$\Phi(g)\left(\sum_i \xi_i e_i + \sum_i E_i(a_i)\right) = \sum_i \xi_i e_i + \sum_i E_i(g_i a_i)$$

is an exact group isomorphism. Thus (1227) *is* Spin(8) *equivariance for the actions on the original Albert coordinates.*

Proof. For unit $u, v \in \mathbb{O}$, reflection in $u^\perp$ is

$$D_u(x) = x - 2h(x, u)u = -u\bar{x}u.$$

To verify the second expression, multiply $x\bar{u} + u\bar{x} = 2h(x, u)$ on the right by u and use $(x\bar{u})u = x$. Define

$$A_{u,v}(x) = v(\bar{u}x), \quad B_{u,v}(y) = (y\bar{u})v, \quad C_{u,v}(z) = v(\bar{u}z\bar{u})v = D_v D_u(z).$$

Repeated outside letters are parenthesized by flexibility. The Moufang identity $(vx)(yv) = v(xy)v$, first for v and then for $\bar{u}$, proves

$$(v(\bar{u}x))((y\bar{u})v) = v(\bar{u}(xy)\bar{u})v.$$

This is a product-related triple. All its coordinates lie in $SO(8)$: multiplication by a unit octonion is orthogonal, and its determinant is continuously $+1$ on the connected unit sphere because it is $+1$ at 1. Every element of $SO(8)$ is a product of an even number of reflections. To obtain such a decomposition, reflect the image of the first vector of an orthonormal basis onto that vector, then repeat inside its orthogonal complement. After finitely many steps the map is the identity; the determinant records the parity of the reflections. Projection of $\mathcal{R} \cap SO(8)^3$ to C is therefore surjective.

Its kernel contains exactly two elements. If $(Ax)(By) = xy$, put $p = A1$, $q = B1$. Then $pq = 1$, $By = \bar{p}y$, and $Ax = xp$. The equation becomes $(xp)(\bar{p}y) = xy$. Replacing y by py gives $(xp)y = x(py)$ for every x, y. Hence p lies in the middle nucleus of $\mathbb{O}$, which is $\mathbb{R}$. The latter fact can be checked without an implicit associativity assumption. In the Cayley–Dickson presentation $\mathbb{O} = \mathbb{H} \oplus \mathbb{H}\ell$ with

$$(a + b\ell)(c + d\ell) = (ac - \bar{d}b) + (da + b\bar{c})\ell,$$

write $p = a + b\ell$ and use quaternion units $\mathsf{ij} = \mathsf{k}$. The ℓ coefficient of $(\mathsf{i}p)\mathsf{j} - \mathsf{i}(p\mathsf{j})$ is $-2b\mathsf{k}$, so $b = 0$. The ℓ coefficient of $(\ell a)x - \ell(ax)$ is $\bar{a}\bar{x} - \bar{x}\bar{a}$. Its vanishing for $x = \mathsf{i}, \mathsf{j}$ forces a real. Since $|p| = 1$, the kernel is $\{(I, I, I), (-I, -I, I)\}$.

There are no orientation-reversing related triples. First, the cyclic invariance of τ gives surjectivity onto $SO(8)$ in each of its three coordinates, and hence the same holds for the three projections of $\mathcal{R} \cap SO(8)^3$. If a product-related triple had $\det A = -1$, compose it with an orientation-preserving product-related triple whose first coordinate is κA^{-1}. The result has first coordinate κ, so satisfies $\bar{x}(By) = C(xy)$. Setting $x = 1$ yields $B = C$. Setting $y = 1$, and writing $p = B1$, yields $Bx = \bar{x}p$. Hence

$$\bar{x}(\bar{y}p) = \overline{xy}\, p \qquad \text{for all } x, y.$$

Set $y = p$ and multiply on the right by $\bar{p}$. The inverse property gives $\bar{x}\bar{p} = \bar{p}\bar{x}$ for all x. The center of $\mathbb{O}$ is $\mathbb{R}$: in the displayed Cayley–Dickson presentation, commuting $a + b\ell$ with i forces $b = 0$, and commuting a with i, j forces a real. Thus $p = \pm 1$. The preceding equation would imply $\bar{x}\bar{y} = \overline{xy} = \bar{y}\bar{x}$ for all x, y, contradicting $\mathsf{ij} = -\mathsf{ji}$. Thus $\det A = 1$. Cyclic permutation of H_τ excludes negative determinants in the other two coordinates too.

The group is connected. The reflection-pair triples are continuously parametrized by $S^7 \times S^7$, contain the identity at $u = v = 1$, and contain $(-I, -I, I)$ at $u = 1, v = -1$. Their third coordinates generate $SO(8)$, so every related triple is a product of these triples and a kernel element. Every such element therefore lies in the identity component. Projection to C is a two-sheeted covering: it is a surjective homomorphism of compact Lie groups with finite kernel, hence a local diffeomorphism with those two-element fibres. Here $\pi_1(SO(8)) = \mathbb{Z}/2$. One proof starts with the unit quaternion conjugation action on imaginary quaternions, which identifies $SO(3)$ with $S^3/\{\pm 1\}$, and uses the fibrations $SO(n-1) \to SO(n) \to S^{n-1}$ for $n \geq 4$; their bases have zero π_1 and π_2, so the fundamental group remains $\mathbb{Z}/2$. A connected two-sheeted cover of this group is simply connected. It is Spin(8), the simply connected cover of $SO(8)$.

The map Φ preserves every product in (1215), by orthogonality and (1227); bilinearity proves preservation of all Jordan products. Its inverse is $\Phi(g^{-1})$. Conversely, a Jordan automorphism fixing all e_i preserves

$$J_i = \{X \in J : e_i \circ X = 0, \quad e_j \circ X = e_k \circ X = X/2\}.$$

Its restrictions define linear maps g_i in the original E_i coordinates. Applying the automorphism to the same-sector Peirce product proves $h(g_i a, g_i b) = h(a, b)$; applying it to the mixed product proves (1228). Thus the restrictions belong to H_τ. The direct-sum decomposition of J proves injectivity and surjectivity of Φ with no further kernel. □

Proposition 30.34. *Without the metrics, the stabilizer of the same typed tensor in $GL(V_1) \times GL(V_2) \times GL(V_3)$ is exactly*

$$\{(s_1 g_1, s_2 g_2, s_3 g_3) : s_i > 0, \quad s_1 s_2 s_3 = 1, \quad (g_1, g_2, g_3) \in H_\tau\}.$$

Thus the metric-preserving stabilizer is Spin(8)*; the unrestricted typed linear stabilizer has two additional independent positive scale parameters.*

Proof. Tensor invariance for (A, B, C) and the h adjoint imply $\mu_{12}(Aa, Bb) = C^{-T}\mu_{12}(a, b)$. For nonzero a, Proposition 30.31 gives $|\det M_a| = N(a)^4$ on dimension eight. Taking determinants yields $N(Aa)^4 = |\det B \det C|^{-1} N(a)^4$. Positivity therefore gives $A = s_1 g_1$, where $s_1 > 0$ and g_1 is orthogonal. The cyclic equations give $B = s_2 g_2$, $C = s_3 g_3$ likewise. For unit a, b, the two sides of the adjoint equation have norms $s_1 s_2$ and s_3^{-1}, so $s_1 s_2 s_3 = 1$. Dividing by those explicit scalars proves $(g_1, g_2, g_3) \in H_\tau$. The converse follows by trilinearity. □

The three actions can also be read directly as the vector and two real half-spin actions. On $S = V_2 \oplus V_3$ define

$$c(a)(b, c_0) = (-\mu_{31}(c_0, a), \mu_{12}(a, b)), \qquad a \in V_1. \tag{1229}$$

The adjoint and inverse identities prove $c(a)^* = -c(a)$ and $c(a)^2 = -N(a)I$. Polarization gives

$$c(a)c(a') + c(a')c(a) = -2h(a,a')I.$$

This is the real Clifford action with negative norm convention. Even products preserve each of the two summands. Equation (1227) gives the exact intertwining map

$$\operatorname{diag}(g_2,g_3)c(a) = c(g_1 a)\operatorname{diag}(g_2,g_3).$$

Conversely, an even product of unit Clifford operators covers the corresponding product of reflections on V_1, is orthogonal on S, and satisfies this equation. Its two diagonal blocks, together with its vector action, therefore form a triple in H_τ. This map from the spin double cover is onto: it covers $SO(V_1)$ and includes both elements over the identity, because the Clifford scalar -1 acts as $-I$ on both summands. The two groups are connected double covers and the map is an isomorphism. This identifies V_1 as the vector representation and V_2, V_3 as the two half-spin modules in the displayed Clifford realization, without conjugating any q_i. Their names S^+, S^- can be fixed by the parity summands; when the names are instead defined by an oriented volume element, that orientation must be specified as additional notation.

There is no nonzero Spin(8) equivariant linear map $V_i \to V_j$ for $i \neq j$. The four triples

$$(I,I,I), \quad (I,-I,-I), \quad (-I,I,-I), \quad (-I,-I,I)$$

belong to H_τ by the product equation. For any two different sectors, one of these acts as $+I$ on the source and $-I$ on the target. An equivariant linear map then equals its own negative and vanishes. Their exact positive relation is the three nonzero equivariant bilinear maps (1216); fixing any nonzero argument gives the intersector isomorphisms and inverses already proved in Proposition 30.31.

An explicit tangent comparison and global bundle descent Put $F_4 = \operatorname{Aut}(J,\circ)$ and use Theorem 30.33 to identify H_τ with the ordered frame stabilizer. Then $M = F_4/H_\tau$ parametrizes the orbit of the ordered frame (e_1,e_2,e_3).

Every Jordan automorphism preserves tr and G. Indeed the Peirce table gives $\operatorname{Tr}_{\operatorname{End}(J)} L_{e_i} = 1+16/2 = 9$ and $\operatorname{Tr}_{\operatorname{End}(J)} L_{E_i(a)} = 0$. The latter operator has zero diagonal blocks in the Peirce decomposition, as each product in its table changes summand. Thus $\operatorname{Tr} L_X = 9\operatorname{tr} X$. Conjugating L_X by a Jordan automorphism gives L_{gX}, proving $\operatorname{tr}(gX) = \operatorname{tr} X$. Product preservation now gives $G(gX,gY) = G(X,Y)$.

Write $L_X(Y) = X \circ Y$. In any Jordan algebra $[L_X, L_Y]$ is a derivation, with the following algebraic proof. The coefficient of three distinct parameters in the Jordan identity gives

$$\begin{aligned} x\circ((z\circ w)\circ y) + z\circ((w\circ x)\circ y) + w\circ((x\circ z)\circ y) = (z\circ w)\circ(y\circ x) + (w\circ x)\circ(y\circ z) \\ + (x\circ z)\circ(y\circ w). \end{aligned}$$

Subtracting this identity with x, y interchanged cancels the right sides and gives exactly

$$[L_x,L_y](z\circ w) = ([L_x,L_y]z)\circ w + z\circ([L_x,L_y]w).$$

For cyclic (i,j,k) define

$$D_i(a) = 4[L_{E_i(a)}, L_{e_j}]. \tag{1230}$$

Its complete values follow from the Peirce product table:

$$\begin{aligned} &D_i(a)e_i = 0, && D_i(a)e_j = E_i(a), && D_i(a)e_k = -E_i(a), \\ &D_i(a)E_i(b) = 2h(a,b)(e_k - e_j), \\ &D_i(a)E_j(b) = -E_k(\overline{ab}), && D_i(a)E_k(c) = E_j(\overline{ca}). \end{aligned} \tag{1231}$$

For example on e_j the commutator before multiplication by 4 is $E_i(a)/2 - E_i(a)/4 = E_i(a)/4$, and on $E_j(b)$ it is $0 - E_k(\overline{ab})/4$. The same product table supplies the other entries. Exponentiating a derivation gives Jordan automorphisms, since the derivation rule and the exponential power series give $e^{tD}(X \circ Y) = (e^{tD}X) \circ (e^{tD}Y)$.

Let $\mathfrak{f}_4$ be the derivation algebra and $\mathfrak{h}$ its subalgebra fixing each e_i. Put $\mathfrak{m}_i = \{D_i(a) : a \in V_i\}$ and $\mathfrak{m} = \bigoplus_i \mathfrak{m}_i$. Then

$$\mathfrak{f}_4 = \mathfrak{h} \oplus \mathfrak{m}, \qquad \mathfrak{m}_i \cong V_i, \qquad \mathrm{Ad}(h)D_i(a) = D_i(h_i a). \tag{1232}$$

To check that no tangent directions are missing, differentiate $e_i^2 = e_i$ to get $De_i = 2e_i \circ De_i$. Thus De_i has only the components $J_j \oplus J_k$. Differentiating $I^2 = I$ gives $DI = 0$, so the two possible occurrences of any J_i component in the triple (De_1, De_2, De_3) are opposite. Equation (1231) therefore supplies unique a_i such that $D - \sum_i D_i(a_i)$ fixes all three e_i. That equation also proves the sum direct. Conjugation of the commutator defining D_i by an automorphism fixing e_j proves the displayed equivariance.

Retain arbitrary distinct original spectral parameters $\lambda_1, \lambda_2, \lambda_3 \in \mathbb{R}$, with $\lambda_1 + \lambda_2 + \lambda_3 = 0$, and set

$$d_\lambda = \sum_i \lambda_i e_i, \qquad \Delta_1 = \lambda_2 - \lambda_3, \quad \Delta_2 = \lambda_3 - \lambda_1, \quad \Delta_3 = \lambda_1 - \lambda_2.$$

Every Δ_i is nonzero and

$$D_i(a)d_\lambda = \Delta_i E_i(a). \tag{1233}$$

The equivariant map $\Xi_\lambda : M \to F_4 d_\lambda$, $[g] \mapsto g d_\lambda$, is a diffeomorphism. Its inverse recovers the ordered frame from the Jordan polynomials

$$p_i(z) = \frac{(z - \lambda_j I) \circ (z - \lambda_k I)}{(\lambda_i - \lambda_j)(\lambda_i - \lambda_k)}.$$

For $z = g d_\lambda$ these equal $g e_i$: evaluate on d_λ first, then apply the Jordan automorphism g. In particular the stabilizer of d_λ is exactly H_τ, and both maps are smooth. Equation (1233) is the differential of this comparison with all scale factors retained.

Use the associated-bundle convention $(g, v) \sim (gh, h^{-1}v)$ for $F_4 \times_{H_\tau} V_i$, and denote this rank-eight real vector bundle by $\mathcal{V}_i$. Define

$$\begin{aligned} K_i &: \mathcal{V}_i \longrightarrow TM, & K_i[g, a] &= \left.\frac{d}{dt}\right|_0 g \exp(tD_i(a))H_\tau, \\ P_i &: \mathcal{V}_i \longrightarrow M \times J, & P_i[g, a] &= (gH_\tau, gE_i(a)). \end{aligned} \tag{1234}$$

Both maps are well defined by the proved equivariance. Equation (1232) shows that $\bigoplus_i K_i : \bigoplus_i \mathcal{V}_i \to TM$ is an isomorphism. The image of P_i above the frame $(p_1, p_2, p_3) = (ge_1, ge_2, ge_3)$ is precisely the intrinsic Peirce space

$$\{X \in J : p_i \circ X = 0, \quad p_j \circ X = p_k \circ X = X/2\}.$$

Suppressing the recorded base-point component of P_i when viewing it as a tangent vector to the orbit, the exact comparison is

$$d\Xi_\lambda \circ K_i = \Delta_i P_i. \tag{1235}$$

Thus the original Albert slots and frame tangent slots are related by an explicit map, rather than an unrecorded change of coordinates.

The tensor and products descend by the formulas

$$\begin{aligned} \boldsymbol{\tau}([g, a], [g, b], [g, c]) &= \tau(a, b, c), \\ \boldsymbol{\mu}_{ij}([g, a], [g, b]) &= [g, \mu_{ij}(a, b)] \in \mathcal{V}_k. \end{aligned} \tag{1236}$$

Representatives over one point can always be chosen with common g. Replacing it by gh replaces the arguments by $h_i^{-1}a, h_j^{-1}b, h_k^{-1}c$. Tensor invariance and (1227) give exactly the same scalar and the same

associated class. This proves well-defined global descent, beyond invariance at a single fibre. The associated metrics induced by h carry all norm, adjoint, and inverse formulas of Proposition 30.31 to identities of bundle maps.

Transport by K_i gives the three subbundles $\mathcal{T}_i = K_i(\mathcal{V}_i)$, the direct sum $TM = \mathcal{T}_1 \oplus \mathcal{T}_2 \oplus \mathcal{T}_3$, and global bilinear maps $\mathcal{T}_i \otimes \mathcal{T}_j \to \mathcal{T}_k$. Define a second, orbit-coordinate realization explicitly by $S_i = K_i/\Delta_i$. Then $d\Xi_\lambda S_i = P_i$. Transport of $\boldsymbol{\mu}$ by S_i is exactly the original coordinate map $E_i(a), E_j(b) \mapsto E_k(\overline{ab})$. These transports obey the following exact comparison. For $u_i \in J_i, u_j \in J_j$ at the reference orbit point, the K-transported product has orbit-coordinate value

$$\frac{\Delta_k}{\Delta_i \Delta_j}\, B_{ij}(u_i, u_j), \tag{1237}$$

whereas the S-transported product has value $B_{ij}(u_i, u_j)$. Indeed the K-coordinate arguments are $E_i^{-1}u_i/\Delta_i$ and $E_j^{-1}u_j/\Delta_j$; bilinearity and the output multiplier Δ_k give the displayed coefficient. At every other point the same identity is transported by $g \in F_4$. The K-transported scalar in orbit coordinates is

$$\frac{\tau(E_1^{-1}u_1, E_2^{-1}u_2, E_3^{-1}u_3)}{\Delta_1\Delta_2\Delta_3},$$

and the S-transported scalar is its numerator. For the pullback $G_\lambda = \Xi_\lambda^* G$ of the original trace metric on the orbit, trace invariance and (1212) give

$$G_\lambda(K_i[g,a], K_i[g,b]) = 2\Delta_i^2 h(a,b), \qquad G_\lambda(S_i[g,a], S_i[g,b]) = 2h(a,b).$$

No Δ_i is set to 1 and no coordinate is conjugated to absorb a sign or a metric factor.

There is a specified connection for which these descended tensors are parallel. On $F_4 \to F_4/H_\tau$ take

$$\omega = \mathrm{pr}_{\mathfrak{h}}(g^{-1}dg) \quad \text{for the decomposition (1232).}$$

Because the projection commutes with $\mathrm{Ad}(H_\tau)$, this form is right equivariant and sends each fundamental vertical vector to its generating element of $\mathfrak{h}$. These are exactly the principal-connection conditions. Upstairs, the descended tensor and products are represented by the constant H_τ-invariant tensors τ, μ_{ij}. Their covariant derivatives are the ordinary derivatives of those constant coefficients plus the infinitesimal $\mathfrak{h}$ action. Both terms vanish. They are therefore parallel for this explicitly defined homogeneous connection and for the connections transported to TM by K or S. No assertion about Levi–Civita parallelism follows from this argument.

Source conventions and scope The matrix convention, related-triple equation, and Jordan frame action agree entry for entry with Ichiro Yokota, *Exceptional Lie groups*, arXiv:0902.0431, Theorem 1.14.2, Lemmas 1.14.3–1.14.4, Theorem 1.16.2, and Theorem 2.7.1 [23]. The present proof supplies the coordinate calculations, norm and inverse maps, determinant coefficients, and global bundle maps in that convention. Its algebraic starting objects are the real alternative composition algebra $\mathbb{O}$ and its Albert Jordan algebra J; those defining structures are retained.

The proved result is an ordered three-sector tangent structure on $F_4/\operatorname{Spin}(8)$ with a normed triality tensor and three exact bilinear morphisms. It does not assert toroidal filling, an integrable complex structure, or physical dynamics. The obstruction to equivariant unary identifications and the positive bilinear relations between sectors were both proved. The dependence on the diagonal-orbit parameters is the exact morphism (1235), with its product and metric consequences; it is not a missing comparison.

30.5.4 A coordinate-preserving triality map onto the determinant boundary

Retain the ordered matrix in (1207) and write

$$E(q) = Q_{\mathbb{O}}(0; q_1, q_2, q_3), \qquad s(q) = |q_1|^2 + |q_2|^2 + |q_3|^2, \qquad \tau(q) = \mathrm{Re}((q_1q_2)q_3).$$

Here the eigenvalues are the three real Jordan eigenvalues, with their multiplicities. They are the roots of $z^3 - s(q)z - 2\tau(q)$. Define

$$a(q) = -\lambda_{\min}(E(q)).$$

Thus $a(q)$ is exactly the largest real root of $\lambda^3 - s(q)\lambda + 2\tau(q) = 0$. This prescription keeps the complete cubic, including its mixed term and its factor two.

Theorem 30.35 (The spectral graph and its exact inverse)**.** *The map*

$$\mathcal{B} : \mathbb{O}^3 \longrightarrow \partial\mathcal{C}(\mathbb{O}), \qquad q \longmapsto Q_{\mathbb{O}}(a(q); q) \tag{1238}$$

is a proper homeomorphism. Its inverse is the coordinate projection

$$\mathcal{P}\left(\begin{pmatrix} \lambda & q_3 & \overline{q}_2 \\ \overline{q}_3 & \lambda & q_1 \\ q_2 & \overline{q}_1 & \lambda \end{pmatrix}\right) = (q_1, q_2, q_3).$$

It extends to a based homeomorphism

$$\mathcal{B}^+ : (\mathbb{O}^3)^+ \xrightarrow{\cong} S^{24}_{\det,\mathbb{O}}.$$

Let $K = \mathrm{Spin}(8) \subset F_4$ be the pointwise stabilizer of the ordered diagonal Jordan frame. On $\mathbb{O}^3$, use the three representations defined by $kE_i(q_i) = E_i(\rho_i(k)q_i)$, with E_i the three ordered off-diagonal inclusions. Then $\mathcal{B}$ and $\mathcal{P}$ are K-equivariant. The norm and the cubic interaction are recovered on the boundary without discarding any coordinate:

$$2\tau(q) = a(q)s(q) - a(q)^3. \tag{1239}$$

Proof. Use the Euclidean Jordan trace inner product $\langle u, v\rangle_J = \mathrm{tr}(u \circ v)$. For $q, p \in \mathbb{O}^3$, direct multiplication gives

$$\|E(q) - E(p)\|_J^2 = 2\sum_{i=1}^{3} |q_i - p_i|^2.$$

Let $\mathcal{I}$ be the compact set of primitive idempotents of trace one. The spectral theorem for the Euclidean Albert algebra gives

$$\lambda_{\min}(u) = \min_{e \in \mathcal{I}} \langle u, e\rangle_J.$$

For completeness, in a spectral Jordan frame write $u = \sum_i \mu_i e_i$. For every $e \in \mathcal{I}$, the numbers $\langle e_i, e\rangle_J$ are nonnegative and have sum one: positivity uses self-duality of the cone of squares, and the sum is $\langle I_3, e\rangle_J = 1$. Thus $\langle u, e\rangle_J \geq \min_i \mu_i$, with equality at an idempotent for a least eigenvalue. This proves the displayed formula. Every such e has $\|e\|_J^2 = \mathrm{tr}(e^2) = 1$. Cauchy–Schwarz and the minimum formula now give

$$|a(q) - a(p)| \leq \|E(q) - E(p)\|_J = \sqrt{2\sum_{i=1}^{3} |q_i - p_i|^2}. \tag{1240}$$

In particular a is continuous.

Since $E(q)$ has trace zero, its least eigenvalue is nonpositive. If $q \neq 0$, all eigenvalues cannot be zero, since the spectral theorem would then give $E(q) = 0$. Their zero sum consequently forces a strictly negative least eigenvalue. Hence $a(0) = 0$ and $a(q) > 0$ for $q \neq 0$. Adding λI_3 adds λ to each of the three Jordan eigenvalues. Therefore

$$Q_{\mathbb{O}}(\lambda; q) \succeq 0 \quad \Longleftrightarrow \quad \lambda \geq a(q),$$

and within this positive cone the determinant vanishes exactly at $\lambda = a(q)$. This is the relative boundary established in theorem 30.29. It proves both inverse compositions in (1238). Continuity of the graph and coordinate projection proves the homeomorphism. Moreover

$$\|Q_{\mathbb{O}}(a(q); q)\|_J^2 = 3a(q)^2 + 2s(q) \geq 2s(q),$$

so unbounded q leaves every compact subset of the boundary. This proves properness directly and the extension at infinity.

A member of K fixes I_3, preserves the Jordan product and trace, and sends each $E_i(q_i)$ to $E_i(\rho_i(k)q_i)$. It preserves the characteristic polynomial and the ordered eigenvalues. Thus $a(\rho_1(k)q_1, \rho_2(k)q_2, \rho_3(k)q_3) = a(q)$, which proves both equivariance identities. Substitution of $\lambda = a(q)$ into the unchanged determinant formula gives (1239). □

The exact tangent map from the ordered flag. For pairwise different real $\lambda_1, \lambda_2, \lambda_3$ with zero sum, retain Ξ_λ of (1193). At $x_0 = (d_1, d_2)$, the inverse of its differential, written in the original off-diagonal coordinates, is

$$\begin{aligned} A(q) &= E_2\left(\frac{q_2}{\lambda_1 - \lambda_3}\right) + E_3\left(\frac{q_3}{\lambda_1 - \lambda_2}\right), \\ B(q) &= E_1\left(\frac{q_1}{\lambda_2 - \lambda_3}\right) - E_3\left(\frac{q_3}{\lambda_1 - \lambda_2}\right), \\ (d\Xi_\lambda)_{x_0}(A(q), B(q)) &= E(q). \end{aligned} \tag{1241}$$

Indeed, differentiate the three idempotents in a Jordan frame. The idempotence equations and the identity that their sum is I_3 force A to have only sectors two and three, B to have only sectors one and three, and their sector-three parts to be negatives. To check that these are actual tangent vectors, the differential of the already proved diffeomorphism Ξ_λ is an isomorphism onto $E_1(\mathbb{O}) \oplus E_2(\mathbb{O}) \oplus E_3(\mathbb{O})$. Its formula is $(\lambda_1 - \lambda_3)A + (\lambda_2 - \lambda_3)B$. Solving its three sector equations gives precisely (1241), so no extra tangent solutions or denominators are suppressed. Composing this tangent coordinate isomorphism with $\mathcal{B}$ gives an explicit equivariant homeomorphism from the complete tangent space at x_0 onto $\partial\mathcal{C}(\mathbb{O})$.

30.5.5 An explicit equivariant flag cell and its degree-one collapse

Write $J = \mathfrak{h}_3(\mathbb{O})$, $J_0 = \ker \operatorname{tr}$, $e = I_3$, and $d_i = \operatorname{diag}(\delta_{i1}, \delta_{i2}, \delta_{i3})$. The inner product on J is

$$\langle x, y\rangle_J = \operatorname{tr}(x \circ y), \qquad L_x(y) = x \circ y.$$

All operator products and brackets below belong to $\operatorname{End}_{\mathbb{R}}(J)$; thus $[A, B] = AB - BA$ is a commutator of real linear operators. An adjoint A^* in these formulas is the adjoint for $\langle\ ,\ \rangle_J$.

We preserve the ordered coordinates of the equal-diagonal slice by putting

$$\begin{aligned} E_1(r) &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & r \\ 0 & \bar{r} & 0 \end{pmatrix}, \qquad E_2(s) = \begin{pmatrix} 0 & 0 & \bar{s} \\ 0 & 0 & 0 \\ s & 0 & 0 \end{pmatrix}, \\ E_3(t) &= \begin{pmatrix} 0 & t & 0 \\ \bar{t} & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad W = \mathbb{O} \oplus \mathbb{O} \oplus \mathbb{O}, \\ E(q) &= E_1(q_1) + E_2(q_2) + E_3(q_3) = Q_{\mathbb{O}}(0; q_1, q_2, q_3). \end{aligned} \tag{1242}$$

In particular the second coordinate is the lower $(3,1)$ entry, and conjugating it in the upper $(1,3)$ entry remains part of every formula. Let $K = \operatorname{Aut}(J) = F_4$ and $M = \{m \in K : md_i = d_i\ (i = 1,2,3)\} = \operatorname{Spin}(8)$. The action of M on W is defined, without choosing three unrelated orthogonal actions, by

$$E_i((m \cdot q)_i) = mE_i(q_i), \qquad i = 1, 2, 3. \tag{1243}$$

The three summands are the vector and the two real half-spin modules in this specified coordinate convention.

We use the real symmetric pair

$$G = E_{6(-26)}, \qquad \mathfrak{g} = \operatorname{Der}(J) \oplus L_{J_0}, \qquad \mathfrak{k} = \operatorname{Der}(J), \qquad \mathfrak{s} = L_{J_0}.$$

Here G is the connected reduced structure group in its real linear representation on J. Its Cartan involution on the Lie algebra is $\theta(A) = -A^*$; hence $\theta(D + L_x) = D - L_x$ for $D \in \mathrm{Der}(J)$ and $x \in J_0$. The symmetric-pair realization and restricted-root spaces are as in [10, Section 2.6, pp. 491–493]. The real-flag theorem we use is precisely [10, Appendix B, Theorem B.2, Proposition B.3 and Corollary B.4, p. 519]: for $P = MAN$ the space $G/P = K/M$ has cells Nw, and $X \mapsto \exp(X)wP$ is a diffeomorphism from the sum of the positive root spaces made negative by w^{-1} onto that cell. Its Cartan comparison is $\Theta(X) = X - \theta X$. We now determine that comparison, the exponential, its flag coordinates, its inverse, and the collapse with every coefficient specified.

Lemma 30.36 (The Cartan inverse with its exact coefficient)**.** *For $h = \mathrm{diag}(h_1, h_2, h_3) \in J_0$, set*

$$\rho_1(h) = \frac{h_3 - h_2}{2}, \qquad \rho_2(h) = \frac{h_1 - h_3}{2}, \qquad \rho_3(h) = \frac{h_1 - h_2}{2} = \rho_1(h) + \rho_2(h).$$

Fix any $h_* = \mathrm{diag}(h_{*,1}, h_{*,2}, h_{*,3}) \in J_0$ with $h_{*,1} > h_{*,3} > h_{*,2}$. For $u \in E_i(\mathbb{O})$ define

$$\nu_i(u) = \frac{1}{2}\left(L_u + \frac{[L_{h_*}, L_u]}{\rho_i(h_*)}\right). \tag{1244}$$

Then $\nu_i(u)$ is independent of the permitted choice of h_*, belongs to $\mathfrak{g}_{\rho_i}$, and satisfies

$$[L_h, \nu_i(u)] = \rho_i(h)\nu_i(u), \qquad \nu_i(u) - \theta\nu_i(u) = L_u, \qquad \nu_i(u)e = \frac{1}{2}u. \tag{1245}$$

Thus ν_i is exactly the inverse of Θ on the ith summand; the factor $1/2$ in (1244) *is necessary. The isomorphism*

$$Z : W \longrightarrow \mathfrak{n} = \mathfrak{g}_{\rho_1} \oplus \mathfrak{g}_{\rho_2} \oplus \mathfrak{g}_{\rho_3}, \qquad Z(q) = \sum_{i=1}^{3} \nu_i(E_i(q_i))$$

is M-equivariant.

Proof. We first record the Jordan identity needed to ensure that the operators are in the correct Lie algebra. For any $a, b \in J$, $D_{a,b} = [L_a, L_b]$ is a derivation, and therefore

$$[D_{a,b}, L_c] = L_{D_{a,b}c}. \tag{1246}$$

Here is the polarization proving this assertion. The coefficient of the product of three distinct real parameters in $(x^2 \circ y) \circ x = x^2 \circ (y \circ x)$ gives

$$\begin{aligned} &d \circ (b \circ (a \circ c)) + a \circ (b \circ (c \circ d)) + c \circ (b \circ (d \circ a)) \\ &\quad = (a \circ c) \circ (b \circ d) + (c \circ d) \circ (b \circ a) + (d \circ a) \circ (b \circ c). \end{aligned}$$

Subtract the same identity with a and b exchanged. The three terms on the right cancel, by commutativity of the Jordan product. The result, rearranged, is

$$D_{a,b}(c \circ d) = (D_{a,b}c) \circ d + c \circ (D_{a,b}d),$$

which proves the assertion and (1246).

Let u have its two off-diagonal entries in positions (j,k) and (k,j), and write $D_h = [L_h, L_u]$ for real diagonal h. If h' is another real diagonal element, direct use of the binary Jordan product gives

$$\begin{aligned} D_h h' &= \left(\frac{(h_j + h_k)(h'_j + h'_k)}{4} - \frac{h_j h'_j + h_k h'_k}{2}\right) u \\ &= -\frac{(h_j - h_k)(h'_j - h'_k)}{4} u = -\rho_i(h)\rho_i(h')u. \end{aligned} \tag{1247}$$

For completeness, the proportionality of D_h in h can also be checked on every $z \in J$ with no associativity assumption on three octonions. Expanding the two Jordan products, the fact that every entry of h is real gives the restricted identity

$$4[L_h, L_u]z = (hu - uh)z - z(hu - uh).$$

In this displayed identity only, the products are binary matrix products; the expansion is valid because every reassociation it uses contains a real entry from h. Since u has only the indicated two off-diagonal entries, $hu - uh$ depends linearly on the single real scalar $h_j - h_k$. Consequently

$$[L_h, L_u] = \frac{\rho_i(h)}{\rho_i(h_*)}[L_{h_*}, L_u]. \tag{1248}$$

This assertion includes $\rho_i(h) = 0$. We use no corresponding identity with arbitrary octonionic h, and no octonionic matrix conjugation is used to define a group action.

Put $D_* = [L_{h_*}, L_u]$. Equations (1246) and (1247) give

$$[L_h, D_*] = -L_{D_* h} = \rho_i(h_*)\rho_i(h)L_u.$$

Combining this with (1248) proves $[L_h, \nu_i(u)] = \rho_i(h)\nu_i(u)$ and also proves independence of h_*. Since $\theta L_u = -L_u$ and $\theta D_* = D_*$, subtracting $\theta\nu_i(u)$ from (1244) gives L_u exactly. Every derivation kills the unit, so $\nu_i(u)e = u/2$. The map $u \mapsto L_u$ is injective because $L_u e = u$. The positive restricted-root spaces have the three stated eight-dimensional Cartan images; thus these injective maps exhaust them. Finally, an $m \in M$ fixes h_* and satisfies $mL_u m^{-1} = L_{mu}$ and $m[L_{h_*}, L_u]m^{-1} = [L_{h_*}, L_{mu}]$. These formulas prove the asserted equivariance. □

Lemma 30.37 (Finite exponential and the coefficient in the nilpotent law)**.** *For every $q \in W$,*

$$Z(q)^5 = 0, \qquad \exp Z(q) = I + Z(q) + \frac{Z(q)^2}{2} + \frac{Z(q)^3}{6} + \frac{Z(q)^4}{24}. \tag{1249}$$

The same formula holds with $Z(q)$ replaced by $-Z(q)^*$. Writing $\nu_i(r)$ temporarily for $\nu_i(E_i(r))$, the brackets are

$$[\nu_1(r), \nu_2(s)] = \nu_3\left(-\frac{1}{2}\overline{rs}\right), \qquad [\nu_i(r), \nu_i(s)] = 0, \qquad [\nu_3(t), \nu_j(r)] = 0. \tag{1250}$$

The last equality holds for $j = 1, 2, 3$. In exponential coordinates the group law on N is

$$\begin{aligned}(q_1, q_2, q_3) \star (q_1', q_2', q_3') = \Big(&q_1 + q_1',\ q_2 + q_2',\\ &q_3 + q_3' - \frac{1}{4}\overline{q_1 q_2'} + \frac{1}{4}\overline{q_1' q_2}\Big).\end{aligned} \tag{1251}$$

Proof. The auxiliary operator $H_0 = L_{\mathrm{diag}(2,-2,0)}$ has the following eigenspaces in its representation on J:

eigenvalue	-2	-1	0	1	2
eigenspace	$\mathbb{R}d_2$	$E_1(\mathbb{O})$	$\mathbb{R}d_3 \oplus E_3(\mathbb{O})$	$E_2(\mathbb{O})$	$\mathbb{R}d_1$.

This table follows from $L_h d_j = h_j d_j$ and $L_h u = (h_j + h_k)u/2$ for an off-diagonal (j, k) element. The actual positive roots evaluated on $\mathrm{diag}(2, -2, 0)$ are $1, 1, 2$. Therefore every term of $Z(q)$ strictly raises these weights by at least one. Five successive applications give zero. Adjoints strictly lower the weights, proving the analogous assertion for $-Z(q)^*$. This proves both finite exponential formulas.

Brackets add restricted roots. There is no root $2\rho_i$, $\rho_1 + \rho_3$, or $\rho_2 + \rho_3$, so the brackets claimed to vanish in (1250) do vanish. To compute the remaining coefficient, put $u = E_1(r)$, $v = E_2(s)$. Binary matrix multiplication gives, in the retained coordinate order,

$$u \circ v = \frac{1}{2}E_3(\overline{rs}) =: w.$$

The weight of u, v, and w under L_h is respectively $-h_1/2$, $-h_2/2$, and $-h_3/2$. Hence

$$[L_h, L_u]v = -\rho_1(h)w, \qquad [L_h, L_v]u = \rho_2(h)w,$$

and (1244) gives $\nu_1(r)v = 0$ and $\nu_2(s)u = w$. Applying the commutator to e,

$$[\nu_1(r), \nu_2(s)]e = \frac{1}{2}(\nu_1(r)v - \nu_2(s)u) = -\frac{1}{4}E_3(\overline{rs}).$$

On the third root space the map $A \mapsto 2Ae$ is inverse to ν_3. This proves the first formula of (1250). All iterated brackets of length three vanish. The exact Baker–Campbell–Hausdorff formula is therefore $\log(\exp X \exp Y) = X + Y + [X, Y]/2$; substituting the just-computed bracket gives (1251), including both signs and the coefficient $1/4$. □

Theorem 30.38 (A finite operator formula for the entire open cell)**.** *For $q \in W$ define*

$$\begin{aligned} A(q) &= \left(I + Z + \frac{Z^2}{2} + \frac{Z^3}{6} + \frac{Z^4}{24}\right) d_2, \\ B(q) &= \left(I - Z^* + \frac{(Z^*)^2}{2} - \frac{(Z^*)^3}{6} + \frac{(Z^*)^4}{24}\right) d_1, \qquad Z = Z(q), \\ \Psi(q) &= \left(\frac{A(q)}{\operatorname{tr} A(q)}, \frac{B(q)}{\operatorname{tr} B(q)}\right). \end{aligned} \tag{1252}$$

Both traces in these formulas are strictly positive. The map Ψ is an M-equivariant diffeomorphism

$$W \xrightarrow{\Psi} U = \{(a, b) \in \mathrm{Fl}(\mathbb{O}) : a_{22} > 0, b_{11} > 0\} = C_{w_0}, \qquad w_0 = (12). \tag{1253}$$

The inverse $q = \Psi^{-1}(a, b)$ is the exact formula

$$\boxed{q_1 = \frac{2a_{23}}{a_{22}}, \qquad q_2 = -\frac{2b_{31}}{b_{11}}, \qquad q_3 = \frac{2a_{12}}{a_{22}} - \frac{1}{4}\overline{q_1 q_2}.} \tag{1254}$$

Every denominator in (1254) *is a positive real scalar. In particular $\Psi(0) = (d_2, d_1)$.*

Proof. Let $\widehat{w}_0$ be the Jordan automorphism that interchanges the first and second real diagonal positions, obtained by conjugation with the real permutation matrix for (12). For this real matrix the conjugation is a well-defined Jordan automorphism. Its action on the retained positive roots is

$$w_0\rho_1 = -\rho_2, \qquad w_0\rho_2 = -\rho_1, \qquad w_0\rho_3 = -\rho_3.$$

It is consequently the longest Weyl element, and the real-flag cell theorem gives a diffeomorphism

$$W \longrightarrow N\widehat{w}_0 P/P, \qquad q \longmapsto \exp Z(q)\widehat{w}_0 P. \tag{1255}$$

Here are the exact Iwasawa factors that identify the flag pair in (1255). Write $g = kan$ uniquely with $k \in K$, $a = \exp L_h \in A$, $n \in N$. Each positive root operator kills d_1. This follows either by applying (1244) to d_1, or from the H_0 weight table, whose highest weight is $\mathbb{R}d_1$. Each operator $-\nu_i^*$ kills d_2 by the lowest-weight statement. Thus

$$nd_1 = d_1, \qquad (n^*)^{-1}d_2 = d_2,$$

and consequently

$$gd_1 = e^{h_1}kd_1, \qquad (g^*)^{-1}d_2 = e^{-h_2}kd_2. \tag{1256}$$

Since Jordan automorphisms preserve the trace, the traces on the right are exactly $e^{h_1} > 0$ and $e^{-h_2} > 0$. It follows that the flag corresponding to gP is

$$\left(\frac{gd_1}{\operatorname{tr}(gd_1)}, \frac{(g^*)^{-1}d_2}{\operatorname{tr}((g^*)^{-1}d_2)}\right) = (kd_1, kd_2).$$

For $g = \exp Z(q)\widehat{w}_0$, its two numerators are $\exp Z(q)d_2$ and $\exp(-Z(q)^*)d_1$, respectively. The finite exponential identities prove (1252), all its positivity assertions, and the identification with (1255).

We next verify the image description, so that no membership criterion is left implicit. For any $n \in N$, the H_0 weight table shows that the coefficient of d_2 in nd_j is 1 if $j = 2$ and 0 otherwise. Indeed n strictly raises weights apart from its identity term. Similarly the coefficient of d_1 in $(n^*)^{-1}d_j$ is 1 if $j = 1$ and 0 otherwise. A flag in $N\widehat{w}P/P$ is represented by the positive-trace multiples of $nd_{w(1)}$ and $(n^*)^{-1}d_{w(2)}$. It satisfies $a_{22} > 0$ precisely when $w(1) = 2$, and $b_{11} > 0$ precisely when $w(2) = 1$. Both conditions hold precisely for $w = w_0 = (12)$. The six-cell decomposition covers the flag, proving (1253).

For the inverse we record the required coefficients of the finite exponentials. The same weights show

$$\begin{gathered} A(q)_{22} = 1, \qquad B(q)_{11} = 1, \qquad A(q)_{23} = \frac{1}{2}q_1, \qquad B(q)_{31} = -\frac{1}{2}q_2, \\ A(q)_{12} = \frac{1}{2}q_3 + \frac{1}{8}\overline{q_1 q_2}. \end{gathered} \tag{1257}$$

To see also the last coefficient directly, the relevant first-order terms are

$$Zd_2 = \frac{1}{2}E_1(q_1) + \frac{1}{2}E_3(q_3), \qquad -Z^*d_1 = -\frac{1}{2}E_2(q_2) - \frac{1}{2}E_3(q_3).$$

An E_3 contribution to Z^2d_2 can arise only from $\nu_2(q_2)\nu_1(q_1)d_2$ or $\nu_1(q_1)\nu_2(q_2)d_2$; the latter is zero, and the former is $\frac{1}{4}E_3(\overline{q_1 q_2})$ by the calculation in Lemma 30.37. The other weight-two contribution $\nu_1(q_1)^2 d_2$ lies in $\mathbb{R}d_3$, and $\nu_2(q_2)^2 d_2 = 0$. The coefficient $1/2$ in the exponential gives $1/8$. Terms of degree three or four raise weight too far to have an E_3 component. The other four identities in (1257) follow from the unique lowest, highest, and adjacent weights.

For $(a, b) = \Psi(q)$, the first two identities imply

$$A(q) = \frac{a}{a_{22}}, \qquad B(q) = \frac{b}{b_{11}}.$$

The remaining identities solve for q_1, q_2, q_3 in precisely the order (1254). Since (1255) is onto U, this proves both inverse compositions everywhere on U, not only a formal local inverse. The displayed formulas show their smoothness.

Finally, $Z(mq) = mZ(q)m^{-1}$ and m is orthogonal for the trace inner product. It fixes d_1, d_2 and preserves the trace. Each expression in (1252) therefore commutes with the M action. This proves equivariance of Ψ and, by its proved bijectivity, of its inverse. □

Proposition 30.39 (The orientation sign, including the conjugated coordinate)**.** *Fix an orientation of $\mathbb{O}$, and give W the ordered product orientation (q_1, q_2, q_3). Give the flag the orientation specified in the integral Euler-class calculation: orient its first root summand by the chosen orientation of $E_1(\mathbb{O})$, its third by $n_2E_1(\mathbb{O})$, and its second by $n_1^{-1}n_2E_1(\mathbb{O})$, where $n_1 = (23)$ and $n_2 = (13)$ are real-permutation lifts. The total orientation is the ordered sum of root summands $(1, 2, 3)$, transported by K. Then $\Psi : W \to U$ is orientation preserving.*

Proof. These are the orientation transport conventions of [10, Section 3, pp. 495–496]. They are independent of the lifts within their cosets because M is connected and each of its orthogonal representations has positive real determinant. We compute the sign explicitly. For this computation only, call the product orientation of the three upper entries (r, q, p) in positions $(23), (13), (12)$ the upper-entry orientation.

Octonionic conjugation fixes the real line and acts by -1 on the seven imaginary directions, so its real determinant is $(-1)^7 = -1$. The lift $n_2 = (13)$ takes an upper (23) entry r to the upper (12) entry $\bar{r}$. The lift $n_1^{-1} = (23)$ then takes that upper (12) entry to the upper (13) entry with the same value. The source orientations of summands $1, 2, 3$ thus have signs $+1, -1, -1$ relative to the upper-entry coordinates. Their product is $+1$, so the flag tangent orientation at d_λ agrees with the upper-entry orientation. Here d_λ has the original pairwise distinct parameters λ_i and the trace-zero relation; no ordering or rescaling of these parameters is needed.

The lift $\widehat{w}_0 = (12)$ interchanges the upper-entry (23) and (13) blocks and conjugates the (12) block. Interchanging two eight-dimensional blocks has determinant $(-1)^{8\cdot 8} = +1$, and conjugation has determinant -1. Thus the transported flag orientation at $\widehat{w}_0 d_\lambda$ is the negative of its upper-entry orientation. This transport preserves the flag orientation because it is the action of an element of connected K.

In the retained q coordinates, E conjugates exactly the second upper-entry block; its determinant relative to the upper-entry orientation is therefore -1. Differentiate (1252) and the original orbit comparison $\Xi_\lambda(a,b) = \lambda_1 a + \lambda_2 b + \lambda_3(e-a-b)$ at $q=0$. The traces have zero first derivative, and the result is

$$\begin{aligned} d(\Xi_\lambda \Psi)_0(q) &= \frac{\lambda_1-\lambda_3}{2}E_1(q_1) + \frac{\lambda_3-\lambda_2}{2}E_2(q_2) + \frac{\lambda_1-\lambda_2}{2}E_3(q_3) \\ &= -\sum_{i=1}^{3} \rho_i(\widehat{w}_0 d_\lambda) E_i(q_i). \end{aligned} \tag{1258}$$

Its three real scalar factors are nonzero; each acts on eight real coordinates, so their determinant contribution is positive. The total differential has sign -1 relative to upper-entry coordinates, exactly the sign of the transported flag orientation at this point. Therefore its sign as an oriented map is $+1$. The domain is connected and the map is a diffeomorphism, so this sign is $+1$ everywhere. □

Theorem 30.40 (The exact equivariant quotient onto the representation sphere)**.** *Let $F = \mathrm{Fl}(\mathbb{O})$ and put*

$$A = F \setminus U = \{(a,b) \in F : a_{22} = 0 \text{ or } b_{11} = 0\}.$$

It is the union of the five cells other than C_{w_0}, and is a closed 16*-dimensional CW subcomplex. Define*

$$c : F \longrightarrow W^+, \qquad c(a,b) = \begin{cases} \Psi^{-1}(a,b), & (a,b) \in U, \\ \infty, & (a,b) \in A. \end{cases} \tag{1259}$$

Then c is continuous, surjective, and M-equivariant and induces a based M-equivariant homeomorphism

$$F/A \xrightarrow{\cong} W^+. \tag{1260}$$

With the orientations in Proposition 30.39, c has degree $+1$. If $\omega_W \in \widetilde{H}^{24}(W^+;\mathbb{Z})$ is its positive generator, then

$$c_*[F] = [W^+], \qquad c^*\omega_W = u_{\mathbb{O}}, \qquad u_{\mathbb{O}} = \beta_1^2\beta_2 = \beta_1\beta_2^2. \tag{1261}$$

On integral homology c_* is the identity in degree zero, an isomorphism in degree 24*, and the zero map on the two copies of $\mathbb{Z}$ in each of degrees* 8 *and* 16*. On fixed points its exact formula is*

$$c(w_0 x_{\mathbb{O},0}) = 0, \qquad c(w x_{\mathbb{O},0}) = \infty \quad (w \neq w_0).$$

These are the six points of F^M mapping onto the two points $(W^+)^M = \{0, \infty\}$.

Proof. All diagonal entries of primitive positive idempotents are nonnegative. The open-cell locus of Theorem 30.38 consequently gives the stated closed-set formula for A. The restricted roots have multiplicity eight. The six cell dimensions are $0, 8, 8, 16, 16, 24$, with the unique top cell C_{w_0}. The CW property of the

real-flag decomposition implies that their union below degree 24 is the closed 16-skeleton, which proves the subcomplex assertion.

Continuity of c on U is already proved. Let $K_0 \subset W$ be compact. Its image $\Psi(K_0)$ is compact in the Hausdorff compact manifold F, and hence closed in F. Thus

$$c^{-1}(W^+ \setminus K_0) = F \setminus \Psi(K_0)$$

is open and contains A. Complements of compact subsets of W form a neighborhood basis of ∞. This proves continuity at every point of A, including all its lower-dimensional strata. It also proves the exact escape statement: a net in U approaching A eventually leaves $\Psi(K_0)$ for every compact $K_0 \subset W$. Surjectivity and the description of the fibers are immediate from the two inverse compositions of the chart: every finite point has one preimage, and $c^{-1}(\infty) = A$. Therefore the induced map $F/A \to W^+$ is a continuous bijection from a compact space to a Hausdorff space and hence a homeomorphism. The open cell is M-invariant by the proved chart equivariance; its complement is invariant too. The formula for c is therefore M-equivariant, with M fixing ∞.

Since A has dimension 16, its integral homology in degrees 23 and 24 is zero. The exact sequence of the pair gives an isomorphism

$$H_{24}(F;\mathbb{Z}) \xrightarrow{\cong} H_{24}(F, A;\mathbb{Z}).$$

Because A is a subcomplex, quotienting identifies this relative group with $\widetilde{H}_{24}(F/A;\mathbb{Z})$. At any point of U, the local map is the inverse of the orientation-preserving chart. It has local degree $+1$. There is exactly one preimage of that finite regular value. This fixes the sign of the preceding homology isomorphism and proves $c_*[F] = [W^+]$. Pairing with cohomology proves $c^*\omega_W = u_{\mathbb{O}}$ for the already fixed positive top generator of the flag. The remaining homology assertions follow because W^+ is a 24-sphere and F is connected with the stated six-cell homology. Each of the three nontrivial irreducible eight-dimensional M-modules has zero invariant subspace, so $W^M = \{0\}$. The point $w_0 x_{\mathbb{O},0}$ is $\Psi(0)$ and belongs to the open cell; the other five Weyl points lie in A. Substitution in the defining formula for c proves the fixed-point assertions. □

Corollary 30.41 (Composition with the retained spectral boundary graph)**.** *For $q \in W$, let*

$$\alpha(q) = -\lambda_{\min}(E(q)), \qquad \Gamma(q) = Q_{\mathbb{O}}(\alpha(q); q_1, q_2, q_3).$$

The exact boundary graph has inverse the off-diagonal coordinate projection $Q_{\mathbb{O}}(\lambda; q) \mapsto q$. It extends to an M-equivariant based homeomorphism

$$\Gamma^+ : W^+ \longrightarrow (\partial\mathcal{C}(\mathbb{O}))^+, \qquad \Gamma^+(\infty) = \infty.$$

Consequently the fully specified map

$$C_{\det} = \Gamma^+ c : F \longrightarrow S^{24}_{\det,\mathbb{O}} \tag{1262}$$

is continuous, M-equivariant, and has degree $+1$ when the boundary sphere carries the orientation transported from its retained ordered coordinates by Γ^+. Its exact fibers are

$$C_{\det}^{-1}(\Gamma(q)) = \{\Psi(q)\} \quad (q \in W), \qquad C_{\det}^{-1}(\infty) = A.$$

Proof. The eigenvalues of $\lambda e + E(q)$ are $\lambda + \lambda_j(E(q))$, with their multiplicities retained. It is positive semidefinite precisely when $\lambda \geq -\lambda_{\min}(E(q))$, and lies on the relative boundary precisely when equality holds. This proves both compositions with the stated coordinate projection. The minimum eigenvalue is continuous, so Γ is continuous, and the projection is continuous. A homeomorphism is proper, because the inverse maps compact sets to compact sets; therefore it extends to the indicated one-point compactifications. An element of M fixes e, preserves eigenvalues as a Jordan automorphism, and sends $E(q)$ to $E(mq)$. Thus $\alpha(mq) = \alpha(q)$ and $\Gamma(mq) = m\Gamma(q)$. This proves equivariance also at infinity. The asserted properties, degree, and fibers of $C_{\det}$ now follow from Theorem 30.40 by composition with the explicitly oriented homeomorphism Γ^+. □

Corollary 30.42 (The exact limit on a reverse sphere map)**.** *Every continuous map $f : S^{24} \to F$ has degree zero. In particular neither c nor $C_{\det}$ admits a continuous section or a homotopy section. These assertions coexist with the degree-one equivariant maps just constructed.*

Proof. The cohomology groups $H^8(S^{24};\mathbb{Z})$ vanish, so $f^*\beta_1 = f^*\beta_2 = 0$. Hence

$$f^*u_{\mathbb{O}} = f^*(\beta_1^2\beta_2) = (f^*\beta_1)^2 f^*\beta_2 = 0.$$

This is exactly the statement that the degree is zero. A homotopy section of c would satisfy $f^*c^*\omega_W = \omega_W$, contradicting the equality just proved and (1261). The same argument applies after the homeomorphism Γ^+. □

Proposition 30.43 (Exact maps among the three marked 24-dimensional constructions)**.** *The following constructions are related by the explicit spectral graph, flag-cell collapse, and smash-coordinate maps proved here.*

(i) *The octonionic flag is the homogeneous* 24*-manifold* $F_4/\operatorname{Spin}(8)$*, with* $H^8 \cong H^{16} \cong \mathbb{Z}^2$ *and* $\chi = 6$*.*

(ii) *After a based diffeomorphism* $X \cong S^6$*, the fourth smash power* $X^{\wedge 4}$ *is a based sphere* $S^{24}_{\wedge}$*; it has* $\widetilde{H}^k = 0$ *for* $k \neq 24$ *and* $\chi = 2$*, and its quotient maps depend on the chosen based candidate and cusp collapses.*

(iii) *The octonionic equal-diagonal cone has the determinant-boundary compactification* $S^{24}_{\det,\mathbb{O}} = (\partial\mathcal{C}(\mathbb{O}))^+$*. It is produced by the explicit cubic* (1209)*, not by a smash product.*

The first package is not homotopy equivalent to either sphere. The two sphere carriers in (ii) and (iii) are related by the based homeomorphism displayed below. The flag maps to the determinant sphere by the explicit degree-one map of Corollary 30.41; its entire infinite fibre is the stated 16*-skeleton. The obstruction to a reverse homotopy section is Corollary 30.42.*

The lower complex determinant orbit pair (S^{10}, S^8) *from theorem 30.15 is a lower rung: its* S^8 *is the compactified complex rank-loss locus and its complement has determinant-phase fundamental group* $\mathbb{Z}$*. It is not a third* 24*-dimensional object.*

Proof. Item (i) is theorems 30.24 and 30.25. The based smash identity $(S^6)^{\wedge 4} \cong S^{24}$ proves the sphere and cohomology assertions in (ii). Item (iii) is theorem 30.29. The nonzero degree-eight and degree-sixteen cohomology of the flag precludes a homotopy equivalence with either sphere; the Euler characteristics give an independent obstruction. For an explicit choice-dependent comparison, let $\eta : (X, x_{\text{base}}) \to (\mathbb{R}^6)^+$ be a based homeomorphism, write the four six-coordinate vectors consecutively as $(x_0, \dots, x_{23})$, and fix the explicit linear isometry $L : (\mathbb{R}^6)^4 \to \mathbb{O}^3$ given by $q_j = \sum_{a=0}^{7} x_{8(j-1)+a}u_a$ in the fixed ordered orthonormal octonion basis. Let $b = \mathcal{P}$ be the off-diagonal coordinate inverse of the spectral graph in Theorem 30.35. The smash compactification map

$$\kappa : ((\mathbb{R}^6)^+)^{\wedge 4} \longrightarrow ((\mathbb{R}^6)^4)^+$$

sends a tuple of finite vectors to their ordered tuple and sends the collapsed wedge to infinity. It is a continuous bijection from a compact space to a Hausdorff space, hence a homeomorphism. Thus

$$(b^+)^{-1} \circ L^+ \circ \kappa \circ \eta^{\wedge 4} : X^{\wedge 4} \xrightarrow{\cong} (\partial\mathcal{C}(\mathbb{O}))^+$$

is a based homeomorphism with inverse obtained by reversing these four maps. It depends explicitly on η and L; neither marked construction specifies them. The final lower-rung assertions are (1167) and (1168). □

30.6 What the mod–12 and mod–24 arithmetic actually is

30.6.1 The integral peripheral representation

The matrices C_1, C_2', C_0 in (1151) are not merely of orders three, four, and infinite. Since C_2' and C_0 generate $SL_2(\mathbb{Z})$, their abelianization detects one exact modulus.

Theorem 30.44 (Peripheral modulus and the mod–24 boundary)**.** *Let*

$$\mathrm{ab} : SL_2(\mathbb{Z}) \twoheadrightarrow SL_2(\mathbb{Z})_{\mathrm{ab}}$$

be the abelianization map. There is a unique isomorphism

$$\overline{\nu} : SL_2(\mathbb{Z})_{\mathrm{ab}} \xrightarrow{\cong} \mathbb{Z}/12$$

for which the marked character

$$\chi = \overline{\nu} \circ \mathrm{ab} : SL_2(\mathbb{Z}) \longrightarrow \mathbb{Z}/12 \tag{1263}$$

sends C_0 to -1. Its values on the three peripheral matrices are

$$\chi(C_1) = 4, \qquad \chi(C_2') = -3, \qquad \chi(C_0) = -1 \pmod{12}. \tag{1264}$$

The residues 4 and $-3 = 9$ are complementary idempotents in $\mathbb{Z}/12$; additively they have exact orders 3 and 4.

Let

$$j : \mathbb{Z}/12 \hookrightarrow \mathbb{Z}/24, \qquad j(a) = 2a.$$

Every group homomorphism $\psi : SL_2(\mathbb{Z}) \to \mathbb{Z}/24$ has image in the even subgroup and factors uniquely through j. In particular

$$\chi_{24} = j\chi, \qquad \big(\chi_{24}(C_1), \chi_{24}(C_2'), \chi_{24}(C_0)\big) = (8, -6, -2) \pmod{24} \tag{1265}$$

is the unique mod–24 re-encoding with

$$\exp\!\left(\frac{2\pi i\, \chi_{24}(M)}{24}\right) = \exp\!\left(\frac{2\pi i\, \chi(M)}{12}\right) \qquad (M \in SL_2(\mathbb{Z})).$$

Its image has order 12, not 24. If $r : \mathbb{Z}/24 \to \mathbb{Z}/12$ is ordinary reduction, then $r \circ j = [2]$, and there is no group homomorphism $\widetilde{\chi} : SL_2(\mathbb{Z}) \to \mathbb{Z}/24$ with $r \circ \widetilde{\chi} = \chi$.

Proof. Put

$$S_0 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad T = C_0.$$

Then $C_2' = S_0^3$, $C_1 = T^{-1}S_0$, and the standard integral modular presentation is

$$SL_2(\mathbb{Z}) = \langle S_0, T \mid S_0^4 = 1,\ S_0^2 = (S_0T)^3 \rangle.$$

Here is why these relations suffice. With $R = S_0T$, the presented group is $G = \langle S_0, R : S_0^4 = 1, S_0^2 = R^3 \rangle$. The element $c = S_0^2 = R^3$ is central and has order at most two; $G/\langle c \rangle = C_2 * C_3$. The modular normal-form theorem identifies $PSL_2(\mathbb{Z})$ with this same free product on the images of S_0, R [3, Theorem C.1, pp. 18–19]. The matrices S_0, T generate $SL_2(\mathbb{Z})$ [3, Theorem 1.1, pp. 1–2]; hence the map from G onto $SL_2(\mathbb{Z})$ is surjective and its kernel lies in $\langle c \rangle$. Since c maps to $-I_2 \neq I_2$, the kernel is trivial. This proves the presentation (equivalently $C_4 *_{C_2} C_6$) with the stated matrix generators. After abelianization, with additive generators s, t, its two relations are

$$4s = 0, \qquad 2s = 3(s + t),$$

or $4s = 0$ and $s+3t = 0$. The relation matrix $\left(\begin{smallmatrix} 4 & 0 \\ 1 & 3 \end{smallmatrix}\right)$ has Smith normal form $\mathrm{diag}(1, 12)$, so the abelianization is cyclic of exact order 12, generated by t. Choosing $t \mapsto -1$ gives

$$s \mapsto 3, \qquad C_2' = S_0^3 \mapsto 9 = -3, \qquad C_1 = T^{-1}S_0 \mapsto 1 + 3 = 4,$$

which proves (1264). Direct calculation in $\mathbb{Z}/12$ gives

$$4^2 = 4, \qquad 9^2 = 9, \qquad 4 + 9 = 1, \qquad 4 \cdot 9 = 0.$$

Every group homomorphism from $SL_2(\mathbb{Z})$ to an abelian group factors through its abelianization. An element $x \in \mathbb{Z}/24$ satisfies $12x = 0$ exactly when x is even, proving the unique factorization through j and excluding a surjective character. Also $rj(a) = 2a \pmod{12}$. Finally a reduction-compatible lift would send C_0 to either 11 or 23 modulo 24, both of which have order 24; an image of the order-12 abelianized generator cannot have that order. This proves the final assertion. □

Proposition 30.45 (The primitive cusp quotient singles out mod–3, not mod–12 or mod–24)**.** *For $N \geq 1$, let*

$$d_N : \mathbb{Z}^3 \longrightarrow \mathbb{Z}/N, \qquad d_N(a_1, a_2, a_3) = a_1 + a_2 + a_3 \pmod{N}.$$

There is a homomorphism $\overline{d}_N : \mathbb{Z}^2 \to \mathbb{Z}/N$ satisfying $d_N = \overline{d}_N P$ if and only if $N \mid 3$. For $N = 3$, it is explicitly

$$\overline{d}_3(y_1, y_2) = y_1 - y_2 \pmod{3}.$$

Similarly, d_N factors through Q if and only if $N \mid 3$; for $N = 3$ the exact formula is

$$d_3 = (1, 1)Q.$$

Thus neither primitive lattice quotient in theorem 30.7 produces a diagonal-sum character modulo 12 *or modulo* 24*.*

Proof. Both $\ker P$ and $\ker Q$ are generated by $k = (1, 1, 1)^{\mathsf{T}}$. A homomorphism out of $\mathbb{Z}^3$ factors through either quotient precisely when it kills that kernel. Since $d_N(k) = 3 \pmod{N}$, this occurs precisely when $N \mid 3$. At $N = 3$, direct multiplication with the matrices in (1134) gives

$$(1, -1)P = (1, -2, 1) \equiv (1, 1, 1) \pmod{3}, \qquad (1, 1)Q = (1, -2, 1) \equiv (1, 1, 1) \pmod{3},$$

which proves both formulas and the claimed boundary. □

Proposition 30.46 (Coefficient reduction and determinant winding do not select a modulus)**.** *For every $N \geq 1$, coefficient reduction gives a graded-ring isomorphism*

$$H^*(\mathrm{Fl}(\mathbb{O}); \mathbb{Z}/N) \cong H^*(\mathrm{Fl}(\mathbb{O}); \mathbb{Z}) \otimes \mathbb{Z}/N,$$

and its reduced Euler class

$$e_N = 6u_{\mathbb{O}} \bmod N \in H^{24}(\mathrm{Fl}(\mathbb{O}); \mathbb{Z}/N)$$

has additive order $N/\gcd(N, 6)$. In particular e_{12} has order two and e_{24} has order four. Ordinary coefficient reduction is defined for every N; this operation alone makes no choice of $N = 12$ or $N = 24$.

Likewise, in the lower complex orbit pair (S^{10}, S^8), the determinant phase identifies the fundamental group of the rank-full complement in (1168) *with $\mathbb{Z}$. Reducing its winding number modulo N gives a surjection to $\mathbb{Z}/N$ for every N, so this plain winding-number reduction makes no choice of* 12 *or* 24*. This phase statement is not being transferred to the real octonionic determinant in* (1209)*.*

Proof. All integral cohomology groups in (1201) are free. The universal coefficient sequence therefore has no Tor term, and multiplication is preserved by coefficient change. Equation (1200) gives the stated reduced Euler class. Since the reduction of the primitive $u_{\mathbb{O}}$ generates $H^{24}(\mathrm{Fl}(\mathbb{O}); \mathbb{Z}/N) \cong \mathbb{Z}/N$, the least positive integer m annihilating e_N is the least m with $N \mid 6m$, namely $N/\gcd(N, 6)$. For the determinant complement, (1168) is a deformation retraction onto the phase circle, and $\arg\det$ is that retraction coordinate. Its winding is an isomorphism on π_1, whose reduction modulo any N is surjective. □

30.6.2 The distinct stable $\mathbb{Z}/12 \hookrightarrow \mathbb{Z}/24$

The injection $j(a) = 2a$ is realized, after compatible generator choices, by the universal sixfold suspension homomorphism below. The cusp quotient supplies one particular element in its domain; it does not supply the injection itself. Neither construction comes from the octonionic flag or from a 24-dimensional smash source. We state all data needed to type these assertions.

Let v_0 be the suspension vertex to which the central cusp fibre is collapsed, write $W = q_2^{-1}(v_0)$, and choose explicitly $x_{\mathrm{base}} \in W$. The exact cusp collapse in the canonical audit then gives a based quotient

$$q_2 : (X, x_{\mathrm{base}}) \longrightarrow (\mathcal{A}, v_0), \qquad \mathcal{A} = \mathrm{Susp}(S^1 \times \overline{T}^2).$$

Let

$$K^{(1)} = (S^1 \times \{0\}) \cup (\{0\} \times S^1) \subset \overline{T}^2,$$

and let $c : \overline{T}^2 \to \overline{T}^2/K^{(1)} \cong S^2$ be the degree-one cell quotient, with $* = c(K^{(1)})$. Write

$$\Sigma_r(S^2, *) = \frac{[0, \epsilon] \times S^2}{(\{0, \epsilon\} \times S^2) \cup ([0, \epsilon] \times \{*\})}$$

for reduced suspension, the entire denominator being collapsed to one basepoint. The formula

$$\widehat{p}_2([r, e^{i\phi}, u]) = [r, c(u)] \in \Sigma_r(S^2, *) \cong S^3$$

defines a based map: it sends both suspension vertices of $\mathcal{A}$ and the entire cylinder over $K^{(1)}$ to that one basepoint. Thus

$$g = \widehat{p}_2 q_2 : (X, x_{\mathrm{base}}) \longrightarrow (S^3, *). \tag{1266}$$

After the fixed based identification $X \cong S^6$, this determines $[g] \in \pi_6(S^3)$. The construction and its endpoint continuity are part of the cited candidate audit [2]. The exact research-file locators are the reduced-suspension construction in

`cusp_sphere_route_strengthening.tex`

and the one-copy map in

`cusp_s12_carrier_homotopy.tex`.

Theorem 30.47 (The stable even-subgroup morphism and its exact scope)**.** *Sixfold reduced suspension is an injection*

$$E^6 : \pi_6(S^3) \cong \mathbb{Z}/12 \hookrightarrow \pi_{12}(S^9) \cong \mathbb{Z}/24 \tag{1267}$$

whose image is the even subgroup. With compatible generators α and ν_9,

$$E^6(a\alpha) = 2a\nu_9.$$

Consequently there is a unique $a \in \mathbb{Z}/12$ for which the particular map in (1266) *satisfies*

$$[E^6 g] = 2a\nu_9. \tag{1268}$$

The coefficient a is not determined by point-set surjectivity or by the ordinary homology map.

For every pair of based maps $x, y : S^6 \to S^3$,

$$[x \wedge y] = 0 \in \pi_{12}(S^6). \tag{1269}$$

In particular,

$$g^{\wedge 4} : S^{24} \longrightarrow S^{12}$$

is based null-homotopic. Thus the $\mathbb{Z}/24$ in (1267) *is the ambient target group of the sixfold-suspended one-copy map $S^{12} \to S^9$. The separate 24-dimensional fourth-smash map is null-homotopic.*

Proof. The relevant initial suspension chain is

$$\pi_6(S^3) \xrightarrow{E} \pi_7(S^4) \xrightarrow{E} \pi_8(S^5).$$

Hatcher computes these groups as $\mathbb{Z}/12$, $\mathbb{Z} \oplus \mathbb{Z}/12$, and $\mathbb{Z}/24$, respectively. The Hopf fibration identifies the torsion summand of the middle group with the suspended first group. In the quotient defining the second suspension, twice the Hopf generator of the free summand is identified with a generator of this torsion summand [5, Theorem 5.41(b), p. 573, and pp. 577–578]. Choose compatible signs and generators $\alpha \in \pi_6(S^3)$ and $\nu_5 = E\nu_4 \in \pi_8(S^5)$. The displayed quotient relation gives $E^2\alpha = 2\nu_5$. Multiplication by two from $\mathbb{Z}/12$ to $\mathbb{Z}/24$ is injective: $24 \mid 2a$ implies $12 \mid a$. For $n \geq 5$, Freudenthal gives an isomorphism

$$E : \pi_{n+3}(S^n) \longrightarrow \pi_{n+4}(S^{n+1}), \qquad n + 3 \leq 2n - 2.$$

Applying it for $n = 5, 6, 7, 8$ and putting $\nu_9 = E^4\nu_5$ gives $E^6\alpha = 2\nu_9$ and $\pi_{12}(S^9) \cong \pi_3^{\mathrm{S}} \cong \mathbb{Z}/24$. The stable group is independently tabulated in [8, Table 1, p. 113]. This proves (1267) and the unique formula (1268).

For (1269), the ordered external smash has the exact factorization

$$[x \wedge y] = (E^3[x]) \circ (E^6[y])$$

by the reduced-join formula [11, Proposition 3.1]; its ordered-coordinate sign is $(-1)^{3\cdot 6} = +1$. The integral low-dimensional table gives $\pi_{12}(S^6) \cong \mathbb{Z}/2$ [6, opening table, p. 339]. The class $E^6[y]$ is even in $\pi_{12}(S^9) \cong \mathbb{Z}/24$, and postcomposition by $E^3[x]$ is a group homomorphism into the exponent-two group $\pi_{12}(S^6)$. Its value on an even element is therefore zero. Hence $g^{\wedge 2}$ is null-homotopic, and smashing its based null-homotopy with $g^{\wedge 2}$ proves the assertion for $g^{\wedge 4}$. □

Proposition 30.48 (The generator-free bridge between the two $\mathbb{Z}/12$-objects)**.** *The marked one-copy class defines a homomorphism*

$$\Theta_g : \mathbb{Z}/12 \longrightarrow \pi_6(S^3), \qquad \Theta_g([n]) = n[g]. \tag{1270}$$

Therefore

$$\Theta_g \circ \chi : SL_2(\mathbb{Z}) \longrightarrow \pi_6(S^3)$$

is a well-defined homomorphism from the marked peripheral data to the one-copy homotopy group. If α and ν_9 are compatible generators as in theorem 30.47 and $[g] = a\alpha$, then

$$E^6\Theta_g\chi(M) = 2a\,\chi(M)\nu_9 \in \pi_{12}(S^9) \cong \mathbb{Z}/24. \tag{1271}$$

For an integer lift $\widetilde{a}$ put $d = \gcd(\widetilde{a}, 12)$. Then

$$\ker \Theta_g = \langle [12/d] \rangle \subset \mathbb{Z}/12, \qquad \operatorname{im} \Theta_g = \langle [g] \rangle = \langle d\alpha \rangle, \qquad |\operatorname{im} \Theta_g| = 12/d.$$

The first group has order d. Since χ is surjective and E^6 injective, both composites have image order $12/d$, and $\ker(E^6\Theta_g\chi) = \chi^{-1}(\langle [12/d] \rangle)$. This bridge is canonical once the marked character χ, the based map g, and the based identification $S^6 \cong X$ are fixed. It does not prove $d = 1$, equivalently that $[g]$ is a generator or that Θ_g is an isomorphism, and it does not define an equivariant map of degenerations.

Proof. The source of Θ_g is well-defined because $\pi_6(S^3) \cong \mathbb{Z}/12$, so $12[g] = 0$. It is a homomorphism by the abelian group law on $\pi_6(S^3)$. Composition with the already defined χ is therefore a well-defined homomorphism. Writing $[g] = a\alpha$ and applying the homomorphism E^6 gives

$$E^6\Theta_g\chi(M) = \chi(M)E^6(a\alpha) = 2a\,\chi(M)\nu_9,$$

which is (1271) for every $M \in SL_2(\mathbb{Z})$. The congruence $n\widetilde{a} \equiv 0 \pmod{12}$ is equivalent to $12/d \mid n$, giving the kernel. Bézout's identity gives $\langle a\alpha \rangle = \langle d\alpha \rangle$, of order $12/d$. Surjectivity of χ preserves that image, and injectivity of E^6 preserves its order and the composite kernel. This group calculation by itself does not compute a. The explicit framed calculation in theorem 30.49 below does: $a = 0$, so $d = 12$, $\ker \Theta_g = \mathbb{Z}/12$, and every image displayed here is zero. □

30.6.3 The original framed cusp coefficient is zero

Let X be the smooth compact source of the original construction, $f : X \to B$ its map, and t the specified cusp coordinate. Put $W = f^{-1}(0)$, choose the original $x_0 \in W$, and retain the ordered quotient lattice basis induced by $(\widehat{\gamma}, \widehat{u})$, with kernel generated by $(\widehat{w}, \widehat{\delta})$. Thus the quotient torus is $\overline{T}^2 = \mathbb{R}^2/\mathbb{Z}^2$. The original periods are

$$Z(s) = sB_0 + C(t), \qquad e^{2\pi i s} = t, \qquad B_0 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad C(t) = \begin{pmatrix} 6\mu(t) & h(t) \\ b(t) - h(t) & \mu(t) \end{pmatrix}. \tag{1272}$$

Every entry of C is holomorphic at $t = 0$. The dense torus of the original toric cusp has coordinates (x_1, x_2, t), with $x = \exp(2\pi i \zeta)$. The residual lattice action is exactly

$$\Psi_\lambda(x, t) = \big(t^{B_0\lambda} \exp(2\pi i C(t)\lambda)x, t\big), \qquad \lambda \in \mathbb{Z}^2. \tag{1273}$$

It extends freely and properly discontinuously across the original unimodular toric filling. In particular the quotient projection is a local biholomorphism at every toric fixed point. These are the reconstructed data and proved local properties of the original source [2, §4, source pp. 24–29].

Let

$$K^{(1)} = (\mathbb{R}/\mathbb{Z} \times \{0\}) \cup (\{0\} \times \mathbb{R}/\mathbb{Z}), \qquad c : \overline{T}^2 \longrightarrow \overline{T}^2/K^{(1)} \cong S^2$$

be the specified degree-one cell quotient. Its restriction to the open two-cell is taken with the smooth coordinates used in the original regular-fibre description. Both radial endpoints and the complete basepoint cylinder are collapsed in

$$\Sigma_r(S^2, *) = \frac{[0, \epsilon] \times S^2}{(\{0, \epsilon\} \times S^2) \cup ([0, \epsilon] \times \{*\})} \cong S^3.$$

The original based map is

$$g(x) = [|t|, c(\overline{\pi}_f(x))] \quad (0 < |t| < \epsilon), \qquad g(W) = g(X \setminus \operatorname{int} N_0) = *. \tag{1274}$$

Theorem 30.49 (The original cusp map is null-homotopic)**.** *The based map $g : (X, x_0) \to (S^3, *)$ in* (1274) *is based null-homotopic. More precisely, a regular inverse image with its actual ordered normal framing induced by (r, c_1, c_2) is framed nullbordant inside one original cusp chart times $[0, 1]$, disjoint from $\{x_0\} \times [0, 1]$.*

Proof. The proof constructs the complete inverse image, compares its actual framing with an explicitly extendible framing, and gives the extension.

1. The exact complete inverse image. Retain the lift

$$u^0 = \begin{pmatrix} -1/4 \\ 1/4 \end{pmatrix}, \qquad \overline{u}^0 = (3/4, 1/4) \pmod{\mathbb{Z}^2}, \qquad B_0 u^0 = \begin{pmatrix} 1/4 \\ 1/4 \end{pmatrix}.$$

Thus $\overline{u}^0 \notin K^{(1)}$, and $B_0 u^0$ lies in the interior of the original triangle $\operatorname{conv}(0, e_1, e_2)$. Use its original unimodular chart

$$z_0 = t/(x_1 x_2), \qquad z_1 = x_1, \qquad z_2 = x_2, \qquad t = z_0 z_1 z_2. \tag{1275}$$

Set

$$\ell = \log|t| < 0, \qquad R(t) = -2\pi \operatorname{Im} C(t), \qquad A(t) = \ell B_0 + R(t).$$

For a representative $\zeta = Z(s)u + k$, where $u, k \in \mathbb{R}^2$, $\operatorname{Im} s = -\ell/(2\pi)$ gives the exact identity

$$\begin{pmatrix} \log|x_1| \\ \log|x_2| \end{pmatrix} = A(t)u. \tag{1276}$$

The quotient value is $u \bmod \mathbb{Z}^2$. Since B_0 is orthogonal, for all $v \in \mathbb{R}^2$,

$$\|A(t)v\| \geq (|\ell| - \|R(t)\|)\|v\|. \tag{1277}$$

The boundedness of R therefore makes $A(t)$ invertible on a sufficiently small punctured disc.

For any point of the fibre quotient over $\overline{u}^0$, subtracting the appropriate original period, equivalently applying $\Psi_{-\lambda}$, gives a representative with the exact lift $u = u^0$. This representative is unique: another residual translate has logarithmic moduli differing by $A(t)\lambda$, which vanishes only for $\lambda = 0$. Conversely every dense-torus point satisfying $\log|x| = A(t)u^0$ has this quotient value, since the remaining real parts of ζ are exactly the kernel coordinates k. For each fixed t, varying $k \bmod \mathbb{Z}^2$ gives all independent phases of x_1, x_2.

The following change of coordinates uses the full original period units:

$$h_j^0(t) = \exp(-2\pi i(C(t)u^0)_j), \qquad j = 1, 2, \tag{1278}$$

$$\mathcal{F}(z_0, z_1, z_2) = (w_0, w_1, w_2) = \left(\frac{z_0}{h_1^0(t)h_2^0(t)},\, h_1^0(t)z_1,\, h_2^0(t)z_2\right). \tag{1279}$$

The functions h_j^0 are nowhere-zero holomorphic units, including at $t = 0$. The product $w_0w_1w_2$ is exactly t. The inverse of $\mathcal{F}$ is

$$(w_0, w_1, w_2) \longmapsto \left(w_0h_1^0(t)h_2^0(t),\, \frac{w_1}{h_1^0(t)},\, \frac{w_2}{h_2^0(t)}\right), \qquad t = w_0w_1w_2.$$

Thus this is a biholomorphism of the original chart restricted to the cusp disc, not a replacement of $C(t)$.

For $r_0 \in (0, \epsilon)$, the entire inverse image of $y_0 = [r_0, c(\overline{u}^0)]$ becomes exactly

$$T_{r_0} = \{|w_0| = \rho_0,\ |w_1| = \rho_1,\ |w_2| = \rho_2\}, \qquad (\rho_0, \rho_1, \rho_2) = (r_0^{1/2}, r_0^{1/4}, r_0^{1/4}). \tag{1280}$$

Indeed, $\log|h^0(t)| = 2\pi \operatorname{Im} C(t)u^0 = -R(t)u^0$, so (1276) gives $\log|w_j| = \ell(B_0u^0)_j$ for $j = 1, 2$; the product identity gives the exponent $1/2$ for w_0. Conversely these moduli recover $u = u^0$ by invertibility of $A(t)$. The angular map is explicitly

$$(\arg t, \arg w_1, \arg w_2) \longleftrightarrow (\arg w_0, \arg w_1, \arg w_2), \qquad \arg w_0 = \arg t - \arg w_1 - \arg w_2 \pmod{2\pi},$$

with inverse $\arg t = \arg w_0 + \arg w_1 + \arg w_2$. Its integral matrix has determinant one. This includes the complete angular mapping torus and all kernel phases, with their original signs.

Let o be the origin of (1275). The original free properly discontinuous action supplies a neighborhood V of o on which the quotient projection is injective. To see the local injectivity directly, first choose a relatively compact neighborhood; proper discontinuity leaves only finitely many translates that can intersect it. None of those nonidentity translates fixes o. Shrinking finitely many times separates the neighborhood from all of them. The quotient map is a local biholomorphism, so its restriction to V is a chart of the original source. Choose a polydisc about 0 whose closure lies in $\mathcal{F}(V)$. For sufficiently small r_0, both T_{r_0} and

$$\{|w_0| \leq \rho_0,\ |w_1| = \rho_1,\ |w_2| = \rho_2\} \tag{1281}$$

lie in this polydisc. This proves that the whole inverse image embeds in one chart and supplies space for the bounding manifold below.

2. The actual normal framing, including its angle dependence. Put $L_j = \log|w_j|$, and use the unique real lift of the quotient coordinate near u^0. From the exact change of coordinates,

$$\begin{pmatrix} L_1 \\ L_2 \end{pmatrix} = \ell B_0u^0 + A(t)(u - u^0), \qquad L_0 = \ell - L_1 - L_2. \tag{1282}$$

Differentiating on $u = u^0$ annihilates the term $(dA(t))(u - u^0)$. Hence no derivative of $C(t)$ has been omitted: the exact differential there is

$$\begin{pmatrix} dL_0 \\ dL_1 \\ dL_2 \end{pmatrix} = J(t) \begin{pmatrix} d\ell \\ du_1 \\ du_2 \end{pmatrix}, \quad J(t) = \begin{pmatrix} 1/2 & -A_{11} - A_{21} & -A_{12} - A_{22} \\ 1/4 & A_{11} & A_{12} \\ 1/4 & A_{21} & A_{22} \end{pmatrix}. \tag{1283}$$

Adding the second and third rows to the first gives $(1, 0, 0)$, and expansion in that row proves $\det J(t) = \det A(t)$.

Let $D = Dc_{\overline{u}^0}$ be the actual two-by-two derivative in the ordered open-cell target coordinates (c_1, c_2). It is a fixed invertible matrix; all its entries and signs are retained. Let

$$H = \begin{pmatrix} r_0 & 0 & 0 \\ 0 & D_{11} & D_{12} \\ 0 & D_{21} & D_{22} \end{pmatrix}.$$

Let n_j be the real radial vector w_j in the j-th complex coordinate, with other components zero. Then $dL_i(n_j) = \delta_{ij}$. Writing $N = (n_0, n_1, n_2)$ as the matrix with these vector columns, the actual dg-dual normal framing is, in the normal quotient bundle,

$$\mathfrak{n}(t) = NJ(t)H^{-1}. \tag{1284}$$

Indeed, $dg(N) = HJ(t)^{-1}$, so multiplication by $J(t)H^{-1}$ gives the identity. In the Euclidean metric of this chart the radial vectors are perpendicular to the product torus; a different source metric changes their representatives by the canonical projection to its normal bundle and preserves this normal-quotient framing.

Put $\ell_0 = \log r_0$. Decrease r_0 further, if necessary, so that $|\ell_0| > 2 \sup_{|t| \le r_0} \|R(t)\|$. For $0 \le \sigma \le 1$, define only a homotopy of frames by

$$A_\sigma(t) = \ell_0 B_0 + \sigma R(t),$$

and let $J_\sigma(t)$ be (1283) with A_σ in place of A. For every $v \in \mathbb{R}^2$,

$$\|A_\sigma(t)v\| \ge (|\ell_0| - \|R(t)\|)\|v\| > 0 \quad (v \ne 0).$$

Thus all matrices are invertible. At $\sigma = 0$,

$$J_0 = \begin{pmatrix} 1/2 & \ell_0 & -\ell_0 \\ 1/4 & 0 & \ell_0 \\ 1/4 & -\ell_0 & 0 \end{pmatrix}, \qquad \det J_0 = \ell_0^2 > 0. \tag{1285}$$

Continuity also proves $\det J_\sigma(t) > 0$ for every σ, t. The homotopy $NJ_\sigma(t)H^{-1}$, run from $\sigma = 1$ to $\sigma = 0$, compares the actual frame with the exact constant-matrix frame NP, where

$$P = J_0 H^{-1}. \tag{1286}$$

The original embedded torus and original functions $C(t)$ have not changed. A framing homotopy is a framed cylinder: choose a smooth parameter function constant near its endpoints and use the displayed frames on $T_{r_0} \times [0, 1]$.

3. A collared framed nullbordism with the exact constant matrix. Use a new time coordinate $v \in [0,1]$. Define

$$q(v) = \begin{cases} 0, & 0 \le v \le 1/4, \\ \rho_0^2 \exp(4 - 1/(v - 1/4)), & 1/4 < v \le 1. \end{cases} \tag{1287}$$

This function is smooth: all right derivatives of $\exp(-1/(v-1/4))$ at $1/4$ vanish. It is strictly increasing for $v > 1/4$, satisfies $q(1/2) = \rho_0^2$, and $q'(1/2) = 16\rho_0^2 > 0$. In the chart times $[0,1]$, put

$$\mathcal{W} = \{|w_0|^2 + q(v) = \rho_0^2, \quad |w_1|^2 = \rho_1^2, \quad |w_2|^2 = \rho_2^2\}. \tag{1288}$$

No point has $v > 1/2$. The three defining differentials are independent: the latter two have nonzero gradients in their separate complex factors; the first has nonzero w_0-gradient whenever $w_0 \neq 0$, and at $w_0 = 0$ it has the nonzero v-derivative $q'(1/2)$. Thus $\mathcal{W}$ is a smooth compact four-manifold. Its only boundary is $T_{r_0} \times \{0\}$, and on $0 \le v \le 1/4$ it is the exact product collar $T_{r_0} \times [0, 1/4]$. The north locus $w_0 = 0, v = 1/2$ is an interior cap: the implicit function theorem expresses v smoothly in terms of $(\operatorname{Re} w_0, \operatorname{Im} w_0)$ there. The first factor is a disk obtained by rotating its radial profile with this smooth cap and one boundary circle. Therefore $\mathcal{W} \cong D^2 \times T^2$. Its spatial coordinates lie in (1281), so the complete manifold lies in the injective original chart.

In the Euclidean chart product metric define

$$V_0 = (w_0, 0, 0, q'(v)/2), \qquad V_1 = (0, w_1, 0, 0), \qquad V_2 = (0, 0, w_2, 0).$$

These are one half of the gradients of the three defining equations in (1288). They are mutually orthogonal and never zero. For example, if $F_0 = |w_0|^2 + q(v) - \rho_0^2$, then

$$dF_0(V_0) = 2|w_0|^2 + \frac{1}{2} q'(v)^2 > 0$$

at every point of $\mathcal{W}$. On the entire collar $q' = 0$, so $(V_0, V_1, V_2) = (N, 0)$ exactly. Consequently

$$(V_0, V_1, V_2)P \tag{1289}$$

is a framing of the normal three-plane bundle of $\mathcal{W}$, and its restriction to the collar is exactly the framing NP from (1286), with every original target-coordinate sign retained. Concatenate the framing cylinder of the preceding paragraph with this nullbordism, rescaling the two interval lengths. Their framings agree exactly on the common product collar, so the concatenation is a smooth framed nullbordism of the actual inverse image.

4. The based map represented by this framed inverse image. Choose the injective chart to exclude x_0 if its center is not x_0. If its center is x_0, the constructed nullbordism still excludes x_0, since $|w_1| = \rho_1 > 0$ and $|w_2| = \rho_2 > 0$ everywhere on it. Compactness therefore gives tubular neighborhoods for the framed cylinder and cap disjoint from $\{x_0\} \times [0,1]$.

The original continuous quotient map is smooth and a submersion near its entire inverse image of y_0. First make it constant in a neighborhood of x_0, by a based homotopy that changes nothing near that inverse image. Explicitly, take a target chart $\kappa : O \to \mathbb{R}^3$ with $\kappa(*) = 0$, whose image is a ball, and $y_0 \notin O$. By continuity choose a neighborhood U of x_0 with $g(U) \subset O$, and a smooth bump function β supported in U and equal to one near x_0. For $x \in U$, replace $g(x)$ during this homotopy by $\kappa^{-1}((1 - a\beta(x))\kappa(g(x)))$, $0 \le a \le 1$, and leave g fixed off U. Convexity of the target chart image makes the formula well-defined, the vanishing of β near the boundary of its support proves continuity of the pasted formula, and $\kappa(g(x_0)) = 0$ makes the homotopy based. It never introduces an inverse image of y_0. Its endpoint is smooth near x_0 and agrees with g near the entire old inverse image.

Denote the endpoint of the preceding homotopy by g_b. Choose one closed tubular neighborhood U_1 of the inverse image, contained in the smooth submersion region and with the entire inverse image contained in $\operatorname{int} U_1$. The compact set $X \setminus \operatorname{int} U_1$ has image under g_b missing y_0. In a fixed target metric put

$$d_0 = \min_{x \in X \setminus \operatorname{int} U_1} d(g_b(x), y_0) > 0.$$

Relative smooth approximation, fixed on the whole closed set U_1 and on a neighborhood of x_0, gives a smooth based representative g_{sm} whose error on $X \setminus \operatorname{int} U_1$ is less than $d_0/2$. It therefore has the same inverse image and differential on U_1, and $d(g_{\mathrm{sm}}(x), y_0) > d_0/2$ on its complement. Thus there is no uncontrolled intervening region and no additional inverse image. Make the approximation also closer than a fixed target injectivity radius; its short geodesics give the based homotopy.

The relative Pontryagin construction now applies to the explicit framed nullbordism. It produces a based homotopy to a map missing y_0; stereographic contraction of $S^3 \setminus \{y_0\} \cong \mathbb{R}^3$ to the image of $*$ is based and contracts that last map. For completeness, the dependence on the framed inverse image in this argument is precisely the local product-neighborhood construction and the interpolation lemma of [1, §7, Theorem B and Lemma 4, pp. 43, 48–50]. Their relative versions fix the source basepoint because all tubular modifications are away from it and stereographic interpolation between equal values keeps that value fixed. A framed cylinder realizes a framing homotopy; the empty Pontryagin manifold realizes a map missing the chosen regular value. This proves the claimed based null-homotopy of the exact map g. □

Corollary 30.50. *Under the original fixed based identification $X \cong S^6$, the one-copy coefficient is*

$$[g] = 0 \in \pi_6(S^3) \cong \mathbb{Z}/12, \qquad a = 0 \pmod{12}, \qquad [E^6 g] = 0 \in \pi_{12}(S^9) \cong \mathbb{Z}/24.$$

The previously defined homomorphism $\Theta_g : \mathbb{Z}/12 \to \pi_6(S^3)$, $\Theta_g([n]) = n[g]$, and its composite with the marked peripheral character are zero.

Proof. The theorem states that the actual based map is null-homotopic. Precomposition with the fixed based identification preserves a based null-homotopy. Suspension preserves it as well. Thus the unique coefficient of $[g]$ in any generator of $\mathbb{Z}/12$ is zero. For every $[n]$, $\Theta_g([n]) = n \cdot 0 = 0$, and every composite with this homomorphism is zero. □

Exact scope. The proof computes this sphere-valued postcomposition. It does not compute the homotopy class of the earlier conical-target map $q_2 : X \to \operatorname{Susp}(S^1 \times \overline{T}^2)$, and it does not assert a null-homotopy for that different target. Point-set surjectivity of g and all of its stated metric restrictions remain properties of the original map: a homotopy-class computation does not alter the defining formula (1274).

30.6.4 The retained-angle cusp quotient and its nonzero top-cell class

The exact object. The notation refers to the specified threefold X, its fixed based smooth identification with S^6, and its cusp filling from [12, §§3.7,4.1–4.5]. The native quotient and its compact extension are the maps in [13]; their full formulas are recalled below. This calculation uses the original period matrix and native period lattices. It computes a new postcomposition of that particular quotient; it does not use the homotopy class of the angle-forgetting map $g : X \to S^3$.

Original coordinates and the specified sphere-valued map The ordered quotient lattice is

$$\overline{\Lambda} = \mathbb{Z}\langle \overline{\widehat{\gamma}}, \overline{\widehat{u}} \rangle, \qquad K_\Lambda = \mathbb{Z}\langle \widehat{w}, \widehat{\delta} \rangle, \qquad \overline{T}^2 = \mathbb{R}^2/\mathbb{Z}^2.$$

In these ordered bases the original period data are

$$B_0 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad Z(s) = sB_0 + C(t), \qquad t = e^{2\pi i s}, \qquad C(t) = \begin{pmatrix} 6\mu(t) & h_{\mathrm{per}}(t) \\ b(t) - h_{\mathrm{per}}(t) & \mu(t) \end{pmatrix}. \quad (1290)$$

Here $b = \beta + \tau$, $s = \tau - h_{\mathrm{per}}(t)$, and every entry of C is holomorphic at $t = 0$, exactly as in the original source. The subscript in h_{per} distinguishes that period function from the sphere-valued map below. Write

$$R(t) = -2\pi \operatorname{Im} C(t), \qquad \ell = \log|t|, \qquad A(t) = \ell B_0 + R(t). \tag{1291}$$

Choose a closed cusp disc $|t| \le \epsilon$ inside the original coordinate disc and small enough for the quotient construction. On a slightly larger closed disc put $M = \sup \|R(t)\| < \infty$. We will choose a regular value with $0 < r_0 < \epsilon$ and $|\log r_0| > M$. This chooses a point of the already specified map; it does not change that map or the cutoff radius ϵ.

The original covering model on its dense torus is

$$x_j = e^{2\pi i \zeta_j}, \qquad \Psi_n(x,t) = \big(e^{2\pi i C(t) n} t^{B_0 n} x, t\big), \qquad n \in \mathbb{Z}^2. \tag{1292}$$

For a branch of s, writing $\zeta = Z(s)u + k$ with $u, k \in \mathbb{R}^2$ gives

$$\log|x| = A(t)u. \tag{1293}$$

Indeed, $\operatorname{Im} s = -\ell/(2\pi)$, so $-2\pi \operatorname{Im}(Z(s)u + k) = \ell B_0 u + R(t)u$. Since B_0 is orthogonal,

$$\big|(\ell B_0 + vR(t))\xi\big| \ge (|\ell| - vM)|\xi| > 0 \quad (0 \le v \le 1,\ \xi \ne 0) \tag{1294}$$

whenever $|\ell| > M$. Thus $A(t)$ is invertible there. Equation (1292) changes $A(t)^{-1}\log|x|$ by n, so the native quotient is exactly

$$\overline{\pi}_f([x,t]) = [A(t)^{-1}\log|x|] \in \mathbb{R}^2/\mathbb{Z}^2. \tag{1295}$$

This also follows from the flat marking in [13].

Let

$$T^3 = S^1_\phi \times \overline{T}^2, \qquad S^1_\phi = \mathbb{R}/(2\pi\mathbb{Z}),$$

with coordinate basepoints $\phi = 0$, $u_1 = 0$, $u_2 = 0$. Its coordinate two-skeleton is the following specified closed subset:

$$K^{(2)} = (\{0\} \times \overline{T}^2) \cup (S^1_\phi \times \{u_1 = 0\}) \cup (S^1_\phi \times \{u_2 = 0\}). \tag{1296}$$

The product CW structure has one open three-cell, with coordinates $(\phi, u_1, u_2) \in (0, 2\pi) \times (0,1)^2$. Consequently

$$c_3 : T^3 \longrightarrow T^3/K^{(2)} \cong S^3$$

is the degree-one quotient, where the orientation of S^3 is fixed by the displayed coordinate order. One explicit identification of its open cell with $\mathbb{R}^3$ is

$$(\phi, u_1, u_2) \longmapsto \big(-\cot(\phi/2), -\cot(\pi u_1), -\cot(\pi u_2)\big).$$

Each derivative in its own variable is strictly positive, and approach to any boundary face leaves every compact subset of $\mathbb{R}^3$. Extending by its point at infinity gives the stated based sphere identification and verifies its continuity on the collapsed skeleton.

The exact conical target and map are

$$\mathcal{A} = \frac{[0,\epsilon] \times T^3}{\{0\} \times T^3 \sim v_0, \quad \{\epsilon\} \times T^3 \sim v_1}, \tag{1297}$$

$$q_2(x) = [r, e^{i\phi}, \overline{\pi}_f(x)] \qquad (t(f(x)) = re^{i\phi},\ 0 < r < \epsilon),$$

$$q_2(W) = v_0, \qquad q_2(X \setminus \operatorname{int} N_0) = v_1. \tag{1298}$$

Fix the same source basepoint $x_{\mathrm{base}} \in W$. Define

$$p_3 : \mathcal{A} \longrightarrow \Sigma_r(S^3, *), \qquad p_3([r,z]) = [r, c_3(z)], \qquad H = p_3 q_2 : (X, x_{\mathrm{base}}) \longrightarrow (S^4, *). \tag{1299}$$

The reduced suspension here collapses both endpoint copies of S^3 and the entire interval over $*$ to one point. Thus p_3 sends both v_0 and v_1 to $*$ and is well-defined and based. It is continuous by the defining quotient topology. An explicit identification of its open four-cell with $\mathbb{R}^4$ is

$$(r, \phi, u_1, u_2) \longmapsto \big(-\cot(\pi r/\epsilon), -\cot(\phi/2), -\cot(\pi u_1), -\cot(\pi u_2)\big). \tag{1300}$$

This fixes the target orientation. Replacing r locally by $\ell = \log r$ preserves this orientation, since $dr/d\ell = r > 0$. The ordered coordinates in (1300) specify the sphere identification used throughout this calculation.

Theorem 30.51. *For the exact map* (1299)*,*

$$[H] = \eta_4 \circ \eta_5 \neq 0 \quad \textit{in} \quad \pi_6(S^4) \cong \mathbb{Z}/2, \tag{1301}$$

where $\eta_n : S^{n+1} \to S^n$ is the $(n-2)$-fold reduced suspension of the complex Hopf map. In particular q_2 is not based null-homotopic. Its stabilized top-cell postcomposition is the element $\eta^2 \in \pi_2^{\mathrm{S}}$.

The entire regular inverse image lies in one original chart Choose the actual lift

$$u_0 = (-1/4, 1/4)^{\mathsf{T}}, \qquad p_0 = B_0 u_0 = (1/4, 1/4)^{\mathsf{T}}, \qquad \phi_0 = \pi. \tag{1302}$$

Modulo $\mathbb{Z}^2$, u_0 is $(3/4, 1/4)$ in the open cell. Put $t_0 = r_0 e^{i\phi_0}$ and $y_0 = [r_0, c_3(e^{i\phi_0}, [u_0])]$. The numbers r_0 and t_0 will remain nonzero throughout the framing calculation.

The original triangle $\{0, e_1, e_2\}$ has toric chart

$$\mathcal{U}_\sigma \cong \mathbb{C}^3, \qquad (z_0, z_1, z_2) = (t/(x_1 x_2), x_1, x_2), \qquad t = z_0 z_1 z_2. \tag{1303}$$

These characters are dual to $(0,0,1), (1,0,1), (0,1,1)$, respectively, as direct pairing verifies. This is the chart of [12, Lemma 4.2(iii)], with its original coordinates.

For $j = 1, 2$, introduce the nowhere vanishing holomorphic functions

$$h_j(t) = \exp\big(-2\pi i (C(t)u_0)_j\big). \tag{1304}$$

The following is an exact coordinate map, with the exact inverse displayed alongside it:

$$F(z) = (w_0, w_1, w_2) = \big(z_0/(h_1(t)h_2(t)),\ h_1(t)z_1,\ h_2(t)z_2\big), \tag{1305}$$

$$F^{-1}(w) = \big(w_0 h_1(t) h_2(t),\ w_1/h_1(t),\ w_2/h_2(t)\big), \quad t = w_0 w_1 w_2. \tag{1306}$$

Both maps are holomorphic wherever $|t|$ is in the period disc, both preserve t, and substitution proves that both composites are the identity. In particular the maps extend across the origin and across every coordinate divisor. No period matrix is replaced by a different matrix in this coordinate change.

Lemma 30.52. *For sufficiently small $r_0 > 0$, the whole inverse image $M = H^{-1}(y_0)$ is represented injectively inside one contractible coordinate neighborhood in X. In its w-coordinates it is exactly the embedded two-torus*

$$w(a,b) = \big(\rho_0 e^{i(\phi_0 - a - b)}, \rho_1 e^{ia}, \rho_2 e^{ib}\big), \qquad \rho_0 = r_0^{1/2}, \quad \rho_1 = \rho_2 = r_0^{1/4}, \quad (a,b) \in \big(\mathbb{R}/(2\pi\mathbb{Z})\big)^2. \tag{1307}$$

Proof. The point y_0 lies off the collapsed subset of the target, so (1299) and (1298) imply

$$M = \{x : t(f(x)) = t_0,\ \overline{\pi}_f(x) = [u_0]\}.$$

Any lift of such a point to $(\mathbb{C}^*)^2 \times \{t_0\}$ has $A(t_0)^{-1} \log|x| = u_0 + n$ for one $n \in \mathbb{Z}^2$. Applying the actual transformation Ψ_{-n} from (1292) produces a lift with $\log|x| = A(t_0)u_0$. If two such lifts represent the same

point, Ψ_n relates them and changes their lifted quotient coordinate by n; hence $n = 0$, and the lifts are identical. This proves both completeness and injectivity of this chosen representative.

Since $\log|h_j(t)| = 2\pi \operatorname{Im}(C(t)u_0)_j = -(R(t)u_0)_j$, its w-coordinates satisfy

$$(\log|w_1|, \log|w_2|)^{\mathsf{T}} = A(t_0)u_0 - R(t_0)u_0 = \ell_0 p_0, \qquad \ell_0 = \log r_0.$$

Thus $|w_1| = |w_2| = r_0^{1/4}$, and the unchanged product $w_0 w_1 w_2 = t_0$ forces $|w_0| = r_0^{1/2}$ and the displayed phase. Conversely every pair (a, b) in (1307) gives one such lift. The original kernel coordinates k_j had period one in $x_j = e^{2\pi i \zeta_j}$; the phase coordinates a, b have period 2π, with no cover or change of lattice.

The covering $Y_\epsilon \to N_0$ is injective on a neighborhood of the origin in (1303), because the original action is free and properly discontinuous. Choose a ball V centered at that origin with closure inside such a neighborhood and the period disc. Choose a ball V_w about the origin with closure contained in $F(V)$. Every coordinate in (1307) tends uniformly to zero as $r_0 \downarrow 0$. Therefore for sufficiently small r_0 the entire torus lies in V_w. The inverse image of V_w under F, followed by the injective covering projection, is the required contractible chart in X. □

The full differential and its ordered normal framing We calculate in a neighborhood of the compact torus (1307). The quotient coordinates admit a unique local lift near u_0. Set

$$v = (\log|w_1|, \log|w_2|)^{\mathsf{T}}, \quad \ell = \log|w_0 w_1 w_2|, \quad \phi = \arg(w_0 w_1 w_2)$$

using the branch of ϕ near $\phi_0 = \pi$. Equations (1293) and (1305) give the exact identity

$$v = \ell B_0 u + R(t)(u - u_0), \qquad u = u_0 + A(t)^{-1}(v - \ell p_0). \tag{1308}$$

Differentiate this identity before restricting to the inverse image:

$$du = -A(t)^{-1}(dA(t))A(t)^{-1}(v - \ell p_0) + A(t)^{-1}(dv - p_0\, d\ell). \tag{1309}$$

On M the vector $v - \ell p_0$ is exactly zero. Therefore

$$du|_M = A_0^{-1}(dv - p_0\, d\ell), \qquad A_0 = A(t_0). \tag{1310}$$

This explicitly accounts for all derivatives of the original bounded period terms; none are omitted or estimated away.

Identify $T_w\mathbb{C}^3$ with $\mathbb{C}^3$ in the given coordinate chart, and define real tangent vectors

$$L_j = (0, \dots, w_j, \dots, 0), \qquad Q_j = (0, \dots, iw_j, \dots, 0) \qquad (0 \le j \le 2). \tag{1311}$$

Thus $d\log|w_i|(L_j) = \delta_{ij}$, $d\arg w_i(Q_j) = \delta_{ij}$, and the crossed evaluations vanish. The ordered tangent vectors of the parametrization (1307) are

$$T_a = -Q_0 + Q_1, \qquad T_b = -Q_0 + Q_2. \tag{1312}$$

They are independent because their w_1 and w_2 components, respectively, are nonzero. For the target order (ℓ, ϕ, u_1, u_2), define the following four lifts of its coordinate vectors:

$$N_\ell = \tfrac{1}{2}L_0 + \tfrac{1}{4}L_1 + \tfrac{1}{4}L_2, \qquad N_\phi = Q_0, \tag{1313}$$

$$N_i = -((A_0)_{1i} + (A_0)_{2i})L_0 + (A_0)_{1i}L_1 + (A_0)_{2i}L_2 \qquad (i = 1, 2). \tag{1314}$$

Let $P = (\ell, \phi, u_1, u_2)$. Evaluation using (1310) gives the entire differential table

	N_ℓ	N_ϕ	N_1	N_2	T_a	T_b
$d\ell$	1	0	0	0	0	0
$d\phi$	0	1	0	0	0	0
du_1	0	0	1	0	0	0
du_2	0	0	0	1	0	0

. (1315)

In particular P, and therefore H in its open target cell, is a submersion along M. The classes of $(N_\ell, N_\phi, N_1, N_2)$ in $TX|_M/TM$ give the specified normal four-framing. Orthogonal representatives are not required for this quotient-bundle definition; projecting the displayed lifts along TM to any chosen orthogonal complement preserves the quotient frame.

If the normal frame is specified using the literal four real coordinates in (1300), their differential at the chosen value is

$$dH_{\text{cell}} = K\,dP, \qquad K = \operatorname{diag}\left(\frac{\pi r_0}{\epsilon}\csc^2\frac{\pi r_0}{\epsilon}, \frac{1}{2}, 2\pi, 2\pi\right).$$

The first factor includes $dr = r_0\,d\ell$, the second uses $\phi_0 = \pi$, and the final two use the actual representatives $(3/4, 1/4)$ of u_0. Thus, writing $\mathcal{N} = (N_\ell, N_\phi, N_1, N_2)$, the literal dual frame is $\mathcal{N}K^{-1}$. All four entries of K are positive constants on M. The explicitly positive diagonal matrices $(1-v)K^{-1} + vI_4$, $0 \le v \le 1$, give a homotopy from that literal frame to $\mathcal{N}$. This retains the original target coordinates and establishes that either coordinate description gives the same framed bordism class.

For completeness, the original period matrix admits the following explicit homotopy of this normal framing. Put

$$A_{0,v} = \ell_0 B_0 + vR(t_0), \qquad 0 \le v \le 1, \tag{1316}$$

and replace A_0 by $A_{0,v}$ only in (1314). Equation (1294) proves invertibility for every v. Their images under the actual (du_1, du_2) are the columns of $A_0^{-1}A_{0,v}$, so these are normal frames throughout the homotopy. This is a homotopy of the computed frame on the same embedded torus, not a replacement of the original degeneration by a different one.

Lemma 30.53 (The exact orientation sign)**.** *Use the complex real-coordinate orientation, whose ordered coordinates are*

$$(\operatorname{Re} w_0, \operatorname{Im} w_0, \operatorname{Re} w_1, \operatorname{Im} w_1, \operatorname{Re} w_2, \operatorname{Im} w_2).$$

Then one has

$$\det(N_\ell, N_\phi, N_1, N_2, T_a, T_b) = -\det(A_0)|w_0 w_1 w_2|^2 < 0. \tag{1317}$$

Hence the tangent orientation induced by the ordered normal frame is the orientation of $(-T_a, T_b)$.

Proof. The ordered frame $(L_0, Q_0, L_1, Q_1, L_2, Q_2)$ has determinant $|w_0|^2|w_1|^2|w_2|^2 > 0$ in real coordinates. Regroup its rows as $(L_0, L_1, L_2, Q_0, Q_1, Q_2)$; this permutation has three inversions and sign -1. Regroup the columns of the frame in (1317) as $(N_\ell, N_1, N_2, N_\phi, T_a, T_b)$; this has two inversions and sign $+1$. The angular block is $\left(\begin{smallmatrix} 1 & -1 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{smallmatrix}\right)$, of determinant one. The radial block has first column $(1/2, 1/4, 1/4)^{\mathsf{T}}$ and the other columns given by (1314). Adding its last two rows to its first row gives first row $(1, 0, 0)$, with remaining lower-right block A_0. Its determinant is $\det A_0$. This proves the equality. Equation (1316) joins A_0 to $\ell_0 B_0$ through invertible matrices, so $\operatorname{sign}\det A_0 = \operatorname{sign}\det(\ell_0 B_0) = +1$. The inequality and the tangent orientation follow. □

An explicit contraction of the full frame Define a complex-linear matrix, also regarded as a real-linear matrix,

$$D(a,b) = \operatorname{diag}(e^{-i(a+b)}, e^{ia}, e^{ib}) \in \mathrm{SU}(3). \tag{1318}$$

By (1307), every L_j and Q_j at (a,b) is $D(a,b)$ times its value at $(0,0)$. The coefficients in (1313), (1314), and (1312) are constant on M, since $t = t_0$. It follows that the positively oriented full frame is exactly

$$(N_\ell, N_\phi, N_1, N_2, -T_a, T_b)_{(a,b)} = D(a,b)G_0, \tag{1319}$$

where $G_0 \in \mathrm{GL}^+(6,\mathbb{R})$ is its single value at $(0,0)$. Thus neither an unspecified phase nor a position-dependent real-linear matrix remains.

Here is a periodic contraction of D, with all parameters specified. For $v \in [0,1]$ put

$$s_v = \sin(\pi v/2), \qquad c_v = \cos(\pi v/2), \qquad Q_v(\theta) = \begin{pmatrix} s_v e^{-i\theta} & -c_v \\ c_v & s_v e^{i\theta} \end{pmatrix}, \qquad K_v(\theta) = Q_v(\theta) Q_v(0)^*.$$

The two columns of Q_v have norm one and zero Hermitian inner product, and $\det Q_v = s_v^2 + c_v^2 = 1$. Thus $Q_v, K_v \in \mathrm{SU}(2)$. Direct substitution gives

$$K_0(\theta) = I, \quad K_1(\theta) = \mathrm{diag}(e^{-i\theta}, e^{i\theta}), \quad K_v(0) = I.$$

Let $\iota_{01}, \iota_{02} : \mathrm{SU}(2) \to \mathrm{SU}(3)$ act in the ordered coordinate planes $(0,1)$ and $(0,2)$, respectively, and as the identity in the unused coordinate. Then

$$D_v(a,b) = \iota_{01}(K_v(a))\, \iota_{02}(K_v(b)) \tag{1320}$$

is a well-defined continuous map on $[0,1] \times (\mathbb{R}/(2\pi\mathbb{Z}))^2$, with $D_0 = I$ and $D_1 = D$. This proves the needed null-homotopy directly, including both phase seams. In particular $D_v(a,b)G_0$ contracts the comparison matrix in (1319) to one constant positive matrix.

Proposition 30.54. *The stable tangential framing induced by the actual normal four-frame of $H^{-1}(y_0)$ agrees, up to homotopy and the orientation sign* (1317)*, with the Lie framing (∂_a, ∂_b) on the native two-torus.*

Proof. In the coordinate chart, the stable tangent trivialization is the isomorphism

$$\mathbb{R}^4 \oplus TM \longrightarrow T\mathbb{C}^3|_M, \qquad (v, V) \longmapsto v_1 N_\ell + v_2 N_\phi + v_3 N_1 + v_4 N_2 + V.$$

Using the intrinsic frame $(-\partial_a, \partial_b)$ on TM, its matrix is precisely (1319). The homotopy (1320) therefore makes this comparison constant. A constant positive matrix changes no homotopy class of stable framing: Gram–Schmidt deforms it to an element of $SO(6)$, and an orthogonal matrix of determinant one can be reduced to the identity by finitely many oriented plane rotations, each joined to the identity by varying its angle. This proves the asserted comparison within the chart.

It remains to compare the chart with the ambient sphere framing. Pull back the usual stable tangent trivialization $TS^6 \oplus \mathbb{R} \cong \mathbb{R}^7$ under the fixed based diffeomorphism $X \cong S^6$. On the contractible w-ball containing M, it differs from the coordinate trivialization plus one real line by a continuous map into $\mathrm{GL}(7,\mathbb{R})$. Contracting the ball to its center contracts this comparison map to a constant. Thus its restriction to M contributes no varying stable framing class. If the chosen sphere orientation disagrees with the complex orientation, the remaining constant has negative determinant and reverses the resulting framed bordism class. Both that sign and the explicit sign (1317) disappear in the order-two class computed below. This argument compares actual ambient trivializations on a contractible chart; it does not suppose that $\mathbb{C}^3$ globally trivializes TX. □

Nonzero framed bordism and the exact unstable class We give the nonvanishing test for the resulting torus explicitly. A stable framing gives a spin structure by lifting its oriented frame to the trivial Spin-bundle. On the Lie-framed circle this is the disconnected double cover, whereas the spin structure induced from the boundary of a framed disc is the connected double cover: as the boundary tangent turns once, its 2π-rotation lifts from 1 to -1 in $\mathrm{Spin}(2)$. We call the former nonbounding and the latter bounding.

For a closed oriented spin surface S, the spin quadratic refinement

$$q_S : H_1(S; \mathbb{F}_2) \longrightarrow \mathbb{F}_2$$

takes a class represented by an embedded circle to zero or one according to whether the induced spin structure, using its normal line in the surface, is bounding or nonbounding. It satisfies

$$q_S(x+y) = q_S(x) + q_S(y) + (x \cdot y). \tag{1321}$$

This is the usual spin quadratic form: resolving transverse intersections adds one tangent turn per intersection, and the 2π-spin-lift calculation above gives the last term. The well-defined correspondence between spin structures and such quadratic refinements is given in [18, Theorem 3.2 and Remark 3.3]; its original surface proof is [17].

For a symplectic basis $(e_i, f_i)_{i=1}^{g}$, put

$$\operatorname{Arf}(S) = \sum_{i=1}^{g} q_S(e_i) q_S(f_i) \in \mathbb{F}_2. \tag{1322}$$

The basis independence can be seen without selecting a preferred basis: factor the finite sum over the symplectic planes to obtain

$$\sum_{x \in H_1(S;\mathbb{F}_2)} (-1)^{q_S(x)} = \prod_{i=1}^{g} \sum_{a,b \in \mathbb{F}_2} (-1)^{a q_S(e_i) + b q_S(f_i) + ab} = 2^g (-1)^{\sum_i q_S(e_i) q_S(f_i)}.$$

The left side has no choice of basis, proving the assertion.

Here is also the bordism-invariance argument needed for the present obstruction. A spin bordism has a handle decomposition. An index zero or three handle creates or removes a spin sphere, whose first homology and Arf invariant are zero. An index one handle either connects two components, in which case the first-homology quadratic forms add orthogonally, or increases the genus of one component. In the latter case the new symplectic pair contains the meridian of the handle. That meridian bounds the cocore disc with its induced spin structure, so its quadratic value is zero; the pair contributes zero to (1322), regardless of the value on its dual. Existing cycles can be represented outside the attaching discs, so their quadratic values are preserved. An index two handle is the reverse of an index one handle. Therefore each handle preserves the total Arf invariant of the boundary, and a spin boundary has Arf invariant zero. This also proves that a stably framed boundary has Arf invariant zero, because forgetting the framing gives a spin bordism. The full spin bordism statement is recorded in [18, Theorem 3.6].

For the intrinsic Lie framing on $M = (\mathbb{R}/(2\pi\mathbb{Z}))^2$, take

$$e = [a \mapsto (a, 0)], \qquad f = [b \mapsto (0, b)].$$

Both the tangent and normal vectors in the surface are constant in the Lie frame along either circle. Thus both circles have the nonbounding spin structure, and

$$e \cdot f = 1, \qquad q_M(e) = q_M(f) = 1, \qquad q_M(xe + yf) = x + xy + y, \qquad \operatorname{Arf}(M) = 1. \tag{1323}$$

Reversing the first Lie tangent vector changes the orientation but does not change these mod-two values. In particular the specifically framed inverse image computed above is not a framed boundary.

To identify the element, let $\eta \in \pi_1^{\mathrm{S}}$ denote the stable class of the complex Hopf map. Its Pontryagin–Thom inverse image is the circle with its Lie framing. This can be verified directly on the Hopf map $S^3 \to \mathbb{CP}^1$: near the fibre $(z_1, z_2) = (e^{ia}, 0)$, the target coordinate is z_2/z_1, so the two horizontal normal vectors are $(0, e^{ia})$ and $(0, ie^{ia})$. Add the tangent vector $(ie^{ia}, 0)$ and the outward radial vector $(e^{ia}, 0)$. In $\mathbb{C}^2$ the resulting full frame makes one complete rotation in each of its two real coordinate planes. This loop is contractible explicitly: for the fixed real-linear involution $J(z_1, z_2) = (z_1, \overline{z}_2)$,

$$J \operatorname{diag}(e^{ia}, e^{ia}) J = \operatorname{diag}(e^{ia}, e^{-ia}) = K_1(-a),$$

and $JK_v(-a)J$ contracts the original loop to the identity using the periodic homotopy already constructed. Thus the induced stable tangent frame on the circle is the Lie frame. This is the basic framed-circle identification in [16].

The product of two normally framed inverse images represents the product in stable homotopy: its tubular collapse is the smash of the two tubular collapse maps, as follows immediately from the product tubular neighborhood $\nu_1 \times \nu_2$ and its ordered normal coordinates. Consequently the product of the two Lie circles represents η^2. Equation (1323) proves $\eta^2 \neq 0$ by an explicit bordism invariant. The stable group $\pi_2^{\mathrm{S}} = \mathbb{Z}/2$ is Pontryagin's computation [15, 16]; the stable table in [19, pp. 384–385] gives the same group. Thus the sign from (1317) satisfies $-\eta^2 = \eta^2$.

Proof of the theorem. The globally continuous quotient H is smooth on a neighborhood of its compact inverse image M, by (1308) and (1300). First make it constant near the source basepoint, before choosing any approximation bound. Choose a target chart $\kappa : O \to \mathbb{R}^4$ about $*$ with $\kappa(*) = 0$, convex ball image, and $y_0 \notin O$. Choose a source neighborhood U of x_{base}, disjoint from M, with $H(U) \subset O$. Let β be a smooth function supported in U, equal to one near x_{base}, with $0 \leq \beta \leq 1$. On U use the based homotopy

$$H_s(x) = \kappa^{-1}\big((1 - s\beta(x))\kappa(H(x))\big), \qquad 0 \leq s \leq 1,$$

and leave H unchanged off U. Its endpoint H_b is smooth and constant near the basepoint; it agrees with H near M and has no new inverse image of y_0.

Choose one closed tubular neighborhood U_1 of M, with $M \subset \operatorname{int} U_1$, contained in the unchanged smooth region and disjoint from the basepoint neighborhood. Compactness gives the positive bound

$$d_0 = \min_{x \in X \setminus \operatorname{int} U_1} d\big(H_b(x), y_0\big) > 0.$$

Relative smooth approximation, fixing the same closed set U_1 and a neighborhood of the basepoint, produces a smooth based map H_{sm} whose error from H_b is less than $d_0/2$ on its complement. Choose the approximation closer also than a fixed target injectivity radius; the short target geodesics give a based homotopy. Thus the exact inverse image and its differential are unchanged on U_1, and the complement acquires no new inverse image. The relative Pontryagin construction applies as in [14, §7, Theorem B and Lemma 4, pp. 43, 48–50].

Pontryagin–Thom therefore identifies the stabilization of $[H]$ with the bordism class of M carrying the four normal vectors (1313)–(1314). The comparison (1319) and its explicit contraction identify that class with the Lie torus, with the harmless order-two orientation sign already computed. Hence

$$E^{\infty}[H] = \eta^2 \neq 0.$$

Freudenthal suspension is an isomorphism

$$E : \pi_{n+2}(S^n) \longrightarrow \pi_{n+3}(S^{n+1}) \qquad (n \geq 4),$$

because $n + 2 \leq 2n - 2$. Therefore stabilization from $\pi_6(S^4)$ is an isomorphism onto π_2^{S}. The composite $\eta_4 \circ \eta_5 : S^6 \to S^4$ stabilizes to η^2, so injectivity gives the exact equality (1301). Finally, a based null-homotopy of q_2 would postcompose with p_3 to a based null-homotopy of H, contradicting (1301). □

Exact scope of the comparison with the angle-forgetting map. The map here retains the angular coordinate before collapsing the coordinate two-skeleton. Its regular inverse image is the specific normally four-framed T^2 above. The angle-forgetting postcomposition has a normally three-framed T^3 inverse image. The calculation above is independent of its class and supplies an actual obstruction to null-homotopy of the original q_2, even when that other postcomposition is null-homotopic. The exact morphism carrying this obstruction is

$$(p_3)_* : [X, \mathcal{A}]_* \longrightarrow [X, S^4]_*, \qquad [q_2] \longmapsto [H] = \eta_4\eta_5.$$

No conclusion about an arbitrary earlier quotient follows merely from the diffeomorphism type of one inverse image; the specified four-framing is what proves this value.

Corollary 30.55 (The exact order-two quotient detected by the original cusp map)**.** *Let $d : (S^6, *) \to (X, x_{\mathrm{base}})$ be the fixed based diffeomorphism, and put*

$$\alpha = [q_2 \circ d] \in \pi_6(\mathcal{A}, v_0).$$

The original top-cell postcomposition induces a surjective homomorphism

$$\varepsilon = (p_3)_* : \pi_6(\mathcal{A}, v_0) \longrightarrow \pi_6(S^4, *) \cong \mathbb{Z}/2, \qquad \varepsilon(\alpha) = \eta_4 \circ \eta_5.$$

It gives the exact quotient isomorphism

$$\pi_6(\mathcal{A}, v_0)/\ker \varepsilon \xrightarrow{[\beta]\mapsto\varepsilon(\beta)} \mathbb{Z}/2.$$

Every odd multiple of α is nonzero. Its order is either infinite or an even positive integer; this calculation does not determine that order or assert a splitting of the displayed quotient.

Proof. Postcomposition distributes over the pinch sum on the source sphere, so $(p_3)_*$ is a group homomorphism. Theorem 30.51 gives $(p_3)_*(\alpha) = [p_3 q_2 d] = \eta_4\eta_5 \neq 0$. This is the nonzero element of the cyclic group of order two, proving surjectivity. The first isomorphism theorem gives the stated quotient with exactly the displayed map. For every integer n,

$$\varepsilon(n\alpha) = (n \bmod 2)\eta_4\eta_5.$$

An odd multiple therefore has nonzero image and cannot vanish. If $m\alpha = 0$, applying ε forces m to be even. These arguments prove all the assertions without inferring the exact order of α. □

30.6.5 The retained cusp class as a three-dimensional bundle inside the triality carrier

Retain the exact based map $H = p_3 q_2 : X \to S^4$, the fixed based diffeomorphism $d : S^6 \to X$, the original cutoff ϵ, the ordered native coordinates (r, ϕ, u_1, u_2), and every period appearing in (1290). Theorem 30.51 has proved $[Hd] = \eta_4\eta_5 \neq 0$ in $\pi_6(S^4) \cong \mathbb{Z}/2$. We construct an exact bundle-valued image of this specific map and then its action on the original three Peirce sectors. A quaternion subalgebra and its ordered basis are explicit choices in this construction.

The literal Hopf pullback and its transition. Fix $\mathbb{H} = \mathbb{R}\langle 1, i, j, k\rangle$ with $i^2 = j^2 = k^2 = ijk = -1$. Identify the ordered open four-cell of the target with $\mathbb{H}$ by

$$z = -\cot(\pi r/\epsilon) - \cot(\phi/2)i - \cot(\pi u_1)j - \cot(\pi u_2)k. \tag{1324}$$

This is exactly the ordered coordinate map (1300), followed by the real linear map carrying the four standard basis vectors to $(1, i, j, k)$. Each of the four coordinate derivatives in the original open cell is positive. The based homeomorphism

$$\chi : S^4 = \mathbb{H} \cup \{\infty\} \longrightarrow \mathbb{HP}^1, \qquad \chi(z) = [1 : z], \qquad \chi(*) = [0 : 1] \tag{1325}$$

uses right quaternionic lines. The orientation on this presentation of $\mathbb{HP}^1$ is the one induced by the ordered affine basis $(1, i, j, k)$. Thus no orientation convention on another presentation of $\mathbb{HP}^1$ is silently substituted.

Let

$$S^7 = \{(v_0, v_1) \in \mathbb{H}^2 : |v_0|^2 + |v_1|^2 = 1\}, \qquad \pi(v_0, v_1) = [v_0 : v_1].$$

The right action is $(v_0, v_1)u = (v_0u, v_1u)$ for $u \in \mathrm{Sp}(1) = \{u \in \mathbb{H} : |u| = 1\}$. It is free: at least one v_j is nonzero, and $v_ju = v_j$ implies $u = 1$. Two unit vectors on the same right line differ by a unique unit

quaternion. Hence π is the quaternionic Hopf principal bundle. Its local sections in the affine coordinates $z = v_1 v_0^{-1}$ and $w = v_0 v_1^{-1}$ are

$$s_0(z) = \frac{(1, z)}{\sqrt{1 + |z|^2}}, \qquad s_\infty(w) = \frac{(w, 1)}{\sqrt{1 + |w|^2}}. \tag{1326}$$

On $z \neq 0$, the coordinate change is $w = z^{-1}$ and

$$s_\infty(z^{-1}) = s_0(z)\, g(z), \qquad g(z) = \frac{\overline{z}}{|z|}. \tag{1327}$$

Indeed, multiplying the right side gives $(\overline{z}/|z|, |z|)/\sqrt{1 + |z|^2}$, and the same expression results from the left side because $\sqrt{1 + |z^{-1}|^2} = \sqrt{1 + |z|^2}/|z|$. This calculation retains both the inverse and the conjugation.

Define

$$P_H = \{(x, v) \in X \times S^7 : \pi(v) = \chi(H(x))\} \longrightarrow X, \qquad (x, v)u = (x, vu). \tag{1328}$$

The preimages under χH of the two affine charts cover X. Equations (1326)–(1327) are local sections and transitions for (1328) after composition with χH. In particular, wherever both charts apply, the transition is the exact $g(z)$ from (1327), with z given by the original four-coordinate expression (1324). The angle ϕ has not been dropped.

Theorem 30.56. *The principal bundle P_H is nontrivial. Its clutching class is the nonzero element*

$$c(P_H) = \partial[\chi H d] \in \pi_5(\mathrm{Sp}(1)) \cong \mathbb{Z}/2, \tag{1329}$$

where ∂ is the boundary of the specified right Hopf fibration. The associated real three-plane bundle

$$\mathcal{A}_H = P_H \times_{\mathrm{Ad}} \mathrm{Im}\,\mathbb{H}, \qquad \mathrm{Ad}_u(v) = uv\overline{u}, \tag{1330}$$

is nontrivial, has nonzero order-two clutching class in $\pi_5(SO(3))$, and admits no nowhere-zero continuous section. Its oriented orthonormal frame bundle has P_H as its unique spin lift up to isomorphism over X. The maps from the retained homotopy group to these two clutching groups are isomorphisms:

$$\pi_6(S^4) \xrightarrow{\chi_*} \pi_6(\mathbb{HP}^1) \xrightarrow{\partial} \pi_5(\mathrm{Sp}(1)) \xrightarrow{\mathrm{Ad}_*} \pi_5(SO(3)). \tag{1331}$$

Proof. The first map is an isomorphism because χ is a based homeomorphism. In the long exact sequence of the displayed Hopf bundle, the segment is

$$\pi_6(S^7) \longrightarrow \pi_6(\mathbb{HP}^1) \xrightarrow{\partial} \pi_5(S^3) \longrightarrow \pi_5(S^7).$$

Both outside groups vanish: cellular approximation homotopes a map from a sphere of dimension less than seven into the zero-cell of the CW structure on S^7. Exactness therefore makes ∂ an isomorphism. To fix its clutching convention, write the oriented source $S^6 = D^6_+ \cup D^6_-$, give the equator the boundary orientation of D^6_+, and choose local lifts $\sigma_\pm$ of the pulled-back bundle. On the equator define g_{+-} by $\sigma_- = \sigma_+ g_{+-}$. Choose the values of the two lifts equal at its fixed basepoint. We use the convention $\partial[\chi H d] = [g_{+-}]$. Changing disk lifts changes this map by null-homotopic factors and does not change its homotopy class. This is the defining boundary construction of the fibration exact sequence. The already proved nonzero class $[Hd]$ therefore gives the nonzero order-two class in (1329).

One can also see nontriviality directly. A section of P_H gives a map $\widetilde{h} : S^6 \to S^7$ with $\pi\widetilde{h} = \chi H d$. Since $\pi_6(S^7) = 0$, this would make $\chi H d$ null-homotopic, contrary to Theorem 30.51. All spaces at issue are simply connected, so based and free null-homotopy agree here.

Conjugation by a unit quaternion preserves $\mathrm{Im}\,\mathbb{H}$ and its Euclidean norm. Its determinant is $+1$ by continuity in u and its value at $u = 1$. The kernel consists of the real unit quaternions $\{1, -1\}$: a quaternion

commuting with i, j, k is real. For a unit imaginary quaternion n and $v \in \operatorname{Im}\mathbb{H}$, write $v = v_{\parallel} + v_{\perp}$ relative to the line $\mathbb{R}n$. Direct multiplication with $u = \cos\alpha + n\sin\alpha$ gives

$$uv\overline{u} = v_{\parallel} + \cos(2\alpha)v_{\perp} + \sin(2\alpha)(n \times v_{\perp}), \tag{1332}$$

where $n \times v = (nv - vn)/2$. Every orientation-preserving orthogonal map of a real three-space fixes an axis and rotates its orthogonal plane: its real odd-degree characteristic polynomial has a real eigenvalue, norm preservation gives ± 1, and determinant $+1$ forces an eigenvalue $+1$. Thus (1332) proves that $\mathrm{Ad} : S^3 \to SO(3)$ is onto with kernel $\{\pm 1\}$. It is the two-sheeted covering and induces an isomorphism on π_5. Its value on (1329) is the clutching class of (1330), proving nontriviality and order two.

The bundle $P_H \to P_H/\{\pm 1\}$ is the spin double covering of the oriented orthonormal frames of $\mathcal{A}_H$. If that frame bundle had a section, its inverse image would be a double covering of the simply connected space $X \cong S^6$. This covering is trivial, so one of its two sections would give a section of P_H, again a contradiction. Any two lifts of an $SO(3)$ bundle differ by a principal $\mathbb{Z}/2$ bundle: locally the ratio between two lifts of each transition lies in the central kernel $\{\pm 1\}$, and the cocycle identity gives a $\mathbb{Z}/2$ cocycle. Every such bundle on the simply connected X is trivial. This proves the claimed uniqueness of the spin lift.

If $\mathcal{A}_H$ had a nowhere-zero section, divide that section by its positive length to obtain a unit section. Orthogonal complement gives $\mathcal{A}_H \cong \mathbb{R} \oplus \mathcal{B}$ for an oriented two-plane bundle $\mathcal{B}$. A two-plane bundle on S^6 has clutching map $S^5 \to SO(2) = S^1$. Every such map lifts to $\mathbb{R}$ and is null-homotopic, so $\mathcal{B}$ is trivial. This would trivialize $\mathcal{A}_H$, contradicting its nonzero clutching class. □

The fibrewise scalar product, orientation and cross product of $\operatorname{Im}\mathbb{H}$ descend to $\mathcal{A}_H$. For example,

$$(uv\overline{u})(uw\overline{u}) = u(vw)\overline{u}$$

by associativity in $\mathbb{H}$, so conjugation preserves $v \times w = (vw - wv)/2$. The induced fibrewise Lie bracket is exactly $[v, w] = 2(v \times w)$. No factor of two is discarded.

An explicit subgroup of the three-sector triality group. Fix the quaternion subalgebra above inside $\mathbb{O} = \mathbb{H} \oplus \mathbb{H}\ell$. Its ordered real basis is

$$(1, i, j, k, \ell, i\ell, j\ell, k\ell),$$

and its multiplication is

$$(a, b)(c, d) = (ac - \overline{d}b,\ da + b\overline{c}), \qquad K(a, b) = \overline{(a, b)} = (\overline{a}, -b). \tag{1333}$$

This is the literal convention of Yokota, §1.10. In his cyclic related-triple presentation the group is

$$\mathrm{RT}(\mathbb{O}) = \{(T_1, T_2, T_3) \in SO(\mathbb{O})^3 : (T_1 x)(T_2 y) = KT_3K(xy) \text{ for all } x, y \in \mathbb{O}\}. \tag{1334}$$

Yokota, Theorem 1.16.2, identifies this group with Spin(8); Theorem 2.7.1 [23] gives its diagonal-fixing Jordan action. In particular the three projection representations in (1334) are the vector and two half-spin sectors with that fixed marking.

For each $u \in \mathrm{Sp}(1)$ define

$$\begin{aligned} B_u(a, b) &= (ua\overline{u}, b), \\ A_u(a, b) &= (ua, b\overline{u}), \\ D_u(a, b) &= (a\overline{u}, b\overline{u}) = KA_uK(a, b). \end{aligned} \tag{1335}$$

Theorem 30.57. *The formula*

$$\rho : \mathrm{Sp}(1) \longrightarrow \mathrm{Spin}(8) = \mathrm{RT}(\mathbb{O}), \qquad \rho(u) = (B_u, A_u, D_u) \tag{1336}$$

is an injective continuous group homomorphism. It satisfies

$$(B_u x)(A_u y) = A_u(xy), \qquad \mathrm{Re}(((B_u x)(A_u y))(D_u z)) = \mathrm{Re}((xy)z) \tag{1337}$$

for all octonions x, y, z and every unit quaternion u.

Proof. All three maps are real linear norm isometries by the quaternionic norm identity. Their determinants are $+1$: they vary continuously with u on the connected S^3 and equal the identity at $u = 1$. For $u, v \in \mathrm{Sp}(1)$, associativity in $\mathbb{H}$ and $\overline{uv} = \overline{v}\,\overline{u}$ give

$$B_u B_v = B_{uv}, \qquad A_u A_v = A_{uv}, \qquad D_u D_v = D_{uv}.$$

For example, the second coordinate of $A_u A_v(a, b)$ is $(b\overline{v})\overline{u} = b\overline{uv}$, retaining the order.

For $x = (a, b)$ and $y = (c, d)$, expansion of (1333) gives

$$\begin{aligned}(B_u x)(A_u y) &= (ua\overline{u}, b)(uc, d\overline{u}) \\ &= \big((ua\overline{u})(uc) - \overline{d\overline{u}}\, b, (d\overline{u})(ua\overline{u}) + b\overline{uc}\big) \\ &= \big(u(ac - \overline{d}b), (da + b\overline{c})\overline{u}\big) = A_u(xy).\end{aligned}$$

This uses only quaternionic associativity in the displayed components. Since $A_u = K D_u K$, it proves membership in (1334). If $\rho(u)$ is the identity, evaluate A_u at $(1, 0)$ to get $(u, 0) = (1, 0)$, so $u = 1$. Thus ρ is injective.

The other two cyclic pair identities, with every conjugation retained, are

$$\begin{aligned}(A_u x)(D_u y) &= (ua, b\overline{u})(c\overline{u}, d\overline{u}) \\ &= \big(u(ac - \overline{d}b)\overline{u},\ da + b\overline{c}\big) = B_u(xy) = K B_u K(xy), \\ (D_u x)(B_u y) &= (a\overline{u}, b\overline{u})(uc\overline{u}, d) \\ &= \big((ac - \overline{d}b)\overline{u}, (da + b\overline{c})\overline{u}\big) = D_u(xy) = K A_u K(xy).\end{aligned}$$

Put $\langle v, w\rangle = \mathrm{Re}(v\overline{w})$ on $\mathbb{O}$. Then $\mathrm{Re}(vz) = \langle v, Kz\rangle$. The already proved product relation and $K D_u = A_u K$ imply

$$\begin{aligned}\mathrm{Re}(((B_u x)(A_u y))(D_u z)) &= \langle A_u(xy), K D_u z\rangle \\ &= \langle A_u(xy), A_u K z\rangle = \langle xy, Kz\rangle = \mathrm{Re}((xy)z).\end{aligned}$$

No reassociation of a non-real octonionic triple product was used. □

The exact original Jordan coordinates and all three bundles. Write the original Hermitian matrix with upper entries (p, q, r) as

$$J(\lambda; p, q, r) = \begin{pmatrix} \lambda_1 & p & q \\ \overline{p} & \lambda_2 & r \\ \overline{q} & \overline{r} & \lambda_3 \end{pmatrix}, \qquad (q_1, q_2, q_3) = (r, \overline{q}, p). \tag{1338}$$

Thus q_2 here is the cyclic Jordan coordinate, not the previously defined cusp quotient map; only the explicitly displayed ordered triple in this paragraph uses this coordinate notation. Under (1336), the cyclic coordinates transform by

$$(q_1, q_2, q_3) \longmapsto (B_u q_1, A_u q_2, D_u q_3).$$

Since $K A_u K = D_u$, the action in the original upper entries is

$$(r, q, p) \longmapsto (B_u r, D_u q, D_u p), \qquad J(\lambda; p, q, r) \longmapsto J(\lambda; D_u p, D_u q, B_u r). \tag{1339}$$

In particular this is an action on the three specific spaces $\mathfrak{h}_{\gamma_1}, \mathfrak{h}_{\gamma_2}, \mathfrak{h}_{\gamma_3}$ of (1195), in their original order.

The determinant and trace form are

$$\det J = \lambda_1\lambda_2\lambda_3 - \lambda_1|r|^2 - \lambda_2|q|^2 - \lambda_3|p|^2 + 2\,\mathrm{Re}((r\overline{q})p), \tag{1340}$$

$$\mathrm{tr}(J^2) = \lambda_1^2 + \lambda_2^2 + \lambda_3^2 + 2(|p|^2 + |q|^2 + |r|^2). \tag{1341}$$

Theorem 30.57 and norm preservation prove that (1339) preserves every term, with its displayed sign and coefficient. Its being a Jordan automorphism also follows directly from Yokota's diagonal-fixing action in Theorem 2.7.1, with exactly the identification (1338). Equivalently, the Peirce multiplication rules $F_i(x) \circ F_i(y) = \langle x, y\rangle(E_{i+1} + E_{i+2})$ and $F_i(x) \circ F_{i+1}(y) = \frac{1}{2}F_{i+2}(\overline{xy})$, where indices are cyclic modulo three, are preserved by the norm identity and the three cyclic pair identities proved above. Here $F_1(x)$ has upper entry x at $(2,3)$, $F_2(x)$ has upper entry $\overline{x}$ at $(1,3)$, and $F_3(x)$ has upper entry x at $(1,2)$; all other entries are zero apart from their Hermitian conjugates. Direct matrix multiplication gives the displayed rules, with coefficient $1/2$ from the Jordan product. Finally $E_i \circ E_j = \delta_{ij}E_i$, $E_i \circ F_i(x) = 0$, and $E_i \circ F_j(x) = F_j(x)/2$ for $i \neq j$, and these rules are preserved because all E_i are fixed. The displayed identities specify the Jordan product on every pair by real bilinearity.

For a right principal bundle our associated-bundle convention is $(pu, v) \sim (p, \sigma(u)v)$. Let

$$\mathcal{L}_H = P_H \times_{u:a\mapsto ua} \mathbb{H}.$$

This is the tautological right quaternionic line pulled back by χH: the fibre map $[(x,v),a] \mapsto (x, va)$ is well-defined, right quaternionic linear and invertible onto the line $\chi H(x)$. Its unit sphere bundle is P_H, by $p \mapsto [p, 1]$. Consequently it is nontrivial as a quaternionic line. It is also nontrivial as a real four-plane bundle: if it had a nowhere-zero real section, dividing by length would give a section of its unit sphere bundle and hence of P_H.

The original ordered Peirce representation produces the real rank-24 bundle

$$\mathcal{W}_H = \mathcal{V}_{r,H} \oplus \mathcal{V}_{q,H} \oplus \mathcal{V}_{p,H}, \quad \mathcal{V}_{r,H} = P_H \times_B \mathbb{O}, \quad \mathcal{V}_{q,H} = \mathcal{V}_{p,H} = P_H \times_D \mathbb{O}. \tag{1342}$$

The equality between the last two names specifies isomorphic representations with two separately retained labelled copies. Their explicit decompositions are

$$\begin{aligned} &\mathcal{V}_{r,H} \cong \mathbb{R} \oplus \mathcal{A}_H \oplus \mathbb{R}^4, && (\mathrm{Re}\,a, \mathrm{Im}\,a, b), \\ &\mathcal{V}_{q,H} \cong \mathcal{L}_H \oplus \mathcal{L}_H, && (a,b) \longmapsto (\overline{a}, \overline{b}), \\ &\mathcal{V}_{p,H} \cong \mathcal{L}_H \oplus \mathcal{L}_H, && (a,b) \longmapsto (\overline{a}, \overline{b}). \end{aligned} \tag{1343}$$

For the last two maps, $\overline{a\overline{u}} = u\overline{a}$ proves equivariance, and similarly for b. Each quaternionic conjugation has real determinant -1 in $(1, i, j, k)$, so their ordered pair has determinant $+1$. The first line keeps the actual coordinate order $(1, \mathrm{Im}\,\mathbb{H}, \mathbb{H}\ell)$; moving the one-dimensional first factor across the imaginary three-space would contribute the sign $(-1)^3 = -1$, and no such reorder is being made. In cyclic coordinates the middle action is A; the equivariant map to two left standard representations is $(a,b) \mapsto (a, \overline{b})$, of determinant -1. This is why the original $\overline{q}$ is retained explicitly.

All fibrewise maps in (1337) and (1340) descend to the associated bundles: replacing a local frame p by pu changes inputs by precisely the three displayed actions, whose product and scalar identities have been proved. In particular

$$\mathbb{R}^3 \oplus \mathcal{W}_H = P_H \times_\rho J_3(\mathbb{O})$$

is a bundle of Jordan algebras with its three fixed diagonal idempotents, its full product, trace form and determinant. The three-sector structure on (1342) includes the explicit nonzero invariant trilinear map $(r,q,p) \mapsto 2\,\mathrm{Re}((r\overline{q})p)$; it is nonzero because its value at $r = q = p = 1$ is exactly 2.

What extension to the full triality group forgets. Form the principal bundle

$$Q_H = P_H \times_\rho \mathrm{Spin}(8) \longrightarrow X, \quad (pu, s) \sim (p, \rho(u)s). \tag{1344}$$

It contains the reduction P_H by $p \mapsto [p, 1]$. This is equivariant because $[pu, 1] = [p, \rho(u)]$.

Proposition 30.58. *The principal bundle Q_H is trivial. All three rank-eight bundles in* (1342) *are therefore trivial as real vector bundles, while their specified subbundles $\mathcal{A}_H$ and $\mathcal{L}_H$ remain nontrivial. After any chosen trivialization of Q_H, its reduction determines a continuous map*

$$f_H : X \longrightarrow \mathrm{Spin}(8)/\rho(\mathrm{Sp}(1)) \tag{1345}$$

whose pullback of the principal bundle $\mathrm{Spin}(8) \to \mathrm{Spin}(8)/\rho(\mathrm{Sp}(1))$ *is P_H. This map is non-null-homotopic.*

Proof. Let $g : S^5 \to \mathrm{Sp}(1)$ be a based clutching representative for P_H in the convention used in (1329). Its homotopy class has order two. The transition D_g on an eight-dimensional sector is conjugate by $C(a, b) = (\overline{a}, \overline{b})$ to $\mathrm{diag}(L_g, L_g)$ on $\mathbb{H}^2$. For $0 \le t \le \pi/2$ put

$$R_t = \begin{pmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{pmatrix} \quad \text{on } \mathbb{H}^2.$$

This is an orthogonal path commuting with quaternionic scalar right multiplication. Its conjugation carries $\mathrm{diag}(L_g, I)$ at $t = 0$ to $\mathrm{diag}(I, L_g)$ at $t = \pi/2$. The maps into $SO(8)$ represented by these two matrices are therefore based homotopic. Pointwise product of maps from a based sphere into a topological group represents addition in its homotopy group: concatenation in one loop coordinate and pointwise multiplication have the same unit and satisfy interchange, yielding the same operation. Consequently

$$[\mathrm{diag}(L_g, L_g)] = 2[\mathrm{diag}(L_g, I)] = \iota_*(2[g]) = 0 \quad \text{in } \pi_5(SO(8)),$$

where $\iota(u) = \mathrm{diag}(L_u, I)$. Conjugating back by C gives $[D_g] = 0$.

The third projection of (1334) is a double covering $\mathrm{Spin}(8) \to SO(8)$: this is the triality covering in the third marking, with the same conjugation K as in (1334). Coverings induce isomorphisms on π_n for $n \ge 2$ by unique lifting of based sphere maps and their homotopies. The projection sends $[\rho g]$ to $[D_g] = 0$, so $[\rho g] = 0$. A bundle obtained by gluing two trivial bundles over disks with a null-homotopic clutching map is trivial: extend that clutching map across a disk and change its local frame by the extension. This proves triviality of Q_H directly. Every associated vector bundle of a trivial principal bundle is trivial; applying the three projections gives the asserted rank-eight trivializations. The earlier proofs of nontriviality of $\mathcal{A}_H$ and $\mathcal{L}_H$ still apply to these actual subbundles, so forgetting their inclusions loses information.

Choose an isomorphism $\tau : Q_H \to X \times G$, where $G = \mathrm{Spin}(8)$, and put $K_0 = \rho(\mathrm{Sp}(1))$. For $p \in (P_H)_x$ write $\tau([p, 1]) = (x, a_p)$. Then $a_{pu} = a_p \rho(u)$, so $f_H(x) = a_p K_0$ is well-defined, independent of the choice of p, and continuous in local sections. The explicit principal bundle isomorphism to the pullback is

$$P_H \longrightarrow f_H^*(G \to G/K_0), \qquad p \longmapsto (x, a_p),$$

using the identification of $\mathrm{Sp}(1)$ with K_0 by the proved injective map ρ. The formula is fibrewise bijective and has inverse in each local trivialization. If f_H were null-homotopic, its pullback bundle would be isomorphic to the pullback along a constant map and hence trivial, contradicting Theorem 30.56. Equivalently, in the fibration exact sequence its clutching boundary is precisely $c(P_H) \neq 0$. □

There is also an explicit map into the space of oriented three-planes in the first eight-dimensional sector. If $T_1 : G \to SO(\mathbb{O})$ is the first projection of (1334), define

$$\begin{aligned} \Gamma : G/K_0 &\longrightarrow \mathrm{Gr}_3^+(\mathbb{O}), \\ gK_0 &\longmapsto \mathrm{span}_{\mathbb{R}}^+(T_1(g)i, T_1(g)j, T_1(g)k). \end{aligned} \tag{1346}$$

Here the superscript records the orientation of the displayed ordered basis. This formula is well-defined: multiplication of g on the right by $\rho(u)$ changes that ordered basis by the orientation-preserving map Ad_u of $\mathrm{Im}\,\mathbb{H}$, because $T_1\rho(u) = B_u$. Continuity follows from local sections of $G \to G/K_0$. Let $\mathcal{T}_3$ be the tautological oriented three-plane bundle on this Grassmannian. The explicit fibre map

$$G \times_{K_0} \mathrm{Im}\,\mathbb{H} \longrightarrow \Gamma^*\mathcal{T}_3, \qquad [g, v] \longmapsto (gK_0, T_1(g)v)$$

is well-defined since $T_1(g\rho(u))v = T_1(g)\,\mathrm{Ad}_u\, v$. It is a linear isometry on every fibre, hence a bundle isomorphism. Pulling back by (1345) therefore gives

$$(\Gamma f_H)^*\mathcal{T}_3 \cong \mathcal{A}_H.$$

In particular $\Gamma f_H : X \to \mathrm{Gr}_3^+(\mathbb{O})$ is non-null-homotopic, since pullback along a null-homotopic map would make the tautological bundle trivial. This locates the three-plane in the actual first triality sector, with its orientation and its nonzero clutching class retained.

Thus the exact bridge is the sequence of explicit constructions

$$H \;\longmapsto\; P_H \;\longmapsto\; \mathcal{A}_H \;\longmapsto\; (\mathcal{W}_H, \rho\text{-reduction}, 2\,\mathrm{Re}((r\overline{q})p)).$$

The first two steps retain the entire nonzero order-two class by (1331). In the last step the specified reduction, or equivalently the distinguished three-plane with its spin frame data, retains that class. The triple of unrestricted rank-eight bundles alone does not. The space G/K_0 parametrizes reductions acting on this very same rank-24 carrier; no claim that $\dim(G/K_0) = 24$ is made.

The precise half-angle sign law and its relation to the cusp. For the fixed imaginary unit i define the path

$$u(\theta) = \cos(\theta/2) + i\sin(\theta/2), \qquad 0 \le \theta \le 2\pi. \tag{1347}$$

Equation (1332) gives

$$\mathrm{Ad}_{u(\theta)}\, i = i, \quad \mathrm{Ad}_{u(\theta)}\, j = \cos\theta\, j + \sin\theta\, k, \quad \mathrm{Ad}_{u(\theta)}\, k = -\sin\theta\, j + \cos\theta\, k.$$

This is one full 2π rotation in the oriented (j, k) plane. Its lift starts at 1 and ends at -1. It is therefore the nontrivial loop in $SO(3)$: a null-homotopy of the loop would lift over a disk and force its lifted endpoints to agree. Substitution in (1335) gives exactly

$$\rho(-1) = (I, -I, -I). \tag{1348}$$

It is central in (1334) because each component is scalar. Thus a full rotation in the three-dimensional adjoint sector acts by minus the identity on both associated spinor sectors while acting by the identity on the vector sector. This conclusion follows from actual related-triple maps and their product identity.

The parameter θ in (1347) is a structure-group parameter; it has not been equated with the original cusp phase ϕ. The actual dependence on that cusp phase is already specified, with all four coordinates, by (1324) and (1327). No connection or holonomy is part of the definition of P_H. In particular the order-two class proved here lies in $\pi_5(\mathrm{Sp}(1))$ through degree-six clutching, and is not being identified with a degree-one line-bundle class. Every real line bundle on $X \cong S^6$ is trivial: its unit sphere bundle is a double covering of a simply connected space, so it has a section. The exact relation established here is instead the faithful order-two detection (1331), the nontrivial real three-plane (1330), its explicit three-sector triality action, and the structural sign law (1348).

30.6.6 A nonzero Hopf–cusp morphism to the stable mod–24 group

The retained-angle input used below is the complete Theorem 30.51 above.

The exact source maps and conventions Retain the original marked source (X, x_0), its fixed based diffeomorphism $d : (S^6, *) \to (X, x_0)$, the cusp coordinate t, the fixed cutoff ϵ, and the native quotient $\overline{\pi}_f$ of the cusp fibres by $\mathbb{Z}\langle\widehat{w}, \widehat{\delta}\rangle$. The quotient coordinates are ordered by $(\overline{\widehat{\gamma}}, \overline{\widehat{u}})$. Thus the fixed continuous based map is

$$q_2 : X \longrightarrow \mathcal{A} = \mathrm{Susp}(S^1 \times \overline{T}^2), \qquad q_2(x) = [|t|, t/|t|, \overline{\pi}_f(x)] \quad (0 < |t| < \epsilon).$$

It sends the entire central fibre W to the first suspension vertex and $X \setminus \operatorname{int} N_0$ to the second. The unreduced suspension $\mathcal{A}$ is based at its first vertex. In the original product torus set

$$K^{(2)} = (\{1\} \times \overline{T}^2) \cup (S^1 \times \{u_1 = 0\}) \cup (S^1 \times \{u_2 = 0\}), \qquad c_3 : T^3 \longrightarrow T^3/K^{(2)} \cong S^3.$$

The retained-angle sphere map and its sphere-source representative are

$$p_3([r, z]) = [r, c_3(z)], \qquad H = p_3 q_2 : X \longrightarrow \Sigma_r(S^3, *) \cong S^4, \qquad h = H \circ d : S^6 \longrightarrow S^4. \tag{1349}$$

Both suspension vertices and the whole cylinder over the basepoint are identified in $\Sigma_r(S^3, *)$. The preceding complete retained-angle framing theorem proves

$$[h] = \eta_4 \circ \eta_5 \neq 0 \quad \text{in} \quad \pi_6(S^4) \cong \mathbb{Z}/2. \tag{1350}$$

Its proof identifies the actual derivative-framed inverse torus with the Lie-framed torus, with Arf invariant one. We use this already proved class, retaining the particular map (1349); see [20].

Fix the complex Hopf map

$$\eta_2 : (S^3, (1, 0)) \longrightarrow (\mathbb{CP}^1, [1 : 0]) \cong (S^2, *), \qquad (z_1, z_2) \longmapsto [z_1 : z_2].$$

Use ordered smash coordinates on spheres. In this note $E^k f = f \wedge \mathrm{id}_{S^k}$ denotes right reduced suspension; the usual ordered sphere homeomorphisms are understood. Put $\eta_n = E^{n-2}\eta_2$, and write η for the stable Hopf class. The order of each factor is retained below.

The Hopf–cusp homomorphism and its nonzero value

Theorem 30.59. *The assignment*

$$\mu_\eta : \pi_6(S^4) \longrightarrow \pi_9(S^6), \qquad \mu_\eta([x]) = [\eta_2 \wedge x] \tag{1351}$$

is an injective group homomorphism.

Its literal representative has the complete domain and codomain

$$\eta_2 \wedge x : S^3 \wedge S^6 \cong S^9 \longrightarrow S^2 \wedge S^4 \cong S^6.$$

Choose any generator ν_6 of $\pi_9(S^6) \cong \mathbb{Z}/24$. Then the particular original cusp map supplies

$$\kappa_H = [\eta_2 \wedge h] = \mu_\eta([h]) = 12\nu_6 \neq 0. \tag{1352}$$

It has exact order two. Its value is independent of orientation reversals in the sphere identifications and of the choice of the generator ν_6.

Proof. A based homotopy of x smashes with the fixed based Hopf map, so (1351) is well-defined. The exact ordered point-set factorization is

$$\eta_2 \wedge x = (\eta_2 \wedge \mathrm{id}_{S^4}) \circ (\mathrm{id}_{S^3} \wedge x) : S^3 \wedge S^6 \longrightarrow S^3 \wedge S^4 \longrightarrow S^2 \wedge S^4. \tag{1353}$$

The first applied map has source dimension nine and target dimension seven; the second has source dimension seven and target dimension six. The second map is $E^4\eta_2$. To compare the first with $E^3x = x \wedge \mathrm{id}_{S^3}$, interchange its sphere factors at source and target. The interchange degrees are respectively $(-1)^{3\cdot 6} = +1$ and $(-1)^{3\cdot 4} = +1$. A based map of a sphere of degree one is based homotopic to the identity, so

$$\mu_\eta([x]) = (E^4\eta_2)_*(E^3[x]). \tag{1354}$$

Suspension is a group homomorphism. Postcomposition is also a group homomorphism on homotopy groups, since it distributes over the source pinch map defining the sum. Thus (1354) proves additivity, including zero and inverses, before passage to stable homotopy.

Freudenthal makes

$$E : \pi_{n+3}(S^n) \longrightarrow \pi_{n+4}(S^{n+1})$$

an isomorphism for $n \geq 5$, because $n + 3 < 2n - 1$. In particular stabilization $\pi_9(S^6) \to \pi_3^{\mathrm{S}}$ is an isomorphism. Stabilization from $\pi_6(S^4)$ is likewise an isomorphism onto π_2^{S}, because $n + 2 < 2n - 1$ for $n \geq 4$. The exact factorization (1353) shows that its stable class is the stable composition product of η with $[x]$. Using the proved original class (1350),

$$E^\infty\kappa_H = \eta\, E^\infty[h] = \eta\,\eta^2 = \eta^3 = 12\nu \neq 0 \quad \text{in} \quad \pi_3^{\mathrm{S}} \cong \mathbb{Z}/24. \tag{1355}$$

The integral relation here is the one explicitly stated in [22, p. 385], not the coefficient 4 in the two-primary quotient. More precisely, the two-primary relation is $\eta^3 = 4\nu$ modulo eight. Bilinearity and $2\eta = 0$ give $2\eta^3 = 0$ integrally. In $\mathbb{Z}/24$ the only elements killed by two are 0 and 12ν, and only the second has nonzero image 4ν modulo eight. This recovers the exact integral coefficient and its nonvanishing.

The inverse of the stabilization isomorphism therefore gives (1352). Since the source of μ_η is cyclic of order two, generated by $[h]$, its nonzero value proves injectivity and identifies its image as $\{0, 12\nu_6\}$.

An orientation reversal at a sphere source or target multiplies the corresponding stable homotopy class by -1; here $-12\nu_6 = 12\nu_6$. A sign change in the chosen Hopf representative also leaves the stable product unchanged since $-\eta = \eta$. Finally, replacing the generator by $u\nu_6$, with u a unit modulo twenty-four, does not change the element: every such u is odd, and $12u \equiv 12 \pmod{24}$. All orientation and generator assertions consequently hold for the actual order-two element. □

Corollary 30.60 (The same nonzero class in the twelve-dimensional receiver)**.** *Put* $\nu_9 = E^3\nu_6$. *The particular based map*

$$E^3(\eta_2 \wedge h) : S^{12} \longrightarrow S^9$$

has class $12\nu_9 \neq 0$ *in* $\pi_{12}(S^9) \cong \mathbb{Z}/24$.

Proof. Each of these three suspensions is an isomorphism in the preceding stable range. Apply their composite to (1352); it preserves the exact order two. □

The actual marked peripheral character Retain the original three elliptic-kernel matrices, with their original signs and ordering:

$$C_0 = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad C_1 = \begin{pmatrix} -1 & -1 \\ 1 & 0 \end{pmatrix}, \qquad C_2' = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}. \tag{1356}$$

The existing peripheral theorem [21] defines the surjective marked abelianization character

$$\chi : SL_2(\mathbb{Z}) \longrightarrow \mathbb{Z}/12, \qquad (\chi(C_0), \chi(C_1), \chi(C_2')) = (-1, 4, -3). \tag{1357}$$

For an explicit check of this normalization, let

$$S_0 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad T = C_0.$$

Then $C_2' = S_0^3$, $C_1 = T^{-1}S_0$, and direct matrix multiplication gives $C_1C_2' = C_0^{-1}$. In the modular presentation $S_0^4 = 1$, $S_0^2 = (S_0T)^3$, the abelianized generators satisfy $4s = 0$ and $s + 3t = 0$. Thus $s = -3t$ and $12t = 0$. The assignment $t \mapsto -1$, $s \mapsto 3$ satisfies both relations and maps onto $\mathbb{Z}/12$; consequently the order of the abelianized generator is exactly twelve. It gives, without a change of peripheral matrices, precisely $\chi(C_0) = -1$, $\chi(C_1) = 1 + 3 = 4$, and $\chi(C_2') = 3 \cdot 3 = -3 \pmod{12}$.

Theorem 30.61 (An exact nonzero peripheral homotopy morphism)**.** *Define*

$$\Theta_H : \mathbb{Z}/12 \longrightarrow \pi_6(S^4), \qquad \Theta_H([n]) = n[h], \qquad \Psi_H = \mu_\eta \circ \Theta_H \circ \chi : SL_2(\mathbb{Z}) \longrightarrow \pi_9(S^6). \tag{1358}$$

These are group homomorphisms. The complete coefficient formula is

$$\Psi_H(M) = 12\,\widetilde{\chi(M)}\,\nu_6, \tag{1359}$$

where $\widetilde{\chi(M)}$ *is any integer lift of the marked residue. In the original matrix order of* (1356)*, its exact values are*

$$(\Psi_H(C_0), \Psi_H(C_1), \Psi_H(C_2')) = (12\nu_6, 0, 12\nu_6). \tag{1360}$$

Its image and kernel are exactly

$$\operatorname{im}\Psi_H = \{0, 12\nu_6\}, \qquad \ker\Psi_H = \chi^{-1}(2\mathbb{Z}/12). \tag{1361}$$

In particular its kernel has index two in $SL_2(\mathbb{Z})$*.*

Proof. The element $[h]$ has exact order two, so replacing n by $n + 12k$ leaves $n[h]$ unchanged. Addition in its abelian homotopy group proves that Θ_H is a group homomorphism. Moreover

$$\ker\Theta_H = 2\mathbb{Z}/12, \qquad \operatorname{im}\Theta_H = \pi_6(S^4).$$

The composite in (1358) is consequently a group homomorphism, and (1352) gives $\mu_\eta(n[h]) = 12n\nu_6$. Changing the integer lift by $12k$ changes its coefficient by $144k$, which is a multiple of twenty-four. This proves (1359) with its independence of lifts. The three coefficients are respectively $-12 \equiv 12$, $48 \equiv 0$, and $-36 \equiv 12 \pmod{24}$, proving (1360). Since $12n \equiv 0 \pmod{24}$ holds exactly when n is even, the asserted kernel is exact. Surjectivity of χ and nonvanishing at C_0 prove that the image is the full order-two subgroup, so the first isomorphism theorem gives the index assertion. □

Here is the complete parity factorization. Let $\mathrm{red}_2 : \mathbb{Z}/12 \to \mathbb{Z}/2$ be ordinary parity reduction, let $\iota_2([e]) = 12e \pmod{24}$, and let $c_{\nu_6} : \pi_9(S^6) \to \mathbb{Z}/24$ send ν_6 to one. Every arrow and its codomain in the diagram are fixed:

$$\begin{array}{ccc} SL_2(\mathbb{Z}) & \xrightarrow{\Psi_H} & \pi_9(S^6) \\ {\scriptstyle \mathrm{red}_2 \circ \chi}\downarrow & & \downarrow{\scriptstyle c_{\nu_6}} \\ \mathbb{Z}/2 & \xrightarrow{\iota_2} & \mathbb{Z}/24. \end{array} \tag{1362}$$

Commutativity is exactly the coefficient calculation above; the bottom arrow is injective because $12e$ is zero modulo twenty-four exactly when e is zero modulo two.

Comparison with the earlier maps, with their exact types Write $\psi_H = c_{\nu_6} \circ \Psi_H$. Ordinary reduction $r : \mathbb{Z}/24 \to \mathbb{Z}/12$ gives

$$r \circ \psi_H = 0, \qquad (r \circ \psi_H)(C_0) = 0 \neq -1 = \chi(C_0). \tag{1363}$$

Thus this nonzero character is not a reduction-compatible lift of χ. Its natural commuting square is (1362). The existing nonexistence theorem for a reduction-compatible lift remains valid: every homomorphism

from $SL_2(\mathbb{Z})$ to an abelian group factors through its abelianization $\mathbb{Z}/12$, whereas a lift of $\chi(C_0) = -1$ would require C_0 to map to 11 or 23 modulo twenty-four, each of order twenty-four. The abelianized generator has order twelve and cannot have either image.

The universal even-subgroup injection is the separate map

$$j : \mathbb{Z}/12 \hookrightarrow \mathbb{Z}/24, \qquad j([a]) = [2a].$$

The new coefficient morphism factors through it by the exact identity

$$\psi_H = j \circ [6] \circ \chi, \qquad [6] : \mathbb{Z}/12 \longrightarrow \mathbb{Z}/12, \quad [a] \longmapsto [6a]. \tag{1364}$$

Indeed $2(6a) = 12a \pmod{24}$. The map j is injective with image of order twelve; the composite (1364) has image of order two because [6] retains precisely parity. The map j is realized by the already proved universal suspension $E^6 : \pi_6(S^3) \to \pi_{12}(S^9)$ after compatible generator choices. Equation (1364) is a coefficient identity; the actual new geometric representative is $\eta_2 \wedge h$, followed, when desired, by three suspensions as above.

For the original angle-forgetting map $g = \widehat{p}_2 q_2 : X \to S^3$, the separate framed calculation has proved $[g \circ d] = 0$. Therefore the previous morphism

$$\Theta_g : \mathbb{Z}/12 \longrightarrow \pi_6(S^3), \qquad [n] \longmapsto n[g \circ d]$$

and $E^6\Theta_g\chi$ are zero. In contrast, $E^3\Psi_H(C_0) = 12\nu_9 \neq 0$ in this same stable receiver $\pi_{12}(S^9)$. These statements agree because the exact postcompositions from $\mathcal{A}$ are different: p_3 retains the angular coordinate in its top-cell map, while $\widehat{p}_2$ forgets it. The new homomorphism uses the already proved value of $p_3 q_2$ and the specified Hopf map, with all maps displayed in (1349)–(1358).

The constructed representative has source $S^3 \wedge X \cong S^9$; its threefold suspension has source S^{12}. The number twenty-four is the order of the receiving stable homotopy group. This calculation does not identify either source with a twenty-four-dimensional carrier, does not supply new periods or analytic overlap maps, and does not complete the global higher torus family. It supplies the exact nonzero homotopy and peripheral morphisms just proved.

Source checks. The three matrices and the marked values were checked directly against equations *elliptic-peripheral-matrices* and *peripheral-mod12-values* in the existing higher-rung draft [21]. The calculation above reproduces their signs using the same S_0, T. The retained-angle input is the complete derivative-framing theorem in [20]. For the stable facts, the author-hosted text [22] was read at Corollary 4.24, printed p. 360, and at printed pp. 384–385. Page 385 explicitly distinguishes the two-primary coefficient four from the integral coefficient twelve. Proposition 4.56 on that page, with its proof on p. 388, gives the stable composition product and its smash description. The fixed-factor proof (1353)–(1354) establishes the needed homomorphism here directly.

30.6.7 Boundary of the three tested numerical mechanisms

Theorem 30.62 (Boundary of the three tested numerical mechanisms)**.** *For the chamber indexing, peripheral abelianization, stable suspension, and the three additional mechanisms tested below, the verified relations are the following.*

(i) *The six Kuhn zero-strata and the six octonionic fixed flags have the marked S_3-equivariant finite-set bijection* (1204)*. Their common labels also index the flag cells, and both Euler characteristics are* 6*.*

(ii) *The elliptic-kernel peripheral matrices produce the surjective character $\chi : SL_2(\mathbb{Z}) \to \mathbb{Z}/12$, with no reduction-compatible or surjective mod–*24 *lift.*

(iii) *The separate stable suspension map embeds* $\mathbb{Z}/12$ *as* $2(\mathbb{Z}/24)$*, and proposition 30.48 gives the exact composite determined by* g*. The actual coefficient is* $a = 0$ *by theorem 30.49; the composite is zero, and the associated* S^{24} *fourth-smash map is null-homotopic.*

(iv) *The ambient* F_4 *root system has* $h = 12$*, and* $W(F_4)/W(D_4) \cong S_3$*. The three specific tests give: diagonal-sum factorization through* P *or* Q *exactly for* $N \mid 3$*; ordinary cohomology reduction for every* N*, with* e_N *of order* $N/\gcd(N, 6)$*; and plain determinant-winding reduction onto* $\mathbb{Z}/N$ *for every* N*. None of these specified operations selects* $N = 12$ *or* $N = 24$ *from the dimension or Coxeter number alone.*

No arrow between these four lines is defined merely by equality of their integers.

Proof. Item (i) is theorem 30.27; item (ii) is theorem 30.44; item (iii) is theorem 30.47 and proposition 30.48; and the Coxeter assertion in (iv) is proposition 30.26. The three stated tests for an additional arithmetic arrow are evaluated by propositions 30.45 and 30.46: the primitive lattice diagonal-sum factorization allows only $N \mid 3$, cohomology reduces for every modulus, and determinant winding reduces onto every modulus. These computations prove the stated boundary for those mechanisms without asserting that arbitrary future structures cannot be added. □

30.7 The full Albert orbit quotient and the Cartan family

The fixed-spectrum comparison of theorem 30.24 and section 30.5.3 extends to the whole Albert algebra with all three eigenvalue variables retained. We give the invariant quotient, its smooth regular part, the exact normal framing, and the two focal strata. The diagonalization and flag geometry are established in [23, Proposition 2.8.2 and Theorems 2.7.4, 2.8.3] and [10, Definition 2.2 and Sections 2.3–2.6]; the calculations below specify their maps in the retained coordinates.

30.7.1 Original space, invariants, and direct diagonalization

Let $\mathbb{O}$ be the real octonion division algebra with its retained conjugation, norm N, and inner product $h(a, b) = \mathrm{Re}(a\bar{b})$. Set $J = \mathfrak{h}_3(\mathbb{O})$, with Jordan product $X \circ Y = (XY + YX)/2$, and retain the cyclic coordinates of (1211):

$$X = \begin{pmatrix} \xi_1 & q_3 & \bar{q}_2 \\ \bar{q}_3 & \xi_2 & q_1 \\ q_2 & \bar{q}_1 & \xi_3 \end{pmatrix} = \sum_{i=1}^{3} \xi_i e_i + \sum_{i=1}^{3} E_i(q_i). \tag{OCB1}$$

Here e_i are the original diagonal idempotents, E_i are the original off-diagonal inclusions, and $I = e_1 + e_2 + e_3$. Matrix products in the Jordan product involve only products of two octonions. Powers of a single Jordan element are unambiguous by the Jordan identity; all subsequent polynomial formulas use these powers.

The positive trace metric and determinant are

$$\begin{aligned} G(X, Y) &= \mathrm{tr}(X \circ Y), \\ G(X, X) &= \xi_1^2 + \xi_2^2 + \xi_3^2 + 2\sum_i N(q_i), \\ d(X) &= \xi_1\xi_2\xi_3 - \sum_i \xi_i N(q_i) + 2\,\mathrm{Re}((q_1 q_2)q_3). \end{aligned} \tag{OCB2}$$

Thus $\dim_{\mathbb{R}} J = 27$, with every diagonal coordinate, metric factor, and mixed cubic retained. Define

$$t(X) = \mathrm{tr}\, X, \qquad u(X) = G(X, X), \qquad e_2(X) = \frac{t(X)^2 - u(X)}{2}, \qquad \Phi(X) = (t(X), u(X), d(X)). \tag{OCB3}$$

Let $F_4 = \mathrm{Aut}_{\mathbb{R}}(J, \circ)$. An automorphism fixes I, preserves trace and G, and therefore preserves the determinant by the trace identity

$$d(X) = \tfrac{1}{3}\,\mathrm{tr}(X^3) - \tfrac{1}{2}t(X)u(X) + \tfrac{1}{6}t(X)^3. \tag{OCB4}$$

For completeness, trace preservation follows without choosing a preferred matrix realization: the Peirce table gives $\mathrm{Tr}_{\mathrm{End}(J)}(L_X) = 9\,\mathrm{tr}\,X$; conjugation of L_X by an automorphism proves the assertion. Product preservation is a closed polynomial condition, so F_4 is closed in the orthogonal group of (J, G) and hence compact.

Proposition 30.63 (Diagonalization and the exact invariant fibres)**.** *Every $X \in J$ is F_4-conjugate to a real diagonal matrix. The fibres of Φ are exactly the F_4 orbits, including repeated-eigenvalue and scalar orbits. The map $\Phi : J \to \Phi(J)$ is proper.*

Proof. On the compact orbit of X, maximize $\xi_1^2 + \xi_2^2 + \xi_3^2$. If $q_i \neq 0$ at a maximizing point, put $a = q_i/|q_i|$, let (i, j, k) be cyclic, and use the already proved derivation

$$D_i(a) = 4[L_{E_i(a)}, L_{e_j}]. \tag{OCB5}$$

Its action satisfies $D_i(a)(e_j - e_k) = 2E_i(a)$ and $D_i(a)E_i(a) = -2(e_j - e_k)$. The remaining off-diagonal slots do not contribute diagonal entries under this action. Write the relevant component as $v(e_j - e_k) + wE_i(a)$, with $v = (\xi_j - \xi_k)/2$ and $w = |q_i| > 0$. Exponentiation rotates (v, w) through angle $2s$. Some rotation gives squared diagonal-difference coordinate $v^2 + w^2$, increasing the diagonal-square sum by $2w^2$. This contradicts maximality. Hence every off-diagonal coordinate vanishes at a maximum, proving diagonalization with the diagonal mean retained.

If $X = g\sum_i \lambda_i e_i$, then (OCB3)–(OCB4) show that the multiset of real eigenvalues is exactly the multiset of roots of

$$p_X(z) = z^3 - t(X)z^2 + e_2(X)z - d(X). \tag{OCB6}$$

It is independent of the diagonalizing automorphism. Every permutation of a real diagonal is realized by conjugation with a real orthogonal permutation matrix. Such conjugation preserves the Jordan product; a commuting sign adjustment can be used if determinant $+1$ is desired. Equal invariant triples thus give the same eigenvalue multiset and the same orbit. Conversely, the invariants are constant on orbits, as proved above. This includes repeated roots and scalar matrices. Finally, a compact set of invariant triples bounds $u = G(X, X)$, and its inverse image is closed and bounded in J, hence compact. □

This is the compact-orbit diagonalization argument of [23, Proposition 2.8.2, pp. 56–57], reproduced in the retained derivation coordinates. It also proves the polynomial identities used below by transporting their diagonal versions.

30.7.2 Exact projectors, smooth orbit inverse, and regular family

Fix labelled, pairwise distinct real numbers $\lambda_1, \lambda_2, \lambda_3$, with arbitrary sum. Put $d_\lambda = \sum_i \lambda_i e_i$ and $\mathcal{O}_\lambda = F_4 d_\lambda$. The ordered frame space is

$$\mathcal{F} = \{(p_1, p_2, p_3) : p_i^2 = p_i,\ \mathrm{tr}\,p_i = 1,\ p_i \circ p_j = 0\ (i \neq j),\ \textstyle\sum_i p_i = I\}.$$

The correspondence $(p_1, p_2, p_3) \mapsto (p_1, p_2)$ and its inverse $(p_1, p_2) \mapsto (p_1, p_2, I - p_1 - p_2)$ identify this space with the flag space of (1192); in particular $\mathcal{F} = F_4/\mathrm{Spin}(8)$ by theorem 30.24. Define

$$\Xi_\lambda : \mathcal{F} \longrightarrow \mathcal{O}_\lambda, \qquad (p_i) \longmapsto \sum_i \lambda_i p_i, \tag{OCB7}$$

and, for $\{i, j, k\} = \{1, 2, 3\}$, define

$$P_i^\lambda(X) = \frac{(X - \lambda_j I) \circ (X - \lambda_k I)}{(\lambda_i - \lambda_j)(\lambda_i - \lambda_k)}. \tag{OCB8}$$

Proposition 30.64 (The ordered spectral diffeomorphism)**.** *The map Ξ_λ is an F_4-equivariant diffeomorphism, with inverse $(P_1^\lambda, P_2^\lambda, P_3^\lambda)$. In particular* $\dim \mathcal{O}_\lambda = 24$*. If $\mathcal{C} = \{\lambda_1 > \lambda_2 > \lambda_3\}$ and J_{reg} is the set of elements with distinct eigenvalues, then $\mathcal{F} \times \mathcal{C} \to J_{\mathrm{reg}}$, given by the same spectral sum, is a diffeomorphism.*

Proof. For $X = \sum_l \lambda_l p_l$, orthogonality and idempotency give

$$\begin{aligned}(X - \lambda_j I) \circ (X - \lambda_k I) &= \sum_l (\lambda_l - \lambda_j)(\lambda_l - \lambda_k) p_l \\ &= (\lambda_i - \lambda_j)(\lambda_i - \lambda_k) p_i.\end{aligned}$$

Thus (OCB8) recovers every ordered projector and proves both inverse compositions. It also proves that the stabilizer of d_λ is the pointwise diagonal-frame stabilizer Spin(8). The quotient orbit map is an injective immersion of the compact homogeneous space $F_4/\operatorname{Spin}(8)$ into the Hausdorff vector space J, hence an embedding. Its inverse is polynomial in X divided by nonzero fixed real constants, proving smoothness of the inverse and the asserted equivariant diffeomorphism.

The differential at fixed λ is explicitly

$$dP_i^\lambda|_X(H) = \frac{2X \circ H - (\lambda_j + \lambda_k)H}{(\lambda_i - \lambda_j)(\lambda_i - \lambda_k)}. \tag{OCB9}$$

At d_λ, the sector $E_i(\mathbb{O})$ joins eigenspaces j, k. On this sector $dP_j(H) = H/(\lambda_j - \lambda_k)$, $dP_k(H) = -H/(\lambda_j - \lambda_k)$, and $dP_i(H) = 0$. Substitution into $\sum_l \lambda_l dP_l(H)$ returns H, verifying the differential inverse with its signs.

The original derivations satisfy

$$\begin{gathered}D_i(a) d_\lambda = (\lambda_j - \lambda_k) E_i(a) = \Delta_i E_i(a), \\ (\Delta_1, \Delta_2, \Delta_3) = (\lambda_2 - \lambda_3, \lambda_3 - \lambda_1, \lambda_1 - \lambda_2), \qquad (i,j,k) \text{ cyclic.}\end{gathered} \tag{OCB10}$$

All three multipliers are nonzero, so the three eight-dimensional off-diagonal sectors constitute the tangent space. Their transported Peirce subbundles at every orbit point have metric $2\Delta_i^2 h$ in the homogeneous tangent coordinates of (1235).

For the full map

$$\mathcal{F} \times \mathcal{C} \longrightarrow J_{\mathrm{reg}}, \qquad ((p_i), (\lambda_i)) \longmapsto \sum_i \lambda_i p_i, \tag{OCB11}$$

surjectivity follows from diagonalization and sorting. Injectivity follows from the unique ordered roots and (OCB8). At a simple root, $\partial p_X / \partial z = \prod_{j \neq i} (\lambda_i - \lambda_j) \neq 0$; the implicit function theorem therefore gives a smooth local root function. Ordering glues these functions globally. Equation (OCB8), with the smoothly varying roots, provides the smooth inverse of (OCB11). □

30.7.3 Retained mean, radius, and angular parameter

For every original X, put

$$\mu = \frac{t(X)}{3}, \qquad Y = X - \mu I, \qquad s = G(Y,Y) = u(X) - \frac{t(X)^2}{3}, \qquad d_0 = d(Y). \tag{OCB12}$$

Write $J_0 = \mathfrak{h}_3^0(\mathbb{O}) = \{Y \in J : \operatorname{tr} Y = 0\}$. The displayed change of variables has inverse $X = \mu I + Y$; μ and s retain their actual values. The eigenvalues $a_i = \lambda_i - \mu$ of Y satisfy $\sum_i a_i = 0$, so expansion gives

$$d(X) = \mu^3 - \frac{\mu s}{2} + d_0. \tag{OCB13}$$

For $s = r^2 > 0$, the ordered eigenvalues have the exact parametrization

$$\begin{aligned} \lambda_1 &= \mu + \sqrt{2/3}\, r\cos\theta, \\ \lambda_2 &= \mu + \sqrt{2/3}\, r\cos(\theta - 2\pi/3), \qquad 0 \le \theta \le \pi/3. \\ \lambda_3 &= \mu + \sqrt{2/3}\, r\cos(\theta + 2\pi/3), \end{aligned} \tag{OCB14}$$

The cosines sum to zero, their squared sum is $3/2$, and their product is $\frac{1}{4}\cos 3\theta$. Consequently

$$\begin{aligned} t(X) &= 3\mu, & u(X) &= 3\mu^2 + r^2, \\ d_0 &= \frac{r^3}{3\sqrt{6}}\cos 3\theta, & d(X) &= \mu^3 - \frac{\mu r^2}{2} + \frac{r^3}{3\sqrt{6}}\cos 3\theta. \end{aligned} \tag{OCB15}$$

For $0 < \theta < \pi/3$ the ordered inequalities are strict. The inverse is

$$\begin{aligned} \mu &= t/3, \qquad r = \sqrt{u - t^2/3}, \\ \theta &= \tfrac{1}{3}\arccos\left(\frac{3\sqrt{6}\,[d - \mu^3 + \mu r^2/2]}{r^3}\right). \end{aligned} \tag{OCB16}$$

To prove that every ordered triple is listed, put $a_3 = -a_1 - a_2$ and expand the discriminant:

$$\mathcal{D} = \prod_{i<j}(a_i - a_j)^2 = \frac{s^3}{2} - 27d_0^2 \ge 0. \tag{OCB17}$$

Thus the arccos argument lies in $[-1, 1]$. The triple in (OCB14) has the same characteristic polynomial and the correct ordering, hence equals the original triple. Equality in (OCB17) means repeated eigenvalues. If $r = 0$, positivity of G gives $Y = 0$ and $X = \mu I$; this scalar stratum requires no angle.

The full invariant image is therefore exactly

$$\left\{(t, u, d) : s = u - t^2/3 \ge 0, \quad 27(d - \mu^3 + \mu s/2)^2 \le s^3/2, \quad \mu = t/3\right\}. \tag{OCB18}$$

Every triple in this set is attained by (OCB14), including the scalar case just described. This gives the whole three-dimensional orbit quotient; specified mean and radius select its corresponding fibre.

The full regular metric also records these parameters. Let h_i be the metric h on the ith associated bundle, expressed by the homogeneous tangent coordinate $K_i[g, a]$ of (1234). The pullback of G by (OCB11)–(OCB14) is

$$3\,d\mu^2 + dr^2 + r^2 d\theta^2 + \sum_i 2\Delta_i(\mu, r, \theta)^2 h_i. \tag{OCB19}$$

Indeed the diagonal differential has coefficient vectors $\mathbf{1}$, a/r, and $\partial_\theta a$. Their pairwise inner products vanish, and their squared norms are $3, 1, r^2$. Every diagonal differential is orthogonal to every off-diagonal Peirce sector. The sector terms follow from (OCB10) and $E_i^* G = 2h$, with every eigenvalue difference and radial factor retained.

30.7.4 Exact codimension and global normal framing

The same 24-dimensional regular orbit has codimension three in J, two in the affine fixed-trace space $\mu I + J_0$, and one in the fixed-trace, fixed-radius sphere $\mu I + S_r^{25}$. The change of variables (OCB12) and the radius-level inclusion give the actual maps between these ambient descriptions.

For a fixed distinct triple, its normal space in J is

$$\nu_X \mathcal{O}_\lambda = \operatorname{span}_{\mathbb{R}}\{P_1^\lambda(X), P_2^\lambda(X), P_3^\lambda(X)\}. \tag{OCB20}$$

At a diagonal point this is the orthogonal complement of the three off-diagonal tangent sectors; transport by F_4 proves the equality everywhere. The projectors are pairwise G-orthogonal with squared norm one.

The differential of the invariant map has gradients

$$\nabla t = I, \qquad \nabla u = 2X, \qquad \nabla d = X^{\#} = X^2 - tX + e_2 I. \tag{OCB21}$$

To check the determinant gradient, diagonalize X. By (OCB2), its derivative on a diagonal variation is $\sum_i \lambda_j \lambda_k H_{ii}$, where $\{i,j,k\} = \{1,2,3\}$, and its derivative on an off-diagonal variation is zero. The right side of (OCB21) is the diagonal vector of these coefficients; automorphism invariance proves the formula for every X, H. In the projector basis the coefficient determinant is

$$\det \begin{pmatrix} 1 & 1 & 1 \\ 2\lambda_1 & 2\lambda_2 & 2\lambda_3 \\ \lambda_2\lambda_3 & \lambda_1\lambda_3 & \lambda_1\lambda_2 \end{pmatrix} = -2(\lambda_1 - \lambda_2)(\lambda_1 - \lambda_3)(\lambda_2 - \lambda_3). \tag{OCB22}$$

Hence Φ has rank three exactly on the regular stratum, where its tangent kernel is the orbit tangent space.

In the fixed-trace hyperplane define the two normal vectors

$$Y = X - \mu I, \qquad Z = Y^2 - \frac{s}{3} I. \tag{OCB23}$$

They have trace zero and are normal by (OCB20). The trace-zero eigenvalue identities give

$$\operatorname{tr} Y^3 = 3d_0, \qquad \operatorname{tr} Y^4 = s^2/2, \qquad \operatorname{Gram}(Y,Z) = \begin{pmatrix} s & 3d_0 \\ 3d_0 & s^2/6 \end{pmatrix}. \tag{OCB24}$$

Specifically, $\sum_i a_i^3 = 3a_1a_2a_3$ follows from $\sum_i a_i = 0$. Squaring $\sum_{i<j} a_i a_j = -s/2$ gives $\sum_{i<j} a_i^2 a_j^2 = s^2/4$, since the remaining cross term is $2a_1a_2a_3 \sum_i a_i = 0$. Expanding s^2 now gives $\sum_i a_i^4 = s^2/2$. These identities and $Z = Y^2 - sI/3$ prove every Gram entry. Therefore

$$\det \operatorname{Gram}(Y,Z) = s^3/6 - 9d_0^2 = \mathcal{D}/3 > 0 \tag{OCB25}$$

for distinct eigenvalues. Thus (Y,Z) is a smooth ordered global two-framing of the orbit normal bundle in $\mu I + J_0$.

The corresponding explicitly specified orthonormal frame is

$$n_1 = \frac{Y}{\sqrt{s}}, \qquad n_2 = \frac{Z - (3d_0/s)Y}{\sqrt{s^2/6 - 9d_0^2/s}}. \tag{OCB26}$$

Every square root here is the positive real root; its argument is positive by (OCB25). In J, prepend $n_0 = I/\sqrt{3}$ to obtain (n_0, n_1, n_2). In the radius sphere the normal direction is n_2. These formulas identify the three normal bundles explicitly.

These normal fields are parallel for the normal connection along the fixed orbit. Each is a linear combination of the P_i^λ with coefficients constant on that orbit. Along $g(s)X$, its ambient derivative is $g'(s)$ applied to the corresponding diagonal normal vector, and lies in the transported off-diagonal tangent sectors by (OCB10). Its normal component is zero. Thus the normal bundle is flat with the displayed global parallel orthonormal frame.

For any transported normal $\eta = \sum_i \nu_i P_i^\lambda$ with fixed real ν_i, use the convention $A_\eta H = -(D_H \eta)^T$. Then

$$A_\eta\big|_{E_i(\mathbb{O})} = -\frac{\nu_j - \nu_k}{\lambda_j - \lambda_k}\ \mathrm{id}, \qquad (i,j,k) \text{ cyclic}, \tag{OCB27}$$

with the sectors transported by g at other points. Indeed a path generated by $D_i(a)$ has tangent $(\lambda_j - \lambda_k)E_i(a)$, whereas the derivative of its normal field is $(\nu_j - \nu_k)E_i(a)$. This proves the formula and its minus sign. The shape operator for n_0 is zero, and the one for the radial normal n_1 is $-\mathrm{id}/r$.

30.7.5 Cartan equations and spherical principal curvatures

On the whole trace-zero space J_0, with its original trace metric, define the explicitly scaled cubic

$$C(Y) = 3\sqrt{6}\, d(Y). \tag{OCB28}$$

The inverse relation is $d(Y) = C(Y)/(3\sqrt{6})$, and (OCB13) recovers the full determinant with its mean and radius. Orthogonal projection of (OCB21) to J_0 gives $\nabla_{J_0} d = Y^2 - sI/3 = Z$. Hence (OCB24) gives

$$\|\nabla C\|_G^2 = 54\|Z\|_G^2 = 9s^2, \qquad \Delta_{J_0} C = 0. \tag{OCB29}$$

For the second equality, the full J Laplacian of (OCB2) is $-8\,\mathrm{tr}\, X$. Each eight-dimensional octonion slot has metric $2h$ and contributes $-8\xi_i$; each pure diagonal second derivative is zero, and the mixed cubic is separately linear in each real coordinate. Restricting to J_0 subtracts the second derivative in the unit scalar direction $I/\sqrt{3}$. The identity

$$d(Y + zI/\sqrt{3}) = d(Y) - \frac{sz}{2\sqrt{3}} + \frac{z^3}{3\sqrt{3}}$$

shows that this second derivative is zero at $z = 0$. The full Laplacian also vanishes at trace zero, proving (OCB29) with the original metric factor two.

On each specified radius sphere $S_r^{25} \subset J_0$, $r > 0$, set $f_r = C/r^3$. Then (OCB15) gives $f_r = \cos 3\theta$, and homogeneity yields

$$\|\nabla_{S_r} f_r\|^2 = \frac{9}{r^2}(1 - f_r^2), \qquad \Delta_{S_r} f_r = -\frac{81}{r^2} f_r. \tag{OCB30}$$

Indeed the radial component of ∇C is $(3C/r)n_1$. The polar Laplacian on the 26-dimensional Euclidean space, applied to the degree-three homogeneous harmonic function, is $r[3(3 + 24)f_1 + \Delta_{S_1} f_1]$. These calculations prove both constants in (OCB30). Its regular level hypersurfaces are therefore isoparametric: both the squared gradient and the Laplacian depend only on the function value. By (OCB15)–(OCB18), the regular levels are precisely the fixed-mean, fixed-radius regular $F_4/\,\mathrm{Spin}(8)$ orbits.

The principal curvatures admit a direct calculation. On the diagonal section let $N_\theta = (1/r)\partial_\theta Y$, a unit vector perpendicular to Y, and transport it by the projectors. The exact eigenvalue differences in (OCB14) are

$$\Delta_1 = \sqrt{2} r \sin\theta, \qquad \Delta_2 = -\sqrt{2} r \sin(\theta + \pi/3), \qquad \Delta_3 = -\sqrt{2} r \sin(\theta - \pi/3). \tag{OCB31}$$

Apply (OCB27) with $\nu_i = (\partial_\theta \lambda_i)/r$. For the declared normal N_θ, the three principal curvatures are

$$-\frac{1}{r}\cot\theta, \qquad -\frac{1}{r}\cot(\theta + \pi/3), \qquad -\frac{1}{r}\cot(\theta - \pi/3), \tag{OCB32}$$

each on its original eight-dimensional Peirce sector. They are distinct for $0 < \theta < \pi/3$. In the invariant frame (OCB26), one has $n_2 = -N_\theta$: differentiating $C = r^3 \cos 3\theta$ gives $\nabla_{S_r} C = -3r^2 \sin 3\theta\, N_\theta$, whereas n_2 is the positive-length version of the projected determinant gradient. The principal curvatures for n_2 are consequently the opposites of (OCB32). This proves the three-curvature, multiplicity-eight Cartan family with both normal orientations, mean, and radius specified.

30.7.6 Focal projective planes and their endpoint maps

At $\theta = 0$ one has $\lambda_2 = \lambda_3 = \mu - r/\sqrt{6}$ and $\lambda_1 = \mu + 2r/\sqrt{6}$; at $\theta = \pi/3$ one has $\lambda_1 = \lambda_2 = \mu + r/\sqrt{6}$ and $\lambda_3 = \mu - 2r/\sqrt{6}$. The two singular orbits in $\mu I + S_r^{25}$ are exactly

$$\begin{aligned} \mathcal{P}^+_{\mu,r} &= \{\mu I + \tfrac{r}{\sqrt{6}}(3p - I) : p^2 = p,\ \mathrm{tr}\, p = 1\}, \\ \mathcal{P}^-_{\mu,r} &= \{\mu I - \tfrac{r}{\sqrt{6}}(3p - I) : p^2 = p,\ \mathrm{tr}\, p = 1\}. \end{aligned} \tag{OCB33}$$

Their inverse maps are, respectively,

$$p = \tfrac{1}{3}\left(I + \tfrac{\sqrt{6}}{r}(X - \mu I)\right), \qquad p = \tfrac{1}{3}\left(I - \tfrac{\sqrt{6}}{r}(X - \mu I)\right). \tag{OCB34}$$

These affine maps are smooth, have the displayed two-sided inverses on the specified sets, and are F_4-equivariant. The primitive idempotent space is the sixteen-dimensional Cayley projective plane $\mathbb{OP}^2 = F_4/\operatorname{Spin}(9)$ [23, Theorems 2.7.4 and 2.8.3] [10, Section 2.3]. Hence the formulas give embedded projective-plane strata.

The endpoint maps of the regular family are the explicit projections

$$\begin{aligned}(p_1, p_2, p_3) &\longmapsto \mu I + \tfrac{r}{\sqrt{6}}(3p_1 - I),\\ (p_1, p_2, p_3) &\longmapsto \mu I - \tfrac{r}{\sqrt{6}}(3p_3 - I).\end{aligned} \tag{OCB35}$$

Each fibre is $\operatorname{Spin}(9)/\operatorname{Spin}(8) \cong S^8$: fixing one primitive idempotent leaves two ordered complementary idempotents, the octonionic projective line. These frame bundles are identified in [10, Sections 2.3–2.6]. They are also the focal maps. Indeed the diagonal path $Y(\theta)$ is a great circle because $\partial_\theta^2 Y = -Y$ and $|\partial_\theta Y| = r$. Transport by F_4 therefore gives the normal geodesics to the regular hypersurfaces. At each endpoint one of the tangent multipliers in (OCB31) vanishes with its eight-dimensional sector, giving exactly the collapse in (OCB35). Each focal stratum has codimension nine in the ambient radius sphere and a normal sphere of dimension eight.

For $r = 0$, the construction gives the scalar line $X = \mu I$, each point fixed by F_4. For $r > 0$ and endpoint angle, the full repeated-eigenvalue stratum retains the two parameters (μ, r) in addition to its projective-plane coordinate, and has dimension eighteen in J. These descriptions retain every scalar, radial, and mean stratum.

30.7.7 Exact map from the equal-diagonal triality slice

For the original matrix $Q_{\mathbb{O}}(\lambda; q) = \lambda I + E(q)$, with $E(q) = \sum_i E_i(q_i)$ and $\tau(q) = \operatorname{Re}((q_1 q_2) q_3)$, the maps above give

$$\mu = \lambda, \qquad Y = E(q), \qquad r^2 = 2\sum_i N(q_i), \qquad d_0 = 2\tau(q), \qquad C(Y) = 6\sqrt{6}\,\tau(q). \tag{OCB36}$$

Thus its invariant triple is

$$\begin{aligned}\Phi(Q_{\mathbb{O}}(\lambda; q)) = \big(&3\lambda,\ 3\lambda^2 + 2\sum_i N(q_i),\\ &\lambda^3 - \lambda\sum_i N(q_i) + 2\tau(q)\big).\end{aligned} \tag{OCB37}$$

This is the explicit invariant morphism from the original triality coordinates to the full orbit quotient. At every simple spectrum, its ordered-projector lift is (OCB8) applied to $Q_{\mathbb{O}}(\lambda; q)$. The positive boundary graph of theorem 30.35 chooses the original scalar $\lambda = -\lambda_{\min}(E(q))$. In the full orbit coordinates, this says that the least eigenvalue is zero and the other two are nonnegative. Equations (OCB36)– (OCB37) prove the relation to the whole Albert orbit family with all coefficients and this scalar choice retained.

Attribution and scope. Yokota's diagonalization is Proposition 2.8.2, pp. 56–57; his primitive-idempotent stabilizer and Cayley-plane identification are Theorems 2.7.4 and 2.8.3, pp. 54 and 57 [23]. Mare and Willems give the primitive idempotents, full real diagonal space, frame stabilizer, projective plane, S^8 frame projections, and the polar isotropy representation in Definition 2.2 and Sections 2.3–2.6, pp. 488–493 [10]. Their description identifies J_0 with the isotropy representation of $E_{6(-26)}/F_4$ and displays its diagonal two-plane and three eight-dimensional root spaces. The Cartan-family equations and focal maps used here are proved in (OCB29)–(OCB35). The orbit and flag geometry is established literature; these calculations assert neither complex integrability nor physical dynamics.

30.8 The exact spectral map of the common marked lattice

The literal common-lattice embedding of theorem 36.6 has a matrix-valued and frame-valued lift into the Albert orbit family of section 30.7. We calculate this map with the three octonionic slots separately labelled, the trace metric, the arbitrary scalar diagonal, and the original determinant coefficient two.

30.8.1 Domain and the original matrix

Use the ordered quaternion subalgebra

$$\mathbb{H} = \mathbb{R}\langle 1, i, j, k\rangle \subset \mathbb{O}, \qquad i^2 = j^2 = k^2 = ijk = -1.$$

For $a = (a_0, a_1, a_2, a_3)$, write

$$\begin{gathered} v = a_1 i + a_2 j + a_3 k, \qquad t = \sqrt{a_1^2 + a_2^2 + a_3^2}, \qquad a = a_0 + v, \\ Q = a_0^2 + t^2, \qquad P = a_0(a_0^2 - 3t^2). \end{gathered} \tag{NCS1}$$

Every square root in this section means the nonnegative real root. Here t denotes the length of the imaginary quaternion, while the trace invariant will always be written $\operatorname{tr} X$. The integral domain and its retained embedding are

$$D_4 = \{a \in \mathbb{Z}^4 : a_0 + a_1 + a_2 + a_3 \in 2\mathbb{Z}\}, \qquad j(a) = \sqrt{2}(a, a, a) \in \mathbb{O}^3. \tag{NCS2}$$

The literal intersection theorem theorem 36.6 identifies $j(D_4)$ with the common lattice of the two specified marked Niemeier/Leech embeddings and the specified Wilson embedding. Its original Niemeier coordinates are

$$x_{i,m} = (-1)^m a_{i-1} \quad (1 \le i \le 4,\ 0 \le m \le 5), \qquad d = 0. \tag{NCS3}$$

The six coordinates in each A_5 block sum to zero. Pairing with the retained Weyl vector gives $3\sum_i a_i$, which is divisible by six for $a \in D_4$; hence these vectors belong to the retained neighbor sublattice K. The full Wilson membership certificate and both intersection equalities are proved in theorem 36.6.

For arbitrary $\mu \in \mathbb{R}$, define

$$Y(a) = \begin{pmatrix} 0 & \sqrt{2}a & \sqrt{2}\bar{a} \\ \sqrt{2}\bar{a} & 0 & \sqrt{2}a \\ \sqrt{2}a & \sqrt{2}\bar{a} & 0 \end{pmatrix}, \qquad X(\mu, a) = \mu I + Y(a). \tag{NCS4}$$

This is the original cyclic convention $(q_1, q_2, q_3) = (\sqrt{2}a, \sqrt{2}a, \sqrt{2}a)$, including the conjugated upper-right entry. On $J = \mathfrak{h}_3(\mathbb{O})$, the product is $A \circ B = (AB + BA)/2$ and the metric is $G(A, B) = \operatorname{tr}(A \circ B)$. Consequently

$$\|j(a)\|^2 = 6Q, \qquad r^2 := G(Y, Y) = 12Q. \tag{NCS5}$$

This retains the factor two between the two metrics. The map $(\mu, a) \mapsto X(\mu, a)$ from $\mathbb{R} \times D_4$ to J is injective: its inverse on its image is $\mu = \operatorname{tr} X/3$ and $a = X_{12}/\sqrt{2}$.

30.8.2 Complete determinant and invariant map

The quaternion identity $v^2 = -t^2$ and associativity give

$$\operatorname{Re}(a^3) = a_0^3 - 3a_0 t^2 = P.$$

Thus the triality cubic and determinant of Y are exactly

$$\begin{aligned} \tau(j(a)) &= \operatorname{Re}((\sqrt{2}a\sqrt{2}a)\sqrt{2}a) = 2\sqrt{2}P, \\ \det Y &= 2\tau(j(a)) = 4\sqrt{2}P. \end{aligned} \tag{NCS6}$$

The complete invariant map is therefore

$$\boxed{(\operatorname{tr} X, \operatorname{tr} X^2, \det X) = (3\mu,\ 3\mu^2 + 12Q,\ \mu^3 - 6\mu Q + 4\sqrt{2}P).} \tag{NCS7}$$

Indeed each of the three squared off-diagonal norms is $2Q$. The negative terms in the original determinant formula thus sum to $-6\mu Q$, while the mixed term is precisely (NCS6). This proves the displayed invariant map with every diagonal variable and original lattice scale retained. The characteristic polynomial of Y is

$$p_Y(z) = z^3 - 6Qz - 4\sqrt{2}P. \tag{NCS8}$$

30.8.3 Exact eigenvalues, projectors, and directional inverse

The three labelled eigenvalues of Y are

$$\nu_0 = 2\sqrt{2}a_0, \qquad \nu_1 = -\sqrt{2}a_0 - \sqrt{6}t, \qquad \nu_2 = -\sqrt{2}a_0 + \sqrt{6}t. \tag{NCS9}$$

Their sum is zero, their squared sum is $12Q$, and their product is $4\sqrt{2}P$. Direct multiplication of $\prod_{j=0}^{2}(z - \nu_j)$ gives (NCS8), proving that these are the eigenvalues. The eigenvalues of X are $\mu + \nu_j$, with μ arbitrary.

For $t > 0$, let $n = v/t$, so $n^2 = -1$ and $v = tn$. The plane $\mathbb{C}_n = \mathbb{R}\langle 1, n\rangle$ is a commutative associative subalgebra. Set

$$T = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, \qquad T^3 = I, \tag{NCS10}$$

$$\omega_n = -\frac{1}{2} + \frac{\sqrt{3}}{2}n, \qquad \omega_n^3 = 1, \qquad 1 + \omega_n + \omega_n^2 = 0.$$

Every entry of (NCS4) lies in $\mathbb{C}_n$, so associative matrix multiplication is legitimate throughout this calculation. Define

$$\Pi_j(n) = \frac{1}{3}\left(I + \omega_n^{-j}T + \omega_n^{-2j}T^2\right), \qquad j = 0, 1, 2. \tag{NCS11}$$

These matrices are Hermitian: $T^* = T^2$, $\bar{\omega}_n = \omega_n^{-1}$, and the two coefficients are conjugates. Each trace is one. To verify all projector identities, use

$$\sum_{k=0}^{2} \omega_n^{mk} = \begin{cases} 3, & 3 \mid m, \\ 0, & 3 \nmid m. \end{cases}$$

Expanding (NCS11) gives

$$\Pi_i\Pi_j = \delta_{ij}\Pi_j, \qquad \sum_{j=0}^{2}\Pi_j = I, \qquad T\Pi_j = \omega_n^j\Pi_j. \tag{NCS12}$$

Furthermore $Y = \sqrt{2}(aT + \bar{a}T^2)$, whence

$$\begin{aligned} Y\Pi_j &= \sqrt{2}(a\omega_n^j + \bar{a}\omega_n^{2j})\Pi_j = \nu_j\Pi_j, \\ X &= \sum_{j=0}^{2}(\mu + \nu_j)\Pi_j. \end{aligned} \tag{NCS13}$$

Multiplication in $\mathbb{C}_n$ gives exactly the three signs in (NCS9). Thus (NCS11) gives an actual map into the ordered Jordan-frame space $F_4/\operatorname{Spin}(8)$.

When the eigenvalues are pairwise distinct, these are also the intrinsic spectral projectors:

$$\Pi_j = \frac{(Y-\nu_k I)\circ(Y-\nu_l I)}{(\nu_j-\nu_k)(\nu_j-\nu_l)}, \qquad \{j,k,l\}=\{0,1,2\}. \tag{NCS14}$$

Substitution of (NCS13), together with (NCS12), proves this formula. Its denominators are retained; their exceptional loci are computed below.

The labelled frame retains the direction n. Specifically,

$$(\Pi_1)_{12} = -\frac{1}{6} - \frac{\sqrt{3}}{6}n, \qquad n = -2\sqrt{3}\,\mathrm{Im}(\Pi_1)_{12}. \tag{NCS15}$$

Together with $a_0 = \nu_0/(2\sqrt{2})$, $t = (\nu_2-\nu_1)/(2\sqrt{6})$, and $v = tn$, this recovers the original quaternion from the labelled spectrum and frame. The scalar parameter is recovered from $\mathrm{tr}\,X/3$. Thus the matrix and frame lift has an exact inverse on its labelled image. The invariant triple (NCS7) alone loses n; the lift (NCS11) records it explicitly.

At $t=0$, one has $Y = \sqrt{2}a_0(T+T^2)$. The rank-one projector $\Pi_0 = (I+T+T^2)/3$ remains fixed, and $I-\Pi_0$ is the rank-two projector of the repeated eigenvalue $-\sqrt{2}a_0$ when $a_0 \neq 0$. No direction n is assigned to a zero imaginary part. At $a=0$, the matrix is the scalar μI.

30.8.4 Singular loci and the exact Cartan angle

The discriminant of the original polynomial (NCS8), with its full scalar factor, is

$$\boxed{\prod_{i<j}(\nu_i-\nu_j)^2 = 864(Q^3-P^2) = 864t^2(3a_0^2-t^2)^2.} \tag{NCS16}$$

Indeed expansion of $(a_0^2+t^2)^3 - a_0^2(a_0^2-3t^2)^2$ gives $t^2(3a_0^2-t^2)^2$. The first equality also follows by multiplying the three differences in (NCS9). The spectrum is simple exactly when $t>0$ and $t^2 \neq 3a_0^2$.

For nonzero a, use the Cartan polynomial of (OCB28) in the original trace metric. Equations (NCS5)–(NCS6) give

$$C(Y) = 3\sqrt{6}\det Y = 24\sqrt{3}P, \qquad r^2 = 12Q,$$
$$\boxed{C(Y) = r^3\cos 3\theta, \qquad \theta = \frac{1}{3}\arccos\left(\frac{P}{Q^{3/2}}\right) \in [0,\pi/3].} \tag{NCS17}$$

The discriminant identity proves that the arccos argument belongs to $[-1,1]$. This is an identity for C with the whole amplitude retained: Y still has radius $r=\sqrt{12Q}$, and its scalar-shifted determinant is still (NCS7).

To link this to the quaternionic argument, define $\varphi \in [0,\pi]$ by

$$a_0 = \sqrt{Q}\cos\varphi, \qquad t = \sqrt{Q}\sin\varphi.$$

When $t>0$, the inverse is $a = \sqrt{Q}(\cos\varphi + n\sin\varphi)$, retaining Q,φ,n. The identity $4\cos^3\varphi - 3\cos\varphi = \cos 3\varphi$ gives

$$\cos 3\theta = \cos 3\varphi. \tag{NCS18}$$

The ordered Cartan chamber thus folds the quaternionic angular variable by this explicit triple-angle map. The labelled frame, (NCS9), and (NCS15) retain the inverse information. This quaternionic angle is defined by (NCS1); its definition makes no identification with a cusp angular coordinate.

For $t>0$, the decreasing orders of the labelled spectrum are

$$\begin{array}{c|l} a_0 > t/\sqrt{3} & \nu_0>\nu_2>\nu_1 \\ -t/\sqrt{3} < a_0 < t/\sqrt{3} & \nu_2>\nu_0>\nu_1 \\ a_0 < -t/\sqrt{3} & \nu_2>\nu_1>\nu_0. \end{array} \tag{NCS19}$$

The differences $\nu_0 - \nu_2 = \sqrt{2}(3a_0 - \sqrt{3}t)$ and $\nu_0 - \nu_1 = \sqrt{2}(3a_0 + \sqrt{3}t)$ prove each row. At $a_0 = t/\sqrt{3}$, precisely $\nu_0 = \nu_2$; at $a_0 = -t/\sqrt{3}$, precisely $\nu_0 = \nu_1$. Thus these rows specify how to order the projectors before applying (OCB7). At the two double-eigenvalue loci, the formulas (NCS11) remain defined, and (NCS12)– (NCS13) give repeated-eigenspace projectors $\Pi_0 + \Pi_2$ and $\Pi_0 + \Pi_1$, respectively. Their complementary simple-eigenvalue projectors are Π_1 and Π_2. This follows directly by grouping the equal coefficients in (NCS13); no vanishing denominator in (NCS14) is used.

30.8.5 Actual integral witnesses in the orbit family

The image proof of theorem 37.1 supplies three original-coordinate lattice lines:

$$\begin{aligned} L_\varepsilon(h) &= (-3h - \varepsilon, h, h, h + \varepsilon), \qquad \varepsilon = \pm 1, \\ L_0(h) &= (-3h, h + 1, h - 1, h), \qquad h \in \mathbb{Z}. \end{aligned} \tag{NCS20}$$

Their coordinate sums are zero. Direct expansion gives

a	$P(a)$	$Q(a)$
$L_\varepsilon(h)$	$6h + 2\varepsilon$	$12h^2 + 8\varepsilon h + 2$
$L_0(h)$	$18h$	$12h^2 + 2.$

(NCS21)

The eigenvalues, projectors, full radius, and scalar-shifted determinant of each actual common-lattice vector are therefore given by (NCS7)– (NCS17), evaluated at (NCS21) and its original imaginary vector v. The Cartan angles satisfy

$$\begin{aligned} \cos 3\theta_\varepsilon(h) &= \frac{6h + 2\varepsilon}{(12h^2 + 8\varepsilon h + 2)^{3/2}}, \\ \cos 3\theta_0(h) &= \frac{18h}{(12h^2 + 2)^{3/2}}. \end{aligned} \tag{NCS22}$$

As $|h| \to \infty$, both ratios tend to zero: the numerators have degree one, while their positive denominators grow as $|h|^3$. Hence the angles tend to $\pi/6$, and simultaneously $r^2 = 12Q \to \infty$. The original matrices and their radii remain those in (NCS4)– (NCS5).

On the shared direction $a = h(-3, 1, 1, 1)$ one has $P = 0$ and $Q = 12h^2$. For $h \neq 0$, the decreasing spectrum is exactly

$$(6\sqrt{2}|h|,\ 0,\ -6\sqrt{2}|h|), \qquad r = 12|h|, \qquad \theta = \pi/6. \tag{NCS23}$$

The discriminant (NCS16) is strictly positive, so this zero-cubic direction lies on a regular Cartan orbit. The spectrum of $X(\mu, a)$ is obtained by adding μ to each displayed eigenvalue.

Sources and scope of the bridge. The literal-intersection theorem theorem 36.6 supplies the marked Niemeier/Leech and Wilson membership proofs for (NCS2)–(NCS3); the mixed cubic is (1482). Wilson's construction is [81]. The full invariant quotient, retained mean and radius, ordered spectral diffeomorphism, and focal maps are proved in section 30.7, using the flag geometry of [10, Sections 2.3–2.6] and [23, Proposition 2.8.2 and Theorems 2.7.4, 2.8.3]. The complete image theorem theorem 37.1 proves

$$P(D_4) = 18\mathbb{Z} \cup (6\mathbb{Z} + 2) \cup (6\mathbb{Z} - 2)$$

using exactly (NCS20). This section provides its matrix-valued and frame-valued lift with an inverse on the labelled image; it makes no assertion that every Cartan orbit meets this rank-four lattice. The construction is independent of the assertion that the original completed complex threefold is diffeomorphic to S^6. No interacting quantum Hamiltonian is specified here. The retained cusp bundle has its separate explicit triality-subgroup map; its source angular coordinate and the quaternionic angular coordinate above retain their respective definitions.

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30.9 The Koide cone and the canonical period discriminant

For $r = (r_1, r_2, r_3) \in \mathbb{R}^3$, set

$$Q_K(r) := \frac{2}{3}(r_1 + r_2 + r_3)^2 - (r_1^2 + r_2^2 + r_3^2) \tag{1365}$$

and introduce the orthogonal coordinates

$$\begin{pmatrix} h \\ x \\ y \end{pmatrix} = \mathcal{F} \begin{pmatrix} r_1 \\ r_2 \\ r_3 \end{pmatrix}, \qquad \mathcal{F} = \begin{pmatrix} 1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} \\ 1/\sqrt{2} & 0 & -1/\sqrt{2} \\ 1/\sqrt{6} & -2/\sqrt{6} & 1/\sqrt{6} \end{pmatrix}. \tag{1366}$$

We use exactly the canonical period convention

$$Z = \begin{pmatrix} 6\mu & \tau \\ \beta & \mu \end{pmatrix}, \qquad \Pi = [Z \mid I_2]. \tag{1367}$$

Write

$$t = \operatorname{Im}\tau, \qquad m = \operatorname{Im}\mu, \qquad \varrho = \operatorname{Im}\beta.$$

Relative to the period-lattice basis

$$(\widehat{\gamma}, \widehat{u}, \widehat{w}, \widehat{\delta})$$

and the interleaved real codomain basis

$$(\operatorname{Re} z_1, \operatorname{Im} z_1, \operatorname{Re} z_2, \operatorname{Im} z_2),$$

the realification of Π is

$$P(\tau,\mu,\beta)=\begin{pmatrix} 6\operatorname{Re}\mu & \operatorname{Re}\tau & 1 & 0\\ 6m & t & 0 & 0\\ \operatorname{Re}\beta & \operatorname{Re}\mu & 0 & 1\\ \varrho & m & 0 & 0\end{pmatrix}. \tag{1368}$$

For comparison with the Lorentz cone, define the auxiliary symmetric coefficient matrix

$$S(t,m,\varrho):=\begin{pmatrix} t & \sqrt{6}\,m\\ \sqrt{6}\,m & \varrho\end{pmatrix}. \tag{1369}$$

Thus S represents the same coefficient discriminant, but it is not the canonical 4×4 real period matrix P.

Theorem 30.65 (Ambient coefficient-space discriminant isomorphism)**.** *The matrix $\mathcal{F}$ is orthogonal, and*

$$Q_K(r)=h^2-x^2-y^2. \tag{1370}$$

The real-linear map

$$\mathcal{K}(h,x,y)=(\operatorname{Im}\tau,\operatorname{Im}\mu,\operatorname{Im}\beta)=\left(h+y,\frac{x}{\sqrt{6}},h-y\right) \tag{1371}$$

is an isomorphism from $\mathbb{R}^3$ onto the ambient real coefficient space, with inverse

$$\mathcal{K}^{-1}(t,m,\varrho)=\left(\frac{t+\varrho}{2},\sqrt{6}\,m,\frac{t-\varrho}{2}\right). \tag{1372}$$

For every choice of the real parts of τ,μ,β,

$$\det P(\tau,\mu,\beta)=\det S(t,m,\varrho)=t\varrho-6m^2. \tag{1373}$$

Consequently, if $(t,m,\varrho)=\mathcal{K}(h,x,y)$, then

$$\det P(\tau,\mu,\beta)=\det S(\mathcal{K}(h,x,y))=Q_K(r). \tag{1374}$$

Moreover, for $r\in\mathbb{R}^3_{\geq 0}\setminus\{0\}$, the equation $Q_K(r)=0$ is equivalent to the assertion that

$$S(\mathcal{K}(\mathcal{F}(r)))=\begin{pmatrix} h+y & x\\ x & h-y\end{pmatrix} \tag{1375}$$

is positive semidefinite of rank one.

Proof. The rows of $\mathcal{F}$ have norm one and pairwise scalar product zero, so $\mathcal{F}\mathcal{F}^{\mathsf{T}}=I_3$. Hence

$$r_1^2+r_2^2+r_3^2=h^2+x^2+y^2,\qquad r_1+r_2+r_3=\sqrt{3}\,h.$$

Substitution in equation (1365) proves equation (1370).

Adding and subtracting the first and third entries of equation (1371), and multiplying the second by $\sqrt{6}$, proves equation (1372). Expanding equation (1368) along its last two columns gives

$$\det P=-\det\begin{pmatrix} 6m & t\\ \varrho & m\end{pmatrix}=t\varrho-6m^2.$$

Equivalently, in the exact canonical convention, $\det P=-\det\operatorname{Im}Z$. Taking the determinant in equation (1369) proves the other equality in equation (1373), while

$$(h+y)(h-y)-6\left(\frac{x}{\sqrt{6}}\right)^2=h^2-x^2-y^2$$

proves equation (1374).

Finally, let $r\in\mathbb{R}^3_{\geq 0}\setminus\{0\}$. Then $h=(r_1+r_2+r_3)/\sqrt{3}>0$. If $Q_K(r)=0$, the symmetric matrix in equation (1375) has determinant zero and trace $2h>0$; its eigenvalues are therefore 0 and $2h$, so it is positive semidefinite of rank one. Conversely, rank one forces its determinant, and hence $Q_K(r)$, to vanish. □

Corollary 30.66 (The Koide cone is not a family of regular tori)**.** *The identification in theorem 30.65 is only a real-linear isomorphism of ambient coefficient spaces together with their quadratic discriminants. It neither supplies the real parts of the periods nor imposes the holomorphic and equivariance conditions of the canonical period family. At every point of the Koide cone, every matrix of the canonical form equation* (1368) *with coefficient triple equation* (1371) *satisfies* $\det P = 0$*. Its columns therefore do not span* $\mathbb{R}^4$*, do not generate a full lattice, and do not define a regular compact complex two-torus. Thus the ambient discriminant identity is not a map from Koide triples to regular fibres of the canonical* S^6 *family.*

Corollary 30.67 (Separation from the actual canonical cusp)**.** *In the notation*

$$L_T = T + H_T, \qquad q_T = L_T - B_T, \qquad D_T = L_T q_T + 6m_T^2$$

of the positive radial cusp, the already-proved determinant formula equation (28) *gives*

$$\det P_T = -D_T < 0. \tag{1376}$$

Moreover, with $H = \operatorname{Im} h(0)$ *and* $B = \operatorname{Im} b(0)$*, equation* (47) *gives*

$$D_T = T^2 + (2H - B)T + O(1), \qquad \det P_T \sim -T^2. \tag{1377}$$

Thus every actual radial cusp fibre is regular and has nonzero oriented period determinant, whereas every point in the Koide image has determinant zero.

Proof. The first assertion is exactly equation (28); the stated asymptotic follows from equation (47). Since $D_T > 0$, the matrix P_T is invertible and its columns form the full period lattice of the corresponding regular fibre. □

31 The modulo-840 lattice, triality, and exact shell maps

This section constructs the original modulo-840 lattice N with root system $A_5^4 D_4$, its full integral isometry group, and the exact maps from denominator shells and finite prime samples to the retained residue coordinates. The original data and source labels are recorded in [78]. The shared continuation proves the index-six Leech neighbour Λ and the marked isometry $\mathcal{L}$ into the three octonion sectors. In the specified Cayley frame, put $\tau(q) = \operatorname{Re}((q_1 q_2) q_3)$ and $F = \tau \circ \mathcal{L}$; the Albert determinant is $\lambda^3 - \lambda(y, y) + 2F(y)$. The value-generated groups have quotient $\mathcal{V}_F(\Lambda)/\mathcal{V}_F(N) \cong \mathbb{Z}/12$ (Theorem 33.3). The three residue-cycled neighbours meet in J, with determinant 81 and index 9 in each; the fixed fourth extension has exactly 72 roots of type A_2^{12} (Theorems 34.3 and 34.5). The common lattice and this fourth extension have the exact same value-generated group $\frac{1}{27}\mathbb{Z} \oplus \frac{\sqrt{2}}{216}\mathbb{Z} \oplus \frac{\sqrt{3}}{18}\mathbb{Z}$. The inclusions from $\mathcal{V}_F(N)$ and into $\mathcal{V}_F(\Lambda)$ have matrices $\operatorname{diag}(1, 4, 1)$ and $\operatorname{diag}(1, 1, 3)$ and quotients $\mathbb{Z}/4$, $\mathbb{Z}/3$ (Theorems 35.2 and 35.3). Wilson's construction [81] is realized with both metrics and the direction $c_{\mathrm{W}} = c^{-1}$ retained. Its fixed Gram matrix is $3G_{E_8}$, its complement is explicitly isometric to $A_2 \otimes E_8$, and the index is 3^8, with quotient $\mathbb{F}_3^8$. Its literal intersections with $\mathcal{L}(N)$ and $\mathcal{L}(\Lambda)$ are the same rank-four lattice of Gram matrix $6G_{D_4}$, minimum squared norm 12 and value-generated subgroup $4\sqrt{2}\mathbb{Z}$ for τ ($8\sqrt{2}\mathbb{Z}$ for 2τ). This is not the actual image: $12\sqrt{2}$ is not attained by τ on that intersection. All coordinates, multiplication conventions, domain lattices and maps are specified in the complete proofs below.

31.1 The finite multiplicative carrier and its four fibres

For an odd prime q and a unit r modulo q, write $\left(\frac{r}{q}\right) = 1$ when r is a square modulo q, and -1 otherwise. Let

$$\begin{aligned} V_{840} &= \{r \in (\mathbb{Z}/840\mathbb{Z})^\times : r \equiv 1 \pmod{24}\}, \\ \beta(r) &= \left(\frac{1 - \left(\frac{r}{5}\right)}{2}, \frac{1 - \left(\frac{r}{7}\right)}{2}\right) \in \mathbb{F}_2^2, \\ S_{840} &= \ker \beta, \qquad F_\eta = \beta^{-1}(\eta) \quad (\eta \in \mathbb{F}_2^2). \end{aligned} \tag{1378}$$

Multiplication is used in V_{840} and addition in $\mathbb{F}_2^2$.

Theorem 31.1 (The marked cyclic fibre)**.** *The map β is an onto group homomorphism, each F_η has six elements, and*

$$\begin{aligned} S_{840} &= \{1, 121, 169, 289, 361, 529\} = \langle 289 \rangle, \\ (289^j)_{j=0}^{5} &= (1, 289, 361, 169, 121, 529) \pmod{840}. \end{aligned} \tag{1379}$$

Each fibre is a torsor under multiplication by S_{840}. In particular the marking

$$\kappa : S_{840} \xrightarrow{\cong} \mathbb{Z}/6\mathbb{Z}, \qquad r = 289^{\kappa(r)} \pmod{840}, \tag{1380}$$

is an isomorphism of groups.

Proof. The Chinese remainder map modulo $24, 5, 7$, whose moduli are pairwise coprime, identifies V_{840} with $(\mathbb{Z}/5)^\times \times (\mathbb{Z}/7)^\times$. The powers of 2 modulo 5 are $1, 2, 4, 3$, and the powers of 3 modulo 7 are $1, 3, 2, 6, 4, 5$; these prove the respective cyclic orders four and six. In a cyclic group of either even order, the squares are exactly the even powers. Therefore each Legendre character is the homomorphism recording exponent parity and is onto. Their product is onto $\mathbb{F}_2^2$. Its kernel has $(4/2)(6/2) = 6$ elements and every fibre has that same cardinality.

Successive multiplication of the six displayed residues by 289 gives the next entry and then returns from 529 to 1. They are distinct. Moreover $289 \equiv 1 \pmod{24}$, $289 \equiv 4 \pmod{5}$, and $289 \equiv 2 \pmod{7}$, so it belongs to the simultaneous square kernel. Its six powers therefore exhaust that kernel. If $r, s \in F_\eta$, then $sr^{-1} \in S_{840}$ and $s = (sr^{-1})r$, uniquely. This proves the torsor assertion and all claims about κ. □

Fix the coordinate order

$$(\eta_1, \eta_2, \eta_3, \eta_4) = (00, 10, 01, 11)$$

and choose one origin $f_i \in F_{\eta_i}$ in each fibre, with $f_1 = 1$. The six ordered elements of its fibre are $f_i 289^j$, $0 \le j \le 5$. These origins are part of the coordinate identifications used below; no selection of them is inferred from an individual denominator certificate.

31.2 The root lattice and its exact discriminant form

For a finite set F, give the free abelian group $\mathbb{Z}[F] = \bigoplus_{r \in F} \mathbb{Z} e_r$ the metric $(e_r, e_s) = 1$ for $r = s$ and zero otherwise. Put

$$A(F) = \left\{ \sum_{r \in F} n_r e_r : \sum_{r \in F} n_r = 0 \right\}.$$

The ordered fibre coordinates identify each $A(F_{\eta_i})$ isometrically with

$$A = \{(n_0, \ldots, n_5) \in \mathbb{Z}^6 : \textstyle\sum_j n_j = 0\}.$$

In a separate Euclidean space with orthonormal basis e_1, e_2, e_3, e_4, define

$$D = \{x \in \mathbb{Z}^4 : x_1 + x_2 + x_3 + x_4 \equiv 0 \pmod{2}\}, \qquad R_{\mathrm{ES}} = \left(\bigoplus_{i=1}^{4} A(F_{\eta_i}) \right) \perp D. \tag{1381}$$

All five summands are orthogonal with precisely their displayed metrics. The symbol R_{ES} denotes this lattice; the unadorned R later used for an integer shell modulus has a different declared domain.

For a full integral lattice L in its real span, let

$$L^\vee = \{x : (x, L) \subset \mathbb{Z}\}, \qquad D_L = L^\vee / L.$$

If L is even integral, its discriminant quadratic form is $q_L(x + L) = (x, x) \pmod{2\mathbb{Z}}$. This is well-defined because replacing x by $x + \ell$ changes its squared norm by $2(x, \ell) + (\ell, \ell) \in 2\mathbb{Z}$.

Theorem 31.2 (Root metric and marked discriminant)**.** *The lattice R_{ES} is even integral of rank 24 and has Gram matrix*

$$G_{R_{\mathrm{ES}}} = G_A^{\oplus 4} \oplus G_D,$$

where

$$G_A = \begin{pmatrix} 2 & -1 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 \\ 0 & 0 & -1 & 2 & -1 \\ 0 & 0 & 0 & -1 & 2 \end{pmatrix}, \qquad G_D = \begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & -1 \\ 0 & -1 & 2 & 0 \\ 0 & -1 & 0 & 2 \end{pmatrix}.$$

Their determinants are 6 and 4, respectively, and $\det R_{\mathrm{ES}} = 6^4 \cdot 4 = 5184$.

In each identified A summand use $\lambda = (5,-1,-1,-1,-1,-1)/6$ as the discriminant generator. For D, use the representatives

$$v = (1,0,0,0), \qquad s = \tfrac{1}{2}(1,1,1,1), \qquad c = \tfrac{1}{2}(-1,1,1,1) \tag{1382}$$

and the exact marking $10 \mapsto v + D$, $01 \mapsto s + D$, $11 \mapsto c + D$. Then

$$\begin{aligned} D_{R_{\mathrm{ES}}} &= (\mathbb{Z}/6\mathbb{Z})^4 \oplus \mathbb{F}_2^2, \\ q_{R_{\mathrm{ES}}}(x_1,x_2,x_3,x_4;y) &= \frac{5}{6}\sum_{i=1}^{4} x_i^2 + \mathbf{1}_{y\neq 0} \pmod{2\mathbb{Z}}. \end{aligned} \tag{1383}$$

Proof. For a six-element ordered fibre the five differences $e_j - e_{j+1}$, $0 \le j \le 4$, form a basis of its augmentation lattice: the coefficients of a zero-sum vector in this basis are its first five successive partial sums. Dot products give G_A. The determinant d_n of the analogous tridiagonal n-by-n matrix satisfies $d_n = 2d_{n-1} - d_{n-2}$, $d_0 = 1, d_1 = 2$, by first-row expansion; induction gives $d_n = n + 1$. The four vectors $e_1 - e_2, e_2 - e_3, e_3 - e_4, e_3 + e_4$ lie in D and their coordinate determinant has absolute value 2. Their integer span therefore has index 2 in $\mathbb{Z}^4$. The parity homomorphism $\mathbb{Z}^4 \to \mathbb{F}_2$, $x \mapsto \sum x_i \pmod 2$, is onto and has kernel D, so this span equals D. Their dot products give G_D, whose determinant is the square of the coordinate determinant, namely 4. Integer coordinates make all pairings integral. In either lattice the squared norm is congruent modulo two to the coordinate sum, hence is even. Rank and determinant now follow from the orthogonal direct sum.

The vector λ has sum zero and integral pairing with each augmentation root. Its norm is $5/6$, and $k\lambda \in A$ holds exactly when 6 divides k. Thus it has exact order six in D_A. In an integral basis of an integral lattice with Gram matrix G, the dual coordinates are $G^{-1}\mathbb{Z}^n$; therefore $|L^\vee/L| = |\det G|$. Since $|D_A| = 6$, this element generates the full discriminant group and gives its stated form.

A vector pairing integrally with all $e_i - e_j$ has common fractional parts in its four coordinates. Pairing also with $e_i + e_j$ forces this common fractional part to be either zero or one half. Conversely every entirely integral or entirely half-integral vector pairs integrally with every root of D. There are two integer classes according to sum parity, and two half-integer classes according to the parity of their difference from s. The four representatives $0, v, s, c$ are therefore all the classes, and their squared norms are $0, 1, 1, 1$. Their addition is as marked because $v + s - c = (2,0,0,0) \in D$. The dual and quadratic form of an orthogonal sum are the direct sums of those of its summands, which proves (1383). □

31.3 The index-72 glue and all glue-class minima

Define the binary and ternary codes

$$\begin{aligned} C_2 &= \mathrm{span}_{\mathbb{F}_2}\{(1111;00), (1100;10), (1010;01)\}, \\ C_3 &= \mathrm{span}_{\mathbb{F}_3}\{z_1 = (1,1,1,0),\ z_2 = (1,2,0,1)\}. \end{aligned} \tag{1384}$$

The semicolon distinguishes the four binary A5 coordinates from the two D4 discriminant coordinates. Let

$$\begin{aligned}\mathcal{H} &= \{(3u+2z;y) : (u;y) \in C_2, z \in C_3\} \subset D_{R_{\mathrm{ES}}},\\ N &= \{x \in R_{\mathrm{ES}}^\vee : x + R_{\mathrm{ES}} \in \mathcal{H}\}.\end{aligned} \tag{1385}$$

Here $3u+2z$ is computed coordinatewise modulo six using the integer representatives of u and z.

Theorem 31.3 (Even unimodular completion with exactly 144 roots)**.** *The subgroup $\mathcal{H}$ is isotropic of order 72. Its inverse image N is a positive-definite even unimodular lattice of rank 24, with $[N : R_{\mathrm{ES}}] = 72$. Among the 71 nonzero glue classes, 46 have minimum squared norm 4 and 25 have minimum squared norm 6. The roots of N are exactly those of R_{ES}, with root system $A_5^4D_4$: 30 roots in each A5 summand and 24 in D4.*

Proof. The binary generators are independent, hence give eight words. Listing their sums shows that the seven nonzero words comprise one word $(1111;00)$ and six words whose first part has weight two and whose second part is nonzero. A ternary word is

$$az_1 + bz_2 = (a+b, a+2b, a, b).$$

Its last two coordinates prove injectivity of the parameter pair (a,b), so there are nine words. If $a = 0 \neq b$ the third coordinate alone vanishes; if $b = 0 \neq a$ the fourth alone vanishes; and if both are nonzero exactly one of $a+b, a+2b$ vanishes. Thus every nonzero word has weight three, and each of the four coordinates is the unique vanishing coordinate for two words.

The map $(u_i, z_i) \mapsto 3u_i + 2z_i \pmod 6$ is bijective: reduction modulo two recovers u_i, and reduction modulo three recovers $2z_i$, from which multiplication by 2 recovers z_i. It is a homomorphism of additive groups, since changing binary or ternary representatives by their moduli changes $3u_i + 2z_i$ by a multiple of six. Its product with the retained binary y-coordinate therefore maps $C_2 \times C_3$ injectively onto the subgroup $\mathcal{H}$. It follows that $|\mathcal{H}| = 8 \cdot 9 = 72$. For $u_i \in \{0,1\}$ and $z_i \in \{0,1,2\}$,

$$\frac{5}{6}(3u_i + 2z_i)^2 = \frac{15}{2}u_i + 10u_iz_i + \frac{10}{3}z_i^2 \equiv \frac{3}{2}u_i + \frac{4}{3}z_i^2 \pmod{2\mathbb{Z}}.$$

The cross-term is even; a nonzero z_i has square 1 or 4, whose contributions in the last expression differ by 4. The binary contribution including D4 is therefore zero, 6, or $3+1=4$, and the ternary contribution is zero or $(4/3)3 = 4$ modulo $2\mathbb{Z}$. This proves isotropy.

The inverse image is a lattice because it is a subgroup between the full lattices R_{ES} and $R_{\mathrm{ES}}^\vee$. Isotropy gives $(x,x) \in 2\mathbb{Z}$ for every $x \in N$, and

$$(x,y) = \frac{(x+y,x+y) - (x,x) - (y,y)}{2} \in \mathbb{Z} \qquad (x,y \in N).$$

Thus it is even integral. Its index is $|\mathcal{H}| = 72$. For a finite-index inclusion of equal-rank lattices, the covolume changes by that index, and the Gram determinant is the square of the covolume in orthonormal coordinates. Hence

$$\det N = \frac{6^4 \cdot 4}{72^2} = 1.$$

The dual-index formula proved above implies $N = N^\vee$. Its rank and positive definiteness are inherited from the unchanged real span and metric.

For $0 \le k \le 5$, every vector in the A5 class $k\lambda + A$ has the form

$$m - \frac{k}{6}(1,1,1,1,1,1), \qquad m \in \mathbb{Z}^6, \quad \sum m_i = k.$$

Its squared norm is $\sum m_i^2 - k^2/6$. Since $m_i^2 \ge m_i$ for every integer m_i, the minimum is at least $k - k^2/6$. It is attained by choosing exactly k of the six entries equal to one and the others zero. Thus

$$\mu_A(k) = \frac{k(6-k)}{6}. \tag{1386}$$

Every nonzero integral D4 class has minimum one, attained by v; every half-integral vector has norm at least $4(1/2)^2 = 1$, attained by s, c in their respective classes. The four A5 minima and the D4 minimum add and are simultaneously attained because the metric is an orthogonal sum.

For a coordinate with $k_i = 3u_i + 2z_i \pmod 6$, its minimum contribution is 0 for $(u_i, z_i) = (0, 0)$, 3/2 for $(1, 0)$, 4/3 for $(0, z_i \neq 0)$, and 5/6 for $(1, z_i \neq 0)$. The complete nonzero case table is

$\mathrm{wt}(u)$	$\mathrm{wt}(z)$	$\lvert\mathrm{supp}(u) \cap \mathrm{supp}(z)\rvert$	classes	minimum squared norm
2	0	0	6	$2(3/2) + 1 = 4$
4	0	0	1	$4(3/2) = 6$
0	3	0	8	$3(4/3) = 4$
2	3	1	24	$(3/2) + 2(4/3) + (5/6) + 1 = 6$
2	3	2	24	$(4/3) + 2(5/6) + 1 = 4$
4	3	3	8	$(3/2) + 3(5/6) = 4.$

(1387)

For each of the six weight-two binary words, four nonzero ternary words omit a coordinate inside its support, and four omit a coordinate outside. This proves the two counts $6 \cdot 4 = 24$; all other counts follow from the enumerated binary and ternary words. Thus the total number at minimum 4 is $6+8+24+8 = 46$, and the total at minimum 6 is $1 + 24 = 25$. Their sum is 71, leaving exactly the zero glue class.

No nonzero glue class can contain a norm-two vector. In the zero class, an A5 norm-two vector has exactly one coordinate 1 and another -1, so is $e_i - e_j$, with $6 \cdot 5 = 30$ possibilities. A D4 norm-two vector has exactly two coordinates of absolute value one, so is $\pm e_i \pm e_j$, with $4\binom{4}{2} = 24$ possibilities. An orthogonal sum of two nonzero root-lattice vectors has norm at least 4 because each summand is positive even. These are therefore all $4 \cdot 30 + 24 = 144$ roots of N. □

A Niemeier lattice means a positive-definite even unimodular lattice of rank 24. The theorem constructs one with root system $A_5^4 D_4$ directly. Its root type appears with lambency 6 in the primary umbral table [74, Table 3, p. 21]. This table reference is not needed to prove the inverse-image construction or any minimum above.

For comparison with the earlier labeled glue presentation in the source [77, 78], put

$$h_1 = (1, 3, 1, 1; 00), \qquad h_2 = (1, 4, 2, 3; 11), \qquad h_3 = (3, 0, 3, 0; 01),$$

and let $\mathcal{H}_{\mathrm{old}}$ be their subgroup. The exact transport is

$$P(x_1, x_2, x_3, x_4; y) = (x_3, x_4, x_1, x_2; y), \qquad P(\mathcal{H}_{\mathrm{old}}) = \mathcal{H}. \tag{1388}$$

Indeed, if $rh_1 + th_2 + wh_3 = 0$ with $r, t \in \mathbb{Z}/6$ and $w \in \mathbb{Z}/2$, the final two coordinates imply that t is even and $w = 0$. The first and third coordinates then give $r + t = 0$ and $r + 2t = 0$ modulo six, whence $t = r = 0$. Thus $|\mathcal{H}_{\mathrm{old}}| = 6 \cdot 6 \cdot 2 = 72$. Directly,

$$\begin{aligned} P(h_1) &= 3(1111; 00) + 2(2, 2, 2, 0; 00), \\ P(h_2) &= 3(0110; 11) + 2(1, 0, 2, 2; 00), \\ P(h_3) &= 3(1010; 01). \end{aligned}$$

The displayed ternary parts are $2z_1$ and $2z_1 + 2z_2$. Consequently $P(\mathcal{H}_{\mathrm{old}}) \subset \mathcal{H}$, and the equal orders prove equality. This transporter is realized by the corresponding isometry permuting the full identified A5 summands.

31.4 The common D4 root lattice, shortest weights, and triality

Use orthonormal coordinates $\varepsilon_1, \dots, \varepsilon_4$ for the ordinary F4 root realization. Its long roots and the three parts of its short roots are

$$\begin{aligned} \Phi_D &= \{\pm\varepsilon_i \pm \varepsilon_j : 1 \le i < j \le 4\}, \\ \mathcal{V} &= \{\pm\varepsilon_i : 1 \le i \le 4\}, \\ \mathcal{S}_\pm &= \left\{ \frac{1}{2}\sum_{i=1}^{4} \delta_i \varepsilon_i : \delta_i \in \{1,-1\},\ \prod_i \delta_i = \pm 1 \right\}, \\ \Phi_F &= \Phi_D \cup \mathcal{V} \cup \mathcal{S}_+ \cup \mathcal{S}_-. \end{aligned} \tag{1389}$$

For a nonzero vector a, reflection means the exact map $s_a(x) = x - 2(x,a)a/(a,a)$. Write W_D and W_F for the groups generated by the reflections in Φ_D and Φ_F, respectively.

Theorem 31.4 (Integral D4 and the marked triality quotient)**.** *The map*

$$\iota_D : D \longrightarrow \mathbb{Z}\langle \Phi_D \rangle, \qquad (x_1,x_2,x_3,x_4) \longmapsto \sum_{i=1}^{4} x_i \varepsilon_i \tag{1390}$$

is an integral isometry onto the long-root lattice. It extends to duals and discriminant forms, sending the shortest-vector sets of the marked classes v, s, c respectively to $\mathcal{V}, \mathcal{S}_+, \mathcal{S}_-$.

The two matrices

$$R_D = \operatorname{diag}(-1,1,1,1), \qquad H_D = \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix} \tag{1391}$$

are the short-root reflections s_{ε_1} and s_c, where $c = (-1,1,1,1)/2$ is read in the ε coordinates. They induce $(s\ c)$ and $(v\ s)$ on the three nonzero discriminant classes. There is a split exact sequence

$$1 \longrightarrow W_D \longrightarrow W_F \longrightarrow S_3 \longrightarrow 1, \qquad W_D \cong \{\textit{even signed permutations of four coordinates}\}, \tag{1392}$$

whose displayed section is $\langle R_D, H_D \rangle \cong S_3$. In particular $|W_D| = 192$ and $|W_F| = 1152$.

Proof. The four D4 basis vectors in Theorem 31.2 are among Φ_D, and every vector of Φ_D has even coordinate sum. Thus the displayed identity of coordinates has exactly the stated image, preserves the metric, and extends to duals by the integer-pairing definition. In a nonzero integral dual class, norm one forces one coordinate ± 1 and all others zero. In a half-integral class, norm one forces all four coordinates to be $\pm 1/2$. The difference from s has even coordinate sum exactly when the number of negative signs is even. This proves the three shortest-vector identifications with the stated marking.

The reflection identity for R_D is immediate, and $H_D = I - 2cc^T$ follows by multiplying the four coordinates of c. Both are orthogonal involutions. The first preserves the parity of an integer coordinate sum. For $x \in D$, the four coordinates of $H_D x$ are integers because each numerator is congruent to $\sum x_i$ modulo two, and their sum is $2x_1$. Thus both preserve D. Their effects on the representatives show that R_D fixes the class v and exchanges s, c, whereas $H_D v = s$, $H_D s = v$, and $H_D c = -c \equiv c \pmod{D}$. They therefore preserve all of Φ_F by the shortest-vector description. On the plane with ordered basis ε_1, c, their product has matrix

$$\begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix},$$

whose cube is the identity, while both reflections fix the orthogonal complement. Consequently $R_D^2 = H_D^2 = (R_D H_D)^3 = 1$. Every word in two such involutions reduces to $(R_D H_D)^k$ or $(R_D H_D)^k R_D$, $0 \le k \le 2$.

Their discriminant permutations are the six elements of S_3, so these six words are distinct and give the stated section.

Reflection in $\varepsilon_i - \varepsilon_j$ swaps coordinates i, j. Reflection in $\varepsilon_i + \varepsilon_j$ swaps them and changes both signs. Hence the long-root reflections generate every permutation and every even sign change, and each long-root reflection is of this form. This proves the assertion about W_D and its order $2^3 \cdot 4! = 192$. Such a reflection acts trivially on $D^\vee/D$: for a root a of squared norm two, $s_a(x) - x = -(x, a)a \in D$ for all $x \in D^\vee$.

Let $K = \langle W_D, R_D, H_D \rangle$. Each of its generators preserves the long roots, so W_D is normal in K by $g s_a g^{-1} = s_{ga}$. The even signed permutations act transitively on each of $\mathcal{V}, \mathcal{S}_+, \mathcal{S}_-$, and the displayed section permutes these three sets transitively. Thus every short root is $g\varepsilon_1$ for some $g \in K$, and its reflection $g R_D g^{-1}$ belongs to K. All generators of W_F therefore belong to K; the reverse inclusion follows from their definitions, proving $W_F = K$. An element in the kernel of its permutation action on the three short sets preserves $\mathcal{V}$, so it is a signed permutation matrix. Preservation of $\mathcal{S}_+$ forces an even number of sign changes, so the kernel is W_D. This proves exactness, splitting, and the order $6 \cdot 192 = 1152$. □

The exact geometric datum and weight marking. Let $\mathbb{O}$ denote the real normed octonion algebra and let $J = \mathfrak{h}_3(\mathbb{O})$ be the real vector space of Hermitian three-by-three matrices, with Jordan product $a \circ b = (ab + ba)/2$. Define

$$\mathbb{OP}^2 = \{a \in J : a^2 = a,\ \operatorname{tr} a = 1\}, \qquad \mathrm{Fl}(\mathbb{O}) = \{(a, b) \in (\mathbb{OP}^2)^2 : \operatorname{Re}\operatorname{tr}(ab) = 0\}.$$

For the compact group $F_4 = \mathrm{Aut}_{\mathbb{R}}(J, \circ)$, Mare and Willems prove the homogeneous identification $\mathrm{Fl}(\mathbb{O}) = F_4/\operatorname{Spin}(8)$ and identify its tangent representation as $V_8 \oplus S_8^+ \oplus S_8^-$ [75, Definition 2.2, Proposition 2.4, Proposition 2.7, and pp. 491–492]. These are the prior geometric results used here. With their torus coordinate forms L_i, their equations (5.4)–(5.6) assign the vector, positive half-spin, and negative half-spin weights respectively the patterns $\pm L_i$, even half-sign sums, and odd half-sign sums [75, pp. 510–511]. The coordinate map $L_i \mapsto \varepsilon_i$ retains that sign marking and identifies these three weight sets with those of (1389).

For completeness the half-spin weight patterns admit the following direct calculation. In the complex exterior algebra $\Lambda^* \mathbb{C}^4$, let a_i be exterior multiplication by the i-th basis vector and b_i contraction by its dual. On a basis wedge, insertion and deletion give

$$a_i a_j + a_j a_i = b_i b_j + b_j b_i = 0, \qquad a_i b_j + b_j a_i = \delta_{ij} I.$$

Thus $C_i = a_i + b_i$ and $D_i = \mathrm{i}(b_i - a_i)$ square to one and mutually anticommute for distinct Clifford generators. Writing $N_i = a_i b_i$, direct multiplication gives

$$\tfrac{1}{2} C_i D_i = \mathrm{i}(N_i - \tfrac{1}{2} I).$$

The four commuting infinitesimal torus operators therefore have weights $\sum_i (\mathbf{1}_{i \in I} - 1/2)\varepsilon_i$ on the basis wedge indexed by $I \subset \{1, 2, 3, 4\}$. The even exterior part has an even number of negative signs, and the odd part has an odd number. There are eight one-dimensional weight spaces in each part. For the vector representation the four real coordinate planes complexify into their two rotation eigenspaces, with weights $\varepsilon_i, -\varepsilon_i$. This proves the displayed weight formulas directly in the stated marking.

The map proved here is consequently the actual chain

$$D \xrightarrow{\iota_D} \mathbb{Z}\langle \Phi_D \rangle, \qquad \{v, s, c\} \longleftrightarrow \{\mathcal{V}, \mathcal{S}_+, \mathcal{S}_-\} = \{\mathrm{wt}(V_8), \mathrm{wt}(S_8^+), \mathrm{wt}(S_8^-)\}.$$

The same S_3 acts through the explicitly proved permutations of these three sets. The companion higher-rung treatment records the geometric datum and root-coordinate comparison at the respective labels

h24:thm:octonionic-triality
h24:prop:two-Coxeter-systems.

The exact source pointers and passages are included in [78]. The integral metric map in this section is the map on D4 and its dual. The definitions make no map from the whole rank-24 lattice N into the flag tangent space.

31.5 The full indicated glue stabilizer and its triality quotient

All vector spaces over finite fields in this subsection retain their displayed bases. Write

$$j : \mathbb{F}_3^2 \longrightarrow \mathbb{F}_3^4, \qquad j(a,b) = (a+b, a+2b, a, b), \qquad C_3 = j(\mathbb{F}_3^2).$$

The four row forms defining this injection are

$$\ell_1 = (1,1), \quad \ell_2 = (1,2), \quad \ell_3 = (1,0), \quad \ell_4 = (0,1).$$

They represent all four one-dimensional subspaces of the dual of $\mathbb{F}_3^2$. Indeed each nonzero row has either first coordinate zero, in which case its line is represented by ℓ_4, or a unique scalar multiple with first coordinate one, namely one of $(1,0), (1,1), (1,2)$.

Let $E \subset \mathbb{F}_2^4$ be the even-weight subspace and define

$$\tau : E \longrightarrow \mathbb{F}_2^2, \qquad \tau(u_1,u_2,u_3,u_4) = (u_2+u_4, u_3+u_4).$$

The binary code in the index-72 glue is exactly

$$C_2 = \{(u, \tau(u)) : u \in E\}.$$

To verify this, the independent vectors $1111, 1100, 1010$ form a basis of the three-dimensional space E, and their images under τ are respectively $00, 10, 01$. These are precisely the three stated binary code generators. In particular τ is onto and

$$\ker \tau = \langle 1111 \rangle.$$

Explicitly, $\tau(u) = 0$ gives $u_2 = u_3 = u_4$, and the even-weight equation then gives $u_1 = u_4$.

For a permutation p of $\{1,2,3,4\}$, use the coordinate convention

$$(P_p u)_i = u_{p(i)}.$$

The permutation preserves E and $\langle 1111 \rangle$, so it induces a unique invertible map B_p satisfying

$$B_p \tau(u) = \tau(P_p u) \qquad (u \in E). \tag{1393}$$

For an entirely explicit formula, the two columns of B_p are $\tau(P_p(1100))$ and $\tau(P_p(1010))$. The convention implies $P_p P_q = P_{q \circ p}$. Accordingly $B_p B_q = B_{q \circ p}$; no reversal is suppressed in what follows.

The subgroup of (1385) can therefore be written

$$\mathcal{H} = \{(3u + 2z; \tau(u)) : u \in E,\ z \in C_3\} \subset (\mathbb{Z}/6)^4 \oplus \mathbb{F}_2^2.$$

Let G be its stabilizer among the transformations

$$g = (\varepsilon, p, B), \qquad (x;y) \longmapsto \big((\varepsilon_i x_{p(i)})_{i=1}^4; By\big),$$

where $\varepsilon_i \in \{1,-1\}$, $p \in S_4$, and $B \in GL_2(\mathbb{F}_2)$. Denote the signed permutation matrix on the first four coordinates by M_g.

Theorem 31.5 (An explicit isomorphism to $GL_2(\mathbb{F}_3)$)**.** *Restriction to the ternary code gives an isomorphism*

$$\Theta : G \xrightarrow{\cong} GL_2(\mathbb{F}_3), \qquad M_g j = j\Theta(g).$$

For every $A \in GL_2(\mathbb{F}_3)$, its inverse image is specified uniquely by

$$\ell_i A = \varepsilon_i \ell_{p(i)} \quad (1 \le i \le 4), \qquad B = B_p. \tag{1394}$$

In particular $|G| = 48$.

Proof. The elements of $\mathcal{H}$ of order dividing two are precisely $(3u;\tau(u))$, and those of order dividing three are precisely $(2z;0)$. Every indicated ambient transformation preserves these primary parts, so an element of G satisfies $M_gC_3 = C_3$. The injection j therefore yields one and only one $\Theta(g) \in GL_2(\mathbb{F}_3)$. The binary part is preserved exactly when $B\tau(u) = \tau(P_pu)$, because all signs become one modulo two. Surjectivity of τ forces $B = B_p$.

If $\Theta(g) = I$, the equality $M_gj = j$ implies $\varepsilon_i\ell_{p(i)} = \ell_i$ for every i. The four row lines are distinct, so $p(i) = i$. Since each ℓ_i is nonzero and $1 \neq -1$ in $\mathbb{F}_3$, also $\varepsilon_i = 1$ for every i. It follows that $B_p = I$ and $g = 1$. Thus Θ is injective.

Conversely fix $A \in GL_2(\mathbb{F}_3)$. The rows ℓ_iA are nonzero and belong to four distinct row lines. Since the ℓ_i represent every row line, each ℓ_iA has a unique expression $\varepsilon_i\ell_{p(i)}$, where the nonzero scalar in $\mathbb{F}_3$ is represented by the exact integer sign $\varepsilon_i = 1$ or -1. Distinctness of the row lines makes p a permutation. The resulting signed permutation satisfies $M_gj = jA$, hence preserves C_3. Choosing $B = B_p$ preserves C_2 as proved above, hence preserves $\mathcal{H}$. This constructs an element of G mapping to A, proving surjectivity and the stated inverse formula.

Finally

$$M_{gh}j = M_gM_hj = M_gj\Theta(h) = j\Theta(g)\Theta(h),$$

so $\Theta(gh) = \Theta(g)\Theta(h)$. An invertible 2-by-2 matrix over $\mathbb{F}_3$ has eight choices for its first nonzero column and six choices for its second column outside the span of the first. Thus its group order is $8 \cdot 6 = 48$. □

The map to the permutation of the four component labels is more precisely $g \mapsto p^{-1}$, because a vector supported on component k is sent to component $p^{-1}(k)$. Its kernel under Θ is $\{I, -I\}$. In fact a matrix preserving all four row lines is diagonal by preservation of ℓ_3 and ℓ_4, and its diagonal entries are equal by preservation of ℓ_1. Conversely scalar matrices preserve all four lines. Therefore the component permutation image has order $48/2 = 24$, so is all S_4, with

$$G/\Theta^{-1}(\{\pm I\}) \xrightarrow{\overline{\Theta}} PGL_2(\mathbb{F}_3) \cong S_4.$$

The center of $GL_2(\mathbb{F}_3)$ is exactly $\{\pm I\}$: a central matrix commutes with $\mathrm{diag}(1,-1)$, forcing its off-diagonal entries to vanish, and commuting with $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ forces the diagonal entries to agree.

Theorem 31.6 (The exact triality quotient and its quaternion kernel)**.** *The projection $\pi : G \to GL_2(\mathbb{F}_2)$, $(\varepsilon, p, B) \mapsto B$, is onto, and under Θ its kernel is the quaternion subgroup generated by*

$$I_q = \begin{pmatrix} 0 & 1 \\ 2 & 0 \end{pmatrix}, \qquad J_q = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}.$$

There is a split exact sequence

$$1 \longrightarrow Q_8 \longrightarrow GL_2(\mathbb{F}_3) \xrightarrow{\pi\Theta^{-1}} GL_2(\mathbb{F}_2) \longrightarrow 1.$$

The splitting is the subgroup generated by

$$A_R = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}, \qquad A_H = \begin{pmatrix} 1 & 2 \\ 0 & 2 \end{pmatrix}.$$

It identifies $GL_2(\mathbb{F}_3)$ with $Q_8 \rtimes S_3$ for the conjugation action of this displayed subgroup on this displayed Q_8.

Proof. The nonzero elements of $E/\langle 1111\rangle$ are represented by the complementary pairs of two-element subsets

$$\{12 \mid 34\}, \qquad \{13 \mid 24\}, \qquad \{14 \mid 23\},$$

marked respectively by $10, 01, 11$ via τ. Thus B_p is exactly the permutation action on these three partitions, expressed in the displayed binary marking.

Using (1394) gives

$$M(A_R)x = (x_2, x_1, x_3, -x_4), \quad B_R = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix},$$

$$M(A_H)x = (x_1, x_3, x_2, -x_4), \quad B_H = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

These binary matrices act respectively as the transpositions (01 11) and (10 01), so they generate all permutations of the three nonzero binary vectors. There are $(4-1)(4-2) = 6$ invertible binary matrices; a matrix fixing all nonzero vectors fixes the basis. Hence this permutation action identifies $GL_2(\mathbb{F}_2)$ with S_3 and proves surjectivity of π. Its kernel therefore has order eight.

For I_q the row rule gives

$$p = (2, 1, 4, 3), \qquad \varepsilon = (-1, 1, 1, -1),$$

and for J_q it gives

$$p = (3, 4, 1, 2), \qquad \varepsilon = (-1, -1, 1, 1).$$

Both permutations fix each of the three displayed complementary partitions, so both matrices lie in the kernel. Direct matrix multiplication over $\mathbb{F}_3$ gives

$$I_q^2 = J_q^2 = -I, \qquad I_q J_q = -J_q I_q, \qquad I_q J_q = \begin{pmatrix} 1 & 2 \\ 2 & 2 \end{pmatrix}.$$

The eight matrices $I, -I, \pm I_q, \pm J_q, \pm I_q J_q$ are pairwise distinct by their displayed entries. The relations show this set is a group: powers reduce using $I_q^2 = J_q^2 = -I$, and factors can be reordered using $J_q I_q = -I_q J_q$. These relations identify it with the quaternion group Q_8. Since the kernel has order eight, this is the whole kernel.

The matrices A_R, A_H satisfy

$$A_R^2 = A_H^2 = I, \qquad A_R A_H = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}, \qquad (A_R A_H)^3 = I.$$

Every word in the two involutions reduces to one of $(A_R A_H)^k$ or $(A_R A_H)^k A_R$, $k = 0, 1, 2$, by alternation and the cubic relation. Their images under π are the six distinct permutations generated by B_R, B_H. Hence the subgroup has exactly six elements and maps isomorphically onto $GL_2(\mathbb{F}_2)$. This gives the splitting. For any g, the unique splitting element with image $\pi(g)$ leaves g times its inverse in the kernel; the intersection of kernel and splitting is trivial. Multiplication therefore realizes the asserted semidirect product with the indicated conjugation action. □

The matrices in the kernel have an additional marked interpretation: their projective component permutations are the four permutations $1, (12)(34), (13)(24), (14)(23)$. Each fixes all three partitions, and these are exactly the kernel of the onto action $S_4 \to S_3$, whose kernel has order $24/6 = 4$. Thus the preceding exact sequence is the precise map from the order-48 ternary-code symmetry to the three D_4 discriminant classes.

Theorem 31.7 (A faithful action on the actual index-72 lattice)**.** *Fix the four identifications of the A_5 coordinate lattices used in the glue presentation, and the discriminant marking $10 \mapsto v, 01 \mapsto s, 11 \mapsto c$ of D_4. The group $G \cong GL_2(\mathbb{F}_3)$ has a faithful action by integral isometries of the index-72 overlattice. Write $\varphi_i : A \longrightarrow A(F_{\eta_i})$ for the restriction of the fixed free-coordinate isometry $e_j \mapsto e_{f_i 289^j}$. For $X_i \in A(F_{\eta_i})$, the i-th output component is exactly $\varepsilon_i \varphi_i \varphi_{p(i)}^{-1} X_{p(i)}$. In the identified A coordinates this is the action*

$$L_g(X_1, X_2, X_3, X_4; d) = \big(\varepsilon_1 X_{p(1)}, \ldots, \varepsilon_4 X_{p(4)}; \delta(B)d\big),$$

where $\delta : GL_2(\mathbb{F}_2) \to O(D_4)$ is the section given by

$$\delta(B_R) = R_D = \mathrm{diag}(-1,1,1,1), \qquad \delta(B_H) = H_D = \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}.$$

Under Θ, the two displayed full triality lifts are exactly A_R and A_H.

Proof. The four identified A_5 summands carry the same Euclidean metric, so the displayed signed permutation is a root-lattice isometry. The matrices R_D, H_D preserve D_4: R_D negates one integral coordinate and preserves parity of the sum; for $x \in D_4$, each coordinate of $H_D x$ is integral because its numerator has the same parity as $\sum x_i$, and the coordinate sum of $H_D x$ is $2x_1$. Both matrices are orthogonal involutions by multiplication. Furthermore $R_D = s_{e_1}$, $H_D = s_c$, with $c = (-1,1,1,1)/2$, and $(e_1, c) = -1/2$. On the plane with basis e_1, c, their product has matrix $\left(\begin{smallmatrix} 0 & -1 \\ 1 & -1 \end{smallmatrix}\right)$ and has cube the identity; both reflections fix the orthogonal complement pointwise. The discriminant actions are the two distinct transpositions B_R, B_H. The same word reduction as in the preceding proof therefore shows that $\langle R_D, H_D \rangle$ has exactly six elements and that its discriminant action is an isomorphism onto $GL_2(\mathbb{F}_2)$. Its inverse is the homomorphism δ.

Thus $g \mapsto L_g$ is a homomorphism of root-lattice isometries. It extends to the dual lattice because an isometry preserves the integer-pairing condition defining the dual. Its discriminant action is the specified (ε, p, B), which preserves $\mathcal{H}$. Consequently it preserves the inverse image

$$N = \{x \in R_{\mathrm{ES}}^\vee : x + R_{\mathrm{ES}} \in \mathcal{H}\}.$$

It is therefore an integral isometry of N. If L_g is the identity on N, it is the identity on the full real span, because N contains the full-rank lattice R_{ES}. It is therefore also the identity on $R_{\mathrm{ES}}^\vee / R_{\mathrm{ES}}$, in particular on the ternary code. Hence $\Theta(g) = I$, and injectivity of Θ gives $g = 1$. This proves faithfulness. The exact formulas for A_R, A_H were established above, so their lifts are the stated full triality transformations. □

This constructs an explicit finite subgroup of $O(N)$, including its action on all five root summands and on their glue, and its exact D_4-triality quotient. The group and its projective quotient agree with the $A_5^4 D_4$ row of [74, Table 3, p. 21]; the isomorphisms here have been proved by their actual code actions.

For clarity, the two full triality elements, denoted $\widehat{R} = L_{\Theta^{-1}(A_R)}$ and $\widehat{H} = L_{\Theta^{-1}(A_H)}$, act in the fixed coordinates by

$$\widehat{R}(X_1, X_2, X_3, X_4; d) = (X_2, X_1, X_3, -X_4; R_D d),$$
$$\widehat{H}(X_1, X_2, X_3, X_4; d) = (X_1, X_3, X_2, -X_4; H_D d).$$

Their squares and the cube of their product are the identity on the entire lattice, by the proved faithful group homomorphism. The product on the first three A5 summands is the indicated three-cycle; its two negations on the fourth summand cancel.

31.6 The exact reduced-shell divisor bijection

Let p be a positive prime with $p \equiv 1 \pmod{24}$, and let the positive integer shell R satisfy $R \equiv 3 \pmod 4$. Retain

$$a = a_R(p) = \frac{p+R}{4}, \qquad N_{\mathrm{sh}} = pa, \qquad g = \gcd(R, N_{\mathrm{sh}}), \qquad R_0 = \frac{R}{g}, \quad N_0 = \frac{N_{\mathrm{sh}}}{g}. \tag{1395}$$

The integer N_{sh} is the source's scalar $N = pa$; the subscript distinguishes it from the lattice N without changing either object. In particular a, R_0, N_0 are positive integers, $\gcd(R_0, N_0) = 1$, and the prescribed

first denominator equation is

$$\frac{4}{p}-\frac{1}{a}=\frac{4a-p}{pa}=\frac{R}{N_{\mathrm{sh}}}=\frac{R_0}{N_0}=\frac{1}{b}+\frac{1}{c}. \tag{1396}$$

Here b, c are positive integer denominators, and their order is initially retained. Define the finite set

$$\mathcal{D}_{R,p}=\{d\in\mathbb{Z}_{>0} : d \mid N_0^2,\ d\equiv -N_0 \pmod{R_0}\},\qquad A_R(p)=|\mathcal{D}_{R,p}|.$$

Theorem 31.8 (Certificates, ordered pairs, and complementary pairs)**.** *There is an exact bijection*

$$\begin{aligned}\mathcal{D}_{R,p} &\longrightarrow \{(b,c)\in\mathbb{Z}_{>0}^2 : 4/p=1/a+1/b+1/c\},\\ d &\longmapsto \left(\frac{N_0+d}{R_0},\frac{N_0+N_0^2/d}{R_0}\right),\end{aligned} \tag{1397}$$

with inverse $d=R_0b-N_0$. The involution $d\mapsto N_0^2/d$ corresponds exactly to $(b,c)\mapsto(c,b)$. Consequently the number of unordered pairs of remaining denominators is

$$U_R(p)=\frac{A_R(p)+\delta_{R,p}}{2},\qquad \delta_{R,p}=\begin{cases}1, & R_0\mid 2,\\ 0, & R_0\nmid 2.\end{cases} \tag{1398}$$

The unique possible fixed certificate is $d=N_0$, giving $b=c=2N_0/R_0$.

Proof. For a positive solution of (1396), $R_0b/N_0=1+b/c>1$, so $d=R_0b-N_0>0$; likewise $e=R_0c-N_0>0$. Multiplying the fraction equation by N_0bc gives $R_0bc=N_0(b+c)$, and therefore

$$(R_0b-N_0)(R_0c-N_0)=N_0^2.$$

Thus $d\mid N_0^2$, $d\equiv -N_0 \pmod{R_0}$, and $e=N_0^2/d$.

Conversely, let $d\in\mathcal{D}_{R,p}$ and $e=N_0^2/d$. For $R_0>1$, the coprimality $\gcd(R_0,N_0)=1$ makes $d\equiv -N_0$ a unit, and hence

$$e\equiv N_0^2(-N_0)^{-1}\equiv -N_0 \pmod{R_0}.$$

For $R_0=1$, this congruence is vacuous and holds automatically. Thus both expressions in (1397) are positive integers. Substitution into $(R_0b-N_0)(R_0c-N_0)=N_0^2$ and expansion give the original fraction equation. Each construction recovers exactly $d=R_0b-N_0$ and $e=R_0c-N_0$, so they are inverse, proving the bijection and the involution statement.

For a positive divisor, $d=N_0^2/d$ holds exactly when $d=N_0$. This divisor is a certificate exactly when $N_0\equiv -N_0 \pmod{R_0}$, that is, $R_0\mid 2N_0$. Coprimality makes this equivalent to $R_0\mid 2$. Every other orbit contains two certificates, so counting the fixed orbits and the two-element orbits gives $U_R(p)=\delta_{R,p}+(A_R(p)-\delta_{R,p})/2$, exactly (1398). □

Factor the exact integer N_0 as $N_0=\prod_{j=1}^m q_j^{e_j}$, with distinct positive primes q_j and positive exponents e_j, and define

$$\begin{aligned}B(N_0)&=\prod_{j=1}^m\{-e_j,-e_j+1,\dots,e_j\},\\ \lambda_{R_0,N_0}&: B(N_0)\longrightarrow(\mathbb{Z}/R_0\mathbb{Z})^\times,\qquad \lambda_{R_0,N_0}(\beta)=\prod_{j=1}^m q_j^{\beta_j}\pmod{R_0}.\end{aligned} \tag{1399}$$

The group for modulus one is taken to be the one-element group, so that the same notation also expresses the vacuous congruence in that case. For $R_0>1$, every q_j is a unit by coprimality, so all negative powers in this formula are defined.

Theorem 31.9 (The exact exponent-box map and the two raw channels)**.** *The map*

$$\beta \longmapsto d = \prod_{j=1}^{m} q_j^{e_j+\beta_j} \tag{1400}$$

bijects $\lambda_{R_0,N_0}^{-1}(-1)$ *with* $\mathcal{D}_{R,p}$*. It retains each valuation and both original integers* R_0, N_0*.*

In addition, suppose $\gcd(R, pa) = 1$ *and* $p \nmid a$*. Write*

$$a = \prod_{j=1}^{s} q_j^{e_j}, \qquad B(a) = \prod_{j=1}^{s} \{-e_j, \dots, e_j\},$$

and define, for each unit t *modulo* R*,*

$$c_R(a;t) = \#\left\{\beta \in B(a) : \prod_j q_j^{\beta_j} \equiv t \pmod{R}\right\}.$$

Then the number of raw divisor certificates, equivalently ordered denominator pairs, is

$$A_R(p) = c_R(a;-1) + 2c_R(a;-p^{-1}). \tag{1401}$$

Its complementary-divisor quotient is the count in (1398)*, with* $R_0 = R$ *under these additional coprimality hypotheses.*

Proof. A divisor of N_0^2 has one and only one exponent vector $\alpha_j \in \{0, \dots, 2e_j\}$. The exact change $\beta_j = \alpha_j - e_j$ gives the entire displayed box and $dN_0^{-1} \equiv \prod_j q_j^{\beta_j} \pmod{R_0}$. Multiplication of $d \equiv -N_0$ by the unit N_0^{-1} gives precisely the target -1; modulo one both statements are vacuous. This proves the first bijection and its inverse.

Under the additional hypotheses, $R_0 = R$, $N_0 = pa$, and the exponent of p in pa is exactly one. Every positive integer divisor d of p^2a^2 therefore has exactly one valuation-index representation

$$d = p^{t+1} \prod_j q_j^{e_j+\beta_j}, \qquad (t, \beta) \in \{-1, 0, 1\} \times B(a).$$

After division by pa, its unit-group residue is

$$d(pa)^{-1} \equiv p^t \prod_j q_j^{\beta_j} \pmod{R}.$$

Distinct valuation-index pairs may have the same unit-group residue; they remain distinct integer divisors in the count. The three conditions for this to equal -1 are

t	$\prod_j q_j^{\beta_j} \pmod{R}$
-1	$-p$
0	-1
1	$-p^{-1}$.

The negation $\beta \mapsto -\beta$ is a bijection of $B(a)$ and inverts the unit-group value, so it bijects the fibres at $-p$ and $-p^{-1}$. Summing the sizes of these three disjoint raw exponent classes proves (1401). The complementary divisor sends (t, β) to $(-t, -\beta)$, so identifying it changes the raw count as proved in Theorem 31.8; it does not leave the factor two in (1401) unchanged. □

This counting convention corrects the wording attached to the source's two-channel formula: the displayed sum is a raw count, whereas a count after identifying complementary divisors is (1398). The exact source locator and formula are retained in [76, 78].

31.7 A fixed-shell count does not descend to the residue fibre

Theorem 31.10 (Two primes in the same residue with different shell counts)**.** *The two primes* 2521 *and* 3361 *both reduce to* $1 \in S_{840}$*, but for the same shell* $R = 3$ *their raw and unordered counts are*

$$A_3(2521) = 0, \qquad A_3(3361) = 6, \qquad U_3(2521) = 0, \quad U_3(3361) = 3. \tag{1402}$$

One actual denominator certificate in the second case is

$$\frac{4}{3361} = \frac{1}{841} + \frac{1}{974690} + \frac{1}{28266010}. \tag{1403}$$

Thus the raw count, its unordered quotient, and the indicator of completion of this fixed shell each fail to factor through prime reduction modulo 840.

Proof. The residue equations are the literal equalities $2521 = 3 \cdot 840 + 1$ and $3361 = 4 \cdot 840 + 1$. For an explicit finite primality verification, the primes at most 57, in increasing order, are

$$\mathcal{P} = (2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53).$$

The remainders against its first 15 entries for 2521, all 16 entries for 3361, and its first nine entries for 631, are respectively

$$(1, 1, 1, 1, 2, 12, 5, 13, 14, 27, 10, 5, 20, 27, 30),$$
$$(1, 1, 1, 1, 6, 7, 12, 17, 3, 26, 13, 31, 40, 7, 24, 22),$$
$$(1, 1, 1, 1, 4, 7, 2, 4, 10).$$

All are nonzero, and the respective integer square roots are 50, 57, and 25. The prime 29 has remainders $1, 2, 4$ against $2, 3, 5$, the primes at most its integer square root 5. Every composite positive integer greater than one has a prime divisor at most its square root: in a factorization into two factors greater than one, the smaller factor is at most the square root and has a prime divisor no larger than itself. These nonzero remainder lists therefore prove each stated primality.

For $R = 3$,

$$a_3(2521) = 631, \qquad a_3(3361) = 841 = 29^2.$$

Both primes are 1 modulo three; also $631 \equiv 1 \pmod 3$ and $841 \equiv 1 \pmod 3$. Thus $\gcd(3, pa) = 1$ in both cases. Both a values are smaller than their respective primes, so $p \nmid a$. Every additional hypothesis of Theorem 31.9 is now verified. The two targets -1 and $-p^{-1}$ coincide modulo three.

For $a = 631$, all exponents $-1, 0, 1$ give residue one, so the target -1 has count zero. For $a = 29^2$, among the five exponents $-2, -1, 0, 1, 2$, exactly $-1, 1$ give -1 modulo three because $29 \equiv -1 \pmod 3$. Both target counts are therefore 2 and the raw count is $2 + 2 \cdot 2 = 6$. Here $R_0 = 3 \nmid 2$, so both unordered counts are half the corresponding raw counts. This proves (1402).

For the second prime, retain

$$N_{\mathrm{sh}} = 3361 \cdot 841 = 2826601, \qquad d = 3361 \cdot 29 = 97469, \qquad \frac{N_{\mathrm{sh}}^2}{d} = 3361 \cdot 29^3 = 81971429.$$

Then

$$\frac{N_{\mathrm{sh}} + d}{3} = 974690, \qquad \frac{N_{\mathrm{sh}} + N_{\mathrm{sh}}^2/d}{3} = 28266010.$$

The bijection of Theorem 31.8 now proves (1403). The first prime has no completion with this prescribed first denominator by its zero certificate count. Two arguments with the same residue and unequal function values cannot be the pullback of a single residue function: if $f(p) = h(p \bmod 840)$, equality of the residues forces equality of the values. Applying this observation to the three displayed functions proves the claimed failures of factorization. □

31.8 An actual residue cocycle and a retained-sample Fourier map

For $r \in S_{840}$, multiplication by r permutes each fibre. Let $\rho(r)$ act by

$$\rho(r)e_s = e_{rs} \quad (s \in V_{840}), \qquad \rho(r)|_D = I, \qquad b(r) = e_r - e_1 \in A(S_{840}) \subset N. \tag{1404}$$

The two free coordinates in $b(r)$ belong to the principal fibre.

Theorem 31.11 (The full-lattice residue action and its exact cocycle)**.** *The permutations $\rho(r)$ extend to a faithful homomorphism $\rho : S_{840} \to O(N)$. The map $b : S_{840} \to N$ satisfies*

$$b(rs) = b(r) + \rho(r)b(s) \qquad (r, s \in S_{840}), \qquad b(1) = 0, \qquad (b(r), b(r)) = 2 \quad (r \neq 1). \tag{1405}$$

Thus reduction of any prime into S_{840}, followed by b, is an explicit lattice-valued map on that prime domain.

Proof. Multiplication permutes the orthonormal free coordinates, preserves their augmentation kernels, and obeys $\rho(r)\rho(s) = \rho(rs)$. On the i-th fibre its dual discriminant generator is

$$\lambda_i = e_{f_i} - \frac{1}{6} \sum_{t \in F_{\eta_i}} e_t.$$

Its image differs from it by $e_{rf_i} - e_{f_i} \in A(F_{\eta_i})$. Hence $\rho(r)$ acts trivially on the entire A5 discriminant group, since λ_i generates it. It fixes D and therefore the whole discriminant group, in particular $\mathcal{H}$. It follows from the inverse-image definition that it preserves N. If $\rho(r)$ were the identity, applying it to $e_1 - e_{289}$ would give $e_r - e_{289r} = e_1 - e_{289}$. Since $289 \neq 1$, the positive coordinate on the left is exactly r, so comparison of free-coordinate coefficients forces $r = 1$. This proves faithfulness.

Finally

$$b(r) + \rho(r)b(s) = (e_r - e_1) + (e_{rs} - e_r) = e_{rs} - e_1 = b(rs).$$

For $r \neq 1$, the two basis vectors in the difference are orthogonal unit vectors, giving squared norm two. The stated prime map is the composition $p \mapsto p \bmod 840 \mapsto b(p \bmod 840)$, whose domains have all been specified. □

Fix a positive shell $R \equiv 3 \pmod 4$, and let P be any finite set of positive primes whose reduction lies in S_{840}. For every $p \in P$, use its actual count $A_R(p)$ defined by the reduced divisor set, including the full data of (1395). Define

$$\begin{aligned} w_{R,P}(r) &= \sum_{\substack{p \in P \\ p \equiv r \pmod{840}}} A_R(p), \\ \widehat{w}_{R,P}(k) &= \sum_{r \in S_{840}} w_{R,P}(r) \exp\left(-\frac{2\pi \mathrm{i} k \kappa(r)}{6}\right), \qquad k \in \mathbb{Z}/6\mathbb{Z}. \end{aligned} \tag{1406}$$

This is the pushforward along prime reduction, followed by the finite Fourier transform with the already fixed generator 289.

Theorem 31.12 (Inversion and an integral augmentation-valued count map)**.** *The function $w_{R,P}$ is recovered by*

$$w_{R,P}(r) = \frac{1}{6} \sum_{k=0}^{5} \widehat{w}_{R,P}(k) \exp\left(\frac{2\pi \mathrm{i} k \kappa(r)}{6}\right). \tag{1407}$$

Let $u = \sum_{r \in S_{840}} e_r$, $W_{R,P} = \sum_r w_{R,P}(r) e_r$, and $M_{R,P} = \sum_r w_{R,P}(r) = \widehat{w}_{R,P}(0)$. Then

$$Q_{R,P} = 6W_{R,P} - M_{R,P}u \in A(S_{840}) \subset N \tag{1408}$$

is a well-defined integral lattice-valued count map. Put $\zeta_6 = e^{2\pi \mathrm{i}/6}$ *and*

$$f_k = \sum_{j=0}^{5} \zeta_6^{kj} e_{289^j} \qquad (0 \le k \le 5).$$

The five vectors $f_1, \dots, f_5$ *form a basis of* $A(S_{840}) \otimes \mathbb{C}$*, with*

$$\rho(289) f_k = \zeta_6^{-k} f_k, \qquad Q_{R,P} = \sum_{k=1}^{5} \widehat{w}_{R,P}(k) f_k. \tag{1409}$$

Both the prime sample P *and the shell* R *are retained in the domain of these maps.*

Proof. For an integer m, the sum $\sum_{k=0}^{5} \zeta_6^{km}$ equals 6 if $6 \mid m$. Otherwise it is the geometric sum $(1-\zeta_6^{6m})/(1-\zeta_6^m) = 0$. Substituting (1406) into the right side of (1407) therefore gives

$$\sum_{s \in S_{840}} w_{R,P}(s) \frac{1}{6} \sum_{k=0}^{5} \zeta_6^{k(\kappa(r)-\kappa(s))} = w_{R,P}(r),$$

because κ is bijective. This proves inversion.

The coefficient sum of $6W_{R,P} - M_{R,P}u$ is $6M_{R,P} - 6M_{R,P} = 0$, and its coefficients are integers, so it belongs to the exact augmentation lattice. The same geometric sum shows that f_k has coefficient sum zero for $1 \le k \le 5$ and $f_0 = u$. Moreover the six f_k are independent: the coefficient Fourier transform of f_k is six in coordinate k and zero in all other coordinates, by that same identity. Since the augmentation space has dimension five, its five nonzero-frequency vectors form a basis. Replacing j by $j+1$ in $\rho(289) f_k$ gives the multiplier ζ_6^{-k}, including its displayed sign.

Inversion in each free coordinate gives $W_{R,P} = (1/6) \sum_{k=0}^{5} \widehat{w}_{R,P}(k) f_k$. Multiplication by 6 and subtraction of $M_{R,P}u = \widehat{w}_{R,P}(0) f_0$ prove the second identity in (1409). All sums were taken over the specified finite set P and its residue image. The transform therefore recovers these aggregate counts; it makes no identification between the separate counts of distinct primes with the same residue. Their possible inequality is proved in Theorem 31.10. □

31.9 The entire lattice automorphism group and its Weyl quotient

Cheng, Duncan and Harvey define the umbral group as the quotient of the Niemeier lattice automorphism group by its root-reflection subgroup; their Table 3 identifies this quotient as $GL_2(3)$ for $A_5^4 D_4$ [74, Section 2.4 and Table 3]. The argument here proves that identification in the present residue, metric, discriminant and glue coordinates, and locates the arithmetic six-cycle in the full group. It does not claim the abstract group as new. The displayed complement is the particular lift already constructed here; no equality of its representation with a simple-root-preserving section is assumed.

Retain the exact lattice N, the root lattice R_{ES}, the four fibre identifications φ_i, the discriminant marking, the glue subgroup $\mathcal{H}$, and the lift $g \mapsto L_g$ constructed above. For a lattice K in a Euclidean space, $O(K)$ means the group of real linear isometries of its real span that map K onto itself. For the present root lattice define

$$W(R_{\mathrm{ES}}) = \langle s_\alpha : \alpha \in R_{\mathrm{ES}},\ (\alpha,\alpha) = 2 \rangle, \qquad s_\alpha(x) = x - (x,\alpha)\alpha.$$

All actions below retain the previously fixed six-coordinate order in each A_5 summand and the marking $10 \mapsto v$, $01 \mapsto s$, $11 \mapsto c$ in $D^\vee/D$.

Theorem 31.13 (All isometries of the displayed A_5 lattice)**.** *Let* $A = \{x \in \mathbb{Z}^6 : \sum_{j=0}^{5} x_j = 0\}$ *with its displayed Euclidean metric, and for a permutation* σ *of* $\{0,1,2,3,4,5\}$ *let* $Q_\sigma e_j = e_{\sigma(j)}$*. Every element of* $O(A)$ *has exactly one expression* $\epsilon Q_\sigma|_{\mathbb{R}A}$*, where* $\epsilon \in \{1,-1\}$*. Thus*

$$O(A) = \langle Q_\sigma|_{\mathbb{R}A} : \sigma \in S_6 \rangle \times \{I, -I\}, \qquad W(A) = \langle Q_\sigma|_{\mathbb{R}A} : \sigma \in S_6 \rangle \cong S_6.$$

On the marked discriminant group $D_A = (\mathbb{Z}/6)\lambda$, $\lambda = e_0 - \frac{1}{6}\sum_{j=0}^{5} e_j$, this isometry acts by $k\lambda + A \mapsto \epsilon k\lambda + A$. The kernel of this discriminant action is exactly $W(A)$.

Proof. The norm-two vectors are precisely $\alpha_{ij} = e_i - e_j$ for $i \neq j$. For two distinct such roots,

$$(\alpha_{ij}, \alpha_{kl}) = \delta_{ik} - \delta_{il} - \delta_{jk} + \delta_{jl}.$$

Their inner product is one exactly when they have the same first index or the same second index. Indeed equality of first indices, with distinct second indices, makes the displayed expression one; equality of second indices, with distinct first indices, does the same. If both positive Kronecker terms vanish the expression is nonpositive, and if both are one then the roots are equal, the excluded case.

Put

$$C_i = \{\alpha_{ij} : j \neq i\}, \qquad -C_i = \{\alpha_{ji} : j \neq i\} \qquad (0 \leq i \leq 5).$$

These twelve sets are exactly the maximal sets of roots in which every two distinct members have inner product one. To prove this, a singleton can be enlarged by keeping its first index and choosing a third index, so a maximal such set contains a pair. If the pair is α_{ij}, α_{ik}, with $j \neq k$, a further root α_{lm} has inner product one with both only if

$$(l = i \text{ or } m = j) \quad \text{and} \quad (l = i \text{ or } m = k).$$

Since $j \neq k$, these conditions force $l = i$. Thus the whole set is contained in C_i, and maximality makes it equal to C_i. For a pair with common second index the same argument, after negating both roots and every possible further member, gives $-C_i$. Each C_i and $-C_i$ has five roots and is maximal by this argument.

Form a graph whose vertices are these twelve maximal sets, with an edge when the two sets have nonempty intersection. Its intersections are exactly

$$C_i \cap C_j = (-C_i) \cap (-C_j) = \varnothing \quad (i \neq j), \qquad C_i \cap (-C_j) = \begin{cases} \{\alpha_{ij}\}, & i \neq j, \\ \varnothing, & i = j. \end{cases}$$

Consequently it is the bipartite graph on the two displayed six-element families with every cross-edge except the six pairs $(C_i, -C_i)$. It is connected: two distinct vertices on the same side have a common neighbor indexed by any index different from both; vertices on opposite sides with different indices are adjacent; and C_i and $-C_i$ are joined by the path $C_i, -C_j, C_k, -C_i$ for distinct i, j, k. A connected bipartite graph has only two choices for its ordered bipartition, because the side of a fixed vertex determines the side of each other vertex along any connecting path. Hence an isometry of A, which permutes roots and preserves their inner products and set intersections, either preserves both displayed families or exchanges them.

In the first case write $C_i \mapsto C_{\sigma(i)}$ and $-C_i \mapsto -C_{\tau(i)}$. The unique opposite-family vertex not adjacent to C_i is $-C_i$. Preservation of that missing edge forces $\tau(i) = \sigma(i)$ for each i. The singleton intersection formula then gives $\alpha_{ij} \mapsto \alpha_{\sigma(i)\sigma(j)}$. In the case exchanging the families, composition with $-I$ reduces to the first case, so the root action is $\alpha_{ij} \mapsto -\alpha_{\sigma(i)\sigma(j)}$. The roots $e_j - e_5$, $0 \leq j \leq 4$, span $\mathbb{R}A$, so the root action determines the whole linear isometry. These cases give precisely the asserted expressions.

The permutation σ is determined by the permutation of the six vertices C_i, after the sign is determined by whether the two families are preserved or exchanged. Thus the expression is unique. Every Q_σ and $-I$ preserves A, and $-I$ commutes with every linear map; uniqueness proves the direct product. Reflection in $e_i - e_j$ interchanges free coordinates i, j and fixes the others, as follows on substituting this root in $x \mapsto x - (x, e_i - e_j)(e_i - e_j)$. Every permutation is a product of transpositions: each cycle $(i_1\ i_2\ \cdots\ i_m)$ is $(i_1\ i_m)(i_1\ i_{m-1}) \cdots (i_1\ i_2)$. These reflections therefore generate precisely the displayed S_6. Finally

$$Q_\sigma \lambda - \lambda = e_{\sigma(0)} - e_0 \in A.$$

Thus the discriminant action is the stated sign. Since $\lambda + A$ has exact order six, its negative is distinct from itself; the action is trivial exactly when $\epsilon = 1$. □

Theorem 31.14 (All isometries of the displayed D_4 lattice)**.** *For the exact lattice $D = \{x \in \mathbb{Z}^4 : \sum_i x_i \equiv 0 \pmod 2\}$, the action on its marked discriminant group gives the split exact sequence*

$$1 \longrightarrow W_D \longrightarrow O(D) \xrightarrow{d_D} GL_2(\mathbb{F}_2) \longrightarrow 1, \qquad O(D) = W_D \rtimes \delta(GL_2(\mathbb{F}_2)).$$

Here W_D consists of all signed coordinate permutations with an even number of minus signs, and δ is exactly the section generated by the earlier matrices R_D, H_D. In particular $O(D) = W_F$ in the root and weight coordinates of Theorem 31.4.

Proof. Every isometry of D preserves its dual: if $T \in O(D)$, $x \in D^\vee$, and $a \in D$, then $(Tx, a) = (x, T^{-1}a) \in \mathbb{Z}$. It therefore induces an additive automorphism of $D^\vee/D \cong \mathbb{F}_2^2$, giving the homomorphism d_D. The three nonzero classes have the exact norm-one representative sets

$$\begin{aligned} \mathcal{V} &= \{\pm e_i : 1 \le i \le 4\}, \\ \mathcal{S}_+ &= \{\tfrac{1}{2}\textstyle\sum_i \delta_i e_i : \prod_i \delta_i = 1\}, \\ \mathcal{S}_- &= \{\tfrac{1}{2}\textstyle\sum_i \delta_i e_i : \prod_i \delta_i = -1\}, \qquad \delta_i \in \{1, -1\}. \end{aligned}$$

in the respective marked classes v, s, c. For the integral nonzero class, norm one forces exactly one coordinate ± 1. A half-integral vector has norm at least $4(1/2)^2 = 1$, with equality exactly when every coordinate is $\pm 1/2$. The difference of such a vector from $s = (1,1,1,1)/2$ has one coordinate -1 per minus sign, so it belongs to D exactly when their number is even. This proves the three sets and their marking directly.

If $d_D(T) = I$, then T preserves $\mathcal{V}$. The vectors Te_i are orthonormal members of $\mathcal{V}$, so $Te_i = \delta_i e_{\sigma(i)}$ for a permutation σ of four indices and signs δ_i. Preservation of $\mathcal{S}_+$ forces $Ts \in \mathcal{S}_+$, hence $\prod_i \delta_i = 1$. Thus the kernel consists only of even signed permutations.

Conversely, reflection in $e_i - e_j$ interchanges coordinates i, j, and reflection in $e_i + e_j$ interchanges them and negates both. Their composition produces a sign change in exactly those two coordinates. Any even set of sign changes is a product of such disjoint pairs, and transpositions generate all permutations. The root reflections therefore generate every even signed permutation, and each root reflection is itself such a matrix. If $\alpha \in D$ has norm two and $x \in D^\vee$, then $s_\alpha x - x = -(x, \alpha)\alpha \in D$. Every root reflection acts trivially on the discriminant group. This proves $\ker d_D = W_D$ in both directions.

For completeness the section has the original exact matrices

$$R_D = \operatorname{diag}(-1, 1, 1, 1), \qquad H_D = \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}.$$

They preserve D, are orthogonal involutions, and act on (v, s, c) by $(s\ c)$ and $(v\ s)$, respectively, as calculated in Theorem 31.4. Their binary matrices in the retained marking are

$$B_R = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad B_H = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

The same earlier coordinate calculation gives $R_D^2 = H_D^2 = (R_D H_D)^3 = I$; thus their subgroup has at most six elements, by reducing words to $(R_D H_D)^k$ or $(R_D H_D)^k R_D$, $k = 0, 1, 2$. The induced two transpositions generate the six permutations of the three nonzero binary vectors. There are exactly six invertible binary matrices, by the $3 \cdot 2$ choices of their independent columns, and their action on the three nonzero vectors is faithful. Hence the six words have distinct discriminant actions, and the restriction of d_D to $\langle R_D, H_D \rangle$ is an isomorphism onto $GL_2(\mathbb{F}_2)$. Its inverse is the specified δ. For every $T \in O(D)$, $T\delta(d_D(T))^{-1} \in \ker d_D = W_D$. The two factors intersect trivially because $d_D\delta = I$. This proves the asserted splitting and exhausts $O(D)$. The earlier theorem proved $W_F = \langle W_D, R_D, H_D \rangle$, so it also proves the last equality. □

Theorem 31.15 (The exact full automorphism quotient of N)**.** *Every isometry of N has a unique factorization*

$$T = wL_g, \qquad w \in W(R_{\mathrm{ES}}), \quad g \in G.$$

The Weyl group is normal, and the induced discriminant action gives the split exact sequence and the indicated isomorphisms

$$\begin{aligned} 1 \longrightarrow W(R_{\mathrm{ES}}) \longrightarrow O(N) &\xrightarrow{d_{\mathrm{ES}}} G \longrightarrow 1, \\ O(N) &= W(R_{\mathrm{ES}}) \rtimes L(G), \\ L(G) &\xrightarrow{\cong} O(N)/W(R_{\mathrm{ES}}) \cong G \xrightarrow{\Theta} GL_2(\mathbb{F}_3), \\ W(R_{\mathrm{ES}}) &\cong S_6^4 \times W_D, \qquad |O(N)| = 48 \cdot (6!)^4 \cdot (2^3 \cdot 4!). \end{aligned} \tag{1410}$$

In identified coordinates, if $w = (Q_{\sigma_1}, Q_{\sigma_2}, Q_{\sigma_3}, Q_{\sigma_4}; w_D)$ and $g = (\varepsilon, p, B)$, the exact semidirect-product action is

$$L_g w L_g^{-1} = (Q_{\sigma_{p(1)}}, Q_{\sigma_{p(2)}}, Q_{\sigma_{p(3)}}, Q_{\sigma_{p(4)}}; \delta(B) w_D \delta(B)^{-1}). \tag{1411}$$

In the original fibre coordinates the i-th permutation factor on the right means $\varphi_i Q_{\sigma_{p(i)}} \varphi_i^{-1}$.

Proof. Theorem 31.3 proves that the norm-two vectors of N are exactly the 144 roots of its five original orthogonal summands. Join two distinct roots by an edge when their inner product is nonzero. The five summands are exactly the connected components of this graph. There are no edges between different summands by orthogonality. Within an A_5 summand, the simple roots $e_0 - e_1, e_1 - e_2, e_2 - e_3, e_3 - e_4, e_4 - e_5$ have consecutive inner product -1, forming a connected graph, and span its real space. Every root not already among them has nonzero inner product with at least one of them; otherwise it is orthogonal to its entire real space and has norm zero, contrary to norm two. Thus its whole root graph is connected. In D, the four roots

$$e_1 - e_2, \quad e_2 - e_3, \quad e_3 - e_4, \quad e_3 + e_4$$

span $\mathbb{R}D$, and the second has inner product -1 with each of the other three. The same spanning argument proves connectedness of the D_4 root graph.

An element $T \in O(N)$ permutes the norm-two vectors and preserves these edges, hence permutes their connected components. Their cardinalities are $30, 30, 30, 30, 24$, so T preserves the D_4 component and permutes the four A_5 components. The roots generate each original lattice summand: the five and four displayed simple roots are their integral bases by Theorem 31.2. Consequently T preserves their integer span R_{ES}. The same component argument applies to every element of $O(R_{\mathrm{ES}})$, since its norm-two vectors are the same 144 roots. It follows that every such element has, in the fixed identified coordinates, the form

$$T(X_1, X_2, X_3, X_4; d) = \left(\varepsilon_i Q_{\sigma_i} X_{p(i)}\right)_{i=1}^{4} ; T_D d, \tag{1412}$$

where $p \in S_4$, $\varepsilon_i \in \{1, -1\}$, $\sigma_i \in S_6$, and $T_D \in O(D)$. This follows by transporting each isometry from its input A_5 summand to its output summand using the fixed φ_i, then applying Theorem 31.13. All choices in (1412) are unique. The convention is precisely the earlier convention: a vector supported on component k is sent to component $p^{-1}(k)$.

Let $\mathcal{A}_{\mathrm{ES}}$ be the group of transformations of the exact discriminant group $(\mathbb{Z}/6)^4 \oplus \mathbb{F}_2^2$ given by

$$(x; y) \longmapsto ((\varepsilon_i x_{p(i)})_{i=1}^4; By), \qquad \varepsilon_i \in \{1, -1\}, \quad p \in S_4, \quad B \in GL_2(\mathbb{F}_2).$$

Each tuple gives a distinct transformation: the image of the order-six generator in each individual summand determines its output summand and its sign, and the images of the two binary basis vectors determine B. The discriminant action of (1412), calculated by the two preceding theorems, is exactly this tuple with $B = d_D(T_D)$. This defines a homomorphism

$$d_{\mathrm{ES}} : O(R_{\mathrm{ES}}) \longrightarrow \mathcal{A}_{\mathrm{ES}}.$$

It is a homomorphism because extending an isometry to the dual and then passing to the quotient commutes with composition. In particular the component permutation rule remains $P_pP_q = P_{q\circ p}$; there is no change of the earlier convention.

This homomorphism is onto. Given any tuple $a = (\varepsilon, p, B) \in \mathcal{A}_{\mathrm{ES}}$, define

$$\widetilde{L}_a(X_1, X_2, X_3, X_4; d) = \left(\varepsilon_i X_{p(i)}\right)_{i=1}^{4} ; \delta(B)d.$$

It is a root-lattice isometry and has discriminant action a. The map $a \mapsto \widetilde{L}_a$ is a homomorphism: if $a = (\varepsilon, p, B)$ and $a' = (\varepsilon', q, B')$, its composite has i-th output $\varepsilon_i\varepsilon'_{p(i)}X_{q(p(i))}$ and D-component $\delta(B)\delta(B')d = \delta(BB')d$, which are exactly the tuple product in the stated convention.

The kernel of d_{ES} consists exactly of the tuples in (1412) with $p = 1$, every $\varepsilon_i = 1$, and $T_D \in W_D$. It is therefore the direct product $S_6^4 \times W_D$. Every root reflection acts on its own orthogonal summand and fixes the other four pointwise, and the preceding two theorems identify the groups generated inside those summands. Hence this direct product is exactly $W(R_{\mathrm{ES}})$.

Every root reflection also preserves N. Indeed $\alpha \in N$ and integrality of N give $s_\alpha x = x - (x, \alpha)\alpha \in N$ for every $x \in N$. The reflection is an involution, so inclusion is equality. Thus $W(R_{\mathrm{ES}}) \subset O(N)$. It is normal there, since for $T \in O(N)$

$$Ts_\alpha T^{-1}(x) = x - (x, T\alpha)T\alpha = s_{T\alpha}(x),$$

and $T\alpha$ is again a norm-two root.

It remains to impose the actual glue, which completely determines the remaining isometries. For $T \in O(R_{\mathrm{ES}})$ and $x \in R_{\mathrm{ES}}^\vee$, the inverse-image definition gives

$$Tx \in N \quad \Longleftrightarrow \quad d_{\mathrm{ES}}(T)(x + R_{\mathrm{ES}}) \in \mathcal{H}.$$

Consequently $T(N) = N$ if and only if $d_{\mathrm{ES}}(T)(\mathcal{H}) = \mathcal{H}$. For the forward implication, represent each class of $\mathcal{H}$ by a vector of N and apply both T and T^{-1}. For the reverse implication, the displayed equivalence applies to T and T^{-1} because the action on $\mathcal{H}$ is bijective. The group preserving $\mathcal{H}$ inside precisely $\mathcal{A}_{\mathrm{ES}}$ is the group G of Theorem 31.5, proved there to be exactly $GL_2(\mathbb{F}_3)$ through the code map Θ. We have therefore proved the equality of subgroups

$$O(N) = d_{\mathrm{ES}}^{-1}(G) \quad \text{inside } O(R_{\mathrm{ES}}).$$

The section $\widetilde{L}$ restricts on G to the already specified lift L. This proves surjectivity of the restricted map $d_{\mathrm{ES}} : O(N) \to G$, with exactly the asserted kernel. For $T \in O(N)$, set $g = d_{\mathrm{ES}}(T)$. Then $w = TL_g^{-1}$ lies in the kernel and $T = wL_g$. If $wL_g = w'L_{g'}$, applying d_{ES} gives $g = g'$, then cancellation gives $w = w'$. In particular $L(G) \cap W(R_{\mathrm{ES}}) = \{I\}$. Normality and this unique factorization prove the semidirect product and the quotient isomorphism, including its surjectivity. The four S_6 factors have order $(6!)^4$; W_D has 2^3 sign choices and $4!$ permutation choices, and $|G| = 48$. Unique factorization proves the displayed order.

Finally put $Z = L_g^{-1}X$. From the i-th output equation $X_i = \varepsilon_i Z_{p(i)}$ follows $Z_{p(i)} = \varepsilon_i X_i$, with each vector read in the fixed identified A coordinates. Therefore the i-th output of $L_g w L_g^{-1} X$ is

$$\varepsilon_i Q_{\sigma_{p(i)}} Z_{p(i)} = \varepsilon_i Q_{\sigma_{p(i)}} \varepsilon_i X_i = Q_{\sigma_{p(i)}} X_i.$$

The D-component is $\delta(B)w_D\delta(B)^{-1}d$. It belongs to W_D by normality of $\ker d_D$. These are exactly the components of (1411), proving the asserted action with all signs and component indices retained. □

Theorem 31.16 (The residue six-cycle in the full automorphism group)**.** *For $r \in S_{840} = \langle 289 \rangle$, let $\rho(r)e_s = e_{rs}$ on each of the four fibres and $\rho(r)|_D = I$, with the exact fibre action used in* (1404)*. Then*

$$\rho(S_{840}) \subset W(R_{\mathrm{ES}}), \qquad d_{\mathrm{ES}}(\rho(r)) = 1, \qquad L_g\rho(r)L_g^{-1} = \rho(r) \quad (g \in G,\ r \in S_{840}). \tag{1413}$$

In particular the map

$$S_{840} \times G \longrightarrow O(N), \qquad (r, g) \longmapsto \rho(r)L_g$$

is an injective homomorphism with image of order $6 \cdot 48 = 288$, isomorphic to $C_6 \times GL_2(\mathbb{F}_3)$.

Proof. Put $Ce_j = e_{j+1 \bmod 6}$ on the free six-coordinate space and restrict it to $\mathbb{R}A$. If $r = 289^m$, with $m \in \{0,1,2,3,4,5\}$, the exact fibre identification gives

$$\rho(r)e_{f_i 289^j} = e_{f_i 289^{j+m}}, \qquad \rho(r)|_{A(F_{\eta_i})} = \varphi_i C^m \varphi_i^{-1}.$$

Thus in all four identified summands $\rho(r)$ is the same coordinate permutation C^m, and it is the identity on D. Theorem 31.13 puts it in the corresponding product of Weyl groups, giving the first two assertions. Each sign $\varepsilon_i I$ commutes with C^m, and the same operator C^m occurs in every component. Directly the i-th outputs of $L_g\rho(r)$ and $\rho(r)L_g$ are respectively $\varepsilon_i C^m X_{p(i)}$ and $C^m \varepsilon_i X_{p(i)}$, which agree; the D-outputs also agree since $\rho(r)|_D = I$. This proves the conjugation identity without changing the chosen origins.

The six restrictions $C^m|_{\mathbb{R}A}$ are distinct: if $C^m(e_0 - e_1) = e_0 - e_1$, comparison of the unique coordinate with coefficient $+1$ gives $m = 0 \pmod 6$. Thus ρ is faithful. Commutation makes the displayed product map a homomorphism. If $\rho(r)L_g = I$, then $L_g = \rho(r)^{-1}$ lies in $L(G) \cap W(R_{\mathrm{ES}}) = \{I\}$. Faithfulness of both factors gives $r = 1$ and $g = 1$. This proves injectivity and the exact image order. □

32 The exact Leech neighbor of the retained index-72 lattice

Retain the lattice, metric, discriminant marking, and glue literally:

$$\begin{aligned}
A &= \{(x_0,\dots,x_5) \in \mathbb{Z}^6 : \textstyle\sum x_j = 0\},\\
D &= \{(d_1,\dots,d_4) \in \mathbb{Z}^4 : \textstyle\sum d_i \equiv 0 \pmod 2\}, \qquad R = A^{\perp 4} \perp D,\\
C_2 &= \langle b_0 = (1111;00), b_1 = (1100;10), b_2 = (1010;01)\rangle_{\mathbb{F}_2},\\
C_3 &= \langle z_1 = (1,1,1,0), z_2 = (1,2,0,1)\rangle_{\mathbb{F}_3},\\
H &= \{(3u + 2z; y) : (u;y) \in C_2,\ z \in C_3\},\\
N &= \{x \in R^\vee : x + R \in H\}.
\end{aligned}$$

The coordinates in each A remain the six ordered elements $f_i 289^j$ of its original residue fibre. The A discriminant generator is $\lambda = (5,-1,-1,-1,-1,-1)/6$. The D markings $10, 01, 11$ remain represented by $v_D = (1,0,0,0)$, $s_D = (1,1,1,1)/2$, $c_D = (-1,1,1,1)/2$. The preceding proof establishes that N is even unimodular, $[N : R] = 72$, $\det R = 5184$, and its roots are exactly those of $A_5^4 D_4$. Every root has squared norm two. No metric rescaling occurs anywhere below.

32.1 The Weyl vector and the exact neighbor

Use the original simple roots $a_j = e_j - e_{j+1}$, $0 \le j \le 4$, in each A component and

$$d_1 = e_1 - e_2, \quad d_2 = e_2 - e_3, \quad d_3 = e_3 - e_4, \quad d_4 = e_3 + e_4$$

in D. Their highest roots are $\theta_A = e_0 - e_5 = \sum_j a_j$ and $\theta_D = e_1 + e_2 = d_1 + 2d_2 + d_3 + d_4$. Both highest-root heights are five, so the common Coxeter number is six. Define

$$\rho_A = (5,3,1,-1,-3,-5)/2, \qquad \rho_D = (3,2,1,0), \qquad \rho = \rho_A^{\oplus 4} \perp \rho_D.$$

Each simple root pairs to one with ρ. Moreover $\rho_A - 3\lambda = (0,2,1,0,-1,-2) \in A$ and $\rho_D \in D$. Thus the exact glue class of ρ is $(3,3,3,3;00)$, the image of b_0, and $\rho \in N$. Directly

$$\rho_A^2 = 35/2, \quad \rho_D^2 = 14, \quad \rho^2 = 4(35/2) + 14 = 84 = 2 \cdot 6 \cdot 7.$$

Consequently $(\rho, N) \subset \mathbb{Z}$. Fix $\alpha = a_0$ in the first A component and put

$$\ell_\rho : N \to \mathbb{Z}/6, \quad x \mapsto (\rho, x) \bmod 6, \qquad K = \ker \ell_\rho, \quad R_0 = R \cap K, \quad v = \rho/6 - \alpha, \quad \Lambda = K + \mathbb{Z}v. \tag{NL1}$$

The pairing $(\rho,\alpha)=1$ proves $[N:K]=[R:R_0]=6$. For $x\in K$, the number $(v,x)=(\rho,x)/6-(\alpha,x)$ is integral, and

$$v^2=84/36-2/6+2=4,\qquad 6v=\rho-6\alpha\in K,\qquad (\rho,6v)=78.$$

Hence Λ is positive definite even integral of rank 24. If $mv\in N$, its pairing after adding $m\alpha$ with α is $m/6\in\mathbb{Z}$, so $6\mid m$. It follows that

$$N\cap\Lambda=K,\qquad [\Lambda:K]=6,\qquad \det K=36,\qquad \det\Lambda=36/6^2=1. \tag{NL2}$$

Changing α to another root of ρ-pairing one changes v by an element of K, and so does not change Λ. We shall use the equivalent exact congruence description

$$\Lambda=\{x+k\rho/6 : x\in N,\ k\in\mathbb{Z},\ (\rho,x)+k\equiv 0 \pmod 6\}. \tag{NL3}$$

Indeed write $x+k\rho/6=(x+k\alpha)+kv$; the first summand is in K exactly under this congruence. Changing k by six can be absorbed in x using $\rho\in N$, and preserves the congruence since $\rho^2=84$.

32.2 A complete rootlessness proof

Theorem 32.1 (The Leech neighbor in the retained coordinates)**.** *The lattice $\Lambda=K+\mathbb{Z}(\rho/6-\alpha)$ is positive definite even unimodular of rank 24 and has minimum squared norm exactly four. It is the holy-construction Leech lattice of the retained $A_5^4D_4$ glue, and $N\cap\Lambda=K$ with $[N:K]=[\Lambda:K]=6$.*

Proof. The rank, parity, unimodularity and indices were proved in (NL1)–(NL2). The holy-presentation identification is proved in (NL8)–(NL11) below. We now prove the required minimum directly. For $k=0$ in (NL3), a possible root is an original root of N. An A root pairs with ρ to one of $\pm1,\dots,\pm5$. For D, all roots $\pm e_i\pm e_j$ have nonzero ρ_D-pairing of absolute value at most five. Thus none belongs to K. Negation and the reduction of k modulo six leave $k=1,2,3$.

Define the exact shifted class minima by

$$m_A(k,t)=\min_{x\in t\lambda+A}|x+k\rho_A/6|^2,\qquad m_D(k,y)=\min_{x+D=y}|x+k\rho_D/6|^2.$$

Their values are

$k\backslash t$	0	1	2	3	4	5
1	35/72	35/72	35/72	35/72	35/72	35/72
2	11/18	4/9	11/18	4/9	11/18	4/9
3	3/8	17/24	17/24	3/8	17/24	17/24

$k\backslash y$	00	10	01	11
1	7/18	7/18	7/18	7/18
2	2/9	5/9	5/9	5/9
3	1/2	1/2	1/2	1/2

. (NL4)

We prove the table and its needed equality information. The shifted A coordinates are precisely

$$\left(n_j-\frac{t}{6}+\frac{k(5-2j)}{12}\right)_{j=0}^{5},\qquad n_j\in\mathbb{Z},\quad \sum n_j=t. \tag{NL5}$$

At $k=1$, coordinatewise nearest representatives are the six numbers $\pm1/12,\pm3/12,\pm5/12$ in an order depending on t. They sum to zero and their squares sum to 35/72. At $k=2$ and even t, the nearest coordinates consist of two copies of each sign of 1/6 and one copy of each sign of 1/2; the half-integer ties admit exactly the needed zero-sum choice and the squared sum is 11/18. At odd t, there are two each of $0,1/3,-1/3$, giving 4/9. For $t=1,3,5$ their actual ordered shifted vectors are

$$(-1,1,0,-1,1,0)/3,\quad (1,0,-1,1,0,-1)/3,\quad (0,-1,1,0,-1,1)/3. \tag{NL6}$$

Subtracting $\rho_A/3$ gives original squared norms $17/6,3/2,17/6$, respectively. At $k=3,t=0,3$, the nearest coordinates alternate $\pm1/4$, giving 3/8 with zero sum. At each other t, the coordinatewise nearest

representatives have three absolute values $1/12$ and three $5/12$, square sum $13/24$, and coordinate sum 1 or -1. Meeting the zero-sum constraint requires an integer change. Every nonzero coordinate change increases the squared sum by at least $(7/12)^2 - (5/12)^2 = 1/6$. Changing one suitable $5/12$ coordinate attains this increase and restores the zero sum. Thus the constrained minimum is $17/24$. At $t = 0, 3$, the original minimizing vectors are

$$(-1,-1,0,0,1,1), \qquad (-3,-1,-1,1,1,3)/2, \tag{NL7}$$

of squared norms $4, 11/2$.

For D and $k = 1$ the shift is $(1/2, 1/3, 1/6, 0)$. The nearest integer coordinate distances are $(1/2, 1/3, 1/6, 0)$; the first-coordinate tie meets either integer parity. The nearest half-integer distances are $(0, 1/6, 1/3, 1/2)$; the last-coordinate tie meets either half-integer class. Both squared sums are $7/18$. At $k = 2$ the shift is $(1, 2/3, 1/3, 0)$. The unique nearest integer vector before shifting is $(-1,-1,0,0) \in D$, giving $2/9$. Changing parity costs at least $1/3$, attained by changing the second or third coordinate, giving $5/9$. The half-integer coordinate distances are $(1/2, 1/6, 1/6, 1/2)$, giving $5/9$; ties meet either marking. At $k = 3$ the shift is $(3/2, 1, 1/2, 0)$. Integer-class minimum distances are $(1/2, 0, 1/2, 0)$, and half-integer-class distances $(0, 1/2, 0, 1/2)$. Ties meet every marking and all square sums are $1/2$. In class 00, the two original minimizing vectors are

$$(-1,-1,0,0), \qquad (-2,-1,-1,0),$$

of squared norms $2, 6$. In class 10, they are

$$(-1,-1,-1,0), \qquad (-2,-1,0,0),$$

of squared norms $3, 5$. In either half-integer class they have the shapes $(-3/2,-1/2,-1/2,\pm 1/2)$ and $(-3/2,-3/2,-1/2,\pm 1/2)$. Exactly one last-coordinate sign in each shape meets each original marking; their squared norms are $3, 5$. This proves every entry and the relevant equality cases in (NL4).

For $k = 1$, every full shifted vector has squared norm at least $4(35/72) + 7/18 = 7/3 > 2$. For $k = 2$, the lower bound from (NL4) is two. Equality forces all t_i odd and $y = 00$. In the actual glue this forces $u = 1111$. If $z = 0$, all $t_i = 3$ and (NL6) gives $x^2 = 4(3/2) + 2 = 8$. If $z \neq 0$, its weight is three, so exactly one $t_i = 3$ and three $t_i \in \{1, 5\}$; then $x^2 = 3/2 + 3(17/6) + 2 = 12$. The norm-two equation is

$$2 = x^2 + \tfrac{2}{3}(\rho, x) + \tfrac{28}{3}, \qquad (\rho, x) = -11 - \tfrac{3}{2}x^2.$$

Its pairing is -23 or -29, always 1 modulo six. Condition (NL3) requires residue 4, so all nine equality candidates, one per ternary word, are excluded.

For $k = 3$, the lower bound two requires $z = 0$: every nonzero ternary word raises the minimum by $3(17/24 - 3/8) = 1$. If $u = 0$, the possible original squared norms are $18, 22$. If $u = 1111$, they are $24, 28$. If u has weight two, its D class is nonzero and they are $2(4) + 2(11/2) + 3 = 22$ or $2(4) + 2(11/2) + 5 = 24$. These are all sixteen equality candidates. But

$$2 = x^2 + (\rho, x) + 21, \qquad (\rho, x) = -19 - x^2,$$

and (NL3) would require $x^2 \equiv 2 \pmod 6$. None of $18, 22, 24, 28$ has that residue. Thus Λ has no roots. Together with (NL2), this proves that it is positive definite even unimodular of rank 24, with minimum squared norm exactly four, attained by v. □

32.3 The exact holy-construction presentation

For $0 \le t \le 5$ set

$$\omega_t = (\underbrace{1, \dots, 1}_{t}, \underbrace{0, \dots, 0}_{6-t}) - \tfrac{t}{6}(1, \dots, 1).$$

Then $\omega_t + A = t\lambda + A$ and $(\rho_A, \omega_t) = t(6-t)/2 = 3\omega_t^2$. For the D markings set

$$\omega_{00} = 0, \quad \omega_{10} = e_1, \quad \omega_{01} = (1,1,1,1)/2, \quad \omega_{11} = (1,1,1,-1)/2.$$

The last differs from the original c_D by $(1,0,0,-1) \in D$, so retains its exact marking. For $a = (t_1,t_2,t_3,t_4;y) \in H$ define

$$u_a = (\omega_{t_1}, \omega_{t_2}, \omega_{t_3}, \omega_{t_4}; \omega_y), \qquad g_a = u_a - \rho/6. \tag{NL8}$$

These give all 72 representatives of N/R. Each nonzero D representative has norm one and pairing three; hence

$$(\rho, u_a) = 3u_a^2 \in 6\mathbb{Z}, \quad u_a \in K, \quad g_a^2 = u_a^2 - (\rho, u_a)/3 + 84/36 = 7/3. \tag{NL9}$$

Let $\mathcal{F}$ be the 29 vectors consisting, in each original component, of the negative original simple roots and the positive highest root. Their pairings with ρ are -1 or 5. There are positive integral zero relations of coefficient sum six:

$$\theta_A + \sum_{j=0}^{4}(-a_j) = 0, \qquad \theta_D + (-d_1) + 2(-d_2) + (-d_3) + (-d_4) = 0. \tag{NL10}$$

Every component of g_a pairs to $1/6$ with the corresponding vertices except one tip, where the pairing is $-5/6$. For A this follows because ω_t has simple-root pairing one only at a_{t-1} when $t \neq 0$, and has $(\omega_t, \theta_A) = 1$ for $t \neq 0$. For D the three nonzero representatives have simple-root pairing one only at d_1, d_4, d_3, respectively; all have highest-root pairing one. The zero representative points to the highest root. These computations fix all orientations in the holy generator diagram.

The following coefficient presentations are exact equalities:

$$\begin{aligned} N &= \left\{ \sum_f a_f f + \sum_a b_a g_a : a_f, b_a \in \mathbb{Z},\ \sum_a b_a = 0 \right\}, \\ \Lambda &= \left\{ \sum_f a_f f + \sum_a b_a g_a : a_f, b_a \in \mathbb{Z},\ \sum_f a_f + \sum_a b_a = 0 \right\}. \end{aligned} \tag{NL11}$$

For the first, the f span R and $g_a - g_0 = u_a$ generate its given extension N; the zero coefficient sum cancels $\rho/6$. For the second, write $A = \sum a_f$, $B = \sum b_a$, $r = \sum a_f f$, and $x = r + \sum b_a u_a$. The vector is $x - B\rho/6$, and by (NL9) $(\rho, x) \equiv -A \equiv B$, exactly (NL3) with $k = -B$. Conversely, for any $x + k\rho/6$ in (NL3), write $x = r + \sum b_a u_a$. The coefficient of $u_0 = 0$ can be set so $B = -k$. Write r with the negative simple-root vertices. The congruence gives $A + B \equiv 0 \pmod 6$. Adding a suitable multiple of a zero relation (NL10) makes $A + B = 0$ exactly. Thus both directions of (NL11) are proved.

These are exactly the holy coefficient rules of Conway–Sloane [79, pp. 277–280]. Borcherds gives the uniform proof of the holy-construction theorem [80, Section 7, especially the coefficient rules (i)–(ii) and Lemma 7.3]. Thus (NL11) is an explicit realization of that Leech construction from the actual 72 marked glue words; all its metric, determinant, integrality and rootlessness assertions have also been independently proved here.

32.4 The indices, discriminant morphism, and cyclic order twelve

Reduction modulo R gives an isomorphism $K/R_0 \cong H$: its kernel is R_0, and (NL9) supplies representatives in K for every H class. Consequently

$$[K : R_0] = 72, \quad [N : R_0] = [\Lambda : R_0] = 432, \quad \det R_0 = 6^2 \cdot 5184 = 186624. \tag{NL12}$$

Thus the original and new inclusions, with every index, are

$$R_0 \subset R \subset N, \quad [R : R_0] = 6,\ [N : R] = 72; \qquad R_0 \subset K \subset \Lambda, \quad [K : R_0] = 72,\ [\Lambda : K] = 6.$$

In particular $R \cap \Lambda = R_0$, an exact common sublattice.

Theorem 32.2 (The exact order-twelve quotient generator)**.** *The quotient of Λ by the original height-zero root sublattice R_0 is $\Lambda/R_0 \cong \mathbb{Z}/12 \oplus (\mathbb{Z}/6)^2$. The class of $v = \rho/6 - \alpha$ has exact order twelve.*

Proof. Let $\bar{b}_i$ denote the image in H of the binary generator b_i, namely its first four coordinates multiplied by three and its last two coordinates retained. Since $6v = \rho - 6\alpha$ and $6\alpha \in R_0$,

$$6[v] = \bar{b}_0 \quad \text{in } \Lambda/R_0. \tag{NL13}$$

This is a nonzero order-two element. The classes $\bar{b}_0, \bar{b}_1, \bar{b}_2, 2z_1, 2z_2$ form the displayed independent binary and ternary generators of H. Eliminate $\bar{b}_0$ using (NL13). The resulting surjective presentation is

$$\Lambda/R_0 \cong \mathbb{Z}/12 \oplus (\mathbb{Z}/2)^2 \oplus (\mathbb{Z}/3)^2 \cong \mathbb{Z}/12 \oplus (\mathbb{Z}/6)^2. \tag{NL14}$$

There are no further relations: this presentation has order $12 \cdot 4 \cdot 9 = 432$, equal to (NL12). Thus $[v]$ has exact order twelve. This particular occurrence of twelve is proved from the Coxeter denominator six and the actual order-two Weyl-vector glue class. □

Set $M = N + \Lambda = N + \mathbb{Z}(\rho/6)$. Then

$$M = K^\vee, \qquad K^\vee/K = \langle [\alpha], [v] \rangle \cong (\mathbb{Z}/6)^2. \tag{NL15}$$

Indeed $M \subset K^\vee$, its index over N is six by pairing with α, and $[K^\vee : N] = [N : K] = 6$. The cyclic images of N/K and Λ/K have order six and trivial intersection, proving the stated generators. Their exact discriminant data are

$$q_K([\alpha]) = q_K([v]) = 0, \quad b_K([\alpha], [v]) = 1/6 \pmod{\mathbb{Z}}, \quad q_K(a[\alpha] + b[v]) = ab/3 \pmod{2\mathbb{Z}}. \tag{NL16}$$

These follow from $\alpha^2 = 2$, $v^2 = 4$ and $(\alpha, v) = 1/6 - 2 = -11/6$. The two original lattices are therefore the inverse images of the two indicated order-six isotropic subgroups in this specified discriminant form.

32.5 An exact Lorentzian quotient and symmetry transport

Take $\mathcal{I} = N \perp U$ in coordinates (x, m, n) with

$$\langle (x, m, n), (x', m', n') \rangle = (x, x') - mn' - m'n.$$

Thus U has Gram matrix $\left(\begin{smallmatrix} 0 & -1 \\ -1 & 0 \end{smallmatrix}\right)$. The vectors $z = (0, 0, 1)$ and $w = (\rho, 6, 7)$ are primitive isotropic: $w^2 = 84 - 2 \cdot 6 \cdot 7 = 0$, and six and seven are coprime. There are integral quotient isometries

$$\begin{aligned} z^\perp/\mathbb{Z}z &\longrightarrow N, \quad [(x, 0, n)] \mapsto x, \\ w^\perp/\mathbb{Z}w &\longrightarrow \Lambda, \quad [(x, m, n)] \mapsto x - m\rho/6. \end{aligned} \tag{NL17}$$

For the second map, orthogonality means $(\rho, x) = 7m + 6n$, giving (NL3) with $k = -m$. Conversely take $m = -k$ and $n = ((\rho, x) + 7k)/6$ for a vector in (NL3). If the image vanishes, $x = m\rho/6$; pairing with α forces $m = 6j$, and then $n = 7j$, so the kernel is exactly $\mathbb{Z}w$. Finally

$$|x - m\rho/6|^2 = x^2 - \tfrac{m}{3}(\rho, x) + \tfrac{7}{3}m^2 = x^2 - 2mn.$$

Polarization gives the full pairing, proving the isometry.

For every $T \in O(N)$ the exact transport laws are

$$T(K_\rho) = K_{T\rho}, \qquad T(\Lambda_\rho) = \Lambda_{T\rho}, \qquad (x, m, n) \mapsto (Tx, m, n) : w_\rho \mapsto w_{T\rho}. \tag{NL18}$$

They follow by substitution in (NL3). In particular the stabilizer of ρ acts on the fixed Λ_ρ. General isometries act on this specified family of neighbors; no assertion that an arbitrary chosen automorphism complement fixes this Weyl vector is required.

32.6 The exact F_4 Coxeter action of order twelve

The order twelve in (NL14) is a lattice quotient invariant of this explicitly marked construction. There is also the ambient F_4 Coxeter action on the same original D_4 real root space. Use the precise F_4 simple roots

$$\beta_1 = e_2 - e_3, \quad \beta_2 = e_3 - e_4, \quad \beta_3 = e_4, \quad \beta_4 = (e_1 - e_2 - e_3 - e_4)/2, \qquad C = s_{\beta_1} s_{\beta_2} s_{\beta_3} s_{\beta_4},$$

where $s_\beta(x) = x - 2(x,\beta)\beta/\beta^2$ and the rightmost factor acts first. The first two roots have squared norm two, and the last two squared norm one. No rescaling of the D_4 roots has occurred.

Theorem 32.3 (Coxeter twelve and its marked triality quotient)**.** *In the retained four-coordinate metric,*

$$C = \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ -1 & 1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \end{pmatrix}, \qquad C^3 = \begin{pmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix}. \tag{NL19}$$

It satisfies $C^4 - C^2 + I = 0$, $C^6 = -I$ and has exact order twelve. Its marked discriminant action is $v_D + D \mapsto c_D + D \mapsto s_D + D \mapsto v_D + D$. The restriction of the original triality quotient is the exact sequence

$$1 \longrightarrow \langle C^3 \rangle \cong \mathbb{Z}/4 \longrightarrow \langle C \rangle \cong \mathbb{Z}/12 \longrightarrow \langle (v_D\, c_D\, s_D) \rangle \cong \mathbb{Z}/3 \longrightarrow 1, \tag{NL20}$$

with $\langle C^3 \rangle \subset W(D_4)$.

Proof. The first reflection exchanges coordinates two and three; the second exchanges coordinates three and four; the third changes the sign of coordinate four; the last is $I - 2\beta_4\beta_4^T$. Their product gives the first matrix in (NL19). Multiplying it three times gives the second, whose square is $-I$. Multiplying the first matrix also gives $C^4 - C^2 + I = 0$, and hence $C^{12} = I$. Its first column gives $Cv_D = c_D + (1,-1,0,0)$, with the displayed correction in D. Directly $Cs_D = v_D$ and $Cc_D = (0,0,-1,-1) + s_D$. The displayed integer correction lies in D and proves the stated three-cycle with the original v_D, s_D, c_D markings. The first two reflections preserve D as root reflections. The last two preserve D because a sign change preserves integer sum parity, and reflection in β_4 sends an integer even-sum vector to an integer vector of even sum. The inverse reflections have the same property, so these maps are isometries of D. Consequently their induced action on $D^\vee/D$ is defined and has the exact order-three image calculated above. The matrix C^3 is an even signed permutation of exact order four, and thus lies in the already identified group $W(D_4)$. If $C^m = I$, its discriminant image forces $3 \mid m$, and the exact order of C^3 then forces $4 \mid m/3$. Thus C has exact order twelve and the discriminant kernel on its cyclic subgroup is exactly $\langle C^3 \rangle$. This proves every map and group order in (NL20). □

In particular C^2 still has nontrivial order-three discriminant action, while C^3 is in the long-root Weyl group. The Coxeter denominator six in (NL1), the cyclic order twelve in (NL14), and the F_4 Coxeter twelve in (NL20) therefore have their stated, distinct domains and explicit relationships through the original D_4 marking.

32.7 An exact cyclic-triality isometry of the entire Niemeier space

Retain the ordered fibres $f_i 289^j$, $1 \le i \le 4$, $0 \le j \le 5$, the Euclidean augmentation coordinates of $A(F_{\eta_i})$, the ordered orthonormal coordinates of $D \subset \mathbb{R}^4$, and the 72-element glue $\mathcal{H}$ of (1385). Thus the actual real span is

$$V_N = N \otimes_{\mathbb{Z}} \mathbb{R} = \left\{ (x_{i,j}; d_k) \in \mathbb{R}^{24} \oplus \mathbb{R}^4 : \sum_{j=0}^{5} x_{i,j} = 0 \ (1 \le i \le 4) \right\},$$

with squared norm $\sum_{i,j} x_{i,j}^2 + \sum_k d_k^2$. The four displayed linear constraints, rather than an implicit change of metric, give its dimension 24. Choose an ordered orthonormal octonion basis $(u_0, \dots, u_7)$ with $u_0 = 1$. Every octonion product below is the product of the chosen real octonion algebra, without a change of multiplication convention.

For $p \in \{0,1\}$ and $j \in \{0,1,2\}$, set

$$m_{i,p} = \frac{1}{3}\sum_{j=0}^{2} x_{i,p+2j}, \qquad t_{i,p,j} = x_{i,p+2j} - m_{i,p}, \tag{1414}$$

$$\alpha_i = \frac{\sum_{j=0}^{2} x_{i,2j} - \sum_{j=0}^{2} x_{i,1+2j}}{\sqrt{6}}, \qquad (b_0, \dots, b_7) = (\alpha_1, \alpha_2, \alpha_3, \alpha_4, d_1, d_2, d_3, d_4). \tag{1415}$$

In particular $m_{i,0} = \alpha_i/\sqrt{6}$, $m_{i,1} = -\alpha_i/\sqrt{6}$, and $\sum_j t_{i,p,j} = 0$. The ordered map is

$$\begin{aligned} &\mathcal{L} : V_N \longrightarrow \mathbb{O}^3, \\ &(x; d) \longmapsto (q_1, q_2, q_3), \\ &q_{j+1} = \sum_{i=1}^{4}\sum_{p=0}^{1} \left(t_{i,p,j} + \frac{b_{2(i-1)+p}}{\sqrt{3}} \right) u_{2(i-1)+p}. \end{aligned} \tag{1416}$$

Theorem 32.4 (Actual rank-24 map, inverse, pairing, and residue action)**.** *The map $\mathcal{L}$ is a real linear isometry onto $\mathbb{O}^3$ with its product norm. Write $q_{j+1} = \sum_{a=0}^{7} q_{j,a} u_a$, and define*

$$b_a = \frac{1}{\sqrt{3}}\sum_{j=0}^{2} q_{j,a}, \qquad t_{i,p,j} = q_{j,2(i-1)+p} - \frac{1}{3}\sum_{h=0}^{2} q_{h,2(i-1)+p}. \tag{1417}$$

Its inverse is exactly

$$x_{i,p+2j} = t_{i,p,j} + \frac{(-1)^p b_{i-1}}{\sqrt{6}}, \qquad d_k = b_{k+3}. \tag{1418}$$

If $r = \rho(289^2) = \rho(361)$, with the residue action of Theorem 31.16, and $c(q_1, q_2, q_3) = (q_3, q_1, q_2)$, then

$$\mathcal{L} r = c\mathcal{L}, \qquad r^3 = c^3 = I, \qquad rN = N. \tag{1419}$$

Thus $\mathcal{L}|_N$ embeds the original even unimodular lattice, with every pairing and glue class retained, into the actual three octonion sectors and intertwines an actual order-three lattice isometry with cyclic triality.

Proof. The two means sum to zero because each six-coordinate fibre has total sum zero. The equations for $m_{i,p}$ follow from $\alpha_i = 6m_{i,0}/\sqrt{6}$. Orthogonality to constant triplets gives

$$\sum_{j=0}^{5} x_{i,j}^2 = \sum_{p=0}^{1}\sum_{j=0}^{2} t_{i,p,j}^2 + 3m_{i,0}^2 + 3m_{i,1}^2 = \sum_{p,j} t_{i,p,j}^2 + \alpha_i^2.$$

For each $a = 2(i-1) + p$, the same zero-sum identity gives

$$\sum_{j=0}^{2} \left(t_{i,p,j} + \frac{b_a}{\sqrt{3}} \right)^2 = \sum_{j=0}^{2} t_{i,p,j}^2 + b_a^2.$$

Summing over all eight values of a proves $\sum_{h=1}^{3} |q_h|^2 = \sum_{i,j} x_{i,j}^2 + \sum_k d_k^2$. Polarization proves equality of all pairings. Equations (1417) recover the means and residuals of each output triplet. Equation (1418) then recovers the original means, the original fibre coordinates, and d. Conversely those formulas applied to arbitrary q yield $\sum_j x_{i,j} = 0$ and return the prescribed q. This proves bijectivity without a dimension-only argument.

The residue action sends the coordinate vector $e_{f_i 289^h}$ to $e_{f_i 289^{h+2}}$, so the coefficient in output position $p+2j$ is the old coefficient in position $p+2(j-1)$, with j read modulo three. Both means, all α_i, and all d_k are fixed. Hence the three output octonions become (q_3, q_1, q_2), proving the intertwining formula. Each coordinate permutation acts trivially on $A^\vee/A$: for $\lambda = e_0 - \frac{1}{6}\sum_h e_h$, its difference from λ is an augmentation root. The D-coordinates are fixed. Thus r acts trivially on the full discriminant group, preserves $\mathcal{H}$, and preserves its exact inverse image N. Its order is three, as seen on $e_0 - e_1$ in any fibre. □

The exact octonion-coordinate glue test. This embedding has the following fully explicit image, denoted $\Lambda_{\mathbb{O}} = \mathcal{L}(N)$. Given q, first recover $(x; d)$ by (1417)–(1418). For each fibre require $x_{i,h} - x_{i,k} \in \mathbb{Z}$ for every h, k, and put

$$k_i = -6x_{i,0} \pmod 6 \in \mathbb{Z}/6\mathbb{Z}.$$

This is defined because $6x_{i,0} = \sum_h (x_{i,0} - x_{i,h}) \in \mathbb{Z}$. Require also that the four d_k are all integral or all half-integral. Let $y(d) \in \mathbb{F}_2^2$ be the unique class with

$$d + D = 0 + D,\ v + D,\ s + D,\ c + D \quad \text{marked respectively by } 00, 10, 01, 11.$$

Then the final and exact test is

$$(k_1, k_2, k_3, k_4; y(d)) = \big(3u + 2(a+b, a+2b, a, b); \tau(u)\big) \tag{1420}$$

for $u \in \mathbb{F}_2^4$ of even weight, $(a, b) \in \mathbb{F}_3^2$, with the first four coordinates read modulo six, and $\tau(u) = (u_2 + u_4, u_3 + u_4)$. Indeed the integral-difference conditions are precisely membership in $A^\vee$, and the common fractional part equals $-k_i/6 \pmod{\mathbb{Z}}$, which is the fractional part of $k_i\lambda$. The half-integral test is precisely membership in $D^\vee$, and (1420) is the original $\mathcal{H}$. This proves both directions of the image description. No new lattice or rescaled glue is substituted.

The determinant and the factor two in the Albert metric. The matrix convention, also specifying this construction independently of its host document, is

$$Q_{\mathbb{O}}(\lambda; q) = \begin{pmatrix} \lambda & q_3 & \overline{q}_2 \\ \overline{q}_3 & \lambda & q_1 \\ q_2 & \overline{q}_1 & \lambda \end{pmatrix} \in \mathfrak{h}_3(\mathbb{O}), \qquad X \circ Y = (XY + YX)/2.$$

Put

$$E(q) = Q_{\mathbb{O}}(0; q_1, q_2, q_3), \qquad \mathfrak{t}(q_1, q_2, q_3) = \operatorname{Re}((q_1 q_2) q_3), \qquad \mathfrak{t}_N(X) = \mathfrak{t}(\mathcal{L}X).$$

The original ordered Albert determinant becomes the exact polynomial

$$\det_{\mathbb{O}}\big(\lambda I + E(\mathcal{L}X)\big) = \lambda^3 - \lambda(X, X) + 2\mathfrak{t}_N(X). \tag{1421}$$

Moreover

$$\operatorname{tr}\big(E(\mathcal{L}X) \circ E(\mathcal{L}Y)\big) = 2(X, Y). \tag{1422}$$

For completeness, direct binary multiplication gives $\operatorname{tr} E = 0$, $\operatorname{tr}(E \circ E) = 2\sum_i |q_i|^2$, and $\operatorname{tr}(E \circ (E \circ E)) = 6\mathfrak{t}(q)$. For the last equality, the three cyclic off-diagonal coordinates of $E \circ E$ are $\overline{q_2 q_3}, \overline{q_3 q_1}, \overline{q_1 q_2}$; their trace pairings with the corresponding q_i each give $2\mathfrak{t}(q)$. The Jordan trace formula $\det X = \operatorname{tr}(X \circ (X \circ X))/3 - \operatorname{tr} X \operatorname{tr}(X \circ X)/2 + (\operatorname{tr} X)^3/6$ therefore gives the first identity after expanding $X = \lambda I + E$ and substituting the proved norm equality. For the second, each off-diagonal octonion occurs in both transposed matrix entries, giving exactly $2\sum_i \operatorname{Re}(q_i \overline{p_i})$. Real associativity and cyclicity of octonion real parts give $\mathfrak{t}(q_3, q_1, q_2) = \mathfrak{t}(q_1, q_2, q_3)$:

$$\operatorname{Re}((q_3 q_1) q_2) = \operatorname{Re}(q_3 (q_1 q_2)) = \operatorname{Re}((q_1 q_2) q_3).$$

Consequently $\mathfrak{t}_N(rX) = \mathfrak{t}_N(X)$, and the whole determinant in (1421) is r-invariant with λ fixed. Equivalently, for the real permutation matrix $Pe_i = e_{i+1}$, indices modulo three,

$$P(\lambda I + E(q))P^T = \lambda I + E(q_3, q_1, q_2).$$

This is also a Jordan automorphism: multiplication by a real permutation matrix merely reindexes the two-factor matrix products, so it preserves $(AB + BA)/2$. It therefore carries the positive cone and its determinant boundary to themselves. The cyclic bridge is simultaneously linear, isometric on $\mathbb{O}^3$, integral on the stated lattice image, and compatible with the actual Albert cubic and positive cone.

Dependence and the four-dimensional lattice summand. The map uses precisely the four specified origins f_i, the generator 289, the even/odd triplet order, the displayed order of b, the ordered D-coordinates, and the chosen orthonormal octonion basis. Changing one of these changes the displayed map; the construction makes no claim of independence from these data. Its restriction to the actual D-summand is

$$d \longmapsto \frac{1}{\sqrt{3}}(z(d), z(d), z(d)), \qquad z(d) = d_1u_4 + d_2u_5 + d_3u_6 + d_4u_7. \tag{1423}$$

This is an isometry on D, and on $D^\vee$ in its real span. In particular its 24 shortest nonzero dual vectors become 24 specified points in this four-dimensional diagonal subspace. The separate Cartan-weight map $d \mapsto \sum_k d_k\varepsilon_k$ is related to it by the exact isometry $\sum_k d_k\varepsilon_k \mapsto (z(d), z(d), z(d))/\sqrt{3}$.

32.8 The Leech neighbour in the same three octonion coordinates

The preceding map also transports the entire neighbour construction, not only the real span of N. Put $N_{\mathbb{O}} = \mathcal{L}(N)$ and retain the actual ρ, α, K, Λ of Theorem 32.1. In the already fixed basis let

$$U = \sum_{a=0}^{7} u_a, \qquad V = \frac{u_0 + u_1 + u_2 + u_3}{\sqrt{2}} + \sqrt{3}u_4 + \frac{2u_5 + u_6}{\sqrt{3}}.$$

Substitution in the original isometry gives the explicit vectors

$$\begin{aligned} \varrho &= \mathcal{L}\rho = (V + 2U, V, V - 2U), \\ \delta &= \mathcal{L}\alpha = \left(\frac{(2+\sqrt{2})u_0 - 2u_1}{3}, \frac{(\sqrt{2}-1)u_0 + u_1}{3}, \frac{(\sqrt{2}-1)u_0 + u_1}{3} \right). \end{aligned} \tag{1424}$$

Indeed each even triplet of ρ_A has mean $1/2$, each odd triplet has mean $-1/2$, and both residual triplets are $(2, 0, -2)$. The four mean-difference coordinates are $\sqrt{6}/2$, while the four D coordinates are $(3, 2, 1, 0)$. For α, the two residual triplets in the first fibre are $(2, -1, -1)/3$ and $(-2, 1, 1)/3$; its only nonzero mean-difference coordinate is $2/\sqrt{6}$. These facts prove every coefficient in (1424). In particular $|\varrho|^2 = 84$, $|\delta|^2 = 2$, and $\langle \varrho, \delta \rangle = 1$ in the product octonion metric.

Define

$$\begin{aligned} K_{\mathbb{O}} &= \{q \in N_{\mathbb{O}} : \langle \varrho, q \rangle \equiv 0 \pmod 6\}, \\ \Lambda_{\mathrm{Leech},\mathbb{O}} &= K_{\mathbb{O}} + \mathbb{Z}(\varrho/6 - \delta). \end{aligned} \tag{1425}$$

The isometry and its inverse prove, in both directions, $\mathcal{L}(K) = K_{\mathbb{O}}$ and $\mathcal{L}(\Lambda) = \Lambda_{\mathrm{Leech},\mathbb{O}}$. Thus (1425) is a full coordinate description of the Leech lattice inside the very same three sectors. The glue test above, followed by its displayed height congruence, is an exact membership test for $K_{\mathbb{O}}$; adjoining the six cosets $k(\varrho/6 - \delta)$, $0 \le k < 6$, gives all of the Leech lattice. The common-sublattice indices, discriminant form, and order-twelve quotient are transported without alteration.

The cyclic action also acts on the marked family of neighbours. Retain $r = \rho(361)$ and its corresponding map $c(q_1, q_2, q_3) = (q_3, q_1, q_2)$. For any transported Weyl vector $\rho' = T\rho$, $T \in O(N)$, the neighbour is defined using $T\alpha$, so that $\langle \rho', T\alpha \rangle = 1$. Then

$$c\,\mathcal{L}(K_{\rho'}) = \mathcal{L}(K_{r\rho'}), \qquad c\,\mathcal{L}(\Lambda_{\rho'}) = \mathcal{L}(\Lambda_{r\rho'}).$$

This follows directly from $\mathcal{L}r = c\mathcal{L}$ and (NL18). The inverse maps use r^{-1} and c^{-1}. It specifies exactly which marked neighbour is the target; it does not silently assume that the chosen ρ is fixed by r. Finally the common cubic evaluates on every one of these lattice points as

$$\det Q_{\mathbb{O}}(\lambda; \mathcal{L}X) = \lambda^3 - \lambda(X, X) + 2\,\mathrm{Re}(((\mathcal{L}X)_1(\mathcal{L}X)_2)(\mathcal{L}X)_3).$$

This identity holds for all $X \in N_{\mathbb{R}}$, hence also on the explicitly embedded N, K and Λ.

32.9 Exactly which root isometries preserve this glue

Retain the coordinate convention $(P_p x)_i = x_{p(i)}$. An arbitrary root-lattice isometry has the original form

$$T(X_1, X_2, X_3, X_4; d) = \big((\varepsilon_i Q_{\sigma_i} X_{p(i)})_{i=1}^4;\ T_D d\big),$$

where $\varepsilon_i \in \{\pm 1\}$, $\sigma_i \in S_6$, $p \in S_4$, and $T_D \in O(D)$. If $B = d_D(T_D)$, the necessary and sufficient condition for $T(N) = N$ is

$$B\tau(u) = \tau(P_p u)\ (u \in E), \qquad (\varepsilon_i \ell_{p(i)})_{i=1}^4 = (\ell_i A)_{i=1}^4 \quad \text{for some } A \in GL_2(\mathbb{F}_3), \tag{1426}$$

where $\ell_1 = (1,1), \ell_2 = (1,2), \ell_3 = (1,0), \ell_4 = (0,1)$. The A, and hence $B = B_p$, are unique. Indeed the two-primary part of the glue is the graph of τ, and signs are trivial modulo two. Its three-primary part is the image of the injective map $j(a,b) = (a+b, a+2b, a, b)$, so preserving it is equivalent to $M_\varepsilon P_p j = jA$. These are exactly the two conditions in (1426); the primary decomposition proves their sufficiency as well as necessity. Since the four ℓ_i represent the four distinct projective row lines over $\mathbb{F}_3$, each A gives exactly one signed permutation by this equation. This recovers the precise 48-element glue quotient, not merely its action on three labels.

The canonical full-automorphism theorem proves, with these same coordinates and section, that

$$O(N) = \big(S_6^4 \times W(D_4)\big) \rtimes L(G), \qquad G \cong GL_2(\mathbb{F}_3).$$

Conjugation by the actual map $\mathcal{L}$ now gives the entire orthogonal stabilizer of the embedded lattice, with explicit matrices obtained from (1416) and its inverse:

$$O(\Lambda_{\mathbb{O}}) = \mathcal{L}O(N)\mathcal{L}^{-1}, \qquad \Gamma_G(g) = \mathcal{L}L_g\mathcal{L}^{-1}, \qquad \mathcal{L}L_g = \Gamma_G(g)\mathcal{L}. \tag{1427}$$

Both inclusions in the first equality follow by conjugating an isometry preserving the specified lattice. This is a faithful action on the whole 24-dimensional space. Each $\Gamma_G(g)$ commutes with c, because the canonical equation $L_g\rho(289^2) = \rho(289^2)L_g$ conjugates to that equation.

For completeness, the previously displayed D_4-triality section and geometric frame permutations have a precise comparison even though that particular proposed linear conjugacy fails. On V_N, each of $\widehat{R}, \widehat{H}$ has trace $5 - 5 + 2 = 2$: the transposed A_5 pair contributes zero, the other two components contribute $5, -5$, and the D_4 reflection contributes 2. Their three-cycle has trace $5 + 1 = 6$. A geometric Albert-frame transposition acts on $\mathbb{O}^3$ by swapping two sectors with conjugation and by conjugating the fixed sector; its trace is $1 - 7 = -6$. Its three-cycle has trace zero. Therefore an invertible map cannot conjugate these two particular S_3 representations, since trace is invariant under conjugation. The complete real character decompositions are, respectively,

$$V_N|_{\langle \widehat{R}, \widehat{H}\rangle} \cong \mathbf{7}\mathbf{1} \oplus 5\,\mathrm{sgn} \oplus 6\,\mathrm{Std}, \qquad \mathbb{O}^3|_{S_3^{\mathrm{frame}}} \cong \mathbf{1} \oplus 7\,\mathrm{sgn} \oplus 8\,\mathrm{Std}\,.$$

These multiplicities follow by taking inner products with the characters $(1,1,1), (1,-1,1), (2,0,-1)$, on classes of sizes $1, 3, 2$. The exact whole-space relation for the original section is (1427); the exact determinant-compatible cyclic relation is (1419). In the latter relation the residue element has trivial discriminant action while its octonion action cycles the sectors. Those are two explicitly calculated actions

of the same operator on two explicitly different constructions; no identification between their quotient homomorphisms is assumed.

Finally, the connected group Spin(8) acts on the ambient triality module, and its subgroup preserving $\Lambda_{\mathbb{O}}$ is exactly its intersection with (1427). It is finite: the columns of a lattice isometry in a fixed lattice basis have prescribed bounded lengths, so each column has finitely many possible lattice values. A connected group preserving a discrete full lattice pointwise as a set acts trivially, since each orbit of a lattice vector is connected and discrete and the lattice spans. The nontrivial connected triality action therefore does not preserve the whole lattice. The proved cyclic subgroup $\langle c\rangle$ does preserve both the lattice and its Albert cubic. Equations (1426) and (1427) specify the full glue stabilizer; they do not assert that every element of that finite orthogonal group also preserves the cubic.

33 The arithmetic of the retained Albert cubic on both lattices

33.1 Original data and the exact polynomial

Retain the ordered coordinates $X = (x_{i,j}; d_k)$, $1 \le i \le 4$, $0 \le j \le 5$, $1 \le k \le 4$, with $\sum_{j=0}^{5} x_{i,j} = 0$, and their original Euclidean metric. Let

$$A_5 = \{x \in \mathbb{Z}^6 : \sum_j x_j = 0\}, \qquad D_4 = \{d \in \mathbb{Z}^4 : \sum_k d_k \equiv 0 \pmod 2\}, \qquad R = A_5^{\perp 4} \perp D_4.$$

Write $\lambda = (5,-1,-1,-1,-1,-1)/6$, $v_D = (1,0,0,0)$, and $s_D = (1,1,1,1)/2$. The original discriminant glue is generated by

$$\begin{aligned} &(3,3,3,3;00), \quad (3,3,0,0;10), \quad (3,0,3,0;01),\\ &(2,2,2,0;00), \quad (2,4,0,2;00) \end{aligned}$$

in $(\mathbb{Z}/6)^4 \oplus \mathbb{F}_2^2$, with the same marking $10 = v_D + D_4$, $01 = s_D + D_4$. Its order is 72: the first three generators have independent two-primary reductions, and the last two independent three-primary reductions. The lattice N is its inverse image in $R^\vee$.

Retain

$$\rho_A = (5,3,1,-1,-3,-5)/2, \qquad \rho_D = (3,2,1,0), \qquad \rho = \rho_A^{\oplus 4} \perp \rho_D, \qquad \alpha = (e_0 - e_1, 0, 0, 0; 0),$$

and the exact neighbour

$$K = \{X \in N : (\rho, X) \equiv 0 \pmod 6\}, \qquad v = \rho/6 - \alpha, \qquad \Lambda = K + \mathbb{Z}v.$$

Here $\rho \in N$, $\rho^2 = 84$, and $(\rho, \alpha) = 1$, as follows immediately from these coordinates and the first glue generator. The name Λ always denotes this neighbour.

Fix an ordered orthonormal octonion basis $u_0 = 1, u_1, \dots, u_7$. For the first part of the calculation no multiplication table is imposed beyond the actual multiplication of these basis elements. Define, in the original coordinates,

$$s_i = \sum_{j=0}^{5} (-1)^j x_{i,j}, \tag{1428}$$

$$A_{2(i-1)+p} = x_{i,p} - \frac{x_{i,p} + x_{i,p+2} + x_{i,p+4}}{3}, \qquad p = 0, 1, \tag{1429}$$

$$B_{2(i-1)+p} = x_{i,p+2} - \frac{x_{i,p} + x_{i,p+2} + x_{i,p+4}}{3}, \qquad p = 0, 1. \tag{1430}$$

Write $A = A_0 + \mathbf{a}$, $B = B_0 + \mathbf{b}$, where $\mathbf{a} = \sum_{h=1}^{7} A_h u_h$ and $\mathbf{b} = \sum_{h=1}^{7} B_h u_h$. The common mean octonion is

$$M = \frac{\sqrt{2}}{6} \sum_{a=0}^{3} s_{a+1} u_a + \frac{\sqrt{3}}{3} \sum_{k=1}^{4} d_k u_{k+3}. \tag{1431}$$

The retained isometry therefore has precisely the ordered form

$$\mathcal{L}X = (A + M, B + M, -A - B + M).$$

This is an identity with its original three triplets: the residuals sum to zero, the first four common mean coefficients are $s_i/\sqrt{18} = \sqrt{2}s_i/6$, and the last four are $d_k/\sqrt{3} = \sqrt{3}d_k/3$. For completeness the original squared norm and the output norm are both

$$|A|^2 + |B|^2 + |A + B|^2 + \frac{1}{6} \sum_i s_i^2 + \sum_k d_k^2.$$

In the original fibres this follows by separating the means $s_i/6, -s_i/6$ from their zero-sum triplets; in the output it follows from $A + B + (-A - B) = 0$ and $3|M|^2 = \sum_i s_i^2/6 + \sum_k d_k^2$. The inverse recovers $M = (q_1 + q_2 + q_3)/3$, then $s_i = 3\sqrt{2}M_{i-1}$, $d_k = \sqrt{3}M_{k+3}$, and $A = q_1 - M, B = q_2 - M$. Adding $(-1)^p s_i/6$ to the p-th residual triplet in the i-th fibre recovers each original $x_{i,p+2j}$. Thus the map is onto and preserves the original pairing by polarization.

For imaginary octonions put

$$\mathbf{a} \times \mathbf{b} = \frac{\mathbf{ab} - \mathbf{ba}}{2}, \qquad \phi(\mathbf{a}, \mathbf{b}, \mathbf{c}) = \langle \mathbf{a} \times \mathbf{b}, \mathbf{c} \rangle.$$

Let $\phi_{ijk} = \phi(u_i, u_j, u_k)$. Thus ϕ is the actual alternating octonion three-form in the chosen frame, including all signs. Define

$$D = A_0^2 + A_0 B_0 + B_0^2 - \langle \mathbf{a}, \mathbf{a} \rangle - \langle \mathbf{a}, \mathbf{b} \rangle - \langle \mathbf{b}, \mathbf{b} \rangle, \tag{1432}$$

$$Z = (2A_0 + B_0)\mathbf{a} + (A_0 + 2B_0)\mathbf{b} - 3\mathbf{a} \times \mathbf{b} = \sum_{h=1}^{7} Z_h u_h. \tag{1433}$$

Equivalently, the component in the final formula is the exact quadratic polynomial

$$Z_h = (2A_0 + B_0)A_h + (A_0 + 2B_0)B_h - 3 \sum_{i,j=1}^{7} \phi_{ijh} A_i B_j.$$

Theorem 33.1 (Exact pullback with every radical and coefficient)**.** *The original cubic* $F(X) = \mathrm{Re}(((\mathcal{L}X)_1(\mathcal{L}X)_2)(\mathcal{L}X)_3)$ *has the complete polynomial expression*

$$F(X) = P_0(X) + \sqrt{2}P_2(X) + \sqrt{3}P_3(X), \tag{1434}$$

where every variable is the displayed linear function of the original $x_{i,j}, d_k$*, and*

$$P_0 = -A_0 B_0 (A_0 + B_0) + B_0 \langle \mathbf{a}, \mathbf{a} \rangle + A_0 \langle \mathbf{b}, \mathbf{b} \rangle + 2(A_0 + B_0)\langle \mathbf{a}, \mathbf{b} \rangle, \tag{1435}$$

$$P_2 = \frac{-s_1 D + s_2 Z_1 + s_3 Z_2 + s_4 Z_3}{6} + \frac{s_1(4s_1^2 - 3\sum_{i=1}^{4} s_i^2)}{108} - \frac{s_1 \sum_{k=1}^{4} d_k^2}{6}, \tag{1436}$$

$$P_3 = \frac{d_1 Z_4 + d_2 Z_5 + d_3 Z_6 + d_4 Z_7}{3}. \tag{1437}$$

These identities hold throughout the original real vector space, in particular on N*,* K*, and* Λ*. They retain the ordered Albert determinant as*

$$\det Q_{\mathbb{O}}(\lambda; \mathcal{L}X) = \lambda^3 - \lambda(X, X) + 2F(X).$$

Proof. For $a = a_0 + \mathbf{a}$, $b = b_0 + \mathbf{b}$, $c = c_0 + \mathbf{c}$, binary octonion multiplication gives

$$\operatorname{Re}((ab)c) = a_0b_0c_0 - a_0\langle \mathbf{b}, \mathbf{c}\rangle - b_0\langle \mathbf{a}, \mathbf{c}\rangle - c_0\langle \mathbf{a}, \mathbf{b}\rangle - \phi(\mathbf{a}, \mathbf{b}, \mathbf{c}).$$

Call the first four, symmetric, terms $S(a, b, c)$. Substitute $(a, b, c) = (A + M, B + M, -A - B + M)$. Terms with exactly two copies of M in S sum to $S(M, M, A + B - A - B) = 0$. Terms with exactly one copy sum to $-S(M, A, A) - S(M, A, B) - S(M, B, B)$. The term with no copy is $S(A, B, -A - B) = P_0$, by multiplying its four displayed terms. The alternating term with no copy vanishes; terms with two or three copies vanish. Its three terms with one copy all equal $\phi(\mathbf{a}, \mathbf{b}, \operatorname{Im} M)$, since

$$\phi(\mathbf{a}, \operatorname{Im} M, -\mathbf{a} - \mathbf{b}) = \phi(\mathbf{a}, \mathbf{b}, \operatorname{Im} M), \quad \phi(\operatorname{Im} M, \mathbf{b}, -\mathbf{a} - \mathbf{b}) = \phi(\mathbf{a}, \mathbf{b}, \operatorname{Im} M).$$

Thus all one-copy terms together are $-M_0D + \langle \operatorname{Im} M, Z\rangle$. Finally the term with three copies is

$$\operatorname{Re}(M^3) = 4M_0^3 - 3M_0|M|^2, \qquad |M|^2 = \frac{\sum_i s_i^2}{18} + \frac{\sum_k d_k^2}{3}, \qquad M_0 = \frac{\sqrt{2}s_1}{6}.$$

Substitution proves (1435)–(1437) with their displayed factors. In particular it proves the absence of a $\sqrt{6}$ term without discarding any mixed term. The determinant formula uses the original isometry and its unchanged squared norm. □

33.2 Finite generators of both original lattices

Order the 24 original simple roots as

$$g_{5(i-1)+j+1} = a_{i,j} = e_{i,j} - e_{i,j+1} \quad (1 \le i \le 4,\ 0 \le j \le 4),$$

followed by

$$g_{21} = e_1 - e_2, \quad g_{22} = e_2 - e_3, \quad g_{23} = e_3 - e_4, \quad g_{24} = e_3 + e_4$$

in the original D_4 coordinates. All unspecified components are zero. Adjoin the five actual glue lifts

$$\begin{aligned} g_{25} &= (3\lambda, 3\lambda, 3\lambda, 3\lambda; 0), & g_{26} &= (3\lambda, 3\lambda, 0, 0; v_D),\\ g_{27} &= (3\lambda, 0, 3\lambda, 0; s_D), & g_{28} &= (2\lambda, 2\lambda, 2\lambda, 0; 0),\\ g_{29} &= (2\lambda, 4\lambda, 0, 2\lambda; 0). \end{aligned}$$

Then

$$N = \sum_{i=1}^{29} \mathbb{Z} g_i, \qquad H_i = (\rho, g_i) = \begin{cases} 1, & 1 \le i \le 24,\\ 30, 18, 18, 15, 20, & i = 25, 26, 27, 28, 29 \text{ respectively.} \end{cases} \tag{1438}$$

Indeed the first 24 vectors span R, the last five generate the original glue, and subtraction of the corresponding glue lift leaves every element of N in R. The height calculation uses $(\rho_A, \lambda) = 5/2$, $(\rho_D, v_D) = (\rho_D, s_D) = 3$, and the simple-root heights one.

Define the following 30, fully specified neighbour generators:

$$h_1 = 6\alpha, \qquad h_i = g_i - H_i\alpha \quad (2 \le i \le 29), \qquad h_{30} = v = \rho/6 - \alpha. \tag{1439}$$

Then $K = \sum_{i=1}^{29} \mathbb{Z} h_i$ and $\Lambda = \sum_{i=1}^{30} \mathbb{Z} h_i$. For if $X = \sum_i n_i g_i \in K$, then

$$X = \sum_{i=2}^{29} n_i(g_i - H_i\alpha) + \left(\sum_{i=1}^{29} n_i H_i\right)\alpha,$$

and the last coefficient is divisible by six. Conversely every displayed generator before h_{30} has height divisible by six. Adjoining h_{30} is exactly the original neighbour definition.

33.3 A complete finite calculation of the cubic value modules

For a lattice L and a homogeneous real cubic F, define its additive value module by

$$\mathcal{V}_F(L) = \sum_{X \in L} \mathbb{Z} F(X) \subset \mathbb{R}.$$

This definition retains every value and takes its generated additive group. It does not claim that every element of the group is one individual value of F.

Theorem 33.2 (Finite cubic-value generators, including negative integers)**.** *If $L = \sum_{i=1}^{m} \mathbb{Z} b_i$, allowing dependent generators, then $\mathcal{V}_F(L)$ is generated by the following finite list:*

$$\begin{aligned}
A_i &= F(b_i), && 1 \le i \le m, && (1440)\\
D_{ij}^{+} &= F(b_i + b_j) - F(b_i) - F(b_j), && i < j, && (1441)\\
D_{ij}^{-} &= F(b_i - b_j) - F(b_i) + F(b_j), && i < j, && (1442)\\
D_{ijk} &= F(b_i + b_j + b_k) - F(b_i + b_j) - F(b_i + b_k) \\
&\quad - F(b_j + b_k) + F(b_i) + F(b_j) + F(b_k), && i < j < k. && (1443)
\end{aligned}$$

Consequently (1438) *gives exactly 4495 specified generators of $\mathcal{V}_F(N)$, and* (1439) *gives exactly 4960 specified generators of $\mathcal{V}_F(\Lambda)$, for every fixed orthonormal octonion frame and its actual ϕ_{ijk}.*

Proof. Let B_F be the symmetric trilinear polarization, with $B_F(X, X, X) = F(X)$; explicitly its value is $1/6$ times the seven-term expression in (1443), with its three inputs replacing the three generators. For every integer tuple n, expansion gives

$$F\left(\sum_i n_i b_i\right) = \sum_i n_i^3 A_i + \sum_{i<j} (n_i^2 n_j C_{ij} + n_i n_j^2 E_{ij}) + \sum_{i<j<k} n_i n_j n_k D_{ijk},$$

where $C_{ij} = 3B_F(b_i, b_i, b_j)$ and $E_{ij} = 3B_F(b_i, b_j, b_j)$. The two pair generators are $D_{ij}^{+} = C_{ij} + E_{ij}$ and $D_{ij}^{-} = -C_{ij} + E_{ij}$. Therefore their contribution is

$$\frac{n_i n_j (n_i + n_j)}{2} D_{ij}^{+} + \frac{n_i n_j (n_j - n_i)}{2} D_{ij}^{-}.$$

Both coefficients are integers, for all positive, zero, and negative integers n_i, n_j: if either integer is even the claim is immediate, and if both are odd their sum and difference are even. This proves that the finite list generates all values. Conversely each entry of the list is an integer linear combination of actual values on L. The proof does not use linear independence of the generators. Finally $29 + 2\binom{29}{2} + \binom{29}{3} = 4495$ and $30 + 2\binom{30}{2} + \binom{30}{3} = 4960$. □

For numerical arithmetic now specify the additional octonion frame choice completely. Use $\mathbb{O} = \mathbb{H} \oplus \mathbb{H}\ell$ in ordered coordinates (a, b), with

$$\begin{gathered}
(a, b)(c, d) = (ac - \overline{d}\, b,\ da + b\overline{c}),\\
u_0 = (1, 0), \quad u_1 = (i, 0), \quad u_2 = (j, 0), \quad u_3 = (k, 0), \qquad (1444)\\
u_4 = (0, 1), \quad u_5 = (0, i), \quad u_6 = (0, j), \quad u_7 = (0, k).
\end{gathered}$$

Here $ij = k$, $jk = i$, $ki = j$, and the reversed products are negative. The positive oriented three-form triples are exactly

$$(123),\ (145),\ (176),\ (246),\ (257),\ (347),\ (365). \qquad (1445)$$

Every ϕ_{ijk} is obtained from these by alternation, or is zero. This is a specified frame for the retained, previously frame-dependent isometry, without a change to any lattice coordinates or to the Albert matrix convention.

Theorem 33.3 (Exact value modules and their quotient of order twelve)**.** *For the frame* (1444)*, the exact additive modules are*

$$\mathcal{V}_F(N) = \frac{1}{27}\mathbb{Z} \ \oplus \ \frac{\sqrt{2}}{54}\mathbb{Z} \ \oplus \ \frac{\sqrt{3}}{18}\mathbb{Z}, \tag{1446}$$

$$\mathcal{V}_F(\Lambda) = \frac{1}{27}\mathbb{Z} \ \oplus \ \frac{\sqrt{2}}{216}\mathbb{Z} \ \oplus \ \frac{\sqrt{3}}{54}\mathbb{Z}. \tag{1447}$$

In particular the inclusion of these two additive subgroups of $\mathbb{R}$ *has the exact quotient*

$$\mathcal{V}_F(\Lambda)/\mathcal{V}_F(N) \cong \mathbb{Z}/4 \oplus \mathbb{Z}/3 \cong \mathbb{Z}/12. \tag{1448}$$

This is an inclusion of cubic value modules; it does not assert the false lattice inclusion $N \subset \Lambda$*.*

Proof. The polynomial coefficients in (1435)–(1437) are rational on both lattices for this exact multiplication table. The numbers $1, \sqrt{2}, \sqrt{3}$ are linearly independent over $\mathbb{Q}$. For example, isolating one radical in a rational relation and squaring either gives a rational multiple of another irrational radical, or a rational square equal to two, three, or $3/2$, which is impossible by parity of prime exponents. Thus a value has a unique rational coefficient triple (P_0, P_2, P_3).

Apply the complete finite list in Theorem 33.2 to the displayed original generators. Exact substitution gives the following least common multiples of coefficient denominators. The denominators in each entry are ordered as (P_0, P_2, P_3), and a zero coefficient has denominator one:

	A_i	D^+_{ij}	D^-_{ij}	D_{ijk}
N	$(27, 27, 1)$	$(9, 18, 18)$	$(9, 18, 18)$	$(9, 18, 18)$.
Λ	$(27, 216, 27)$	$(27, 108, 54)$	$(27, 108, 54)$	$(9, 36, 54)$

(1449)

This table is a finite rational calculation, with no bound on lattice points or heuristic sampling. The complete arithmetic certificate is the accompanying `verify_cubic.py`: its `residual_data` and `cubic` functions are exactly (1428)–(1437); it forms every $A_i, D^+_{ij}, D^-_{ij}, D_{ijk}$ for the 29 or 30 explicitly defined vectors, using rational arithmetic. The `cross` function executes the three cyclic positive and three negative entries for each of the seven triples (1445). Hence the full finite calculation is specified by these equations, the finite index bounds in Theorem 33.2, and integer arithmetic. Its 4495 and 4960 entries verify the upper inclusions in (1446)–(1447).

Here are short certificates for the reverse inclusions. Coordinates in the next two tables are relative to the respective three proposed generators on the right sides of (1446) and (1447). In the first table all pair symbols use the g_i:

$F(g_1)$	-4	4	0
$F(g_2)$	-5	-4	0
$F(g_7)$	0	18	0
$F(g_{28})$	-80	52	0 .
$D^+_{21,28}$	0	-36	16
$D^+_{2,27}$	-27	27	3
$D^+_{2,25}$	-81	45	0

(1450)

The determinants of the three columns obtained by transposing rows $(1, 4, 5)$, $(2, 3, 6)$, and $(2, 7, 6)$ are, respectively, $1792, -270, -1647$. Their greatest common divisor is one. In the second table all pair symbols use the h_i:

$F(h_2)$	14	-272	0
$F(h_4)$	5	-128	0
$F(h_{28})$	13780	-74672	0 .
$F(h_{30})$	14	-63	-2
$D^+_{2,27}$	7425	-60624	-9

(1451)

Rows $(1,3,4)$, $(1,4,5)$, and $(2,4,5)$ give determinants $-5405504, 2315394, 1281267$, whose greatest common divisor is also one.

For clarity, the elementary integer argument using these determinants does not require a normal-form theorem. Let $W \subset \mathbb{Z}^3$ be the group generated by one displayed table. For any three of its columns forming a matrix C, the adjugate identity $C\operatorname{adj}(C) = \det(C)I$ implies $\det(C)\mathbb{Z}^3 \subset W$. Bezout's identity for the three displayed determinants, whose gcd is one, therefore gives $\mathbb{Z}^3 \subset W$. The opposite inclusion is immediate from the integer entries. Every table entry is an integer combination of actual values by its definition, proving the two reverse inclusions. Finally the coordinates of the inclusion of the first module in the second are $(a,b,c) \mapsto (a,4b,3c)$. Its cokernel is the displayed product of cyclic groups, with order twelve. This also proves the asserted exact quotient map. □

33.4 Frame dependence and exact symmetry transport

The finite-list theorem is valid for the actual coefficients of every orthonormal frame. The numerical modules (1446)–(1447) refer to the frame (1444). This dependence has an exact comparison map. If $u'_a = Tu_a$ for a real isometry T fixing 1, then

$$\mathcal{L}_{u'} = (T,T,T)\mathcal{L}_u.$$

The symmetric part S of the trilinear tensor is unchanged, because T preserves the scalar coordinate and inner product. Writing $\mathbf{m} = \operatorname{Im} M$, its exact change is

$$F_{u'}(X) - F_u(X) = -3\big(\phi(T\mathbf{a}, T\mathbf{b}, T\mathbf{m}) - \phi(\mathbf{a},\mathbf{b},\mathbf{m})\big). \tag{1452}$$

This follows from the complete expansion in the polynomial proof. If T is an octonion automorphism it preserves ϕ, so the cubic and its value modules are unchanged.

The dependence on a general orthonormal frame is substantive. Take the actual lattice vector with the only nonzero fibre coordinates

$$x_{1,1} = 1,\ x_{1,5} = -1, \quad x_{2,2} = 1,\ x_{2,4} = -1, \qquad d = (1,1,0,0).$$

It lies in $R \subset N$, has $A = u_1$, $B = u_2$, and $M = (u_4 + u_5)/\sqrt{3}$. In the Cayley frame its value is zero. Replace only

$$u'_3 = (2u_3 - \sqrt{5}u_4)/3, \qquad u'_4 = (\sqrt{5}u_3 + 2u_4)/3.$$

This is an exact orthonormal frame change; the same original lattice vector now has value $-\sqrt{15}/3$, because $\phi(u_1,u_2,u'_4) = \sqrt{5}/3$ and $\phi(u_1,u_2,u_5) = 0$. This value is outside $\mathbb{Q} + \sqrt{2}\mathbb{Q} + \sqrt{3}\mathbb{Q}$. Indeed the independent sign changes of $\sqrt{2}, \sqrt{3}$ are automorphisms of $\mathbb{Q}(\sqrt{2},\sqrt{3})$. If an element z in that field had $z^2 = 15$, each sign change would send it to z or $-z$. In the basis $1, \sqrt{2}, \sqrt{3}, \sqrt{6}$ this makes z a rational multiple of a single basis vector. Its square would force one of $15, 15/2, 5, 5/2$ to be a rational square, impossible by parity of its prime exponents. Thus the numerical specialization is not silently extended to an unspecified orthonormal frame.

Return to the explicit Cayley frame for the following numerical witnesses. Let $\sigma = \rho(289)$ act simultaneously by $e_{i,j} \mapsto e_{i,j+1 \bmod 6}$, and trivially on D_4. The exact isometry gives $\mathcal{L}\sigma^2 = c\mathcal{L}$ for $c(q_1,q_2,q_3) = (q_3,q_1,q_2)$. Cyclic real associativity therefore proves $F(\sigma^{2k}X) = F(X)$ for every real X. These maps preserve N by their original discriminant action. The stabilizer of this cubic within the six-element residue group is exactly its three-element subgroup: direct substitution at $\alpha = g_1$ gives

$$F(\alpha) = \frac{-4 + 2\sqrt{2}}{27}, \qquad F(\sigma\alpha) = \frac{-5 - 2\sqrt{2}}{27}, \tag{1453}$$

which differ. All odd powers have the same effect on the cubic as σ, by its proved invariance under even powers.

The original reflection exchanging $e_{1,0}, e_{1,1}$ sends α to $-\alpha$, hence changes its nonzero cubic value to its negative. The original full glue-preserving lifts

$$\widehat{R}(X_1, X_2, X_3, X_4; d) = (X_2, X_1, X_3, -X_4; R_D d),$$
$$\widehat{H}(X_1, X_2, X_3, X_4; d) = (X_1, X_3, X_2, -X_4; H_D d).$$

also have explicit witnesses:

$$F(\widehat{R}\alpha) = 0, \qquad F(g_7) = \frac{\sqrt{2}}{3}, \qquad F(\widehat{H}g_7) = 0.$$

Their original matrices remain

$$R_D = \mathrm{diag}(-1, 1, 1, 1), \qquad H_D = \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}.$$

These maps preserve the original lattice, as can be checked on its displayed generators. Both R_D and H_D preserve D_4: the former changes one sign, while the latter has integral coordinates on every even-sum integer vector and has output sum twice its first input coordinate. They are orthogonal involutions. Their marked discriminant actions fix v_D and exchange s_D, c_D, and exchange v_D, s_D while fixing c_D, respectively. On the binary glue generators, $\widehat{R}$ fixes the first two and sends the third to their second-plus-third sum; $\widehat{H}$ fixes the first and exchanges the second and third. On the ternary code generators $z_1 = (1, 1, 1, 0)$, $z_2 = (1, 2, 0, 1)$, the two signed component permutations act as $(z_1, z_2) \mapsto (z_1, 2z_2)$ and $(z_1, z_2) \mapsto (z_1, 2z_1 + 2z_2)$, respectively. Thus both preserve the full original glue and its inverse image. Every within-fibre coordinate permutation also preserves that inverse image, since its action on $\lambda + A_5$ is trivial: $Q\lambda - \lambda = Qe_0 - e_0 \in A_5$. These are concrete failures to preserve this cubic, without discarding their actual lattice actions or glue.

Negation preserves each lattice N and Λ, and its exact polynomial law is $F(-X) = -F(X)$. In the neighbour the nonzero witness is its displayed norm-four generator:

$$F(\rho) = 15\sqrt{2} + 16\sqrt{3}, \qquad F(v) = \frac{14}{27} - \frac{7\sqrt{2}}{24} - \frac{\sqrt{3}}{27}. \tag{1454}$$

Both equalities follow by substituting the original coordinates of ρ and v into the proved polynomial.

Every original lattice isometry $T \in O(N)$, whether or not it preserves the cubic, has the exact replacement transport

$$\Gamma_T = \mathcal{L}T\mathcal{L}^{-1}, \qquad F^T(Y) = F(T^{-1}Y), \qquad F^T(TX) = F(X), \tag{1455}$$

and

$$T\Lambda_\rho = \Lambda_{T\rho}, \qquad \mathcal{V}_{F^T}(\Lambda_{T\rho}) = \mathcal{V}_F(\Lambda_\rho), \qquad \mathcal{V}_{F^T}(N) = \mathcal{V}_F(N).$$

For the neighbour transport, write its exact elements as $x + k\rho/6$, where $x \in N$ and $(\rho, x) + k \equiv 0 \pmod 6$; application of T gives the identical congruence with $(T\rho, Tx)$. The inverse is T^{-1}. The value-module equalities then follow by the bijections and the displayed identity $F^T(TX) = F(X)$. In particular for $T = \sigma^{2k}$, one has $F^T = F$, so the same retained cubic has exactly the same value module on every neighbour $\Lambda_{\sigma^{2k}\rho}$ in this cyclic orbit. This statement names its target neighbour and does not assume the chosen ρ is fixed.

34 The residue cubic action and the common integral Leech correspondence

Retain the exact lattices $R = A^{\perp 4} \perp D \subset N$ and the binary and ternary codes from the index-72 construction. The four A coordinates remain the ordered six-element residue fibres. To avoid confusing a vector with a representation, write ϱ for the Weyl vector

$$\varrho_A = (5,3,1,-1,-3,-5)/2, \qquad \varrho_D = (3,2,1,0), \qquad \varrho = \varrho_A^{\oplus 4} \perp \varrho_D.$$

Thus $\varrho \in N$ and $\varrho^2 = 84$. The original residue representation is still denoted by ρ. The specific element studied here is

$$r = \rho(361) = \rho(289)^2, \qquad r|_{A_i} = C^2, \quad Ce_j = e_{j+1 \bmod 6}, \qquad r|_D = I. \tag{RF1}$$

In particular $r^3 = I$. This is the actual residue coordinate permutation, with no conjugation or replacement.

For each $\sigma \in \Omega = O(N)\varrho$, choose an original root α_σ with $(\sigma, \alpha_\sigma) = 1$ and set

$$\begin{aligned} \ell_\sigma(x) &= (\sigma, x) \bmod 6, \qquad K_\sigma = \ker(\ell_\sigma : N \to \mathbb{Z}/6), \\ \Lambda_\sigma &= K_\sigma + \mathbb{Z}(\sigma/6 - \alpha_\sigma) \\ &= \{x + k\sigma/6 : x \in N,\ k \in \mathbb{Z},\ (\sigma, x) + k \equiv 0 \pmod 6\}. \end{aligned} \tag{RF2}$$

The preceding neighbor proof shows that the definition is independent of α_σ, every Λ_σ is the specified rootless even unimodular lattice, and $N \cap \Lambda_\sigma = K_\sigma$ with both indices six. The family considered in this section is precisely $\mathscr{F} = \{\Lambda_\sigma : \sigma \in \Omega\}$.

34.1 A necessary and sufficient equality criterion

Theorem 34.1 (Equality of the marked Weyl neighbors)**.** *For $\sigma, \tau \in \Omega$, the following three statements are equivalent:*

$$\Lambda_\sigma = \Lambda_\tau, \qquad K_\sigma = K_\tau, \qquad \tau - \epsilon\sigma \in 6N \text{ for some } \epsilon \in \{1, -1\}. \tag{RF3}$$

Proof. Equality of the two neighbors gives equality of the kernels by intersection with N. Each ℓ_σ is onto because $(\sigma, \alpha_\sigma) = 1$. If the kernels are equal, their induced isomorphisms $N/K_\sigma \to \mathbb{Z}/6$ differ by an automorphism of $\mathbb{Z}/6$. Its two units are $1, -1$. Therefore $(\tau - \epsilon\sigma, N) \subset 6\mathbb{Z}$. Unimodularity $N^\vee = N$ gives $(\tau - \epsilon\sigma)/6 \in N$. Conversely this congruence makes the two kernels equal.

It remains to show that this kernel criterion actually gives equality of the chosen unimodular extensions. The exact formula (RF2), after replacing k by $-k$, gives $\Lambda_{-\sigma} = \Lambda_\sigma$. We may consequently suppose $\tau = \sigma + 6t$ with $t \in N$. Both Weyl vectors have squared norm 84, so

$$0 = \tau^2 - \sigma^2 = 12(\sigma, t) + 36t^2, \qquad (\sigma, t) = -3t^2 \in 6\mathbb{Z}. \tag{RF4}$$

The last inclusion uses the actual evenness of N. For $x + k\tau/6$ in (RF2), put $x' = x + kt \in N$. Then $x + k\tau/6 = x' + k\sigma/6$, and

$$(\sigma, x') + k \equiv (\sigma, x) + k \equiv (\tau, x) + k \equiv 0 \pmod 6.$$

This proves $\Lambda_\tau \subset \Lambda_\sigma$. Interchanging σ, τ proves the reverse inclusion. Thus no additional unverified choice of an extension is being identified merely from equality of its common kernel. □

It follows in particular that the map

$$\Omega/\!\sim \ \longrightarrow \ \mathscr{F}, \qquad [\sigma] \longmapsto \Lambda_\sigma, \qquad \sigma \sim \tau \iff \tau \equiv \pm\sigma \pmod{6N}, \tag{RF5}$$

is a bijection. It is $O(N)$-equivariant: the original formula $T\Lambda_\sigma = \Lambda_{T\sigma}$ follows by substitution in (RF2), and T preserves the congruence relation since $T(N) = N$.

34.2 The exact tetracode obstruction to a fixed neighbor

Theorem 34.2 (No fixed member of this specified family)**.** *The actual residue element $r = \rho(361)$ fixes no lattice of $\mathscr{F}$. Every orbit of its action on $\mathscr{F}$ has exactly three members.*

Proof. Suppose first that $r\sigma \equiv -\sigma \pmod{6N}$. Applying r successively, using $r(N) = N$ and $r^3 = I$, gives $\sigma \equiv -\sigma \pmod{6N}$. Then $\sigma/3 \in N$, contradicting its pairing $1/3$ with the root $\alpha_\sigma \in N$. Thus (RF3) permits fixedness only if $t = (r\sigma - \sigma)/6 \in N$.

Every isometry of N preserves the original root lattice, permutes its four A_5 components, and acts in each of them by a signed coordinate permutation, as proved in the full automorphism theorem. Negation permutes the six entries of ϱ_A. Consequently each A block $s = (s_0, \dots, s_5)$ of every $\sigma \in \Omega$ is a permutation of the six distinct numbers $5/2, 3/2, 1/2, -1/2, -3/2, -5/2$. In this block put $\Delta_j = s_{j-2 \bmod 6} - s_j$. If the corresponding block $\Delta/6$ of t belongs to $A^\vee$, its coordinate differences are integral, so

$$\Delta_j \equiv c \pmod 6 \qquad \text{for every } j \tag{RF6}$$

for one integer c modulo six. On either three-cycle of the index permutation, the three Δ_j sum to zero. Hence $3c \equiv 0 \pmod 6$ and $c \in \{0, 2, 4\}$. Each Δ_j is an integer between -5 and 5. If $c = 0$, (RF6) forces all $\Delta_j = 0$, contrary to the distinctness of the entries on each three-cycle. Therefore $c = 2$ or 4.

In the original discriminant marking, a vector of the A class $t_i\lambda + A$ has common fractional part $-t_i/6$. The corresponding class of $\Delta/6$ is therefore $t_i = -c \pmod 6$, also in $\{2, 4\}$. The difference on the D block is exactly zero because $r|_D = I$. Thus membership of the full vector $(r\sigma - \sigma)/6$ in N would require a glue word

$$(t_1, t_2, t_3, t_4; 00) \in H, \qquad t_i \in \{2, 4\} \text{ for every } i. \tag{RF7}$$

In $H = \{(3u + 2z; y)\}$ this forces $u = 0$ and all four z_i nonzero. But the exact ternary code is

$$C_3 = \{(a + b, a + 2b, a, b) : a, b \in \mathbb{F}_3\}. \tag{RF8}$$

If exactly one parameter is zero, exactly one coordinate is zero. If both are nonzero, precisely one of $a + b, a + 2b$ is zero. Thus every nonzero word has weight three, and the zero word has weight zero. No word in (RF7) belongs to H. This contradicts $t \in N$.

Neither sign in (RF3) can occur. Since $r^3 = I$, every orbit therefore has cardinality three. □

The conclusion concerns exactly $\mathscr{F}$ and the action (RF1). The integral correspondence for these three-member orbits is constructed next; no assertion about other embeddings or constructions of a Leech lattice follows from this obstruction.

34.3 The canonical orbit and its common integral lattice

Put $\sigma_j = r^j\varrho$, $\Lambda_j = \Lambda_{\sigma_j}$ and $K_j = K_{\sigma_j}$, with j read modulo three. Keep the same original root $\alpha = e_0 - e_1$ in the first A block. Directly $(\sigma_j, \alpha) = 1$ for all three j. Define

$$v_j = \sigma_j/6 - \alpha, \quad v = v_0, \qquad q_1 = (\sigma_1 - \sigma_0)/6, \quad q_2 = (\sigma_2 - \sigma_0)/6. \tag{RF9}$$

The complete original coordinates are

$$\begin{aligned} q_1 &= \big((-2, -2, 1, 1, 1, 1)/3\big)^{\oplus 4} \perp 0, \\ q_2 &= \big((-1, -1, -1, -1, 2, 2)/3\big)^{\oplus 4} \perp 0. \end{aligned} \tag{RF10}$$

Their classes in $R^\vee/R$ are respectively $(4, 4, 4, 4; 00)$ and $(2, 2, 2, 2; 00)$. Neither is in H, while $3q_1, 3q_2 \in R$. In particular each $q_i + N$ has exact order three. Direct pairings give

$$q_1^2 = q_2^2 = 16/3, \quad (q_1, q_2) = 8/3, \quad (\varrho, q_1) = (\varrho, q_2) = -16, \qquad q_1 + q_2 \in R. \tag{RF11}$$

For $x \in N$, write its original glue class as $(3u + 2z; y)$, with $z = (a + b, a + 2b, a, b) \in C_3$, and define the homomorphism

$$\beta : N \longrightarrow \mathbb{F}_3, \qquad \beta(x) = b = z_4. \tag{RF12}$$

This map is defined through the original quotient N/R. This definition uses the retained residue marking without changing it. The sum of the four z coordinates is also b modulo three. Write $\ell_j(x) = (\sigma_j, x) \bmod 6$. Then the exact three character formulas are

$$\ell_1(x) = \ell_0(x) + 4\beta(x), \qquad \ell_2(x) = \ell_0(x) + 2\beta(x) \quad \text{in } \mathbb{Z}/6. \tag{RF13}$$

The maps $b \mapsto 2b, 4b$ from $\mathbb{F}_3$ to $\mathbb{Z}/6$ are well-defined homomorphisms. To prove (RF13), the discriminant bilinear form gives

$$(q_1, x) \equiv \tfrac{5}{6}\, 4 \sum t_i \equiv \tfrac{1}{3} \sum t_i \equiv \tfrac{2}{3}\beta(x) \pmod{\mathbb{Z}}, \qquad t_i = 3u_i + 2z_i \pmod 6.$$

For q_2 the same calculation gives $(q_2, x) \equiv \beta(x)/3 \pmod{\mathbb{Z}}$. Multiplication by six and (RF9) prove both formulas.

Using the original dominant representatives $\omega_t = (1^t, 0^{6-t}) - \frac{t}{6}(1, \ldots, 1)$, put

$$u_* = (\omega_2, \omega_4, 0, \omega_2; 0) \in N. \tag{RF14}$$

Its ternary class is exactly the original generator $z_2 = (1, 2, 0, 1)$. It has $u_*^2 = 4$, $(\varrho, u_*) = 12$ and $\beta(u_*) = 1$. The map

$$(\ell_0, \beta) : N \longrightarrow \mathbb{Z}/6 \oplus \mathbb{F}_3 \tag{RF15}$$

is onto: α maps to $(1, 0)$ and u_* to $(0, 1)$. Consequently

$$\begin{gathered} I = \ker(\ell_0, \beta) = K_0 \cap K_1 = K_1 \cap K_2 = K_2 \cap K_0 = \bigcap_j K_j, \\ [N : I] = 18, \qquad [K_j : I] = 3, \qquad \det I = 324. \end{gathered} \tag{RF16}$$

The kernel equalities follow from (RF13), because either $2b = 0$ or $4b = 0$ in $\mathbb{Z}/6$ forces $b = 0$ in $\mathbb{F}_3$. The action r permutes the K_j, so it preserves I. Also $3q_i \in I$: these are in R, hence have $\beta = 0$, and $(\varrho, 3q_i) = -48$ is divisible by six.

Define the full-rank lattice

$$J = I + \mathbb{Z}(3v). \tag{RF17}$$

It is contained in Λ_0. The element $6v = \varrho - 6\alpha$ lies in I: its ℓ_0 value is zero, and its glue has zero ternary part. The element $3v$ is not in N, because $(3v, \alpha) = -11/2$ is not an integer. Thus

$$N \cap J = I, \qquad [J : I] = 2, \qquad \det J = 324/4 = 81, \qquad [\Lambda_0 : J] = 9. \tag{RF18}$$

This lattice is positive definite, even, and rootless, because it is a sublattice of Λ_0.

All exact root corrections in the cyclic action are retained. Put $\kappa_j = \alpha - r^j\alpha \in R$. The vectors $r\alpha = e_2 - e_3$ and $r^2\alpha = e_4 - e_5$ are original simple roots with ϱ-pairing one. Hence $\kappa_j \in I$. The precise formulas, rather than literal identification of different generators, are

$$r^j v = v + q_j + \kappa_j \quad (j = 1, 2), \qquad r(3v) = 3v + 3q_1 + 3\kappa_1. \tag{RF19}$$

Since $rI = I$, $3q_1 \in I$ and $\kappa_1 \in I$, this proves $rJ \subset J$. Applying $r^3 = I$ gives equality $rJ = J$.

Theorem 34.3 (All three pairwise Leech intersections)**.** *The actual three-member orbit satisfies*

$$\Lambda_0 \cap \Lambda_1 = \Lambda_1 \cap \Lambda_2 = \Lambda_2 \cap \Lambda_0 = \bigcap_j \Lambda_j = J, \qquad [\Lambda_j : J] = 9. \tag{RF20}$$

The element r is an integral isometry of J, and induces isometries $\Lambda_j \to \Lambda_{j+1}$ with this common restriction.

Proof. First $J \subset \Lambda_0 \cap \Lambda_1$: I lies in both, and $3v = 3v_1 - 3q_1$ with $3q_1 \in I$. Suppose $y \in \Lambda_0 \cap \Lambda_1$. Using the same α, write $y = x_0 + k_0 v_0 = x_1 + k_1 v_1$, $x_j \in K_j$. The fractional part of (v_j, α) is $1/6$ for both j, so pairing with $\alpha \in N$ gives $k_0 \equiv k_1 \pmod 6$. Absorbing multiples of $6v_j \in K_j$, take the same $k \in \{0, \dots, 5\}$ in both expressions. Then $x_0 - x_1 = kq_1 \in N$. The exact order three of $q_1 + N$ forces $k = 0$ or 3. For these values $kq_1 \in I \subset K_1$, so $x_0 = x_1 + kq_1 \in K_0 \cap K_1 = I$. Thus $y = x_0 + kv \in J$, proving the first equality. Apply r and r^2, using $rJ = J$, to prove the other pairwise equalities. Their common index is nine by (RF18) and the isometries induced by r. □

For comparison with the original index-72 glue, let $R_0 = R \cap K_0$. Since $\beta|_R = 0$, (RF13) gives $R \cap K_j = R_0$ for every j, and $R \cap I = R_0$. The reduction map $K_0/R_0 \cong H$ identifies I/R_0 with the kernel of the original ternary coefficient b. Thus

$$[I : R_0] = 24, \qquad [J : R_0] = 48, \qquad [K_0 : R_0] = 72, \qquad [\Lambda_j : R_0] = 432. \tag{RF21}$$

All these are inclusions of the displayed original lattices.

34.4 The exact discriminant form and its cyclic three planes

Theorem 34.4 (The integral correspondence over $\mathbb{F}_3^4$)**.** *One has*

$$J^\vee = \Lambda_i + \Lambda_j \quad (i \neq j), \qquad J^\vee/J \cong \mathbb{F}_3^4. \tag{RF22}$$

In the ordered basis

$$e_1 = u_* + J, \quad e_2 = v + J, \quad e_3 = ru_* + J, \quad e_4 = rv + J, \tag{RF23}$$

the discriminant quadratic form is

$$q_J(a, b, c, d) = \frac{2}{3}(2ad + bc + bd) \pmod{2\mathbb{Z}}, \tag{RF24}$$

and three times its discriminant bilinear form has matrix

$$B = \begin{pmatrix} 0 & 0 & 0 & 2 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 \\ 2 & 1 & 0 & 0 \end{pmatrix} \quad \text{over } \mathbb{F}_3. \tag{RF25}$$

The three original Leech extensions are exactly the inverse images in $J^\vee$ of the isotropic planes

$$\begin{aligned} P_0 &= \{(a, b, 0, 0) : a, b \in \mathbb{F}_3\}, \\ P_1 &= \{(0, 0, c, d) : c, d \in \mathbb{F}_3\}, \\ P_2 &= \{(a, b, a - b, b) : a, b \in \mathbb{F}_3\}. \end{aligned} \tag{RF26}$$

The induced action of the actual r is the matrix

$$\mathcal{R} = \begin{pmatrix} 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 2 \\ 1 & 0 & 2 & 2 \\ 0 & 1 & 0 & 2 \end{pmatrix}, \qquad \mathcal{R}^3 = I, \quad \mathcal{R}^T B \mathcal{R} = B, \quad \mathcal{R} P_j = P_{j+1}. \tag{RF27}$$

All entries in (RF23)–(RF27) use the original glue generator z_2, root α, pairing, and residue permutation.

Proof. Since $J \subset \Lambda_i = \Lambda_i^\vee$, each Λ_i lies in $J^\vee$. By (RF20), the two subgroups Λ_i/J and Λ_j/J have order nine and intersect trivially for $i \neq j$. Their sum has order 81, equal to $|J^\vee/J| = \det J = 81$, proving the first equality in (RF22).

The homomorphism $\beta : K_0/I \to \mathbb{F}_3$ is an isomorphism, with generator $u_* + I$. Hence Λ_0/J is generated by $u_* + J, v + J$. Both have order dividing three because $3u_* \in I$ and $3v \in J$. They are independent: if $au_* + bv \in J$, reduce first modulo N; the subgroup $J + N$ has the two cosets represented by $0, 3v$, whereas $v + N$ has order six. For $a, b \in \{0, 1, 2\}$ this forces $b = 0$. Then membership of au_* in $I = N \cap J$ forces $a = 0$. Thus $\Lambda_0/J \cong \mathbb{F}_3^2$ with the first two generators in (RF23). Applying r proves the corresponding statement for Λ_1/J and the last two generators. Their trivial intersection and total order 81 prove that (RF23) is indeed a basis of the full discriminant group.

Here are the full Euclidean Gram entries before any passage to a quotient:

$$\operatorname{Gram}(u_*, v, ru_*, rv) = \begin{pmatrix} 4 & 2 & -2 & -4/3 \\ 2 & 4 & -2/3 & -2/3 \\ -2 & -2/3 & 4 & 2 \\ -4/3 & -2/3 & 2 & 4 \end{pmatrix}. \tag{RF28}$$

For an explicit verification, each of the three nonzero A blocks of u_* pairs to $-2/3$ with its shifted block, giving $(u_*, ru_*) = -2$. One has $(u_*, \alpha) = (u_*, r\alpha) = (ru_*, \alpha) = 0$, $(\varrho, u_*) = 12$, $(\sigma_1, u_*) = -8$, and $(\sigma_2, u_*) = -4$, directly from (RF10) and (RF14). These give respectively $(u_*, v) = 2$, $(u_*, rv) = -4/3$, and $(v, ru_*) = -2/3$. Also $(\varrho, \sigma_1) = -12$, $(\varrho, r\alpha) = (\sigma_1, \alpha) = 1$, and $(\alpha, r\alpha) = 0$. Thus

$$(v, rv) = -12/36 - 1/6 - 1/6 = -2/3.$$

The four diagonal entries equal four, and $(ru_*, rv) = (u_*, v) = 2$ by the exact isometry r. This proves (RF28). Reducing its entries modulo $\mathbb{Z}$ gives (RF25). The diagonal and integral cross terms in the squared norm are even. Reducing the remaining cross terms modulo $2\mathbb{Z}$ gives exactly (RF24). Changing any coordinate by three changes (RF24) by an even integer, so it is well-defined on the asserted field basis.

It remains to calculate the action without identifying a family permutation with a fixed Leech automorphism. By definition $re_1 = e_3$, $re_2 = e_4$. The vector $u_* + ru_* + r^2u_*$ lies in I: its β value is $1 + 1 + 1 = 0$, because r acts trivially on N/R, and its ℓ_0 value is $12 - 4 - 8 = 0$. Consequently

$$r^2u_* \equiv -u_* - ru_* \pmod{J}. \tag{RF29}$$

By (RF19),

$$v + rv + r^2v = 3v + q_1 + q_2 + \kappa_1 + \kappa_2.$$

The κ_j belong to I, and $3v \in J$. Furthermore $q_1 + q_2 - u_* + ru_* \in I$. Indeed $q_1 + q_2 \in R$, so its β value is zero; subtracting u_* and adding ru_* keeps that value zero. Its pairing with ϱ is $-32 - 12 - 4 = -48$, divisible by six. Thus

$$r^2v \equiv u_* - v - ru_* - rv \pmod{J}. \tag{RF30}$$

The four images in (RF29)–(RF30) give the four columns of $\mathcal{R}$ in (RF27). The matrix has cube the identity and preserves B either by multiplication, or by the already proved facts $r^3 = I$ and $rJ = J$ with its original isometric pairing.

The first two planes in (RF26) are respectively Λ_0/J and Λ_1/J by their proved bases. Their next image is the span of $(-1, 0, -1, 0)$ and $(1, -1, -1, -1)$, from (RF29)–(RF30). Solving their two linear parameters gives exactly $(a, b, a - b, b)$, proving the stated $P_2 = \Lambda_2/J$. Each plane is isotropic: the first two vanish in (RF24), and the third has $2ab + b(a - b) + b^2 = 3ab = 0$ in $\mathbb{F}_3$. They have nine elements and pairwise intersection zero, also following directly from (RF26). Finally each Λ_j is exactly the inverse image of its quotient plane because it contains J. This proves the claimed integral correspondence and all its maps, indices, pairings, and cyclic actions. □

34.5 The fixed fourth plane and its exact A_2^{12} extension

The same finite discriminant form has the additional plane

$$P_\infty = \{(a, b, -a + b, -b) : a, b \in \mathbb{F}_3\}. \tag{RF31}$$

Put

$$p = u_* - ru_*, \qquad q = v + ru_* - rv, \qquad M = J + \mathbb{Z}p + \mathbb{Z}q. \tag{RF32}$$

These are literal vectors in the original Euclidean space.

Theorem 34.5 (The fixed fourth extension)**.** *The plane P_∞ is isotropic and is fixed by the induced residue action. Its inverse image M is a positive definite even unimodular lattice with exactly 72 roots, of root system A_2^{12} and minimum squared norm two. The original residue element r is an integral isometry of M. It fixes four of these A_2 components pointwise and acts as a coordinate three-cycle on each of the other eight.*

The exact intersection and index relations are

$$\begin{gathered} M \cap \Lambda_j = J, \qquad [M : J] = [\Lambda_j : J] = 9, \\ M \cap N = L_0 := \{x \in N : \beta(x) = 0,\ (\varrho, x) \equiv 0 \pmod 2\}, \\ [M : L_0] = [N : L_0] = 6. \end{gathered} \tag{RF33}$$

If R_M denotes the lattice generated by all its roots, then

$$R_M \cong A_2^{\perp 12}, \qquad \det R_M = 3^{12} = 531441, \qquad [M : R_M] = 3^6 = 729. \tag{RF34}$$

Proof. Substitution of (RF31) in (RF24) gives the numerator $2a(-b) + b(-a + b) + b(-b) = -3ab = 0$ in $\mathbb{F}_3$. Thus P_∞ is isotropic. The matrix (RF27) maps its vector with parameters (a, b) to

$$(a + b, b, -a, -b), \tag{RF35}$$

which is again in P_∞, now with parameters $(a + b, b)$. The images of p, q in (RF23) are $(1, 0, -1, 0)$ and $(0, 1, 1, -1)$, a basis of P_∞. Consequently (RF32) is its exact inverse image under $J^\vee \to J^\vee/J$, and $[M : J] = 9$. Isotropy gives even norms on M; polarization gives integral pairings. Its determinant is $\det J/9^2 = 1$, and its positive definiteness and rank are inherited from the unchanged ambient real space. Equation (RF35), together with $rJ = J$, proves $rM = M$. Each of P_0, P_1, P_2 intersects P_∞ in zero, as follows by equating their coordinates in (RF26) and (RF31). Taking inverse images proves $M \cap \Lambda_j = J$.

For its relation with N, (RF19) gives the exact expression

$$q = ru_* - q_1 - \kappa_1. \tag{RF36}$$

Therefore

$$N + M = N + \mathbb{Z}(\varrho/2) + \mathbb{Z}q_1. \tag{RF37}$$

The two indicated cosets modulo N have orders two and three. For the first, pair with α; for the second, use (RF10). Subgroups of coprime orders intersect trivially, so the quotient in (RF37) has order six. Also $p \in N$ has $\beta(p) = 0$ and $(\varrho, p) = 12 - (-4) = 16$, hence image $(4, 0)$ in (RF15). Thus $p + I$ has order three. The vector $3q$ lies in J because $q + J$ has order three, and lies in N by (RF36) and $3q_1 \in N$. It is therefore in $N \cap J = I$. We also have $6v \in I$. Write an element of M using generators $I, 3v, p, q$ and reduce the coefficient of $3v$ modulo two and of q modulo three. If this element is in N, their two coprime-order cosets in (RF37) force both reduced coefficients to vanish. Hence

$$M \cap N = I + \mathbb{Z}p. \tag{RF38}$$

Its image in (RF15) is precisely $2(\mathbb{Z}/6) \oplus 0$, which proves the displayed formula for L_0 and $[N : L_0] = 6$. Since $\det N = \det M = 1$, the equal covolumes give $[M : L_0] = 6$, proving (RF33).

We next prove that the following root count is exhaustive. Every vector in $N + M$ has one of the six forms

$$x + a\varrho/2 - bq_1, \qquad x \in N,\ a \in \{0,1\},\ b \in \{0,1,2\}. \tag{RF39}$$

When $a = 0$, the vector is in $R^\vee$. Every norm-two vector of this particular $R^\vee$ is already a root of R. Indeed, the A class minima are $0, 5/6, 4/3, 3/2, 4/3, 5/6$, and the three nonzero D classes have minimum one. In a nonzero class of $R^\vee/R$ with minimum at most two, the possible minima are exactly among

$$\{5/6, 1, 4/3, 3/2, 5/3, 11/6\}. \tag{RF40}$$

This exhausts them because at most two A classes can be nonzero under the bound; if two are nonzero, each has minimum $5/6$; with a nonzero D class there is at most one nonzero A class and its minimum must be $5/6$. No number in (RF40) is congruent to two modulo $2\mathbb{Z}$. All norms within a fixed discriminant class differ by an even integer, so no such class has a norm-two vector. This proves the claim about $R^\vee$.

All original roots have $\beta = 0$, so (RF33) says that those lying in M are exactly the ones with even ϱ height. In each original A block they are $e_i - e_j$ with $i \neq j$ and $i \equiv j \pmod 2$. These give two orthogonal A_2 systems per block, on the coordinate triples $\{0,2,4\}$ and $\{1,3,5\}$, and 48 roots in total. In D they are the eight roots $\pm e_1 \pm e_3,\ \pm e_2 \pm e_4$. They form four pairwise orthogonal A_1 systems. This accounts for exactly 56 roots with $a = 0$ in (RF39).

For $a = 1$, absorb the A dual vector $-bq_1$ into the marked A discriminant classes. If the original class of x is $(t_1, t_2, t_3, t_4; y) \in H$, the shifted class in each A is $t_i - 4b$. The exact half-Weyl-shift minima proved in the neighbor calculation are

$$\min_{w \in t\lambda + A} |w + \varrho_A/2|^2 = \begin{cases} 3/8 & t \equiv 0 \pmod 3, \\ 17/24 & t \not\equiv 0 \pmod 3, \end{cases} \qquad \min_{w+D=y} |w + \varrho_D/2|^2 = 1/2. \tag{RF41}$$

For completeness the first line follows by choosing centered quarter coordinates when $t = 0, 3$; for other t, the nearest twelfth coordinates have square sum $13/24$ and coordinate sum ± 1, and restoring zero sum costs $1/6$. The second line follows from the integer coordinate distances $(1/2, 0, 1/2, 0)$ and half-integer distances $(0, 1/2, 0, 1/2)$, with their ties meeting every D marking. Thus the total minimum in (RF39) with $a = 1$ is at least two. Equality requires $t_i - 4b \equiv 0 \pmod 3$ in every component. Since $t_i \equiv 2z_i \pmod 3$, this says $z_i = 2b$ for all four coordinates. For $b = 1, 2$ this is excluded by the exact code (RF8). For $b = 0$ it requires $z = 0$. The eight binary codewords and the two D ties then give exactly sixteen equality vectors $y = x + \varrho/2$. Every one belongs to M: the norm equation gives

$$2 = x^2 + (\varrho, x) + 21, \qquad (\varrho, x) = -19 - x^2, \tag{RF42}$$

which is odd because N is even. Also $\beta(x) = 0$ because $z = 0$. Therefore $x + 3\alpha \in L_0$ and $y = (x + 3\alpha) + 3v \in M$. This proves exhaustively that M has exactly $56 + 16 = 72$ roots.

Here are the full root coordinates and the resulting exact orthogonal decomposition. Let ζ_i be supported in the ith original A block with entries

$$\zeta = (1, -1, 1, -1, 1, -1)/4, \qquad \zeta^2 = 3/8.$$

In the following table every Z_y is an actual sum of those four mutually orthogonal vectors, and every δ_y is an original D root:

y	Z_y	δ_y
00	$\zeta_1 + \zeta_2 + \zeta_3 + \zeta_4$	$e_1 + e_3$
10	$-\zeta_1 - \zeta_2 + \zeta_3 + \zeta_4$	$e_1 - e_3$
01	$-\zeta_1 + \zeta_2 - \zeta_3 + \zeta_4$	$e_2 + e_4$
11	$\zeta_1 - \zeta_2 - \zeta_3 + \zeta_4$	$e_2 - e_4$

. (RF43)

The sixteen new roots are exactly

$$\{\epsilon Z_y + \eta\delta_y/2 : y \in \mathbb{F}_2^2,\ \epsilon,\eta \in \{1,-1\}\}. \tag{RF44}$$

To verify the signs from the original glue, the A half-shift minimizer for binary coordinate $u_i = 0$ is ζ_i and for $u_i = 1$ it is $-\zeta_i$. For $y = 00$ the two codewords are $0000, 1111$; for 10 they are $1100, 0011$; for 01 they are $1010, 0101$; and for 11 they are $0110, 1001$. These give respectively the four opposite pairs $\pm Z_y$ in (RF43). The two shifted D minimizers in these exact original markings are respectively $\pm(e_1+e_3)/2, \pm(e_1-e_3)/2, \pm(e_2+e_4)/2, \pm(e_2-e_4)/2$. This proves (RF44) with every sign and marking retained.

The four Z_y are mutually orthogonal and have squared norm $3/2$, as follows by dotting their four displayed sign rows. The δ_y are mutually orthogonal of squared norm two. The Z_y are orthogonal to the entire original D space, and to the eight old A_2 systems because their coordinates are constant on each parity triple. For each y, the six roots

$$\{\pm\delta_y\} \ \cup\ \{\epsilon Z_y + \eta\delta_y/2 : \epsilon,\eta = \pm 1\} \tag{RF45}$$

are an A_2 system: take simple roots δ_y and $Z_y - \delta_y/2$, whose Gram matrix is $\left(\begin{smallmatrix} 2 & -1 \\ -1 & 2 \end{smallmatrix}\right)$; their sum is $Z_y + \delta_y/2$. The four such systems are mutually orthogonal and orthogonal to the eight old systems. They exhaust the 72 proved roots. Thus their root lattice is $A_2^{\perp 12}$ of determinant 3^{12}, and unimodularity of M gives the index 3^6. This proves (RF34) and the Coxeter number three of this rooted Niemeier lattice directly from its twelve components.

Finally r fixes each ζ_i because it permutes coordinates within their parity triples; it fixes each δ_y because it is the identity on D. Hence it fixes the four systems (RF45) pointwise. On each of the eight old A_2 systems it cyclically permutes its three coordinate positions, giving its exact order-three Coxeter action. This completes the asserted classification of the fixed fourth extension and its precise relation to the three Leech lattices through J. □

35 The cubic on the common lattice and the fourth Niemeier extension

Retain the original coordinates, Cayley multiplication, F, the generators g_i and their heights H_i from Section 33, and the exact common lattice and fourth extension of Section 34. The symbol u below is the unchanged vector u_* in (RF32).

35.1 Integral generators of the common lattice and the fixed extension

Define the unchanged residue action by

$$(rX)_{i,j} = x_{i,j-2 \pmod 6}, \qquad (rX)_{D,k} = d_k.$$

The two primary parts of the original glue have the exact descriptions

$$H_2 = \{(\varepsilon_1+\varepsilon_2+\varepsilon_3, \varepsilon_1+\varepsilon_2, \varepsilon_1+\varepsilon_3, \varepsilon_1; \varepsilon_2, \varepsilon_3) : \varepsilon_i \in \mathbb{F}_2\},$$
$$H_3 = \{(t+s, t+2s, t, s; 00) : t, s \in \mathbb{F}_3\}.$$

In the original $(\mathbb{Z}/6)^4$ coordinates this means $k = 3w + 2z$, with w the four binary entries of H_2 and z the four ternary entries of H_3. The binary D_4 marking is retained. Chinese remainders recover w, z uniquely, and the displayed parameterizations are injective. Consequently

$$\beta : N \longrightarrow \mathbb{F}_3, \qquad \beta(X) = z_4 = s$$

is a well-defined additive homomorphism. The actual lifts (1438) satisfy

$$\beta(g_i) = 0 \quad (1 \le i \le 28), \qquad \beta(g_{29}) = 1.$$

Indeed roots have zero glue, the first three lifts are two-primary, and the last two have $(t,s)=(1,0),(0,1)$, respectively.

Set, with no change of representatives,

$$I=\{X\in N:(\rho,X)\equiv 0 \pmod 6,\ \beta(X)=0\}, \qquad J=I+\mathbb{Z}(3v), \tag{1456}$$

$$\omega_t=(\underbrace{1,\dots,1}_{t},0,\dots,0)-\frac{t}{6}(1,1,1,1,1,1), \qquad u=(\omega_2,\omega_4,0,\omega_2;0),$$

$$p=u-r(u), \qquad q=v+r(u)-r(v),$$

$$M=J+\mathbb{Z}p+\mathbb{Z}q. \tag{1457}$$

Thus J and M are precisely the common-lattice and residue-fixed extension presentations being evaluated here; the following value-group proof needs no root count, lattice determinant, or classification label for M. For explicitness, the two extra vectors in the original five blocks are

$$p=((1,1,-1,-1,0,0),(1,1,0,0,-1,-1),0,(1,1,-1,-1,0,0);0),$$
$$q=\tfrac{1}{3}((-2,4,4,-2,-2,-2),0,(2,2,-1,-1,-1,-1),(1,1,1,1,-2,-2);0).$$

The zero blocks have their original sizes six and four. These coordinates follow by substituting $v=\rho/6-\alpha$ into (1457); in particular the shifted α term in $r(v)$ is retained. Also $u\in N$, $\beta(u)=1$, $(\rho,u)=12$, and $(u,u)=4$: its class is $(2,4,0,2;00)$, $(\rho_A,\omega_t)=t(6-t)/2$, and $(\omega_t,\omega_t)=t(6-t)/6$.

Theorem 35.1 (Integral generating lists with both kernel conditions)**.** *Define the following 32 ordered vectors:*

$$b_1=6\alpha, \qquad b_i=g_i-H_i\alpha \quad (2\le i\le 28),$$
$$b_{29}=3g_{29}-60\alpha, \qquad b_{30}=3v=\rho/2-3\alpha,$$
$$b_{31}=p, \qquad b_{32}=q. \tag{1458}$$

Then the integral, not merely rational, equalities are

$$I=\sum_{i=1}^{29}\mathbb{Z}b_i, \qquad J=\sum_{i=1}^{30}\mathbb{Z}b_i, \qquad M=\sum_{i=1}^{32}\mathbb{Z}b_i.$$

Proof. Every one of the first 29 vectors has $\beta=0$ and height divisible by six, by the exact heights and characters already calculated. Conversely take any integral expression $X=\sum_{i=1}^{29}n_ig_i$ for $X\in I$. Applying β shows $n_{29}=3a$ for an integer a, regardless of which expression was chosen. The exact equality

$$X=\sum_{i=2}^{28}n_i(g_i-H_i\alpha)+a(3g_{29}-60\alpha)+\left(n_1+\sum_{i=2}^{28}n_iH_i+60a\right)\alpha$$

has last coefficient (ρ,X), which is divisible by six. It therefore expresses X as an integer combination of the first 29 b_i. The reverse containment was proved for each generator. The last two equalities follow by adjoining exactly the vectors in (1456) and (1457). No assertion of linear independence is used. □

35.2 Complete finite certificates for the two new value groups

Theorem 35.2 (Equality of the common-lattice and fixed-extension value-generated groups)**.** *For the retained Cayley multiplication and the original ordered coordinates, the two additive cubic-value groups are exactly*

$$\mathcal{V}_F(J)=\mathcal{V}_F(M)=\frac{1}{27}\mathbb{Z}\ \oplus\frac{\sqrt{2}}{216}\mathbb{Z}\ \oplus\frac{\sqrt{3}}{18}\mathbb{Z}. \tag{1459}$$

The equality concerns generated additive groups, not equality of individual-value sets $F(J)$ and $F(M)$.

Proof. Apply Theorem 33.2 to the first 30 and all 32 vectors in (1458). The resulting list lengths are exactly

$$30 + 2\binom{30}{2} + \binom{30}{3} = 4960, \qquad 32 + 2\binom{32}{2} + \binom{32}{3} = 5984.$$

Write all pair and triple symbols in the next tables using these b_i, not the g_i or h_i. Substitution in the full polynomial gives the complete denominator certificate

	A_i	D^+_{ij}	D^-_{ij}	D_{ijk}
J	$(27, 216, 1)$	$(9, 36, 18)$	$(9, 36, 18)$	$(9, 36, 18)$
M	$(27, 216, 1)$	$(9, 36, 18)$	$(9, 36, 18)$	$(9, 36, 18)$

. (1460)

As before each triple is the least common multiple of all reduced rational denominators in its full category, in the order (P_0, P_2, P_3); a zero has denominator one. The finite arithmetic is completely specified by (1428)–(1437), (1458), and all indices in (1440)–(1443). The accompanying standard-library checker `check_jm_values.py`, using the included `cubic_engine.py`, executes every entry with exact fractions and asserts this table. It reconstructs all 72 marked glue classes, checks all 29 $\beta(g_i)$, and checks both kernel conditions on all the displayed I generators. These finite computations establish the upper containment in (1459), including all mixed polarization terms involving p, q, not just their individual cubic values.

Here is a short certificate for the reverse containment, using only J. Coordinates of each row are taken in the ordered proposed value-group basis $(1/27, \sqrt{2}/216, \sqrt{3}/18)$:

$F(b_2)$	14	-272	0
$F(b_4)$	5	-128	0
$F(b_{27})$	23220	-111672	-90
$F(b_{28})$	13780	-74672	0
$F(b_{30})$	378	-1701	-18
$D^+_{2,27}$	7425	-60624	-3
$D^+_{11,21}$	36	48	-16

. (1461)

Let C_1, C_2, C_3 have, respectively, columns obtained by transposing rows $(2, 3, 5)$, $(2, 5, 6)$, and $(1, 4, 7)$ of this table. Direct three-by-three determinants are

$$d_1 = -39859290, \qquad d_2 = 11531403, \qquad d_3 = -43244032.$$

The following explicit integer identity certifies their gcd one:

$$(-6882196252)d_1 + (-23788908901)d_2 + (-907)d_3 = 1. \tag{1462}$$

Put $(c_1, c_2, c_3) = (-6882196252, -23788908901, -907)$. For each standard basis column $e_t \in \mathbb{Z}^3$ the exact integer reconstruction is

$$e_t = \sum_{j=1}^{3} C_j\big(c_j \operatorname{adj}(C_j)e_t\big). \tag{1463}$$

All columns of every C_j represent integer combinations of actual cubic values on J; their integer coefficient vectors in (1463) therefore give each of the three claimed generators as such a combination. This is an explicit reverse-containment morphism, not an index inference. The checker also expands every adjugate and verifies all three axis reconstructions, saving their integer coefficients in `EXACT_RESULTS.json`. Thus $\mathcal{V}_F(J)$ contains the right-hand side of (1459). Since $J \subset M$, the same certificate proves its containment in $\mathcal{V}_F(M)$. The upper bounds already established finish both equalities. □

For additional exact checks on the unchanged representatives, the three individual values are

$$F(3v) = 14 - \frac{63}{8}\sqrt{2} - \sqrt{3}, \qquad F(p) = -2, \qquad F(q) = \frac{64}{27}.$$

They agree with the corresponding rows of the complete calculation; they are not substituted for the mixed-term certificate.

35.3 Exact inclusions, quotient maps, and the factor of the Albert determinant

Theorem 35.3 (The exact four-to-three chain of value groups)**.** *There is a strict chain of additive subgroups of the original real coefficient field*

$$\mathcal{V}_F(N) \subsetneq \mathcal{V}_F(J) = \mathcal{V}_F(M) \subsetneq \mathcal{V}_F(\Lambda).$$

The inclusion matrices in the ordered bases already displayed are

$$\mathcal{V}_F(N) \longrightarrow \mathcal{V}_F(J) = \mathcal{V}_F(M) : (a,b,c) \longmapsto (a,4b,c),$$

$$\mathcal{V}_F(J) = \mathcal{V}_F(M) \longrightarrow \mathcal{V}_F(\Lambda) : (a,b,c) \longmapsto (a,b,3c).$$

Their exact quotient maps are

$$\frac{a}{27} + \frac{b\sqrt{2}}{216} + \frac{c\sqrt{3}}{18} \longmapsto b \pmod 4,$$
$$\frac{a}{27} + \frac{b\sqrt{2}}{216} + \frac{c\sqrt{3}}{54} \longmapsto c \pmod 3.$$

In particular

$$\mathcal{V}_F(J)/\mathcal{V}_F(N) \cong \mathbb{Z}/4, \qquad \mathcal{V}_F(M)/\mathcal{V}_F(N) \cong \mathbb{Z}/4,$$
$$\mathcal{V}_F(\Lambda)/\mathcal{V}_F(J) \cong \mathbb{Z}/3, \qquad \mathcal{V}_F(\Lambda)/\mathcal{V}_F(M) \cong \mathbb{Z}/3,$$
$$\mathcal{V}_F(M)/\mathcal{V}_F(J) = 0.$$

Proof. The first basis comparison uses $\sqrt{2}/54 = 4(\sqrt{2}/216)$, with the other basis entries equal. The second uses $\sqrt{3}/18 = 3(\sqrt{3}/54)$, with its other entries equal. Uniqueness of rational radical coefficients proves that the displayed kernels are exactly the corresponding subgroups. Surjectivity is witnessed by $\sqrt{2}/216$ and $\sqrt{3}/54$, respectively, both actual elements of the proved additive groups. Their orders modulo the smaller groups are exactly four and three: the coefficient divisibility conditions are necessary and sufficient. This also proves strictness. The last quotient is zero by the equality of the two proved groups. In particular the composite inclusion is $(a,b,c) \mapsto (a,4b,3c)$, recovering the full quotient $\mathbb{Z}/4 \oplus \mathbb{Z}/3$, with explicit map

$$\frac{a}{27} + \frac{b\sqrt{2}}{216} + \frac{c\sqrt{3}}{54} \longmapsto (b \bmod 4, c \bmod 3).$$

Its identification with $\mathbb{Z}/12$ is $(\bar{b},\bar{c}) \mapsto 9b + 4c \pmod{12}$, whose inverse is reduction modulo four and modulo three. Every assertion follows from the proved cubic coefficient groups, not from an index of the original lattices. □

The factor two in the original Albert determinant is unchanged:

$$\det Q_{\mathbb{O}}(\lambda; \mathcal{L}X) = \lambda^3 - \lambda(X,X) + 2F(X).$$

Thus, if one instead asks for the additive group generated by the actual cubic term $2F$, it is exactly $2\mathcal{V}_F(L)$. For J, M this is

$$\frac{2}{27}\mathbb{Z} \oplus \frac{\sqrt{2}}{108}\mathbb{Z} \oplus \frac{\sqrt{3}}{9}\mathbb{Z}.$$

Multiplication by two is an injective additive map of the real coefficient field and restricts to the exact isomorphism $\mathcal{V}_F(L) \to \mathcal{V}_{2F}(L) = 2\mathcal{V}_F(L)$, $x \mapsto 2x$, with inverse $y \mapsto y/2$. It intertwines the same two inclusion matrices and the same quotients; no factor has been absorbed into F.

35.4 Reproduction and scope of the exact certificate

The accompanying evidence folder is

`supporting_materials/workbench/evidence/`
`period_lattice_continuation_20260906/jm_cubic/`

It contains both full source files `cubic_engine.py` and `check_jm_values.py`. Running the latter with Python's standard library performs the following finite proof certificate: all 20399 polarization entries for N, Λ, J, M, all displayed denominator and determinant assertions, the integer Bezout and adjugate axis reconstructions, and the exact inclusion coefficient comparisons. The output `EXACT_RESULTS.json` retains each selected column, determinant, Bezout coefficient, and every expanded integer linear combination reconstructing a value-group axis.

There is additionally an independent coefficient-identity check: the original 24 simple roots form a real basis of the original 24-dimensional space. The direct Cayley–Dickson implementation and the rational polynomial implementation are compared on the 24 basis vectors, their 552 ordered-by-index plus/minus pairs, and their 2024 three-element sums, a total of 2600 exact inputs. The finite polarization proof shows these evaluations determine every coefficient of a homogeneous cubic in those 24 variables. Consequently this is a full polynomial-identity certificate on the real ambient space, not sampling of lattice points. The field implementation retains all four rational coefficients in $\mathbb{Q}(\sqrt{2}, \sqrt{3})$, and verifies that the $\sqrt{6}$ coefficient is identically zero. A further 65 direct checks cover the displayed new generating lists and u, p, q.

This computation proves the stated additive value groups and their morphisms. It asserts neither equality of individual-value sets nor that the cubic group reconstructs a lattice or classifies all cubic-preserving lattice isometries.

36 A cyclic Wilson Leech lattice in the original triality space

36.1 The two retained metrics and the octonion convention

Let $V = \mathbb{O}_1 \oplus \mathbb{O}_2 \oplus \mathbb{O}_3$ be the original ordered octonion space. Retain its metric and cubic

$$h_0(x, x) = N(x_1) + N(x_2) + N(x_3),$$
$$\tau(x_1, x_2, x_3) = \mathrm{Re}((x_1 x_2) x_3), \qquad d(Q_{\mathbb{O}}(0; x)) = 2\tau(x_1, x_2, x_3),$$

where

$$Q_{\mathbb{O}}(\lambda; x) = \begin{pmatrix} \lambda & x_3 & \bar{x}_2 \\ \bar{x}_3 & \lambda & x_1 \\ x_2 & \bar{x}_1 & \lambda \end{pmatrix}.$$

Wilson's norm on the same underlying ordered space is

$$q_{\mathrm{W}}(x) = \tfrac{1}{2} h_0(x, x).$$

The map

$$\iota : (V, q_{\mathrm{W}}) \longrightarrow (V, h_0), \qquad \iota(x_1, x_2, x_3) = (x_1/\sqrt{2}, x_2/\sqrt{2}, x_3/\sqrt{2}) \tag{1464}$$

is an explicit isometry. Both metrics remain specified. In particular, for the original λ, which is not rescaled,

$$d(Q_{\mathbb{O}}(\lambda; \iota x)) = \lambda^3 - \lambda q_{\mathrm{W}}(x) + \frac{1}{\sqrt{2}} \tau(x_1, x_2, x_3). \tag{1465}$$

Indeed the quadratic coordinates scale by $1/2$, while the trilinear term scales by $(1/\sqrt{2})^3 = 1/(2\sqrt{2})$ and its original coefficient is 2. Thus the off-diagonal cubic after ι is $1/(2\sqrt{2})$ times its value before ι.

For an explicit realization inside the same $\mathbb{O}$, choose a Cayley–Dickson basis $(1, \mathsf{i}, \mathsf{j}, \mathsf{k}, \ell, \mathsf{i}\ell, \mathsf{j}\ell, \mathsf{k}\ell)$, with

$$(a + b\ell)(c + d\ell) = (ac - \bar{d}b) + (da + b\bar{c})\ell, \qquad \mathsf{ij} = \mathsf{k}.$$

Wilson's ordered orthonormal basis is realized by

$$(i_\infty, i_0, i_1, i_2, i_3, i_4, i_5, i_6) = (1, \mathsf{i}, \mathsf{j}, \ell, \mathsf{k}, \mathsf{j}\ell, -\mathsf{k}\ell, \mathsf{i}\ell). \tag{1466}$$

The seven oriented Fano triples are $(t, t+1, t+3)$ modulo 7: $i_t i_{t+1} = i_{t+3}$, with cyclic products within each triple and opposite signs on reversal. To verify (1466), its seven corresponding products, in order $t = 0, \dots, 6$, are

$$\mathsf{ij} = \mathsf{k}, \quad \mathsf{j}\ell = \mathsf{j}\ell, \quad \ell\mathsf{k} = -\mathsf{k}\ell, \quad \mathsf{k}(\mathsf{j}\ell) = \mathsf{i}\ell, \quad (\mathsf{j}\ell)(-\mathsf{k}\ell) = \mathsf{i}, \quad (-\mathsf{k}\ell)(\mathsf{i}\ell) = \mathsf{j}, \quad (\mathsf{i}\ell)\mathsf{i} = \ell.$$

Alternativity supplies cyclic products on each quaternionic line. The seven lines cover all pairs of different imaginary basis units; squares are -1 and 1 is the unit. This proves the algebra isomorphism on the full multiplication table, hence by bilinearity on all octonions. It also preserves conjugation and the norm. No conjugation of a coordinate of $Q_{\mathbb{O}}$ is performed.

36.2 The explicit eight-dimensional lattice and two sublattices

In the displayed Wilson basis write $e_0 = i_\infty, e_{t+1} = i_t$ and put

$$\begin{aligned} D_8 &= \{(z_0, \dots, z_7) \in \mathbb{Z}^8 : \textstyle\sum z_i \equiv 0 \pmod 2\}, \\ s &= \tfrac{1}{2}(-1 + i_0 + i_1 + i_2 + i_3 + i_4 + i_5 + i_6), \qquad L_{\mathrm{W}} = D_8 + \mathbb{Z}s. \end{aligned}$$

This is Wilson's lattice L, renamed L_{W} to distinguish it from the project's original rank-24 Niemeier lattice. Since $2s \in D_8$ and $s \notin D_8$, it is exactly $D_8 \cup (s + D_8)$.

Lemma 36.1. *The lattice L_{W} is even and unimodular, has minimum norm 2, and has the explicit basis*

$$b_1 = s, \quad b_j = e_{j-1} - e_j \ (2 \le j \le 7), \quad b_8 = e_6 + e_7.$$

Its Gram matrix for the unchanged octonion norm is

$$G_8 = \begin{pmatrix} 2 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 2 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 2 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 2 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 2 & -1 & -1 \\ 0 & 0 & 0 & 0 & 0 & -1 & 2 & 0 \\ 1 & 0 & 0 & 0 & 0 & -1 & 0 & 2 \end{pmatrix}, \qquad \det G_8 = 1. \tag{1467}$$

Its norm-two vectors are precisely the 112 vectors $\pm e_i \pm e_j$ with $i < j$ and the 128 vectors $\frac{1}{2}(\pm e_0 \pm \cdots \pm e_7)$ with an odd number of minus signs.

Proof. For $z \in D_8$, $N(z) \equiv \sum z_i = 0 \pmod 2$ and $h(s, z) = \frac{1}{2}\sum z_i - z_0 \in \mathbb{Z}$. As $N(s) = 2$, every norm in $D_8 + \mathbb{Z}s$ is even. The covolume of D_8 is 2, and its displayed extension has index 2, so L_{W} has covolume 1. The integral bilinear form follows by polarizing the even norm, and covolume one proves self-duality. The columns b_i all belong to L_{W}. Their determinant has absolute value one: expansion along the first coordinate gives $1/2$ times the determinant 2 of the indicated D_7 basis. They therefore form a basis. Their scalar products give the displayed matrix. Positivity and evenness give minimum at least 2, and $N(b_i) = 2$ attains it. Integer vectors of norm two have exactly two nonzero entries ± 1. Half-integer vectors have norm at least $8/4 = 2$, with equality exactly when all eight entries are $\pm 1/2$. Their membership in $s + D_8$ is exactly the stated odd-minus parity. □

Define, with the bars in the positions printed in Wilson,

$$\mathsf{M} = L_{\mathrm{W}}\bar{s}, \qquad \mathsf{N} = L_{\mathrm{W}}s.$$

All products are right products. Let B be the basis matrix with columns b_i. Right multiplication by s in this basis is the following integer matrix:

$$S = B^{-1}R_sB = \begin{pmatrix} 3 & 0 & 0 & 0 & 0 & 0 & 0 & 2 \\ -2 & -1 & 0 & 1 & -1 & 1 & 0 & -2 \\ -4 & -1 & -1 & 1 & 0 & 0 & 1 & -3 \\ -6 & -1 & -1 & 0 & 0 & 1 & 0 & -4 \\ -8 & 0 & -1 & 0 & -1 & 1 & 1 & -5 \\ -10 & -1 & 0 & 0 & -1 & 0 & 1 & -5 \\ -5 & 0 & -1 & 1 & -1 & 0 & 0 & -2 \\ -7 & 0 & 0 & 0 & 0 & 0 & 0 & -4 \end{pmatrix}. \tag{1468}$$

This is a direct multiplication-table calculation using (1466); it can be checked by multiplying each column b_j by the displayed s. Set $T = -I_8 - S$. Since $s + \bar{s} = -1$, $s\bar{s} = 2$, and alternativity allows repeated s and $\bar{s} = -1 - s$, one has

$$T = B^{-1}R_{\bar{s}}B, \qquad S + T = -I_8, \qquad ST = TS = 2I_8, \qquad S^{\mathsf{T}}G_8S = T^{\mathsf{T}}G_8T = 2G_8. \tag{1469}$$

Lemma 36.2. *Both* M, N *are sublattices of* L_{W} *of index* 16 *and minimum norm* 4*, and*

$$2L_{\mathrm{W}} \subset \mathsf{M}, \mathsf{N} \subset L_{\mathrm{W}}, \qquad \mathsf{M} + \mathsf{N} = L_{\mathrm{W}}, \qquad \mathsf{M} \cap \mathsf{N} = 2L_{\mathrm{W}}.$$

Their images $\overline{\mathsf{M}}, \overline{\mathsf{N}}$ *in* $\mathcal{E} = L_{\mathrm{W}}/2L_{\mathrm{W}}$ *are complementary four-dimensional* $\mathbb{F}_2$ *subspaces.*

Proof. The integer matrices S, T give the inclusions in L_{W}; $ST = 2I$ gives the inclusions of $2L_{\mathrm{W}}$. Norm composition shows that each right multiplication scales all squared lengths by $N(s) = N(\bar{s}) = 2$; thus its determinant has absolute value $(\sqrt{2})^8 = 16$, proving the indices and minimum norms. The identity $S + T = -I$ gives $\mathsf{M} + \mathsf{N} = L_{\mathrm{W}}$. Modulo $2L_{\mathrm{W}}$, each image has size $2^8/16 = 2^4$. Their sum is the eight-dimensional space $\mathcal{E}$, so their intersection is zero. This proves the asserted intersection upstairs. □

36.3 The binary glue code and its exact generators

Wilson's original lattice is exactly

$$\Lambda_{\mathrm{W}} = \{(x, y, z) \in L_{\mathrm{W}}^3 : x + y, x + z, y + z \in \mathsf{M}, \quad x + y + z \in \mathsf{N}\}. \tag{1470}$$

It carries Wilson's norm q_{W}, as printed. Define the lattice inside the original triality metric by

$$\Lambda_{\mathrm{cyc}} = \iota(\Lambda_{\mathrm{W}}) \subset (V, h_0).$$

The simultaneous use of $\mathsf{M} = L_{\mathrm{W}}\bar{s}$ for pair sums and $\mathsf{N} = L_{\mathrm{W}}s$ for the total sum is essential.

Write any class in $\mathcal{E}$ uniquely as $m + n$, with $m \in \overline{\mathsf{M}}$ and $n \in \overline{\mathsf{N}}$. Then the image of (1470) in $\mathcal{E}^3$ is the explicit binary code

$$\mathcal{C} = \{(m_1 + n, m_2 + n, m_1 + m_2 + n) : m_1, m_2 \in \overline{\mathsf{M}},\ n \in \overline{\mathsf{N}}\}. \tag{1471}$$

Indeed the pair conditions say that the three N components are equal; the total-sum condition says that the sum of the three M components is zero. This proves both directions of the code description. Its dimension is $4 + 4 + 4 = 12$.

For a concrete lift, the first four columns of S and of T are bases of their images modulo 2. The determinant of the eight-column matrix $(T_1, T_2, T_3, T_4, S_1, S_2, S_3, S_4)$ is -7, so these columns are independent modulo 2; each of the two images already has dimension four. Put $m_j = b_j\bar{s}$, $n_j = b_j s$ for $1 \le j \le 4$. A complete set of 36 integer generators of Λ_{W} is

$$\begin{aligned}&(2b_i,0,0),\ (0,2b_i,0),\ (0,0,2b_i) \quad (1\le i\le 8),\\ &(m_j,m_j,0),\ (m_j,0,m_j),\ (n_j,n_j,n_j) \quad (1\le j\le 4).\end{aligned} \tag{1472}$$

These vectors are all in (1470). Their reductions span precisely (1471); the first 24 generate its full kernel $(2L_{\mathrm{W}})^3$. This proves that they generate the entire lattice, not only a sublattice.

Theorem 36.3. *The explicitly generated lattice Λ_{W} with q_{W}, and therefore Λ_{cyc} with the unaltered h_0, is an even unimodular rank-24 lattice of minimum norm 4. In particular it is rootless. This is the actual Wilson Leech construction, through the explicit isometry* (1464).

Proof. The code proves $[L_{\mathrm{W}}^3 : \Lambda_{\mathrm{W}}] = 2^{24-12} = 2^{12}$. Since L_{W} has covolume one for its original octonion metric, Λ_{W} has covolume 2^{12} for h_0. Changing to the separately recorded metric q_{W} multiplies volume in dimension 24 by $(1/\sqrt{2})^{24} = 2^{-12}$, so its covolume is exactly one.

For every triple of vectors in a real inner product space,

$$N(x)+N(y)+N(z) = N(x+y)+N(x+z)+N(y+z)-N(x+y+z).$$

For a vector of Λ_{W} the four terms on the right are norms in M or N, so each is divisible by 4: these are twice norms in the even lattice L_{W}. It follows that $q_{\mathrm{W}}(x,y,z)$ is even. Polarization of these even norms gives an integral bilinear form. An integral lattice of covolume one equals its dual, proving unimodularity.

If a nonzero vector had $q_{\mathrm{W}} = 2$, its three coordinate norms would sum to 4. Each nonzero coordinate in L_{W} has norm at least 2, so one coordinate is zero. The other two then lie in M, each of whose nonzero vectors has norm at least 4. At most one can be nonzero. That coordinate also lies in N by the total-sum condition, so it lies in $\mathsf{M} \cap \mathsf{N} = 2L_{\mathrm{W}}$ and has norm at least 8, a contradiction. Hence all nonzero norms are at least 4. The vectors $(2b_i, 0, 0)$ have $q_{\mathrm{W}} = 4$, so the minimum is exactly 4. The isometry ι transfers every statement to the original metric. □

36.4 Cyclic symmetry, the fixed lattice, and exact three-primary glue

Retain the canonical cycle $c(x,y,z) = (z,x,y)$. Denote the opposite cycle used in Wilson coordinates by $c_{\mathrm{W}}(x,y,z) = (y,z,x)$. Direct substitution gives

$$c_{\mathrm{W}} = c^{-1} = c^2, \qquad c_{\mathrm{W}}^2 = c, \qquad \langle c_{\mathrm{W}}\rangle = \langle c\rangle.$$

Both actions are retained with these directions; no action is renamed by silently reversing it. The defining three coordinate memberships, three pair conditions, and total-sum condition are all unchanged by c_{W}; hence both cycles preserve Λ_{W} and Λ_{cyc}. They preserve their metrics and the original cubic:

$$\mathrm{Re}((xy)z) = \mathrm{Re}((yz)x).$$

This follows from the octonion adjoint identities $h(ab,d) = h(b,\bar{a}d) = h(a,d\bar{b})$, which imply real associativity, and $\mathrm{Re}(ab) = \mathrm{Re}(ba)$. In the Albert algebra c_{W} is realized by conjugation with the *real* permutation matrix for (1 3 2). It sends the coordinates (x_1,x_2,x_3) to (x_2,x_3,x_1) with no conjugations, and permutes the ordered diagonal frame accordingly. The inverse real permutation matrix, for (1 2 3), realizes the canonical c and sends (x_1,x_2,x_3) to (x_3,x_1,x_2). Thus the fixed lattices of the two cycles are identical. For the retained residue isometry $\mathcal{L}r = c\mathcal{L}$, the opposite action satisfies $\mathcal{L}r^{-1} = c_{\mathrm{W}}\mathcal{L}$.

Theorem 36.4. *Before applying ι, the cyclic fixed lattice and its orthogonal complement in Λ_{W} are exactly*

$$F = \{(n,n,n) : n \in \mathsf{N}\}, \qquad C = \{(m_1, m_2, -m_1 - m_2) : m_1, m_2 \in \mathsf{M}\}.$$

Their ranks are 8 and 16. For the retained Wilson metric, and therefore after ι for the original triality metric, their Gram matrices in the displayed bases are

$$G_F = 3G_8, \qquad G_C = \begin{pmatrix} 2G_8 & G_8 \\ G_8 & 2G_8 \end{pmatrix}. \tag{1473}$$

The fixed-lattice basis is $(b_j s, b_j s, b_j s)$, $1 \le j \le 8$; the coinvariant bases are $(b_j \bar{s}, 0, -b_j \bar{s})$ and $(0, b_j \bar{s}, -b_j \bar{s})$. In particular the fixed lattice has minimum norm 6 and determinant 3^8. The coinvariant lattice has determinant 3^8, and the map which negates its second L_{W} parameter identifies its displayed Gram matrix with the usual $A_2 \otimes E_8$ Gram matrix.

Proof. A fixed triple has the form (x,x,x). Its pair condition holds for every $x \in L_{\mathrm{W}}$ because $2L_{\mathrm{W}} \subset \mathsf{M}$. Its total condition is $3x \in \mathsf{N}$. As $L_{\mathrm{W}}/\mathsf{N}$ has exponent two, $3x \in \mathsf{N}$ is equivalent to $x \in \mathsf{N}$. This proves the first description. Orthogonality to the whole diagonal real eight-space says $x + y + z = 0$. The pair conditions are then exactly $x, y, z \in \mathsf{M}$, and the total condition is automatic. This proves the description of C.

Right multiplication by either s or $\bar{s}$ multiplies the octonion metric by 2. Thus

$$q_{\mathrm{W}}(as, as, as) = 3N(a),$$

and

$$q_{\mathrm{W}}(a\bar{s}, b\bar{s}, -(a+b)\bar{s}) = N(a) + N(b) + N(a+b) = 2N(a) + 2N(b) + 2h(a,b).$$

Polarization gives both full Gram matrices. Their determinants are

$$\det G_F = 3^8 \det G_8 = 3^8, \qquad \det G_C = \det \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}^8 (\det G_8)^2 = 3^8.$$

Minimum norm 6 for F follows from the exact factor 3 and the minimum 2 in L_{W}. The sign map $b \mapsto -b$ changes both off-diagonal blocks from G_8 to $-G_8$, proving the stated exact comparison to $A_2 \otimes E_8$. □

There is a complete integral comparison, including the glue:

$$0 \longrightarrow F \oplus C \longrightarrow \Lambda_{\mathrm{W}} \xrightarrow{\Sigma} \mathsf{N}/3\mathsf{N} \longrightarrow 0, \qquad \Sigma(x,y,z) = x + y + z \pmod{3\mathsf{N}}. \tag{1474}$$

To prove surjectivity, for every $a \in L_{\mathrm{W}}$ the explicit vector

$$v(a) = (as, a, -a)$$

lies in Λ_{W} and has sum as. Its pair sums are $a(s+1) = -a\bar{s} \in \mathsf{M}$, $a(s-1) = -a\bar{s} - 2a \in \mathsf{M}$, and 0; its total lies in N. If a lattice vector has total $3n$, subtracting $(n,n,n) \in F$ leaves a vector of C. This proves the kernel assertion. Thus

$$\Lambda_{\mathrm{W}}/(F \oplus C) \cong \mathsf{N}/3\mathsf{N} \cong L_{\mathrm{W}}/3L_{\mathrm{W}} \cong (\mathbb{Z}/3)^8, \qquad [\Lambda_{\mathrm{W}} : F \oplus C] = 3^8.$$

All maps commute with c_{W} and with its inverse c; both cycles act trivially on the quotient.

The discriminant anti-isometry is also explicit. Use the quadratic discriminant convention $q_D(u+D) = h_D(u,u) \pmod{2\mathbb{Z}}$ for an even lattice D with its retained metric. The orthogonal projections of the lift $v(a)$ are

$$\begin{aligned} f(a) &= \tfrac{1}{3}(as, as, as), \\ \gamma(a) &= \left(\tfrac{2}{3}as,\ a - \tfrac{1}{3}as,\ -a - \tfrac{1}{3}as\right). \end{aligned} \tag{1475}$$

They belong to F^* and C^*, respectively: their pairings with F or C are the pairings of the integral vector $v(a)$ with the same lattice, because the other projection is orthogonal. The classes $f(a)+F$ identify $L_{\mathrm{W}}/3L_{\mathrm{W}}$ with F^*/F. If $\gamma(a)\in C$, then $f(a)=v(a)-\gamma(a)\in\Lambda_{\mathrm{W}}\cap F_{\mathbb{R}}=F$, so $a\in 3L_{\mathrm{W}}$. Since C^*/C has order $\det G_C=3^8$, the map

$$f(a)+F\longmapsto\gamma(a)+C$$

is an isomorphism of the two discriminant groups. Its quadratic sign is determined without an unspecified convention:

$$q_F(f(a))=\tfrac{1}{3}N(a),\qquad q_C(\gamma(a))=q_{\mathrm{W}}(v(a))-\tfrac{1}{3}N(a)=2N(a)-\tfrac{1}{3}N(a).$$

As $N(a)$ is even, their sum $2N(a)$ is zero modulo $2\mathbb{Z}$. This is the required anti-isometry. Equations (1474) and (1475) specify the overlattice itself, including its actual coset representatives.

36.5 Finite lattice symmetries and exact cubic tests

Independent sign changes of the three sectors preserve Λ_{W}. Changing the sign of x changes a pair sum by $-2x\in 2L_{\mathrm{W}}\subset\mathsf{M}$ and changes the total by $-2x\in 2L_{\mathrm{W}}\subset\mathsf{N}$. The same argument applies to each sector. All six sector permutations preserve the defining conditions too. Under signs $(\epsilon_1,\epsilon_2,\epsilon_3)$ the cubic τ is multiplied by $\epsilon_1\epsilon_2\epsilon_3$; hence exactly the even sign changes in this subgroup preserve it. Cyclic permutations preserve τ as already proved.

The odd permutation $(x,y,z)\mapsto(y,x,z)$ need not preserve it, even on the lattice itself. In Wilson's basis put

$$a=2(1+i_0),\qquad b=2(1+i_1),\qquad d=2(i_2+i_3).$$

Every coordinate belongs to $2L_{\mathrm{W}}$, so $(a,b,d)\in\Lambda_{\mathrm{W}}$. The real part of $((1+i_0)(1+i_1))(i_2+i_3)$ is -1, because $i_0i_1=i_3$ and the i_3^2 term is the only real contribution. Reversing the first two factors changes this contribution to $+1$. Consequently

$$\tau(a,b,d)=-8,\qquad\tau(b,a,d)=8.\tag{1476}$$

After the specified scalar isometry the original determinant cubic has values $-4\sqrt{2}$ and $4\sqrt{2}$.

The simultaneous Fano-index operations

$$A:i_t\longmapsto i_{t+1},\qquad B_0:i_t\longmapsto i_{2t},\qquad i_\infty\longmapsto i_\infty$$

are octonion algebra automorphisms, as the defining oriented Fano products show. They are orthogonal permutations preserving D_8 and the odd-minus half coset, and they fix the displayed s and $\bar{s}$. They consequently preserve $\mathsf{M},\mathsf{N},\Lambda_{\mathrm{W}}$. Applied simultaneously to all three sectors, they preserve τ. They satisfy $A^7=B_0^3=1$ and $B_0AB_0^{-1}=A^2$; their permutations are the distinct maps $t\mapsto 2^jt+k$ with $j=0,1,2$ and $k\in\mathbb{F}_7$. Thus they form an explicitly realized $C_7\rtimes C_3$ of order 21.

The even sign changes and the cycle c_{W} form $(\mathbb{Z}/2)^2\rtimes C_3\cong A_4$. They commute with the simultaneous Fano operations, and their intersection is trivial: a simultaneous Fano operation fixes the real unit in every sector and does not permute sectors. Hence an explicit finite group

$$A_4\times(C_7\rtimes C_3)$$

of order 252 preserves both the literal lattice and the original cubic. This is a proved subgroup; no assertion that it is the full cubic stabilizer is needed.

Wilson also gives the lattice symmetries

$$r_t(x,y,z)=(x,yi_t,zi_t)\qquad(t=0,\dots,6).$$

For completeness their lattice action can be checked solely from the explicit finite matrices above. Let $U_t = B^{-1}R_{i_t}B$. The signed permutation multiplication table and (1468) give

$$U_t \in GL_8(\mathbb{Z}), \qquad T^{-1}(U_t - I_8) \in M_8(\mathbb{Z}), \qquad \tfrac{1}{2}(U_t - I_8)T \in M_8(\mathbb{Z}) \quad (0 \le t \le 6). \tag{1477}$$

These are finite exact matrix assertions, not assumptions about octonionic matrix associativity. The full multiplication rule and the exact rational algorithm checking every entry for all seven values are included in the accompanying `check_wilson.py`. Concretely, start with the 64 products $1i_t = i_t 1 = i_t$, $i_t^2 = -1$, and the seven oriented Fano triples, form the eight columns $R_{i_t}(e_j)$, conjugate by the explicitly displayed B, and substitute the integer $T = -I_8 - S$. For each of the three matrices in (1477) the reduced denominators of every entry are 1, for each $t = 0, 1, 2, 3, 4, 5, 6$. Also $\det U_t = 1$. This is a reproducible $7 \cdot 3 \cdot 64$-entry certificate. The first condition is also immediate from the signed permutation: right multiplication by i_t flips four coordinate signs, preserving the odd-minus parity of L_{W}.

The second matrix assertion says $L_{\mathrm{W}}(i_t - 1) \subset \mathsf{M}$, and the third says $\mathsf{M}(i_t - 1) \subset 2L_{\mathrm{W}}$. Thus $x + yi_t$ and $x + zi_t$ differ from the original pair sums by elements of M. The pair $(y + z)i_t$ remains in M, and the transformed total differs from $x + y + z$ by $(y + z)(i_t - 1) \in 2L_{\mathrm{W}} \subset \mathsf{N}$. All defining conditions are preserved. The inverse follows from $r_t^2(x, y, z) = (x, -y, -z)$ and the proved sign symmetries. Norm composition proves that r_t is an isometry.

Nevertheless none of these seven r_t preserves the original cubic. For $a_t = 2(1 + i_t)$, the triple (a_t, a_t, a_t) lies in $(2L_{\mathrm{W}})^3 \subset \Lambda_{\mathrm{W}}$. All products below occur in the associative real subalgebra generated by i_t, and give

$$\tau(a_t, a_t, a_t) = 8\,\mathrm{Re}((1 + i_t)^3) = -16,$$

whereas $(1 + i_t)i_t = -1 + i_t$ and

$$\tau(a_t, a_t i_t, a_t i_t) = 8\,\mathrm{Re}((1 + i_t)(-1 + i_t)^2) = 16.$$

Therefore the original determinant cubic on the isometric lattice Λ_{cyc} changes from $-8\sqrt{2}$ to $8\sqrt{2}$. This supplies actual lattice vectors as counterexamples, not only ambient real vectors outside the lattice.

36.6 Exact comparison with coordinate lattices and the marked lane

The construction has the explicit integral sandwich

$$(\sqrt{2}L_{\mathrm{W}})^3 \;\subset\; \Lambda_{\mathrm{cyc}} \;\subset\; (L_{\mathrm{W}}/\sqrt{2})^3, \qquad [\Lambda_{\mathrm{cyc}} : (\sqrt{2}L_{\mathrm{W}})^3] = [(L_{\mathrm{W}}/\sqrt{2})^3 : \Lambda_{\mathrm{cyc}}] = 2^{12}. \tag{1478}$$

This is (1471) transported by the exact isometry (1464). The inclusions are literal inclusions in the fixed three octonion sectors, and each commutes with both c_{W} and the canonical $c = c_{\mathrm{W}}^{-1}$. Together with the explicit bases, the code, and the three-primary glue (1475), it determines actual commensurability maps and cosets rather than only an abstract isometry class.

Under (1466), the single negative column takes the Wilson odd-minus half roots to even-minus half roots in the current ordered Cayley–Dickson basis. The element s remains exactly

$$s = \tfrac{1}{2}(-1 + \mathsf{i} + \mathsf{j} + \mathsf{k} + \ell + \mathsf{i}\ell + \mathsf{j}\ell - \mathsf{k}\ell),$$

with coordinate vector $(-1, 1, 1, 1, 1, 1, 1, -1)/2$. The signed permutation has determinant $+1$ and all 64 basis products agree, as independently checked in the accompanying code. Thus the lattice, its cyclic action, and the triality cubic live in one specified multiplication convention.

36.7 Literal intersections with both retained lattice embeddings

Let $U = \mathrm{span}_{\mathbb{R}}(1, \mathsf{i}, \mathsf{j}, \mathsf{k})$ be the first four coordinates in the *current* ordered Cayley–Dickson basis, and let

$$D_4 = \{a = (a_0, a_1, a_2, a_3) \in \mathbb{Z}^4 : a_0 + a_1 + a_2 + a_3 \in 2\mathbb{Z}\} \subset U.$$

This quaternionic D_4 denotes the displayed four-coordinate lattice; the original Niemeier D summand below remains in its original place.

Let J_{W} denote the signed permutation matrix in (1466). If $U_0 : \mathbb{R}^4 \to \mathbb{R}^8$ is the coordinate inclusion in the current basis, then

$$S^{-1}B^{-1}J_{\mathrm{W}}^{-1}U_0 = \begin{pmatrix} 1/2 & -1/2 & -1/2 & -1/2 \\ -1/2 & 0 & 0 & 1/2 \\ -1 & 1/2 & 0 & 1/2 \\ -3/2 & 1 & 1/2 & 1/2 \\ -2 & 1 & 1 & 1/2 \\ -5/2 & 3/2 & 1 & 1 \\ -5/4 & 3/4 & 3/4 & 3/4 \\ -7/4 & 3/4 & 3/4 & 3/4 \end{pmatrix}. \tag{1479}$$

All matrices on the left have already been explicitly specified. Thus the equation is a direct rational multiplication certificate.

Lemma 36.5. *In this fixed coordinate frame,*

$$\mathsf{N} \cap U = 2D_4.$$

Proof. Since $\mathsf{N} \subset L_{\mathrm{W}}$, an element of $\mathsf{N} \cap U$ has integral first four coordinates: the half-integer coset of the current E_8 description has no zero coordinate among its last four. Write those integral coordinates as k_0, k_1, k_2, k_3. Membership in N is exactly integrality of the matrix (1479) applied to k. Subtracting its last two rows gives $k_0/2$, so k_0 is even. The second row then forces k_3 even, the third forces k_1 even, and the fourth forces k_2 even. Write $k = 2a$. The first six rows are now integral. The seventh and eighth rows are integral precisely when $-5a_0 + 3a_1 + 3a_2 + 3a_3$ is even, equivalently $a_0 + a_1 + a_2 + a_3$ is even. Conversely these conditions make all eight rows integral. This proves both inclusions with no unstated lattice identification. □

We now retain the original lattices and their actual map $\mathcal{L}$. The original coordinates are $X = (x_{i,j}; d_k)$, $1 \le i \le 4$, $0 \le j \le 5$, $1 \le k \le 4$, with $\sum_j x_{i,j} = 0$. The root lattice is $R = A_5^{\perp 4} \perp D$, with its unchanged Euclidean metric. Its index-72 overlattice N is the inverse image of the original glue

$$(k_1, k_2, k_3, k_4; y) = \big(3u + 2(a+b, a+2b, a, b);\ (u_2 + u_4, u_3 + u_4)\big),$$

where $u \in \mathbb{F}_2^4$ has even weight and $a, b \in \mathbb{F}_3$; the first four entries are modulo 6. Here $k_i = -6x_{i,0} \pmod 6$ and y is the original $D^\vee/D$ marking, so $d = 0$ has $y = 00$. Retain the original neighbour

$$\rho_A = (5, 3, 1, -1, -3, -5)/2, \quad \rho_D = (3, 2, 1, 0), \quad \rho = \rho_A^{\oplus 4} \perp \rho_D,$$

$$K = \{X \in N : (\rho, X) \in 6\mathbb{Z}\}, \qquad \alpha = (e_0 - e_1, 0, 0, 0; 0), \qquad \Lambda = K + \mathbb{Z}(\rho/6 - \alpha).$$

Thus $\rho \in N$, $\rho^2 = 84$, $(\rho, \alpha) = 1$. Every vector of N or Λ has rational original coordinates.

For exact use of the retained isometry, set

$$\sigma_i = \sum_{j=0}^{5} (-1)^j x_{i,j}, \qquad t_{i,p,j} = x_{i,p+2j} - \frac{1}{3}\sum_{h=0}^{2} x_{i,p+2h}.$$

Write $u_0 = 1, u_1 = \mathsf{i}, u_2 = \mathsf{j}, u_3 = \mathsf{k}$ and retain the remaining current Cayley–Dickson basis vectors $u_4, \dots, u_7$. The already specified map has the exact coordinate form

$$(\mathcal{L}X)_{j+1,a} = \begin{cases} t_{i,p,j} + \sqrt{2}\,\sigma_{a+1}/6, & 0 \le a \le 3, \\ t_{i,p,j} + \sqrt{3}\,d_{a-3}/3, & 4 \le a \le 7, \end{cases} \qquad a = 2(i-1) + p.$$

The placement of σ_{a+1} is retained, including its difference from the index i of the residual. No coordinate is reassigned.

Theorem 36.6. *The literal intersections in the original metric space are exactly*

$$\Lambda_{\mathrm{cyc}} \cap \mathcal{L}(N) = \Lambda_{\mathrm{cyc}} \cap \mathcal{L}(\Lambda) = \mathcal{I} := \{\sqrt{2}(a,a,a) : a \in D_4 \subset U\}. \tag{1480}$$

In original Niemeier coordinates the inverse image of this common lattice is

$$\mathcal{L}^{-1}(\mathcal{I}) = \{x_{i,j} = (-1)^j a_{i-1},\ d = 0 :\ a \in D_4\} \subset K.$$

It has rank four, minimum norm 12*, and Gram matrix*

$$G_{\mathcal{I}} = 6\begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & -1 \\ 0 & -1 & 2 & 0 \\ 0 & -1 & 0 & 2 \end{pmatrix}, \qquad \det G_{\mathcal{I}} = 4 \cdot 6^4 = 5184. \tag{1481}$$

An explicit basis is $\sqrt{2}(\delta_j, \delta_j, \delta_j)$*, where* $\delta_1 = (1,-1,0,0)$*,* $\delta_2 = (0,1,-1,0)$*,* $\delta_3 = (0,0,1,-1)$*,* $\delta_4 = (0,0,1,1)$*.*

Proof. Each coordinate of Λ_{cyc} belongs to $\mathbb{Q}/\sqrt{2} = \sqrt{2}\mathbb{Q}$ in the current frame. For a rational original vector X, the first four coordinate formulas above therefore force $t_{i,p,j} = 0$ at the corresponding positions, by linear independence of $1, \sqrt{2}$ over $\mathbb{Q}$. The last four force both $t_{i,p,j} = 0$ and $d_{a-3} = 0$, by linear independence of $1, \sqrt{2}, \sqrt{3}$. For the latter independence, a relation with nonzero $\sqrt{3}$ coefficient would place $\sqrt{3}$ in $\mathbb{Q}(\sqrt{2})$; squaring $A + B\sqrt{2}$ forces $AB = 0$ and then would make 3 or 3/2 a rational square, a contradiction. All residuals therefore vanish and $d = 0$. The zero fibre sums give

$$x_{i,j} = (-1)^j \sigma_i/6.$$

First let $X \in N$. Its $A_5^\vee$ integral-difference conditions are exactly $k_i := \sigma_i/3 \in \mathbb{Z}$, yielding $x_{i,j} = (-1)^j k_i/2$. Their discriminant labels are $-3k_i = 3k_i \pmod 6$. Their three-primary parts are zero, so the injectivity of $(a,b) \mapsto (a+b, a+2b, a, b)$ forces $a = b = 0$. Since $d = 0$, the binary condition is $u_2 + u_4 = u_3 + u_4 = 0$ and $\sum u_i = 0$. These equations give $u_1 = u_2 = u_3 = u_4$. Thus all four k_i have the same parity. Conversely these conditions satisfy every original glue requirement. The image is the diagonal triple

$$\mathcal{L}X = (k/\sqrt{2}, k/\sqrt{2}, k/\sqrt{2}), \qquad k \in U.$$

By the fixed-lattice description of Wilson's construction it belongs to Λ_{cyc} exactly when $k \in \mathsf{N}$. The previous lemma makes this equivalent to $k = 2a$, $a \in D_4$, which proves the first intersection equality.

For the second equality, a vector of Λ can be written $X = n + m\rho/6$ with $n \in N$, $m \in \mathbb{Z}$. The already proved necessary condition $d_X = 0$ gives $d_n = -m(3,2,1,0)/6 \in D^\vee$. All coordinates of a vector of $D^\vee$ are integral or all half-integral; the fourth coordinate is zero, so all are integral. Its third coordinate forces $6 \mid m$, and hence $X \in N$. In fact $\Lambda \cap N = K$: in $k + m(\rho/6 - \alpha) \in N$ the pairing with the original root α forces $6 \mid m$, and $(\rho, \rho - 6\alpha) = 84 - 6 = 78 \in 6\mathbb{Z}$. Thus an intersection vector belongs to K. For the diagonal-fibre vectors just found,

$$(\rho, X) = \frac{3}{2}\sum_{i=1}^{4} k_i.$$

Consequently the condition $X \in K$ is $\sum k_i \equiv 0 \pmod 4$. Every $k = 2a$ with $a \in D_4$ satisfies it, proving equality of the two intersections and the displayed inverse image.

Finally $h_0(\sqrt{2}(a,a,a), \sqrt{2}(b,b,b)) = 6h(a,b)$. The four δ_j form a basis of D_4, since their determinant has absolute value two and D_4 has index two in $\mathbb{Z}^4$. Their scalar products give (1481). The least nonzero norm in D_4 is two: a norm-one integer vector has odd coordinate sum, while $(1,-1,0,0)$ has norm two. This proves the exact minimum and determinant. □

The two slice indices are also literal. Define the common four-dimensional real subspace $\mathcal{U} = \{(u,u,u) : u \in U\}$. The proof gives

$$\mathcal{L}(N) \cap \mathcal{U} = \{(k,k,k)/\sqrt{2} : k \in P\}, \quad P = 2\mathbb{Z}^4 + \mathbb{Z}(1,1,1,1),$$
$$\mathcal{L}(\Lambda) \cap \mathcal{U} = \{(k,k,k)/\sqrt{2} : k \in P_0\}, \quad P_0 = \{k \in P : \textstyle\sum k_i \equiv 0 \pmod 4\} = 2D_4 + \mathbb{Z}(1,1,1,1).$$

The first proof does not need Wilson membership to obtain these slice equations. The lattice P has basis $2e_0, 2e_1, 2e_2, (1,1,1,1)$ and covolume 8. The lattice $2D_4$ has covolume $2^4 \cdot 2 = 32$, so $[P : 2D_4] = 4$. The class of $(1,1,1,1)$ has exact order two modulo $2D_4$, yielding $[P_0 : 2D_4] = 2$. Therefore

$$[\mathcal{L}(N) \cap \mathcal{U} : \mathcal{I}] = 4, \qquad [\mathcal{L}(\Lambda) \cap \mathcal{U} : \mathcal{I}] = 2.$$

The rank-four intersection, strictly below rank 24, also proves that these full lattices are not commensurable *in the retained embedding*: a common finite-index subgroup would have full rank. Their exact existing relation is the common lattice with the explicit embeddings, inverse map, metrics, and indices just proved.

36.8 The cubic value group on the literal common lattice

Retain $F(y) = \tau(y_1, y_2, y_3)$, so the Albert off-diagonal determinant is $2F$. For $y = \sqrt{2}(a,a,a) \in \mathcal{I}$ write $a = a_0 + \mathbf{a}$ in the same quaternionic U. Since $\mathbf{a}^2 = -N(\mathbf{a})$ and a_0 is real, direct multiplication inside this associative subalgebra gives

$$\mathrm{Re}(a^3) = a_0^3 - 3a_0 N(\mathbf{a}) = 4a_0^3 - 3a_0 N(a).$$

Every scalar and factor is therefore

$$F(\sqrt{2}(a,a,a)) = 2\sqrt{2}(4a_0^3 - 3a_0 N(a)). \tag{1482}$$

For $a \in D_4$, $N(a)$ is even, because $N(a) \equiv \sum a_i = 0 \pmod 2$. The integer in parentheses is therefore even, so every value belongs to $4\sqrt{2}\mathbb{Z}$. For the actual lattice vector $a = (1,1,0,0)$ it equals $4 - 3 \cdot 2 = -2$, giving $F = -4\sqrt{2}$. Consequently the *additive subgroup generated by all values* is exactly

$$\langle F(\mathcal{I}) \rangle_{\mathbb{Z}} = 4\sqrt{2}\mathbb{Z}, \qquad \langle d(Q_{\mathbb{O}}(0;\mathcal{I})) \rangle_{\mathbb{Z}} = 8\sqrt{2}\mathbb{Z}.$$

This assertion concerns the generated additive groups; it does not claim that every element of either group is an individually attained cubic value. For an explicit distinction, put $P(a) = a_0(a_0^2 - 3(a_1^2 + a_2^2 + a_3^2))$. If $3 \mid P(a)$, reduction modulo three forces $3 \mid a_0$, and substitution $a_0 = 3b$ gives $P(a) = 9b(3b^2 - a_1^2 - a_2^2 - a_3^2)$. Thus $3 \mid P(a)$ implies $9 \mid P(a)$. The value $P(a) = 6$ is impossible, so $F = 12\sqrt{2}$ is not attained, although it belongs to the proved additive value group.

36.9 Primary source and exact reproducibility data

Wilson's octonionic construction is the source of the pair and total conditions [81, Sections 2–4, pp. 2–6]. His page 2 uses the odd-minus lattice L and $R = \bar{L}$; page 3 gives pair sums in $L\bar{s}$, the total in Ls, and the metric equal to one half of the sum of octonion norms. The displayed basis map and (1464) retain these conjugations and the exact scale in the present coordinates. The distinct coordinate change on Wilson's page 6 is not substituted for this isometry.

The complete integer-arithmetic program and its output are supplied in the following directory (the two displayed lines form one path):

```
supporting_materials/workbench/evidence/
lattice_foundation_20260906/wilson/
```

The files are `check_wilson.py` and `EXACT_CHECKS.json`. They record the generating-lattice index 4096, Gram determinant 1, even diagonal, integral cyclic action, the fixed and coinvariant Gram matrices, and the index 6561 of their sum. The same program checks all 64 multiplication-table products for (1466) and the stated exact cubic tests. The proofs above establish rootlessness and the global descriptions; these conclusions are not inferred from a finite search.

37 The exact cubic image of the original common lattice

Retain the coordinates a_0, a_1, a_2, a_3 and the original lattice

$$D_4 = \{a \in \mathbb{Z}^4 : a_0 + a_1 + a_2 + a_3 \in 2\mathbb{Z}\}. \tag{D4I1}$$

The polynomial already used in section 36.8 is

$$P(a) = a_0(a_0^2 - 3(a_1^2 + a_2^2 + a_3^2)). \tag{D4I2}$$

Here and throughout this section F is the original triality cubic $F = \tau$, with its ordered octonionic arguments as in sections 36 and 36.8. In particular, on the literal common lattice of theorem 36.6,

$$\mathcal{I} = \{\sqrt{2}(a, a, a) : a \in D_4\}, \qquad F(\sqrt{2}(a, a, a)) = 2\sqrt{2}\, P(a). \tag{D4I3}$$

No coordinate permutation, scaling of P, or alteration of the lattice is made. The witnesses below are actual points of this original lattice. The image proof is elementary and requires no external arithmetic existence theorem.

37.1 Necessary restrictions in the original coordinates

For every integer b, one has $b^2 \equiv b \pmod 2$. Consequently, for $a \in D_4$,

$$a_0^2 + a_1^2 + a_2^2 + a_3^2 \equiv a_0 + a_1 + a_2 + a_3 \equiv 0 \pmod 2.$$

Writing $s = a_1^2 + a_2^2 + a_3^2$, the factor $a_0^2 - 3s$ is congruent to $a_0^2 + s \equiv 0 \pmod 2$. Thus $P(a)$ is even.

Modulo three the original polynomial satisfies $P(a) \equiv a_0^3$. If $3 \mid P(a)$, the field $\mathbb{Z}/3\mathbb{Z}$ has no nonzero element with zero cube, so $3 \mid a_0$. Substitution of $a_0 = 3b$ into the original formula gives

$$P(a) = 3b(9b^2 - 3s) = 9b(3b^2 - s). \tag{D4I4}$$

Therefore

$$P(a) \in 2\mathbb{Z}, \qquad 3 \mid P(a) \implies 9 \mid P(a). \tag{D4I5}$$

37.2 Three explicit affine lines and their exact values

Define the common direction

$$v = (-3, 1, 1, 1). \tag{D4I6}$$

Its coordinate sum is zero, and direct substitution gives

$$P(v) = -3\big(9 - 3(1 + 1 + 1)\big) = 0.$$

For every $r \in \mathbb{Z}$ and $\varepsilon \in \{-1, 1\}$, put

$$L_\varepsilon(r) = (-3r - \varepsilon, r, r, r + \varepsilon) = rv + (-\varepsilon, 0, 0, \varepsilon). \tag{D4I7}$$

Its coordinate sum is $-3r - \varepsilon + r + r + r + \varepsilon = 0$, so it belongs to D_4. The original last-three-coordinate square sum and first-coordinate square are respectively

$$s = r^2 + r^2 + (r + \varepsilon)^2 = 3r^2 + 2r\varepsilon + 1, \qquad (-3r - \varepsilon)^2 = 9r^2 + 6r\varepsilon + 1.$$

Thus the factor in the original cubic is exactly

$$a_0^2 - 3s = (9r^2 + 6r\varepsilon + 1) - (9r^2 + 6r\varepsilon + 3) = -2,$$

and therefore

$$P(L_\varepsilon(r)) = (-3r-\varepsilon)(-2) = 6r+2\varepsilon. \tag{D4I8}$$

For every $h \in \mathbb{Z}$, define the third line

$$L_0(h) = (-3h, h+1, h-1, h) = hv + (0,1,-1,0). \tag{D4I9}$$

Its coordinate sum is zero, so it also belongs to D_4. Here

$$s = (h+1)^2 + (h-1)^2 + h^2 = 3h^2 + 2,$$

and consequently

$$P(L_0(h)) = (-3h)(9h^2 - 3(3h^2+2)) = (-3h)(-6) = 18h. \tag{D4I10}$$

These identities hold for all signed integer parameters. In particular, $L_0(0) = (0,1,-1,0)$ has cubic value zero, and the zero vector also has cubic value zero.

37.3 The global image and a proved reconstruction map

Set

$$S = \{n \in \mathbb{Z} : n \in 2\mathbb{Z} \text{ and } (3 \nmid n \text{ or } 9 \mid n)\}. \tag{D4I11}$$

Theorem 37.1 (Complete cubic image and explicit section)**.** *The original polynomial has the exact image*

$$P(D_4) = 18\mathbb{Z} \ \cup \ (6\mathbb{Z}+2) \ \cup \ (6\mathbb{Z}-2) = S. \tag{D4I12}$$

Equivalently,

$$P(D_4) = \{n \in \mathbb{Z} : n \bmod 18 \in \{0,2,4,8,10,14,16\}\}. \tag{D4I13}$$

The following map is an actual section $\sigma : S \to D_4$:

$$\sigma(n) = \begin{cases} (-3h, h+1, h-1, h), & n = 18h, \\ (-3r-\varepsilon, r, r, r+\varepsilon), & n = 2(3r+\varepsilon), \quad \varepsilon \in \{-1,1\}. \end{cases} \tag{D4I14}$$

Its displayed parameters are uniquely determined in their respective cases, and $P \circ \sigma = \mathrm{id}_S$.

Proof. Equation (D4I5) proves $P(D_4) \subseteq S$. For the reverse inclusion let $n \in S$. If $3 \mid n$, then $9 \mid n$, and evenness gives $n = 18h$ for a unique $h \in \mathbb{Z}$. Equation (D4I10) gives $P(L_0(h)) = n$.

If $3 \nmid n$, write $n = 2q$. Then $3 \nmid q$, so there is a unique $\varepsilon \in \{-1,1\}$ with $q \equiv \varepsilon \pmod 3$. The number $r = (q-\varepsilon)/3$ is consequently a unique integer. Equation (D4I8) gives

$$P(L_\varepsilon(r)) = 6r + 2\varepsilon = 2q = n.$$

The two displayed cases are disjoint and cover S, and both witnesses lie in D_4 by section 37.2. These calculations prove the asserted section identity and both inclusions in (D4I12). The first arithmetic progression contributes residue zero modulo eighteen; the second contributes $2, 8, 14$; and the third contributes $16, 4, 10$. This proves (D4I13). □

The classification includes every positive, zero, and negative value. The full excluded set is the union of the odd integers with $18\mathbb{Z}+6$ and $18\mathbb{Z}+12$. Let

$$A = \{a \in \mathbb{Z}^4 : a_0 + a_1 + a_2 + a_3 = 0\} \tag{D4I15}$$

be the literal sum-zero sublattice in the original coordinates. The three witnesses lie in $A \subseteq D_4$, so the two inclusions $S \subseteq P(A) \subseteq P(D_4) = S$ prove

$$P(A) = P(D_4) = S. \tag{D4I16}$$

The image of the original cubic $F = \tau$ on the common lattice is therefore exactly

$$\begin{aligned} F(\mathcal{I}) &= 2\sqrt{2}\, S \\ &= 4\sqrt{2}((9\mathbb{Z}) \cup (3\mathbb{Z}+1) \cup (3\mathbb{Z}-1)). \end{aligned} \tag{D4I17}$$

Because the original off-diagonal Albert determinant is $2F$, its corresponding image is $4\sqrt{2}\, S$. The additive group generated by that determinant image is $8\sqrt{2}\mathbb{Z}$, while the additive group generated by $F(\mathcal{I})$ is $4\sqrt{2}\mathbb{Z}$. Indeed every P value is even, and the first two lines at parameter zero attain $P = 2$ and $P = -2$. Thus the generated additive group of the P values is exactly $2\mathbb{Z}$; applying the two original scalar factors proves both group assertions without identifying an image set with its generated group.

37.4 Exact local images with explicit local sections

For a prime p, let

$$D_{4,p} = \{a \in \mathbb{Z}_p^4 : a_0 + a_1 + a_2 + a_3 \in 2\mathbb{Z}_p\}. \tag{D4I18}$$

This is precisely $D_4 \otimes_{\mathbb{Z}} \mathbb{Z}_p$ in the original coordinates. To verify the equality directly, use the integer basis

$$\begin{aligned} b_0 &= (1,-1,0,0), \quad b_1 = (0,1,-1,0), \\ b_2 &= (0,0,1,-1), \quad b_3 = (0,0,1,1). \end{aligned} \tag{D4I19}$$

For any a in the displayed definition of $D_{4,p}$, the unique coefficients of these basis vectors are

$$\begin{aligned} c_0 &= a_0, & c_1 &= a_0 + a_1, \\ c_3 &= (a_0 + a_1 + a_2 + a_3)/2, & c_2 &= c_3 - a_3. \end{aligned} \tag{D4I20}$$

All belong to $\mathbb{Z}_p$, and substitution gives the original four coordinates. Conversely a $\mathbb{Z}_p$-linear combination of the four vectors has coordinate sum $2c_3$. These two direct calculations prove the tensor-product identification, including at $p = 2$. Taking integral coefficients gives the same basis description of the original D_4.

The three line identities of section 37.2 hold over $\mathbb{Z}_p$, since they are integer polynomial identities. Each line stays in the sum-zero submodule of $D_{4,p}$.

Theorem 37.2 (All local images and the exact local-to-global statement)**.** *For the unchanged polynomial P,*

$$\begin{aligned} P(D_{4,2}) &= 2\mathbb{Z}_2, \\ P(D_{4,3}) &= \mathbb{Z}_3^{\times} \cup 9\mathbb{Z}_3, \\ P(D_{4,p}) &= \mathbb{Z}_p \qquad (p \geq 5). \end{aligned} \tag{D4I21}$$

An integer n belongs to every local image if and only if $n \in S$, and hence if and only if $n \in P(D_4)$. The global section is exactly (D4I14)*; explicit local sections are given in the proof.*

Proof. For $p = 2$, the parity calculation of section 37.1, reduced modulo two, gives $P(D_{4,2}) \subseteq 2\mathbb{Z}_2$. Conversely, given an arbitrary $n = 2q \in 2\mathbb{Z}_2$, put

$$r = (q-1)/3 \in \mathbb{Z}_2.$$

Division by three is valid since three is a unit in $\mathbb{Z}_2$. Then $L_1(r) \in D_{4,2}$ and

$$P(L_1(r)) = 6r + 2 = 2q = n.$$

Thus $n \mapsto L_1((n/2-1)/3)$ is an explicit section on $2\mathbb{Z}_2$, and proves the first equality.

For $p = 3$, reduction of the original polynomial modulo three and the substitution $a_0 = 3b$ in (D4I4) prove that any value divisible by three belongs to $9\mathbb{Z}_3$. Every unit $n \in \mathbb{Z}_3^\times$ is congruent modulo three to exactly one of 2ε, with $\varepsilon \in \{-1, 1\}$. Set

$$r = (n - 2\varepsilon)/6 \in \mathbb{Z}_3.$$

This quotient is integral because the numerator is divisible by three and two is a unit. The line $L_\varepsilon(r)$ then has P value $6r + 2\varepsilon = n$. Every $n \in 9\mathbb{Z}_3$ can be written uniquely as $18h$ with $h = n/18 \in \mathbb{Z}_3$, again because two is a unit. The third line has $P(L_0(h)) = 18h = n$. These two disjoint cases give the explicit local section

$$n \longmapsto \begin{cases} L_\varepsilon((n - 2\varepsilon)/6), & n \in \mathbb{Z}_3^\times, \quad n \equiv 2\varepsilon \pmod 3, \\ L_0(n/18), & n \in 9\mathbb{Z}_3. \end{cases} \tag{D4I22}$$

They prove the second equality in (D4I21).

For every prime $p \geq 5$, six is a unit. Given any $n \in \mathbb{Z}_p$, choose $r = (n - 2)/6 \in \mathbb{Z}_p$. The point $L_1(r)$ belongs to $D_{4,p}$ and has $P(L_1(r)) = 6r + 2 = n$. Thus $n \mapsto L_1((n - 2)/6)$ is a section on all of $\mathbb{Z}_p$, proving the third equality.

For integers n, the first local condition is precisely evenness; the second is precisely $3 \nmid n$ or $9 \mid n$; and the other primes impose no condition. Membership in every local image is therefore exactly $n \in S$. The explicit integral section in theorem 37.1 proves that those conditions suffice for global integral representability. This proves the local-to-global assertion without approximation or inference from a finite modular calculation. □

37.5 The exact factorization and quotient metric on the sum-zero slice

Define the integral map into the original sublattice A of (D4I15) by

$$\Phi : \mathbb{Z}^3 \longrightarrow A, \qquad \Phi(w, u, v) = (-3w - u - v, w + u, w + v, w). \tag{D4I23}$$

Its coordinate sum is zero. For $a \in A$, its proposed inverse is

$$\Phi^{-1}(a) = (a_3, a_1 - a_3, a_2 - a_3). \tag{D4I24}$$

Substitution returns the original second, third, and fourth coordinates immediately; the first becomes $-a_1 - a_2 - a_3 = a_0$ by the defining equation of A. Substituting Φ into the inverse returns (w, u, v). These two equalities prove that Φ is an exact integral-module isomorphism.

For $a = (a_0, y, z, w) \in A$, put $T = y + z + w = -a_0$. Direct expansion of the original polynomial gives

$$\begin{aligned} P(a) &= -T\big(T^2 - 3(y^2 + z^2 + w^2)\big) \\ &= 2T(y^2 + z^2 + w^2 - yz - yw - zw) \\ &= T\big((y - z)^2 + (y - w)^2 + (z - w)^2\big). \end{aligned} \tag{D4I25}$$

Under Φ, one has $T = 3w + u + v$, and the three differences are $u - v, u, v$. Consequently

$$P(\Phi(w, u, v)) = 2(3w + u + v)(u^2 - uv + v^2). \tag{D4I26}$$

This identity explains the three original lines without changing their coordinates. On $L_\varepsilon(r)$, the quotient differences are $(u, v) = (-\varepsilon, -\varepsilon)$, so $u^2 - uv + v^2 = 1$; on $L_0(h)$, they are $(u, v) = (1, -1)$, so $u^2 - uv + v^2 = 3$. The original first coordinates remain $-3r - \varepsilon$ and $-3h$, respectively.

The exact quotient map is

$$\chi : A \longrightarrow \mathbb{Z}^2, \qquad \chi(a) = (a_1 - a_3, a_2 - a_3). \tag{D4I27}$$

It is onto because $\chi\Phi(0,u,v)=(u,v)$. An element lies in its kernel exactly when its last three coordinates all equal a common integer w; its first coordinate is then $-3w$. Thus $\ker\chi=\mathbb{Z}(-3,1,1,1)$. The section $(u,v)\mapsto\Phi(0,u,v)$ proves the split exact sequence

$$0\longrightarrow\mathbb{Z}(-3,1,1,1)\longrightarrow A\xrightarrow{\chi}\mathbb{Z}^2\longrightarrow 0. \tag{D4I28}$$

For $\omega=(-1+\sqrt{-3})/2$, one has $\omega+\overline{\omega}=-1$ and $\omega\overline{\omega}=1$. Direct multiplication therefore proves

$$(u+v\omega)(u+v\overline{\omega})=u^2-uv+v^2. \tag{D4I29}$$

The quadratic factor in the original cubic is exactly this Eisenstein norm, with the preceding coefficient two unchanged.

The original Euclidean metric on these same coordinates is, by expansion,

$$\|\Phi(w,u,v)\|^2=12w^2+8w(u+v)+2(u^2+uv+v^2). \tag{D4I30}$$

Write $d=(-3,1,1,1)$ for the direction when used in the metric calculation. Its squared norm is twelve, and the exact pairing is

$$\langle\Phi(w,u,v),d\rangle=12w+4(u+v). \tag{D4I31}$$

Hence orthogonal projection away from the real line $\mathbb{R}d$ is given in the original coordinates by

$$\begin{aligned}\pi\Phi(w,u,v)&=\Phi(w,u,v)-\left(w+\frac{u+v}{3}\right)d\\&=\left(0,\frac{2u-v}{3},\frac{2v-u}{3},-\frac{u+v}{3}\right).\end{aligned} \tag{D4I32}$$

This depends only on the quotient coordinate $\chi(a)=(u,v)$. Its squared norm is

$$\begin{aligned}\|\pi\Phi(w,u,v)\|^2&=\frac{(2u-v)^2+(2v-u)^2+(u+v)^2}{9}\\&=\frac{2}{3}(u^2-uv+v^2).\end{aligned} \tag{D4I33}$$

The quotient metric on $A/\mathbb{Z}d$, realized by this exact projection, thus has Gram matrix

$$\frac{1}{3}\begin{pmatrix}2&-1\\-1&2\end{pmatrix} \tag{D4I34}$$

in the displayed quotient coordinates (u,v). The retained common-lattice map $a\mapsto\sqrt{2}(a,a,a)$ multiplies every squared norm by six, as proved in theorem 36.6. Consequently the corresponding quotient Gram matrix on the image of A in $\mathcal{I}$ is

$$\begin{pmatrix}4&-2\\-2&4\end{pmatrix}. \tag{D4I35}$$

This is the exact quotient-metric calculation on the indicated rank-three slice. It neither identifies that slice with the full D_4 nor changes the original common-lattice metric.

38 The marked Niemeier lattice, its twelve theta indices, and the umbral shadow

The original objects are

$$A=\{(x_0,\dots,x_5)\in\mathbb{Z}^6:\sum_j x_j=0\},\qquad D=\{d\in\mathbb{Z}^4:\sum_i d_i\equiv 0\pmod 2\},\qquad R=A^{\perp 4}\perp D.$$

The metrics are the displayed Euclidean metrics. The four copies of A retain their original fibre order $f_i 289^j$ for $0 \le j < 6$, with fibre labels $(00, 10, 01, 11)$. Their discriminant generator is $\lambda = (5, -1, -1, -1, -1, -1)/6$. The discriminant of D retains $10 \mapsto v = (1, 0, 0, 0)$, $01 \mapsto s = (1, 1, 1, 1)/2$, and $11 \mapsto c = (-1, 1, 1, 1)/2$. Consequently the original norm-valued quadratic refinement, with codomain $\mathbb{Q}/2\mathbb{Z}$, is

$$q_R(k_1, k_2, k_3, k_4; y) = \frac{5}{6} \sum_i k_i^2 + \mathbf{1}_{y \neq 0} \pmod{2\mathbb{Z}}.$$

The original glue and Niemeier lattice are

$$\begin{aligned} E &= \{u \in \mathbb{F}_2^4 : \sum_i u_i = 0\}, \qquad \tau(u) = (u_2 + u_4, u_3 + u_4), \\ j(a, b) &= (a + b, a + 2b, a, b), \qquad C_3 = j(\mathbb{F}_3^2), \\ H &= \{(3u + 2j(a, b); \tau(u)) : u \in E,\ (a, b) \in \mathbb{F}_3^2\}, \\ N &= \{x \in R^\vee : x + R \in H\}. \end{aligned}$$

Thus $|H| = 8 \cdot 9 = 72$ and $N/R = H$, with the root system $X = A_5^4 D_4$ proved in the retained lattice calculation. In particular, no norm-two vectors are introduced by the nonzero glue classes: their original minima are four or six. We use this proved literal root system, rather than a rank-only identification, throughout the construction below.

38.1 The primary prescription and its complete coefficient matrix

Write $e(t) = \exp(2\pi i t)$ and $q = e(\tau)$ for $\tau \in \mathbb{H}$. Cheng–Duncan–Harvey, arXiv:1307.5793v2, Section 3.2 [74, Sections 3.2–3.3], source labels `thetexp`, `transf_theta`, and `transf_theta_2`, use

$$\theta_{m,r}(\tau, z) = \sum_{k \equiv r \ (2m)} q^{k^2/(4m)} e(kz), \qquad r \in \mathbb{Z}/(2m).$$

Their Section 3.3, source label `def:OmegaMatrices`, defines

$$\Omega_m(n)_{r,s} = \begin{cases} 1, & r + s \equiv 0 \pmod{2n}, \quad r - s \equiv 0 \pmod{2m/n}, \\ 0, & \text{otherwise}, \end{cases} \qquad n \mid m.$$

Their ADE table, source label `ADE1`, assigns

$$\Omega^{A_{m-1}} = \Omega_m(1), \qquad \Omega^{D_{m/2+1}} = \Omega_m(1) + \Omega_m(m/2),$$

and extends by addition over components with the same Coxeter number. These are the primary authors' prescribed maps. The following calculation specializes and proves their values in the present marked coordinates; it does not assert a new ADE classification.

The six-cycle $e_j \mapsto e_{j+1 \bmod 6}$ on an A fibre has characteristic polynomial $(x^6 - 1)/(x - 1)$ on its augmentation space. Its order is six. In D, take the original simple roots $\alpha_1 = e_1 - e_2$, $\alpha_2 = e_2 - e_3$, $\alpha_3 = e_3 - e_4$, $\alpha_4 = e_3 + e_4$. Their reflection product $s_{\alpha_1} s_{\alpha_2} s_{\alpha_3} s_{\alpha_4}$ sends

$$e_1 \mapsto e_2, \quad e_2 \mapsto e_3, \quad e_3 \mapsto -e_1, \quad e_4 \mapsto -e_4.$$

It has order six and characteristic polynomial

$$(x^3 + 1)(x + 1) = (x^2 - x + 1)(x + 1)^2 = \frac{(x^2 - 1)(x^6 - 1)}{(x - 1)(x^3 - 1)}.$$

Thus the original five components have Coxeter number $h = m = 6$, and the primary theta prescription has precisely $2h = 12$ indices and denominator $4h = 24$ in its exponents. The root rank remains $4 \cdot 5 + 4 = 24$; the glue index remains 72.

Put $J = \Omega_6(3)$. Its two congruences are

$$r + s \equiv 0 \pmod 6, \qquad r - s \equiv 0 \pmod 4.$$

The simultaneous solutions modulo twelve consist of the single value $s \equiv 5r \pmod{12}$: this value satisfies both congruences, and the difference of any two solutions is divisible by both six and four, hence by twelve. Therefore

$$\boxed{\Omega^X = 5I_{12} + J, \qquad J_{r,s} = \mathbf{1}_{s \equiv 5r \ (12)}.}$$

In the full original index order $0, 1, \ldots, 11$ this is

$$\Omega^X = \begin{pmatrix} 6 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 5 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 5 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 6 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 5 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 5 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 6 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 5 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 5 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 6 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 5 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 5 \end{pmatrix}.$$

The unary series and the actual shadow are

$$S_{6,r}(\tau) = \frac{1}{2\pi \mathrm{i}} \frac{\partial \theta_{6,r}}{\partial z}\Big|_{z=0} = \sum_{k \equiv r \ (12)} k q^{k^2/24}, \qquad S^X = \Omega^X S_6.$$

All series and their derivatives converge normally on compact subsets of their stated domains, by Gaussian decay. Substitution $k \mapsto -k$ proves $S_{6,-r} = -S_{6,r}$, so $S_{6,0} = S_{6,6} = 0$. Let $\mathcal{A} : \mathbb{C}^5 \to \mathbb{C}^{12}$ insert zero in positions zero and six and put $(\mathcal{A}u)_{12-r} = -u_r$ for $1 \le r \le 5$. This retains, and explicitly reconstructs, all twelve components. Then

$$\Omega^X \mathcal{A} = \mathcal{A} M^X, \qquad \boxed{M^X = \begin{pmatrix} 5 & 0 & 0 & 0 & 1 \\ 0 & 4 & 0 & 0 & 0 \\ 0 & 0 & 6 & 0 & 0 \\ 0 & 0 & 0 & 4 & 0 \\ 1 & 0 & 0 & 0 & 5 \end{pmatrix}.}$$

Indeed its (r, s) entry is $\Omega^X_{r,s} - \Omega^X_{r,12-s}$; direct substitution gives the matrix. In particular the fourth A-type multiplicity and the full D-type contribution have both been retained. The diagonal entries $(5, 4, 6, 4, 5)$ are exactly the Coxeter exponent multiplicities: four copies of $(1, 2, 3, 4, 5)$ and one copy of $(1, 3, 3, 5)$.

38.2 Quadratic refinements and the finite theta transformations

Let $L = \mathbb{Z}\ell$ with $(\ell, \ell) = 12$. Its dual is $L^\vee = \mathbb{Z}\ell/12$. Hence

$$A_L = L^\vee/L = \mathbb{Z}/12, \qquad q_L(r) = r^2/12 \pmod{2\mathbb{Z}}, \qquad Q_L(r) = q_L(r)/2 = r^2/24 \pmod{\mathbb{Z}},$$

and the associated bilinear form is $B_L(r, s) = rs/12 \pmod{\mathbb{Z}}$. The symbols q_L and Q_L deliberately have different codomains and retain the factor two.

Define the unitary Fourier matrices and reflection by

$$(F_\pm)_{r,s} = 12^{-1/2} e(\pm rs/12), \qquad R_{r,s} = \mathbf{1}_{s \equiv -r \ (12)}.$$

Finite geometric-series summation gives

$$F_\pm^2 = R, \qquad F_- = RF_+ = F_+R, \qquad F_+F_- = I.$$

The substitution $k \mapsto k+12$ shows that $e(k^2/24)$ depends only on $r = k \bmod 12$. Thus, with $D_{r,s} = e(r^2/24)\delta_{r,s}$,

$$\theta_6(\tau+1, z) = D\theta_6(\tau, z).$$

Poisson summation of the Gaussian gives

$$\theta_6(-1/\tau, z/\tau) = \sqrt{-\mathrm{i}\tau}\, e(6z^2/\tau) F_- \theta_6(\tau, z),$$
$$\theta_6(-1/\tau, -z/\tau) = \sqrt{-\mathrm{i}\tau}\, e(6z^2/\tau) F_+ \theta_6(\tau, z).$$

Here $\sqrt{-\mathrm{i}\tau}$ is the holomorphic square root positive when τ is positive imaginary. For completeness, apply the Fourier transform convention $\widehat{f}(u) = \int_{\mathbb{R}} f(x) e(-ux)\, dx$ to $f(x) = \exp(\pi \mathrm{i} \tau x^2/12 + 2\pi \mathrm{i} z x)$. Completing the square evaluates its transform; the real positive Gaussian integral is obtained by squaring the integral and using polar coordinates, and holomorphic continuation gives the expression for $\Im\tau > 0$. Periodizing $f(r+12x)$, its n-th Fourier coefficient is $12^{-1} e(nr/12) \widehat{f}(n/12)$. Summing these coefficients at zero and grouping n by its residue modulo twelve gives the first displayed transformation. Absolute Gaussian convergence justifies the integration, grouping, and Fourier evaluation. Replacing z by $-z$ and using $\theta_{6,-r}(\tau, z) = \theta_{6,r}(\tau, -z)$ gives the second. The second, with its plus-sign Fourier matrix, is precisely CDH's convention; the first uses the usual generator $S = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ with elliptic argument z/τ.

The matrices carrying the finite multiplier for this first convention are

$$U(T) = D, \qquad U(S) = e(-1/8) F_-.$$

The square-root weight factor has not been included in U. Differentiation of the theta transformation at $z = 0$ gives

$$S_6(-1/\tau) = \tau \sqrt{-\mathrm{i}\tau} F_- S_6(\tau), \qquad S_6(\tau+1) = D S_6(\tau).$$

Equivalently, CDH's reflected convention gives $S_6(-1/\tau) = -\tau\sqrt{-\mathrm{i}\tau} F_+ S_6(\tau)$. The equality follows from $RS_6 = -S_6$. Thus the unary series has weight $3/2$ with the same finite multiplier; theta has weight $1/2$.

The permutation J preserves Q_L because $((5r)^2 - r^2)/24 = r^2 \in \mathbb{Z}$. Since $5^{-1} = 5$ modulo twelve,

$$(JF_\pm)_{r,s} = 12^{-1/2} e(\pm 5rs/12) = (F_\pm J)_{r,s}.$$

Therefore Ω^X commutes with both finite multiplier matrices. The shadow transformation follows by applying this proved intertwiner. Its component series vanish at infinity since every nonzero summand has strictly positive exponent; the modular transformation shows the same cuspidal behavior at every cusp of the full modular group.

There is also an exact comparison with the original A5 discriminant, without replacing its form. Reduction

$$p : \mathbb{Z}/12 \longrightarrow \mathbb{Z}/6, \qquad p(r) = r \bmod 6$$

has kernel $\{0, 6\}$ and satisfies

$$q_A(p(r)) = \frac{5}{6} r^2 = 10 q_L(r) \pmod{2\mathbb{Z}}.$$

Thus it is a surjective quadratic similitude with precisely multiplier ten, and its bilinear forms satisfy the same multiplier relation. It is not asserted to be an isometry. Applying p to four coordinates and the identity to the retained D discriminant gives

$$\widetilde{D} = (\mathbb{Z}/12)^4 \oplus \mathbb{F}_2^2 \longrightarrow D_R, \qquad (r; y) \longmapsto (p(r_1), \dots, p(r_4); y).$$

The pullback quadratic refinement is exactly $10 \sum_i q_L(r_i) + \mathbf{1}_{y \neq 0}$, its radical is $\{0, 6\}^4 \oplus 0$, and its quotient by that radical is the original (D_R, q_R). To prove the radical claim, pairing with each coordinate generator forces $5r_i/6 \in \mathbb{Z}$, hence $6 \mid r_i$; the nondegenerate D form forces $y = 0$. The vectors so obtained have quadratic value 30 in each nonzero coordinate, hence zero modulo two. The inverse image of the original H has order $16 \cdot 72 = 1152$ and is isotropic for this pullback form. This gives the exact relation of the theta indices to all original glue coordinates, with the form and multiplier specified on both sides.

38.3 The full Fourier determinant and the retained order-twelve character

Let $\chi : \mathrm{SL}_2(\mathbb{Z}) \to \mathbb{Z}/12$ be the retained marking

$$\chi(T) = -1, \qquad \chi(S) = 3.$$

These values respect the presentation $S^4 = I$, $(ST)^3 = S^2$: in the abelian target the required relations are $4 \cdot 3 = 0$ and $3(3 - 1) = 2 \cdot 3$ modulo twelve. They therefore define a character, and its value at T makes it onto. We now calculate the determinant of the entire twelve-dimensional finite theta representation, rather than that of a reduced component list.

Put $\zeta = e(1/12)$. The Vandermonde determinant gives

$$\det F_+ = 12^{-6} \prod_{0 \le j < k \le 11} (\zeta^k - \zeta^j).$$

For every such pair,

$$\zeta^k - \zeta^j = 2\mathrm{i} \sin\!\left(\frac{\pi(k-j)}{12}\right) \exp\!\left(\frac{\pi \mathrm{i}(j+k)}{12}\right),$$

whose sine is strictly positive. There are $\binom{12}{2} = 66$ pairs, and each index occurs in eleven of them, so

$$\sum_{j<k} (j + k) = 11 \sum_{j=0}^{11} j = 11 \cdot 66 = 726.$$

The phase of the determinant is consequently

$$\mathrm{i}^{66} \exp(726\pi\mathrm{i}/12) = (-1)\mathrm{i} = -\mathrm{i}.$$

Its remaining factor is positive. Unitarity, proved by the finite geometric sum above, makes its absolute value one. It follows exactly that

$$\boxed{\det F_+ = -\mathrm{i}, \qquad \det F_- = \mathrm{i}.}$$

In particular retaining CDH's reflection changes the Fourier determinant by $\det R = (-1)^5 = -1$; it cannot be dropped.

The sum of the squares of all twelve indices is

$$\sum_{r=0}^{11} r^2 = \frac{11 \cdot 12 \cdot 23}{6} = 506.$$

Consequently

$$\boxed{\det U(T) = e(506/24) = e(1/12), \qquad \det U(S) = e(-12/8)\det F_- = -\mathrm{i} = e(-3/12).}$$

For an explicit check that the finite matrices have the relevant metaplectic relations, let $G = \sum_{r=0}^{11} e(r^2/24)$. Listing the squares modulo twenty-four gives

$$G = 1 + 4e(1/24) + 2e(1/6) + 2e(3/8) + 2e(2/3) + e(1/2).$$

The first and last terms cancel, as do the terms with arguments $1/6$ and $2/3$. The elementary sine and cosine values at $\pi/12$ and $3\pi/4$ give

$$4e(1/24) + 2e(3/8) = \sqrt{6}(1+\mathrm{i}) = \sqrt{12}\,e(1/8).$$

Completing the square in the finite sum proves

$$(F_-D)^2 = e(1/8)D^{-1}F_-, \qquad (F_-D)^3 = e(1/8)R.$$

For the first equality, its (r,s) entry before evaluation is $12^{-1}e(s^2/24)\sum_t e(t^2/24 - t(r+s)/12)$; shifting t by $r+s$ produces exactly the stated Gauss sum and matrix entry. The second follows by multiplication and $F_-^2 = R$. It follows that

$$(U(S)U(T))^3 = U(S)^2 = -\mathrm{i}R, \qquad U(S)^4 = -I_{12}.$$

Taking determinants kills this metaplectic central sign because the dimension twelve is even. The determinant therefore descends to an ordinary character of $\mathrm{SL}_2(\mathbb{Z})$. Its two computed generator values, together with generation by S, T, prove

$$\boxed{\det U(\gamma) = e(-\chi(\gamma)/12).}$$

The dual determinant is its inverse and is exactly $e(\chi(\gamma)/12)$, with the retained signs $T \mapsto e(-1/12)$, $S \mapsto \mathrm{i}$.

This is a statement about the determinant multiplier, with its domain and codomain specified. It is not a statement that a vector of weight $1/2$ has scalar weight one after taking a determinant. The separate Dedekind-eta identities

$$\eta(\tau+1) = e(1/24)\eta(\tau), \qquad \eta(-1/\tau) = \sqrt{-\mathrm{i}\tau}\,\eta(\tau)$$

show that η^2 has weight one and multiplier $\det U$, whereas η^{-2} has weight minus one and multiplier $(\det U)^{-1}$. Indeed $\eta^2(S\tau) = (-\mathrm{i})\tau\eta^2(\tau)$ and $\eta^{-2}(S\tau) = \mathrm{i}\tau^{-1}\eta^{-2}(\tau)$. The integer-weight factors $(c\tau+d)^{\pm 1}$ obey the cocycle identity by matrix multiplication, so the two generator formulas prove the statements for every γ. At $-I$ the character and odd integer weight each contribute a minus sign, whose product is one.

38.4 The residue Fourier map and what the Weyl quotient retains

The original marking is $\kappa(289^j) = j \pmod 6$, with the full ordered residue list

$$(289^j)_{j=0}^{5} = (1, 289, 361, 169, 121, 529) \pmod{840}.$$

On the six-dimensional complex coordinate carrier define

$$E_2 : \mathbb{C}[S_{840}] \longrightarrow \mathbb{C}[\mathbb{Z}/12], \qquad E_2(e_{289^j}) = e_{2j},$$

and on a separately indexed $\mathbb{C}^6$ define

$$B(e_k) = \frac{e_k + e_{k+6}}{\sqrt{2}}, \qquad 0 \le k < 6.$$

Both are injective and preserve the indicated coordinate Hermitian products. Retain the original negative-sign normalized Fourier matrix $F_6^- = (6^{-1/2} e(-kj/6))_{k,j}$. Entry-by-entry calculation gives

$$\boxed{F_- E_2 = B F_6^-.}$$

Indeed the (r, j) entry of the left side is $12^{-1/2} e(-rj/6)$, and the right side has the same value, using r mod 6. Thus the original unnormalized sample transform is carried into this formula by its exact factor $\sqrt{6}$. Translation $j \mapsto j + 1$ intertwines under E_2 with translation $r \mapsto r + 2$ modulo twelve. This gives a literal Fourier and translation map, not an identification of different quadratic forms. In particular $Q_L(6) = 3/2 = 1/2 \pmod{\mathbb{Z}}$, so the quotient $\mathbb{Z}/12 \to \mathbb{Z}/6$ does not carry Q_L itself to a quadratic form on the quotient. Its exact multiplier-ten comparison with the original A5 form was proved above.

For the original lattice action, $\rho(289)$ is the same six-cycle in each of the four A coordinate sets and is the identity on D. Each coordinate transposition of A is reflection in the corresponding root $e_i - e_j$, so the full residue action is in $W(R) = W(A)^4 \times W(D)$. It acts trivially on every A discriminant because $Q_\sigma \lambda - \lambda = e_{\sigma(0)} - e_0 \in A$, and it fixes the D discriminant. Consequently

$$S_{840} \xrightarrow{\rho} O(N) \longrightarrow O(N)/W(R)$$

is the trivial homomorphism. This proves exactly which residue-group action is killed, without discarding its already constructed Fourier map.

The original marked quotient is

$$G = O(N)/W(R) \cong \mathrm{GL}_2(\mathbb{F}_3).$$

Here its actual coordinate map uses $\ell_1 = (1, 1)$, $\ell_2 = (1, 2)$, $\ell_3 = (1, 0)$, $\ell_4 = (0, 1)$. For $A \in \mathrm{GL}_2(\mathbb{F}_3)$ determine the unique signs and permutation by

$$\ell_i A = \varepsilon_i \ell_{p(i)}, \qquad (P_p u)_i = u_{p(i)}, \qquad B_p \tau(u) = \tau(P_p u).$$

The four row lines exhaust the projective line, so the signs and permutation exist uniquely. The binary equation defines B_p uniquely since τ is onto. Then $(k; y) \mapsto (\varepsilon_i k_{p(i)}; B_p y)$ preserves precisely the original two glue codes, and its ternary restriction is $M_g j = jA$. Conversely a marked glue stabilizer gives this same A. Multiplication of these identities proves the group isomorphism, with the original row/column convention retained.

To compare its chosen signed lifts with CDH's simple-root permutation section, let $Q_{\mathrm{rev}} e_j = e_{5-j}$ in an A fibre. When $\varepsilon_i = -1$, replace the representative $-I$ by $-Q_{\mathrm{rev}}$; their quotient is $Q_{\mathrm{rev}} \in W(A)$ and $-Q_{\mathrm{rev}}(e_j - e_{j+1}) = e_{4-j} - e_{5-j}$. Thus the resulting map permutes the original simple roots by diagram reversal. When $\varepsilon_i = 1$ no change is needed. For D, put $\Delta = (\alpha_1\ \alpha_2\ \alpha_3\ \alpha_4)$. Associate the original classes v, s, c with outer nodes $1, 4, 3$, respectively; their fundamental weights are e_1, $(1, 1, 1, 1)/2$, and $(1, 1, 1, -1)/2$, the last of which differs from the retained c by $(1, 0, 0, -1) \in D$. For B_p, let P_{B_p} permute these outer nodes exactly as B_p permutes v, s, c and fix node two. Then

$$d_{B_p} = \Delta P_{B_p} \Delta^{-1}$$

is the exact original-coordinate diagram representative. Its orthogonality follows because permuting the outer nodes preserves the original Gram matrix. It preserves D, and its action on fundamental weights proves its discriminant action is exactly B_p. Its quotient with the originally chosen D lift has trivial

discriminant action, hence belongs to the proved Weyl kernel $W(D)$. This specifies the actual change of section on every summand.

Define the following four class functions using those retained maps:

$$a_g = \#\{i : p(i) = i\}, \quad b_g = \sum_{p(i)=i} \varepsilon_i, \quad s_g = \operatorname{sgn}(B_p|_{\{v,s,c\}}), \quad c_g = \#\operatorname{Fix}(B_p|_{\{v,s,c\}}) - 1.$$

The first two are the traces on the four-component permutation and signed permutation representations. The last is the trace on the two-dimensional sum-zero subspace of the three outer-node permutation representation, and s_g is its sign character. Thus these are precisely CDH's $(\bar{\chi}^{X_A}, \chi^{X_A}, \chi^{X_D}, \check{\chi}^{X_D})$; the remaining $\bar{\chi}^{X_D}$ is one because the single D component is fixed. The comparison of sections above proves this identification for the actual marked quotient.

CDH Section 5.1 [74, Section 5.1] defines the full twisted coefficient matrix by

$$(\Omega_g^{X_A})_{r,s} = \big(b_g \mathbf{1}_{r \text{ even}} + a_g \mathbf{1}_{r \text{ odd}}\big)\delta_{r,s},$$

and, for its exceptional $m = 6$ D4 case, by

$$(\Omega_g^{X_D})_{r,s} = \begin{cases} c_g\delta_{r,s}, & r \equiv 0,3 \pmod 6, \\ \delta_{r,s} + s_g J_{r,s}, & r \equiv 1,5 \pmod 6, \\ s_g\delta_{r,s} + s_g J_{r,s}, & r \equiv 2,4 \pmod 6. \end{cases}$$

All deltas in this formula have indices modulo twelve. The exact maps from the marked glue quotient to the prescribed twisted shadows are

$$g \longmapsto \Omega_g^X = \Omega_g^{X_A} + \Omega_g^{X_D}, \qquad S_g^X = \Omega_g^X S_6.$$

They are class-function constructions from traces of representations; the assignment $g \mapsto \Omega_g^X$ is not asserted to be a group representation. On the odd five-component presentation the explicit matrix is

$$\boxed{M_g^X = \begin{pmatrix} a_g+1 & 0 & 0 & 0 & s_g \\ 0 & b_g & 0 & 0 & 0 \\ 0 & 0 & a_g+c_g & 0 & 0 \\ 0 & 0 & 0 & b_g & 0 \\ s_g & 0 & 0 & 0 & a_g+1 \end{pmatrix}.}$$

This follows by the same signed component reconstruction $\mathcal{A}$: in rows two and four the two D terms cancel, in row three the contribution is c_g, and rows one and five pair with one another. For reference, all possible character tuples are

$[g]$	a_g	b_g	s_g	c_g	number of elements
$1A$	4	4	1	2	1
$2A$	4	−4	1	2	1
$2B$	2	0	−1	0	12
$4A$	0	0	1	2	6
$3A$	1	1	1	−1	8
$6A$	1	−1	1	−1	8
$8A, 8B$	0	0	−1	0	12

with the CDH class names of their table `tab:chars:eul:6`. The formulas apply to each actual A, so their validity does not depend on a choice of conjugacy-class names. The finite checker obtains the table from all 48 invertible matrices over $\mathbb{F}_3$ and the original row forms.

The distinguished residue action has quotient $g = 1$ and therefore produces the untwisted shadow, while a product $\rho(r)L_g$ produces the same S_g^X for every r. The four-component signed action and the D4

triality quotient survive and enter the displayed matrix. The central element $-I \in \mathrm{GL}_2(\mathbb{F}_3)$, for example, changes b_g from four to minus four while fixing a_g, s_g, c_g; it changes the even-index odd-space entries with exactly that sign.

The already marked composite

$$\mathrm{SL}_2(\mathbb{Z}) \xrightarrow{\chi} \mathbb{Z}/12 \longrightarrow \mathbb{Z}/6 \xrightarrow{j\mapsto 289^j} S_{840}$$

is onto, has kernel $\chi^{-1}(\{0,6\})$, and sends T to $289^5 = 529$. Following it by the Weyl quotient kills it, by the proved residue calculation. Following χ into the dual determinant multiplier retains its complete order twelve, by the determinant theorem. These are explicit different outgoing morphisms from the same marking, including their kernels and signs.

38.5 The shadow-to-Fricke map with the original scalar seventy-two

The Coxeter calculation above gives the original formal frame exponent

$$\pi_X = -5[1] + [2] - [3] + 5[6].$$

Let $\mathcal{E}(\sum a_d[d]) = \prod_d \eta(d\tau)^{a_d}$. It is a group homomorphism from addition of formal exponents to multiplication of nonzero holomorphic functions, by the exponent laws. The retained scalar function is

$$\boxed{T_6(\tau) = \mathcal{E}(-\pi_X) = \frac{\eta(\tau)^5\eta(3\tau)}{\eta(2\tau)\eta(6\tau)^5}.}$$

The minus sign on π_X is essential: $\mathcal{E}(\pi_X) = T_6^{-1}$.

Here is a direct coefficient map from the actual shadow matrix to this eta function. Its diagonal difference vector is $\alpha = (5,4,6,4,5)$, and set $\alpha_0 = 0$ modulo six. Define the Lambert-series map at this marked matrix by

$$\mathscr{L}(\Omega^X)(\tau) = 1 + \sum_{n\geq 1} \alpha_{n \bmod 6} \frac{nq^n}{1-q^n}.$$

Convergence is absolute and locally uniform since $|q| < 1$ and α is bounded. The coefficient of q^N is exactly $\sum_{d|N} d\alpha_{d \bmod 6}$; thus this specifies every coefficient, not only its initial terms. For all positive integers n the retained multiplicities satisfy

$$\alpha_{n \bmod 6} = 5 - \mathbf{1}_{2|n} + \mathbf{1}_{3|n} - 5\mathbf{1}_{6|n}.$$

Checking the six residues proves this identity. The product definition $\eta(\tau) = q^{1/24}\prod_{n\geq 1}(1-q^n)$ and termwise logarithmic differentiation then give

$$\boxed{f^X := \mathscr{L}(\Omega^X) = -q\frac{d}{dq}\log T_6 = 5\lambda_6 + \lambda_2 - \lambda_3,}$$

where the original functions on the right are

$$\lambda_d = q\frac{d}{dq}\log\frac{\eta(d\tau)}{\eta(\tau)} = \frac{d-1}{24} + \sum_{n\geq 1}\sigma_1(n)(q^n - dq^{dn}).$$

The constant is $[5(6-1) + (2-1) - (3-1)]/24 = 1$, and the other coefficients agree by the six-residue identity. This proves the shadow-to-eta calculation directly for every coefficient.

The precise primary connecting map is CDH Section 4.2 [74, Section 4.2]:

$$\sigma^X(\tau,z) = \sum_{r \ (12)} \overline{S_r^X(\tau)}\theta_{6,r}(\tau,z), \qquad \mathscr{S}_{1,1}(\sigma^X) = f^X.$$

Its source labels are `eqn:sigmaX`, `def:weight2`, `weight2:Aseries`, and `weight2_EZ`. The displayed Lambert map is its explicit specialization at this root-system shadow. It verifies the resulting function independently of the general Shimura-lift theorem. In particular the shadow and the retained Fricke function are connected by this complete coefficient map.

The original modular subgroup here is $\Gamma_0(6)$, and $[\mathrm{SL}_2(\mathbb{Z}) : \Gamma_0(6)] = 12$. To compute the index, reduction modulo six gives $\mathrm{SL}_2(\mathbb{Z}/6) \cong \mathrm{SL}_2(\mathbb{F}_2) \times \mathrm{SL}_2(\mathbb{F}_3)$ of order $6 \cdot 24 = 144$. Reduction is onto: elementary upper and lower unipotent matrices generate over each field, and CRT choices of their entries realize each factor while acting trivially on the other. The upper triangular subgroup modulo six has $\varphi(6) \cdot 6 = 12$ elements (choose its first unit diagonal entry and its upper-right entry; the other diagonal entry is its inverse). Its inverse image is $\Gamma_0(6)$, proving index $144/12 = 12$. This modular-group index and the twelve theta indices have their specified constructions; neither replaces the lattice index seventy-two.

Under the exact Fricke transformation $W_6\tau = -1/(6\tau)$, the eta inversion formula sends each $\eta(dW_6\tau)$ to $\sqrt{-\mathrm{i}(6/d)\tau}\,\eta((6/d)\tau)$. The retained exponent vector at $d = 1, 2, 3, 6$ is $(5, -1, 1, -5)$; its reversed vector is its negative and its exponent sum is zero. The τ factors therefore cancel. Since each $6/d$ is positive, the consistent square-root branch gives the positive scalar

$$\prod_{d|6} (6/d)^{a_d/2} = 6^{5/2} 3^{-1/2} 2^{1/2} = 72.$$

Consequently

$$\boxed{T_6(W_6\tau) = 72/T_6(\tau).}$$

For the weight-two slash action of the determinant-six representative,

$$(f|_2 W_6)(\tau) = (6\tau^2)^{-1} f(-1/(6\tau)).$$

Differentiating the last identity and using $d(W_6\tau)/d\tau = (6\tau^2)^{-1}$ gives

$$\boxed{f^X|_2 W_6 = -f^X.}$$

On the Jacobi side, the primary Eichler–Zagier operator $\mathcal{W}_6(6)$ acts on theta by $\Omega_6(6) = R$. This is verified by its defining congruences, $r + s = 0 \pmod{12}$, $r - s = 0 \pmod 2$. On the odd shadow and on the corresponding skew-Jacobi form its eigenvalue is minus one. The specialized coefficient map just proved carries that negative eigenline to the negative Fricke eigenline of f^X. This proves an exact equivariant linear comparison on these lines; it does not evaluate an $\mathrm{SL}_2(\mathbb{Z})$ character on the determinant-six matrix.

Finally, the original sheet coordinates remain

$$J_6 = T_6 + 5 + 72/T_6, \qquad Y_6 = T_6 - 72/T_6.$$

Direct substitution and differentiation prove

$$J_6(W_6\tau) = J_6(\tau), \quad Y_6(W_6\tau) = -Y_6(\tau), \quad Y_6^2 = (J_6 - 5)^2 - 288, \quad dJ_6 = Y_6\, dT_6/T_6 = -2\pi\mathrm{i} Y_6 f^X\, d\tau.$$

Thus the original scalar 72, the additive shift 5, the branch constant 288, and the differential factor $-2\pi\mathrm{i}$ all survive the complete shadow-to-sheet calculation.

38.6 Established scope and reproducibility

The constructed maps are the ADE coefficient specialization, the exact finite-theta intertwiner, the multiplier determinant and its dual, the quadratic similitude to the original discriminants, the marked glue-group trace prescription, the original six-to-twelve Fourier map, and the coefficient-level shadow-to-Fricke map. Their kernels, weights, signs, and constants were specified and proved above. They establish concrete relations even where a direct isometry or an identical group action is absent. They do not construct a

new graded moonshine module or a new moonshine theorem. The name "umbral shadow" here refers to precisely the CDH prescription, with its source provenance retained.

The standard-library program `check_shadow.py` uses exact arithmetic in $\mathbb{Q}[z]/(z^8 - z^4 + 1)$ with $z = e(1/24)$. It verifies both full and odd matrices, the Fourier determinant by the Vandermonde product, the Gauss sum, the character signs, the Fourier embedding, all 48 original-code character records, the first 240 Lambert coefficients, and the squared positive Fricke scalar 72^2. The proofs above establish the unbounded identities independently of those finite checks.

39 Exact arithmetic of the Koide defect-one shell

This section isolates an arithmetic result that does not use the existence of the proposed complex structure on S^6, any period-map assertion, or any Yang–Mills interpretation. All coordinates, chambers, local factors, and normalizations used in the final constant are fixed explicitly below.

For $r = (r_1, r_2, r_3)$, define the Koide quadratic form

$$Q_K(r_1, r_2, r_3) := \frac{2}{3}(r_1 + r_2 + r_3)^2 - (r_1^2 + r_2^2 + r_3^2). \tag{1483}$$

For integral coordinates put

$$\Delta(x, y, z) := 4(xy + xz + yz) - (x^2 + y^2 + z^2). \tag{1484}$$

Expanding (1483) gives the exact identity

$$\Delta(x, y, z) = 3Q_K(x, y, z). \tag{1485}$$

Thus the equation studied here, $\Delta = -1$, is the nonzero level $Q_K = -1/3$; it is not an affine translate of the null cone $Q_K = 0$.

39.1 Vieta reduction and exact ternary coordinates

Regarded as a monic quadratic equation in z, the defect equation is

$$z^2 - 4(x + y)z + x^2 + y^2 - 4xy - 1 = 0. \tag{1486}$$

Consequently the other root is

$$z' = 4(x + y) - z. \tag{1487}$$

The analogous substitutions in x and y are integral involutions preserving Δ.

Let e_1, e_2, e_3 be the standard basis and let B_K be the integral bilinear form with Gram matrix

$$A_K = \begin{pmatrix} -1 & 2 & 2 \\ 2 & -1 & 2 \\ 2 & 2 & -1 \end{pmatrix}, \qquad B_K(u, v) = u^{\mathsf{T}} A_K v. \tag{1488}$$

Thus $\Delta(u) = B_K(u, u)$. The eigenvalue of A_K on $\mathbb{R}(1, 1, 1)$ is 3, and its eigenvalue on the orthogonal complement is -3, so this integral presentation has signature $(1, 2)$.

Proposition 39.1 (Vieta moves are Lorentzian reflections)**.** *For $i = 1, 2, 3$, define*

$$s_i(u) = u - \frac{2B_K(u, e_i)}{B_K(e_i, e_i)} e_i. \tag{1489}$$

Then $s_i \in \mathrm{O}(B_K, \mathbb{Z})$, $s_i^2 = 1$, *and* $\det s_i = -1$. *Explicitly,*

$$s_1(x, y, z) = (-x + 4y + 4z, y, z), \tag{1490}$$
$$s_2(x, y, z) = (x, -y + 4x + 4z, z), \tag{1491}$$
$$s_3(x, y, z) = (x, y, -z + 4x + 4y). \tag{1492}$$

Consequently these are exactly the three Vieta root substitutions for $\Delta = -1$, *now identified as reflections of the unchanged Lorentzian lattice* $(\mathbb{Z}^3, B_K)$.

Proof. Since $B_K(e_i, e_i) = -1$, formula (1489) is integral and direct evaluation gives (1490)– (1492). The standard reflection calculation gives

$$B_K(s_i(u), s_i(v)) = B_K(u, v), \qquad s_i^2 = 1.$$

The map fixes the hyperplane $e_i^{\perp}$ pointwise and sends e_i to $-e_i$, hence has determinant -1. Finally, when the equation is viewed as a monic quadratic in one coordinate, the displayed replacement is the sum of its two roots minus the original root, proving the last assertion. □

Proposition 39.2 (Strict Vieta reduction)**.** *Let* $0 < x \le y \le z$ *be integers satisfying* $\Delta(x, y, z) = -1$. *If* $z < 4(x + y)$, *then the other root in* (1487) *satisfies* $0 < z' < y$. *If* $z \ge 4(x + y)$, *then* $z' \le 0$.

Proof. In the first case $z' > 0$ follows immediately from (1487). Let $P(Z)$ be the polynomial on the left of (1486). Since $x \le y$,

$$P(y) = x^2 - 8xy - 2y^2 - 1 \le xy - 8xy - 2y^2 - 1 < 0.$$

But $P(y) = (y - z)(y - z')$, and $y - z \le 0$; hence $y - z' > 0$, so $z' < y$. The second assertion again follows directly from (1487). □

Define the reduced-root set and its height count by

$$\mathcal{R} := \{(x, y, z) \in \mathbb{Z}^3_{>0} : x \le y \le z,\ \Delta(x, y, z) = -1,\ z \ge 4(x + y)\}, \tag{1493}$$
$$R(X) := \#\{(x, y, z) \in \mathcal{R} : z \le X\}. \tag{1494}$$

Proposition 39.2 shows that the wall $z \ge 4(x + y)$ is exactly the terminal side of the decreasing Vieta move after the coordinates have been ordered.

Proposition 39.3 (The positive Vieta graph is a rooted forest)**.** *Let* $\mathcal{S}^+$ *be the set of unordered triples of positive integers satisfying* $\Delta = -1$. *Join two vertices in a simple graph when one is obtained from the other by a Vieta reflection* s_i *and the reflected coordinate is positive. Every connected component of this graph is a tree with exactly one root, and the roots are precisely the unordered triples whose ordered representatives lie in* $\mathcal{R}$. *Equivalently, every positive integral solution has a unique reduced representative, and it is obtained by repeatedly ordering the coordinates and reflecting the largest coordinate whenever it lies below the terminal wall.*

Proof. First, no Vieta reflection fixes a point of $\Delta = -1$. Indeed, a fixed reflection in a coordinate w, with the other two coordinates denoted by u, v, would give $w = 2(u + v)$, and substitution in the unchanged form gives

$$\Delta(u, v, 2(u + v)) = 3(u^2 + 4uv + v^2) > 0.$$

Thus the sum of the three coordinates changes strictly across every edge.

Now order a vertex as $0 < x \le y \le z$. In fact $y < z$, because $y = z$ would give

$$\Delta(x, y, y) = -x^2 + 8xy + 2y^2 > 0.$$

Reflecting either of the two smaller coordinates produces a new coordinate strictly larger than z:

$$4(y+z)-x>z, \qquad 4(x+z)-y>z.$$

If $z<4(x+y)$, Proposition 39.2 says that reflection in the largest coordinate produces $0<z'<y$ and strictly decreases the coordinate sum. If $z\geq 4(x+y)$, that reflection is nonpositive and hence is not an edge of $\mathcal{S}^+$. Therefore every nonterminal vertex has exactly one neighbour of smaller coordinate sum, whereas a terminal vertex has none.

Iteration of the unique decreasing move terminates because the coordinate sum is a positive integer. The endpoint is represented by an element of $\mathcal{R}$. The graph has no cycle: on a putative simple cycle choose a vertex of largest coordinate sum; its two neighbours on the cycle would be two distinct smaller neighbours, contrary to uniqueness. Thus each component is a tree. A tree in which following the unique decreasing edge always terminates has exactly one terminal vertex, proving both uniqueness and the characterization of the roots. □

For an ordered triple define

$$n=z-4(x+y), \qquad q=y-x, \qquad a=x+y-4n, \tag{1495}$$

and let

$$\mathcal{T}:=\left\{(a,q,n)\in\mathbb{Z}^3:\ \begin{array}{c} a>0,\ q\geq 0,\ n\geq 0,\ a+4n>q,\\ a^2-3q^2-18n^2=-2\end{array}\right\}. \tag{1496}$$

Theorem 39.4 (Integral ternary-coordinate bijection)**.** *The map in* (1495) *restricts to a bijection* $\Phi:\mathcal{R}\to\mathcal{T}$*. Its inverse is*

$$(a,q,n)\longmapsto\left(\frac{a+4n-q}{2},\frac{a+4n+q}{2},4a+17n\right). \tag{1497}$$

For arbitrary integral (x,y,z) *and the associated* (a,q,n)*,*

$$2\Delta(x,y,z)=a^2-3q^2-18n^2, \qquad z=4a+17n. \tag{1498}$$

In particular,

$$R(X)=\#\{(a,q,n)\in\mathcal{T}:4a+17n\leq X\}. \tag{1499}$$

Proof. Put $p=x+y$. Since $q=y-x$, one has $2x=p-q$ and $2y=p+q$, and direct substitution into (1484) gives

$$2\Delta=p^2-3q^2+8pz-2z^2.$$

The definition $n=z-4p$ gives $z=4p+n$, whence

$$2\Delta=p^2-3q^2-8pn-2n^2.$$

Finally $a=p-4n$, equivalently $p=a+4n$, gives both identities in (1498).

Suppose first that $(x,y,z)\in\mathcal{R}$. Then $q,n\geq 0$, while $a+4n=p>q$ because $p-q=2x>0$. The defect equation and (1498) give $a^2-3q^2-18n^2=-2$. If $n=0$, then $a=p>0$. If $n\geq 1$, then

$$a^2=3q^2+18n^2-2\geq 18n^2-2\geq 16n^2.$$

Were $a\leq 0$, this inequality would force $a\leq -4n$, contrary to $a+4n>q\geq 0$. Hence $a>0$, and the forward map lands in $\mathcal{T}$.

Conversely, reducing the ternary equation modulo 2 gives $a\equiv q \pmod 2$, so all three coordinates in (1497) are integral. Put $p=a+4n$. The chamber inequality gives

$$x=\frac{p-q}{2}>0, \qquad y=\frac{p+q}{2}\geq x.$$

Moreover

$$z = 4a + 17n = 4p + n \geq 4p > p > y.$$

The identities (1498) now give $\Delta = -1$ and $z - 4(x + y) = n \geq 0$. Thus the inverse map lands in $\mathcal{R}$. The displayed formulae show directly that the two compositions are the identity. The last assertion follows from the height identity in (1498). □

Proposition 39.5 (Complete Pell classification of the chamber faces)**.** *The integral points of $\mathcal{T}$ on its two coordinate faces are the following two disjoint families.*

On the terminal Vieta wall $n = 0$, define integers a_k, q_k by

$$a_k + q_k\sqrt{3} = (1 + \sqrt{3})(2 + \sqrt{3})^k \qquad (k \geq 0). \tag{1500}$$

Then the points of $\mathcal{T}$ with $n = 0$ are exactly $(a_k, q_k, 0)$ for $k \geq 1$, and their reduced roots are

$$\left(\frac{a_k - q_k}{2}, \frac{a_k + q_k}{2}, 4a_k\right). \tag{1501}$$

The first is $(1, 4, 20)$.

On the symmetry face $q = 0$, define integers A_k, N_k by

$$A_k + N_k\sqrt{18} = (4 + \sqrt{18})(17 + 4\sqrt{18})^k \qquad (k \geq 0). \tag{1502}$$

Then the points of $\mathcal{T}$ with $q = 0$ are exactly $(A_k, 0, N_k)$ for $k \geq 0$, and their reduced roots are

$$\left(\frac{A_k + 4N_k}{2}, \frac{A_k + 4N_k}{2}, 4A_k + 17N_k\right). \tag{1503}$$

The first is $(4, 4, 33)$.

In the projectivized real quadric these families tend, respectively, to the two chamber-boundary rays

$$(a, q, n) = \mathbb{R}_{>0}(\sqrt{3}, 1, 0), \qquad (a, q, n) = \mathbb{R}_{>0}(3\sqrt{2}, 0, 1). \tag{1504}$$

Each family contributes only $O(\log X)$ points to $R(X)$.

Proof. On $n = 0$, the equation of $\mathcal{T}$ is $a^2 - 3q^2 = -2$. Here $q = 0$ is impossible. If $q = 1$, positivity gives $a = 1$, which fails the strict chamber inequality $a > q$. For any positive solution with $q \geq 2$, multiplication by the norm-one unit $2 - \sqrt{3}$ gives

$$a' + q'\sqrt{3} = (a + q\sqrt{3})(2 - \sqrt{3}), \qquad a' = 2a - 3q, \quad q' = 2q - a. \tag{1505}$$

The unchanged equation implies

$$a < 2q, \qquad a > \frac{3}{2}q, \qquad a > q \quad (q \geq 2),$$

because its squares differ from $4q^2$, $9q^2/4$, and q^2 by, respectively, $q^2 + 2$, $3q^2/4 - 2 > 0$, and $2q^2 - 2 > 0$. Thus $a', q' > 0$ and $q' < q$. Repeated descent reaches $(1, 1)$, and reversing the descent proves (1500) is the complete positive solution set. The recurrence

$$a_{k+1} = 2a_k + 3q_k, \qquad q_{k+1} = a_k + 2q_k$$

shows inductively that a_k, q_k are both odd. The chamber excludes only $k = 0$, and Theorem 39.4 now gives (1501).

On $q = 0$, one has $a^2 - 18n^2 = -2$. Necessarily $n > 0$. Direct substitution for $1 \leq n \leq 5$ gives the sole solution $(a, n) = (4, 1)$. If $n \geq 6$, multiplication by the norm-one unit $17 - 4\sqrt{18}$ gives

$$a' + n'\sqrt{18} = (a + n\sqrt{18})(17 - 4\sqrt{18}), \qquad a' = 17a - 72n, \quad n' = 17n - 4a. \tag{1506}$$

Both new coordinates are positive. Indeed,

$$\frac{a}{n} = \sqrt{18 - \frac{2}{n^2}} > \frac{72}{17} \quad (n \geq 6), \qquad \frac{a}{n} < \sqrt{18} < \frac{17}{4}.$$

Moreover $a > 4n$ and hence $0 < n' < n$. Descent therefore reaches $(4, 1)$, and reversal proves (1502) is the complete positive solution set. Its recurrence,

$$A_{k+1} = 17A_k + 72N_k, \qquad N_{k+1} = 4A_k + 17N_k,$$

keeps A_k even and N_k odd, so (1503) follows from the inverse coordinate formula.

Conjugating the two identities defining the sequences shows

$$\frac{a_k}{q_k} \longrightarrow \sqrt{3}, \qquad \frac{A_k}{N_k} \longrightarrow \sqrt{18} = 3\sqrt{2},$$

which proves (1504). Finally, the dominant factors $2 + \sqrt{3} > 1$ and $17 + 4\sqrt{18} > 1$ give exponential growth in k, whereas the heights in (1501) and (1503) are positive linear forms in the displayed coordinates. Hence either face contains only $O(\log X)$ points of height at most X. □

39.2 The affine quadric and its local densities

Put

$$F(a, q, n) := a^2 - 3q^2 - 18n^2, \qquad \mathscr{X} := \{F = -2\} \subset \mathbb{A}^3. \tag{1507}$$

The matrix of F is $A = \operatorname{diag}(1, -3, -18)$, so $\det A = 54$, and $-(-2)\det A = 108$ is not a square. The integral point $x_0 = (1, 1, 0)$ lies on $\mathscr{X}$.

For every prime p, define

$$N_{p,k} := \#\{(a, q, n) \bmod p^k : F(a, q, n) \equiv -2 \pmod{p^k}\}, \qquad \sigma_p := \lim_{k\to\infty} p^{-2k} N_{p,k}. \tag{1508}$$

Theorem 39.6 (Exact local densities)**.** *The limits in* (1508) *exist and are*

$$\sigma_2 = \frac{3}{2}, \qquad \sigma_3 = 2, \qquad \sigma_p = \frac{1 - p^{-2}}{1 - \left(\frac{3}{p}\right) p^{-1}} \quad (p \geq 5). \tag{1509}$$

Writing χ_{12} for the primitive real character modulo 12, the singular series, with the primes taken in increasing order (equivalently, with Borovoi's convergence-factor prescription), is therefore

$$\mathfrak{S}(\mathscr{X}) := \prod_p \sigma_p = \frac{9}{2} \frac{L(1, \chi_{12})}{\zeta(2)} = \frac{9}{2} \frac{\log(2 + \sqrt{3})}{\sqrt{3}\,\zeta(2)}. \tag{1510}$$

Proof. We compute the two bad primes first.

For $p = 3$, once $q, n \bmod 3^k$ have been chosen, the equation for a is

$$a^2 \equiv -2 + 3q^2 + 18n^2 \equiv 1 \pmod 3.$$

The two roots modulo 3 are nonsingular, and each has a unique Hensel lift modulo every 3^k. Thus

$$N_{3,k} = 2 \cdot 3^{2k}, \tag{1511}$$

which gives $\sigma_3 = 2$.

For $p = 2$, reduction modulo 4 leaves exactly two parity branches:

$$(a, q, n) \equiv (0, 0, 1) \pmod 2, \qquad (a, q, n) \equiv (1, 1, 0) \pmod 2. \tag{1512}$$

Fix $k \ge 4$ and put $j = k-1$. On the first branch write $a = 2A$, $q = 2Q$, and put $r = 3n$. Then

$$F(a,q,n) + 2 = -2\big(r^2 - 2A^2 + 6Q^2 - 1\big). \tag{1513}$$

Hence we must solve

$$r^2 \equiv 1 + 2A^2 - 6Q^2 \pmod{2^j}.$$

An odd unit is a square modulo 2^j, $j \ge 3$, precisely when it is $1 \pmod 8$, in which case it has four roots. Here that condition is equivalent to A, Q both being even. There are 2^{2j-2} such pairs $(A,Q) \bmod 2^j$, four choices of $r \bmod 2^j$, and, because 3 is a unit, two odd choices of $n \bmod 2^k$ above each such r. This branch therefore contributes

$$2^{2j-2} \cdot 4 \cdot 2 = 2^{2k-1} \tag{1514}$$

solutions modulo 2^k.

On the second branch write $n = 2m$ and set

$$v = \frac{a-q}{2}, \qquad r = \frac{-a+3q}{2}. \tag{1515}$$

Then r, v have opposite parity, the inverse formula is $a = r + 3v$, $q = r + v$, and

$$F(a,q,n) + 2 = -2\big(r^2 - 3v^2 + 36m^2 - 1\big). \tag{1516}$$

First suppose $v = 2V$ and r is odd. For fixed $V \bmod 2^{j-1}$ and $m \bmod 2^j$, the equation for r is

$$r^2 \equiv 1 + 12V^2 - 36m^2 \pmod{2^j}.$$

Its right side is $1 \pmod 8$ exactly when V, m have the same parity. There are $2 \cdot 2^{j-2} 2^{j-1} = 2^{2j-2}$ such pairs, and four roots r, so this subbranch gives 2^{2j} triples $(r, v, m) \bmod 2^j$.

Now suppose $r = 2R$ and v is odd. The congruence in (1516), solved for v^2, has an odd square on the left; its right side is $1 \pmod 8$ exactly when R, m have opposite parity. Again there are 2^{2j-2} admissible pairs and four square roots, hence 2^{2j} transformed triples.

It remains to account for the divided-coordinate map. Each class $(r, v) \bmod 2^j$ of opposite parity has exactly two inverse classes $(a, q) \bmod 2^k$: shifting either r or v by 2^j changes both $a = r + 3v$ and $q = r + v$ by 2^j, while shifting both changes neither modulo 2^k. The class $m \bmod 2^j$ gives one even class $n = 2m \bmod 2^k$. Thus the second branch contributes

$$2\big(2^{2j} + 2^{2j}\big) = 2^{2k}. \tag{1517}$$

Combining (1514) and (1517) gives

$$N_{2,k} = 3 \cdot 2^{2k-1} \qquad (k \ge 4), \tag{1518}$$

and hence $\sigma_2 = 3/2$.

For $p \ge 5$, one has $p \nmid 108$. Borovoi's good-prime formula [52, Section 1.3] gives

$$\sigma_p = \frac{1 - p^{-2}}{1 - \chi(p) p^{-1}}, \qquad \chi(p) = \left(\frac{-(-2)\det A}{p}\right) = \left(\frac{108}{p}\right) = \left(\frac{3}{p}\right),$$

which proves the remaining formula in (1509). Since $\chi(p) = \chi_{12}(p)$ for $p \ge 5$,

$$\prod_{p \ge 5} \sigma_p = L(1, \chi_{12}) \prod_{p \ge 5} (1 - p^{-2}) = \frac{3}{2} \frac{L(1, \chi_{12})}{\zeta(2)}.$$

Multiplication by $\sigma_2\sigma_3 = 3$ gives the first equality in (1510). Finally, the residue classes on which χ_{12} is nonzero give

$$\begin{aligned} L(1,\chi_{12}) &= \sum_{m\ge 0}\left(\frac{1}{12m+1}-\frac{1}{12m+5}-\frac{1}{12m+7}+\frac{1}{12m+11}\right) \\ &= \int_0^1 \frac{1-x^2}{x^4-x^2+1}\,\mathrm{d}x = \frac{\log(2+\sqrt{3})}{\sqrt{3}}, \end{aligned}$$

where the middle equality follows by summing the geometric series and the last by an elementary partial-fraction integration. This completes the local computation. □

39.3 The archimedean chamber factor

The Poincaré-residue gauge form on the whole smooth affine quadric is the unique regular 2-form ω_∞ whose local representatives satisfy

$$\omega_\infty \wedge \mathrm{d}F = \mathrm{d}a \wedge \mathrm{d}q \wedge \mathrm{d}n, \tag{1519}$$

Explicitly, on the three standard charts it is

$$\omega_\infty = \frac{\mathrm{d}q\wedge\mathrm{d}n}{2a} \quad (a\neq 0), \qquad \omega_\infty = \frac{\mathrm{d}a\wedge\mathrm{d}n}{6q} \quad (q\neq 0), \qquad \omega_\infty = -\frac{\mathrm{d}a\wedge\mathrm{d}q}{36n} \quad (n\neq 0). \tag{1520}$$

These expressions agree on overlaps after restriction to the tangent bundle of $F=-2$. Since the gradient $(2a,-6q,-36n)$ cannot vanish on that level, the three charts cover $\mathscr{X}$, so ω_∞ is globally defined and nowhere vanishing. Let $m_\infty := |\omega_\infty|$ be its associated positive measure. On the sheet $a>0$, used for the chamber count, the first expression in (1520) applies. Let

$$\mathcal{C} := \{(a,q,n)\in\mathbb{R}^3 : a>0,\ q\ge 0,\ n\ge 0,\ a+4n>q\}. \tag{1521}$$

Theorem 39.7 (Exact archimedean factor)**.** *As $X\to\infty$,*

$$m_\infty\{u\in\mathscr{X}(\mathbb{R})\cap\mathcal{C} : 4a+17n\le X\} \sim I_\infty X, \qquad I_\infty = \frac{1}{2\sqrt{3}}\log\!\left(\frac{3}{2\sqrt{2}}\right). \tag{1522}$$

Proof. First, the wall $a+4n>q$ removes only a bounded part of the positive orthant. Indeed, if $q\ge a+4n$, then $a\le q-4n$; using $a^2 = 3q^2+18n^2-2$ gives

$$3q^2+18n^2-2\le (q-4n)^2, \qquad \text{hence} \qquad q^2+4qn+n^2\le 1.$$

Thus the removed real region has bounded m_∞-measure. Its only integral point on $\mathscr{X}$ with $a>0, q,n\ge 0$ is $(a,q,n)=(1,1,0)$.

Introduce polar coordinates in the negative plane by

$$\sqrt{3}\,q = r\cos\theta, \qquad 3\sqrt{2}\,n = r\sin\theta, \qquad 0\le\theta\le\frac{\pi}{2}. \tag{1523}$$

On $F=-2$, one has $a=\sqrt{r^2-2}$ and $r\,\mathrm{d}r = a\,\mathrm{d}a$. The Jacobian in (1523) therefore gives

$$\mathrm{d}m_\infty = \frac{1}{6\sqrt{6}}\,\mathrm{d}a\,\mathrm{d}\theta. \tag{1524}$$

At fixed θ, the height boundary is

$$4a+\frac{17}{3\sqrt{2}}\sqrt{a^2+2}\sin\theta = X. \tag{1525}$$

Let $A_X(\theta)$ be its nonnegative solution when one exists, and put it equal to zero otherwise. Then $0 \le A_X(\theta)/X \le 1/4$, and for every fixed θ,

$$\frac{A_X(\theta)}{X} \longrightarrow \frac{1}{4 + \frac{17}{3\sqrt{2}}\sin\theta}.$$

Dominated convergence and (1524) give

$$I_\infty = \frac{1}{6\sqrt{6}} \int_0^{\pi/2} \frac{\mathrm{d}\theta}{4 + \frac{17}{3\sqrt{2}}\sin\theta}. \tag{1526}$$

For completeness, set $t = \tan(\theta/2)$. If $\alpha = 2\sqrt{2}/3$ and $\beta = 3/(2\sqrt{2})$, then $\alpha\beta = 1$, and

$$\int_0^{\pi/2} \frac{\mathrm{d}\theta}{4 + \frac{17}{3\sqrt{2}}\sin\theta} = \frac{1}{2}\int_0^1 \frac{\mathrm{d}t}{(t+\alpha)(t+\beta)} = 3\sqrt{2}\log\Bigl(\frac{3}{2\sqrt{2}}\Bigr).$$

Substitution in (1526) proves (1522). □

39.4 A genuine norm and the homogeneous counting theorem

Lemma 39.8 (A genuine norm realizing the height)**.** *Define*

$$\|(a,q,n)\|_* := \max\{|4a+17n|, |q|, |a|\}. \tag{1527}$$

Then $\|\cdot\|_*$ is a norm on $\mathbb{R}^3$, and on the chamber $\mathcal{C}$ it satisfies

$$\|(a,q,n)\|_* = 4a + 17n. \tag{1528}$$

For every integer $X \ge 0$,

$$R(X) = \#\{u \in \mathscr{X}(\mathbb{Z}) \cap \mathcal{C} : \|u\|_* < X+1\}. \tag{1529}$$

Proof. The linear map

$$(a,q,n) \longmapsto (4a+17n, q, a)$$

has determinant -17. Thus (1527) is the pullback of the supremum norm by an invertible linear map and is a genuine norm. If $(a,q,n) \in \mathcal{C}$, then

$$0 < a < 4a+17n, \qquad 0 \le q < a+4n < 4a+17n,$$

which proves (1528). Since $4a+17n$ is integral on $\mathscr{X}(\mathbb{Z})$, equation (1529) follows from (1499). □

Lemma 39.9 (Verification of the Gorodnik–Oh–Shah hypotheses)**.** *Let $\mathbf{G} = \mathrm{Spin}(F)$, acting through its standard three-dimensional orthogonal representation on $W = \mathbb{R}^3$, and let $x_0 = (1,1,0)$. Then*

$$\mathscr{X}(\mathbb{R}) = \mathbf{G}(\mathbb{R})^\circ x_0.$$

This orbit is a closed affine symmetric variety; $\mathbf{G}$ is connected and absolutely almost simple over $\mathbb{Q}$, is isotropic over $\mathbb{R}$, and acts irreducibly on W over $\mathbb{R}$; and the stabilizer $\mathbf{H} = \mathrm{Stab}_{\mathbf{G}}(x_0)$ has no nontrivial $\mathbb{Q}$-characters. The cone $\mathcal{C}$ in (1521) *is admissible and generic for [53, Theorem 1.6].*

Proof. The form F has signature $(1,2)$. Hence $\mathbf{G}$ is a connected absolutely almost simple group of type A_1, it is $\mathbb{R}$-isotropic, and its standard three-dimensional representation is $\mathbb{R}$-irreducible. The nonzero level $\mathscr{X}$ is Zariski closed. With

$$Q = \sqrt{3}\,q, \qquad N = 3\sqrt{2}\,n,$$

its real equation becomes

$$Q^2 + N^2 - a^2 = 2.$$

The parametrization

$$(a, \theta) \longmapsto (a, \sqrt{a^2+2}\cos\theta, \sqrt{a^2+2}\sin\theta), \qquad (a, \theta) \in \mathbb{R} \times (\mathbb{R}/2\pi\mathbb{Z}),$$

shows that $\mathscr{X}(\mathbb{R})$ is connected. Witt extension first produces an element $t \in \mathrm{O}(F)(\mathbb{R})$ taking any prescribed point of the level set to any other. If $\det t = -1$, choose an anisotropic vector in the orthogonal complement of the target point and compose t on the left with its reflection. That reflection fixes the target point and has determinant -1; hence the composite lies in $\mathrm{SO}(F)(\mathbb{R})$. Thus $\mathrm{SO}(F)(\mathbb{R})$ is transitive on the level set. The identity-component orbits have the full dimension of the level surface and are open; because their complements are unions of open orbits, each is also closed. Connectedness therefore leaves only one such orbit, proving $\mathscr{X}(\mathbb{R}) = \mathbf{G}(\mathbb{R})^\circ x_0$. The stabilizer of a nonisotropic vector is the fixed subgroup of the involution induced by reflection in that vector, so the orbit is affine symmetric.

The orthogonal complement of x_0 is

$$x_0^\perp = \{(3s, s, t) : s, t \in \mathbb{Q}\}, \qquad F(3s, s, t) = 6(s^2 - 3t^2). \tag{1530}$$

Thus $\mathbf{H}$ is the norm-one torus of $K = \mathbb{Q}(\sqrt{3})$. Its character lattice is $\mathbb{Z}$, with the nontrivial element of $\mathrm{Gal}(K/\mathbb{Q})$ acting by -1; consequently it has no nonzero Galois-invariant character and no nontrivial $\mathbb{Q}$-character. The integral point x_0 supplies the required nonempty integral locus.

It remains to check the sector rather than merely the affine orbit. For the given representation and any Euclidean sphere in W, the Satake boundary is

$$\mathscr{X}^\infty = \{F = 0\} \cap S(W) = \mathcal{O}_+ \sqcup \mathcal{O}_-, \qquad \mathcal{O}_\pm := \{F = 0,\ \pm a > 0\} \cap S(W), \tag{1531}$$

and $\mathbf{G}(\mathbb{R})^\circ$ is transitive on each displayed circle. For a nonzero null vector with $a, q, n \geq 0$,

$$a = \sqrt{3q^2 + 18n^2} > q - 4n.$$

Hence both the closure and the relative interior of $\mathcal{C}$ meet exactly $\mathcal{O}_+$, not $\mathcal{O}_-$. The intersection of $\partial\mathcal{C}$ with the Satake boundary consists of the two radial directions

$$(a, q, n) = (\sqrt{3}, 1, 0), \qquad (a, q, n) = (3\sqrt{2}, 0, 1). \tag{1532}$$

These are exactly the two projective limits of the integral Pell families in Proposition 39.5. This two-point intersection has measure zero for the smooth measure class on the boundary orbit. This is precisely the admissibility and genericity condition required by the cited sector theorem. □

39.5 The 2-adic orbit sign and Borovoi's coefficient

Write $\widehat{\mathbb{Z}} = \prod_p \mathbb{Z}_p$. For the nonsquare ternary quadric under consideration, Borovoi associates to every local point an orbit sign $\nu_v \in \{\pm 1\}$, constant on $\mathrm{Spin}(F)(\mathbb{Q}_v)$-orbits and equal to $+1$ at almost every finite place. Thus $\nu_f((x_p)_p) = \prod_p \nu_p(x_p)$ is defined on $\mathscr{X}(\widehat{\mathbb{Z}})$. Because $\mathscr{X}(\mathbb{R})$ is connected, his normalization reads

$$c_{\mathscr{X}} = \frac{2m_f\{x \in \mathscr{X}(\widehat{\mathbb{Z}}) : \nu_f(x) = +1\}}{m_f(\mathscr{X}(\widehat{\mathbb{Z}}))}, \tag{1533}$$

where m_f is the finite Tamagawa measure attached to the gauge form (1519) and Borovoi's convergence factors.

Proposition 39.10 (Finite adelic sign balance)**.** *For the affine quadric $\mathscr{X} = \{F = -2\}$, the coefficient $c_{\mathscr{X}}$ in Borovoi's counting formula [52, Theorem 1.4.1 and Proposition 1.5.1] is exactly*

$$c_{\mathscr{X}} = 1. \tag{1534}$$

Proof. Continue to write $A = \mathrm{diag}(1, -3, -18)$. Put $e = (1, 0, 0)$, $v = (1, 0, 1)$, and for every anisotropic vector w let

$$r_w(u) = u - \frac{2u^{\mathsf{T}}Aw}{w^{\mathsf{T}}Aw}\, w$$

be its orthogonal reflection. The element $g = r_v r_e$ is

$$g = \begin{pmatrix} -19/17 & 0 & -36/17 \\ 0 & 1 & 0 \\ -2/17 & 0 & -19/17 \end{pmatrix}. \tag{1535}$$

Direct multiplication gives

$$g^{\mathsf{T}}Ag = A, \qquad \det g = 1. \tag{1536}$$

Because $17 \in \mathbb{Z}_2^{\times}$, both g and g^{-1} have entries in $\mathbb{Z}_2$, so $g \in \mathrm{SO}(F, \mathbb{Z}_2)$. Its spinor norm is

$$\vartheta(g) = F(v)F(e) = -17 \quad \text{in } \mathbb{Q}_2^{\times}/\mathbb{Q}_2^{\times 2}. \tag{1537}$$

We identify the stabilizer's spinor-norm image. By (1530), over $\mathbb{Q}_2$ the orthogonal complement of x_0 is

$$F|_{x_0^{\perp}} = 6(s^2 - 3t^2) = 6N_{K_2/\mathbb{Q}_2}(s + t\sqrt{3}), \qquad K_2 = \mathbb{Q}_2(\sqrt{3}). \tag{1538}$$

The odd unit 3 is not a square in $\mathbb{Q}_2$, so K_2 is a quadratic field and this binary plane is anisotropic. Put $u_0 = (3, 1, 0)$, so $F(u_0) = 6$. If $h \in \mathrm{SO}(F|_{x_0^{\perp}})(\mathbb{Q}_2)$, then hr_{u_0} is a two-dimensional orthogonal transformation of determinant -1, hence is a reflection r_w in some nonisotropic vector $w = (3s, s, t)$. Therefore $h = r_w r_{u_0}$ and

$$\vartheta(h) = F(w)F(u_0) = 36N_{K_2/\mathbb{Q}_2}(s + t\sqrt{3}) \mod \mathbb{Q}_2^{\times 2}.$$

Conversely these products realize every norm square class as w ranges through the nonzero vectors of the anisotropic plane. Hence

$$\vartheta\Big(\mathrm{Stab}_{\mathrm{SO}(F)}(x_0)(\mathbb{Q}_2)\Big) = N_{K_2/\mathbb{Q}_2}(K_2^{\times}) \mod \mathbb{Q}_2^{\times 2}. \tag{1539}$$

For a quadratic extension, its norm group is the kernel of the Hilbert symbol with the defining square class [60, Corollary 13.4.6]. The odd-unit formula [60, Equation 12.4.14] gives

$$(-17, 3)_2 = (-1)^{((-17-1)/2)((3-1)/2)} = -1. \tag{1540}$$

Thus $-17 \notin N_{K_2/\mathbb{Q}_2}(K_2^{\times})$.

By Witt extension, every $x \in \mathscr{X}(\mathbb{Q}_2)$ is tx_0 for some $t \in \mathrm{O}(F)(\mathbb{Q}_2)$. If $\det t = -1$, choose an anisotropic vector in $x^{\perp}$ and compose on the left with its reflection; this fixes x and changes the determinant. Hence every such x has the form $x = sx_0$ with $s \in \mathrm{SO}(F)(\mathbb{Q}_2)$. The image of $\mathrm{Spin}(F)(\mathbb{Q}_2)$ in $\mathrm{SO}(F)(\mathbb{Q}_2)$ is the kernel of the spinor norm, as used in [52, Lemma 3.1]. Consequently the $\mathrm{Spin}(F)(\mathbb{Q}_2)$-orbit of $x = sx_0$ is determined by

$$\vartheta(s) \in \mathbb{Q}_2^{\times}/(\mathbb{Q}_2^{\times 2}N_{K_2/\mathbb{Q}_2}(K_2^{\times})), \tag{1541}$$

a quotient of order two. Equations (1537) and (1540) show that multiplication by g exchanges these two orbits. In Borovoi's convention, the orbit of x_0 has sign $\nu_2 = +1$ and the other orbit has sign $\nu_2 = -1$. Hence

$$\nu_2(gx) = -\nu_2(x) \qquad (x \in \mathscr{X}(\mathbb{Q}_2)). \tag{1542}$$

Apply g at the prime 2 and the identity at every other finite prime. This gives a bijection of $\mathscr{X}(\widehat{\mathbb{Z}})$. It preserves the finite Tamagawa measure: at 2 this follows from invariance of the local measure, equivalently from $g^*F = F$ and $\det g = 1$, and at all other primes the map is the identity. Equation (1542) shows that

it exchanges the finite sign loci $\nu_f = +1$ and $\nu_f = -1$. They therefore have equal measure. Since $\mathscr{X}(\mathbb{R})$ is connected, Borovoi's formula becomes

$$c_{\mathscr{X}} = \frac{2m_f(\mathscr{X}(\widehat{\mathbb{Z}})_+)}{m_f(\mathscr{X}(\widehat{\mathbb{Z}}))} = 1.$$

The argument uses a measure-preserving, sign-reversing bijection; it does not require g to be an involution (indeed $g^2 \neq I$). □

39.6 Calibration and the counting constant

Theorem 39.11 (Calibrated reduced-root asymptotic)**.** *The number of reduced positive defect-one roots satisfies*

$$R(X) \sim C_0 X \qquad (X \to \infty), \tag{1543}$$

where

$$\begin{aligned} C_0 &= \mathfrak{S}(\mathscr{X}) I_\infty \\ &= \frac{9}{2\pi^2} \log(2+\sqrt{3}) \log\!\left(\frac{3}{2\sqrt{2}}\right) \\ &= 0.035362047799731853\ldots. \end{aligned} \tag{1544}$$

Proof. Let μ_{GOS} denote the invariant measure on $\mathscr{X}(\mathbb{R})$ with the normalization appearing in the Gorodnik–Oh–Shah sector theorem [53, Theorem 1.6]. The gauge measure m_∞ in (1519) is also a nonzero $\mathbf{G}(\mathbb{R})^\circ$-invariant Radon measure. Because $\mathscr{X}(\mathbb{R})$ is a single homogeneous orbit, uniqueness of invariant measure gives

$$\mu_{\mathrm{GOS}} = \kappa m_\infty \tag{1545}$$

for some $\kappa > 0$.

The scalar is fixed by applying both counting theorems to the same Euclidean balls and the full cone W. Gorodnik–Oh–Shah give

$$\#\{x \in \mathscr{X}(\mathbb{Z}) : |x|_2 < T\} \sim \kappa\, m_\infty\{x \in \mathscr{X}(\mathbb{R}) : |x|_2 < T\}.$$

Borovoi's Theorem 1.4.1 and Proposition 1.5.1, together with Proposition 39.10, give for the same balls

$$\#\{x \in \mathscr{X}(\mathbb{Z}) : |x|_2 < T\} \sim \mathfrak{S}(\mathscr{X})\, m_\infty\{x \in \mathscr{X}(\mathbb{R}) : |x|_2 < T\}.$$

The real volume here is asymptotic to a positive constant times T. Comparison therefore gives the invariant-measure calibration

$$\kappa = \mathfrak{S}(\mathscr{X}). \tag{1546}$$

Lemma 39.8 and Lemma 39.9 now allow the sector theorem to be applied with the actual norm and chamber of the reduced-root problem:

$$\begin{aligned} R(X) &\sim \mu_{\mathrm{GOS}}\{u \in \mathscr{X}(\mathbb{R}) \cap \mathcal{C} : \|u\|_* \le X\} \\ &= \mathfrak{S}(\mathscr{X})\, m_\infty\{u \in \mathscr{X}(\mathbb{R}) \cap \mathcal{C} : 4a + 17n \le X\} \\ &\sim \mathfrak{S}(\mathscr{X}) I_\infty X. \end{aligned}$$

The switch between strict and weak inequalities is harmless by a $(1 \pm \varepsilon)X$ squeeze; on integral points it also follows directly from (1529). Since $R(X) = R(\lfloor X \rfloor)$, the integer-height asymptotic extends to all real $X \to \infty$. Finally, substituting (1510) and (1522), and using $\zeta(2) = \pi^2/6$, gives exactly (1544). □

39.7 Exact computation and the remaining open problem

The first five reduced roots are

$$(1,4,20),\quad (4,4,33),\quad (4,8,49),\quad (4,15,76),\quad (8,24,129). \tag{1547}$$

The first root is the first admissible term of the $n=0$ Pell family and the second is the first term of the $q=0$ Pell family in Proposition 39.5. Exact integer enumeration in the ternary coordinates gives the following values. The residual column uses the full constant in (1544), not a fitted constant.

X	$R(X)$	$R(X)-C_0X$
10,000	353	−0.62048
20,000	715	7.75904
50,000	1,755	−13.10239
75,000	2,654	1.84642
100,000	3,535	−1.20478
125,000	4,460	39.74403
150,000	5,294	−10.30717
175,000	6,195	6.64164
200,000	7,109	36.59044

The enumeration loops over $n \geq 0$ and $a > 0$ with $4a + 17n \leq X$, tests whether

$$q^2 = \frac{a^2 - 18n^2 + 2}{3}$$

is the square of a nonnegative integer satisfying $q < a+4n$, and then reconstructs (x, y, z) with (1497). Every retained triple is rechecked in the original polynomial (1484) and against all ordering and wall inequalities. Separate exhaustive congruence enumerations agree with the formulae above at $p = 2, 3, 5, 7, 11, 13$ through the computed powers. These finite checks verify the implementation; the asymptotic theorem rests on the proofs above, not on numerical fitting.

The reproducibility script is

`supporting_materials/workbench/reproduction/t1_koide_exact/verify_koide_narain_root_density.py`.

Running it with `-max-z 200000` emits the coordinate, local-density, 2-adic isometry, and bounded-enumeration certificates recorded in

`supporting_materials/workbench/reproduction/t1_koide_exact/koide_verification_200000.json`.

The script uses vectorized square detection only to propose candidates; it rechecks every retained candidate with integer arithmetic in both coordinate systems and in the original polynomial Δ.

Validation, repair, or disproof test 39.12 (Square-root discrepancy)**.** **[OPEN]**Determine whether, for every $\varepsilon > 0$,

$$R(X) - C_0X = O_\varepsilon(X^{1/2+\varepsilon}). \tag{1548}$$

The homogeneous-dynamics argument above and the exact effective-well-roundedness verification in theorem 46.2 below prove, for some fixed $\eta > 0$,

$$R(X) = C_0X + O(X^{1-\eta}),$$

but do not prove (1548). No unproved spectral gap, GRH assumption, or implicit "conditional theorem" is being used here: the sharper error term is an open problem in this workbench; see the primary-source comparison in section 51.1.

40 Exact ideal and unit-orbit parametrization of the unchanged sharp Koide count

Throughout this section the variables, equation, chamber, and height are exactly

$$a^2 - 3q^2 - 18n^2 = -2, \qquad a > 0, \quad q \geq 0, \quad n \geq 0, \quad a + 4n > q, \qquad 4a + 17n \leq X. \tag{1549}$$

Every auxiliary symbol introduced below is connected to these same integer points by an explicitly proved map and inverse. No count for a normalized form, a rescaled lattice, or a substituted cutoff is used in place of (1549). Thus

$$R(X) = \#\left\{(a, q, n) \in \mathbb{Z}^3 : \text{all conditions in (1549) hold}\right\}. \tag{1550}$$

The already calibrated leading constant is

$$C_0 = \frac{9}{2\pi^2} \log(2 + \sqrt{3}) \log\left(\frac{3}{2\sqrt{2}}\right). \tag{1551}$$

The exact parametrization proved below has four pieces of data for each fixed $n \geq 1$: a positive integer C, a positive integer D, a square root E of 3 modulo D, and a nonnegative Pell exponent k. There is no quotient left implicit: equations (1569), (1570), and (1572) construct (a, q, n) by finite integer operations, while (1564) and the descent (1568) recover (C, D, E, k) from (a, q, n). The face $n = 0$ is recorded separately and exactly.

This parametrization gives the exact sharp-cutoff formula (1586). It also separates the discrepancy into the radial shifted-divisor sum and the unit-phase discrepancy in (1593). The square-root estimate

$$R(X) - C_0 X = O_\varepsilon(X^{1/2+\varepsilon}) \tag{1552}$$

is not proved or refuted here. What is proved unconditionally, by applying the effective symmetric-variety theorem to the exact chamber, is that there is a constant $\eta > 0$ for which

$$R(X) = C_0 X + O(X^{1-\eta}). \tag{1553}$$

The cited general theorem does not supply $\eta \geq 1/2$, and its proof does not justify replacing (1553) by (1552).

41 Arithmetic of $\mathbb{Z}[\sqrt{3}]$

Put

$$K = \mathbb{Q}(\sqrt{3}), \qquad \mathcal{O} = \mathbb{Z}[\sqrt{3}], \qquad \epsilon = 2 + \sqrt{3}. \tag{1554}$$

For $\alpha = u + v\sqrt{3} \in K$, write

$$\alpha' = u - v\sqrt{3}, \qquad \mathrm{N}(\alpha) = \alpha\alpha' = u^2 - 3v^2. \tag{1555}$$

Lemma 41.1 (Ring, class number, and totally positive units)**.** *The ring of integers of K is $\mathcal{O}$. Every nonzero integral ideal of $\mathcal{O}$ is principal. Every unit has norm $+1$, and the totally positive unit group is*

$$\mathcal{O}^{\times,+} = \{\epsilon^j : j \in \mathbb{Z}\}. \tag{1556}$$

Proof. Since $3 \equiv 3 \pmod 4$, the standard integral-basis calculation for a quadratic field gives $\mathcal{O}_K = \mathbb{Z}[\sqrt{3}]$ and discriminant 12. Minkowski's bound says that every ideal class in a real quadratic field of discriminant 12 has an integral representative of norm at most $\frac{2!}{2^2}\sqrt{12} = \sqrt{3} < 2$. Its norm must therefore be 1, so it is $\mathcal{O}$ and the ordinary class number is one.

The norm of a unit is ± 1. The equation $u^2 - 3v^2 = -1$ is impossible modulo 3, because it would give $u^2 \equiv 2 \pmod 3$. Hence every unit has norm $+1$. Dirichlet's unit theorem makes the positive units an infinite cyclic group. To identify its least generator, suppose $\gamma = u + v\sqrt{3} > 1$ and $\gamma' > 0$. Replacing γ by its inverse if necessary makes $v \geq 0$. If $v = 0$, then $\gamma = 1$. If $v \geq 1$, then $u^2 - 3v^2 = 1$ forces $u \geq 2$, and consequently $\gamma \geq 2 + \sqrt{3} = \epsilon$. Since ϵ itself has norm one, it is the least positive unit greater than one and generates the group. □

For $m \geq 1$ define

$$\chi_{12}(d) = \left(\frac{12}{d}\right), \qquad \rho(m) = \sum_{d \mid m} \chi_{12}(d). \tag{1557}$$

Here the symbol is Kronecker's symbol; in particular

$$\chi_{12}(d) = \begin{cases} 0, & 2 \mid d \text{ or } 3 \mid d, \\ 1, & d \equiv 1, 11 \pmod{12}, \\ -1, & d \equiv 5, 7 \pmod{12}. \end{cases} \tag{1558}$$

Lemma 41.2 (Ideal number)**.** *The number of integral ideals of $\mathcal{O}$ having absolute norm m is $\rho(m)$.*

Proof. The Dedekind zeta function factors as $\zeta_K(s) = \zeta(s)L(s, \chi_{12})$. Comparing Dirichlet coefficients gives the assertion. Equivalently, the equality follows prime by prime. At 2 and 3 there is one ramified prime ideal, so there is one ideal of norm p^e for every $e \geq 0$. If $\chi_{12}(p) = 1$, then p splits and there are $e+1$ ideals of norm p^e. If $\chi_{12}(p) = -1$, then p is inert, so there is one such ideal when e is even and none when e is odd. These are respectively the local values of $\sum_{j=0}^{e} \chi_{12}(p)^j$. Consequently, if $m = 2^{e_2} 3^{e_3} \prod_{p \geq 5} p^{e_p}$, then

$$\rho(m) = \begin{cases} \displaystyle\prod_{\substack{p \mid m,\ p \geq 5 \\ p \equiv 1, 11 \ (12)}} (e_p + 1), & e_p \text{ is even for every } p \equiv 5, 7 \pmod{12}, \\ 0, & \text{otherwise.} \end{cases} \tag{1559}$$

□

42 Hermite-normal-form parameters for every ideal

For $m \geq 1$ let

$$\mathcal{P}(m) = \left\{ (C, D, E) : \begin{array}{ll} C, D \in \mathbb{Z}_{>0}, & C^2 D = m, \\ 0 \leq E < D, & E^2 \equiv 3 \pmod D \end{array} \right\}. \tag{1560}$$

Proposition 42.1 (Exact HNF parametrization of ideals)**.** *The map*

$$(C, D, E) \longmapsto \mathfrak{a}(C, D, E) := C\big(D\mathbb{Z} + (E + \sqrt{3})\mathbb{Z}\big) \tag{1561}$$

is a bijection from $\mathcal{P}(m)$ to the integral ideals of $\mathcal{O}$ of norm m. Consequently

$$\#\mathcal{P}(m) = \sum_{C^2 D = m} \#\{E \bmod D : E^2 \equiv 3 \pmod D\} = \rho(m). \tag{1562}$$

Proof. Relative to the basis $1, \sqrt{3}$ of $\mathcal{O}$, the lattice in (1561) has column basis

$$\begin{pmatrix} CD & CE \\ 0 & C \end{pmatrix}.$$

Its index is $C^2D = m$. It is stable under multiplication by $\sqrt{3}$ because

$$\sqrt{3}(CD) = D\,C(E+\sqrt{3}) - E(CD),$$
$$\sqrt{3}\,C(E+\sqrt{3}) = E\,C(E+\sqrt{3}) + \frac{3-E^2}{D}(CD),$$

and the last coefficient is integral. It is therefore an integral ideal of norm m.

Conversely, write the unique column Hermite normal form of an integral ideal $\mathfrak{a}$ as

$$\mathfrak{a} = A\mathbb{Z} + (B + C\sqrt{3})\mathbb{Z}, \qquad A, C > 0, \quad 0 \le B < A.$$

Stability of A under multiplication by $\sqrt{3}$ implies $C \mid A$ and $C \mid B$. Write $A = CD$ and $B = CE$. Stability of $B + C\sqrt{3}$ then implies $D \mid 3 - E^2$. The HNF inequality gives $0 \le E < D$, and the index is $AC = C^2D$. HNF uniqueness proves injectivity and surjectivity.

For completeness, the last equality in (1562) can be checked without zeta functions. Let $r_3(D) = \#\{E \bmod D : E^2 \equiv 3 \pmod{D}\}$. For $p \ge 5$, Hensel's lemma gives $r_3(p^j) = 2$ for every $j \ge 1$ if $\chi_{12}(p) = 1$, and $r_3(p^j) = 0$ if $\chi_{12}(p) = -1$. Also $r_3(2) = r_3(3) = 1$ and $r_3(2^j) = r_3(3^j) = 0$ for $j \ge 2$. Thus, for every prime power p^e,

$$\sum_{2c+j=e} r_3(p^j) = \sum_{j=0}^{e} \chi_{12}(p)^j.$$

Multiplicativity proves (1562). □

The HNF parameters can be recovered directly from a generator. Suppose $\alpha = a + q\sqrt{3}$ has positive norm m, with $a > 0$ and $q \ge 0$. Put

$$C = \gcd(a, q), \qquad A = a/C, \qquad B = q/C, \qquad D = m/C^2. \tag{1563}$$

If $D = 1$, put $E = 0$. If $D > 1$, then

$$\gcd(B, D) = 1, \qquad E \equiv AB^{-1} \pmod{D}, \qquad 0 \le E < D. \tag{1564}$$

Indeed, $\gcd(B, D) = \gcd(B, A^2 - 3B^2) = \gcd(B, A^2) = 1$, and $E^2 \equiv 3 \pmod{D}$. The principal ideal (α) is exactly $\mathfrak{a}(C, D, E)$: the element $A + B\sqrt{3}$ lies in $D\mathbb{Z} + (E + \sqrt{3})\mathbb{Z}$, so the principal ideal is contained in the HNF ideal, while both ideals have norm D after division by C.

43 Complete integral parametrization of the original equation

For $n \ge 1$ retain the exact integer

$$m_n = 18n^2 - 2. \tag{1565}$$

It satisfies $m_n > 0$ and $m_n \equiv 1 \pmod 3$.

Lemma 43.1 (Every ideal of norm m_n has positive-norm generators)**.** *If $\mathfrak{a}$ is an integral ideal of $\mathcal{O}$ of norm m_n, then every generator has norm $+m_n$, rather than $-m_n$.*

Proof. Lemma 41.1 makes $\mathfrak{a}$ principal. A generator $u + v\sqrt{3}$ has norm $\pm m_n$. Negative norm would give

$$u^2 - 3v^2 = -m_n,$$

and reduction modulo 3 would give $u^2 \equiv -m_n \equiv 2 \pmod 3$, which is impossible. Thus the norm is $+m_n$. □

For each $\mathbf{p} = (C, D, E) \in \mathcal{P}(m_n)$, choose a generator of $\mathfrak{a}(C, D, E)$. By Lemma 43.1 its norm is m_n. Change its sign so that both real embeddings are positive, and multiply by a unique power of ϵ so as to obtain the unique generator

$$\alpha_{n,\mathbf{p}} = a_{n,\mathbf{p},0} + q_{n,\mathbf{p},0}\sqrt{3}, \qquad \sqrt{m_n} \le \alpha_{n,\mathbf{p}} < \epsilon\sqrt{m_n}. \tag{1566}$$

Existence and uniqueness follow from (1556), because the half-open intervals $[\epsilon^j\sqrt{m_n}, \epsilon^{j+1}\sqrt{m_n})$ partition $\mathbb{R}_{>0}$. Since $\alpha'_{n,\mathbf{p}} = m_n/\alpha_{n,\mathbf{p}}$, the lower inequality in (1566) implies $q_{n,\mathbf{p},0} \ge 0$.

Lemma 43.2 (Purely integral reduced seed and descent)**.** *For an integral pair (a, q) satisfying*

$$a^2 - 3q^2 = m_n, \qquad a > 0, \qquad q \ge 0,$$

the half-open inequalities

$$\sqrt{m_n} \le a + q\sqrt{3} < \epsilon\sqrt{m_n}$$

are equivalent to the purely integral inequalities

$$a > 2q \ge 0. \tag{1567}$$

If $a \le 2q$, multiplication by $\epsilon^{-1} = 2 - \sqrt{3}$ is the integral inverse step

$$(a, q) \longmapsto (a^-, q^-) = (2a - 3q, 2q - a). \tag{1568}$$

It preserves the norm, has $a^- > 0$ and $0 \le q^- < q$, and can therefore be iterated until the unique pair satisfying (1567) *is reached.*

Proof. Put $u = (a + q\sqrt{3})/\sqrt{m_n}$. Norm positivity and the sign conditions give $u \ge 1$. Moreover

$$\frac{a}{\sqrt{m_n}} = \frac{u + u^{-1}}{2}, \qquad \frac{q\sqrt{3}}{\sqrt{m_n}} = \frac{u - u^{-1}}{2}.$$

For $u \ge 1$, the ratio

$$\frac{a}{q} = \sqrt{3}\,\frac{u^2 + 1}{u^2 - 1}$$

(with value $+\infty$ when $q = 0$) is strictly decreasing. At $u = \epsilon$ it equals 2, since $\epsilon + \epsilon^{-1} = 4$. Thus $u < \epsilon$ is equivalent to $a > 2q$; the strict inequality exactly matches the half-open upper endpoint.

If $a \le 2q$, norm positivity gives $a > \sqrt{3}q$, and hence

$$a^- = 2a - 3q > (2\sqrt{3} - 3)q > 0, \qquad q^- = 2q - a \ge 0.$$

Also $a > \sqrt{3}q > q$ gives $q^- = 2q - a < q$. Thus the positive integer q strictly decreases at every inverse step, so the process terminates. A terminal pair has $a > 2q$. Conversely, for such a pair the second coordinate after one inverse step is negative, so no further inverse step stays in the nonnegative-q region. This proves uniqueness as well as termination. □

The preceding seed can be reconstructed from the HNF data by a bounded integer search; thus the parametrization does not require an unspecified choice of a generator. More precisely, for $\mathbf{p} = (C, D, E) \in \mathcal{P}(m_n)$ there is a unique pair (A, B) with

$$\begin{gathered} A, B \in \mathbb{Z}, \qquad A > 0, \qquad 0 \le B, \qquad B^2 < D, \\ A^2 - 3B^2 = D, \qquad A \equiv EB \pmod{D}. \end{gathered} \tag{1569}$$

It gives the generator in (1566) by the exact integer formula

$$(a_{n,\mathbf{p},0}, q_{n,\mathbf{p},0}) = (CA, CB). \tag{1570}$$

Indeed, existence follows by writing the reduced generator as $C(A+B\sqrt{3})$. The seed inequality $A>2B$ gives $D=A^2-3B^2>B^2$. Conversely, the conditions in (1569) imply $A>2B$ and put $A+B\sqrt{3}$ in $D\mathbb{Z}+(E+\sqrt{3})\mathbb{Z}$. Its principal ideal is contained in that HNF ideal, and both have norm D, so they are equal. Lemma 43.2 and HNF uniqueness give uniqueness. In particular one needs to test only the integers $0\le B\le\lfloor\sqrt{D-1}\rfloor$ and the positive square $A=\sqrt{D+3B^2}$. The phase introduced below is therefore also explicit in the same integer data:

$$\tau_{n,(C,D,E)}=\frac{\log((A+B\sqrt{3})/\sqrt{D})}{\log\epsilon}. \tag{1571}$$

Here the factor C cancels because $m_n=C^2D$; no ideal generator remains to be chosen.

For $k\ge 0$ define integers $a_{n,\mathbf{p},k}$ and $q_{n,\mathbf{p},k}$ by

$$a_{n,\mathbf{p},k}+q_{n,\mathbf{p},k}\sqrt{3}=\alpha_{n,\mathbf{p}}\epsilon^k. \tag{1572}$$

Equivalently,

$$\begin{pmatrix}a_{n,\mathbf{p},k+1}\\ q_{n,\mathbf{p},k+1}\end{pmatrix}=\begin{pmatrix}2&3\\1&2\end{pmatrix}\begin{pmatrix}a_{n,\mathbf{p},k}\\ q_{n,\mathbf{p},k}\end{pmatrix}. \tag{1573}$$

Theorem 43.3 (Integral parametrization, with inverse and multiplicity)**.** *The map*

$$(n,\mathbf{p},k)\longmapsto(a_{n,\mathbf{p},k},q_{n,\mathbf{p},k},n),\qquad n\ge 1,\quad \mathbf{p}\in\mathcal{P}(m_n),\quad k\ge 0, \tag{1574}$$

is a bijection onto all integral solutions of

$$a^2-3q^2-18n^2=-2,\qquad a>0,\ q\ge 0,\ n\ge 1. \tag{1575}$$

Every such solution automatically satisfies $a+4n>q$.

The inverse is as follows. From (a,q,n) form $\alpha=a+q\sqrt{3}$ and recover (C,D,E) by (1563)–(1564)*. There is a unique $k\ge 0$ such that*

$$\alpha=\alpha_{n,(C,D,E)}\epsilon^k. \tag{1576}$$

Thus every point occurs with multiplicity exactly one.

Proof. Equation (1572) preserves the norm, so its image satisfies

$$a_{n,\mathbf{p},k}^2-3q_{n,\mathbf{p},k}^2=m_n,$$

which is exactly the equation in (1575). Both embeddings of the generator are positive, hence $a>0$; and $\alpha_{n,\mathbf{p}}\ge\sqrt{m_n}$ and $k\ge 0$ imply $q\ge 0$.

Conversely, a solution in (1575) has $\mathrm{N}(a+q\sqrt{3})=m_n>0$. Since $a>0$ and $q\ge 0$, the inequality $a^2-3q^2=m_n>0$ gives $a>\sqrt{3}q$; hence both embeddings are positive. Its principal ideal has unique HNF data (C,D,E). The quotient of $a+q\sqrt{3}$ by the canonical generator of that ideal is a totally positive unit, hence is ϵ^k for a unique $k\in\mathbb{Z}$. Lemma 43.2 supplies the completely integral recovery of k: iterate (1568) precisely while $a\le 2q$; the number of steps is k, and the terminal pair is $(a_{n,\mathbf{p},0},q_{n,\mathbf{p},0})$. In particular $k\ge 0$. This proves surjectivity and the stated inverse. HNF uniqueness, seed uniqueness, and unit-exponent uniqueness prove multiplicity one.

It remains to prove the chamber wall. If a point with $n\ge 1$ had $q\ge a+4n$, then $a\le q-4n$, and the unchanged equation would give

$$3q^2+18n^2-2=a^2\le(q-4n)^2.$$

After moving terms and dividing by 2, this is

$$q^2+4qn+n^2\le 1.$$

For nonnegative integers with $n\ge 1$, equality could occur only at $(q,n)=(0,1)$, but that pair does not satisfy $q\ge a+4n$ with $a>0$. Thus the assumed wall inequality is impossible. □

The congruence needed by the original inverse map to the (x, y, z) coordinates is retained automatically: reducing $a^2 - 3q^2 - 18n^2 = -2$ modulo 2 gives

$$a \equiv q \pmod 2. \tag{1577}$$

No parity class has been lost or added in Theorem 43.3.

Proposition 43.4 (The face $n = 0$)**.** *All points satisfying the original equation and chamber with $n = 0$ are*

$$a_k + q_k\sqrt{3} = (1 + \sqrt{3})\epsilon^k, \qquad k \geq 1. \tag{1578}$$

They have height $4a_k$ and occur with multiplicity one.

Proof. When $n = 0$, the norm is -2. Its parity implies $a \equiv q \pmod 2$, and therefore

$$\frac{a + q\sqrt{3}}{1 + \sqrt{3}} = \frac{3q - a}{2} + \frac{a - q}{2}\sqrt{3} \in \mathcal{O}.$$

The quotient has norm one, so it is $\pm\epsilon^k$ by Lemma 41.1. The conditions $a > 0$ and $q \geq 0$ select the positive sign. At $k = 0$ one gets $(a, q) = (1, 1)$, which is excluded by the strict chamber wall $a > q$. At every $k \geq 1$, the recurrence (1573) gives $a_{k+1} - q_{k+1} = a_k + q_k > 0$. For $k \leq -1$, direct use of $a = (t - 2/t)/2$ with $t = (1 + \sqrt{3})\epsilon^k \leq \sqrt{3} - 1 < \sqrt{2}$ gives $a < 0$. Hence precisely the exponents $k \geq 1$ occur. □

44 Exact divisor and sharp-height formulas

The number of unit-orbits at a fixed positive n has an especially simple divisor formula.

Proposition 44.1 (Orbit divisor sum and shifted factorization)**.** *For $n \geq 1$, the number of orbits under $(a, q) \mapsto (2a + 3q, a + 2q)$ among the solutions with $a > 0$ and $q \geq 0$ is*

$$\rho(m_n) = \sum_{d \mid 18n^2 - 2} \chi_{12}(d) = \rho(3n - 1)\rho(3n + 1). \tag{1579}$$

Proof. The first equality is Theorem 43.3 and Proposition 42.1. For the second, write

$$m_n = 2(3n - 1)(3n + 1).$$

The two displayed linear factors have greatest common divisor dividing 2. The function $\rho = 1 * \chi_{12}$ is multiplicative and $\rho(2^j) = 1$ for all $j \geq 0$. Removing all powers of 2 therefore makes the two factors coprime without changing any of the three ρ values, proving the product identity. □

For $\mathbf{p} \in \mathcal{P}(m_n)$ define its exact unit phase by

$$\tau_{n,\mathbf{p}} = \frac{\log(\alpha_{n,\mathbf{p}}/\sqrt{m_n})}{\log \epsilon}, \qquad 0 \leq \tau_{n,\mathbf{p}} < 1. \tag{1580}$$

For $n \geq 1$ put

$$Y_n(X) = \frac{X - 17n}{4}. \tag{1581}$$

Whenever $Y_n(X) \geq \sqrt{m_n}$, put

$$L_n(X) = \frac{1}{\log \epsilon} \log\left(\frac{Y_n(X) + \sqrt{Y_n(X)^2 - m_n}}{\sqrt{m_n}}\right). \tag{1582}$$

Let

$$\mathcal{N}(X) = \{n \in \mathbb{Z}_{\geq 1} : Y_n(X) \geq \sqrt{m_n}\}. \tag{1583}$$

This is the exact finite range

$$X \geq 17n + 4\sqrt{18n^2 - 2}; \tag{1584}$$

no asymptotic replacement of this boundary is made.

For the $n = 0$ face define

$$R_0(X) = \max\left\{0, \left\lfloor \frac{\log\left(\dfrac{X/4 + \sqrt{(X/4)^2 + 2}}{1 + \sqrt{3}}\right)}{\log \epsilon} \right\rfloor\right\}. \tag{1585}$$

Theorem 44.2 (Exact sharp-height floor formula)**.** *For every real $X \geq 0$,*

$$R(X) = R_0(X) + \sum_{n \in \mathcal{N}(X)} \sum_{\mathbf{p} \in \mathcal{P}(m_n)} \left(\lfloor L_n(X) - \tau_{n,\mathbf{p}} \rfloor + 1\right). \tag{1586}$$

Every equality case in $4a + 17n \leq X$ is included by the floor in (1586)*.*

Proof. For a fixed positive n and parameter $\mathbf{p}$, put $t_k = \alpha_{n,\mathbf{p}} \epsilon^k$. Since $t_k \geq \sqrt{m_n}$,

$$a_{n,\mathbf{p},k} = \frac{1}{2}\left(t_k + \frac{m_n}{t_k}\right) \tag{1587}$$

is nondecreasing, and is strictly increasing after the possible initial case $q = 0$. The unchanged height inequality is $a_{n,\mathbf{p},k} \leq Y_n(X)$. It has no solution unless $Y_n(X) \geq \sqrt{m_n}$. In that range, solving the quadratic inequality in t_k gives

$$t_k \leq Y_n(X) + \sqrt{Y_n(X)^2 - m_n}. \tag{1588}$$

After division by $\sqrt{m_n}$ and logarithms, this is precisely

$$k + \tau_{n,\mathbf{p}} \leq L_n(X). \tag{1589}$$

Because $L_n(X) \geq 0$ and $0 \leq \tau_{n,\mathbf{p}} < 1$, the number of integers $k \geq 0$ satisfying (1589) is exactly $\lfloor L_n(X) - \tau_{n,\mathbf{p}} \rfloor + 1$; this expression is zero, not negative, when $L_n(X) < \tau_{n,\mathbf{p}}$. Equality is retained. Theorem 43.3 now proves the positive-n sum.

For $n = 0$, put $t = (1 + \sqrt{3})\epsilon^k$. Since the norm is -2, $a_k = (t - 2/t)/2$. The inequality $4a_k \leq X$ is equivalent to

$$t \leq X/4 + \sqrt{(X/4)^2 + 2}.$$

Counting the integers $k \geq 1$ gives (1585) and completes the proof. □

Write

$$J_n(X) = \lfloor L_n(X) \rfloor, \qquad \lambda_n(X) = L_n(X) - J_n(X) \in [0, 1), \tag{1590}$$

and define the exact phase distribution

$$A_n(t) = \#\{\mathbf{p} \in \mathcal{P}(m_n) : \tau_{n,\mathbf{p}} \leq t\}, \qquad 0 \leq t < 1, \tag{1591}$$

with discrepancy

$$\Delta_n(t) = A_n(t) - t\rho(m_n). \tag{1592}$$

Since $\lfloor J + \lambda - \tau \rfloor + 1 = J + \mathbf{1}_{\{\tau \leq \lambda\}}$, Theorem 44.2 gives the exact identity

$$R(X) - C_0 X = R_0(X) + \left[\sum_{n \in \mathcal{N}(X)} \rho(m_n) L_n(X) - C_0 X\right] + \sum_{n \in \mathcal{N}(X)} \Delta_n(\lambda_n(X)). \tag{1593}$$

This identity includes the two unbounded coordinate faces and every sharp height boundary point. Formula (1585) also gives $R_0(X) = O(\log(2 + X))$, so the $n = 0$ face is below the requested $X^{1/2+\varepsilon}$ scale; the two positive-n brackets are the genuine analytic terms.

45 The phase term as exact Hecke-character data

The last sum in (1593) is not determined by the divisor count $\rho(m_n)$. It records where the generator of each ideal lies inside a fundamental interval for the positive units.

For $\ell \in \mathbb{Z}$ and a nonzero integral ideal $\mathfrak{a} = (\gamma)$, define

$$\xi_\ell(\mathfrak{a}) = \exp\Big(\frac{\pi i \ell}{\log \epsilon} \log\Big|\frac{\gamma}{\gamma'}\Big|\Big). \tag{1594}$$

This is independent of the generator: replacing γ by $\pm\epsilon^j\gamma$ changes the logarithm by $2j\log\epsilon$, and the exponential by $e^{2\pi i \ell j} = 1$. It is multiplicative, hence is the unramified ideal part of a unitary Hecke character of K. For $m = m_n$ and the canonical generator,

$$\xi_\ell(\mathfrak{a}(C,D,E)) = e^{2\pi i \ell \tau_{n,(C,D,E)}}. \tag{1595}$$

Define its norm-m coefficient by

$$S_\ell(m) = \sum_{\mathrm{N}\,\mathfrak{a}=m} \xi_\ell(\mathfrak{a}). \tag{1596}$$

Then $S_0(m) = \rho(m)$, whereas the nonzero $S_\ell(m_n)$ are the Fourier coefficients of the phase multiset.

The coefficient $S_\ell(m)$ also has a completely explicit Euler-factor description. It is multiplicative in m. If $p \geq 5$ is split, choose one prime ideal $\mathfrak{p}$ above p and put $\omega_{\ell,p} = \xi_\ell(\mathfrak{p})$. Since $\mathfrak{p}\bar{\mathfrak{p}} = (p)$ and $\xi_\ell((p)) = 1$, one has

$$S_\ell(p^e) = \sum_{j=0}^{e} \omega_{\ell,p}^{2j-e} \qquad (\chi_{12}(p) = 1). \tag{1597}$$

If $p \geq 5$ is inert, then

$$S_\ell(p^e) = \begin{cases} 1, & 2 \mid e, \\ 0, & 2 \nmid e. \end{cases} \qquad (\chi_{12}(p) = -1). \tag{1598}$$

Finally the unique primes above 2 and 3 are generated by $1+\sqrt{3}$ and $\sqrt{3}$, respectively. Since

$$\left|\frac{1+\sqrt{3}}{1-\sqrt{3}}\right| = \epsilon, \qquad \left|\frac{\sqrt{3}}{-\sqrt{3}}\right| = 1,$$

their factors are

$$S_\ell(2^e) = (-1)^{\ell e}, \qquad S_\ell(3^e) = 1. \tag{1599}$$

These formulas enumerate every ideal factor with multiplicity one. At $\ell = 0$ they reduce prime by prime to the divisor sum $\rho(m)$; at $\ell \neq 0$ they show exactly what angular information the unweighted divisor sum omits.

They also retain the shifted factorization of m_n at every Fourier mode. Writing $3n-1 = 2^r u$ and $3n+1 = 2^s v$ with u, v odd, one has $\gcd(u,v) = 1$. Multiplicativity and (1599) therefore give the exact identity

$$S_\ell(18n^2-2) = (-1)^\ell S_\ell(3n-1) S_\ell(3n+1). \tag{1600}$$

Indeed, the exponent of 2 on the left is $1+r+s$, while the product on the right has total exponent $r+s$; all odd prime factors are disjoint. Thus the angular obstruction is itself a shifted correlation, now of nontrivial Hecke coefficients rather than merely of their dimensions.

Here is an endpoint-exact Fourier statement. Suppose $0 < \lambda < 1$, and let

$$b_n(t) = \#\{\mathbf{p} \in \mathcal{P}(m_n) : \tau_{n,\mathbf{p}} = t\}. \tag{1601}$$

Fejér summation of the Fourier series of $\mathbf{1}_{[0,\lambda]} - \lambda$ gives

$$\Delta_n(\lambda) = \frac{1}{2}\big(b_n(0) + b_n(\lambda)\big) + \lim_{M\to\infty} \sum_{0<|\ell|\le M} \left(1 - \frac{|\ell|}{M+1}\right) \frac{1 - e^{-2\pi i \ell \lambda}}{2\pi i \ell} S_\ell(m_n). \tag{1602}$$

The two half-weights correct the midpoint convention at the circular endpoint 0 and at λ. When $\lambda = 0$, the exact value is simply $\Delta_n(0) = b_n(0)$.

Combining this with (1600) makes the aggregate obstruction a literal shifted correlation. Define the exact endpoint contribution

$$\mathcal{E}(X) = \sum_{\substack{n\in\mathcal{N}(X)\\ \lambda_n(X)=0}} b_n(0) + \frac{1}{2} \sum_{\substack{n\in\mathcal{N}(X)\\ \lambda_n(X)>0}} \big(b_n(0) + b_n(\lambda_n(X))\big). \tag{1603}$$

Since $\mathcal{N}(X)$ is finite, summation over n commutes with the Fejér limit and gives

$$\sum_{n\in\mathcal{N}(X)} \Delta_n(\lambda_n(X)) = \mathcal{E}(X) + \lim_{M\to\infty} \sum_{0<|\ell|\le M} \left(1 - \frac{|\ell|}{M+1}\right) \frac{(-1)^\ell}{2\pi i \ell} \times \sum_{\substack{n\in\mathcal{N}(X)\\ \lambda_n(X)>0}} \big(1 - e^{-2\pi i \ell \lambda_n(X)}\big) S_\ell(3n-1) S_\ell(3n+1). \tag{1604}$$

Every coefficient and endpoint in this identity comes from the original weak height inequality; no smoothing has been inserted.

Equations (1593) and (1602) identify the analytic problem without discarding the original height. Its radial part is the sharply weighted shifted correlation

$$\sum_{n\in\mathcal{N}(X)} \rho(3n-1)\rho(3n+1) L_n(X), \tag{1605}$$

and its angular part requires cancellation in the polynomially sampled Hecke coefficients

$$S_\ell(18n^2 - 2), \tag{1606}$$

uniformly over the Fourier modes needed to approximate the discontinuous threshold $\tau \le \lambda_n(X)$. The endpoint (1584) gives $L_n(X) = 0$ and has a square-root singularity in its derivative; it is not a fixed smooth weight. A bound for the unweighted orbit total $\rho(m_n)$ therefore does not bound the actual sharp-height count.

46 An unconditional effective exponent for the exact sector

This section records the power saving that is presently justified without assuming the square-root estimate.

Let

$$F(a,q,n) = a^2 - 3q^2 - 18n^2, \qquad \mathscr{X} = \{F = -2\}, \qquad \mathbf{G} = \mathrm{Spin}(F). \tag{1607}$$

Let $x_0 = (1,1,0)$. Its orthogonal complement is

$$x_0^\perp = \{(3s, s, t) : s, t \in \mathbb{Q}\}, \qquad F(3s,s,t) = 6(s^2 - 3t^2), \tag{1608}$$

so the identity component of its stabilizer is the norm-one torus of $\mathbb{Q}(\sqrt{3})$. Its rational character group is zero because Galois conjugation acts by -1 on its geometric character lattice.

The form F is anisotropic over $\mathbb{Q}$. Indeed, suppose a primitive integral triple satisfies $F(a,q,n)=0$. Modulo 3 one first gets $3 \mid a$. Writing $a=3a_1$ and dividing the equation by 3 gives $3a_1^2-q^2-6n^2=0$, hence $3 \mid q$. Write $q=3q_1$ and divide by 3 again to get

$$a_1^2-3q_1^2-2n^2=0. \tag{1609}$$

Modulo 3, the sum $a_1^2+n^2$ is zero, which forces $3 \mid a_1$ and $3 \mid n$. Thus 3 divides a,q,n, contradicting primitivity. In particular the associated arithmetic Fuchsian quotient is cocompact.

For $X>0$ let B_X be the exact subset of $\mathscr{X}(\mathbb{R})$ cut out by

$$a>0, \quad q\geq 0, \quad n\geq 0, \quad a+4n>q, \quad 4a+17n\leq X. \tag{1610}$$

With the gauge measure

$$dm_\infty=\frac{dq\,dn}{2a} \qquad (a>0), \tag{1611}$$

put

$$I_\infty=\frac{1}{2\sqrt{3}}\log\!\left(\frac{3}{2\sqrt{2}}\right). \tag{1612}$$

Lemma 46.1 (Exact-sector volume with a power-saving remainder)**.** *As $X\to\infty$,*

$$m_\infty(B_X)=I_\infty X+O(1). \tag{1613}$$

Moreover, the family B_X is effectively well-rounded in the sense of Benoist–Oh, Definition 13.1 [54, p. 1931].

Proof. Introduce exact polar coordinates on the negative plane by

$$\sqrt{3}q=r\cos\theta, \qquad 3\sqrt{2}n=r\sin\theta, \qquad 0\leq\theta\leq\pi/2. \tag{1614}$$

On $F=-2$ one has $r=\sqrt{a^2+2}$ and

$$dm_\infty=\frac{1}{6\sqrt{6}}\,da\,d\theta. \tag{1615}$$

Write $b(\theta)=17\sin\theta/(3\sqrt{2})$. In the positive orthant the outer boundary is the unique nonnegative solution $A_X(\theta)$ of

$$4A_X(\theta)+b(\theta)\sqrt{A_X(\theta)^2+2}=X. \tag{1616}$$

For all sufficiently large X, uniformly in θ,

$$A_X(\theta)=\frac{X}{4+b(\theta)}+O(X^{-1}). \tag{1617}$$

To see this without a formal expansion, subtract $(4+b)A_X$ from (1616) and use

$$\sqrt{A_X^2+2}-A_X=\frac{2}{\sqrt{A_X^2+2}+A_X}=O(X^{-1}),$$

where (1616) uniformly gives $A_X\asymp X$. Integration of (1617) in (1615) gives

$$\frac{X}{6\sqrt{6}}\int_0^{\pi/2}\frac{d\theta}{4+\frac{17}{3\sqrt{2}}\sin\theta}+O(X^{-1})=I_\infty X+O(X^{-1}).$$

For the asserted exact value, set $c = 17/(3\sqrt{2})$, $d = \sqrt{c^2 - 16} = 1/(3\sqrt{2})$, and $t = \tan(\theta/2)$. Then

$$\begin{aligned}\int_0^{\pi/2} \frac{d\theta}{4 + c\sin\theta} &= \int_0^1 \frac{dt}{2t^2 + ct + 2} \\ &= \frac{1}{d}\left[\log\left(\frac{4t + c - d}{4t + c + d}\right)\right]_{t=0}^{t=1} \\ &= 3\sqrt{2}\log\left(\frac{3}{2\sqrt{2}}\right).\end{aligned}$$

Multiplication by $1/(6\sqrt{6})$ is exactly the constant in (1612). If the strict chamber wall fails, then the calculation in the proof of Theorem 43.3, now over $\mathbb{R}$, gives $q^2 + 4qn + n^2 \le 1$. The removed real subset is therefore compact. For all large X it contributes a fixed finite measure, proving (1613).

It remains to check effective well-roundedness for the actual sector. Let U_δ be the image in the fixed vector representation of the group ball used by Benoist–Oh [54, Definition 13.1]. For every linear functional ℓ on $\mathbb{R}^3$ there is a constant, independent of x and δ, such that

$$|\ell(xg) - \ell(x)| \le C_\ell \delta \|x\| \qquad (g \in U_\delta). \tag{1618}$$

Apply this to the five literal boundary functionals

$$a, \qquad q, \qquad n, \qquad a + 4n - q, \qquad 4a + 17n.$$

On the unbounded part of the positive sheet, $\|x\| \asymp a + 1$, and the last functional is also comparable to $\|x\|$. Thus membership in the outer height inequality can change only in a shell of radial width $O(\delta X)$. Membership in $q \ge 0$ or $n \ge 0$ can change only in angular strips of width $O(\delta)$ about $\theta = \pi/2$ or $\theta = 0$. The boundary $a = 0$ is compact. The boundary $a + 4n - q = 0$ is compact by the calculation preceding (1613); moreover its restriction to $\mathscr{X}$ is regular, because proportionality of $(1, -1, 4)$ to $(2a, -6q, -36n)$ is incompatible with $a, q, n \ge 0$. Consequently its group-action boundary tube has measure $O(\delta)$. By (1615), all these regions together have measure $O(\delta X)$. The same analysis applies to the inner intersection that defines $B^-_{X,\delta}$. Hence

$$m_\infty(B^+_{X,\delta} \setminus B^-_{X,\delta}) \ll \delta\, m_\infty(B_X), \tag{1619}$$

which is precisely Definition 13.1(2), with boundary exponent 1, for the group-action thickenings rather than merely Euclidean tubes.

For Definition 13.1(3) of [54], take the archimedean height to be the Euclidean norm in the displayed integral basis. It is bounded below on $\mathscr{X}$. Uniformly for $0 < \delta < 1$, the thickening $B^+_{X,\delta}$ lies in a fixed finite enlargement of the polar sector; there $H_\infty(x) \asymp 1 + a$ and the measure in a radial interval is $O(da)$. Hence for every $s > 0$, uniformly in δ,

$$\int_{B_X^+} \|u\|^{-s}\, dm_\infty(u) \ll \begin{cases} X^{1-s}, & 0 < s < 1, \\ \log X, & s = 1, \\ 1, & s > 1. \end{cases}$$

Each line is $O(m_\infty(B_X)^{1-\delta_s})$, for example with $\delta_s = \min(s/2, 1/2) > 0$. This is the remaining condition. Since $S = \{\infty\}$, Definition 13.1(1), concerning a finite-place compact open subgroup, is void. The choices of strict or weak inequality on the finitely many piecewise smooth boundary arcs do not change these neighborhood estimates. □

Theorem 46.2 (Unconditional effective conclusion)**.** *There exists $\eta > 0$, depending only on the fixed form, lattice, chamber, and height in* (1549)*, such that*

$$R(X) = C_0 X + O(X^{1-\eta}). \tag{1620}$$

Proof. Use the standard vector representation $\rho : \mathrm{Spin}(F) \to \mathrm{SO}(F) \subset \mathrm{GL}_3$, whose kernel is finite, and the displayed integral basis of the exact lattice $\mathbb{Z}^3$. Let

$$\mathbf{G}_{\mathbb{Z}} = \{g \in \mathbf{G}(\mathbb{Q}) : \rho(g)\mathbb{Z}^3 = \mathbb{Z}^3\}.$$

This is the integral group attached to the chosen representation and lattice in the Benoist–Oh setup [54, pp. 1892–1893]; take their congruence subgroup Γ_S to be the level-one group $\mathbf{G}_{\mathbb{Z}}$ itself. The group $\mathbf{G}$ is connected, simply connected, and almost $\mathbb{Q}$-simple. It is noncompact over $\mathbb{R}$ because F has signature $(1, 2)$. Thus $S = \{\infty\}$ contains every noncompact archimedean place and G_S is noncompact. The orbit $x_0\mathbf{G}$ is the affine symmetric variety $\mathscr{X}$, and its stabilizer has no nontrivial rational character by (1608). These verify every standing hypothesis on pp. 1892–1893 of [54]. Their Theorem 12.5 supplies the required effective equidistribution of the symmetric-stabilizer translates.

The Borel–Harish-Chandra finiteness theorem quoted in [54, p. 1892] p. 1892 says that $\mathscr{X}(\mathbb{Z})$ is a finite union of $\mathbf{G}_{\mathbb{Z}}$-orbits. A rational representative of any such orbit again has a symmetric stabilizer with no rational character: its orthogonal complement is anisotropic over $\mathbb{Q}$, since an isotropic vector in it would be an isotropic vector for F. Apply Benoist–Oh, Theorem 13.2 [54, pp. 1931–1932], to each orbit and to the exact effectively well-rounded family B_X of Lemma 46.1. For each orbit it gives its invariant-volume main term with error $O(m_\infty(B_X)^{1-\eta_j})$ for some $\eta_j > 0$. There are finitely many orbits, so their sum has the same form with $\eta = \min_j \eta_j > 0$.

All invariant measures on the same real homogeneous variety are proportional. Hence the sum of the finitely many orbit main terms is $K\, m_\infty(B_X)$ for one fixed $K > 0$. Lemma 46.1 and the already established asymptotic $R(X) \sim C_0X$ force $KI_\infty = C_0$. Therefore

$$K\, m_\infty(B_X) = K(I_\infty X + O(1)) = C_0X + O(1).$$

Since $m_\infty(B_X) \asymp X$, summing the effective errors proves (1620); the $O(1)$ term is absorbed after decreasing η if necessary. The theorem is applied first to the sequence of positive integral X. For real X, the original height $4a + 17n$ is an integer, so $R(X) = R(\lfloor X \rfloor)$; replacing X by $\lfloor X \rfloor$ changes C_0X by $O(1)$ and proves the same estimate. □

Remark 46.3 (What exponent has actually been obtained). Theorem 46.2 is an unconditional exponent strictly smaller than 1: the error is $O(X^{1-\eta})$ for some $\eta > 0$. Neither the statement nor the proof of [54, Theorem 13.2] identifies an exponent $1 - \eta \le 1/2$ for this sharp sector. Tracing their Sobolev and mixing constants would make a positive η effective in principle, but that calculation has not been carried out in this section and cannot be replaced by the assertion $\eta = 1/2$.

47 Exact obstruction to the square-root claim

There are two complementary descriptions of the remaining obstruction; the first is an exact identity, while the second explains the relevant spectral mechanisms without asserting an unproved intertwining formula.

First, the elementary ideal description gives the literal identity (1593). Thus the square-root claim is equivalent to a bound of that size for the displayed sum of the radial bracket, the aggregate phase discrepancy, and $R_0(X)$. Controlling the radial and phase terms separately is a sufficient route, though the exact identity also permits cancellation between them. Formula (1602) shows that a separate estimate of the latter entails uniform cancellation in the nonzero archimedean Hecke modes (1606). Merely using $\#\mathcal{P}(m_n) = \rho(m_n)$, or replacing each phase distribution by Lebesgue measure, deletes a term that is exactly present in $R(X)$.

Second, F is rationally anisotropic, so $\mathbf{G}$ gives a cocompact arithmetic Fuchsian quotient. A negative-norm vector represents an oriented geodesic, and its stabilizer is the split real torus appearing in (1608). In the standard radial torus-orbit counting kernel, the spectral expansion has two possible sources larger than the square-root scale:

1. an exceptional eigenvalue $\lambda = s(1-s) < 1/4$ can contribute $c_s X^s$ with $s > 1/2$, unless the exact torus period and exact sector transform make $c_s = 0$;

2. after exceptional terms are absent or removed, estimating the tempered spectral sum for a discontinuous cutoff at the conjectural $X^{1/2+\varepsilon}$ scale requires cancellation beyond the general effective-mixing argument.

For the present nonradial chamber, one must first construct a remainder-preserving sector spectral expansion before these candidate terms can even be asserted with their exact coefficients. An actual refutation by a secondary term would then require proving both $s > 1/2$ and $c_s \neq 0$ for this fixed arithmetic quotient and exact chamber. No such construction or nonvanishing has been established here. Conversely, absence of exceptional terms alone would not estimate the full tempered sharp-cutoff sum. This is why the exact parametrization neither proves nor refutes (1552), while it does isolate the data a future proof must bound.

For comparison only, not as a theorem about the present nonradial sector, the classical hyperbolic lattice-point problem for a conjugacy class has an unconditional $O(X^{2/3})$ remainder due to Good [55, Theorem 4], with the modern proof of Chatzakos–Petridis [56, Theorem 1.1 and Conjecture 5.7], and a conjectural $O_\varepsilon(X^{1/2+\varepsilon})$ scale. That published $O(X^{2/3})$ theorem counts a different radial kernel. An exact intertwining of its kernel with the simultaneous chamber and height in (1549) has not been proved, so $O(X^{2/3})$ is not claimed here for $R(X)$.

The independent finite replay used to check the HNF multiplicities, seed reduction, inverse descent, and divisor factorization for $1 \leq n \leq 200$ is retained at `supporting_materials/workbench/evidence/koide_exact_checkpoint_20260904/parametrization/verify_parametrization.py`; its SHA–256 digest is `061c5e22258628169fd1ecb8bcc5e7cf8b8d19bfdb918650668c3e90f2f9730d`. That replay is a finite consistency check. The proofs above, rather than the finite range, establish the parametrization for every admissible integer.

48 Boundary ledger

For clarity, every boundary used above has the following exact treatment.

- $a > 0$ is strict. There is no omitted integral solution with $a = 0$, because the original equation would give a multiple of 3 equal to 2.

- $q = 0$ is included. It corresponds to phase $\tau = 0$, is included by $\tau \leq \lambda$, and is retained in the endpoint term $b_n(0)/2$ in (1602).

- $n = 0$ is included and is counted exactly by $R_0(X)$; the sole solution $(a, q, n) = (1, 1, 0)$ on $a + 4n = q$ is excluded.

- $a + 4n = q$ is strict. Its real intersection is bounded and its sole integral point in the nonnegative chamber is $(1, 1, 0)$.

- $4a + 17n = X$ is included. Equality is included in (1588), (1589), and every floor in (1586).

- The canonical unit interval is half-open, $[\sqrt{m_n}, \epsilon\sqrt{m_n})$. Consequently no generator is counted twice at a unit boundary.

- Conjugate ideals are distinct unless the ideal is fixed by conjugation. They yield distinct canonical phases and distinct $q \geq 0$ unit-orbits; no division by 2 is made. When $q = 0$, the ideal is self-conjugate and still occurs once by HNF uniqueness.

49 Exact unit-orbit census through height 10^8

The census retains exactly the same equation, chamber, strict wall, and height as the preceding theorems:

$$(a,q,n)\in\mathbb{Z}^3,\qquad a^2-3q^2-18n^2=-2,\qquad a>0,\quad q,n\ge 0,\quad a+4n>q,\qquad 4a+17n\le X. \tag{1621}$$

It does not infer an asymptotic from finite data. Its exact fixed-n equation and factorization are

$$a^2-3q^2=m_n,\qquad m_n=18n^2-2=2(3n-1)(3n+1). \tag{1622}$$

For each $n\ge 1$, every ideal of norm m_n is enumerated by its Hermite-normal-form triple

$$(C,D,E)\in\mathbb{Z}_{>0}^2\times\mathbb{Z},\qquad C^2D=m_n,\qquad 0\le E<D,\qquad E^2\equiv 3 \pmod{D}. \tag{1623}$$

The generator is reduced by exact multiplication by the fundamental unit $2+\sqrt{3}$ to the unique seed $a_0>2q_0\ge 0$. Its forward unit orbit is

$$\begin{pmatrix}a_{k+1}\\ q_{k+1}\end{pmatrix}=\begin{pmatrix}2&3\\1&2\end{pmatrix}\begin{pmatrix}a_k\\ q_k\end{pmatrix},\qquad k\ge 0. \tag{1624}$$

Each emitted point is reconstructed by

$$x=\frac{a+4n-q}{2},\qquad y=\frac{a+4n+q}{2},\qquad z=4a+17n, \tag{1625}$$

and is rechecked in both coordinate systems and in the original polynomial $\Delta(x,y,z)=-1$. Thus the accelerated enumeration does not replace the original objects by a floating-point or rescaled surrogate.

49.1 Complete result and checkpoint table

The complete integer-domain audit for $1\le X\le 100{,}000{,}000$ gives

$$\boxed{R(100{,}000{,}000)=3{,}536{,}002.} \tag{1626}$$

The endpoint in n is 2,943,725; the enumeration contains 2,327,513 canonical unit orbits. The following checkpoints are exact integer counts. The residuals use the proved constant C_0 in (1544).

X	$R(X)$	$R(X)-C_0X$	$(R(X)-C_0X)/\sqrt{X}$
200,000	7,109	36.590440053629	0.081818711287
250,000	8,854	13.488050067037	0.026976100134
500,000	17,695	13.976100134073	0.019765190359
1,000,000	35,381	18.952200268147	0.018952200268
2,000,000	70,637	−87.095599463706	−0.061585888992
5,000,000	176,943	132.761001340734	0.059372524752
10,000,000	353,598	−22.477997318532	−0.007108166877
20,000,000	707,458	217.044005362937	0.048532515010
50,000,000	1,768,230	127.610013407342	0.018046781166
100,000,000	3,536,002	−202.779973185317	−0.020277997319

49.2 Certified extrema over every integer

Write $E(X) = R(X) - C_0 X$. Between consecutive occupied heights, $R(X)$ is constant and both $E(X)$ and $E(X)/\sqrt{X}$ are strictly decreasing, since

$$\frac{d}{dX}\left(\frac{R}{\sqrt{X}} - C_0\sqrt{X}\right) = -\frac{R}{2X^{3/2}} - \frac{C_0}{2\sqrt{X}} < 0. \tag{1627}$$

Checking both integer endpoints of every constant-count interval therefore certifies extrema over every integer in the audited range, rather than at selected checkpoints. On $1 \le X \le 10^8$, the maximum signed residual is

$$\begin{aligned} E(77{,}078{,}404) &= 721.79342495712975696566815535386910933 79\ldots, \\ R(77{,}078{,}404) &= 2{,}726{,}372, \end{aligned} \tag{1628}$$

and the minimum signed, hence largest absolute, residual is

$$\begin{aligned} E(32{,}461{,}294) &= -812.83006914880688970919660067974916559 33\ldots, \\ R(32{,}461{,}294) &= 1{,}147{,}085. \end{aligned} \tag{1629}$$

The maximum of $|E(X)|/\sqrt{X}$ over the full audited range is attained at $X = 160$, where

$$\frac{|E(160)|}{\sqrt{160}} = 0.18515707693408214138401018551531723381 6\ldots.$$

On $10{,}000 \le X \le 10^8$, the maximum is

$$\begin{aligned} \frac{|E(867{,}240)|}{\sqrt{867{,}240}} &= 0.17140019280881455972894965803234555250 2\ldots, \\ R(867{,}240) &= 30{,}827, \end{aligned} \tag{1630}$$

and the root entering at that height is

$$(a, q, n) = (199997, 115061, 3956), \qquad (x, y, z) = (50380, 165441, 867240). \tag{1631}$$

The bound $|E(X)| \le 0.171\sqrt{X}$ first fails on this tail at $X = 867{,}216$, where $R(X) = 30{,}826$ and

$$E(X) = 159.46635530774122203304493532343198171 99\ldots.$$

No counterexample to the finite statement with constant 0.172 occurs through 10^8. These are bounded computations, not a claim that either constant persists.

The audit finds 12,916 sign changes. The 3,536,002 roots occupy 2,472,015 distinct heights; the maximum shell multiplicity is 35, at $z = 92{,}923{,}120$, and the largest gap between consecutive occupied heights is 564, from 71,586,276 to 71,586,840.

49.3 Independent replay and exact algorithm

An independent pass decompressed and rechecked all 3,536,002 six-coordinate rows, recomputed every checkpoint, verified the coordinate maps, equations, ordering, chamber, height, nonduplication, and gzip CRC, and returned `PASS`. It also compared the literal (n, a) scan through $X = 20{,}000$, every generalized-Pell seed for $1 \le n \le 1000$, and the full six-coordinate point set through $X = 200{,}000$.

The reproducibility sources and receipts are retained project-relatively under `supporting_materials/workbench/evidence/koide_exact_checkpoint_20260904/census`. The two 10^8-range outputs and the full coordinate receipt are in its `final_100000000` subdirectory. The principal digests are

artifact	SHA-256
`koide_fast_enum.py`	`dddb73b828b63be08e2ff0b5346edb32 31662005f83d5b2448c06fef6ba31017`
`koide_discrepancy_100000000.json`	`da92f0866074e530b8174a48915caa0f 65cd4cd8957e209306adc200739d1961`
`koide_roots_through_100000000.csv.gz`	`dfc313492aedb396dcfa10520bfefc699 ba601357bc9933d921dfa3884310bd4`
`coordinate_audit_receipt.json`	`3010363108e7b460b7331a92f3263bbd 77992aedfc3064973750e16d707da80f`

The exact algorithm first finds the largest admissible n by integer binary search, then factors the unchanged integers $3n-1$ and $3n+1$ with a smallest-prime-factor sieve. It enumerates every HNF triple in (1623), constructs all roots of 3 modulo D by prime-power lifting and the Chinese remainder theorem, reduces each generator to the unique seed, and iterates (1624) until the literal height fails. For each n, the independently checked identity

$$\#\mathcal{P}(m_n) = \rho(m_n) = \rho(3n-1)\rho(3n+1)$$

certifies the orbit multiplicity. The work is

$$X^{1+o(1)} + O(R(X))$$

integer operations before the optional height sort. This complexity and the finite census use no conjectural discrepancy estimate.

50 Exact quaternion bridge: order, shell, and hyperbolic conjugacy classes

The proposed quaternionic realization is correct. The displayed lattice is an order in an indefinite rational quaternion division algebra of discriminant 6. The parity-restricted ternary shell is exactly the reduced-trace-4, reduced-norm-1 locus in that order. Every resulting element is primitive hyperbolic, with eigenvalues $2 \pm \sqrt{3}$ and translation length $2\log(2+\sqrt{3})$.

The centralizer statement is correct when written intrinsically as

$$\mathcal{O} \cap \mathbb{Q}(\gamma) = \mathbb{Z}[\gamma - 2] \cong \mathbb{Z}[\sqrt{3}].$$

The entire integral shell is a finite union of conjugacy classes under $\Gamma = \mathcal{O}^1$. This gives an exact reduction of the Koide count to a finite sum of sharp, non-radial sector counts inside primitive hyperbolic conjugacy classes of a cocompact arithmetic Fuchsian group.

It does **not** prove the open estimate

$$R(X) - C_0 X = O_\varepsilon(X^{1/2+\varepsilon}).$$

Rather, it converts the count exactly into a sharp sector problem on a finite union of hyperbolic conjugacy classes. This places it beside, but does not prove it equivalent to, the standard hyperbolic conjugacy-class lattice-point problem. Even for the radial Huber cutoff, the general pointwise theorem gives an $O(X^{2/3})$ remainder after all exceptional spectral terms are included, while the $O(X^{1/2+\varepsilon})$ remainder is conjectural. The Koide cutoff is a different sharp polyhedral-sector height, so the standard radial theorem cannot be imported without a new transfer or trace-formula argument.

50.1 Quaternion algebra and reduced invariants

Let

$$B = (18, -6)_{\mathbb{Q}} = \mathbb{Q}\langle I, J\rangle/(I^2 - 18,\ J^2 + 6,\ IJ + JI), \qquad K = IJ.$$

Then

$$K^2 = -I^2J^2 = 108.$$

For

$$x = x_0 + x_1 I + x_2 J + x_3 K,$$

the standard involution, reduced trace, and reduced norm are

$$\bar{x} = x_0 - x_1 I - x_2 J - x_3 K, \qquad \mathrm{trd}(x) = x + \bar{x} = 2x_0,$$

and

$$\mathrm{nrd}(x) = x\bar{x} = x_0^2 - 18x_1^2 + 6x_2^2 - 108x_3^2. \tag{1632}$$

All signs in (1632) are forced by $\mathrm{nrd}(I) = -18$, $\mathrm{nrd}(J) = 6$, and $\mathrm{nrd}(K) = -108$.

50.2 The proposed lattice is an order

Put

$$E = \frac{I + J}{2}, \qquad F_0 = \frac{I - J}{2}, \qquad G = \frac{K}{2},$$

and let

$$\mathcal{O} = \mathbb{Z}1 \oplus \mathbb{Z}E \oplus \mathbb{Z}F_0 \oplus \mathbb{Z}G. \tag{1633}$$

The displayed elements are a rational basis of B, because

$$I = E + F_0, \qquad J = E - F_0, \qquad K = 2G.$$

Direct multiplication gives the complete nontrivial table

	E	F_0	G
E	3	$6 - G$	$6E - 3F_0$
F_0	$6 + G$	3	$3E - 6F_0$
G	$-6E + 3F_0$	$-3E + 6F_0$	$27.$

(1634)

Every structure constant in (1634) is integral. Thus (1633) is a unital subring and a full rank-four $\mathbb{Z}$-lattice in B, hence an order. In particular, the unital ring sometimes denoted $\mathbb{Z}[1, E, F_0, G]$ is exactly the direct sum in (1633).

An arbitrary order element is

$$x = r + sE + tF_0 + uG = r + \frac{s + t}{2}I + \frac{s - t}{2}J + \frac{u}{2}K \qquad (r, s, t, u \in \mathbb{Z}), \tag{1635}$$

and (1632) gives

$$\mathrm{trd}(x) = 2r, \qquad \mathrm{nrd}(x) = r^2 - 3s^2 - 12st - 3t^2 - 27u^2. \tag{1636}$$

The reduced-trace pairing on the basis $(1, E, F_0, G)$ is

$$\left(\mathrm{trd}(b_i b_j)\right)_{i,j} = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 6 & 12 & 0 \\ 0 & 12 & 6 & 0 \\ 0 & 0 & 0 & 54 \end{pmatrix},$$

with determinant

$$-11664 = -108^2. \tag{1637}$$

Thus the reduced discriminant of this order is 108, not 6. The algebra has discriminant 6, proved below, so $\mathcal{O}$ is nonmaximal and has index 18 in a maximal order containing it. This index is not an Eichler level: the order is nonmaximal at the ramified primes 2 and 3, so it is not an Eichler order in the standard sense. Any optimal-embedding or trace-formula result that assumes a maximal order must account for this exact nonmaximal order instead of replacing $\mathcal{O}$.

50.3 Trace-zero lattice and exact shell bijection

Equation (1636) gives

$$\mathcal{O}^0 := \{x \in \mathcal{O} : \mathrm{trd}(x) = 0\} = \mathbb{Z}E \oplus \mathbb{Z}F_0 \oplus \mathbb{Z}G.$$

Writing

$$q = s + t, \qquad a = s - t, \qquad n = u$$

gives the exact description

$$\boxed{\mathcal{O}^0 = \left\{ \frac{qI + aJ + nK}{2} : a, q, n \in \mathbb{Z},\ a \equiv q \pmod 2 \right\}.} \tag{1638}$$

The inverse formulas are

$$s = \frac{q + a}{2}, \qquad t = \frac{q - a}{2}, \qquad u = n,$$

which are integral exactly when $a \equiv q \pmod 2$.
Define

$$F(a, q, n) = a^2 - 3q^2 - 18n^2, \qquad v(a, q, n) = \frac{qI + aJ + nK}{2},$$

and

$$\gamma(a, q, n) = 2 + v(a, q, n). \tag{1639}$$

Distinct pure quaternion basis elements anticommute, so

$$v(a, q, n)^2 = \frac{18q^2 - 6a^2 + 108n^2}{4} = -\frac{3}{2}F(a, q, n), \tag{1640}$$

and

$$\mathrm{nrd}(v) = \frac{3}{2}F(a, q, n).$$

Since v has reduced trace zero,

$$\mathrm{trd}(\gamma) = 4, \qquad \mathrm{nrd}(\gamma) = 4 + \mathrm{nrd}(v) = 4 + \frac{3}{2}F(a,q,n). \tag{1641}$$

Therefore (1639) defines a bijection

$$\boxed{\left\{(a,q,n) \in \mathbb{Z}^3 : F(a,q,n) = -2,\ a \equiv q \pmod 2\right\} \xrightarrow{\sim} \{\gamma \in \mathcal{O} : \mathrm{trd}(\gamma) = 4,\ \mathrm{nrd}(\gamma) = 1\}.} \tag{1642}$$

The inverse sends

$$\gamma = 2 + sE + tF_0 + uG$$

to

$$(a,q,n) = (s - t, s + t, u).$$

There is no hidden congruence loss. On the integral equation $F = -2$, reduction modulo 2 gives

$$a^2 - q^2 \equiv 0 \pmod 2,$$

so $a \equiv q \pmod 2$ is automatic. The parity condition in (1642) records the order lattice exactly but removes no integral shell point.

Finally,

$$(\gamma - 2)^2 = 3, \qquad \gamma^2 - 4\gamma + 1 = 0, \qquad \gamma^{-1} = 4 - \gamma = 2 - v \in \mathcal{O}. \tag{1643}$$

Thus every element on the right of (1642) lies in $\Gamma = \mathcal{O}^1$.

50.4 Division over $\mathbb{Q}$ and splitting over $\mathbb{R}$

Square factors may be removed from a Hilbert-symbol presentation, so

$$(18, -6)_{\mathbb{Q}} \simeq (2, -6)_{\mathbb{Q}}.$$

At $p = 3$, write

$$18 = 3^2 \cdot 2, \qquad -6 = 3 \cdot (-2).$$

The odd-prime Hilbert-symbol formula gives

$$(18, -6)_3 = (-1)^{2\cdot 1\cdot (3-1)/2} \left(\frac{2}{3}\right)\left(\frac{-2}{3}\right)^2 = -1. \tag{1644}$$

Thus $B \otimes \mathbb{Q}_3$ is division, and B is a division algebra over $\mathbb{Q}$.

At $p = 2$, write

$$18 = 2 \cdot 9, \qquad -6 = 2 \cdot (-3).$$

The 2-adic formula

$$(2^\alpha u, 2^\beta w)_2 = (-1)^{((u-1)/2)((w-1)/2) + \alpha(w^2-1)/8 + \beta(u^2-1)/8}$$

gives

$$(18, -6)_2 = -1. \tag{1645}$$

At every odd prime other than 3, both valuations are zero and the Hilbert symbol is $+1$. At the real place, $18 > 0$, so the real Hilbert symbol is $+1$. Consequently

$$\mathrm{Ram}(B) = \{2, 3\}, \qquad \mathrm{disc}(B) = 6, \tag{1646}$$

and

$$B \otimes_{\mathbb{Q}} \mathbb{R} \simeq M_2(\mathbb{R}).$$

An explicit real splitting is

$$\rho(I) = \begin{pmatrix} 3\sqrt{2} & 0 \\ 0 & -3\sqrt{2} \end{pmatrix}, \qquad \rho(J) = \begin{pmatrix} 0 & 6 \\ -1 & 0 \end{pmatrix}. \tag{1647}$$

These matrices square to 18 and -6 and anticommute. Therefore

$$\rho(K) = \begin{pmatrix} 0 & 18\sqrt{2} \\ 3\sqrt{2} & 0 \end{pmatrix}.$$

For the element associated with (a, q, n),

$$\rho(\gamma) = \begin{pmatrix} 2 + \frac{3\sqrt{2}}{2} q & 3a + 9\sqrt{2}\, n \\ -\frac{1}{2} a + \frac{3\sqrt{2}}{2} n & 2 - \frac{3\sqrt{2}}{2} q \end{pmatrix}. \tag{1648}$$

Its determinant is the reduced norm in (1641).

Let $\mathcal{M}$ be a maximal overorder containing $\mathcal{O}$. Since $[\mathcal{M} : \mathcal{O}] = 18$, the finite abelian group $\mathcal{M}/\mathcal{O}$ has order 18. Lagrange's theorem therefore shows that multiplication by 18 annihilates every element of this additive quotient; equivalently, the ideal $(18) = 18\mathbb{Z}$ annihilates $\mathcal{M}/\mathcal{O}$, and $18\mathcal{M} \subset \mathcal{O}$. Hence the principal congruence subgroup

$$\mathcal{M}^1(18\mathcal{M}) := \{u \in \mathcal{M}^1 : u - 1 \in 18\mathcal{M}\}$$

satisfies the exact inclusions

$$\mathcal{M}^1(18\mathcal{M}) \subset \mathcal{O}^1 \subset \mathcal{M}^1.$$

Thus $\mathcal{O}^1$ has finite index in the norm-one group of a maximal order and contains a principal congruence subgroup. This is the required finite-index and congruence bridge; it does not replace the local type of $\mathcal{O}$ by the local type of $\mathcal{M}$.

The image

$$\bar{\Gamma} = \rho(\mathcal{O}^1)/\{\pm 1\} \subset PSL_2(\mathbb{R})$$

is a discrete congruence arithmetic Fuchsian group. Since the defining rational quaternion algebra is division, $\bar{\Gamma}\backslash\mathbb{H}$ is compact; see [60, Lemma 38.1.2, Proposition 38.1.3, and Definition 38.7.2].

50.5 Centralizer, optimal quadratic order, and primitivity

Fix a shell point and set

$$v = \gamma - 2.$$

Equation (1643) gives $v^2 = 3$, hence

$$\mathbb{Q}(\gamma) = \mathbb{Q}(v) \cong \mathbb{Q}(\sqrt{3}). \tag{1649}$$

Since γ is noncentral, the centralizer theorem for quaternion algebras gives

$$C_B(\gamma) = \mathbb{Q}(\gamma). \tag{1650}$$

Moreover,

$$\mathbb{Z}[v] \subseteq \mathcal{O} \cap \mathbb{Q}(v).$$

Every element of $\mathcal{O}$ is integral over $\mathbb{Z}$, so every element of the intersection is an algebraic integer in $\mathbb{Q}(\sqrt{3})$. Because

$$\mathcal{O}_{\mathbb{Q}(\sqrt{3})} = \mathbb{Z}[\sqrt{3}]$$

and $\mathbb{Z}[v]$ is already isomorphic to this maximal quadratic order, both inclusions force

$$\boxed{\mathcal{O} \cap \mathbb{Q}(\gamma) = \mathbb{Z}[v] = \mathbb{Z}[\gamma - 2] \cong \mathbb{Z}[\sqrt{3}].} \tag{1651}$$

Thus each shell point gives an optimal embedding

$$\iota_\gamma : \mathbb{Z}[\sqrt{3}] \hookrightarrow \mathcal{O}, \qquad \sqrt{3} \longmapsto \gamma - 2. \tag{1652}$$

Conversely, every embedding (1652) gives a trace-4, norm-1 element $2 + \iota(\sqrt{3})$, hence one integral ternary point. The shell is exactly the set of these optimal embeddings before quotienting by conjugation.

The norm-one units of $\mathbb{Z}[\sqrt{3}]$ are

$$\{\pm(2 + \sqrt{3})^k : k \in \mathbb{Z}\}. \tag{1653}$$

For completeness, $2 + \sqrt{3}$ is the smallest positive norm-one unit greater than 1. If $x + y\sqrt{3} > 1$ has norm 1 and $y > 0$, then $y \geq 1$, $x^2 = 1 + 3y^2$, and $x + y\sqrt{3} \geq 2 + \sqrt{3}$. Multiplication by a suitable power of $2 - \sqrt{3}$ reduces any positive norm-one unit to $[1, 2 + \sqrt{3})$, proving (1653).

Equations (1650)–(1653) imply

$$C_\Gamma(\gamma) = \{\pm\gamma^k : k \in \mathbb{Z}\}. \tag{1654}$$

Hence γ is primitive in Γ and its image is primitive in $\bar{\Gamma}$. Indeed, a root of γ, or a root of its class modulo the center, would commute with γ and therefore be a unit from (1653); the exponent equation $rk = 1$ has no solution with $k \geq 2$.

The reduced characteristic polynomial is

$$T^2 - 4T + 1,$$

so the real eigenvalues are

$$2 + \sqrt{3}, \qquad 2 - \sqrt{3}. \tag{1655}$$

For a hyperbolic element of $PSL_2(\mathbb{R})$,

$$2\cosh(\ell/2) = |\operatorname{tr}(\rho(\gamma))| = 4.$$

Thus every shell element has primitive translation length

$$\boxed{\ell_0 = 2\operatorname{arcosh}(2) = 2\log(2 + \sqrt{3}).} \tag{1656}$$

50.6 Integral adjoint action and finitely many classes

Let

$$\mathscr{S} = \{\gamma \in \Gamma : \mathrm{trd}(\gamma) = 4\}.$$

By (1642), $\mathscr{S}$ is exactly the entire integral ternary shell. By (1656), every element of $\mathscr{S}$ is hyperbolic with translation length ℓ_0.

Put

$$L = \{(a, q, n) \in \mathbb{Z}^3 : a \equiv q \pmod 2\}.$$

Conjugation preserves $\mathcal{O}$, $\mathcal{O}^0$, and reduced norm. Because

$$\mathrm{nrd}(v(a, q, n)) = \frac{3}{2} F(a, q, n),$$

it gives an exact orthogonal representation on the actual parity lattice

$$\mathrm{Ad} : \Gamma/\{\pm 1\} \hookrightarrow SO^+(L, F). \tag{1657}$$

The kernel before dividing by the center is $\Gamma \cap \mathbb{Q}^\times = \{\pm 1\}$. No action on the larger lattice $\mathbb{Z}^3$, and no finite-index equality with its full integral orthogonal group, is asserted.

There are only finitely many Γ-conjugacy classes in $\mathscr{S}$. Choose a compact fundamental set $D \subset \mathbb{H}$ for $\bar{\Gamma}$, and fix $z_0 \in \mathbb{H}$. For each conjugacy class, conjugate a representative so that its invariant axis meets D. If $z \in D$ lies on that axis, then

$$d(z, \gamma z) = \ell_0.$$

Writing

$$R_D = \max_{w \in D} d(z_0, w),$$

the triangle inequality gives

$$d(z_0, \gamma z_0) \le 2R_D + \ell_0.$$

Proper discontinuity leaves only finitely many elements of $\bar{\Gamma}$ moving z_0 by at most this amount. Hence only finitely many $\bar{\Gamma}$-classes occur. This also proves finiteness of the corresponding Γ-classes: if two images are conjugate in $\bar{\Gamma}$, lifting a conjugator gives

$$\delta \gamma_1 \delta^{-1} = \pm \gamma_2.$$

Both sides before the possible sign have reduced trace 4; the minus sign would have reduced trace -4, so it is impossible.

Choose representatives $\gamma_1, \dots, \gamma_h$. Then

$$\boxed{\mathscr{S} = \coprod_{j=1}^{h} \{\delta \gamma_j \delta^{-1} : [\delta] \in \Gamma/C_\Gamma(\gamma_j)\}, \qquad C_\Gamma(\gamma_j) = \{\pm \gamma_j^k : k \in \mathbb{Z}\}.} \tag{1658}$$

This proves finiteness but does not compute h or a representative list. Those finite arithmetic data are needed by a class-by-class spectral formula whose coefficients contain geodesic periods.

50.7 Exact reformulation of the Koide chamber count

For $\gamma \in \mathscr{S}$, let

$$c(\gamma) = (a(\gamma), q(\gamma), n(\gamma))$$

be the inverse coordinate map in section 50.3. Let

$$\mathcal{C} = \{(a, q, n) \in \mathbb{R}^3 : a > 0,\ q \geq 0,\ n \geq 0,\ a + 4n > q\},$$

and retain the canonical height

$$H(a, q, n) = 4a + 17n.$$

The integral bijections in theorems 39.4 and 43.3 give

$$\boxed{R(X) = \sum_{j=1}^{h} \# \left\{ [\delta] \in \Gamma / C_\Gamma(\gamma_j) : \begin{array}{l} c(\delta\gamma_j\delta^{-1}) \in \mathcal{C}, \\ H(c(\delta\gamma_j\delta^{-1})) \leq X \end{array} \right\}.} \tag{1659}$$

There is no hidden finite-to-one factor in (1659): cosets by the centralizer parametrize the elements of each class exactly once, and (1642) parametrizes shell elements by triples exactly once. Put $C_j = C_\Gamma(\gamma_j)$. Here Γ/C_j is the set of left cosets δC_j, not the set of right cosets $C_j\delta$, and the exact orbit bijection is

$$\Phi_j : \Gamma/C_j \longrightarrow \{\delta\gamma_j\delta^{-1} : \delta \in \Gamma\}, \qquad \delta C_j \longmapsto \delta\gamma_j\delta^{-1}.$$

Indeed, right multiplication of δ by an element of C_j does not change its conjugate, so Φ_j is well defined and surjective; and

$$\Phi_j(\delta_1 C_j) = \Phi_j(\delta_2 C_j) \iff \delta_2^{-1}\delta_1 \in C_j \iff \delta_1 C_j = \delta_2 C_j,$$

so it is injective. If $\pi : \Gamma \to \bar{\Gamma}$ is the quotient map, then

$$\Gamma/C_j \cong \bar{\Gamma}/C_{\bar{\Gamma}}(\bar{\gamma}_j)$$

via $\delta C_j \mapsto \pi(\delta) C_{\bar{\Gamma}}(\bar{\gamma}_j)$. To prove the required equality of stabilizers, observe first that $\pi(C_j) \subseteq C_{\bar{\Gamma}}(\bar{\gamma}_j)$. Conversely, if $\pi(\delta)$ centralizes $\bar{\gamma}_j$, then $\delta\gamma_j\delta^{-1} = \pm\gamma_j$. Reduced-trace invariance and $\operatorname{trd}(\gamma_j) = 4$ exclude the minus sign, so $\delta \in C_j$. Thus $\pi(C_j) = C_{\bar{\Gamma}}(\bar{\gamma}_j)$; together with $\ker\pi = \{\pm 1\} \subset C_j$, this proves the displayed bijection and shows that passage to PSL_2 introduces no factor 2. Positive trace selects the lift sign, but the chamber is not invariant under $\gamma \mapsto \gamma^{-1}$; inverse classes and their orientations must still be retained separately.

Under the splitting (1647),

$$\|\rho(\gamma)\|_{\mathrm{F}}^2 = 8 + 9q^2 + \frac{37}{4}a^2 + \frac{105\sqrt{2}}{2}an + \frac{333}{2}n^2. \tag{1660}$$

For $i \in \mathbb{H}$,

$$2\cosh d(i, \rho(\gamma)i) = \|\rho(\gamma)\|_{\mathrm{F}}^2.$$

Since $\sinh(\ell_0/2) = \sqrt{3}$, the standard Huber parameter is

$$X_i(\gamma)^2 = \frac{\|\rho(\gamma)\|_{\mathrm{F}}^2 - 2}{12}. \tag{1661}$$

The shell equation simplifies (1660) exactly:

$$\|\rho(\gamma)\|_{\mathrm{F}}^2 = 14 + \frac{(7a + 15\sqrt{2}\,n)^2}{4},$$

and hence

$$\boxed{X_i(\gamma)^2 = 1 + \frac{(7a + 15\sqrt{2}\,n)^2}{48}.} \tag{1662}$$

Thus, in the positive chamber, the radial cutoff has the exact irrational linear height $7a + 15\sqrt{2}\,n$. The Koide height $H = 4a + 17n$ is a different linear height. The two sizes are comparable:

$$a \le \frac{H}{4}, \qquad n \le \frac{H}{17}, \qquad 0 \le q < a + 4n \le \frac{33}{68}H,$$

and conversely

$$H \le \sqrt{305}\sqrt{a^2 + q^2 + n^2}.$$

Norm equivalence in the trace-zero space then compares H coarsely with X_i. Such comparability identifies the growth scale but does not transfer a main coefficient or a remainder between the sharply bounded regions.

50.8 Consequence for the open square-root discrepancy

The quaternion construction proves:

1. the ternary shell is exactly a finite union of primitive hyperbolic conjugacy classes in one explicit cocompact arithmetic Fuchsian group;
2. $\log(2 + \sqrt{3}) = \ell_0/2$, so its occurrence in the main constant is tied to the primitive translation length;
3. the conjectural exponent $1/2$ is the same exponent proposed for the standard hyperbolic lattice-point problem in conjugacy classes, giving a precise analytic analogy but not an equivalence of the two cutoff problems.

The third item classifies the remaining analytic problem; it does not prove the error estimate. Chatzakos–Petridis [56, Theorem 1.1] give

$$E(\mathcal{H}, X; z) = O(X^{2/3})$$

for the standard radial count after the main expression, including every small-eigenvalue term, is subtracted. Their Conjecture 5.7 proposes

$$E(\mathcal{H}, X; z) = O_\varepsilon(X^{1/2+\varepsilon}).$$

The cutoff in (1659) uses both a sharp chamber and $4a + 17n$, not the radial height $7a + 15\sqrt{2}\,n$ in (1662). Their symmetric difference can have main-order size, so even the conjectured radial bound would not by itself prove the Koide bound. Therefore the status of

$$R(X) - C_0 X = O_\varepsilon(X^{1/2+\varepsilon})$$

remains **OPEN**. Closing it requires a spectral or automorphic argument for the exact finite sum (1659), with its exact sector and height.

50.9 Missing mathematics before an error theorem can be claimed

The bridge does not supply the following results.

1. **Class data.** Determine h, representatives γ_j, and orientation/inversion conventions. Finiteness does not compute the periods in a spectral expansion.

2. **Exact nonmaximal order.** The order has reduced discriminant 108, is nonmaximal at the ramified primes, and is not an Eichler order in the standard sense. A result for maximal or specified Eichler orders must be checked at this order or transferred through finite-index groups with all coset multiplicities retained. Index and discriminant alone do not determine the precise local order type required by a trace formula.

3. **Central quotient.** Track the passage from $\Gamma \subset SL_2(\mathbb{R})$ to $\bar{\Gamma} \subset PSL_2(\mathbb{R})$, including ± 1, inverse classes, and possible orbifold torsion.

4. **Exact cutoff.** Prove a trace formula or effective orbital theorem for all four inequalities defining $\mathcal{C}$ and for $4a + 17n \leq X$. The radial Huber cutoff is different.

5. **Sharp-boundary control.** Remove smoothing with an explicit estimate at the required scale. The two Pell faces contribute only $O(\log X)$, as already proved canonically, but this alone does not control every smoothing strip adjacent to all chamber walls.

6. **Exceptional spectrum.** Once an appropriate sector spectral expansion is constructed, one must determine whether it contains X^{s_j} terms for $\lambda_j = s_j(1 - s_j) < 1/4$, weighted by closed-geodesic periods. To subtract only $C_0 X$ with a $O(X^{1/2+\varepsilon})$ remainder, every relevant exceptional term must be shown to vanish or cancel, or it must be displayed and subtracted. No such vanishing is proved here.

7. **Pointwise square-root estimate.** After secondary terms and the exact sector are handled, the needed pointwise $1/2 + \varepsilon$ bound remains at the conjectural strength of the classical problem. Mean-square results do not give it for every X.

8. **Main coefficient.** A spectral derivation must reproduce

$$C_0 = \frac{9}{2\pi^2} \log(2 + \sqrt{3}) \log\left(\frac{3}{2\sqrt{2}}\right)$$

   using the same lattice, chamber, strict walls, height, and Haar measure. The existing local-density proof fixes this constant but does not calibrate a new trace formula automatically.

9. **Transfer between the two exact heights.** Formula (1662) already gives the exact standard radial parameter. What is missing is a remainder-level transfer between its height $7a + 15\sqrt{2}\, n$ and the canonical height $4a + 17n$, together with the chamber walls. Coarse norm comparability does not preserve a discrepancy estimate.

These are concrete mathematical tasks. None invalidates the quaternion bridge; together they explain why it does not close the open claim.

50.10 Reproducibility and references

The canonical quaternion replay is `supporting_materials/workbench/evidence/koide_exact_checkpoint_20260904/quaternion/verify_quaternion_bridge.py` (SHA-256 `6675ba2fc4fa8c2b10e14558e412178b6b5b579c9c96afca78e1db9bcdf5c6b4`). Its recorded successful output is `supporting_materia`

ls/workbench/evidence/koide_exact_checkpoint_20260904/quaternion/verification.log (SHA-256 66853c6191a000c7c827aacd1e0e33dcf34dbe3dd61d746f66fa4d805f34c91e), and its local manifest is supporting_materials/workbench/evidence/koide_exact_checkpoint_20260904/quaternion/SHA256SUMS.txt (SHA-256 f9d044db0d4b003e7aef7649b3394e6b299c09a6b2eb7adca095f8f9100ba623). The checkpoint-wide manifest is supporting_materials/workbench/evidence/koide_exact_checkpoint_20260904/SHA256SUMS (SHA-256 8d108eb928e5d07093badde169f443a4a38b6fcf3eb59d5192bdd33ecef8d7a4). The replay checks the basis multiplication table; the trace, norm, and coordinate formulas for $(r,s,t,u) \in \{-4,-3,\dots,4\}^4$; the shell identities for $(a,q,n) \in \{-24,-23,\dots,24\}^2 \times \{-8,-7,\dots,8\}$, containing 48 shell points; the inverse identity on those points; the radial-height identity on that box; and the Hilbert-symbol formulas at 2 and 3. These finite ranges are consistency checks, not proofs outside the stated boxes. The exact canonical sources used for the counting statements imported into this bridge are:

- supporting_materials/workbench/research/koide_counting_repair.tex, SHA-256 965b0a3dca8bfe4f22b9e9208368fa04697070fd25bd06bd67f2481d17cb9d9d;
- supporting_materials/workbench/research/koide_exact_parametrization.tex, SHA-256 a457709dcce0fce0774eb9326b84b00d5d8144c2087f480da704974e499bcdd3;
- supporting_materials/workbench/research/koide_exact_census.tex, SHA-256 18ed5968b7e67d9a0b23d8675b40c37ba46170ad29efdbb9ce10ed632c99600a.

The bounded coordinate-bijection replay, whose scope does not include the quaternion multiplication table, Hilbert symbols, centralizer, conjugacy finiteness, or spectral conclusions, is supporting_materials/workbench/reproduction/t1_koide_exact/verify_koide_narain_root_density.py (SHA-256 e4bf2982b26b07add6057b5d2d5c771a52ef52c629b5ad5f0aab24e440e41986). Its stored height-200000 output is supporting_materials/workbench/reproduction/t1_koide_exact/koide_verification_200000.json (SHA-256 b907f0ee5bdd3ad80622b669722d218b79a11a5baab0d4571e141936100706b7); the exact invocation and scope are stated in supporting_materials/workbench/reproduction/t1_koide_exact/README.md (SHA-256 5e9cdcb9e67954c37d8fde69e13549172aa03245674b358e4a1d25f8d9d3bff0). The quaternion identities and deductions in this section are proved in the displayed algebraic calculations themselves, not inferred from that bounded replay.

References:

- John Voight, *Quaternion Algebras*, Graduate Texts in Mathematics 288, especially Section 10.1, Lemma 38.1.2, and Proposition 38.1.3; DOI.
- Dimitrios Chatzakos and Yiannis N. Petridis, *The hyperbolic lattice point problem in conjugacy classes*, arXiv:1504.01307v2, Theorem 1.1 and Conjecture 5.7; arXiv.

- The exact Koide chamber, height, main term, and Pell-face estimates are in Sections 39–49 of this workbench.

51 Primary-source theorem audit of effective counting for the fixed Koide ternary quadric sector

Date: 4 September 2026
Exact counting function:

$$R(X) = \#\left\{(a,q,n) \in \mathbb{Z}^3 : \begin{array}{l} a^2 - 3q^2 - 18n^2 = -2, \\ a > 0,\ q \geq 0,\ n \geq 0,\ a + 4n > q, \\ 4a + 17n \leq X \end{array}\right\}.$$

The leading constant proved in theorem 39.11 is

$$C_0 = \frac{9}{2\pi^2}\log(2+\sqrt{3})\log\left(\frac{3}{2\sqrt{2}}\right).$$

The canonical project-relative mathematical sources used by this audit are:

- `supporting_materials/workbench/research/koide_counting_repair.tex`, which proves the coordinate bijection, local densities, invariant-volume factor, Gorodnik–Oh–Shah hypotheses, and calibrated constant, SHA-256 `965b0a3dca8bfe4f22b9e9208368fa04697070fd25bd06bd67f2481d17cb9d9d`;
- `supporting_materials/workbench/research/koide_exact_parametrization.tex`, which proves the exact parametrization and the Benoist–Oh power-saving application, SHA-256 `a457709dcce0fce0774eb9326b84b00d5d8144c2087f480da704974e499bcdd3`;
- `supporting_materials/workbench/research/koide_quaternion_bridge.tex`, which proves the exact order, shell, centralizer, conjugacy-class, and radial-height statements used below, SHA-256 `93759f7ba99870c84c9144a373a53d2e9a8bc3ae909518e47991373ace0b47bb`;
- `supporting_materials/workbench/research/koide_exact_census.tex`, which records the separately bounded finite census, SHA-256 `18ed5968b7e67d9a0b23d8675b40c37ba46170ad29efdbb9ce10ed632c99600a`.

The primary publications themselves are identified by exact versions, dates where version-sensitive, locators, and DOI or arXiv links in section 51.11. They are not claimed to be stored in a separate canonical evidence manifest.

51.1 Determination

Among the source versions enumerated in section 51.11, no theorem proves, for this exact fixed level, full integral lattice, sharp chamber, and sharp polyhedral height, the unconditional estimate

$$R(X) - C_0X = O_\varepsilon(X^{1/2+\varepsilon}).$$

The strongest exact-domain conclusion obtained from the general theorems enumerated there is

$$\boxed{R(X) = C_0X + O(X^{1-\eta}) \quad \text{for some unspecified } \eta > 0.}$$

It follows by applying Benoist–Oh, Definition 13.1 and Theorem 13.2, orbit by orbit after directly verifying effective well-roundedness for the present family. Benoist–Oh do not give a numerical lower bound for η, and their proof does not imply $\eta \geq 1/2$.

The superficially stronger exponents in nearby problems do not answer the question:

- Good's sharp hyperbolic conjugacy-class theorem, as formulated and reproved by Chatzakos–Petridis, gives $O(X^{2/3})$ only for a different radial cutoff and only after every small-eigenvalue term X^{s_j} has been placed in the main expression. The same paper states $O_\varepsilon(X^{1/2+\varepsilon})$ as Conjecture 5.7.
- Liu–Sarnak obtain a smooth radial error $O_\varepsilon(X^{1/2+\theta+\varepsilon})$, hence $O_\varepsilon(X^{39/64+\varepsilon})$ from $\theta = 7/64$, and square-root strength only under the generalized Selberg/Ramanujan-at-infinity conjecture for the relevant Jacquet–Langlands transfers. Their paper's global affine-sieve setup assumes $t\det(f)$ square-free, but the analytic Proposition 3.2 itself is formulated for a variable cocompact lattice with a spectral gap. After the exact quaternionic identification it applies class by class to this lattice. What it still does not supply is a sharp cutoff, the coordinate chamber, or the nonradial height $4a + 17n$.

- Gorodnik–Nevo's published sharp polar-sector theorem loses powers through set thickening. For $\dim G = 3$, it gives $O(X^{19/20+\epsilon})$ even with tempered spectrum; the Kim–Sarnak complementary-series threshold, which does apply to the exact congruence arithmetic lattice by Jacquet–Langlands, gives the still weaker $O(X^{39/40+\epsilon})$. This remains a comparison theorem because Gorodnik–Nevo's fixed polar-sector family has not been identified with the exact polyhedron, including all integral orbits.
- Square-root-scale statements in Chatzakos–Petridis, Chatzakos, Voskou, and Lekkas–Petridis are mean-square, averaged, Omega, or conjectural statements. They are not pointwise sharp-sector upper bounds.
- Recent split-stabilizer theorems assume $-m\det(F)$ is a square and have a $T\log T$-type main term. Here $-(-2)\det(F) = 108$ is not a square, so they concern the opposite arithmetic regime.

The two independent missing inputs are therefore (i) a sharp-boundary spectral or arithmetic estimate for the actual height and chamber, and (ii) control or elimination of any complementary-spectrum secondary terms for the actual sector. The Pell faces $q = 0$ and $n = 0$, which contribute only $O(\log X)$, do not solve the moving outer-boundary problem.

51.2 Exact arithmetic and geometric classification

51.2.1 The rational group and stabilizer

Let

$$F(a,q,n) = a^2 - 3q^2 - 18n^2, \qquad \mathscr{X} = \{F = -2\}.$$

The matrix of F is $\mathrm{diag}(1,-3,-18)$, so $\det(F) = 54$ and

$$-(-2)\det(F) = 108 \notin \mathbb{Q}^{\times 2}.$$

The homogeneous form F is anisotropic over $\mathbb{Q}$. Indeed, suppose a primitive integral triple satisfies

$$a^2 - 3q^2 - 18n^2 = 0.$$

Modulo 3, one gets $3 \mid a$; write $a = 3a_1$. Dividing by 3 and reducing modulo 3 gives $3 \mid q$; write $q = 3q_1$. The resulting equation is

$$a_1^2 = 3q_1^2 + 2n^2.$$

Modulo 3, this says $a_1^2 + n^2 \equiv 0 \pmod 3$, whence $3 \mid a_1, n$. Thus $3 \mid a, q, n$, contradicting primitivity. Put $G = \mathrm{Spin}(F)$ and $x_0 = (1,1,0)$. The orthogonal complement of x_0 is

$$x_0^{\perp} = \{(3s,s,t) : s,t \in \mathbb{Q}\}, \qquad F(3s,s,t) = 6(s^2 - 3t^2).$$

Consequently the connected stabilizer $H = \mathrm{Stab}_G(x_0)^{\circ}$ is the norm-one torus

$$\mathrm{Res}^1_{\mathbb{Q}(\sqrt{3})/\mathbb{Q}}\,\mathbb{G}_m.$$

It is split over $\mathbb{R}$, anisotropic over $\mathbb{Q}$, and has no nontrivial $\mathbb{Q}$-character. Under the quaternionic identification in section 50.5, x_0 corresponds to $v_0 = (I+J)/2$, with $v_0^2 = 3$, and

$$\mathcal{O} \cap \mathbb{Q}(v_0) = \mathbb{Z}[v_0], \qquad H(\mathbb{Q}) \cap \Gamma = \{\pm(2+v_0)^k : k \in \mathbb{Z}\}.$$

Thus $(H(\mathbb{Q}) \cap \Gamma)/\{\pm 1\}$ is infinite cyclic. Under $H(\mathbb{R})^{\circ} \cong \mathbb{R}_{>0}$, the logarithm sends its positive part to $\log(2+\sqrt{3})\mathbb{Z} \subset \mathbb{R}$; the quotient is compact, and in particular $(H \cap \Gamma)\backslash H(\mathbb{R})$ has finite volume. Since F is

$\mathbb{Q}$-anisotropic, the arithmetic Fuchsian quotient attached to G is cocompact as well. There is no ambient hyperbolic cusp.

This is the finite-volume stabilizer regime associated with a closed geodesic. It is not the rationally split regime with divergent stabilizer in which one obtains $T \log T$ growth.

51.2.2 The exact height really is a norm-sector height

Define

$$\|(a,q,n)\|_* = max\{|4a+17n|, |q|, |a|\}.$$

This is a genuine norm because it is the pullback of the sup norm under the invertible linear map

$$(a,q,n) \longmapsto (4a+17n, q, a),$$

whose determinant is -17. On the target chamber,

$$0 < a < 4a+17n, \qquad 0 \le q < a+4n < 4a+17n,$$

and therefore $\|(a,q,n)\|_* = 4a+17n$. Thus the outer cutoff is not merely comparable to a norm: it is exactly the restriction of a fixed ambient norm.

The additional wall $a+4n > q$ removes only a bounded real set. If $q \ge a+4n$, then

$$a^2+2 = 3q^2+18n^2 \ge 3(a+4n)^2+18n^2,$$

so $2a^2+24an+66n^2 \le 2$. Its only integral point in the positive orthant on $\mathscr{X}$ is $(1,1,0)$. The unbounded faces $q=0$ and $n=0$ are Pell families and contribute $O(\log X)$ integral points, but their real boundary strips still have to be included in a sharp-domain argument.

51.2.3 Invariant volume with a strong enough secondary estimate

Use

$$\sqrt{3}q = r\cos\theta, \qquad 3\sqrt{2}n = r\sin\theta, \qquad 0 \le \theta \le \frac{\pi}{2}.$$

On $F=-2$, $r=\sqrt{a^2+2}$, and the gauge measure is

$$dm_\infty = \frac{1}{6\sqrt{6}}\, da\, d\theta.$$

Writing $c = 17/(3\sqrt{2})$, the outer boundary is

$$4a + c\sqrt{a^2+2}\sin\theta = X.$$

If $A_X(\theta)$ is its positive solution, then uniformly in θ,

$$A_X(\theta) = \frac{X}{4+c\sin\theta} + O(X^{-1}).$$

Indeed $A_X(\theta) \asymp X$ uniformly and $\sqrt{A^2+2} = A + O(A^{-1})$. Once X contains the bounded wall excision, it follows that

$$m_\infty(B_X) = I_\infty X + O(1), \qquad I_\infty = \frac{1}{2\sqrt{3}}\log\left(\frac{3}{2\sqrt{2}}\right).$$

This $O(1)$ refinement matters: a theorem whose main term is the exact invariant volume can be converted to the displayed C_0X main term without spending a power of X.

51.2.4 The full integral locus is not assumed to be one orbit

The exact lattice must not be silently replaced by a convenient single orbit or by a commensurable maximal-order lattice. Borel–Harish-Chandra finiteness, in the form quoted by Benoist–Oh on journal p. 1892, says that $G(\mathbb{Z})$ has finitely many orbits on $\mathscr{X}(\mathbb{Z})$. Thus

$$\mathscr{X}(\mathbb{Z}) = \bigsqcup_{j=1}^{h} x_j \Gamma$$

for the integral spin group Γ attached to the chosen lattice and finitely many integral representatives x_j. An effective theorem for one orbit must be applied to every $x_j\Gamma$, and the finitely many main-measure scalars must be summed. The full-lattice scalar is calibrated from Borovoi's local/Brauer–Manin density formula in theorems 39.10 and 39.11; it is not obtained by declaring that all integral points form the orbit of x_0.

51.2.5 Exact quaternionic lattice and its congruence status

For the spectral comparisons below, the precise arithmetic lattice can be retained. Put

$$B = (18,-6)_{\mathbb{Q}} = \mathbb{Q}\langle I, J\rangle/(I^2 - 18,\ J^2 + 6,\ IJ + JI), \qquad K = IJ,$$

and

$$\mathcal{O} = \mathbb{Z}1 \oplus \mathbb{Z}\frac{I+J}{2} \oplus \mathbb{Z}\frac{I-J}{2} \oplus \mathbb{Z}\frac{K}{2}.$$

The complete integral multiplication table is (1634), and its reduced-trace Gram determinant is -108^2 by (1637); hence $\operatorname{discrd}(\mathcal{O}) = 108$. The Hilbert-symbol calculations (1644) and (1645), together with the unramified-place and real-place calculation summarized in (1646), show that the ramification set is exactly $\{2, 3\}$. Thus B is the indefinite quaternion division algebra of discriminant 6. The map

$$(a, q, n) \longmapsto \gamma(a, q, n) = 2 + \frac{qI + aJ + nK}{2}$$

is the bijection (1642) from $F(a, q, n) = -2$ to the reduced-trace-4, reduced-norm-one locus in $\mathcal{O}$. Indeed, the parity condition $a \equiv q \pmod 2$, which is exactly what membership in $\mathcal{O}$ requires, follows automatically from $F = -2$ modulo 2, and

$$\operatorname{nrd}(\gamma) = 4 + \frac{3}{2}F(a, q, n) = 1.$$

Thus the acting Fuchsian lattice is exactly $\Gamma = \mathcal{O}^1$, not a substituted maximal-order lattice. If $\mathcal{M} \supset \mathcal{O}$ is maximal, the reduced-discriminant index formula gives

$$[\mathcal{M} : \mathcal{O}] = \frac{108}{6} = 18, \qquad 18\mathcal{M} \subset \mathcal{O},$$

and consequently

$$\mathcal{M}^1(18\mathcal{M}) \subset \mathcal{O}^1 \subset \mathcal{M}^1.$$

Here $\mathcal{M}^1(18\mathcal{M}) = \{u \in \mathcal{M}^1 : u - 1 \in 18\mathcal{M}\}$. Therefore $\mathcal{O}^1$ is a congruence arithmetic Fuchsian group in the exact sense used by Chu–Li. The number 108 is the order's reduced discriminant, not an Eichler level: the order is nonmaximal at the ramified primes $2, 3$. This non-Eichler feature changes the finite compact-open subgroup and hence which automorphic representations have fixed vectors and the dimensions of those fixed-vector spaces; it does not change the Jacquet–Langlands/Kim–Sarnak archimedean bound.

51.3 Result matrix

Source and exact locator	What is actually proved	Status for the target
Duke–Rudnick–Sarnak (1993), Theorem 1.2, p. 145; Example 1.5, pp. 146–147	One arithmetic orbit on an affine symmetric variety: count $\sim$ invariant volume; ternary anisotropic stabilizer gives linear rather than $T \log T$ growth	Correct qualitative regime, no rate and no exact sharp-sector theorem
Eskin–McMullen (1993), Theorem 1.4, author PDF p. 3	Every well-rounded sequence has count asymptotic to volume	Qualitative only; no exponent
Gorodnik–Oh–Shah (2009), definitions before Theorem 1.6 and Theorem 1.6, p. 5; Theorem 1.18, pp. 8–9	Any norm and an admissible generic cone: full integral count $\sim$ volume	Direct qualitative theorem for the exact norm and cone, whose hypotheses are proved in theorems 39.8 and 39.9; no error term
Gorodnik–Oh–Shah (2010), Theorem 1.13, author PDF pp. 4–5	Well-rounded polar sectors and their asymptotic	Qualitative only
Benoist–Oh (2012), Theorem 1.3, p. 1894	Standard product/Euclidean height ball: relative $O(T^{-\delta})$ for some $\delta > 0$	Theorem 1.3 alone is not the exact polyhedron
Benoist–Oh (2012), Definition 13.1, p. 1931, and Theorem 13.2, pp. 1931–1932	Any independently verified effectively well-rounded family: $\mathrm{vol}(B)(1 + O(\mathrm{vol}(B)^{-\delta_0}))$	Applies to the exact family after the proof in section 51.4.2; yields only unspecified $O(X^{1-\eta})$
Gorodnik–Nevo (2012), Definition 3.1, p. 13; Theorem 8.1 and Corollary 8.3, pp. 34–35	Sharp fixed polar sectors: $O(\mathrm{vol}^{1-\zeta+\epsilon})$, $\zeta = (2n_e(p))^{-1}(1 + 3\dim G)^{-1}$	Explicit but much weaker; target polyhedron and finite orbit union require an additional reduction
Liu–Sarnak (2010), equations (2.3)–(2.8), pp. 4–5; Proposition 3.2, pp. 10–11; equation (3.65), p. 16	Smooth radial count for a cocompact lattice containing the fixed cyclic stabilizer: $O_\epsilon(T^{1/2+\theta+\epsilon})$	Exact lattice and finite class sum are available; unconditional 39/64, conditional 1/2, but fixed-width smoothing and the radial height do not match the sharp target
Good (1983), Theorem 4, p. 116, as explicitly restated and reproved in Chatzakos–Petridis (2016), Theorem 1.1, p. 3	Sharp radial hyperbolic conjugacy-class count: $O(X^{2/3})$ after all X^{s_j} small-spectrum terms	Closest sharp spectral model, not the exact polyhedron; square root is CP Conjecture 5.7
Bourgain–Kontorovich–Sarnak (2010), Theorem 1.1, p. 2	Fixed angular Fourier modes in a sharp group norm ball: $O_{n,k}(T^{4/3})$ after T^{2s_j} terms	With affine height $X \asymp T^2$, again $X^{2/3}$; constants are not uniform enough to recover a sharp interval for free
Browning–Gorodnik (2017), Theorem 2.1, p. 11; Corollaries 2.5–2.6, pp. 24–25	Full Euclidean balls, finite adelic/orbit sums, and congruence conditions: unspecified $O(\mathrm{vol}^{1-\varrho})$	Correct nonsquare homogeneous regime, wrong height geometry and no explicit exponent

Source and exact locator	What is actually proved	Status for the target
Cao–Huang (2022), Theorem 1.5, p. 4; Theorem 3.4, pp. 21–22	Arithmetic purity for anisotropic nonsquare ternary quadrics; effective input for Euclidean balls has an unspecified power	Qualitative full-lattice theorem, not a square-root sector result
Cao–Huang–Zhang (arXiv:2509.23320), Setting 2.1 and Theorem 2.1	Product Euclidean/max height balls with congruence-uniform $O(\mathbf{N}(\mathfrak{l})^{\kappa}\,\mathrm{vol}^{1-\delta})$	Current extension, but δ is unspecified and the growing real factor is the standard height ball, not this cone
Oh–Shah (2014), Theorem 1.2, p. 3; Theorem 6.1, pp. 13–14	Divergent-geodesic/split-stabilizer $T\log T$ count; smooth error $O(T)$, sharp relative logarithmic error	Excluded by the finite-volume anisotropic stabilizer
Xu–Zhang (2023), main theorem; Huang (arXiv:2412.03350), Theorem 1.1; Zhang (arXiv:2603.21080), Theorem 1.1	Split-stabilizer results assuming $-m\det(F)$ is a square, with smooth or radial $T\log T$-type expansions	108 is nonsquare; wrong main-term regime
Huang (arXiv:2412.05903), Theorems 1.1–1.2	Smooth compact real weight while the level varies as $m_0p_0^{2h}$; error of logarithmic relative strength	Varying level and smooth weight, not the fixed-level sharp count
Li–Liu (arXiv:2501.15033v2), Theorems 1.1–1.2, pp. 2–3; Lemma 2.3, pp. 5–6	Almost-prime results using a smooth radial orbit count with $O_\epsilon(d^{1+\epsilon}T^{1/2+\theta+\epsilon})$	Current use of the Liu–Sarnak input; globally assumes $t\det(f)$ square-free and supplies no sharp-sector count
Voskou (arXiv:2407.03134v3), Theorem 3 and Conjecture 1, p. 3	Four sharp orientation sectors in a closed-geodesic double-coset problem: $O(X^{2/3})$; square root conjectured	Current sharp comparator, not the target
Lekkas–Petridis (arXiv:2509.12902, version dated 24 Aug. 2026), Theorem 1.2, Theorem 1.5, Conjecture 1.6	Sharp closed-geodesic double-coset $O(X^{2/3})$, square-root mean square, pointwise square-root conjecture	Version-specific confirmation that mean square does not furnish the requested pointwise bound

51.4 Benoist–Oh: exact applicability and exact limitation

Among the general effective theorems enumerated in section 51.3, this is the one whose stated hypotheses can be verified for the actual family without changing its height or chamber.

51.4.1 What the theorem assumes

Benoist–Oh work with a connected semisimple simply connected almost K-simple group G, a finite-kernel rational representation, a symmetric subgroup H, and the hypothesis that H° has no nontrivial K-character; see journal pp. 1892–1893. For a congruence subgroup Γ_S, Theorem 12.5, journal p. 1929, supplies the effective equidistribution of the translates of $(\Gamma_S\cap H_S)\backslash H_S$ used in Theorem 13.2. Example (A), journal p. 1896, explicitly treats ternary quadratic forms provided the homogeneous form does not represent zero over $\mathbb{Q}$.

All of these hypotheses hold here:

- $G=\mathrm{Spin}(F)$ is simply connected and absolutely almost simple of type A_1, and $G(\mathbb{R})$ is noncompact.

- The standard three-dimensional orthogonal representation has finite kernel.
- The stabilizer of a nonisotropic vector is symmetric.
- The mod-3 descent in section 51.2.1 proves that F does not represent zero over $\mathbb{Q}$.
- H° is the $\mathbb{Q}$-anisotropic norm-one torus of $\mathbb{Q}(\sqrt{3})$, so it has no nontrivial $\mathbb{Q}$-character and its arithmetic quotient is compact.
- With $S = \{\infty\}$, the full integral spin group is itself a congruence subgroup in Benoist–Oh's sense.

There is no omitted wavefront or properness condition: symmetry of H is exactly what invokes their effective wavefront argument in Lemma 12.3 and Theorem 12.5. Cocompactness makes escape in the automorphic quotient simpler, but does not remove Definition 13.1(3), which still has to be checked on the growing affine set.

Here "congruence" is an exact property of the chosen integral group, not an inference from mere commensurability. Under the quaternionic realization of G, the finite-integrality conditions define a compact-open subgroup of $G(\mathbb{A}_f)$; equivalently, after placing the corresponding order $\mathcal{O}$ in a maximal order $\mathcal{O}_{\max}$, some principal subgroup $\{u \in \mathcal{O}^1_{\max} : u \equiv 1 \pmod{N\mathcal{O}_{\max}}\}$ lies in the exact integral spin group. This distinction matters for spectral-gap transfer: an arbitrary finite-index noncongruence subgroup need not inherit the Kim–Sarnak gap.

51.4.2 Definition 13.1 for the actual B_X

Let $B_X \subset \mathscr{X}(\mathbb{R})$ be the exact chamber cut out by the five linear conditions and $4a + 17n \le X$. In Benoist–Oh's notation, $B^+_{X,\epsilon} = B_X B_\epsilon$ and $B^-_{X,\epsilon} = \bigcap_{u \in B_\epsilon} B_X u$.

Condition (1) is void because there is no finite-place factor.

For condition (2), if u lies in an ϵ-ball about the identity, every fixed linear functional L defining the chamber or height satisfies

$$|L(xu) - L(x)| \ll \epsilon \|x\|_2.$$

On the unbounded portion of this chamber, $\|x\|_2 \asymp 1 + a \asymp 1 + (4a + 17n)$. Hence a membership change under u is confined to:

1. an angular strip of width $O(\epsilon)$ at $q = 0$ or $n = 0$;
2. an $O(\epsilon X)$-thick radial shell at $4a + 17n = X$; or
3. an $O(\epsilon)$-neighborhood of one of the compact boundary pieces $a = 0$ or $a + 4n = q$.

Since $dm_\infty = (6\sqrt{6})^{-1} da\, d\theta$, their total measure is $O(\epsilon X)$. Thus

$$m_\infty(B^+_{X,\epsilon} \setminus B^-_{X,\epsilon}) \ll \epsilon\, m_\infty(B_X),$$

so Definition 13.1(2) holds with $\kappa = 1$.

For condition (3), Benoist–Oh's Euclidean height is comparable to $1 + a$ throughout a fixed group-neighborhood enlargement of the chamber. For every $k > 0$, uniformly in $0 < \epsilon < 1$,

$$\int_{B^+_{X,\epsilon}} H_\infty(x)^{-k}\, dm_\infty(x) \ll_k \int_0^{O(X)} (1 + a)^{-k}\, da.$$

The right side is $O(X^{1-k})$ for $0 < k < 1$, $O(\log X)$ for $k = 1$, and $O(1)$ for $k > 1$. As $m_\infty(B_X) \asymp X$, this is $O(m_\infty(B_X)^{1-\delta(k)})$ for some $\delta(k) > 0$ in every case. Definition 13.1(3) is therefore satisfied.

Proposition 14.2 of Benoist–Oh verifies only their standard product height balls. It should not be cited as though it automatically covers this polyhedron. The preceding direct verification is the necessary bridge to Theorem 13.2.

51.4.3 Finite-orbit passage and main-term normalization

Benoist–Oh state on journal p. 1892 that $G(\mathcal{O}_S)$ has finitely many orbits on $\mathscr{X}(\mathcal{O}_S)$. Apply Theorem 13.2 to each exact integral orbit. If $\eta_j > 0$ is the exponent for the j-th orbit, take $\eta = \min_j \eta_j > 0$. The finite sum has the form

$$R(X) = \kappa_{\mathrm{orb}}\, m_\infty(B_X) + O(X^{1-\eta}).$$

Here κ_{orb} is the sum of the orbit-volume normalization constants. The comparison with Borovoi's Theorem 1.4.1 and Proposition 1.5.1 is carried out in theorem 39.11; specifically, (1510), (1522), and (1544) give

$$\kappa_{\mathrm{orb}} = \mathfrak{S}(\mathscr{X}) = \frac{9\sqrt{3}}{\pi^2}\log(2+\sqrt{3}), \qquad \kappa_{\mathrm{orb}} I_\infty = C_0.$$

Together with $m_\infty(B_X) = I_\infty X + O(1)$, this yields

$$R(X) = C_0 X + O(X^{1-\eta}).$$

This counts the actual integral lattice. No assertion that $\mathscr{X}(\mathbb{Z})$ is one orbit, and no replacement by a maximal order, is involved.

51.4.4 Why the exponent remains unspecified

The proof of Benoist–Oh Theorem 13.2 balances an effective-equidistribution/Sobolev error and a boundary-thickening error:

$$c_1 \epsilon^{-p} v_X^{1-\delta} + c_2 \epsilon^{\kappa} v_X, \qquad v_X = \mathrm{vol}(B_X).$$

The resulting saving is of the form

$$\eta = \frac{\delta\kappa}{p+\kappa} > 0.$$

Neither the Sobolev exponent p nor the effective equidistribution exponent δ is made numerically strong enough in the theorem to show $\eta \geq 1/2$. Thus Benoist–Oh supplies a genuine exact-family power saving, but not the requested square-root bound.

51.5 The sharp hyperbolic frontier

51.5.1 Good and Chatzakos–Petridis

For a cocompact or cofinite Fuchsian group and a fixed hyperbolic conjugacy class $\mathcal{H}$, Chatzakos–Petridis define, in equations (1.4)–(1.6),

$$M(\mathcal{H}, X; z) = \sum_{1/2 < s_j \leq 1} A(s_j)\widehat{u}_j u_j(z) X^{s_j},$$

where $\lambda_j = s_j(1-s_j) < 1/4$ and $\widehat{u}_j$ is the period of u_j along the invariant closed geodesic. Their Theorem 1.1, author manuscript p. 3, restates Good's Theorem 4, p. 116, and proves

$$N(\mathcal{H}, X; z) - M(\mathcal{H}, X; z) = O(X^{2/3}).$$

The 2/3 arises from the sharp majorant/minorant error

$$O(XY^{-1/2} + Y + X^{1/2}\log X)$$

after choosing $Y = X^{2/3}$. Theorems 1.2–1.3 are mean-square results, not pointwise improvements. Conjecture 5.7, author manuscript p. 22, explicitly asks for

$$O_\varepsilon(X^{1/2+\varepsilon}).$$

There are two independent failures of direct transfer to $R(X)$:

1. the Huber cutoff is radial, whereas $4a + 17n \le X$ plus the chamber is a nonradial sharp polyhedron; norm comparability does not preserve the leading constant or error because the symmetric difference can have volume of order X;

2. even in the radial problem, the theorem subtracts every nonzero small-eigenvalue contribution, not only the linear term.

For the exact Koide sector no spectral expansion establishing actual X^{s_j} terms has yet been produced. They are therefore a necessary obstruction to check, not terms that this audit claims are already known to occur. A proof of $R(X) - C_0X = O(X^{1/2+\varepsilon})$ would have to show absence or vanishing of all relevant complementary-spectrum coefficients, or explicitly subtract them and then prove they vanish in the target formulation.

51.5.2 Current sharp refinements remain at 2/3

Voskou, Theorem 3 of arXiv:2407.03134v3, proves $O(X^{2/3})$ for four prescribed orientation/side sectors in a closed-geodesic-to-closed-geodesic problem, with all small-spectrum period terms included in the main expression. Conjecture 1 on the same page proposes $O_\varepsilon(X^{1/2+\varepsilon})$, while Theorem 4 is a mean-square result.

Lekkas–Petridis, in the 24 August 2026 version of arXiv:2509.12902, Theorem 1.2, likewise prove

$$N(X, l) = \sum_{1/2<s_j\le 1} \frac{2}{\pi}\gamma_1(s_j)|\widehat{u}_j|^2 X^{s_j} + O(X^{2/3})$$

for a closed-geodesic double-coset count. Their Theorem 1.5 gives

$$\frac{1}{X}\int_X^{2X} |E(x, l)|^2\, dx \ll X \log^2 X,$$

and Conjecture 1.6 proposes the pointwise $X^{1/2+\epsilon}$ bound. Their introduction states that Good's 2/3 has not been improved for any of the nine point/geodesic/cusp double-coset configurations. This is strong current evidence about the sharp frontier, but neither paper treats the exact Koide height.

51.5.3 Fixed angular harmonics do not give a sharp interval for free

Bourgain–Kontorovich–Sarnak, Theorem 1.1, prove for fixed left and right K-Fourier modes n, k

$$\sum_{\|\gamma\|<T} e^{2in\theta_1(\gamma)}e^{2ik\theta_2(\gamma)} = \mathbf{1}_{n=0}\mathbf{1}_{k=0}c_0T^2 + \sum_j c_j(n, k)T^{2s_j} + O_{n,k}(T^{4/3}).$$

On the adjoint orbit, affine height is quadratic in the group norm, so $T^{4/3}$ again corresponds to $X^{2/3}$. Recovering a sharp angular interval requires summing many Fourier modes, while the theorem's implied constants depend on n, k. It is not a uniform desmoothing theorem for this chamber.

51.6 Smooth spectral counting

Liu–Sarnak's paper is the closest source with a square-root-plus-gap exponent. Its global arithmetic setup and the hypotheses of its analytic proposition must be separated.

The introduction, author PDF p. 2, fixes an indefinite anisotropic integral ternary form f, a nonzero level t, and assumes $t\det(f)$ is square-free. Thus the paper's affine-sieve theorem, as globally stated, does not cover $t\det(f) = -108$. Equations (2.3)–(2.4), author PDF p. 4, fix one arithmetic orbit $\mathcal{O}$ and a smooth K-invariant radial weight F_T satisfying

$$F_T = 1 \quad (|x| \le T/c_0), \qquad F_T = 0 \quad (|x| \ge c_0 T),$$

with $c_0 > 1$ fixed. In Fermi coordinates, Proposition 3.2 uses a smooth even function equal to one for $|\rho| \le R-1$, zero for $|\rho| \ge R+1$, and with uniformly bounded derivatives. Unlike the global sieve statement, Proposition 3.2 is formulated for any variable subgroup $\Gamma < \mathrm{SL}_2(\mathbb{R})$ satisfying its uniform local condition (3.1), the spectral gap (2.7), and containing the fixed primitive cyclic group Γ_ξ. It proves

$$H_R(z) = \frac{X(R)}{\mathrm{Area}(\Gamma\backslash\mathbb{H})} + O_\varepsilon(e^{(1/2+\theta+\varepsilon)R}), \qquad X(R) \asymp e^R.$$

Equation (3.60) has $T = \sinh R$, and equation (3.65) transfers this to the one-sheeted affine orbit. Thus the smooth affine error is

$$O_\varepsilon(T^{1/2+\theta+\varepsilon}).$$

Equation (2.7) assumes $\lambda_1 \ge 1/4 - \theta^2$. The paper invokes Jacquet–Langlands and Kim–Sarnak to take $\theta = 7/64$, giving

$$O_\varepsilon(T^{39/64+\varepsilon})$$

unconditionally in its congruence-arithmetic setting; $\theta = 0$, and hence square root, is conditional on the generalized Selberg/Ramanujan-at-infinity conjecture for the relevant Jacquet–Langlands transfers.

The numerical input itself is Kim–Sarnak, Appendix 2, Proposition 2, printed p. 176. It applies to an arbitrary cuspidal automorphic representation of $\mathrm{GL}_2/\mathbb{Q}$: for a spherical real component, $|\Re\mu_{j,\infty}| \le 7/64$, independently of finite conductor. Jacquet–Langlands preserves the real component because the quaternion algebra is split at infinity. Chu–Li, proof of Corollary 1.2, equation (5.2), printed p. 5126, records the resulting

$$\lambda_1 \ge \frac{975}{4096}$$

for every congruence arithmetic Fuchsian group. Thus the nonmaximal or non-Eichler structure of the exact finite order does not invalidate $s \le 39/64$. It changes the finite compact-open subgroup and hence which automorphic representations have fixed vectors and the dimensions of those fixed-vector spaces; it does not change this archimedean bound. This validates the spectral-gap input for the exact lattice. It does not provide sharp-boundary control or force the periods of possible complementary-spectrum eigenfunctions to vanish.

For the exact order lattice $\Gamma = \mathcal{O}^1$, each primitive trace-four conjugacy class supplies the cyclic subgroup in Proposition 3.2, and there are only finitely many such classes. Hence the proposition and equation (3.65) do give an unconditional 39/64 estimate after summing the corresponding **smooth standard-radial** counts. The square-free hypothesis is not an obstruction to this isolated analytic application.

This still does not desmooth to the target. A transition of fixed width in R is a fixed multiplicative annulus in T, whose invariant volume is of order T, not $T^{1/2}$. Shrinking the transition makes the derivative/Sobolev constants in Proposition 3.2 grow, and the paper gives no uniform estimate that balances this loss at square-root scale. It also has no sharp coordinate chamber, and its radial height is

not $4a + 17n$. These geometric failures remain even after the exact lattice and finite class sum have been supplied.

The same distinction applies to any informal statement that a tempered spectral component has square-root size. Duke–Rudnick–Sarnak's discussion on p. 155 estimates individual tempered frequencies; it does not control the complete spectral sum evaluated at a point with a shrinking approximation to a sharp characteristic function.

51.7 General effective sector and full-lattice theorems

51.7.1 Gorodnik–Nevo's explicit generic exponent

Gorodnik–Nevo Definition 3.1 defines

$$n_e(2) = 1, \qquad n_e(p) = \text{the least even integer at least } p/2 \quad (p > 2).$$

Their Section 8 considers

$$S_t(\Phi, \Psi) = \{g : \log \|gv_0\| \le t\} \cap \Phi A^+ \Psi.$$

Theorem 8.1, author PDF p. 34, assumes L^{p+} matrix coefficients and Lipschitz-well-rounded fixed angular sets; Corollary 8.3, p. 35, passes to an orbit when $(\Gamma \cap H)\backslash H$ is compact. The exponent is

$$\zeta = \frac{1}{2n_e(p)(1 + 3\dim G)}.$$

For $\dim G = 3$,

$$\zeta = \frac{1}{20n_e(p)}.$$

Even tempered spectrum has $p = 2$, $n_e(p) = 1$, and hence absolute error $O(X^{19/20+\epsilon})$. For the exact congruence arithmetic lattice, Jacquet–Langlands and Kim–Sarnak give $s \le 39/64$. The corresponding matrix coefficients are L^{p+} for $p = 64/25$, so $n_e(p) = 2$ and the theorem gives only $O(X^{39/40+\epsilon})$.

This numerical calculation is an upper benchmark for the published generic polar-sector theorem, not a claim that Corollary 8.3 literally counts $R(X)$. A fixed polar sector contains every radial parameter in fixed polar directions. An arbitrary affine linear cone and its finite-radius polyhedral height need an exact polar decomposition or a quantitatively controlled symmetric-difference argument; all integral orbits must then be included. None of those identifications appears in the theorem statement.

Gorodnik–Nevo themselves note after Corollary 5.3, author PDF p. 23, that their sharp-set counting mechanism remains worse than square-root volume error even for tempered spectrum, whereas square-root quality is available there only for a smooth weighted problem.

51.7.2 Browning–Gorodnik and subsequent full-lattice work

Browning–Gorodnik Theorem 2.1 gives, for a Euclidean ball B_H in one congruence orbit,

$$|\Gamma_\ell y \cap B_H| = \frac{m_{Z_{y,\ell}}(Z_{y,\ell})}{m_{X_\ell}(X_\ell)} m_Y(B_H) + O\Big(\ell^{\dim H + \dim G} m_Y(B_H)^{1-\varrho}\Big)$$

for some unspecified $\varrho > 0$. Corollaries 2.5–2.6 sum adelic and integral orbits and retain the Brauer/Kottwitz density, at the cost of $\ell^{\dim H + 2\dim G}$. This is a useful exact-lattice precedent, but its archimedean set is the full norm ball, not the present sector, and no square-root exponent is asserted.

Cao–Huang Theorem 1.5 proves arithmetic purity of the Hardy–Littlewood property for an anisotropic ternary affine quadric with $-m\det(q)$ nonsquare. Its Theorem 3.4 invokes Nevo–Sarnak, Gorodnik–Nevo, and Browning–Gorodnik to furnish an unspecified positive exponent, denoted β_X there with X meaning

the affine variety; write it here as $\beta_{\mathscr{X}} > 0$ to distinguish it from the height variable. It is not a sharp-sector remainder theorem.

The current Cao–Huang–Zhang preprint, arXiv:2509.23320, Setting 2.1 and Theorem 2.1, gives a congruence-uniform power saving

$$O\Big(\mathbf{N}(\mathfrak{l})^{\kappa}\,\mathrm{vol}(\mathbb{B}_S(N))^{1-\delta}\Big)$$

for standard product Euclidean/max height balls and a compact piecewise-smooth factor. The exponent δ is unspecified, and the unbounded real factor in the target is neither compact nor their standard Euclidean ball. This preprint extends the general effective framework but does not improve the conclusion of the direct Benoist–Oh argument in sections 51.4.2 and 51.4.3.

51.8 Results in different arithmetic regimes

51.8.1 Square discriminant and divergent stabilizer

Oh–Shah Theorem 1.2 assumes

$$\mathrm{vol}((H\cap\Gamma)\backslash H) = \infty$$

and obtains a smoothed $cT\log T + O(T)$ count. Its sharp version has only a logarithmic relative error. Theorem 6.1 allows an open cone and arbitrary norm, but retains the logarithmic main-factor characteristic of its divergent-stabilizer regime; that regime is excluded here.

Xu–Zhang, *Quarterly Journal of Mathematics* 74 (2023), 659–685, assumes $-m\det(f)$ is a square over the number field. Huang, arXiv:2412.03350, Theorem 1.1, assumes the same split condition and a fixed smooth compact weight; its principal error is $B(\log B)^{1983/1984}$ in a $B\log B$ count. Zhang, arXiv:2603.21080, Section 1.1 and Theorem 1.1, again assumes $-m\det(q)\in(k^{\times})^2$ and proves a smooth K_0-invariant expansion with an error $O(T^{l_1+l_2-\delta})$, $0<\delta<1/2$; Remark 1.2 obtains an authentic count for $\Gamma=\mathrm{SL}_2(\mathcal{O}_k)$. These are important current results, but all lie in the rationally split torus regime excluded by $108\notin\mathbb{Q}^{\times 2}$, and none uses the target polyhedral height.

51.8.2 Varying level, isotropic forms, and special rotational quadrics

Huang, arXiv:2412.05903, studies $F(x)=m_0N$ with $N=p_0^{2h}\to\infty$, a smooth compactly supported real weight $w(x/\sqrt{N})$, and prescribed local conditions. Theorems 1.1–1.2 have a $\sqrt{N}/(\log N)^A$ error. This is a Linnik-type varying-level theorem, not a fixed affine shell.

Li–Liu, arXiv:2501.15033v2 (18 April 2025), Theorems 1.1–1.2, prove new almost-prime conclusions for anisotropic ternary equations. Their global assumption remains $t\det(f)$ square-free. The quantitative counting input in Lemma 2.3, pp. 5–6, is

$$O_{\varepsilon}(d^{1+\varepsilon}T^{1/2+\theta+\varepsilon})$$

for the same kind of smooth radial orbit weight inherited from Liu–Sarnak. It is a current application of the 7/64 gap, not a new unsmoothed sector theorem.

Lowry-Duda's thesis, arXiv:1704.02376, Theorem 5.5.4, treats the special rotational form $x_1^2+x_2^2=x_3^2+h$ in a full Euclidean ball. Written in ordinary radius T, its three-dimensional sharp error is $O(T^{21/22+\epsilon})$. The homogeneous form is $\mathbb{Q}$-isotropic, and neither the lattice nor the height sector is the target one.

Results on values of irrational ternary forms, shrinking sets while the discriminant varies, or positive-definite ternary representation numbers answer different problems. Lindelöf or generalized Lindelöf hypotheses occurring there do not yield the present fixed-level pointwise sector bound.

51.9 Spectral-gap, GRH, smoothing, and averaging taxonomy

Classification	Source result	Why it does not prove the target
Unconditional, exact sharp family	Benoist–Oh Theorem 13.2 plus the direct Definition 13.1 verification	Gives only $O(X^{1-\eta})$ with $\eta > 0$ unspecified
Unconditional, exact norm/cone, full lattice	Gorodnik–Oh–Shah Theorem 1.6, with the exact hypotheses proved in theorems 39.8 and 39.9	Qualitative $C_0X + o(X)$ only
Unconditional, sharp radial closed-geodesic count	Good / Chatzakos–Petridis	$O(X^{2/3})$, different height, and main expression includes small-spectrum terms
Unconditional, smooth radial count	Liu–Sarnak Proposition 3.2 with $\theta = 7/64$, summed over the exact finite conjugacy-class decomposition	$39/64$ for the exact lattice, but fixed multiplicative smoothing and a different radial height
Selberg/Ramanujan conditional	Liu–Sarnak with $\theta = 0$	Square-root only for the smooth radial problem; still not the exact count
GRH/Lindelöf conditional	None among the source versions in section 51.11 proves this fixed sharp-sector estimate	The occurrences listed there concern varying levels/forms, shrinking-set equidistribution, or lower/Omega results
Smoothed	Liu–Sarnak; Oh–Shah; Huang I; Zhang 2026	Smoothing or split arithmetic changes the theorem; no square-root desmoothing supplied
Averaged/mean square	Chatzakos–Petridis Theorems 1.2–1.3; Voskou Theorem 4; Lekkas–Petridis Theorem 1.5	Does not imply a uniform pointwise estimate at every X
Omega/lower-bound	Chatzakos (2019), including a cocompact geodesic-average $\Omega(X^{1/2}\log\log\log X)$	Evidence for scale, not an upper bound
Isotropic or split	Oh–Shah; Xu–Zhang; Huang I; Zhang 2026; Lowry-Duda	Different stabilizer, quotient, main order, form, or level
Varying level	Huang II	Counts $F = m_0p_0^{2h}$, not $F = -2$

In particular, "assuming GRH" is not a substitute for the missing theorem. The conditional hypothesis that occurs naturally in the closest smooth spectral result is the Selberg/Ramanujan eigenvalue conjecture, and even it only reaches square root before the sharp-boundary unsmoothing problem is addressed.

51.10 Citation-ready conclusion

Subject to the bounded-search qualification in section 51.12, the following paragraph records the conclusion supported by the enumerated source versions:

> Among the source versions enumerated in section 51.11, none implies the unconditional sharp estimate $R(X) - C_0X = O_\varepsilon(X^{1/2+\varepsilon})$ for the fixed ternary quadric $a^2 - 3q^2 - 18n^2 = -2$ in the chamber $a > 0$, $q, n \geq 0$, $a + 4n > q$, with height $4a + 17n \leq X$. Gorodnik–Oh–Shah, Theorem

1.6, applies to the exact admissible generic cone and an arbitrary norm—with those exact hypotheses proved in theorems 39.8 and 39.9—but gives only a qualitative asymptotic. Benoist–Oh, Definition 13.1 and Theorem 13.2, applies after the direct effective-well-roundedness verification in section 51.4.2 and, after summing the finitely many integral spin-group orbits, yields $R(X) = C_0X + O(X^{1-\eta})$ for some unspecified $\eta > 0$, not $\eta = 1/2$. In the closest sharp closed-geodesic problem, Good's theorem, as restated and reproved by Chatzakos–Petridis, Theorem 1.1, gives $O(X^{2/3})$ only after all small-eigenvalue terms X^{s_j} are included; Chatzakos–Petridis, Conjecture 5.7, states the $O_\varepsilon(X^{1/2+\varepsilon})$ improvement as a conjecture. Liu–Sarnak, Proposition 3.2, applies class by class to the exact congruence quaternionic lattice and proves the smooth radial bound $O_\varepsilon(X^{1/2+\theta+\varepsilon})$, unconditionally with $\theta = 7/64$ and conditionally with $\theta = 0$, but its fixed-width smoothing and radial height do not match the exact sharp count. Thus complementary-spectrum terms and sharp nonradial boundary control remain separate unresolved inputs.

51.11 Primary references and exact locators

1. Yves Benoist and Hee Oh, "Effective equidistribution of S-integral points on symmetric varieties," *Annales de l'Institut Fourier* **62** (2012), 1889–1942. DOI, official article/PDF, arXiv. Standing setting pp. 1892–1893; finite orbit statement p. 1892; Theorem 1.3 p. 1894; Theorem 12.5 p. 1929; Definition 13.1 p. 1931; Theorem 13.2 pp. 1931–1932; Proposition 14.2 p. 1933; ternary Example (A) p. 1896.

2. Alexander Gorodnik, Hee Oh, and Nimish Shah, "Integral points on symmetric varieties and Satake compactifications," *American Journal of Mathematics* **131** (2009), 1–57. DOI, arXiv, author PDF. Admissible/generic definitions and Theorem 1.6 pp. 4–5; Theorem 1.18 pp. 8–9.

3. Alexander Gorodnik, Hee Oh, and Nimish Shah, "Strong wavefront lemma and counting lattice points in sectors," *Israel Journal of Mathematics* **176** (2010), 419–444. DOI, arXiv. Theorem 1.13, author PDF pp. 4–5.

4. Alexander Gorodnik and Amos Nevo, "Counting lattice points," *Journal für die reine und angewandte Mathematik* **663** (2012), 127–176. DOI, arXiv, author PDF. Theorem 1.9 pp. 4–5; Definition 3.1 p. 13; Theorem 4.5 pp. 17–18; Theorem 8.1 and Corollary 8.3 pp. 34–35; Proposition 8.4 p. 35.

5. William Duke, Zeév Rudnick, and Peter Sarnak, "Density of integer points on affine homogeneous varieties," *Duke Mathematical Journal* **71** (1993), 143–179. DOI, author PDF. Theorem 1.2 p. 145; Example 1.5 pp. 146–147; discussion of individual tempered frequencies p. 155; Theorem 3.5 p. 161.

6. Alex Eskin and Curtis McMullen, "Mixing, counting, and equidistribution in Lie groups," *Duke Mathematical Journal* **71** (1993), 181–209. DOI, author PDF. Theorem 1.4, author PDF p. 3; Theorems 6.1 and 6.4 pp. 30–31; Open Question 1 p. 35.

7. Jianya Liu and Peter Sarnak, "Integral points on quadrics in three variables whose coordinates have few prime factors," *Israel Journal of Mathematics* **178** (2010), 393–426. DOI, author PDF. Arithmetic hypothesis p. 2; equations (2.3)–(2.8) pp. 4–5; Proposition 3.2 pp. 10–11; equations (3.60) and (3.65) pp. 15–16.

8. Anton Good, *Local Analysis of Selberg's Trace Formula*, Lecture Notes in Mathematics 1040, Springer, 1983. DOI. Theorem 4, p. 116. The theorem statement and a new proof were checked through Chatzakos–Petridis (2016) because the monograph text was not directly available in this audit.

9. Dimitrios Chatzakos and Yiannis N. Petridis, "The hyperbolic lattice point problem in conjugacy classes," *Forum Mathematicum* **28** (2016), 981–1003. DOI, arXiv, author manuscript. Equations (1.4)–(1.6) and Theorem 1.1 p. 3; Theorems 1.2–1.3 pp. 3–4; Conjecture 5.7 p. 22.

10. Dimitrios Chatzakos, "Mean value results and Ω-results for the hyperbolic lattice point problem in conjugacy classes," *Revista Matemática Iberoamericana* **35** (2019), 1123–1152. DOI, arXiv. Mean-value and Omega theorems, not a pointwise upper bound.

11. Jean Bourgain, Alex Kontorovich, and Peter Sarnak, "Sector estimates for hyperbolic isometries," *Geometric and Functional Analysis* **20** (2010), 1175–1200. DOI, arXiv. Theorem 1.1, p. 2.

12. Hee Oh and Nimish Shah, "Limits of translates of divergent geodesics and integral points on one-sheeted hyperboloids," *Israel Journal of Mathematics* **199** (2014), 915–931. DOI, arXiv. Theorem 1.2 p. 3; Corollary 1.4 pp. 3–4; Theorem 6.1 pp. 13–14.

13. Tim D. Browning and Alexander Gorodnik, "Power-free values of polynomials on symmetric varieties," *Proceedings of the London Mathematical Society* **114** (2017), 1044–1080. DOI, arXiv. Theorem 2.1 p. 11; Corollaries 2.5–2.6 pp. 24–25.

14. Yang Cao and Zhizhong Huang, "Arithmetic purity of the Hardy–Littlewood property and geometric sieve for affine quadrics," *Advances in Mathematics* **398** (2022), 108236. DOI, arXiv. Theorem 1.5 p. 4; Proposition 2.6 pp. 12–13; Theorem 3.4 pp. 21–22.

15. Yang Cao, Zhizhong Huang, and Runlin Zhang, "Effective equidistribution, arithmetic purity of strong approximation, and geometric sieve for affine quadrics," arXiv:2509.23320. Setting 2.1 and Theorem 2.1; arXiv HTML retrieved 4 September 2026.

16. Mikhail Borovoi, "On representations of integers by indefinite ternary quadratic forms," *Journal of Number Theory* **90** (2001), 281–293. DOI, arXiv, author PDF. Theorem 1.4.1 and Proposition 1.5.1; qualitative local/Brauer–Manin coefficient, no error exponent.

17. Fei Xu and Runlin Zhang, "Counting integral points on indefinite ternary quadratic equations over number fields," *Quarterly Journal of Mathematics* **74** (2023), 659–685. DOI, arXiv. Square-discriminant hypothesis in the main theorem.

18. Zhizhong Huang, "Quantitative strong approximation for ternary quadratic forms I."
    arXiv:2412.03350; Theorem 1.1; square-discriminant, smooth fixed-weight $B \log B$ count.

19. Zhizhong Huang, "Quantitative strong approximation for ternary quadratic forms II."
    arXiv:2412.05903; Theorems 1.1–1.2; varying level $m_0 p_0^{2h}$ and smooth compact weight.

20. Runlin Zhang, "Effective count of integer points on ternary affine quadrics and effective equidistribution," arXiv:2603.21080. Section 1.1 and Theorem 1.1; split-stabilizer smooth count; Remark 1.2 on the authentic radial count.

21. Marios Voskou, "Refined Counting of Geodesic Segments in the Hyperbolic Plane."
    arXiv:2407.03134v3; Theorems 3–4 and Conjecture 1, pp. 3–4.

22. Dimitrios Lekkas and Yiannis N. Petridis, "A relative trace formula and counting geodesic arcs in the hyperbolic plane," arXiv:2509.12902, version dated 24 August 2026. Theorem 1.2, Theorem 1.5, and Conjecture 1.6.

23. Henry H. Kim, "Functoriality for the exterior square of GL_4 and the symmetric fourth of GL_2," with Appendix 2 by Henry H. Kim and Peter Sarnak, "Refined estimates towards the Ramanujan and Selberg conjectures," *Journal of the American Mathematical Society* **16** (2003), 139–183. DOI, official AMS PDF. Appendix 2, Proposition 2, printed p. 176: $|\Re \mu_{j,\infty}| \le 7/64$ for a spherical real component of any cuspidal $\mathrm{GL}_2/\mathbb{Q}$ representation, and the displayed consequence $\lambda_1 \ge 975/4096$.

24. Michelle Chu and Han Li, "Small generators of cocompact arithmetic Fuchsian groups," *Proceedings of the American Mathematical Society* **144** (2016), 5121–5127. DOI, arXiv. Section 3.1, p. 5122, defines congruence arithmetic Fuchsian groups; proof of Corollary 1.2, equation (5.2), p. 5126, records $\lambda_1 \geq 975/4096$ for every such group via Jacquet–Langlands and Kim–Sarnak.

25. Hervé Jacquet and Robert P. Langlands, *Automorphic Forms on* GL(2), Lecture Notes in Mathematics 114, Springer, 1970. DOI, IAS-hosted text. Chapter III, especially §16, supplies the global quaternionic transfer; at split places, including the real place here, the local component is preserved.

26. John Voight, *Quaternion Algebras*, Graduate Texts in Mathematics 288, Springer, 2021. DOI, open author version. Proposition 15.5.2 (maximal superorders); §23.1.3 (Eichler orders at ramified primes); equations (27.6.6)–(27.6.7) (compact-open finite adelic order/unit group); Theorem 28.2.10 (strong approximation for indefinite B^1); Definition 38.7.2 (principal congruence groups).

27. Jiamin Li and Jianya Liu, "Anisotropic quadratic equations in three variables."

    arXiv:2501.15033v2; Theorems 1.1–1.2, pp. 2–3; Lemma 2.3, pp. 5–6. The global form-level hypothesis is $t\det(f)$ square-free; the quantitative input is a smooth radial, congruence-orbit estimate with

$$O_\varepsilon\Big(d^{1+\varepsilon}T^{1/2+\theta+\varepsilon}\Big).$$

28. David Lowry-Duda, *On Some Variants of the Gauss Circle Problem*, arXiv:1704.02376. Theorem 5.5.4, thesis pp. 97–98.

51.12 Audit limitations

The claim-by-claim source versions and locators used in this bounded audit are enumerated in section 51.11. The canonical tree does not preserve byte-identical snapshots or retrieval hashes for those external publications, so this file does not assert a reproducible local source corpus. Good's 1983 book was the sole exception to direct full-text access: its theorem locator and statement were checked through the explicit restatement and independent proof in Chatzakos–Petridis (2016) and the version-dated Lekkas–Petridis treatment.

The statement that no theorem was located is a result about the source versions enumerated here, not a proof that no unpublished or differently indexed result exists. It does establish that none of the enumerated works may be cited as a proof of the exact unconditional square-root discrepancy.

52 The exact gapless auxiliary operator and the physical boundary

The preceding flat-torus calculation produces eigenvalues tending to zero along the radial cusp. This section identifies a single self-adjoint operator for which those numbers genuinely form a gapless spectrum, proves that complex and Hodge-theoretic data alone cannot determine such a spectrum, and states the exact additional morphism needed to transfer an auxiliary Rayleigh sequence to a physical Hamiltonian. In particular, the word *gapless* below is attached to a specified operator and Hilbert space, not to the complex threefold in isolation.

52.1 A direct-integral scalar Laplacian with zero essential threshold

Use the angular coordinates $y \in \mathbb{R}^4/\mathbb{Z}^4$ of theorem 11.1. The map $y \mapsto P_T y$ identifies this fixed torus with F_T, and normalized Haar measure on F_T pulls back to dy. Write

$$e_k(y) := e^{2\pi i k\cdot y}, \qquad \lambda_k(T) := 4\pi^2 |P_T^{-\mathsf{T}} k|^2, \qquad k \in \mathbb{Z}^4.$$

By theorem 11.1, these are exactly the scalar-Laplacian eigenfunctions and eigenvalues on (F_T, g_T), normalized in L^2 for Haar probability measure.

Fix T_0 large enough that all radial formulas of theorems 11.1 and 11.2 hold, and put

$$\mathscr{H}_{\mathrm{aux}} := L^2\Big([T_0,\infty), dT; L^2_0(\mathbb{R}^4/\mathbb{Z}^4, dy)\Big). \tag{1663}$$

Fourier expansion gives the concrete unitary identification

$$\mathscr{H}_{\mathrm{aux}} \cong \bigoplus_{k\in\mathbb{Z}^4\setminus\{0\}} L^2([T_0,\infty), dT). \tag{1664}$$

Define the multiplication operator

$$(\mathscr{L}_{\mathrm{aux}} u)_k(T) := \lambda_k(T) u_k(T) \tag{1665}$$

on the exact domain

$$D(\mathscr{L}_{\mathrm{aux}}) := \left\{ u \in \mathscr{H}_{\mathrm{aux}} : \sum_{k\neq 0} \int_{T_0}^{\infty} \lambda_k(T)^2 |u_k(T)|^2 \, dT < \infty \right\}. \tag{1666}$$

Thus $\mathscr{L}_{\mathrm{aux}}$ is, by definition and not by analogy, the direct integral of the mean-zero scalar Laplacians Δ_{F_T}.

Theorem 52.1 (A proved gapless auxiliary operator)**.** *The operator $\mathscr{L}_{\mathrm{aux}}$ in equation* (1665) *is nonnegative and self-adjoint. It has no zero eigenvector, but*

$$0 \in \sigma_{\mathrm{ess}}(\mathscr{L}_{\mathrm{aux}}). \tag{1667}$$

In particular, this specified auxiliary operator has no positive spectral gap above zero.

Proof. For each $k \neq 0$, $\lambda_k(T)$ is a positive continuous real function. Multiplication by it on

$$\{v \in L^2([T_0,\infty)) : \lambda_k v \in L^2([T_0,\infty))\}$$

is self-adjoint and nonnegative: its adjoint domain is obtained by testing against compactly supported bounded functions, which forces $\lambda_k v \in L^2$, and the adjoint acts by the same real multiplier. The orthogonal sum of these multiplication operators has precisely the domain in equation (1666); hence it is self-adjoint and nonnegative. Since every $\lambda_k(T)$ is strictly positive, $\mathscr{L}_{\mathrm{aux}} u = 0$ implies $u_k = 0$ almost everywhere for every $k \neq 0$, so its kernel is zero.

Let $k_0 = (1,0,0,0)$. The exact formula equation (37) gives

$$\lambda_{k_0}(T) = 4\pi^2 \frac{L_T^2 + m_T^2}{D_T^2} = \frac{4\pi^2}{T^2}(1 + O(T^{-1})). \tag{1668}$$

Choose integers $n > T_0$, and define $u^{(n)}$ by

$$u^{(n)}_{k_0}(T) = \mathbf{1}_{[n,n+1]}(T), \qquad u^{(n)}_k(T) = 0 \quad (k \neq k_0). \tag{1669}$$

The vectors $u^{(n)}$ lie in $D(\mathscr{L}_{\mathrm{aux}})$, are orthonormal, and converge weakly to zero because their T-supports escape to infinity. Moreover,

$$\|\mathscr{L}_{\mathrm{aux}} u^{(n)}\|^2 = \int_n^{n+1} \lambda_{k_0}(T)^2 \, dT \leq \sup_{T\in[n,n+1]} \lambda_{k_0}(T)^2 \longrightarrow 0$$

by equation (1668). This orthonormal singular sequence proves $0 \in \sigma_{\mathrm{ess}}(\mathscr{L}_{\mathrm{aux}})$. Explicitly, if zero were outside the essential spectrum, then, because the kernel is zero, some interval $[0,\varepsilon)$ would have finite-dimensional spectral projection. Its complement would satisfy $\|\mathscr{L}_{\mathrm{aux}} v\| \geq \varepsilon \|v\|$; weak convergence would make the components of $u^{(n)}$ in the finite-dimensional spectral subspace tend strongly to zero, contradicting $\|\mathscr{L}_{\mathrm{aux}} u^{(n)}\| \to 0$ and $\|u^{(n)}\| = 1$. □

Remark 52.2 (The unit-volume normalization is a different operator family). The theorem uses the unscaled Euclidean quotient metrics g_T. By theorem 11.3, the first positive scalar eigenvalue for $\widehat{g}_T = D_T^{-1/2} g_T$ still tends to zero, but at the different rate

$$\lambda_1(F_T, \widehat{g}_T) = \frac{4\pi^2}{T}(1 + O(T^{-1})),$$

rather than the unscaled T^{-2} rate. Rescaling a metric changes its Laplacian, so the two operator families and their asymptotic scales must not be identified.

52.2 Complex data do not select a spectral gap

Proposition 52.3 (Fixed complex torus, fixed volume, arbitrarily small gap)**.** *Let*

$$A = (\mathbb{C}/(\mathbb{Z} + i\mathbb{Z}))^2$$

with fixed complex coordinates $z_j = x_j + iy_j$. For $R \geq 1$, the metric

$$g_R = R^2(dx_1^2 + dy_1^2) + R^{-2}(dx_2^2 + dy_2^2) \tag{1670}$$

is a flat Kähler metric of volume one on the same complex torus, while

$$\lambda_1(\Delta_{g_R}) = \frac{4\pi^2}{R^2} \longrightarrow 0. \tag{1671}$$

Consequently the complex structure, Hodge decomposition, integral cohomology, and coherent direct-image data of a trivial family do not, even together with total volume, determine a positive lower bound for the scalar Laplacian gap.

Proof. The associated two-form

$$\omega_R = R^2 dx_1 \wedge dy_1 + R^{-2} dx_2 \wedge dy_2$$

is closed, positive, and J-invariant, so g_R is Kähler for the fixed complex structure. In the ordered real coordinates (x_1, y_1, x_2, y_2), its matrix is $\mathrm{diag}(R^2, R^2, R^{-2}, R^{-2})$, whose determinant is one; therefore the volume is one. The Fourier character

$$e_{a,b,c,d} = \exp(2\pi i(ax_1 + by_1 + cx_2 + dy_2))$$

has eigenvalue

$$4\pi^2 \left(R^{-2}(a^2 + b^2) + R^2(c^2 + d^2)\right).$$

For $R \geq 1$, the minimum over nonzero integer quadruples is $4\pi^2 R^{-2}$, attained at the four vectors

$$(a, b, c, d) \in \{(1, 0, 0, 0), (-1, 0, 0, 0), (0, 1, 0, 0), (0, -1, 0, 0)\}.$$

This proves equation (1671). None of the listed complex or cohomological objects depends on R, and for a trivial holomorphic family their coherent direct images are likewise unchanged. □

52.3 The exact bridge to a physical Hamiltonian

The mathematical mass gap in the Clay formulation is a property of the Hamiltonian of a constructed four-dimensional quantum Yang–Mills theory, not of an internal scalar Laplacian or of a classical Hessian [64]. Euclidean input reaches such a Hamiltonian only after a positive-metric reconstruction satisfying, in particular, the Osterwalder–Schrader requirements [68, 69]. The following proposition isolates the exact operator-theoretic statement that any proposed transfer must prove.

Proposition 52.4 (Typed Rayleigh-sequence transfer)**.** *Let $H \geq 0$ be self-adjoint on a Hilbert space $\mathcal{H}_{\mathrm{phys}}$. Define*

$$\mathcal{V}_0 := \ker H, \qquad P_0 := \mathbf{1}_{\{0\}}(H), \qquad \mathcal{H}_{\mathrm{exc}} := (I - P_0)\mathcal{H}_{\mathrm{phys}}, \tag{1672}$$

and let

$$\mathfrak{h}[\psi] := \|H^{1/2}\psi\|^2, \qquad \mathcal{Q}(H) := D(H^{1/2}). \tag{1673}$$

For each n, let $\mathcal{K}_n$ be a Hilbert space, $\mathfrak{q}_n$ a nonnegative quadratic form on a domain $D(\mathfrak{q}_n) \subseteq \mathcal{K}_n$, and $0 \neq u_n \in D(\mathfrak{q}_n)$. Suppose there is a linear map

$$J_n : \mathrm{span}\{u_n\} \longrightarrow \mathcal{Q}(H),$$

and put $\psi_n = (I - P_0)J_n u_n$. If there are $c > 0$, $C_n \geq 0$, and $\varepsilon_n \geq 0$ such that

$$\|\psi_n\| \geq c\|u_n\|, \tag{1674}$$

$$\mathfrak{h}[\psi_n] \leq C_n \mathfrak{q}_n[u_n] + \varepsilon_n \|u_n\|^2, \tag{1675}$$

$$C_n \frac{\mathfrak{q}_n[u_n]}{\|u_n\|^2} + \varepsilon_n \longrightarrow 0. \tag{1676}$$

Then the excitation Hamiltonian

$$H_{\mathrm{exc}} := H\big|_{D(H)\cap\mathcal{H}_{\mathrm{exc}}}$$

is self-adjoint and satisfies

$$\inf \sigma(H_{\mathrm{exc}}) = 0, \qquad 0 \in \sigma_{\mathrm{ess}}(H_{\mathrm{exc}}). \tag{1677}$$

Proof. The spectral projection P_0 commutes with H and $H^{1/2}$, so

$$\mathcal{H}_{\mathrm{phys}} = \ker H \oplus \mathcal{H}_{\mathrm{exc}}$$

is a reducing decomposition. Hence H_{exc} is self-adjoint, nonnegative, has form domain $\mathcal{Q}(H) \cap \mathcal{H}_{\mathrm{exc}}$, and has zero kernel. Define $w_n = \psi_n/\|\psi_n\|$. Then $\|w_n\| = 1$, $w_n \in \mathcal{Q}(H) \cap \mathcal{H}_{\mathrm{exc}}$, and

$$\begin{aligned} \mathfrak{h}[w_n] &= \frac{\mathfrak{h}[\psi_n]}{\|\psi_n\|^2} \\ &\leq c^{-2}\left(C_n \frac{\mathfrak{q}_n[u_n]}{\|u_n\|^2} + \varepsilon_n\right) \longrightarrow 0. \end{aligned}$$

The variational characterization of the bottom of a nonnegative self-adjoint operator gives

$$\inf \sigma(H_{\mathrm{exc}}) = \inf_{0\neq\psi\in\mathcal{Q}(H)\cap\mathcal{H}_{\mathrm{exc}}} \frac{\mathfrak{h}[\psi]}{\|\psi\|^2} = 0.$$

Since the spectrum is closed, zero belongs to it; since $\ker H_{\mathrm{exc}} = 0$, zero is not an eigenvalue. It is therefore a continuous spectral point, and hence belongs to the essential spectrum under the standard definition that removes isolated eigenvalues of finite multiplicity. □

Remark 52.5 (Why one vacuum vector is not enough)*.* If the vacuum is unique, choose a normalized vector Ω spanning $\ker H$. Then $P_0 = |\Omega\rangle\langle\Omega|$. Without uniqueness, merely requiring $J_n u_n \perp \Omega$ does not project away every zero-energy state. For example,

$$H = \mathrm{diag}(0, 0, 1), \qquad \Omega = (1, 0, 0), \qquad \psi = (0, 1, 0)$$

has $\psi \perp \Omega$ and zero Rayleigh quotient, although the genuine excitation space $(\ker H)^{\perp}$ is the third coordinate and has gap one. Thus equation (1672) must use the full kernel unless vacuum uniqueness is independently proved.

Audit record 52.6 (What the S^6 calculation proves)**.** The radial period matrix, scalar spectra, classical gauge Hessian, finite cochain operator, magnetic Laplacian, and reducible $SU(2)$ Jacobi operator are explicit auxiliary geometric or classical operators. They prove the operator statements made in theorems 12.6, 12.10, 14.5, 14.8 and 52.1. They do not construct a common gauge-invariant positive-metric physical Hilbert space $\mathcal{H}_{\mathrm{phys}}$, a quantum Hamiltonian H, or maps J_n satisfying equations (1674) to (1676) after continuum, infinite-volume, and renormalization limits. Nor do the period, Hodge, Leray, or direct-image calculations determine those data, as theorem 52.3 already disproves at the level of geometric Laplacians. Therefore the auxiliary gaplessness is **[VERIFIED]**, while a conclusion about the four-dimensional quantum Yang–Mills Hamiltonian is **[OPEN]**.

53 Integral topology without the cusp-collapse argument

This section follows the independent nearby-cycle/Leray route. It does not use the candidate's cellular collapse, its Mayer–Vietoris homology matrix, or the asserted equality between those two constructions. All coefficient groups are integral.

Put

$$B = \mathbb{P}^1, \quad \Sigma = \{p_0, p_1, p_2\}, \quad B^\circ = B \setminus \Sigma, \quad j : B^\circ \hookrightarrow B,$$

and let F be a smooth fibre. Retain

$$V = H^1(F) = \mathbb{Z}\langle \gamma, u, w, \delta \rangle, \quad q = uw + 6\gamma\delta, \quad \mathrm{vol} = \gamma uw\delta.$$

53.1 Integral specialization at the cusp

Theorem 53.1 (nearby-cycle specialization)**.** ***[VERIFIED]*** *For the proper holomorphic cusp filling $f_0 : N_0 \to \Delta$ of theorem 10.6, with its proved local normal-crossings equations and normalization combinatorics, the specialization map of the central fibre W,*

$$\mathrm{sp}^r : H^r(W;\mathbb{Z}) \longrightarrow H^r(F;\mathbb{Z})$$

is injective and

$$\operatorname{im} \mathrm{sp}^r = H^r(F;\mathbb{Z})^{T_0} \tag{5.1}$$

for $0 \le r \le 4$. In particular, the image is saturated and there is no unrecorded finite-index defect at p_0.

Proof. Let $\widetilde{\Delta}^* \to \Delta^*$ be the universal cover, put $\widetilde{N}_0^* = N_0 \times_\Delta \widetilde{\Delta}^*$, and let

$$k : \widetilde{N}_0^* \longrightarrow N_0, \qquad i : W \hookrightarrow N_0$$

be the natural maps. The analytic nearby-cycle complex used here is

$$\psi = \psi_{f_0}\mathbb{Z} := i^* R k_* k^* \mathbb{Z}_{N_0} \in D_c^b(W). \tag{5.0}$$

Dimca defines analytic nearby cycles in [33, Definition 4.2.1]; the Milnor-fibre description of their stalks is [33, Proposition 4.2.2]. The proper-direct-image theorem for nearby cycles is [33, Proposition 4.2.11]; applied to f_0 and the identity map of the disc, it gives

$$\mathbb{H}^n(W;\psi) \simeq H^n(F;\mathbb{Z})$$

and identifies the morphism induced by the unit $\mathbb{Z}_W \to \psi$ with specialization. Proper base change itself is [33, Theorem 2.3.26]. Dimca's coefficient hypothesis is a commutative noetherian ring of finite global dimension, which $\mathbb{Z}$ satisfies. Thus the general integral analytic formalism is available here. The candidate-specific existence and local description of f_0 are proved in theorems 10.6 and 10.7. Kashiwara–Schapira [34, §8.6] corroborates the analytic formalism, while SGA 7, Exposé XIII [32], is its algebraic antecedent.

At a point where $f = z_0 \cdots z_k$, $k = 0, 1, 2$, the Milnor fibre deformation retracts onto $(S^1)^k$: in polar coordinates the radial variables form a contractible open simplex and the phases satisfy one linear relation. Hence

$$\mathcal{H}^0\psi = \mathbb{Z}_W, \qquad \mathcal{H}^1\psi =: \mathscr{V}, \qquad \mathcal{H}^2\psi = \mathbb{Z}_P \oplus \mathbb{Z}_Q, \qquad \mathcal{H}^b\psi = 0 \ (b \geq 3), \tag{5.2}$$

where $\mathscr{V}$ is supported on the three double curves $\bar{C}_1, \bar{C}_2, \bar{C}_3 \simeq \mathbb{P}^1$.

Orient each vanishing circle by the displayed order of its two branches in the normalization. At P, the branches of $(\bar{C}_1, \bar{C}_2, \bar{C}_3)$ are respectively

$$(b_0, b_1), \qquad (b_0, b_2), \qquad (b_1, b_2).$$

The three restrictions are therefore evaluations on

$$e_0 - e_1, \qquad e_0 - e_2, \qquad e_1 - e_2,$$

whose alternating sum is zero. At Q, the ordered branch pairs are

$$(c_2, c_1), \qquad (c_0, c_1), \qquad (c_0, c_2),$$

so the evaluations are

$$e_2 - e_1, \qquad e_0 - e_1, \qquad e_0 - e_2,$$

and again the first minus the second plus the third is zero. The global normalization labels C_k and C_{k+3} distinguish the two branches of each double curve and therefore fix these signs simultaneously. Thus

$$0 \longrightarrow \mathscr{V} \longrightarrow \bigoplus_{k=1}^{3} i_{k*}\mathbb{Z}_{\bar{C}_k} \xrightarrow{\delta_1} \mathbb{Z}_P \oplus \mathbb{Z}_Q \longrightarrow 0,$$

$$\delta_1(n_1, n_2, n_3) = (s, s), \qquad s = n_1 - n_2 + n_3. \tag{5.3}$$

The image $\mathbb{Z}(1, 1)$ is primitive. Taking cohomology of (5.3) gives free groups with ranks

$$\operatorname{rk} H^a(W; \mathscr{V}) = (2, 1, 3, 0) \quad (a = 0, 1, 2, 3). \tag{5.4}$$

The normalization resolution for the constant sheaf is

$$0 \longrightarrow \mathbb{Z}_W \longrightarrow \nu_*\mathbb{Z}_{\mathrm{dP}_6} \longrightarrow \bigoplus_{k=1}^{3} i_{k*}\mathbb{Z}_{\bar{C}_k} \xrightarrow{\delta_1} \mathbb{Z}_P \oplus \mathbb{Z}_Q \longrightarrow 0. \tag{5.5}$$

This is exact stalkwise. At a smooth point it is the identity $\mathbb{Z} \to \mathbb{Z}$. At a generic double point it is the augmented complex

$$0 \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Z}^2 \xrightarrow{(x_0, x_1) \mapsto x_0 - x_1} \mathbb{Z} \longrightarrow 0.$$

At a triple point, in either of the branch orders just fixed, it is

$$0 \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Z}^3 \xrightarrow{d_0} \mathbb{Z}^3 \xrightarrow{d_1} \mathbb{Z} \longrightarrow 0,$$

where at P

$$d_0(x_0, x_1, x_2) = (x_0 - x_1, x_0 - x_2, x_1 - x_2), \qquad d_1(n_1, n_2, n_3) = n_1 - n_2 + n_3,$$

while at Q

$$d_0(x_0, x_1, x_2) = (x_2 - x_1, x_0 - x_1, x_0 - x_2), \qquad d_1(n_1, n_2, n_3) = n_1 - n_2 + n_3.$$

These augmented simplicial complexes are exact, which proves (5.5), including its endpoint signs.

On H^2, the only nontrivial map is

$$d : H^2(\mathrm{dP}_6;\mathbb{Z}) \longrightarrow \mathbb{Z}^3, \qquad d(L) = (L\cdot C_k - L\cdot C_{k+3})_{k=1}^3.$$

Write dP_6 as the blow-up of $\mathbb{P}^2$ at three noncollinear points, with basis H, E_1, E_2, E_3, and order its hexagon as

$$E_1,\ H-E_1-E_2,\ E_2,\ H-E_2-E_3,\ E_3,\ H-E_1-E_3.$$

For $L = aH - \sum b_i E_i$, direct intersection gives

$$d(L) = (s,-s,s), \qquad s = b_1+b_2+b_3-a. \tag{5.6}$$

Thus $\operatorname{im} d = \mathbb{Z}(1,-1,1)$ is primitive, $\ker d \simeq \mathbb{Z}^3$, and $\operatorname{coker} d \simeq \mathbb{Z}^2$. The resolution (5.5) has width two and the cohomology of its smooth strata is concentrated in even degrees. The only possible higher differential would be $d_2 : E_2^{0,q} \to E_2^{2,q-1}$, whose target is an odd row and is zero; differentials with $r \geq 3$ leave the width-two range. Therefore

$$H^a(W;\mathbb{Z}) = (\mathbb{Z},\mathbb{Z}^2,\mathbb{Z}^4,\mathbb{Z}^2,\mathbb{Z}) \quad (a=0,1,2,3,4), \tag{5.7}$$

with no torsion.

The hypercohomology spectral sequence

$$E_2^{a,b} = H^a(W;\mathcal{H}^b\psi) \Longrightarrow H^{a+b}(F;\mathbb{Z}) \tag{5.8}$$

has the following ranks, all on free groups:

$b=2$	2	0	0	0	0
$b=1$	2	1	3	0	0
$b=0$	1	2	4	2	1
	$a=0$	$a=1$	$a=2$	$a=3$	$a=4.$

Let the possible differential ranks be

$$\alpha = \operatorname{rk} d_2^{0,1}, \quad \beta = \operatorname{rk} d_2^{1,1}, \quad \gamma' = \operatorname{rk} d_2^{2,1}, \quad \delta' = \operatorname{rk} d_2^{0,2}, \quad \epsilon = \operatorname{rk} d_3^{0,2}.$$

Comparison with the torus Betti numbers $(1,4,6,4,1)$ in total degrees one, four, and two gives

$$\alpha = 0, \qquad \gamma' = 0, \qquad \beta + \delta' + \epsilon = 1. \tag{5.9}$$

We determine the remaining ambiguity geometrically, keeping the supported specialization map explicit. Choose a compact T_f-orbit $K = T_f\cdot w_0 \simeq T^2$ in the smooth stratum $W^\circ \simeq (\mathbb{C}^*)^2$. On a relatively compact neighbourhood of K, average a Riemannian metric over T_f and horizontally lift $\partial_{\mathrm{Re}\,t}$ and $\partial_{\mathrm{Im}\,t}$. If ξ_1^h, ξ_2^h are those lifts, put

$$\xi_t = (\mathrm{Re}\,t)\xi_1^h + (\mathrm{Im}\,t)\xi_2^h.$$

Choose a relatively compact T_f-invariant open neighbourhood K' with $K \Subset K'$. For some $r_0 > 0$, the time-one flow

$$\Theta : K' \times \Delta_{r_0} \longrightarrow N_0, \qquad \Theta(w,t) = \Phi_1^{\xi_t}(w), \tag{5.9a}$$

is a T_f-equivariant open embedding over Δ_{r_0}, with $\Theta(w,0) = w$. Indeed $df_0(\xi_t) = t$, so $f_0(\Theta(w,t)) = t$; its differential is invertible along $K'\times\{0\}$, and injectivity follows first by applying f_0 and then from uniqueness for the time-one flow. Shrinking K' and r_0 gives the claimed embedding.

For $0 < r < r_0$, set

$$A_r = f_0^{-1}(\Delta_r), \qquad Z_r = \Theta(K\times\Delta_r), \qquad K_t = \Theta(K\times\{t\}).$$

The subset Z_r is closed in A_r. Indeed, if a sequence in Z_r converges in A_r, compactness of K gives a convergent subsequence of its K-coordinates; its base coordinates converge to the image under f_0, which still lies in Δ_r, and the embedding Θ then identifies the limit with a point of Z_r.

For $0 < |t| < r$, functoriality of relative cohomology gives the two commutative squares

$$\begin{array}{ccc} H^3(A_r, A_r \setminus Z_r) & \longrightarrow & H^3(A_r) \\ {\scriptstyle u_0\ (\simeq)}\big\downarrow & & \big\downarrow{\scriptstyle \rho_0} \\ H^3(W, W \setminus K) & \longrightarrow & H^3(W) \end{array} \qquad \begin{array}{ccc} H^3(A_r, A_r \setminus Z_r) & \longrightarrow & H^3(A_r) \\ {\scriptstyle u_t\ (\simeq)}\big\downarrow & & \big\downarrow{\scriptstyle \rho_t} \\ H^3(F_t, F_t \setminus K_t) & \longrightarrow & H^3(F_t). \end{array} \tag{5.9b}$$

The two left vertical arrows are isomorphisms. By excision they are the restrictions to the two slices of

$$H^3\big(K' \times \Delta_r, (K' \setminus K) \times \Delta_r\big),$$

and homotopy invariance makes both restrictions isomorphisms. Taking the direct limit over shrinking r, the maps ρ_0 induce the proper-base-change isomorphism

$$\varinjlim_{r \downarrow 0} H^3(A_r; \mathbb{Z}) \xrightarrow{\sim} H^3(W; \mathbb{Z}).$$

Choose the points t on one radial ray and identify the F_t with the fixed marked fibre F by parallel transport. Restriction to F_t, followed by this identification, is compatible under shrinking r. The nearby-cycle specialization is precisely the composite

$$\mathrm{sp}^3 : H^3(W; \mathbb{Z}) \xleftarrow[\sim]{\rho_0} \varinjlim_{r \downarrow 0} H^3(A_r; \mathbb{Z}) \xrightarrow{\rho_{\mathrm{near}}} H^3(F; \mathbb{Z});$$

this is the neighbourhood realization of the unit $\mathbb{Z}_W \to \psi_{f_0}\mathbb{Z}$ in the comparison cited above. Taking the direct limit in the two functorial squares therefore proves, rather than assumes, that

$$H^3(W, W \setminus K) \longrightarrow H^3(W) \xrightarrow{\mathrm{sp}^3} H^3(F_t) \tag{5.9c}$$

is the transported supported class followed by the Gysin map for $K_t \hookrightarrow F_t$.

Under the marking of the torus fibre,

$$\Lambda_{\mathrm{tor}} = \mathbb{Z}\widehat{w} \oplus \mathbb{Z}\widehat{\delta}$$

is a saturated direct summand of Λ, and K_t is a translate of $(\Lambda_{\mathrm{tor}} \otimes \mathbb{R})/\Lambda_{\mathrm{tor}}$. Choose a complementary summand Λ', and write $F_t \simeq T_f \times T_B$ as oriented real tori, where $T_B = (\Lambda' \otimes \mathbb{R})/\Lambda'$, with the product orientation chosen to agree with that of F_t. The real rank-two normal bundle of the translate K_t is trivial and oriented. The Thom isomorphism gives

$$H^3(F_t, F_t \setminus K_t) \simeq H^1(T_f),$$

and its map to absolute cohomology is

$$\iota_!(a) = a \otimes [T_B]^\vee \in H^1(T_f) \otimes H^2(T_B) \subset H^3(F_t). \tag{5.9d}$$

This is an injective primitive rank-two Künneth direct summand. On the central smooth stratum, $W^\circ \simeq (\mathbb{C}^*)^2 \simeq T_f \times \mathbb{R}^2$ as an oriented real manifold; hence the real rank-two normal bundle of K is also trivial and oriented. The same Thom calculation gives $H^3(W, W \setminus K) \simeq H^1(K) \simeq \mathbb{Z}^2$. Hence (5.9c) has primitive rank-two image. Since $\operatorname{rk} H^3(W) = 2$, specialization in degree three is injective and has precisely that primitive image. In (5.9), this forces $\beta = \epsilon = 0$, and consequently $\delta' = 1$.

We make the integral saturation argument explicit. The unit $\mathbb{Z}_W \to \psi$ is a morphism from the complex concentrated in cohomological degree zero to the Postnikov filtration of ψ. It therefore induces a morphism of hypercohomology spectral sequences from the single row

$$H^a(W;\mathbb{Z}) = E_2^{a,0}(\mathbb{Z}_W)$$

to (5.8). On the $b = 0$ row the induced E_2-map is the identity, because $\mathcal{H}^0\psi = \mathbb{Z}_W$. Consequently, in total degree r, specialization factors through the last filtration step

$$H^r(W;\mathbb{Z}) \xrightarrow{\sim} E_\infty^{r,0} = F^r H^r(F;\mathbb{Z}) \hookrightarrow H^r(F;\mathbb{Z}) \tag{5.9e}$$

whenever no differential $d_k : E_k^{r-k,k-1} \to E_k^{r,0}$ enters on any page. For $r = 1$ this is automatic; for $r = 2$ the only possible entering differential is $d_2^{0,1}$, and $\alpha = 0$. Thus (5.9e) is an isomorphism onto the specialization image in degrees one and two. The quotient by F^r has the finite filtration whose graded groups are the remaining $E_\infty^{a,r-a}$, $a < r$. In degree one these are kernels of maps of free abelian groups. In degree two they are $E_\infty^{1,1} = \mathbb{Z}$ and $E_\infty^{0,2} = \ker d_2^{0,2}$; here the possible subsequent map is $d_3^{0,2}$, and it is zero because its already determined rank ϵ is zero. Both groups are therefore free. Successive extensions of free abelian groups split, so $H^r(F;\mathbb{Z})/F^r$ is free for $r = 1, 2$. Hence the specialization images in those degrees are primitive.

In degree three, (5.9c)–(5.9d) give a primitive rank-two lattice contained in $\operatorname{im}\mathrm{sp}^3$. Since $H^3(W;\mathbb{Z})$ itself has rank two, the specialization image has the same rational span; a primitive lattice has no proper finite-index overlattice in that span inside $H^3(F;\mathbb{Z})$. Therefore the two images coincide, proving both injectivity and primitivity in degree three. In degree four, $E_\infty^{4,0} = H^4(F;\mathbb{Z})$, so specialization is an isomorphism. Thus specialization is injective with primitive image in every degree. Monodromy invariance gives

$$\operatorname{im}\mathrm{sp}^r \subseteq H^r(F;\mathbb{Z})^{T_0}.$$

By theorem 5.2, the invariant ranks are $(1, 2, 4, 2, 1)$, exactly those in (5.7), and each invariant lattice is the kernel of an integral endomorphism, hence primitive. Two primitive sublattices of the same rational subspace coincide: both equal the intersection of that subspace with the ambient lattice. This proves (5.1). □

53.2 The global E_2-page

The Leray spectral sequence used in this subsection is [33, Corollary 2.3.4]. Let $R^r = R^r f_*\mathbb{Z}$ and $\mathcal{L}^r = j^*R^r$. The finite-fibre lattices have already been proved as abstract lattice statements in theorem 5.4. Their identification with the stalks of the proposed family requires the explicit radial-stalk compatibility below. This compatibility is treated separately in theorem 10.10 and is not silently absorbed here.

Proposition 53.2 (all Leray terms used below)**.** ***[VERIFIED]*** *The specialization theorem theorem 53.1, the finite fillings of theorem 10.9, and theorem 10.10 give, for $j = 1, 2$ and $0 \le r \le 4$, the finite-filling compatibility*

$$N_j \text{ strongly deformation retracts onto } S_j, \qquad H^r(S_j;\mathbb{Z}) \longrightarrow H^r(F;\mathbb{Z})^{T_j} \text{ is } \pi_j^* \tag{FF}$$

under the marked punctured-fibre identification. Consequently,

$$H^0(B, R^1) = \mathbb{Z}(12\gamma), \quad H^0(B, R^2) = \mathbb{Z}(2q), \quad H^0(B, R^3) = \mathbb{Z}(2\gamma uw), \tag{5.10}$$

$$H^1(B, R^1) = H^1(B, R^2) = 0, \tag{5.11}$$

and, with ω a generator of $H^2(B;\mathbb{Z})$,

$$H^2(B, R^1) = \mathbb{Z}\omega[\delta], \qquad H^2(B, R^2) = \mathbb{Z}\omega[\gamma\delta], \qquad H^2(B, R^4) = \mathbb{Z}\omega[\mathrm{vol}]. \tag{5.12}$$

These identifications are compatible with exterior products in V.

Proof. Under the stated hypotheses, proper base change identifies the stalk of R^r at p_0 with $H^r(W;\mathbb{Z})$, while $(j_*\mathcal{L}^r)_{p_0} = H^r(F;\mathbb{Z})^{T_0}$. The stalk map is specialization and is an isomorphism by theorem 53.1; hence its cokernel at p_0 is zero. At p_j, condition (FF) identifies the stalk with $H^r(S_j;\mathbb{Z})$, the target with $H^r(F;\mathbb{Z})^{T_j}$, and the stalk map with π_j^*. The finite-fibre calculation theorem 5.4 makes each of these maps injective. Away from the three special points the map is the identity of the local system. Stalkwise kernels and cokernels therefore give the exact sequence

$$0 \longrightarrow R^r \longrightarrow j_*\mathcal{L}^r \longrightarrow Q^r \longrightarrow 0, \tag{5.13}$$

with

$$Q^1 = (\mathbb{Z}/3)_{p_1} \oplus (\mathbb{Z}/4)_{p_2}, \quad Q^2 = (\mathbb{Z}/2)_{p_2}, \quad Q^3 = (\mathbb{Z}/2)_{p_2}, \quad Q^4 = (\mathbb{Z}/3)_{p_1} \oplus (\mathbb{Z}/4)_{p_2}.$$

The degree-one quotients have orders three and four and are generated by the class of γ. In degree two the p_1-quotient is zero, while

$$(\textstyle\bigwedge^2 V)^{T_2}/\{a\gamma\eta_2 + bq : a \equiv b \pmod 2\} \simeq \mathbb{Z}/2$$

is generated by the class of q. In degree three the p_1-quotient is zero and the p_2-quotient has order two, generated by the class of γuw. This last identification uses the actual twist $v_2 = -\varepsilon'$, for which the source's admissibility quantity $\eta_2(v_2)$ is zero and hence divisible by four. In degree four, pullback multiplies the orientation class by the covering degree m_j, giving the displayed Q^4. Hence the skyscraper sheaves have exactly the asserted stalks; no claim for an arbitrary twist is being imported. By theorem 5.3,

$$V^G = \mathbb{Z}\gamma, \quad (\textstyle\bigwedge^2 V)^G = \mathbb{Z}q, \quad (\textstyle\bigwedge^3 V)^G = \mathbb{Z}\gamma uw, \tag{5.14}$$

and

$$V_G = \mathbb{Z}[\delta], \qquad (\textstyle\bigwedge^2 V)_G = \mathbb{Z}[\gamma\delta], \qquad [q] = 12[\gamma\delta]. \tag{5.15}$$

Since $\det T_1 = \det T_2 = 1$, the action on $\bigwedge^4 V = \mathbb{Z}\,\mathrm{vol}$ is trivial and

$$(\textstyle\bigwedge^4 V)_G = \mathbb{Z}[\mathrm{vol}]. \tag{5.15a}$$

The maps from the invariant groups in (5.14) onto the corresponding skyscraper fibres are surjective, with kernels $12\mathbb{Z}\gamma$, $2\mathbb{Z}q$, and $2\mathbb{Z}\gamma uw$. This proves (5.10).

Because each Q^r is a skyscraper, it has no positive cohomology; the long exact sequence of (5.13) therefore identifies $H^2(B, R^r)$ with $H^2(B, j_*\mathcal{L}^r)$. For any free local system $\mathcal{L}$ on B°, with stalk M, the quotient $j_*\mathcal{L}/j_!\mathcal{L}$ is a skyscraper. Hence

$$H^2(B, j_*\mathcal{L}) \simeq H^2_c(B^\circ, \mathcal{L}) \simeq H_0(B^\circ, \mathcal{L}) = M_G, \tag{5.16}$$

where the middle isomorphism is Poincaré duality with local coefficients [10, §3.H]; see also G. E. Bredon, *Sheaf Theory*, 2nd ed., Chapter VI. Cap product with the locally finite fundamental class makes (5.16) natural for coefficient pairings, proving (5.12) and the product assertion.

It remains to prove (5.11), for which we spell out the parabolic-cohomology quotient. The low-degree Leray sequence for $j : B^\circ \hookrightarrow B$ contains

$$0 \longrightarrow H^1(B, j_*\mathcal{L}) \longrightarrow H^1(B^\circ, \mathcal{L}) \longrightarrow H^0(B, R^1 j_*\mathcal{L}).$$

If $I_i = \langle t_i \rangle$ is the local inertia group at p_i, then

$$(R^1 j_*\mathcal{L})_{p_i} = H^1(I_i, M) = M/(T_i - I)M.$$

Consequently the last arrow sends the class of a cocycle c to the three classes of $c(t_i)$, and $H^1(B, j_*\mathcal{L})$ is exactly the kernel defined by the three parabolic conditions below. Use left cocycles, so $c(gh) = c(g) + gc(h)$.

A cocycle on $\pi_1(B^\circ) = F(t_1, t_2)$ is determined by $x = c(t_1)$ and $y = c(t_2)$. Extension across the three punctures is the parabolic condition

$$x \in (T_1 - I)M, \qquad y \in (T_2 - I)M, \qquad c(t_0) \in (T_0 - I)M, \qquad t_0 = (t_1 t_2)^{-1}.$$

Write $y = (T_2 - I)m$ and subtract the coboundary of m; this normalizes $y = 0$. A further coboundary preserves the normalization precisely when its defining element lies in M^{T_2}, and then changes x by an element of $(T_1 - I)M^{T_2}$. With $y = 0$, $c(t_0) = -T_0 x$. Since T_0 preserves $(T_0 - I)M$, the last parabolic condition is equivalent to $x \in (T_0 - I)M$. Therefore

$$H^1(B, j_*\mathcal{L}) \simeq \frac{(T_1 - I)M \cap (T_0 - I)M}{(T_1 - I)M^{T_2}}. \tag{5.17}$$

For $M = V$,

$$(T_1 - I)V = \langle 2\gamma + u + w, 2\gamma - u\rangle, \qquad (T_0 - I)V = \langle \gamma, u\rangle,$$

and their intersection is $\mathbb{Z}(2\gamma - u) = (T_1 - I)\mathbb{Z}\eta_2$. For $M = \bigwedge^2 V$,

$$\begin{aligned}(\textstyle\bigwedge^2 T_1 - I)\bigwedge^2 V = \langle &\gamma u, \gamma w, 2u\delta + w\delta,\\ &uw + u\delta - w\delta - 6\gamma\delta\rangle,\end{aligned}$$

whereas

$$(\textstyle\bigwedge^2 T_0 - I)\bigwedge^2 V = \langle \gamma u, \gamma w + u\delta\rangle.$$

Their intersection is $\mathbb{Z}\gamma u$, equal to $(\bigwedge^2 T_1 - I)\mathbb{Z}(\gamma\eta_2)$. Thus both groups in (5.17) vanish. The maps $H^0(B, j_*\mathcal{L}^r) \to H^0(B, Q^r)$ were proved surjective above for $r = 1, 2$, and skyscraper sheaves have no positive cohomology. The long exact sequence of (5.13) therefore gives (5.11) for R^1, R^2. □

53.3 The three transgressions

Lemma 53.3 (the primitive first transgression)**.** ***[VERIFIED]*** *For the constructed map and the based meridians of theorem 53.6, one may orient ω so that*

$$d_2^{0,1}(12\gamma) = \omega. \tag{5.18}$$

Proof. Choose disjoint discs D_i about the special points and a pair-of-pants neighbourhood V_0 of their complement. The class $\theta = 12\gamma$ is a section of $R^1 f_*\mathbb{Z}$ over each D_i: at p_0 this is theorem 53.1, and at p_j it is the degree-one finite-fibre lattice. It is also a section over V_0, because it is globally monodromy invariant. For every member $U \in \{D_0, D_1, D_2, V_0\}$, the local Leray five-term sequence contains

$$H^1(f^{-1}(U); \mathbb{Z}) \longrightarrow H^0(U, R^1 f_*\mathbb{Z}) \longrightarrow H^2(U, R^0 f_*\mathbb{Z}) = H^2(U; \mathbb{Z}) = 0.$$

Thus the section θ has a cohomological lift $\Theta_U \in H^1(f^{-1}(U); \mathbb{Z})$ on every member of the cover. On an overlap $D_i \cap V_0$, the two lifts have the same image in $H^0(R^1 f_*\mathbb{Z})$; local five-term exactness therefore identifies their difference with a class in $H^1(D_i \cap V_0, R^0 f_*\mathbb{Z}) = H^1(D_i \cap V_0; \mathbb{Z})$. These three differences form the Čech connecting cocycle used below.

Use clockwise meridians. The affine deck relation at p_j is

$$\widehat{\rho}_j^{\,m_j} = t_{v_j}. \tag{5.19}$$

Put $\Theta_j = \Theta_{D_j}$. Since Θ_j is integral and restricts to θ on the fibre, (5.19) gives $m_j\Theta_j(\widehat{\rho}_j) = \Theta_j(t_{v_j}) = \theta(v_j)$. Therefore

$$\Theta_1(\widehat{\rho}_1) = 12\gamma(v_1)/3 = 4, \qquad \Theta_2(\widehat{\rho}_2) = 12\gamma(v_2)/4 = -3. \tag{5.20}$$

By theorem 53.6, the toric meridian $\widetilde{a}_0$ is the boundary of the holomorphic disc $\{(1,1,t) : |t| \le \epsilon\}$. It is therefore null-homotopic and the local lift evaluates to zero.

Write $\alpha_i = \Theta_{V_0}(\widehat{\rho}_i)$. By the exact based cusp relation (5.23a), the three annular discrepancies have meridian values

$$4 - \alpha_1, \qquad -3 - \alpha_2, \qquad \alpha_1 + \alpha_2, \tag{5.21}$$

whose sum is one. Mayer–Vietoris on the base identifies

$$H^1\left(\coprod_i (D_i \cap V_0)\right) / \operatorname{im} H^1(V_0) \simeq H^2(B;\mathbb{Z}).$$

Under meridian evaluation the numerator is $\mathbb{Z}^3$, the image is the sum-zero lattice, and the connecting map is the signed sum of the three coordinates.

We now identify this connecting class with the Leray differential rather than taking that identification as an extra hypothesis. Put $K = Rf_*\mathbb{Z}_X$. The canonical truncation triangle [37, Tag 08J5] is

$$R^0 f_*\mathbb{Z}_X \longrightarrow \tau_{\le 1} K \longrightarrow R^1 f_*\mathbb{Z}_X[-1] \xrightarrow{\kappa} R^0 f_*\mathbb{Z}_X[1]. \tag{5.21a}$$

After applying $R\Gamma(B,-)$, the connecting homomorphism

$$H^0(B, R^1 f_*\mathbb{Z}) \xrightarrow{H^0(\kappa)} H^2(B, R^0 f_*\mathbb{Z}) \tag{5.21b}$$

is precisely $d_2^{0,1}$ in the hypercohomology, hence Leray, spectral sequence; this is the canonical-truncation construction of that spectral sequence [37, Tag 0FLJ]. On the chosen disc–pair-of-pants cover, functorial Mayer–Vietoris for the triangle [37, Tag 0F16] computes $H^0(\kappa)$ by choosing the local degree-one lifts Θ_U and taking their differences on the annular overlaps. Those differences are exactly the cocycle (5.21). Thus the signed sum just computed is $d_2^{0,1}(\theta)$. Equation (5.21) therefore represents a primitive generator; orient ω to obtain (5.18). □

Lemma 53.4 (the higher transgressions)**.** ***[VERIFIED]*** *In the integral Leray spectral sequence of the constructed map,*

$$d_2^{0,2}(2q) = \omega[\delta], \qquad d_2^{0,3}(2\gamma uw) = \omega[\gamma\delta]. \tag{5.22}$$

Proof. Use the multiplicative Leray spectral sequence obtained from a filtered multiplicative cochain resolution. Its E_2-product is the natural cup product

$$H^p(B, R^q) \otimes H^{p'}(B, R^{q'}) \longrightarrow H^{p+p'}(B, R^{q+q'}),$$

and every d_r is a graded derivation. This is the standard filtered-DGA construction; compare the filtered-complex spectral sequence [37, Tags 012K and 0FLJ] with the derived cup product [37, §20.31]. The generator identifications and their products are natural by the last sentence of theorem 53.2. Put $\theta = 12\gamma$, $a = 2q$, and $b = 2\gamma uw$. By theorem 53.2, write

$$d_2 a = r\omega[\delta], \qquad d_2 b = s\omega[\gamma\delta].$$

In the E_2-ring,

$$\theta a = 12b, \qquad ab = 0, \qquad [q] = 12[\gamma\delta].$$

The derivation rule and $|\theta| = 1$ give

$$12 d_2 b = d_2(\theta a) = d_2\theta \cdot a - \theta \cdot d_2 a,$$

hence $s = 2 - r$. Since $|a| = 2$, $0 = d_2(ab) = d_2 a \cdot b + a \cdot d_2 b$. The exterior products are

$$\delta \wedge 2\gamma uw = -2\,\mathrm{vol}, \qquad 2q \wedge \gamma\delta = 2\,\mathrm{vol}\,.$$

Therefore $-2r + 2s = 0$, so $r = s$. Together with $s = 2 - r$, this gives $r = s = 1$, proving (5.22). □

Theorem 53.5 (integral cohomology in low degrees)**.** ***[VERIFIED]*** *For the constructed holomorphic map,*

$$H^1(X;\mathbb{Z}) = H^2(X;\mathbb{Z}) = H^3(X;\mathbb{Z}) = 0, \qquad H_2(X;\mathbb{Z}) = 0. \tag{5.23}$$

Proof. The Leray terms in total degree two are

$$E_2^{2,0} = \mathbb{Z}\omega, \qquad E_2^{1,1} = 0, \qquad E_2^{0,2} = \mathbb{Z}(2q).$$

By theorems 53.3 and 53.4, the incoming map to the first term and the outgoing map from the last term are isomorphisms. Thus all three E_∞-pieces vanish and $H^2(X) = 0$. The first map has zero kernel, giving $H^1(X) = 0$. In total degree three, the quotient of $E_2^{2,1}$ by the image of $E_2^{0,2}$ is zero, $E_2^{1,2} = 0$, and the outgoing map from $E_2^{0,3}$ is injective by (5.22). Hence $H^3(X) = 0$. No later differential can meet these terms because the base has cohomological dimension two.

The universal coefficient sequences show both $\mathrm{Hom}(H_2(X),\mathbb{Z}) = 0$ and $\mathrm{Ext}(H_2(X),\mathbb{Z}) = 0$. Since $H_2(X)$ is finitely generated, it has neither a free nor a torsion summand, so it is zero. □

53.4 Based van Kampen and sphere recognition

Proposition 53.6 (the geometric fundamental-group presentation)**.** ***[VERIFIED]*** *For the punctured-overlap gluings of theorems 10.7 and 10.9, the based van Kampen presentation is*

$$\pi_1(X) \simeq \langle c, x, y \mid c \text{ central},\ xy = 1,\ x^3 = c,\ y^4 = c^{-1} \rangle \simeq 1. \tag{5.24}$$

Proof. Choose b_* in a pair-of-pants core of B°, put x_* on the holomorphic zero section above it, and choose three nonintersecting base whiskers from b_* to clockwise boundary circles. Lift all whiskers by the zero section. With the ordering compatible with $g_1 g_2 g_0 = 1$, the resulting based loops obey

$$\widehat{\rho}_1 \widehat{\rho}_2 \widetilde{a}_0 = 1. \tag{5.23a}$$

The period-family action is linear and hence preserves the zero section. It splits the homotopy sequence over the pair of pants, giving

$$\pi_1(f^{-1}(B^\circ)) = \Lambda \rtimes F(\widehat{\rho}_1, \widehat{\rho}_2),$$

with conjugation actions A_1, A_2. The clockwise meridian at p_j lifts to the affine generator in (2.9), with its exact sign and based translation. To verify this in the actual real marking, put

$$\alpha(s) = \frac{\log s_j}{2\pi i}, \qquad J_s = \Pi(s)^{-1} i \Pi(s), \qquad a_j(s) = \mathrm{Re}\,\alpha(s)\, v_j + \mathrm{Im}\,\alpha(s)\, J_s v_j.$$

The overlap translation by $\alpha(s)\Pi(s)v_j$ is translation by $a_j(s)$ in the real marking. Thus the inverse image of the period-family zero section is $x(s) = -a_j(s)$.

The covariance of J_s is a consequence of the proved period identity (2.8e), not an additional assumption. That identity is

$$R_{g_j}(s)\Pi(s) = \Pi(g_j s) A_j.$$

The map $R_{g_j}(s)$ is complex linear. Hence

$$\begin{aligned} J_{g_j s} A_j &= \Pi(g_j s)^{-1} i\, \Pi(g_j s) A_j \\ &= \Pi(g_j s)^{-1} i\, R_{g_j}(s) \Pi(s) \\ &= \Pi(g_j s)^{-1} R_{g_j}(s) i\, \Pi(s) = A_j J_s. \end{aligned}$$

The original vectors are

$$v_1 = (1, 2, -4, 0)^{\mathsf{t}}, \qquad v_2 = (-1, -3, 3, 0)^{\mathsf{t}},$$

in the basis $(\widehat{\gamma}, \widehat{u}, \widehat{w}, \widehat{\delta})$. Multiplication by the printed matrices gives

$$A_1 v_1 = (1,\, 6-4,\, -6-2+4,\, -2+2)^{\mathsf{t}} = (1, 2, -4, 0)^{\mathsf{t}},$$

$$A_2 v_2 = (-1,\, -3,\, 6-3,\, -3+3)^{\mathsf{t}} = (-1, -3, 3, 0)^{\mathsf{t}}.$$

Thus $A_j v_j = v_j$ exactly. Clockwise continuation by g_j changes the chosen logarithm to $\alpha(g_j s) = \alpha(s) - 1/m_j$. Using this real change, the unchanged imaginary part, and $J_{g_j s} A_j = A_j J_s$, one obtains

$$a_j(g_j s) = A_j a_j(s) - v_j/m_j.$$

Therefore the affine generator sends the initial lift of the zero section to its actual continued endpoint:

$$A_j(-a_j(s)) + v_j/m_j = -a_j(g_j s).$$

Changing the logarithm branch by an integer changes the initial lift by an integer multiple of v_j. Such a translation commutes with the affine generator because $A_j v_j = v_j$, so the endpoint identification does not introduce a translated conjugate. The fixed zero-section whiskers retain the based generator. After m_j clockwise iterations, the coordinate returns to the same point and α has decreased by one. Thus the endpoint of $-a_j$ has increased by v_j, in agreement with (2.8f). Consequently

$$\widehat{\rho}_j^{\,m_j} = t_{v_j}. \tag{5.23b}$$

The radial model of theorem 10.10 shows that attaching N_j imposes exactly this relation. Since the global and local whiskers are the same zero-section path under the overlap, conjugating back to x_* adds no fibre drift.

For completeness, the cusp covering has no hidden fundamental group. A loop in the infinite toric variety has compact image and therefore lies in the open $X_{\Sigma_{\mathrm{tor}}}$ associated with a finite subfan Σ_{tor}. Enlarge that subfan by the rays through $(0,1), (e_1,1), (e_2,1)$. Write

$$N_{\Sigma_{\mathrm{tor}}} := \big\langle |\Sigma_{\mathrm{tor}}| \cap N' \big\rangle_{\mathbb{Z}}.$$

The three displayed primitive generators span N', so the classical formula $\pi_1(X_{\Sigma_{\mathrm{tor}}}) = N'/N_{\Sigma_{\mathrm{tor}}}$ [24, Theorem 12.1.10] makes the resulting finite toric open simply connected; the original loop therefore has a null-homotopy in that open. Thus Y is simply connected. The one-parameter subgroup in the third torus coordinate acts by $t \mapsto rt$, $r > 0$. Given the compact image of a null-homotopy, choose r so small that its translate lies in $|t| < \epsilon$; the path $r_s = (1-s) + sr$ gives a homotopy from the original loop to its translate inside Y_ϵ. Hence Y_ϵ is also simply connected. The verified free quotient $N_0 = Y_\epsilon/\bar{\Lambda}$ therefore has universal cover Y_ϵ, so $\pi_1(N_0) = \bar{\Lambda}$. Equation (2.8iii) identifies a fibre-lattice loop $\lambda \in \Lambda$ with the deck transformation $\bar{\lambda}$; hence the inclusion of a nearby fibre is exactly

$$\Lambda \longrightarrow \bar{\Lambda}, \qquad \lambda \longmapsto \bar{\lambda}, \tag{5.23c}$$

with kernel $\Lambda_{\mathrm{tor}} = \langle \widehat{w}, \widehat{\delta} \rangle$. On the zero section the same overlap sends $(z, 0)$ to $(1, 1, t_c)$. In the toric chart of the triangle $\{0, e_1, e_2\}$, whose coordinates may be written $(t_c/(x_1 x_2), x_1, x_2)$, this is the holomorphic disc $t_c \mapsto (t_c, 1, 1)$. Thus the cusp meridian $\widetilde{a}_0$ is null-homotopic, with no fibre translation. Combining this fact with (5.23a), the cusp attachment imposes precisely $\widehat{\rho}_1 \widehat{\rho}_2 = 1$.

Van Kampen now gives

$$\left\langle \Lambda, \widehat{\rho}_1, \widehat{\rho}_2 \;\middle|\; \widehat{\rho}_j \lambda \widehat{\rho}_j^{-1} = A_j \lambda, \widehat{\rho}_1^3 = t_{v_1}, \widehat{\rho}_2^4 = t_{v_2}, \widehat{\rho}_1 \widehat{\rho}_2 = 1, \Lambda_{\mathrm{tor}} = 1 \right\rangle.$$

The normal closure of Λ_{tor} under A_1, A_2 is $\ker \gamma = \langle \widehat{u}, \widehat{w}, \widehat{\delta} \rangle$; for example $A_1 \widehat{w} = \widehat{u} - \widehat{w}$. The image of Λ is therefore generated by $c = [\widehat{\gamma}]$, which is central because γ is invariant. Since $\gamma(v_1) = 1$ and $\gamma(v_2) = -1$, the presentation becomes (5.24). It is the case $(\ell_0, \ell_1, \ell_2) = (0, 1, -1)$ of theorem 5.5, where $p = -1$; hence the group is trivial. □

Corollary 53.7 (topology and the third Chern number)**.** ***[VERIFIED]*** *The underlying smooth manifold of the constructed threefold is diffeomorphic to the standard six-sphere, and*

$$\langle c_3(T_X), [X] \rangle = 2. \tag{5.25}$$

Proof. Combine theorems 53.5 and 53.6. The universal coefficient theorem and Poincaré duality give the integral homology of S^6, exactly as in theorem 58.15. The complete recognition argument of theorem 58.16 then gives $X \cong_{\mathrm{diff}} S^6$. Its Euler characteristic is two. On a complex threefold the top Chern class of the complex tangent bundle is the Euler class of the underlying oriented real tangent bundle, proving (5.25). □

Independent role of the collapse route. The proof above is independent of the cellular route. The candidate's cusp collapse, its integral homology map, and its equality with nearby-cycle specialization are now independently verified in theorems 58.5 to 58.7. The actual finite-band and cusp Mayer–Vietoris maps, their primitive blocks, and their commuting comparison with the differentials proved in this section are now theorems 58.10, 58.11 and 58.13. This supplies a second complete integral topology certificate; it remains logically independent of the sphere-recognition conclusion already obtained in this section.

54 The direct algebraic-reduction proof

The calculation in this section is independent of the integral topology of X. It uses the period family of theorem 10.2 and the proper connected fibration of theorem 10.11; it does not use $H^2(X;\mathbb{Z})$, the sphere-recognition argument, or any theorem of Campana–Demailly–Peternell.

Lemma 54.1 (algebraic reduction of a two-dimensional torus)**.** *Let T be a complex torus of dimension two. Suppose*

$$\mathrm{NS}(T) = \mathbb{Z}\eta, \qquad \eta^2 \neq 0,$$

and the Appell–Humbert Hermitian form H_η is nondegenerate and indefinite. Then $a(T) = 0$.

Proof. The algebraic reduction of a compact complex torus is a quotient $q : T \to A = T/K$ by a connected complex subtorus, with A an abelian variety and $\dim A = a(T)$ [19, Chapter 2, §6]; see also [22, §8, p. 2437].

Assume $a(T) > 0$. If $a(T) = 2$, then T is an abelian surface. An ample line bundle has first Chern class $n\eta$, $n \neq 0$, and its Appell–Humbert form nH_η is positive definite. This contradicts the indefiniteness of H_η.

If $a(T) = 1$, then A is an elliptic curve. For a point $p \in A$, put

$$\varphi = q^* c_1(\mathcal{O}_A(p)) \in \mathrm{NS}(T).$$

The class is nonzero: for any translation-invariant Kahler form ω on T, the projection formula gives $\int_T \varphi \wedge \omega = \int_K \omega > 0$. But $\varphi^2 = 0$, since A is a curve. Writing $\varphi = n\eta$, $n \neq 0$, yields $0 = n^2\eta^2 \neq 0$, a contradiction. Both possible positive algebraic dimensions are excluded. □

Let

$$V = \mathbb{Z}\langle \gamma, u, w, \delta \rangle, \qquad \eta = u \wedge w + 6\gamma \wedge \delta.$$

For an integral two-class $E \in \bigwedge^2 V$, write its coordinates in the ordered basis $(\gamma u, \gamma w, \gamma\delta, uw, u\delta, w\delta)$ as $E_{\gamma u}, \dots, E_{w\delta}$, and set

$$\begin{aligned} P_E = {} & E_{\gamma u} - E_{\gamma w}\tau - E_{\gamma\delta}\mu + 6E_{uw}\mu + E_{u\delta}\beta \\ & + E_{w\delta}(6\mu^2 - \tau\beta). \end{aligned} \tag{3.1}$$

Proposition 54.2 (very general Neron–Severi group)**.** ***[VERIFIED]*** *For the descended period family of theorem 10.2, outside a countable subset of B° the smooth fibre F_b satisfies*

$$\mathrm{NS}(F_b) = \mathbb{Z}\eta, \qquad a(F_b) = 0. \tag{3.2}$$

Proof. The two holomorphic one-forms defining the Hodge subspace are

$$\sigma_1 = 6\mu\gamma + \tau u + w, \qquad \sigma_2 = \beta\gamma + \mu u + \delta.$$

The determinant inequality (2.2) gives $V_{\mathbb{C}} = F^1 \oplus \overline{F^1}$, where $F^1 = \langle \sigma_1, \sigma_2 \rangle$. For a real two-class E, the Hodge decomposition therefore gives

$$E \in H^{1,1}(F_b) \quad \Longleftrightarrow \quad E \wedge \sigma_1 \wedge \sigma_2 = 0.$$

Expanding the wedge product gives

$$E \wedge \sigma_1 \wedge \sigma_2 = P_E\, \gamma \wedge u \wedge w \wedge \delta. \tag{3.3}$$

We next determine exactly which P_E vanish identically. Suppose

$$a + b\tau + c\mu + d\beta + e(6\mu^2 - \tau\beta) \equiv 0. \tag{3.4}$$

Apply the cusp transformation $(\tau, \mu, \beta) \mapsto (\tau - 1, \mu, \beta + 1)$ and subtract. This gives

$$-b + d + e(\beta - \tau + 1) = 0.$$

Because $\beta - \tau$ is translated by two, $e = 0$, and then $d = b$. Apply the order-four transformation in (2.1), subtract (3.4), and multiply by $-\tau$. One obtains

$$6b\mu^2 + c(\tau - 1)\mu + b(\tau^2 + 3\tau + 1) - c\tau = 0.$$

Subtract its cusp translate:

$$c\mu + 2b(\tau + 1) - c = 0. \tag{3.5}$$

Translate (3.5) once around the cusp. Since μ is invariant and $\tau \mapsto \tau - 1$, subtracting the translated identity from (3.5) gives $2b = 0$. Thus $b = 0$, and (3.5) becomes $c(\mu - 1) = 0$. If $c \neq 0$, then $\mu \equiv 1$, whereas the order-four law would give $1 = \mu \circ g_2 = 1 + 1/\tau$, an impossibility. Hence $c = 0$, then $b = d = 0$, and finally $a = 0$. Thus the five functions

$$1, \quad \tau, \quad \mu, \quad \beta, \quad 6\mu^2 - \tau\beta$$

are linearly independent over $\mathbb{C}$.

Reading the coefficients in (3.1) now gives

$$P_E \equiv 0 \quad \Longleftrightarrow \quad E \in \mathbb{Z}\eta. \tag{3.6}$$

For each other integral class, the zero set of P_E on the connected Riemann surface $\mathfrak{h}_z$ is discrete, hence countable. There are only countably many integral classes. The class $\eta = q = uw + 6\gamma\delta$ is monodromy invariant by theorem 5.3, so it defines a class on every marked fibre after descent. The image in B° of the countable union of the remaining discrete zero sets is still countable. A base point outside that image has no additional integral $(1,1)$-class on any lift, proving the Neron–Severi statement on the quotient family.

Finally,

$$\eta^2 = 12\, \gamma \wedge u \wedge w \wedge \delta. \tag{3.7}$$

For a nondegenerate alternating form E, write $E^{\vee}$ for its inverse alternating form on the dual real vector space. If $G_\eta = (i\eta^{\vee}(\sigma_a, \overline{\sigma_b}))_{a,b}$, direct inversion of the alternating-form matrix and substitution give

$$G_\eta = \begin{pmatrix} 2\operatorname{Im}\tau & 2\operatorname{Im}\mu \\ 2\operatorname{Im}\mu & \frac{1}{3}\operatorname{Im}\beta \end{pmatrix},$$

$$\det G_\eta = \frac{2}{3}\operatorname{Im}\tau\left(\operatorname{Im}\beta - \frac{6(\operatorname{Im}\mu)^2}{\operatorname{Im}\tau}\right) < 0. \tag{3.8}$$

Here is the Appell–Humbert bridge needed to read the inertia of this matrix. With the convention $E = \operatorname{Im} H_E$, one has

$$E = \frac{1}{2i}\sum_{a,b}(H_E)_{ab}\,\sigma_a \wedge \overline{\sigma_b}.$$

Thus, in the ordered basis $(\sigma_1, \sigma_2, \overline{\sigma_1}, \overline{\sigma_2})$, the upper-right block of the alternating-form matrix of E is $A = -(i/2)H_E$. Inverting the block matrix gives

$$E^{\vee}(\sigma_a, \overline{\sigma_b}) = [-(A^t)^{-1}]_{ab} = -2i(\overline{H_E}^{-1})_{ab}, \qquad G_E = 2\overline{H_E}^{-1}. \tag{3.8a}$$

The displayed G_η is real symmetric, so $H_\eta = 2G_\eta^{-1}$. Inversion and multiplication by a positive scalar preserve inertia. Hence H_η is nondegenerate and indefinite, while (3.7) is nonzero. Apply theorem 54.1. □

Lemma 54.3 (meromorphic descent)**.** *Let $f : X \to B$ be a proper surjective holomorphic map from a connected complex threefold to a connected compact Riemann surface, with connected fibres, and a submersion over $B^\circ = B \setminus \Sigma$, where Σ is finite and $B^\circ \neq \varnothing$. If $g \in \mathcal{M}(X)$ is constant on the fibres over an uncountable subset of B°, then there is a unique $\varphi \in \mathcal{M}(B)$ such that $g = f^*\varphi$.*

Proof. Let P be the polar divisor of g, and put $\Omega = f^{-1}(B^\circ) \setminus |P|$. A proper analytic subset of a connected complex manifold has real codimension at least two and does not disconnect it. First note that $f^{-1}(B^\circ)$ is connected. Indeed, each of its connected components is open, and its image is open because f is a submersion and closed because the base change $f^{-1}(B^\circ) \to B^\circ$ is proper. Since B° is connected, every component would meet every fibre; connectedness of the fibres therefore permits only one component. Removing $|P|$ now leaves Ω connected. Only finitely many fibres are components of $|P|$. For every other b, the complement $F_b \cap \Omega$ of a proper analytic subset in the connected manifold F_b is connected.

On Ω, define the analytic set

$$Z = \{dg \wedge f^*(dt) = 0\},$$

where t is any local coordinate on B. If $g|_{F_b}$ is constant, then $F_b \cap \Omega \subset Z$. If $Z \neq \Omega$, then, because Ω is second countable, Z has at most countably many irreducible components. Each irreducible two-dimensional fibre contained in Z is one of those components, and distinct fibres give distinct components. This contradicts the uncountable hypothesis. Therefore $Z = \Omega$.

It follows that dg annihilates $\ker df$. Connectedness of $F_b \cap \Omega$, together with local holomorphic sections of the submersion, gives $g = \varphi \circ f$ on Ω, for a holomorphic φ away from finitely many points of B.

Fix such a point t_0. Choose a general holomorphic curve germ $\kappa : (\Delta, 0) \to (X, x)$, not contained in $|P|$, with

$$f \circ \kappa(s) = t_0 + s^k u(s), \qquad u(0) \neq 0.$$

If $g \circ \kappa$ has pole order at most M, then $|g \circ \kappa(s)| \leq C|s|^{-M}$. After shrinking the disc, boundedness above and below of $|u|$ gives constants $c, C_1 > 0$ such that

$$c|t - t_0|^{1/k} \leq |s| \leq C_1|t - t_0|^{1/k}$$

on each local inverse branch. Therefore

$$|\varphi(t)| \leq C_2|t - t_0|^{-M/k}$$

away from the image of a discrete exceptional set, and then everywhere on the punctured disc by holomorphy. For any integer $N > M/k$, the function $(t - t_0)^N \varphi(t)$ is bounded and extends holomorphically. Thus φ is meromorphic at t_0. Repeating at the finitely many exceptional points produces $\varphi \in \mathcal{M}(B)$. Equality on the dense open set Ω gives equality of meromorphic functions on X, and surjectivity of f gives uniqueness. □

Theorem 54.4 (algebraic dimension and algebraic reduction)**.** ***[VERIFIED]*** *For the global holomorphic family of theorem 10.11,*

$$\mathcal{M}(X) = f^*\mathcal{M}(\mathbb{P}^1) = f^*\mathbb{C}(t), \qquad a(X) = 1, \tag{3.9}$$

and f is the algebraic reduction.

Proof. By theorem 54.2, all but countably many smooth fibres have algebraic dimension zero. Let $g \in \mathcal{M}(X)$. Only finitely many fibres are components of its polar divisor, so on uncountably many of the preceding fibres the restriction of g is a meromorphic function on a torus of algebraic dimension zero and is therefore constant. By theorem 54.3, $g = f^*\varphi$ for a unique $\varphi \in \mathcal{M}(\mathbb{P}^1)$. The reverse inclusion is immediate, and (3.9) follows. Since f is proper, surjective, has connected fibres, and induces the equality of meromorphic function fields, it is the algebraic reduction. □

Disposition. The earlier label "algebraic reduction open" was too broad. The direct proof above validates the assertion independently of Section 7 and of the CDP comparison. Its provenance is the candidate's Lemmas 9.2–9.6 and Proposition 9.7, reconstructed here with an explicit Appell–Humbert matrix bridge, a proof that the smooth locus remains connected, and the omitted root estimate in the final meromorphic-extension step. This does not by itself prove the claimed topology of X; that is a separate calculation.

55 Structure-sheaf direct images and uniform non-torsion

This section reconstructs the analytic inputs used by the candidate's Proposition 10.8. The proof differs at one important point from the candidate's route: extension across the cusp is obtained from the intrinsic nearby-cycle isomorphism of theorem 53.1, not from the cellular collapse. The two cusp maps are now independently proved equal in theorem 58.7; the argument below nevertheless retains its logically independent intrinsic formulation.

Put

$$e = d\zeta_1 \wedge d\zeta_2$$

on the marked torus family. Retain the local coordinates and notation of section 10:

$$s_j \circ g_j = \zeta_j s_j, \qquad (m_1, m_2) = (3, 4), \qquad t_j = s_j^{m_j}, \qquad f^{-1}(p_j) = m_j S_j.$$

55.1 The Hodge line and the finite normal characters

Lemma 55.1 (determinant characters and the Hodge line)**.** ***[VERIFIED]*** *Let*

$$\mathfrak{K} = f_*\omega_{X/B}.$$

Then $\mathfrak{K}$ is invertible and

$$\mathfrak{K} \simeq \mathcal{O}_B(1). \tag{9.1}$$

More precisely,

$$\det R_{g_1} = -\frac{1}{\tau}, \qquad \det R_{g_2} = \frac{1}{\tau}, \qquad \det R_{g_0} = 1, \tag{9.2}$$

and at the fixed points

$$\det R_{g_j}(z_j) = \zeta_j^{a_j}, \qquad (a_1, a_2) = (2, 1). \tag{9.3}$$

The form e generates $\mathfrak{K}$ at the cusp, while a unit times $s_j^{-a_j} e$ generates it at p_j.

Proof. Equation (2.3d) in theorem 10.2 gives

$$R_g(z)\Pi(z)\lambda = \Pi(gz)A_g\lambda.$$

Using

$$A_1\widehat{w} = \widehat{u} - \widehat{w}, \quad A_1\widehat{\delta} = \widehat{\delta}, \qquad A_2\widehat{w} = -\widehat{u} + \widehat{\delta}, \quad A_2\widehat{\delta} = \widehat{\delta},$$

and writing $\tau^\sharp = \tau(g_j z)$, $\mu^\sharp = \mu(g_j z)$, direct multiplication gives

$$R_{g_1} = \begin{pmatrix} \tau^\sharp - 1 & 0 \\ \mu^\sharp & 1 \end{pmatrix}, \qquad R_{g_2} = \begin{pmatrix} -\tau^\sharp & 0 \\ 1 - \mu^\sharp & 1 \end{pmatrix}.$$

The transformation laws $\tau \circ g_1 = (\tau - 1)/\tau$ and $\tau \circ g_2 = -1/\tau$ prove (9.2) at g_1, g_2; the cusp formulas give $R_{g_0} = I$. Since $\widehat{\delta}$ is fixed, one eigenvalue is 1. At $\tau(z_1) = \rho = e^{\pi i/3}$, the other is $\rho - 1 = e^{2\pi i/3} = \zeta_1^2$; at $\tau(z_2) = i$, it is $-i = \zeta_2$. This proves (9.3).

We determine the extension over a finite filling without suppressing the ramification. On J_j', the nowhere-zero form $ds_j \wedge e$ transforms by

$$(\widetilde{g}_j^{\log})^*(ds_j \wedge e) = \chi_j(s_j)\, ds_j \wedge e, \qquad \chi_j = \zeta_j \det R_{g_j}. \tag{9.4}$$

Iteration m_j times shows $\prod_{r=0}^{m_j-1} \chi_j(\zeta_j^r s) = 1$. Write $\chi_j(s) = \chi_j(0)e^{\psi(s)}$, with $\psi(0) = 0$. The product identity says that the power series of ψ has no terms whose degree is divisible by m_j. Hence

$$\varphi(s) - \varphi(\zeta_j s) = \psi(s)$$

has a holomorphic solution, obtained coefficientwise by division by $1 - \zeta_j^n$. Therefore

$$s^{k_j} e^{\varphi(s)} ds \wedge e, \qquad k_j = m_j - 1 - a_j, \tag{9.5}$$

is invariant: $\chi_j(0) = \zeta_j^{1+a_j}$, so invariance is equivalent to $k_j + 1 + a_j \equiv 0 \pmod{m_j}$. We now distinguish the descended canonical section from the generator of its direct image. Let

$$q_j : J_j' \longrightarrow N_j, \qquad \widetilde{f}_j : J_j' \longrightarrow \Delta_j, \qquad \varpi_j : \Delta_j \longrightarrow D_j$$

denote the finite quotient, the torus-family projection, and the cyclic base quotient, respectively, so that

$$f_j \circ q_j = \varpi_j \circ \widetilde{f}_j, \qquad t_j = s_j^{m_j}.$$

Put $\mathcal{A}_j = \widetilde{f}_j^{-1}(0) = \{s_j = 0\}$ and $G_j = \langle \widetilde{g}_j^{\log} \rangle$. The action is free by theorem 10.9; hence q_j is finite étale and $q_j^{-1}(S_j) = \mathcal{A}_j$.

The invariant form in (9.5), now denoted

$$\sigma_j = s_j^{k_j} e^{\varphi(s_j)} ds_j \wedge e, \qquad k_j = m_j - 1 - a_j, \tag{9.5a}$$

descends uniquely to $\overline{\sigma}_j \in H^0(N_j, K_{N_j})$ with $q_j^* \overline{\sigma}_j = \sigma_j$. Since $e^{\varphi(s_j)}$ is a unit and $ds_j \wedge e$ is nowhere vanishing,

$$\operatorname{div}_{J_j'}(\sigma_j) = k_j \mathcal{A}_j.$$

Because q_j is étale,

$$q_j^* \operatorname{div}_{N_j}(\overline{\sigma}_j) = \operatorname{div}_{J_j'}(q_j^* \overline{\sigma}_j) = k_j \mathcal{A}_j = q_j^*(k_j S_j).$$

Faithfulness of finite surjective pullback on Cartier divisors therefore gives

$$\operatorname{div}_{N_j}(\overline{\sigma}_j) = k_j S_j. \tag{9.5b}$$

Thus (9.5a) is a canonical section with that divisor, not a nowhere-vanishing frame when $k_j > 0$; explicitly, $k_1 = 0$ and $k_2 = 2$.

For the direct-image generator, let

$$\vartheta_j = (ds_j \wedge e) \otimes (dt_j)^{-1} \tag{9.5c}$$

be the nowhere-vanishing frame of

$$q_j^* \omega_{N_j/D_j} = K_{J'_j} \otimes (f_j \circ q_j)^* K_{D_j}^{-1}.$$

Here $(dt_j)^{-1}$ is the pulled-back frame of the inverse canonical line of D_j, not division by the differential of $s_j^{m_j}$. Put $\vartheta'_j = e^{\varphi(s_j)} \vartheta_j$. The defining equation for φ, together with $\chi_j(s) = \chi_j(0) e^{\psi(s)}$ and $\chi_j(0) = \zeta_j^{1+a_j}$, gives

$$(\widetilde{g}_j^{\log})^* \vartheta'_j = \zeta_j^{1+a_j} \vartheta'_j. \tag{9.5d}$$

Every holomorphic coefficient of a relative top form is pulled back from Δ_j, because the fibres of $\widetilde{f}_j$ are connected compact complex tori. Hence

$$\widetilde{f}_{j*} q_j^* \omega_{N_j/D_j} = \mathcal{O}_{\Delta_j} \vartheta'_j.$$

For $F(s_j) \in \mathcal{O}_{\Delta_j}$, the section $F(s_j)\vartheta'_j$ is G_j-invariant exactly when

$$F(\zeta_j s_j) \zeta_j^{1+a_j} = F(s_j). \tag{9.5e}$$

Writing $F(s_j) = \sum_{n \geq 0} c_n s_j^n$, this says that $c_n = 0$ unless

$$n + 1 + a_j \equiv 0 \pmod{m_j}, \qquad \text{equivalently} \qquad n \equiv k_j \pmod{m_j}.$$

Consequently,

$$\Big((\varpi_j)_* \widetilde{f}_{j*} q_j^* \omega_{N_j/D_j} \Big)^{G_j} = \mathcal{O}_{D_j} \cdot s_j^{k_j} \vartheta'_j,$$

and finite descent gives the precise invariant-module identity

$$f_{j*} \omega_{N_j/D_j} = \mathcal{O}_{D_j} \cdot s_j^{k_j} \vartheta'_j. \tag{9.5f}$$

Finally, over Δ_j^*, the differential identity

$$dt_j = m_j s_j^{m_j - 1} ds_j$$

identifies this generator with

$$s_j^{k_j} \vartheta'_j = \frac{e^{\varphi(s_j)}}{m_j} s_j^{k_j - (m_j - 1)} e = \frac{e^{\varphi(s_j)}}{m_j} s_j^{-a_j} e. \tag{9.5g}$$

Thus a unit times $s_j^{-a_j} e$ generates $f_{j*} \omega_{N_j/D_j} = (f_* \omega_{X/B})|_{D_j}$, whereas the canonical section (9.5a) itself has divisor $k_j S_j$.

At the cusp, in every smooth toric chart the unimodular monomial change of coordinates turns

$$\Omega_0 = \frac{dx_1}{x_1} \wedge \frac{dx_2}{x_2} \wedge dt_c$$

into a unit times the ordinary coordinate volume form. Thus it generates the canonical line. It is invariant under translations and under the unimodular shears defining the quotient. Under the punctured-overlap map, $\Omega_0 = (2\pi i)^2 dt_c \wedge e$, so e generates $\mathfrak{K}$ at p_0.

It remains to compute the degree. The modular-form calculation already used in the proof of theorem 10.2 gives

$$\Theta = \left(\frac{E_4^2 E_6^{1/2}}{\Delta_{\mathrm{mod}}}\right) \circ \tau.$$

The square root exists because the zeros of $E_6 \circ \tau$ have even order. The weights $8, 3, 12$ give the same automorphy factor as e; the sign at z_2 is fixed by its linearizing coordinate. The orders are

$$\mathrm{ord}_{z_1} \Theta = 2 = a_1, \qquad \mathrm{ord}_{z_2} \Theta = 1 = a_2, \qquad \Theta = t_c^{-1} \cdot (\text{unit}) \tag{9.6}$$

at the cusp. Consequently $\mathsf{e} = \Theta^{-1} e$ is single-valued. At p_j it is a unit times $s_j^{-a_j} e$; at p_0 it is a unit times $t_c e$. Hence it is a global section of $\mathfrak{K}$ with the single simple zero p_0. This proves $\mathfrak{K} \simeq \mathcal{O}_B(p_0) \simeq \mathcal{O}_B(1)$. Finally, $\mathfrak{K}$ is torsion free of generic rank one on the smooth curve B, hence is invertible. □

Lemma 55.2 (exact order of the finite-fibre normal line)**.** ***[VERIFIED]*** *For $j = 1, 2$,*

$$\mathcal{O}_X(S_j)|_{S_j} = N_{S_j/X}$$

has exact order m_j in $\mathrm{Pic}(S_j)$.

Proof. On the finite cover of theorem 10.9, the central torus $A_0^{(j)} = \{s_j = 0\}$ maps to S_j as a connected principal $G_j \simeq \mathbb{Z}/m_j$-cover. Its normal line is trivial, and a generator of G_j acts on the fibre by $s_j \mapsto \zeta_j s_j$. Thus $N_{S_j/X}$ is associated to this principal cover by the character ζ_j, and its m_j-th power is trivial.

Suppose its k-th power were trivial for some $0 < k < m_j$. Pulling a nowhere-zero trivializing section back to $A_0^{(j)}$ would give a nowhere-zero holomorphic function h satisfying

$$h(gx) = \zeta_j^k h(x).$$

Every holomorphic function on the compact connected torus $A_0^{(j)}$ is constant. The displayed equation would force $\zeta_j^k = 1$, contrary to $0 < k < m_j$. Hence the order is exactly m_j. □

55.2 Exact product covers, normalization branches, and peripheral kernels of the finite fillings

We retain the actual period functions and finite fillings of theorems 5.1, 10.2 and 10.9. The following explicit product maps describe the original central tori and surfaces, not replacements of their periods. We also identify the complete varying family as the normalization of the actual ramified base change, compute the entire peripheral inclusion kernel, and give a smooth trivialization of the finite normal line. These calculations use no sphere identification, global integral cohomology, or CDP theorem. The separate period and cusp constructions retain their own proofs and are not inferred from these finite-local calculations.

55.2.1 The original lattice, periods, and their complex summands

Use the ordered marked lattice and its dual basis

$$\Lambda = \mathbb{Z}\langle \widehat{\gamma}, \widehat{u}, \widehat{w}, \widehat{\delta} \rangle, \qquad \Lambda^* = \mathbb{Z}\langle \gamma, u, w, \delta \rangle. \tag{FF1}$$

In these original column coordinates the matrices and translations are

$$A_1 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 6 & 0 & 1 & 0 \\ -6 & -1 & -1 & 0 \\ -2 & 1 & 0 & 1 \end{pmatrix}, \qquad A_2 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ -6 & 1 & 0 & 0 \\ 3 & 0 & 1 & 1 \end{pmatrix}, \tag{FF2}$$

$$\varepsilon = (1, 2, -4, 0)^{\mathsf{t}}, \quad \varepsilon' = (1, 3, -3, 0)^{\mathsf{t}}, \quad v_1 = \varepsilon, \quad v_2 = -\varepsilon', \quad (m_1, m_2) = (3, 4).$$

They satisfy $A_j^{m_j} = I$, $A_j v_j = v_j$, and $\gamma A_j = \gamma$, with $\gamma(v_1) = 1$ and $\gamma(v_2) = -1$. Retain the entire period matrix

$$\Pi(z) = \begin{pmatrix} 6\mu(z) & \tau(z) & 1 & 0 \\ \beta(z) & \mu(z) & 0 & 1 \end{pmatrix}, \qquad D(z) = \operatorname{Im}\beta(z) - \frac{6(\operatorname{Im}\mu(z))^2}{\operatorname{Im}\tau(z)} < 0. \tag{FF3}$$

At the original elliptic centres let

$$\begin{gathered} \rho = e^{\pi i/3}, \quad \tau(z_1) = \rho, \quad \tau(z_2) = i, \quad \mu_1 := \mu(z_1) = \frac{2-\rho}{3}, \quad \mu_2 := \mu(z_2) = \frac{1-i}{2}, \\ \beta_j = \beta(z_j), \qquad \kappa_1 = \beta_1 + 2\mu_1, \qquad \kappa_2 = \beta_2 + 3\mu_2. \end{gathered} \tag{FF4}$$

Neither β_j nor either of its real components is adjusted. Since $\operatorname{Im}\mu_1 = -\operatorname{Im}\rho/3$ and $\operatorname{Im}\mu_2 = -1/2$, the exact inequalities are

$$\operatorname{Im}\kappa_1 = \operatorname{Im}\beta_1 - \tfrac{2}{3}\operatorname{Im}\rho = D(z_1) < 0, \qquad \operatorname{Im}\kappa_2 = \operatorname{Im}\beta_2 - \tfrac{3}{2} = D(z_2) < 0. \tag{FF5}$$

Thus $(1, \kappa_j)$ is real-linearly independent with its original negative imaginary orientation. No conjugation or positive-imaginary replacement is made.

Put $\delta_* := \widehat{\delta}$ and define

$$\begin{aligned} e_1 &= (0,1,-2,0)^{\mathsf{t}}, \quad f_1 = (0,0,-3,1)^{\mathsf{t}}, \\ e_2 &= (0,1,-1,0)^{\mathsf{t}}, \quad f_2 = (0,0,-2,1)^{\mathsf{t}}. \end{aligned} \tag{FF6}$$

Proposition 55.3 (The full fixed and moving integral lattices)**.** *The exact fixed and image lattices are*

$$\begin{aligned} \Lambda^{A_1} &= \mathbb{Z}\varepsilon \oplus \mathbb{Z}\delta_*, \quad (A_1 - I)\Lambda = \mathbb{Z}e_1 \oplus \mathbb{Z}f_1, \\ \Lambda^{A_2} &= \mathbb{Z}\varepsilon' \oplus \mathbb{Z}\delta_*, \quad (A_2 - I)\Lambda = \mathbb{Z}e_2 \oplus \mathbb{Z}f_2. \end{aligned} \tag{FF7}$$

The image lattices are saturated, with complete annihilators

$$\begin{aligned} \operatorname{Ann}((A_1 - I)\Lambda) &= \mathbb{Z}\gamma \oplus \mathbb{Z}\psi_1, \quad \psi_1 = 2u + w + 3\delta, \\ \operatorname{Ann}((A_2 - I)\Lambda) &= \mathbb{Z}\gamma \oplus \mathbb{Z}\psi_2, \quad \psi_2 = u + w + 2\delta, \\ \det[\varepsilon, \delta_*, e_1, f_1] &= -3, \qquad \det[\varepsilon', \delta_*, e_2, f_2] = -2. \end{aligned} \tag{FF8}$$

In particular the sum of the fixed and moving lattices is not being asserted to be all of Λ*.*

Proof. Solving $(A_1 - I)(a,b,c,d)^{\mathsf{t}} = 0$ gives $b = 2a, c = -4a$; the second equation gives $b = 3a, c = -3a$. In both cases a, d are arbitrary integers. The columns of the two image matrices are, in order,

$$\begin{aligned} A_1 - I: &\quad 6e_1 - 2f_1,\ -e_1 + f_1,\ e_1,\ 0, \\ A_2 - I: &\quad 3f_2,\ -e_2,\ -e_2 + f_2,\ 0. \end{aligned} \tag{FF9}$$

These columns generate precisely the two stated image lattices. The first is exactly the set of integer columns satisfying $a = 0$ and $2b + c + 3d = 0$: such a column is $be_1 + df_1$. The second is exactly the set satisfying $a = 0$ and $b + c + 2d = 0$, with the analogous expression $be_2 + df_2$. They are therefore saturated. Their annihilators are the displayed full integer spans: in the first case an annihilating row has coefficients $(a, 2c, c, 3c)$, and in the second $(a, c, c, 2c)$. Expansion of the two four-by-four determinants gives -3 and -2. □

The complex images under the unchanged periods are

$$\begin{aligned} &\Pi(z_1)\varepsilon = (0, \kappa_1), && \Pi(z_2)\varepsilon' = (0, \kappa_2), \quad \Pi(z_j)\delta_* = (0,1), \\ &\Pi(z_1)e_1 = (\rho - 2, -(\rho-2)/3), && \Pi(z_1)f_1 = (-3, 1), \\ &\Pi(z_2)e_2 = (i - 1, -(i-1)/2), && \Pi(z_2)f_2 = (-2, 1). \end{aligned} \tag{FF10}$$

For example the first fixed coordinate vanishes because $6(2-\rho)/3 + 2\rho - 4 = 0$ or $6(1-i)/2 + 3i - 3 = 0$, respectively. The fixed real plane consequently maps onto the complex line $\{(0, \xi) : \xi \in \mathbb{C}\}$. The moving real planes map onto $\{(\eta, -\eta/3)\}$ and $\{(\eta, -\eta/2)\}$, respectively, because their displayed period pairs are real-linearly independent. These are complementary complex lines in the original $\mathbb{C}^2$.

55.2.2 The exact elliptic-product covers of the central tori and surfaces

Define the one-dimensional complex tori

$$\begin{aligned} E_j &= \mathbb{C}/(\mathbb{Z}+\kappa_j\mathbb{Z}),\\ F_1 = \mathbb{C}/(\mathbb{Z}(\rho-2)+\mathbb{Z}(-3)), &\qquad F_2 = \mathbb{C}/(\mathbb{Z}(i-1)+\mathbb{Z}(-2)). \end{aligned} \tag{FF11}$$

Each period pair is real-linearly independent. Translation by its rank-two lattice is free and properly discontinuous; the image of a closed real parallelotope covers the quotient, proving compactness. Thus these are smooth elliptic curves with the class of zero distinguished.

Theorem 55.4 (Product maps onto the original finite central surfaces)**.** *The complex-linear maps*

$$L_1(\xi,\eta) = (\eta, \xi-\eta/3), \qquad L_2(\xi,\eta) = (\eta, \xi-\eta/2) \tag{FF12}$$

induce surjective local biholomorphic group homomorphisms

$$\ell_j : E_j \times F_j \longrightarrow A_0^{(j)} := \mathbb{C}^2/\Pi(z_j)\Lambda, \quad \ker \ell_1 = \langle k_1\rangle, \quad \ker \ell_2 = \langle k_2\rangle, \tag{FF13}$$

where

$$k_1 = (2/3, \rho), \quad \operatorname{ord}(k_1) = 3, \qquad k_2 = (1/2, i), \quad \operatorname{ord}(k_2) = 2. \tag{FF14}$$

The following automorphisms lift the original affine generators:

$$g_1^{\mathrm{prod}}(\xi,\eta) = (\xi+\kappa_1/3, (\rho-1)\eta), \qquad g_2^{\mathrm{prod}}(\xi,\eta) = (\xi-\kappa_2/4, -i\eta). \tag{FF15}$$

Writing k_j also for translation by that point, their joint groups are $\langle k_1, g_1^{\mathrm{prod}}\rangle \cong C_3 \times C_3$ and $\langle k_2, g_2^{\mathrm{prod}}\rangle \cong C_2 \times C_4$. They act freely, and the original reduced central surfaces are exactly

$$S_1 = (E_1 \times F_1)/\langle k_1, g_1^{\mathrm{prod}}\rangle, \qquad S_2 = (E_2 \times F_2)/\langle k_2, g_2^{\mathrm{prod}}\rangle. \tag{FF16}$$

The finite étale product-cover degrees are 9 *and* 8*.*

Proof. Each L_j has determinant -1 in the displayed product and original (ζ_1,ζ_2) coordinates. Equation (FF10) shows that the product period lattice maps exactly to the sum of the fixed and moving sublattices inside $\Pi(z_j)\Lambda$. Thus L_j induces the stated surjective local biholomorphism, and its kernel is identified with $\Lambda/(\Lambda^{A_j} + (A_j - I)\Lambda)$, of order 3 or 2 by (FF8). Explicitly,

$$\widehat{u} = \tfrac{2}{3}\delta_* + e_1 - \tfrac{2}{3}f_1, \qquad \widehat{u} = \tfrac{1}{2}\delta_* + e_2 - \tfrac{1}{2}f_2. \tag{FF17}$$

Consequently $L_j(k_j) = \Pi(z_j)\widehat{u}$. The first kernel point has order exactly three, since its fixed coordinate does, and $3\rho = 3(\rho-2) - 2(-3)$ is a moving period. The second has order exactly two, since its fixed coordinate does, and $2i = 2(i-1) - (-2)$ is a moving period. These orders equal the full kernel cardinalities, proving the complete kernel assertions.

At the centres the original covariance matrices are

$$R_1 = \begin{pmatrix} \rho-1 & 0 \\ (2-\rho)/3 & 1 \end{pmatrix}, \qquad R_2 = \begin{pmatrix} -i & 0 \\ (1+i)/2 & 1 \end{pmatrix}. \tag{FF18}$$

They act by 1 on the fixed lines and by $\rho - 1 = e^{2\pi i/3}$ or $-i$ on the corresponding moving line. Substitution gives the full equality of affine maps on $\mathbb{C}^2$,

$$L_j g_j^{\mathrm{prod}} = R_j L_j + \Pi(z_j) v_j/m_j. \tag{FF19}$$

In particular the negative translation $-\kappa_2/4$ is forced by $v_2 = -\varepsilon'$. It has not been replaced by its opposite.

The moving rotations preserve their exact period lattices. For example, using $\rho^2 = \rho - 1$, their images of the two period generators are

$$\begin{aligned} (\rho-1)(\rho-2) &= -2(\rho-2) + (-3), \quad (\rho-1)(-3) = -3(\rho-2) + (-3), \\ (-i)(i-1) &= (i-1) - (-2), \qquad (-i)(-2) = 2(i-1) - (-2). \end{aligned} \tag{FF20}$$

The first matrix in these period bases has columns $(-2,1)$ and $(-3,1)$, determinant 1; the second has columns $(1,-1)$ and $(2,-1)$, determinant 1. Equivalently, invariance and invertibility follow by applying the finite orders of the rotations to these integral period images. The kernel classes are fixed by the moving rotations because

$$(\rho-1)\rho - \rho = -1 - \rho = -(\rho-2) + (-3), \qquad (-i)i - i = 1 - i = -(i-1). \tag{FF21}$$

Hence the product automorphism commutes with translation by k_j on the product torus. It has exact order m_j: its moving rotation has that order; its m_j-th fixed-coordinate translation is κ_1 or $-\kappa_2$; and no smaller positive multiple of that translation is a fixed-coordinate period, since κ_j is nonreal.

To prove both the direct-product assertion and freeness of the entire joint group, put $(d_1, d_2) = (3, 2)$, $(c_1, c_2) = (2/3, 1/2)$ and $(\epsilon_1, \epsilon_2) = (1, -1)$. A fixed point for $k_j^a (g_j^{\mathrm{prod}})^b$, with $0 \le a < d_j$ and $0 \le b < m_j$, would require

$$ac_j + \epsilon_j b\kappa_j/m_j = n + q\kappa_j, \qquad n, q \in \mathbb{Z}. \tag{FF22}$$

Imaginary parts give $\epsilon_j b/m_j = q$, hence $b = 0$. The real equality then gives $ac_j \in \mathbb{Z}$, hence $a = 0$ in its stated range. Every nonidentity pair is therefore fixed-point free; in particular it is not the identity transformation. This proves the claimed joint groups. Quotienting first by k_j and then by the lifted affine generator gives exactly the original S_j of theorem 10.9, by (FF19). The total covering degrees are $d_j m_j = 9, 8$ and the actions are free, so the covers are étale. □

55.2.3 The varying quotient and its full local analytic coordinates

Retain the original linearizing coordinate s_j on Δ_j, with its actual round-disc radius $r_j > 0$, and the original phase

$$s_j(g_j z) = \zeta_j s_j(z), \quad \zeta_j = e^{-2\pi i/m_j}, \quad t_j = s_j^{m_j}, \qquad |s_j| < r_j, \quad |t_j| < r_j^{m_j}. \tag{FF23}$$

The radius is not replaced by one. On the universal marked cover the holomorphic affine map and the exact period covariance are

$$\begin{aligned} \widehat{G}_j(z,\zeta) &= (g_j z, R_{g_j}(z)\zeta + \Pi(g_j z)v_j/m_j), \\ R_{g_j}(z)\Pi(z) &= \Pi(g_j z)A_j, \qquad t_\lambda(z,\zeta) = (z, \zeta + \Pi(z)\lambda). \end{aligned} \tag{FF24}$$

Let G_j denote the induced map of the original varying torus family $J_j' = (\Delta_j \times \mathbb{C}^2)/\Lambda$ and put $N_j = J_j'/\langle G_j \rangle$, with quotient map q_j. The real-analytic marking $\Xi(z,x) = (z, \Pi(z)x)$ converts its powers to the exact formulas

$$\Xi^{-1}\widehat{G}_j^k \Xi(z,x) = (g_j^k z, A_j^k x + kv_j/m_j), \qquad \widehat{G}_j^{m_j} = t_{v_j}. \tag{FF25}$$

Proposition 55.5 (Smooth quotient charts and the exact multiplicities)**.** *The map $q_j : J_j' \to N_j$ is a finite étale covering of degree m_j. The quotient is a Hausdorff smooth complex threefold, and its proper map $f_j : N_j \to D_j$ has exact scheme fibre $m_j S_j$ at the centre. In its local quotient coordinates the transverse map is literally $t_j = s_j^{m_j}$.*

Proof. At $s_j = 0$ the freeness is proved in theorem 55.4. Away from zero, a power with $0 < k < m_j$ has base coordinate multiplied by $\zeta_j^k \neq 1$, so has no fixed point. Equation (FF25) shows that the m_j-th power is the original lattice translation, hence is the identity on J_j'. This gives a free action of exact order m_j.

For $y \in J_j'$, its finitely many distinct orbit points can be separated by pairwise disjoint coordinate neighbourhoods. Intersecting their inverse translates and shrinking gives a coordinate neighbourhood U of y with the $G_j^a U$ pairwise disjoint. Then $q_j|_U : U \to q_j(U)$ is a biholomorphism. Two distinct finite orbits have disjoint saturated neighbourhoods by the same finite separation argument. These charts prove the Hausdorff and smooth assertions, and the quotient is a finite étale covering. They give directly the trivial-stabilizer case of the quotient analytic structure in [39, conditions (I)–(II), Theorem 1]; smoothness is not inferred from a theorem about arbitrary finite quotients.

At a point of S_j, use the quotient chart to retain s_j as a transverse coordinate, together with two fibre coordinates. The full map is $t_j = s_j^{m_j}$. Thus S_j is a smooth reduced Cartier divisor, and its scheme multiplicity is exactly m_j.

For compact $C \subset \Delta_j$, the map induced by Ξ from $C \times [0,1]^4$ onto the inverse image of C in J_j' is continuous and surjective, so that inverse image is compact. The actual power map between the discs of radii r_j and $r_j^{m_j}$ is proper: the inverse image of a compact target set is closed, bounded, and stays strictly inside the radius-r_j disc. For compact $K \subset D_j$, the inverse image in J_j' of all roots of its t_j-values is therefore compact. Its finite quotient image is exactly $f_j^{-1}(K)$, proving properness of the full restriction. □

55.2.4 Every branch in the normalization of the actual base change

Let Δ_j^{base} be the same radius-r_j disc with coordinate u, mapped to D_j by $t_j = u^{m_j}$, and form

$$P_j = N_j \times_{D_j} \Delta_j^{\mathrm{base}}, \qquad \nu_j : J_j' \longrightarrow P_j, \quad \nu_j(y) = (q_j(y), s_j(y)). \tag{FF26}$$

Theorem 55.6 (The original varying family is the normalization)**.** *The map ν_j is exactly the normalization of P_j. It is a biholomorphism away from $u = 0$; at the centre it has all m_j local branches prescribed by the original cyclic action. Its central fibre is the connected original torus $A_0^{(j)}$, and*

$$\nu_j(G_j y) = (q_j(y), \zeta_j s_j(y)). \tag{FF27}$$

Thus the original negative-phase action lifts $u \mapsto \zeta_j u$, permutes the local branches cyclically, and has quotient N_j.

Proof. The definition uses no logarithm choice. If $u \neq 0$, the s_j-values in a covering orbit are exactly the m_j distinct roots of its t_j-value, so there is exactly one representative with value u. If $u = 0$, all m_j covering points have $s_j = 0$. This proves surjectivity and describes all fibres. A compact subset of P_j projects to a compact subset of N_j, whose inverse image under q_j is compact. Its inverse image under ν_j is a closed subset of that compact set, hence is compact. Thus ν_j is proper with finite fibres. Away from the centre the unique orbit representative and quotient charts give the holomorphic inverse.

More precisely, around $n \in S_j$ choose a chart $q_j(U)$ as in theorem 55.5, with inherited transverse coordinate s and fibre coordinates z_1, z_2. The fibre product is locally

$$s^{m_j} - u^{m_j} = \prod_{a=0}^{m_j-1} (s - e^{2\pi i a/m_j} u) = 0. \tag{FF28}$$

The factors are distinct smooth prime hypersurface factors in the regular convergent-power-series ring $\mathbb{C}\{z_1, z_2, s, u\}$. Their product ideal is radical: in this factorial ring, divisibility of a power by every distinct prime factor implies divisibility of the element by their product. Thus the fibre-product germ is reduced. Its total ring of fractions is the product of the branch fraction fields. Its integral closure is the product of the regular branch rings $\mathbb{C}\{z_1, z_2, u\}$: adjoining the branch idempotents is integral, since each satisfies $e^2 - e = 0$, and each component is integrally closed. This proves the complete local normalization, including all central branches.

On $G_j^a U$ the second component of ν_j is $u = \zeta_j^a s$, or equivalently

$$s = \zeta_j^{-a} u = e^{2\pi i a/m_j} u. \tag{FF29}$$

Each of the m_j disjoint covering charts therefore maps isomorphically onto exactly one branch of (FF28); these exhaust the branches. The local identifications glue by the specified map ν_j, giving the global normalization. The source is the original varying family, not an isotrivial product. Restricting its coordinate to zero gives its connected torus $A_0^{(j)}$. Equation (FF27) follows directly from $s_j(G_j y) = \zeta_j s_j(y)$ and $q_j G_j = q_j$. □

55.2.5 Universal covers and the full peripheral inclusion kernel

Theorem 55.7 (Exact finite-filling deck groups and attachment kernel)**.** *The universal cover $\Delta_j \times \mathbb{C}^2 \to N_j$ has the following complete deck-group presentation:*

$$\Gamma_j = \left\langle \Lambda, \widehat{G}_j \;\middle|\; \begin{array}{l} \Lambda \text{ abelian}, \\ \widehat{G}_j t_\lambda \widehat{G}_j^{-1} = t_{A_j\lambda}, \\ \widehat{G}_j^{m_j} = t_{v_j} \end{array} \right\rangle. \tag{FF30}$$

This is also the universal-cover group of S_j. On the punctured filling the complete group is $\Lambda \rtimes_{A_j} \mathbb{Z}$, with the generator specified below, and the entire inclusion kernel is

$$\ker\{\pi_1(N_j \setminus S_j) \to \pi_1(N_j)\} = \langle \widetilde{G}_j^{m_j} t_{-v_j} \rangle \cong \mathbb{Z}. \tag{FF31}$$

It is a central subgroup, not merely a necessary power relation.

Proof. In the real marking the original universal-cover transformations are

$$t_\lambda(s,x) = (s, x+\lambda), \qquad \widehat{G}_j(s,x) = (\zeta_j s, A_j x + v_j/m_j). \tag{FF32}$$

They satisfy the two relations in (FF30) by $A_j v_j = v_j$. Moving translations to the left and using the power relation expresses every word as $t_\lambda \widehat{G}_j^k$ with $\lambda \in \Lambda$ and $0 \le k < m_j$. These forms are unique: the base action determines k, and the remaining translation determines λ. Therefore the abstract presented group acts faithfully and no additional relation is implicit.

For $0 < k < m_j$, a fixed-point equation would imply, on applying the original integral covector γ to the real fibre coordinate,

$$\gamma(\lambda) + k\gamma(v_j)/m_j = 0. \tag{FF33}$$

This is impossible because $\gamma(\lambda)$ is integral and $\gamma(v_j) = 1$ or -1. A nonzero translation with $k = 0$ has no fixed point either. The lattice action is properly discontinuous: on compact base sets the real inverses $\Pi(z)^{-1}$ are uniformly bounded, leaving finitely many lattice translations that can meet two fixed compact sets. The full group is a finite union of lattice cosets, so the same property holds for it. Thus the simply connected $\Delta_j \times \mathbb{C}^2$ is the universal cover. Its restriction to $s = 0$ is the simply connected $\mathbb{C}^2$ covering S_j with the same action and group.

On the punctured disc retain exactly

$$w = \frac{\log s_j}{2\pi i}, \qquad s_j = e^{2\pi i w}, \qquad \operatorname{Im} w > -\frac{\log r_j}{2\pi}, \qquad \widetilde{G}_j(w,x) = (w - 1/m_j, A_j x + v_j/m_j). \tag{FF34}$$

The displayed half-plane times $\mathbb{C}^2$, with its pulled-back holomorphic coordinates, is simply connected. Along with the lattice translations, $\widetilde{G}_j$ generates its entire punctured deck group: the ordinary logarithm deck map is already

$$Q(w,x) = (w+1, x) = t_{v_j} \widetilde{G}_j^{-m_j}(w,x). \tag{FF35}$$

There is no finite-power relation for $\widetilde{G}_j$, since its k-th power has w-displacement $-k/m_j$. Thus the exact normal forms are $t_\lambda \widetilde{G}_j^k$ for unrestricted $k \in \mathbb{Z}$, and the group is precisely the stated semidirect product with action A_j.

The inclusion sends t_λ to t_λ and $\widetilde{G}_j$ to $\widehat{G}_j$. Write $k = qm_j + r$ with $0 \le r < m_j$. Its image has normal form $t_{\lambda+qv_j}\widehat{G}_j^r$, which is the identity exactly when $r = 0$ and $\lambda = -qv_j$. This proves (FF31) in both directions. Its generator has w-displacement -1, so has infinite order. It commutes with all translations because $A_j^{m_j} = I$, and with $\widetilde{G}_j$ because $A_j v_j = v_j$, proving centrality. □

55.2.6 The original punctured overlap and integral cohomology pullback

On the affine punctured model choose a local logarithm and put

$$\alpha(z) = \frac{\log s_j(z)}{2\pi i}, \qquad h_j[z,\zeta] = [z, \zeta + \alpha(z)\Pi(z)v_j]. \tag{FF36}$$

A branch change adds $n\Pi(z)v_j$, an original lattice period, so the map is well-defined and holomorphic on the punctured torus quotient; subtraction of the same term gives its inverse. Under the clockwise step $s_j \mapsto \zeta_j s_j$, the compatible branch is $\alpha(g_j z) = \alpha(z) - 1/m_j$. The covariance (FF24) and $A_j v_j = v_j$ therefore give the exact conjugacy to the original linear lift $\widehat{G}_j^0(z,\zeta) = (g_j z, R_{g_j}(z)\zeta)$:

$$h_j \widehat{G}_j h_j^{-1} = \widehat{G}_j^0. \tag{FF37}$$

Indeed the two translations in $h_j\widehat{G}_j$ add to $\Pi(g_j z)v_j(1/m_j + \alpha(g_j z)) = \alpha(z)\Pi(g_j z)v_j$, exactly the translation in $\widehat{G}_j^0 h_j$.

The corresponding real translation retains the full complex structure:

$$\begin{aligned} J_z = \Pi(z)^{-1} i \Pi(z), \qquad & a_j(z) = \operatorname{Re}\alpha(z) v_j + \operatorname{Im}\alpha(z) J_z v_j, \\ J_{g_j z} A_j = A_j J_z, \qquad & a_j(g_j z) = A_j a_j(z) - v_j/m_j. \end{aligned} \tag{FF38}$$

This follows by applying $\Pi(z)^{-1}$ to $\alpha(z)\Pi(z)v_j$ and then using covariance; complex scalar multiplication is not being imposed on the marked real lattice. The inverse image of the middle zero section is consequently $-a_j(z)$, whose endpoint after one clockwise step is exactly $A_j(-a_j(z)) + v_j/m_j$. Iterating m_j such steps increases this marked endpoint by precisely v_j, not by an unspecified translated generator. Thus the overlap map realizes the based peripheral convention in (FF31).

The map from the fibre over z to its deck-translated fibre over $g_j z$ has marked linear part A_j; its inverse has part A_j^{-1}. The original contragredient convention is

$$(A_j^{-1})^{\mathsf{t}} = T_j, \qquad A_j = (T_j^{-1})^{\mathsf{t}}. \tag{FF39}$$

These directions keep a based loop distinct from its inverse deck map. No matrix or generator is reversed to obtain the attachment relation.

Corollary 55.8 (The exact integral first-cohomology covering image)**.** *For the original covering $\pi_j : A_0^{(j)} \to S_j$,*

$$H_1(S_j,\mathbb{Z}) \cong \mathbb{Z}^2, \qquad \pi_j^* H^1(S_j,\mathbb{Z}) = \mathbb{Z}(m_j\gamma) \oplus \mathbb{Z}\psi_j \subset H^1(A_0^{(j)},\mathbb{Z}). \tag{FF40}$$

The index m_j in the invariant covector lattice is retained.

Proof. By theorem 55.3, the map

$$\Lambda \longrightarrow \mathbb{Z}^2, \qquad \lambda \longmapsto (\gamma(\lambda), \psi_j(\lambda)) \tag{FF41}$$

has kernel exactly $(A_j - I)\Lambda$. It is onto: $\widehat{\gamma}$ gives $(1,0)$, and $\widehat{w}$ for $j = 1$, or $\widehat{u}$ for $j = 2$, gives $(0,1)$. Thus the full coinvariant group is $\mathbb{Z}^2$. In these coordinates v_j is $(1,0)$ or $(-1,0)$. Abelianizing (FF30) adjoins one generator with relation $m_j\widehat{G}_j = v_j$, eliminating a primitive generator and leaving $\mathbb{Z}^2$. This computes $H_1(S_j,\mathbb{Z})$ without global topology. A homomorphism from this group to $\mathbb{Z}$ restricts on Λ to $a\gamma + b\psi_j$ precisely when $m_j \mid a$, since it must assign the integer $a\gamma(v_j)/m_j$ to $\widehat{G}_j$. This necessary and sufficient condition proves the exact covering pullback. □

55.2.7 The finite-to-global analytic gluing interface

The global construction of theorem 10.11 glues complete punctured restrictions of the original middle family and the three local fillings. The finite identifications are precisely (FF36). The special discs are pairwise disjoint, and a middle point over each punctured disc has exactly one representative in its filling. Thus an equivalence class meets any given summand at most once. Saturations of open subsets are open, because the overlap identifications are biholomorphisms, so each summand embeds openly and injectively.

Two points at one special base value have representatives in its same filling; two points at one ordinary base value have representatives in the middle piece. They have disjoint neighbourhoods in that Hausdorff open summand, and their images remain disjoint because no equivalence class meets that summand twice. Points at distinct base values are separated by the base. This proves the gluing's Hausdorff assertion using the actual full restrictions, not an unrestricted gluing principle.

In particular $f^{-1}(D_j)$ is the entire corresponding filling: every middle point over D_j^* is identified with its unique filling representative. Thus the properness in theorem 55.5 is properness of the full inverse image. For a compact base set K, finitely many relatively compact V_a whose closures lie inside members U_a of the original four-piece base cover can be chosen to cover K. The compact sets $K \cap \overline{V}_a$ have compact preimages under the corresponding full proper restrictions, and these preimages cover exactly $f^{-1}(K)$. This is the compact-cover argument for global properness. The separate cusp and middle constructions supply their own full proper restrictions; the finite-local proof neither replaces those constructions nor implies smooth sphere recognition.

55.2.8 The exact normal character and a smooth trivialization

Theorem 55.9 (Holomorphic normal order and the explicit smooth product)**.** *Let $\mathcal{N}_j = N_{S_j/N_j}$, equivalently $\mathcal{O}_X(S_j)|_{S_j}$ after the specified open embedding into the original global analytic construction. This line has exact order m_j in the holomorphic Picard group, but has the following explicit smooth trivialization and hence zero integral first Chern class.*

Proof. Differentiating the actual transverse coordinate in (FF23) identifies the normal line as the quotient of the trivial line over $A_0^{(j)}$ by

$$(x, n) \longmapsto (A_j x + v_j/m_j, \zeta_j n), \qquad \zeta_j = e^{-2\pi i/m_j}. \tag{FF42}$$

Here x is the original real marking of the central complex torus; the action on the torus and line is holomorphic. A section of its k-th power therefore pulls back to a function satisfying

$$h(A_j x + v_j/m_j) = \zeta_j^k h(x). \tag{FF43}$$

The positive exponent follows from the quotient itself: the representative $(x, h(x))$ is identified at the image point with $(A_j x + v_j/m_j, \zeta_j^k h(x))$, which must equal the representative of the section there. The negative exponent would instead describe the dual normal line under the declared action.

The m_j-th power has the trivial character, so the constant function one descends to a nowhere-zero holomorphic section of that power. If a power $0 < k < m_j$ were holomorphically trivial, its trivializing section would pull back to a nowhere-zero holomorphic function on the compact connected torus $A_0^{(j)}$. Such a function is constant, and (FF43) would force $\zeta_j^k = 1$, impossible in this range. This proves the exact holomorphic order in theorem 55.2, with the normal tangent character fully specified.

For the smooth trivialization put

$$\epsilon_j = \gamma(v_j) \in \{1, -1\}, \qquad H_j(x) = \exp(-2\pi i \epsilon_j \gamma(x)) \quad (x \in \Lambda \otimes \mathbb{R}). \tag{FF44}$$

Integrality of $\gamma(\lambda)$ gives $H_j(x + \lambda) = H_j(x)$, so this is a well-defined nowhere-zero smooth function on the actual marked central torus. Moreover $\gamma A_j = \gamma$ and $\epsilon_j^2 = 1$ give

$$\begin{aligned} H_j(A_j x + v_j/m_j) &= \exp(-2\pi i \epsilon_j \gamma(x)) \exp(-2\pi i \epsilon_j^2/m_j) \\ &= \zeta_j H_j(x). \end{aligned} \tag{FF45}$$

It consequently descends to a nowhere-zero section of the actual normal line. Explicitly the smooth fibrewise-linear trivialization and its inverse are

$$\begin{aligned} \Theta_j &: \mathcal{N}_j \longrightarrow S_j \times \mathbb{C}, \qquad [x,n] \longmapsto ([x], n/H_j(x)), \\ \Theta_j^{-1} &: S_j \times \mathbb{C} \longrightarrow \mathcal{N}_j, \qquad ([x],z) \longmapsto [x, zH_j(x)]. \end{aligned} \tag{FF46}$$

Replacing x by a lattice translate leaves both formulas unchanged. Replacing (x,n) by $(A_j x + v_j/m_j, \zeta_j n)$ also leaves $n/H_j(x)$ unchanged by (FF45); the inverse has the same equivariance. The two maps are therefore well-defined, smooth, and inverse, retaining the original signs for each j. Thus $c_1(\mathcal{N}_j) = 0$ in integral cohomology while its holomorphic order is exactly m_j. This conclusion uses only the original affine action and integral covector, not the topology of a global X. □

55.3 The cusp structure sheaf

Lemma 55.10 (structure-sheaf cohomology of the cusp fibre)**.** ***[VERIFIED]*** *The reduced normal-crossings fibre W satisfies*

$$h^q(W, \mathcal{O}_W) = (1, 2, 1) \qquad (q = 0, 1, 2). \tag{9.7}$$

The natural map

$$H^1(W; \mathbb{C}) \longrightarrow H^1(W, \mathcal{O}_W) \tag{9.8}$$

is an isomorphism. In degree two the corresponding map is surjective and is injective on the kernel of pullback to the normalization $\nu : \mathrm{dP}_6 \to W$. Finally,

$$\bigwedge^2 H^1(W, \mathcal{O}_W) \longrightarrow H^2(W, \mathcal{O}_W) \tag{9.9}$$

is an isomorphism.

Proof. Let $D_1, D_2, D_3 \simeq \mathbb{P}^1$ be the double curves and P, Q the two triple points, with the branch order fixed in the proof of theorem 53.1. The normalization complex is

$$0 \longrightarrow \mathcal{O}_W \longrightarrow \nu_* \mathcal{O}_{\mathrm{dP}_6} \xrightarrow{\delta_0} \bigoplus_{k=1}^{3} \mathcal{O}_{D_k} \xrightarrow{\delta_1} \mathbb{C}_P \oplus \mathbb{C}_Q \longrightarrow 0, \tag{9.10}$$

where δ_0 is the difference of the two branch restrictions and

$$\delta_1(g_1, g_2, g_3) = (g_1(P) - g_2(P) + g_3(P), g_1(Q) - g_2(Q) + g_3(Q)).$$

For completeness, exactness is stalkwise. At a generic double point it is the fibre-product sequence

$$0 \to \mathbb{C}\{x,y,z\}/(xy) \to \mathbb{C}\{y,z\} \oplus \mathbb{C}\{x,z\} \to \mathbb{C}\{z\} \to 0.$$

At a triple point it is the augmented branch complex for $\mathbb{C}\{x,y,z\}/(xyz)$: three branch rings, three pairwise intersection rings, and the common residue field, with alternating differences. The augmented two-simplex complex is exact; successively lifting compatible power series proves exactness of the analytic rings. Away from the singular locus the first arrow is an isomorphism.

The same formulas give an exact complex with $\mathbb{C}$ in place of $\mathcal{O}$. Since dP_6 is rational and each D_k is a rational curve, the positive structure-sheaf cohomology of every term on the right of (9.10) vanishes. Hypercohomology therefore reduces to

$$\mathbb{C} \xrightarrow{0} \mathbb{C}^3 \xrightarrow{d_1} \mathbb{C}^2, \qquad d_1(a_1, a_2, a_3) = (s, s), \quad s = a_1 - a_2 + a_3. \tag{9.11}$$

Its cohomology has dimensions $1, 2, 1$, proving (9.7).

For the constant-sheaf complex, the row of degree zero is again (9.11), the row of degree one vanishes, and the remaining degree-two row begins with $H^2(\mathrm{dP}_6;\mathbb{C})$. The comparison induced by $\mathbb{C}_W \hookrightarrow \mathcal{O}_W$ is the identity on (9.11) and zero on the positive rows of the target. It follows that (9.8) is an isomorphism, while in degree two it maps the dual-complex summand $\ker \nu^*$ isomorphically onto $H^2(W,\mathcal{O}_W)$.

Choose the basis $\gamma_W, u_W \in H^1(W;\mathbb{Z})$ whose intrinsic specializations are γ, u, respectively; it exists and is unique by theorem 53.1. Their cup product is nonzero, because its specialization is $\gamma \wedge u \neq 0 \in H^2(F;\mathbb{Z})$. Its pullback to dP_6 is zero since $H^1(\mathrm{dP}_6;\mathbb{C}) = 0$. The degree-two comparison therefore sends it to a nonzero element of $H^2(W,\mathcal{O}_W)$. The images of γ_W, u_W form a basis by (9.8), and both sides of (9.9) are one-dimensional. Hence (9.9) is an isomorphism. □

55.4 Direct images, base change, and the Picard line

Theorem 55.11 (structure-sheaf direct images)**.** ***[VERIFIED]*** *For the constructed map $f : X \to B = \mathbb{P}^1$,*

$$f_*\mathcal{O}_X = \mathcal{O}_B, \qquad R^1 f_*\mathcal{O}_X \simeq \mathcal{O}_B \oplus \mathcal{O}_B(-1), \qquad R^2 f_*\mathcal{O}_X \simeq \mathcal{O}_B(-1). \tag{9.12}$$

All three direct images are locally free and every base-change morphism

$$(R^q f_*\mathcal{O}_X) \otimes k(b) \longrightarrow H^q(X_b, \mathcal{O}_{X_b}), \qquad 0 \le q \le 2, \tag{9.13}$$

is an isomorphism, including at the two nonreduced fibres. Moreover the invariant class γ defines the trivial summand in $R^1 f_*\mathcal{O}_X$ and is nowhere zero on B.

Proof. The proper map has connected fibres and B is normal, so Stein factorization gives $f_*\mathcal{O}_X = \mathcal{O}_B$.

The map is flat. A local parameter on B pulls back to a nonzero element of the regular local domain $\mathcal{O}_{X,x}$; if it vanished identically near one point, analytic continuation on the connected manifold would make f constant. Thus $\mathcal{O}_{X,x}$ is torsion-free, hence flat, over the discrete valuation ring $\mathcal{O}_{B,f(x)}$.

Over B°, the fibres are two-dimensional complex tori and their structure-sheaf cohomology dimensions are $1, 2, 1$. At the cusp the same dimensions hold by theorem 55.10. Grauert's theorem therefore gives local freeness of ranks $1, 2, 1$ and base change on a neighbourhood of $B^\circ \cup \{p_0\}$; the precise analytic statement is [7, Chapter III, §4, Theorem 4.12], originally due to Grauert [8].

At p_j, use the finite cover $J_j' \to N_j$. If $\widetilde{f} : J_j' \to \Delta_j$ is the smooth torus family and $\pi(s) = s^{m_j}$, exactness of finite-group invariants in characteristic zero gives

$$R^q f_*\mathcal{O}_{N_j} = \left(\pi_* R^q \widetilde{f}_*\mathcal{O}_{J_j'}\right)^{G_j}. \tag{9.14}$$

After averaging, a free eigenframe v may be chosen for each character ν. The invariant sections are exactly $s^r v\,\mathcal{O}_{D_j}$, where $0 \le r < m_j$ is determined by $\nu\zeta_j^r = 1$. Hence (9.14) is free of ranks $1, 2, 1$. Higher direct images vanish by the fibre-dimension bound, so these are all the nonzero direct images. Since all direct images are locally free, the analytic cohomology-and-base-change theorem, applied by descending induction from $q = 2$, proves (9.13) at the finite fibres as well. In the notation of [7, Chapter III, §3, Theorem 3.4, Corollaries 3.5, 3.7, and 3.10]. Corollary 3.10 says that, for each fixed q, freeness of $R^p f_*\mathcal{O}_X$ for every $p \ge q$ is equivalent to cohomological flatness in all dimensions $p \ge q$. Applying this successively with $q = 2, 1, 0$ gives the displayed base-change maps in every relevant degree. This is a local statement on the base and so applies equally to the nonreduced scheme-theoretic fibres.

Write $G = R^1 f_*\mathcal{O}_X$. We construct its distinguished section without using a cellular collapse. On B°, the monodromy-invariant integral class γ maps to a holomorphic section $\gamma^{0,1}$ of G. It is nowhere zero there: if its image in $H^{0,1}(F)$ vanished, then $\gamma \in H^{1,0}(F) \cap H^1(F;\mathbb{R}) = 0$. On a finite cover it is a constant invariant class and remains nonzero at $s_j = 0$, so it descends across p_j.

At the cusp, let $j : \Delta^* \hookrightarrow \Delta$. The natural specialization morphism

$$R^1 f_*\mathbb{C} \longrightarrow j_* R^1 f_*\mathbb{C}|_{\Delta^*}$$

is an isomorphism: away from 0 this is tautological, while at 0 it is exactly the isomorphism from $H^1(W;\mathbb{C})$ onto the T_0-invariant subspace proved in theorem 53.1. Hence the invariant classes γ, u extend as sections of $R^1f_*\mathbb{C}$ on the whole disc. The natural morphism induced by $\mathbb{C} \hookrightarrow \mathcal{O}$,

$$R^1f_*\mathbb{C} \otimes \mathcal{O}_\Delta \longrightarrow G|_\Delta,$$

sends them at the central point to the two classes in $H^1(W, \mathcal{O}_W)$. These form a basis by theorem 55.10. Thus $\gamma^{0,1}$ extends across the cusp and is nowhere zero everywhere. In particular $\mathcal{O}_B\gamma^{0,1} \subset G$ is a line subbundle.

Relative duality applies to the proper flat Gorenstein map $f\colon X$ is smooth, the base is a smooth curve, and $\omega_{X/B}$ is invertible. Using theorem 55.1, it gives

$$R^2f_*\mathcal{O}_X \simeq \mathfrak{K}^\vee \simeq \mathcal{O}_B(-1). \tag{9.15}$$

The relative theorem is the one of Ramis–Ruget–Verdier [6, pp. 261–283]; the absolute singular dualizing-complex comparison is independently recorded in [5, Theorems 1–2 and Proposition 2].

The relative cup product

$$\bigwedge^2 G \longrightarrow R^2f_*\mathcal{O}_X \tag{9.16}$$

is an isomorphism. It is so on every smooth torus fibre. At the cusp it is so by (9.13) and theorem 55.10. At p_j, choose on the cover an eigenframe v_1, v_2, with $v_1 = \gamma^{0,1}$ invariant. Its invariant exponent is $r_1 = 0$. The action on $v_1 \wedge v_2$ is dual to the Hodge character in (9.3), so its invariant generator is $s_j^{a_j}(v_1 \wedge v_2)$. The invariant generator for the second line is consequently $s_j^{a_j}v_2$, because $0 < a_j < m_j$. Wedge product sends $v_1 \wedge (s_j^{a_j}v_2)$ to exactly the invariant generator of the right side of (9.16), with no wrap modulo m_j.

Consequently $\det G \simeq R^2f_*\mathcal{O}_X \simeq \mathcal{O}_B(-1)$. The quotient of G by its trivial subbundle is $\mathcal{O}_B(-1)$, and the extension splits because

$$\operatorname{Ext}^1(\mathcal{O}_B(-1), \mathcal{O}_B) = H^1(B, \mathcal{O}_B(1)) = 0.$$

This proves (9.12) and identifies the trivial summand. □

Corollary 55.12 (the Picard parameter and fibrewise restriction)**.** ***[VERIFIED]*** *One has*

$$H^1(X, \mathcal{O}_X) = \mathbb{C}, \qquad \operatorname{Pic}(X) = \operatorname{Pic}^0(X) \simeq (\mathbb{C}, +). \tag{9.17}$$

For every scheme-theoretic fibre X_b, restriction induces an injective complex-linear map

$$\iota_b : H^1(X, \mathcal{O}_X) \longrightarrow H^1(X_b, \mathcal{O}_{X_b}). \tag{9.18}$$

On a smooth fibre its image is the complex line generated by the $(0,1)$-image of γ.

Proof. The Leray spectral sequence and $H^1(B, \mathcal{O}_B) = 0$ identify

$$H^1(X, \mathcal{O}_X) = H^0(B, R^1f_*\mathcal{O}_X) = \mathbb{C}$$

by theorem 55.11. Since the verified integral topology gives $H^1(X;\mathbb{Z}) = H^2(X;\mathbb{Z}) = 0$, the analytic exponential sequence

$$0 \to \mathbb{Z} \to \mathcal{O}_X \xrightarrow{u \mapsto e^{2\pi i u}} \mathcal{O}_X^* \to 1$$

identifies $H^1(X, \mathcal{O}_X)$ with $\operatorname{Pic}(X)$ and shows that every line bundle is topologically trivial. Under Leray, restriction is evaluation of a global section of $R^1f_*\mathcal{O}_X$, followed by base change. The only global section is a scalar multiple of the nowhere-zero section $\gamma^{0,1}$, so evaluation is injective at every point and has the stated image on a smooth fibre. □

55.5 One line bundle non-torsion on every fibre

Theorem 55.13 (uniform non-torsion)**.** ***[VERIFIED]*** *There is a line bundle $L \in \operatorname{Pic}(X)$ such that $L|_{X_b}$ is non-torsion for every scheme-theoretic fibre $X_b = f^*(b)$, and $L|_{S_j}$ is non-torsion for $j = 1, 2$. More precisely, identify $\operatorname{Pic}(X)$ with $\mathbb{C}$, $z \mapsto L_z$. Every z outside the union of $\mathbb{R}$ with a countable set has these properties.*

Proof. The pullback $f^*\mathcal{O}_B(1)$ is nontrivial: by the projection formula and $f_*\mathcal{O}_X = \mathcal{O}_B$,

$$h^0(X, f^*\mathcal{O}_B(1)) = h^0(B, \mathcal{O}_B(1)) = 2,$$

whereas $h^0(X, \mathcal{O}_X) = 1$. Rescale the additive coordinate in theorem 55.12 so that

$$L_{12} = f^*\mathcal{O}_B(1). \tag{9.19}$$

Fix a smooth fibre F_q. Put

$$\Lambda_q = \iota_q^{-1}\Big(\operatorname{im}[H^1(F_q; \mathbb{Z}) \to H^1(F_q, \mathcal{O}_{F_q})]\Big) \subset \mathbb{C}. \tag{9.20}$$

The integral image is a lattice in the underlying real vector space of $H^1(F_q, \mathcal{O}_{F_q})$, so Λ_q is discrete. It contains 12, because (9.19) is trivial on every fibre. The exponential sequence on F_q shows

$$L_z|_{F_q} \text{ is torsion} \quad \Longleftrightarrow \quad z \in \mathbb{Q}\Lambda_q. \tag{9.21}$$

If $\operatorname{rk}\Lambda_q = 1$, its nonzero element 12 forces $\Lambda_q \subset \mathbb{R}$; all its bad parameters are real.

Suppose $\operatorname{rk}\Lambda_q = 2$. By theorem 55.12, the image of ι_q is the line generated by the image of γ. Thus there is an integral class

$$\lambda = a\gamma + bu + cw + d\delta \notin \mathbb{Q}\gamma \tag{9.22}$$

whose $(0,1)$-image lies on that line. On the covering family, the four marked lattice columns are

$$\pi_{\widehat{\gamma}} = (6\mu, \beta), \quad \pi_{\widehat{u}} = (\tau, \mu), \quad \pi_{\widehat{w}} = (1, 0), \quad \pi_{\widehat{\delta}} = (0, 1). \tag{9.23}$$

Every real cohomology class has a unique expression

$$2\operatorname{Re}(\bar{\xi}\, d\zeta_1 + \bar{v}\, d\zeta_2).$$

Evaluating it on (9.23) gives

$$\begin{aligned} 2\operatorname{Re}\bar{\xi} &= c, & 2\operatorname{Re}\bar{v} &= d, \\ 2\operatorname{Re}(\bar{\xi}\tau + \bar{v}\mu) &= b, & 2\operatorname{Re}(6\bar{\xi}\mu + \bar{v}\beta) &= a. \end{aligned} \tag{9.24}$$

For γ, the first two right sides vanish. The third equation then shows that its $(0,1)$-line is represented, up to a nonzero scalar, by $(\operatorname{Im}\mu, -\operatorname{Im}\tau)$; the fourth equation only fixes the scalar. Write $\bar{\xi} = c/2 + ir$, $\bar{v} = d/2 + is$. The condition that the class in (9.22) have $(0,1)$-image on the γ-line is

$$\bar{\xi}\operatorname{Im}\tau + \bar{v}\operatorname{Im}\mu = 0.$$

Its real part, together with the third equation of (9.24), is equivalent to the single holomorphic equation

$$c\tau + d\mu - b = 0. \tag{9.25}$$

Neither a nor β occurs, as adding a multiple of γ must not change the condition.

The functions $1, \tau, \mu$ are linearly independent. Applying the cusp transformation $(\tau, \mu) \mapsto (\tau - 1, \mu)$ to a constant linear relation first kills the coefficient of τ. If the remaining relation made μ constant,

the order-four law $\mu \circ g_2 = 1 + \mu/\tau$, with nonconstant τ, would force an impossible identity. Hence all coefficients vanish.

For every integral $\lambda \notin \mathbb{Q}\gamma$, equation (9.25) is therefore a nonzero holomorphic function on the covering Riemann surface. Its zero set is discrete and countable. There are countably many λ, so their union, and its image in B°, are countable. This image contains every q for which $\operatorname{rk} \Lambda_q = 2$. Each such fibre contributes the countable bad set $\mathbb{Q}\Lambda_q$; all rank-one fibres together contribute only parameters in $\mathbb{R}$. Thus the smooth fibres exclude only $\mathbb{R}$ together with a countable set.

There remain the three scheme-theoretic special fibres. By theorems 55.11 and 55.12, restriction

$$\mathbb{C} \longrightarrow H^1(X_b, \mathcal{O}_{X_b})$$

is complex-linear and injective for each of them. The exponential sequence and countability of $H^1(X_b; \mathbb{Z})$ show that the parameters for which $L_z|_{X_b}$ is torsion form a countable set.

Finally consider the reductions S_j. The restriction $\varphi_j : \mathbb{C} \to H^1(S_j, \mathcal{O}_{S_j})$ is complex-linear. It is nonzero. Write $\mathcal{O}_X(S_j) = L_{z_j}$. Since

$$\mathcal{O}_X(m_j S_j) = f^*\mathcal{O}_B(1) = L_{12},$$

the additive parameter gives $z_j = 12/m_j$, while theorem 55.2 says that

$$L_{12/m_j}|_{S_j} = \mathcal{O}_X(S_j)|_{S_j}$$

is nontrivial. If φ_j were zero, the exponential sequence and the vanishing of the restricted first Chern class would make every $L_z|_{S_j}$ trivial, a contradiction. Thus φ_j is injective, and the same exponential-sequence argument makes its torsion parameters countable. Removing these two additional countable sets proves the theorem. □

Remark 55.14 (provenance of the repair). The period equation $c\tau + d\mu - b = 0$ and the countability strategy are the candidate's. The exact normal-character proof above makes the order of $\mathcal{O}_X(S_j)|_{S_j}$ explicit, while the cusp extension in theorem 55.11 replaces the candidate's collapse-dependent argument by the independently verified intrinsic nearby-cycle map. Identifying the restriction line with the global $\gamma^{0,1}$-summand also removes the candidate's separate continuity argument for a moving integral lift.

55.6 A local analytic obstruction to a cusp-only Picard argument

The relative Picard space of the cusp family cannot by itself force the normalised central restriction to vanish. The obstruction is visible directly in the toric model of theorem 10.6. We give the complete factor-of-automorphy calculation, including the descent cocycle and the sign of the generic Chern class.

Retain the cusp notation of theorems 10.2 and 10.6:

$$B_0 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad Z(s) = sB_0 + C(t), \qquad C(t) = \begin{pmatrix} 6\mu(t) & h(t) \\ b(t) - h(t) & \mu(t) \end{pmatrix}, \qquad t = e^{2\pi i s}, \tag{5.17t}$$

$$c_\lambda(t) = \exp(2\pi i C(t)\lambda), \qquad \exp(2\pi i Z(s)\lambda) = t^{B_0\lambda} c_\lambda(t) \quad (\lambda \in \bar{\Lambda} \simeq \mathbb{Z}^2), \tag{5.17u}$$

and

$$\Phi_\lambda(x,t) = (t^{B_0\lambda}x, t), \qquad \Psi_\lambda(p) = (c_\lambda(t(p)), 1) \cdot \Phi_\lambda(p), \qquad \Psi_\lambda\Psi_\mu = \Psi_{\lambda+\mu}. \tag{5.17v}$$

Write E_v, $v \in \mathbb{Z}^2$, for the toric prime divisors of Y_ϵ, so that $E_0 \simeq \mathrm{dP}_6$, and let

$$\pi : Y_\epsilon \longrightarrow N_0 = Y_\epsilon/\bar{\Lambda}, \qquad \nu = \pi|_{E_0} : E_0 \longrightarrow W$$

be the quotient and the normalization of the central fibre. Put

$$N' = \mathbb{Z}^2 \oplus \mathbb{Z}, \qquad M' = \operatorname{Hom}(N', \mathbb{Z}).$$

For $m=(\ell,k)\in M'=\mathbb{Z}^2\oplus\mathbb{Z}$, fix the character convention

$$\chi^m(x,t)=x^\ell t^k,\qquad \operatorname{div}(\chi^m)=\sum_{v\in\mathbb{Z}^2}\langle m,(v,1)\rangle E_v. \tag{5.17w}$$

Proposition 55.15 (an explicit nonzero local cusp restriction)**.** ***[VERIFIED]*** *Let*

$$H=\begin{pmatrix}1&0\\0&-6\end{pmatrix},\qquad P=B_0^{\mathsf t}H=\begin{pmatrix}0&6\\1&0\end{pmatrix},\qquad Q(v)=v^{\mathsf t}Hv=v_1^2-6v_2^2. \tag{5.17x}$$

Then

$$\mathcal{D}_Q=\sum_{v\in\mathbb{Z}^2}Q(v)E_v \tag{5.17y}$$

is a locally finite Cartier divisor on Y_ϵ. The line bundle $\mathcal{O}_{Y_\epsilon}(\mathcal{D}_Q)$ has an explicit $(\bar\Lambda,\Psi)$-equivariant structure and descends to a holomorphic line bundle $\mathcal{L}_Q$ on N_0. If

$$(a_1,a_2,a_3,a_4,a_5,a_6)=(e_1,e_2,e_2-e_1,-e_1,-e_2,e_1-e_2),\qquad C_i=E_0\cap E_{a_i},$$

then

$$\nu^*(\mathcal{L}_Q|_W)\simeq\mathcal{O}_{E_0}\Big(\sum_{i=1}^6 Q(a_i)C_i\Big),\qquad \big(\deg_{C_i}\nu^*(\mathcal{L}_Q|_W)\big)_{i=1}^6=(-12,2,0,-12,2,0). \tag{5.17z}$$

In particular $\nu^*(\mathcal{L}_Q|_W)$ is nontrivial.

On every nonzero fibre

$$F_t=\mathbb{C}^2/(Z(s)\mathbb{Z}^2+\mathbb{Z}^2),$$

the first Chern class of $\mathcal{L}_Q|_{F_t}$, in the ordered lattice basis $(\widehat\gamma,\widehat u,\widehat w,\widehat\delta)$ indexing the columns of $[Z\mid I_2]$, is exactly

$$c_1(\mathcal{L}_Q|_{F_t})=-2q,\qquad q=u\wedge w+6\gamma\wedge\delta. \tag{5.17aa}$$

Proof. Local finiteness and Cartier status. The fan of Y consists of the cones over the cells of the locally finite A_2-triangulation at height one. If $U_\sigma\simeq\mathbb{C}^3$ is the affine chart belonging to a triangle $\sigma=\{v_0,v_1,v_2\}$, then U_σ meets precisely E_{v_0},E_{v_1},E_{v_2}. Indeed, by the orbit–cone correspondence, $U_\sigma\cap E_v\neq\varnothing$ exactly when $\mathbb{R}_{\ge0}(v,1)$ is a face of the cone over σ. Thus $\{E_v\}_{v\in\mathbb{Z}^2}$ is locally finite and $\mathcal{D}_Q$ is a well-defined locally finite Weil divisor.

Every triangle of the fan is unimodular, as proved in theorem 10.6. If z_0,z_1,z_2 are the characters dual to $(v_0,1),(v_1,1),(v_2,1)$, then

$$\mathcal{D}_Q|_{U_\sigma}=\operatorname{div}\Big(z_0^{Q(v_0)}z_1^{Q(v_1)}z_2^{Q(v_2)}\Big). \tag{5.17ab}$$

Negative exponents merely make the displayed monomial meromorphic, as permitted for a local equation of a Cartier divisor. Hence $\mathcal{D}_Q$ is Cartier on every chart and therefore on Y_ϵ.

The signed divisor pullback identity. For $\lambda\in\bar\Lambda=\mathbb{Z}^2$, set

$$b_\lambda=B_0\lambda,\qquad \ell_\lambda=2Hb_\lambda,\qquad k_\lambda=Q(b_\lambda),\qquad m_\lambda=(\ell_\lambda,k_\lambda)\in M'. \tag{5.17ac}$$

The fan automorphism sends E_v to E_{v+b_λ}. Consequently $\Phi_\lambda^*E_w=E_{w-b_\lambda}$, and

$$\begin{aligned}\Phi_\lambda^*\mathcal{D}_Q-\mathcal{D}_Q&=\sum_{v\in\mathbb{Z}^2}\big(Q(v+b_\lambda)-Q(v)\big)E_v\\&=\sum_{v\in\mathbb{Z}^2}\big(2v^{\mathsf t}Hb_\lambda+b_\lambda^{\mathsf t}Hb_\lambda\big)E_v=\operatorname{div}(\chi^{m_\lambda}).\end{aligned}$$

The t-dependent horizontal torus translation $p \mapsto (c_\lambda(t(p)), 1) \cdot p$ preserves every torus-invariant prime divisor E_v. Hence

$$\boxed{\Psi_\lambda^* \mathcal{D}_Q = \mathcal{D}_Q + \operatorname{div}(\chi^{m_\lambda})}. \tag{5.17ad}$$

The complete Ψ-cocycle. The matrices in (5.17t) and (5.17x) give

$$PC(t) = \begin{pmatrix} 6(b(t) - h(t)) & 6\mu(t) \\ 6\mu(t) & h(t) \end{pmatrix}, \tag{5.17ae}$$

which is symmetric. Thus

$$A(t) := 2PC(t) \tag{5.17af}$$

is a symmetric holomorphic matrix on the full cusp disc. Define the holomorphic units h_λ and meromorphic functions f_λ by

$$h_\lambda(t) = \exp\bigl(\pi i \lambda^{\mathsf{t}} A(t) \lambda\bigr), \qquad f_\lambda = h_\lambda(t) \chi^{m_\lambda}. \tag{5.17ag}$$

Equation (5.17ad) gives

$$\operatorname{div}(f_\lambda) = \Psi_\lambda^* \mathcal{D}_Q - \mathcal{D}_Q. \tag{5.17ah}$$

The cocharacter map underlying Φ_μ is

$$\varphi_\mu(y, y_3) = (y + y_3 b_\mu, y_3),$$

so its pullback on characters is

$$\varphi_\mu^*(\ell, k) = (\ell, k + \ell^{\mathsf{t}} b_\mu). \tag{5.17ai}$$

The quadratic identity for Q gives

$$m_{\lambda+\mu} = m_\mu + \varphi_\mu^* m_\lambda. \tag{5.17aj}$$

Indeed, the first coordinates add, while the last coordinate on the right is

$$Q(b_\mu) + Q(b_\lambda) + 2 b_\lambda^{\mathsf{t}} H b_\mu = Q(b_\lambda + b_\mu).$$

Using (5.17u), (5.17v), and (5.17ai), one obtains

$$\begin{aligned} \Psi_\mu^* \chi^{m_\lambda} &= \chi^{m_\lambda}(c_\mu(t), 1) \chi^{\varphi_\mu^* m_\lambda} \\ &= \exp\bigl(2\pi i \ell_\lambda^{\mathsf{t}} C(t) \mu\bigr)\, \chi^{\varphi_\mu^* m_\lambda} \\ &= \exp\bigl(2\pi i \lambda^{\mathsf{t}} A(t) \mu\bigr)\, \chi^{\varphi_\mu^* m_\lambda}. \end{aligned} \tag{5.17ak}$$

The last equality is exact because

$$\ell_\lambda^{\mathsf{t}} C \mu = (2 H B_0 \lambda)^{\mathsf{t}} C \mu = 2 \lambda^{\mathsf{t}} B_0^{\mathsf{t}} H C \mu = 2 \lambda^{\mathsf{t}} P C \mu = \lambda^{\mathsf{t}} A \mu.$$

Since $A = A^{\mathsf{t}}$, (5.17ag), (5.17aj), and (5.17ak) imply

$$\begin{aligned} f_\mu\, \Psi_\mu^* f_\lambda &= h_\mu h_\lambda \exp(2\pi i \lambda^{\mathsf{t}} A \mu) \chi^{m_\mu + \varphi_\mu^* m_\lambda} \\ &= \exp\bigl(\pi i (\lambda + \mu)^{\mathsf{t}} A (\lambda + \mu)\bigr)\, \chi^{m_{\lambda+\mu}} = f_{\lambda+\mu}. \end{aligned} \tag{5.17al}$$

Use the divisor-sheaf convention

$$\mathcal{O}(D)(U) = \{ g \in \mathcal{M}_Y(U) : \operatorname{div}(g) + D|_U \geq 0 \}.$$

If $D' = D + \operatorname{div}(f)$, multiplication by f maps $\mathcal{O}(D')$ isomorphically to $\mathcal{O}(D)$. Thus (5.17ad) and (5.17ag) define

$$\theta_\lambda : \Psi_\lambda^* \mathcal{O}(\mathcal{D}_Q) \xrightarrow{\times f_\lambda} \mathcal{O}(\mathcal{D}_Q). \tag{5.17am}$$

Because $(\Psi_\lambda\Psi_\mu)^* = \Psi_\mu^*\Psi_\lambda^*$, equation (5.17al) is exactly

$$\theta_{\lambda+\mu} = \theta_\mu \circ \Psi_\mu^*\theta_\lambda. \tag{5.17an}$$

This proves a complete equivariance datum, not merely invariance of the divisor class.

Analytic descent. By theorem 10.6, the action on Y_ϵ is free and properly discontinuous. Every point therefore has an open neighbourhood U whose distinct translates are disjoint and on which $\pi|_U$ is a biholomorphism onto $\pi(U)$. Transport $\mathcal{O}(\mathcal{D}_Q)|_U$ to $\pi(U)$, and on overlaps arising from translates glue by (5.17am). Equation (5.17an) is precisely the triple-overlap condition. The resulting holomorphic line bundle $\mathcal{L}_Q$ on N_0 satisfies

$$\pi^*\mathcal{L}_Q \simeq \mathcal{O}_{Y_\epsilon}(\mathcal{D}_Q). \tag{5.17ao}$$

The normalized central restriction. Let $j_0 : E_0 \hookrightarrow Y_\epsilon$. Since $\nu = \pi \circ j_0$, (5.17ao) gives

$$\nu^*(\mathcal{L}_Q|_W) = j_0^*\pi^*\mathcal{L}_Q \simeq \mathcal{O}_{E_0}(\mathcal{D}_Q|_{E_0}). \tag{5.17ap}$$

The only E_v, $v \neq 0$, meeting E_0 are those indexed by the six a_i in the statement, and the intersections are the six toric (-1)-curves C_i. Since $Q(0) = 0$, there is no $E_0|_{E_0}$ term, so

$$\mathcal{D}_Q|_{E_0} = \sum_{i=1}^{6} Q(a_i)C_i, \qquad (Q(a_1), \dots, Q(a_6)) = (1, -6, -5, 1, -6, -5). \tag{5.17aq}$$

The hexagonal intersection table is

$$C_i^2 = -1, \qquad C_i \cdot C_{i\pm1} = 1, \qquad C_i \cdot C_j = 0 \quad (j \notin \{i, i \pm 1\}), \tag{5.17ar}$$

with indices modulo six. Therefore

$$\deg_{C_i} \mathcal{O}_{E_0}\left(\sum_j Q(a_j)C_j\right) = -Q(a_i) + Q(a_{i-1}) + Q(a_{i+1}),$$

and direct substitution gives

i	1	2	3	4	5	6
$-Q(a_i) + Q(a_{i-1}) + Q(a_{i+1})$	-12	2	0	-12	2	0.

(5.17as)

The degree on $C_1 = C_{e_1}$ is -12, proving nontriviality.

The generic-fibre factor and its exact Chern class. Fix $t \neq 0$, choose s with $t = e^{2\pi i s}$, and write $x = e^{2\pi i\zeta}$, $\zeta \in \mathbb{C}^2$. Equation (5.17u) identifies the deck transformation indexed by $\lambda \in \mathbb{Z}^2$ with

$$\zeta \longmapsto \zeta + Z(s)\lambda$$

modulo the horizontal translations $\zeta \mapsto \zeta + n$, $n \in \mathbb{Z}^2$. The matrix identities

$$HB_0 = P^{\mathsf{t}}, \qquad B_0^{\mathsf{t}}HB_0 = PB_0 \tag{5.17at}$$

give

$$\chi^{m_\lambda}(x,t) = x^{2P^{\mathsf{t}}\lambda}t^{\lambda^{\mathsf{t}}PB_0\lambda}, \qquad h_\lambda(t) = \exp(2\pi i\lambda^{\mathsf{t}}PC(t)\lambda).$$

Since $Z = sB_0 + C$, it follows that

$$f_\lambda(\zeta) = \exp\!\left(2\pi i\lambda^{\mathsf{t}}PZ\lambda + 4\pi i\lambda^{\mathsf{t}}P\zeta\right). \tag{5.17au}$$

The matrix $PZ = sPB_0 + PC$ is symmetric:

$$PB_0 = \operatorname{diag}(-6, 1),$$

and PC is symmetric by (5.17ae).

Under the descent convention (5.17am), a pulled-back local section written as $\varphi(\zeta)$ in the canonical frame satisfies

$$\varphi(\zeta + Z\lambda) = f_\lambda(\zeta)^{-1}\varphi(\zeta).$$

Thus the factor of automorphy for the lattice translation $\zeta \mapsto \zeta + Z\lambda + n$ is

$$a_{(\lambda,n)}(\zeta) = \exp(2\pi i g_{(\lambda,n)}(\zeta)), \qquad g_{(\lambda,n)}(\zeta) = -\lambda^{\mathsf{t}} PZ\lambda - 2\lambda^{\mathsf{t}} P\zeta. \tag{5.17av}$$

This is independent of n as a factor of automorphy: replacing ζ by $\zeta + n$ changes g by the integer $-2\lambda^{\mathsf{t}} Pn$.

For $l = (\lambda, n)$, $l' = (\mu, m)$, the exponential connecting cocycle is

$$\begin{aligned} c(l, l') &= g_l(\zeta + Z\mu + m) + g_{l'}(\zeta) - g_{l+l'}(\zeta) \\ &= -2\lambda^{\mathsf{t}} Pm \in \mathbb{Z}, \end{aligned} \tag{5.17aw}$$

where the symmetry of PZ cancels the two mixed Z-terms. The corresponding alternating integral form is

$$E(l, l') = c(l, l') - c(l', l) = -2\lambda^{\mathsf{t}} Pm + 2\mu^{\mathsf{t}} Pn. \tag{5.17ax}$$

This is $c_1(\mathcal{L}_Q|_{F_t})$. In the ordered column basis of $[Z \mid I_2]$, the cross-block of the class $q = u \wedge w + 6\gamma \wedge \delta$ from $(\widehat{\gamma}, \widehat{u})$ to $(\widehat{w}, \widehat{\delta})$ is

$$\begin{pmatrix} q(\widehat{\gamma}, \widehat{w}) & q(\widehat{\gamma}, \widehat{\delta}) \\ q(\widehat{u}, \widehat{w}) & q(\widehat{u}, \widehat{\delta}) \end{pmatrix} = \begin{pmatrix} 0 & 6 \\ 1 & 0 \end{pmatrix} = P. \tag{5.17ay}$$

Hence (5.17ax) is exactly $-2q$, as asserted. □

Remark 55.16 (the exact obstruction to the purely analytic route). The proposition proves that the local homomorphism

$$\mathrm{Pic}(N_0) \longrightarrow \mathrm{Pic}(E_0), \qquad \mathcal{L} \longmapsto \nu^*(\mathcal{L}|_W),$$

is not zero. Thus the toric cusp geometry, conductor descent on W, or local relative-Picard representability cannot alone prove the desired global vanishing

$$\nu^* i^* : \mathrm{Pic}(X) \longrightarrow \mathrm{Pic}(E_0).$$

There is no contradiction with theorems 60.14 and 60.18: the local bundle above has nonzero generic class $-2q$, whereas the two global proofs use the order-three and order-four fillings and their common boundary relation to exclude every nonzero multiple of q.

More generally, Bingener proves that if a holomorphic map is proper, flat, and cohomologically flat in dimension zero, then its relative Picard functor is represented by a *not necessarily separated* complex space [38, Introduction, p. 318; Satz 6.2, pp. 336–337]. The same introduction explains that the exponential-sequence quotient works directly under topological local triviality and cohomological flatness in dimension one, but that passage to the quotient fails in general. The cusp map of theorem 10.6 is not topologically locally trivial across 0. Bingener's theorem therefore supplies representability but neither separatedness nor the component-specialization assertion needed here.

Consequently a purely analytic completion must add global compatibility with both finite fillings and must control the boundary support function; neither relative-Picard representability nor conductor descent supplies those statements. The later calculation in theorems 60.19 to 60.21 now supplies them: it first forces the generic Appell–Humbert form to vanish using the two finite fillings, and then proves that its mixed block is the polarization of the three normalized central degrees. Thus this proposition remains the sharp cusp-local obstruction, while the global purely analytic route is complete.

55.7 The canonical divisor and the exact anticanonical ring

The analytic construction admits a further explicit calculation which does not use its claimed identification with a sphere. Retain the actual proper map $f : X \to B = \mathbb{P}^1$, the cusp p_0, the finite points p_1, p_2, the reduced finite fibres S_1, S_2, and $f^*(p_1) = 3S_1$, $f^*(p_2) = 4S_2$. All line-bundle isomorphisms below refer to this constructed analytic space. No replacement period matrix or new compactification is introduced.

Theorem 55.17 (Canonical bundle and recovery of the original fibration)**.** *The evaluation map*

$$f^*(f_*\omega_{X/B}) \longrightarrow \omega_{X/B}$$

has zero divisor exactly $2S_2$. Consequently

$$\omega_X \simeq f^*\mathcal{O}_B(-p_2) \otimes \mathcal{O}_X(2S_2) \simeq \mathcal{O}_X(-2S_2), \qquad \omega_X^{-2} \simeq f^*\mathcal{O}_B(p_2). \tag{CR1}$$

For every integer k the actual direct image of the vertical divisor is

$$f_*\mathcal{O}_X(kS_2) = \mathcal{O}_B(\lfloor k/4 \rfloor p_2) \tag{CR2}$$

as subsheaves of the meromorphic-function sheaf, using pullback to identify their sections. In particular, for each integer $m \geq 0$,

$$h^0(X, \omega_X^{-m}) = \lfloor m/2 \rfloor + 1, \qquad h^0(X, \omega_X^m) = 0 \quad (m > 0). \tag{CR3}$$

There is a graded algebra isomorphism

$$\mathbb{C}[U, V] \xrightarrow{\sim} \bigoplus_{m \geq 0} H^0(X, \omega_X^{-m}), \qquad \deg U = 1, \quad \deg V = 2. \tag{CR4}$$

The degree-two complete linear system has no base points and its map is the original f, followed by a specified projective-coordinate map of B. The degree-one system consists of the single divisor $2S_2$.

Proof. Write $\mathfrak{K} = f_*\omega_{X/B}$. The map is the natural adjunction evaluation, not an identification inferred from the ranks of its two terms. On a smooth torus fibre the invariant top form is nowhere zero. At the cusp, the original form $(dx_1/x_1) \wedge (dx_2/x_2) \wedge dt_c$ is a nowhere-zero canonical frame in each of the unimodular toric charts. Its relative version is the local generator of $\mathfrak{K}$ used in theorem 55.1, so evaluation has no zero there.

For a finite filling retain $t_j = s_j^{m_j}$, $(m_1, m_2) = (3, 4)$ and $(a_1, a_2) = (2, 1)$. On its finite étale total-space cover $q_j : J_j' \to N_j$, the actual direct-image generator pulls back, by (9.5c)–(9.5f), to

$$s_j^{m_j - 1 - a_j} e^{\varphi_j(s_j)} (ds_j \wedge e) \otimes (dt_j)^{-1}. \tag{CR5}$$

Here $(dt_j)^{-1}$ is the pulled-back inverse canonical frame on the base, not division by $m_j s_j^{m_j - 1} ds_j$. The coefficient e^{φ_j} is a unit. Thus the zero divisor on the cover is $(m_j - 1 - a_j)\{s_j = 0\}$. Since q_j is étale, faithful pullback of Cartier divisors makes the downstairs divisor $(m_j - 1 - a_j)S_j$. The two coefficients are respectively 0 and 2. These neighbourhoods and the smooth locus cover X, proving the stated global zero divisor with no omitted component.

The evaluation therefore identifies

$$\omega_{X/B} \simeq f^*\mathfrak{K} \otimes \mathcal{O}_X(2S_2).$$

By theorem 55.1, $\mathfrak{K} \simeq \mathcal{O}_B(p_0)$. In the original affine base coordinate t, with $t(p_1) = 0$, $t(p_2) = 1$, $t(p_0) = \infty$, retain the explicit choices

$$h = \frac{1}{t - 1}, \qquad \eta_B = \frac{dt}{(t - 1)^2}.$$

Their divisors are $p_0 - p_2$ and $-2p_2$, respectively: at infinity, with $w = 1/t$, they are $w/(1-w)$ and $-dw/(1-w)^2$. Thus η_B specifies $\omega_B \simeq \mathcal{O}_B(-2p_2)$. Tensoring with $f^*\omega_B$ gives

$$\omega_X \simeq \mathcal{O}_X\big(f^*(p_0) - 2f^*(p_2) + 2S_2\big).$$

Multiplication by f^*h, with its displayed divisor, identifies the base factor $\mathcal{O}_B(p_0 - 2p_2)$ with $\mathcal{O}_B(-p_2)$. Using the actual multiplicity $f^*(p_2) = 4S_2$ proves (CR1).

To prove (CR2), a section of $\mathcal{O}_X(kS_2)$ is represented by a meromorphic function on X with the indicated vertical order bound. Away from p_2 it is holomorphic. Its restriction to each connected compact fibre there is constant; $f_*\mathcal{O}_X = \mathcal{O}_B$ therefore descends it uniquely to a holomorphic function on that base open set. Near p_2, pull back to the original cover J_2', including its zero section. The descended function becomes a function $a(s_2)$, and $s_2^k a(s_2)$ is holomorphic at zero (with the evident vanishing interpretation when $k < 0$). Thus a is meromorphic and has a Laurent series bounded below. Invariance under the actual quotient, whose base coordinate is $s_2 \mapsto \zeta_2 s_2$, says $a(\zeta_2 s_2) = a(s_2)$. Only exponents divisible by four occur, so $a(s_2) = b(s_2^4)$ for a meromorphic b on the original base disc. The order inequality is exactly

$$4\,\mathrm{ord}_{p_2}(b) + k \geq 0 \quad \Longleftrightarrow \quad \mathrm{ord}_{p_2}(b) \geq -\lfloor k/4 \rfloor.$$

Conversely every such b pulls back to an allowed meromorphic function. This proves equality in both directions, as well as compatibility on overlaps. No assumption about the field of all meromorphic functions on X is used. Applying (CR2) with $k = 2m$ and $k = -2m$ and using $h^0(\mathbb{P}^1, \mathcal{O}(d)) = \max\{d+1, 0\}$ proves (CR3).

Here is the exact algebra map in (CR4). Fix the isomorphism $\omega_X^{-1} \simeq \mathcal{O}_X(2S_2)$ obtained above. Let u be the section corresponding to the canonical divisor section of $\mathcal{O}_X(2S_2)$, so $\mathrm{div}(u) = 2S_2$. Its square, under the induced degree-two identification in (CR1), is $f^*\sigma_{p_2}$, where σ_{p_2} is the canonical divisor section of $\mathcal{O}_B(p_2)$. Choose a second section ℓ of this line bundle with $\ell(p_2) \neq 0$, retaining that choice, and let v correspond to $f^*\ell$ under the same degree-two identification. Define $U \mapsto u$, $V \mapsto v$.

For $m = 2r$, the monomials $u^{2r-2j}v^j$, $0 \leq j \leq r$, correspond exactly to $f^*(\sigma_{p_2}^{r-j}\ell^j)$, a basis of $H^0(B, \mathcal{O}_B(rp_2))$. For $m = 2r+1$, multiplication by u identifies the same basis with the full space: in the meromorphic description (CR2), both $4rS_2$ and $(4r+2)S_2$ allow precisely the base pole orders at most r. Thus the displayed map is an isomorphism in every degree, proving both surjectivity and absence of polynomial relations. Replacing ℓ by $a\ell + b\sigma_{p_2}$, $a \neq 0$, changes the second generator by exactly $v \mapsto av + bu^2$; no scalar is silently discarded. Replacing the chosen isomorphism $\omega_X^{-1} \to \mathcal{O}_X(2S_2)$ by c times itself changes the generators by $(u, v) \mapsto (c^{-1}u, c^{-2}v)$, with $c \in \mathbb{C}^*$ retained.

The two degree-two generators have no common zero because σ_{p_2}, ℓ do not. Their map is $x \mapsto [\sigma_{p_2}(f(x)) : \ell(f(x))]$, which is exactly f followed by that chosen coordinate map $B \to \mathbb{P}^1$. Explicitly, writing $\ell = A/(t-1) + B$ with $A \neq 0$, it is $t \mapsto [t-1 : A + B(t-1)]$, including the values $[0 : A]$ at p_2 and $[1 : B]$ at p_0. Degree one has basis u and its divisor is $2S_2$, as claimed. □

In the already specified additive Picard coordinate (9.19), with $L_{12} = f^*\mathcal{O}_B(1)$, the identities retain the original constants:

$$\mathcal{O}_X(S_1) = L_4, \qquad \mathcal{O}_X(S_2) = L_3, \qquad \omega_X = L_{-6}, \qquad \omega_X^{-2} = L_{12}.$$

This last expression uses the previously proved Picard identification; the divisor, direct-image and ring proofs above do not depend on it.

55.8 The retained period constant is an effective deformation parameter

Keep the functions, lattice marking, base coordinate and local fillings of section 10. In particular, fix the actual cusp-regular solution β^{part}, not a replacement with a cusp pole, and write

$$\beta_c = \beta^{\mathrm{part}} + c, \qquad \mathcal{P} = \{c \in \mathbb{C} : \mathrm{Im}\, c < -M\}, \quad M = \max_{B \setminus \{p_0\}} D_{\mathrm{part}}. \tag{PD1}$$

The maximum and the exact admissible domain were proved in theorem 10.3. The following calculation uses the analytic construction, not its integral cohomology or sphere recognition. Its purpose is to determine whether the freedom in c actually changes the constructed complex manifold.

Theorem 55.18 (Exact period deformation)**.** *For every open disc Q with $\overline{Q} \subset \mathcal{P}$, the retained construction gives a proper holomorphic submersion*

$$\pi : \mathcal{X}_Q \longrightarrow Q, \qquad F : \mathcal{X}_Q \longrightarrow B \times Q, \qquad \pi = \mathrm{pr}_Q \circ F, \tag{PD2}$$

whose fibre over c is the analytically constructed threefold X_c and whose restriction there is the original $f_c : X_c \to B$. The three distinguished values remain the original $t(p_1) = 0$, $t(p_2) = 1$, $t(p_0) = \infty$.

Fix a regular base point and one of its original lifts z, and fix $c_* \in Q$. With all functions below evaluated at that z, set

$$T = \operatorname{Im}\tau > 0, \quad m = \operatorname{Im}\mu, \quad D = \operatorname{Im}\beta_{c_*} - \frac{6m^2}{T} < 0, \quad a = \frac{m}{T}, \qquad P = \begin{pmatrix} 0 & 0 \\ -a & 1 \end{pmatrix}. \tag{PD3}$$

For $\delta = c - c_*$, the exact marked real-linear map from the universal cover of the fibre at c_* to that at c is

$$w = A_\delta \zeta + B_\delta \overline{\zeta}, \qquad A_\delta = I + \frac{\delta}{2iD} P, \qquad B_\delta = -\frac{\delta}{2iD} P. \tag{PD4}$$

It maps the original period lattice onto the new period lattice, and therefore descends to a real-analytic marked diffeomorphism of the two tori. In the explicit convention that the transported $(0,1)$ tangent space is spanned by $\partial_{\overline{\zeta}_k} + \sum_j \nu_k^j \partial_{\zeta_j}$, its exact coefficient matrix and its first derivative are

$$\nu_\delta = \frac{\delta}{2iD + \delta} P, \qquad \dot{\nu}_z = \frac{1}{2iD}\, \partial_{\zeta_2} \otimes (d\overline{\zeta}_2 - a\, d\overline{\zeta}_1). \tag{PD5}$$

The class of $\dot{\nu}_z$ in $H^{0,1}(f_{c_*}^{-1}(t(z)), T^{1,0})$ is nonzero. With the tangent-exact-sequence convention $\mathrm{KS}(\partial_c) = [\bar{\partial} h]$ for a smooth $(1,0)$ lift h of ∂_c, the actual fibre class is $\mathrm{KS}_z(\partial_c) = -[\dot{\nu}_z]$. Moreover the first-order deformation of the entire X_{c_*} obtained from (PD2) is nontrivial: its Kodaira–Spencer class in $H^1(X_{c_*}, T_{X_{c_*}})$ is nonzero. In particular $h^1(X_{c_*}, T_{X_{c_*}}) \geq 1$.

Proof. Construction with the parameter retained. On an ordinary lifted base open set use the original period map

$$\Pi(z,c) = \begin{pmatrix} 6\mu(z) & \tau(z) & 1 & 0 \\ \beta^{\mathrm{part}}(z) + c & \mu(z) & 0 & 1 \end{pmatrix} : \Lambda \otimes \mathbb{R} \longrightarrow \mathbb{C}^2. \tag{PD6}$$

Its real determinant is $\operatorname{Im}\tau\, D_c$, which never vanishes on Q. Thus $(z,c,u) \mapsto [z, c, \Pi(z,c)u]$ identifies the quotient, as a real-analytic family, with the original marked real torus over this base open set times Q. The lattice action is holomorphic in (z,c,ζ); it is free and properly discontinuous locally uniformly on compact base-parameter sets, since the inverse real period matrices are uniformly bounded there. The quotient is therefore a complex manifold. Projection to Q is a submersion because it is so before the covering quotient. Its map to the lifted base times Q is proper: over any compact subset the image of that subset times a fixed closed lattice parallelotope is the full inverse image.

The deck matrices are the original integral M_g and the original matrices $R_g(z)$, with $\Pi(gz,c) = R_g(z)\Pi(z,c)M_g$. The R_g do not depend on c. Consequently the ordinary base quotient is jointly holomorphic. Proper discontinuity there follows from proper discontinuity on the base, exactly as in theorem 10.8, with compact parameter sets carried along.

At a finite filling retain $m_1 = 3$, $m_2 = 4$, $s_j(g_j z) = e^{-2\pi i/m_j} s_j(z)$, and the actual logarithmic lift

$$(z,c,\zeta) \longmapsto \left(g_j z, c, R_{g_j}(z)\zeta + \Pi(g_j z, c)\frac{v_j}{m_j} \right). \tag{PD7}$$

In the real marking this is $(z, c, u) \mapsto (g_j z, c, A_j u + v_j/m_j)$, independent of c. Every nonidentity power remains free: the invariant integral covector γ takes the translation of its k-th power to $k/3$ or $-k/4$, which is not an integer for $0 < k < m_j$. The finite quotient is therefore a local biholomorphism and is a submersion to Q. The same compact-parallelotope argument proves properness over $D_j \times Q$. Its map is still exactly $t_j = s_j^{m_j}$. On the punctured overlap the translating section is

$$\frac{\log s_j(z)}{2\pi i}\Pi(z, c)v_j. \tag{PD8}$$

A change of logarithm adds an actual period. It is therefore a branch-independent holomorphic torus section, jointly in z, c. The calculation (2.9e) applies with c retained and proves the punctured-overlap conjugacy, not merely equality of the monodromy orders.

For the cusp retain the same infinite unimodular fan and the full matrix

$$C(t_c, c) = \begin{pmatrix} 6\mu(t_c) & h(t_c) \\ b^{\mathrm{part}}(t_c) - h(t_c) + c & \mu(t_c) \end{pmatrix}, \qquad B_0 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}. \tag{PD9}$$

Choose a closed cusp disc of radius r_1 on which these functions are holomorphic. Put

$$K_Q = \max_{|t_c| \le r_1,\ c \in \overline{Q}} \| - 2\pi \operatorname{Im} C(t_c, c)\|, \qquad 0 < \epsilon < \min(r_1, 1), \quad |\log \epsilon| > 2K_Q. \tag{PD10}$$

The action on $Y_\epsilon \times Q$ is the original one,

$$(x, t_c, c) \longmapsto \left(e^{2\pi i C(t_c,c)\lambda} t_c^{B_0\lambda} x, t_c, c\right), \quad \lambda \in \mathbb{Z}^2. \tag{PD11}$$

It is holomorphic across the boundary by the same toric automorphism and nowhere-zero torus multiplier. The estimates in (2.8) now hold uniformly on Q:

$$B_{t_c,c} = B_0 + \frac{-2\pi \operatorname{Im} C(t_c, c)}{\log |t_c|}, \quad \|B_{t_c,c} - B_0\| < \tfrac{1}{2}, \quad \|B_{t_c,c}\lambda\| > \tfrac{1}{2}\|\lambda\| \quad (\lambda \neq 0). \tag{PD12}$$

They exclude stabilizers on the dense torus and bound the finite set of translations that can join each pair of bounded charts, uniformly in c. At $t_c = 0$ a nonzero translation moves the bounded cell, so there is no stabilizer there either. A compact set is contained in finitely many of the open charts $\mathcal{U}_\sigma(2, \epsilon) \times Q$ from theorem 10.5. The chart-pair estimate proves proper discontinuity on the whole product, including sequences tending to its boundary.

For clarity properness is also uniform, not only pointwise in c. For $\epsilon_1 < \epsilon$, let K_0 be the finite union of closed unit polydiscs associated to triangles meeting the closed radius $\sqrt{2}$ ball, intersected with $|t_c| \le \epsilon_1$, exactly as in theorem 10.6. For every c and $t_c \neq 0$, choose λ so that $\|B_{t_c,c}^{-1} y + \lambda\|_\infty \le 1/2$. The translated logarithmic position has norm at most $3/(2\sqrt{2}) < \sqrt{2}$, and hence lies in a chart defining K_0. At $t_c = 0$, the six closed bidiscs covering E_0 give the same representatives after the exact translation by $B_0 v$, with multiplier $\exp(2\pi i C(0, c)B_0 v)$ retained. Therefore, for compact $K \subset Q$, the image of the compact $K_0 \times K$ is the full inverse image of $\{|t_c| \le \epsilon_1\} \times K$. Every compact set in the cusp base times Q is closed in a set of this form. This proves properness. The quotient is a submersion to Q because its covering charts are open subsets of $Y_\epsilon \times Q$.

Exponentiating the original logarithmic torus coordinates gives the jointly holomorphic cusp overlap, with period identity $Z = sB_0 + C(t_c, c)$. Glue it and (PD8) to the middle family. The special base discs are disjoint. Each resulting inverse image of a member of this base cover is its entire Hausdorff local piece. Two points with different (t, c) values are separated on the base; two with the same value lie in one Hausdorff open piece and can be separated there. This proves Hausdorffness. Local complex charts give a complex fourfold and the submersion π. Properness of F follows by covering any compact set of $B \times Q$ by finitely many relatively compact sets inside these base pieces and using the just proved local

properness. Since B is compact, projection $B \times Q \to Q$ is proper, proving (PD2). The fibres and overlaps are precisely those in the original analytic construction.

The exact variation on a regular fibre. Write $u = (u_1, u_2, u_3, u_4)^t$ in the original ordered lattice basis and $\zeta = \Pi(z, c_*)u$. Taking imaginary parts gives the two exact equations

$$\operatorname{Im}\zeta_1 = 6mu_1 + Tu_2, \qquad \operatorname{Im}\zeta_2 = (D + 6m^2/T)u_1 + mu_2. \tag{PD13}$$

Subtracting a times the first equation from the second yields

$$u_1 = \frac{\operatorname{Im}\zeta_2 - a\operatorname{Im}\zeta_1}{D}. \tag{PD14}$$

In the changed periods the same u maps to $w = \Pi(z, c)u = \zeta + \delta e_2 u_1$. Substitution of $\operatorname{Im}\zeta = (\zeta - \bar{\zeta})/(2i)$ proves (PD4). This also proves its asserted lattice map and real invertibility: it is exactly $\Pi(z, c)\Pi(z, c_*)^{-1}$ as a real-linear map. Its real determinant is $(D + \operatorname{Im}\delta)/D > 0$. Neither the nonholomorphic part nor the original marking is dropped.

Here $P^2 = P$, and

$$\det A_\delta = 1 + \frac{\delta}{2iD}, \qquad A_\delta^{-1} = I - \frac{\delta}{2iD + \delta}P. \tag{PD15}$$

The denominator can vanish only for $\delta = -2iD$. At that parameter $D + \operatorname{Im}\delta = -D > 0$, which is outside the admissible domain. Thus this inverse is defined at every $c \in Q$. Annihilating w by the declared new $(0,1)$ fields gives $A_\delta \nu_\delta + B_\delta = 0$, which proves (PD5), including its sign. All its coefficients are constant in the fibre coordinates. Thus its first derivative is a globally defined $\bar{\partial}$-closed tangent-valued $(0,1)$ form on the torus.

It cannot be $\bar{\partial}V$ for a globally defined smooth vector field V. Indeed the invariant holomorphic frame $\partial_{\zeta_1}, \partial_{\zeta_2}$ writes V with periodic smooth coefficients. The integral of each of their derivatives $\partial_{\bar{\zeta}_k} V^j$ against the translation-invariant probability volume is zero: real constant vector fields preserve this volume, and integration of their derivatives on the compact torus is zero. But the coefficient of $\partial_{\zeta_2} \otimes d\bar{\zeta}_2$ in $\dot{\nu}_z$ is the nonzero constant $1/(2iD)$. Its integral is that same constant. This contradiction proves nonzero Dolbeault class without a dimension count or an appeal to numerical period variation.

The conversion to the tangent-exact-sequence convention retains a minus sign. In the moving coordinates the real-marking lift has $(1,0)$ part $h = \partial_c + u_1\partial_{w_2}$ at c_*. Since (PD14) gives

$$\bar{\partial}u_1 = -\frac{d\bar{\zeta}_2 - a\, d\bar{\zeta}_1}{2iD},$$

the connecting class $[\bar{\partial}h]$ is exactly $-[\dot{\nu}_z]$. Both conventions therefore give a nonzero class; they are not identified without this sign conversion.

The intrinsic pencil in the parameter family. Let $\mathcal{S}_2 \subset \mathcal{X}_Q$ be the reduced finite divisor over $p_2 \times Q$, so $F^*(p_2 \times Q) = 4\mathcal{S}_2$. The proof of theorem 55.17 works for relative forms over Q, as can be checked on each original chart. On the middle family the frame is $d\zeta_1 \wedge d\zeta_2$ relative to $B \times Q$. At the cusp the relative canonical frame over Q is $(dx_1/x_1) \wedge (dx_2/x_2) \wedge dt_c$. Extra dc terms in transition differentials vanish in relative forms. The finite determinant characters χ_j, their correcting units $e^{\varphi_j(s_j)}$, and the modular function Θ used in (9.4)–(9.6) are independent of c, since R_g, τ, μ are. Thus their coefficient identities hold jointly over Q. The evaluation exponents remain $m_j - 1 - a_j = 0, 2$. It follows by the actual same transition functions and divisors that

$$\omega_{\mathcal{X}_Q/Q} \simeq \mathcal{O}_{\mathcal{X}_Q}(-2\mathcal{S}_2), \qquad \omega_{\mathcal{X}_Q/Q}^{-2} \simeq F^* \operatorname{pr}_B^* \mathcal{O}_B(p_2). \tag{PD16}$$

Here the base identifications retain $h = 1/(t-1)$ and $\eta_B = dt/(t-1)^2$ from (CR1); these functions also do not depend on c. In particular (PD16) is a relative line-bundle isomorphism, not a collection of separately chosen isomorphisms of its fibres.

We need its complete degree-two system after an infinitesimal base change, and prove this explicitly. Put $A = \mathbb{C}[\epsilon]/(\epsilon^2)$, base-change by $c = c_* + \epsilon$, write X_A for the resulting deformation and $L_A = \omega_{X_A/A}^{-2}$. For $L_* = \omega_{X_{c_*}}^{-2}$, flatness of the deformation and local freeness of L_A give

$$0 \longrightarrow L_* \xrightarrow{\epsilon} L_A \longrightarrow L_* \longrightarrow 0. \tag{PD17}$$

The two sections of $\mathcal{O}_B(p_2)$ used in (CR4), namely σ_{p_2} and $\ell = A_0/(t-1) + B_0'$ with $A_0 \neq 0$, pull back by (PD16) to sections s_0, s_1 of L_A whose reductions are a basis of $H^0(X_{c_*}, L_*)$. This basis assertion is exactly the proved degree-two case of (CR2)–(CR4). Any section of L_A reduces to a complex linear combination of these two reductions; subtract that lifted combination. Exactness of (PD17) then writes the remainder as ϵ times a section of L_*, again a combination of that basis. Conversely, if an A-linear combination of s_0, s_1 is zero, reduce modulo ϵ and then use the injective first arrow of (PD17) to see that all four complex coefficients vanish. Therefore

$$H^0(X_A, L_A) = As_0 \oplus As_1. \tag{PD18}$$

The system is basepoint-free and its morphism is f_A followed by the fixed coordinate map $t \mapsto [t-1 : A_0 + B_0'(t-1)]$. This proves the required base-change statement without presuming that pointwise dimensions alone imply it.

Nontriviality for the complete threefold. Suppose its first-order deformation were trivial, with an isomorphism $X_A \simeq X_{c_*} \times \operatorname{Spec} A$ reducing to the identity. Intrinsic relative canonical bundles and their complete spaces of sections are preserved by that isomorphism. By (PD18), it would therefore identify f_A with $a_A \circ f_{c_*}$ for a projective linear change $a_A \in \mathrm{PGL}_2(A)$ reducing to the identity.

This change fixes all three distinguished points even to first order. Here is a scheme-sensitive check. In the finite covering charts f_A is $t_j = s_j^{m_j}$, so its relative critical ideal is $(s_j^{m_j-1})$, and its restriction to that critical scheme has $t_j = 0$. The map from the parameter ring A to this critical scheme is injective; it contains a constant-parameter section given by $s_j = 0$ and the zero section of the covering torus. At the cusp, in a chart containing a triple point, $t_c = z_0 z_1 z_2$ and the critical ideal is $(z_1 z_2, z_0 z_2, z_0 z_1)$. Again $t_c = 0$ in its ring and A injects, using $z_0 = z_1 = z_2 = 0$. At a double chart the same assertion follows from $t_c = z_0 z_1$; a smooth chart has no critical point. These critical ideals are intrinsic to the differential of the map over A, and the charts are local biholomorphisms onto X_A. Away from the three distinguished base values the map is a submersion, so it has no critical point. Consequently the scheme-theoretic critical-value sections are exactly the three constant sections p_1, p_2, p_0. An automorphism reducing to the identity cannot permute these three distinct closed points. Its value at each is therefore the same point over A, not a nonzero infinitesimal displacement: such a displacement would be a nonzero element of $\epsilon\mathbb{C}$ in an injective copy of A on the just displayed critical chart.

For completeness write $a_A(t) = t + \epsilon(b_0 + b_1 t + b_2 t^2)$. Fixing 0 gives $b_0 = 0$, fixing 1 gives $b_0 + b_1 + b_2 = 0$, and fixing infinity, in $u = 1/t$, gives $b_2 = 0$. Hence a_A is the identity. The hypothesized trivialization thus preserves f, and restricting it to any regular base value would trivialize the torus deformation (PD4). Its nonzero class (PD5) excludes this.

Finally, the zero Kodaira–Spencer class is precisely the condition for such a first-order trivialization. One may see the needed direction directly: choose local product charts for the holomorphic submersion π and differentiate their transition functions at c_*. The resulting tangent-valued overlap functions form a Cech 1-cocycle. If its class vanishes, after refinement it is the coboundary of local holomorphic vector fields. Changing each chart by that vector field times ϵ cancels the first-order overlap terms, producing the identity-reducing trivialization. This is the usual Kodaira–Spencer connecting class (for the standard deformation-space interpretation see [1, Section 67, p. 15]); the explicitly recorded conversion from the plus-graph convention in (PD5) changes its sign, not its vanishing. The contradiction proves the final assertion. □

The theorem establishes a genuinely varying analytic family with its original marked fibration and its intrinsic degree-two canonical pencil. It does not assert that every two distinct parameters give

nonisomorphic unmarked manifolds, or that this family exhausts their deformation space. The global integral topology and the CDP argument are not inputs to this deformation calculation.

The global identifications within this retained family are computed in theorem 55.19: they are precisely $c' - c \in 2\mathbb{Z}$, including the finite-filling obstruction and every cusp overlap.

55.9 Exact global identifications of the retained period parameter

Keep the original base B, coordinate t, values $t(p_1) = 0, t(p_2) = 1, t(p_0) = \infty$, and the original functions $\tau, \mu, \beta^{\mathrm{part}}$. Write

$$\beta_c = \beta^{\mathrm{part}} + c, \qquad \mathcal{P} = \{c \in \mathbb{C} : \operatorname{Im} c < -M\}, \qquad \Pi_c(z) = \begin{pmatrix} 6\mu(z) & \tau(z) & 1 & 0 \\ \beta_c(z) & \mu(z) & 0 & 1 \end{pmatrix}. \tag{PS1}$$

All assertions below concern the analytic pieces and their actual gluing. No integral cohomology, sphere recognition, or CDP assertion is an input. In particular, the global parameter classification is not inferred from an equality between two fibre lattices.

Put $e = (0,1)^{\mathsf{t}} \in \mathbb{C}^2$ and retain the ordered lattice

$$\Lambda = \mathbb{Z}\langle \widehat{\gamma}, \widehat{u}, \widehat{w}, \widehat{\delta} \rangle, \qquad E = \widehat{\delta} \otimes \gamma. \tag{PS2}$$

Thus E is the matrix with its only nonzero entry, equal to one, in row four and column one; $E^2 = 0$ and $\Pi_c E = e\gamma$. The finite data, with every sign retained, are

$$A_1 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 6 & 0 & 1 & 0 \\ -6 & -1 & -1 & 0 \\ -2 & 1 & 0 & 1 \end{pmatrix}, \qquad A_2 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ -6 & 1 & 0 & 0 \\ 3 & 0 & 1 & 1 \end{pmatrix}, \tag{PS3}$$

$$v_1 = (1,2,-4,0)^{\mathsf{t}}, \quad v_2 = (-1,-3,3,0)^{\mathsf{t}}, \quad (m_1, m_2) = (3,4), \quad (\epsilon_1, \epsilon_2) = (1,-1), \quad \epsilon_j = \gamma(v_j).$$

The covariance is $R_g \Pi_c = \Pi_c(gz) A_g$, with

$$R_{g_1} = \begin{pmatrix} -1/\tau & 0 \\ (1-\mu)/\tau & 1 \end{pmatrix}, \qquad R_{g_2} = \begin{pmatrix} 1/\tau & 0 \\ -\mu/\tau & 1 \end{pmatrix}, \qquad R_{g_0} = I_2. \tag{PS4}$$

In particular $R_g e = e$ for every g.

Theorem 55.19 (The exact identification subgroup is $2\mathbb{Z}$)**.** *For $c, c' \in \mathcal{P}$, the retained analytic threefolds are biholomorphic if and only if*

$$c' - c \in 2\mathbb{Z}. \tag{PS5}$$

Every such biholomorphism preserves the original map to B. For every even integer n, the proof below constructs an isomorphism $X_c \longrightarrow X_{c+n}$ whose complex-linear part on the original universal fibre coordinates is I_2. Its map on the marked fibre lattice is $I - nE$.

The equality of the middle lattices for all integer shifts loses a nonzero order-two finite-filling invariant: the shift n changes that invariant by $n \bmod 2$. The cusp contributes no additional obstruction to the displayed even-shift construction.

55.9.1 The middle family and its complete rational centralizer

For an integer n, put $S_n = I + nE$. Since $S_n^{-1} = S_{-n}$,

$$\Pi_{c+n} = \Pi_c S_n, \qquad \Pi_{c+n} \Lambda = \Pi_c \Lambda. \tag{PS6}$$

Both original matrices fix $\widehat{\delta}$ and preserve γ. Consequently $A_j E = E A_j = E$, and the same commutation holds for every monodromy word. The identity of $\mathbb{C}^2$ therefore identifies the two full middle torus families with their actual R_g-action. Its source-to-target lattice map is S_{-n}, since $\Pi_{c+n} S_{-n} = \Pi_c$.

Lemma 55.20 (All middle-family linear parts)**.** *The simultaneous rational centralizer of A_1, A_2 is exactly*

$$\{aI + bE : a, b \in \mathbb{Q}\}. \tag{PS7}$$

A biholomorphism between the middle families for c, c' over the identity of B° forces $c' - c \in \mathbb{Z}$. Its complex-linear part on every fibre is either I_2 or $-I_2$.

Proof. Solving the two fixed-vector systems from (PS3) gives

$$(\Lambda \otimes \mathbb{Q})^{A_1} = \mathbb{Q}(1, 2, -4, 0)^{\mathsf{t}} + \mathbb{Q}\widehat{\delta}, \quad (\Lambda \otimes \mathbb{Q})^{A_2} = \mathbb{Q}(1, 3, -3, 0)^{\mathsf{t}} + \mathbb{Q}\widehat{\delta}.$$

Their intersection is $\mathbb{Q}\widehat{\delta}$. The fixed covectors are respectively

$$\mathbb{Q}\gamma + \mathbb{Q}(2u + w + 3\delta), \qquad \mathbb{Q}\gamma + \mathbb{Q}(u + w + 2\delta).$$

Their intersection is $\mathbb{Q}\gamma$: comparing the u and w coefficients of a common multiple gives $2x = y$ and $x = y$, hence $x = y = 0$. Every commuting endomorphism therefore preserves $\mathbb{Q}\widehat{\delta}$ and $\ker \gamma$.

On $\ker \gamma / \mathbb{Q}\widehat{\delta}$, in the classes of $(\widehat{u}, \widehat{w})$, the two matrices are

$$Q_1 = \begin{pmatrix} 0 & 1 \\ -1 & -1 \end{pmatrix}, \qquad Q_2 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

A matrix commuting with Q_2 has the form $\left(\begin{smallmatrix} b & d \\ -d & b \end{smallmatrix}\right)$. Commutation with Q_1 gives $d = 0$. Thus a commuting endomorphism H has columns

$$H\widehat{\gamma} = (a, r, s, k)^{\mathsf{t}}, \quad H\widehat{u} = b\widehat{u} + u_0\widehat{\delta}, \quad H\widehat{w} = b\widehat{w} + w_0\widehat{\delta}, \quad H\widehat{\delta} = d_0\widehat{\delta}.$$

Use the actual columns $A_1\widehat{u} = -\widehat{w} + \widehat{\delta}$, $A_1\widehat{w} = \widehat{u} - \widehat{w}$, and $A_2\widehat{u} = \widehat{w}$. Commutation on these columns gives

$$d_0 - w_0 = b + u_0, \qquad u_0 = 2w_0, \qquad w_0 = u_0.$$

Hence $u_0 = w_0 = 0$ and $d_0 = b$. Commutation on $\widehat{\gamma}$ with A_1 gives

$$r = 2(a - b), \qquad s = -4(a - b).$$

Commutation on that column with A_2 also gives $r = -s$. Therefore $a = b$, $r = s = 0$, and $H = aI + kE$. Conversely these matrices commute, by the fixed-vector and fixed-covector identities preceding (PS6). This proves (PS7).

For the geometric assertion, a holomorphic map between two compact complex tori lifts to their universal covers. The derivative of the lift is periodic, so its entries descend to holomorphic functions on the compact source torus and are constant. The lift is consequently affine, with a complex-linear part L. In a holomorphic family these linear parts vary holomorphically. Their lattice maps are integral and locally constant, hence give an isomorphism of the two marked local systems. Because the map is over the identity of the base, that integral map H commutes with both A_j. Thus $H = aI + bE$, with $a, b \in \mathbb{Z}$. Its determinant is a^4; invertibility over $\mathbb{Z}$ gives $a = 1$ or $a = -1$.

The complete period equation is

$$L(z)\Pi_c(z) = \Pi_{c'}(z)(aI + bE). \tag{PS8}$$

Its third and fourth columns give $L(z) = aI_2$. Its first column then gives $a(c' - c) + b = 0$. Therefore $c' - c = -b/a \in \mathbb{Z}$, and the asserted two linear parts exhaust the possibilities. □

55.9.2 The exact finite obstruction, including normalization

The two cyclic norm matrices are, by multiplication of (PS3),

$$N_1 = I + A_1 + A_1^2 = \begin{pmatrix} 3 & 0 & 0 & 0 \\ 6 & 0 & 0 & 0 \\ -12 & 0 & 0 & 0 \\ 0 & 2 & 1 & 3 \end{pmatrix},$$

$$N_2 = I + A_2 + A_2^2 + A_2^3 = \begin{pmatrix} 4 & 0 & 0 & 0 \\ 12 & 0 & 0 & 0 \\ -12 & 0 & 0 & 0 \\ 0 & 2 & 2 & 4 \end{pmatrix}. \tag{PS9}$$

For example these formulas give $N_1\widehat{w} = \widehat{\delta}$, $N_2\widehat{u} = N_2\widehat{w} = 2\widehat{\delta}$, $N_1\widehat{\delta} = 3\widehat{\delta}$, and $N_2\widehat{\delta} = 4\widehat{\delta}$.

We first justify why a local isomorphism must satisfy the exact affine equations on the original finite cover. The actual ramified base change $t_j = u^{m_j}$ has local equation

$$s_j^{m_j} - u^{m_j} = \prod_{a=0}^{m_j-1} (s_j - e^{2\pi i a/m_j} u) = 0.$$

Its normalization separates all these smooth branches, each with coordinates (z_1, z_2, u). The original map $y \mapsto (q_j(y), s_j(y))$ identifies the original torus family J_j' with this normalization, as proved by the explicit branches in theorem 55.6. A biholomorphism over the identity of D_j base-changes over the identity of the u-disc and lifts uniquely to the normalization: on local rings it carries integral elements in the total fraction ring to integral elements, and its inverse does the same. This lift commutes with the original deck action because both composites lift the same map $(x, u) \mapsto (F(x), \zeta_j u)$. In particular one may not replace the generator by an arbitrary coprime power.

Let $c' = c + n$, where n is an integer, and suppose the linear part of the lifted map is aI_2, $a \in \{1, -1\}$. The lifted map on the torus family is $[z, \zeta] \mapsto [z, a\zeta + B_j(z)]$. Its translating section has a holomorphic lift $B_j : \Delta_j \to \mathbb{C}^2$: $\Delta_j \times \mathbb{C}^2 \to J_j'$ is a covering map and the disc is simply connected. Local inverses of this covering show that the lift is holomorphic. The deck-equivariance identity holds modulo an actual lattice vector, so, after choosing this lift, it is

$$B_j(g_j z) - R_{g_j}(z) B_j(z) = \Pi_c(g_j z) \left(\frac{(1-a)v_j + n\epsilon_j \widehat{\delta}}{m_j} + \ell_j \right), \qquad \ell_j \in \Lambda. \tag{PS10}$$

The integer column ℓ_j is constant: it is locally constant by discreteness of the marked lattice and the disc is connected. Iterating m_j times, with $A_j v_j = v_j$ and $A_j\widehat{\delta} = \widehat{\delta}$, yields the necessary equality

$$N_j \ell_j = -(1-a)v_j - n\epsilon_j \widehat{\delta}. \tag{PS11}$$

Here the left side of the iterated equation is zero because the base point and the original vector coordinate return after m_j steps; real invertibility of Π_c makes the lattice equality exact.

If $a = -1$, applying γ to (PS11) gives

$$m_j \gamma(\ell_j) = -2\epsilon_j.$$

This is impossible already at $j = 1$, since 3 does not divide 2. Thus a global isomorphism cannot have negative linear part.

For $a = 1$, the first row of (PS11) gives $\gamma(\ell_j) = 0$. Equation (PS9) then proves that the order-three filling imposes no restriction on n, whereas the order-four filling imposes precisely $n \in 2\mathbb{Z}$. More intrinsically, the finite obstruction is the class of $-n\epsilon_j\widehat{\delta}$ in

$$\Lambda^{A_j}/N_j\Lambda. \tag{PS12}$$

In the original fixed bases this quotient is $\mathbb{Z}/3$ at $j=1$, with $\widehat{\delta}$ zero, and $\mathbb{Z}/4\oplus\mathbb{Z}/2$ at $j=2$, with $\widehat{\delta}$ generating the second summand. Indeed (PS9) gives

$$N_1\Lambda = 3\mathbb{Z}v_1\oplus\mathbb{Z}\widehat{\delta},\quad N_2\Lambda = 4\mathbb{Z}\varepsilon'\oplus 2\mathbb{Z}\widehat{\delta},\qquad \varepsilon' = (1,3,-3,0)^{\mathsf{t}}.$$

This proves the exact parity obstruction rather than merely failure of the untranslated identity.

55.9.3 Local maps for every even shift

Let $n\in 2\mathbb{Z}$. Define the following original rational columns, integral cocycles, and holomorphic translation vectors:

$$\begin{aligned} r_1 &= -\frac{n}{3}(\widehat{u}+\widehat{w}),\quad \ell_1 = -n\widehat{w},\quad b_1(z)=\Pi_c(z)r_1 = -\frac{n}{3}(\tau(z)+1,\mu(z))^{\mathsf{t}},\\ r_2 &= \frac{n}{4}(\widehat{u}+\widehat{w}),\qquad \ell_2 = \frac{n}{2}\widehat{u},\qquad b_2(z)=\Pi_c(z)r_2 = \frac{n}{4}(\tau(z)+1,\mu(z))^{\mathsf{t}}. \end{aligned}\tag{PS13}$$

Direct multiplication of the complete matrices in (PS3) gives

$$(I-A_j)r_j = \frac{n\epsilon_j}{m_j}\widehat{\delta}+\ell_j.\tag{PS14}$$

For $j=1$ the four entries are $(0,0,-n,n/3)$, and for $j=2$ they are $(0,n/2,0,-n/4)$. Pairing with the full $\Pi_c(g_jz)$ shows that b_j satisfies (PS10) for $a=1$. Translation by b_j is therefore already an isomorphism of the two finite local fillings. The remaining issue is its exact overlap with the fixed middle identification.

Keep the original linearizing coordinate, radius, and phase $s_j(g_jz)=e^{-2\pi i/m_j}s_j(z)$. On a logarithm branch put

$$w_j(z)=\frac{\log s_j(z)}{2\pi i},\qquad a_j(z)=b_j(z)+n\epsilon_j w_j(z)e.\tag{PS15}$$

Changing the logarithm adds $n\epsilon_j\widehat{\delta}$ under the period map, so a_j is branch independent as a torus section. For the specified negative generator, $w_j(g_jz)=w_j(z)-1/m_j$. Equations (PS4) and (PS14) consequently give the exact equality

$$a_j(g_jz)=R_{g_j}(z)a_j(z)+\Pi_c(g_jz)\ell_j.\tag{PS16}$$

This is why the logarithmic term in (PS15) cannot be omitted. In fact the original finite overlap maps are translations by $\sigma_{j,c}=w_j\Pi_c v_j$, and

$$\sigma_{j,c+n}-\sigma_{j,c}=n\epsilon_j w_j e.\tag{PS17}$$

Thus a_j is exactly the middle translation induced by the local translation b_j, with the original gluing maps retained.

At the cusp retain $s=\tau-h$, $e^{2\pi is}=t_c$, $s(g_0z)=s(z)-1$, and the complete period matrix

$$Z_c = sB_0+C_c(t_c),\quad B_0=\begin{pmatrix}0&1\\-1&0\end{pmatrix},\quad C_c=\begin{pmatrix}6\mu & h\\ b^{\mathrm{part}}-h+c & \mu\end{pmatrix}.\tag{PS18}$$

For $\lambda=(\lambda_1,\lambda_2)^{\mathsf{t}}\in\mathbb{Z}^2$, $C_{c+n}\lambda-C_c\lambda=(0,n\lambda_1)^{\mathsf{t}}$. Therefore

$$\exp(2\pi iC_{c+n}\lambda)t_c^{B_0\lambda}=\exp(2\pi iC_c\lambda)t_c^{B_0\lambda}.\tag{PS19}$$

The entire cusp actions, including their nonconstant holomorphic units, are identical for an integer shift.

The affine cocycle convention in (PS16) has multiplication law

$$\ell_{gh}=\ell_g+A_g\ell_h.\tag{PS20}$$

Since $g_0=(g_1g_2)^{-1}$, it gives

$$\begin{aligned} \ell_1+A_1\ell_2 &= -\frac{3n}{2}\widehat{w}+\frac{n}{2}\widehat{\delta},\\ \ell_0 = -A_{g_0}(\ell_1+A_1\ell_2) &= \frac{3n}{2}\widehat{w}-\frac{n}{2}\widehat{\delta}. \end{aligned}\tag{PS21}$$

Here A_{g_0} fixes both toric lattice vectors, as is also immediate from $A_{g_0} = \left(\begin{smallmatrix} I & 0 \\ B_0 & I \end{smallmatrix}\right)$. Put

$$q_n = (-3n/2, n/2)^{\mathsf{t}} \in \mathbb{Z}^2, \qquad a_0(z) = s(z)q_n. \tag{PS22}$$

Then $a_0(g_0 z) - a_0(z) = -q_n = \Pi_c(g_0 z)\ell_0$, exactly as required by (PS16). Exponentiating gives the map $(x, t_c) \mapsto (t_c^{q_n} x, t_c)$ on the dense cusp torus. It extends holomorphically and invertibly to the original infinite fan: the integral map

$$(y, y_3) \longmapsto (y + y_3 q_n, y_3)$$

translates each height-one triangle by the integer vector q_n and therefore permutes the actual cones, with inverse given by $-q_n$. It commutes with the cusp quotient action in (PS19). This proves extension across every cusp stratum, not only across its dense torus.

55.9.4 The global additive correction and all overlap identities

We give the global correction explicitly as an additive gluing construction and prove its solvability. Let $\widetilde{B}^\circ$ be the universal cover of the three-punctured base. Its fundamental group is free on the original two negative peripheral generators. Pull back the original Π_c, R to this cover. Assign ℓ_1, ℓ_2 from (PS13) to its free generators and extend by (PS20). Freeness makes this an actual integral cocycle; its third peripheral value is precisely (PS21). The maps

$$(z, y) \longmapsto (gz, R_g(z)y + \Pi_c(gz)\ell_g) \tag{PS23}$$

form an affine action. To check the action law, covariance gives $R_g(hz)\Pi_c(hz) = \Pi_c(ghz)A_g$; substituting this equality in the composition gives exactly (PS20). The quotient is an affine holomorphic bundle over B°, since the deck action on the base is free and properly discontinuous and one may use evenly covered base opens as affine charts.

On a finite punctured disc, (PS16) says that a_j is a section of this affine bundle. On the cusp punctured disc, (PS21)–(PS22) say the same for a_0. Define a sheaf $\mathscr{H}_n$ on B as follows. Away from the special points it is the sheaf of sections of (PS23). Near p_j, a section is required to have the form $a_j + \eta_j$, where $\eta_j : \Delta_j \to \mathbb{C}^2$ is holomorphic and obeys $\eta_j(g_j z) = R_{g_j}(z)\eta_j(z)$. Near the cusp it is required to have the form $a_0 + \eta_0$, with $\eta_0(t_c)$ holomorphic on the full cusp disc. These are compatible definitions on the punctured overlaps by the equal affine cocycles just proved. They also prove local nonemptiness, including at all three special points.

The differences of these local sections form a vector sheaf $\mathscr{V}$. Its ordinary sections have covariance $\eta(gz) = R_g\eta(z)$; at the two finite points and the cusp its extensions are precisely the holomorphic extensions specified above. Projection to the first coordinate gives an exact sequence on the original compact base

$$0 \longrightarrow \mathcal{O}_B e \longrightarrow \mathscr{V} \longrightarrow \mathcal{L} \longrightarrow 0, \qquad \mathcal{L} \simeq \mathcal{O}_B(-p_0). \tag{PS24}$$

Here $\mathcal{L}$ is exactly the homogeneous period sheaf from theorem 10.2, not a newly chosen extension. Indeed the first diagonal entries in (PS4) are its factors $-1/\tau$ and $1/\tau$, and at the cusp both extensions require ordinary holomorphicity. The original function

$$\Theta = \frac{(E_4 \circ \tau)^2 (E_6 \circ \tau)^{1/2}}{\Delta_{\mathrm{mod}} \circ \tau}$$

has those exact factors, orders two and one in s_1, s_2, and a simple pole in t_c. Dividing by Θ identifies the ordinary and finite stalks with $\mathcal{O}_B$, and the cusp stalk with $\mathfrak{m}_{p_0}$. This recovers the displayed $\mathcal{L}$ with its original cusp condition. These transformation and order calculations are the ones used to construct the retained μ; no topological conclusion about X_c is involved.

For completeness, exactness on the finite stalks in (PS24) retains the off-diagonal entry of R_g. Given an equivariant first component $u(z)$, begin with the vector $(u(z), 0)^{\mathsf{t}}$ and average it under the finite group

action on vector-valued holomorphic functions. The resulting vector is equivariant, and its first component is still u. Thus projection is surjective there. Its kernel is exactly the invariant second-component functions, which are holomorphic functions of $s_j^{m_j} = t_j$, multiplied by e. The ordinary and cusp stalks are immediate. This also proves that $\mathscr{V}$ is locally free of rank two: lift a local frame of $\mathcal{L}$ and join it to e.

Both $H^1(B, \mathcal{O}_B)$ and $H^1(B, \mathcal{O}_B(-p_0))$ vanish. One can compute these two groups without a degree approximation: on the two standard affine charts of $\mathbb{P}^1$, split a holomorphic Laurent series on their overlap into its nonnegative and negative powers. For $\mathcal{O}_B$ these are the two coboundary contributions, and for $\mathcal{O}_B(-p_0)$ its transition factor shifts the second contribution by one, still leaving no missing exponent. The positive part converges on the full first affine chart and the transformed negative part on the full second affine chart. This proves the two vanishings, equivalently the usual additive Cousin calculation on these charts. The exact sequence (PS24) now gives

$$H^1(B, \mathscr{V}) = 0. \tag{PS25}$$

Here is how to choose the correction entering the maps. Choose local sections h_α of $\mathscr{H}_n$, taking the displayed a_j, a_0 on sufficiently small special discs and affine charts elsewhere. Their differences $d_{\alpha\beta} = h_\beta - h_\alpha$ are a Cech 1-cocycle with values in $\mathscr{V}$. Project it to $\mathcal{L}$ and solve that additive cocycle, using its just proved H^1 vanishing. Lift the resulting local $\mathcal{L}$ sections to $\mathscr{V}$ by the local lifting construction in (PS24). Subtracting their coboundary from d leaves a scalar $\mathcal{O}_B e$ cocycle. Solve that scalar additive cocycle by its H^1 vanishing. This produces sections v_α with $d_{\alpha\beta} = v_\beta - v_\alpha$. The local sections $h_\alpha - v_\alpha$ agree and define a global section $a \in \mathscr{H}_n(B)$. This is a two-stage construction of the required correction, not an unproved additional gluing condition. It gives holomorphic vectors η_j, η_0 with

$$a = a_j + \eta_j \quad (j = 1, 2), \qquad a = a_0 + \eta_0 \quad \text{at the cusp.} \tag{PS26}$$

Any choices made in the additive solutions give maps with the same asserted properties. Their differences as sections of this affine bundle lie in $H^0(B, \mathscr{V}) = \mathbb{C}e$: this follows from (PS24), $H^0(B, \mathcal{L}) = 0$, and $H^0(B, \mathcal{O}_B) = \mathbb{C}$.

Projecting a modulo $\Pi_c\Lambda$ gives a holomorphic section of the middle torus family because every ℓ_g in (PS23) is integral. The promised three kinds of local maps are therefore

$$\begin{aligned} F_{\mathrm{mid}}[z, \zeta] &= [z, \zeta + a(z)], \\ F_j[z, \zeta] &= [z, \zeta + b_j(z) + \eta_j(z)] \quad (j = 1, 2), \\ F_0[x, t_c] &= [\exp(2\pi i \eta_0(t_c)) t_c^{q_n} x, t_c]. \end{aligned} \tag{PS27}$$

The first descends by (PS23) and equality of the integer-shift lattices. The finite maps descend by (PS10), (PS14), and the exact covariance of η_j. The cusp map descends by (PS19); its holomorphic-unit factor extends by torus multiplication on every original chart, and its integer factor extends by the fan map in (PS22). Each map is a biholomorphism, with inverse given by the negative translation or the inverse torus multiplier and fan map.

On a finite overlap, write the original gluing as $h_{j,c}(\zeta) = \zeta + \sigma_{j,c}$. Then (PS17), (PS26), and (PS27) give the equality of actual torus maps

$$h_{j,c+n} F_j = F_{\mathrm{mid}} h_{j,c}. \tag{PS28}$$

Both sides add $\sigma_{j,c} + a = \sigma_{j,c+n} + b_j + \eta_j$. On the cusp overlap the original map is exponentiation of ζ; (PS22) and (PS26) give $\exp(2\pi i a) = t_c^{q_n} \exp(2\pi i \eta_0)$, exactly the third line of (PS27). Thus that overlap also agrees, including all nonconstant period units. The maps and their inverses glue to a global biholomorphism $X_c \to X_{c+n}$ over B.

The choices can be made coherently for every parameter and every even shift. Both initial cocycle columns in (PS13) have $\gamma = 0$. Because every A_g preserves γ, (PS20) shows that $\gamma(\ell_g) = 0$ for every word. Therefore the affine term $\Pi_c(gz)\ell_g$ in (PS23) is independent of c. The displayed a_j, a_0, and the vector

sheaf $\mathscr{V}$ are also independent of c. Perform the additive construction once for $n = 2$, keeping one resulting section $a^{(2)}$. For $n = 2k$ choose $a^{(2k)} = ka^{(2)}$ and multiply its local correcting vectors by k. All their affine cocycles and local formulas are then exactly those for $n = 2k$. The resulting maps satisfy the full group law

$$F_{2h,c+2k} \circ F_{2k,c} = F_{2(h+k),c}, \qquad F_{0,c} = \mathrm{id}_{X_c}. \tag{PS28a}$$

This follows by adding the actual translation vectors in the first two lines of (PS27), and by multiplying their units and adding the integer exponents in its third line. The formulas are jointly holomorphic in c, since their corrections are independent of it.

55.9.5 Intrinsic necessity and the resulting parameter space

The analytically computed complete linear system $|\omega_{X_c}^{-2}|$ is the original map f_c, by theorem 55.17. A biholomorphism preserves its canonical bundle, the complete space of sections, and its evaluation map. Consequently it induces $\alpha \in \mathrm{PGL}_2(\mathbb{C})$ with $f_{c'}F = \alpha f_c$. The three special fibres have distinct intrinsic types: the cusp is reduced and nonnormal, while the other two have multiplicities three and four. All remaining fibres are smooth reduced tori. Thus α fixes p_0, p_1, p_2 individually. In the original coordinate, a fractional-linear map fixing $\infty, 0, 1$ has zero lower-left and upper-right entries and equal diagonal entries; hence $\alpha = \mathrm{id}_B$.

Now theorem 55.20 forces $c' - c = n \in \mathbb{Z}$ and linear part $\pm I_2$. Equations (PS10)–(PS11), obtained on the original normalized finite covers, exclude $-I_2$ and then exclude odd n for $+I_2$. Conversely (PS27)–(PS28) construct the global map for every even n. This proves theorem 55.19 without converting a failed particular map into a nonisomorphism assertion.

As a complex parameter space for the isomorphism classes represented by this retained family, the quotient $\mathcal{P}/(2\mathbb{Z})$ has the explicit coordinate

$$w = \exp(-\pi i c), \qquad 0 < |w| < \exp(-\pi M). \tag{PS29}$$

Indeed $|w| = \exp(\pi \operatorname{Im} c)$, and two complex numbers have the same value of this exponential exactly when they differ by an even integer. Every point in the displayed punctured disc has a logarithm giving a $c \in \mathcal{P}$. The exponential has nonzero derivative, so these local inverse logarithms give the claimed quotient complex structure. The integer-period quotient seen by the middle family instead has coordinate w^2; it is covered two-to-one by (PS29). Its two lifts are distinguished by the exact order-two summand in (PS12). This statement classifies the parameters of the retained family; it does not assert that the family exhausts all deformations or all complex threefolds with the same singular-fibre types.

56 Hodge numbers, cotangent direct images, and the Frölicher page

This section first derives the one-parameter Hodge table and the complete Frölicher spectral sequence from the candidate-specific structure-sheaf calculation and the verified sphere topology. It then performs the missing logarithmic direct-image calculation through all three special fibres. The cyclic invariant lattices, logarithmic Kodaira–Spencer divisor, residue cancellation, and one remaining global scalar give $h^{1,2} = 1$ and $h^{1,1} = 2$; thus no Hodge number remains undetermined.

56.1 Vanishing of holomorphic forms

We first place the holomorphic-form calculation inside the workbench rather than using it as an unexpanded assertion from the reconstructed source.

Proposition 56.1 (candidate-specific holomorphic-form vanishing)**.** ***[VERIFIED]*** *For the constructed threefold,*

$$H^0(X, \Omega_X^p) = 0 \qquad (p = 1, 2, 3).$$

Proof. Over the marked cover of the smooth part, write the fibre coordinates as $\zeta = (\zeta_1, \zeta_2)$ and

$$\Pi = [Z \mid I_2], \qquad Z = \begin{pmatrix} 6\mu & \tau \\ \beta & \mu \end{pmatrix}.$$

Translation by $\lambda \in \Lambda$ sends $d\zeta$ to $d\zeta + \dot{\Pi}\lambda\, dz$. Compactness of each torus fibre therefore puts every translation-invariant holomorphic form into one of the following normal forms:

$$\begin{aligned} &p = 1: \quad a(z)\, dz + c(z) \cdot d\zeta, && c(z)\dot{Z}(z) = 0, \\ &p = 2: \quad dz \wedge (b(z) \cdot d\zeta), \\ &p = 3: \quad c(z)\, dz \wedge d\zeta_1 \wedge d\zeta_2. \end{aligned}$$

Indeed, fibrewise periodic coefficients are constant. A possible affine part in a coefficient vanishes after restriction to the real span of the two constant columns of Π; in degree two this also kills the putative $d\zeta_1 \wedge d\zeta_2$ term because $\dot{\tau}$ is nonzero on the smooth base.

The finite fillings are obtained from smooth torus families by free cyclic quotients, so the displayed coefficients extend holomorphically across the two elliptic lifts. At the cusp use the verified toric charts $t = w_0 w_1 w_2$. Substitution of

$$d\zeta_i = (2\pi i)^{-1} dx_i/x_i, \qquad ds = (2\pi i)^{-1} dt/t$$

and restriction successively to the coordinate lines and planes shows that the coefficients of $d\zeta_i$, ds, $ds \wedge d\zeta_i$, and $ds \wedge d\zeta_1 \wedge d\zeta_2$ vanish to order at least one in t. Indeed, on every unimodular monomial chart the triple $(d\log x_1, d\log x_2, d\log t)$ is a constant invertible linear combination of the $d\log w_j$. A coefficient $F(t)$ multiplying a nonzero logarithmic residue is holomorphic only if its constant term vanishes: restricting to $w_j = 0$ gives that residue as $F(0)$, so $F(t) = tG(t)$. Invertibility of the monomial change of basis applies this argument to each displayed coefficient. Thus every coefficient used below descends with the asserted regularity at all three points of the base.

The linear part of a deck transformation has the form

$$R_g = \begin{pmatrix} r_g & 0 \\ * & 1 \end{pmatrix}, \qquad \jmath_g := r_g^{-1} = c_g\tau + d_g, \qquad g^*(d\tau) = \jmath_g^{-2} d\tau.$$

For a one-form, the second component c_2 is a holomorphic invariant function on the base and vanishes at the cusp, hence is zero. The second column of $c\dot{Z} = 0$ then gives $c_1\dot{\tau} = 0$, so $c_1 = 0$. What remains is a holomorphic one-form on $\mathbb{P}^1$, and hence vanishes.

For a two-form, $b_2\, dz$ descends to a holomorphic one-form on $\mathbb{P}^1$ and is zero. The remaining form $\psi = b_1\, dz$ satisfies $g^*\psi = \jmath_g \psi$. Consequently

$$\psi^{\otimes 2} \otimes d\tau$$

is invariant. At the order-three and order-four points it descends as a cubic differential with a pole of order at most two, while the toric cusp estimate gives a pole of order at most one. It is therefore a section of

$$(\Omega^1_{\mathbb{P}^1})^{\otimes 3}(2p_1 + 2p_2 + p_0) \simeq \mathcal{O}_{\mathbb{P}^1}(-1),$$

so it is zero. Since $\dot{\tau}$ is nonzero on a dense open set, $\psi = 0$. The coefficient of a three-form obeys the same weight-one law and the same argument applies. Hence all three spaces vanish. □

56.2 The exact one-parameter Hodge diamond

Theorem 56.2 (Hodge diamond determined up to one integer)**.** ***[VERIFIED]*** *Put $m = h^{1,2}(X)$. Then $m \in \mathbb{Z}_{\geq 0}$ and the only nonzero Hodge numbers are*

$$h^{0,0} = h^{0,1} = h^{3,2} = h^{3,3} = 1, \qquad h^{1,1} = h^{2,2} = m + 1, \qquad h^{1,2} = h^{2,1} = m.$$

Equivalently,

$q \backslash p$	0	1	2	3
3	0	0	0	1
2	0	m	$m+1$	1
1	1	$m+1$	m	0
0	1	0	0	0

where the row index is q and the column index is p.

Proof. The independently reconstructed structure-sheaf direct images are

$$f_*\mathcal{O}_X = \mathcal{O}, \qquad R^1 f_*\mathcal{O}_X = \mathcal{O} \oplus \mathcal{O}(-1), \qquad R^2 f_*\mathcal{O}_X = \mathcal{O}(-1)$$

by theorem 55.11. Coherent Leray on $\mathbb{P}^1$ therefore gives

$$(h^{0,0}, h^{0,1}, h^{0,2}, h^{0,3}) = (1, 1, 0, 0).$$

Together with theorem 56.1 and Serre duality [44, §13, Theorem 4, p. 22], this fixes every entry except the four middle entries, with $h^{1,2} = h^{2,1} = m$ and $h^{1,1} = h^{2,2}$.

The verified integral sphere topology gives $c_1(X) = c_2(X) = 0$ and $\int_X c_3(X) = 2$. In complex dimension three,

$$\operatorname{td}(X) = 1 + \frac{1}{2}c_1 + \frac{1}{12}(c_1^2 + c_2) + \frac{1}{24}c_1c_2,$$
$$\operatorname{ch}(\Omega_X^1) = 3 - c_1 + \frac{1}{2}(c_1^2 - 2c_2) - \frac{1}{6}(c_1^3 - 3c_1c_2 + 3c_3).$$

Hirzebruch–Riemann–Roch for arbitrary compact complex manifolds [43, Theorem 3, p. 427] hence gives

$$\chi(\Omega_X^1) = -\frac{1}{2}\int_X c_3 = -1.$$

The already known zero entries reduce this identity to $-h^{1,1} + h^{1,2} = -1$, proving $h^{1,1} = m + 1$. □

This specializes the general constraints of Gray and Ugarte; a convenient published synthesis is Angella [48], while the second-page symmetry and inequalities originate in [51]. The argument above uses the stronger candidate-specific direct images and form vanishings, rather than assuming a general Hodge pattern for a hypothetical complex S^6.

Theorem 56.3 (the complete Frölicher page)**.** ***[VERIFIED]*** *The Frölicher spectral sequence of X does not degenerate at E_1, but it does degenerate at E_2. Apart from the endpoint copies of $\mathbb{C}$, the two nonzero $d_1 = \partial$ rows are exact complexes*

$$0 \longrightarrow \mathbb{C} \longrightarrow \mathbb{C}^{m+1} \longrightarrow \mathbb{C}^m \longrightarrow 0,$$

$$0 \longrightarrow \mathbb{C}^m \longrightarrow \mathbb{C}^{m+1} \longrightarrow \mathbb{C} \longrightarrow 0.$$

Their differential ranks are respectively $(1, m)$ and $(m, 1)$. Consequently

$$\dim_{\mathbb{C}} E_1 = 6 + 4m, \qquad \dim_{\mathbb{C}} E_2 = 2, \qquad E_2 = E_\infty.$$

Proof. Frölicher's original construction identifies E_1 with Dolbeault cohomology and E_∞ with the associated graded de Rham cohomology [42, p. 643, Lemmas 1–2 and Theorem 3]. The table in theorem 56.2 leaves only the displayed $q = 1$ and $q = 2$ row complexes between the endpoint classes. Since $H^k_{\mathrm{dR}}(X;\mathbb{C}) = 0$ for $1 \le k \le 5$, every middle $E_\infty^{p,q}$ is zero. There is only one a priori possible higher differential between middle positions, $d_2 : E_2^{0,2} \to E_2^{2,1}$, but already $E_1^{0,2} = H^2(X, \mathcal{O}_X) = 0$. Every other d_r with $r \ge 2$ has zero source or target by the same Hodge table. Thus $E_2 = E_\infty$, so the two rows must be exact. The ranks follow from their dimensions. □

The theorem explains why neither topology nor the Frölicher sequence alone can select a value of m: the two exact row complexes exist numerically for every $m \ge 0$. The singular-fibre calculation below supplies the missing analytic information.

56.3 Generic maximal variation and its elliptic zero

The most tempting shortcut to the remaining Hodge number is to treat the Kodaira–Spencer matrix as invertible everywhere. The following calculation shows precisely what is true and what is false.

Proposition 56.4 (determinant of the printed period derivative)**.** ***[VERIFIED]*** *With dots denoting differentiation in the coordinate on $\mathfrak{h}_z$, set*

$$K = \det(\dot Z) = 6\dot\mu^{\,2} - \dot\tau\dot\beta.$$

Then K is not identically zero, so the Kodaira–Spencer map has rank two on a dense open set. However $K(z_1) = 0$ at the chosen order-three elliptic point. In particular, the assertion $\dot Z \neq 0$ does not imply that $\dot Z$ is invertible, and direct images across the finite fibres require their actual cyclic invariant lattices.

Proof. At the cusp, theorem 10.2 gives

$$t = e^{2\pi i s}, \qquad Z = sB_0 + C(t), \qquad B_0 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix},$$

with C holomorphic at $t = 0$. Hence

$$\frac{dZ}{ds} = B_0 + 2\pi i\, tC'(t), \qquad \det\!\left(\frac{dZ}{ds}\right) = 1 + O(t).$$

The cusp coordinate s is locally biholomorphic on the marked cover, so K is nonzero near the punctured cusp and therefore is not identically zero.

Now put $\rho = e^{\pi i/3} = \tau(z_1)$. The difference between μ and the local affine solution

$$\mu_1 = \frac{2 - \tau}{3}$$

vanishes to order at least two at z_1: the homogeneous line has local generator of order two there. Since $\tau - \rho$ has a simple zero,

$$\dot\mu(z_1) = -\frac{1}{3}\dot\tau(z_1).$$

Write

$$\phi_1 = 2 - \frac{6(1-\mu)^2}{\tau}, \qquad \beta \circ g_1 = \beta + \phi_1.$$

Using $\rho^2 - \rho + 1 = 0$ and the preceding derivative gives

$$\dot\phi_1(z_1) = -\frac{2}{3}(1+\rho)\dot\tau(z_1).$$

The derivative of the linearized order-three action at its fixed point is $g_1'(z_1) = -\rho$. Differentiating the transformation law for β therefore yields

$$-(1+\rho)\dot\beta(z_1) = \dot\phi_1(z_1), \qquad \dot\beta(z_1) = \frac{2}{3}\dot\tau(z_1).$$

Substitution gives

$$K(z_1) = 6\left(-\frac{1}{3}\dot\tau(z_1)\right)^2 - \dot\tau(z_1)\left(\frac{2}{3}\dot\tau(z_1)\right) = 0.$$

□

The zero divisor of K cannot be read from the single equality $K(z_1) = 0$. It is determined below only after the invariant lattices at the two multiple fibres have been inserted. This distinction is essential: direct image does not commute with taking the dual lattice at a ramified coarse fibre.

56.4 Logarithmic cotangent direct images

Put

$$B = \mathbb{P}^1, \qquad \Sigma = p_0 + p_1 + p_2, \qquad D = f^{-1}(\Sigma)_{\mathrm{red}} = W + S_1 + S_2,$$

and set

$$\mathcal{L} = \Omega^1_B(\log\Sigma) \simeq \mathcal{O}_B(1), \qquad \mathcal{A} = \Omega^1_X(\log D), \qquad \mathcal{E} = \mathcal{A}/f^*\mathcal{L}. \tag{11.1}$$

At the cusp the local equations $t = z_1\cdots z_r$, $1 \le r \le 3$, show directly that $\mathcal{A}$ and $\mathcal{E}$ are locally free. At a finite fibre the covering family has divisor $s = 0$, the cyclic action is free on the total space, and $dt/t = m\,ds/s$; hence its locally free logarithmic sequence descends through the finite ètale quotient. Thus everywhere one has the logarithmic conormal sequence

$$0 \longrightarrow f^*\mathcal{L} \longrightarrow \mathcal{A} \longrightarrow \mathcal{E} \longrightarrow 0. \tag{11.2}$$

The local $d\log$ bases and their relative form, including $t = \prod z_i^{e_i}$ and analytic base change, are the logarithmic conormal formalism of [41, Chapter II, §3.2–3.3, pp. 72–75]; the display above also follows directly from the stated local equations.

Proposition 56.5 (the three logarithmic relative direct images)**.** ***[VERIFIED]*** *One has*

$$\begin{aligned} f_*\mathcal{E} &\simeq \mathcal{O} \oplus \mathcal{O}(-1),\\ R^1f_*\mathcal{E} &\simeq \mathcal{O}^{\oplus 2} \oplus \mathcal{O}(-1)^{\oplus 2},\\ R^2f_*\mathcal{E} &\simeq \mathcal{O} \oplus \mathcal{O}(-1). \end{aligned}$$

All three sheaves are locally free and commute with base change.

Proof. At the cusp, the toric coordinates give

$$\mathcal{E}|_{N_0} = \mathcal{O}_{N_0}[d\log x_1] \oplus \mathcal{O}_{N_0}[d\log x_2]. \tag{11.3}$$

Indeed a quotient shear changes each $d\log x_i$ only by a holomorphic multiple of $d\log t$, which is zero in $\mathcal{E}$. The already proved structure-sheaf calculation therefore gives free local direct images of ranks $2, 4, 2$ at p_0.

We now keep the ramification at p_j. Write

$$A = \mathbb{C}\{t\}, \qquad C = \mathbb{C}\{s\}, \qquad t = s^m, \qquad (m,a) = (3,2) \text{ or } (4,1).$$

Cartan's analytic quotient theorem identifies local holomorphic functions on the quotient with invariants [39, §1, Theorem 1, pp. 1–2]. The action on the total space is free, so the quotient map is finite étale; the corresponding locally free sheaves and their equivariant structures descend by finite étale descent. This

sheaf statement is not being attributed to Cartan's theorem on invariant functions. Finite pushforward is exact, and Reynolds averaging $v \mapsto |G_j|^{-1} \sum_{g \in G_j} gv$ makes the invariant functor exact in characteristic zero. Hence

$$R^q f_* \mathcal{E}|_{D_j} = \left(\pi_* R^q \widetilde{f}_* \widetilde{\mathcal{E}}\right)^{G_j}. \tag{11.3a}$$

Choose equivariant frames e_0, e_1 of the relative cotangent and b_0, b_1 of $R^1 \widetilde{f}_* \mathcal{O}$ whose semilinear characters are respectively

$$(1, \zeta^a), \qquad (1, \zeta^{-a});$$

b_0 is the nowhere-zero class $\gamma^{0,1}$. These may be chosen as holomorphic constant-character frames, rather than merely as eigenvectors on the central fibre. Indeed, if w lifts a ν-eigenvector at $s = 0$, then

$$P_\nu(w) = \frac{1}{m} \sum_{k=0}^{m-1} \nu^{-k} (g^*)^k w$$

satisfies $g^* P_\nu(w) = \nu P_\nu(w)$ and has the same nonzero central value; after shrinking the disc, the projected eigenvectors form a frame. On the marked torus family the projection formula gives

$$R^q \widetilde{f}_* \widetilde{\mathcal{E}} \simeq (\mathcal{O} e_0 \oplus \mathcal{O} e_1) \otimes \bigwedge^q (\mathcal{O} b_0 \oplus \mathcal{O} b_1), \qquad 0 \le q \le 2.$$

Solving $\nu \zeta^r = 1$ for the invariant exponent gives the following *actual* A-bases:

$$\begin{aligned} f_* \mathcal{E} :&\quad e_0,\ s^{m-a} e_1, \\ R^1 f_* \mathcal{E} :&\quad e_0 b_0,\ s^a e_0 b_1,\ s^{m-a} e_1 b_0,\ e_1 b_1, \\ R^2 f_* \mathcal{E} :&\quad s^a e_0 (b_0 \wedge b_1),\ e_1 (b_0 \wedge b_1). \end{aligned} \tag{11.4}$$

For comparison, the variable tangent lattice is generated by $s^a v_1$. Thus evaluation of the descended variable cotangent on the descended variable tangent is $s^m = t$, not a unit. This proves explicitly why

$$f_* \mathcal{E} \longrightarrow (f_* T_{X/B})^\vee$$

is multiplication by $\operatorname{diag}(1, t)$ locally, although $\mathcal{E}^\vee \simeq T_{X/B}$ before direct image. This last identity is an identity of locally free relative sheaves, not an assertion that formation of the kernel commutes with taking a tangent-space fibre. For example, at a triple chart

$$R = \mathbb{C}\{x, y, z\}, \qquad t = xyz,$$

one has

$$T_{X/B} = \ker\left[R^3 \xrightarrow{(yz, xz, xy)} R \right].$$

If $yz\,u + xz\,v + xy\,w = 0$, reduction modulo x and y gives $u = xU$ and $v = yV$, and cancellation in the domain gives $w = -z(U + V)$. Hence

$$T_{X/B} = R(x\partial_x - z\partial_z) \oplus R(y\partial_y - z\partial_z) = \mathcal{E}^\vee.$$

The kernel of the pointwise linear map df_x at the triple point is three-dimensional, whereas the fibre of this sheaf kernel is two-dimensional: tensoring the kernel sequence with the residue field is not left exact because the image ideal (xy, xz, yz) is not flat. Thus a section can be nowhere zero in $T_{X/B} = \mathcal{E}^\vee$ even when its image in the ambient tangent-space fibre vanishes.

The invariant vertical field ξ gives a trivial subline of $V := f_* T_{X/B}$; the quotient has the reciprocal automorphy of the Hodge line $\mathfrak{K} \simeq \mathcal{O}(1)$ from theorem 55.1. Since $H^1(\mathcal{O}(1)) = 0$, the resulting extension splits:

$$V \simeq \mathcal{O}\xi \oplus \mathcal{O}(-1), \qquad V^\vee \simeq \mathcal{O} \oplus \mathcal{O}(1).$$

Evaluation gives a generically invertible map $f_*\mathcal{E} \to V^\vee$. Formula (11.4) shows that its elementary divisors are $(1,t)$ at each of p_1, p_2 and $(1,1)$ elsewhere. Thus its cokernel has total length two and $\deg(f_*\mathcal{E}) = 1-2 = -1$. Evaluation on ξ is stalkwise surjective: the covector dual to the fixed eigenvector is a unit generator at the cusp and in both finite invariant bases. Its kernel is consequently a line bundle of degree -1, and

$$0 \longrightarrow \mathcal{O}(-1) \longrightarrow f_*\mathcal{E} \xrightarrow{\mathrm{ev}_\xi} \mathcal{O} \longrightarrow 0.$$

Because $H^1(\mathcal{O}(-1)) = 0$, this extension also splits, proving

$$f_*\mathcal{E} \simeq \mathcal{O} \oplus \mathcal{O}(-1).$$

Also, by theorem 55.11,

$$G := R^1 f_*\mathcal{O}_X = \mathcal{O} \oplus \mathcal{O}(-1), \qquad R^2 f_*\mathcal{O}_X = \det G = \mathcal{O}(-1). \tag{11.5}$$

Here is the global filtration implicit in those bases. Choose the degree-zero summand in the preceding split sequence so that its generator evaluates to one on ξ; locally this is the fixed eigen-covector e_0. Together with the fixed class $b_0 = \gamma^{0,1}$, take the corresponding saturated sublines in $f_*\mathcal{E}$ and G. Formula (11.3) at the cusp and the bases (11.4) at the finite points show that these sublines are locally generated by e_0, b_0 and that their quotients are locally generated by $s^{m-a}e_1, s^a b_1$; hence both inclusions and both quotients are locally free. The evaluation splitting and Hodge-line calculation identify the two filtrations as

$$0 \subset \mathcal{O} \subset f_*\mathcal{E}, \qquad 0 \subset \mathcal{O} \subset G, \qquad \mathrm{gr}(f_*\mathcal{E}) = \mathrm{gr}(G) = \mathcal{O} \oplus \mathcal{O}(-1).$$

Over the smooth locus the cup-product identification is $R^1 f_*\mathcal{E} = (f_*\mathcal{E}) \otimes G$. Formula (11.4) extends its four character lines as saturated rank-one subsheaves over all of B. Three associated-graded lines are generated locally by $e_0 b_0, s^a e_0 b_1, s^{m-a} e_1 b_0$ and have degrees $0, -1, -1$. On the fourth, the separately descended factors would give

$$(s^{m-a}e_1)(s^a b_1) = t(e_1 b_1),$$

whereas $e_1 b_1$ itself is invariant. Thus that graded line is the upper modification of $\mathcal{O}(-2)$ at $p_1 + p_2$, hence is $\mathcal{O}$. Similarly, the two graded lines of $R^2 f_*\mathcal{E}$ are $\mathcal{O}(-1)$ and $\mathcal{O}(-2)(p_1+p_2) = \mathcal{O}$. Every extension between the resulting degree-zero and degree-minus-one lines splits: its Ext^1 is one of $H^1(\mathcal{O}(-1))$, $H^1(\mathcal{O})$, or $H^1(\mathcal{O}(1))$, all zero. This proves the three asserted splittings. This is also the elementary filtered form of the Birkhoff–Grothendieck classification [46, Theorem 2.1, p. 126]; the divisor modifications and Ext^1 identifications may equivalently be read in [37, Tags 0C6K, 0B3A, and 01XT].

Finally, (11.3a)–(11.4) make every direct image free of ranks $2, 4, 2$ at the finite points; (11.3) does the same at the cusp, and the smooth locus is standard. The multiple-fibre base change can also be read directly from the invariant modules. The Reynolds idempotent is A-linear, so invariants commute with arbitrary finite A-base change. For the scheme-theoretic special fibre,

$$C \otimes_A A/(t) = C/(s^m), \qquad \big((C/(s^m))v\big)^G = \mathbb{C}\, s^r v$$

on a character line whose invariant exponent satisfies $0 \le r < m$. Every basis vector in (11.4) therefore survives reduction modulo t, and the special-fibre dimensions are exactly $2, 4, 2$. No R^q with $q > 2$ occurs. Descending induction in analytic cohomology and base change then makes every base-change map an isomorphism, including for the nonreduced finite fibres; precisely the same criterion, with citation, is recorded in the proof of theorem 55.11; see in particular [7, Chapter III, §3, Theorem 3.4 and Corollary 3.10]. □

Proposition 56.6 (the logarithmic Kodaira–Spencer divisor)**.** ***[VERIFIED]*** *The connecting maps of (11.2) are*

$$\kappa_0 : f_*\mathcal{E} \longrightarrow \mathcal{L} \otimes R^1 f_*\mathcal{O}_X,$$
$$\kappa_1 : R^1 f_*\mathcal{E} \longrightarrow \mathcal{L} \otimes R^2 f_*\mathcal{O}_X \simeq \mathcal{O}.$$

Then

$$\operatorname{div}(\det \kappa_0) = p_1 + p_2, \qquad \kappa_1 \text{ is surjective as a sheaf.} \tag{11.6}$$

In particular κ_0 *is injective and* $\operatorname{coker} \kappa_0 = \mathbb{C}_{p_1} \oplus \mathbb{C}_{p_2}$. *Moreover the raw period determinant has*

$$\operatorname{ord}_{z_1} K = 2, \qquad \operatorname{ord}_{z_2} K = 0. \tag{11.7}$$

Proof. On the smooth locus, the connecting construction and its exterior-product Leibniz rule are the Gauss–Manin/Kodaira–Spencer formalism of [47, §2, equations (5)–(10), pp. 202–204]. Katz–Oda work with a smooth algebraic morphism; the citation is not being used as a logarithmic analytic special-fibre theorem. In the present family, Deligne's local monomial calculation supplies the locally split logarithmic sequence (11.2) [41, Chapter II, §3.3]. Taking a local splitting and a Čech representative, the connecting cochain is the difference of the two splittings. Since exterior product of cochains obeys

$$\delta(\alpha \wedge \beta) = \delta\alpha \wedge \beta + (-1)^{\deg \alpha} \alpha \wedge \delta\beta,$$

the two displayed connecting maps extend across every logarithmic chart and satisfy the same Leibniz rule. Thus the singular-fibre formula below is a direct calculation from (11.2), not an extrapolation of the Katz–Oda hypotheses. The determinant is a section of

$$\det(\mathcal{L} \otimes G) \otimes \det(f_*\mathcal{E})^{-1} = \mathcal{L}^{\otimes 2} \otimes \det G \otimes \det(f_*\mathcal{E})^{-1} \simeq \mathcal{O}(2) \otimes \mathcal{O}(-1) \otimes \mathcal{O}(1) \simeq \mathcal{O}(2). \tag{11.8}$$

At the cusp, $Z = sB_0 + C(t)$ and $dZ/ds = B_0 + O(t)$, so it is a unit. At a finite point use $\ell = dt/t = m\,ds/s$. In invariant determinant frames with no normalization imposed against the raw marked frames, direct calculation gives

$$\det \kappa_0 = \varepsilon(t) m^{-2} s^{m+2-2a} K(s), \qquad \varepsilon(0) \neq 0. \tag{11.9}$$

Here ε is the nowhere-zero invariant holomorphic unit recording the two frame normalizations; compatible eigenframes may be normalized so that $\varepsilon = 1$. The dot in K means d/ds in the chosen linearizing coordinate. Replacing s by another such coordinate changes only the unit, so every divisor and order calculation below is coordinate-independent. Equivariance gives

$$\operatorname{ord}_s K \equiv 2a - 2 \pmod{m}. \tag{11.10}$$

At $(m, a) = (3, 2)$, theorem 56.4 supplies $K(z_1) = 0$; hence the right side of (11.9) has coarse order at least one. At $(m, a) = (4, 1)$, the prefactor is $s^4 = t$, so the coarse order is again at least one. The two forced zeros exhaust the degree in (11.8). They are therefore simple, there are no others, and (11.7) follows from (11.9)–(11.10).

For completeness, we derive the coarse matrices rather than only recording their shape. Let

$$D_{\mathrm{KS}}(s) = (d_{rs}(s))_{0 \le r,s \le 1}$$

be the Kodaira–Spencer matrix on the marked cover after the common factor ds has been removed. Since $t = s^m$ and $ds = (s/m)\ell$, passage from the marked frames to the actual invariant frames in (11.4) gives

$$A = \frac{s}{m} d_{00}, \qquad B = \frac{s^{1-a}}{m} d_{01}, \qquad C = \frac{s^{1-a}}{m} d_{10}, \qquad D_0 = \frac{s^{m+1-2a}}{m} d_{11}. \tag{11.10a}$$

The character of each expression in (11.10a) is trivial, so each is a holomorphic function of t. In the bases $(e_0, s^{m-a}e_1)$ and $(b_0, s^a b_1)$ one consequently has the exact matrix

$$[\kappa_0] = \begin{pmatrix} A & tB \\ C & D_0 \end{pmatrix}. \tag{11.10b}$$

Its determinant is (11.9). Applying the exterior-product Leibniz rule to the four ordered generators

$$e_0b_0, \quad s^a e_0 b_1, \quad s^{m-a} e_1 b_0, \quad e_1 b_1$$

now gives, without a change of basis,

$$[\kappa_1] = (-C,\ A,\ -D_0,\ B) \tag{11.11}$$

into the invariant generator $s^a(b_0 \wedge b_1)\ell$. If (11.11) were not surjective at a finite point, all four entries would lie in (t) and $\det \kappa_0$ would have order at least two, contrary to (11.6). Away from p_1, p_2, the determinant divisor (11.6) makes κ_0 invertible, so the exterior-product row (11.11) contains a unit and is surjective there as well. At the cusp this reduction is not an additional choice. Use the logarithmic base frame $\ell = ds = (2\pi i)^{-1} dt/t$, the relative cotangent frame $(e_0, e_1) = ([d\log x_1], [d\log x_2])$ of (11.3), and the quotient frame (γ, u) of G, which extends across the cusp by theorem 55.11. In these frames the smooth-locus Kodaira–Spencer matrix is the transpose of the derivative of the period matrix: for $h = ae_0 + be_1$,

$$\kappa_0(h) = (6a\dot{\mu} + b\dot{\beta})\gamma + (a\dot{\tau} + b\dot{\mu})u. \tag{11.11a}$$

Here the dot is d/ds. The cusp expansion $Z = sB_0 + C(t)$, $t = e^{2\pi i s}$, gives

$$\frac{dZ}{ds} = B_0 + 2\pi i\, tC'(t), \qquad B_0 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

These logarithmic frames extend over $t = 0$, so (11.11a) reduces exactly to B_0^{t}. Write the cusp matrix, separately from the finite-fibre normalization above, as $[\kappa_0]_{p_0} = \left(\begin{smallmatrix} A & E \\ C & D \end{smallmatrix}\right)$. In the same ordered exterior basis used for (11.11), its Leibniz row is $(-C, A, -D, E)$. Substituting $(A, E, C, D) = (0, -1, 1, 0)$ gives the second reduction. Thus, in the named frames,

$$[\kappa_0] \equiv \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \pmod{t}, \qquad [\kappa_1] \equiv (-1, 0, 0, -1) \pmod{t},$$

exactly. Under arbitrary holomorphic changes of source and target frames the matrices change by invertible left and right factors; in particular invertibility of κ_0 and the existence of a unit entry in the row of κ_1 are unchanged. Thus κ_1 is surjective everywhere. □

56.5 Residues and the absolute cotangent direct images

Let $i_D : D \hookrightarrow X$, and let

$$\nu : \widetilde{D} = \mathrm{dP}_6 \sqcup S_1 \sqcup S_2 \longrightarrow D$$

be the normalization. In formulas below, $\nu_*\mathcal{O}_{\widetilde{D}}$ abbreviates $(i_D \circ \nu)_*\mathcal{O}_{\widetilde{D}}$ as a sheaf on X. The ordinary/logarithmic local calculation for $z_1 \cdots z_r = 0$ glues to

$$0 \longrightarrow \Omega^1_X \longrightarrow \mathcal{A} \xrightarrow{\mathrm{res}} \nu_*\mathcal{O}_{\widetilde{D}} \longrightarrow 0. \tag{11.12}$$

The normalization in the target is indispensable at the self-crossing cusp. This is the degree-one residue sequence associated with the normalized strata of a normal-crossings divisor [41, Chapter II, §3.4–3.7, pp. 76–78]; the displayed local calculation fixes its convention in the present self-crossing model.

Proposition 56.7 (absolute cotangent direct images)**.** ***[VERIFIED]*** *Put $\mathcal{K} = \ker \kappa_1$. Then*

$$f_*\Omega^1_X \simeq \mathcal{O}(-2), \qquad R^1 f_*\Omega^1_X \simeq \mathcal{K}, \qquad R^2 f_*\Omega^1_X \simeq \mathcal{O} \oplus \mathcal{O}(-1). \tag{11.13}$$

None of these sheaves has torsion.

Proof. The three normalized components have

$$R^0 f_*\nu_*\mathcal{O}_{\widetilde{D}} = \mathbb{C}_{p_0} \oplus \mathbb{C}_{p_1} \oplus \mathbb{C}_{p_2}, \quad R^1 f_*\nu_*\mathcal{O}_{\widetilde{D}} = \mathbb{C}_{p_1} \oplus \mathbb{C}_{p_2}, \quad R^2 = 0. \tag{11.14}$$

Indeed dP_6 is rational and S_1, S_2 are bielliptic, with $h^1(\mathcal{O}_{S_j}) = 1$ and $h^2(\mathcal{O}_{S_j}) = 0$.

The long direct-image sequence of (11.2), together with theorem 56.6, gives

$$f_*\mathcal{A} = \mathcal{O}(1), \qquad 0 \longrightarrow T \longrightarrow R^1 f_*\mathcal{A} \longrightarrow \mathcal{K} \longrightarrow 0, \qquad R^2 f_*\mathcal{A} = \mathcal{O} \oplus \mathcal{O}(-1), \tag{11.15}$$

where $T = \mathbb{C}_{p_1} \oplus \mathbb{C}_{p_2}$. The residue of the base logarithmic form is weighted evaluation:

$$\mathcal{O}(1) \longrightarrow \mathbb{C}_{p_0} \oplus \mathbb{C}_{p_1} \oplus \mathbb{C}_{p_2}, \qquad q \longmapsto (q(p_0),\, 3q(p_1),\, 4q(p_2)). \tag{11.16}$$

It is surjective as a sheaf and has kernel $\mathcal{O}(1)(-p_0 - p_1 - p_2) = \mathcal{O}(-2)$.

It remains to identify the torsion in (11.15). At p_j, the residue map on $G \otimes \mathbb{C}_{p_j}$ sends

$$x b_0 + y s^a b_1 \longmapsto m_j x[b_0] \in H^1(S_j, \mathcal{O}_{S_j}). \tag{11.17}$$

More precisely, after trivializing $\mathcal{L}$ by $\ell = dt/t$, the logarithmic conormal and residue maps form the exact commutative diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & f^*\mathcal{L} & \longrightarrow & \mathcal{A} & \longrightarrow & \mathcal{E} & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle \mathrm{res}_B} & & \downarrow{\scriptstyle \mathrm{res}_{S_j}} & & \downarrow{\scriptstyle 0} & & \\ 0 & \longrightarrow & i_{j*}\mathcal{O}_{S_j} & \xrightarrow{=} i_{j*}\mathcal{O}_{S_j} & \longrightarrow & 0 & & & \end{array} \tag{11.16a}$$

with

$$\mathrm{res}_B(q\ell) = m_j q|_{S_j}. \tag{11.16b}$$

Applying Rf_* to (11.16a) gives the typed composite

$$f_*\mathcal{E}|_{p_j} \xrightarrow{\kappa_0} (\mathcal{L} \otimes G)|_{p_j} \xrightarrow{\mathrm{res}_{1,j}} H^1(S_j, \mathcal{O}_{S_j}), \qquad \mathrm{res}_{1,j}\,\kappa_0 = 0, \tag{11.16c}$$

and (11.16b) gives exactly

$$\mathrm{res}_{1,j}(x b_0 + y s^a b_1) = m_j x[b_0].$$

Thus its kernel is the one-dimensional line $\mathbb{C}\, s^a b_1$. The simple zero of $\det \kappa_0$ proved in theorem 56.6 implies that $\kappa_0 \bmod t$ has rank one; the zero composite in (11.16c) therefore forces

$$\mathrm{im}(\kappa_0 \bmod t) = \ker \mathrm{res}_{1,j} = \mathbb{C}\, s^a b_1. \tag{11.17a_{fin}}$$

Consequently the induced map $\operatorname{coker} \kappa_0 \to H^1(S_j, \mathcal{O}_{S_j})$ is an isomorphism at each finite fibre. Thus each summand $T_{p_j} \to H^1(S_j, \mathcal{O}_{S_j})$ is an isomorphism. In the long sequence of (11.12), (11.16) first removes the entire degree-zero residue, and (11.17) then cancels T. What remains is precisely (11.13). □

Remark 56.8 (the cusp is not the torsion locus). At p_0, if $Q = \nu_*\mathcal{O}_{\mathrm{dP}_6}/\mathcal{O}_W$, then

$$0 \longrightarrow \Omega^1_{N_0/\Delta} \longrightarrow \mathcal{E}|_{N_0} \longrightarrow i_*Q \longrightarrow 0, \qquad h^\bullet(Q) = (2, 1, 0). \tag{11.17a}$$

This relative sequence is the right column of the following exact normalization–residue diagram; all sheaves in the bottom row are regarded as sheaves on N_0 by i_*:

$$\begin{array}{ccccccccc} & & 0 & & 0 & & 0 & & \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & f^*\Omega^1_\Delta & \longrightarrow & \Omega^1_{N_0} & \longrightarrow & \Omega^1_{N_0/\Delta} & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & f^*\Omega^1_\Delta(\log 0) & \longrightarrow & \Omega^1_{N_0}(\log W) & \longrightarrow & \mathcal{E}|_{N_0} & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle \mathrm{res}_B} & & \downarrow{\scriptstyle \mathrm{res}_W} & & \downarrow{\scriptstyle \rho} & & \\ 0 & \longrightarrow & i_*\mathcal{O}_W & \longrightarrow & i_*\nu_*\mathcal{O}_{\mathrm{dP}_6} & \longrightarrow & i_*Q & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ & & 0 & & 0 & & 0 & & \end{array} \tag{11.17a$'$}$$

The first two columns are the base and normalized-divisor residue sequences; the bottom row is the definition of Q. Hence the snake lemma identifies the right column with (11.17a), including its displayed map ρ.

The exactness assertion is also visible stalkwise. At a double chart $t = xy$, with one coordinate along the double stratum and logarithmic generator e, the image of ordinary relative differentials in $\mathcal{E}$ contains the stratum differential and has transverse part $(x, y)e$; its cokernel is therefore $R/(x, y)$. At a triple chart $t = xyz$, write $\mathcal{E} = Re_1 \oplus Re_2$ with $d\log x = e_1$, $d\log y = e_2$, and $d\log z = -e_1 - e_2$. The image is generated by

$$xe_1, \qquad ye_2, \qquad z(e_1 + e_2),$$

Writing a normalized branch value as a triple modulo the diagonal, the quotient map is explicitly

$$e_1 \longmapsto [(1, 0, 0)], \qquad e_2 \longmapsto [(0, 1, 0)]. \tag{11.17a$''$}$$

Its kernel is exactly $\langle xe_1, ye_2, z(e_1 + e_2)\rangle$, so this is the standard local presentation of the normalization quotient Q. Away from the double locus the map is an isomorphism. This proves (11.17a) stalkwise. Here are the maps, rather than only their ranks. Quotienting the normalization complex (9.10) by $\mathcal{O}_W$ gives

$$0 \longrightarrow Q \longrightarrow \bigoplus_{k=1}^{3} \mathcal{O}_{D_k} \xrightarrow{\delta_1} \mathbb{C}_P \oplus \mathbb{C}_Q \longrightarrow 0.$$

Because $D_k \simeq \mathbb{P}^1$, its cohomology is the cohomology of

$$\mathbb{C}^3 \longrightarrow \mathbb{C}^2, \qquad (a_1, a_2, a_3) \longmapsto (s, s), \qquad s = a_1 - a_2 + a_3.$$

Thus

$$H^0(Q) = \{(a_1, a_2, a_3) : a_1 - a_2 + a_3 = 0\} \simeq \mathbb{C}^2, \qquad H^1(Q) = \mathbb{C}^2/\mathbb{C}(1, 1) \simeq \mathbb{C}, \tag{11.17b}$$

and $H^2(Q) = 0$. Since dP_6 is rational, the connecting maps for $0 \to \mathcal{O}_W \to \nu_*\mathcal{O}_{\mathrm{dP}_6} \to Q \to 0$ are isomorphisms

$$\delta_0 : H^0(Q) \xrightarrow{\sim} H^1(W, \mathcal{O}_W), \qquad \delta_1 : H^1(Q) \xrightarrow{\sim} H^2(W, \mathcal{O}_W). \tag{11.17c}$$

At a triple chart $t = z_0z_1z_2$, a relative logarithmic form is represented by $\eta = \sum a_i\, d\log z_i$, with holomorphic coefficient triple (a_0, a_1, a_2) taken modulo $\mathcal{O}(1, 1, 1)$. Restricting the branch differences to the three double strata gives

$$\rho(\eta) = (a_0 - a_1,\ a_0 - a_2,\ a_1 - a_2). \tag{11.17d}$$

Formula (11.17d) is written in the P-ordering of the globally labelled double curves. The Q-point sign is fixed by a separate calculation, not by an implicit relabelling. In the common e_1, e_2-plane, use the ordered branch vertices

$$(b_0, b_1, b_2) = (0, e_1, e_2), \qquad (c_0, c_1, c_2) = (e_1, e_1 + e_2, e_2)$$

at P and Q, respectively. The coefficient triples of $d\log x_1, d\log x_2$ are

$$(0, 1, 0),\ (0, 0, 1) \quad \text{at } P, \qquad (1, 1, 0),\ (0, 1, 1) \quad \text{at } Q.$$

The globally labelled branch-difference maps are

$$d_P(a) = (a_0 - a_1, a_0 - a_2, a_1 - a_2), \qquad d_Q(a) = (a_2 - a_1, a_0 - a_1, a_0 - a_2). \tag{11.17d$'$}$$

Direct substitution gives, at both triple points, the same two global sections

$$q_1 = (-1, 0, 1), \qquad q_2 = (0, -1, -1). \tag{11.17d$''$}$$

For constant coefficients this induces an isomorphism $\mathbb{C}^3/\mathbb{C}(1,1,1) \to H^0(Q)$. In the toric chart $(z_0,z_1,z_2) = (t/(x_1x_2), x_1, x_2)$ it sends

$$d\log x_1 \longmapsto (-1,0,1), \qquad d\log x_2 \longmapsto (0,-1,-1),$$

which is a basis of the hyperplane in (11.17b); unimodular changes of cusp chart merely change this basis by $\mathrm{GL}_2(\mathbb{Z})$.

Write $\eta_1 = d\log x_1$, $\eta_2 = d\log x_2$, take the q_i of (11.17d″), and put $\alpha_i = \delta_0(q_i)$. By (11.17c), α_1, α_2 form a basis of $H^1(W, \mathcal{O}_W)$. Naturality of the connecting maps fixes a sign that must be retained. With the standard Čech differential $(\delta c)_{i_0\dots i_{r+1}} = \sum_k (-1)^k c_{i_0\dots\widehat{i_k}\dots i_{r+1}}$, lift q_i to the normalization complex. The total differential of a degree-one cocycle α times that lift is $(-1)^1 \alpha \smile \delta\widetilde{q}_i$. The projection formula therefore gives exactly

$$\delta_1\big(\rho_*(\alpha \otimes \eta_i)\big) = -\alpha \smile \delta_0(q_i) = -\alpha \smile \alpha_i.$$

The cup product in theorem 55.10 is an isomorphism. Define, with this convention,

$$\theta = -\delta_1^{-1}(\alpha_1 \smile \alpha_2).$$

Graded commutativity now gives, with no residual sign ambiguity,

$$\rho_* = (0,1,-1,0) : \mathbb{C}\langle \alpha_1\eta_1, \alpha_1\eta_2, \alpha_2\eta_1, \alpha_2\eta_2\rangle \longrightarrow \mathbb{C}\theta. \tag{11.17e}$$

In particular the map is surjective. Applying direct images to (11.17a), the first map is reduction modulo t followed by the isomorphism (11.17d), and the second is (11.17e). Consequently

$$f_*\Omega^1_{N_0/\Delta} = t\mathcal{O}^2, \quad R^1f_*\Omega^1_{N_0/\Delta} = \ker(\mathcal{O}^4 \to \mathbb{C}_{p_0}), \quad R^2f_*\Omega^1_{N_0/\Delta} = \mathcal{O}^2.$$

For the row in (11.17e), the middle kernel is generated by e_1, e_4, e_2+e_3, te_2. Thus $\Omega^1_{X/B}$ is torsion-free in a neighbourhood of the cusp, but non-locally-free along its xy and xyz strata. Globally its torsion is supported precisely on the two multiple fibres, as the following local module calculation proves. On the smooth cover J_j', let $R = \mathcal{O}_{J_j',x}$ and choose coordinates (s, y_1, y_2) smooth over the s-disc. Relative to the original base B, where $t = s^{m_j}$, one has

$$\Omega^1_{J_j'/B,x} = \frac{R\,ds \oplus R\,dy_1 \oplus R\,dy_2}{(m_j s^{m_j-1} ds)} \simeq R\,dy_1 \oplus R\,dy_2 \oplus \big(R/(s^{m_j-1})\big)ds. \tag{11.17f}$$

Thus the last summand is exactly the torsion submodule. The quotient map $q : J_j' \to N_j$ is finite étale, so the relative cotangent sequence gives

$$q^*\Omega^1_{N_j/B} \simeq \Omega^1_{J_j'/B}.$$

Faithful flatness descends both the presence of the displayed torsion and its absence off $s=0$. Hence ordinary relative torsion is supported exactly on the two multiple fibres and is absent at the cusp. The reconstructed manuscript's assertion that torsion occurs at all three singular fibres is false.

56.6 The scalar and the exact Hodge diamond

Theorem 56.9 (the remaining scalar is nonzero)**.** ***[VERIFIED]*** *The induced map*

$$H^0(B, R^1f_*\mathcal{E}) \simeq \mathbb{C}^2 \xrightarrow{\;H^0(\kappa_1)\;} H^0(B,\mathcal{O}) \simeq \mathbb{C}$$

has rank one.

Proof. Every matrix R_g in (2.3d) has second column $(0,1)^{\mathsf{t}}$. Consequently the marked vertical field

$$\xi = \frac{\partial}{\partial \zeta_2}$$

is invariant under every deck transformation. The affine translations in theorem 10.9 have identity differential in this direction, and the corresponding toric logarithmic direction extends across the cusp charts of theorem 10.6. Thus ξ is a global section nowhere zero in the locally free relative sheaf $T_{X/B} = \mathcal{E}^{\vee}$ and is the section used in the evaluation sequence

$$0 \longrightarrow \mathcal{O}(-1) \longrightarrow f_*\mathcal{E} \xrightarrow{\mathrm{ev}_\xi} \mathcal{O} \longrightarrow 0.$$

This sequence splits, and the lift h of $1 \in H^0(\mathcal{O})$ is unique: the difference of two lifts is a global section of $\mathcal{O}(-1)$, while $H^0(B, \mathcal{O}(-1)) = 0$. On the marked cover write that unique lift as

$$h = (a(z), 1)\begin{pmatrix} d\zeta_1 \\ d\zeta_2 \end{pmatrix}. \tag{11.18}$$

The function a is holomorphic on each finite cover because h is a section of the actual invariant lattice, not merely of its generic fibre. For g_2, invariance of (11.18) and $R_{g_2} = \left(\begin{smallmatrix} 1/\tau & 0 \\ -\mu/\tau & 1 \end{smallmatrix}\right)$ mean exactly

$$(a(z), 1) = (a(g_2 z), 1) R_{g_2}(z), \qquad \text{hence} \qquad a(g_2 z) = \tau(z) a(z) + \mu(z). \tag{11.18a}$$

At the fixed point z_2, the normalized values $\tau(z_2) = i$ and $\mu(z_2) = (1 - i)/2$ therefore give

$$a(z_2) = \frac{1}{2}. \tag{11.18b}$$

The cup-product map $f_*\mathcal{E} \otimes G \to R^1 f_*\mathcal{E}$ sends $h \otimes \gamma$ to the unit generator $e_0 b_0$ of the degree-zero character line in (11.4). This identifies the precise entry of the row (11.11) that must be evaluated.

Normalize the order-four linearizing coordinate by

$$s = (z_2 - \bar{z}_2) s_2, \qquad s \circ g_2 = -is, \qquad \frac{ds}{dz}(z_2) = 1.$$

Write a dot for d/ds, and put $\ell = dt/t = 4\, ds/s$. In the quotient basis (γ, u) of G, the marked period derivative and $ds = (s/4)\ell$ give the exact logarithmic map

$$\kappa_0(h) = \frac{s}{4}\Big[(6a\dot{\mu} + \dot{\beta})\gamma + (a\dot{\tau} + \dot{\mu})u\Big] \otimes \ell. \tag{11.19}$$

For $(m, a) = (4, 1)$, the actual invariant target generator from (11.4) is $s(b_0 \wedge b_1)\ell$. Normalize b_1 so that

$$\gamma \wedge b_1 = u_2(s)(\gamma \wedge u), \qquad u_2(0) = 1;$$

equivalently, use the invariant frame

$$\omega_2 = s\, u_2(s)(\gamma \wedge u)\ell. \tag{11.19f}$$

The uncorrected expression $s(\gamma \wedge u)\ell$ is only invariant at $s = 0$: indeed $g_2^*(\gamma \wedge u) = \tau(z)(\gamma \wedge u)$, so its pullback factor is $(-i)\tau(z)$. Formula (11.11), in the genuine invariant frame (11.19f), sends the first ordered source generator $e_0 b_0$ to minus the second coefficient of $\kappa_0(e_0)$. Since $u_2(0) = 1$, (11.19) therefore gives the exact signed coarse value

$$\kappa_1(h \otimes \gamma)(p_2) = -\frac{1}{4}\big(a(z_2)\dot{\tau}(z_2) + \dot{\mu}(z_2)\big). \tag{11.19a}$$

At the order-four lift z_2, the quotient coordinate has order four while the modular invariant has order two at $\tau = i$; see [26, Chapter VII]. Equivalently, $j(\tau) - 1728$ has a double zero at i and $j(\tau(z)) - 1728$ has order four at z_2. Hence

$$\operatorname{ord}_{z_2}(\tau - i) = 2, \qquad \dot{\tau}(z_2) = 0. \tag{11.20}$$

By (11.7), $K(z_2) \neq 0$; since $K = 6\dot{\mu}^2 - \dot{\tau}\dot{\beta}$, (11.20) gives $K(z_2) = 6\dot{\mu}(z_2)^2 \neq 0$. Substitution in (11.19a) therefore gives

$$\kappa_1(h \otimes \gamma)(p_2) = -\frac{1}{4}\dot{\mu}(z_2) \neq 0. \tag{11.20a}$$

The image is a global section of $\mathcal{L} \otimes R^2 f_* \mathcal{O}_X \simeq \mathcal{O}_B$; because it is nonzero at p_2, it is a nonzero constant. Hence $H^0(\kappa_1)$ has rank one. □

Corollary 56.10 (exact Hodge diamond)**.** ***[VERIFIED]*** *The last parameter in theorem 56.2 is*

$$h^{1,2}(X) = 1.$$

Consequently $h^{1,1} = 2$*, and the Hodge table is*

$q\backslash p$	0	1	2	3
3	0	0	0	1
2	0	1	2	1
1	1	2	1	0
0	1	0	0	0.

The two nontrivial Frölicher d_1 *rows are both exact complexes* $0 \to \mathbb{C} \to \mathbb{C}^2 \to \mathbb{C} \to 0$*, and* $\dim E_1 = 10$*,* $\dim E_2 = 2$*.*

Proof. Put $\mathcal{K} = \ker \kappa_1$. Since

$$0 \longrightarrow \mathcal{K} \longrightarrow \mathcal{O}^{\oplus 2} \oplus \mathcal{O}(-1)^{\oplus 2} \xrightarrow{\kappa_1} \mathcal{O} \longrightarrow 0$$

and the global map has rank one, its restriction to the two trivial summands is a nonzero constant row (the negative summands have no global sections). After a constant change of basis it is the identity on one copy of $\mathcal{O}$. Subtracting the remaining component from that summand makes the map the projection, so its kernel is

$$\mathcal{K} \simeq \mathcal{O} \oplus \mathcal{O}(-1)^{\oplus 2}.$$

By theorem 56.7 and coherent Leray,

$$h^{1,2} = h^1(B, \mathcal{K}) + h^0(B, \mathcal{O} \oplus \mathcal{O}(-1)) = 0 + 1 = 1.$$

The remaining assertions follow from theorems 56.2 and 56.3. □

56.7 Bott–Chern and Aeppli dimensions

For a compact complex manifold write

$$H^{p,q}_{BC}(X) = \frac{\ker \partial \cap \ker \bar{\partial} \subset A^{p,q}(X)}{\operatorname{im} \partial\bar{\partial}}, \qquad H^{p,q}_{A}(X) = \frac{\ker \partial\bar{\partial} \subset A^{p,q}(X)}{\operatorname{im} \partial + \operatorname{im} \bar{\partial}}.$$

Theorem 56.11 (exact Bott–Chern and Aeppli diamonds)**.** ***[VERIFIED]*** *The Bott–Chern dimensions of the candidate are*

$q\backslash p$	0	1	2	3
3	0	0	1	1
2	0	1	2	1
1	0	2	1	0
0	1	0	0	0,

(11.21)

and its Aeppli dimensions are

$q\backslash p$	0	1	2	3
3	0	0	0	1
2	0	1	2	0
1	1	2	1	0
0	1	1	0	0.

(11.22)

In total degree $k = 0, \dots, 6$ this is

$$(h^k_{BC}) = (1, 0, 2, 2, 2, 2, 1), \qquad (h^k_A) = (1, 2, 2, 2, 2, 0, 1). \tag{11.23}$$

Consequently, set

$$\Delta^k = h^k_{BC} + h^k_A - 2b_k$$

for the degree-k non-$\partial\bar\partial$ defects. The Angella–Tomassini inequality [45, Theorem A] gives $\Delta^k \geq 0$ for every compact complex manifold; here the exact values are

$$(\Delta^k) = (0, 2, 4, 4, 4, 2, 0). \tag{11.24}$$

Proof. McHugh associates to a compact complex manifold the finite-dimensional quotient

$$K^{p,q} = \frac{\ker\partial \cap \ker\bar\partial \subset A^{p,q}}{\big(\operatorname{im}\partial \cap \ker\bar\partial\big) + \big(\operatorname{im}\bar\partial \cap \ker\partial\big)}, \qquad k^{p,q} = \dim_{\mathbb{C}} K^{p,q}.$$

Put

$$Z^{p,q} = \ker\partial \cap \ker\bar\partial \subset A^{p,q}, \qquad B^{p,q}_K = \big(\operatorname{im}\partial \cap \ker\bar\partial\big) + \big(\operatorname{im}\bar\partial \cap \ker\partial\big).$$

The natural map $Z^{p,q} \to H^{p+q}_{\mathrm{dR}}(X;\mathbb{C})$ need not descend to an injection from $K^{p,q}$; the dimension estimate has the opposite quotient mechanism. If a pure form $\alpha \in Z^{p,q}$ is d-exact, the (p,q) component of a primitive gives

$$\alpha = \partial\beta^{p-1,q} + \bar\partial\beta^{p,q-1}.$$

Because $\bar\partial\alpha = 0$ and $\partial\alpha = 0$, respectively, $\partial\beta^{p-1,q}$ is $\bar\partial$-closed and $\bar\partial\beta^{p,q-1}$ is ∂-closed. Thus $\ker(Z^{p,q} \to H^{p+q}_{\mathrm{dR}}) \subset B^{p,q}_K$, so the de Rham image of $Z^{p,q}$ surjects onto $K^{p,q}$. Hence

$$k^{p,q} \leq b_{p+q}. \tag{11.25}$$

Because X has the integral cohomology of S^6, (11.25) gives $k^{p,q} = 0$ whenever $0 < p + q < 6$.

For a compact complex threefold, McHugh's Theorem 6.3, Proposition 6.4, equation (7), and the ensuing table give

$$\begin{aligned}
h^{1,1}_{BC} &= 2h^{0,1} - b_1 + \dim_{\mathbb{R}} H^{1,1}(X;\mathbb{R}),\\
h^{1,2}_{BC} &= h^{1,2} + \dim_{\mathbb{C}} E_2^{3,1} - k^{2,0},\\
h^{2,2}_{BC} &= -h^{1,1}_{BC} + h^{0,1} - h^{0,2} - h^{1,0} + h^{1,1} + h^{1,2} + h^{2,0}\\
&\qquad + 2\dim_{\mathbb{C}} E_2^{3,1} + k^{1,1} - k^{1,2} - 2k^{2,0},
\end{aligned}$$

and his threefold table gives the remaining entries [49, Theorem 6.3, Proposition 6.4, equation (7), §7.1]; see also the independent synthesis in [48, §4]. Here $H^{1,1}(X;\mathbb{R})$ denotes the subspace of $H^2_{\mathrm{dR}}(X;\mathbb{R})$ represented by real d-closed forms of type $(1,1)$, and it is zero because $H^2_{\mathrm{dR}}(X;\mathbb{R}) = 0$, while theorem 56.3 gives $E_2 = E_\infty$; since $b_4 = 0$, this in particular gives $E_2^{3,1} = 0$. Substitution of

$$h^{0,1} = 1, \quad h^{0,2} = h^{1,0} = h^{2,0} = 0, \quad h^{1,1} = 2, \quad h^{1,2} = 1$$

therefore gives

$$h^{1,1}_{BC} = 2, \qquad h^{1,2}_{BC} = h^{2,1}_{BC} = 1, \qquad h^{2,2}_{BC} = 2.$$

The same table gives $h^{2,3}_{BC} = h^{3,2}_{BC} = 1$, the two endpoint dimensions equal to one, and every other entry zero. This proves (11.21). The Hodge-star duality of Schweitzer [50, §2.c, pp. 9–10], $h^{p,q}_{A} = h^{3-p,3-q}_{BC}$, proves (11.22). Summing along diagonals and using $b_0 = b_6 = 1$ and $b_k = 0$ for $1 \le k \le 5$ proves (11.23)–(11.24). □

The dimension calculation above is completed by the explicit representative and product theorem in the following section.

57 Period-coordinate Dolbeault classes and the Bott–Chern algebra

57.1 Fixed input and conventions

Let $f\colon X \to B = \mathbb{P}^1$ be the compact complex threefold of theorem 10.11. We use $d = \partial + \bar\partial$ and, following [50, §1.a],

$$H^{p,q}_{BC}(X) = \frac{\ker \partial \cap \ker \bar\partial \subset \mathcal{A}^{p,q}(X)}{\operatorname{im}(\partial\bar\partial)}. \tag{1678}$$

The natural map to Dolbeault cohomology is denoted

$$\pi^{p,q}_{\bar\partial}\colon H^{p,q}_{BC}(X) \longrightarrow H^{p,q}_{\bar\partial}(X). \tag{1679}$$

Exterior multiplication makes the direct sum of the Bott–Chern groups a bigraded algebra, and complex conjugation interchanges bidegrees (p,q) and (q,p) [50, §1.a].

By theorem 56.10, the two nonzero rows of the Frölicher differential are the exact complexes

$$0 \longrightarrow H^{0,1}_{\bar\partial}(X) \xrightarrow{d_1} H^{1,1}_{\bar\partial}(X) \xrightarrow{d_1} H^{2,1}_{\bar\partial}(X) \longrightarrow 0, \tag{1680}$$

$$0 \longrightarrow H^{1,2}_{\bar\partial}(X) \xrightarrow{d_1} H^{2,2}_{\bar\partial}(X) \xrightarrow{d_1} H^{3,2}_{\bar\partial}(X) \longrightarrow 0, \tag{1681}$$

and each row has dimensions $1, 2, 1$. The exact Bott–Chern calculation of theorem 56.11, obtained from the compact-threefold formulae of [49], gives

(p,q)	$(0,0)$	$(1,1)$	$(2,1)$	$(1,2)$	$(2,2)$	$(3,2)$	$(2,3)$	$(3,3)$
$h^{p,q}_{BC}$	1	2	1	1	2	1	1	1.

(1682)

All other Bott–Chern groups vanish. By theorem 53.7, $b_k(X) = 0$ for $1 \le k \le 5$.

On a marked smooth fibre retain the integral basis

$$V = \mathbb{Z}\langle \gamma, u, w, \delta \rangle$$

without renaming its element u, and put

$$\eta = u \wedge w + 6\gamma \wedge \delta, \qquad \eta^2 = 12\gamma \wedge u \wedge w \wedge \delta. \tag{1683}$$

The holomorphic Hodge rows supplied by theorems 10.2 and 54.2 are

$$\sigma_1 = 6\mu\gamma + \tau u + w, \qquad \sigma_2 = \beta\gamma + \mu u + \delta. \tag{1684}$$

Thus no global Dolbeault or Bott–Chern generator below is denoted by u or x; those letters remain reserved for the printed fibre basis and for local real linear functions.

57.2 Exact fibre representatives

Write

$$y = \operatorname{Im}\tau, \qquad m = \operatorname{Im}\mu, \qquad r = \operatorname{Im}\beta, \qquad \Delta = r - \frac{6m^2}{y} < 0. \tag{1685}$$

The four marked lattice columns in the fibre coordinate $\zeta = (\zeta_1, \zeta_2)$ are

$$(6\mu, \beta), \quad (\tau, \mu), \quad (1, 0), \quad (0, 1), \tag{1686}$$

in the order $\widehat{\gamma}, \widehat{u}, \widehat{w}, \widehat{\delta}$. On the universal cover of a marked fibre define the real linear functions

$$x_\gamma = \frac{y\,\operatorname{Im}\zeta_2 - m\,\operatorname{Im}\zeta_1}{y\Delta}, \qquad x_u = \frac{r\,\operatorname{Im}\zeta_1 - 6m\,\operatorname{Im}\zeta_2}{y\Delta}. \tag{1687}$$

For the first two columns in equation (1686), the numerators of the changes in x_γ are respectively $yr - 6m^2 = y\Delta$ and $ym - my = 0$, while the numerators of the changes in x_u are $6rm - 6mr = 0$ and $ry - 6m^2 = y\Delta$. The last two columns are real and change neither imaginary part. Hence

$$\begin{aligned} (\Delta_{\widehat{\gamma}} x_\gamma, \Delta_{\widehat{u}} x_\gamma, \Delta_{\widehat{w}} x_\gamma, \Delta_{\widehat{\delta}} x_\gamma) &= (1, 0, 0, 0), \\ (\Delta_{\widehat{\gamma}} x_u, \Delta_{\widehat{u}} x_u, \Delta_{\widehat{w}} x_u, \Delta_{\widehat{\delta}} x_u) &= (0, 1, 0, 0). \end{aligned} \tag{1688}$$

Their differentials, although not the functions themselves, therefore descend to the fibre. Their $(0,1)$ parts are

$$\Gamma := \bar\partial_F x_\gamma = \frac{i}{2\Delta}\left(d\bar\zeta_2 - \frac{m}{y} d\bar\zeta_1\right), \tag{1689}$$

$$U := \bar\partial_F x_u = \frac{i}{2y\Delta}\left(r\, d\bar\zeta_1 - 6m\, d\bar\zeta_2\right). \tag{1690}$$

They represent respectively the $(0,1)$ images of the integral classes γ and u.

The identity

$$\eta = u \wedge \sigma_1 + 6\gamma \wedge \sigma_2 = -\sigma_1 \wedge u - 6\sigma_2 \wedge \gamma \tag{1691}$$

holds in $\bigwedge^2 V_{\mathbb{C}}$. Since η is of type $(1,1)$ on every marked fibre, equation (1691) gives its exact Dolbeault representative

$$Q_F = -d\zeta_1 \wedge U - 6 d\zeta_2 \wedge \Gamma. \tag{1692}$$

Equivalently, with the Appell–Humbert convention $\eta = (2i)^{-1} \sum (H_\eta)_{ab} \sigma_a \wedge \bar\sigma_b$, the matrix calculation in the proof of theorem 54.2 gives

$$H_\eta = \frac{1}{y\Delta}\begin{pmatrix} r & -6m \\ -6m & 6y \end{pmatrix}, \tag{1693}$$

and therefore

$$Q_F = \frac{1}{2iy\Delta}\Big(r\, d\zeta_1 \wedge d\bar\zeta_1 - 6m\, d\zeta_1 \wedge d\bar\zeta_2 - 6m\, d\zeta_2 \wedge d\bar\zeta_1 + 6y\, d\zeta_2 \wedge d\bar\zeta_2\Big). \tag{1694}$$

Direct exterior multiplication, with holomorphic differentials ordered before antiholomorphic differentials, gives

$$Q_F \wedge \Gamma = \frac{1}{4y\Delta} d\zeta_1 \wedge d\bar\zeta_1 \wedge d\bar\zeta_2 \neq 0, \tag{1695}$$

$$Q_F^2 = \frac{3}{y\Delta} d\zeta_1 \wedge d\zeta_2 \wedge d\bar\zeta_1 \wedge d\bar\zeta_2 \neq 0. \tag{1696}$$

Indeed, $\det H_\eta = (6ry - 36m^2)/(y^2\Delta^2) = 6/(y\Delta)$, and the coefficient in equation (1696) is $\frac{1}{2}\det H_\eta$ in the displayed ordering. The identities Equations (1695) and (1696) will be used only in fibre cohomology.

57.3 Extension of the two Dolbeault inputs

Lemma 57.1 (the degree-one input)**.** ***[VERIFIED]*** *There is a distinguished class $\mathfrak{g} \in H^{0,1}_{\bar\partial}(X)$ whose restriction to every smooth marked fibre is represented by Γ.*

Proof. By theorem 55.11, the monodromy-invariant integral class γ maps to the nowhere-zero holomorphic section $\gamma^{0,1}$ of $R^1 f_*\mathcal{O}_X \simeq \mathcal{O} \oplus \mathcal{O}(-1)$. Coherent Leray gives

$$H^1(X, \mathcal{O}_X) = H^0(B, R^1 f_*\mathcal{O}_X) = \mathbb{C}\gamma^{0,1}.$$

Let $\mathfrak{g}$ be the class mapping to $\gamma^{0,1}$. Equation (1689) is the exact period-coordinate representative of its fibre restriction. This is the distinguished coordinate of theorem 55.12 before any analytic rescaling. □

Lemma 57.2 (extension of the invariant $(1,1)$ class)**.** ***[VERIFIED]*** *The fibre class η extends to a nonzero holomorphic section*

$$\boldsymbol{\eta} \in H^0(B, R^1 f_*\Omega^1_X). \tag{1697}$$

Consequently there is a class $\mathfrak{q} \in H^{1,1}_{\bar\partial}(X)$ whose coherent-Leray edge value is $\boldsymbol{\eta}$. For every smooth fibre inclusion $i_b\colon F_b \to X$,

$$i_b^*\mathfrak{q} = [Q_F]_{\bar\partial} \in H^{1,1}_{\bar\partial}(F_b). \tag{1698}$$

Proof. Over the smooth marked locus, η is flat, monodromy invariant, and of type $(1,1)$ identically. Equation Equation (1691) identifies the corresponding holomorphic section of the relative group as

$$-\sigma_1 \wedge u^{0,1} - 6\sigma_2 \wedge \gamma^{0,1}. \tag{1699}$$

At a finite filling write $t = s^M$, with $(M, a) = (3, 2)$ or $(4, 1)$. The covering family is smooth at $s = 0$, and equation (1691) holds identically there, so equation (1699) remains holomorphic. It is invariant under the cyclic action. In equivariant frames e_0, e_1 for the relative cotangent and b_0, b_1 for $R^1\mathcal{O}$, the actual invariant $\mathbb{C}\{t\}$-basis from theorem 56.5 is

$$e_0b_0, \quad s^a e_0 b_1, \quad s^{M-a} e_1 b_0, \quad e_1 b_1. \tag{1700}$$

Exactness of cyclic invariants in characteristic zero therefore makes equation (1699) a holomorphic descended section at both finite points.

At the cusp, γ and u extend as sections of $R^1 f_*\mathbb{C}$, and their $(0,1)$ images form a basis of $H^1(W, \mathcal{O}_W)$ by theorem 55.11. On the dense torus,

$$\sigma_i = d\zeta_i = (2\pi i)^{-1} d\log x_i.$$

The classes of these logarithmic forms extend in the relative logarithmic sheaf as

$$\mathcal{E}|_{N_0} = \mathcal{O}_{N_0}[d\log x_1] \oplus \mathcal{O}_{N_0}[d\log x_2] \tag{1701}$$

by theorem 56.5. A cusp quotient shear changes $d\log x_i$ only by a holomorphic multiple of $d\log t$, which is zero in the relative logarithmic sheaf. Thus equation (1699) also extends holomorphically through the cusp. This concerns classes in the relative logarithmic sheaf; it does not assert that the $d\log x_i$ are global ordinary one-forms on X.

The Gauss–Manin/Kodaira–Spencer map

$$\kappa_1\colon R^1 f_*\mathcal{E} \longrightarrow \Omega^1_B(\log \Sigma) \otimes R^2 f_*\mathcal{O}_X$$

annihilates equation (1699) on the smooth locus: a flat class which remains of type $(1,1)$ has zero $(0,2)$ Higgs component. Both the section and κ_1 extend holomorphically, so the identity theorem gives $\kappa_1(\boldsymbol{\eta}) = 0$ on B. The ordinary-cotangent calculation in theorems 56.6, 56.7 and 56.9 identifies

$$R^1 f_*\Omega^1_X = \ker \kappa_1 =: \mathcal{K}, \qquad \mathcal{K} \simeq \mathcal{O} \oplus \mathcal{O}(-1)^{\oplus 2}. \tag{1702}$$

This proves equation (1697). Its value on every smooth fibre is nonzero in Dolbeault cohomology, not merely as a differential form: by equation (1683) its square is the integral top class $12\gamma \wedge u \wedge w \wedge \delta$, and therefore

$$\int_{(F_b,\mathrm{or}_{\mathrm{mark}})} Q_F^2 = 12, \qquad \int_{(F_b,\mathrm{or}_{\mathbb{C}})} Q_F^2 = -12 \qquad (b \in B^\circ), \tag{1703}$$

where $\mathrm{or}_{\mathrm{mark}}$ is the orientation determined by (γ, u, w, δ) and $\mathrm{or}_{\mathbb{C}}$ is the complex orientation. They are opposite because the determinant of the four marked lattice columns in real coordinates is $y\Delta < 0$. Thus $[Q_F]_{\bar\partial} \neq 0$, so the holomorphic section $\boldsymbol{\eta}$ is nonzero. Explicitly, if $Q_F = \bar\partial_F \alpha$ for a $(1,0)$-form α, then $Q_F^2 = \bar\partial_F(\alpha \wedge Q_F)$; integration and Stokes' theorem would contradict equation (1703). Here $\alpha \wedge Q_F$ has bidegree $(2,1)$, so its ∂_F is zero in complex dimension two and the displayed $\bar\partial_F$-boundary is also a d-boundary.

Finally, coherent Leray over the curve gives the exact edge sequence

$$0 \longrightarrow H^1(B, f_*\Omega^1_X) \longrightarrow H^1(X, \Omega^1_X) \longrightarrow H^0(B, R^1 f_* \Omega^1_X) \longrightarrow 0, \tag{1704}$$

where $f_*\Omega^1_X \simeq \mathcal{O}(-2)$ by theorem 56.7. Choose $\mathfrak{q}$ over $\boldsymbol{\eta}$. Base change on a smooth fibre gives equation (1698). Any two choices differ by the one-dimensional left term in equation (1704). That term restricts to zero on every smooth fibre. The nonzero class $d_1\mathfrak{g}$ also restricts to zero, because its restriction is $[\partial_F \Gamma]_{\bar\partial} = 0$; since $H^1(B, \mathcal{O}(-2))$ is one-dimensional, the left term is exactly $\mathrm{im}(d_1 \colon H^{0,1}_{\bar\partial} \to H^{1,1}_{\bar\partial})$. □

Remark 57.3 (what is and is not explicit). Choose global $\bar\partial$-closed forms

$$G \in \mathcal{A}^{0,1}(X), \qquad Q \in \mathcal{A}^{1,1}(X) \tag{1705}$$

representing $\mathfrak{g}$ and $\mathfrak{q}$. Equation (1698) says that $i_b^*[Q]_{\bar\partial}$ is the Dolbeault class of the explicit translation-invariant form displayed in equation (1694); it does not say that an arbitrary representative Q restricts literally to that form. A literal global coefficient formula would require additional C^∞ gauge choices and horizontal correction forms on the disc–pair-of-pants cover, compatible with the two affine cyclic conjugacies and the toric cusp. Those choices are not part of the canonical data. No such literal gauge is needed for the class and product calculations below.

57.4 The factor 12 and the Bott–Chern first Chern class

Let

$$\exp \colon H^1(X, \mathcal{O}_X) \longrightarrow \mathrm{Pic}(X)$$

be induced by $z \mapsto e^{2\pi i z}$. By theorem 53.2,

$$H^0(B, R^1 f_* \mathbb{Z}) = \mathbb{Z}(12\gamma). \tag{1706}$$

The exponential–Leray comparison gives a line bundle M on B such that

$$\exp(12\mathfrak{g}) = f^*M, \qquad c_1(M) = d_2(12\gamma). \tag{1707}$$

For completeness, the first assertion follows because equation (1706) says that the restriction of $12\mathfrak{g}$ to every fibre lies in the image of integral cohomology and hence exponentiates to the trivial bundle. The low-degree Leray sequence for $\mathcal{O}^*$ then makes the resulting bundle a pullback. On a disc–pair-of-pants cover of B, choose local integral lifts of 12γ. Their differences on the annular overlaps form a Cech one-cochain, and its Cech coboundary is simultaneously the two-cocycle defining the Leray transgression and the two-cocycle of logarithms defining $c_1(M)$. This proves the second assertion with the same sign convention; it is the exponential-sequence version of the local-lift calculation in the proof of theorem 53.3.

Orient the generator $\omega \in H^2(B; \mathbb{Z})$ by

$$\omega = c_1(\mathcal{O}_B(1)). \tag{1708}$$

Then theorem 53.3 and equation (1707) give

$$\exp(12\mathfrak{g}) = f^*\mathcal{O}_B(1). \tag{1709}$$

If the opposite orientation is used for the primitive generator in theorem 53.3, the corresponding statement is $\exp(-12\mathfrak{g}) = f^*\mathcal{O}_B(1)$. Thus the canonical integral data fixes the absolute factor 12, while its sign is the orientation choice fixed in equation (1708).

Put

$$a = \partial G, \qquad a^\dagger = \bar\partial \bar G, \qquad \mathsf{A} = [a]_{BC}, \qquad \mathsf{B} = [a^\dagger]_{BC}. \tag{1710}$$

Both forms are d-closed. We now compute their Chern-class normalization without choosing local metric weights. For $z \in \mathbb{C}$, equip the smooth trivial line with a global frame s and the holomorphic structure

$$\bar\partial_z s = -2\pi i z G \otimes s. \tag{1711}$$

It is integrable because $\bar\partial G = 0$. If $G = \bar\partial \phi_i$ on a good cover, then $e_i = e^{2\pi i z \phi_i} s$ is $\bar\partial_z$-holomorphic and

$$e_j = e^{2\pi i z(\phi_j - \phi_i)} e_i.$$

The holomorphic cocycle $\phi_j - \phi_i$ represents $\mathfrak{g}$ under the Cech–Dolbeault comparison, so equation (1711) is precisely the line bundle $\exp(z\mathfrak{g})$ for the displayed exponential convention.

Give s unit length. The unit-metric Chern connection is

$$\nabla_z = d - 2\pi i\big(zG + \bar z\, \bar G\big). \tag{1712}$$

Indeed, its $(0,1)$ part is equation (1711), and its connection one-form is purely imaginary, so it is metric compatible; uniqueness identifies it with the Chern connection. Since $\bar\partial G = 0$ and $\partial \bar G = 0$, its curvature is the global $(1,1)$ form

$$F_{\nabla_z} = -2\pi i\big(z\,\partial G + \bar z\,\bar\partial \bar G\big) = -2\pi i\big(z a + \bar z\, a^\dagger\big). \tag{1713}$$

With the Bott–Chern convention

$$c_{1,BC}(L,h) = \left[\frac{i}{2\pi} F_{\nabla^h}\right]_{BC} \tag{1714}$$

of [50, §8.a], equation (1713) gives, for every $z \in \mathbb{C}$,

$$c_{1,BC}\big(\exp(z\mathfrak{g})\big) = z\mathsf{A} + \bar z\,\mathsf{B}. \tag{1715}$$

Consequently, if $\varepsilon = 1$ for the orientation fixed by equation (1708) and $\varepsilon = -1$ for its opposite, then

$$c_{1,BC}\big(f^*\mathcal{O}_B(1)\big) = 12\varepsilon(\mathsf{A} + \mathsf{B}). \tag{1716}$$

This is the convention-consistent form of the factor-12 identity; no analytic rescaling of the Picard coordinate changes its integral content.

57.5 All Bott–Chern generators

Using equation (1705), define the following d-closed forms and their classes:

class	representative	bidegree
A	$a = \partial G$	$(1,1)$
B	$a^\dagger = \bar\partial \bar G$	$(1,1)$
C	$c = \partial Q$	$(2,1)$
$\overline{\mathsf{C}}$	$c^\dagger = \bar\partial \bar Q$	$(1,2)$
D	$d_0 = \partial(G \wedge Q)$	$(2,2)$
$\overline{\mathsf{D}}$	$d_0^\dagger = \bar\partial(\bar G \wedge \bar Q)$	$(2,2)$
E	$e = \partial(Q^2)$	$(3,2)$
$\overline{\mathsf{E}}$	$e^\dagger = \bar\partial(\bar Q^2)$	$(2,3)$
T	$\nu, \quad \int_X \nu = 1$	$(3,3)$.

(1717)

Here ν is any smooth positive top-degree form divided by its integral. The chosen Q cannot be real: its fibre class is real, but the proof below shows that ∂Q is nonzero globally, whereas a real $\bar\partial$-closed Q would also be ∂-closed.

Proposition 57.4 (nonvanishing and bases)**.** ***[VERIFIED]*** *Together with the unit* $\mathbf{1} \in H^{0,0}_{BC}(X)$*, the entries in equation* (1717) *give bases of all the nonzero Bott–Chern groups in equation* (1682)*:* A, B *form the basis in bidegree* $(1,1)$ *and* $\mathsf{D}, \overline{\mathsf{D}}$ *form the basis in bidegree* $(2,2)$*; each other listed bidegree is one-dimensional and is generated by its displayed class.*

Proof. Because X is compact and connected, a smooth function annihilated by both ∂ and $\bar\partial$ is constant. There is no $\partial\bar\partial$ image in bidegree $(0,0)$, so $\mathbf{1} := [1]_{BC}$ generates $H^{0,0}_{BC}(X) = \mathbb{C}\mathbf{1}$.

The Dolbeault image of A is $d_1\mathfrak{g} \neq 0$ by injectivity in equation (1680). Thus ∂G is a nonzero Bott–Chern class. The Dolbeault image of B is zero because $a^\dagger = \bar\partial\bar G$, while $\mathsf{B} = \overline{\mathsf{A}} \neq 0$. Hence A, B are independent, and equation (1682) makes them a basis.

The classes $d_1\mathfrak{g}$ and $\mathfrak{q}$ are independent in $H^{1,1}_{\bar\partial}(X)$: the first restricts to zero on every smooth torus fibre, while the second restricts to the nonzero class by equations (1698) and (1703). Exactness of equation (1680) therefore gives

$$d_1\mathfrak{q} = [\partial Q]_{\bar\partial} \neq 0. \tag{1718}$$

Thus $\mathsf{C} \neq 0$, and conjugation gives $\overline{\mathsf{C}} \neq 0$.

Set

$$v = [G \wedge Q]_{\bar\partial} \in H^{1,2}_{\bar\partial}(X). \tag{1719}$$

Its fibre restriction is the class represented by $\Gamma \wedge Q_F$. This class is nonzero, rather than merely represented by a nonzero invariant form: wedging with the holomorphic one-form $d\zeta_2$ and using equation (1695) gives

$$\int_{F_b} Q_F \wedge \Gamma \wedge d\zeta_2 = \frac{1}{4y\Delta} \int_{F_b} d\zeta_1 \wedge d\zeta_2 \wedge d\bar\zeta_1 \wedge d\bar\zeta_2 = -1 \tag{1720}$$

for the complex orientation: the middle integral equals $-4y\Delta > 0$. Thus the displayed constant top form has nonzero integral. Dolbeault–Serre duality on F_b therefore gives $[\Gamma \wedge Q_F]_{\bar\partial} \neq 0$. Equivalently, if $Q_F \wedge \Gamma = \bar\partial_F \Psi$ with Ψ of bidegree $(1,1)$, then, since $\bar\partial_F d\zeta_2 = 0$,

$$Q_F \wedge \Gamma \wedge d\zeta_2 = \bar\partial_F(\Psi \wedge d\zeta_2),$$

and Stokes' theorem would make the integral in equation (1720) zero. Here $Q_F \wedge \Gamma$ has bidegree $(1,2)$, $d\zeta_2$ has bidegree $(1,0)$, and $\Psi \wedge d\zeta_2$ has bidegree $(2,1)$; its ∂_F is zero in complex dimension two, so the displayed $\bar\partial_F$-boundary is a d-boundary. Thus $v \neq 0$, and it spans $H^{1,2}_{\bar\partial}(X)$. Injectivity in equation (1681) gives

$$d_1 v = [\partial(G \wedge Q)]_{\bar\partial} \neq 0. \tag{1721}$$

Hence $\mathsf{D} \neq 0$, and conjugation gives $\overline{\mathsf{D}} \neq 0$. The latter has zero Dolbeault image because its representative is $\bar\partial$-exact, so it is independent of D; equation (1682) gives the asserted $(2,2)$ basis.

The class $[Q^2]_{\bar\partial}$ has nonzero fibre restriction by equation (1703). Since v spans $H^{1,2}_{\bar\partial}(X)$, the image of d_1 is spanned by $d_1 v$. On a smooth fibre,

$$i_b^*(d_1 v) = d_1[\Gamma \wedge Q_F]_{\bar\partial} = [\partial_F(\Gamma \wedge Q_F)]_{\bar\partial} = 0,$$

because Γ and Q_F have constant fibre coefficients. Thus every class in $\operatorname{im}(d_1 \colon H^{1,2}_{\bar\partial} \to H^{2,2}_{\bar\partial})$ restricts to zero. Therefore $[Q^2]_{\bar\partial}$ is independent of $d_1 v$. Exactness of equation (1681) yields

$$d_1[Q^2]_{\bar\partial} = [\partial(Q^2)]_{\bar\partial} \neq 0. \tag{1722}$$

Hence $\mathsf{E} \neq 0$, and conjugation gives $\overline{\mathsf{E}} \neq 0$. Finally, $\int_X \nu = 1$ proves $\mathsf{T} \neq 0$. The dimensions in equation (1682) complete the proof. □

57.6 Every Bott–Chern product

Theorem 57.5 (square-zero positive ideal)**.** ***[VERIFIED]*** *Let $N \subset H^{*,*}_{BC}(X)$ be the span of the nine positive-degree generators in equation* (1717)*. Then*

$$H^{*,*}_{BC}(X) = \mathbb{C}\mathbf{1} \oplus N, \qquad N^2 = 0. \tag{1723}$$

Equivalently, every product of two positive-degree Bott–Chern classes is zero.

Proof. We first calculate the products landing below top degree. Since $\bar\partial G = \bar\partial Q = 0$ and, after conjugation, $\partial \bar G = \partial \bar Q = 0$, the graded Leibniz rule gives the exact identities

$$\partial\bar\partial(G \wedge \bar G) = -a \wedge a^\dagger, \tag{1724}$$

$$\partial\bar\partial(G \wedge \bar Q) = -a \wedge c^\dagger, \qquad \partial\bar\partial(\bar G \wedge Q) = a^\dagger \wedge c. \tag{1725}$$

Consequently

$$\mathsf{AB} = 0, \qquad \mathsf{A}\overline{\mathsf{C}} = 0, \qquad \mathsf{BC} = 0. \tag{1726}$$

Choose the pullback Fubini–Study metric on $f^*\mathcal{O}_B(1)$. Its Chern form is the pullback of a $(1,1)$ form on the curve B, so its square vanishes pointwise. Squaring equation (1716) and using equation (1726) gives

$$\mathsf{A}^2 + \mathsf{B}^2 = 0. \tag{1727}$$

Equation (1727) alone does not separate the two squares. Apply $\pi^{2,2}_{\bar\partial}$. The class B^2 has zero Dolbeault image because

$$(a^\dagger)^2 = \bar\partial(\bar G \wedge a^\dagger). \tag{1728}$$

It follows from equation (1727) that $[a^2]_{\bar\partial} = 0$. On the other hand,

$$[a^2]_{\bar\partial} = d_1[G \wedge a]_{\bar\partial}, \qquad \partial(G \wedge a) = a^2. \tag{1729}$$

The first arrow in equation (1681) is injective, so equation (1729) implies $[G \wedge a]_{\bar\partial} = 0$. There is therefore a form $\Phi \in \mathcal{A}^{1,1}(X)$ such that

$$G \wedge a = \bar\partial\Phi. \tag{1730}$$

Applying ∂ gives

$$a^2 = \partial\bar\partial\Phi, \tag{1731}$$

and hence $\mathsf{A}^2 = 0$. Conjugation gives $\mathsf{B}^2 = 0$.

It remains among the products below top bidegree to prove $\mathsf{AC} = 0$; its conjugate will give $\mathsf{B}\overline{\mathsf{C}} = 0$. First, the natural map

$$\pi^{3,2}_{\bar\partial} \colon H^{3,2}_{BC}(X) \longrightarrow H^{3,2}_{\bar\partial}(X) \tag{1732}$$

is an isomorphism. Indeed, both spaces have dimension one by equation (1682) and theorem 56.10, and equation (1722) says that E has nonzero image. Pair $\pi^{3,2}_{\bar\partial}(\mathsf{AC})$ with the generator $[G]_{\bar\partial} \in H^{0,1}_{\bar\partial}(X)$. By equation (1730), $\bar\partial c = 0$, and Stokes' theorem,

$$\int_X a \wedge c \wedge G = -\int_X G \wedge a \wedge c = -\int_X \bar\partial\Phi \wedge c = -\int_X \bar\partial(\Phi \wedge c) = 0. \tag{1733}$$

Dolbeault–Serre duality makes

$$H^{3,2}_{\bar\partial}(X) \times H^{0,1}_{\bar\partial}(X) \longrightarrow \mathbb{C}, \qquad ([\alpha],[\rho]) \longmapsto \int_X \alpha \wedge \rho, \tag{1734}$$

a perfect pairing. Since $H^{0,1}_{\bar\partial}(X) = \mathbb{C}[G]_{\bar\partial}$, equation (1733) implies $\pi^{3,2}_{\bar\partial}(\mathsf{AC}) = 0$; injectivity in equation (1732) gives

$$\mathsf{AC} = 0, \qquad \mathsf{B}\overline{\mathsf{C}} = 0. \tag{1735}$$

This step uses Dolbeault–Serre duality together with the independently proved injectivity of equation (1732). The Bott–Chern–Aeppli pairing of [50, §2.c] by itself would supply a perfect pairing but would not annihilate this product.

The only products not yet considered which can have bidegree at most $(3,3)$ are

$$\mathsf{AD}, \quad \mathsf{A}\overline{\mathsf{D}}, \quad \mathsf{BD}, \quad \mathsf{B}\overline{\mathsf{D}}, \quad \mathsf{C}\overline{\mathsf{C}}, \tag{1736}$$

all of bidegree $(3,3)$. Every non-top generator in equation (1717) is de Rham exact:

$$\begin{gathered} a = dG, \qquad a^\dagger = d\bar{G}, \qquad c = dQ, \qquad c^\dagger = d\bar{Q}, \\ d_0 = d(G \wedge Q), \qquad d_0^\dagger = d(\bar{G} \wedge \bar{Q}), \qquad e = d(Q^2), \qquad e^\dagger = d(\bar{Q}^2). \end{gathered} \tag{1737}$$

Each factor paired with an exact factor is d-closed, so every product in equation (1736) is d-exact and has integral zero. Integration induces an isomorphism

$$H^{3,3}_{BC}(X) \xrightarrow{\int_X} \mathbb{C}. \tag{1738}$$

It is well defined by Stokes' theorem, the source has dimension one by equation (1682), and it is nonzero on T. Therefore all five products in equation (1736) vanish in Bott–Chern cohomology.

All other products of positive-degree generators have $p > 3$ or $q > 3$ and vanish already as differential forms. Equations (1726), (1731), (1735), (1736) and (1738) exhaust every possible positive-degree product and prove equation (1723). □

Remark 57.6 (literal global gauge boundary). The Bott–Chern algebra equation (1723) and the fibre-coordinate restrictions displayed in equations (1689), (1690), (1692) and (1694) to (1696) are complete. The canonical data do not select a preferred literal global coefficient atlas for the smooth forms G and Q. Such an atlas would require explicit smooth local trivializations and horizontal correction functions through both finite affine fillings and the cusp. After fixing $\mathfrak{g}$ and one coherent-Leray lift $\mathfrak{q}$, changing those smooth gauges changes G and Q by Dolbeault coboundaries but changes none of the cohomology classes, nonvanishing arguments, or products proved above. This is a limitation only on a preferred global coefficient presentation, not on the representatives as cohomology classes or on the Bott–Chern algebra.

Audit record 57.7 (H2: independent replication of the cotangent calculation)**.** **[OPEN]** A further context-isolated AI derivation has independently recovered the cyclic invariant lattices (11.4), the determinant divisor (11.6), the residue cancellation (11.17), the scalar test (11.19)–(11.20), and the splittings

$$R^0 f_* \Omega^1_X = \mathcal{O}(-2), \quad R^1 f_* \Omega^1_X = \mathcal{O} \oplus \mathcal{O}(-1)^{\oplus 2}, \quad R^2 f_* \Omega^1_X = \mathcal{O} \oplus \mathcal{O}(-1).$$

It also independently gives $h^{1,2} = 1$ and $h^{1,1} = 2$. Independent human specialist replication remains outstanding, so the record stays **[OPEN]**; this external-replication boundary is not an unresolved premise of the theorem above.

58 The alternate collapse and Mayer–Vietoris route

The nearby-cycle/Leray proof in section 53 gives one verified route to the topology of X. This section audits the manuscript's independent cellular-collapse/Mayer–Vietoris route. It now closes the entire alternate certificate: the equivariant cusp retraction, its primitive integral cellular map, equality with analytic nearby-cycle specialization, both geometrically realized Mayer–Vietoris stages, and the comparison of their obstruction maps with Leray.

We first correct a conflation in the earlier audit. For any two-set decomposition $Y = A \cup B$, our Mayer–Vietoris convention is

$$\Delta_q = (i_{A*}, -i_{B*}) : H_q(A \cap B; \mathbb{Z}) \longrightarrow H_q(A; \mathbb{Z}) \oplus H_q(B; \mathbb{Z}). \tag{4.1}$$

The manuscript does not identify its printed α_q with a one-shot map for three collars. It uses two stages. First,

$$X^\circ = X_1 \cup X_2, \qquad X_1 \cap X_2 \simeq F, \qquad \alpha_q = (\pi_{1*}, -\pi_{2*}). \tag{4.2}$$

Second, it attaches the cusp through

$$X = N_0' \cup X^\circ, \qquad N_0' \cap X^\circ = M, \qquad K_q = \ker(H_q(M) \to H_q(W)). \tag{4.3}$$

The proof below realizes both stages by actual oriented inclusions and then also records the equivalent one-shot degree-one collar matrix.

58.1 The abstract cusp-retraction lemma

Proposition 58.1 (equivariant topological spine)**.** ***[VERIFIED]*** *Let A be a contractible manifold with boundary and let $A_0 \subset \partial A$ be a contractible boundary component. Let a group Γ act freely and properly discontinuously on (A, A_0). Assume $Q = A/\Gamma$ and $H = A_0/\Gamma$ are manifolds, Q is compact, $H \hookrightarrow Q$ is collared, and the two quotient maps are connected coverings with deck group Γ. If the inclusion identifies the two deck groups by the identity automorphism of Γ, then $H \hookrightarrow Q$ is a strong deformation retract. The retraction lifts uniquely from the identity lift to a Γ-equivariant strong deformation retraction of A onto A_0.*

Proof. Because A and A_0 are contractible universal covers, $\pi_n(Q) = \pi_n(H) = 0$ for $n \geq 2$, while $\pi_1(Q) \cong \Gamma \cong \pi_1(H)$. By hypothesis the inclusion induces the identity on Γ, so it is a weak homotopy equivalence. Manifolds have the homotopy type of CW complexes; Whitehead's theorem [10, Theorem 4.5] therefore makes the inclusion a homotopy equivalence.

The collar makes (Q, H) an NDR pair, hence the inclusion is a cofibration. We record the standard cofibration argument. Choose a homotopy inverse $r_0 : Q \to H$. Since $r_0|_H \simeq \mathrm{id}_H$, the homotopy extension property modifies r_0 through maps $Q \to H$ to a retraction r satisfying $r|_H = \mathrm{id}_H$. The maps id_Q and $i \circ r$ are homotopic and agree on H. Applying the homotopy extension property once more makes that homotopy stationary on H. It is therefore a strong deformation retraction.

Let $P : A \to Q$ be the covering. Lift the deformation homotopy from the initial lift id_A. Since the base homotopy fixes H and the initial lift fixes A_0, uniqueness of path lifting makes the lifted homotopy fix A_0. For $\gamma \in \Gamma$, both $\widetilde{H}_t \circ \gamma$ and $\gamma \circ \widetilde{H}_t$ lift the same map and agree at $t = 0$; uniqueness gives equality for every t. The lift is therefore Γ-equivariant and ends in A_0. □

58.2 The candidate-specific equivariant collapse

We now verify every hypothesis of theorem 58.1 in the candidate. Retain the toric threefold Y, the compact torus

$$T_c = (S^1)^3, \qquad T_f = (S^1)^2 \times \{1\} \subset T_c,$$

and the free properly discontinuous action $\Gamma = \{\Psi_{\bar\lambda} : \bar\lambda \in \bar\Lambda\}$ from theorem 10.6. Put

$$Y_{\leq\eta} = \{|t| \leq \eta\}, \qquad Y_{\geq 0} = \overline{(\mathbb{R}_{>0})^3} \subset Y, \qquad A = (Y_{\leq\eta})_{\geq 0}, \qquad A_0 = (Y_0)_{\geq 0}. \tag{C.1}$$

For a torus point write $y = \log|x| / \log|t|$. The period matrix on a punctured cusp fibre is

$$Z_t = sB_0 + C(t), \qquad s = \frac{\log t}{2\pi i}, \qquad B_t = B_0 + \frac{R(t)}{\log|t|}, \quad R(t) = -2\pi \operatorname{Im} C(t). \tag{C.2}$$

Here $B_0 : \bar\Lambda \to \Lambda_{\mathrm{tor}}$ is unimodular.

Lemma 58.2 (straightening the twist)**.** ***[VERIFIED]*** *For all sufficiently small $\eta > 0$ there is a homeomorphism $G : Y_{\leq\eta} \to Y_{\leq\eta}$, fixing Y_0 pointwise and preserving t, which conjugates Γ to the action Γ^0 obtained by replacing $C(t)$ by $C_0 = C(0)$. It commutes with T_f and, on every smooth quotient fibre, induces a real-linear isomorphism which is the identity on the marked lattice Λ.*

Proof. Shrink η until B_t and $B_t^0 = B_0 + R(0)/\log|t|$ are invertible for $0 < |t| \le \eta$. In logarithmic coordinates $\xi = (2\pi i)^{-1}\log x$ define

$$L_t(\xi) = \xi - (C(t) - C_0)(\operatorname{Im} Z_t)^{-1}\operatorname{Im}\xi. \tag{C.3}$$

Every ξ has the unique expression $\xi = Z_t a + b$, with $a, b \in \mathbb{R}^2$, and then $L_t(\xi) = Z_t^0 a + b$. Thus L_t is invertible, fixes $\mathbb{R}^2$ and its sublattice $\mathbb{Z}^2$, and sends each deck translation $Z_t\bar\lambda$ to $Z_t^0\bar\lambda$. It therefore descends on $Y_t = \mathbb{C}^2/\mathbb{Z}^2$ and conjugates the two $\bar\Lambda$-actions. In toric coordinates it is

$$G(x,t) = (\exp(-2\pi i(C(t) - C_0)B_t^{-1}y)x, t), \qquad G|_{Y_0} = \mathrm{id}\,. \tag{C.4}$$

It commutes with T_f because y is T_f-invariant.

Only continuity at Y_0 is nonformal. Around any $p \in Y_0$, choose one of the bounded-position subcharts used in the properness proof of theorem 10.6. On that subchart y stays bounded, B_t^{-1} stays bounded, and $C(t) - C_0 \to 0$. Hence the torus factor in (C.4) tends to 1. The same argument applies to the inverse, obtained by interchanging $C(t)$ and C_0. Thus G is a homeomorphism. Because it fixes the real lattice and sends the two period columns $Z_t\bar\lambda$ to $Z_t^0\bar\lambda$, it induces the identity on the marked $H_1 = \Lambda$. □

Write $c_{\bar\lambda}^0 = u_{\bar\lambda}a_{\bar\lambda}$ for its polar decomposition, with $u_{\bar\lambda} \in T_f$ and $a_{\bar\lambda} = \exp(R(0)\bar\lambda) \in (\mathbb{R}_{>0})^2$. Let Γ^+ be the positive action

$$\Psi_{\bar\lambda}^+ = (a_{\bar\lambda}, 1)\Phi_{\bar\lambda}. \tag{C.5}$$

Lemma 58.3 (positive quotient and honeycomb)**.** ***[VERIFIED]*** *The pair (A, A_0) in (C.1) consists of contractible topological manifolds, A_0 being a boundary component of A. The action Γ^+ is free and properly discontinuous, and*

$$Q^+ = A/\Gamma^+, \qquad H = A_0/\Gamma^+ \tag{C.6}$$

are respectively a compact 3-manifold with boundary and one of its boundary 2-tori. The inclusion $H \hookrightarrow Q^+$ is a strong deformation retract, and its retraction lifts to a Γ^+-equivariant strong deformation retraction P of A onto A_0.

Proof. Polar multiplication $m : T_c \times Y_{\ge 0} \to Y$ is a closed quotient map: in every smooth unimodular toric chart it is the usual map $(S^1)^3 \times [0,\infty)^3 \to \mathbb{C}^3$, and T_c is compact. If q is in the orbit indexed by a face C of the A_2 triangulation, its stabilizer is the connected compact subtorus on

$$N_{\widehat C} = \bigoplus_{v \in V(C)} \mathbb{Z}(v, 1).$$

Unimodularity makes this lattice saturated, and direct intersection with Λ_{tor} gives

$$\operatorname{Stab}_{T_c}(q) \cap T_f = \begin{cases} 1, & C \text{ is a vertex,} \\ S^1_{v'-v}, & C = \{v, v'\} \text{ is an edge,} \\ T_f, & C \text{ is a triangle.} \end{cases} \tag{C.7}$$

There are no hidden finite components. The lattices $N_{\widehat C}$ and Λ_{tor} are saturated, and every face contains a vertex $(v, 1)$, so $N_{\widehat C} + \Lambda_{\mathrm{tor}} = N'$. Thus the displayed intersections are the connected subtori attached to their primitive intersection lattices.

In a toric chart the positive set A is

$$\{(r_0, r_1, r_2) \in [0,\infty)^3 : r_0 r_1 r_2 \le \eta\}.$$

Its corner links are discs, so it is a topological 3-manifold with boundary $A_0 \sqcup \{|t| = \eta\}_{>0}$. Its interior is $\mathbb{R}^2_y \times (0, \eta)$. Brown's collaring theorem [29] makes the inclusion of the interior a homotopy equivalence; therefore A is contractible.

The nonnegative part of each component $E_v \simeq \mathrm{dP}_6$ is its moment hexagon. These hexagons meet along exactly the arcs prescribed by the dual honeycomb $\mathcal{H}$ of the A_2 triangulation. Map the two vertex orbits of $\mathcal{H}$ to the corresponding torus fixed points, choose homeomorphisms on representatives of the three edge orbits, and extend the resulting circle homeomorphism over one hexagon. Propagating by $B_0\bar{\Lambda}$ defines a Γ^+-equivariant continuous bijection

$$\vartheta : \mathbb{R}^2 \xrightarrow{\sim} A_0, \qquad \vartheta(z + B_0\bar{\lambda}) = \Psi^+_{\bar{\lambda}}\vartheta(z). \tag{C.8}$$

Local finiteness of the fan makes this bijection proper, hence a homeomorphism. Thus A_0 is contractible.

The action in (C.5) is the already verified cusp action with constant twist $C^+ = (2\pi i)^{-1}R(0)$, since $e^{2\pi i C^+\bar{\lambda}} = a_{\bar{\lambda}}$; hence it is free and properly discontinuous. Properness of its quotient over the disc, proved by the same bounded fundamental-domain argument as theorem 10.6, makes Q^+ compact. Its two boundary components are $H \simeq \mathbb{R}^2/B_0\bar{\Lambda}$ and $\mathbb{R}^2/B^+_\eta\bar{\Lambda}$, both tori. Since a free properly discontinuous action preserves the boundary charts, Q^+ is a topological 3-manifold with boundary; Brown's collaring theorem makes $H \hookrightarrow Q^+$ collared. The covers $A \to Q^+$ and $A_0 \to H$ have the same deck group, and the equivariant inclusion identifies that group by the identity. The result is now theorem 58.1. □

Lemma 58.4 (spreading the positive retraction by phases). ***[VERIFIED]*** *The lifted retraction P of theorem 58.3 spreads to a Γ^0- and T_c-equivariant strong deformation retraction of $Y_{\le\eta}$ onto Y_0. Conjugating by G gives a Γ- and T_f-equivariant strong deformation retraction, and hence a strong deformation retraction*

$$r_s : N_0' = Y_{\le\eta}/\Gamma \longrightarrow W = Y_0/\Gamma. \tag{C.9}$$

Proof. For $\phi \in T_c$ and $q \in A$ set

$$R_s(\phi q) = \phi P_s(q). \tag{C.10}$$

This is well defined. On the open orbit and the outer boundary the stabilizer is trivial; on A_0 the homotopy fixes q, so (C.7) causes no ambiguity. It is continuous because its pullback along the closed quotient map $m \times \mathrm{id}_{[0,1]}$ is continuous.

The frozen and positive actions satisfy

$$\Psi^0_{\bar{\lambda}}(\phi q) = u_{\bar{\lambda}}\varphi_{\bar{\lambda}}(\phi)\Psi^+_{\bar{\lambda}}(q), \tag{C.11}$$

where $\varphi_{\bar{\lambda}}$ is the compact-torus automorphism induced by $(z, z_3) \mapsto (z + z_3 B_0\bar{\lambda}, z_3)$. Equation (C.11) and Γ^+-equivariance of P prove Γ^0-equivariance of R. The remaining deformation-retraction identities follow directly from those of P. Finally theorem 58.2 conjugates R to the original action and permits descent to (C.9). □

Proposition 58.5 (fixed-fibre cellular collapse). ***[VERIFIED]*** *Fix t_0 with $0 < |t_0| \le \eta$ and put $F = f^{-1}(t_0)$. The retraction in theorem 58.4 can be chosen so that its end map $r : N_0' \to W$ restricts to a cellular collapse $c = r|_F$ with the following description. There is a splitting*

$$F \xrightarrow[\bar{G}]{\sim} F^0_{t_0} \simeq T_B \times T_f, \tag{C.12}$$

where $\bar{G}$ is the identity on the marking Λ, and the quotient honeycomb $\Theta \subset T_B$ has two vertices and three edges e_k. Over a vertex, c collapses all of T_f to a triple point of W; over e_k° it collapses the primitive circle $S^1_{n_k}$ and maps onto the double curve $\bar{C}_k$; over $T_B \setminus \Theta$ it is injective onto $W \setminus D$.

Proof. Let $S \subset A$ be the positive slice $|t| = |t_0|$ and put

$$B^+_\rho = B_0 + \frac{R(0)}{\log\rho}, \qquad \ell_\rho = B_0(B^+_\rho)^{-1}.$$

The map $y : S \to \mathbb{R}^2$ identifies the positive action with translations $y \mapsto y + B^+_{|t_0|}\bar\lambda$, while $\ell_{|t_0|}$ conjugates these to $z \mapsto z + B_0\bar\lambda$. Define on the frozen fibre

$$\widetilde{c}((\phi, \arg t_0)q) = (\phi, \arg t_0)\, \vartheta(\ell_{|t_0|}y(q)), \qquad \phi \in T_f, q \in S. \tag{C.13}$$

The stabilizer on S is trivial, so the formula is well defined. Equations (C.8) and (C.11) make it Γ^0-equivariant, hence it descends to $c_\vartheta : F^0_{t_0} \to W$.

Writing

$$K_{t_0} = \operatorname{Re} C_0 + \frac{\arg t_0}{2\pi} B_0,$$

the frozen action on $T_f \times \mathbb{R}^2_y$ is

$$\bar\lambda(\phi, y) = (\phi e^{2\pi i K_{t_0}\bar\lambda}, y + B^+_{|t_0|}\bar\lambda).$$

The change of variables

$$\Xi(\phi, y) = (\phi e^{-2\pi i K_{t_0}(B^+_{|t_0|})^{-1}y}, y) \tag{C.14}$$

conjugates this to the product action and gives (C.12). Formula (C.7) now proves the asserted cell description, including primitivity of the collapsed circle.

It remains to force the end map of an actual strong deformation retraction to equal (C.13), rather than merely be homotopic to it. In the chart $\vartheta : \mathbb{R}^2 \to A_0$ put

$$\widehat{a} = \vartheta^{-1} P_1.$$

Put $\rho_0 = |t_0|$. If $\rho_0 < \eta$, choose a closed positive band $\{\rho_- \le |t| \le \rho_+\}$, disjoint from A_0, with ρ_0 in its interior. If $\rho_0 = \eta$, instead use the one-sided band $\{\rho_- \le |t| \le \eta\}$. Define $\widehat{b} = \ell_{|t|}y$ only on the chosen band. Wherever they are defined, both maps satisfy $\widehat{a}(\Psi^+_{\bar\lambda} q) = \widehat{a}(q) + B_0\bar\lambda$ and the same identity for $\widehat{b}$. Choose a continuous tent function $\chi(|t|)$, extended by zero off the band, with $\chi(\rho_0) = 1$ and vanishing near every boundary of the band across which it must be pasted; in the one-sided case no vanishing is required at the ambient boundary $|t| = \eta$. With $w(q) = \chi(|t(q)|)$ define

$$K_u(q) = \vartheta\big((1 - uw(q))\widehat{a}(q) + uw(q)\widehat{b}(q)\big) \tag{C.15}$$

inside the band and $K_u = P_1$ outside. Because the transition action is by translations, affine interpolation preserves Γ^+-equivariance. The weight vanishes near every gluing boundary and near A_0, so (C.15) pastes continuously, fixes A_0, and takes values in A_0.

Concatenate the first half of P with K_u. The result P' still satisfies all four properties used in theorem 58.4, and its end map on S is exactly $\vartheta\ell_{|t_0|}y$. Spreading P' by phases and conjugating by G therefore yields a strong deformation retraction whose restriction to F is

$$r|_F = c_\vartheta \circ \bar G.$$

This proves the proposition, including continuity across all toric strata. □

58.3 The integral collapse map

Let $D = \bar C_1 \cup \bar C_2 \cup \bar C_3$ be the double locus. Choose primitive orientations of its three curve classes and of the honeycomb edges so that the annulus degrees below have the displayed signs. These orientations are compared with the clockwise collar convention in theorems 58.10 and 58.11. In $\Lambda_{\mathrm{tor}} = \mathbb{Z}\langle\widehat{w}, \widehat{\delta}\rangle$ put

$$n_1 = \widehat{w}, \qquad n_2 = \widehat{\delta} - \widehat{w}, \qquad n_3 = -\widehat{\delta}, \qquad \nu_k(v) = \det(n_k, v). \tag{C.16}$$

Proposition 58.6 (integral homology of the cellular collapse)**.** ***[VERIFIED]*** *The map c_* of theorem 58.5 is surjective in every degree, and*

$$0 \longrightarrow \operatorname{im}(\textstyle\bigwedge^q M_0 - I) \longrightarrow \bigwedge^q \Lambda \xrightarrow{c_*} H_q(W;\mathbb{Z}) \longrightarrow 0 \tag{C.17}$$

is an exact sequence of free abelian groups. In the splitting $\Lambda = \bar{\Lambda} \oplus \Lambda_{\mathrm{tor}}$,

$$c_* : \Lambda \to H_1(W) = \bar{\Lambda}$$

is projection, and in degree two

$$c_*(\widehat{\gamma}\widehat{u}) = [\mathbb{T}_B], \qquad c_*(\widehat{w}\widehat{\delta}) = 0, \tag{C.18}$$

$$c_*(\beta \otimes v) = \sum_{k=1}^{3} \nu_k(B_0\beta)\nu_k(v)[\bar{C}_k]. \tag{C.19}$$

Equivalently, in the ordered source basis

$$(\widehat{\gamma}\widehat{u}, \widehat{\gamma}\widehat{w}, \widehat{\gamma}\widehat{\delta}, \widehat{u}\widehat{w}, \widehat{u}\widehat{\delta}, \widehat{w}\widehat{\delta})$$

and target basis $([\mathbb{T}_B], [\bar{C}_1], [\bar{C}_2], [\bar{C}_3])$,

$$[c_*]_2 = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & -1 & -1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \end{pmatrix}. \tag{C.20}$$

In degree three,

$$c_*(\widehat{\gamma}\widehat{u}\,v) = v \quad (v \in \Lambda_{\mathrm{tor}}), \qquad c_*(\beta\,\widehat{w}\widehat{\delta}) = 0 \quad (\beta \in \bar{\Lambda}), \tag{C.21}$$

where the first formula fixes the displayed orientation of $H_3(W)$.

Proof. First identify the target integrally. The normalization $\nu : \mathrm{dP}_6 \to W$ identifies opposite sides of the anticanonical hexagon. A plumbing neighbourhood of that hexagon descends to a regular neighbourhood of D. If P_0 is its complement in dP_6, then $P_0 \simeq (\mathbb{C}^*)^2 \simeq T^2$, and excision followed by Lefschetz duality gives

$$H_k(W, D) \simeq H_k(P_0, \partial P_0) \simeq H^{4-k}(T^2). \tag{C.22}$$

The class $[\mathbb{T}_B] = c_*[T_B \times \{*\}]$ pairs once with a compact T_f-orbit in $W \setminus D$, so it lifts the generator of $H_2(W, D)$. Likewise the cycles $c_*(T_B \times S^1_{\widehat{w}})$ and $c_*(T_B \times S^1_{\widehat{\delta}})$ pair unimodularly with the two coordinate circles in T_f. Thus the boundary maps of the pair vanish and

$$H_*(W) = (\mathbb{Z}, \mathbb{Z}^2, \mathbb{Z}^4, \mathbb{Z}^2, \mathbb{Z}), \tag{C.23}$$

with $H_2(W) = \mathbb{Z}\langle[\bar{C}_1], [\bar{C}_2], [\bar{C}_3], [\mathbb{T}_B]\rangle$ and $H_3(W)$ generated by the two cycles in (C.21).

The cell description gives projection in degree one and (C.18). Give T_B the CW structure with one open hexagonal 2-cell and 1-skeleton Θ. A class $\beta \in H_1(T_B)$ has the unique cellular representative

$$\sum_{k=1}^{3} \nu_k(B_0\beta)e_k.$$

On $e_k \times S^1_v$, the collapse kills $S^1_{n_k}$ and maps the quotient circle to $\bar{C}_k$ with degree $\nu_k(v)$. This proves (C.19), without a rank inference. The three choices $(B_0\beta, v) = (\widehat{w}, \widehat{w}), (\widehat{\delta}, \widehat{\delta}), (\widehat{w}, \widehat{\delta})$ give coefficient vectors

$$(0,1,1), \qquad (1,1,0), \qquad (0,1,0),$$

whose determinant is a unit. Hence degree two is surjective. Formula (C.21) follows directly by crossing the base 2-cell with a primitive fibre circle; it proves degree-three surjectivity and records the orientations. In degree four the top-dimensional open stratum $(T_B \setminus \Theta) \times T_f$ maps homeomorphically to the open 4-stratum of W, so the map is multiplication by a unit after choosing the target orientation. Degree zero is immediate.

Let h be geometric monodromy around the cusp. Fibre transport in N_0' is a homotopy from the inclusion of F to its composite with h. Composing with r yields $c \simeq c \circ h$, and therefore

$$\operatorname{im}(\textstyle\bigwedge^q M_0 - I) \subseteq \ker c_*. \tag{C.24}$$

The inclusion is an equality integrally, not only rationally. In the ordered basis $(\widehat{\gamma}, \widehat{u}, \widehat{w}, \widehat{\delta})$, direct exterior multiplication gives the primitive image lattices

q	$\operatorname{im}(\bigwedge^q M_0 - I)$
$0, 4$	0
1	$\langle \widehat{w}, \widehat{\delta} \rangle$
2	$\langle \widehat{w}\widehat{\delta}, \widehat{\gamma}\widehat{w} + \widehat{u}\widehat{\delta} \rangle$
3	$\langle \widehat{\gamma}\widehat{w}\widehat{\delta}, \widehat{u}\widehat{w}\widehat{\delta} \rangle$.

(C.25)

Each displayed subgroup is a direct summand. Its rank is $\binom{4}{q} - b_q(W)$ by (C.23), while surjectivity gives the same rank for $\ker c_*$. The quotient $H_q(W)$ is free, so $\ker c_*$ is saturated. Two saturated sublattices of the same rank and rational span coincide, proving (C.17). □

58.4 Comparison with analytic nearby cycles

Theorem 58.7 (collapse equals nearby-cycle specialization)**.** ***[VERIFIED]*** *With the radial marking of the punctured cusp family, for every $0 \le q \le 4$ one has an equality of integral homomorphisms*

$$\mathrm{sp}^q = i_F^* r^* = c^* : H^q(W; \mathbb{Z}) \longrightarrow H^q(F; \mathbb{Z}). \tag{C.26}$$

Consequently the cellular calculation gives

$$\operatorname{im} c^* = (\textstyle\bigwedge^q V)^{T_0}$$

as a saturated lattice. In particular, the degree-three orientation in the cellular computation is the degree-three orientation of intrinsic specialization; there is no independent sign or index.

Proof. Let $N_\rho = f^{-1}(|t| \le \rho) \subset N_0'$ and let $i_{W,\rho} : W \hookrightarrow N_\rho$ be the central-fibre inclusion. The closed tubes are cofinal among neighbourhoods of W: if an open set $\mathcal{O} \subset N_0'$ contains W, properness makes $f(N_0' \setminus \mathcal{O})$ closed and disjoint from 0, so $N_\rho \subset \mathcal{O}$ for all sufficiently small ρ. For each such ρ, choose t_ρ on the radial ray through the fixed point t_0 whose fibre is F, with $0 < |t_\rho| < \rho$, and write $i_{t_\rho} : F_{t_\rho} \hookrightarrow N_\rho$. Identify $H^q(F_{t_\rho}; \mathbb{Z})$ with $H^q(F; \mathbb{Z})$ by parallel transport along that ray. The neighborhood realization of the unit $\mathbb{Z}_W \to \psi_f \mathbb{Z}$, together with proper base change, identifies specialization with

$$H^q(W; \mathbb{Z}) \xleftarrow[\sim]{\rho_0} \varinjlim_{\rho \downarrow 0} H^q(N_\rho; \mathbb{Z}) \xrightarrow{\rho_{\mathrm{near}}} H^q(F; \mathbb{Z}), \tag{C.27}$$

where ρ_0 is induced by $i_{W,\rho}^*$ and ρ_{near} is induced by the transported maps $i_{t_\rho}^*$. These restriction maps are compatible under shrinking ρ. This is the same analytic integral construction used in theorem 53.1; see [33, Definition 4.2.1 and Propositions 4.2.2, 4.2.11].

Applying the construction of theorem 58.4 at radius ρ shows that $i_{W,\rho}^*$ is an isomorphism; no compatibility of those radius-dependent deformation homotopies is needed. Let $j_\rho : N_\rho \hookrightarrow N_0'$ be inclusion. The single retraction $r : N_0' \to W$ satisfies

$$i_{W,\rho}^* j_\rho^* r^* = (r \circ j_\rho \circ i_{W,\rho})^* = \mathrm{id}_{H^q(W;\mathbb{Z})}.$$

Hence $j_\rho^* r^*$ is the unique inverse of $i_{W,\rho}^*$. These inverses are compatible under shrinking because they are restrictions of the single map $r^* : H^q(W;\mathbb{Z}) \to H^q(N_0';\mathbb{Z})$. Consequently $\rho_0^{-1}(\alpha)$ in (C.27) is represented, for every sufficiently small ρ, by $j_\rho^* r^* \alpha$.

The restrictions of one class on N_0' to the fibres along the punctured radial ray form a section of the Gauss–Manin local system. Therefore, after the parallel-transport identifications used to define ρ_{near}, $i_{t_\rho}^* j_\rho^* r^* \alpha$ equals $i_F^* r^* \alpha$. It follows that the composite in (C.27) is $i_F^* r^*$. Since $c = r|_F$, functoriality gives $i_F^* r^* = (r \circ i_F)^* = c^*$.

Changing the lift of the radial marking composes the target with monodromy. By (C.24), c^* lands in the monodromy-invariant lattice, so the equality is independent of that harmless choice. Finally theorem 53.1 identifies the common image with the full saturated invariant lattice. □

Validation, repair, or disproof test 58.8 (resolved cusp-collapse falsifiers)**.** **[VERIFIED]** The audit tested the failure modes previously attached to C1–C3: continuity at every toric stratum, finite stabilizer components, equivariance, primitive edge circles, degree-two and degree-three orientations, torsion and saturation, and functorial equality with the nearby-cycle unit. The proofs above exclude each one. This status concerns the cusp collapse only; it does not infer the global Mayer–Vietoris matrices.

Audit record 58.9 (V1: based van Kampen inclusions)**.** **[VERIFIED]** The actual overlap coordinate maps, with the zero section as base locus and clockwise meridians, show that the finite fillings impose $\widehat{\rho}_j^{\,m_j} = t_{v_j}$ with the same sign, the cusp filling kills Λ_{tor} and its toric meridian, and the tautological gluing has zero translation drift. Thus the abstract terminal presentation is the actual $\pi_1(X)$. The deck generators, common clockwise convention, zero-section disc, normal closure, boundary inclusion $\Lambda \to \Lambda/\Lambda_{\mathrm{tor}}$, $\widetilde{a}_0 \mapsto 1$, and absence of extra whisker drift are proved in theorem 53.6.

58.5 The actual finite-band Mayer–Vietoris maps

Fix the ordered orientation

$$H_1(F) = \Lambda = \mathbb{Z}\langle \widehat{\gamma}, \widehat{u}, \widehat{w}, \widehat{\delta} \rangle$$

dual to (γ, u, w, δ), use the complex orientations of F, S_1, S_2, and orient every base meridian clockwise. Put

$$\eta_1 = 2u + w + 3\delta, \qquad \eta_2 = u + w + 2\delta, \qquad q = uw + 6\gamma\delta. \tag{MV.1}$$

Proposition 58.10 (actual finite-band maps in primitive bases)**.** ***[VERIFIED]*** *In the decomposition $X^\circ = X_1 \cup X_2$, the maps denoted α_r in the manuscript are the actual Mayer–Vietoris maps*

$$\alpha_r = (\pi_{1*}, -\pi_{2*}) : H_r(F;\mathbb{Z}) \longrightarrow H_r(S_1;\mathbb{Z}) \oplus H_r(S_2;\mathbb{Z}). \tag{MV.2}$$

There is no conjugacy, unrecorded index, or independent orientation sign in (MV.2).

Choose primitive bases

$$H_1(S_1) = \mathbb{Z}\langle g_1, h_1 \rangle, \quad [\widehat{\gamma}]_1 = 3g_1, \quad h_1 = \pi_{1*}\widehat{w},$$

$$H_1(S_2) = \mathbb{Z}\langle g_2, h_2 \rangle, \quad [\widehat{\gamma}]_2 = -4g_2, \quad h_2 = \pi_{2*}\widehat{w}.$$

Then, in the source basis $(\widehat{\gamma}, \widehat{u}, \widehat{w}, \widehat{\delta})$ and the target basis (g_1, h_1, g_2, h_2),

$$[\alpha_1] = \begin{pmatrix} 3 & 0 & 0 & 0 \\ 0 & 2 & 1 & 3 \\ 4 & 0 & 0 & 0 \\ 0 & -1 & -1 & -2 \end{pmatrix}. \tag{MV.3}$$

It has Smith form $\mathrm{diag}(1,1,1,0)$ *and*

$$\ker \alpha_1 = \mathbb{Z}\nu_0, \qquad \nu_0 = -\widehat{u} - \widehat{w} + \widehat{\delta}. \tag{MV.4}$$

Define cohomology bases (a_j, b_j) by

$$\pi_1^* a_1 = \gamma\eta_1, \qquad \pi_1^* b_1 = q - 2\gamma\eta_1,$$
$$\pi_2^* a_2 = 2\gamma\eta_2, \qquad \pi_2^* b_2 = q - 3\gamma\eta_2, \tag{MV.5}$$

and let $(a_j^\sharp, b_j^\sharp)$ be the Kronecker-dual homology bases. In the ordered source basis

$$(\widehat{\gamma}\widehat{u}, \widehat{\gamma}\widehat{w}, \widehat{\gamma}\widehat{\delta}, \widehat{u}\widehat{w}, \widehat{u}\widehat{\delta}, \widehat{w}\widehat{\delta})$$

and target basis $(a_1^\sharp, b_1^\sharp, a_2^\sharp, b_2^\sharp)$,

$$[\alpha_2] = \begin{pmatrix} 2 & 1 & 3 & 0 & 0 & 0 \\ -4 & -2 & 0 & 1 & 0 & 0 \\ -2 & -2 & -4 & 0 & 0 & 0 \\ 3 & 3 & 0 & -1 & 0 & 0 \end{pmatrix}. \tag{MV.6}$$

This matrix also has Smith form $\operatorname{diag}(1,1,1,0)$*; its primitive left kernel is*

$$\Phi = (4, 2, 3, 2). \tag{MV.7}$$

The class $\mathcal{A} \in H^2(X^\circ; \mathbb{Z})$ represented by Φ restricts to $2q$, because

$$2q = 4\pi_1^* a_1 + 2\pi_1^* b_1 = 3\pi_2^* a_2 + 2\pi_2^* b_2. \tag{MV.8}$$

If the degree-three bases are Poincaré dual to the degree-one cohomology bases just fixed, the projection formula makes $[\alpha_3]$ the same matrix as (MV.3). Finally

$$[\alpha_0] = \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \qquad [\alpha_4] = \begin{pmatrix} 3 \\ -4 \end{pmatrix}. \tag{MV.9}$$

Every image in (MV.3), (MV.6), and (MV.9) is saturated.

Proof. Choose closed discs D_1', D_2' whose intersection is a contractible band disjoint from the three special points. Over the annular shell between D_j' and the smaller filling disc, f is a proper submersion. An Ehresmann connection lifts a base deformation retraction that is stationary near the smaller disc; adjoining the identity there gives a deformation retraction $X_j \to N_j$. The verified local radial retraction of theorem 10.10 keeps the marked flat coordinate x fixed and sends its endpoint to the orbit $\pi_j(x) \in S_j$. Since the band is contractible, choose its marking by the same radial paths. The composite of the actual band inclusion with $X_j \to N_j \to S_j$ is therefore literally π_j. The standard chain-level Mayer–Vietoris injection is $z \mapsto (i_{1*}z, -i_{2*}z)$ [10, §2.2], proving (MV.2) with its sign.

The affine fundamental-group presentations of the finite fibres and the exact finite pullback lattices of theorem 5.4 give the stated primitive degree-one bases and (MV.3). Its 3×3 minors include 3 and 4, so their gcd is one; direct elimination gives (MV.4). The two bases in (MV.5) are unimodular changes of

$$\langle \gamma\eta_1, q \rangle, \qquad \langle 2\gamma\eta_2, \gamma\eta_2 + q \rangle,$$

the exact integral images of π_j^* in degree two. Evaluating them on the six source tori gives (MV.6). Its rank-three minors include 2 and -3, so the image is saturated; multiplication verifies (MV.7) and (MV.8).

For degree three, if $x \in H_3(F)$ and $\theta \in H^1(S_j)$, then

$$\langle \theta, \pi_{j*} x \rangle = \langle \pi_j^* \theta, x \rangle.$$

Thus in the Poincaré-dual bases the degree-three matrix equals the degree-one matrix. In degree four, the holomorphic quotient coverings have positive degrees 3 and 4; the sole minus sign is the Mayer–Vietoris minus sign. This proves (MV.9) and completes the integral certificate. □

58.6 The cusp attachment and its primitive blocks

Proposition 58.11 (actual cusp kernels and global blocks)**.** ***[VERIFIED]*** *For the second decomposition $X = N_0' \cup X^\circ$, the map $H_r(M) \to H_r(W)$ is surjective and*

$$K_1 = \mathbb{Z}[\widetilde{a}_0], \qquad K_2 = \mathbb{Z}\langle \xi_{\widehat{w}}, \xi_{\widehat{\delta}} \rangle, \tag{MV.10}$$

where

$$\xi_\nu = a_*([S^1_\nu] \times [\widetilde{a}_0])$$

uses the oriented fibrewise translation action and the clockwise toric meridian. Both displayed subgroups are primitive.

Let ζ be the generator of $H_1(X^\circ)$ normalized by $\Phi_1(\zeta) = 1$. In the source basis $(\widehat{\gamma}, \widehat{u}, \widetilde{a}_0)$ of $H_1(M)$ and target basis $(\bar{\gamma}, \bar{u}, \zeta)$ of $H_1(W) \oplus H_1(X^\circ)$, the actual map $(c_*, -j_*)$ is

$$D_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -12 & 0 & -p \end{pmatrix}, \qquad \det D_1 = -p. \tag{MV.11}$$

Orient the connecting map by

$$\partial(j_* \xi_\nu) = (I - A_2)\nu = (I - A_1^{-1})\nu \quad (\nu \in \Lambda_{\mathrm{tor}}). \tag{MV.12}$$

Let e be the primitive generator of $C_2 := \operatorname{coker} \alpha_2$ with $\langle \mathcal{A}, e \rangle = 1$, and put $z = j_* \xi_{\widehat{w}}$. Then (e, z) is a basis of $H_2(X^\circ)$ and

$$j_* \xi_{\widehat{\delta}} = -p\, e, \qquad [j_*|_{K_2}]^{(e,z)}_{(\xi_{\widehat{\delta}}, \xi_{\widehat{w}})} = \begin{pmatrix} -p & 0 \\ 0 & 1 \end{pmatrix}. \tag{MV.13}$$

After splitting the surjections onto $H_r(W)$, the full degree-one and degree-two Mayer–Vietoris maps are integrally equivalent to

$$I_2 \oplus [p], \qquad I_4 \oplus \begin{pmatrix} -p & 0 \\ 0 & 1 \end{pmatrix}, \tag{MV.14}$$

respectively. They are unimodular for the candidate $p = -1$.

Proof. The composite $H_r(F) \to H_r(M) \to H_r(W)$ is the verified collapse c_*, which is surjective by theorem 58.6; hence the second arrow is surjective. Under the Wang projection, the base class $[\widetilde{a}_0]$ is primitive. It bounds the explicit zero-section disc in the toric filling, proving the first formula in (MV.10).

For K_2, use the toric chart $t = z_1 z_2 z_3$. The literal solid tori

$$\{|z_1| \le \rho,\ |z_2| = r,\ z_3 = c\} \quad \text{and} \quad \{|z_2| \le \rho,\ |z_1| = r,\ z_3 = c\}$$

have oriented boundaries $\xi_{\widehat{w}}$ and $\xi_{\widehat{\delta}}$. Under

$$H_2(M) \twoheadrightarrow \ker(M_0 - I|\Lambda) = \Lambda_{\mathrm{tor}},$$

these classes map to $\widehat{w}$ and $\widehat{\delta}$. Their span is therefore primitive; the Wang ranks and the verified homology of W show that it is all of K_2.

The actual meridian calculation gives

$$j_* \widehat{\gamma} = 12\zeta, \qquad j_* \widehat{u} = 0, \qquad j_*[\widetilde{a}_0] = p\zeta,$$

while c_* is projection onto $\mathbb{Z}\langle \bar{\gamma}, \bar{u} \rangle$. This is (MV.11). Because $A_1 A_2 = M_0^{-1}$ acts trivially on Λ_{tor}, the two expressions in (MV.12) agree. Since $A_2 \widehat{w} = -\widehat{u} + \widehat{\delta}$ and $A_2 \widehat{\delta} = \widehat{\delta}$,

$$\partial(j_* \xi_{\widehat{w}}) = -\nu_0, \qquad \partial(j_* \xi_{\widehat{\delta}}) = 0. \tag{MV.15}$$

Thus z is a primitive lift, up to the fixed unit sign, of the generator of $\ker \alpha_1$.

It remains to determine the coefficient of e. Normalize contraction by

$$\iota_v(\theta_1 \wedge \cdots \wedge \theta_k) = \sum_r (-1)^{r-1} \theta_r(v) \theta_1 \wedge \cdots \widehat{\theta_r} \cdots \wedge \theta_k.$$

The globally free $S^1_{\widehat{\delta}}$-action and the sweep/slant identity give

$$\langle \mathcal{A}, j_* \xi_{\widehat{\delta}} \rangle = \langle \iota_{\widehat{\delta}} \mathcal{A}, j_*[\widetilde{a}_0] \rangle, \qquad \iota_{\widehat{\delta}}(2q) = -12\gamma. \tag{MV.16}$$

Let $\Theta \in H^1(X^\circ)$ be the primitive class restricting to 12γ. Restriction to F is injective, so (MV.16) implies $\iota_{\widehat{\delta}} \mathcal{A} = -\Theta$. The based meridian relation gives

$$\Theta(j_*[\widetilde{a}_0]) = p.$$

Hence the left side of (MV.16) is $-p$, which proves (MV.13). Splitting the surjective cusp blocks gives (MV.14). □

Remark 58.12 (one-shot H_1 check). The preceding two-stage calculation is the manuscript's argument. There is also a useful one-shot check. Let U be the torus bundle over the pair-of-pants core and $N = N_0' \sqcup N_1' \sqcup N_2'$. In the target basis

$$(c, \rho_1, \rho_2; \bar{\gamma}, \bar{u}; g_1, h_1; g_2, h_2)$$

and domain basis

$$(\gamma_0, u_0, a_0; \gamma_1, w_1, \rho_1; \gamma_2, w_2, \rho_2),$$

the actual map $(i_{U*}, -i_{N*})$ for the candidate is

$$\Delta_1 = \begin{pmatrix} 1 & 0 & \ell_0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 1 \\ -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -3 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 4 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \end{pmatrix}. \tag{MV.17}$$

For example, a_0 maps to $\ell_0 c - \rho_1 - \rho_2$ on the pair-of-pants side and to zero on the cusp side; γ_1 maps to c and $3g_1$, while γ_2 maps to c and $-4g_2$. Thus every column is an actual primitive collar cycle. Exact elimination gives

$$\det \Delta_1 = 1 - 12\ell_0 = -p$$

for $(\ell_1, \ell_2) = (1, -1)$. At $\ell_0 = 0$, this determinant is one.

58.7 Exact comparison with the Leray transgressions

Theorem 58.13 (Mayer–Vietoris/Leray comparison)**.** ***[VERIFIED]*** *With the preceding primitive bases and clockwise meridians, the actual Mayer–Vietoris obstruction maps commute with the intrinsic Leray transgressions in degrees one, two, and three. If $\omega_{\mathrm{MV}} \in H^2(B; \mathbb{Z})$ is the generator fixed below, then*

$$d_2^{0,1}(12\gamma) = -p\, \omega_{\mathrm{MV}}, \tag{L.1}$$

$$d_2^{0,2}(2q) = -p\, \omega_{\mathrm{MV}}[\delta], \qquad d_2^{0,3}(2\gamma u w) = -p\, \omega_{\mathrm{MV}}[\gamma\delta]. \tag{L.2}$$

For the candidate $p = -1$, all three maps are multiplication by $+1$ in these bases.

Proof. Let V be the pair-of-pants neighbourhood in the base and let D_i be the three special discs. Order the annuli as $(D_1 \cap V, D_2 \cap V, D_0 \cap V)$ and orient their meridians clockwise. Define

$$\omega_{\mathrm{MV}} = \partial_B[(1,0,0)].$$

The chain-level Mayer–Vietoris sequence on the base gives

$$\partial_B(n_1,n_2,n_0) = (n_1+n_2+n_0)\omega_{\mathrm{MV}}, \tag{L.3}$$

because the image of $H^1(V)$ is the sum-zero sublattice.

Choose a lift Θ_V of 12γ over the pair of pants and write $b_i = \Theta_V(\widehat{\rho}_i)$. The finite local lifts have meridian values $4\ell_1$ and $3\ell_2$, while the cusp lift vanishes on $\widetilde{a}_0$. For the general affine cusp regluing used in this comparison, let t_μ denote translation by the class $\mu \in \Lambda/\Lambda_{\mathrm{tor}}$ and set $\ell_0 = \gamma(\mu)$. Thus $\Theta_V(t_\mu) = 12\ell_0$. The tautological gluing of the constructed candidate has $\mu = 0$ and $\ell_0 = 0$; in that case the following relation has no vertical drift. In general,

$$\widetilde{a}_0 = t_\mu \widehat{\rho}_2^{-1}\widehat{\rho}_1^{-1},$$

the local-minus-pair-of-pants discrepancy is

$$(4\ell_1 - b_1,\ 3\ell_2 - b_2,\ -(12\ell_0 - b_1 - b_2)). \tag{L.4}$$

Its signed sum is $-p$. The canonical truncation triangle identifies this very cocycle, functorially on the induced total-space cover, with Leray d_2; this is the construction proved in theorem 53.3. Equations (L.3)–(L.4) prove (L.1).

This is more than equality of the terminal integer. The transpose of the oriented collar map applied to the tuple of local lifts has meridian values

$$(b_1 - 4\ell_1,\ b_2 - 3\ell_2,\ 12\ell_0 - b_1 - b_2),$$

the negative of (L.4). Consequently the square

$$\begin{array}{ccc} \mathbb{Z}\langle 12\gamma\rangle & \xrightarrow{d_2^{0,1}} & \mathbb{Z}\langle \omega_{\mathrm{MV}}\rangle \\ \cong\downarrow & & \downarrow\cong \\ H^1(X^\circ;\mathbb{Z}) & \xrightarrow{(j_*|_{K_1})^\vee} & K_1^\vee \end{array} \tag{L.5}$$

commutes when the left vertical map sends 12γ to Θ and the right sends ω_{MV} to $-\widetilde{a}_0^\vee$. The lower map is multiplication by p, while the upper map is multiplication by $-p$.

For degree two, quotient the primitive lift in (MV.15):

$$\overline{K}_2 = K_2/\mathbb{Z}\xi_{\widehat{w}}, \qquad C_2 = \operatorname{coker}\alpha_2.$$

Equation (MV.13) gives an actual reduced map

$$\overline{j} : \overline{K}_2 \longrightarrow C_2, \qquad [\xi_{\widehat{\delta}}] \longmapsto -p\,e. \tag{L.6}$$

Under the primitive identifications

$$2q \longmapsto \Phi \in C_2^\vee, \qquad \omega_{\mathrm{MV}}[\delta] \longmapsto \xi_{\widehat{\delta}}^\vee \in \overline{K}_2^\vee,$$

the square

$$\begin{array}{ccc} \mathbb{Z}\langle 2q\rangle & \xrightarrow{d_2^{0,2}} & \mathbb{Z}\langle \omega_{\mathrm{MV}}[\delta]\rangle \\ \cong\downarrow & & \downarrow\cong \\ C_2^\vee & \xrightarrow{\overline{j}^\vee} & \overline{K}_2^\vee \end{array} \tag{L.7}$$

commutes, and both horizontal maps are multiplication by $-p$. This proves the first formula in (L.2).

The final formula has no new sign choice. Write

$$d_2(2q) = r\,\omega_{\mathrm{MV}}[\delta], \qquad d_2(2\gamma uw) = s\,\omega_{\mathrm{MV}}[\gamma\delta].$$

Using

$$(12\gamma)(2q) = 12(2\gamma uw), \qquad (2q)(2\gamma uw) = 0,$$

the derivation rule and the already fixed degree-one sign give

$$s = 2(-p) - r.$$

The identities

$$\delta \wedge 2\gamma uw = -2\,\mathrm{vol}, \qquad 2q \wedge \gamma\delta = 2\,\mathrm{vol}$$

applied to the second product give $r = s$. Hence $r = s = -p$, proving the second formula in (L.2).

Changing every meridian orientation reverses p and ω_{MV} simultaneously. Reversing the $S^1_{\widehat{\delta}}$ orientation reverses both $\xi_{\widehat{\delta}}$ and its dual coefficient generator. Thus all remaining convention changes are common units and cannot change an index. □

Validation, repair, or disproof test 58.14 (resolved global topological falsifiers)**. [VERIFIED]** The actual finite-band maps are (MV.2), their degree-one and degree-two images are saturated, the multiple-fibre top class is $(3, -4)$, the cusp kernels are literal primitive collar cycles, the reduced cusp blocks are (MV.11) and (MV.13), and the comparison is the pair of commuting squares (L.5) and (L.7), not a comparison of absolute determinants. The former MV1 and L1 failure modes are therefore excluded for the candidate.

58.8 Complete recognition theorem

Lemma 58.15 (cohomological closure in dimension six)**.** ***[VERIFIED]*** *Let X be a smooth, closed, connected, oriented six-manifold. If*

$$\pi_1(X) = 0, \qquad H^2(X;\mathbb{Z}) = H^3(X;\mathbb{Z}) = 0, \tag{4.4}$$

then $H_k(X;\mathbb{Z})$ is $\mathbb{Z}$ for $k = 0, 6$ and zero for $1 \le k \le 5$.

Proof. The Hurewicz map in degree one identifies $H_1(X;\mathbb{Z}) = \pi_1(X)^{\mathrm{ab}} = 0$. The universal coefficient sequence

$$0 \longrightarrow \mathrm{Ext}(H_1(X),\mathbb{Z}) \longrightarrow H^2(X;\mathbb{Z}) \longrightarrow \mathrm{Hom}(H_2(X),\mathbb{Z}) \longrightarrow 0$$

shows that $\mathrm{Hom}(H_2(X),\mathbb{Z}) = 0$, so the finitely generated group $H_2(X)$ is torsion. The next universal coefficient sequence contains an injection

$$\mathrm{Ext}(H_2(X),\mathbb{Z}) \hookrightarrow H^3(X;\mathbb{Z}) = 0.$$

For a finitely generated torsion group $A = \bigoplus_i \mathbb{Z}/n_i$, one has $\mathrm{Ext}(A,\mathbb{Z}) = \bigoplus_i \mathbb{Z}/n_i$; hence $H_2(X) = 0$. Poincaré duality gives

$$H_3(X) \simeq H^3(X) = 0, \quad H_4(X) \simeq H^2(X) = 0, \quad H_5(X) \simeq H^1(X) = 0.$$

Finally connectedness and orientation give $H_0(X) = H_6(X) = \mathbb{Z}$. □

Theorem 58.16 (sphere recognition from the exact inputs)**.** ***[VERIFIED]*** *Let X be a smooth, closed, connected six-manifold. If*

$$\pi_1(X) = 0, \qquad H_k(X;\mathbb{Z}) \cong \begin{cases} \mathbb{Z}, & k = 0, 6, \\ 0, & 1 \le k \le 5, \end{cases} \tag{4.5}$$

then X is diffeomorphic to the standard S^6.

Proof. A closed smooth manifold has the homotopy type of a finite CW complex. We first prove successively that $\pi_i(X) = 0$ for $1 \le i \le 5$. The case $i = 1$ is assumed. Suppose $2 \le n \le 5$ and $\pi_i(X) = 0$ for $i < n$. Then X is $(n-1)$-connected, so the Hurewicz homomorphism

$$h_n : \pi_n(X) \longrightarrow H_n(X;\mathbb{Z})$$

is an isomorphism by the Hurewicz theorem [10, Theorem 4.32]. The target vanishes by (4.5), hence $\pi_n(X) = 0$. Induction proves the assertion.

Now X is 5-connected. Hurewicz in degree six gives $\pi_6(X) \cong H_6(X;\mathbb{Z}) \cong \mathbb{Z}$. Choose a map $g : S^6 \to X$ whose homotopy class maps to a chosen orientation generator. It has degree one. Hence g_* is an isomorphism in degrees zero and six; in every other degree both integral homology groups vanish. Thus g is an integral homology equivalence. Both spaces are simply connected CW complexes, so the homological Whitehead theorem [10, Corollary 4.33] makes g a homotopy equivalence. Therefore X is a smooth homotopy 6-sphere.

Smale's generalized Poincaré theorem in dimensions greater than four [15, Theorem A] identifies the underlying topological manifold with S^6. The smooth endpoint is stated directly by Smale: every homotopy 6-sphere is diffeomorphic to S^6 [16, Corollary 1.2]. Equivalently, smooth homotopy 6-spheres up to oriented h-cobordism form Θ_6; Kervaire–Milnor's opening table gives $|\Theta_6| = 1$, and their following remark identifies h-cobordism with diffeomorphism in dimension six [17, opening table and following Remark, pp. 504–505]. Hence there is no exotic smooth structure in dimension six, and X is diffeomorphic to the standard sphere. □

Corollary 58.17 (abstract application theorem)**.** ***[VERIFIED]*** *Let X be a smooth, closed, connected, oriented six-manifold. Suppose V1 identifies the actual based van Kampen quotient with $G(\ell_0,\ell_1,\ell_2)$ of theorem 5.5, and suppose an independently constructed MV1/L1 comparison proves*

$$H^2(X;\mathbb{Z}) \cong H^3(X;\mathbb{Z}) \cong \mathbb{Z}/|p|\mathbb{Z}, \qquad p = 12\ell_0 - 4\ell_1 - 3\ell_2. \tag{4.6}$$

If $|p| = 1$, then $X \cong_{\mathrm{diff}} S^6$.

Proof. For $|p| = 1$, theorem 5.5 and the assumed geometric identification give $\pi_1(X) = 0$, while (4.6) gives $H^2(X) = H^3(X) = 0$. Theorem 58.15 converts these inputs into the homology condition (4.5), and theorem 58.16 gives the claimed diffeomorphism. □

Corollary 58.18 (independent Mayer–Vietoris certificate for the candidate)**.** ***[VERIFIED]*** *For the tautological candidate $(\ell_0,\ell_1,\ell_2) = (0,1,-1)$, the two-stage geometric Mayer–Vietoris calculation gives*

$$H_k(X;\mathbb{Z}) \cong \begin{cases} \mathbb{Z}, & k = 0, 6, \\ 0, & 1 \le k \le 5. \end{cases}$$

Together with the independently verified based van Kampen calculation, this proves $X \cong_{\mathrm{diff}} S^6$ without using the Leray computation of the same groups.

Proof. Here $p = -1$. Thus (MV.11) and the reduced block (MV.13) are unimodular. The two Mayer–Vietoris sequences give $H_1(X) = H_2(X) = 0$ integrally, with no torsion hidden by a rank calculation. The verified collapse homology (C.23) gives $e(W) = 1 - 2 + 4 - 2 + 1 = 2$. Away from W, the map to the punctured base is stratified by smooth four-tori and the two bielliptic fibres, all of Euler characteristic zero. Additivity therefore gives $e(X) = e(W) = 2$.

Poincaré duality and universal coefficients give $H_5(X) = H_4(X) = 0$. Moreover

$$H_3(X) \cong H^3(X) \cong \mathrm{Hom}(H_3(X),\mathbb{Z}),$$

because $H_2(X) = 0$; hence $H_3(X)$ is free. The equality $e(X) = 2$ forces its rank to be zero. The based van Kampen result theorem 53.6 gives $\pi_1(X) = 1$, and theorem 58.16 completes the proof. □

Thus C1–C3, V1, MV1, and L1 are all verified for the candidate. The nearby-cycle/Leray route and the collapse/Mayer–Vietoris route are now two separate integral certificates for the same sphere-recognition input.

59 Fibre degree and the cyclic-six centre

This section supplies the theorem–proof module behind the fibre-degree and cyclic-centre formulas recorded in the orientation material. There are three different objects in the calculation:

$$H^4(U;\mathbb{Z}), \qquad H^4(U;\mathbb{Z}/6), \qquad Z(SU(6)) = \mu_6.$$

The first is a free rank-six characteristic lattice, the second has six independent residue coordinates, and the third is a single cyclic group. The candidate determines one particular surjection from the first object to the third. It does not identify all three objects, and it does not construct a gauge connection or a spectral operator.

59.1 Spaces, markings, and the finite-band presentation

Retain the reconstructed map

$$f : X \longrightarrow \mathbb{P}^1, \qquad X \cong_{\mathrm{diff}} S^6,$$

and write

$$D_0 = W, \qquad D_1 = S_1, \qquad D_2 = S_2, \qquad D = D_0 \sqcup D_1 \sqcup D_2, \qquad U = X \setminus D.$$

Here W is the reduced non-normal fibre and $3S_1, 4S_2$ are the two multiple fibres. The verified local homology computations give

$$H_1(W;\mathbb{Z}) = \mathbb{Z}\langle a_0, b_0\rangle, \qquad H_1(S_j;\mathbb{Z}) = \mathbb{Z}\langle a_j, b_j\rangle \quad (j = 1, 2).$$

The marking is

$$a_0 = [\widehat{\gamma}], \quad b_0 = [\widehat{u}], \quad a_j = [\widehat{g}_j],$$

where $\widehat{g}_j$ is the clockwise affine deck generator. The second generator b_j is the invariant coinvariant coordinate. With the marking used in the finite-band calculation one may take $b_1 = \pi_{1*}[\widehat{w}]$ and $b_2 = \pi_{2*}[\widehat{w}]$; in the second coinvariant quotient $\pi_{2*}[\widehat{u}] = \pi_{2*}[\widehat{w}]$, so this agrees with the alternative representative used elsewhere. The two specializations of the marked $\widehat{\gamma}$-cycle are

$$\pi_{1*}[\widehat{\gamma}] = 3a_1, \qquad \pi_{2*}[\widehat{\gamma}] = -4a_2. \tag{1739}$$

Fix Alexander duality using the complex orientation of X, outward-first boundary orientation, and the convention that the Poincare dual of a positive nearby fibre is the negative covector of its clockwise meridian. It identifies

$$L := H^4(U;\mathbb{Z}) \xrightarrow{\sim} H_1(D;\mathbb{Z}) = \mathbb{Z}\langle a_0, b_0; a_1, b_1; a_2, b_2\rangle \cong \mathbb{Z}^6. \tag{1740}$$

Indeed, D is compact and locally contractible, so

$$\widetilde{H}^q(S^6 \setminus D;\mathbb{Z}) \cong \widetilde{H}_{5-q}(D;\mathbb{Z}).$$

Take $q = 4$ and use the disjoint-union decomposition of D.

Let X° be the space obtained after the two finite fillings and before the central toric filling. For logarithmic twist integers ℓ_1, ℓ_2, its verified degree-one Mayer–Vietoris calculation is the presentation

$$H_1(X^\circ;\mathbb{Z}) = \frac{\mathbb{Z}\langle c, \bar{g}_1, \bar{g}_2\rangle}{\mathbb{Z}\langle 3\bar{g}_1 - \ell_1 c, 4\bar{g}_2 - \ell_2 c\rangle}. \tag{1741}$$

Thus $c, \bar{g}_1, \bar{g}_2$ are three marked generators, not a basis. The relation matrix is

$$R_1 = \begin{pmatrix} -\ell_1 & 3 & 0 \\ -\ell_2 & 0 & 4 \end{pmatrix}.$$

For admissible twists $3 \nmid \ell_1$ and $2 \nmid \ell_2$, the gcd of its rank-two minors

$$3\ell_2, \qquad -4\ell_1, \qquad 12$$

is one. Therefore the presented group is torsion-free of rank one. The homomorphism

$$\begin{aligned} &\Phi_1 : H_1(X^\circ;\mathbb{Z}) \longrightarrow \mathbb{Z}, \\ &\qquad \Phi_1(c) = 12, \quad \Phi_1(\bar{g}_1) = 4\ell_1, \quad \Phi_1(\bar{g}_2) = 3\ell_2 \end{aligned} \tag{1742}$$

annihilates both relations. Its three displayed values have gcd one under the same admissibility hypotheses; hence it is an isomorphism. For the candidate

$$(\ell_0, \ell_1, \ell_2) = (0, 1, -1), \qquad \Phi_1(c) = 12, \quad \Phi_1(\bar{g}_1) = 4, \quad \Phi_1(\bar{g}_2) = -3. \tag{1743}$$

Put

$$\zeta := \bar{g}_1 + \bar{g}_2. \tag{1744}$$

Then $\Phi_1(\zeta) = 1$. Since Φ_1 is an isomorphism, ζ is the positive basis element of $H_1(X^\circ;\mathbb{Z})$ selected by Φ_1, and

$$c = 12\zeta, \qquad \bar{g}_1 = 4\zeta, \qquad \bar{g}_2 = -3\zeta. \tag{1745}$$

Let

$$\overline{q}_{\mathrm{col}} : H_1(D;\mathbb{Z}) \longrightarrow H_1(X^\circ;\mathbb{Z})$$

be the homomorphism sending a_0, a_1, a_2 to $c, \bar{g}_1, \bar{g}_2$, respectively, and sending every b_i to zero. Define $q_{\mathrm{col}} = \overline{q}_{\mathrm{col}} \circ \mathrm{AD}$, where AD is the fixed Alexander isomorphism above. Thus the marked collar homomorphism is well typed:

$$\begin{aligned} &q_{\mathrm{col}} : L \longrightarrow H_1(X^\circ;\mathbb{Z}), \\ &q_{\mathrm{col}}(a_0) = c, && q_{\mathrm{col}}(b_0) = 0, \\ &q_{\mathrm{col}}(a_1) = \bar{g}_1, && q_{\mathrm{col}}(b_1) = 0, \\ &q_{\mathrm{col}}(a_2) = \bar{g}_2, && q_{\mathrm{col}}(b_2) = 0. \end{aligned} \tag{1746}$$

The central-band projection sends $a_0 = [\widehat{\gamma}]$ to c and $b_0 = [\widehat{u}]$ to zero. In the two finite-band cokernels, the actual Mayer–Vietoris map is $(\pi_{1*}, -\pi_{2*})$; it sends the affine generators to $\bar{g}_1, \bar{g}_2$ and kills the invariant coordinates b_1, b_2. These statements derive the six displayed values from the marked collar calculation. The map is surjective because its images contain the three presentation generators in (1741). It is not an inclusion-induced map $H_1(D) \to H_1(X^\circ)$: no inclusion of the deleted central fibre W into X° exists.

59.2 The exact fibre-degree row

Let $F \cong T^4$ be a complex-oriented smooth fibre and choose $\omega_F \in H^4(F;\mathbb{Z})$ with $\langle \omega_F, [F] \rangle = 1$. If $i_F : F \hookrightarrow U$ is the inclusion, put $\mathrm{res}_F = i_F^* : H^4(U;\mathbb{Z}) \to H^4(F;\mathbb{Z})$. Define

$$\mathrm{res}_F(\alpha) = \deg_F(\alpha)\omega_F, \qquad \deg_F : L \longrightarrow \mathbb{Z}. \tag{1747}$$

Theorem 59.1 (Fibre-degree row)**.** *With the fixed Alexander-duality convention above, one has*

$$\boxed{\deg_F(n_0 a_0 + r_0 b_0 + n_1 a_1 + r_1 b_1 + n_2 a_2 + r_2 b_2) = 12n_0 + 4n_1 - 3n_2.} \tag{1748}$$

Equivalently,

$$[\deg_F]_{(a_0,b_0,a_1,b_1,a_2,b_2)} = (12, 0, 4, 0, -3, 0).$$

Moreover,

$$\boxed{\deg_F = \Phi_1 \circ q_{\mathrm{col}}} \tag{1749}$$

for the candidate twists, and

$$\boxed{\ker \deg_F = \mathbb{Z}\langle b_0, b_1, b_2, a_0 - 3a_1, 3a_1 + 4a_2\rangle.} \tag{1750}$$

This kernel is saturated of rank five.

Proof. We retain every sign in the one-shot collar calculation. In the ordered boundary-source basis

$$(\gamma_0, u_0, \widetilde{a}_0; \gamma_1, w_1, \rho_1; \gamma_2, w_2, \rho_2)$$

and ordered target basis

$$(c, \bar{\rho}_1, \bar{\rho}_2; a_0, b_0; a_1, b_1; a_2, b_2),$$

the actual map $(i_{U*}, -i_{N*})$ at $(\ell_0, \ell_1, \ell_2) = (0, 1, -1)$ is

$$\Delta_1 = \begin{pmatrix} 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 1 \\ -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -3 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 4 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \end{pmatrix}. \tag{1751}$$

For example, the columns say

$$\gamma_0 \mapsto c - a_0, \quad u_0 \mapsto -b_0, \quad \widetilde{a}_0 \mapsto -\bar{\rho}_1 - \bar{\rho}_2,$$

$$\gamma_1 \mapsto c - 3a_1, \quad w_1 \mapsto -b_1, \quad \rho_1 \mapsto \bar{\rho}_1 - a_1,$$

$$\gamma_2 \mapsto c + 4a_2, \quad w_2 \mapsto -b_2, \quad \rho_2 \mapsto \bar{\rho}_2 - a_2.$$

Thus the entries $-3, +4$ are exactly (1739); all other displayed minus signs are the second-inclusion signs in the Mayer–Vietoris convention. Direct integral elimination gives $\det \Delta_1 = 1$.

Let $\widetilde{a}_0^\vee$ be the integral covector on the ordered boundary-source group that is one on $\widetilde{a}_0$ and zero on the other eight basis elements. Pullback of target covectors is represented by Δ_1^{T}. By the fixed orientation convention, the target covector extending the Poincare dual of a complex-oriented nearby fibre must pull back to $-\widetilde{a}_0^\vee$. Since Δ_1 is unimodular, there is a unique integral extension. Its last six entries are precisely the Alexander coordinates of the restriction functional on $H_1(D)$. Consequently they are obtained from

$$\Delta_1^{\mathsf{T}} y = -\widetilde{a}_0^\vee. \tag{1752}$$

Writing

$$y = (A, B, C; X_0, Y_0; X_1, Y_1; X_2, Y_2),$$

the nine equations are

$$A = X_0, \quad Y_0 = Y_1 = Y_2 = 0, \quad A = 3X_1, \quad A = -4X_2,$$

$$B = X_1, \qquad C = X_2, \qquad -B - C = -1.$$

Their unique solution is

$$y = (12, 4, -3; 12, 0; 4, 0; -3, 0).$$

The last six entries prove (1748). Equations (1742), (1743), and (1746) give the same row on the six basis elements, proving the typed equality (1749).

The row is primitive because $\gcd(12, 4, 3) = 1$; explicitly, $\deg_F(a_1 + a_2) = 1$. Each of the five displayed elements in (1750) lies in the kernel. Conversely, if $12n_0 + 4n_1 - 3n_2 = 0$, reduction modulo four gives $n_2 = 4v$, and then $n_1 = 3v - 3n_0$. Therefore

$$n_0a_0 + n_1a_1 + n_2a_2 = n_0(a_0 - 3a_1) + v(3a_1 + 4a_2).$$

The three b_i-coordinates are free, and the five displayed generators are independent. They therefore form a basis of the kernel. Finally, a kernel of a surjection from a free abelian group to $\mathbb{Z}$ is saturated, proving the last assertion. □

Reversing the global Alexander-duality convention negates all six entries of (1748) simultaneously. It cannot change the sign of only one component. In particular, with the convention fixed here, $q_{\mathrm{col}}(a_2) = +\bar{g}_2$ while the coordinate of $\bar{g}_2$ in the positive generator of $H_1(X^\circ; \mathbb{Z})$ is -3.

For completeness, if $j : M \hookrightarrow X^\circ$ denotes the central collar inclusion, the toric attaching meridian obeys

$$j_*[\widetilde{a}_0] = \ell_0 c - \bar{g}_1 - \bar{g}_2,$$

and hence

$$\Phi_1(j_*[\widetilde{a}_0]) = 12\ell_0 - 4\ell_1 - 3\ell_2 = p.$$

At the candidate values this is $p = -1$. Thus the same correctly typed functional controls the smooth-fibre degree and the terminal meridian, but the two statements use the maps displayed above rather than an untyped identification of their source groups.

59.3 Normal-line holonomy and the two finite reductions

The transverse disc in the order-m_j local quotient is acted on by $e^{-2\pi i/m_j}$, with $m_1 = 3, m_2 = 4$. Therefore the normal-line holonomy characters, in the marked bases, are

$$\begin{aligned} h_1 &: H_1(S_1; \mathbb{Z}) \longrightarrow \mu_3, \quad h_1(a_1) = e^{-2\pi i/3}, \quad h_1(b_1) = 1, \\ h_2 &: H_1(S_2; \mathbb{Z}) \longrightarrow \mu_4, \quad h_2(a_2) = e^{-2\pi i/4}, \quad h_2(b_2) = 1. \end{aligned} \tag{1753}$$

Via (1740), write each $x \in L$ uniquely as $x = (x_0, x_1, x_2)$ with $x_i \in H_1(D_i; \mathbb{Z})$. Regard μ_3 and μ_4 as their standard subgroups of μ_{12}. The inverse on the second factor below records the candidate twist $\ell_2 = -1$, rather than a discretionary change of sign. Define

$$\begin{aligned} \chi_{12} &: L \longrightarrow \mu_{12}, \quad \chi_{12}(x_0, x_1, x_2) = h_1(x_1)h_2(x_2)^{-1}, \\ \chi_6 &: L \longrightarrow \mu_6, \qquad\qquad \chi_6(x) = \chi_{12}(x)^2. \end{aligned} \tag{1754}$$

Proposition 59.2 (The order-twelve and order-six reductions)**.** *Put*

$$\rho_m : \mathbb{Z} \longrightarrow \mu_m, \qquad \rho_m(k) = e^{-2\pi ik/m}.$$

Then

$$\boxed{\chi_{12} = \rho_{12} \circ \deg_F, \qquad \chi_6 = \rho_6 \circ \deg_F\,.} \tag{1755}$$

Both characters are surjective, and

$$L/\ker \chi_{12} \cong \mu_{12}, \qquad L/\ker \chi_6 \cong \mu_6. \tag{1756}$$

Proof. For $x = \sum_{i=0}^{2}(n_i a_i + r_i b_i)$, (1753) gives, with $\zeta_{12} = e^{-2\pi i/12}$,

$$\chi_{12}(x) = \zeta_{12}^{4n_1 - 3n_2}.$$

Equation (1748) differs in its exponent only by $12n_0$, which vanishes modulo twelve. This proves the first identity. Squaring it is reduction of the same integer modulo six and proves the second. Since $\deg_F(a_1 + a_2) = 1$, both reductions are surjective. The first isomorphism theorem gives (1756). □

The kernels can be read without suppressing any coordinate:

$$\ker \chi_{12} = \mathbb{Z}a_0 \oplus \mathbb{Z}b_0 \oplus 3\mathbb{Z}a_1 \oplus \mathbb{Z}b_1 \oplus 4\mathbb{Z}a_2 \oplus \mathbb{Z}b_2,$$

$$\ker \chi_6 = \mathbb{Z}a_0 \oplus \mathbb{Z}b_0 \oplus 3\mathbb{Z}a_1 \oplus \mathbb{Z}b_1 \oplus 2\mathbb{Z}a_2 \oplus \mathbb{Z}b_2.$$

The first follows by reducing $4n_1 - 3n_2$ modulo three and four; the second follows by reducing it modulo three and two. In particular, the natural combined character has order twelve before its order-six quotient.

59.4 Coefficient reduction is not the distinguished quotient

Alexander duality also gives

$$H^5(U;\mathbb{Z}) \cong \widetilde{H}_0(D;\mathbb{Z}) \cong \mathbb{Z}^2.$$

The cohomological universal-coefficient sequence with coefficients $\mathbb{Z}/6$ is therefore

$$0 \longrightarrow H^4(U;\mathbb{Z}) \otimes \mathbb{Z}/6 \longrightarrow H^4(U;\mathbb{Z}/6) \longrightarrow \operatorname{Tor}_1^{\mathbb{Z}}(H^5(U;\mathbb{Z}),\mathbb{Z}/6) \longrightarrow 0,$$

whose last term is zero. Thus the marked coefficient group is

$$\boxed{H^4(U;\mathbb{Z}/6) \cong L/6L \cong (\mathbb{Z}/6)^6.} \tag{1757}$$

If $r_6 : L \to L/6L$ is coefficient reduction, then χ_6 factors as

$$L \xrightarrow{r_6} (\mathbb{Z}/6)^6 \xrightarrow{\overline{d}_F} \mathbb{Z}/6 \xrightarrow{k \mapsto e^{-2\pi i k/6}} \mu_6,$$

where, in the six marked residue coordinates,

$$[\overline{d}_F] = (0,0,4,0,3,0) \pmod 6. \tag{1758}$$

Thus

$$\overline{d}_F(\bar{n}_0,\bar{r}_0,\bar{n}_1,\bar{r}_1,\bar{n}_2,\bar{r}_2) = 4\bar{n}_1 + 3\bar{n}_2 \in \mathbb{Z}/6.$$

It is surjective because $\overline{d}_F(0,0,1,0,1,0) = 1$ in $\mathbb{Z}/6$. Consequently

$$L/\ker \chi_6 \cong (L/6L)/\ker \overline{d}_F \cong \mathbb{Z}/6.$$

This is one distinguished cyclic quotient of the six-coordinate group in (1757); it is not that group itself.

59.5 The tensor subgroup and the genuine common-centre quotient

Theorem 59.3 (Tensor subgroup and centres)**.** *Tensor product defines a faithful closed homomorphism*

$$\iota : SU(2) \times SU(3) \hookrightarrow SU(6), \qquad \iota(A,B) = A \otimes B. \tag{1759}$$

Its restriction to the centres is the isomorphism

$$\mu_2 \times \mu_3 \xrightarrow{\sim} \mu_6 = Z(SU(6)), \qquad (\lambda,\mu) \longmapsto \lambda\mu. \tag{1760}$$

The image of ι is a proper subgroup of $SU(6)$, of real dimension $3 + 8 = 11$.

Proof. The matrix $A \otimes B$ is unitary and

$$\det(A \otimes B) = (\det A)^3 (\det B)^2 = 1,$$

so ι has the stated codomain. Suppose more generally that $A \otimes B = sI_6$. Choose an eigenvector w of B, with eigenvalue β. Then

$$(A \otimes B)(v \otimes w) = \beta Av \otimes w = sv \otimes w$$

for every v, so $A = (s/\beta)I_2$; substitution shows that B is scalar as well. Applying this scalar–tensor lemma with $s = 1$ gives

$$A = \lambda I_2, \qquad B = \mu I_3, \qquad \lambda\mu = 1.$$

The determinant conditions give $\lambda^2 = 1$ and $\mu^3 = 1$. Hence $\lambda = \mu^{-1} \in \mu_2 \cap \mu_3 = \{1\}$, proving injectivity. A continuous injective homomorphism from a compact group into a Hausdorff group has closed image.

On centres the tensor matrix is $(\lambda\mu)I_6$. Multiplication is injective because $\mu_2 \cap \mu_3 = \{1\}$, and both its domain and target have six elements, proving (1760). Explicitly, the inverse sends $s \in \mu_6$ to (s^3, s^{-2}). Finally the dimension of the image is $11 < 35 = \dim SU(6)$, so it is proper. □

The isomorphisms proved above give the explicit chain

$$L/\ker\chi_6 \xrightarrow{\overline{\chi}_6} \mu_6 \xrightarrow{\lambda \mapsto \lambda I_6} Z(SU(6)). \tag{1761}$$

This is an isomorphism of finite groups. It does not provide transition functions for an $SU(6)$-bundle. Also, ι does not factor through the quotient of $SU(2) \times SU(3)$ by its full centre: that centre maps isomorphically, rather than trivially, to $Z(SU(6))$.

Put $\zeta_6 = e^{2\pi i/6}$. In the integer $U(1)$-charge normalization, the common-centre generator used for the Standard-Model global form is

$$(\zeta_6, -I_2, e^{2\pi i/3} I_3).$$

If $\omega = e^{2\pi i/3} = \zeta_6^2$, then

$$\iota(-I_2, \omega I_3) = (-\omega)I_6 = \zeta_6^{-1} I_6;$$

the first coordinate ζ_6 cancels this scalar under the tensor–scalar map below. This agrees with the convention in David Tong, *Line Operators in the Standard Model*, JHEP 07 (2017) 104, arXiv:1705.01853v2. The following matrix theorem proves the quotient directly and does not rely on a physical interpretation of that reference.

Theorem 59.4 (Common-centre quotient with $U(1)$)**.** *Define*

$$\delta : \mu_6 \hookrightarrow U(1) \times SU(2) \times SU(3), \qquad \delta(\xi) = (\xi, \xi^3 I_2, \xi^2 I_3), \tag{1762}$$

and

$$\widehat{\iota} : U(1) \times SU(2) \times SU(3) \longrightarrow U(6), \qquad \widehat{\iota}(z, A, B) = z(A \otimes B). \tag{1763}$$

Then

$$\ker \widehat{\iota} = \delta(\mu_6), \tag{1764}$$

so the induced map is a faithful closed embedding

$$\frac{U(1) \times SU(2) \times SU(3)}{\delta(\mu_6)} \hookrightarrow U(6). \tag{1765}$$

Moreover,

$$\boxed{\det \widehat{\iota}(z, A, B) = z^6,} \qquad \boxed{\operatorname{im} \widehat{\iota} \cap SU(6) = \iota(SU(2) \times SU(3)).} \tag{1766}$$

Proof. For $\xi \in \mu_6$, one has

$$\det(\xi^3 I_2) = \xi^6 = 1, \qquad \det(\xi^2 I_3) = \xi^6 = 1.$$

Thus δ has the stated codomain. It is central, is a homomorphism, and is injective because its first coordinate is ξ.

For $\xi \in \mu_6$,

$$\widehat{\iota}(\delta(\xi)) = \xi(\xi^3 I_2 \otimes \xi^2 I_3) = \xi^6 I_6 = I_6,$$

so $\delta(\mu_6)$ lies in the kernel. Conversely, suppose $z(A \otimes B) = I_6$. The eigenvalue argument in the preceding proof, now applied to $A \otimes B = z^{-1} I_6$, gives

$$A = \lambda I_2, \qquad B = \mu I_3, \qquad \lambda^2 = 1, \quad \mu^3 = 1, \quad z\lambda\mu = 1.$$

The Chinese-remainder isomorphism

$$\mu_6 \longrightarrow \mu_2 \times \mu_3, \qquad \xi \longmapsto (\xi^3, \xi^2)$$

provides a unique $\xi \in \mu_6$ with $\lambda = \xi^3$ and $\mu = \xi^2$. Then $z = (\lambda\mu)^{-1} = \xi$, because $\xi^{-5} = \xi$. Hence the kernel is exactly $\delta(\mu_6)$. Compactness makes the induced faithful image closed.

The determinant formula is

$$\det\bigl(z(A \otimes B)\bigr) = z^6 (\det A)^3 (\det B)^2 = z^6.$$

Every tensor matrix lies in the intersection by taking $z = 1$. Conversely, if $z(A \otimes B)$ lies in $SU(6)$, then $z^6 = 1$. By (1760), choose $(\lambda, \mu) \in \mu_2 \times \mu_3$ with $\lambda\mu = z$. Then

$$z(A \otimes B) = (\lambda A) \otimes (\mu B),$$

and $\lambda A \in SU(2)$, $\mu B \in SU(3)$. This proves the reverse inclusion in (1766). □

Writing the quotient in (1765) as G_6, determinant therefore gives the exact sequence

$$1 \longrightarrow SU(2) \times SU(3) \xrightarrow{\ \iota\ } G_6 \xrightarrow{\ \det \circ \widehat{\iota}\ } U(1) \longrightarrow 1.$$

Here the middle arrow means the tensor embedding followed by the quotient identification with $\operatorname{im}\widehat{\iota}$; surjectivity on the right follows because $z \mapsto z^6$ is surjective on $U(1)$, and its kernel is (1766).

The quotient in (1765) is the genuine common-centre quotient: the $U(1)$ factor cancels the nontrivial scalar produced by the two special-unitary centres. The displayed tensor–scalar image in $U(6)$ is not contained in $SU(6)$; by (1766), its intersection with $SU(6)$ is exactly the tensor subgroup. The quotient is not isomorphic to $SU(6)$: their real dimensions are 12 and 35.

59.6 Exact boundary of the result

The construction proves the primitive fibre-degree map, its saturated kernel, its order-twelve and order-six finite reductions, the distinction between the full coefficient group and the selected cyclic quotient, and the two matrix-group embeddings. It does not produce a degree-changing morphism from

$$H^4(U; \mathbb{Z}) \longrightarrow H^2(M; \mu_6)$$

for a four-dimensional spacetime M. In particular, χ_6 is not, merely by its codomain, the obstruction to lifting a common-centre quotient bundle. No physical gauge bundle, connection, Yang–Mills Hamiltonian, confinement statement, or mass-gap conclusion is inferred here.

60 The non-normal fibre and the CDP obstruction

Campana–Demailly–Peternell prove that a compact complex threefold with $H^1(X;\mathbb{Z}) = H^2(X;\mathbb{Z}) = 0$, algebraic dimension one, and holomorphic algebraic reduction has $c_3(X) \leq 0$ [3, Theorem 2.2, p. 680]. Their corollary gives $a(X) = 0$ for a compact complex threefold homeomorphic to S^6 [3, Corollary 2.3, p. 680]. The candidate asserts all those hypotheses together with $c_3(X) = 2$ and $a(X) = 1$. This section separates:

(i) validity of one printed inference on p. 692 of [3];

(ii) validity of the published theorem by any other argument;

(iii) validity of the candidate's proposed non-normal-fibre mechanism; and

(iv) truth of the candidate's upstream construction and topology.

60.1 The exact p. 692 inference

Let S be the reduction of a fibre in a smooth threefold $\mathcal{X}$, and let A denote the line-bundle twist after duality. The two relevant sequences are

$$0 \longrightarrow N^*_{S/\mathcal{X}} \otimes A \longrightarrow \Omega^1_{\mathcal{X}}|_S \otimes A \xrightarrow{\rho} \Omega^1_S \otimes A \longrightarrow 0 \tag{5.1}$$

and

$$0 \longrightarrow \mathcal{T}(\Omega^1_S) \otimes A \longrightarrow \Omega^1_S \otimes A \longrightarrow \widetilde{\Omega}^1_S \otimes A \longrightarrow 0, \tag{5.2}$$

where $\mathcal{T}$ denotes sections supported on a nowhere dense analytic set. The opening reduction of [3, §7, p. 692] treats the vanishing of the global sections of the left end of (5.1) and the right end of (5.2) as sufficient for the vanishing of the middle term of (5.1). That inference leaves untreated a section whose image under ρ is a nonzero torsion section. The omitted sheaf can be identified exactly. Put

$$\mathscr{T}_S := \ker(\Omega^1_S \to j_*\Omega^1_{S_{\mathrm{reg}}}), \qquad \widetilde{\Omega}^1_S = \Omega^1_S/\mathscr{T}_S,$$

and

$$\mathscr{K}_S := \ker(\Omega^1_X|_S \longrightarrow \widetilde{\Omega}^1_S).$$

Proposition 60.1 (the missing cotangent filtration)**.** *Let S be a reduced effective Cartier divisor in a smooth complex threefold X. There are natural short exact sequences*

$$0 \longrightarrow N^*_{S/X} \longrightarrow \mathscr{K}_S \longrightarrow \mathscr{T}_S \longrightarrow 0,$$

$$0 \longrightarrow \mathscr{K}_S \longrightarrow \Omega^1_X|_S \longrightarrow \widetilde{\Omega}^1_S \longrightarrow 0.$$

Proof. The conormal sequence is right exact. It is also left exact here. Locally write $S = (g = 0)$ with g reduced. If $h\,dg|_S = 0$, then h vanishes at the generic smooth point of every irreducible component, because dg is nonzero there. The ring $\mathcal{O}_S$ is reduced, so the intersection of its minimal primes is zero; hence $h = 0$. Thus

$$0 \to N^*_{S/X} \to \Omega^1_X|_S \to \Omega^1_S \to 0$$

is exact. By definition, $\mathscr{K}_S$ is the inverse image of $\mathscr{T}_S$ under the last surjection. Restriction to that inverse image is onto $\mathscr{T}_S$ and has kernel $N^*_{S/X}$, proving the first sequence. Composing with the quotient by $\mathscr{T}_S$ proves the second. □

Proposition 60.2 (the exact global-section criterion)**.** *Let A be a line bundle on S. If*

$$H^0(S, \widetilde{\Omega}^1_S \otimes A) = 0,$$

then

$$H^0(S, \Omega^1_X|_S \otimes A) = H^0(S, \mathscr{K}_S \otimes A).$$

If also $H^0(S, N^*_{S/X} \otimes A) = 0$, then this group is canonically

$$\ker\Big[H^0(S, \mathscr{T}_S \otimes A) \xrightarrow{\delta_A} H^1(S, N^*_{S/X} \otimes A)\Big].$$

Hence the two vanishings printed in the p. 692 reduction imply the desired ambient vanishing if and only if δ_A is injective.

Proof. Tensoring the second sequence of theorem 60.1 by the locally free sheaf A preserves exactness; its global-section sequence gives the first equality. Tensor the first sequence and take global sections. If the conormal H^0 vanishes, exactness identifies the middle group with the displayed kernel. No condition on the torsion-free quotient controls that kernel. □

60.2 Local algebra of a reduced normal-crossings fibre

Let $g : \mathcal{X} \to \Delta$ be a proper holomorphic map from a smooth complex threefold to a disc, and put $S = g^{-1}(0)$. Assume S is reduced and has normal crossings: at every point there are coordinates (x, y, z) in which

$$g = ux, \qquad g = uxy, \qquad \text{or} \qquad g = uxyz \tag{5.3}$$

for a unit u. Let $\eta : \widetilde{S} \to S$ be the normalization, $D = S_{\mathrm{sing,red}}$, and $\widetilde{D} = \eta^{-1}(D)_{\mathrm{red}}$.

Lemma 60.3 (the conormal map is injective)**.** ***[VERIFIED]*** *In the setting above, $N^*_{S/\mathcal{X}} \simeq \mathcal{O}_S$, and its generator maps to $dg|_S$. Multiplication by $dg|_S$ is injective, so*

$$0 \longrightarrow \mathcal{O}_S \xrightarrow{h \mapsto h\,dg} \Omega^1_{\mathcal{X}}|_S \longrightarrow \Omega^1_S \longrightarrow 0 \tag{5.4}$$

is exact.

Proof. The fibre ideal is the principal ideal (g) and is reduced, so $(g)/(g^2) \simeq \mathcal{O}_S$. Multiplication by a unit does not affect injectivity, and the term $f\,du$ vanishes after restriction to $f = 0$. It therefore suffices to use $f = x, xy, xyz$.

For $f = xy$, if a lift $\widetilde{h}$ satisfies $\widetilde{h}(y\,dx + x\,dy) = 0$ modulo (xy), then $\widetilde{h}y, \widetilde{h}x \in (xy)$. Hence $\widetilde{h} \in (x) \cap (y) = (xy)$ in the regular local ring, so $h = 0$ on S. For $f = xyz$, the three coefficient conditions say $\widetilde{h}yz, \widetilde{h}xz, \widetilde{h}xy \in (xyz)$; hence $\widetilde{h} \in (x) \cap (y) \cap (z) = (xyz)$. The case $f = x$ is immediate. Thus the conormal map is injective in every local model, proving (5.4). □

Lemma 60.4 (normalization–conductor exact sequence)**.** ***[VERIFIED]*** *The sequence*

$$0 \longrightarrow \mathcal{O}_S \xrightarrow{\alpha} \eta_*\mathcal{O}_{\widetilde{S}} \oplus \mathcal{O}_D \xrightarrow{\beta} \eta_*\mathcal{O}_{\widetilde{D}} \longrightarrow 0, \tag{5.5}$$

$$\alpha(h) = (\eta^* h, h|_D), \qquad \beta(\widetilde{h}, k) = \widetilde{h}|_{\widetilde{D}} - \eta^* k,$$

is exact.

Proof. The assertion is local. Away from D, η is an isomorphism. At a double point $S = \{xy = 0\}$, a member of $\ker \beta$ is

$$\big((h_1(y,z), h_2(x,z)), k(z)\big), \qquad h_1(0,z) = h_2(0,z) = k(z).$$

The function

$$h = h_1(y,z) + h_2(x,z) - h_1(0,z)$$

on $\{xy = 0\}$ restricts to h_1, h_2 on the two branches and to k on their intersection. This proves middle exactness in the double model.

At a triple point $S = \{xyz = 0\}$, write the branch functions as $h_1(y,z)$, $h_2(x,z)$, $h_3(x,y)$. Membership in $\ker \beta$ is exactly

$$h_1(0,z) = h_2(0,z), \quad h_1(y,0) = h_3(0,y), \quad h_2(x,0) = h_3(x,0).$$

Then

$$\begin{aligned} h = {} & h_1(y,z) + h_2(x,z) + h_3(x,y) \\ & - h_1(0,z) - h_1(y,0) - h_2(x,0) + h_1(0,0) \end{aligned}$$

restricts to the prescribed function on every branch. Injectivity of α follows because S is reduced and η is an isomorphism over a dense open set. Surjectivity of β follows locally by lifting a function on the branch conductor to the smooth branch and taking $k = 0$. This proves exactness in every local model. □

Theorem 60.5 (conductor-supported ambient differential)**.** ***[VERIFIED]*** *In the setting (5.3), let A be a line bundle on S with $\eta^* A \simeq \mathcal{O}_{\widetilde{S}}$, and choose a nowhere-vanishing section e of $\eta^* A$. There is a unique nonzero*

$$s \in H^0(S, \Omega^1_{\mathcal{X}}|_S \otimes A) \tag{5.6}$$

such that

$$\eta^* s = \eta^*(dg|_S) \otimes e, \qquad s|_D = 0. \tag{5.7}$$

Its image $\rho(s)$ in $\Omega^1_S \otimes A$ is supported on D and therefore torsion; its image in $\widetilde{\Omega}^1_S \otimes A$ is zero. Moreover:

- *(a) $\rho(s) = 0$ if and only if $e = \eta^* \sigma$ for a nowhere-vanishing $\sigma \in H^0(S, A)$;*
- *(b) if S is connected and $A \not\simeq \mathcal{O}_S$, then $H^0(S, A) = 0$ and s does not come from the conormal term in (5.1).*

Proof. In the double model, $d(xy) = y\,dx + x\,dy$ has coefficients in the reduced ideal (x, y) of D. In the triple model,

$$d(xyz) = yz\,dx + xz\,dy + xy\,dz$$

has coefficients in (xy, xz, yz), the reduced ideal of the union of the three double curves. Unit factors in (5.3) do not change this after restriction to S. Hence $\eta^*(dg|_S)$ restricts to zero on $\widetilde{D}$. Tensor (5.5) by the locally free sheaf $E = \Omega^1_{\mathcal{X}}|_S \otimes A$. The compatible pair

$$\big(\eta^*(dg|_S) \otimes e, 0\big)$$

therefore descends uniquely to s. On the smooth locus of S, $dg|_S$ is a nonzero conormal vector and e is nowhere zero, so $s \neq 0$.

On $S \setminus D$, the normalization is an isomorphism and $s = dg|_S \otimes e$. By (5.4) its intrinsic image vanishes there. Thus $\rho(s)$ is supported on D and lies in the torsion subsheaf; its image modulo torsion is zero.

If $\rho(s) = 0$, exactness of (5.4) after tensoring by A gives $s = dg|_S \otimes \sigma$ for a section σ of A. Pulling back and comparing with (5.7) on the dense open set where $dg \neq 0$ gives $\eta^* \sigma = e$ there, hence everywhere by holomorphic continuation. Since e never vanishes and η is surjective, σ never vanishes and trivializes A. The converse is immediate.

Finally suppose S is connected and $\sigma \in H^0(S, A)$. On each compact connected component of $\widetilde{S}$, write $\eta^*\sigma = c_i e$; the holomorphic function c_i is constant. If one c_i is zero and another is nonzero, connectedness of the finite union of irreducible components of S provides two such components meeting at a point. The value of σ at that point would be both zero and nonzero, a contradiction. Hence either every $c_i = 0$, so $\sigma = 0$, or every $c_i \neq 0$, so σ is nowhere zero and A is trivial. When A is nontrivial only the first case occurs. Since $N^*_{S/\mathcal{X}} \simeq \mathcal{O}_S$, the conormal global-section space is $H^0(S, A) = 0$, while $s \neq 0$. □

60.3 A proper counterexample to the printed inference

Theorem 60.6 (nodal-cubic test family)**.** ***[REFUTED]*** *The p. 692 implication described after (5.2) is false as a formal sheaf-theoretic statement, even for a proper family with smooth total space and an irreducible reduced normal-crossings central fibre.*

Proof. Let Δ be a sufficiently small disc with coordinate t and put

$$\mathcal{C} = \{([x : y : z], t) \in \mathbb{P}^2 \times \Delta : y^2 z = x^2(x + z) + tz^3\}. \tag{5.8}$$

This is proper over Δ. Its total space is smooth. On $z \neq 0$, the partial derivative of the defining equation with respect to t is $-z^3 \neq 0$. On $z = 0$, the equation forces $x = 0$; then $y \neq 0$ and the z-derivative is $y^2 \neq 0$. The central fibre C_0 is the irreducible nodal cubic $y^2 z = x^2(x + z)$, with node $[0 : 0 : 1]$.

Set

$$\mathcal{X} = \mathcal{C} \times \mathbb{P}^1, \qquad S = C_0 \times \mathbb{P}^1,$$

and let $g : \mathcal{X} \to \Delta$ be projection. Then $\mathcal{X}$ is a smooth threefold, g is proper, and S is an irreducible reduced normal-crossings surface. Its normalization is $\widetilde{S} \simeq \mathbb{P}^1 \times \mathbb{P}^1$.

Let p, q be the two inverse images of the node in the normalization of C_0. Fix $\lambda \in \mathbb{C}^*$ which is not a root of unity (for example, $\lambda = 2$). Glue the fibres of the trivial line bundle on the normalization at p and q by the scalar λ. This defines a line bundle $A_\lambda \in \mathrm{Pic}^0(C_0)$ whose pullback is trivial. It has infinite order: its nth tensor power has gluing scalar λ^n, which is never one. It is not trivial: a global section pulls back to a constant c satisfying $c = \lambda c$, so

$$H^0(C_0, A_\lambda) = 0. \tag{5.9}$$

Put $A = \mathrm{pr}_1^* A_\lambda$ on S. Then $\eta^* A \simeq \mathcal{O}_{\widetilde{S}}$ and $H^0(S, A) = 0$. Because S is the full fibre, $N^*_{S/\mathcal{X}} \simeq \mathcal{O}_S$, so the conormal-twist global sections vanish.

The natural map $\widetilde{\Omega}^1_S \to \eta_* \Omega^1_{\widetilde{S}}$ is injective: its kernel is supported where normalization is not an isomorphism, while the source is torsion-free by definition. Projection formula and $\eta^* A \simeq \mathcal{O}_{\widetilde{S}}$ therefore give

$$H^0(S, \widetilde{\Omega}^1_S \otimes A) \hookrightarrow H^0(\widetilde{S}, \Omega^1_{\widetilde{S}}) = 0, \tag{5.10}$$

the last equality following from $\widetilde{S} = \mathbb{P}^1 \times \mathbb{P}^1$. Nevertheless theorem 60.5 produces

$$0 \neq s \in H^0(S, \Omega^1_{\mathcal{X}}|_S \otimes A).$$

Thus both vanishings invoked in the printed reduction hold while the desired ambient vanishing fails. This test chooses the line bundle A directly on S; it does not assert that A extends to a line bundle on the total space $\mathcal{X}$. It therefore refutes the claimed *formal implication between the three sheaf-cohomology groups* for a non-torsion A, but does not by itself instantiate the narrower global-line-bundle situation on p. 692. The family (5.8) also does not satisfy all hypotheses of CDP Theorem 2.2, so this is not a counterexample to that theorem. □

60.4 Cartier-divisor duality without a normality hypothesis

Theorem 60.7 (complete Cartier-fibre duality)**.** ***[VERIFIED]*** *Let X be a compact smooth complex threefold, $i : W \hookrightarrow X$ an effective Cartier divisor, and E a holomorphic vector bundle. Put $F = E|_W$ and*

$$\omega_W = (\omega_X \otimes \mathcal{O}_X(W))|_W.$$

For $q = 0, 1, 2$ there are natural perfect pairings

$$H^q(W, F)^\vee \simeq H^{2-q}(W, F^\vee \otimes \omega_W). \tag{5.11}$$

Moreover $H^q(W, F) = 0$ for $q > 2$. No normality or reducedness hypothesis on W is used.

Proof. Let $i : W \hookrightarrow X$ be the given Cartier inclusion. The length-one locally free resolution

$$0 \longrightarrow \mathcal{O}_X(-W) \xrightarrow{\cdot g} \mathcal{O}_X \longrightarrow i_*\mathcal{O}_W \longrightarrow 0$$

is locally defined by a nonzerodivisor g; changing g by a unit gives the usual transition functions for $\mathcal{O}_X(W)$. Applying sheaf Hom into ω_X, the degree-zero map is injective and its cokernel is

$$\mathcal{E}xt^1_X(i_*\mathcal{O}_W, \omega_X) = i_*((\omega_X \otimes \mathcal{O}_X(W))|_W) = i_*\omega_W.$$

The other sheaf Ext groups vanish. Equivalently, W is a two-dimensional Cohen–Macaulay Gorenstein complex space and its analytic dualizing complex is

$$K^\bullet_W \simeq \omega_W[2].$$

This is the hypersurface calculation in [5, Proposition 2 and pp. 90–91]; no normality or reducedness of the Cartier divisor is used.

For the locally free sheaf $F = E|_W$, the analytic duality trace of [5, Theorem 2, pp. 82–83] supplies the actual functorial pairing, and identifies the dual of $H^q_c(W, F)$ with

$$\operatorname{Ext}^{-q}_W(F, K^\bullet_W) = \operatorname{Ext}^{2-q}_W(F, \omega_W) = H^{2-q}(W, F^\vee \otimes \omega_W).$$

The last equality follows because the positive sheaf Ext groups of a locally free first argument vanish. Since W is compact, $H^q_c(W, F) = H^q(W, F)$; coherent cohomology is finite-dimensional, so this pairing is perfect in the ordinary algebraic sense. It is this trace pairing, rather than an unspecified arrow between two matching five-term sequences, that constructs the isomorphism (5.11). Its contravariance in F is precomposition in the Ext argument.

For $q > 2$, the displayed Ext degree $2 - q$ is negative. The derived Hom of the locally free sheaf F, concentrated in degree zero, into the sheaf ω_W, concentrated in degree zero, has no negative cohomology. Hence the duality theorem gives $H^q(W, F) = 0$. This also follows directly from coherent cohomological dimension on the compact surface W. □

Proposition 60.8 (absolute duality on a compact Cartier fibre)**.** ***[VERIFIED]*** *Let X be a smooth complex threefold, not necessarily compact, and let $i : W \hookrightarrow X$ be a compact effective Cartier divisor that is the full scheme-theoretic fibre of a holomorphic map $f : X \to B$ to a smooth curve. For every vector bundle E on X there is a functorial perfect pairing*

$$H^2(W, E|_W) \times H^0(W, E^\vee|_W \otimes \omega_X|_W) \longrightarrow \mathbb{C}.$$

Equivalently,

$$H^2(W, E|_W)^\vee \simeq H^0(W, E^\vee|_W \otimes \omega_X|_W). \tag{5.15a}$$

Neither normality of W nor compactness of the ambient threefold is required.

Proof. An effective Cartier divisor in a smooth threefold is locally a two-dimensional hypersurface quotient of a regular local ring. Hence W is Cohen–Macaulay and Gorenstein. If $K_W^\bullet$ denotes its analytic dualizing complex, the complete-intersection calculation of Ramis–Ruget gives

$$K_W^\bullet \simeq \omega_W[2]$$

with ω_W invertible [5, Proposition 2 and the complete-intersection calculation, pp. 90–91]. Cartier adjunction gives

$$\omega_W \simeq (\omega_X \otimes \mathcal{O}_X(W))|_W.$$

Because W is the full fibre over $b \in B$, one has $\mathcal{O}_X(W) \simeq f^*\mathcal{O}_B(b)$ locally along W; its restriction to W is the pullback of a one-dimensional vector space. Choosing a nonzero vector identifies $\mathcal{O}_X(W)|_W \simeq \mathcal{O}_W$ and $\omega_W \simeq \omega_X|_W$. A different choice rescales the final trace pairing by a nonzero scalar.

Ramis–Ruget duality pairs $H_c^p(W, \mathcal{G})$ with $\operatorname{Ext}_W^{-p}(\mathcal{G}, K_W^\bullet)$ [5, Theorem 2, pp. 82–83]. Since W is compact, $H_c^p = H^p$. Taking $p = 2$ and $\mathcal{G} = E|_W$, the shift and local freeness give

$$\operatorname{Ext}_W^{-2}(\mathcal{G}, K_W^\bullet) \simeq \operatorname{Hom}_W(\mathcal{G}, \omega_W) = H^0(W, E^\vee|_W \otimes \omega_W).$$

Grauert finiteness makes both coherent cohomology spaces finite-dimensional [8]; the strong analytic duality is therefore the ordinary algebraic perfect pairing (5.15a). Its variance is contravariant in $E|_W$ by precomposition on the Hom term. This proves the assertion. □

60.5 Flatness and top-degree analytic base change

Lemma 60.9 (automatic properness, flatness, and Cartier fibres)**.** ***[VERIFIED]*** *Let $f : X \to B$ be a nonconstant holomorphic map from a connected complex manifold to a Riemann surface. Then f is flat, every holomorphic vector bundle on X is f-flat, and every full scheme-theoretic fibre is an effective Cartier divisor of pure codimension one. If X is compact, then f is proper.*

Proof. Fix $x \in X$, $c = f(x)$, and put $R = \mathcal{O}_{B,c}$, $S = \mathcal{O}_{X,x}$. The ring R is a discrete valuation ring and S is a regular local domain. The homomorphism $R \to S$ is injective. Indeed, if a nonzero germ $h \in R$ mapped to zero, $h \circ f$ would vanish on a neighbourhood of x. The zero set of a nonzero one-variable holomorphic germ is isolated, so f would be constant on that neighbourhood and therefore, by the identity theorem on connected X, constant everywhere.

Thus every nonzero element of R acts on the domain S as a nonzero element and hence as a nonzerodivisor. Therefore S is torsion-free as an R-module. A module over a discrete valuation ring is flat if and only if it is torsion-free [37, Tag 0AUW]; hence S is R-flat. This holds at every point, so f is flat. A vector bundle is locally a finite free $\mathcal{O}_X$-module, and a finite direct sum of R-flat modules is flat; hence it is f-flat. If t is a local parameter at c, then f^*t is a nonzero element of the domain S and thus a nonzerodivisor. The full scheme-theoretic fibre is therefore locally cut out by one nonzerodivisor, so it is an effective Cartier divisor of pure codimension one. Finally, when X is compact, the underlying continuous map to the Hausdorff surface B is proper. □

Proposition 60.10 (the exact analytic base-change input)**.** ***[VERIFIED]*** *Let $f : X \to B$ be proper between complex spaces, let $\mathcal{F}$ be coherent and f-flat, and fix $c \in B$. If*

$$H^3(X_c, \mathcal{F}|_{X_c}) = 0, \tag{5.16}$$

then the canonical fibre map

$$\beta_c^2 : (R^2 f_* \mathcal{F})_c \otimes_{\mathcal{O}_{B,c}} \mathbb{C}(c) \longrightarrow H^2(X_c, \mathcal{F}|_{X_c}) \tag{5.17}$$

is an isomorphism.

Proof. Apply Bănică–Stănăşilă, Chapter III, §3, Corollary 3.5, with $q = 3$ [7, pp. 120–121]. Its hypotheses are precisely properness, coherence, and flatness. Condition (5.16) first forces the degree-three direct image to vanish near c. The corollary then applies the equivalent base-change conditions of its preceding Theorem 3.4 in degree $q-1 = 2$ and states that the canonical map (5.17) is bijective. This is the top-degree argument used in [4, Lemma 1.5]; it does not require the stronger, false assertion that all direct images are locally free. □

Corollary 60.11 (uniform top-degree base change for surface fibres)**.** ***[VERIFIED]*** *Let $f : X \to B$ be a proper flat holomorphic map to a Riemann surface whose full scheme-theoretic fibres have pure complex dimension two, and let E be a holomorphic vector bundle on X. Then*

$$R^q f_* E = 0 \quad (q > 2) \tag{5.17-V}$$

and, for every $b \in B$, including a point with nonreduced fibre, the canonical map

$$(R^2 f_* E)_b \otimes_{\mathcal{O}_{B,b}} \mathbb{C}(b) \xrightarrow{\sim} H^2(X_b, E|_{X_b}) \tag{5.17-T}$$

is an isomorphism. No constancy of h^2, local freeness of $R^2 f_* E$, or reducedness of X_b is required.

Proof. On a sufficiently small neighbourhood U of b, the finite-complex theorem used in analytic cohomology and base change represents $Rf_*E|_U$ by a bounded complex $K^\bullet$ of finite free $\mathcal{O}_U$-modules [7, Chapter III, §3, Theorem 3.4]. Flatness of E over B identifies its derived fibre with the ordinary restriction to the full scheme-theoretic fibre. Over the discrete valuation ring $A = \mathcal{O}_{B,b}$, the universal-coefficient sequence is

$$0 \longrightarrow H^q(K^\bullet)_b \otimes_A \mathbb{C}(b) \longrightarrow H^q(K^\bullet \otimes_A \mathbb{C}(b)) \longrightarrow \operatorname{Tor}_1^A(H^{q+1}(K^\bullet)_b, \mathbb{C}(b)) \longrightarrow 0. \tag{5.17-U}$$

If some $R^q f_* E$ with $q > 2$ were nonzero, choose the largest such q and a point of its support. Nakayama's lemma makes the left term of (5.17-U) nonzero there, while maximality kills the Tor term. This would give nonzero coherent cohomology in degree $q > 2$ on a compact two-dimensional complex space, a contradiction. Thus (5.17-V) holds. For $q = 2$, the only obstruction in (5.17-U) is $\operatorname{Tor}_1^A((R^3 f_* E)_b, \mathbb{C}(b))$, which is zero by (5.17-V), proving (5.17-T). This is the explicit obstruction calculation behind theorem 60.10; it was independently supplied, with eight named local certificates, in Reviewer B's candidate-specific follow-up [36]. □

60.6 A topology-independent Pic^0 route and three all-Pic routes

Let $i : W \hookrightarrow X$ denote the cusp-fibre inclusion and $\eta : \widetilde{W} \simeq \mathrm{dP}_6 \to W$ its normalization.

Proposition 60.12 (topologically trivial twists pull back trivially)**.** ***[VERIFIED]*** *Assume that ω_X is trivial on a neighbourhood of W. For every $L \in \mathrm{Pic}^0(X)$, put*

$$A_L = (L^\vee \otimes \omega_X)|_W.$$

Then

$$\eta^* A_L \simeq \mathcal{O}_{\mathrm{dP}_6}. \tag{5.17a}$$

No assertion about $H^2(X;\mathbb{Z})$ is used.

Proof. Every line bundle in $\mathrm{Pic}^0(X)$ has $c_1(L) = 0$. Naturality therefore gives $c_1(\eta^* i^* L) = 0$. The degree-six del Pezzo surface is the blow-up of $\mathbb{P}^2$ in three noncollinear points, so $H^1(\mathrm{dP}_6, \mathcal{O}) = 0$. The exponential sequence consequently makes

$$c_1 : \mathrm{Pic}(\mathrm{dP}_6) \longrightarrow H^2(\mathrm{dP}_6; \mathbb{Z})$$

injective. Hence $\eta^* i^* L$ is trivial. The assumed neighbourhood trivialization of ω_X also pulls back trivially, which proves (5.17a). □

Proposition 60.13 (the exact all-line-bundle restriction obstruction)**.** ***[VERIFIED]*** *Define*

$$r = \eta^* i^* : \mathrm{Pic}(X) \longrightarrow \mathrm{Pic}(\mathrm{dP}_6).$$

Under the same local canonical trivialization,

$$\eta^* A_L \simeq r(L)^\vee \qquad (L \in \mathrm{Pic}(X)). \tag{5.17b}$$

Thus the trivial-pullback hypothesis used by theorem 60.5 holds simultaneously for every line bundle exactly when $r = 0$. It always holds on $\mathrm{Pic}^0(X)$*.*

Proof. Pullback commutes with tensor product and duals, while the local trivialization gives $\eta^* i^* \omega_X \simeq \mathcal{O}$. This is (5.17b), and its uniform equivalence with $r = 0$ is immediate. The final assertion is theorem 60.12. □

Proposition 60.14 (Leray-edge vanishing of normalized cusp restrictions)**.** ***[VERIFIED]*** *For the constructed map $f : X \to B$ and every $L \in \mathrm{Pic}(X)$, the degree-two Leray edge section satisfies*

$$e(c_1(L)) = 0 \in H^0(B, R^2 f_* \mathbb{Z}), \qquad \eta^* i^* L \simeq \mathcal{O}_{\mathrm{dP}_6}. \tag{5.17c}$$

This proof does not invoke the packaged conclusion $H^2(X; \mathbb{Z}) = 0$.

Proof. First record explicitly the compatibility used here. For every $b \in B$, proper base change and functoriality of the filtered-complex spectral sequence give a commutative square

$$\begin{array}{ccc} H^2(X;\mathbb{Z}) & \xrightarrow{\;e\;} & H^0(B, R^2 f_* \mathbb{Z}) \\ {\scriptstyle i_b^*}\big\downarrow & & \big\downarrow{\scriptstyle \mathrm{ev}_b} \\ H^2(X_b;\mathbb{Z}) & = & H^2(X_b;\mathbb{Z}). \end{array} \tag{5.17d}$$

Indeed, pull the Leray filtered complex back along $\{b\} \to B$; the spectral sequence over a point has only column zero, so its edge is the identity, while the right vertical map is the proper-base-change identification. This uses the functoriality and convergence statements in [37, Tags 012O, 012U and 012W] and analytic proper base change [33, Theorem 2.3.26]. The connecting homomorphism in the exponential sequence is natural under i_b^*, so (5.17d) sends $c_1(L)$ to $c_1(L|_{X_b})$.

Put $s_L = e(c_1(L))$. The image of an edge morphism consists of permanent cycles, hence

$$s_L \in E_\infty^{0,2} \subseteq \ker(d_2^{0,2} : E_2^{0,2} \to E_2^{2,1}); \tag{5.17e}$$

see [37, Tags 011N, 011O and 012U]. For the constructed candidate, theorems 53.2 and 53.4 give

$$E_2^{0,2} = \mathbb{Z}(2q), \qquad E_2^{2,1} = \mathbb{Z}\,\omega[\delta], \qquad d_2^{0,2}(2q) = \omega[\delta].$$

Thus $d_2^{0,2}$ is an isomorphism and (5.17e) forces $s_L = 0$. Taking $b = p_0$ in (5.17d) gives $c_1(i^* L) = 0$, and hence $c_1(\eta^* i^* L) = 0$. Finally, the injectivity of $c_1 : \mathrm{Pic}(\mathrm{dP}_6) \to H^2(\mathrm{dP}_6; \mathbb{Z})$ proved in theorem 60.12 makes $\eta^* i^* L$ trivial. □

Lemma 60.15 (the pair-of-pants defect class)**.** ***[VERIFIED]*** *Choose the compact pair-of-pants core $P \subset B^\circ$ and the zero-section whiskers used in theorem 53.6, and put $J_P = f^{-1}(P)$. Orient its boundary meridians clockwise. Write ρ_0 for the projection of the zero-section cusp meridian $\widetilde{a}_0$. The constructed candidate uses the tautological cusp gluing, whose affine regluing class is $\mu = 0$ (equivalently $\ell_0 = \gamma(\mu) = 0$); hence the flat-marked cusp lift has no vertical translation. The base loops and their chosen lifts therefore satisfy*

$$\rho_1 \rho_2 \rho_0 = 1, \qquad \widehat{\rho}_1 \widehat{\rho}_2 \widetilde{a}_0 = 1. \tag{5.17g}$$

For $L \in \mathrm{Pic}(X)$, write $E = c_1(L|_F) \in \bigwedge^2 V$. Then $E \in (\bigwedge^2 V)^G$, and there is a crossed homomorphism

$$z : \pi_1(P) \longrightarrow V_{\mathbb{R}}, \qquad z(gh) = z(g) + T_g z(h), \tag{5.17h}$$

whose class records $c_1(L|_{J_P}) - [\Omega_E]$. Here Ω_E is the closed, translation- invariant vertical two-form induced by E in the flat marking. If $d_i = z(\rho_i)$ and a bar denotes the image in $V_G \otimes \mathbb{R}$, then

$$\bar{d}_1 + \bar{d}_2 + \bar{d}_0 = 0. \tag{5.17i}$$

The three barred classes, and the number $\langle d_0, \widehat{\delta}\rangle$, do not depend on the chosen crossed- homomorphism representative. More precisely, use the base-first convention for the edge injection in (5.17j). For any oriented loop $\rho \subset P$ and any A_ρ-invariant $\nu \in \Lambda$, its flat-marked sweep satisfies

$$\langle z(\rho), \nu\rangle = -\langle c_1(L|_{J_P}) - [\Omega_E], a_*([S^1_\nu] \times [\widetilde{\rho}])\rangle. \tag{5.17j'}$$

Here a is the fibrewise translation action and $\widetilde{\rho}$ is the flat-marked lift of ρ.

Proof. The flat marking identifies the pullback of J_P to the universal cover of P with $\widetilde{P} \times (\Lambda_{\mathbb{R}}/\Lambda)$. Since E is the restriction of a class on the total space, it is fixed by every monodromy T_g. The constant vertical form representing E is therefore invariant under the deck action and descends to the asserted form Ω_E on J_P.

The pair of pants retracts onto a graph. Hence the Serre spectral sequence of the marked real torus bundle has only columns zero and one and gives the short exact edge sequence

$$0 \longrightarrow H^1(P; V_{\mathbb{R}}) \longrightarrow H^2(J_P; \mathbb{R}) \longrightarrow H^0(P; \textstyle\bigwedge^2 V_{\mathbb{R}}) \longrightarrow 0. \tag{5.17j}$$

Both $c_1(L|_{J_P})$ and $[\Omega_E]$ map to E, so their difference is the image of a unique class in $H^1(P; V_{\mathbb{R}})$. The standard group- cohomology model for a graph identifies this group with crossed homomorphisms modulo principal crossed homomorphisms [28, Chapter I, §5]; this gives (5.17h).

For completeness, the slant convention in (5.17j') is part of this identification, not an additional geometric assertion. Give an oriented edge of a graph spine of P a coordinate s increasing with its based loop and represent the corresponding $V_{\mathbb{R}}$-valued cellular one-cochain by a base-first form $ds \wedge \beta_s$. On the sweep of an A_ρ-invariant fibre circle S^1_ν, Fubini's formula gives

$$\int_{S^1_s \times S^1_\nu} ds \wedge \beta_s = \int_{S^1_s} \langle \beta_s, \nu\rangle, ds = \langle z(\rho), \nu\rangle.$$

The sweep in (5.17j') is written in the fibre-first order $[S^1_\nu] \times [\widetilde{\rho}]$, which differs from this base-first cross-product orientation by $(-1)^{1\cdot 1} = -1$; this is the displayed minus sign. Changing the cellular representative by a principal cochain changes both sides by the same boundary term, which vanishes because $A_\rho \nu = \nu$. This proves (5.17j').

Applying z to (5.17g) gives

$$d_1 + T_1 d_2 + T_1 T_2 d_0 = 0.$$

All T_g act trivially on the global coinvariants, proving (5.17i). A change of representative has the form $z(g) \mapsto z(g) + (T_g - I)a$. Its image in $V_G \otimes \mathbb{R}$ is zero. Moreover the displayed matrices in theorem 5.1 fix $\widehat{\delta}$ under every dual action A_g. Therefore

$$\langle (T_g - I)a, \widehat{\delta}\rangle = \langle a, (A_g^{-1} - I)\widehat{\delta}\rangle = 0,$$

which proves the final assertion. □

Lemma 60.16 (the finite-cover norm identity)**.** ***[VERIFIED]*** *Retain the notation of theorems 10.9 and 60.15. For $j = 1, 2$, let*

$$N_{T_j} = I + T_j + \cdots + T_j^{m_j - 1}.$$

With the clockwise conventions of (5.17g), the pullback to the m_j-fold s_j-cover gives

$$N_{T_j} d_j = \iota_{v_j} E \quad \text{in } V_{\mathbb{R}}, \qquad m_j \bar{d}_j = [\iota_{v_j} E] \quad \text{in } V_G \otimes \mathbb{R}. \tag{5.17k}$$

Proof. Restriction of the crossed homomorphism to the subgroup generated by $\rho_j^{m_j}$ has value

$$z(\rho_j^{m_j}) = \sum_{r=0}^{m_j - 1} T_j^r z(\rho_j) = N_{T_j} d_j. \tag{5.17l}$$

This is the cohomological pullback of the annular defect to the s_j-cover; the norm, rather than unqualified multiplication by m_j, is essential before passage to global coinvariants. A principal change $d_j \mapsto d_j + (T_j - I)a$ does not alter this value, since $N_{T_j}(T_j - I)a = (T_j^{m_j} - I)a = 0$.

Let $Q_j : J_j' \to N_j$ be the finite quotient map. The global flat marking Ξ over the full disc Δ_j makes the constant vertical form Ω_E well defined on J_j'. Radial retraction onto the central torus makes

$$H^2(J_j'; \mathbb{R}) \longrightarrow H^2(A_0^{(j)}; \mathbb{R})$$

an isomorphism. On the central fibre the quotient is exactly π_j, and naturality gives

$$(Q_j|_{A_0^{(j)}})^* c_1(L|_{S_j}) = \pi_j^* c_1(L|_{S_j}) = E. \tag{5.17l'}$$

The second equality is the actual radial/stalk identification in theorem 10.10; on a nearby fibre the gluing map h_j is a translation and therefore acts trivially on H^2 of that fibre. Consequently the two classes $c_1(Q_j^*(L|_{N_j}))$ and $[\Omega_E]$ have the same central restriction, and the retraction isomorphism proves

$$c_1(Q_j^*(L|_{N_j})) - [\Omega_E] = 0 \quad \text{in } H^2(J_j'; \mathbb{R}). \tag{5.17l"}$$

On Δ_j^*, the source identity $h_j \widetilde{g}_j^{\log} h_j^{-1} = \widetilde{g}_j^0$ shows that h_j descends from the logarithmic quotient N_j^* to the linear/global quotient J^*. Thus

$$Q_j^*(L|_{N_j})|_{\Delta_j^*} \simeq h_j^*(L|_{J^*}). \tag{5.17l"'}$$

The logarithmic overlap is the holomorphic translation

$$h_j[s, \zeta] = [s, \zeta + \alpha(s)\Pi(s)v_j], \qquad \alpha(s) = \frac{\log s_j}{2\pi i}.$$

To compute its effect on a real cohomology class, retain the exact real marking

$$\Xi(s, x + \Lambda) = [s, \Pi(s)x], \qquad J_s = \Pi(s)^{-1} \circ i \circ \Pi(s) \quad \text{on } \Lambda_{\mathbb{R}}.$$

The period matrix $\Pi(s)$ is a real isomorphism by (2.2)–(2.4); thus J_s is a well-defined smooth family of real complex structures. On a branch of the logarithm the actual real translation vector is

$$a_j(s) = \operatorname{Re}\alpha(s)\, v_j + \operatorname{Im}\alpha(s)\, J_s v_j. \tag{5.17m}$$

Indeed, applying $\Pi(s)$ gives precisely $\alpha(s)\Pi(s)v_j$. A different logarithm branch changes a_j by an integral multiple of v_j, so its torus translation is branch-independent. Complex multiplication by α is not multiplication by a real scalar in this marking.

Write $E = \frac{1}{2}\sum_{a,b} E_{ab}\,dx^a \wedge dx^b$ in the ordered real coordinates dual to the marked lattice; the matrix (E_{ab}) is alternating and is constant over the covering disc. Direct substitution in the real map $h_j(s,x) = (s, x + a_j(s))$ gives

$$\begin{aligned} h_j^*\Omega_E - \Omega_E = &\sum_{a,b} E_{ab}\,da_j^a \wedge dx^b \\ &+ \frac{1}{2}\sum_{a,b} E_{ab}\,da_j^a \wedge da_j^b. \end{aligned} \tag{5.17m'}$$

The last term has both arguments in the base directions. In particular, it restricts to zero when the base is an oriented circle.

Take one full clockwise circle in the s_j-cover,

$$s_j(\theta) = re^{-2\pi i\theta}, \qquad 0 \le \theta \le 1.$$

Choose the logarithm continuously along that circle, so $\alpha(1) = \alpha(0) - 1$. The endpoints represent the same point of the covering disc. The marking is defined on the full disc, and therefore $J_{s_j(1)} = J_{s_j(0)}$. Consequently

$$a_j(1) - a_j(0) = -v_j. \tag{5.17m"}$$

For any $\lambda \in \Lambda$, restrict (5.17m') to the sweep of the marked circle $x(u) = u\lambda$, using the base-first orientation $d\theta \wedge du$. Its integral is exactly

$$\int_0^1 E\Big(\frac{da_j}{d\theta}, \lambda\Big)\,d\theta = E(a_j(1) - a_j(0), \lambda) = -E(v_j, \lambda).$$

The horizontal term contributes zero. Since marked lattice circles span $H_1(F;\mathbb{R})$, these integrals determine the mixed component in

$$H^1(S^1;\mathbb{R}) \otimes H^1(F;\mathbb{R}).$$

Thus the mixed component of $h_j^*\Omega_E - \Omega_E$ is $-\iota_{v_j}E$, with $(\iota_{v_j}E)(\lambda) = E(v_j, \lambda)$.

The pullback of the defect $c_1(L|_{J^*}) - [\Omega_E]$ has zero fibre component and mixed component $N_{T_j}d_j$ by (5.17l). Translation by h_j does not change this mixed component: on the covering circle such a fibre-zero class is represented by $d\theta \wedge \beta$, with β a constant fibre one-form, and substituting $dx \mapsto dx + da_j$ adds only a base two-form, which vanishes on the circle. Equations (5.17l") and (5.17l"') state that the total pulled-back class extends over the covering disc and has zero mixed component. It follows that

$$N_{T_j}d_j - \iota_{v_j}E = 0.$$

In the global coinvariants every T_j acts as the identity, so $m_j\bar{d}_j = [\iota_{v_j}E]$. This proves (5.17k), with the original clockwise convention and no change to either translation vector. □

Lemma 60.17 (the cusp boundary defect and Chern-class specialization)**.** ***[VERIFIED]*** *In the notation of theorem 60.15, one has*

$$\bar{d}_0 = 0 \quad \text{in } V_G \otimes \mathbb{R}. \tag{5.17n}$$

Moreover, for every line bundle L on the cusp neighbourhood N_0, nearby-cycle specialization is compatible with first Chern classes:

$$\mathrm{sp}^2(c_1(L|_W)) = c_1(L|_F). \tag{5.17o}$$

Proof. Let $\widetilde{a}_0$ be the flat-marked clockwise zero-section meridian in the lifted relation of (5.17g), and let

$$\xi_{\widehat{\delta}} = a_*([S^1_{\widehat{\delta}}] \times [\widetilde{a}_0])$$

be its sweep under fibrewise translation by $\widehat{\delta}$. The boundary ρ_0 of P was chosen to be its projection. This is the zero-section sweep: there is no hidden vertical drift in $\widetilde{a}_0$, by the explicit overlap calculation in theorem 53.6. Hence the vertical form Ω_E vanishes on the swept torus, and the slant identity (5.17j') gives

$$\langle d_0, \widehat{\delta}\rangle = -\langle c_1(L|_{J_P}) - [\Omega_E], \xi_{\widehat{\delta}}\rangle = -\langle c_1(L|_{J_P}), \xi_{\widehat{\delta}}\rangle. \tag{5.17p}$$

Combine the exponential overlap coordinates (x_1, x_2, t) of (2.8ii) with the unimodular cusp chart

$$(u, x_1, x_2) = \left(\frac{t}{x_1 x_2}, x_1, x_2\right).$$

The literal solid torus

$$\mathbb{S}_{\widehat{\delta}} = \{|u| = \rho,\ x_1 = 1,\ |x_2| \le 1\}$$

has boundary parametrization

$$(\theta, \phi) \longmapsto (\rho e^{i(\theta-\phi)}, 1, e^{i\phi}) \tag{5.17p0}$$

and hence $t = \rho e^{i\theta}$; its ϕ-circle is translation by $\widehat{\delta}$. Direct evaluation in the proved cusp marking therefore identifies $\partial\mathbb{S}_{\widehat{\delta}}$ with $\xi_{\widehat{\delta}}$. Reversing θ supplies the fixed clockwise convention and only changes the common orientation sign. No reduced Mayer–Vietoris block or Leray comparison is used here. Since L extends across N_0, its Chern class evaluates to zero on that boundary. Thus the left side of (5.17p) is zero. By theorem 5.3, $V_G = \mathbb{Z}[\delta]$, and evaluation on the globally invariant dual vector $\widehat{\delta}$ identifies $V_G \otimes \mathbb{R}$ with $\mathbb{R}$. This proves (5.17n).

For (5.17o), let $\ell = c_1(L|_{N_0})$, let $\iota_0 : W \hookrightarrow N_0$, and let $\iota_t : F_t \hookrightarrow N_0$ be a sufficiently near smooth cusp fibre. The neighbourhood realization of the nearby-cycle unit $\mathbb{Z}_W \to \psi_{f_0}\mathbb{Z}$ recalled in the proof of theorem 53.1 gives the commutative square

$$\begin{array}{ccc} H^2(N_0;\mathbb{Z}) & \xrightarrow{\iota_0^*} & H^2(W;\mathbb{Z}) \\ {\scriptstyle \iota_t^*}\big\downarrow & & \big\downarrow{\scriptstyle \mathrm{sp}^2} \\ H^2(F_t;\mathbb{Z}) & = & H^2(F_t;\mathbb{Z}). \end{array} \tag{5.17p'}$$

Applying this square to ℓ gives $\mathrm{sp}^2(c_1(L|_W)) = c_1(L|_{F_t})$. Flat parallel transport in the marked smooth family identifies the latter with $c_1(L|_F) = E$. Equivalently, c_1 is the connecting morphism in the exponential sequence, and restriction to the shrinking neighbourhoods and to F_t commutes with that morphism before passage to the direct limit. This proves the claimed naturality without using any global Leray differential. □

Proposition 60.18 (direct boundary-Picard vanishing)**.** ***[VERIFIED]*** *For the constructed threefold and every analytic line bundle $L \in \mathrm{Pic}(X)$,*

$$c_1(L|_F) = 0, \qquad \eta^* i^* L \simeq \mathcal{O}_{\mathrm{dP}_6}. \tag{5.17q}$$

This proof is independent of the global Leray transgressions $d_2^{0,q}$ and of the packaged conclusion $H^2(X;\mathbb{Z}) = 0$; it uses only the actual finite-filling lattices, their logarithmic overlap maps, the based boundary relation, the local cusp solid torus, and the local integral specialization theorem.

Proof. The common invariant calculation in theorem 5.3 first gives

$$E = c_1(L|_F) = kq.$$

Because L extends across the order-four filling, its fibre class belongs to the actual specialization lattice

$$\pi_2^* H^2(S_2;\mathbb{Z}) = \{a\gamma\eta_2 + bq : a \equiv b \pmod 2\}$$

of theorems 5.4 and 10.10. Taking coefficient independence in the displayed invariant basis forces $a = 0$ and $b = k$; hence k is even. Write $E = 2nq$.

Contraction with the two affine translation vectors is an exact integral calculation:

$$\iota_{v_1}(2q) = 4\eta_1, \qquad \iota_{v_2}(2q) = -6\eta_2. \tag{5.17r}$$

Indeed $v_1 = (1, 2, -4, 0)^t$, $v_2 = (-1, -3, 3, 0)^t$ in the basis dual to (γ, u, w, δ), while $q = uw + 6\gamma\delta$. In the global coinvariants,

$$[\eta_1] = 3[\delta], \qquad [\eta_2] = 2[\delta].$$

Applying theorem 60.16 with $m_1 = 3$, $m_2 = 4$ therefore gives

$$\bar{d}_1 = 4n[\delta], \qquad \bar{d}_2 = -3n[\delta]. \tag{5.17s}$$

Now combine the balance relation (5.17i) with the cusp vanishing (5.17n):

$$0 = \bar{d}_1 + \bar{d}_2 + \bar{d}_0 = n[\delta].$$

The coinvariant group is torsion-free, so $n = 0$ and hence $E = 0$.

Finally (5.17o) and the injectivity of the local integral specialization in theorem 53.1 imply $c_1(L|_W) = 0$. Pulling back to the normalization gives $c_1(\eta^* i^* L) = 0$. The injectivity of $c_1 : \mathrm{Pic}(\mathrm{dP}_6) \to H^2(\mathrm{dP}_6; \mathbb{Z})$ proved in theorem 60.12 yields the second assertion of (5.17q). □

60.7 An analytic completion of the all-Picard certificate

The local example in theorem 55.15 shows that the cusp neighbourhood alone cannot annihilate normalized central restrictions. We now close the stronger analytic route by retaining the order-three and order-four fillings. The proof has two independent parts. First, logarithms of the actual $\mathcal{O}^*$-factors give the boundary cocycle and its two finite-filling norm identities. Second, a candidate-specific support-function calculation reads every possible central restriction directly from the mixed Appell–Humbert block on a nearby torus. No global Leray differential, nearby-cycle specialization, or Chern-class evaluation on a cusp bounding chain occurs.

Proposition 60.19 (analytic Appell–Humbert boundary cocycle)**.** ***[VERIFIED]*** *Let $P \subset B^\circ$ be the compact pair-of-pants core, orient its boundary meridians clockwise, and retain the zero-section lifts of theorem 53.6. Thus*

$$\rho_1\rho_2\rho_0 = 1, \qquad \widehat{\rho}_1\widehat{\rho}_2\widetilde{a}_0 = 1, \tag{5.17-A1}$$

and the second relation has no terminal vertical translation. For $L \in \mathrm{Pic}(X)$, let

$$E = c_1(L|_F) \in \mathrm{Alt}^2(\Lambda, \mathbb{Z}) = \bigwedge^2 V$$

be its fibrewise Appell–Humbert form. Put $V_{\mathbb{R}} = \mathrm{Hom}(\Lambda, \mathbb{R})$, and use

$$A_g = (T_g^{-1})^{\mathsf{t}}, \qquad \langle T_g\alpha, \lambda\rangle = \langle \alpha, A_g^{-1}\lambda\rangle. \tag{5.17-A2}$$

There is a crossed homomorphism, unique up to a principal crossed homomorphism,

$$d : \pi_1(P) \longrightarrow V_{\mathbb{R}}, \qquad d(gh) = d(g) + T_g d(h), \qquad d(g) \sim d(g) + (T_g - I)a. \tag{5.17-A3}$$

If $d_i = d(\rho_i)$, then

$$d_1 + T_1 d_2 + T_1T_2 d_0 = 0, \qquad \bar{d}_1 + \bar{d}_2 + \bar{d}_0 = 0 \quad \text{in } V_G \otimes \mathbb{R}. \tag{5.17-A4}$$

For the finite filling of order m_j, $j = 1, 2$,

$$N_{T_j} d_j = \iota_{v_j} E, \qquad m_j \bar{d}_j = [\iota_{v_j} E], \qquad N_{T_j} = I + T_j + \cdots + T_j^{m_j - 1}, \tag{5.17-A5}$$

where $(\iota_v E)(\lambda) = E(v,\lambda)$. *At the cusp,*

$$\langle d_0, \widehat{\delta}\rangle = 0, \qquad \bar{d}_0 = 0 \quad \text{in } V_G \otimes \mathbb{R}. \tag{5.17-A6}$$

These identities are obtained from holomorphic transition factors and the argument principle.

Proof. Choose an open Riemann-surface neighbourhood $P^{\mathrm{an}} \Subset B^\circ$ which deformation retracts onto P, and pull the period family to its universal cover. Before quotienting by the fibre lattice, the resulting space is

$$\widetilde{\mathcal{J}} = \widetilde{P}^{\mathrm{an}} \times \mathbb{C}^2.$$

It is Stein and contractible, so the pullback of L is holomorphically trivial. In a nowhere-zero frame, write the factor belonging to a deck transformation α as

$$a_\alpha \in \mathcal{O}^*(\widetilde{\mathcal{J}}), \qquad a_{\alpha\beta}(x) = a_\alpha(\beta x)\, a_\beta(x). \tag{5.17-A7}$$

The space is simply connected, hence $a_\alpha = e^{2\pi i h_\alpha}$ for a holomorphic logarithm. Therefore

$$c(\alpha,\beta) = h_\alpha(\beta x) + h_\beta(x) - h_{\alpha\beta}(x) \in \mathbb{Z} \tag{5.17-A8}$$

is independent of x, and associativity makes c an integral group two-cocycle.

Let τ_λ denote fibre translation by $\Pi(z)\lambda$, $\lambda \in \Lambda$, and let $\widehat{g}$ be the zero-section lift of a base deck transformation. Period equivariance gives

$$\widehat{g}\,\tau_\lambda\,\widehat{g}^{-1} = \tau_{A_g\lambda}. \tag{5.17-A9}$$

On fibre translations, the commutator of (5.17-A8) is

$$E(\lambda,\mu) = c(\tau_\lambda,\tau_\mu) - c(\tau_\mu,\tau_\lambda). \tag{5.17-A10}$$

The four oriented sides of the (λ,μ)-parallelogram identify this integer with the exponential connecting class of $L|_F$; this is the Appell–Humbert convention used in (5.17av)–(5.17ax), and agrees with the standard factor-of-automorphy calculation [19, Chapter 1, §2, pp. 5–8, especially Theorem 2.1, p. 7][20, equations (3.3)–(3.10), pp. 3–7]. Conjugating the commutator by $\widehat{g}$ yields

$$E(A_g\lambda, A_g\mu) = E(\lambda,\mu). \tag{5.17-A11}$$

Push the logarithmic central extension defined by c out along $\mathbb{Z} \hookrightarrow \mathbb{R}$, and denote its central element of weight r by $\mathbf{c}(r)$. Choose lifts U_λ of the fibre translations satisfying the real Appell–Humbert normalization

$$U_\lambda U_\mu = \mathbf{c}\Big(\frac{1}{2}E(\lambda,\mu)\Big)\, U_{\lambda+\mu}. \tag{5.17-A12}$$

To see that this normalization exists, start with arbitrary lifts. The difference between their real cocycle and $E/2$ has zero commutator, so it defines an abelian extension of the free abelian group Λ by $\mathbb{R}$. Choose lifts of an integral basis and extend multiplicatively; this splits that abelian extension and gives (5.17-A12). If U'_λ is a second normalized system, then $U'_\lambda U_\lambda^{-1}$ is central, say $\mathbf{c}(\langle a,\lambda\rangle)$. Comparing the two copies of (5.17-A12) proves additivity in λ, so $a \in V_{\mathbb{R}}$. Thus every two normalized systems differ by exactly the character change used below.

For $g \in \pi_1(P)$, choose a lift G_g of $\widehat{g}$ and define $d(g) \in V_{\mathbb{R}}$ by

$$G_g U_{A_g^{-1}\lambda} G_g^{-1} = \mathbf{c}(\langle d(g),\lambda\rangle) U_\lambda. \tag{5.17-A13}$$

The exponent is additive in λ: conjugate (5.17-A12) and use (5.17-A11). The lifts of gh and G_gG_h differ centrally, so

$$\begin{aligned}\langle d(gh),\lambda\rangle &= \langle d(g),\lambda\rangle + \langle d(h), A_g^{-1}\lambda\rangle \\ &= \langle d(g) + T_g d(h),\lambda\rangle.\end{aligned}$$

This proves the crossed-homomorphism law. Replacing U_λ by $\mathbf{c}(\langle a,\lambda\rangle)U_\lambda$ changes (5.17-A13) by $(T_g - I)a$. Replacing G_g by a central multiple does not change (5.17-A13), so this proves the stated independence and uniqueness. Applying the crossed law to (5.17-A1) proves the first identity in (5.17-A4); passage to coinvariants proves the second.

For the order-m_j filling, distinguish every covering level. Put

$$\widetilde{J}_j = \Delta_j \times \mathbb{C}^2, \qquad r_j : \widetilde{J}_j \longrightarrow J'_j$$

for the lattice quotient, whose deck translations are τ_λ. On the vector cover define

$$Q_{j,\log} := \widetilde{g}_j^{\,v_j/m_j}, \qquad Q_{j,\mathrm{lin}} := \widetilde{g}_j^{\,0}.$$

They satisfy

$$\begin{aligned} Q_{j,\log}\tau_\lambda Q_{j,\log}^{-1} &= \tau_{A_j\lambda}, \quad (Q_{j,\log})^{m_j} = \tau_{v_j}, \\ Q_{j,\mathrm{lin}}\tau_\lambda Q_{j,\mathrm{lin}}^{-1} &= \tau_{A_j\lambda}, \quad (Q_{j,\mathrm{lin}})^{m_j} = 1. \end{aligned} \tag{5.17-A14}$$

The positive sign is fixed by $s_j \circ g_j = e^{-2\pi i/m_j} s_j$ and (2.8f): after exactly m_j clockwise iterations, the logarithmic vector-cover lift translates by $+v_j$.

Let $q_{j,\log}, q_{j,\mathrm{lin}}$ be the descended automorphisms of J'_j, and put $J'^*_j = J'_j|_{\Delta_j^*}$. Both descended automorphisms have order dividing m_j, because τ_{v_j} is trivial after the lattice quotient; each covers g_j, whose order is exactly m_j, so each descended order is exactly m_j. The branch-independent torus translation

$$h_j : J'^*_j \longrightarrow J'^*_j$$

constructed in (2.9d) satisfies, on this quotient level,

$$h_j q_{j,\log} h_j^{-1} = q_{j,\mathrm{lin}}.$$

Define

$$\begin{aligned} p_{j,\log} : J'^*_j &\longrightarrow N_j^* := J'^*_j/\langle q_{j,\log}\rangle, \\ p_{j,\mathrm{lin}} : J'^*_j &\longrightarrow J_j^* := J'^*_j/\langle q_{j,\mathrm{lin}}\rangle. \end{aligned}$$

The conjugacy induces a biholomorphism

$$\bar{h}_j : N_j^* \longrightarrow J_j^*, \qquad p_{j,\mathrm{lin}} \circ h_j = \bar{h}_j \circ p_{j,\log}.$$

The gluing of X identifies $L|_{N_j^*} \simeq \bar{h}_j^*(L|_{J_j^*})$; hence the exactly typed line-bundle square gives

$$p_{j,\log}^*(L|_{N_j^*}) \simeq h_j^* p_{j,\mathrm{lin}}^*(L|_{J_j^*}). \tag{5.17-A14a}$$

For the factor extension return to the filled vector cover. Let

$$P_{j,\log} : J'_j \longrightarrow N_j, \qquad \pi_j^{\mathrm{vec}} := P_{j,\log} \circ r_j : \widetilde{J}_j \longrightarrow N_j.$$

Here $P_{j,\log}$ is the full quotient and $P_{j,\log}|_{J'^*_j} = p_{j,\log}$. The deck group is

$$\Gamma_j^{\log} = \langle \tau_\lambda, Q_{j,\log} : \lambda \in \Lambda\rangle.$$

The pullback $(\pi_j^{\mathrm{vec}})^*(L|_{N_j})$ is trivial on the contractible Stein manifold $\widetilde{J}_j$; its factors define an integral central extension, which we push out along $\mathbb{Z} \hookrightarrow \mathbb{R}$. Over Δ_j^*, the line-bundle square (5.17-A14a) identifies each fibre with the global factor extension by a translation, so its commutator form is E. That form is integer-valued and locally constant on the connected disc Δ_j; therefore it is E on the filled cover as well. Choose vertical lifts $U_\lambda^{(j)}$ normalized by

$$U_\lambda^{(j)} U_\mu^{(j)} = \mathbf{c}\Big(\frac{1}{2}E(\lambda,\mu)\Big)\, U_{\lambda+\mu}^{(j)}.$$

For $\alpha \in \Gamma_j^{\log}$, define A_α, T_α by

$$\alpha \tau_\lambda \alpha^{-1} = \tau_{A_\alpha \lambda}, \qquad T_\alpha = (A_\alpha^{-1})^{\mathsf{t}}.$$

For a lift $\widetilde{G}_\alpha$, define $\widetilde{d}_j(\alpha) \in V_{\mathbb{R}}$ through

$$\widetilde{G}_\alpha U^{(j)}_{A_\alpha^{-1}\lambda} \widetilde{G}_\alpha^{-1} = \mathbf{c}\Big(\langle \widetilde{d}_j(\alpha), \lambda \rangle\Big)\, U^{(j)}_\lambda. \tag{5.17-A14c}$$

Indeed, before naming $\widetilde{d}_j(\alpha)$, the quotient of the two sides of (5.17-A14c) is central because both cover the same translation. Conjugating the normalized product and using

$$E(A_\alpha \lambda, A_\alpha \mu) = E(\lambda, \mu)$$

shows that its exponent $r_\alpha(\lambda)$ satisfies

$$r_\alpha(\lambda + \mu) = r_\alpha(\lambda) + r_\alpha(\mu).$$

Thus it is the pairing with a unique element of $V_{\mathbb{R}}$, as asserted. Since $A_{\alpha\beta} = A_\alpha A_\beta$, choose the lift of $\alpha\beta$ to be $\widetilde{G}_\alpha \widetilde{G}_\beta$; any other choice differs centrally. Then

$$\begin{aligned} &\widetilde{G}_\alpha \widetilde{G}_\beta U^{(j)}_{A_\beta^{-1} A_\alpha^{-1} \lambda} \widetilde{G}_\beta^{-1} \widetilde{G}_\alpha^{-1} \\ &\quad = \mathbf{c}\Big(\langle \widetilde{d}_j(\beta), A_\alpha^{-1}\lambda \rangle + \langle \widetilde{d}_j(\alpha), \lambda \rangle\Big)\, U^{(j)}_\lambda. \end{aligned}$$

The defining dual action $\langle T_\alpha v, \lambda \rangle = \langle v, A_\alpha^{-1} \lambda \rangle$ therefore proves the crossed law

$$\widetilde{d}_j(\alpha\beta) = \widetilde{d}_j(\alpha) + T_\alpha \widetilde{d}_j(\beta). \tag{5.17-A14d}$$

Put

$$\widetilde{J}_j^* := \Delta_j^* \times \mathbb{C}^2, \qquad r_j^* := r_j|_{\widetilde{J}_j^*} : \widetilde{J}_j^* \longrightarrow J_j'^*$$

and define

$$\begin{aligned} \Phi_{j,\log} &:= p_{j,\log} \circ r_j^*, \\ \Phi_{j,\mathrm{glob}} &:= p_{j,\mathrm{lin}} \circ h_j \circ r_j^* = \bar{h}_j \circ \Phi_{j,\log}. \end{aligned}$$

Both maps are invariant under $\Gamma_j^{\log}$. This is immediate for every τ_λ, while

$$\begin{aligned} \Phi_{j,\log} Q_{j,\log} &= p_{j,\log} q_{j,\log} r_j^* = \Phi_{j,\log}, \\ \Phi_{j,\mathrm{glob}} Q_{j,\log} &= p_{j,\mathrm{lin}} h_j q_{j,\log} r_j^* = p_{j,\mathrm{lin}} q_{j,\mathrm{lin}} h_j r_j^* = \Phi_{j,\mathrm{glob}}. \end{aligned}$$

Thus $\widetilde{J}_j^*$ is the same $\Gamma_j^{\log}$-cover of the two biholomorphic quotients. Pulling the gluing isomorphism to this cover gives

$$\Theta_j : \Phi_{j,\log}^*(L|_{N_j^*}) \xrightarrow{\sim} \Phi_{j,\mathrm{glob}}^*(L|_{J_j^*}).$$

The filled factor extension restricts to this cover, and the global factor extension pulls back to it through the fixed boundary whisker. Because Θ_j intertwines descent, it induces an isomorphism of the resulting pushed-out $\Gamma_j^{\log}$-factor extensions which sends $\mathbf{c}(r)$ to $\mathbf{c}(r)$ for every $r \in \mathbb{R}$.

Let $\gamma_j^J : [0,1] \to J_j^*$ be the zero-section image of the unwhiskered clockwise boundary meridian, and let ω_j be the fixed zero-section whisker from the global basepoint to $\gamma_j^J(0)$. Thus the based total-space loop retained in (5.17-A1) is

$$\widehat{\rho}_j = \omega_j * \gamma_j^J * \omega_j^{-1}.$$

Choose $x_j \in J_j'^*$ such that $p_{j,\mathrm{lin}}(h_j(x_j)) = \gamma_j^J(0)$, put $x_{j,\mathrm{lin}} = h_j(x_j)$, and let $\widetilde{\gamma}_{j,\mathrm{lin}}$ be the $p_{j,\mathrm{lin}}$-lift of γ_j^J beginning at $x_{j,\mathrm{lin}}$. For the chosen clockwise generator,

$$\widetilde{\gamma}_{j,\mathrm{lin}}(1) = q_{j,\mathrm{lin}}(x_{j,\mathrm{lin}}).$$

Set

$$\gamma_j^N := \bar{h}_j^{-1} \circ \gamma_j^J, \qquad \widetilde{\gamma}_{j,\log} := h_j^{-1} \circ \widetilde{\gamma}_{j,\mathrm{lin}}.$$

Then the quotient square gives the point-valued identities

$$\begin{aligned} p_{j,\log} \circ \widetilde{\gamma}_{j,\log} &= \gamma_j^N, \\ \widetilde{\gamma}_{j,\log}(0) &= x_j, \\ \widetilde{\gamma}_{j,\log}(1) &= q_{j,\log}(x_j). \end{aligned}$$

Choose $\widetilde{x}_j \in \widetilde{J}_j^*$ with $r_j^*(\widetilde{x}_j) = x_j$, and let $\widetilde{\gamma}_{j,\mathrm{vec}}$ be the r_j^*-lift of $\widetilde{\gamma}_{j,\log}$ beginning at $\widetilde{x}_j$. On adjacent sheets choose compatible logarithm branches ℓ, ℓ' with

$$\ell'(w) = \ell(g_j^{-1} w) - \frac{2\pi i}{m_j}.$$

The calculation in (2.9e) gives the point-valued endpoint formula

$$\widetilde{\gamma}_{j,\mathrm{vec}}(1) = Q_{j,\log}(\widetilde{x}_j),$$

excluding both $Q_{j,\log}^{-1}(\widetilde{x}_j)$ and every $\tau_\lambda Q_{j,\log}(\widetilde{x}_j)$ with $\lambda \neq 0$. Fibre loops remain τ_λ, since h_j is a fibre translation.

The factor of L along the fixed lift of ω_j identifies the target factor over the local boundary segment with the globally based factor. Let $G_{\rho_j}^{(j)}$ be the target lift of the local boundary deck element whose whiskered image is the chosen global lift G_{ρ_j}. Thus it covers $\widehat{\rho}_j$ after whiskering and, by (5.17-A13), satisfies

$$G_{\rho_j}^{(j)} U_{A_j^{-1}\lambda} (G_{\rho_j}^{(j)})^{-1} = \mathbf{c}(\langle d_j, \lambda \rangle) U_\lambda, \qquad d_j = d(\rho_j).$$

Consequently, for some $a_j \in V_{\mathbb{R}}$ and $b_j \in \mathbb{R}$,

$$\Theta_{j,*}(U_\lambda^{(j)}) = \mathbf{c}(\langle a_j, \lambda \rangle) U_\lambda, \qquad \Theta_{j,*}(\widetilde{G}_{Q_{j,\log}}) = \mathbf{c}(b_j) G_{\rho_j}^{(j)}.$$

Here U_λ is the normalized global system in (5.17-A12), and $\mathbf{c}(b_j)$ disappears under conjugation. Therefore

$$\begin{aligned} \langle \widetilde{d}_j(Q_{j,\log}), \lambda \rangle + \langle a_j, \lambda \rangle &= \langle d_j, \lambda \rangle + \langle a_j, A_j^{-1} \lambda \rangle \\ &= \langle d_j + T_j a_j, \lambda \rangle, \end{aligned}$$

so

$$\widetilde{d}_j(Q_{j,\log}) = d_j + (T_j - I) a_j. \tag{5.17-A14e}$$

For a translation choose $\widetilde{G}_{\tau_v} = U_v^{(j)}$. The filled normalization above gives

$$U_v^{(j)} U_\lambda^{(j)} (U_v^{(j)})^{-1} = \mathbf{c}(E(v, \lambda)) U_\lambda^{(j)},$$

and hence

$$\widetilde{d}_j(\tau_v) = \iota_v E, \qquad (\iota_v E)(\lambda) = E(v, \lambda). \tag{5.17-A14f}$$

Here

$$A_{Q_{j,\log}} = A_j, \qquad T_{Q_{j,\log}} = T_j.$$

Since $N_{T_j}(T_j - I) = T_j^{m_j} - I = 0$, equations (5.17-A14), (5.17-A14d), and (5.17-A14e) give

$$\begin{aligned} N_{T_j} d_j &= N_{T_j} \widetilde{d}_j(Q_{j,\log}) \\ &= \widetilde{d}_j((Q_{j,\log})^{m_j}) \\ &= \widetilde{d}_j(\tau_{v_j}) \\ &= \iota_{v_j} E. \end{aligned} \tag{5.17-A15}$$

Changing a line-bundle frame gives an isomorphic factor extension; changing $\widetilde{G}_\alpha$ by a central factor does not change conjugation; and every second normalized vertical system differs by a character, changing $\widetilde{d}_j(\alpha)$ by $(T_\alpha - I)a$. Finally, $\ell \mapsto \ell + 2\pi i n$ changes a vector-cover lift of h_j by

$$H_j \longmapsto H_j' = \tau_{n v_j} H_j.$$

At the factor-extension level this means

$$\Theta'_{j,*} = \mathrm{Ad}(U_{n v_j}) \circ \Theta_{j,*}.$$

Consequently the comparison character changes by $n\iota_{v_j}E$, because

$$U_{n v_j}(\mathbf{c}(\langle a_j, \lambda\rangle)U_\lambda)U_{n v_j}^{-1} = \mathbf{c}(\langle a_j + n\iota_{v_j}E, \lambda\rangle)U_\lambda.$$

Since $A_j v_j = v_j$ and E is A_j-invariant,

$$\langle T_j(\iota_{v_j}E), \lambda\rangle = E(v_j, A_j^{-1}\lambda) = E(A_j v_j, \lambda) = E(v_j, \lambda),$$

so this branch change leaves both (5.17-A14e) and (5.17-A15) unchanged. The translation $\tau_{n v_j}$ also commutes with $Q_{j,\mathrm{lin}}$ on the vector cover because $A_j v_j = v_j$; on the lattice quotient it is already the identity. Hence it changes only the central lift choice there. The descended $h_j, \bar{h}_j$, and the quotient square are branch independent. Thus (5.17-A15) is choice-independent and proves (5.17-A5), including its sign.

It remains to obtain (5.17-A6) analytically. Since $A_0\widehat{\delta} = \widehat{\delta}$, equation (5.17-A13) says that

$$[G_0, U_{\widehat{\delta}}] = \mathbf{c}(\langle d_0, \widehat{\delta}\rangle). \tag{5.17-A16}$$

This logarithmic commutator is the clutching winding with the base loop first. Thus, if b and f denote the oriented base and fibre loops, respectively,

$$\mathrm{wind}_{b\wedge f}(L) = \langle d_0, \widehat{\delta}\rangle, \qquad \mathrm{wind}_{\xi_{\widehat{\delta}}}(L) = -\langle d_0, \widehat{\delta}\rangle. \tag{5.17-A17}$$

The second equality uses the fibre-first convention $\xi_{\widehat{\delta}} = [S^1_{\widehat{\delta}}] \times [\rho_0]$, hence the single sign change. Use the actual unimodular cusp chart

$$(u, x_1, x_2) = \left(\frac{t}{x_1 x_2}, x_1, x_2\right), \qquad t = u x_1 x_2. \tag{5.17-A18}$$

For $0 < \rho \ll 1$, the boundary parametrization

$$(\theta, \phi) \longmapsto (\rho e^{i(\theta-\phi)}, 1, e^{i\phi}), \qquad t = \rho e^{i\theta}, \tag{5.17-A19}$$

has base coordinate θ and fibre translation $\widehat{\delta}$ in the ϕ-direction. The latter fixed-base loop does *not* itself bound the x_2-disc. To identify the correct meridian, put $\alpha = \theta - \phi$, let a_u be the positively oriented u-longitude, and let m_{x_2} be the positively oriented x_2-meridian at fixed u. In the basis (a_u, m_{x_2}), the base and fibre loops are

$$b = a_u, \qquad f = m_{x_2} - a_u, \qquad b \wedge f = a_u \wedge m_{x_2}. \tag{5.17-A19a}$$

The change-of-basis determinant is $+1$, so this preserves the base-first orientation in (5.17-A17). Now m_{x_2} does bound the analytic disc

$$z \longmapsto (\rho e^{i\alpha_0}, 1, z), \qquad |z| \le 1. \tag{5.17-A19b}$$

Choose a thin annulus A_u about $|u| = \rho$, a disc D_{x_1} about 1, and a disc D_{x_2} containing $\{|x_2| \le 1\}$, small enough that

$$A_u \times D_{x_1} \times D_{x_2} \subset \{|u x_1 x_2| < \epsilon\}. \tag{5.17-A19c}$$

This neighbourhood is Stein and retracts onto S^1, so its second integral cohomology vanishes. Cartan's theorem B and the exponential sequence therefore give trivial Picard group [40, §2, fundamental theorem, p. 81][9, Proposition 6.28 and equations (6.35)–(6.36), pp. 180–182]. Pull $L|_{N_0}$ back to the cusp cover and choose one holomorphic frame on this entire neighbourhood. Compute the clutching cocycle with this extending frame on both sides of a cut in the u-longitude. Its transition unit on D_{x_2} is then identically $F = 1$, so by (5.17-A19a) its base–fibre commutator winding is

$$\frac{1}{2\pi i}\int_{|x_2|=1}\frac{dF}{F} = 0. \tag{5.17-A20}$$

This computes the same alternating logarithmic cocycle as (5.17-A16). Indeed, changing from the global Appell–Humbert frame to the extending chart frame changes c by a group coboundary. Put

$$g = \widehat{g}_0, \qquad h = \tau_{\widehat{\delta}}$$

in the underlying deck group. Equation (5.17-A9) and $A_0\widehat{\delta} = \widehat{\delta}$ give $gh = hg$. If $c' = c + \delta\beta$ is the cocycle after a frame change, then

$$[c'(g,h) - c'(h,g)] - [c(g,h) - c(h,g)] = \beta(hg) - \beta(gh) = 0.$$

Thus the alternating cocycle is frame-invariant without assuming that its central-extension lifts commute. Equations (5.17-A17), (5.17-A19a), and (5.17-A20) therefore give $\langle d_0, \widehat{\delta}\rangle = 0$. This is a calculation in $\mathcal{O}^*$, not an evaluation of a Chern class on a bounding chain. Finally, the matrices A_1, A_2 fix $\widehat{\delta}$, so the evaluation is unchanged by every principal change in (5.17-A3). Since $V_G = \mathbb{Z}[\delta]$ and $\delta(\widehat{\delta}) = 1$, its vanishing is equivalent to $\bar{d}_0 = 0$. This proves (5.17-A6). □

Lemma 60.20 (the toric support function detects central restriction)**.** ***[VERIFIED]*** *Let $M \in \mathrm{Pic}(N_0)$, let $\pi : Y_\epsilon \to N_0$ be the cusp covering, and let $\nu : E_0 \to W$ be the normalization. If the Appell–Humbert form of $M|_{F_t}$, $t \neq 0$, is E_M, then the three independent degrees of $\nu^*(M|_W)$ determine an integral symmetric form $B : \mathbb{Z}^2 \times \mathbb{Z}^2 \to \mathbb{Z}$ satisfying*

$$E_M\big((\lambda,0),(0,m)\big) = -B(B_0\lambda, m) \qquad (\lambda, m \in \mathbb{Z}^2). \tag{5.17-A21}$$

Consequently

$$E_M = 0 \quad\Longrightarrow\quad \nu^*(M|_W) \simeq \mathcal{O}_{E_0}. \tag{5.17-A22}$$

Proof. Put $\mathcal{M} = \pi^*M$. The identity $\pi \circ \Psi_\lambda = \pi$ gives $\mathcal{M}$ its genuine Ψ-equivariant descent structure. In the notation of theorem 55.15, order the six rays in the star of the origin as

$$(r_1, r_2, r_3, r_4, r_5, r_6) = (e_1, e_2, e_2 - e_1, -e_1, -e_2, e_1 - e_2), \qquad C_i = E_0 \cap E_{r_i}. \tag{5.17-A23}$$

The quotient identifies opposite conductor branches. Therefore, for $A = \mathcal{M}|_{E_0} = \nu^*(M|_W)$,

$$d_i = \deg_{C_i} A, \qquad (\deg_{C_1} A, \dots, \deg_{C_6} A) = (d_1, d_2, d_3, d_1, d_2, d_3). \tag{5.17-A24}$$

Define the integer-valued quadratic function

$$a(x,y) = (d_2 + d_3)\binom{x}{2} + d_3xy + (d_1 + d_3)\binom{y}{2}. \tag{5.17-A25}$$

Its polarization is

$$B(v,w) = a(v+w) - a(v) - a(w), \qquad [B] = \begin{pmatrix} d_2 + d_3 & d_3 \\ d_3 & d_1 + d_3 \end{pmatrix}. \tag{5.17-A26}$$

The locally finite divisor

$$D_a = \sum_{v\in\mathbb{Z}^2} a(v)E_v \tag{5.17-A27}$$

is Cartier: on every maximal chart, the three height-one ray generators form an integral basis, so the three assigned coefficients are the values of one integral character. In the standard toric convention the associated support function has ray values $-a(v)$; the divisor coefficients here are the values $a(v)$, consistent with $\mathrm{div}(\chi^m) = \sum_\rho \langle m, u_\rho\rangle D_\rho$ [23, §3.3, pp. 60–63; §3.4, pp. 63–72][24, Propositions 4.1.1–4.1.2, p. 171; Theorems 4.2.8, p. 179 and 4.2.12, p. 182]. This is the same chartwise Cartier argument as in (5.17ab), and it applies to an arbitrary integer-valued a.

Direct substitution gives

$$(a(r_1),\dots,a(r_6)) = (0,0,d_2,d_2+d_3,d_1+d_3,d_1). \tag{5.17-A28}$$

Using $C_i^2 = -1$ and $C_iC_{i\pm1} = 1$, the degree vector of $\mathcal{O}(D_a)|_{E_0}$ is exactly the vector in (5.17-A24). More generally, restrict the principal identity $\mathrm{div}(t) = \sum_w E_w$ to E_v. Since E_v meets precisely E_{v+r_i}, it gives

$$\mathcal{O}(D_a)|_{E_v} \simeq \mathcal{O}_{E_v}\left(\sum_{i=1}^{6}(a(v+r_i)-a(v))C_{v,i}\right). \tag{5.17-A29}$$

The coefficient difference is

$$a(v+r_i) - a(v) = a(r_i) + B(v,r_i). \tag{5.17-A30}$$

The second summand in (5.17-A30) is the divisor of the toric character with exponent $B(v,-)$. Thus every restriction in (5.17-A29) has the same six degrees as $\mathcal{O}(D_a)|_{E_0}$.

Because $B_0\bar{\Lambda} = \mathbb{Z}^2$, the deck action is transitive on the components E_v. The genuine descent structure on $\mathcal{M}$ therefore transports (5.17-A24), with its ordered six boundary directions, to every E_v. It follows that

$$\mathcal{R} := \mathcal{M}\otimes\mathcal{O}(-D_a) \quad \text{satisfies} \quad \mathcal{R}|_{E_v} \simeq \mathcal{O}_{E_v} \quad (v\in\mathbb{Z}^2). \tag{5.17-A31}$$

Indeed, the six toric curves detect $\mathrm{Pic}(\mathrm{dP}_6)$: in the blow-up basis H, E_1, E_2, E_3, vanishing of the degrees on the three exceptional curves kills the three E_i-coefficients, and vanishing on any strict transform $H - E_i - E_j$ kills the remaining H-coefficient. Since $H^1(\mathrm{dP}_6,\mathcal{O}) = 0$, equality of those degrees implies isomorphism, not merely numerical equivalence.

For $b_\lambda = B_0\lambda$, the torus-translation part of Ψ_λ preserves each toric prime divisor, while its fan part translates the index by b_λ. Hence

$$\Psi_\lambda^* D_a - D_a = \sum_v (a(v+b_\lambda)-a(v))E_v = \mathrm{div}(\chi^{m_\lambda}), \tag{5.17-A32}$$

where

$$m_\lambda = (\ell_\lambda, k_\lambda), \qquad \ell_\lambda = B(b_\lambda,-), \qquad k_\lambda = a(b_\lambda). \tag{5.17-A33}$$

Thus multiplication by χ^{m_λ} gives an individual isomorphism $\Psi_\lambda^*\mathcal{O}(D_a)\simeq\mathcal{O}(D_a)$. No cocycle assertion for these individual isomorphisms is needed: we use them only to separate the horizontal winding of the genuine descent factor of $\mathcal{M}$.

For a ray $\mathbb{R}_{\ge0}(v,1)$, its toric open set in Y_ϵ is

$$U_v \simeq \Delta_\epsilon \times (\mathbb{C}^*)^2, \qquad E_v^\circ = \{0\}\times(\mathbb{C}^*)^2. \tag{5.17-A34}$$

Indeed $(e_1,0),(e_2,0),(v,1)$ is an integral basis and t is the character dual to the last vector; compare [23, §2.1, pp. 28–29] and [24, Example 1.2.21, p. 31]. The set U_v is Stein and retracts onto E_v°. By (5.17-A31), $c_1(\mathcal{R}|_{U_v}) = 0$; the exponential sequence and Cartan's theorem B therefore trivialize $\mathcal{R}|_{U_v}$. Choose a global nowhere-zero frame s_v of $\mathcal{R}|_{E_v}$. A frame on U_v can be multiplied by a unit pulled back from $(\mathbb{C}^*)^2$ so that its restriction to E_v° agrees with $s_v|_{E_v^\circ}$; denote the resulting frame by e_v.

We also need to compare these frames on the generic torus. The direction of the toric affine-open inclusions is essential here. If v, w are adjacent vertices, choose a triangle $\sigma = \{v, w, u\}$ containing them, write z_v, z_w, z_u for the coordinates dual to its three height-one ray generators, and put

$$S_\sigma := U_\sigma \cap Y_\epsilon = \{(z_v, z_w, z_u) \in \mathbb{C}^3 : |z_v z_w z_u| < \epsilon\}. \tag{5.17-A34a}$$

Because $\rho_v < \sigma$, localization of the maximal affine chart gives

$$U_{\rho_v} = D(z_w z_u) \subset U_\sigma, \qquad U_{\rho_w} = D(z_v z_u) \subset U_\sigma, \qquad U_{\rho_v} \cap U_{\rho_w} = D(z_v z_w z_u) = U_0. \tag{1767}$$

Thus U_σ is not contained in $U_{\rho_v} \cap U_{\rho_w}$, and the coordinate discs in S_σ cannot be applied directly to the ratio e_w/e_v.

The space S_σ is nevertheless the required comparison chart. It is the closed graph

$$\{(z_v, z_w, z_u, t) \in \mathbb{C}^3 \times \Delta_\epsilon : t = z_v z_w z_u\}$$

in the Stein space $\mathbb{C}^3 \times \Delta_\epsilon$, and hence is Stein. Radial scaling of the three z-coordinates contracts it to the origin. The exponential sequence and Cartan's theorem B therefore give $\mathrm{Pic}(S_\sigma) = 0$. Choose a frame e_σ of $\mathcal{R}|_{S_\sigma}$.

Let s_v be the global frame of $\mathcal{R}|_{E_v} \simeq \mathcal{O}_{E_v}$ used to normalize e_v. On $U_{\rho_v} \cap Y_\epsilon$ put

$$A_{\sigma,v} = e_v/e_\sigma.$$

At $t = 0$, this is the restriction to $(\mathbb{C}^*)^2$ of the holomorphic unit

$$G_{\sigma,v} = \frac{s_v}{e_\sigma|_{E_v \cap S_\sigma}} \in \mathcal{O}^*(E_v \cap S_\sigma), \qquad E_v \cap S_\sigma = \{z_v = 0\} \simeq \mathbb{C}^2.$$

Consequently its winding on every horizontal cocharacter is zero at $t = 0$. Under $U_{\rho_v} \cap Y_\epsilon \simeq \Delta_\epsilon \times (\mathbb{C}^*)^2$, that winding is an integer locally constant in t, and is therefore zero on every nearby nonzero fibre. The same argument applies to $A_{\sigma,w}$. On their actual common domain $U_0 \cap Y_\epsilon$,

$$e_w/e_v = A_{\sigma,w}/A_{\sigma,v},$$

so this ratio has zero horizontal winding. The one-skeleton of the A_2-triangulation is connected, and multiplication along an edge path now gives

$$\mathrm{wind}_m(e_w/e_v) = 0 \qquad (v, w, m \in \mathbb{Z}^2) \tag{5.17-A34b}$$

on every nonzero fibre.

Let $\theta_\lambda^{\mathcal{M}} : \Psi_\lambda^* \mathcal{M} \to \mathcal{M}$ be the genuine descent isomorphism, let $\mu_\lambda^+ : \Psi_\lambda^* \mathcal{O}(D_a) \to \mathcal{O}(D_a)$ be multiplication by χ^{m_λ} from (5.17-A32), and define the inverse-dual isomorphism

$$\mu_\lambda^- := ((\mu_\lambda^+)^{-1})^\vee : \Psi_\lambda^* \mathcal{O}(-D_a) \longrightarrow \mathcal{O}(-D_a), \qquad s \longmapsto \chi^{-m_\lambda} s.$$

The ordinary dual of μ_λ^+ has the opposite arrow; the inverse-dual above is the map required by the pullback tensor product. Then

$$\theta_\lambda^{\mathcal{R}} := \theta_\lambda^{\mathcal{M}} \otimes \mu_\lambda^- : \Psi_\lambda^* \mathcal{R} \longrightarrow \mathcal{R}.$$

Since $\Psi_\lambda(U_v) = U_{v+b_\lambda}$, define $r_{\lambda,v} \in \mathcal{O}^*(U_v)$ by the typed identity

$$\theta_\lambda^{\mathcal{R}}(\Psi_\lambda^* e_{v+b_\lambda}) = r_{\lambda,v} e_v. \tag{1768}$$

At $t = 0$ it extends to an isomorphism between two trivial line bundles on the compact surface E_v. Its factor in the chosen global frames is therefore a nonzero constant. The winding of $r_{\lambda,v}(t, -)$ around every

horizontal one-parameter subgroup is an integer locally constant in t; at $t = 0$ it is zero, hence it is zero for every nearby $t \neq 0$.

To express this isomorphism in one frame on the generic torus, put $q_{\lambda,v} = e_{v+b_\lambda}/e_v$. Equation (5.17-A34b) gives zero horizontal winding for $q_{\lambda,v}$; because $\Psi_\lambda|_{Y_t}$ is multiplication by a fixed torus point, it induces the identity on horizontal H_1, and $\Psi_\lambda^* q_{\lambda,v}$ also has zero winding. Relative to the single frame e_v, the residual multiplier is therefore

$$h_{\lambda,v} = \frac{r_{\lambda,v}}{\Psi_\lambda^* q_{\lambda,v}}, \qquad \mathrm{wind}_m(h_{\lambda,v}) = 0. \tag{5.17-A34c}$$

On U_0 let $\mathbf{1}_a$ be the canonical toric frame of $\mathcal{O}(D_a)$, and put $F_v = e_v \otimes \mathbf{1}_a$. Equations equation (1768) and (5.17-A32) give

$$\theta_\lambda^{\mathcal{M}}(\Psi_\lambda^* F_v) = h_{\lambda,v}\chi^{m_\lambda} F_v.$$

On a nonzero fibre, $\chi^{m_\lambda} = x^{\ell_\lambda} t^{k_\lambda}$. With the section-descent convention (5.17am), the factor of automorphy is therefore

$$\alpha_\lambda = (h_{\lambda,v}\chi^{m_\lambda})^{-1}.$$

The residual factor has zero horizontal winding by (5.17-A34c), and hence

$$\mathrm{wind}_m(\alpha_\lambda) = -\ell_\lambda(m) = -B(B_0\lambda, m). \tag{1769}$$

This winding is exactly the mixed Chern-class entry. Pull F_v back to the universal cover

$$p_t : \mathbb{C}^2 \longrightarrow Y_t, \qquad \zeta \longmapsto (e^{2\pi i\zeta}, t).$$

The period indexed by $(\lambda, n) \in \mathbb{Z}^2 \oplus \mathbb{Z}^2$ acts by $\zeta \mapsto \zeta + Z(s)\lambda + n$. The horizontal factor is one, because F_v is already defined on Y_t. Choose a logarithm $g_\lambda \in \mathcal{O}(\mathbb{C}^2)$ with

$$p_t^* \alpha_\lambda = e^{2\pi i g_\lambda}.$$

For every horizontal period n,

$$g_\lambda(\zeta + n) - g_\lambda(\zeta) = \mathrm{wind}_n(\alpha_\lambda).$$

In the exponential connecting cocycle

$$c(l, l') = g_l(\zeta + l') + g_{l'}(\zeta) - g_{l+l'}(\zeta),$$

take the horizontal logarithm to be zero and the logarithm for the combined period (λ, n) to be g_λ. Then

$$c((\lambda, 0), (0, n)) = \mathrm{wind}_n(\alpha_\lambda), \qquad c((0, n), (\lambda, 0)) = 0.$$

Therefore the alternating exponential connecting integer between the vertical deck translation indexed by λ and the horizontal loop indexed by m is

$$E_M\big((\lambda, 0), (0, m)\big) = -\ell_\lambda(m) = -B(B_0\lambda, m),$$

which proves (5.17-A21). This sign also recovers the explicit check in theorem 55.15: for $a(v) = v^{\mathrm{t}} H v$, one has $B = 2H$, so (5.17-A21) is the cross-block $-2B_0^{\mathrm{t}} H = -2P$, namely $-2q$.

If $E_M = 0$, equation (5.17-A21) and the unimodularity of B_0 force $B = 0$. The matrix in (5.17-A26) then gives $d_3 = 0$, $d_2 = 0$, and $d_1 = 0$. Thus all six degrees of A vanish, and the Picard detection argument following (5.17-A31) gives $A \simeq \mathcal{O}_{E_0}$. This is (5.17-A22). □

Theorem 60.21 (purely analytic vanishing of the global restriction)**.** ***[VERIFIED]*** *For the constructed threefold,*

$$r = \eta^* i^* : \mathrm{Pic}(X) \longrightarrow \mathrm{Pic}(\mathrm{dP}_6)$$

is the zero homomorphism. This proof uses only holomorphic $\mathcal{O}^*$-factors, the two finite logarithmic fillings, the cusp argument principle, and the Ψ-equivariant toric support function. It uses neither a global integral Leray transgression nor local integral nearby-cycle specialization.

Proof. Let $L \in \mathrm{Pic}(X)$. By theorem 5.3, its invariant fibre class is

$$E = c_1(L|_F) = kq, \qquad k \in \mathbb{Z}, \qquad q = uw + 6\gamma\delta. \tag{5.17-A35}$$

The exact contractions, now used without an integral specialization lattice, are

$$\iota_{v_1} q = 2\eta_1, \qquad \iota_{v_2} q = -3\eta_2, \qquad [\eta_1] = 3[\delta], \qquad [\eta_2] = 2[\delta]. \tag{5.17-A36}$$

Apply (5.17-A5) with $(m_1, m_2) = (3, 4)$. In the real global coinvariants,

$$\bar{d}_1 = 2k[\delta], \qquad \bar{d}_2 = -\frac{3}{2}k[\delta]. \tag{5.17-A37}$$

Equations (5.17-A4) and (5.17-A6) therefore give

$$0 = \bar{d}_1 + \bar{d}_2 + \bar{d}_0 = \frac{k}{2}[\delta]. \tag{5.17-A38}$$

Since $V_G \otimes \mathbb{R} = \mathbb{R}[\delta]$, one has $k = 0$, so $E = 0$. Apply theorem 60.20 to $M = L|_{N_0}$. It gives

$$\eta^* i^* L = \nu^*(L|_W) \simeq \mathcal{O}_{E_0}.$$

This holds for every L, proving $r = 0$. □

Corollary 60.22 (candidate nonvanishing for Pic^0 without topology)**.** ***[VERIFIED]*** *For the constructed threefold and every $L \in \mathrm{Pic}^0(X)$,*

$$H^0(W, \Omega^1_X|_W \otimes A_L) \neq 0, \qquad H^2(W, (T_X \otimes L)|_W) \neq 0,$$

and

$$R^2 f_*(T_X \otimes L) \neq 0. \tag{5.17f}$$

This proof uses the analytic construction of the cusp fibre but no global homology, Mayer–Vietoris, or Leray calculation.

Proof. The cusp theorem theorem 10.6 supplies the full compact reduced normal-crossings fibre, its dP_6 normalization, and the trivialization of ω_X near W. By theorem 60.12, $\eta^* A_L$ is trivial, so theorem 60.5 gives the first nonvanishing. Apply theorem 60.8 to $E = T_X \otimes L$; its dual space is exactly the preceding nonzero section space, giving the fibrewise H^2 nonvanishing. The fibre has complex dimension two, so its coherent H^3 vanishes. The global construction makes X compact and f nonconstant; theorem 60.9 therefore gives properness and f-flatness of $T_X \otimes L$. The base-change isomorphism of theorem 60.10 now identifies the fibre of the R^2 stalk with the nonzero H^2 group. Hence the stalk, and therefore the direct-image sheaf, is nonzero. □

Remark 60.23 (reconciliation of the three restriction routes). The Pic^0 argument proves (5.17f) without global homology, Mayer–Vietoris, or Leray. The all-Pic statement theorem 60.14 avoids citing the packaged theorem $H^2(X; \mathbb{Z}) = 0$, but it reuses the same verified global integral Leray and specialization machinery from which that theorem is deduced. By contrast, theorem 60.18 supplies a second all-Pic proof without any global Leray differential: it recovers the primitive $4 - 3 = 1$ obstruction directly from the finite logarithmic gluing and the cusp boundary. It still uses the local integral cusp-specialization theorem. The third proof in theorem 60.21 replaces both that specialization and the cusp Chern-class bounding-chain evaluation by logarithmic $\mathcal{O}^*$ factors, the argument principle, and the exact Ψ-equivariant support function. Thus the purely analytic strengthening is now complete; the three proofs have genuinely different dependency sets.

60.8 Abstract conductor theorem for the candidate fibre

Theorem 60.24 (the candidate conductor mechanism, with all hypotheses)**.** ***[VERIFIED]*** *Assume:*

(H1) *X is a connected compact smooth complex threefold and $f : X \to B$ is a nonconstant holomorphic map to a smooth compact curve;*

(H2) *$W = f^{-1}(p_0)$ is the full scheme-theoretic fibre, is reduced, and has the normal-crossings local models (5.3);*

(H3) *the normalization $\eta : \widetilde{W} \to W$ is the degree-six del Pezzo surface dP_6;*

(H4) *$H^2(X;\mathbb{Z}) = 0$.*

Then, for every $L \in \mathrm{Pic}(X)$,

$$H^2(W, (T_X \otimes L)|_W) \neq 0 \quad \textit{and} \quad R^2 f_*(T_X \otimes L) \neq 0. \tag{5.18}$$

Proof. Fix L and put

$$\mathcal{F}_L = T_X \otimes L, \qquad A_L = (L^\vee \otimes \omega_X)|_W.$$

Hypothesis (H4) gives $c_1(L^\vee \otimes \omega_X) = 0$. Pullback therefore gives $c_1(\eta^* A_L) = 0$. The surface dP_6 is the blow-up of $\mathbb{P}^2$ at three noncollinear points; the standard blow-up calculation gives $H^1(\mathrm{dP}_6, \mathcal{O}) = 0$ [18, Chapter V]. In the exponential sequence, the kernel of

$$c_1 : \mathrm{Pic}(\mathrm{dP}_6) \longrightarrow H^2(\mathrm{dP}_6; \mathbb{Z})$$

is the image of $H^1(\mathrm{dP}_6, \mathcal{O})$ and is therefore zero. Consequently

$$\eta^* A_L \simeq \mathcal{O}_{\widetilde{W}}. \tag{5.19}$$

Theorem 60.5 now gives

$$H^0(W, \Omega^1_X|_W \otimes A_L) \neq 0. \tag{5.20}$$

Because $W = f^*(p_0)$, one has $\mathcal{O}_X(W) = f^*\mathcal{O}_B(p_0)$, whose restriction to the fibre is trivial. Adjunction therefore gives $\omega_W \simeq \omega_X|_W$. Apply theorem 60.7 to $E = \mathcal{F}_L$:

$$\begin{aligned} H^2(W, \mathcal{F}_L|_W)^\vee &\simeq H^0(W, (\mathcal{F}_L|_W)^\vee \otimes \omega_W) \\ &= H^0(W, \Omega^1_X|_W \otimes A_L). \end{aligned}$$

The right side is nonzero by (5.20), proving the first assertion of (5.18). The same duality theorem gives $H^3(W, \mathcal{F}_L|_W) = 0$.

Since X is compact, f is proper. By theorem 60.9, f is flat and the vector bundle $\mathcal{F}_L$ is f-flat. Thus every hypothesis of theorem 60.10 is satisfied at p_0, and

$$(R^2 f_* \mathcal{F}_L)_{p_0} \otimes \mathbb{C}(p_0) \simeq H^2(W, \mathcal{F}_L|_W) \neq 0.$$

The stalk, and hence the coherent direct-image sheaf, is nonzero. This is the second assertion of (5.18). □

Corollary 60.25 (application to the constructed threefold)**.** ***[VERIFIED]*** *For the global construction of theorem 10.11 and every $L \in \mathrm{Pic}(X)$,*

$$H^2(W, (T_X \otimes L)|_W) \neq 0, \qquad R^2 f_*(T_X \otimes L) \neq 0.$$

Proof. The cusp construction gives the compact full Cartier fibre, its dP_6 normalization, and the local trivialization of ω_X. By the local canonical trivialization and theorem 60.18, $\eta^* A_L$ is trivial for every L. The conductor proposition then gives $H^0(W, \Omega^1_X|_W \otimes A_L) \neq 0$; absolute Cartier-fibre duality turns this into the displayed H^2 nonvanishing, and the uniform top-degree base-change isomorphism turns it into a nonzero R^2 stalk. This is exactly the chain proved in theorem 60.22, now with the all-Pic restriction supplied directly by theorem 60.18. The global Leray-edge proof in theorem 60.14 is an independent certificate of the same restriction statement. □

60.9 A second printed issue: abelianization of split extensions

Lemma 60.26 (abelianization of a semidirect product)**.** ***[VERIFIED]*** *If* $1 \to K \to G \to Q \to 1$ *is split, then*

$$G^{\mathrm{ab}} \simeq Q^{\mathrm{ab}} \oplus (K^{\mathrm{ab}})_Q, \tag{5.21}$$

where the subscript denotes coinvariants for the conjugation action of Q*.*

Proof. Choose a splitting and write $G = K \rtimes Q$. Let $N \triangleleft K$ be generated by $[K, K]$ and by all elements $qkq^{-1}k^{-1}$ with $q \in Q$, $k \in K$. It is normal in G: $[K, K]$ makes it stable under conjugation by K, while conjugation by Q permutes the displayed cross-commutators. By construction $K/N = (K^{\mathrm{ab}})_Q$, and Q acts trivially on this quotient. Hence

$$G/N \simeq (K^{\mathrm{ab}})_Q \times Q.$$

Since $N \subseteq [G, G]$, abelianizing the last display gives (5.21). □

The sentence in [4, Lemma 3.2, pp. 84–85] that passes from a split fundamental-group extension to a direct sum with $H_1(F; \mathbb{Z})$ is therefore valid only when monodromy acts trivially after abelianization. A minimal counterexample retaining a free quotient group is

$$G = \mathbb{Z} \rtimes_{-1} \mathbb{Z}.$$

The extension splits, but $G^{\mathrm{ab}} \simeq \mathbb{Z} \oplus \mathbb{Z}/2$, whereas the printed direct-product formula gives $\mathbb{Z}^2$. For the candidate's printed monodromy, theorem 5.3 gives

$$H^1(F; \mathbb{Q})^{\pi_1(B^\circ)} = \mathbb{Q}\gamma,$$

so the rational coinvariant rank is one, not $b_1(F) = 4$.

If C° is a sphere punctured at s points, the correct rank formula is

$$b_1(X^\circ) = s - 1 + \dim_{\mathbb{Q}} H_1(F; \mathbb{Q})_{\pi_1(C^\circ)}. \tag{5.22}$$

Thus the proof of CDP98 Theorem 3.1 reaches only the coinvariant rank unless an additional theorem proves trivial monodromy. CDP20 Lemma 4.2 repeats the direct-sum formula, and CDP20 Corollary 5.2 imports CDP98 Theorem 3.1. Repairing the local-freeness issue discussed there does not repair this distinct monodromy error. The direct-product step in the torus branch of the CDP20 proof is therefore refuted, independently of the singular-fibre defect above.

60.10 The corrected theorem that survives both defects

The original hypotheses do not force the CDP conclusion: the constructed threefold above is a counterexample. There is nevertheless a sharp useful replacement. It isolates the exact extra global vanishing needed by the intact part of the argument, and the following fibrewise criterion replaces the invalid p. 692 reduction.

Lemma 60.27 (generic endpoint vanishing)**.** ***[VERIFIED]*** *Let* Y *be a connected compact complex manifold, let* D *be a nonzero effective Cartier divisor, and suppose* $\mathcal{O}_Y(D) \in \mathrm{Pic}^0(Y)$*. For every holomorphic vector bundle* B*, a general* $P \in \mathrm{Pic}^0(Y)$ *satisfies*

$$H^0(Y, B \otimes P) = 0, \qquad H^{\dim Y}(Y, B \otimes P) = 0. \tag{5.23}$$

Proof. The inclusions $\mathcal{O}_Y(-(k+1)D) \hookrightarrow \mathcal{O}_Y(-kD)$ give a descending chain

$$V_k = H^0(Y, B \otimes \mathcal{O}_Y(-kD)) \subseteq H^0(Y, B).$$

It stabilizes because the last vector space is finite-dimensional. A section in the stable value belongs to every V_k. In a local equation $z = 0$ for D, every coefficient of that section is divisible by z^k for all k, and hence is zero. The section vanishes on a neighbourhood of D, and therefore on connected Y by analytic continuation. Thus $V_k = 0$ for some k. Since $\mathcal{O}_Y(-kD) \in \mathrm{Pic}^0(Y)$, upper semicontinuity in a local universal family of topologically trivial line bundles shows that

$$\{P : h^0(Y, B \otimes P) > 0\}$$

is a proper analytic subset of $\mathrm{Pic}^0(Y)$. This proves the first vanishing for general P. Serre duality identifies the dual of the second group in (5.23) with

$$H^0(Y, B^\vee \otimes \omega_Y \otimes P^\vee).$$

Apply the first part to $B^\vee \otimes \omega_Y$ and use that inversion is an automorphism of $\mathrm{Pic}^0(Y)$. □

Theorem 60.28 (corrected CDP global-vanishing criterion)**.** ***[VERIFIED]*** *Let X be a compact connected complex threefold such that*

$$H^1(X;\mathbb{Z}) = H^2(X;\mathbb{Z}) = 0, \qquad a(X) = 1,$$

and suppose its algebraic reduction is a holomorphic map $f : X \to C$. If there exists $L_0 \in \mathrm{Pic}(X)$ with

$$H^2(X, T_X \otimes L_0) = 0, \tag{5.24}$$

then, for M in a dense open subset of $\mathrm{Pic}(X)$,

$$H^i(X, T_X \otimes M) = 0 \quad (i \neq 1), \tag{5.25}$$

$$\chi(X, T_X \otimes M) \leq 0, \qquad \langle c_3(T_X), [X] \rangle \leq 0. \tag{5.26}$$

Proof. The exponential sequence and the two integral vanishings give

$$\mathrm{Pic}(X) = \mathrm{Pic}^0(X) \simeq H^1(X, \mathcal{O}_X). \tag{5.27}$$

Thus this Picard group is a connected irreducible complex vector group. Any fibre divisor D of f is nonzero and effective, and its line bundle lies in $\mathrm{Pic}^0(X)$ by $H^2(X;\mathbb{Z}) = 0$. Applying theorem 60.27 to $B = T_X$ shows that the degree-zero and degree-three groups in (5.25) vanish away from two proper analytic subsets of $\mathrm{Pic}(X)$.

The cohomology-jump locus

$$\{M \in \mathrm{Pic}(X) : h^2(X, T_X \otimes M) \geq 1\}$$

is analytic by upper semicontinuity. It is proper because (5.24) gives a point outside it. Its complement therefore meets the two endpoint vanishing loci in a dense open set. This proves (5.25), and on that set

$$\chi(X, T_X \otimes M) = -h^1(X, T_X \otimes M) \leq 0. \tag{5.28}$$

Every $M \in \mathrm{Pic}(X)$ has rationally trivial first Chern class. The hypotheses also give $c_1(T_X) = 0$ in rational cohomology and, by Poincaré duality, $H^4(X;\mathbb{Q}) = 0$, so $c_2(T_X) = 0$ rationally. The degree-six term of $\mathrm{ch}(T_X \otimes M)\,\mathrm{td}(X)$ is consequently $\frac{1}{2}c_3(T_X)$. Hirzebruch–Riemann–Roch turns (5.28) into

$$\chi(X, T_X \otimes M) = \frac{1}{2}\,\langle c_3(T_X), [X] \rangle \leq 0,$$

which proves (5.26). □

Corollary 60.29 (intact relative criterion)**.** ***[VERIFIED]*** *In the setting of theorem 60.28, it is enough that one line bundle $L \in \mathrm{Pic}(X)$ satisfy*

$$R^2 f_*(T_X \otimes L) = 0. \tag{5.29}$$

This is the valid content of the sufficient criterion printed as [3, Proposition 2.4, p. 680].

Proof. Let A be a positive-degree line bundle on the compact Riemann surface C, and put $E = T_X \otimes L$. Projection formula gives

$$R^2 f_*(E \otimes f^* A^m) = R^2 f_* E \otimes A^m = 0.$$

For $m \gg 0$, Serre vanishing on C gives

$$H^1(C, R^1 f_* E \otimes A^m) = 0.$$

The Leray spectral sequence has only columns zero and one, so its total-degree-two terms yield

$$H^2(X, E \otimes f^* A^m) = 0.$$

Thus $L_0 = L \otimes f^* A^m$ satisfies (5.24), and theorem 60.28 applies. □

Theorem 60.30 (exact reduced-fibre repair of the p. 692 step)**.** ***[VERIFIED]*** *Retain the setting of theorem 60.28, assume every full scheme-theoretic fibre $W_c = f^{-1}(c)$ is reduced, and fix $L \in \mathrm{Pic}(X)$. Put*

$$A_c = (\omega_X \otimes L^\vee)|_{W_c}, \qquad \mathscr{T}_c = \ker\Big(\Omega^1_{W_c} \longrightarrow j_* \Omega^1_{(W_c)_{\mathrm{reg}}}\Big).$$

For each c, let

$$\delta_c : H^0(W_c, \mathscr{T}_c \otimes A_c) \longrightarrow H^1(W_c, N^*_{W_c/X} \otimes A_c) \tag{5.30}$$

be the connecting map from the missing cotangent filtration. If, for every c,

$$H^0(W_c, \widetilde{\Omega}^1_{W_c} \otimes A_c) = 0, \qquad H^0(W_c, N^*_{W_c/X} \otimes A_c) = 0, \qquad \delta_c \text{ is injective}, \tag{5.31}$$

then (5.29), (5.25), and (5.26) hold.

Proof. Apply theorem 60.2 to the whole reduced Cartier fibre W_c, not separately to its components. Conditions (5.31) give

$$H^0(W_c, \Omega^1_X|_{W_c} \otimes A_c) = \ker \delta_c = 0. \tag{5.32}$$

The map f is flat: a local parameter at c pulls back to a non-zero-divisor in every regular local ring of the connected manifold X, and torsion-freeness over the base DVR is flatness. Moreover, the normal line of the full fibre is the pullback of the one-dimensional vector space $T_c C$, hence is trivial. Adjunction therefore identifies $\omega_{W_c} \simeq \omega_X|_{W_c}$. With $E = T_X \otimes L$, absolute Cartier-fibre duality now gives

$$H^2(W_c, E|_{W_c})^\vee \simeq H^0(W_c, \Omega^1_X|_{W_c} \otimes A_c) = 0.$$

The uniform top-degree base-change isomorphism of theorem 60.11 therefore gives

$$(R^2 f_* E)_c \otimes_{\mathcal{O}_{C,c}} \mathbb{C}(c) = 0$$

for every c. Nakayama's lemma kills every stalk of the coherent sheaf $R^2 f_* E$, proving (5.29). The remaining conclusions follow from theorem 60.29. □

Remark 60.31 (scope and sharpness of the repair). Under the two endpoint vanishings in (5.31), injectivity of δ_c is equivalent to the desired ambient vanishing (5.32); it is not a politeness assumption hiding the original claim. For a reducible reduced fibre the map must be computed on the whole Cartier fibre. A nonreduced fibre requires additional nilpotent-thickening layers and is not covered by theorem 60.30. The global criterion theorem 60.28 and the relative criterion theorem 60.29 do allow nonreduced fibres.

The reconstructed candidate violates both added conditions rather than being silently excluded by terminology. Since $c_3(X) = 2$, the contrapositive of theorem 60.28 proves

$$H^2(X, T_X \otimes L_0) \neq 0 \quad \text{for every } L_0 \in \mathrm{Pic}(X).$$

Independently, theorem 60.25 proves

$$R^2 f_*(T_X \otimes L) \neq 0 \quad \text{for every } L \in \mathrm{Pic}(X),$$

and its cusp conductor supplies a nonzero element of the relevant $\ker \delta_c$. The first conclusion rules out the global hypothesis; the second independently rules out the stronger relative sufficient condition. Thus the corrected theorem is compatible with, and precisely separates, the verified counterexample to CDP20 Theorem 2.2 as stated.

The exact monodromy repair. The split-extension argument can reach only the invariant integral lattice $H^1(F;\mathbb{Z})^{\pi_1(C^\circ)}$. Its final complex-linear inference is valid precisely when

$$\mathrm{Span}_{\mathbb{C}}\,\mathrm{im}\Big(H^1(F;\mathbb{Z})^{\pi_1(C^\circ)} \longrightarrow H^1(F,\mathcal{O}_F)\Big) = H^1(F,\mathcal{O}_F). \tag{5.33}$$

Trivial monodromy on $H_1(F;\mathbb{Z})$ is a stronger sufficient hypothesis. Neither (5.33) nor trivial monodromy follows from the hypotheses of CDP20 Theorem 2.2. The corrected theorem above bypasses this rank route entirely; it does not relabel the false direct-sum formula as proved.

60.11 Exact present disposition

1. **[REFUTED]** The p. 692 reduction in [3] is not a formal consequence of the two cotangent sequences. Theorem 60.6 is a proper geometric counterexample to that exact sheaf-level inference for a non-torsion bundle on the central fibre; no extension of that bundle to the displayed total space is claimed.

2. **[REFUTED]** The direct-product homology formula in [4, Lemma 3.2] omits monodromy coinvariants. This leaves the torus branch imported by [3] without its printed rank argument.

3. **[VERIFIED]** The normalization–conductor construction, Cartier-fibre duality, flatness over the base curve, and top-degree analytic base change are proved above under their stated hypotheses.

4. **[VERIFIED]** The period-family descent, toric and finite fillings, global gluing, and integral analytic nearby-cycle/Leray bridges are proved in sections 10 and 53. Therefore the candidate's hypotheses (H1)–(H4) hold and $R^2 f_*(T_X \otimes L) \neq 0$ for every L by theorem 60.25.

5. **[VERIFIED]** Those same construction and topology results, together with theorem 54.4, hold for the constructed compact complex threefold. Hence $H^1(X;\mathbb{Z}) = H^2(X;\mathbb{Z}) = 0$, $a(X) = 1$, f is a holomorphic algebraic reduction, and $c_3(X) = 2$. The resulting X satisfies every stated hypothesis of CDP Theorem 2.2 and contradicts its conclusion; it is therefore a counterexample to that theorem as stated. This theorem-level conclusion uses the complete construction and topology proofs; it is not inferred from the two printed proof defects alone.

6. **[VERIFIED]** The corrected global criterion theorem 60.28 bypasses both refuted dependencies: existence of one L_0 with $H^2(X, T_X \otimes L_0) = 0$ implies generic off-degree-one vanishing and $c_3(X) \leq 0$. The intact relative sufficient condition is $R^2 f_*(T_X \otimes L) = 0$, and theorem 60.30 replaces the invalid reduced-fibre step by the exact injectivity condition on δ_c. The original theorem without these added hypotheses remains refuted by the constructed threefold. Independent specialist replication remains desirable as external adjudication, but it is neither a mathematical hypothesis nor a gate on the present proof program.

Validation, repair, or disproof test 60.32 (decisive theorem-level outcomes)**.** The current workbench records two exact defects in the printed CDP chain, a completed route from the constructed threefold to a theorem-level counterexample, and the strongest corrected conclusion justified by the intact argument. The global vanishing in theorem 60.28 is false for every L_0 by contraposition from $c_3(X) = 2$. Independently, the cusp-conductor calculation forces $R^2 f_*(T_X \otimes L) \neq 0$ for every L, so the stronger relative sufficient condition fails as well. Thus the repair does not contradict the counterexample or recover the original theorem by renaming its conclusion. Any further attempt to prove the original statement must falsify a named step in the candidate reconstruction or derive one of the new vanishing hypotheses from genuinely stronger geometry. Merely invoking the published theorem does neither.

61 Confirmed local repairs kept separate from the theorem

These corrections are mathematically definite but do not settle the central construction.

1. **[VERIFIED]** Proposition 4.7(iv)'s closed-image conclusion is correct, although its printed justification is not. If $s : B \to E$ is the section of $f : E \to B$, then

$$s(B) = \{x \in E : s(f(x)) = x\},$$

the equalizer of id_E and $s \circ f$. Since E is Hausdorff, its diagonal is closed, so this equalizer is closed. Thus the explicit holomorphic section, its cusp disc, and the null-homotopy used in theorems 53.3 and 53.6 require no additional chartwise or local-finiteness hypothesis.

2. **[REFUTED]** Remark 7.28 is false for a fixed triple. For example,

$$(\ell_0, \ell_1, \ell_2) = (100, 1, -1), \qquad p = 12(100) - 4(1) - 3(-1) = 1199 \neq \pm 1.$$

The intended statement must quantify existentially over ℓ_0 for fixed admissible (ℓ_1, ℓ_2).

3. **[VERIFIED]** Appendix Lemma A.3 prints cohomological arrows while labelling them as homology. The sequence must be written

$$H^k(W) \longrightarrow H^k(\widetilde{W}) \oplus H^k(D) \longrightarrow H^k(\widetilde{D}) \longrightarrow H^{k+1}(W),$$

followed by universal coefficients when homology ranks are required.

4. **[VERIFIED]** Appendix Remark A.6 must read $\pi_1(N_0) = \bar{\Lambda} = \pi_1(W)$, not $\pi_1(N_0) = \Lambda$.

5. **[VERIFIED]** The modular assembler has a page-71 portability defect under the tested MiKTEX route; the canonical monolithic source renders correctly. This is a build defect, not a mathematical counterexample.

6. **[REFUTED]** The manuscript's assertion that $\Omega^1_{X/B}$ has torsion at all three singular fibres is false literally. The explicit $t = xy$ and $t = xyz$ modules show that it is torsion-free but non-locally-free at the cusp; torsion occurs only at the two multiple fibres.

7. **[VERIFIED]** In the finite-filling Hodge-line computation, the invariant canonical section

$$\sigma_j = s_j^{m_j-1-a_j} e^{\varphi(s_j)} ds_j \wedge e$$

has divisor $(m_j - 1 - a_j)S_j$ after descent and therefore is not a nowhere-vanishing frame when $j = 2$. The correct direct-image statement is the invariant-module identity

$$f_{j*}\omega_{N_j/D_j} = \mathcal{O}_{D_j} \cdot s_j^{m_j-1-a_j} \vartheta'_j, \qquad \vartheta'_j = e^{\varphi(s_j)}(ds_j \wedge e) \otimes (dt_j)^{-1}.$$

On the punctured cover its generator is a unit times $s_j^{-a_j} e$. This repairs the local wording while leaving $f_*\omega_{X/B} \simeq \mathcal{O}_{\mathbb{P}^1}(1)$, the structure-sheaf direct images, and uniform non-torsion unchanged.

62 Tool and computational resource disclosure

The source manuscript available for this audit was reconstructed from a PDF in ten consecutive source strips—eight strips of eleven pages and two of ten pages—in ChatGPT Professional-mode sessions and then consolidated into a single LaTeX source. Two separate ChatGPT Professional-mode sessions produced the first full-manuscript proof audits. The present consolidation, citation audit, exact-arithmetic reproduction, and validate/repair/disprove workbench were prepared with Codex using ChatGPT 5.6 Sol in Ultra mode. The exact computations reported here use integer and rational arithmetic; their mathematical certificates are reproduced in the body rather than delegated to software output.

These systems are disclosed as tools, not treated as mathematical authorities or authors. Every imported claim is checked against a publicly resolvable human source, and every new assertion remains subject to human verification. This disclosure and the point-of-use citation policy follow the recommendations of the Leiden Declaration on Artificial Intelligence and Mathematics [2]. Responsibility for any version submitted or published, including the adequacy of its proofs and references, belongs to the human author or editor who releases that version.

The explicit candidate and its original theorem claims remain attributed to the reported Claude/Fabel source. The purpose of the present work is narrower but substantive: to make the construction's load-bearing maps and calculations explicit, reproduce them independently where possible, repair identified proof gaps when the geometry permits, and isolate the exact boundary of what still requires outside adjudication. This is offered as a verification aid that may save expert reconstruction time; it is not a claim of priority for the candidate, a transfer of authorship, or a substitute for specialist review.

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